Software Requirements Specification

Inventory Management System (IMS)

Version 1.0 | July 7, 2026 | SRS-IMS-MASTER-001

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Software Requirements Specification (SRS)

Inventory Management System (IMS)

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Document ID	SRS-IMS-MASTER-001
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Document Purpose

This master Software Requirements Specification consolidates the complete IMS requirements package into a single authoritative document. It covers business rules, use cases, domain model, system architecture, database design, REST API specifications, UI/UX specifications, and frontend architecture mapping.

Scope

The Inventory Management System is an enterprise web application providing authoritative inventory truth across a multi-location network — covering product master data, procurement, inbound receiving, inventory control, order fulfillment, transfers, returns, reporting, governance, and administration.

Delivery Phasing

Phase	Timeline	Focus
MVP (Phase 1)	Months 0–6	Core visibility, movement recording, fulfillment support, audit
Phase 2	Months 6–12	Network maturity, procurement automation, commercial campaigns
Phase 3	Months 12–18	Finance alignment, executive analytics, advanced integrations

Package Contents

Part	Content
I	Introduction & System Context
II	Business Rules & Constraints
III	Use Cases (48 use cases)
IV	Domain Model (13 bounded contexts)
V	Event Storming
VI	System Architecture
VII	Database Design
VIII	API Design (174 REST endpoints)
IX	UI/UX Specification (50 screens, 60 components)

Part	Content
X	Frontend Architecture

Traceability Summary

Artifact	Count
Business Rules	32
Use Cases	48
Bounded Contexts	13
Domain Events	81
REST APIs	174
UI Screens	UI-001 to UI-050
UI Components	C-001 to C-060
User Roles	19

Part I — Introduction & System Context

Architecture Vision

Inventory Management System

Document Information

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Scope	High-level architectural direction only — no module design, API specification, database schema, or technology selection

Source Documents

This Architecture Vision is derived from the following approved and reviewed artifacts:

Artifact	Document ID	Role in This Vision
Business Requirements Document	BRD-IMS-001 (v1.0)	Business intent, objectives, scope, constraints, phased vision

Artifact	Document ID	Role in This Vision
Bounded Contexts	SRS-DM-007	Strategic domain decomposition and context relationships
Domain Model Review	SRS-DM-009	Pre-architecture gate findings and core domain identification
Domain Aggregates, Events, Services	SRS-DM-003, SRS-DM-005, SRS-DM-006	Consistency boundaries and cross-context coordination patterns
Event Storming Board & Hotspots	SRS-ES-008, SRS-ES-007	End-to-end lifecycles and unresolved architectural uncertainties
Non-Functional Requirements Suite	NFR-PERF, NFR-SCAL, NFR-AVL/REL, NFR-SEC, NFR-LOG, NFR-MON, NFR-BDR, NFR-COM (204 requirements)	Quality attribute targets and design constraints
NFR Review Report	NFR Review (2026-07-03)	Cross-NFR conflicts, gaps, and design-readiness gate

Document Purpose

This document establishes the **architectural direction** for the Inventory Management System — the principles, drivers, quality priorities, risks, and recommended style that will guide subsequent solution architecture, technology decisions, and detailed design. It answers *why* and *what kind of system* we are building at a strategic level, not *how each component is built*.

Intended Audience

Audience	Primary Interest
Steering Committee / Sponsor	Investment alignment, risk posture, phased delivery rationale
Product Owner	Scope phasing and architectural trade-offs affecting roadmap
Solution & DDD Architects	Design principles, context boundaries, integration patterns
Delivery Leads	Team structure implications, decomposition readiness
Security / DevOps / QA	Quality attribute priorities and NFR-driven constraints

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1. Architectural Goals

Architectural goals translate business intent into enduring design principles. They guide trade-off decisions throughout the 18-month initiative and beyond.

AG-01 — Single Authoritative Source of Inventory Truth

Establish **Inventory Balance** as the system's quantity authority — one coherent answer to *what do we have, where is it, and what is available?* — consumable by all operational and management stakeholders without manual reconciliation across locations or parallel tools.

Grounding: BRD Executive Summary; BO-01, BO-03, BO-22; SRS-DM-009 (core domain identification); 13 bounded contexts with Inventory Balance at center of context map.

AG-02 — Domain-Aligned Modularity

Organize the solution around **13 bounded contexts** identified in strategic domain design, preserving distinct ubiquitous languages, ownership boundaries, and change velocities — so catalog governance, warehouse execution, and financial controls can evolve independently without corrupting quantity invariants.

Grounding: SRS-DM-007; independence principles (distinct language, separate ownership, autonomous consistency).

AG-03 — Governed Correctness Over Raw Speed

Enforce business rules — ATP commitment, put-away before availability, segregation of duties, approval thresholds, audit trail — as **architectural invariants**, not optional validations. Operational throughput must not come at the cost of ungoverned stock mutations.

Grounding: 32 business rules (BR-001 – BR-032); 20 SoD rules; NFR-COM-008, NFR-COM-009, NFR-SEC-007; event storming policies (POL-xxx).

AG-04 — Phased Delivery With Credible Control

Deliver **measurable business control in Phase 1 (Months 0–6)** — core visibility, movement recording, fulfillment support, and audit — before pursuing network maturity, commercial campaigns, and financial alignment in later phases. Architecture must support incremental location rollout without operational shutdown (OC-01).

Grounding: BRD phased vision; MVP functional requirements; MoSCoW prioritization; ~122 Must Have NFRs.

AG-05 — Event-Transparent Operations

Make every material inventory state change **observable, attributable, and reconstructable** through domain events and audit projections — enabling movement history, KPI dashboards, compliance evidence, and finance reconciliation without coupling reporting queries to transactional write paths.

Grounding: 81 domain events; Inventory Intelligence as event-consuming read context; BO-19; NFR-LOG, NFR-COM-010.

AG-06 — Integration-Ready Enterprise Capability

Position IMS as the inventory system of record while **respecting external system boundaries** — Sales/CRM for customer orders, Finance/GL for valuation and payment — through explicit anti-corruption layers rather than absorbing foreign domain models.

Grounding: BRD scope exclusions; SRS-DM-007 context map (EXT Sales/CRM, EXT Finance); HS-014 Finance System Boundary.

AG-07 — Scale Without Architectural Replacement

Design for **15% near-term and 30% medium-term growth** in orders, locations, SKUs, and users without forcing a rewrite — through capacity-aware core design, separable read workloads, and decomposition-ready module boundaries.

Grounding: NFR-SCAL-001 – NFR-SCAL-003; long-term business vision (3–5 years); NFR-SCAL-019 (no major replacement within 36 months).

AG-08 — Resolve Domain Uncertainty Before Implementation

Treat **16 open event-storming hotspots** and **6 Priority-1 domain model findings** as architecture gate items — not implementation deferrals. Structural forks (lot/batch model, freeze authority, reservation coordination) must be closed before detailed design of affected capabilities.

Grounding: SRS-ES-007; SRS-DM-009 P1 items; Domain Model Review readiness gate.

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2. Architectural Drivers

Architectural drivers are the forces that most strongly shape structural and technology decisions. They originate from domain complexity, NFR pressure, and delivery constraints.

2.1 Domain Structure Drivers

Driver ID	Driver	Architectural Implication
AD-01	**Central quantity authority with many consumers** — Stock Position is mutated by receiving, fulfillment, transfers, adjustments, returns, and campaigns	Inventory Balance context must be isolated, highly consistent, and optimized for write contention; other contexts interact via commands, domain services, and events — not shared mutable state
AD-02	**81 domain events across 12 transactional lifecycles** — event storming reveals rich cross-aggregate choreography (receipt→put-away→availability, confirm→reserve→pick→ship)	Internal **event-driven coordination** between bounded contexts; policies react to events rather than synchronous cross-module calls for non-immediate outcomes
AD-03	**13 bounded contexts with heterogeneous change velocity** — catalog changes quarterly; balance changes continuously; governance changes on audit cycles	**Modular boundaries** aligned to contexts; avoid monolithic data model that couples master data to high-frequency transactions
AD-04	**24 domain services spanning aggregates** — availability confirmation, reservation coordination, freeze guard, network availability evaluation	Cross-context **domain services and sagas** for workflows that cannot complete within a single aggregate transaction
AD-05	**Inventory Intelligence as read-only projection context** — dashboards, KPIs, movement history, audit trail, valuation snapshots	**CQRS separation** — transactional write models distinct from read-optimized projections fed by domain events
AD-06	**Governance and SoD as first-class concerns** — counter ≠ approver; adjustment thresholds; reason codes mandatory	**Policy enforcement layer** consulting Inventory Governance and Identity & Access before operational mutations; not ad hoc per-screen checks
AD-07	**External system boundaries** — customer orders from Sales/CRM; financial authority potentially in ERP	**Anti-corruption layers** at Order Fulfillment and Inventory Intelligence boundaries; published interchange language (SKU, Location Code, User Identifier)

2.2 Non-Functional Drivers

Driver ID	Driver	Architectural Implication
AD-08	**Sub-3-second operational response** (NFR-PERF-001) and **sub-2-second ATP lookup** (NFR-PERF-002)	Hot-path read and write paths must minimize cross-context synchronous chains; ATP queries served from optimized read path or balance context with minimal fan-out
AD-09	**99.5% business-hours uptime** for essential functions (NFR-AVL-001); peak 99.7% (NFR-AVL-003)	Resilience patterns for core transactional paths; graceful degradation for reporting; no single non-critical dependency blocking dock operations
AD-10	**≤15-minute recovery point** for transactions (NFR-BDR-001, NFR-REL-007)	Durable event and transaction log; backup architecture aligned to stock balance and open reservations as critical data
AD-11	**204 NFRs with ~122 Must Have** — security, logging, monitoring, compliance intertwined	**Cross-cutting platform capabilities** (auth, audit, observability) as shared infrastructure serving all contexts — not reimplemented per module
AD-12	**7-year retention and immutable audit trail** (NFR-COM-006, NFR-COM-010, NFR-COM-020)	Tiered data architecture — hot operational store, warm audit store, cold archive — influencing projection and logging design
AD-13	**Sequential multi-location rollout** (NFR-SCAL-007, OC-01)	Location-scoped configuration and phased feature enablement; architecture supports per-location go-live without network-wide cutover
AD-14	**Concurrent reservation contention on high-velocity SKUs** (HS-007, NFR-PERF-007)	Optimistic or serialized reservation strategy on Stock Position; explicit saga compensation when order confirmation and reservation diverge

2.3 Delivery and Constraint Drivers

Driver ID	Driver	Architectural Implication
AD-15	**Fixed initiative budget** (BC-01)	Favor **modular monolith or minimal service count** for MVP; defer distributed-system complexity until proven scale or team capacity demands it
AD-16	**Limited internal IT capacity** (ORC-04)	Reduce operational surface area — fewer deployable units, standardized observability, managed platform services where feasible

Driver ID	Driver	Architectural Implication
AD-17	**No offline MVP** (TEC-03)	Online-first design; warehouse mobility via connected clients; no sync-engine architecture in Phase 1
AD-18	**No operational shutdown for go-live** (OC-01)	Parallel run, opening balance certification (HS-002), and rollback-capable deployment; architecture supports data migration and cutover modes
AD-19	**Open hotspots blocking MVP rules** — ad-hoc receipt (HS-001), opening balance (HS-002), lot/batch model (HS-005)	Architecture sign-off gated on hotspot resolution workshops; assumptions documented where Phase 2 deferral is accepted

3. Business Drivers

Business drivers articulate *why* the organization is investing and what outcomes architecture must enable. They trace directly to BRD objectives and problem statements.

3.1 Strategic Business Drivers

Driver ID	Business Driver	Current Pain	Architectural Expectation
BD-01	**Inventory as strategic balance-sheet asset** — working capital, margin, and resilience	~78% record accuracy; ~8% valuation variance vs. GL	Authoritative quantity and movement history enabling financial reconciliation (Phase 3)
BD-02	**Growth has outpaced control** — expanded SKUs, channels, locations	Fragmented visibility; tribal knowledge; parallel spreadsheets	Network-wide single source of truth scalable to additional locations and channels
BD-03	**Customer trust through reliable fulfillment**	~84% fill rate; sales commits without ATP verification	Real-time ATP with reservation discipline (BR-001, BR-002); sub-2-second availability inquiry
BD-04	**Operational excellence as market differentiator**	Inconsistent processes across locations; reactive replenishment	Standardized movement recording (BO-03); proactive alerts; network rebalancing capability
BD-05	**Accountability and audit readiness**	~40% audit trail completeness; weak shrinkage governance	100% attributable transactions (BO-19); immutable audit projections; SoD enforcement
BD-06	**18-month credible control horizon**	Manual reconciliation consuming operations capacity	Phased architecture delivering P1 objectives by Month 6, operational performance by Month 12, decision intelligence by Month 18

3.2 Operational Business Drivers

Driver ID	Business Driver	Key Processes	Phase
BD-07	**Establish control at first locations**	BP-01, BP-02, BP-07–BP-09, BP-10, BP-16–BP-18	Phase 1 (MVP)
BD-08	**Improve fulfillment and replenishment performance**	BP-17–BP-19, BP-21–BP-22, BP-24–BP-26	Phase 2
BD-09	**Enable management and financial alignment**	BP-29–BP-30, BP-32	Phase 3
BD-10	**Support promotional and commercial readiness**	BP-31	Phase 2
BD-11	**Retire parallel manual tracking**	Cross-cutting	Phase 1 per location (BO-22)
BD-12	**Multi-location B2B and B2C service**	BP-16, BP-48 (multi-location split fulfillment)	Phase 1 foundation; Phase 2 maturity

3.3 Stakeholder-Driven Priorities

Stakeholder Group	Primary Need	Architectural Response
Warehouse / Receiving	Fast, reliable transaction recording at dock and pick floor	Low-latency write path; clear confirmation; mobile-ready operational UI layer

Stakeholder Group	Primary Need	Architectural Response
Customer Service / Sales	Immediate ATP answers without warehouse escalation	Optimized availability read model; self-service inquiry (BO-16)
Purchasing	Network visibility before buying; PO-to-receipt accountability	Procurement & Inbound contexts with event handoff; network availability evaluation (BR-011)
Finance / Audit	Traceable adjustments; valuation reconciliation; SoD evidence	Event-sourced audit projections; governance workflow; ERP ACL (HS-014)
Operations Leadership	Dashboards, KPIs, exception alerts	Inventory Intelligence CQRS projections; monitoring integration (NFR-MON)
Executive / Sponsor	Predictable ROI; controlled risk; phased value	MVP-first modular delivery; measurable KPI gates per phase

4. Quality Attributes Influencing the Design

Quality attributes are derived from the 204 NFRs and BRD success metrics. They are ranked by **design influence** — how strongly they constrain architectural shape.

4.1 Quality Attribute Priority Matrix

Priority	Quality Attribute	Key Requirements	Design Influence
1 — Critical	**Data Integrity & Consistency**	NFR-REL-006–010; BR-001, BR-002, BR-005; DE-011–DE-022	Strong consistency within Stock Position aggregate; eventual consistency across contexts with explicit compensation; no dual writers of ATP
2 — Critical	**Security & Compliance**	NFR-SEC (32); NFR-COM (22); FR-081–FR-090	Centralized authN/authZ; location-scoped RBAC; immutable audit; GDPR and financial retention
3 — Critical	**Availability & Recoverability**	NFR-AVL-001 (99.5%); NFR-BDR-001 (15-min RPO); NFR-REL-001–005	HA for essential functions; backup of balances and reservations; defined RTO scenarios
4 — High	**Performance (Operational)**	NFR-PERF-001 (3s screens); NFR-PERF-002 (2s ATP); NFR-PERF-003 (5s txn confirm)	Separate hot read paths; minimize cross-context sync in user-facing flows; index strategy for SKU/location lookup
5 — High	**Auditability & Traceability**	NFR-LOG (28); NFR-COM-010–014; BO-19	Append-only movement and audit projections; 7-year retention tier; 15-minute auditor export (NFR-PERF-026)
6 — High	**Scalability**	NFR-SCAL-001–003; NFR-SCAL-010–015	Horizontal scaling readiness for read tier; product-location aggregate partitioning; growth headroom without replacement
7 — Medium	**Observability**	NFR-MON (26); NFR-LOG-026	Unified metrics, alerts, and runbooks; reservation failure and ATP contention visibility
8 — Medium	**Maintainability & Evolvability**	AD-03, AD-15; bounded context boundaries	Modular codebase aligned to contexts; published integration contracts; hotspot resolution before entrenchment
9 — Medium	**Usability (Operational)**	BO-21 (90% adoption); warehouse role differentiation	Role-appropriate UX; sub-3s response to prevent manual workaround reversion
10 — Lower (Phase 2–3)	**Reporting Performance**	NFR-PERF-023–027; NFR-AVL-004	Async report generation; materialized KPI views; reporting outage must not block dock

4.2 Quality Attribute Scenarios (Architecturally Significant)

Scenario	Stimulus	Response Expectation	Primary Attributes
QA-S01	Customer service queries ATP during live call	Result within 2 seconds for single SKU	Performance, Availability
QA-S02	Two orders compete for last unit of Class A SKU	Deterministic reservation winner; no oversell	Integrity, Performance
QA-S03	Put-away completed at receiving dock	Stock becomes available within transaction SLA; event published to intelligence	Integrity, Performance, Auditability

Scenario	Stimulus	Response Expectation	Primary Attributes
QA-S04	Physical inventory count initiated	Location freeze enforced; non-count mutations blocked (BR-024)	Integrity, Availability
QA-S05	Order confirmed but reservation fails on Stock Position	Compensating action defined; no orphaned commitment (HS-007)	Integrity, Maintainability
QA-S06	Peak season — 2× normal transaction rate	Essential functions remain within response targets; uptime ≥99.7%	Performance, Scalability, Availability
QA-S07	Finance auditor requests Q2 adjustment evidence	Exportable package within 15 minutes	Auditability, Performance
QA-S08	Primary site failure mid-business-day	Recovery within defined RTO; ≤15 minutes data loss	Availability, Recoverability
QA-S09	New location go-live with imported opening balances	Certified balances before transactional activation (HS-002)	Integrity, Maintainability
QA-S10	Marketing ring-fences stock for campaign	ATP segregation honored; general orders blocked unless approved (BR-029)	Integrity, Evolvability

4.3 NFR-Driven Architectural Constraints

The NFR review identified constraints that architecture must explicitly address:

Constraint	Source	Architectural Response
Numeric conflicts between availability budgets and uptime percentages	NFR Review C-01	Harmonize SLA definitions in capacity model before build
Aggressive RPO/RTO vs. fixed budget	NFR Review; BC-01	Scenario-based DR tiers — application failure vs. site disaster
Duplicate audit trail requirements across NFR-LOG, NFR-SEC, NFR-COM	NFR Review	Single authoritative audit projection in Inventory Intelligence
Missing integration and bulk-import performance NFRs	NFR Review PERF-R-02	Add architecture capacity assumptions for FR-090 go-live import
Tiered data architecture needed	NFR Review §8.2	Hot/warm/cold storage strategy spanning operational, audit, and archive

5. Major Architectural Risks

Risks are classified by impact on architectural integrity, delivery, and business outcomes. Mitigations are architectural — not project-management — actions.

5.1 Critical Risks

Risk ID	Risk	Source	Impact	Mitigation Direction
AR-01	**Unresolved domain hotspots block correct aggregate design** — 2 Critical (HS-001, HS-002) and 5 High hotspots open	SRS-ES-007	Wrong persistence model, unenforceable BR-007/BR-010, failed MVP go-live	Architecture sign-off gated on Session 1 workshop outcomes; document assumptions for deferred items
AR-02	**Split-brain ATP ownership** — promotional ring-fence (HS-004, P1-02) and reservation coordination (HS-007) span contexts	SRS-DM-009; SRS-ES-007	Overselling, inconsistent availability, customer trust failure	Confirm event ownership: Commercial Campaign emits intent; Stock Position owns ATP mechanics; saga for order-reservation
AR-03	**Stock Position identity fork** — lot/batch scoping undecided (HS-005, P1-05)	SRS-DM-009	Fundamental data model divergence affecting reservation, pick, count, transfer	Adopt single Stock Position per product-location with lot child entities before detailed design
AR-04	**Finance system boundary ambiguity** (HS-014)	SRS-ES-007	Duplicate three-way match logic or missing financial controls	Early integration workshop; ERP ACL decision for MVP vs. Phase 2; IMS owns operational facts, Finance owns payment authority

Risk ID	Risk	Source	Impact	Mitigation Direction
AR-05	**Reservation contention and partial failure** under peak load	HS-007; NFR-PERF-007	Oversell or stale reservations; customer service incidents	Define reservation saga with compensation; prototype Class A SKU contention; reservation TTL policy (HS-016)

5.2 High Risks

Risk ID	Risk	Source	Impact	Mitigation Direction
AR-06	**Fixed budget vs. NFR ambition** — 122 Must Have NFRs including 15-min RPO, 99.5% uptime, full security suite	BC-01; NFR Review	Over-engineering or under-delivery	MVP NFR subset (~40 requirements); phased compliance program; modular monolith to reduce ops cost
AR-07	**Location freeze dual authority** (HS-003, P1-03)	SRS-DM-009	Race conditions during physical count; unauthorized mutations	Physical Count initiates freeze; Inventory Location holds flag; Freeze Guard reads Location only
AR-08	**Opening balance go-live without governing workflow** (HS-002)	SRS-ES-007	Transactions on uncertified stock; accuracy KPI failure at launch	Opening Balance Establishment workflow; certify-before-activate chain
AR-09	**Premature microservices decomposition** — 13 contexts mistaken for 13 services	SRS-DM-007 scope note	Distributed complexity exceeding ORC-04 capacity; operational failure	Context ≠ service; modular monolith first; extract only on proven need (Balance vs. Intelligence)
AR-10	**Reporting load corrupting transactional performance**	SRS-DM-007 Context 13; NFR-PERF	Dock slowdown during month-end	Strict CQRS; no reporting queries against write store
AR-11	**Channel integration lag causing oversell**	BRD risk register; TEC-04	Revenue loss; customer trust damage	ACL at fulfillment boundary; manual sync interim; ATP as integration contract
AR-12	**NFR internal conflicts undetected until UAT**	NFR Review	Rework, delayed go-live	Capacity model, harmonized SLA register, traceability matrix before build

5.3 Medium Risks

Risk ID	Risk	Source	Impact	Mitigation Direction
AR-13	**Consignment ownership trigger undefined** (HS-006)	SRS-ES-007	Incorrect valuation and ATP inclusion	Defer to Phase 2 if out of MVP scope; else formalize MR-08 trigger catalog
AR-14	**Ad-hoc receipt and three-way match conflict** (HS-001, CF-007)	Business Rule Dependencies	Audit finding; payment control gap	BR-033 exception matrix with Finance sign-off
AR-15	**Behavioral specification weakness** — entities defined without behavioral contracts	SRS-DM-009	Anemic domain model in implementation	Enforce command/event contracts per aggregate before coding
AR-16	**Geographic and retention compliance** — GDPR, 7-year financial retention	NFR-COM	Regulatory exposure	Privacy by design; tiered retention; legal hold capability (NFR-COM-021)
AR-17	**Team Conway mismatch** — context boundaries not reflected in team ownership	SRS-DM-007	Cross-team coupling, integration defects	Align squads to core contexts for MVP (Balance, Receiving, Fulfillment, Intelligence)

5.4 Risk Heat Map

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                    IMPACT →
                    Low     Medium     High     Critical
               ┌──────────┬──────────┬──────────┬──────────┐
        High   │ AR-17    │ AR-13    │ AR-06    │ AR-01    │
               │          │ AR-14    │ AR-07    │ AR-02    │
LIKELIHOOD     │          │ AR-15    │ AR-09    │ AR-03    │
               ├──────────┼──────────┼──────────┼──────────┤
        Med    │          │ AR-16    │ AR-10    │ AR-04    │
               │          │          │ AR-11    │ AR-05    │
               │          │          │ AR-12    │ AR-08    │
               └──────────┴──────────┴──────────┴──────────┘
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6. Recommended Architecture Style

6.1 Recommended Style

Domain-Driven Design (DDD) with a Modular Monolith core, internal event-driven integration, and CQRS for operational analytics — evolving toward selective service extraction as scale and organizational maturity demand.

This is a **compound architectural style**, not a single pattern. Each element addresses specific drivers:

Style Element	Purpose	Addresses
Strategic DDD (bounded contexts, context map, ubiquitous language)	Align software structure to business domains	AD-01–AD-07; AG-02
Modular Monolith (context-aligned modules, single deployable unit for MVP)	Control cost and operational complexity within BC-01 and ORC-04	AD-15, AD-16; AR-09
Hexagonal / Ports & Adapters (per context module)	Isolate domain logic from UI, persistence, and external integrations	AD-07; AG-06
Event-Driven Architecture (internal)	Choreograph cross-context reactions via domain events	AD-02; AG-05
CQRS (command/query separation for Intelligence context)	Protect write-path performance from reporting load	AD-05; AR-10
Saga / Process Manager (cross-aggregate workflows)	Coordinate reservation, transfer, and go-live with compensation	AD-04; AR-05, AR-08
Anti-Corruption Layer (external boundaries)	Protect IMS domain from Sales/CRM and Finance/ERP models	AD-07; AR-04, AR-11

6.2 Style Justification

Why Domain-Driven Design

The domain analysis produced **13 bounded contexts**, **18 aggregates**, **81 domain events**, **24 domain services**, and **72 ubiquitous language terms** — indicating high domain complexity with distinct consistency boundaries. DDD is not optional decoration; it is the most direct mapping from the approved domain model to software structure.

Inventory Balance as the **core domain** warrants the richest model and protection. Supporting contexts (Catalog, Network, Governance) are upstream suppliers. Generic subdomains (Identity & Access, cross-cutting audit) may leverage commodity solutions.

Why Modular Monolith (Phase 1) Over Microservices

Factor	Modular Monolith	Microservices
Fixed budget (BC-01)	Single deployment, one ops pipeline	Multiplied infrastructure and monitoring cost
Limited IT capacity (ORC-04)	One runtime to manage	Service mesh, distributed tracing, contract versioning overhead

Factor	Modular Monolith	Microservices
13 contexts / phased delivery	Modules can be developed in parallel within one repo	Premature network boundaries before volume proof
Stock Position consistency	In-process domain events; ACID within aggregate	Distributed transaction complexity for reservation
NFR-AVL-001 99.5% uptime	Fewer failure domains	Cascading failure risk across services

SRS-DM-007 explicitly states bounded contexts are **strategic design, not deployment**. The context map defines integration patterns (events, ACL, published language) that remain valid whether modules are co-located or distributed.

Extraction candidates (Phase 2+) if metrics justify:

- **Inventory Intelligence** — read-heavy, different scaling profile, eventual consistency acceptable
- **Identity & Access** — commodity capability, likely enterprise IdP integration
- **Integration gateway** — external Sales/CRM and Finance adapters

Why Internal Event-Driven Integration

Event storming revealed **12 end-to-end lifecycles** with policy chains (e.g., `PutAwayCompleted` → `StockReceived` → availability update → backorder fulfillment policy). Synchronous coupling between all contexts would violate performance drivers (AD-08) and entangle lifecycles.

Domain events provide:

- **Loose coupling** between context modules
- **Natural audit trail** feeding Inventory Intelligence
- **Extensibility** for Phase 2 policies (campaign readiness, replenishment recommendation) without modifying upstream aggregates

Events are **internal** in Phase 1 (in-process event bus or transactional outbox within the monolith). External message broker introduction is deferred until cross-system async integration volume requires it.

Why CQRS for Inventory Intelligence

Inventory Intelligence owns **no write aggregates** — only projections (movement history, audit trail, KPIs, alerts, valuation snapshots). Mixing these queries into Stock Position or Order Fulfillment is the "reporting database inside the transaction" anti-pattern identified in SRS-DM-007.

CQRS enables:

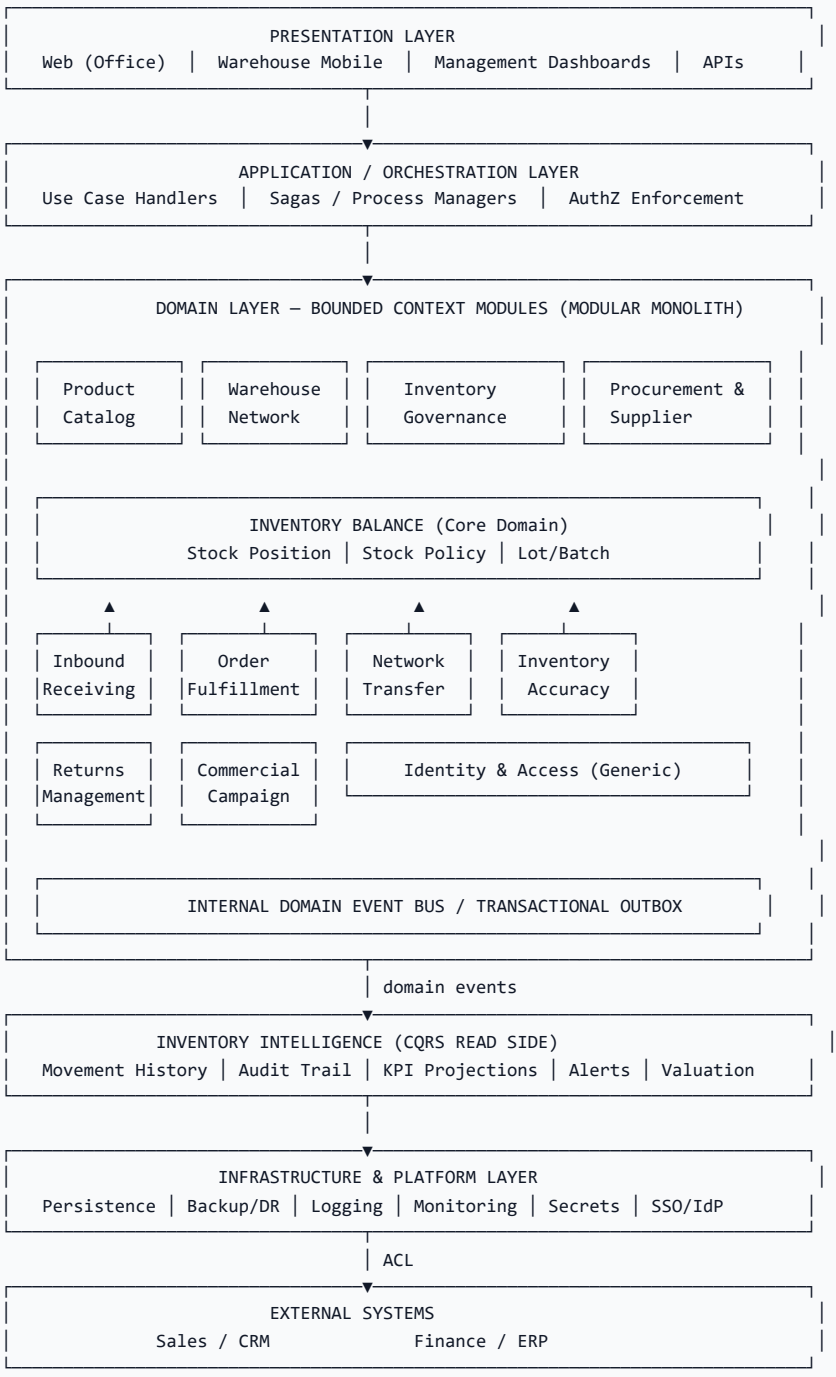
- Optimized read models for dashboards (NFR-PERF-004, NFR-PERF-023)
- 7-year audit retention without bloating operational tables (NFR-COM-019, NFR-COM-020)
- Rebuild projections from event history after corruption or schema evolution

Why Saga Pattern for Cross-Aggregate Workflows

Single-aggregate ACID is sufficient **within** Stock Position, Goods Receipt, or Customer Order. Cross-context workflows — confirm order then reserve stock, dispatch transfer then deduct source and increase destination, certify opening balance then activate location — require **long-running coordination with compensation**.

Event storming hotspot HS-007 (reservation coordination) and policies POL-001/002 explicitly demand a documented saga: success path and failure compensation must be architectural first-class concerns, not implementation afterthoughts.

6.3 Logical Architecture Overview (High-Level)



6.4 Phased Architecture Evolution

Phase	Period	Architectural Focus
Phase 1 — Stabilize	Months 0–6	Modular monolith with core contexts: Catalog, Network, Balance, Inbound, Fulfillment, Accuracy (cycle count), Governance, Identity, Intelligence (basic projections). Internal events. Single-location then sequential rollout.
Phase 2 — Perform	Months 6–12	Full Procurement, Network Transfer, Returns, Commercial Campaign modules. Enhanced sagas (backorder priority, transfer-before-buy). Evaluate Intelligence read-tier scaling. External integration hardening.
Phase 3 — Decide	Months 12–18	Finance ACL and valuation snapshots. Advanced KPI and network analytics. Selective extraction if volume metrics warrant. Tiered data architecture fully operational.

6.5 Alternatives Considered and Rejected

Alternative	Reason for Rejection
Traditional layered monolith (no domain modules)	Cannot preserve 13 context boundaries; high coupling risk; violates AG-02
Microservices from day one	Disproportionate ops cost under BC-01/ORC-04; distributed consistency risk for Stock Position (AR-09)
CRUD-centric data model	Cannot express 32 business rules, SoD, and event choreography; leads to anemic validation scattered in UI
Single shared database schema without context ownership	Encourages cross-context joins and dual ATP writers; contradicts SRS-DM-007 integration patterns
Offline-first / sync architecture	Explicitly out of scope (TEC-03); adds conflict resolution complexity without MVP requirement

6.6 Architecture Decision Records (Preview)

The following decisions will be formalized as ADRs in subsequent architecture documents:

ADR Topic	Decision Direction
ADR-001 Deployment topology	Modular monolith for Phase 1
ADR-002 Cross-context integration	Domain events with transactional outbox
ADR-003 Read/write separation	CQRS for Inventory Intelligence
ADR-004 Stock Position identity	Single aggregate per product-location (pending HS-005 confirmation)
ADR-005 Reservation coordination	Saga with defined compensation (pending HS-007 workshop)
ADR-006 External finance boundary	ACL with IMS as operational SoR (pending HS-014 workshop)
ADR-007 Authentication	Enterprise SSO with location-scoped RBAC
ADR-008 Data retention tiers	Hot / warm / cold aligned to NFR-COM-019

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7. Architecture Vision Summary

The Inventory Management System will be built as a **domain-driven, modular platform** centered on **Inventory Balance** as the authoritative quantity core. Cross-context collaboration will follow **event-driven choreography** and **explicit sagas** where business workflows span aggregates. Operational analytics and compliance evidence will flow through a **CQRS read side** (Inventory Intelligence), protecting transactional integrity and performance.

Architecture prioritizes **governed correctness, auditability, and phased credible control** over premature distribution. The modular monolith approach honors the **fixed budget and limited IT capacity** while preserving **decomposition readiness** as the organization scales locations, channels, and transaction volume toward the 3–5 year vision.

Gate to proceed to Solution Architecture: Resolution of Critical hotspots (HS-001, HS-002), confirmation of Stock Position identity model (HS-005), and documented integration contracts for freeze authority (HS-003) and reservation coordination (HS-007).

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8. Traceability

Vision Element	Primary Sources
Architectural Goals AG-01 – AG-08	BRD §3, §6, §7; SRS-DM-007; SRS-DM-009; SRS-ES-007
Architectural Drivers AD-01 – AD-19	SRS-DM-003/005/006/007; NFR suite; BRD constraints
Business Drivers BD-01 – BD-12	BRD §4–§7, §10; stakeholder analysis
Quality Attributes §4	NFR-PERF, NFR-SCAL, NFR-AVL/REL, NFR-SEC, NFR-LOG, NFR-MON, NFR-BDR, NFR-COM; NFR Review
Risks AR-01 – AR-17	SRS-ES-007; SRS-DM-009; NFR Review; BRD risk register

Vision Element	Primary Sources
Recommended Style §6	SRS-DM-007; SRS-ES-008; BC-01; ORC-04; TEC-03

Document Approval

Role	Name	Signature	Date
Senior Solution Architect			
DDD Solution Architect			
Product Owner			
Steering Committee Sponsor			
Security Engineer			
DevOps Engineer			

End of Architecture Vision

Document ID: SRS-SA-001 | Version 1.0 | July 6, 2026

System Context

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-SA-002
Document Title	System Context — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Solution Architect
Source Documents	BRD-IMS-001, SRS-ES-004, SRS-DM-007, SRS-UC-001, Business Constraints (TEC-01 – TEC-06)
Date	July 6, 2026
Scope	C4 Level 1 — system boundary, users, external systems, and third-party services; no internal module design

Purpose

This document defines the **System Context** for the Inventory Management System (IMS) — the single boundary within which inventory truth is established, governed, and consumed. It identifies who and what interacts with IMS, what data crosses each boundary, and why each interaction exists.

System Purpose (Context Summary)

The IMS is the organization's **inventory system of record** — authoritative for stock quantities, movements, reservations, replenishment state, and inventory audit evidence. It is **not** the enterprise system of record for sales, finance, HR, or channel commerce (BRD §8.2; TEC-01).

1. Primary Users

IMS serves **19 business user roles** (BRD §11). External trading partners do **not** hold standing IMS accounts in MVP scope. Users access IMS via standard web browser on company-approved devices (TEC-03, TEC-06).

1.1 User Groups

Group	User Roles	Primary Purpose
Operations & Warehouse	Inventory Controller, Receiving Clerk, Warehouse Staff, Picker/Packer, Warehouse Manager, Regional/Store Manager, Operations Director	Record and control physical inventory — receive, store, count, transfer, pick, ship, quarantine
Supply & Procurement	Buyer, Purchasing Manager	Authorize inbound supply — requisitions, purchase orders, inbound planning, replenishment response
Commercial & Customer-Facing	Sales Representative, Customer Service Representative, Marketing Manager	Confirm availability, reserve and fulfill orders, manage returns, ring-fence promotional stock
Finance & Governance	Finance Manager, Internal Auditor, External Auditor *(temporary)*	Reconcile valuation, three-way match, audit trail review, control testing
Executive Oversight	CEO, CFO	Review inventory KPIs, valuation alignment, strategic performance
Technology Administration	System Administrator	Provision users, assign location-scoped roles, enforce SoD at access layer

1.2 Primary User Detail

ID	Role	Key Interactions With IMS
UR-01	Inventory Controller	Product/location master, policies, counts, opening balances, governance
UR-02	Receiving Clerk	Goods receipt verification, discrepancies, return receipt
UR-03	Warehouse Staff	Put-away, cycle count, transfer dispatch/receipt
UR-04	Picker/Packer	Pick verification, shipment dispatch
UR-05	Warehouse Manager	Approvals, quarantine, adjustments, exceptions, returns disposition
UR-06	Regional/Store Manager	Transfer requests, local stock monitoring
UR-07	Operations Director	Transfer/adjustment approval, network rebalancing, executive operations view
UR-08	Buyer	Purchase orders, inbound scheduling, replenishment execution
UR-09	Purchasing Manager	Requisition approval, low-stock response
UR-10	Sales Representative	ATP inquiry, order availability confirmation
UR-11	Customer Service Representative	Order confirm/cancel, reservation, backorder, RMA, fulfillment status
UR-12	Marketing Manager	Promotional stock allocation, campaign readiness
UR-13	Finance Manager	Supplier invoice, three-way match, valuation snapshot, GL reconciliation
UR-14	Internal Auditor	Read-only audit trail, movement history, SoD evidence
UR-15	External Auditor	Temporary read-only access during statutory audit
UR-16	CEO / CFO	Executive dashboards and KPI reports
UR-17	System Administrator	User accounts, role assignments, location scope

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2. External Systems

External systems are organization or partner systems **outside the IMS boundary** that exchange business facts or receive inventory-derived information. Integration depth varies by phase (TEC-04, TEC-05).

ID	External System	MVP Integration	Phase
EXT-01	**Sales / Order Management (CRM)**	Manual or batch order intake; CS/Sales enter fulfillment commands	MVP: indirect; Phase 2+: API/ACL
EXT-02	**Finance ERP / General Ledger**	Scheduled export; manual reconciliation (TEC-05)	MVP: manual; Phase 3: automated ACL
EXT-03	**Sales Channel / E-Commerce / Marketplace**	No real-time feed (TEC-04); manual stock updates	Phase 2+
EXT-04	**Supplier / Vendor**	PO communicated outside IMS; physical delivery triggers receipt	MVP: human-mediated
EXT-05	**Logistics / Carrier Partner**	Tracking and delay info via operational workflow	MVP: human-mediated
EXT-06	**Customer (B2B / B2C)**	No direct IMS access; interacts via sales channels and CS	External participant
EXT-07	**Marketing / Campaign Planning**	Campaign plans may originate externally; IMS holds ring-fence	Phase 2
EXT-08	**HR / Enterprise Directory**	User identity source for provisioning	MVP: manual or sync
EXT-09	**Customs / Regulatory Authority**	Clearance decisions affect quarantine release	Phase 2 if bonded in scope

Trading partners (EXT-04, EXT-05, EXT-06) do not execute commands inside IMS — internal actors record outcomes.

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3. Third-Party Services

Third-party services are **managed capabilities** consumed by IMS infrastructure. They are not business participants but enable security, notification, resilience, and operations.

ID	Third-Party Service	Purpose	MVP
SVC-01	**Enterprise Identity Provider (SSO / Directory)**	Authenticate users; optional federated identity (LDAP/Entra ID/Okta)	Should Have
SVC-02	**Email / Notification Gateway**	Deliver low-stock alerts, approval requests, exception escalations (NFR-PERF-014)	Must Have
SVC-03	**SMS Gateway** *(optional)*	Urgent operational alerts for warehouse supervisors	Could Have
SVC-04	**Cloud Hosting Platform**	Run IMS application and data services	Must Have
SVC-05	**Backup & Disaster Recovery Service**	Protect stock balances, transactions, audit logs; ≤15-min RPO (NFR-BDR-001)	Must Have
SVC-06	**Monitoring & Alerting Platform**	Availability, performance, error, and KPI monitoring (NFR-MON)	Must Have
SVC-07	**Certificate / Secrets Management**	TLS, API keys, credential rotation	Must Have
SVC-08	**Time Synchronization (NTP)**	Consistent timestamps for audit trail and movement ordering	Must Have

Excluded from business context (infrastructure only): report rendering engine, internal job scheduler — treated as IMS internal components.

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4. Data Exchanged & Communication Purpose

4.1 User ↔ IMS

User Group	Data Sent to IMS	Data Received From IMS	Communication Purpose
Operations & Warehouse	Receipt lines, put-away confirmations, count results, transfer dispatch/receipt, pick/ship confirmations, quarantine actions, adjustments	Stock positions, ATP, location status, freeze alerts, pick lists, transfer authorizations, operational alerts	Execute and control physical inventory with governed accuracy
Supply & Procurement	Requisitions, POs, inbound schedules, delay notices	Low-stock alerts, network ATP, inbound pipeline status, replenishment recommendations	Commit and track inbound supply
Commercial	Availability checks, order confirm/cancel, reservations, backorders, RMAs, promotional allocations	ATP by SKU/location, reservation status, shipment status, backorder pipeline, campaign readiness	Sell and fulfill without overselling
Finance & Governance	Supplier invoices, match decisions, valuation reconciliation inputs	Movement history, receipt facts, match results, valuation snapshots, write-off records, audit exports	Financial control and statutory audit
Executive	Report filters, dashboard scope	KPIs — fill rate, accuracy, turnover, stockout, valuation variance	Strategic oversight
System Administrator	User profiles, role assignments, location scope	SoD violation alerts, access audit log	Secure, scoped access

4.2 External System ↔ IMS

External System	Direction	Data Exchanged	Communication Purpose
Sales / Order Management	→ IMS	Order header, line items (SKU, qty), ship-from preference, customer ref	Initiate availability check and reservation
	← IMS	ATP confirmation, reservation ID, fulfillment status, shipment notice	Prevent oversell; synchronize order execution state
Finance ERP / GL	→ IMS	GL inventory balance, cost method parameters, period status	Month-end reconciliation
	← IMS	Valuation snapshot, adjustment/write-off facts, three-way match outcome, consignment transfer events	Align operational and financial inventory records
Sales Channel / E-Commerce	← IMS *(Phase 2+)*	ATP or available qty feed by SKU/channel	Reduce channel oversell (TEC-04 risk mitigation)
Supplier / Vendor	→ IMS *(via clerk)*	ASN/delivery notice, invoice, discrepancy response	Trigger receipt verification and invoice match
	← IMS *(via buyer)*	PO, PO change/cancel, discrepancy claim	Commercial commitment and claims
Logistics / Carrier	→ IMS *(via staff)*	Pickup confirmation, delivery proof, delay reason	Complete shipment/transfer; record in-transit delays
	← IMS *(via staff)*	Shipment/transfer dispatch notice	Coordinate pickup
Customer	→ IMS *(via CS/channel)*	Order, cancellation, backorder ack, return request	Demand and reverse logistics intake
	← IMS *(via CS/channel)*	Availability answer, shipment tracking, RMA, backorder notice	Customer commitment and service
Marketing / Campaign	→ IMS	Campaign SKU list, qty, channel, period	Ring-fence stock for promotional readiness
	← IMS	Allocation status, readiness gap alerts	Campaign launch confidence
HR / Directory	→ IMS	Employee ID, name, department, status	Provision/deprovision user accounts
Customs / Regulatory	→ IMS *(via compliance)*	Clearance grant, hold notice, recall directive	Release or hold bonded/quarantined stock

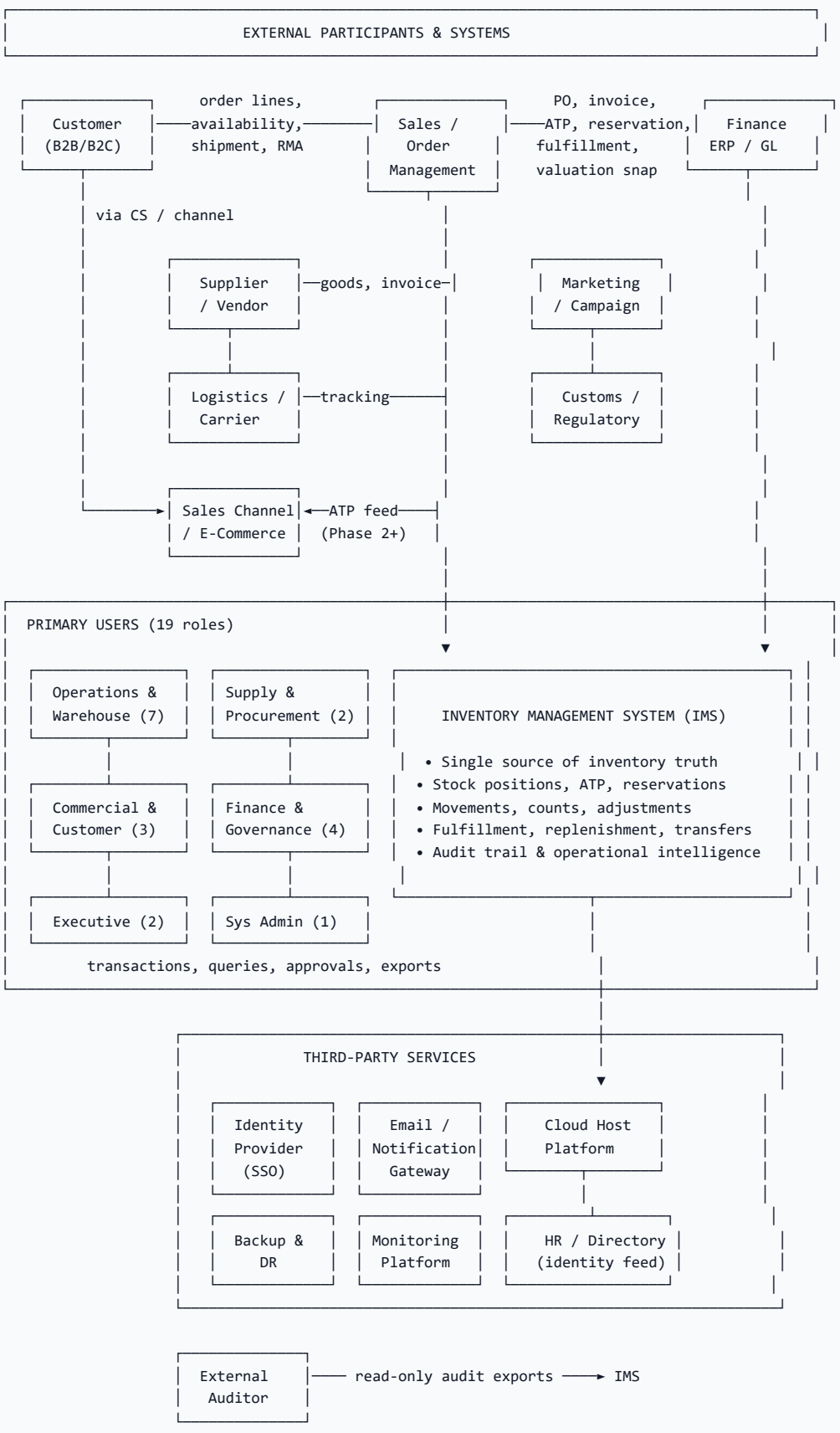
External System	Direction	Data Exchanged	Communication Purpose
External Auditor	← IMS	Audit trail export, movement history, count certification, SoD evidence	Independent control testing

4.3 Third-Party Service ↔ IMS

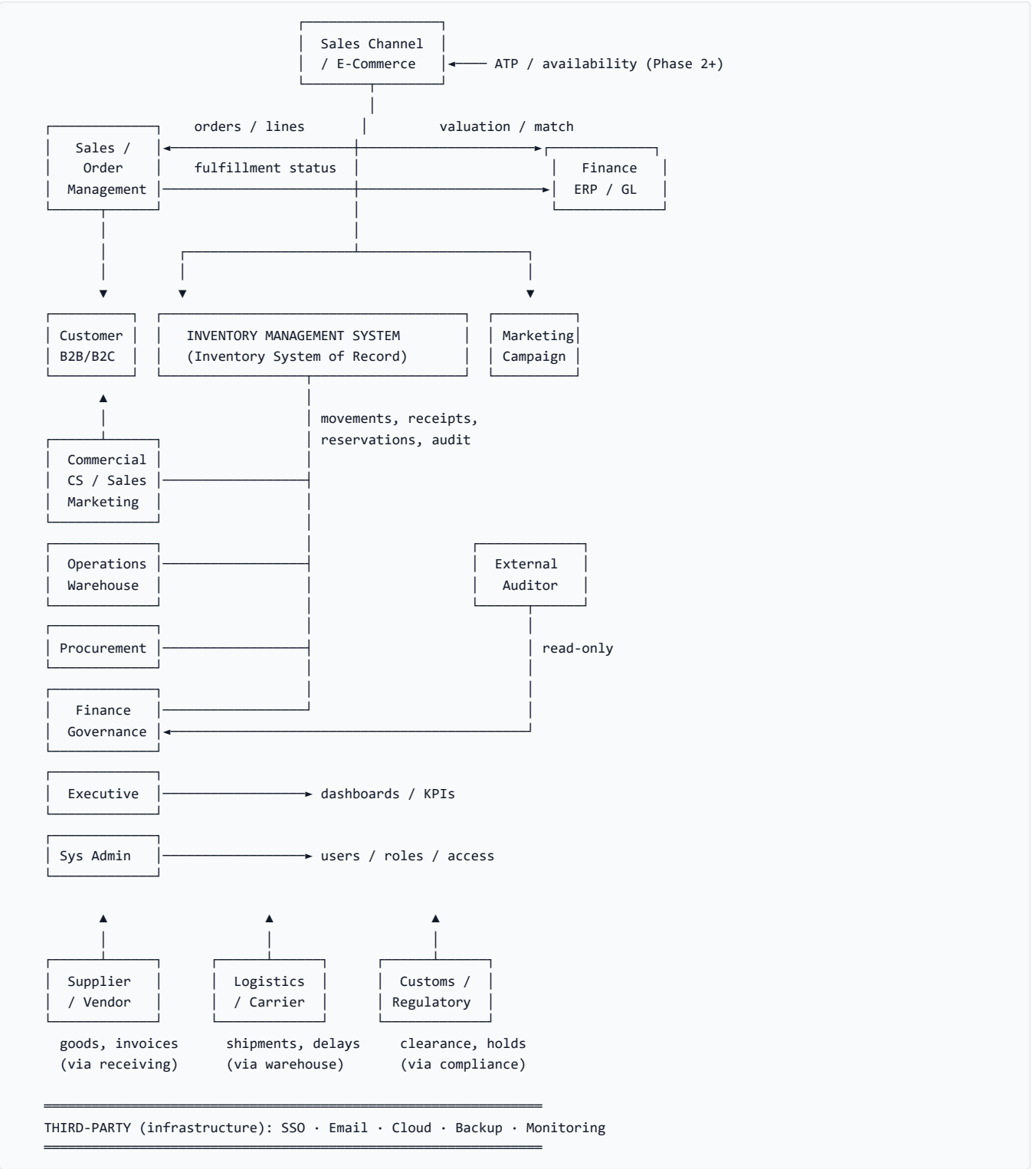
Service	Data Exchanged	Communication Purpose
Identity Provider	Auth tokens, user claims, group membership	Single sign-on; unique user attribution (NFR-SEC-001)
Email / SMS Gateway	Alert payloads, recipient, delivery status	Timely operational notification (NFR-PERF-014)
Cloud Platform	Compute, storage, network	Host application and data
Backup / DR	Database snapshots, transaction logs, config	Business continuity; ≤15-min RPO (NFR-BDR-001)
Monitoring	Metrics, logs, traces, alert events	Detect outage within 5 min (NFR-MON-001)

5. Text-Based Context Diagram

5.1 C4 System Context (Primary View)



5.2 Simplified Context Diagram (Relationships Only)



5.3 Interaction Legend

Symbol	Meaning
`⟶`	Primary data or command flow
`⟵`	Response, query result, or outbound notification
`(Phase 2+)`	Not in MVP integration scope per TEC-04 / TEC-05
`via clerk/staff`	Human-mediated — no direct system integration in MVP

6. System Boundary Rules

Rule	Statement
SB-01	IMS owns inventory quantity state, movement history, and inventory audit evidence
SB-02	IMS does **not** own general ledger, customer master, pricing, or sales pipeline
SB-03	Customer orders may **originate** in Sales/Order Management; IMS owns **fulfillment-relevant subset** via ACL
SB-04	Finance ERP remains **financial system of record**; IMS publishes operational valuation facts
SB-05	External trading partners interact through **business workflows**, not standing IMS accounts (MVP)
SB-06	Notifications are **delivered** via third-party gateways; alert **content and routing** are IMS responsibility
SB-07	Channel ATP sync is **out of MVP scope** (TEC-04); oversell risk managed by manual process until Phase 2

7. Phase Integration Summary

Integration	MVP (Months 0–6)	Phase 2 (6–12)	Phase 3 (12–18)
Sales / Order Management	Manual entry / batch import	API with ACL	Bi-directional status sync
Finance ERP	Export + manual reconcile (TEC-05)	Scheduled file / API	Automated valuation alignment
Sales Channel	Manual stock updates	ATP feed	Real-time availability
Supplier	Human workflow	Optional ASN import	Supplier portal (long-term)
Marketing Campaign	In-app only	External campaign feed ACL	—
HR Directory	Manual provisioning	Directory sync	Federated SSO

Document Approval

Role	Name	Signature	Date
Solution Architect			
Product Owner			
Domain Expert			
Security Engineer			

End of System Context

Document ID: SRS-SA-002 | Version 1.0 | July 6, 2026

Architecture Style Recommendation

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-SA-002

Attribute	Detail
Document Title	Architecture Style Recommendation — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Solution Architect
Prepared For	Steering committee, product owner, delivery leads, and technical architects
Date	July 6, 2026
Classification	Internal — Architecture Confidential
Scope	Architecture style evaluation and recommendation — no technology stack, module design, or implementation detail

Related Documents

Document	ID	Relationship
Architecture Vision	SRS-SA-001	Parent strategic direction
Bounded Contexts	SRS-DM-007	Domain decomposition input
Domain Model Review	SRS-DM-009	Core domain and gate findings
Business Requirements Document	BRD-IMS-001	Constraints, objectives, phased scope
Non-Functional Requirements Suite	204 NFRs	Quality attribute drivers

Document Purpose

This document evaluates six candidate architecture styles against the Inventory Management System's business drivers, domain complexity, non-functional requirements, and delivery constraints. It provides a structured comparison and a final recommendation to guide solution architecture and implementation planning.

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1. Evaluation Context

1.1 Project Characteristics Relevant to Style Selection

Characteristic	Detail	Style Implication
Domain complexity	13 bounded contexts, 18 aggregates, 81 domain events, 32 business rules	Favors domain-centric styles over purely technical layering
Core domain	Inventory Balance (Stock Position, ATP, reservations) — high write contention	Requires strong consistency within core; careful cross-boundary integration
Integration pattern	Event-driven choreography, sagas, ACLs to Sales/CRM and Finance/ERP	Needs ports/adapters and explicit boundary protection
Read/write profile	Transactional operations + heavy reporting/audit (Inventory Intelligence)	CQRS-friendly; reporting must not burden write path
Delivery horizon	18-month phased initiative; MVP in Months 0–6	Style must support incremental delivery, not big-bang
Budget	Fixed initiative budget (BC-01)	Limits operational overhead of distributed architectures
IT capacity	Limited internal IT operations (ORC-04)	Fewer deployable units and simpler runtime preferred for MVP
Scale trajectory	15% near-term / 30% medium-term growth; multi-location rollout	Style must scale without forced rewrite
Quality priorities	Data integrity, security/compliance, availability (99.5%), auditability	Governance and traceability are non-negotiable

1.2 Evaluation Criteria

Each architecture style is assessed against:

Criterion	Weight	Description
Domain alignment	High	Supports 13 bounded contexts and DDD strategic design
Correctness & integrity	High	Protects ATP invariants, SoD, and audit trail
MVP feasibility	High	Deliverable within Months 0–6 under BC-01 and ORC-04
Operational simplicity	High	Runnable and maintainable by limited IT capacity
Evolvability	Medium	Supports Phase 2–3 without architectural replacement
Performance fit	Medium	Meets sub-2s ATP and sub-3s operational response targets
Team scalability	Medium	Aligns with squad/team structure as delivery grows
Testability	Medium	Enables isolated testing of domain rules and integrations

1.3 Important Distinction — Deployment Style vs. Internal Structure

Several options in this comparison operate at **different levels of abstraction**:

Level	Styles
Deployment / system topology	Modular Monolith, Microservices
Internal code organization	Layered, Clean, Hexagonal, Onion

These are **not always mutually exclusive**. A modular monolith can implement hexagonal or clean architecture inside each module. The final recommendation combines styles at the appropriate level rather than selecting a single label.

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2. Architecture Style Comparison

2.1 Layered Architecture

Description: Organizes the system into horizontal tiers — typically Presentation → Application/Business Logic → Data Access → Database — with dependencies flowing downward.

Advantages

Advantage	Benefit for IMS
Familiar and widely understood	Low learning curve for mixed-skill delivery teams
Clear separation of UI, logic, and persistence	Straightforward CRUD screens for master data (products, locations)
Simple deployment model	Single application unit aligns with budget constraints
Easy to bootstrap	Fast initial scaffolding for MVP timelines
Tooling support	Most frameworks default to layered structure

Disadvantages

Disadvantage	Risk for IMS
Domain logic leaks across layers	32 business rules and SoD policies scatter into services, controllers, and repositories
No natural bounded context boundaries	13 contexts collapse into shared services and tables
Database-centric design tendency	Stock Position invariants driven by schema rather than aggregate behavior
High coupling over time	Cross-cutting changes (e.g., ATP formula, freeze rules) ripple through all layers

Disadvantage	Risk for IMS
Reporting couples to transactional path	Dashboard queries against same tables degrade dock performance (AR-10)
Poor fit for event-driven coordination	81 domain events have no first-class home; added as afterthought

Suitability for This Project

Dimension	Rating	Rationale
Domain alignment	**Low**	Cannot express 13 bounded contexts or Inventory Balance as core domain
MVP feasibility	**High**	Fast to start, but accrues rework as rules accumulate
Long-term evolvability	**Low**	Becomes monolithic "big ball of mud" under cross-location growth
NFR compliance	**Medium**	Achievable with discipline, but audit and CQRS require bolt-on patterns

Overall suitability: Low — acceptable only for trivial CRUD subsets, not as the system-wide style.

Selected or Rejected?

Rejected as the primary architecture style.

Layered architecture does not provide the domain boundaries, event transparency, or integration isolation that the approved domain model requires. It may be used **locally within a module's infrastructure slice** (e.g., repository implementations beneath domain core) but must not define the system's top-level structure.

2.2 Clean Architecture

Description: Robert Martin's concentric dependency rule — entities at the center, use cases around them, interface adapters outside, frameworks at the edge. Dependencies point inward; business rules are independent of UI, database, and external systems.

Advantages

Advantage	Benefit for IMS
Domain-centric dependency rule	Stock Position, Goods Receipt, and Order aggregates stay free of framework code
Testable use cases	Command handlers for ReserveStock, ReceiveGoods testable without database
Framework independence	UI swap (web → mobile warehouse) without touching domain rules
Explicit use case layer	Maps cleanly to 48 use cases and application services
Supports ports and adapters	Natural fit for ACL to Sales/CRM and Finance/ERP

Disadvantages

Disadvantage	Risk for IMS
Ceremony and boilerplate	Many interfaces and DTO mappings across 13 contexts
No deployment guidance	Clean Architecture is internal structure only — does not resolve monolith vs. microservices
No modular boundary enforcement	Teams can still create cross-context imports without compile-time module boundaries
Learning curve	Requires discipline to avoid "clean in name only" anemic entities
Event-driven patterns not prescribed	Domain events and CQRS must be added explicitly

Suitability for This Project

Dimension	Rating	Rationale
Domain alignment	**High**	Aligns with DDD aggregates and domain services
MVP feasibility	**Medium–High**	Achievable with templates; overhead manageable per module
Long-term evolvability	**High**	Use cases and entities stable as infrastructure changes
NFR compliance	**High**	Supports testability, ACL, and separation of concerns

Overall suitability: High — as **internal structure within each bounded context module**.

Selected or Rejected?

Selected as an internal organizing principle (per bounded context module), **not as the sole architecture label**.

Clean Architecture provides the dependency discipline needed to protect domain rules (BR-001, BR-002, BR-005, SoD) from infrastructure concerns. It does not address cross-module integration, deployment topology, or CQRS — those require complementary patterns.

2.3 Hexagonal Architecture (Ports and Adapters)

Description: Alistair Cockburn's model placing the application core at the center with **ports** (interfaces) and **adapters** (implementations) for driving actors (UI, API) and driven actors (database, external systems, message bus).

Advantages

Advantage	Benefit for IMS
Explicit integration boundaries	ACL for Sales/CRM orders and Finance/ERP valuation as outbound adapters
Multiple driving adapters	Web office UI, warehouse mobile, and integration API share same core ports
Swappable infrastructure	Change persistence or messaging without altering Stock Position logic
Natural anti-corruption layers	External order model translated at adapter, not in domain
Testability via port mocks	Simulate ERP responses for three-way match scenarios
Aligns with event storming	Commands enter via driving ports; events exit via driven ports

Disadvantages

Disadvantage	Risk for IMS
Adapter proliferation	13 contexts × multiple adapters (DB, events, external) increases surface area
Mapping overhead	DTO ↔ domain object translation at every boundary
No strategic domain map	Hexagonal is tactical — does not define bounded context relationships
Risk of anemic ports	Ports become CRUD pass-throughs if domain model is weak
Distributed hexagons = microservices	Without modular monolith discipline, each hexagon drifts to separate service

Suitability for This Project

Dimension	Rating	Rationale
Domain alignment	**High**	Protects core domain from external system models (HS-014)
MVP feasibility	**High**	Adapters can start simple (in-process) and evolve
Long-term evolvability	**High**	External integrations change frequently; adapters absorb change

Dimension	Rating	Rationale
NFR compliance	**High**	Supports security (auth adapter), logging (observability adapter), ERP ACL

Overall suitability: High — especially for **Inventory Balance**, **Order Fulfillment**, and **Inbound Receiving** contexts with external touchpoints.

Selected or Rejected?

Selected as the integration and isolation pattern within each bounded context module.

Hexagonal architecture is the primary mechanism for implementing AG-06 (integration-ready enterprise capability) and mitigating AR-04 (finance boundary) and AR-11 (channel integration). It complements Clean/Onion rather than replacing them.

2.4 Onion Architecture

Description: Jeffrey Palermo's variant of layered architecture with the domain model at the center, domain services surrounding it, application services outside, and infrastructure on the outermost ring. Like Clean Architecture, dependencies point inward.

Advantages

Advantage	Benefit for IMS
Domain model at the center	Reinforces Inventory Balance as core with rich entity behavior
Clear ring boundaries	Domain → Application → Infrastructure progression is teachable
Repository interfaces in domain ring	Persistence abstraction without ORM leaking into aggregates
Consistent with DDD	Aggregates, domain services, and repositories have natural homes
Incremental adoption	Teams can onion-wrap one context at a time during phased delivery

Disadvantages

Disadvantage	Risk for IMS
Substantial overlap with Clean Architecture	Teams may debate nomenclature (use case vs. application service) without design benefit
No cross-context guidance	13 onions in one codebase need modular monolith rules to prevent coupling
Application service bloat risk	Complex sagas (reservation coordination) can inflate the application ring
Event publishing not explicit	Domain events need deliberate placement — domain ring or application ring
Shared infrastructure temptation	Shared outer-ring components (logging, auth) can create hidden coupling

Suitability for This Project

Dimension	Rating	Rationale
Domain alignment	**High**	Supports rich Stock Position aggregate with invariants
MVP feasibility	**Medium–High**	Similar effort to Clean Architecture
Long-term evolvability	**High**	Infrastructure rings swappable per context
NFR compliance	**High**	Domain-centric testing supports rule verification

Overall suitability: High — largely **equivalent to Clean Architecture** for IMS purposes.

Selected or Rejected?

Selected in principle, merged with Clean Architecture — use one consistent internal ring model across all context modules to avoid team confusion.

Onion and Clean Architecture solve the same problem. The IMS architecture standard will use **"Clean/Hexagonal"** as the documented term: domain core (entities, aggregates, domain services) → application layer (commands, queries, sagas) → adapters (persistence, messaging, external systems).

2.5 Modular Monolith

Description: A single deployable application composed of **loosely coupled modules** aligned to domain boundaries (bounded contexts). Modules communicate through well-defined interfaces or domain events — not shared database tables or direct internal API calls across boundaries.

Advantages

Advantage	Benefit for IMS
Single deployment unit	One runtime, one pipeline — fits BC-01 fixed budget and ORC-04 limited ops
Bounded context alignment	13 modules map to SRS-DM-007 without network overhead
ACID within module	Stock Position aggregate transactions remain in-process and strongly consistent
In-process domain events	81 events published via internal bus or transactional outbox — no broker required for MVP
Phased module activation	Enable Procurement, Transfer, Campaign modules in Phase 2 without new deployments
Simpler NFR achievement	99.5% uptime easier with fewer failure domains than microservices
Decomposition readiness	Modules with clear interfaces can be extracted to services later (Intelligence, Identity)
Faster cross-module refactoring	Compiler enforces module boundaries; rename/extract safer than distributed refactor

Disadvantages

Disadvantage	Risk for IMS
Requires discipline	Without enforced module boundaries, devolves to traditional monolith
Scaling limitations	Cannot independently scale Stock Position writes vs. reporting reads without read-side separation
Shared runtime blast radius	Defect in one module can affect entire application availability
Technology homogeneity	All modules share same language, framework, and database stack
Team coordination at scale	Large teams on one codebase need strong ownership rules to avoid merge conflicts
Not a complete architecture	Still needs internal structure (Clean/Hexagonal) and integration patterns (events, CQRS)

Suitability for This Project

Dimension	Rating	Rationale
Domain alignment	**High**	Direct mapping to 13 bounded contexts
MVP feasibility	**High**	Fastest path to credible control by Month 6
Long-term evolvability	**High**	Extraction path for Intelligence and integration gateway
NFR compliance	**High**	Meets availability, performance, and budget constraints
Operational simplicity	**High**	Best fit for limited IT capacity

Overall suitability: Very High — strongest **deployment topology** choice for Phase 1–2.

Selected or Rejected?

Selected as the primary deployment and system topology style for Phase 1 and Phase 2.

Modular monolith is the only deployment style that simultaneously satisfies fixed budget, limited IT capacity, strong Stock Position consistency, phased delivery, and the strategic domain decomposition already approved. SRS-DM-007 explicitly states bounded contexts are strategic design — not microservice boundaries.

2.6 Microservices

Description: Decomposes the system into **independently deployable services**, each owning its data, communicating via network protocols (REST, gRPC, messaging). Often aligned to bounded contexts — one service per context.

Advantages

Advantage	Benefit for IMS
Independent scaling	Scale Order Fulfillment and Inventory Balance separately under peak load
Technology heterogeneity	Different stacks per service (e.g., analytics service in specialized engine)
Team autonomy	Each squad owns full service lifecycle — deploy without coordinating other teams
Fault isolation	Reporting service failure does not take down receiving dock
Bounded context mapping	Natural 1:1 with 13 contexts — attractive on paper
Clear data ownership	Each service owns its database — no shared tables

Disadvantages

Disadvantage	Risk for IMS
Distributed consistency complexity	Reservation saga (HS-007) across Order and Balance services requires eventual consistency, compensation, and idempotency — high defect risk
Operational overhead	Service mesh, distributed tracing, contract testing, multiple CI/CD pipelines — exceeds ORC-04 capacity
Infrastructure cost	Multiple runtimes, databases, brokers strain BC-01 fixed budget
Latency on hot paths	ATP lookup and reservation cross service boundaries — threatens NFR-PERF-002 (2s)
Premature decomposition risk	16 open domain hotspots (SRS-ES-007) make service boundaries unstable before implementation
Cross-service transactions	Put-away → stock available → backorder fulfill chain becomes fragile distributed workflow
Integration testing burden	End-to-end tests require full service topology — slows MVP delivery
Data duplication and sync	Product SKU and Location Code replicated across services or queried synchronously

Suitability for This Project

Dimension	Rating	Rationale
Domain alignment	**Medium** (theoretical) / **Low** (practical)	Contexts are clear, but hotspots and sagas make boundaries premature
MVP feasibility	**Low**	Months 0–6 delivery at risk from distributed complexity
Long-term evolvability	**High** (at maturity)	Valid at 3–5 year scale with mature ops — not at initiative start
NFR compliance	**Medium**	Achievable but expensive; 99.5% uptime harder across many services
Operational simplicity	**Very Low**	Conflicts directly with ORC-04

Overall suitability: Low for Phase 1–2; Medium for selective Phase 3 extraction.

Selected or Rejected?

Rejected as the initial system topology. Approved as a future evolution path for proven extraction candidates.

Microservices address scale and team autonomy problems the organization does not yet have at MVP. The risk of distributed data inconsistency on Stock Position — the system's core value — outweighs theoretical benefits. Selective extraction of **Inventory Intelligence** (read-only, CQRS) and **integration gateway** may be reconsidered in Phase 3 when volume metrics and ops maturity justify the cost.

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3. Comparative Summary

3.1 Rating Matrix

Criterion (Weight)	Layered	Clean	Hexagonal	Onion	Modular Monolith	Microservices
Domain alignment (High)	●○○	●●●	●●●	●●●	●●●	●●○
Correctness & integrity (High)	●○○	●●●	●●●	●●●	●●●	●●○
MVP feasibility (High)	●●●	●●○	●●●	●●○	●●●	●○○
Operational simplicity (High)	●●●	●●○	●●○	●●○	●●●	●○○
Evolvability (Medium)	●○○	●●●	●●●	●●●	●●●	●●●
Performance fit (Medium)	●●○	●●○	●●○	●●○	●●●	●●○
Team scalability (Medium)	●○○	●●○	●●○	●●○	●●○	●●●
Testability (Medium)	●●○	●●●	●●●	●●●	●●●	●●○
Weighted overall	**Low**	**High***	**High***	**High***	**Very High**	**Low**

** High as internal module structure, not standalone deployment style.*

Legend: ●●● Strong | ●●○ Adequate | ●○○ Weak

3.2 Decision Summary

Architecture Style	Verdict	Role in IMS
Layered Architecture	**Rejected**	Not suitable as system-wide style; infrastructure sub-layer only
Clean Architecture	**Selected**	Internal structure within each bounded context module
Hexagonal Architecture	**Selected**	Ports/adapters for persistence, events, and external systems
Onion Architecture	**Selected** (merged)	Equivalent to Clean — use unified internal ring model
Modular Monolith	**Selected**	Primary deployment topology for Phase 1–2
Microservices	**Rejected** (initially)	Selective extraction in Phase 3+ if metrics justify

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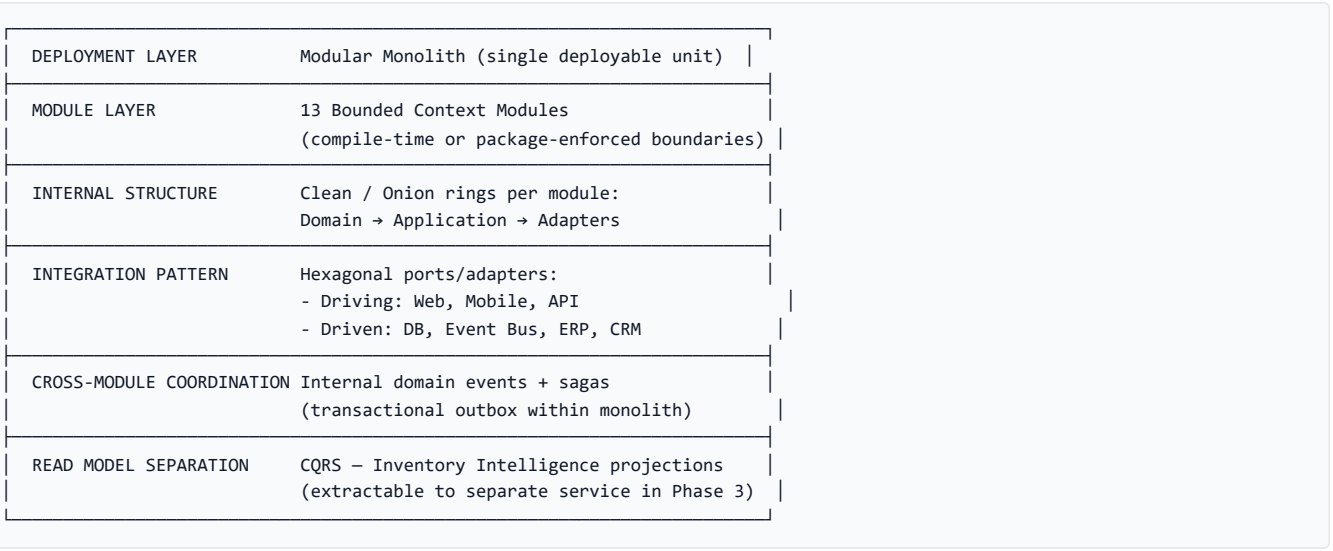
4. Final Recommendation

4.1 Recommended Architecture Style

Modular Monolith with Clean/Hexagonal internal architecture, domain-driven bounded context modules, internal event-driven integration, and CQRS for Inventory Intelligence.

This is a **deliberate composition** of complementary styles at the correct abstraction level:



4.2 Why This Combination Wins

Project Requirement	How the Recommended Style Delivers
Single source of inventory truth (BO-01)	Inventory Balance module owns Stock Position; other modules interact via commands and events — no shared mutable state
13 bounded contexts (SRS-DM-007)	One module per context with published integration contracts
32 business rules + SoD (BR-001 – BR-032)	Clean/Hexagonal domain core enforces invariants; governance consulted via ports
81 domain events (SRS-ES-001)	Internal event bus with transactional outbox ensures reliable cross-module reactions
Sub-2s ATP lookup (NFR-PERF-002)	In-process reads and optimized query side; no cross-service network hops in hot path
99.5% uptime (NFR-AVL-001)	Single runtime reduces failure domains vs. microservices
Fixed budget (BC-01)	One deployment pipeline, one operational surface
Limited IT capacity (ORC-04)	No service mesh, no multi-broker ops for MVP
Phased delivery (Months 0–18)	Modules activated per phase; Phase 2 contexts added without topology change
Finance/CRM integration (HS-014)	Hexagonal ACL adapters at module boundary
Reporting without dock slowdown (AR-10)	CQRS projections in Intelligence module, separate read store
Future scale (NFR-SCAL-001)	Intelligence and integration gateway extractable when proven necessary

4.3 What We Explicitly Avoid

Anti-Pattern	Why
Layered monolith (UI → Service → Repository → DB for entire system)	Collapses 13 contexts; rules scatter; no event-first design
Microservices day one (13 services)	Distributed consistency risk on ATP; ops cost exceeds constraints
Clean Architecture without modules	Dependency rule alone does not prevent cross-context coupling
Modular monolith without events	Cross-context workflows become synchronous spaghetti across modules
Shared database tables across modules	Violates context ownership; enables bypass of Stock Position invariants

4.4 Phased Style Evolution

Phase	Period	Architecture Emphasis
Phase 1	Months 0–6	Modular monolith; core modules (Balance, Inbound, Fulfillment, Catalog, Network, Governance, Identity); basic Intelligence projections; in-process events

Phase	Period	Architecture Emphasis
Phase 2	Months 6–12	Add Procurement, Transfer, Returns, Campaign modules; formalize sagas; strengthen event catalog; evaluate read-store scaling
Phase 3	Months 12–18	Finance ACL hardening; advanced Intelligence; **evaluate extraction** of Intelligence read service and integration gateway if load metrics require

4.5 Module Boundary Rules (Mandatory)

To prevent modular monolith erosion, the following rules apply from Phase 1:

1. **No cross-module database access** — modules own their persistence; other modules reference by identifier only (SKU, Location Code, Order ID).
2. **No direct cross-module domain object imports** — integration via application commands, queries, or domain events only.
3. **Inventory Balance is the sole ATP mutator** — no other module modifies reserved, on-hand, or quarantined quantities directly.
4. **External systems integrate through ACL adapters** — foreign models are translated at the adapter, never in domain entities.
5. **Inventory Intelligence is read-only** — projections consume events; never write back to source aggregates.
6. **Sagas own cross-module workflows** — reservation, transfer, and go-live coordination live in application orchestration, not buried in adapters.

4.6 Architecture Decision Record

Field	Decision
ADR ID	ADR-001
Title	Primary Architecture Style Selection
Status	Proposed
Context	IMS requires domain-rich inventory platform under fixed budget, limited ops capacity, and 18-month phased delivery with 13 bounded contexts
Decision	Adopt **Modular Monolith** as deployment topology with **Clean/Hexagonal** internal structure per bounded context module, **internal event-driven** cross-module integration, and **CQRS** for Inventory Intelligence
Consequences	Single deployable unit; enforced module boundaries required; selective service extraction deferred to Phase 3; team must adopt DDD and ports/adapters discipline
Alternatives rejected	Layered (insufficient domain alignment); Microservices (premature ops and consistency cost)

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5. Conclusion

No single architecture style from the comparison list is sufficient on its own for the Inventory Management System.

- **Layered Architecture** is too weak for this domain complexity and is **rejected**.
- **Microservices** is too heavy for MVP constraints and core consistency needs and is **rejected** as the initial topology.
- **Clean Architecture**, **Hexagonal Architecture**, and **Onion Architecture** are **selected together** as the internal organization of each bounded context module — they protect domain rules and external integrations.
- **Modular Monolith** is **selected** as the deployment and system topology — it honors the approved bounded context map, budget, IT capacity, and phased delivery while preserving a credible path to selective service extraction.

The recommended style is documented in SRS-SA-001 (Architecture Vision) and should guide all subsequent solution architecture, technology selection, and detailed design work.

Document Approval

Role	Name	Signature	Date
Solution Architect			

Role	Name	Signature	Date
DDD Solution Architect			
Product Owner			
Delivery Lead			
DevOps Engineer			

End of Architecture Style Recommendation

Document ID: SRS-SA-002 | Version 1.0 | July 6, 2026

Part II — Business Rules & Constraints

Business Rules Specification

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-BR-001
Document Title	Business Rules — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Senior Business Analyst
Source Document	BRD-IMS-001 (Business Requirements Document, Version 1.0)
Date	July 5, 2026
Scope	Business policies only — no technical validation logic, system constraints, or implementation detail

Purpose

This document extracts and elaborates all **32 industry business rules** defined in the approved BRD (Section 13). Each rule is expressed as an enforceable business policy governing how the organization manages inventory, independent of how any system implements it.

Rule Category Legend

Category	Description
Inventory	Stock availability, commitment, replenishment, and network balance
Purchasing	Procurement, inbound supply, and supplier accountability
Warehousing	Receiving, storage, movement, and fulfillment execution
Products	Product master data, lifecycle, classification, and commercial positioning
Users	Governance, accountability, segregation of duties, and authorization
Reporting	Performance measurement, audit, and management oversight

Business Rules Catalog

Availability and Customer Commitment

BR-001 — ATP-Only Customer Commitment

Attribute	Detail
Rule Name	ATP-Only Customer Commitment
Business Rule Statement	The organization must sell and promise stock only against available-to-promise (ATP) quantity. On-hand quantity that is reserved, committed, quarantined, or otherwise unavailable must not be quoted or committed to customers.
Business Reason	On-hand quantity includes stock already reserved or committed to other orders. Quoting or selling against total on-hand causes overselling — confirming orders the business cannot fulfill. ATP is the only honest measure of what can still be promised to new customers.
Related Business Process	BP-10 (Inventory Status Monitoring), BP-16 (Order Availability Confirmation)
Related Functional Requirements	FR-024 (Display ATP quantity), FR-041 (Confirm order availability before commitment), FR-025 (Display reserved and committed stock)
Rule Category	Inventory

BR-002 — Reservation at Order Confirmation

Attribute	Detail
Rule Name	Reservation at Order Confirmation
Business Rule Statement	Stock must be reserved at the point of order confirmation. Once a customer order is confirmed for fulfillment, the required quantity must be committed against on-hand stock and removed from ATP until fulfilled or released.
Business Reason	The gap between order confirmation and physical picking creates a window where the same stock can be sold to multiple customers. Reservation converts a commercial commitment into an operational hold, preventing double-allocation.
Related Business Process	BP-17 (Inventory Reservation)
Related Functional Requirements	FR-042 (Reserve inventory for confirmed orders), FR-025 (Display reserved and committed stock), FR-043 (Release inventory reservations)
Rule Category	Inventory

BR-003 — Verified Pick Before Shipment

Attribute	Detail
Rule Name	Verified Pick Before Shipment
Business Rule Statement	The organization must not ship or deduct inventory until pick verification is complete. Orders may only be marked shipped after the correct items are physically retrieved, verified against the order, packed, and handed to the carrier.
Business Reason	Inventory should not be deducted and orders should not be marked shipped until the correct items are physically retrieved, verified, packed, and handed to the carrier. Premature shipment recording creates inaccurate stock and false delivery expectations.
Related Business Process	BP-18 (Pick, Pack & Ship)
Related Functional Requirements	FR-044 (Support pick, pack, and ship workflow), FR-029 (Record all inventory movements with full history)
Rule Category	Warehousing

BR-028 — Backorder Customer Acknowledgment

Attribute	Detail
Rule Name	Backorder Customer Acknowledgment
Business Rule Statement	A backorder requires explicit customer acknowledgment before it is recorded. Customers must be informed of the delay, expected fulfillment timing, and their option to cancel or accept partial delivery.
Business Reason	A backorder is a mutual agreement to wait — not an internal note that an order cannot ship. Customers must be informed of the delay, the expected fulfillment date, and their right to cancel. Silent backorders destroy trust.
Related Business Process	BP-19 (Partial Fulfillment & Backorder Management)
Related Functional Requirements	FR-045 (Handle partial fulfillment), FR-046 (Manage backorders and replenishment priority)
Rule Category	Inventory

Inbound and Procurement

BR-004 — Verify Before Inventory Acceptance

Attribute	Detail
Rule Name	Verify Before Inventory Acceptance
Business Rule Statement	Inbound goods must be verified before acceptance into inventory. Quantity, product identity, and condition must be confirmed against the purchase order or authorized receipt documentation before stock is recorded as on-hand.
Business Reason	Arrival at the dock is not receipt. Until quantity, product identity, and condition are verified against the purchase order, the business does not know what it actually received. Recording unverified quantities corrupts inventory from the point of entry.
Related Business Process	BP-07 (Goods Receipt & Verification)
Related Functional Requirements	FR-015 (Receive and verify inbound goods), FR-003 (Support product identifier lookup)
Rule Category	Warehousing

BR-005 — Put-Away Before Availability

Attribute	Detail
Rule Name	Put-Away Before Availability
Business Rule Statement	Received stock is not available for sale or fulfillment until put-away is complete. Stock verified at the receiving dock does not increase ATP until it is placed in its designated storage location and put-away is confirmed.
Business Reason	Stock sitting in the receiving area is physically in the building but not operationally available. Pickers cannot find it efficiently, it may be damaged in transit areas, and it is not in its designated storage location. Availability begins when put-away is confirmed.
Related Business Process	BP-09 (Stock Put-Away & Storage)
Related Functional Requirements	FR-018 (Manage put-away and stock availability timing), FR-024 (Display ATP quantity)
Rule Category	Warehousing

BR-006 — Document Receipt Discrepancies

Attribute	Detail
Rule Name	Document Receipt Discrepancies
Business Rule Statement	Receipt discrepancies must be documented before acceptance. Short shipments, over-shipments, wrong items, and damage must be recorded at the point of receipt with supporting evidence before quantities are accepted into inventory.
Business Reason	Short shipments, over-shipments, wrong items, and damage must be recorded at the point of receipt — when evidence exists and the carrier or supplier is present. Undocumented acceptance waives the business's right to claim and hides supplier performance problems.
Related Business Process	BP-08 (Receipt Discrepancy Resolution)
Related Functional Requirements	FR-016 (Document and resolve receipt discrepancies), FR-011 (Maintain standard reason codes)
Rule Category	Purchasing

BR-007 — Purchase Order Required for Supplier Receipt

Attribute	Detail
Rule Name	Purchase Order Required for Supplier Receipt
Business Rule Statement	A valid purchase order is required before receiving supplier goods. Supplier shipments must be matched to an authorized purchase order before acceptance, except where an explicitly approved ad-hoc receipt exception applies.
Business Reason	A purchase order is the authorized commercial commitment that defines what was ordered, at what price, and from whom. Receiving without a PO breaks financial controls, enables unauthorized purchases, and prevents three-way matching.
Related Business Process	BP-05 (Purchase Order Placement), BP-07 (Goods Receipt & Verification)
Related Functional Requirements	FR-017 (Enforce PO reference for supplier receipts), FR-013 (Record purchase order commitments)
Rule Category	Purchasing

BR-008 — Three-Way Match Before Payment

Attribute	Detail
Rule Name	Three-Way Match Before Payment
Business Rule Statement	Supplier payment must not be authorized until a three-way match is confirmed among the purchase order, goods receipt, and supplier invoice within defined tolerance. Payment for unordered, unreceived, or mispriced goods is prohibited.
Business Reason	Payment must be authorized only when the purchase order (what was ordered), goods receipt (what was received), and supplier invoice (what is billed) agree within defined tolerance. This prevents paying for unordered, unreceived, or mispriced goods.
Related Business Process	BP-05 (Purchase Order Placement), BP-08 (Receipt Discrepancy Resolution), BP-30 (Inventory Valuation & Financial Reconciliation)
Related Functional Requirements	FR-013 (Record purchase order commitments), FR-015 (Receive and verify inbound goods), FR-016 (Document and resolve receipt discrepancies), FR-089 (Support financial inventory reconciliation)
Rule Category	Purchasing

Replenishment and Stock Policy

BR-009 — Reorder at Reorder Level

Attribute	Detail
Rule Name	Reorder at Reorder Level
Business Rule Statement	Replenishment action must be triggered at or before the defined reorder level. When available stock reaches the reorder threshold, a replenishment review or purchase action must be initiated the same business day.
Business Reason	The reorder level is calculated to allow enough time for replenishment to arrive before stock reaches zero, accounting for lead time and demand during that period. Delaying action past the reorder level guarantees a stockout during lead time.
Related Business Process	BP-03 (Stock Policy & Threshold Definition), BP-21 (Low-Stock Alert Review), BP-22 (Replenishment Planning & Requisition)
Related Functional Requirements	FR-008 (Define stock policy and replenishment thresholds), FR-051 (Generate low-stock and out-of-stock alerts), FR-052 (Review and acknowledge stock alerts)
Rule Category	Inventory

BR-010 — Safety Stock Protection

Attribute	Detail
Rule Name	Safety Stock Protection
Business Rule Statement	Safety stock must not be bypassed without an authorized exception. Safety stock buffers may not be consumed for routine fulfillment or reallocated without documented approval and a replenishment plan.
Business Reason	Safety stock is a deliberate buffer protecting against demand and supply uncertainty. Consuming safety stock to fulfill routine orders or reallocating it without replenishment plan removes the organization's protection against the next disruption.
Related Business Process	BP-03 (Stock Policy & Threshold Definition), BP-17 (Inventory Reservation), BP-24 (Inter-Location Transfer Request)
Related Functional Requirements	FR-056 (Enforce safety stock protection rules), FR-008 (Define stock policy and replenishment thresholds)
Rule Category	Inventory

Network Balance

BR-011 — Network Check Before Purchase

Attribute	Detail
Rule Name	Network Check Before Purchase
Business Rule Statement	Network availability must be checked before placing a new purchase order. When a location experiences shortage, excess stock at other locations must be evaluated for transfer before initiating new procurement.
Business Reason	A local stockout may be solvable with a transfer from another location that holds excess. Purchasing new stock when the network already owns sufficient quantity wastes capital, increases carrying cost, and worsens network imbalance.
Related Business Process	BP-21 (Low-Stock Alert Review), BP-24 (Inter-Location Transfer Request), BP-26 (Network Stock Rebalancing)
Related Functional Requirements	FR-028 (Display expected inbound and network-wide availability), FR-053 (Generate replenishment suggestions), FR-058 (Request inter-location stock transfers)
Rule Category	Inventory

BR-012 — Protect Source Location Minimum

Attribute	Detail
Rule Name	Protect Source Location Minimum
Business Rule Statement	An inter-location transfer must not deplete the source location below its minimum threshold. Transfer quantity must be limited to stock the source can spare while maintaining its own service level.
Business Reason	Sending stock to fix one location's shortage must not create a shortage at the source. Transfers require verification that the sending location can spare the quantity while maintaining its own service level.
Related Business Process	BP-24 (Inter-Location Transfer Request), BP-25 (Inter-Location Transfer Execution)
Related Functional Requirements	FR-063 (Enforce source location minimum on transfers), FR-058 (Request inter-location stock transfers), FR-060 (Execute and record inter-location transfers)
Rule Category	Warehousing

Quality and Rotation

BR-013 — Quarantine Exclusion from Sellable Stock

Attribute	Detail
Rule Name	Quarantine Exclusion from Sellable Stock
Business Rule Statement	Quarantined stock must be excluded from sellable availability. Inventory under quality hold, investigation, or recall must not be included in ATP or committed to customer orders.
Business Reason	Quarantined inventory is held pending quality investigation, regulatory clearance, or recall response. Including quarantined stock in sellable availability risks shipping defective, contaminated, or non-compliant products to customers.
Related Business Process	BP-14 (Stock Adjustment & Write-Off), BP-20 (Outbound Exception Handling)
Related Functional Requirements	FR-038 (Manage quarantine and hold stock status), FR-024 (Display ATP quantity)
Rule Category	Inventory

BR-014 — FEFO Rotation for Shelf-Life Products

Attribute	Detail
Rule Name	FEFO Rotation for Shelf-Life Products
Business Rule Statement	First-expired, first-out (FEFO) rotation is required for products with defined shelf life. Fulfillment must prioritize stock with the earliest expiration date.
Business Reason	Products with expiration dates must ship in order of earliest expiry first. Shipping newer stock while older stock expires in the warehouse causes write-offs, customer safety issues, and regulatory violations.
Related Business Process	BP-09 (Stock Put-Away & Storage), BP-18 (Pick, Pack & Ship)
Related Functional Requirements	FR-044 (Support pick, pack, and ship workflow), FR-029 (Record all inventory movements with full history)
Rule Category	Products

BR-015 — FIFO Rotation for Standard Goods

Attribute	Detail
Rule Name	FIFO Rotation for Standard Goods

Attribute	Detail
Business Rule Statement	First-in, first-out (FIFO) rotation is required for non-perishable standard goods. Oldest received stock must be picked and shipped before newer stock for the same SKU.
Business Reason	Oldest stock should move first to minimize obsolescence, packaging degradation, and model-year expiration. Consistent FIFO prevents aged inventory from sitting indefinitely while new stock cycles normally.
Related Business Process	BP-09 (Stock Put-Away & Storage), BP-18 (Pick, Pack & Ship)
Related Functional Requirements	FR-044 (Support pick, pack, and ship workflow), FR-029 (Record all inventory movements with full history)
Rule Category	Products

BR-016 — Minimum Remaining Shelf Life at Receipt

Attribute	Detail
Rule Name	Minimum Remaining Shelf Life at Receipt
Business Rule Statement	Products with shelf-life requirements must meet minimum remaining shelf life (MRSL) at receipt. Inbound deliveries that do not meet MRSL policy must be rejected or dispositioned before acceptance into sellable inventory.
Business Reason	Receiving product too close to expiration guarantees write-off before sale. Minimum remaining shelf life requirements protect the business from accepting stock with insufficient selling window.
Related Business Process	BP-07 (Goods Receipt & Verification), BP-08 (Receipt Discrepancy Resolution)
Related Functional Requirements	FR-015 (Receive and verify inbound goods), FR-016 (Document and resolve receipt discrepancies)
Rule Category	Products

BR-017 — Lot or Batch Capture for Traceable Products

Attribute	Detail
Rule Name	Lot or Batch Capture for Traceable Products
Business Rule Statement	Lot or batch identity must be captured for traceable products at receipt and through all subsequent inventory movements. Traceable products must not be received, transferred, or shipped without lot or batch attribution.
Business Reason	Lot capture at every transaction point enables targeted recall, quality investigation, and regulatory compliance. Without lot data, the only option in a defect scenario is a full product-line recall.
Related Business Process	BP-07 (Goods Receipt & Verification), BP-18 (Pick, Pack & Ship), BP-25 (Inter-Location Transfer Execution)
Related Functional Requirements	FR-029 (Record all inventory movements with full history), FR-015 (Receive and verify inbound goods)
Rule Category	Products

Inventory Control and Governance

BR-018 — Blind Cycle Counting

Attribute	Detail
Rule Name	Blind Cycle Counting

Attribute	Detail
Business Rule Statement	Cycle counts must be performed blind. Counters must not be shown the system's expected quantity before recording the physical count result.
Business Reason	Counters who see the system's expected quantity before counting may consciously or unconsciously match it, defeating the purpose of verification. Blind counting produces unbiased accuracy measurement.
Related Business Process	BP-11 (Cycle Counting)
Related Functional Requirements	FR-031 (Perform blind cycle counts), FR-030 (Schedule and assign cycle counts)
Rule Category	Inventory

BR-019 — Independent Count Approval

Attribute	Detail
Rule Name	Independent Count Approval
Business Rule Statement	The person who performed a count must not approve adjustments arising from their own count. Count variances require review and approval by an independent authorized person.
Business Reason	Segregation of duties prevents fraud and undetected error. When the same person counts, investigates, and approves, there is no independent verification of the variance or its cause.
Related Business Process	BP-13 (Count Variance Investigation & Approval)
Related Functional Requirements	FR-034 (Investigate and approve count adjustments), FR-084 (Enforce segregation of duties)
Rule Category	Users

BR-020 — Mandatory Adjustment Reason Codes

Attribute	Detail
Rule Name	Mandatory Adjustment Reason Codes
Business Rule Statement	Every inventory adjustment requires a reason code. No stock correction, increase, or decrease may be applied without a categorized business reason.
Business Reason	Adjustments without categorized reasons are analytically useless. The business knows quantity changed but not why — preventing shrinkage analysis, trend identification, and targeted corrective action.
Related Business Process	BP-14 (Stock Adjustment & Write-Off), BP-15 (Shrinkage & Loss Management)
Related Functional Requirements	FR-035 (Process stock adjustments with mandatory reason codes), FR-011 (Maintain standard reason codes)
Rule Category	Inventory

BR-021 — Independent Approval for Material Adjustments

Attribute	Detail
Rule Name	Independent Approval for Material Adjustments
Business Rule Statement	Adjustments above the defined financial or quantity threshold require independent approval by an authorized person who did not initiate the underlying transaction.
Business Reason	High-value or high-quantity adjustments carry greater financial and fraud risk. Independent approval by an authorized person who did not perform the underlying transaction provides oversight proportional to the impact.
Related Business Process	BP-14 (Stock Adjustment & Write-Off), BP-13 (Count Variance Investigation & Approval)

Attribute	Detail
Related Functional Requirements	FR-037 (Enforce adjustment approval thresholds), FR-084 (Enforce segregation of duties)
Rule Category	Users

BR-022 — Investigate Before Adjusting

Attribute	Detail
Rule Name	Investigate Before Adjusting
Business Rule Statement	Count and inventory variances must be investigated before an adjustment is applied. Adjustments may only be approved after root cause analysis or documented investigation effort.
Business Reason	Adjustments fix the record; investigation fixes the process. Applying adjustments without understanding why the variance occurred guarantees the same discrepancy will reappear on the next count.
Related Business Process	BP-13 (Count Variance Investigation & Approval)
Related Functional Requirements	FR-033 (Identify and report count variances), FR-034 (Investigate and approve count adjustments)
Rule Category	Inventory

BR-023 — Documented Write-Off Authorization

Attribute	Detail
Rule Name	Documented Write-Off Authorization
Business Rule Statement	Inventory write-off requires documented evidence and authorization. Write-offs must include supporting documentation and approval by an authorized person before inventory value is removed.
Business Reason	Write-offs permanently remove inventory value from the business. Unauthorized or undocumented write-offs enable fraud, inflate remaining inventory inaccurately, and create audit exposure.
Related Business Process	BP-14 (Stock Adjustment & Write-Off), BP-15 (Shrinkage & Loss Management)
Related Functional Requirements	FR-036 (Process inventory write-offs with authorization), FR-011 (Maintain standard reason codes)
Rule Category	Users

BR-024 — Physical Inventory Freeze

Attribute	Detail
Rule Name	Physical Inventory Freeze
Business Rule Statement	A physical inventory freeze applies during a full wall-to-wall count. Receiving, shipping, transfers, and adjustments must be suspended for the counted location until the count is completed and verified.
Business Reason	Movements during a full count change the quantity being counted, making the snapshot unreliable. A freeze ensures the counted quantity is stable and auditable.
Related Business Process	BP-12 (Full Physical Inventory Count)
Related Functional Requirements	FR-032 (Conduct full physical inventory count), FR-033 (Identify and report count variances)
Rule Category	Warehousing

BR-032 — Class A Priority Monitoring

Attribute	Detail
Rule Name	Class A Priority Monitoring
Business Rule Statement	Class A products require the highest count frequency and monitoring intensity. Priority SKUs must receive more frequent cycle counts, availability monitoring, and replenishment attention than lower-classification products.
Business Reason	Not all products warrant equal operational effort. Class A items drive the majority of revenue and customer impact. Higher count frequency and monitoring intensity on A items prevents stockouts and accuracy failures where they matter most.
Related Business Process	BP-04 (ABC Product Classification), BP-11 (Cycle Counting), BP-21 (Low-Stock Alert Review)
Related Functional Requirements	FR-009 (Classify products by ABC priority), FR-030 (Schedule and assign cycle counts), FR-051 (Generate low-stock and out-of-stock alerts)
Rule Category	Products

Returns and Commercial

BR-025 — Return Authorization Required

Attribute	Detail
Rule Name	Return Authorization Required
Business Rule Statement	Customer returns require authorization before acceptance. Returned goods must not be received at the warehouse without a valid return authorization linked to the original order and return policy.
Business Reason	Unauthorized returns cannot be linked to original orders, may violate return policy, and give the warehouse no advance notice. Authorization connects the physical return to the commercial transaction and expected resolution.
Related Business Process	BP-27 (Customer Return Intake)
Related Functional Requirements	FR-064 (Authorize and record customer returns), FR-067 (Track return processing status)
Rule Category	Inventory

BR-026 — Inspect Returns Before Restocking

Attribute	Detail
Rule Name	Inspect Returns Before Restocking
Business Rule Statement	Returned goods must be inspected before restocking. Products may only be returned to sellable inventory after condition assessment confirms they meet resale quality standards.
Business Reason	Not all returns are resalable. Used, damaged, or defective products placed back in sellable inventory will be shipped to the next customer as new — generating repeat returns, complaints, and brand damage.
Related Business Process	BP-28 (Return-to-Stock or Disposal Decision)
Related Functional Requirements	FR-065 (Receive and inspect returned goods), FR-066 (Restock or write off returned inventory)
Rule Category	Warehousing

BR-027 — No Reorder of Discontinued Products

Attribute	Detail
Rule Name	No Reorder of Discontinued Products
Business Rule Statement	Discontinued products must not be reordered. Once a product is marked discontinued, new purchase orders and replenishment requisitions for that SKU are prohibited.
Business Reason	Discontinued status means the business has decided to exit the product. Reordering discontinued SKUs creates obsolete inventory that must eventually be written off, and conflicts with supersession plans.
Related Business Process	BP-01 (Product Master Data Management), BP-22 (Replenishment Planning & Requisition)
Related Functional Requirements	FR-002 (Manage product status lifecycle), FR-054 (Create and track purchase requisitions)
Rule Category	Products

BR-029 — Promotional Stock Ring-Fencing

Attribute	Detail
Rule Name	Promotional Stock Ring-Fencing
Business Rule Statement	Promotional stock must be ring-fenced during a campaign. Inventory allocated to a promotion or launch must not be consumed by non-promotional orders without commercial leadership approval.
Business Reason	Promotional inventory is positioned for a specific campaign period and channel. Allowing non-promotional orders to consume promotional stock defeats the campaign plan and leaves promoted channels without supply.
Related Business Process	BP-31 (Promotional & Launch Stock Readiness)
Related Functional Requirements	FR-057 (Support promotional stock readiness review), FR-080 (Deliver campaign and launch readiness alerts)
Rule Category	Products

Ownership and Trade

BR-030 — Bonded Inventory Customs Clearance

Attribute	Detail
Rule Name	Bonded Inventory Customs Clearance
Business Rule Statement	Bonded inventory must not be released into domestic sellable stock without customs clearance. Duties and regulatory release must be confirmed before bonded goods are made available for sale or fulfillment.
Business Reason	Bonded warehouse stock is under customs control. Duties and taxes have not been paid. Releasing bonded goods into domestic commerce without customs authorization is illegal and creates financial and regulatory liability.
Related Business Process	BP-07 (Goods Receipt & Verification), BP-09 (Stock Put-Away & Storage)
Related Functional Requirements	FR-038 (Manage quarantine and hold stock status), FR-018 (Manage put-away and stock availability timing)
Rule Category	Purchasing

BR-031 — Consignment Ownership at Consumption

Attribute	Detail
Rule Name	Consignment Ownership at Consumption
Business Rule Statement	Consignment stock is not owned by the organization until a defined consumption or sale event occurs per supplier agreement. Consignment inventory must not be treated as owned stock for valuation or fulfillment until ownership transfers.
Business Reason	Consignment inventory is physically at the buyer's location but owned by the supplier until the buyer consumes, sells, or issues it per agreement. Treating consignment as owned inventory overstates assets and triggers incorrect financial recording.
Related Business Process	BP-30 (Inventory Valuation & Financial Reconciliation), BP-07 (Goods Receipt & Verification)
Related Functional Requirements	FR-075 (Support inventory valuation summary views), FR-089 (Support financial inventory reconciliation)
Rule Category	Purchasing

Rules Summary by Category

Category	Rule IDs	Count
Inventory	BR-001, BR-002, BR-009, BR-010, BR-011, BR-013, BR-018, BR-020, BR-022, BR-025, BR-028	11
Purchasing	BR-006, BR-007, BR-008, BR-030, BR-031	5
Warehousing	BR-003, BR-004, BR-005, BR-012, BR-024, BR-026	6
Products	BR-014, BR-015, BR-016, BR-017, BR-027, BR-029, BR-032	7
Users	BR-019, BR-021, BR-023	3
Reporting	—	0
Total	BR-001 through BR-032	**32**

Note: Reporting as a category applies to business oversight processes (BP-29, BP-32) that consume rule compliance data rather than standalone inventory policy rules.

Rules by Business Process Domain

Process Domain	Related Rules
Master Data & Governance (BP-01 – BP-04)	BR-027, BR-032
Inbound & Receiving (BP-05 – BP-09)	BR-004, BR-005, BR-006, BR-007, BR-008, BR-016, BR-017, BR-030
Inventory Control (BP-10 – BP-15)	BR-013, BR-018, BR-019, BR-020, BR-021, BR-022, BR-023, BR-024
Order Fulfillment (BP-16 – BP-20)	BR-001, BR-002, BR-003, BR-028
Replenishment (BP-21 – BP-23)	BR-009, BR-010, BR-011
Network Balance (BP-24 – BP-26)	BR-011, BR-012
Returns (BP-27 – BP-28)	BR-025, BR-026
Performance & Financial (BP-29 – BP-30)	BR-008, BR-031
Commercial & Governance (BP-31 – BP-32)	BR-029, BR-019, BR-021, BR-023

Interdependent Rule Combinations

The BRD identifies the following rule combinations that must operate together to be effective:

Combination	Business Purpose
BR-001 + BR-002	ATP defines what is available; reservation locks it against confirmed commitments
BR-004 + BR-006	Verification confirms what arrived; documentation preserves supplier accountability
BR-018 + BR-019	Blind counting detects variance; independent approval prevents fraud
BR-020 + BR-022	Investigation finds root cause; reason code captures it for analysis
BR-014 + BR-016	FEFO manages rotation; MRSL prevents accepting stock too close to expiry
BR-011 + BR-012	Network check enables transfer; source protection prevents new shortages

Traceability Reference

Source	Identifier
Business Requirements Document	BRD-IMS-001, Section 13
Industry Business Rules (Domain Knowledge)	32 operational rules
Functional Requirements	FR-001 – FR-090
Business Processes	BP-01 – BP-32
Separation of Duties	SOD-003, SOD-004, SOD-006, SOD-007 (support BR-019, BR-021)

Document Approval

Role	Name	Signature	Date
Senior Business Analyst			
Product Owner			
Operations Director			
Inventory Controller			

End of Business Rules Specification

Document ID: SRS-BR-001 | Version 1.0 | July 5, 2026

Validation Rules Specification

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-VR-001
Document Title	Validation Rules — Inventory Management System
Version	1.0

Attribute	Detail
Status	Draft
Document Owner	Senior Business Analyst
Source Documents	BRD-IMS-001, Functional Requirements (FR-001–FR-090), Business Rules (BR-001–BR-032), Acceptance Criteria
Date	July 5, 2026
Scope	Data validation and business validation only — no implementation, API, or database design

Purpose

This document defines validation rules that must be satisfied when users interact with inventory management capabilities. Each rule is tied to a functional requirement and feature. Rules express *what* must be validated from a business perspective, not *how* the system enforces it technically.

Validation Rule Fields

Field	Description
Validation ID	Unique identifier (VR-001, VR-002, ...)
Related Feature	Primary feature from Feature Catalog (F-01–F-30)
Related FR	Functional requirement identifier
Validation Rule	The condition that must be satisfied
Error Condition	When validation fails
User Message	Plain-language message presented to the user
Business Reason	Why the rule exists

Domain 1: Master Data & Governance

FR-001 — Create and Maintain Product Master Records (F-01)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-001	F-01	SKU, product name, and unit of measure are required on every product record.	Any required attribute is missing or blank.	Required product information is missing. Enter SKU, product name, and unit of measure before saving.	Incomplete product records cause ordering, receiving, and fulfillment errors across the network.
VR-002	F-01	SKU must be unique across the product catalog.	Submitted SKU already exists on another active or inactive product.	This SKU already exists. Use a unique SKU or update the existing product record.	Duplicate SKUs fragment stock visibility and undermine a single product identity (BR-001 context).
VR-003	F-01	Only authorized users may create or modify product master records.	User lacks product maintenance authority.	You are not authorized to create or change product records.	Product master data quality requires controlled ownership by the Inventory Controller function.

FR-002 — Manage Product Status Lifecycle (F-01)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-004	F-01	Discontinued products must not be eligible for new replenishment or purchase.	User attempts to reorder or replenish a discontinued product.	This product is discontinued and cannot be reordered.	Prevents obsolete inventory accumulation and conflicting assortment decisions (BR-027).

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-005	F-01	Product status changes must record prior status, new status, user, and date.	Status change submitted without auditable attribution.	Product status could not be saved. A valid status and authorization are required.	Status history supports lifecycle accountability and audit review.
VR-006	F-01	Inactive or discontinued products must be blocked from new reservations unless policy explicitly permits.	Reservation or fulfillment attempted on non-active product without exception.	This product is not active and cannot be committed to new orders.	Prevents customer commitments against products the business has withdrawn from sale.

FR-003 — Support Product Identifier Lookup (F-01)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-007	F-01	Lookup by SKU, barcode, or assigned identifier must match an existing product record.	Identifier does not match any product.	No product found for this identifier. Verify the SKU or barcode and try again.	Prevents implicit creation of products during operational transactions.
VR-008	F-01	Stock context shown during lookup must respect the user's authorized location scope.	User requests stock detail for an unauthorized location.	You do not have access to inventory at this location.	Aligns visibility with operational accountability and location ownership.

FR-004 — Detect and Resolve Duplicate Product Records (F-01)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-009	F-01	New product submissions must be checked for matching or conflicting SKU, barcode, or highly similar name.	Potential duplicate detected and user proceeds without resolution.	A similar product already exists. Review the existing record before creating a new product.	Prevents parallel identities for the same physical product (PP-07).
VR-010	F-01	Merging duplicate products requires Inventory Controller authorization when both records have on-hand stock.	Unauthorized merge attempted on stocked products.	Product merge requires Inventory Controller approval when stock exists on both records.	Protects stock integrity during catalog consolidation.

FR-005 — Configure and Maintain Inventory Locations (F-02)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-011	F-02	Location name, type, and operational status are required.	Required location attributes missing.	Required location information is missing. Enter name, type, and status before saving.	Locations are the foundation for all stock assignment and reporting.
VR-012	F-02	Deactivated locations must not accept new receipts, reservations, or fulfillments except authorized disposition activity.	Transaction attempted at deactivated location.	This location is not active for new inventory activity.	Prevents stock and transactions at locations that are closed or not yet ready.
VR-013	F-02	Location activation for transactions requires approved opening balance when go-live policy applies.	Operational transaction at location without approved opening balance.	This location cannot process transactions until opening stock is verified and approved.	Go-live with unverified balances embeds errors that undermine trust (BA-16).

FR-006 — Manage Location Hierarchy (F-02)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-014	F-02	A location cannot be assigned as its own parent or ancestor.	Circular hierarchy proposed.	Invalid hierarchy: a location cannot be placed under itself or its own child.	Circular hierarchies break network roll-up and reporting.
VR-015	F-02	Each location may have only one parent in the hierarchy at a time.	Multiple conflicting parent assignments.	This location already belongs to another hierarchy branch. Resolve the assignment before saving.	Ensures unambiguous network structure for accountability and reporting.

FR-007 — Plan Location Closure and Stock Disposition (F-02)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-016	F-02	Location closure must be blocked while on-hand stock remains unless a disposition plan is documented or explicitly waived by an authorized approver.	Closure attempted with remaining stock and no plan.	This location still holds inventory. Complete stock disposition or obtain closure approval before deactivating.	Prevents stranded inventory and orphaned records when locations exit the network.
VR-017	F-02	New inbound and fulfillment activity must be blocked at a location in closure status except authorized disposition transactions.	Standard operational transaction at closing location.	This location is closing. Only authorized disposition activities are permitted.	Protects closure integrity while stock is being cleared.

FR-008 — Define Stock Policy and Replenishment Thresholds (F-03)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-018	F-03	Reorder level, minimum threshold, and safety stock must be zero or positive numeric values.	Negative or non-numeric threshold entered.	Threshold values must be zero or greater. Enter valid numbers for all thresholds.	Invalid thresholds produce false alerts or missed replenishment signals.
VR-019	F-03	Minimum threshold must not exceed reorder level unless business policy explicitly allows inverted configuration.	Illogical threshold combination submitted.	Minimum threshold cannot be higher than reorder level. Adjust values to match stock policy.	Illogical thresholds cause incorrect low-stock detection and replenishment timing (BR-009).
VR-020	F-03	Threshold changes must include an effective date and retain prior values in history.	Threshold saved without effective date or history loss.	Stock policy could not be saved. An effective date is required for threshold changes.	Historical thresholds explain past alert behavior and support audit review.

FR-009 — Classify Products by ABC Priority (F-03)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-021	F-03	ABC classification must be one of: A, B, or C.	Invalid classification value submitted.	Invalid product class. Select A, B, or C.	Standard classification enables consistent operational prioritization (BR-032).
VR-022	F-03	Classification may only be assigned to active or inactive products, not to invalid or unknown product references.	Classification applied to non-existent product.	Product not found. Classification can only be applied to valid product records.	Ensures classification drives count and alert rules for real catalog items.

FR-010 — Establish Opening Stock Balance at Location Go-Live (F-02)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-023	F-02	Opening balance quantities must be non-negative numbers for each counted SKU.	Negative or non-numeric opening quantity entered.	Opening quantity must be zero or greater. Correct invalid count entries before submitting.	Negative opening balances corrupt inventory from go-live.
VR-024	F-02	Every opening balance SKU must exist in the product catalog before location activation.	Opening balance references unknown SKU.	Unknown product in opening balance. Resolve catalog gaps before activating this location.	Opening balances must tie to a single trusted product identity.
VR-025	F-02	Location activation requires authorized approval of the complete opening balance package.	Activation attempted before approver sign-off.	Opening balance must be approved before this location can go live.	Verified baselines are required for credible inventory control at cutover.

FR-011 — Maintain Standard Reason Codes (F-17)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-026	F-17	Adjustments, write-offs, and receipt discrepancies require selection of an active reason code before submission.	Transaction submitted without reason code.	A reason code is required. Select the reason for this inventory change.	Uncategorized adjustments prevent shrinkage analysis and audit substantiation (BR-020).
VR-027	F-17	Retired reason codes must not be selectable for new transactions.	Retired code used on new transaction.	This reason code is no longer active. Select a current reason code.	Retired codes remain on history only; active list reflects current policy.

FR-012 — Manage Inventory Governance Policies (F-03)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-028	F-03	Only one governance policy version may be effective for a given parameter at any point in time.	Overlapping effective policy versions detected.	Conflicting policy versions exist. Publish a single effective policy before proceeding.	Ambiguous policy causes inconsistent approvals and control failures.
VR-029	F-03	Policy parameters referenced by workflows must be defined before dependent transactions are processed.	Transaction evaluated against undefined policy value.	Required business policy is not configured. Contact operations leadership to define policy values.	Prevents silent defaults on financially material decisions.

Domain 2: Inbound & Receiving

FR-013 — Record Purchase Order Commitments (F-07)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-030	F-07	Purchase order must include product, ordered quantity, supplier reference, receiving location, and expected delivery date.	Required PO field missing.	Required purchase order information is missing. Complete all mandatory fields before saving.	Incomplete POs cannot support receiving verification or inbound visibility.
VR-031	F-07	PO lines must reference active products and active receiving locations.	Discontinued product or deactivated location on PO line.	This product or location is not valid for ordering. Select an active product and receiving location.	Prevents procurement against invalid catalog or network entities.

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-032	F-07	Ordered quantity must be greater than zero.	Zero or negative PO quantity.	Order quantity must be greater than zero.	Zero-quantity orders add no inbound supply value and corrupt replenishment planning.

FR-014 — Plan and Communicate Expected Inbound Deliveries (F-07)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-033	F-07	Expected delivery date must be a valid calendar date not in the distant past for new schedules.	Invalid or unreasonably past delivery date entered.	Enter a valid expected delivery date.	Invalid dates undermine receiving preparation and availability projections.
VR-034	F-07	Inbound schedule views must be limited to locations the user is authorized to access.	User views unauthorized location schedule.	You do not have access to inbound schedules for this location.	Protects location-scoped operational data.

FR-015 — Receive and Verify Inbound Goods (F-05)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-035	F-05	Accepted receipt quantity must be greater than zero and must not exceed physically verified quantity.	Zero, negative, or unsubstantiated accepted quantity.	Enter a valid accepted quantity greater than zero.	Recording unverified quantities creates phantom inventory (BR-004).
VR-036	F-05	Received product must match the PO line product identity.	Product on receipt does not match PO line.	Product received does not match the purchase order. Correct the product or document a discrepancy.	Wrong-item receipt corrupts inventory and supplier accountability.
VR-037	F-05	Only verified accepted quantity may increase on-hand balance; rejected or damaged quantity must be recorded separately.	Full PO quantity accepted when portion is damaged or rejected.	Record accepted and rejected quantities separately. Only accepted stock may enter inventory.	Separates sellable stock from disputed or unsellable goods (BR-006).

FR-016 — Document and Resolve Receipt Discrepancies (F-05)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-038	F-05	Open discrepancies must be documented with type, variance quantity, and notes before disputed quantity is finalized.	Discrepancy closed without required documentation.	Document the discrepancy type and quantity before resolving this receipt.	Undocumented discrepancies waive supplier accountability (BR-006).
VR-039	F-05	Disputed quantity must remain excluded from sellable availability until resolution is approved.	Disputed stock made available before resolution.	Disputed quantity cannot be made available until the discrepancy is resolved.	Prevents fulfillment from stock that is not confirmed received.

FR-017 — Enforce Purchase Order Reference for Supplier Receipts (F-05)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-040	F-05	Standard supplier receipts require a valid open purchase order reference.	Receipt attempted without PO or valid ad-hoc exception.	A valid purchase order is required to receive this shipment. Obtain PO confirmation or authorized ad-hoc approval.	Maintains procurement control and supports financial matching (BR-007).

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-041	F-05	Receipt against a PO line must not exceed remaining open quantity plus approved tolerance.	Receipt quantity exceeds allowable open PO quantity.	Received quantity exceeds the remaining purchase order quantity. Adjust quantity or obtain over-receipt approval.	Prevents silent over-receipt that affects payment and inventory accuracy.
VR-042	F-05	Ad-hoc receipts require documented approver and business reason.	Ad-hoc receipt without authorization.	Emergency receipt requires authorized approval and a documented reason.	Authorized exceptions only; unaudited receipts break financial controls.

FR-018 — Manage Put-Away and Stock Availability Timing (F-06)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-043	F-06	Put-away quantity must not exceed received quantity still in receiving status.	Put-away total exceeds received quantity.	Put-away quantity cannot exceed the quantity received. Verify counts before confirming put-away.	Prevents availability inflation beyond physically received stock (BR-005).
VR-044	F-06	ATP must not increase until put-away is confirmed for the corresponding quantity.	Availability updated before put-away confirmation.	Stock is not yet available. Complete put-away before this quantity can be promised or picked.	Stock in the receiving area is not operationally locatable (BR-005).

FR-019 — Receive Return and Non-PO Inbound Stock (F-05)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-045	F-05	Customer return receipts require a valid return authorization reference.	Return received without valid authorization.	A valid return authorization is required to accept this return.	Links physical returns to commercial transactions (BR-025).
VR-046	F-05	Return receipt quantity must not exceed authorized return quantity.	Received return quantity exceeds authorization.	Received quantity exceeds the authorized return amount. Verify the return authorization.	Prevents acceptance of unauthorized or excess returns.
VR-047	F-05	Sellable ATP must not increase from return receipt until inspection passes when inspection is required.	Restock to available before inspection completion.	Returned goods must pass inspection before being added to sellable stock.	Protects customers from receiving used or damaged goods as new (BR-026).

FR-020 — Support Receipt Quantity Tolerance Evaluation (F-05)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-048	F-05	Receipt variances within defined tolerance may be auto-accepted; variances beyond tolerance require approver action.	Over- or under-receipt beyond tolerance accepted without approval.	Receipt variance exceeds allowed tolerance. Supervisor approval is required.	Balances operational flexibility with control over material variances.
VR-049	F-05	Tolerance decision must be recorded on the receipt record.	Tolerance applied without audit record.	Receipt could not be finalized. Tolerance evaluation must be documented.	Supports supplier performance tracking and payment dispute evidence.

FR-021 — Cancel or Amend Open Purchase Orders (F-07)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-050	F-07	Fully received PO lines must not be cancelled or reduced below received quantity.	Amendment would reduce quantity below already received amount.	Cannot amend below quantity already received. Adjust the change to reflect actual receipts.	Protects integrity between PO commitment and receipt history.
VR-051	F-07	PO cancellation or material amendment requires authorized purchasing approval.	Unauthorized user amends or cancels PO.	You are not authorized to change this purchase order.	Prevents unauthorized changes to supplier commitments.

FR-022 — Record Supplier Delivery Performance (F-07)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-052	F-07	Performance metrics must be derived only from completed receipts and resolved discrepancies tied to valid PO records.	Performance calculated from incomplete or unlinked transactions.	Supplier performance cannot be calculated for this period. Complete receipts and resolve discrepancies first.	Ensures supplier scorecards reflect actual delivery outcomes.

Domain 3: Inventory Visibility & Status

FR-023 — Display On-Hand Quantity (F-04)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-053	F-04	On-hand display must reflect only locations within the user's authorized scope.	Unauthorized location data requested.	You do not have access to on-hand data for this location.	Location-scoped access aligns with operational accountability.
VR-054	F-04	Inquiry must reference a valid product or valid filter criteria.	Empty or invalid product/location selection.	Select a valid product or location to view on-hand quantity.	Prevents ambiguous queries that could mislead decisions.

FR-024 — Display Available-to-Promise Quantity (F-04)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-055	F-04	ATP must equal on-hand minus reservations and non-sellable holds including quarantined stock.	ATP displayed inconsistent with underlying availability components.	Availability cannot be displayed. Stock status data is incomplete.	ATP is the only honest quotable quantity (BR-001).
VR-056	F-04	Quarantined and held stock must be excluded from ATP in all availability views.	Quarantined quantity included in ATP.	Quarantined stock is not available for sale.	Prevents promising stock under quality or regulatory hold (BR-013).

FR-025 — Display Reserved and Committed Stock (F-04)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-057	F-04	Reserved quantity must not exceed on-hand quantity at the same product and location.	Reserved balance greater than on-hand.	Reserved quantity cannot exceed on-hand stock. Escalate to operations for investigation.	Excess reservation indicates data integrity failure or overselling risk.

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-058	F-04	Reservation display must include linked order reference for each reserved quantity.	Reservation shown without order linkage.	Reservation detail is incomplete. Order reference is required for committed stock display.	Supports fulfillment planning and dispute resolution between sales and warehouse.

FR-026 — Indicate Stock Status (F-04)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-059	F-04	Stock status must be derived from ATP relative to defined reorder and minimum thresholds.	Status shown without configured thresholds for product-location.	Stock status is unavailable. Define thresholds for this product at this location.	Status indicators require policy thresholds to be meaningful.
VR-060	F-04	Out-of-stock status applies when ATP is zero or below minimum sellable unit.	Positive ATP shown as out of stock, or zero ATP shown as in stock.	Stock status does not match current availability. Refresh or escalate data review.	Incorrect status drives wrong replenishment and sales decisions.

FR-027 — Search and Filter Inventory Views (F-04)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-061	F-04	Search results must respect user location access permissions.	Results include unauthorized locations.	Some results are hidden based on your location access.	Prevents cross-location data exposure to unauthorized users.
VR-062	F-04	Date-bounded filters must have a valid start date on or before end date.	Invalid date range entered.	Enter a valid date range. Start date must be on or before end date.	Invalid ranges produce empty or misleading inventory views.

FR-028 — Display Expected Inbound and Network-Wide Availability (F-04)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-063	F-04	Expected inbound quantities must include only open PO lines not yet fully received.	Closed or fully received PO lines counted as inbound.	Inbound projection unavailable. Verify purchase order status.	Accurate inbound projection supports transfer-before-purchase decisions (BR-011).
VR-064	F-04	Network ATP aggregation must sum only authorized locations.	Unauthorized locations included in network total.	Network availability is limited to locations you are authorized to view.	Network views still require appropriate access scope.

Domain 4: Inventory Control & Accuracy

FR-029 — Record All Inventory Movements (F-05)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-065	F-05	Every balance-changing transaction must record product, quantity, location, transaction type, user, and timestamp.	Movement recorded without required audit attributes.	Transaction could not be completed. Required movement details are missing.	Complete movement history is required for accountability (BO-003, BO-019).
VR-066	F-05	Movement quantity must be non-zero.	Zero-quantity movement submitted.	Quantity must be greater than zero for inventory movements.	Zero movements add noise without business meaning.

FR-030 — Schedule and Assign Cycle Counts (F-14)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-067	F-14	Count assignments must reference valid active products at valid active locations.	Assignment to invalid product or location.	Count assignment is invalid. Select an active product and location.	Counts on invalid entities waste effort and produce no control value.
VR-068	F-14	Class A products must be scheduled at higher frequency than Class C per ABC policy when classification exists.	Class A scheduled less frequently than Class C without authorized exception.	Class A products require more frequent counting per policy. Adjust the count schedule.	Focuses accuracy effort where business impact is highest (BR-032).

FR-031 — Perform Blind Cycle Counts (F-14)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-069	F-14	Counter must not see system expected quantity during an active blind count.	Expected quantity visible before count submission.	Blind count in progress. System quantity is hidden until you submit your count.	Blind counting prevents biased confirmation of bad data (BR-018).
VR-070	F-14	Physical count quantity must be a non-negative number.	Negative or non-numeric count entered.	Enter a valid physical count quantity of zero or greater.	Invalid entries cannot support variance analysis.

FR-032 — Conduct Full Physical Inventory Count (F-15)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-071	F-15	Movement freeze must be in effect at the location during an active full wall-to-wall count.	Receipt, shipment, transfer, or adjustment attempted during freeze.	Inventory count in progress. Movements are frozen until the count is completed.	Movements during count invalidate the snapshot (BR-024).
VR-072	F-15	Full count must cover all SKUs with on-hand or receiving status at the location before variances are finalized.	Count finalized with uncounted SKUs remaining.	Full count is incomplete. All products must be counted before closing the count.	Partial counts cannot establish a reliable location baseline.

FR-033 — Identify and Report Count Variances (F-16)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-073	F-16	Variance must be calculated as physical count minus system quantity after count submission.	Variance calculated before count submission or with visible expected quantity in blind count.	Variance cannot be calculated until the physical count is submitted.	Ensures unbiased variance measurement (BR-018).
VR-074	F-16	Variances exceeding defined tolerance must be routed for investigation before adjustment.	Within-tolerance auto-adjustment applied to material variance, or material variance ignored.	Count variance exceeds tolerance and requires investigation.	Focuses correction effort on material discrepancies (BR-022).

FR-034 — Investigate and Approve Count Adjustments (F-16)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-075	F-16	Counter must not approve adjustments arising from their own count.	Same user submits count and approves adjustment.	You cannot approve an adjustment for a count you performed. An independent approver is required.	Segregation of duties prevents fraud and undetected error (BR-019).

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-076	F-16	Material variances require investigation notes before approval.	Adjustment approved without investigation documentation.	Document investigation findings before approving this count adjustment.	Adjustments fix records; investigation fixes processes (BR-022).

FR-035 — Process Stock Adjustments (F-17)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-077	F-17	Adjustments require product, location, quantity, direction, and active reason code.	Missing mandatory adjustment attribute.	Required adjustment information is missing. Complete product, location, quantity, and reason code.	Incomplete adjustments are unauditable and analytically useless (BR-020).
VR-078	F-17	Decrease adjustments must not exceed available on-hand at the product and location.	Adjustment would drive negative on-hand.	Adjustment quantity exceeds available on-hand stock.	Negative on-hand indicates invalid correction or requires write-off workflow.
VR-079	F-17	User must be authorized for adjustment at the specified location.	Unauthorized location adjustment attempted.	You are not authorized to adjust inventory at this location.	Location-scoped accountability for stock corrections.

FR-036 — Process Inventory Write-Offs (F-17)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-080	F-17	Write-offs require documented evidence and approver per policy threshold.	Write-off submitted without evidence or required approval.	Write-off requires supporting evidence and authorized approval.	Unauthorized write-offs enable fraud and financial misstatement (BR-023).
VR-081	F-17	Write-off quantity must not exceed on-hand quantity at the location.	Write-off exceeds available stock.	Write-off quantity exceeds on-hand stock at this location.	Cannot write off stock that does not exist on record.

FR-037 — Enforce Adjustment Approval Thresholds (F-16)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-082	F-16	Adjustments above quantity or value threshold require independent approver who did not initiate the transaction.	Self-approved above-threshold adjustment.	This adjustment requires approval from an authorized independent approver.	Oversight proportional to financial impact (BR-021).
VR-083	F-16	Submitter must not be the final approver when segregation of duties policy applies.	Initiator and approver are the same person on restricted transaction.	You cannot approve your own adjustment. Route to an independent approver.	Core internal control requirement (BR-019).

FR-038 — Manage Quarantine and Hold Stock (F-18)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-084	F-18	Quarantined quantity must be immediately excluded from ATP.	Quarantined stock still available for reservation or fulfillment.	Quarantined stock cannot be sold or reserved.	Prevents shipping uninspected or defective goods (BR-013).
VR-085	F-18	Release from quarantine to sellable stock requires inspection outcome and authorized disposition decision.	Quarantine released without inspection decision.	Quarantined stock cannot be released without an approved inspection outcome.	Ensures only quality-approved stock returns to sellable inventory.

FR-039 — Analyze Shrinkage and Loss Trends (F-19)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-086	F-19	Shrinkage analysis periods must use valid start and end dates with start on or before end.	Invalid analysis period.	Select a valid reporting period for shrinkage analysis.	Invalid periods produce meaningless loss trends.
VR-087	F-19	Analysis must include only transactions with valid reason codes.	Uncategorized adjustments included in shrinkage totals without flag.	Shrinkage analysis excludes uncategorized adjustments. Review reason code compliance.	Reason codes enable root-cause analysis (BR-020).

FR-040 — Review Obsolete and Slow-Moving Stock (F-19)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-088	F-19	Aging threshold must be a positive number of days when configuring slow-moving criteria.	Invalid aging threshold.	Enter a valid number of days for slow-moving stock criteria.	Invalid thresholds misidentify obsolete inventory.
VR-089	F-19	Discontinued products with remaining on-hand must appear in obsolete review lists.	Discontinued stocked SKU omitted from review.	Discontinued products with remaining stock require disposition review.	Prevents trapped capital in products the business has exited (BR-027).

Domain 5: Order Fulfillment

FR-041 — Confirm Order Availability (F-10)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-090	F-10	Availability confirmation must compare requested quantity to ATP at the fulfillment location, not on-hand alone.	Commitment made using on-hand ignoring reservations.	Insufficient available stock. Only unreserved quantity can be promised to customers.	Prevents overselling against committed stock (BR-001).
VR-091	F-10	Confirmed delivery commitment must not exceed ATP plus approved inbound only when inbound policy allows inclusion.	Commitment exceeds realistic fulfillable quantity.	Requested quantity cannot be confirmed with current availability.	Aligns customer promises with fulfillable stock.

FR-042 — Reserve Inventory for Confirmed Orders (F-10)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-092	F-10	Reservation quantity must not exceed ATP at the reservation location.	Reservation exceeds available stock.	Cannot reserve more than available stock. Reduce quantity or select another location.	Prevents double-allocation and pick failures (BR-002).
VR-093	F-10	Reservation must tie to a confirmed order line with product, quantity, and location.	Reservation without valid order reference.	Reservation requires a confirmed order line with product and quantity.	Links operational holds to commercial commitments.
VR-094	F-10	Safety stock must not be consumed by reservation without authorized exception when safety stock policy applies.	Reservation breaches safety stock without approval.	This reservation would use safety stock. Authorized exception and replenishment plan required.	Preserves buffer against next disruption (BR-010).

FR-043 — Release Inventory Reservations (F-10)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-095	F-10	Released quantity must not exceed reserved quantity for the order line.	Release exceeds active reservation.	Release quantity exceeds reserved amount for this order line.	Prevents artificial inflation of ATP.
VR-096	F-10	Reservation release must record reason such as cancellation, quantity change, fulfillment, or policy expiry.	Release without documented reason.	Select a reason for releasing this reservation.	Supports audit trail and availability restoration analysis.

FR-044 — Pick, Pack, and Ship (F-11)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-097	F-11	Shipment must not be recorded until pick verification confirms correct product and quantity.	Shipment recorded without pick confirmation.	Confirm picked product and quantity before marking this order shipped.	Inventory must not be deducted before verified pick (BR-003).
VR-098	F-11	Shipped quantity must not exceed reserved quantity for the order line.	Ship quantity greater than reservation.	Shipped quantity cannot exceed reserved quantity. Adjust shipment or reservation.	Maintains alignment between commitment, reservation, and fulfillment.
VR-099	F-11	Picked product must match order line product unless authorized substitution is approved.	Wrong product picked and shipped without approval.	Picked product does not match the order line. Obtain authorized substitution or re-pick.	Prevents wrong-item shipments and customer disputes.

FR-045 — Handle Partial Fulfillment (F-12)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-100	F-12	Partial shipment quantity must be greater than zero and less than full order line quantity.	Partial ship of zero or full quantity without backorder handling.	Partial shipment must be between zero and the full order quantity.	Defines true partial fulfillment versus full ship or cancel.
VR-101	F-12	Backorder for remaining quantity requires documented customer acknowledgment before processing.	Backorder created without customer acknowledgment.	Customer acknowledgment is required before placing the remaining quantity on backorder.	Backorder is a mutual agreement, not an internal note (BR-028).

FR-046 — Manage Backorders (F-12)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-102	F-12	Backorder fulfillment must follow defined priority rules when stock becomes available.	Stock reserved for backorder out of priority sequence without override.	Backorder fulfillment must follow priority policy. Escalate for manual override if needed.	Ensures fair and policy-driven delayed order fulfillment.
VR-103	F-12	Backorder quantity must equal unfulfilled order line quantity after partial shipments.	Backorder quantity inconsistent with order remainder.	Backorder quantity does not match the remaining order quantity. Reconcile order lines.	Keeps customer expectation aligned with system records.

FR-047 — Handle Outbound Fulfillment Exceptions (F-13)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-104	F-13	Pick failure or damage exception must be documented with type and quantity before order status is updated.	Exception closed without documentation.	Document the fulfillment exception type and affected quantity.	Enables root-cause analysis and customer communication.

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-105	F-13	Inventory adjustment from pick failure requires reason code and may require approval per policy.	Record adjusted without reason or approval.	Inventory correction from this exception requires a reason code and may require approval.	Maintains accuracy after fulfillment disruptions.

FR-048 — Multi-Location Split Fulfillment (F-10)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-106	F-10	Total reserved quantity across locations must equal order line quantity.	Sum of location reservations does not match order line.	Split fulfillment reservations must total the full order line quantity.	Prevents under- or over-allocation on multi-location orders.
VR-107	F-10	Each location reservation must not exceed ATP at that location.	Location reservation exceeds local ATP.	Insufficient stock at {location}. Reduce reservation or choose another location.	Each fulfillment point must have real available stock.

FR-049 — Display Order Fulfillment Status (F-11)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-108	F-11	Order status inquiry requires valid order reference.	Invalid or unknown order reference.	Order not found. Verify the order reference and try again.	Prevents misleading status responses.
VR-109	F-11	Displayed shipped, reserved, and backordered quantities must reconcile to order line quantity.	Status quantities inconsistent.	Order status is temporarily unavailable. Quantities do not reconcile — escalate to operations.	Customer service requires trustworthy order status.

FR-050 — Authorized Product Substitution (F-13)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-110	F-13	Substitution must use only pre-approved alternate SKUs for the ordered product.	Non-approved substitute selected.	This product is not an approved substitute. Select an authorized alternate or cancel the line.	Controls acceptable replacements and margin impact.
VR-111	F-13	Substitution requires authorized approval and customer acceptance when policy requires.	Substitution without required approvals.	Product substitution requires authorized approval and customer acceptance.	Protects customer expectation and commercial policy.

Domain 6: Replenishment & Inbound Supply

FR-051 — Generate Low-Stock and Out-of-Stock Alerts (F-20)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-112	F-20	Alerts must fire when ATP is at or below reorder level or when ATP is zero.	Alert missing despite threshold breach.	Low-stock condition detected. Review replenishment for this product at this location.	Timely alerts prevent preventable stockouts (BR-009).
VR-113	F-20	Duplicate active alerts for the same product-location-condition must be suppressed until resolved or acknowledged.	Multiple identical open alerts created.	An active alert already exists for this product at this location.	Reduces alert fatigue and duplicate procurement action.

FR-052 — Review and Acknowledge Stock Alerts (F-20)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-114	F-20	Dismissed or deferred alerts require documented rationale.	Alert dismissed without reason.	Provide a reason for dismissing or deferring this stock alert.	Ensures alerts drive deliberate action or accountable deferral.
VR-115	F-20	Acknowledged alerts must record action taken: requisition, transfer, or dismiss.	Acknowledgment without action category.	Select the action taken for this alert before acknowledging.	Links alert response to downstream replenishment or transfer processes.

FR-053 — Generate Replenishment Suggestions (F-20)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-116	F-20	Replenishment suggestion must consider open PO inbound quantity before recommending new purchase.	Purchase suggested when sufficient inbound supply exists.	Open purchase orders already cover this need. Review inbound supply before ordering.	Avoids duplicate procurement investment (BR-011).
VR-117	F-20	Suggested order quantity must be greater than zero and based on defined threshold parameters.	Zero or negative suggestion generated.	Replenishment suggestion cannot be calculated. Verify stock policy thresholds.	Invalid suggestions would misdirect buyers.
VR-118	F-20	Network excess must be evaluated before purchase suggestion is finalized.	Purchase suggested while transferable network stock exists without documented transfer decline.	Stock is available at other locations. Consider a transfer before purchasing.	Transfer-before-buy discipline optimizes network inventory (BR-011).

FR-054 — Create and Track Purchase Requisitions (F-20)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-119	F-20	Requisition must include product, quantity, receiving location, and business justification.	Missing mandatory requisition field.	Required requisition information is missing. Complete product, quantity, location, and justification.	Formalizes replenishment handoff with accountability.
VR-120	F-20	Requisitions for discontinued products must be blocked.	Requisition created for discontinued SKU.	Cannot requisition a discontinued product.	Enforces product lifecycle policy (BR-027).
VR-121	F-20	Requisition approval must follow purchasing authority limits.	Requisition approved by unauthorized role.	This requisition requires approval from an authorized purchasing approver.	Maintains procurement financial controls.

FR-055 — Track Inbound Supply Status (F-07)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-122	F-07	Received quantity on PO lines must not exceed ordered quantity plus approved tolerance.	Cumulative receipt exceeds allowable PO quantity.	Received quantity exceeds the purchase order limit. Resolve over-receipt before closing the PO line.	Keeps inbound tracking aligned with supplier commitment.
VR-123	F-07	Overdue PO lines must be identifiable when expected date has passed without receipt.	Overdue PO not flagged.	Purchase order is past expected delivery. Update status or follow up with supplier.	Enables proactive response to replenishment delays.

FR-056 — Enforce Safety Stock Protection (F-03)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-124	F-03	Reservations and transfers that would reduce stock below safety level must be blocked or require authorized exception.	Safety stock breached without approval.	This action would use safety stock. Authorized exception and replenishment plan required.	Safety stock protects against demand and supply variability (BR-010).
VR-125	F-03	Safety stock exception must document approver, reason, and replenishment plan.	Exception granted without required documentation.	Safety stock exception requires approver, reason, and replenishment plan.	Exceptions without recovery plan guarantee the next stockout.

FR-057 — Promotional Stock Readiness Review (F-25)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-126	F-25	Campaign readiness review requires defined target quantity by SKU, location, and campaign start date.	Readiness review without campaign targets.	Campaign stock targets are not defined. Set promotional quantities before running readiness review.	Cannot assess readiness without defined requirements.
VR-127	F-25	Promotional ring-fenced quantity must not be consumed by non-promotional orders during campaign period.	General order consumes ring-fenced stock.	This quantity is reserved for the promotion and cannot be used for standard orders.	Protects campaign supply (BR-029).

Domain 7: Network Balance & Transfers

FR-058 — Request Inter-Location Transfers (F-08)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-128	F-08	Transfer request must include product, quantity, source location, destination location, and justification.	Missing mandatory transfer field.	Required transfer information is missing. Complete product, quantity, source, destination, and reason.	Incomplete requests cannot be evaluated for network impact.
VR-129	F-08	Source and destination locations must be different active locations.	Same location or inactive location selected.	Source and destination must be different active locations.	Transfers require distinct valid endpoints.
VR-130	F-08	Requested quantity must not exceed source ATP.	Transfer quantity exceeds source available stock.	Transfer quantity exceeds available stock at the source location.	Cannot transfer stock that is not available or is reserved.

FR-059 — Approve or Reject Transfer Requests (F-08)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-131	F-08	Rejection must include documented reason communicated to requester.	Transfer rejected without reason.	Provide a reason for rejecting this transfer request.	Requesters need rationale to pursue alternate replenishment paths.
VR-132	F-08	Approver must revalidate source ATP at approval time when stock may have changed since request.	Approval granted when source no longer has stock.	Source stock has changed. Revalidate availability before approving this transfer.	Prevents approval of transfers that can no longer be fulfilled.

FR-060 — Execute and Record Transfers (F-08)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-133	F-08	Dispatch quantity must not exceed approved transfer quantity.	Dispatch exceeds approval.	Dispatch quantity cannot exceed the approved transfer amount.	Execution must match authorized transfer.
VR-134	F-08	Source on-hand must decrease on dispatch and destination on-hand must increase only on verified receipt.	Destination increased before receipt or source not reduced on dispatch.	Transfer receipt must be confirmed at the destination before stock is added.	Prevents stock appearing in two places or lost in transit.
VR-135	F-08	Receipt variance at destination must trigger investigation before transfer is closed.	Transfer closed with unreconciled receipt variance.	Received quantity does not match dispatched quantity. Investigate before closing this transfer.	Maintains network accuracy and loss accountability.

FR-061 — Monitor In-Transit Inventory (F-08)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-136	F-08	In-transit quantity must be excluded from source ATP and destination on-hand until receipt confirmed.	In-transit counted as available at source or destination.	Stock in transit is not available at either location until receipt is confirmed.	Accurate availability during transfer lead time.
VR-137	F-08	Transfers exceeding defined in-transit age must be flagged as overdue.	Overdue in-transit transfer not escalated.	Transfer shipment is overdue. Investigate in-transit status.	Prevents silent loss of goods between locations.

FR-062 — Network Rebalancing Recommendations (F-09)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-138	F-09	Recommended transfer quantity must respect source minimum threshold and safety stock rules.	Recommendation would breach source minimum.	Recommended transfer would leave the source below its minimum threshold. Reduce quantity or choose another source.	Network optimization must not create new shortages (BR-012).
VR-139	F-09	Recommendations must not be generated for inactive products or locations.	Recommendation includes invalid entities.	Rebalancing recommendation includes invalid products or locations. Refresh network data.	Recommendations must reflect current operational network.

FR-063 — Enforce Source Location Minimum on Transfers (F-08)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-140	F-08	Maximum transferable quantity equals source ATP minus source minimum threshold and safety stock per policy.	Transfer approved above maximum transferable quantity.	Transfer would leave the source below its minimum stock level. Maximum transferable quantity is {maxQty}.	Protects source service level (BR-012).
VR-141	F-08	Modified transfer quantity at approval must be revalidated against source minimum.	Approved quantity exceeds allowable maximum after modification.	Approved quantity exceeds what the source can spare. Reduce transfer quantity.	Approval must respect real-time source constraints.

Domain 8: Returns Management

FR-064 — Authorize and Record Customer Returns (F-21)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-142	F-21	Return authorization must link to a valid original order and return-eligible products per return policy.	Return authorized without valid order or ineligible item.	This return is not eligible. Verify the original order and return policy.	Prevents unauthorized returns and policy violations (BR-025).
VR-143	F-21	Authorized return quantity must be greater than zero and not exceed original shipped quantity for the line.	Return quantity exceeds original order line quantity.	Return quantity cannot exceed the quantity originally shipped on this order.	Protects against fraudulent or excessive return claims.

FR-065 — Receive and Inspect Returned Goods (F-21)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-144	F-21	Return receipt must match a valid, open return authorization.	Receipt against invalid or expired authorization.	Return authorization is invalid or expired. Obtain a valid authorization before receiving.	Warehouse must not accept unauthorized returns (BR-025).
VR-145	F-21	Received product must match authorized return product unless documented exception approved.	Wrong product received against authorization.	Returned product does not match the authorization. Contact customer service.	Prevents incorrect inventory and refund processing.
VR-146	F-21	Inspection outcome is required before restock or write-off disposition.	Disposition applied without inspection result.	Inspection outcome is required before restocking or writing off returned goods.	Only resalable goods re-enter sellable stock (BR-026).

FR-066 — Restock or Write Off Returned Inventory (F-21)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-147	F-21	Restock quantity plus write-off quantity must equal inspected received quantity.	Disposition quantities do not reconcile to received amount.	Restock and write-off quantities must equal the inspected return quantity.	Ensures full accountability for returned goods.
VR-148	F-21	Write-off portion requires reason code and approval per write-off policy.	Return write-off without reason or approval.	Write-off of returned goods requires a reason code and authorized approval.	Returned damaged goods follow same control as other write-offs (BR-023).

FR-067 — Track Return Processing Status (F-21)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-149	F-21	Return status inquiry requires valid return authorization or order reference.	Invalid return reference.	Return not found. Verify the return authorization or order reference.	Enables accurate customer communication.
VR-150	F-21	Displayed status must reflect current stage: authorized, in transit, received, inspecting, restocked, written off, or complete.	Status inconsistent with actual processing stage.	Return status is being updated. Refresh or contact the returns team.	Customers and CS need reliable return progress visibility.

Domain 9: Reporting & Dashboards

FR-068 — Inventory Summary Dashboard (F-22)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-151	F-22	Dashboard aggregates must include only authorized locations.	Unauthorized location data in dashboard totals.	Dashboard shows data for your authorized locations only.	Executive and operational views respect access scope.
VR-152	F-22	Dashboard metrics must use current inventory balances, not stale cached totals beyond defined freshness policy.	Metrics exceed acceptable data freshness.	Dashboard data may be stale. Refresh for current inventory position.	Decisions require timely stock picture (BO-001).

FR-069 — Stock Movement Reports (F-22)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-153	F-22	Report period must have valid start and end dates with start on or before end.	Invalid report date range.	Select a valid report period. Start date must be on or before end date.	Invalid periods produce unusable movement analysis.
VR-154	F-22	Movement report must include only transaction types and locations within user authority.	Unauthorized data in exported report.	Report export excludes locations outside your access scope.	Protects sensitive operational data in exports.

FR-070 — Stockout and Overstock Reports (F-22)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-155	F-22	Stockout list must include only products where ATP is zero or below minimum sellable unit at the location.	Incorrect stockout classification.	Stockout report requires current thresholds and availability. Refresh report parameters.	Directs action to genuine availability failures.
VR-156	F-22	Overstock list must compare on-hand or ATP to defined maximum or policy threshold where configured.	Overstock shown without threshold basis.	Overstock criteria are not defined for some products. Configure thresholds or review policy.	Overstock identification requires policy reference.

FR-071 — Inventory Accuracy and Variance Reports (F-14)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-157	F-14	Accuracy rate must be calculated only from completed counts in the selected period.	Incomplete counts included in accuracy rate.	Accuracy report includes only completed counts for the selected period.	Incomplete counts distort accuracy metrics (BO-002).
VR-158	F-14	Report must exclude count lines still under investigation when calculating finalized accuracy.	Open variances counted as accurate.	Pending variances are excluded until investigation is complete.	Accuracy metrics must reflect resolved position.

FR-072 — Location Performance Snapshots (F-22)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-159	F-22	Comparison view must use consistent metric definitions and period across all locations shown.	Inconsistent metric basis between locations.	Location comparison uses inconsistent periods or definitions. Select a uniform reporting period.	Fair network comparison requires consistent measurement.

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-160	F-22	User must have network or regional authority to view multi-location comparison.	Location-scoped user accesses network comparison.	You do not have access to network-wide location comparisons.	Restricts sensitive comparative performance data.

FR-073 — Inventory Performance KPIs (F-23)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-161	F-23	KPI calculations must use defined formulas and agreed targets from business objectives.	KPI displayed without defined target or formula.	KPI target is not configured for this metric.	KPIs without targets cannot govern performance (BO-005, BO-010).
VR-162	F-23	KPI period selection must be valid and align to available transaction history.	KPI requested for period with insufficient data.	Insufficient data to calculate this KPI for the selected period.	Prevents misleading KPIs from incomplete history.

FR-074 — Basic Replenishment Reporting (F-20, F-22)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-163	F-20	Replenishment report must show only items at or below reorder level with configured thresholds.	Items without thresholds appear as below reorder without basis.	Some products lack thresholds and are excluded from replenishment reporting.	Report must reflect policy-based replenishment signals.
VR-164	F-22	Open alert and requisition status must reconcile to current workflow state, not historical duplicates.	Report shows stale alert status.	Replenishment report refreshed to show current alert and requisition status.	Buyers need current decision information.

FR-075 — Inventory Valuation Summary Views (F-24)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-165	F-24	Valuation view access restricted to authorized finance and executive roles.	Unauthorized user accesses valuation summary.	You are not authorized to view inventory valuation.	Valuation data is financially sensitive (NFR-SEC-017).
VR-166	F-24	Valuation requires valid unit cost for each product included in the summary.	Product missing cost in valuation calculation.	Valuation incomplete. Unit cost is missing for one or more products.	Incomplete costing produces unreliable financial substantiation (BO-018).

Domain 10: Notifications & Alerts

FR-076 — Low-Stock and Out-of-Stock Notifications (F-20, F-26)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-167	F-26	Notifications must be sent only when a new alert condition occurs or escalation timer expires, not on every minor stock movement.	Duplicate notifications for unchanged condition.	Alert already sent for this condition. Acknowledge or resolve the existing alert.	Prevents notification fatigue and ignored critical alerts.
VR-168	F-26	Critical out-of-stock notifications for Class A products must escalate if unacknowledged within policy timeframe.	Class A out-of-stock alert unacknowledged beyond escalation threshold.	Critical stockout alert unacknowledged. Immediate replenishment review required.	Priority SKUs require fastest response (BR-032).

FR-077 — Pending Approval Notifications (F-17, F-26)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-169	F-26	Approval notifications must identify transaction type, requester, quantity, and age.	Notification missing required context.	Approval request details are incomplete. Open the approval task for full information.	Approvers need sufficient context for timely decisions.
VR-170	F-26	Reminder notifications must fire when approval pending exceeds defined escalation age.	Stale approval without reminder.	Approval pending beyond policy timeframe. Review required items awaiting your approval.	Approval bottlenecks delay replenishment and corrections.

FR-078 — Transfer and Fulfillment Exception Alerts (F-13, F-26)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-171	F-26	Exception alerts must be generated only for documented exception types with business impact.	Alert fired without logged exception record.	Exception alert requires a recorded fulfillment or transfer exception.	Alerts must trace to actionable operational events.
VR-172	F-26	Alert recipients must be roles authorized for the affected location and process.	Alert sent to unauthorized recipient.	Alert could not be delivered. Recipient is not authorized for this location or process.	Targeted alerts reduce noise and protect data scope.

FR-079 — Inbound Supply Delay Notifications (F-07, F-26)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-173	F-26	Delay notification requires PO line past expected delivery date with outstanding quantity.	Delay alert for on-time or closed PO.	No overdue inbound supply for this notification.	Alerts must reflect genuine supply delays affecting replenishment.
VR-174	F-26	Delay notification must include product, original ETA, and outstanding quantity.	Incomplete delay notification content.	Inbound delay details are incomplete. Review purchase order status directly.	Purchasing and operations need actionable delay information.

FR-080 — Campaign and Launch Readiness Alerts (F-25, F-26)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-175	F-26	Readiness alert fires when campaign stock gap exceeds tolerance within defined pre-launch window.	Alert without configured campaign target or timeline.	Campaign readiness alert requires defined targets and launch date.	Early warning requires measurable readiness criteria (BO-024).
VR-176	F-26	Alert must identify affected SKUs, locations, gap quantity, and days until campaign start.	Incomplete readiness alert.	Campaign readiness gap detected. Review promotional stock targets before launch.	Marketing and operations need specific gap visibility.

Domain 11: User Access & Accountability

FR-081 — Enforce Role-Based Access Control (F-27)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-177	F-27	Users must be authenticated before any inventory view or transaction.	Unauthenticated access attempt.	Sign in to continue. Your session has expired or you are not authenticated.	Every transaction must be attributable to an individual (NFR-SEC-001).

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-178	F-27	Users may perform only actions permitted by their assigned role.	Action outside role permission.	You are not authorized to perform this action.	Role boundaries protect inventory integrity and segregation of duties.

FR-082 — Restrict Access by Location (F-27)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-179	F-27	Transactional users may view and transact only at assigned locations unless granted network scope.	Cross-location transaction by location-scoped user.	You are not authorized for inventory activity at this location.	Aligns access with warehouse accountability.
VR-180	F-27	Attempts to access unauthorized locations must be logged for security review.	Unauthorized access attempt not recorded.	Access denied. This attempt has been recorded.	Supports investigation of improper access attempts.

FR-083 — Log User Activity (F-28)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-181	F-28	Material inventory actions must log user, action type, timestamp, and affected entity.	Material action without activity log entry.	Action could not be completed. Activity logging is required for this operation.	Accountability and audit depend on complete activity records (BO-019).
VR-182	F-28	Activity log entries must not be altered or deleted by operational users.	User attempts to modify historical activity log.	Activity history cannot be changed. Contact an administrator or auditor for log inquiries.	Log integrity is required for audit and investigation.

FR-084 — Enforce Segregation of Duties (F-28)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-183	F-28	User may not approve a transaction they initiated where policy prohibits self-approval.	Self-approval attempted on restricted workflow.	You cannot approve a transaction you submitted. An independent approver is required.	Prevents fraud and undetected error (BR-019, SOD-003).
VR-184	F-28	Conflicting role assignments that violate segregation rules must be blocked or flagged at assignment time.	User assigned conflicting roles without exception.	This role combination violates segregation of duties policy. Adjust role assignments.	Control design must prevent inherent conflicts (SOD rules).

FR-085 — Manage User Accounts and Role Assignments (F-27)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-185	F-27	Deactivated users must not access the system or perform transactions.	Deactivated account used for login or action.	This account is deactivated. Contact your administrator for access.	Prevents orphaned access after role changes or departure.
VR-186	F-27	Each user account must map to a unique identifiable individual; shared accounts must not be used.	Shared or generic account used for transactions.	Shared accounts are not permitted. Use your individual user account.	Attribution requires unique user identity (NFR-SEC-001).
VR-187	F-27	User must have at least one role and valid location scope when assigned to transactional functions.	Transactional user without role or location.	User account is incomplete. Assign a role and location scope before granting operational access.	Incomplete provisioning causes authorization failures or excessive access.

Domain 12: Governance, Audit & Compliance

FR-086 — Maintain Complete Audit Trail (F-29)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-188	F-29	Every balance-changing event must create an audit entry with before quantity, after quantity, reason, user, and timestamp.	Balance changes without audit entry.	Transaction blocked. Inventory changes must be fully auditable.	Complete audit trail is a P1 business objective (BO-019).
VR-189	F-29	Audit records must not be modified or deleted after creation.	Audit record alteration attempted.	Audit records cannot be changed or deleted.	Immutability is required for audit and dispute resolution.

FR-087 — Support Audit and Control Review Queries (F-29)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-190	F-29	Audit queries must require valid period, location scope, and auditor authorization.	Unauthorized or unbounded audit export attempted.	Audit query requires authorized access and a defined scope.	Protects sensitive audit data while supporting review efficiency.
VR-191	F-29	Exported audit evidence must match query criteria and include required fields per audit policy.	Export missing mandatory audit attributes.	Audit export is incomplete. Required audit fields are missing from the export.	Audit packages must substantiate control testing.

FR-088 — Detect and Record Control Policy Breaches (F-29)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-192	F-29	Policy breaches such as receipt without PO, self-approval, or safety stock override must be logged even when authorized exception applies.	Exception or breach not recorded.	Control exception must be documented for governance review.	Makes control weaknesses visible before audit discovery.
VR-193	F-29	Authorized exceptions must include approver, policy reference, and business justification.	Exception without required documentation.	Policy exception requires approver, policy reference, and documented reason.	Exceptions without documentation cannot be audited.

FR-089 — Support Financial Inventory Reconciliation (F-24)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-194	F-24	Period-end reconciliation export must include all adjustments and write-offs in the reporting period.	Reconciliation export omits period adjustments.	Reconciliation data is incomplete for this period. Include all adjustments before closing.	Finance reconciliation requires complete operational substantiation (BO-018).
VR-195	F-24	Reconciliation snapshot must be tied to a defined period close date and cannot span overlapping open periods.	Overlapping or open period reconciliation snapshot.	Select a closed reporting period for reconciliation snapshot.	Period boundaries must align with finance close discipline.

FR-090 — Support Initial Data Import (F-30)

Validation ID	Related Feature	Validation Rule	Error Condition	User Message	Business Reason
VR-196	F-30	Import file must include required product attributes and non-negative opening quantities for each row.	Import row missing required fields or invalid quantity.	Import row is invalid. Correct missing product fields or negative quantities before importing.	Bad cutover data embeds errors from day one (BA-16).
VR-197	F-30	Import must reject duplicate SKUs within the file and against existing catalog unless merge rules apply.	Duplicate SKU in import batch.	Duplicate SKU found in import file or catalog. Resolve duplicates before activation.	Prevents catalog fragmentation at go-live.
VR-198	F-30	Import activation requires authorized approval after validation preview.	Import activated without approver sign-off.	Initial data import requires authorized approval after validation review.	Controlled cutover protects production inventory integrity.
VR-199	F-30	Opening balance import must reference valid active locations in the network.	Import references unknown or inactive location.	Import contains invalid location references. Use active configured locations only.	Stock must be assigned to real accountable locations.
VR-200	F-30	Import must not activate operational transactions until validation errors are resolved to zero blocking errors.	Activation with unresolved blocking validation errors.	Import cannot be activated until all blocking errors are resolved.	Go-live with validation errors undermines accuracy objectives (BO-002).

Validation Rules Summary

Domain	FR Range	Validation Rules	Rule IDs
Master Data & Governance	FR-001 – FR-012	29	VR-001 – VR-029
Inbound & Receiving	FR-013 – FR-022	23	VR-030 – VR-052
Inventory Visibility & Status	FR-023 – FR-028	12	VR-053 – VR-064
Inventory Control & Accuracy	FR-029 – FR-040	25	VR-065 – VR-089
Order Fulfillment	FR-041 – FR-050	22	VR-090 – VR-111
Replenishment & Inbound Supply	FR-051 – FR-057	16	VR-112 – VR-127
Network Balance & Transfers	FR-058 – FR-063	14	VR-128 – VR-141
Returns Management	FR-064 – FR-067	9	VR-142 – VR-150
Reporting & Dashboards	FR-068 – FR-075	16	VR-151 – VR-166
Notifications & Alerts	FR-076 – FR-080	10	VR-167 – VR-176
User Access & Accountability	FR-081 – FR-085	11	VR-177 – VR-187
Governance, Audit & Compliance	FR-086 – FR-090	13	VR-188 – VR-200
Total	**90 FRs**	**200**	**VR-001 – VR-200**

Traceability Reference

Source	Identifier
Functional Requirements	FR-001 – FR-090
Feature Catalog	F-01 – F-30
Business Rules	BR-001 – BR-032
Acceptance Criteria	AC-001 – AC-090 (Validation scenarios)
Separation of Duties	SOD-001 – SOD-020

Document Approval

Role	Name	Signature	Date
Senior Business Analyst			
Product Owner			
QA Lead			

End of Validation Rules Specification

Document ID: SRS-VR-001 | Version 1.0 | July 5, 2026

Business Constraints Specification

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-BC-001
Document Title	Business Constraints — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Senior Business Analyst
Source Documents	BRD-IMS-001 (Section 18), Business Constraints Discovery Artifact
Date	July 5, 2026
Scope	Business and operational constraints only — not technical architecture, infrastructure, or implementation design

Purpose

This document identifies the **38 business constraints** that bound how the Inventory Management System initiative may be delivered and operated. Constraints are fixed business conditions — not optional preferences. Each constraint is expressed in operational terms: what the organization can and cannot do while running inventory management as a business capability.

Constraint Categories

Prefix	Category	Count
BC	Budget	6
TC	Time	6
LC	Legal	5
OC	Operational	8
ORC	Organizational	7
TEC	Technology — Business Level	6

Severity Classification

Severity	Constraints
Blocking	BC-01, TC-02, OC-01, ORC-07
High	BC-02, BC-05, TC-01, TC-03, TC-04, ORC-01, ORC-03, ORC-05, TEC-01, TEC-04
Medium	BC-03, BC-04, BC-06, TC-05, TC-06, LC-01, LC-02, LC-03, OC-02, OC-04, OC-07, OC-08, ORC-02, ORC-04, ORC-06, TEC-02, TEC-05, TEC-06
Low	LC-04, LC-05, OC-03, OC-05, OC-06

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1. Budget Constraints

BC-01: Fixed Total Initiative Budget

Attribute	Detail
Constraint ID	BC-01
Description	The IMS initiative must be delivered within a fixed total budget approved by executive leadership, covering software, services, training, change management, and first-year operational support. No additional funding is guaranteed if overruns occur.
Business Justification	Inventory improvement must demonstrate return within approved capital and operating investment limits. Executive leadership has committed a defined envelope tied to business case ROI — not an open-ended transformation program.
Affected Business Processes	All processes (BP-01 – BP-32); particularly BP-29 (Performance Reporting), BP-31 (Promotional Readiness), and Phase 2/3 processes deferred by scope prioritization
Consequences if Violated	Scope reduction or project pause; P1 objectives (BO-01, BO-03, BO-19) at risk; Phase 2/3 capabilities (replenishment automation, financial alignment, network rebalancing) indefinitely deferred; stakeholder trust in program governance erodes

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BC-02: No Additional Warehouse Headcount Budget

Attribute	Detail
Constraint ID	BC-02
Description	The business case does not include budget for additional warehouse or inventory control headcount in the first 12 months. Efficiency must come from improved processes and system support, not new hires.
Business Justification	The initiative is funded as a productivity and control investment, not a staffing expansion. Leadership expects existing teams to operate more effectively with better visibility and standardized workflows.
Affected Business Processes	BP-07 (Goods Receipt), BP-09 (Put-Away), BP-11 (Cycle Counting), BP-18 (Pick, Pack & Ship), BP-21 (Low-Stock Alert Review), BP-30 (Financial Reconciliation)
Consequences if Violated	Unfunded headcount creates budget overrun (BC-01 conflict); existing staff overwhelmed during rollout; adoption fails (BO-21); manual parallel tools persist (BO-22); accuracy and movement recording objectives (BO-02, BO-03) not sustained

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BC-03: Capital Expenditure vs. Operating Expenditure Limits

Attribute	Detail
Constraint ID	BC-03
Description	Finance has defined separate caps for capital expenditure (licenses, implementation) and operating expenditure (subscriptions, support, training). Spending cannot be reclassified between categories without CFO approval.
Business Justification	Corporate financial policy requires predictable classification of investment versus recurring cost for balance-sheet and budget planning. IMS spending must fit established fiscal governance.
Affected Business Processes	BP-05 (Purchase Order Placement), BP-22 (Replenishment Planning), BP-30 (Financial Reconciliation) — indirect; primarily affects procurement timing of delivery activities
Consequences if Violated	Delivery delayed to next fiscal period; training or go-live support deferred; misaligned spend blocks vendor payment; project timeline slips against TC-02 and TC-05 milestones

BC-04: Training and Change Management Budget Ceiling

Attribute	Detail
Constraint ID	BC-04
Description	A fixed budget is allocated for training, communications, and change management. This budget cannot be reallocated from development or licensing budgets.
Business Justification	Change investment is necessary but bounded. Leadership requires demonstrable efficiency in rollout — scalable training models rather than unlimited per-location bespoke programs.
Affected Business Processes	All operational processes at rollout; especially BP-07, BP-11, BP-14, BP-18, BP-24 where floor adoption is critical
Consequences if Violated	Incomplete role training; location managers reject standardized processes (BO-03); user adoption below 90% (BO-21); parallel spreadsheets and paper logs continue (BO-22); rework and support costs increase without budget to absorb them

BC-05: No Budget for External System Replacement

Attribute	Detail
Constraint ID	BC-05
Description	The IMS budget does not fund replacement of finance, sales, HR, or ERP systems. IMS must operate within the current enterprise application landscape.
Business Justification	IMS addresses inventory control as a targeted capability gap — not a full enterprise platform replacement. Finance, sales, and HR systems remain authoritative in their domains.
Affected Business Processes	BP-05 (Purchase Order Placement), BP-16 (Order Availability), BP-30 (Financial Reconciliation), BP-31 (Promotional Readiness)
Consequences if Violated	Scope creep into ERP/CRM territory; stakeholder resistance from system owners; delivery failure on inventory-specific objectives; duplicated master data across systems; BO-18 and BO-20 remain unachievable without agreed interim reconciliation

BC-06: Ongoing Operational Cost Limit

Attribute	Detail
Constraint ID	BC-06
Description	Annual recurring cost for licenses, support, and maintenance after go-live must remain within a CFO-approved operating budget threshold, typically capped relative to current inventory carrying cost.
Business Justification	Sustainable inventory management must not impose recurring cost that exceeds the economic value of improved control. Ongoing spend must be defensible against working-capital and carrying-cost benefits.
Affected Business Processes	BP-02 (Location Setup) — future location onboarding; BP-29 (Performance Reporting) — executive visibility scope
Consequences if Violated	New locations cannot be onboarded without separate approval; feature tiers or user counts restricted; network-wide rollout (TC-03) stalls; long-term scalability of BO-01 and BO-14 constrained

2. Time Constraints

TC-01: 18-Month Business Vision Horizon

Attribute	Detail
Constraint ID	TC-01
Description	All SMART business objectives and phased outcomes must be achievable within 18 months of project initiation, aligned with the short-term business vision.

Attribute	Detail
Business Justification	Leadership committed to measurable inventory improvement within a defined planning horizon. Beyond 18 months, priorities may shift and funding is not assured.
Affected Business Processes	All processes; Phase 3 processes especially BP-26 (Network Rebalancing), BP-29 (KPI Reporting), BP-30 (Financial Reconciliation), BP-31 (Promotional Readiness)
Consequences if Violated	Objectives BO-09, BO-10, BO-18, BO-20, BO-23, BO-24 measured as failed despite incomplete delivery; steering committee loses confidence; initiative descoped or restarted

TC-02: MVP Delivery Within 6 Months

Attribute	Detail
Constraint ID	TC-02
Description	Minimum Viable Product must be live at the first primary location within 6 months of project kickoff to meet Phase 1 vision and P1 SMART objectives.
Business Justification	Early credible control is the strategic intent of Phase 1. Delayed go-live prolongs stockouts, inaccurate records, and fulfillment failures that the initiative was created to resolve.
Affected Business Processes	BP-01 – BP-18, BP-24 – BP-25, BP-29 (core reports), BP-32 (audit trail) — MVP process set
Consequences if Violated	P1 objectives missed: trusted view (BO-01), movement recording (BO-03), audit trail (BO-19), sales self-service (BO-16), adoption (BO-21), manual tool retirement (BO-22); executive sponsorship weakens; business continues absorbing current-state losses

TC-03: Sequential Location Rollout Window

Attribute	Detail
Constraint ID	TC-03
Description	Primary locations must be onboarded sequentially over approximately 6 months following first go-live, with a minimum 4-week gap between deployments — not simultaneous network-wide cutover.
Business Justification	Support capacity, change absorption, and lessons-learned application require controlled rollout. Simultaneous deployment would overwhelm operations and SMEs (ORC-01).
Affected Business Processes	BP-02 (Location Setup), BP-10 (Status Monitoring), BP-24 – BP-26 (Network Balance), BP-29 (Reporting)
Consequences if Violated	Inconsistent processes across locations persist (BO-03); network visibility incomplete until ~month 12 (BO-01); cross-location imbalance correction delayed (BO-14); hypercare support fails; accuracy and adoption vary by site

TC-04: Peak Season Blackout Periods

Attribute	Detail
Constraint ID	TC-04
Description	Go-live, major upgrades, and intensive training cannot occur during defined peak business periods (e.g., holiday season, fiscal year-end, major promotional events) at each location.
Business Justification	Peak periods generate disproportionate revenue and customer commitments. Operational disruption during peak directly damages fill rate, on-time delivery, and brand reputation.
Affected Business Processes	BP-16 – BP-19 (Order Fulfillment), BP-18 (Pick, Pack & Ship), BP-31 (Promotional Readiness), BP-07 (Receiving)
Consequences if Violated	Fill rate and on-time delivery drop (BO-05, BO-15); stock-related cancellations increase (BO-08); customer complaints and revenue at risk; rollout calendar slips 3–6 months; staff cannot complete training before peak

TC-05: SMART Objective Measurement Deadlines

Attribute	Detail
Constraint ID	TC-05
Description	Business objectives have fixed measurement milestones at months 3, 6, 9, 12, and 18 post-initiation. Enabling capabilities must be live before their measurement window opens.
Business Justification	Accountability to the business case requires time-bound outcomes. Objectives without aligned delivery dates cannot be governed or rewarded.
Affected Business Processes	Process-to-objective mapping: BP-10/16 (BO-01, BO-16); BP-11/13 (BO-02); BP-07/18/25 (BO-03); BP-18/19 (BO-05, BO-08); BP-21/22 (BO-07, BO-13); BP-30 (BO-18, BO-20)
Consequences if Violated	False "red" KPI status when capability not yet delivered; premature descoping of objectives; steering committee makes decisions on misleading data; team morale and sponsor confidence decline

TC-06: Baseline Measurement Lead Time

Attribute	Detail
Constraint ID	TC-06
Description	Pre-implementation baseline data for SMART objectives must be collected 30–60 days before first go-live, consuming part of the overall project timeline.
Business Justification	Improvement claims require credible before-and-after comparison. Without baseline, ROI and objective achievement cannot be substantiated to leadership or auditors.
Affected Business Processes	BP-29 (Performance Reporting), BP-11/12/13 (Accuracy), BP-18/19 (Fulfillment), BP-15 (Shrinkage)
Consequences if Violated	BO-04, BO-06, BO-07, BO-09, BO-11 targets unprovable; steering committee cannot validate success; external audit challenges benefit claims; late baseline compresses build and test window (conflicts with TC-02)

3. Legal Constraints

LC-01: Data Protection and Privacy Regulations

Attribute	Detail
Constraint ID	LC-01
Description	The business must comply with applicable data protection laws (e.g., GDPR, local privacy regulations) for personal data in user accounts, audit logs, and customer-linked order and return records.
Business Justification	Legal obligation to protect employee and customer personal data. Non-compliance exposes the organization to regulatory penalties, legal action, and reputational harm.
Affected Business Processes	BP-16 – BP-19 (customer-linked fulfillment), BP-27 – BP-28 (Returns), BP-32 (Audit & Control Review)
Consequences if Violated	Regulatory fines and enforcement actions; required data deletion or access restrictions disrupt audit trail (BO-19); customer data breaches damage trust; legal hold on system use pending remediation

LC-02: Financial Record Retention Requirements

Attribute	Detail
Constraint ID	LC-02
Description	Inventory transaction records, adjustment logs, valuation substantiation, and audit evidence must be retained for the minimum period required by tax and corporate law — typically 5–7 years.

Attribute	Detail
Business Justification	Statutory and audit requirements mandate preservation of financial and inventory substantiation records. Early destruction invalidates audit defense and tax compliance.
Affected Business Processes	BP-14 (Adjustments & Write-Offs), BP-30 (Financial Reconciliation), BP-32 (Audit & Control Review), BP-07 (Goods Receipt), BP-25 (Transfer Execution)
Consequences if Violated	Audit findings and qualified opinions; inability to defend inventory valuations in tax review; regulatory penalties; BO-19 (audit trail) and BO-18 (valuation alignment) fail at external audit

LC-03: Employment and Labor Law

Attribute	Detail
Constraint ID	LC-03
Description	Changes to work procedures, performance measurement, and mandatory transaction recording introduced by IMS must comply with employment contracts, collective bargaining agreements, and labor regulations.
Business Justification	Workforce practices are legally governed. Unilateral imposition of new performance or recording requirements without proper consultation creates labor relations and legal risk.
Affected Business Processes	BP-07, BP-09, BP-11, BP-18 (floor work procedures); BP-32 (accountability policies)
Consequences if Violated	Union grievances or work stoppages at affected locations; rollback of mandatory recording policies (BO-03, BO-22 fail); go-live delay at unionized sites; HR and legal remediation costs

LC-04: Supplier and Commercial Contract Terms

Attribute	Detail
Constraint ID	LC-04
Description	Purchase order formats, delivery documentation, and receipt dispute processes must remain compatible with existing supplier contracts. Suppliers cannot be required to adopt new processes without contract amendment.
Business Justification	Supplier relationships are contractually bound. Receiving and procurement workflows must work with current commercial terms — the IMS initiative does not renegotiate the supplier base.
Affected Business Processes	BP-05 (Purchase Order Placement), BP-07 (Goods Receipt), BP-08 (Receipt Discrepancy Resolution)
Consequences if Violated	Supplier refusal to deliver or document shipments; receipt disputes unresolvable; inbound supply disruption; emergency procurement increases (BO-12); BR-006 and BR-007 controls weakened

LC-05: Consumer Protection and Trade Regulations

Attribute	Detail
Constraint ID	LC-05
Description	Customer-facing availability and delivery commitments based on IMS data must comply with consumer protection laws on accurate product representation, cancellation rights, and delivery obligations.
Business Justification	B2B and B2C customers have legal protections against misleading availability and unfair commercial practices. ATP-based commitments must be defensible.
Affected Business Processes	BP-16 (Order Availability Confirmation), BP-19 (Partial Fulfillment & Backorder), BP-18 (Pick, Pack & Ship)
Consequences if Violated	Customer complaints to regulators; chargebacks and compensation claims; contractual penalties with B2B customers; reputational damage; BO-08 (cancellations) and BO-15 (on-time delivery) worsen

4. Operational Constraints

OC-01: No Operational Shutdown for Go-Live

Attribute	Detail
Constraint ID	OC-01
Description	Warehouses and stores must continue receiving, fulfilling, and shipping orders during and after go-live. No full operational shutdown is permitted for system transition.
Business Justification	Customer commitments and revenue depend on continuous fulfillment. A shutdown-based cutover is commercially unacceptable for a distribution and retail operator.
Affected Business Processes	BP-07 (Receiving), BP-17 – BP-18 (Reservation & Fulfillment), BP-24 – BP-25 (Transfers), BP-10 (Status Monitoring)
Consequences if Violated	Lost sales and customer cancellations; carrier and supplier penalties; emergency manual fulfillment at higher cost; opening balance errors from rushed cutover (BO-02); rollback required with extended dual-system operation

OC-02: Daily Fulfillment Service Levels Must Be Maintained

Attribute	Detail
Constraint ID	OC-02
Description	Agreed customer fulfillment service levels (fill rate, dispatch timeliness) must not materially degrade during implementation and rollout.
Business Justification	Customer contracts and brand promise depend on reliable fulfillment. The IMS program must improve control without becoming a cause of service failure.
Affected Business Processes	BP-16 – BP-20 (Order Fulfillment), BP-17 (Reservation), BP-18 (Pick, Pack & Ship)
Consequences if Violated	BO-05 (fill rate), BO-06 (cycle time), BO-15 (on-time delivery) regress; NPS on fulfillment drops (KPI-27); B2B penalties; steering committee halts rollout; revert to manual processes

OC-03: Physical Warehouse Layout and Storage Are Fixed

Attribute	Detail
Constraint ID	OC-03
Description	Warehouse layouts, racking, and storage zones will not be restructured as part of the IMS initiative. Physical infrastructure remains as-is.
Business Justification	Facility investment is outside IMS funding and timeline. Accuracy and efficiency gains must come from process discipline and visibility, not capital warehouse redesign.
Affected Business Processes	BP-09 (Put-Away), BP-18 (Pick, Pack & Ship), BP-11 (Cycle Counting)
Consequences if Violated	Unfunded facility projects distract from IMS; bin-level optimization deferred; pick path inefficiency persists; accuracy improvements slower than planned (BO-02) if layout is root cause

OC-04: Existing Product Portfolio and SKU Count

Attribute	Detail
Constraint ID	OC-04
Description	IMS must accommodate the current active product catalog and projected SKU growth over 18 months without requiring product range reduction.

Attribute	Detail
Business Justification	Commercial strategy continues to expand assortment and channels. Inventory control must scale with the business — not force assortment rationalization as a prerequisite.
Affected Business Processes	BP-01 (Product Master), BP-03 (Stock Policy), BP-04 (ABC Classification), BP-10 (Status Monitoring)
Consequences if Violated	Forced SKU reduction damages sales; catalog cutover errors; threshold and alert programs unmanageable; BO-07 (priority stockouts) on excluded products; data migration failure at go-live

OC-05: Current Supplier and Carrier Relationships

Attribute	Detail
Constraint ID	OC-05
Description	Inbound and outbound logistics continue through existing approved suppliers and carriers. The IMS project does not include supplier renegotiation or carrier replacement.
Business Justification	Supply chain partnerships are long-term commercial relationships. Receiving and shipping workflows must align with current partner capabilities and lead times.
Affected Business Processes	BP-05 – BP-09 (Inbound), BP-18 (Shipment Handoff), BP-23 (Inbound Supply Tracking)
Consequences if Violated	Unplanned partner changes disrupt supply; receiving documentation mismatch; inbound delays invalidate replenishment thresholds (BO-13); carrier tracking gaps persist

OC-06: Returns Volume Within Manual Handling Capacity

Attribute	Detail
Constraint ID	OC-06
Description	Customer returns volume is expected to remain processable through basic return workflows without a dedicated returns operation for the first 12 months. If volume exceeds ~5% of outbound, scope review is required.
Business Justification	Returns handling capacity is sized for current business volume. Full reverse-logistics maturity is Phase 2 — not day-one operational capacity.
Affected Business Processes	BP-27 (Customer Return Intake), BP-28 (Return-to-Stock or Disposal)
Consequences if Violated	Returns backlog; incorrect restock or write-off; customer refund delays; warehouse floor congestion; BO-11 (shrinkage) worsens from mishandled returns; scope exception and unfunded work required

OC-07: Cycle Counting Without Operational Disruption

Attribute	Detail
Constraint ID	OC-07
Description	Physical inventory verification required for accuracy objectives must be performed without shutting down receiving, picking, or shipping at any location.
Business Justification	Continuous operations are non-negotiable (OC-01, OC-02). Accuracy programs must use rolling cycle counts — not full shutdowns — except scheduled full counts planned outside peak.
Affected Business Processes	BP-11 (Cycle Counting), BP-12 (Full Physical Count), BP-13 (Variance Investigation)
Consequences if Violated	Forced shutdown conflicts with OC-01; accuracy targets delayed (BO-02); count variances increase from concurrent movements; local managers refuse count program (ORC-05); audit substantiation weak

OC-08: Multi-Location Network with Existing Transfer Practices

Attribute	Detail
Constraint ID	OC-08
Description	Inter-location transfers follow existing approval practices and freight arrangements. The project does not introduce new logistics infrastructure or transfer service level agreements.
Business Justification	Network redistribution depends on current freight capacity and approval culture. IMS standardizes recording — not physical logistics network redesign.
Affected Business Processes	BP-24 (Transfer Request), BP-25 (Transfer Execution), BP-26 (Network Rebalancing)
Consequences if Violated	Unfunded logistics investment expected; transfer SLAs assumed but not met; BO-14 (imbalance reduction) delayed; emergency transfers continue outside system; source locations stockout from uncontrolled transfers (BR-012 risk)

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5. Organizational Constraints

ORC-01: No Dedicated Full-Time Project Team

Attribute	Detail
Constraint ID	ORC-01
Description	The project relies on part-time contribution from business SMEs (~20% allocation). No dedicated full-time business operations lead is assigned exclusively to the initiative.
Business Justification	Operational leaders must continue running daily business. The organization cannot spare full-time inventory SMEs without compromising fulfillment and replenishment.
Affected Business Processes	All processes requiring SME definition: BP-01, BP-03, BP-07, BP-11, BP-14, BP-21, BP-32
Consequences if Violated	Slow decisions delay TC-02; UAT incomplete; policies undefined at go-live; floor users adopt workarounds; requirements errors discovered post-go-live; accuracy and adoption objectives fail

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ORC-02: Single Product Owner Authority

Attribute	Detail
Constraint ID	ORC-02
Description	Product prioritization is centralized under one Product Owner with steering committee oversight. Department heads cannot unilaterally add scope for their function.
Business Justification	Twenty-nine stakeholder groups create conflicting demands. Central prioritization protects MVP delivery and budget (BC-01) against local optimization.
Affected Business Processes	Scope affects all processes; escalations common for BP-05, BP-31, BP-26, BP-30 when deferred
Consequences if Violated	Scope creep breaches BC-01; MVP date missed (TC-02); fragmented product with inconsistent workflows (BO-03); steering committee loses governance confidence

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ORC-03: Change Resistance in Warehouse Culture

Attribute	Detail
Constraint ID	ORC-03
Description	Warehouse teams rely on informal practices — spreadsheets, paper logs, tribal knowledge — and history shows resistance to mandated system use without management enforcement.

Attribute	Detail
Business Justification	Cultural inertia is a known organizational reality, not a failure of intent. Adoption requires enforcement policy, not only training and tooling.
Affected Business Processes	BP-07, BP-09, BP-11, BP-14, BP-18, BP-25 — all floor-recording processes
Consequences if Violated	Dual systems persist (BO-22 fails); movement recording below 98% (BO-03); accuracy stays at ~78% (BO-02); audit trail incomplete (BO-19); pilot failure poisons network rollout (TC-03)

ORC-04: Limited Internal IT Business Support Capacity

Attribute	Detail
Constraint ID	ORC-04
Description	Internal technology support has limited capacity for business process design support, bulk user provisioning, and post-go-live business hypercare. Primary IT focus remains existing enterprise systems.
Business Justification	IT resources are committed to maintaining current finance, sales, and infrastructure systems. IMS operational support must be business-led at go-live.
Affected Business Processes	BP-32 (user access and audit); rollout of all transactional processes
Consequences if Violated	User provisioning delays location go-live; access errors block floor operations; post-go-live issues unresolved; business hypercare unfunded; adoption stalls (BO-21)

ORC-05: Decentralized Location Management Structure

Attribute	Detail
Constraint ID	ORC-05
Description	Each warehouse and store is managed locally with operational autonomy. Central mandates require regional or executive endorsement for enforcement.
Business Justification	The operating model delegates day-to-day accountability to site managers. Network standardization is a cultural and governance challenge — not automatic with system deployment.
Affected Business Processes	BP-02 (Location Setup), BP-03 (Stock Policy), BP-11 (Cycle Counting), BP-24 (Transfers), BP-29 (Location Performance)
Consequences if Violated	Location-to-location process variation continues (BO-03); network KPIs not comparable; some sites never adopt; BO-14 (imbalance) uncorrected; executive dashboard misleading (BO-01)

ORC-06: Finance and Operations Reporting Alignment Gap

Attribute	Detail
Constraint ID	ORC-06
Description	Finance and operations use different data sources and reconciliation practices for inventory. No established single reconciliation process exists between operational and financial records.
Business Justification	Historical organizational silos separated operational stock records from general ledger inventory. IMS alone cannot create alignment — cross-department process must be designed.
Affected Business Processes	BP-14 (Adjustments & Write-Offs), BP-30 (Financial Reconciliation), BP-29 (Reporting)
Consequences if Violated	BO-18 (valuation variance ≤2%) unachievable; month-end close remains 7+ days (BO-20); finance distrusts operational data; manual reconciliation continues (BO-04 fails); audit findings on inventory substantiation

ORC-07: No Formal Inventory Management Function Today

Attribute	Detail
Constraint ID	ORC-07
Description	The business lacks a mature, centralized inventory management function with defined policies, KPIs, and single-point accountability. Control is distributed across warehouse, purchasing, and finance.
Business Justification	Growth outpaced governance design. The initiative assumes creation of Inventory Controller accountability — a structural gap that must be closed for sustained control.
Affected Business Processes	BP-01, BP-03, BP-04, BP-11, BP-13, BP-15, BP-21, BP-32 — all governance-heavy processes
Consequences if Violated	Policies unwritten and unenforced; no owner for accuracy program (BO-02); replenishment remains reactive (BO-13); shrinkage unanalyzed (BO-11); project delivers software without organizational capability; chronic inaccuracy persists

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6. Technology Constraints — Business Level

Expressed as business operating boundaries, not technical implementation limits.

TEC-01: Must Coexist With Existing Enterprise Systems

Attribute	Detail
Constraint ID	TEC-01
Description	IMS operates alongside existing finance, sales, and HR systems without replacing them. Users continue working across multiple systems daily.
Business Justification	Enterprise systems remain authoritative in their domains per BC-05. IMS is the inventory system of record — not the enterprise platform of record.
Affected Business Processes	BP-05 (PO with finance), BP-16 (orders from sales systems), BP-30 (valuation to finance), BP-27 (returns with CS systems)
Consequences if Violated	Stakeholder conflict and duplicate data entry; users revert to legacy tools; BO-16 limited to IMS-native workflows; finance alignment delayed (BO-18); initiative perceived as failed ERP attempt

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TEC-02: No Budget for Dedicated Warehouse Hardware

Attribute	Detail
Constraint ID	TEC-02
Description	The business will not purchase dedicated warehouse hardware (rugged scanners, handhelds, label printers) as part of IMS. Users access the system via existing company computers and devices.
Business Justification	Hardware investment is a separate capital decision. IMS rollout must be achievable with assets already deployed at sites.
Affected Business Processes	BP-07 (Receiving), BP-11 (Cycle Counting), BP-18 (Pick, Pack & Ship)
Consequences if Violated	Unbudgeted capex request (BC-03 conflict); floor process slower than planned (BO-06); barcode benefits limited; users skip scanning steps; receipt and pick errors persist (BO-02, BO-05)

TEC-03: Standard Web Browser Access Required

Attribute	Detail
Constraint ID	TEC-03
Description	All users must access IMS through a standard web browser on company-approved devices without installing specialized desktop applications.

Attribute	Detail
Business Justification	Corporate IT standardizes on browser-based applications for supportability and security. Proprietary client software is not approved for this initiative.
Affected Business Processes	All user-facing processes; floor operations at BP-07, BP-11, BP-18 most affected
Consequences if Violated	IT blocks deployment; locations without reliable connectivity cannot operate (floor downtime); adoption limited to office roles; warehouse recording compliance drops (BO-03)

TEC-04: No Real-Time External Channel Integration in MVP

Attribute	Detail
Constraint ID	TEC-04
Description	Sales channels (e-commerce, B2B portal, marketplaces) will not receive automated real-time inventory feeds from IMS in the first 12 months. Channel availability updates follow existing manual processes.
Business Justification	Channel integration requires separate business case and partner agreements. MVP focuses on internal operational truth before external synchronization.
Affected Business Processes	BP-10 (Status Monitoring), BP-16 (Availability Confirmation) — external channel context
Consequences if Violated	Unplanned integration work breaches BC-01 and TC-02; channel overselling continues (BO-08); marketplace penalties; BO-16 applies only to internal sales — external channels remain inaccurate

TEC-05: Manual Data Exchange With Finance System

Attribute	Detail
Constraint ID	TEC-05
Description	Inventory valuation and movement data transfer to finance through scheduled export or manual reconciliation in the first 12 months — not automated general ledger posting.
Business Justification	Finance system integration is out of IMS scope (BC-05). Interim operating model must close books using agreed manual reconciliation until Phase 3 automation.
Affected Business Processes	BP-14 (Adjustments & Write-Offs), BP-30 (Financial Reconciliation)
Consequences if Violated	Finance team unprepared for new data volume; month-end delays continue (BO-20); valuation variance stays at ~8% (BO-18); duplicate or lost adjustments; audit trail gaps if exports not archived (LC-02)

TEC-06: User Access Limited to Company Network and Approved Devices

Attribute	Detail
Constraint ID	TEC-06
Description	Access to inventory data — stock levels, valuation, movement history — is restricted to company-managed devices and approved network access per corporate security policy.
Business Justification	Inventory data is commercially and financially sensitive. Corporate security policy prohibits unmanaged device access to operational and valuation information.
Affected Business Processes	BP-10 (Visibility), BP-16 (Sales Availability), BP-29/30 (Management & Finance Reporting)
Consequences if Violated	Security policy breach; data leakage of stock positions or valuation; remote sales and regional managers cannot access ATP without VPN; workaround spreadsheets reappear (BO-22); regulatory exposure (LC-01)

Constraint Interaction Summary

Constraints frequently compound. The following interactions have amplified operational impact:

Interaction	Constraints	Operational Consequence if Ignored
Budget + Time	BC-01 + TC-02	MVP scope inflation causes missed go-live and unfunded overrun
Budget + Organization	BC-02 + ORC-01	Same staff run daily ops and project — burnout and adoption failure
Operational + Time	OC-01 + TC-04	Narrow rollout windows; missed blackout = delayed network benefits 6+ months
Organization + Technology	ORC-03 + TEC-01	Resistant users juggle IMS plus legacy systems without integration
Legal + Operations	LC-02 + TEC-05	Manual finance exports must still meet 5–7 year retention
Operational + Organization	OC-07 + ORC-05	Rolling counts need local manager cooperation at every decentralized site
Governance gap + Time	ORC-07 + TC-02	Go-live without Inventory Controller — policies undefined, accuracy fails

Constraints by Business Process Domain

Process Domain	Primary Constraints
Master Data & Governance (BP-01 – BP-04)	ORC-07, ORC-01, OC-04, TC-02
Inbound & Receiving (BP-05 – BP-09)	LC-04, OC-05, OC-01, ORC-03, TEC-02
Inventory Control (BP-10 – BP-15)	OC-07, ORC-07, ORC-05, LC-02
Order Fulfillment (BP-16 – BP-20)	OC-02, OC-01, LC-05, TEC-04, TC-04
Replenishment (BP-21 – BP-23)	ORC-07, BC-02, OC-05, LC-04
Network Balance (BP-24 – BP-26)	OC-08, TC-03, ORC-05
Returns (BP-27 – BP-28)	OC-06, LC-01, LC-05
Performance & Financial (BP-29 – BP-30)	ORC-06, TEC-05, LC-02, TC-05
Commercial & Governance (BP-31 – BP-32)	ORC-07, LC-01, LC-03, BC-01

Traceability Reference

Source	Identifier
Business Requirements Document	BRD-IMS-001, Section 18
Business Constraints Discovery	BC-01 – BC-06, TC-01 – TC-06, LC-01 – LC-05, OC-01 – OC-08, ORC-01 – ORC-07, TEC-01 – TEC-06
Business Processes	BP-01 – BP-32
SMART Business Objectives	BO-01 – BO-24
Business Assumptions	BA-03, BA-13, BA-24, BA-30, BA-38, BA-43

Document Approval

Role	Name	Signature	Date
Senior Business Analyst			
Product Owner			
Project Sponsor			

End of Business Constraints Specification

Business Exception Scenarios Specification

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-EX-001
Document Title	Business Exception Scenarios — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Senior Business Analyst
Source Documents	BRD-IMS-001, Business Events Catalog (BE-01–BE-59), Business Workflows (WF-01–WF-32), Common Inventory Management Scenarios, Business Rules (BR-001–BR-032)
Date	July 5, 2026
Scope	Business exception behavior only — not technical error handling, system faults, or implementation recovery

Purpose

This document identifies **exception scenarios** — situations that deviate from the standard happy-path inventory process and require defined business judgment, escalation, and response. Exceptions are normal operational events, not system failures. Each scenario describes what the business should do when the exception occurs.

How to Read Each Exception

Field	Meaning
Exception ID	Unique identifier (EX-001, EX-002, ...)
Business Scenario	What is happening from a business perspective
Trigger	Event or condition that causes the exception
Expected Business Response	What the organization should do — policies and actions
Responsible Actor	Primary role accountable for resolution
Business Impact	Consequences if not handled correctly

Related Business Event Reference

Where applicable, exceptions map to the Business Events Catalog (BE-xx) and Business Processes (BP-xx).

Domain 1: Master Data & Governance Exceptions

EX-001 — Duplicate Product Identity Detected

Attribute	Detail
Exception ID	EX-001

Attribute	Detail
Business Scenario	A new product submission matches or closely resembles an existing SKU, barcode, or product name, risking parallel product identities for the same physical item.
Trigger	Inventory Controller or stakeholder attempts to create or update a product record (BE-02, BE-01).
Expected Business Response	Halt new record creation; compare existing and proposed records side by side; merge or correct the existing record rather than create a duplicate; document justification if a genuinely new SKU is confirmed.
Responsible Actor	Inventory Controller
Business Impact	Fragmented stock records; inaccurate ATP; duplicate purchasing; reconciliation effort; unreliable network visibility

EX-002 — Discontinued Product With Open Activity

Attribute	Detail
Exception ID	EX-002
Business Scenario	A product marked for discontinuation still has open purchase orders, active reservations, or remaining on-hand stock requiring disposition.
Trigger	Product discontinuation requested (BE-03).
Expected Business Response	Block new replenishment per BR-027; warn stakeholders of dependent open POs and reservations; define runout timeline for remaining stock; cancel or amend open POs; fulfill or release existing reservations per policy; plan clearance or write-off for residual quantity.
Responsible Actor	Inventory Controller (with Purchasing Manager and Operations Director)
Business Impact	Obsolete inventory accumulation; customer orders for withdrawn products; trapped working capital; conflicting assortment decisions

EX-003 — Location Closure With Remaining Stock

Attribute	Detail
Exception ID	EX-003
Business Scenario	A warehouse or store approved for closure still holds on-hand inventory that must be dispositioned before deactivation.
Trigger	Inventory location change approved for closure (BE-04).
Expected Business Response	List all remaining stock; execute disposition plan (transfer out, sell-through, write-off); block new inbound and fulfillment except authorized disposition; obtain Operations Director sign-off before location deactivation.
Responsible Actor	Operations Director
Business Impact	Stranded inventory; orphaned records; financial misstatement; lost asset value

EX-004 — Opening Balance Dispute at Go-Live

Attribute	Detail
Exception ID	EX-004
Business Scenario	Physical opening count results for a new or cutover location are challenged by local management or finance before location activation.
Trigger	Opening stock balance submitted for approval (BP-02, BP-12).
Expected Business Response	Suspend location activation; conduct recount or independent verification for disputed SKUs; resolve catalog gaps for unknown products; require authorized approver sign-off on final baseline before operational transactions begin.
Responsible Actor	Inventory Controller
Business Impact	Go-live with embedded errors; chronic inaccuracy from day one; distrust in system data; cascading fulfillment failures

Domain 2: Inbound & Procurement Exceptions

EX-005 — Supplier Short-Shipment

Attribute	Detail
Exception ID	EX-005
Business Scenario	Supplier delivers fewer units than ordered on a purchase order, especially for a product already low or committed to customer orders.
Trigger	Receipt verification finds quantity below PO expected quantity (BE-15).
Expected Business Response	Record only verified received quantity; document short-shipment discrepancy; exclude disputed quantity from sellable stock; escalate with supplier for balance; recalculate ATP and customer commitments; hold payment pending three-way match resolution per BR-008.
Responsible Actor	Receiving Clerk (document) → Purchasing Manager (supplier resolution)
Business Impact	Phantom inventory if PO quantity accepted silently; stockouts on committed orders; supplier accountability lost; payment for undelivered goods

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EX-006 — Damaged Goods at Receiving

Attribute	Detail
Exception ID	EX-006
Business Scenario	Inbound shipment arrives with visible damage — crushed cartons, broken units, or temperature excursion — making some or all units unsellable.
Trigger	Receiving clerk inspects packaging and product condition (BE-14, BE-15).
Expected Business Response	Segregate damaged goods; accept only sellable units into inventory; photograph and document damage; initiate supplier or carrier claim; quarantine damaged quantity; exclude damaged stock from ATP per BR-013.
Responsible Actor	Receiving Clerk → Warehouse Manager
Business Impact	Defective goods sold to customers; returns and brand damage; unrecoverable supplier claims without dock evidence; inflated on-hand balances

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EX-007 — Wrong Item Received Against Purchase Order

Attribute	Detail
Exception ID	EX-007
Business Scenario	Physical product delivered does not match the purchase order line — different SKU, variant, or entirely wrong merchandise.
Trigger	Product identity verification fails during goods receipt (BE-15).
Expected Business Response	Reject mismatched product from inventory acceptance; document wrong-item discrepancy; notify purchasing and supplier; accept only correct items if partial shipment; do not record wrong SKU as on-hand.
Responsible Actor	Receiving Clerk → Buyer
Business Impact	Inventory records show stock that cannot be fulfilled; customer orders fail at pick; supplier dispute without documentation; payment for wrong goods

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EX-008 — Receipt Without Valid Purchase Order

Attribute	Detail
Exception ID	EX-008
Business Scenario	Supplier truck arrives with no matching open purchase order, or emergency delivery requires receipt outside standard procurement.

Attribute	Detail
Trigger	Receiving initiated without PO reference (BE-14).
Expected Business Response	Hold shipment until purchasing confirms valid PO or grants authorized ad-hoc exception with documented approver and reason per BR-007; record exception for audit; do not accept standard supplier receipt without authorization.
Responsible Actor	Receiving Clerk → Purchasing Manager
Business Impact	Unauthorized spending; maverick buying; three-way match failure; audit control weakness

EX-009 — Over-Shipment Beyond Tolerance

Attribute	Detail
Exception ID	EX-009
Business Scenario	Supplier delivers more units than ordered beyond the business-defined receipt tolerance.
Trigger	Received quantity exceeds PO open quantity plus tolerance (BE-15).
Expected Business Response	Flag over-receipt; require Warehouse Manager or Purchasing Manager approval before acceptance; document decision to accept excess, return excess, or create supplemental PO; adjust payment only for accepted quantity.
Responsible Actor	Receiving Clerk → Warehouse Manager / Purchasing Manager
Business Impact	Unplanned inventory investment; payment for unapproved goods; storage capacity strain; supplier assumes over-shipment is always accepted

EX-010 — Inbound Delivery Delayed Past Customer Promise

Attribute	Detail
Exception ID	EX-010
Business Scenario	Purchase order or transfer shipment is delayed past the date already communicated to customers waiting on backorders or scheduled deliveries.
Trigger	Supplier or carrier reports delay; expected delivery date passes without receipt (BE-39).
Expected Business Response	Update expected arrival internally; proactively notify affected customers of revised dates; escalate with supplier; evaluate network transfer or alternate supply for critical commitments; reassess backorder queue priorities.
Responsible Actor	Buyer → Customer Service Representative
Business Impact	Silent delay destroys customer trust; backorder cancellations; contract penalties; emergency procurement at premium cost

EX-011 — Supplier Lead Time Extension

Attribute	Detail
Exception ID	EX-011
Business Scenario	Supplier communicates longer lead times due to capacity, material, or logistics disruption, invalidating current reorder points and safety stock.
Trigger	Demand or supply condition changed (BE-06); supplier notification during inbound tracking (BE-39).
Expected Business Response	Recalculate reorder levels and safety stock for affected products; place orders earlier; inform sales of longer delivery windows; evaluate alternate suppliers for critical SKUs; document policy revision effective date.
Responsible Actor	Inventory Controller → Purchasing Manager
Business Impact	Preventable stockouts despite alerts; false confidence in replenishment timing; working capital shock if reactive safety stock increase

Domain 3: Inventory Control & Accuracy Exceptions

EX-012 — Material Cycle Count Variance

Attribute	Detail
Exception ID	EX-012
Business Scenario	Cycle count finds physical quantity differs from recorded balance beyond defined tolerance.
Trigger	Count variance identified after blind count submission (BE-24).
Expected Business Response	Order recount if needed; investigate root cause before adjustment per BR-022; document investigation notes; route to independent approver per BR-019; apply adjustment with mandatory reason code per BR-020.
Responsible Actor	Warehouse Manager → Inventory Controller
Business Impact	Recurring variances if root cause ignored; accuracy metrics meaningless; audit findings; undetected shrinkage or process failure

EX-013 — Phantom Inventory (Record Shows Stock, Shelf Empty)

Attribute	Detail
Exception ID	EX-013
Business Scenario	System shows positive available quantity but product cannot be found physically — empty pick location, bare shelf, or customer cannot locate item.
Trigger	Pick exception reported (BE-33); customer or sales inquiry; verification count flagged (BE-21).
Expected Business Response	Conduct physical search including overflow and mislabeled locations; if not found, adjust inventory with shrinkage or count correction reason code immediately; correct ATP to prevent further overselling; investigate root cause (theft, misplacement, unrecorded movement).
Responsible Actor	Warehouse Manager → Inventory Controller
Business Impact	Continued overselling; pick failures; customer embarrassment; lost sales despite "in stock" status; eroded data trust

EX-014 — Unexplained Shrinkage Trend at Location

Attribute	Detail
Exception ID	EX-014
Business Scenario	A location shows rising shrinkage, write-offs, or negative adjustments exceeding its historical norm and network average without a single large incident.
Trigger	Significant loss or adjustment event; monthly shrinkage review (BE-26, BE-27).
Expected Business Response	Analyze trends by SKU, reason code, and shift; conduct process and security review; increase count frequency for affected categories; assign corrective action owner; escalate to Internal Auditor if suspected fraud.
Responsible Actor	Inventory Controller → Operations Director
Business Impact	Escalating unexplained loss; margin erosion; audit material weakness; chronic accuracy failure at location

EX-015 — Quality Hold on Received Lot

Attribute	Detail
Exception ID	EX-015
Business Scenario	Inbound lot fails inspection, receives supplier quality alert, or is flagged for investigation before release to sellable stock.
Trigger	Quality concern at receipt or post-receipt inspection.

Attribute	Detail
Expected Business Response	Move lot to quarantine; exclude from ATP immediately per BR-013; investigate with supplier; disposition as release, return, rework, or write-off with documented authorization; reassess customer orders depending on quarantined lot.
Responsible Actor	Warehouse Manager → Inventory Controller
Business Impact	Shipping defective or non-compliant product; regulatory exposure; customer safety risk; recall scope expansion

EX-016 — Full Physical Count Material Variance

Attribute	Detail
Exception ID	EX-016
Business Scenario	Full wall-to-wall count produces location-wide variance exceeding acceptable threshold for financial substantiation.
Trigger	Full physical count completed (BE-22, BE-23).
Expected Business Response	Investigate significant line items; escalate to Operations Director and Finance; apply approved adjustments with full documentation; identify systemic causes; agree corrected baseline with finance before close.
Responsible Actor	Inventory Controller → Operations Director / Finance Manager
Business Impact	Financial misstatement; audit qualification; delayed month-end close; loss of inventory baseline credibility

Domain 4: Order Fulfillment Exceptions

EX-017 — Customer Order Exceeds Available-to-Promise

Attribute	Detail
Exception ID	EX-017
Business Scenario	Customer order or sales quote requests quantity greater than current ATP at the fulfillment location, including cumulative effect of multiple orders against limited stock.
Trigger	Customer order or quote received (BE-28); large or strategic order submitted (BE-29).
Expected Business Response	Confirm only fulfillable quantity against ATP per BR-001; offer partial shipment, backorder with customer acknowledgment per BR-028, alternate location, or revised delivery date; do not confirm full order without documented plan; reserve only accepted quantity per BR-002.
Responsible Actor	Sales Representative / Customer Service Representative
Business Impact	Overselling; order cancellations; pick failures; customer complaints; expedited shipping costs; damaged sales credibility

EX-018 — Pick Failure — Stock Not Found at Location

Attribute	Detail
Exception ID	EX-018
Business Scenario	Picker cannot locate reserved quantity at the designated storage location despite positive system record.
Trigger	Pick exception reported during fulfillment (BE-33).
Expected Business Response	Conduct immediate physical search; if unfound, trigger phantom inventory response (EX-013); re-pick if found elsewhere; initiate partial fulfillment or order line cancellation if unrecoverable; do not ship without verified pick per BR-003.
Responsible Actor	Warehouse Picker / Packer → Warehouse Manager
Business Impact	Shipment delay; wrong inventory record persists; customer promise missed; cascading pick queue disruption

EX-019 — Damaged Item Discovered During Pick

Attribute	Detail
Exception ID	EX-019
Business Scenario	Product retrieved for order is damaged, expired, or otherwise unsellable at the pick point.
Trigger	Picker reports condition exception during pick verification (BE-33).
Expected Business Response	Quarantine damaged unit; process stock adjustment or write-off with reason code; attempt re-pick from alternate stock if available; substitute only with authorized approval; notify customer service if order affected.
Responsible Actor	Warehouse Picker / Packer → Warehouse Manager
Business Impact	Wrong or damaged goods shipped if ignored; customer return; brand damage; compounding record inaccuracy

EX-020 — Concurrent Demand for Last Available Units

Attribute	Detail
Exception ID	EX-020
Business Scenario	Multiple customers, channels, or locations simultaneously require the last available units of a product — total demand exceeds remaining ATP.
Trigger	Multiple simultaneous availability confirmations or reservations against depleting ATP (BE-28, BE-30).
Expected Business Response	Apply documented allocation priority rules (contract obligation, customer tier, order date, channel priority); hard-reserve allocated quantity; communicate honestly with deprioritized parties; record allocation decision and rationale; do not award stock ad hoc under pressure.
Responsible Actor	Inventory Controller / Sales Manager
Business Impact	Arbitrary fulfillment; B2B contract breach; channel conflict; customer disputes; revenue loss on strategic accounts

EX-021 — Customer Order Cancelled After Reservation

Attribute	Detail
Exception ID	EX-021
Business Scenario	Customer cancels order or reduces quantity after stock was reserved but before shipment.
Trigger	Customer order cancelled (BE-35).
Expected Business Response	Release reservation to restore ATP; return picked but unshipped goods to stock if applicable; record cancellation reason; notify operations if material quantity restored; update fulfillment status for customer service.
Responsible Actor	Customer Service Representative → Warehouse Manager
Business Impact	Stock artificially locked to dead orders; false stockouts; reduced sellable availability; fulfillment queue clutter

EX-022 — Partial Fulfillment Required

Attribute	Detail
Exception ID	EX-022
Business Scenario	Full order quantity cannot be fulfilled at reservation or pick time, but customer policy allows partial delivery.
Trigger	Insufficient stock at fulfillment (BE-34).
Expected Business Response	Identify fulfillable vs. unfulfillable quantity; obtain customer acknowledgment before partial ship per BR-028; ship available portion; place remainder on backorder with ETA or cancellation option; alert purchasing for replenishment.
Responsible Actor	Customer Service Representative → Warehouse Manager

Attribute	Detail
Business Impact	Complete order hold loses partial revenue; silent partial ship without acknowledgment causes disputes; backorder queue unmanaged

EX-023 — Authorized Product Substitution Required

Attribute	Detail
Exception ID	EX-023
Business Scenario	Primary ordered SKU is unavailable at pick, but an approved alternate product could satisfy the customer need.
Trigger	Pick exception or availability shortfall with defined substitute relationship (BE-33).
Expected Business Response	Present only pre-approved substitute options; obtain customer acceptance and authorized approval; reserve and ship substitute SKU; record substitution on order with original and fulfilled product reference; adjust inventory for actual product shipped.
Responsible Actor	Warehouse Manager → Sales Manager / Customer Service Representative
Business Impact	Unauthorized substitute causes margin loss, warranty issues, or customer rejection; order remains unfulfilled if substitution declined without plan

Domain 5: Replenishment Exceptions

EX-024 — False Positive Low-Stock Alert

Attribute	Detail
Exception ID	EX-024
Business Scenario	Low-stock alert fires but investigation shows adequate stock, inbound supply en route, or data timing issue — no genuine replenishment need.
Trigger	Low-stock alert issued (BE-36); purchasing daily review.
Expected Business Response	Validate against ATP, open POs, and in-transit transfers; dismiss alert with documented reason; review threshold configuration if recurring false positives; do not place duplicate purchase order.
Responsible Actor	Buyer / Purchasing Manager
Business Impact	Duplicate procurement; excess inventory; alert fatigue; ignored genuine alerts later

EX-025 — Emergency Stockout on Priority SKU

Attribute	Detail
Exception ID	EX-025
Business Scenario	Class A or priority product reaches zero ATP with active customer demand or open commitments.
Trigger	Emergency stockout declared (BE-10); stock threshold breached at zero (BE-19).
Expected Business Response	Escalate urgently to purchasing and operations leadership; check network for transfer before new purchase per BR-011; initiate expedited PO or emergency transfer; communicate realistic re-availability to sales and customer service; document root cause for post-incident review.
Responsible Actor	Purchasing Manager → Operations Director
Business Impact	Lost sales; customer cancellations; premium freight costs; revenue at risk on priority products; reputational damage

EX-026 — Safety Stock Override Requested

Attribute	Detail
Exception ID	EX-026
Business Scenario	Reservation, transfer, or fulfillment request would consume safety stock below policy buffer without a replenishment plan.
Trigger	Reservation or transfer evaluation against safety stock (BE-30, BE-41).
Expected Business Response	Block routine consumption of safety stock per BR-010; require authorized exception with approver, documented reason, and replenishment plan; record exception for governance review; deny if no recovery plan exists.
Responsible Actor	Operations Director / Inventory Controller
Business Impact	Next demand or supply shock causes immediate stockout; safety investment wasted; chronic crisis replenishment

Domain 6: Network Balance & Transfer Exceptions

EX-027 — Local Stockout With Network Excess Elsewhere

Attribute	Detail
Exception ID	EX-027
Business Scenario	One location is out of stock on a product while other network locations hold available excess quantity.
Trigger	Network imbalance identified (BE-42); low-stock at one site with surplus elsewhere.
Expected Business Response	Verify network ATP; initiate inter-location transfer request before new purchase per BR-011; approve transfer if source can spare stock per BR-012; communicate revised availability to sales.
Responsible Actor	Regional / Store Manager → Operations Director
Business Impact	Duplicate purchasing; excess at one site while another stockouts; trapped capital; lost local sales

EX-028 — Transfer Would Breach Source Minimum

Attribute	Detail
Exception ID	EX-028
Business Scenario	Approved or requested transfer quantity would deplete the source location below its minimum threshold or safety stock.
Trigger	Inter-location transfer requested or approved (BE-41, BE-43).
Expected Business Response	Reduce transfer to maximum spareable quantity per BR-012; reject or modify request with documented reason; suggest alternate source location if available; do not approve transfer that creates source stockout.
Responsible Actor	Operations Director / Warehouse Manager (source)
Business Impact	Cascading stockouts across network; emergency replenishment at source; net worsening of network availability

EX-029 — In-Transit Transfer Receipt Variance

Attribute	Detail
Exception ID	EX-029
Business Scenario	Destination receives fewer units than dispatched, or goods arrive damaged in transit between locations.
Trigger	Transfer shipment received with variance (BE-46).
Expected Business Response	Document receipt variance; investigate loss or damage in transit; adjust source and destination records accordingly; process write-off or claim if applicable; close transfer only when quantities reconciled.
Responsible Actor	Receiving Clerk (destination) → Warehouse Manager

Attribute	Detail
Business Impact	Stock lost between locations; both sites show incorrect balances; accountability gap; unresolved in-transit aging

EX-030 — Emergency Transfer Request

Attribute	Detail
Exception ID	EX-030
Business Scenario	Location urgently needs stock — key customer order, promotional event, or unexpected spike — faster than supplier replenishment allows.
Trigger	Inter-location transfer requested with urgency flag (BE-41).
Expected Business Response	Verify source ATP and minimum threshold; obtain expedited approval; execute pick and ship with documented freight authorization; track in-transit until receipt confirmed; review why standard replenishment failed.
Responsible Actor	Regional / Store Manager → Operations Director
Business Impact	Customer promise missed if transfer delayed; premium freight unbudgeted; source location risk if minimum not protected

Domain 7: Returns Exceptions

EX-031 — Return Received Without Authorization

Attribute	Detail
Exception ID	EX-031
Business Scenario	Warehouse receives returned goods without a valid return authorization linked to an original order.
Trigger	Returned goods arrive at dock (BE-48) without matching authorization.
Expected Business Response	Hold goods; contact customer service to validate or issue retroactive authorization per policy; refuse acceptance if policy violation or fraud suspected; do not add to sellable stock without valid linkage per BR-025.
Responsible Actor	Receiving Clerk → Customer Service Representative
Business Impact	Returns cannot match orders; fraudulent returns accepted; refund without receipt; inventory records incorrect

EX-032 — Returned Product Fails Inspection

Attribute	Detail
Exception ID	EX-032
Business Scenario	Returned item is used, damaged, or defective and does not meet resale quality standards.
Trigger	Return inspection completed with fail outcome (BE-49).
Expected Business Response	Write off or liquidate non-resalable quantity with reason code per BR-026; do not restock to sellable inventory; process customer refund or credit per original resolution agreement; capture return reason for trend analysis.
Responsible Actor	Warehouse Manager → Inventory Controller
Business Impact	Damaged goods resold as new; repeat returns; customer complaints; inflated sellable ATP

EX-033 — Return Quantity or Product Mismatch

Attribute	Detail
Exception ID	EX-033
Business Scenario	Physical return does not match authorization — wrong product, partial quantity, or excess units.

Attribute	Detail
Trigger	Return receipt verification against authorization (BE-48).
Expected Business Response	Document mismatch; contact customer; hold mismatched items; accept only authorized product and quantity; adjust refund or credit proportionally per policy.
Responsible Actor	Receiving Clerk → Customer Service Representative
Business Impact	Over-refund; inventory credited for wrong items; unresolved return status; customer dispute

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Domain 8: Commercial & Promotional Exceptions

EX-034 — Promotional Demand Exceeds Forecast

Attribute	Detail
Exception ID	EX-034
Business Scenario	Marketing promotion drives demand above planned levels and promotional stock depletes faster than campaign duration.
Trigger	Marketing campaign scheduled with accelerating sell-through (BE-55); stock threshold breached on promoted SKUs.
Expected Business Response	Ring-fence remaining promotional stock per BR-029; allocate to promoted channels; expedite replenishment or network transfer if feasible; inform marketing to adjust campaign intensity; communicate "while supplies last" if stock will not last.
Responsible Actor	Marketing Manager → Purchasing Manager / Operations Director
Business Impact	Campaign stockout during peak visibility; promotional ROI lost; customer disappointment at worst moment; channel conflict

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EX-035 — Campaign Stock Gap Before Launch

Attribute	Detail
Exception ID	EX-035
Business Scenario	Pre-launch readiness review finds current plus inbound stock insufficient for campaign targets within lead time.
Trigger	Marketing campaign scheduled or new product launch date set (BE-55, BE-56).
Expected Business Response	Quantify gap by SKU and location; escalate go/no-go decision to commercial and operations leadership; expedite purchasing if recoverable; delay or reduce campaign scope if not; document risk acceptance if proceeding.
Responsible Actor	Marketing Manager → Operations Director
Business Impact	Public launch without stock; brand embarrassment; wasted marketing spend; BO-24 promotional readiness failure

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EX-036 — Channel Conflict Over Shared Inventory

Attribute	Detail
Exception ID	EX-036
Business Scenario	Online, retail, wholesale, or marketplace channels compete for the same limited ATP — conflicting allocation requests for last units.
Trigger	Concurrent orders from multiple channels against shared pool (BE-28).
Expected Business Response	Apply leadership-defined channel priority rules; hard-allocate per EX-020; update channel-facing availability manually until integration available; escalate recurring conflicts to commercial leadership for policy review.
Responsible Actor	Inventory Controller / Sales Manager
Business Impact	Double-selling across channels; store associates vs. e-commerce conflict; wholesale contract breach; inconsistent customer experience

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Domain 9: Financial, Governance & Compliance Exceptions

EX-037 — Operations and Finance Inventory Mismatch at Month-End

Attribute	Detail
Exception ID	EX-037
Business Scenario	Operational inventory quantities or valuation do not reconcile with general ledger balance at month-end close.
Trigger	Month-end financial close initiated (BE-53).
Expected Business Response	Categorize variances (timing, quantity, valuation); investigate unexplained gaps; correct operational records or agree financial adjustment; complete reconciliation within close calendar; escalate material gaps to CFO.
Responsible Actor	Finance Manager → Inventory Controller
Business Impact	Financial misstatement; audit findings; delayed close; BO-18 and BO-20 objectives fail; operations-finance distrust persists

EX-038 — Adjustment or Write-Off Exceeds Approval Threshold

Attribute	Detail
Exception ID	EX-038
Business Scenario	Proposed inventory adjustment or write-off exceeds the approver's financial or quantity authority.
Trigger	Stock correction required (BE-25); significant write-off recorded (BE-54).
Expected Business Response	Route to independent higher-authority approver per BR-021; require documented evidence for write-offs per BR-023; block application until approved; notify finance for material write-offs.
Responsible Actor	Warehouse Manager (initiator) → Operations Director / CFO
Business Impact	Unauthorized large adjustments; fraud opportunity; financial misstatement; audit material weakness

EX-039 — Segregation of Duties Violation Attempted

Attribute	Detail
Exception ID	EX-039
Business Scenario	Same person attempts to perform and approve a conflicting action — count and approve own variance, receive and approve own PO variance, or self-approve material adjustment.
Trigger	Control incident or policy breach identified (BE-59).
Expected Business Response	Block self-approval; route to independent approver per BR-019; log policy breach or authorized exception; review role assignment for systemic conflicts; inform Internal Auditor if material or recurring.
Responsible Actor	Warehouse Manager / Inventory Controller → Operations Director
Business Impact	Fraud and error undetected; SOX or audit failure; invalid accuracy and adjustment records

EX-040 — Product Recall Requiring Lot Quarantine

Attribute	Detail
Exception ID	EX-040
Business Scenario	Quality defect, contamination, or regulatory notice requires recall of specific lot or batch — stock may be in warehouse, in transit, or already shipped.
Trigger	Supplier quality alert; regulatory notice; internal quality failure discovery.
Expected Business Response	Identify all affected lot quantity through traceability records; quarantine immediately and exclude from ATP per BR-013; trace shipped units to customers; notify regulatory authorities within mandated timelines; disposition quarantined stock after investigation.

Attribute	Detail
Responsible Actor	Inventory Controller → Quality / Regulatory (with Operations Director)
Business Impact	Consumer harm; regulatory penalties; full product-line recall if lot data missing; legal liability; brand catastrophe

EX-041 — Control Policy Breach With Authorized Exception

Attribute	Detail
Exception ID	EX-041
Business Scenario	Business situation requires deviation from standard policy — ad-hoc receipt without PO, safety stock override, or emergency approval outside normal authority — with legitimate operational justification.
Trigger	Control incident identified with exception path (BE-59).
Expected Business Response	Document approver, policy reference, business reason, and replenishment or recovery plan; apply exception only within granted authority; log for governance review; do not normalize exception into standard practice.
Responsible Actor	Operations Director / Purchasing Manager
Business Impact	If undocumented: audit finding and control erosion; if denied when needed: operational paralysis during genuine emergency

Domain 10: Demand Shock & Lifecycle Exceptions

EX-042 — Flash Demand Spike From External Event

Attribute	Detail
Exception ID	EX-042
Business Scenario	Unplanned external event — viral trend, competitor stockout redirect, weather — causes sudden demand depleting stock in hours or days.
Trigger	Rapid ATP depletion; multiple urgent availability inquiries; stock threshold breached on unaffected forecast SKUs.
Expected Business Response	Apply allocation rules immediately; assess fastest replenishment path; communicate realistic availability internally and externally; avoid overcommitment; conduct post-event review for safety stock and supplier capacity.
Responsible Actor	Operations Director → Inventory Controller / Purchasing Manager
Business Impact	Overselling during spike; reputational damage outlasting revenue gain; unfilled strategic customer orders; supplier also caught short

EX-043 — End-of-Season or Obsolete Stock Requiring Write-Off

Attribute	Detail
Exception ID	EX-043
Business Scenario	Seasonal, discontinued, or slow-moving product has no remaining commercial value and must be removed from active inventory.
Trigger	Scheduled aging review; product discontinued with residual stock (BE-03); overstock report review.
Expected Business Response	Segregate physically; execute planned exit (markdown, outlet, donation, destroy); process authorized write-off with evidence per BR-023; update financial records; do not delay write-off inflating reported inventory.
Responsible Actor	Inventory Controller → Finance Manager
Business Impact	Trapped working capital; inflated operational and financial inventory; recurring carrying cost; misleading availability if stock remains "active"

Exception Response Pattern Summary

Pattern	Exception IDs	Core Business Discipline
Verify before recording	EX-005 – EX-009, EX-031 – EX-033	Document at point of occurrence; never accept silent variance
Investigate before adjusting	EX-012, EX-013, EX-014, EX-016	Root cause before number correction
Check network first	EX-025, EX-027, EX-030	Transfer before purchase
ATP-only commitment	EX-017, EX-020, EX-022, EX-036	Promise only fulfillable quantity
Independent approval	EX-038, EX-039	Segregation of duties on material actions
Proactive communication	EX-010, EX-017, EX-022, EX-025	Customer and internal honesty on delays
Quarantine and exclude	EX-006, EX-015, EX-040	Non-sellable stock never in ATP
Governed exception only	EX-008, EX-026, EX-041	Documented override — not normalized bypass

Exception-to-Process Traceability

Business Process	Related Exceptions
BP-01 Product Master	EX-001, EX-002
BP-02 Location Setup	EX-003, EX-004
BP-05 – BP-09 Inbound	EX-005 – EX-011
BP-10 – BP-15 Inventory Control	EX-012 – EX-016
BP-16 – BP-20 Fulfillment	EX-017 – EX-023
BP-21 – BP-23 Replenishment	EX-024 – EX-026
BP-24 – BP-26 Network Balance	EX-027 – EX-030
BP-27 – BP-28 Returns	EX-031 – EX-033
BP-31 Promotional Readiness	EX-034, EX-035
BP-30 Financial Reconciliation	EX-037
BP-14 Adjustments & Write-Offs	EX-038, EX-043
BP-32 Audit & Governance	EX-039, EX-041
Cross-cutting	EX-020, EX-036, EX-040, EX-042

Document Approval

Role	Name	Signature	Date
Senior Business Analyst			
Product Owner			
Operations Director			
Inventory Controller			

End of Business Exception Scenarios Specification

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Business Process Decision Tables

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-DT-001
Document Title	Business Process Decision Tables — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Senior Business Analyst
Source Documents	BRD-IMS-001 (Section 10), Business Workflows (WF-01–WF-32), Business Exception Scenarios (EX-001–EX-043)
Date	July 5, 2026
Scope	Business decision logic only — not system rules, validation logic, or technical branching

Purpose

This document identifies **32 business processes** that contain **multiple decision paths** and expresses key branching logic as **decision tables**. Each table maps business **conditions** to **actions** and **outcomes** so stakeholders can see how the organization should respond when situations differ.

How to Read Decision Tables

Column	Meaning
Rule	Rule identifier within the process (R1, R2, ...). Rules are evaluated in priority order unless noted.
Conditions	Business facts that must be true for the rule to apply. "—" means any value.
Actions	What responsible actors should do when conditions are met.
Outcome	Resulting process state or next workflow triggered.

Notation: Y = Yes / True · N = No / False · — = Don't care / Not applicable

Process Index

BP	Process	Decision Table	Complexity
BP-01	Product Master Data Management	DT-01	Medium
BP-02	Location & Warehouse Setup	DT-02	Medium
BP-03	Stock Policy & Threshold Definition	DT-03	Low
BP-04	ABC Product Classification	DT-04	Low
BP-05	Purchase Order Placement	DT-05	High
BP-06	Inbound Shipment Planning	DT-06	Low
BP-07	Goods Receipt & Verification	DT-07	High
BP-08	Receipt Discrepancy Resolution	DT-08	High
BP-09	Stock Put-Away & Storage	DT-09	Low
BP-10	Inventory Status Monitoring	DT-10	Medium
BP-11	Cycle Counting	DT-11	Medium

BP	Process	Decision Table	Complexity
BP-12	Full Physical Inventory Count	DT-12	Medium
BP-13	Count Variance Investigation & Approval	DT-13	High
BP-14	Stock Adjustment & Write-Off	DT-14	High
BP-15	Shrinkage & Loss Management	DT-15	Medium
BP-16	Order Availability Confirmation	DT-16	High
BP-17	Inventory Reservation	DT-17	Medium
BP-18	Pick, Pack & Ship	DT-18	High
BP-19	Partial Fulfillment & Backorder	DT-19	High
BP-20	Outbound Exception Handling	DT-20	High
BP-21	Low-Stock Alert Review	DT-21	High
BP-22	Replenishment Planning & Requisition	DT-22	Medium
BP-23	Inbound Supply Tracking	DT-23	Medium
BP-24	Inter-Location Transfer Request	DT-24	High
BP-25	Inter-Location Transfer Execution	DT-25	Medium
BP-26	Network Stock Rebalancing	DT-26	Medium
BP-27	Customer Return Intake	DT-27	Medium
BP-28	Return-to-Stock or Disposal	DT-28	High
BP-29	Inventory Performance Reporting	DT-29	Low
BP-30	Inventory Valuation & Financial Reconciliation	DT-30	Medium
BP-31	Promotional & Launch Stock Readiness	DT-31	High
BP-32	Inventory Audit & Control Review	DT-32	Medium

Domain 1: Master Data & Governance

BP-01: Product Master Data Management (DT-01)

Decision context: Product create, update, or discontinuation request received.

Rule	Request complete & valid	Duplicate SKU exists	Dept approval	Requested status	Actions	Outcome
R1	N	—	—	—	Return to requester for missing information	Request pending — not in catalog
R2	Y	Y	—	—	Compare records; merge or correct existing SKU; do not create duplicate	Single product identity maintained
R3	Y	N	N	—	Revise definition; resubmit for purchasing and operations review	Request pending approval
R4	Y	N	Y	Active	Finalize record; publish to all locations; enable receive, replenish, fulfill	Product active in network
R5	Y	N	Y	Inactive	Set inactive; block new replenishment; allow fulfill existing stock per policy	Product inactive
R6	Y	N	Y	Discontinued	Set discontinued; block new orders; flag remaining stock for clearance or write-off plan	Product discontinued — runout planning

BP-02: Location & Warehouse Setup (DT-02)

Decision context: Leadership approves a location change (open, close, or reorganize).

Rule	Strategically approved	Change type	Stock disposition planned (if close)	Actions	Outcome
R1	N	—	—	Defer or reject; document reason	No location change
R2	Y	New	—	Configure location; assign manager; set hierarchy; plan opening balance	New location setup in progress
R3	Y	Reorganize	—	Update hierarchy and accountability; no stock disposition required	Network structure updated
R4	Y	Close	N	Block closure; require disposition plan for all on-hand stock	Closure blocked
R5	Y	Close	Y — Transfer	Execute transfers; deactivate when stock cleared	Location closed — stock redistributed
R6	Y	Close	Y — Sell through	Fulfill remaining via sales; deactivate when cleared	Location closed — sell-through
R7	Y	Close	Y — Write off	Authorized write-off of residual; deactivate location	Location closed — stock removed

BP-03: Stock Policy & Threshold Definition (DT-03)

Decision context: Stock policy review for a product at a location.

Rule	Review trigger	Thresholds aligned with lead time & demand	ABC class	Actions	Outcome
R1	New product	—	—	Set conservative initial thresholds; schedule 90-day review	Interim policy active
R2	Chronic stockout	N	—	Recalculate; increase reorder level or safety stock; notify purchasing	Revised policy — stockout mitigation
R3	Chronic overstock	N	—	Reduce thresholds; consider rebalancing or promotion	Revised policy — overstock mitigation
R4	Any	Y	A	Approve thresholds; set priority alert treatment	Policy published — Class A monitoring
R5	Any	Y	B or C	Approve thresholds; standard alert treatment	Policy published — standard monitoring
R6	Any	N	—	Adjust and recalculate; resubmit to Operations Director	Policy not yet effective

BP-04: ABC Product Classification (DT-04)

Decision context: Semi-annual or ad-hoc ABC classification review.

Rule	Data supports classification	Assigned class	Disputed	Actions	Outcome
R1	Y	A	N	Approve; increase count frequency and alert priority	Class A operational priority
R2	Y	B	N	Approve; standard count and alert rules	Class B standard priority
R3	Y	C	N	Approve; reduced count frequency	Class C low priority
R4	N	—	Y	Escalate to Operations Director with business justification	Director decides final class

Rule	Data supports classification	Assigned class	Disputed	Actions	Outcome
R5	Any	Any	N	Defer operational rule update to next cycle	Classification saved; rules unchanged until next cycle

Domain 2: Inbound & Receiving

BP-05: Purchase Order Placement (DT-05)

Decision context: Replenishment need validated; buyer prepares purchase order.

Rule	Sufficient stock or open PO	Order value vs authority	Receiving capacity	Emergency	Actions	Outcome
R1	Y — sufficient stock	—	—	N	Cancel or reduce requisition; reference existing supply	No new PO
R2	N	Within buyer authority	Y	N	Issue standard PO; record inbound commitment	Open PO — standard lead time
R3	N	Exceeds buyer authority	Y	N	Escalate to Purchasing Manager or Finance for approval; then issue PO	Open PO — approved escalation
R4	N	Any	N	N	Negotiate alternate delivery date with supplier	PO delayed — capacity conflict
R5	N	Approved	Y	Y	Issue expedited PO; notify Operations Director; authorize premium freight if needed	Emergency PO placed

BP-06: Inbound Shipment Planning (DT-06)

Decision context: Supplier confirms shipment; inbound date approaching.

Rule	Dock capacity available	Shipment aligns with PO	Priority SKU or stockout	Actions	Outcome
R1	Y	Y	N	Schedule receiving resources; standard processing	Receiving prepared
R2	Y	Y	Y	Schedule receiving; notify sales and operations of inbound ETA	Stakeholders informed — priority inbound
R3	Y	N	—	Flag discrepancy for receiving; alert buyer	Receiving on alert — content mismatch expected
R4	N	—	—	Request delivery reschedule from supplier	Inbound rescheduled

BP-07: Goods Receipt & Verification (DT-07)

Decision context: Supplier shipment arrives at receiving dock. *High-path process — full verification matrix.*

Rule	Valid PO reference	Quantity matches PO	Product identity correct	Condition acceptable	Actions	Outcome
R1	Y	Y	Y	Y	Record accepted quantity; create receipt; route to put-away	Clean receipt → BP-09
R2	N	—	—	—	Hold shipment; contact purchasing for PO or ad-hoc authorization	Receipt blocked — no PO
R3	Y	N	Y	Y	Document quantity variance; accept verified quantity only	Partial receipt → BP-08

Rule	Valid PO reference	Quantity matches PO	Product identity correct	Condition acceptable	Actions	Outcome
R4	Y	—	N	—	Reject wrong item; document discrepancy; do not add to inventory	Wrong item → BP-08
R5	Y	—	Y	N	Segregate damaged units; accept sound units if any; document damage	Damage discrepancy → BP-08
R6	Y	N	N	N	Document all variances; hold disputed quantity from sellable stock	Multi-issue receipt → BP-08

BP-08: Receipt Discrepancy Resolution (DT-08)

Decision context: Discrepancy identified during goods receipt.

Rule	Discrepancy type	Accept partial now	Supplier accepts claim	Actions	Outcome
R1	Short shipment	Y	Y	Record partial receipt; pursue balance shipment or credit	Inventory updated — short resolved
R2	Short shipment	N	—	Hold entire line; negotiate with supplier before acceptance	Receipt pending — full resolution
R3	Over-shipment	Y	Y	Accept within tolerance or per approval; adjust PO/payment	Excess accepted or returned per agreement
R4	Damage	Y	Y	Accept good units; reject damaged; initiate claim	Partial inventory — damage excluded
R5	Wrong item	N	—	Reject wrong merchandise; return to supplier	No inventory impact — wrong item returned
R6	Any	—	N (disputed)	Escalate to Purchasing Manager; withhold payment pending resolution	Dispute open — supplier negotiation

BP-09: Stock Put-Away & Storage (DT-09)

Decision context: Verified goods await placement in storage.

Rule	Designated location available	Special handling required	Same-day put-away feasible	Actions	Outcome
R1	Y	N	Y	Confirm put-away; increase ATP for quantity put away	Stock available for fulfillment
R2	Y	Y	Y	Place in designated special-handling area; confirm put-away	Stock available — special location
R3	N	—	Y	Assign alternate overflow location; plan rebalancing	Stock available — overflow location
R4	Y	—	N	Leave in receiving status; queue for next shift put-away	Received but not yet available (BR-005)

Domain 3: Inventory Control

BP-10: Inventory Status Monitoring (DT-10)

Decision context: Daily or event-driven inventory status review.

Rule	ATP below reorder	ATP zero (stockout)	Overstock detected	Inbound expected soon	Actions	Outcome
R1	Y	N	N	—	Generate low-stock alert; route to BP-21	Low-stock review triggered
R2	—	Y	N	N	Urgent escalation; emergency replenishment or transfer	Stockout response — BP-21 / BP-24
R3	—	Y	N	Y	Monitor inbound; notify sales of ETA; defer emergency PO if inbound covers gap	Stockout mitigated by inbound
R4	N	N	Y	—	Flag for network rebalancing review — BP-26	Overstock flagged
R5	N	N	N	—	Continue routine monitoring	Normal status

BP-11: Cycle Counting (DT-11)

Decision context: Scheduled or ad-hoc cycle count performed.

Rule	Physical vs system within tolerance	Recount needed	Fulfillment active during count	Actions	Outcome
R1	Y	N	—	Close count task; no adjustment	Count confirmed — accurate
R2	N	Y	—	Perform blind recount before submission	Recount in progress
R3	N	N	Y	Record variance; continue operations; route material variance to BP-13	Variance investigation triggered
R4	N	N	N	Count during low-activity window; record variance	Variance recorded — minimal disruption

BP-12: Full Physical Inventory Count (DT-12)

Decision context: Full wall-to-wall count at a location.

Rule	Location prepared	All SKUs counted	Total variance within range	Financial adjustment needed	Actions	Outcome
R1	N	—	—	—	Reschedule count; complete preparation	Count deferred
R2	Y	N	—	—	Identify gaps; recount missing areas	Count incomplete
R3	Y	Y	Y	N	Certify baseline; lift movement freeze	Location accuracy certified
R4	Y	Y	N	Y	Investigate material variances; approve adjustments — BP-13/BP-14	Baseline corrected — financial impact
R5	Y	Y	N	N	Full investigation before close	Investigation required — BP-13

BP-13: Count Variance Investigation & Approval (DT-13)

Decision context: Material count variance identified.

Rule	Recount confirms variance	Root cause identified	Within approver authority	Segregation OK (counter ≠ approver)	Actions	Outcome
R1	N	—	—	—	Dispute variance; order additional recount	Investigation continues

Rule	Recount confirms variance	Root cause identified	Within approver authority	Segregation OK (counter ≠ approver)	Actions	Outcome
R2	Y	Y	Y	Y	Approve adjustment with reason code and documentation	Inventory corrected — explained
R3	Y	Y	N	Y	Escalate to Operations Director for approval	Pending executive approval
R4	Y	N	—	Y	Escalate unexplained variance; loss prevention review if warranted	Unexplained variance — BP-15
R5	Y	Y	Y	N	Reassign to independent approver per BR-019	Independent approval required
R6	Y	Any	—	—	Reject adjustment; continue investigation	No inventory change

BP-14: Stock Adjustment & Write-Off (DT-14)

Decision context: Inventory balance change required (correction, damage, shrinkage, obsolescence).

Rule	Reason code & evidence sufficient	Value within approver authority	Write-off required	Segregation of duties satisfied	Actions	Outcome
R1	N	—	—	—	Return to initiator for documentation	Adjustment not applied
R2	Y	Y	N	Y	Approve correction; update on-hand and ATP	Routine adjustment applied
R3	Y	N	N	Y	Escalate to Operations Director or COO	Pending high-value approval
R4	Y	Y	Y	Y	Approve write-off; notify finance; remove from sellable stock	Write-off applied — financial treatment
R5	Y	—	—	N	Reassign to independent approver	SOD block — rerouted approval
R6	Y	N	Y	Y	Executive and finance approval for material write-off	Material write-off pending

BP-15: Shrinkage & Loss Management (DT-15)

Decision context: Periodic shrinkage review identifies abnormal loss pattern.

Rule	Shrinkage within threshold	Root cause type	Corrective action defined	Actions	Outcome
R1	Y	—	—	Monitor; document in review cycle	Normal shrinkage — no action
R2	N	Operational	Y	Implement process fix or training; follow up next cycle	Corrective plan active
R3	N	Criminal suspected	—	Escalate to loss prevention and Internal Auditor	Security investigation
R4	N	Supplier-related	—	Initiate supplier quality or receipt review with purchasing	Supplier corrective action
R5	N	—	N	Continue monitoring; escalate to Operations Director	Extended review period

Domain 4: Order Fulfillment

BP-16: Order Availability Confirmation (DT-16)

Decision context: Customer order or quote requires availability check before commitment.

Rule	ATP sufficient at primary location	Network ATP sufficient	Inbound arrives in time	Customer accepts partial/delay	Actions	Outcome
R1	Y	—	—	—	Confirm full order; commit delivery date; proceed to BP-17	Full availability confirmed
R2	N	Y	—	—	Evaluate alternate location or split fulfillment — BP-24 / BP-17	Multi-location fulfillment path
R3	N	N	Y	—	Commit with inbound-backed date; disclose timing to customer	Delayed commitment — inbound dependent
R4	N	N	N	Y	Offer partial ship and/or backorder with acknowledgment — BP-19	Partial or backorder path
R5	N	N	N	N	Decline or revise order; alert replenishment — BP-21	Order not confirmed as requested

BP-17: Inventory Reservation (DT-17)

Decision context: Confirmed order requires stock reservation.

Rule	Full ATP available	Partial ATP only	Priority order	Safety stock impact	Actions	Outcome
R1	Y	N	N	N	Reserve full quantity at selected location; queue for pick	Full reservation — BP-18
R2	Y	N	Y	N	Reserve full quantity; place in priority pick queue	Priority reservation
R3	N	Y	—	N	Reserve available portion; route remainder to BP-19	Partial reservation
R4	N	N	—	Y	Block unless authorized safety stock exception with replenishment plan	Reservation blocked — BR-010
R5	N	N	—	N	Release confirmation; return to BP-16 for renegotiation	Reservation failed — availability changed

BP-18: Pick, Pack & Ship (DT-18)

Decision context: Reserved order released to warehouse for physical fulfillment.

Rule	Full quantity locatable	Product condition OK	Order packed complete	Expedited shipment	Actions	Outcome
R1	Y	Y	Y	N	Verify pick; pack; ship; deduct inventory; confirm shipment	Order shipped — complete
R2	Y	Y	Y	Y	Expedited carrier handoff; document premium freight authorization	Priority shipment complete
R3	Y	Y	N	—	Hold shipment; wait for remaining lines or partial ship per BP-19	Shipment held — partial
R4	N	—	—	—	Report pick exception — BP-20	Pick failure path
R5	Y	N	—	—	Quarantine damaged unit; re-pick or exception — BP-20	Damage exception path

BP-19: Partial Fulfillment & Backorder Management (DT-19)

Decision context: Full order quantity cannot be fulfilled immediately.

Rule	Customer accepts partial	Replenishment ETA known	Network transfer faster than PO	Actions	Outcome
R1	Y	Y	—	Ship available now; backorder remainder with communicated ETA; alert purchasing	Partial shipped — backorder tracked
R2	Y	N	—	Ship partial; provide ETA range or cancel option for remainder	Partial shipped — uncertain backorder
R3	N	—	—	Hold full order or cancel per customer preference; release reservation if cancelled	Order held or cancelled
R4	Y	—	Y	Initiate emergency transfer — BP-24; ship when transfer received	Backorder resolved by transfer
R5	Y	Y	N	Place replenishment priority; auto-reserve inbound to backorder queue — BP-17	Backorder awaiting supplier stock

BP-20: Outbound Exception Handling (DT-20)

Decision context: Fulfillment disruption during pick, pack, or ship.

Rule	Exception type	Order still fulfillable	Record inaccurate	Customer accepts substitute	Actions	Outcome
R1	Stock not found	Y (re-pick)	Y	—	Search location; adjust inventory; re-queue pick	Inventory corrected — order continues
R2	Stock not found	N	Y	—	Adjust inventory; cancel or partial per customer	Order revised — record fixed
R3	Damage at pick	Y	—	—	Quarantine; re-pick from alternate stock	Order continues
R4	Damage at pick	N	—	Y	Ship approved substitute with documentation	Substitute shipped
R5	Customer cancellation	—	—	—	Release reservation; return picked goods to stock if applicable	Order cancelled
R6	Carrier failure	Y	—	—	Rebook carrier; maintain shipment ready state	Shipment delayed — rebooked

Domain 5: Replenishment

BP-21: Low-Stock Alert Review (DT-21)

Decision context: Low-stock or out-of-stock alert requires purchasing response.

Rule	Alert valid (not false positive)	Open PO covers need	Network transfer faster	Priority SKU	Actions	Outcome
R1	N	—	—	—	Dismiss with documented reason; review thresholds if recurring	Alert closed — no action
R2	Y	Y	—	—	Monitor open PO; track inbound — BP-23	Awaiting inbound supply
R3	Y	N	Y	—	Initiate transfer request — BP-24	Transfer path
R4	Y	N	N	Y	Emergency escalation; expedited PO or transfer	Emergency replenishment

Rule	Alert valid (not false positive)	Open PO covers need	Network transfer faster	Priority SKU	Actions	Outcome
R5	Y	N	N	N	Forward to replenishment planning — BP-22	Standard replenishment path

BP-22: Replenishment Planning & Requisition (DT-22)

Decision context: Validated replenishment need converted to purchase requisition.

Rule	Budget available	Requisition within PM authority	Multiple SKUs same supplier	Promotional pre-build	Actions	Outcome
R1	N	—	—	—	Defer or reduce quantity; inform sales and operations	Replenishment deferred
R2	Y	Y	N	N	Approve requisition; hand to BP-05 for PO	Requisition approved — PO path
R3	Y	N	—	—	Escalate to Finance or COO per value	Pending executive approval
R4	Y	Y	Y	—	Consolidate into single supplier order	Combined PO requisition
R5	Y	Y	—	Y	Approve larger pre-build per campaign plan — BP-31	Promotional replenishment

BP-23: Inbound Supply Tracking (DT-23)

Decision context: Open purchase order monitored for delivery performance.

Rule	Delivery on schedule	Delay material to stockout	Alternative supply viable	Actions	Outcome
R1	Y	—	—	Continue routine monitoring until receipt	On track
R2	N	N	—	Follow up with supplier; update ETA	Minor delay — tracked
R3	N	Y	Y	Expedite, alternate supplier, or network transfer; notify sales	Critical delay — mitigation active
R4	N	Y	N	Notify operations and customer service; revise customer commitments	Delay communicated — wait
R5	Received	—	—	Close PO line; trigger backorder fulfillment — BP-17 if applicable	Supply arrived

Domain 6: Network Balance

BP-24: Inter-Location Transfer Request (DT-24)

Decision context: Stock movement requested between locations.

Rule	Source has unreserved excess	Transfer vs purchase justified	Within WM authority	Urgent stockout at destination	Source minimum protected	Actions	Outcome
R1	N	—	—	—	—	Reject; document reason; pursue purchase — BP-22	Transfer rejected
R2	Y	N	—	—	—	Reject; wait for supplier replenishment	Purchase preferred

Rule	Source has unreserved excess	Transfer vs purchase justified	Within WM authority	Urgent stockout at destination	Source minimum protected	Actions	Outcome
R3	Y	Y	Y	N	Y	Approve standard transfer — BP-25	Transfer approved
R4	Y	Y	N	—	Y	Escalate to Operations Director for approval	Pending executive approval
R5	Y	Y	—	Y	Y	Expedite approval and execution	Emergency transfer
R6	Y	Y	—	—	N	Reduce quantity to maximum spareable; approve modified transfer	Partial transfer approved (BR-012)

BP-25: Inter-Location Transfer Execution (DT-25)

Decision context: Approved transfer physically executed.

Rule	Source can ship full qty	Receipt matches dispatch	Condition at receipt	Actions	Outcome
R1	Y	Y	Acceptable	Dispatch; receive; update both locations; close transfer	Transfer complete — balances correct
R2	Y	N	—	Investigate variance; adjust records; possible write-off	Transfer variance — investigation
R3	N	—	—	Ship partial; notify destination; cancel or re-request remainder	Partial transfer executed
R4	Y	Y	Damaged	Accept sound units; write off damaged; document transit loss	Partial receipt — damage adjustment

BP-26: Network Stock Rebalancing (DT-26)

Decision context: Network review identifies persistent imbalance.

Rule	Imbalance warrants transfer cost	Inbound PO resolves soon	Plan size	Actions	Outcome
R1	N	Y	—	Monitor; adjust PO destination allocation	No transfer — inbound fix
R2	N	N	—	Continue monitoring; review next cycle	Defer rebalancing
R3	Y	—	Standard	Operations Director approves; initiate BP-24/BP-25 transfers	Rebalancing executed
R4	Y	—	Large network move	COO approves; coordinated multi-location transfers	Executive-approved rebalancing
R5	Y	—	Any	Update location-specific replenishment to prevent recurrence	Policy adjusted — transfer executed

Domain 7: Returns

BP-27: Customer Return Intake (DT-27)

Decision context: Customer requests return.

Rule	Eligible per policy	Resolution type	Replacement ATP available	Return channel	Actions	Outcome
R1	N	—	—	—	Deny with explanation; document decision	Return denied
R2	Y	Refund	—	Ship	Issue RA; notify warehouse and finance	Return authorized — refund path
R3	Y	Replacement	Y	—	Issue RA; place replacement order — BP-16	Return authorized — replacement shipped
R4	Y	Replacement	N	—	Issue RA; backorder replacement	Return authorized — replacement backordered
R5	Y	Store credit	—	Store drop-off	Issue RA; route to store then warehouse	Store return path

BP-28: Return-to-Stock or Disposal (DT-28)

Decision context: Returned goods received and inspected.

Rule	Matches authorization	Condition for resale	Quarantine required	Actions	Outcome
R1	N	—	—	Contact customer; hold goods; resolve mismatch	Return mismatch — CS intervention
R2	Y	Resalable	N	Restock to sellable inventory; process refund/credit	Return complete — stock recovered
R3	Y	Not resalable	N	Write off with return reason code; process refund per policy	Return complete — stock written off
R4	Y	Pending	Y	Hold in quarantine; inspect; disposition after outcome	Inspection pending — quarantined
R5	Y	Rework possible	—	Route to rework; restock if passes secondary inspection	Rework path

Domain 8: Performance, Financial & Governance

BP-29: Inventory Performance Reporting (DT-29)

Decision context: Scheduled or ad-hoc reporting cycle.

Rule	KPI breach	Audience	Data anomaly found	Actions	Outcome
R1	Y	Executive	N	Escalate; assign corrective action owner	Management intervention triggered
R2	N	Operational	N	Distribute operational detail report	Routine reporting
R3	N	Management	N	Distribute management summary	Standard management review
R4	—	Executive	N	Distribute executive dashboard	Executive visibility
R5	—	—	Y	Investigate before distribution; or distribute with data caveat	Qualified report issued

BP-30: Inventory Valuation & Financial Reconciliation (DT-30)

Decision context: Month-end operational vs finance inventory reconciliation.

Rule	Variance within tolerance	Timing difference explains gap	Operational correction needed	Actions	Outcome
R1	Y	—	—	Accept reconciliation; support financial posting	Clean close
R2	N	Y	N	Document timing explanation; reconcile without ops change	Explained timing variance
R3	N	N	Y	Initiate BP-14 adjustments; re-reconcile	Ops correction before close
R4	N	N	N	Hold financial posting; escalate to CFO and Operations Director	Close blocked — investigation
R5	Y	—	—	CFO approves; archive substantiation for audit	Reconciliation approved

BP-31: Promotional & Launch Stock Readiness (DT-31)

Decision context: Campaign or launch approaching; stock readiness assessed.

Rule	Current + inbound sufficient	Gap closable before launch	Ops capacity for surge	Actions	Outcome
R1	Y	—	Y	Go decision; ring-fence promotional stock — BR-029	Launch ready — go
R2	N	Y	Y	Expedite pre-build PO; proceed with risk note	Launch proceed — mitigated gap
R3	N	N	—	Delay campaign or reduce scope	Launch delayed or reduced
R4	—	—	N	Limit promotion intensity or add fulfillment capacity	Launch conditional — capacity constraint
R5	N	Y	N	Go with fulfillment risk acceptance documented	Launch proceed — ops risk accepted

BP-32: Inventory Audit & Control Review (DT-32)

Decision context: Internal or external audit of inventory controls.

Rule	Controls effective	Finding material	Corrective action needed	External audit risk	Actions	Outcome
R1	Y	—	—	—	Document clean audit; no action	Controls effective
R2	N	N	Process improvement	N	Assign improvement actions; follow up next cycle	Minor remediation
R3	N	Y	Immediate	Y	Escalate to COO and CFO; accelerated remediation	Material finding — executive escalation
R4	N	Y	Policy update	N	Update governance policies — BP-03/BP-12; retrain	Policy remediation
R5	—	—	—	Y	Prepare evidence package; support external auditor	External audit support

Cross-Process Decision Flow Summary

High-frequency decision chains across processes:

```
Goods Receipt (BP-07)
  ├─ Clean → Put-Away (BP-09) → Available
  └─ Variance → Discrepancy (BP-08) → Resolved receipt

Order Intake (BP-16)
  ├─ Full ATP → Reserve (BP-17) → Pick/Ship (BP-18)
  ├─ Partial ATP → Partial/Backorder (BP-19)
  └─ No ATP → Replenish (BP-21) or Transfer (BP-24)

Low Stock (BP-21)
  ├─ False alert → Dismiss
  ├─ Open PO → Track (BP-23)
  ├─ Network stock → Transfer (BP-24)
  └─ No network stock → Requisition (BP-22) → PO (BP-05)

Count Variance (BP-11/12)
  └─ Material variance → Investigate (BP-13)
        ├─ Approved → Adjust (BP-14)
        └─ Rejected → Continue investigation
```

Traceability Reference

Source	Identifier
Business Processes	BP-01 – BP-32
Business Workflows	WF-01 – WF-32
Business Rules	BR-001 – BR-032
Business Exception Scenarios	EX-001 – EX-043

Document Approval

Role	Name	Signature	Date
Senior Business Analyst			
Product Owner			
Operations Director			

End of Business Process Decision Tables

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Business Rule Dependency Analysis

Inventory Management System

Document Information

Attribute	Detail
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Document Title	Business Rule Dependencies — Inventory Management System
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Attribute	Detail
Status	Draft
Document Owner	Senior Domain Expert
Source Documents	SRS-BR-001, BRD-IMS-001 (Section 13), Business Workflows (WF-01–WF-32), SRS-DT-001
Date	July 5, 2026
Scope	Business rule relationships only — not system implementation, validation rules, or technical constraints

Purpose

This document analyzes **dependencies among the 32 business rules** (BR-001–BR-032). For each dependency it identifies the **parent rule**, **dependent rule**, **dependency type**, and **business impact**. It also catalogs **conflicting** and **overlapping** rules that require explicit policy resolution.

Dependency Type Definitions

Type	Meaning
Prerequisite	The parent rule must be satisfied or in force before the dependent rule can be applied meaningfully.
Sequential	The parent rule governs an earlier step in the same business process; the dependent rule applies only after the parent step completes.
Compositional	The dependent rule builds on or is calculated from conditions established by the parent rule.
Constraint	The parent rule limits, bounds, or restricts how the dependent rule may be executed.
Exception	The parent rule defines an authorized exception path to the dependent rule under defined conditions.
Reinforcement	Rules apply to the same control objective from different angles; both must hold simultaneously.

Conflict / Overlap Classification

Classification	Meaning
Scope Overlap	Two rules govern the same business object or decision point with partially duplicate intent.
Operational Tension	Both rules are valid but compete for the same resource, timing, or stock pool.
Priority Ambiguity	Precedence between rules is not explicit; actors may apply them in conflicting order.
Mutual Exclusion	Rules require mutually exclusive actions in the same context without a defined selector.

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1. Dependency Catalog

Dependencies are grouped by business domain. Rule pairs already noted in SRS-BR-001 as interdependent combinations are included and extended.

1.1 Availability, Commitment & Fulfillment

ID	Parent Rule	Dependent Rule	Dependency Type	Business Impact
DEP-001	**BR-005** Put-Away Before Availability	**BR-001** ATP-Only Customer Commitment	Compositional	ATP cannot include received-but-not-put-away stock. Violating BR-005 inflates ATP and causes BR-001 overselling — customers confirmed against stock pickers cannot locate.
DEP-002	**BR-013** Quarantine Exclusion	**BR-001** ATP-Only Customer Commitment	Compositional	Quarantined, hold, and recall stock must be excluded from ATP. If BR-013 is ignored, BR-001 permits commitment to non-sellable inventory — regulatory, quality, and customer risk.

ID	Parent Rule	Dependent Rule	Dependency Type	Business Impact
DEP-003	**BR-030** Bonded Inventory Clearance	**BR-001** ATP-Only Customer Commitment	Compositional	Bonded stock is non-sellable until customs release. Same ATP exclusion logic as BR-013; failure blocks legal domestic sale while appearing available.
DEP-004	**BR-001** ATP-Only Customer Commitment	**BR-002** Reservation at Order Confirmation	Compositional	BR-001 defines what quantity is promiseable; BR-002 locks that quantity at confirmation. Without BR-001's ATP definition, reservations lack a consistent baseline and double-allocation risk returns.
DEP-005	**BR-002** Reservation at Order Confirmation	**BR-001** ATP-Only Customer Commitment	Compositional	Each reservation reduces ATP. BR-002 is the mechanism that keeps BR-001 accurate as orders confirm — circular but mutually defining: ATP = on-hand minus reservations and non-sellable holds.
DEP-006	**BR-001** ATP-Only Customer Commitment	**BR-028** Backorder Customer Acknowledgment	Sequential	When ATP is insufficient at confirmation, BR-028 governs the customer agreement path before backorder is recorded. BR-001 prevents silent over-commitment; BR-028 governs explicit under-commitment.
DEP-007	**BR-002** Reservation at Order Confirmation	**BR-003** Verified Pick Before Shipment	Sequential	Stock must be reserved before pick execution. BR-003 deducts inventory only after verified pick; BR-002 ensures the picked quantity was already committed to the correct order.
DEP-008	**BR-002** Reservation at Order Confirmation	**BR-028** Backorder Customer Acknowledgment	Sequential	Partial reservation (available portion) pairs with BR-028 for the unreserved remainder. Without sequencing, partial ship may occur without customer agreement on the backorder portion.
DEP-009	**BR-010** Safety Stock Protection	**BR-002** Reservation at Order Confirmation	Constraint	Reservations must not consume safety stock without authorized exception and replenishment plan. Violation depletes buffer protecting against demand/supply uncertainty — next disruption causes stockout despite apparent policy compliance.
DEP-010	**BR-029** Promotional Stock Ring-Fencing	**BR-001** ATP-Only Customer Commitment	Constraint	Promotional allocation reduces ATP available to non-promotional demand. BR-029 further restricts which orders may consume ring-fenced stock beyond ATP arithmetic alone.
DEP-011	**BR-029** Promotional Stock Ring-Fencing	**BR-002** Reservation at Order Confirmation	Constraint	Non-promotional orders must not reserve ring-fenced promotional stock without commercial leadership approval. Prevents campaign stock erosion through standard reservation.
DEP-012	**BR-014** FEFO Rotation	**BR-003** Verified Pick Before Shipment	Constraint	Pick verification must confirm correct lot/batch per FEFO priority. Wrong lot picked violates rotation even if quantity matches — expiry write-off and compliance risk.
DEP-013	**BR-015** FIFO Rotation	**BR-003** Verified Pick Before Shipment	Constraint	Same as DEP-012 for non-perishable goods — oldest receipt date must be picked first.
DEP-014	**BR-017** Lot/Batch Capture	**BR-003** Verified Pick Before Shipment	Prerequisite	Traceable products require lot attribution at pick. BR-003 verification is incomplete without lot capture — recall scope cannot be determined.
DEP-015	**BR-017** Lot/Batch Capture	**BR-014** FEFO Rotation	Reinforcement	FEFO requires expiry-aware lot identity; BR-017 provides the data FEFO depends on for shelf-life products.

1.2 Inbound, Receiving & Procurement

ID	Parent Rule	Dependent Rule	Dependency Type	Business Impact
DEP-016	**BR-007** PO Required for Supplier Receipt	**BR-004** Verify Before Inventory Acceptance	Prerequisite	Verification is performed against an authorized PO (or approved ad-hoc exception). Without BR-007, verification lacks a reference baseline — unauthorized receipts enter inventory.
DEP-017	**BR-004** Verify Before Inventory Acceptance	**BR-005** Put-Away Before Availability	Sequential	Only verified quantities proceed to put-away. Put-away without verification propagates unconfirmed stock into the availability chain (DEP-001 risk).
DEP-018	**BR-004** Verify Before Inventory Acceptance	**BR-006** Document Receipt Discrepancies	Sequential	Discrepancies are identified during verification and must be documented before acceptance. BR-006 preserves evidence BR-004 discovers — undocumented acceptance waives supplier claims.

ID	Parent Rule	Dependent Rule	Dependency Type	Business Impact
DEP-019	**BR-006** Document Receipt Discrepancies	**BR-008** Three-Way Match Before Payment	Prerequisite	Documented receipt quantities are the "received" leg of three-way match. Missing discrepancy documentation causes payment for goods not received or failure to withhold payment for shorts.
DEP-020	**BR-007** PO Required for Supplier Receipt	**BR-008** Three-Way Match Before Payment	Prerequisite	PO is the "ordered" leg of three-way match. Receiving without PO (BR-007 violation) makes BR-008 impossible — payment control breakdown.
DEP-021	**BR-004** Verify Before Inventory Acceptance	**BR-008** Three-Way Match Before Payment	Prerequisite	Verified receipt is the "received" leg. Unverified acceptance corrupts the match and financial records simultaneously.
DEP-022	**BR-016** Minimum Remaining Shelf Life at Receipt	**BR-004** Verify Before Inventory Acceptance	Constraint	MRSL check is part of condition verification for shelf-life products. Failed MRSL triggers reject/disposition before acceptance — prevents BR-014 FEFO from managing already-expired inbound stock.
DEP-023	**BR-016** Minimum Remaining Shelf Life at Receipt	**BR-014** FEFO Rotation	Reinforcement	MRSL at receipt ensures sufficient selling window; FEFO during fulfillment minimizes remaining expiry at customer delivery. Jointly protect shelf-life economics.
DEP-024	**BR-017** Lot/Batch Capture	**BR-004** Verify Before Inventory Acceptance	Constraint	Traceable products require lot capture at receipt as part of verification completeness.
DEP-025	**BR-017** Lot/Batch Capture	**BR-005** Put-Away Before Availability	Sequential	Lot identity must be assigned before or during put-away so subsequent movements maintain traceability.
DEP-026	**BR-030** Bonded Inventory Clearance	**BR-005** Put-Away Before Availability	Constraint	Bonded goods may be physically stored but must not enter sellable availability (DEP-003) until customs clearance — put-away to sellable location prohibited until cleared.
DEP-027	**BR-031** Consignment Ownership at Consumption	**BR-004** Verify Before Inventory Acceptance	Constraint	Consignment receipt is physical verification without ownership transfer. Misclassifying as owned at receipt overstates assets and breaks BR-031 valuation treatment.
DEP-028	**BR-031** Consignment Ownership at Consumption	**BR-008** Three-Way Match Before Payment	Exception	Consignment may not trigger standard PO-to-payment flow until consumption event per supplier agreement — BR-008 applies at ownership transfer, not physical receipt.

1.3 Replenishment & Network Balance

ID	Parent Rule	Dependent Rule	Dependency Type	Business Impact
DEP-029	**BR-009** Reorder at Reorder Level	**BR-021** Low-Stock Alert Review (via BP-21)	Sequential	Reorder level breach triggers replenishment review. BR-009 mandates same-business-day action; delay guarantees stockout during lead time.
DEP-030	**BR-011** Network Check Before Purchase	**BR-009** Reorder at Reorder Level	Constraint	Before executing replenishment (BR-009), network transfer must be evaluated. Skipping BR-011 causes unnecessary PO spend and worsens network imbalance while BR-009 appears satisfied.
DEP-031	**BR-011** Network Check Before Purchase	**BR-012** Protect Source Location Minimum	Reinforcement	Network check (BR-011) often resolves shortage via transfer; BR-012 ensures the transfer path does not create a new shortage at source. Both required for effective network balance.
DEP-032	**BR-012** Protect Source Location Minimum	**BR-011** Network Check Before Purchase	Constraint	Transfer feasibility assessment must respect source minimum. BR-011 "network has stock" is insufficient if BR-012 blocks the transferable quantity.
DEP-033	**BR-010** Safety Stock Protection	**BR-012** Protect Source Location Minimum	Reinforcement	Both protect minimum stock buffers — BR-010 at policy level, BR-012 at transfer execution. Source minimum typically includes safety stock component.
DEP-034	**BR-027** No Reorder of Discontinued Products	**BR-009** Reorder at Reorder Level	Constraint	BR-009 replenishment action is prohibited for discontinued SKUs. Alert may still fire but response must be runout/clearance, not reorder.

ID	Parent Rule	Dependent Rule	Dependency Type	Business Impact
DEP-035	**BR-027** No Reorder of Discontinued Products	**BR-011** Network Check Before Purchase	Constraint	Network transfer may still move discontinued stock for clearance; new procurement is blocked. Prevents obsolete inventory creation.
DEP-036	**BR-032** Class A Priority Monitoring	**BR-009** Reorder at Reorder Level	Constraint	Class A SKUs receive heightened monitoring and faster escalation when approaching reorder level. BR-032 intensifies BR-009 response without changing the threshold logic.
DEP-037	**BR-032** Class A Priority Monitoring	**BR-011** Network Check Before Purchase	Reinforcement	Priority SKUs require expedited network evaluation before external purchase — reduces stockout duration on high-impact items.

1.4 Inventory Control & Governance

ID	Parent Rule	Dependent Rule	Dependency Type	Business Impact
DEP-038	**BR-018** Blind Cycle Counting	**BR-022** Investigate Before Adjusting	Sequential	Blind count produces unbiased variance detection that triggers investigation. Showing system quantity before count (BR-018 violation) undermines variance integrity and root cause analysis.
DEP-039	**BR-022** Investigate Before Adjusting	**BR-020** Mandatory Adjustment Reason Codes	Sequential	Investigation findings must be captured as reason code at adjustment. BR-020 without BR-022 produces codes without verified root cause — analytically weak.
DEP-040	**BR-022** Investigate Before Adjusting	**BR-021** Independent Approval for Material Adjustments	Sequential	Investigation completes before approval decision. Approving without investigation (BR-022 violation) repeats same variance on next count.
DEP-041	**BR-018** Blind Cycle Counting	**BR-019** Independent Count Approval	Reinforcement	Blind count (BR-018) detects variance; independent approval (BR-019) prevents counter from self-approving biased or erroneous counts. Joint fraud/error control.
DEP-042	**BR-019** Independent Count Approval	**BR-021** Independent Approval for Material Adjustments	Reinforcement	BR-019 applies to count variances; BR-021 applies to material adjustments generally. Same segregation principle — overlapping scope (see OV-003).
DEP-043	**BR-020** Mandatory Adjustment Reason Codes	**BR-023** Documented Write-Off Authorization	Compositional	Write-off is a subset of adjustment requiring reason code (BR-020) plus additional evidence and authorization (BR-023).
DEP-044	**BR-021** Independent Approval for Material Adjustments	**BR-023** Documented Write-Off Authorization	Reinforcement	Material write-offs require both threshold-based independent approval (BR-021) and documented evidence (BR-023).
DEP-045	**BR-024** Physical Inventory Freeze	**BR-018** Blind Cycle Counting	Constraint	Full count freeze suspends movements that would change quantities during count. Cycle counts during operations (BR-018) continue under BP-11 without full freeze.
DEP-046	**BR-024** Physical Inventory Freeze	**BR-003** Verified Pick Before Shipment	Operational Tension	Freeze suspends shipping until count complete. Conflicts with open order fulfillment — requires freeze scheduling and customer communication (see CF-001).
DEP-047	**BR-024** Physical Inventory Freeze	**BR-005** Put-Away Before Availability	Operational Tension	Freeze suspends receiving/put-away at counted location. Inbound goods queue — delays BR-005 and downstream BR-001 ATP increase.
DEP-048	**BR-013** Quarantine Exclusion	**BR-020** Mandatory Adjustment Reason Codes	Sequential	Quarantine disposition (damage, recall) typically results in adjustment with mandatory reason code — links quality hold to governance trail.
DEP-049	**BR-032** Class A Priority Monitoring	**BR-018** Blind Cycle Counting	Constraint	Class A products receive highest count frequency per BR-032. More frequent BR-018 application on priority SKUs.

1.5 Returns & Commercial

ID	Parent Rule	Dependent Rule	Dependency Type	Business Impact
DEP-050	**BR-025** Return Authorization Required	**BR-026** Inspect Returns Before Restocking	Sequential	Authorization precedes physical receipt; inspection precedes restocking. BR-026 cannot apply to unauthorized returns accepted at dock.
DEP-051	**BR-026** Inspect Returns Before Restocking	**BR-013** Quarantine Exclusion	Sequential	Failed inspection routes to quarantine or write-off — must not enter ATP until disposition. BR-026 is the gate; BR-013 maintains exclusion during hold.
DEP-052	**BR-026** Inspect Returns Before Restocking	**BR-001** ATP-Only Customer Commitment	Compositional	Only inspection-passed returns increase sellable ATP. Restocking without inspection (BR-026 violation) inflates ATP with potentially unsellable goods.
DEP-053	**BR-025** Return Authorization Required	**BR-028** Backorder Customer Acknowledgment	Reinforcement	Both require explicit customer agreement before recording a commercial state (return acceptance / backorder). Shared consent principle across inbound return and outbound delay.
DEP-054	**BR-026** Inspect Returns Before Restocking	**BR-023** Documented Write-Off Authorization	Sequential	Non-resalable returns proceed to write-off path requiring BR-023 authorization and evidence.

1.6 Cross-Domain & Financial

ID	Parent Rule	Dependent Rule	Dependency Type	Business Impact
DEP-055	**BR-004** Verify Before Inventory Acceptance	**BR-031** Consignment Ownership at Consumption	Compositional	Physical receipt verified under BR-004; ownership treatment governed separately by BR-031.
DEP-056	**BR-008** Three-Way Match Before Payment	**BR-031** Consignment Ownership at Consumption	Sequential	Standard three-way match at invoice; consignment payment triggers at consumption per agreement — different timing, same control intent.
DEP-057	**BR-020** Mandatory Adjustment Reason Codes	**BR-008** Three-Way Match Before Payment	Reinforcement	Inventory adjustments from receipt discrepancies (reason codes) must align with received quantities used in BR-008 — financial and operational records must reconcile.
DEP-058	**BR-001** ATP-Only Customer Commitment	**BR-031** Consignment Ownership at Consumption	Constraint	Consignment stock may be physically present but not owned — must be excluded from owned ATP or separately identified to prevent BR-001 misapplication.

2. Dependency Summary Matrix

Reading guide: Row = Parent; Column = Dependent. Cell shows dependency type abbreviation.

Parent ↓ / Dependent →	BR-001	BR-002	BR-003	BR-004	BR-005	BR-008	BR-009	BR-013	BR-018	BR-020	BR-022	BR-026	BR-028
BR-001	—	Comp	—	—	—	—	—	—	—	—	—	—	Seq
BR-002	Comp	—	Seq	—	—	—	—	—	—	—	—	—	Seq
BR-004	—	—	—	—	Seq	Pre	—	—	—	—	—	—	—
BR-005	Comp	—	—	—	—	—	—	—	—	—	—	—	—
BR-007	—	—	—	Pre	—	Pre	—	—	—	—	—	—	—
BR-009	—	—	—	—	—	—	—	—	—	—	—	—	—
BR-010	—	Con	—	—	—	—	—	—	—	—	—	—	—
BR-011	—	—	—	—	—	—	Con	—	—	—	—	—	—

Parent ↓ / Dependent →	BR-001	BR-002	BR-003	BR-004	BR-005	BR-008	BR-009	BR-013	BR-018	BR-020	BR-022	BR-026	BR-028
BR-013	Comp	—	—	—	—	—	—	—	—	Seq	—	Seq	—
BR-018	—	—	—	—	—	—	—	—	—	—	Seq	—	—
BR-022	—	—	—	—	—	—	—	—	—	Seq	—	—	—
BR-025	—	—	—	—	—	—	—	—	—	—	—	Seq	Ref
BR-029	Con	Con	—	—	—	—	—	—	—	—	—	—	—

Abbreviations: Comp = Compositional, Seq = Sequential, Pre = Prerequisite, Con = Constraint, Ref = Reinforcement

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3. Rule Dependency Chains

Critical end-to-end chains where breaking any link undermines the chain:

Chain A — Sellable Availability (Inbound to Customer)

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BR-007 (PO) → BR-004 (Verify) → BR-006 (Document variance)
     → BR-005 (Put-away) → BR-013/BR-030 (Exclude non-sellable)
     → BR-001 (ATP) → BR-002 (Reserve) → BR-003 (Pick/Ship)
```

Business impact of chain failure: Any break causes phantom availability, overselling, or shipment of non-compliant stock.

Chain B — Replenishment Without Waste

```
BR-009 (Reorder trigger) → BR-011 (Network check) → BR-012 (Source minimum)
     → [Transfer] or [BR-007 PO path]
```

Business impact: Skipping network check wastes capital; ignoring source minimum relocates shortage.

Chain C — Count Integrity

```
BR-018 (Blind count) → BR-022 (Investigate) → BR-020 (Reason code)
     → BR-019/BR-021 (Independent approval)
```

Business impact: Skipping investigation or approval perpetuates errors and fraud exposure.

Chain D — Financial Control

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BR-007 (PO) → BR-004 (Verify) → BR-006 (Document) → BR-008 (Three-way match)
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Business impact: Payment for unordered, unreceived, or mispriced goods; audit failure.

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4. Conflicting Rules

Rules that create genuine tension requiring explicit policy, timing, or precedence resolution.

CF-001 — BR-024 Physical Inventory Freeze vs. BR-002 / BR-003 Fulfillment

Attribute	Detail
Rules in Conflict	BR-024 (suspend shipping during full count) vs. BR-002 (honor reservations) vs. BR-003 (ship verified picks)
Conflict Type	Operational Tension
Nature	Full count freeze stops outbound movements; confirmed orders with reservations expect fulfillment.

Attribute	Detail
Business Impact	Customer delivery delays; reservation hold times extend; potential BR-028 backorder triggers if freeze is prolonged.
Recommended Resolution	Schedule freeze during low-activity window; pre-communicate delays; complete freeze before peak; regional freeze rotation so network can fulfill from alternate locations (BR-011).

CF-002 — BR-024 Physical Inventory Freeze vs. BR-005 / BR-004 Inbound

Attribute	Detail
Rules in Conflict	BR-024 vs. BR-004 / BR-005
Conflict Type	Operational Tension
Nature	Freeze suspends receiving and put-away; supplier deliveries may arrive on schedule regardless.
Business Impact	Dock congestion; delayed ATP increase; supplier detention charges; inbound backlog after freeze lifts.
Recommended Resolution	Coordinate inbound schedule with count calendar; stage received goods without put-away to quarantine/staging if policy allows; or reschedule supplier deliveries.

CF-003 — BR-009 Same-Day Replenishment vs. BR-011 Network Evaluation Time

Attribute	Detail
Rules in Conflict	BR-009 (act same business day at reorder level) vs. BR-011 (evaluate network before purchase)
Conflict Type	Priority Ambiguity
Nature	Thorough network transfer evaluation may delay PO placement beyond same day; rushing BR-011 risks missing transferable stock.
Business Impact	Either stockout (BR-009 delayed) or unnecessary PO (BR-011 skipped).
Recommended Resolution	Define SLA: network check within same business day for Class A (BR-032); standard SKUs may place PO same day while transfer evaluation runs in parallel for next cycle.

CF-004 — BR-006 Document Before Acceptance vs. Partial Receipt Urgency

Attribute	Detail
Rules in Conflict	BR-006 (document all discrepancies before acceptance) vs. operational need to accept good units immediately
Conflict Type	Priority Ambiguity
Nature	Damaged/wrong items mixed with good stock — strict interpretation blocks all acceptance until full documentation; operations want good portion in inventory.
Business Impact	Dock hold delays BR-005/BR-001; or premature full acceptance waives claims (BR-006 violation).
Recommended Resolution	Accept verified good quantity with documented discrepancy on remainder — partial acceptance path in BP-08/DT-08; never accept undocumented variance.

CF-005 — BR-029 Promotional Ring-Fence vs. BR-001 / BR-002 General Fulfillment

Attribute	Detail
Rules in Conflict	BR-029 (promo stock reserved for campaign) vs. BR-001/BR-002 (ATP and reservation for any confirmed order)
Conflict Type	Operational Tension
Nature	ATP may show stock available that BR-029 reserves for promotional channel only.
Business Impact	Campaign stock consumed by routine orders; promoted channel stockout; or routine orders rejected despite apparent ATP.

Attribute	Detail
Recommended Resolution	Treat ring-fenced quantity as separate ATP pool or hard reservation at campaign setup; BR-029 overrides general BR-001 for ring-fenced portion.

CF-006 — BR-010 Safety Stock Exception vs. BR-002 Reservation for Urgent Orders

Attribute	Detail
Rules in Conflict	BR-010 (no safety stock consumption without exception) vs. BR-002 (reserve at confirmation for urgent/priority orders)
Conflict Type	Priority Ambiguity
Nature	Urgent order confirmation may require stock below safety threshold; BR-010 blocks unless exception documented.
Business Impact	Order confirmation fails despite physical stock; or safety buffer eroded without replenishment plan.
Recommended Resolution	Explicit safety-stock exception workflow with approver, replenishment plan, and time limit; priority orders do not auto-bypass BR-010.

CF-007 — BR-007 PO Required vs. Emergency Ad-Hoc Receipt

Attribute	Detail
Rules in Conflict	BR-007 (PO required) vs. emergency receiving without pre-issued PO
Conflict Type	Exception Gap
Nature	BR-007 allows "explicitly approved ad-hoc receipt exception" but BR-008 three-way match assumes standard PO flow.
Business Impact	Emergency stock received but payment blocked; or unauthorized receipt bypasses financial controls.
Recommended Resolution	Define ad-hoc exception approver, retroactive PO process, and BR-008 alternate match rules for emergency receipts.

5. Overlapping Rules

Rules with partially duplicate scope that are complementary when coordinated, or redundant/confusing when not.

OV-001 — BR-013 Quarantine Exclusion vs. BR-030 Bonded Inventory Clearance

Attribute	Detail
Overlap Type	Scope Overlap
Shared Scope	Stock excluded from sellable ATP
Distinction	BR-013: quality hold, investigation, recall. BR-030: customs/regulatory hold.
Business Impact	If treated as one status, customs-cleared bonded stock may remain blocked; or quality-hold stock may be released incorrectly.
Recommendation	Maintain separate hold reasons; both exclude from ATP (DEP-002, DEP-003); disposition paths differ.

OV-002 — BR-014 FEFO vs. BR-015 FIFO

Attribute	Detail
Overlap Type	Mutual Exclusion (by product attribute)
Shared Scope	Pick and ship rotation priority
Distinction	BR-014 applies to shelf-life products; BR-015 to non-perishable standard goods.
Business Impact	Applying wrong rule ships suboptimal lot — expiry write-off (BR-014) or aged stock accumulation (BR-015).

Attribute	Detail
Recommendation	Product master must classify rotation rule per SKU; never apply both simultaneously to same product.

OV-003 — BR-019 Independent Count Approval vs. BR-021 Independent Approval for Material Adjustments

Attribute	Detail
Overlap Type	Scope Overlap
Shared Scope	Segregation of duties for inventory adjustments
Distinction	BR-019: count-specific (counter ≠ approver). BR-021: value/quantity threshold for any adjustment.
Business Impact	Low-value count variance may satisfy BR-019 but not BR-021 threshold — approver rules differ. Material non-count adjustment triggers BR-021 only.
Recommendation	Apply both where applicable; BR-021 is superset for materiality; BR-019 always applies to count-originated adjustments regardless of value.

OV-004 — BR-020 Mandatory Reason Codes vs. BR-022 Investigate Before Adjusting

Attribute	Detail
Overlap Type	Scope Overlap
Shared Scope	Adjustment governance
Distinction	BR-022: process requirement (investigate first). BR-020: data requirement (reason code).
Business Impact	Reason code without investigation produces categorized but unexplained adjustments; investigation without code loses analytical value.
Recommendation	Enforce sequence: investigate (BR-022) → reason code reflects finding (BR-020) → approve (BR-019/BR-021).

OV-005 — BR-009 Reorder at Reorder Level vs. BR-032 Class A Priority Monitoring

Attribute	Detail
Overlap Type	Scope Overlap
Shared Scope	Low-stock response intensity
Distinction	BR-009: universal threshold trigger. BR-032: heightened frequency and escalation for Class A.
Business Impact	Class A stockout if treated with same response speed as Class C; or over-procurement if Class A rules applied to all SKUs.
Recommendation	BR-009 sets the trigger; BR-032 modulates response priority and monitoring cadence by classification.

OV-006 — BR-001 ATP-Only Commitment vs. BR-029 Promotional Ring-Fencing

Attribute	Detail
Overlap Type	Scope Overlap
Shared Scope	What quantity may be promised to customers
Distinction	BR-001: general ATP arithmetic. BR-029: commercial allocation restriction on subset of ATP.
Business Impact	See CF-005 — double-counting available stock across channels.
Recommendation	Campaign setup creates explicit ring-fence reservation reducing general ATP.

OV-007 — BR-023 Write-Off Authorization vs. BR-021 Material Adjustment Approval

Attribute	Detail
Overlap Type	Scope Overlap
Shared Scope	Authorization for inventory value removal
Distinction	BR-023: evidence and authorization specific to write-offs. BR-021: independent approver for material adjustments generally.
Business Impact	Write-off may satisfy BR-023 documentation but still need BR-021 approver if above threshold; or low-value write-off may need BR-023 evidence but not BR-021 escalation.
Recommendation	All write-offs require BR-023; material write-offs also require BR-021 independent approver.

OV-008 — BR-025 Return Authorization vs. BR-028 Backorder Acknowledgment

Attribute	Detail
Overlap Type	Scope Overlap (Principle)
Shared Scope	Explicit customer consent before recording commercial inventory state
Distinction	BR-025: inbound return acceptance. BR-028: outbound delay acceptance.
Business Impact	Inconsistent consent standards between returns and backorders erode customer trust policy coherence.
Recommendation	Unified "customer acknowledgment required" policy in training; separate process paths.

6. High-Risk Dependency Failures

Risk ID	Broken Dependency	Likely Symptom	Affected Objectives
RK-001	DEP-001 (BR-005 → BR-001)	Overselling received-but-not-located stock	BO-03 accuracy, BO-08 cancellation reduction
RK-002	DEP-004/005 (BR-001 ↔ BR-002)	Double allocation of same units	BO-08, customer satisfaction
RK-003	DEP-016–021 (Inbound → Financial)	Payment without receipt; receipt without PO	Audit, BO-18 valuation
RK-004	DEP-030–032 (BR-009/011/012)	PO placed while transferable stock sits idle	Carrying cost, BO-17 network balance
RK-005	DEP-038–040 (Count governance chain)	Adjustments without investigation or independent approval	BO-02 accuracy, fraud exposure
RK-006	CF-005 / OV-006 (Promo vs ATP)	Campaign stockout despite warehouse availability	BO-14 promotional readiness
RK-007	DEP-051–052 (Returns → ATP)	Damaged returns in sellable stock	Returns rate, customer complaints

7. Dependency Statistics

Metric	Count
Total dependencies cataloged	58
Prerequisite	6
Sequential	22
Compositional	10
Constraint	14

Metric	Count
Reinforcement	12
Exception	1
Conflicts identified	7
Overlaps identified	8
Rules with no upstream dependency (foundational)	BR-007, BR-009, BR-014, BR-015, BR-018, BR-024, BR-025, BR-027, BR-029, BR-032
Rules with highest fan-out (most dependents)	BR-001 (9), BR-004 (8), BR-022 (4), BR-013 (4)

8. Traceability Reference

Source	Identifier
Business Rules	SRS-BR-001 (BR-001 – BR-032)
Decision Tables	SRS-DT-001 (DT-01 – DT-32)
Business Exception Scenarios	SRS-EX-001
Separation of Duties	SOD-003, SOD-004, SOD-006, SOD-007

Document Approval

Role	Name	Signature	Date
Senior Domain Expert			
Product Owner			
Operations Director			
Inventory Controller			

End of Business Rule Dependency Analysis

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Part III — Use Cases

Use Case Catalog

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-UC-001
Document Title	Use Case Catalog — Inventory Management System
Version	1.0

Attribute	Detail
Status	Draft
Document Owner	Senior Business Analyst
Source Documents	BRD-IMS-001 (v1.0), SRS-BR-001 (Business Rule Catalog, v1.0)
Date	July 5, 2026
Scope	Business interactions only — workflow steps are documented separately

Purpose

This catalog identifies every business use case for the Inventory Management System derived from the approved Business Requirements Document and Business Rule Catalog. Each use case describes a meaningful interaction between business actors and the system to achieve a defined business goal. Detailed workflows are out of scope for this document.

Traceability Legend

Reference	Meaning
BP-xx	Business Process (BRD Section 10)
FR-xxx	Functional Requirement (BRD Section 14)
BR-xxx	Business Rule (SRS-BR-001 / BRD Section 13)

Use Case Index

ID	Use Case Name	Primary Actor	Business Process	Phase
UC-001	Manage Product Master Data	Inventory Controller	BP-01	MVP
UC-002	Configure Inventory Locations	Inventory Controller	BP-02	MVP
UC-003	Define Stock Policies and Thresholds	Inventory Controller	BP-03	MVP
UC-004	Classify Products by ABC Priority	Inventory Controller	BP-04	Phase 2
UC-005	Maintain Governance Policies and Reason Codes	Inventory Controller	BP-01, BP-03	MVP / Phase 2
UC-006	Place and Manage Purchase Orders	Buyer	BP-05	Phase 2
UC-007	Plan Expected Inbound Deliveries	Buyer	BP-06	Phase 2
UC-008	Receive and Verify Supplier Goods	Receiving Clerk	BP-07	MVP
UC-009	Document and Resolve Receipt Discrepancies	Receiving Clerk	BP-08	MVP
UC-010	Perform Stock Put-Away	Warehouse Staff	BP-09	MVP
UC-011	Receive Return and Non-PO Inbound Stock	Receiving Clerk	BP-07, BP-27	MVP
UC-012	Monitor Inventory Status	Inventory Controller	BP-10	MVP
UC-013	Inquire Inventory Availability	Sales Representative	BP-10, BP-16	MVP
UC-014	Perform Cycle Count	Warehouse Staff	BP-11	MVP
UC-015	Conduct Full Physical Inventory Count	Inventory Controller	BP-12	MVP
UC-016	Investigate and Approve Count Variances	Warehouse Manager	BP-13	MVP
UC-017	Process Stock Adjustments	Warehouse Manager	BP-14	MVP
UC-018	Authorize and Process Inventory Write-Offs	Operations Director	BP-14, BP-15	MVP
UC-019	Manage Quarantine and Hold Stock	Warehouse Manager	BP-14, BP-20	MVP
UC-020	Analyze Shrinkage, Loss, and Obsolete Stock	Inventory Controller	BP-15	Phase 2
UC-021	Confirm Order Availability	Sales Representative	BP-16	MVP

ID	Use Case Name	Primary Actor	Business Process	Phase
UC-022	Reserve Inventory for Confirmed Orders	Customer Service Representative	BP-17	MVP
UC-023	Release Inventory Reservations	Customer Service Representative	BP-17	MVP
UC-024	Pick, Pack, and Ship Orders	Warehouse Picker / Packer	BP-18	MVP
UC-025	Handle Partial Fulfillment and Backorders	Customer Service Representative	BP-19	MVP / Phase 2
UC-026	Handle Outbound Fulfillment Exceptions	Warehouse Manager	BP-20	MVP
UC-027	View Order Fulfillment Status	Customer Service Representative	BP-18, BP-19	MVP
UC-028	Review Low-Stock and Out-of-Stock Alerts	Purchasing Manager	BP-21	MVP
UC-029	Plan Replenishment and Create Purchase Requisitions	Buyer	BP-22	Phase 2
UC-030	Track Inbound Supply Status	Buyer	BP-23	Phase 2
UC-031	Request Inter-Location Stock Transfer	Regional / Store Manager	BP-24	MVP
UC-032	Approve or Reject Transfer Requests	Operations Director	BP-24, BP-25	MVP
UC-033	Execute and Record Inter-Location Transfer	Warehouse Staff	BP-25	MVP
UC-034	Analyze and Recommend Network Rebalancing	Operations Director	BP-26	Phase 2
UC-035	Authorize and Record Customer Returns	Customer Service Representative	BP-27	MVP
UC-036	Receive, Inspect, and Disposition Returned Goods	Receiving Clerk	BP-28	MVP / Phase 2
UC-037	Review Inventory Dashboards and Operational Reports	Operations Director	BP-29	MVP / Phase 2
UC-038	Review Inventory Performance KPIs	Chief Executive Officer	BP-29	Phase 3
UC-039	Reconcile Inventory Valuation with Finance	Finance Manager	BP-30	Phase 2
UC-040	Confirm Promotional and Launch Stock Readiness	Marketing Manager	BP-31	Phase 2
UC-041	Conduct Inventory Audit and Control Review	Internal Auditor	BP-32	MVP / Phase 2
UC-042	View Inventory Movement and Transaction History	Inventory Controller	BP-10, BP-29	MVP
UC-043	Query and Review Audit Trail	Internal Auditor	BP-32	MVP
UC-044	Manage User Accounts and Role Assignments	System Administrator	—	MVP
UC-045	Enforce Role-Based and Location-Scoped Access	System Administrator	—	MVP
UC-046	Import Initial Product and Stock Master Data	Inventory Controller	BP-01, BP-02	MVP
UC-047	Receive and Respond to Operational Notifications	Warehouse Manager	BP-21, BP-24	MVP / Phase 2
UC-048	Support Multi-Location Split Fulfillment	Warehouse Manager	BP-18, BP-19	Phase 2

Total Use Cases: 48

Domain 1: Master Data & Governance

UC-001 — Manage Product Master Data

Attribute	Detail
Use Case ID	UC-001
Use Case Name	Manage Product Master Data
Goal	Establish and maintain accurate, unique product identity records so all inventory operations reference a single trusted SKU definition across the network.
Primary Actor	Inventory Controller
Supporting Actors	Operations Director, Warehouse Manager, Purchasing Manager, Sales Manager

Attribute	Detail
Trigger	A new product is introduced, an existing product attribute requires correction, or duplicate product records are detected.
Business Process	BP-01 — Product Master Data Management
Related Functional Requirements	FR-001, FR-002, FR-003, FR-004
Related Business Rules	BR-027 (No Reorder of Discontinued Products)

UC-002 — Configure Inventory Locations

Attribute	Detail
Use Case ID	UC-002
Use Case Name	Configure Inventory Locations
Goal	Define and maintain the inventory location network so stock can be assigned, tracked, and reported by warehouse, store, or distribution point.
Primary Actor	Inventory Controller
Supporting Actors	Operations Director, Warehouse Manager, Regional / Store Manager, Finance Manager
Trigger	A new location is opened, location hierarchy changes, a location requires closure planning, or opening stock balances must be established at go-live.
Business Process	BP-02 — Location & Warehouse Setup
Related Functional Requirements	FR-005, FR-006, FR-007, FR-010
Related Business Rules	—

UC-003 — Define Stock Policies and Thresholds

Attribute	Detail
Use Case ID	UC-003
Use Case Name	Define Stock Policies and Thresholds
Goal	Set reorder levels, minimum thresholds, and safety stock policies so replenishment and availability monitoring reflect business service-level targets.
Primary Actor	Inventory Controller
Supporting Actors	Purchasing Manager, Operations Director, Warehouse Manager
Trigger	A new product is onboarded, demand or lead-time assumptions change, or replenishment performance indicates threshold recalibration is needed.
Business Process	BP-03 — Stock Policy & Threshold Definition
Related Functional Requirements	FR-008
Related Business Rules	BR-009 (Reorder at Reorder Level), BR-010 (Safety Stock Protection)

UC-004 — Classify Products by ABC Priority

Attribute	Detail
Use Case ID	UC-004
Use Case Name	Classify Products by ABC Priority

Attribute	Detail
Goal	Segment products by business importance so operational effort, count frequency, and monitoring intensity focus on high-impact SKUs.
Primary Actor	Inventory Controller
Supporting Actors	Operations Director, Purchasing Manager, Sales Manager, Finance Manager
Trigger	Periodic classification review cycle, significant shift in product velocity or revenue contribution, or new product portfolio expansion.
Business Process	BP-04 — ABC Product Classification
Related Functional Requirements	FR-009
Related Business Rules	BR-032 (Class A Priority Monitoring)

UC-005 — Maintain Governance Policies and Reason Codes

Attribute	Detail
Use Case ID	UC-005
Use Case Name	Maintain Governance Policies and Reason Codes
Goal	Define standardized reason codes and inventory governance policies that govern adjustments, exceptions, and accountability across all locations.
Primary Actor	Inventory Controller
Supporting Actors	Operations Director, Internal Auditor, Finance Manager
Trigger	New exception categories emerge, audit findings require policy updates, or reason code taxonomy needs expansion or consolidation.
Business Process	BP-01 — Product Master Data Management; BP-03 — Stock Policy & Threshold Definition
Related Functional Requirements	FR-011, FR-012
Related Business Rules	BR-020 (Mandatory Adjustment Reason Codes)

Domain 2: Inbound & Receiving

UC-006 — Place and Manage Purchase Orders

Attribute	Detail
Use Case ID	UC-006
Use Case Name	Place and Manage Purchase Orders
Goal	Create authorized commercial commitments to suppliers that define what will be ordered, enabling controlled inbound supply and financial accountability.
Primary Actor	Buyer
Supporting Actors	Purchasing Manager, Supplier / Vendor, Finance Manager, Receiving Clerk
Trigger	Replenishment planning identifies a purchase need, safety stock is breached, or an existing purchase order requires amendment or cancellation.
Business Process	BP-05 — Purchase Order Placement
Related Functional Requirements	FR-013, FR-021, FR-022
Related Business Rules	BR-007 (Purchase Order Required for Supplier Receipt), BR-008 (Three-Way Match Before Payment), BR-027 (No Reorder of Discontinued Products)

UC-007 — Plan Expected Inbound Deliveries

Attribute	Detail
Use Case ID	UC-007
Use Case Name	Plan Expected Inbound Deliveries
Goal	Communicate and track expected supplier deliveries so receiving operations and replenishment planners can anticipate inbound supply.
Primary Actor	Buyer
Supporting Actors	Warehouse Manager, Receiving Clerk, Supplier / Vendor, Logistics / Carrier Partner
Trigger	A purchase order is confirmed, supplier communicates shipment schedule, or inbound delivery timing changes.
Business Process	BP-06 — Inbound Shipment Planning
Related Functional Requirements	FR-014, FR-055
Related Business Rules	—

UC-008 — Receive and Verify Supplier Goods

Attribute	Detail
Use Case ID	UC-008
Use Case Name	Receive and Verify Supplier Goods
Goal	Confirm quantity, product identity, and condition of inbound supplier shipments before stock is accepted into inventory records.
Primary Actor	Receiving Clerk
Supporting Actors	Supplier / Vendor, Warehouse Manager, Buyer, Logistics / Carrier Partner
Trigger	A supplier shipment arrives at the receiving dock with purchase order or authorized receipt documentation.
Business Process	BP-07 — Goods Receipt & Verification
Related Functional Requirements	FR-015, FR-017, FR-003
Related Business Rules	BR-004 (Verify Before Inventory Acceptance), BR-007 (Purchase Order Required for Supplier Receipt), BR-016 (Minimum Remaining Shelf Life at Receipt), BR-017 (Lot or Batch Capture for Traceable Products), BR-030 (Bonded Inventory Customs Clearance)

UC-009 — Document and Resolve Receipt Discrepancies

Attribute	Detail
Use Case ID	UC-009
Use Case Name	Document and Resolve Receipt Discrepancies
Goal	Record and resolve differences between expected and received goods while preserving supplier accountability and evidence for claims.
Primary Actor	Receiving Clerk
Supporting Actors	Warehouse Manager, Buyer, Supplier / Vendor, Finance Manager
Trigger	Verification identifies short shipment, over-shipment, wrong item, damage, or shelf-life non-compliance at receipt.
Business Process	BP-08 — Receipt Discrepancy Resolution
Related Functional Requirements	FR-016, FR-020

Attribute	Detail
Related Business Rules	BR-006 (Document Receipt Discrepancies), BR-008 (Three-Way Match Before Payment), BR-016 (Minimum Remaining Shelf Life at Receipt)

UC-010 — Perform Stock Put-Away

Attribute	Detail
Use Case ID	UC-010
Use Case Name	Perform Stock Put-Away
Goal	Move verified received goods to designated storage locations and confirm availability for sale and fulfillment.
Primary Actor	Warehouse Staff
Supporting Actors	Receiving Clerk, Warehouse Manager, Warehouse Picker / Packer
Trigger	Goods receipt verification is complete and stock awaits placement in storage.
Business Process	BP-09 — Stock Put-Away & Storage
Related Functional Requirements	FR-018
Related Business Rules	BR-005 (Put-Away Before Availability), BR-014 (FEFO Rotation for Shelf-Life Products), BR-015 (FIFO Rotation for Standard Goods), BR-030 (Bonded Inventory Customs Clearance)

UC-011 — Receive Return and Non-PO Inbound Stock

Attribute	Detail
Use Case ID	UC-011
Use Case Name	Receive Return and Non-PO Inbound Stock
Goal	Accept and record inbound stock that arrives without a standard purchase order, including customer returns and approved ad-hoc receipts.
Primary Actor	Receiving Clerk
Supporting Actors	Customer Service Representative, Warehouse Manager, Inventory Controller
Trigger	Customer return shipment arrives, approved ad-hoc receipt is authorized, or non-PO inbound stock requires recording.
Business Process	BP-07 — Goods Receipt & Verification; BP-27 — Customer Return Intake
Related Functional Requirements	FR-019
Related Business Rules	BR-004 (Verify Before Inventory Acceptance), BR-025 (Return Authorization Required)

Domain 3: Inventory Visibility & Status

UC-012 — Monitor Inventory Status

Attribute	Detail
Use Case ID	UC-012
Use Case Name	Monitor Inventory Status
Goal	Continuously observe stock levels and status indicators across locations to detect low stock, out-of-stock, and overstock conditions proactively.
Primary Actor	Inventory Controller

Attribute	Detail
Supporting Actors	Operations Director, Warehouse Manager, Purchasing Manager
Trigger	Scheduled monitoring cycle, threshold breach, or management request for inventory health review.
Business Process	BP-10 — Inventory Status Monitoring
Related Functional Requirements	FR-026, FR-023
Related Business Rules	BR-009 (Reorder at Reorder Level), BR-013 (Quarantine Exclusion from Sellable Stock)

UC-013 — Inquire Inventory Availability

Attribute	Detail
Use Case ID	UC-013
Use Case Name	Inquire Inventory Availability
Goal	Provide accurate on-hand, reserved, and available-to-promise visibility to support sales conversations, customer service, and fulfillment planning without warehouse escalation.
Primary Actor	Sales Representative
Supporting Actors	Customer Service Representative, Sales Manager, B2B Customer, B2C Customer
Trigger	Customer or internal stakeholder requests stock availability, product lookup, or network-wide availability check.
Business Process	BP-10 — Inventory Status Monitoring; BP-16 — Order Availability Confirmation
Related Functional Requirements	FR-023, FR-024, FR-025, FR-027, FR-028
Related Business Rules	BR-001 (ATP-Only Customer Commitment)

Domain 4: Inventory Control & Accuracy

UC-014 — Perform Cycle Count

Attribute	Detail
Use Case ID	UC-014
Use Case Name	Perform Cycle Count
Goal	Verify physical stock against system records through scheduled partial counts without disrupting full warehouse operations.
Primary Actor	Warehouse Staff
Supporting Actors	Inventory Controller, Warehouse Manager
Trigger	Cycle count schedule assigns products or locations for counting, or Class A SKU requires elevated count frequency.
Business Process	BP-11 — Cycle Counting
Related Functional Requirements	FR-030, FR-031
Related Business Rules	BR-018 (Blind Cycle Counting), BR-032 (Class A Priority Monitoring)

UC-015 — Conduct Full Physical Inventory Count

Attribute	Detail
Use Case ID	UC-015

Attribute	Detail
Use Case Name	Conduct Full Physical Inventory Count
Goal	Execute a comprehensive wall-to-wall physical count at a location to establish authoritative inventory accuracy baseline.
Primary Actor	Inventory Controller
Supporting Actors	Warehouse Manager, Warehouse Staff, Operations Director, Finance Manager
Trigger	Annual or periodic full count cycle, location go-live cutover, or audit-mandated physical verification.
Business Process	BP-12 — Full Physical Inventory Count
Related Functional Requirements	FR-032
Related Business Rules	BR-024 (Physical Inventory Freeze)

UC-016 — Investigate and Approve Count Variances

Attribute	Detail
Use Case ID	UC-016
Use Case Name	Investigate and Approve Count Variances
Goal	Investigate root causes of count discrepancies and obtain independent approval before adjusting system records.
Primary Actor	Warehouse Manager
Supporting Actors	Inventory Controller, Operations Director, Warehouse Staff
Trigger	Cycle count or physical count identifies variance between system quantity and physical quantity.
Business Process	BP-13 — Count Variance Investigation & Approval
Related Functional Requirements	FR-033, FR-034
Related Business Rules	BR-019 (Independent Count Approval), BR-021 (Independent Approval for Material Adjustments), BR-022 (Investigate Before Adjusting)

UC-017 — Process Stock Adjustments

Attribute	Detail
Use Case ID	UC-017
Use Case Name	Process Stock Adjustments
Goal	Correct recorded inventory quantities with mandatory reason codes and appropriate authorization when physical reality differs from system records.
Primary Actor	Warehouse Manager
Supporting Actors	Inventory Controller, Operations Director, Warehouse Staff
Trigger	Approved count variance, identified damage, misplacement, or other correction requiring quantity change.
Business Process	BP-14 — Stock Adjustment & Write-Off
Related Functional Requirements	FR-035, FR-037
Related Business Rules	BR-020 (Mandatory Adjustment Reason Codes), BR-021 (Independent Approval for Material Adjustments), BR-022 (Investigate Before Adjusting)

UC-018 — Authorize and Process Inventory Write-Offs

Attribute	Detail
Use Case ID	UC-018
Use Case Name	Authorize and Process Inventory Write-Offs
Goal	Formally remove inventory value for damaged, obsolete, or lost goods with documented evidence and authorized approval.
Primary Actor	Operations Director
Supporting Actors	Finance Manager, Inventory Controller, Warehouse Manager
Trigger	Goods are confirmed damaged beyond resale, obsolete, expired, or lost and require permanent removal from inventory.
Business Process	BP-14 — Stock Adjustment & Write-Off; BP-15 — Shrinkage & Loss Management
Related Functional Requirements	FR-036
Related Business Rules	BR-023 (Documented Write-Off Authorization), BR-020 (Mandatory Adjustment Reason Codes)

UC-019 — Manage Quarantine and Hold Stock

Attribute	Detail
Use Case ID	UC-019
Use Case Name	Manage Quarantine and Hold Stock
Goal	Place inventory under quality hold or regulatory restriction and exclude it from sellable availability until cleared or dispositioned.
Primary Actor	Warehouse Manager
Supporting Actors	Inventory Controller, Operations Director, Regulatory Authority
Trigger	Quality concern, recall, customs hold, investigation requirement, or bonded inventory pending clearance.
Business Process	BP-14 — Stock Adjustment & Write-Off; BP-20 — Outbound Exception Handling
Related Functional Requirements	FR-038
Related Business Rules	BR-013 (Quarantine Exclusion from Sellable Stock), BR-030 (Bonded Inventory Customs Clearance)

UC-020 — Analyze Shrinkage, Loss, and Obsolete Stock

Attribute	Detail
Use Case ID	UC-020
Use Case Name	Analyze Shrinkage, Loss, and Obsolete Stock
Goal	Identify shrinkage patterns, loss trends, and slow-moving or obsolete inventory to drive corrective action and working capital optimization.
Primary Actor	Inventory Controller
Supporting Actors	Finance Manager, Operations Director, Purchasing Manager, Warehouse Manager
Trigger	Periodic shrinkage review, rising adjustment volumes, or aging inventory exceeds policy thresholds.
Business Process	BP-15 — Shrinkage & Loss Management
Related Functional Requirements	FR-039, FR-040
Related Business Rules	BR-020 (Mandatory Adjustment Reason Codes), BR-027 (No Reorder of Discontinued Products)

Domain 5: Order Fulfillment

UC-021 — Confirm Order Availability

Attribute	Detail
Use Case ID	UC-021
Use Case Name	Confirm Order Availability
Goal	Validate that available-to-promise stock can support a customer order before the business commits to fulfillment.
Primary Actor	Sales Representative
Supporting Actors	Customer Service Representative, Sales Manager, B2B Customer, B2C Customer
Trigger	Customer places order or requests delivery commitment; sales or customer service initiates availability check.
Business Process	BP-16 — Order Availability Confirmation
Related Functional Requirements	FR-041
Related Business Rules	BR-001 (ATP-Only Customer Commitment)

UC-022 — Reserve Inventory for Confirmed Orders

Attribute	Detail
Use Case ID	UC-022
Use Case Name	Reserve Inventory for Confirmed Orders
Goal	Commit on-hand stock to confirmed customer orders and remove reserved quantity from available-to-promise to prevent double-allocation.
Primary Actor	Customer Service Representative
Supporting Actors	Sales Representative, Warehouse Manager, Warehouse Picker / Packer
Trigger	Customer order is confirmed for fulfillment after availability validation.
Business Process	BP-17 — Inventory Reservation
Related Functional Requirements	FR-042, FR-025
Related Business Rules	BR-002 (Reservation at Order Confirmation), BR-010 (Safety Stock Protection), BR-029 (Promotional Stock Ring-Fencing)

UC-023 — Release Inventory Reservations

Attribute	Detail
Use Case ID	UC-023
Use Case Name	Release Inventory Reservations
Goal	Return previously reserved stock to available-to-promise when orders are cancelled, amended, or no longer require committed inventory.
Primary Actor	Customer Service Representative
Supporting Actors	Sales Representative, Warehouse Manager
Trigger	Customer cancels order, order line is removed, or reservation is no longer required for fulfillment.
Business Process	BP-17 — Inventory Reservation

Attribute	Detail
Related Functional Requirements	FR-043
Related Business Rules	BR-002 (Reservation at Order Confirmation)

UC-024 — Pick, Pack, and Ship Orders

Attribute	Detail
Use Case ID	UC-024
Use Case Name	Pick, Pack, and Ship Orders
Goal	Physically retrieve, verify, pack, and dispatch ordered goods and record inventory deduction only after verified pick completion.
Primary Actor	Warehouse Picker / Packer
Supporting Actors	Warehouse Manager, Logistics / Carrier Partner, Customer Service Representative
Trigger	Reserved order is released to warehouse for fulfillment execution.
Business Process	BP-18 — Pick, Pack & Ship
Related Functional Requirements	FR-044, FR-029
Related Business Rules	BR-003 (Verified Pick Before Shipment), BR-014 (FEFO Rotation for Shelf-Life Products), BR-015 (FIFO Rotation for Standard Goods), BR-017 (Lot or Batch Capture for Traceable Products)

UC-025 — Handle Partial Fulfillment and Backorders

Attribute	Detail
Use Case ID	UC-025
Use Case Name	Handle Partial Fulfillment and Backorders
Goal	Fulfill available order quantity when full quantity is unavailable and obtain customer acknowledgment for backordered remainder.
Primary Actor	Customer Service Representative
Supporting Actors	Sales Representative, Warehouse Picker / Packer, B2B Customer, B2C Customer, Purchasing Manager
Trigger	Insufficient stock to fulfill entire order quantity at confirmation or pick stage.
Business Process	BP-19 — Partial Fulfillment & Backorder Management
Related Functional Requirements	FR-045, FR-046, FR-050
Related Business Rules	BR-028 (Backorder Customer Acknowledgment), BR-001 (ATP-Only Customer Commitment)

UC-026 — Handle Outbound Fulfillment Exceptions

Attribute	Detail
Use Case ID	UC-026
Use Case Name	Handle Outbound Fulfillment Exceptions
Goal	Resolve fulfillment failures such as pick discrepancies, damaged goods, or shelf-record mismatches without compromising inventory accuracy.
Primary Actor	Warehouse Manager
Supporting Actors	Warehouse Picker / Packer, Inventory Controller, Customer Service Representative

Attribute	Detail
Trigger	Pick verification fails, shelf quantity differs from system record, or outbound quality issue is identified.
Business Process	BP-20 — Outbound Exception Handling
Related Functional Requirements	FR-047
Related Business Rules	BR-003 (Verified Pick Before Shipment), BR-013 (Quarantine Exclusion from Sellable Stock)

UC-027 — View Order Fulfillment Status

Attribute	Detail
Use Case ID	UC-027
Use Case Name	View Order Fulfillment Status
Goal	Provide operations and customer-facing teams visibility into order reservation, picking, shipping, and exception status.
Primary Actor	Customer Service Representative
Supporting Actors	Sales Representative, Sales Manager, B2B Customer, B2C Customer, Warehouse Manager
Trigger	Customer or internal stakeholder inquires about order progress, shipment status, or fulfillment exception.
Business Process	BP-18 — Pick, Pack & Ship; BP-19 — Partial Fulfillment & Backorder Management
Related Functional Requirements	FR-049
Related Business Rules	—

Domain 6: Replenishment & Inbound Supply

UC-028 — Review Low-Stock and Out-of-Stock Alerts

Attribute	Detail
Use Case ID	UC-028
Use Case Name	Review Low-Stock and Out-of-Stock Alerts
Goal	Review and acknowledge replenishment alerts so purchasing and operations respond before stockouts affect fulfillment.
Primary Actor	Purchasing Manager
Supporting Actors	Buyer, Inventory Controller, Operations Director, Warehouse Manager
Trigger	Available stock reaches reorder level, falls below minimum threshold, or reaches zero ATP.
Business Process	BP-21 — Low-Stock Alert Review
Related Functional Requirements	FR-051, FR-052, FR-076
Related Business Rules	BR-009 (Reorder at Reorder Level), BR-032 (Class A Priority Monitoring)

UC-029 — Plan Replenishment and Create Purchase Requisitions

Attribute	Detail
Use Case ID	UC-029
Use Case Name	Plan Replenishment and Create Purchase Requisitions

Attribute	Detail
Goal	Translate replenishment needs into purchase requisitions and procurement actions after evaluating network availability and safety stock rules.
Primary Actor	Buyer
Supporting Actors	Purchasing Manager, Inventory Controller, Operations Director, Supplier / Vendor
Trigger	Acknowledged low-stock alert, replenishment suggestion, or planned restocking cycle.
Business Process	BP-22 — Replenishment Planning & Requisition
Related Functional Requirements	FR-053, FR-054, FR-056, FR-074
Related Business Rules	BR-009 (Reorder at Reorder Level), BR-010 (Safety Stock Protection), BR-011 (Network Check Before Purchase), BR-027 (No Reorder of Discontinued Products)

UC-030 — Track Inbound Supply Status

Attribute	Detail
Use Case ID	UC-030
Use Case Name	Track Inbound Supply Status
Goal	Monitor open purchase orders and expected inbound deliveries to inform replenishment decisions and delay response.
Primary Actor	Buyer
Supporting Actors	Purchasing Manager, Warehouse Manager, Receiving Clerk, Supplier / Vendor
Trigger	Replenishment review requires inbound visibility, supplier delivery is delayed, or receiving planning needs expected stock confirmation.
Business Process	BP-23 — Inbound Supply Tracking
Related Functional Requirements	FR-055, FR-079
Related Business Rules	—

Domain 7: Network Balance & Transfers

UC-031 — Request Inter-Location Stock Transfer

Attribute	Detail
Use Case ID	UC-031
Use Case Name	Request Inter-Location Stock Transfer
Goal	Initiate movement of stock between locations to resolve local shortages using network inventory before new procurement.
Primary Actor	Regional / Store Manager
Supporting Actors	Warehouse Manager, Operations Director, Inventory Controller, Purchasing Manager
Trigger	Local stockout or low-stock condition while excess exists at another location, or network rebalancing need identified.
Business Process	BP-24 — Inter-Location Transfer Request
Related Functional Requirements	FR-058
Related Business Rules	BR-011 (Network Check Before Purchase), BR-012 (Protect Source Location Minimum), BR-010 (Safety Stock Protection)

UC-032 — Approve or Reject Transfer Requests

Attribute	Detail
Use Case ID	UC-032
Use Case Name	Approve or Reject Transfer Requests
Goal	Authorize or deny inter-location transfer requests based on network balance, source location capacity, and service-level impact.
Primary Actor	Operations Director
Supporting Actors	Warehouse Manager, Regional / Store Manager, Inventory Controller
Trigger	Transfer request is submitted and requires management approval per authorization policy.
Business Process	BP-24 — Inter-Location Transfer Request; BP-25 — Inter-Location Transfer Execution
Related Functional Requirements	FR-059, FR-077
Related Business Rules	BR-012 (Protect Source Location Minimum)

UC-033 — Execute and Record Inter-Location Transfer

Attribute	Detail
Use Case ID	UC-033
Use Case Name	Execute and Record Inter-Location Transfer
Goal	Physically dispatch and receive stock between locations with complete transaction history and in-transit visibility.
Primary Actor	Warehouse Staff
Supporting Actors	Warehouse Manager, Regional / Store Manager, Logistics / Carrier Partner
Trigger	Transfer request is approved and stock is ready for dispatch from source location.
Business Process	BP-25 — Inter-Location Transfer Execution
Related Functional Requirements	FR-060, FR-061, FR-063
Related Business Rules	BR-012 (Protect Source Location Minimum), BR-017 (Lot or Batch Capture for Traceable Products)

UC-034 — Analyze and Recommend Network Rebalancing

Attribute	Detail
Use Case ID	UC-034
Use Case Name	Analyze and Recommend Network Rebalancing
Goal	Proactively identify and recommend stock redistribution across the network to reduce imbalance, stockouts, and excess capital.
Primary Actor	Operations Director
Supporting Actors	Inventory Controller, Purchasing Manager, Warehouse Manager, Regional / Store Manager
Trigger	Periodic network review, rising cross-location imbalance events, or executive KPI indicates network inefficiency.
Business Process	BP-26 — Network Stock Rebalancing
Related Functional Requirements	FR-062, FR-028
Related Business Rules	BR-011 (Network Check Before Purchase), BR-012 (Protect Source Location Minimum)

Domain 8: Returns Management

UC-035 — Authorize and Record Customer Returns

Attribute	Detail
Use Case ID	UC-035
Use Case Name	Authorize and Record Customer Returns
Goal	Issue return authorization linked to original order and return policy before goods are accepted at the warehouse.
Primary Actor	Customer Service Representative
Supporting Actors	B2B Customer, B2C Customer, Sales Representative, Receiving Clerk
Trigger	Customer requests return of purchased goods per return policy.
Business Process	BP-27 — Customer Return Intake
Related Functional Requirements	FR-064, FR-067
Related Business Rules	BR-025 (Return Authorization Required)

UC-036 — Receive, Inspect, and Disposition Returned Goods

Attribute	Detail
Use Case ID	UC-036
Use Case Name	Receive, Inspect, and Disposition Returned Goods
Goal	Receive authorized returns, assess condition, and determine restock, quarantine, or write-off disposition.
Primary Actor	Receiving Clerk
Supporting Actors	Warehouse Manager, Inventory Controller, Customer Service Representative, Finance Manager
Trigger	Authorized return shipment arrives at receiving location.
Business Process	BP-28 — Return-to-Stock or Disposal Decision
Related Functional Requirements	FR-065, FR-066, FR-067
Related Business Rules	BR-026 (Inspect Returns Before Restocking), BR-025 (Return Authorization Required)

Domain 9: Reporting, Performance & Financial Alignment

UC-037 — Review Inventory Dashboards and Operational Reports

Attribute	Detail
Use Case ID	UC-037
Use Case Name	Review Inventory Dashboards and Operational Reports
Goal	Access inventory summary, movement, stockout, overstock, accuracy, and location performance views to manage daily operations.
Primary Actor	Operations Director
Supporting Actors	Chief Operating Officer, Inventory Controller, Warehouse Manager, Purchasing Manager, Sales Manager
Trigger	Scheduled management review, operational issue investigation, or stakeholder request for inventory snapshot.
Business Process	BP-29 — Inventory Performance Reporting
Related Functional Requirements	FR-068, FR-069, FR-070, FR-071, FR-072, FR-074

Attribute	Detail
Related Business Rules	—

UC-038 — Review Inventory Performance KPIs

Attribute	Detail
Use Case ID	UC-038
Use Case Name	Review Inventory Performance KPIs
Goal	Monitor executive-level inventory KPIs including turnover, fill rate, accuracy, and working capital metrics for strategic decision-making.
Primary Actor	Chief Executive Officer
Supporting Actors	Chief Operating Officer, Chief Financial Officer, Operations Director, Finance Manager
Trigger	Executive review cycle, steering committee meeting, or KPI threshold breach.
Business Process	BP-29 — Inventory Performance Reporting
Related Functional Requirements	FR-073
Related Business Rules	—

UC-039 — Reconcile Inventory Valuation with Finance

Attribute	Detail
Use Case ID	UC-039
Use Case Name	Reconcile Inventory Valuation with Finance
Goal	Align operational inventory records with financial valuation and general ledger to support accurate month-end close and supplier payment controls.
Primary Actor	Finance Manager
Supporting Actors	Chief Financial Officer, Inventory Controller, Operations Director, Internal Auditor, External Auditor
Trigger	Month-end close cycle, valuation variance identified, or audit reconciliation requirement.
Business Process	BP-30 — Inventory Valuation & Financial Reconciliation
Related Functional Requirements	FR-075, FR-089
Related Business Rules	BR-008 (Three-Way Match Before Payment), BR-031 (Consignment Ownership at Consumption)

Domain 10: Commercial Alignment & Governance

UC-040 — Confirm Promotional and Launch Stock Readiness

Attribute	Detail
Use Case ID	UC-040
Use Case Name	Confirm Promotional and Launch Stock Readiness
Goal	Verify that ring-fenced promotional inventory is available and sufficient before campaign or product launch commitment.
Primary Actor	Marketing Manager
Supporting Actors	Purchasing Manager, Warehouse Manager, Sales Manager, Operations Director

Attribute	Detail
Trigger	Upcoming promotional campaign, product launch, or channel-specific stock commitment deadline.
Business Process	BP-31 — Promotional & Launch Stock Readiness
Related Functional Requirements	FR-057, FR-080
Related Business Rules	BR-029 (Promotional Stock Ring-Fencing)

UC-041 — Conduct Inventory Audit and Control Review

Attribute	Detail
Use Case ID	UC-041
Use Case Name	Conduct Inventory Audit and Control Review
Goal	Test inventory controls, review audit trails, and verify compliance with governance policies and segregation of duties.
Primary Actor	Internal Auditor
Supporting Actors	External Auditor, Finance Manager, Operations Director, Inventory Controller
Trigger	Scheduled internal audit cycle, control breach detection, or external audit engagement.
Business Process	BP-32 — Inventory Audit & Control Review
Related Functional Requirements	FR-087, FR-088
Related Business Rules	BR-019 (Independent Count Approval), BR-021 (Independent Approval for Material Adjustments), BR-023 (Documented Write-Off Authorization)

Domain 11: Cross-Cutting System Interactions

UC-042 — View Inventory Movement and Transaction History

Attribute	Detail
Use Case ID	UC-042
Use Case Name	View Inventory Movement and Transaction History
Goal	Trace complete history of inventory movements by product, location, user, and date range for operational investigation and accountability.
Primary Actor	Inventory Controller
Supporting Actors	Warehouse Manager, Finance Manager, Internal Auditor, Operations Director
Trigger	Discrepancy investigation, customer inquiry, or management request for movement traceability.
Business Process	BP-10 — Inventory Status Monitoring; BP-29 — Inventory Performance Reporting
Related Functional Requirements	FR-029
Related Business Rules	BR-017 (Lot or Batch Capture for Traceable Products)

UC-043 — Query and Review Audit Trail

Attribute	Detail
Use Case ID	UC-043
Use Case Name	Query and Review Audit Trail

Attribute	Detail
Goal	Access immutable record of all material inventory changes with user identity, timestamp, and before/after state for governance and compliance.
Primary Actor	Internal Auditor
Supporting Actors	External Auditor, Finance Manager, Operations Director
Trigger	Audit engagement, control testing, fraud investigation, or regulatory inquiry.
Business Process	BP-32 — Inventory Audit & Control Review
Related Functional Requirements	FR-086, FR-083
Related Business Rules	BR-019 (Independent Count Approval), BR-020 (Mandatory Adjustment Reason Codes), BR-021 (Independent Approval for Material Adjustments)

UC-044 — Manage User Accounts and Role Assignments

Attribute	Detail
Use Case ID	UC-044
Use Case Name	Manage User Accounts and Role Assignments
Goal	Provision, modify, and deactivate user accounts with appropriate role assignments to ensure authorized system access.
Primary Actor	System Administrator
Supporting Actors	Operations Director, Internal Auditor
Trigger	New employee onboarding, role change, location reassignment, or employee departure.
Business Process	User Access & Accountability (BRD Section 8.1, 16)
Related Functional Requirements	FR-085
Related Business Rules	—

UC-045 — Enforce Role-Based and Location-Scoped Access

Attribute	Detail
Use Case ID	UC-045
Use Case Name	Enforce Role-Based and Location-Scoped Access
Goal	Ensure users can only perform permitted functions within assigned locations and that conflicting duties are segregated.
Primary Actor	System Administrator
Supporting Actors	Internal Auditor, Operations Director, Warehouse Manager
Trigger	User attempts system action; role or permission policy is defined or updated; segregation of duties rule applies.
Business Process	User Access & Accountability (BRD Section 16)
Related Functional Requirements	FR-081, FR-082, FR-084
Related Business Rules	BR-019 (Independent Count Approval), BR-021 (Independent Approval for Material Adjustments)

UC-046 — Import Initial Product and Stock Master Data

Attribute	Detail
Use Case ID	UC-046
Use Case Name	Import Initial Product and Stock Master Data
Goal	Load verified product catalog and opening stock balances to establish authoritative system baseline at location go-live.
Primary Actor	Inventory Controller
Supporting Actors	System Administrator, Operations Director, Finance Manager, Warehouse Manager
Trigger	New location or system go-live requires bulk load of product master and verified opening balances.
Business Process	BP-01 — Product Master Data Management; BP-02 — Location & Warehouse Setup
Related Functional Requirements	FR-090
Related Business Rules	—

UC-047 — Receive and Respond to Operational Notifications

Attribute	Detail
Use Case ID	UC-047
Use Case Name	Receive and Respond to Operational Notifications
Goal	Deliver timely alerts for low stock, pending approvals, transfer exceptions, inbound delays, and campaign readiness so responsible actors can act.
Primary Actor	Warehouse Manager
Supporting Actors	Purchasing Manager, Operations Director, Buyer, Marketing Manager, Customer Service Representative
Trigger	System detects threshold breach, pending approval, exception condition, or readiness deadline approaching.
Business Process	BP-21 — Low-Stock Alert Review; BP-24 — Inter-Location Transfer Request; BP-31 — Promotional & Launch Stock Readiness
Related Functional Requirements	FR-076, FR-077, FR-078, FR-079, FR-080
Related Business Rules	BR-009 (Reorder at Reorder Level)

UC-048 — Support Multi-Location Split Fulfillment

Attribute	Detail
Use Case ID	UC-048
Use Case Name	Support Multi-Location Split Fulfillment
Goal	Fulfill a single customer order from multiple inventory locations when no single location holds sufficient available stock.
Primary Actor	Warehouse Manager
Supporting Actors	Customer Service Representative, Warehouse Picker / Packer, Operations Director, Logistics / Carrier Partner
Trigger	Confirmed order quantity exceeds single-location ATP but network-wide stock can satisfy the order.
Business Process	BP-18 — Pick, Pack & Ship; BP-19 — Partial Fulfillment & Backorder Management
Related Functional Requirements	FR-048
Related Business Rules	BR-001 (ATP-Only Customer Commitment), BR-002 (Reservation at Order Confirmation)

Traceability Summary

Use Cases by Business Process

Business Process	Use Case(s)
BP-01	UC-001, UC-005, UC-046
BP-02	UC-002, UC-046
BP-03	UC-003, UC-005
BP-04	UC-004
BP-05	UC-006
BP-06	UC-007
BP-07	UC-008, UC-011
BP-08	UC-009
BP-09	UC-010
BP-10	UC-012, UC-013, UC-042
BP-11	UC-014
BP-12	UC-015
BP-13	UC-016
BP-14	UC-017, UC-018, UC-019
BP-15	UC-018, UC-020
BP-16	UC-013, UC-021
BP-17	UC-022, UC-023
BP-18	UC-024, UC-027, UC-048
BP-19	UC-025, UC-027, UC-048
BP-20	UC-019, UC-026
BP-21	UC-028, UC-047
BP-22	UC-029
BP-23	UC-030
BP-24	UC-031, UC-032, UC-047
BP-25	UC-032, UC-033
BP-26	UC-034
BP-27	UC-011, UC-035
BP-28	UC-036
BP-29	UC-037, UC-038, UC-042
BP-30	UC-039
BP-31	UC-040, UC-047
BP-32	UC-041, UC-043

Functional Requirement Coverage

All 90 functional requirements (FR-001 through FR-090) are covered by at least one use case in this catalog.

Business Rule Coverage

All 32 business rules (BR-001 through BR-032) are referenced by at least one use case in this catalog.

Document Approval

Role	Name	Signature	Date
Senior Business Analyst			
Product Owner			
Operations Director			

End of Use Case Catalog

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Use Case Main Flows

Inventory Management System

Document Information

Attribute	Detail
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Date	July 5, 2026
Scope	Successful scenario only — no alternative or exception paths

Purpose

This document describes the main flow for each of the 48 use cases identified in the Use Case Catalog. Each flow specifies the actor action and corresponding system response for the successful completion of the business interaction.

Flow Legend

Column	Description
Step	Sequential step number within the main flow
Actor Action	Action performed by the primary or supporting actor
System Response	Observable system behavior or outcome in response to the actor action

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Domain 1: Master Data & Governance

UC-001 — Manage Product Master Data

Primary Actor: Inventory Controller

Step	Actor Action	System Response
1	Inventory Controller opens the product master management function.	System displays the product catalog with search and filter options.
2	Inventory Controller selects to create a new product or edit an existing product record.	System presents the product data entry form with required fields (SKU, name, description, unit of measure, category).
3	Inventory Controller enters or updates product attributes and product identifier (barcode) where applicable.	System validates required fields and checks for duplicate SKU or identifier.
4	Inventory Controller submits the product record for saving.	System saves the product master record and confirms successful creation or update.
5	Inventory Controller reviews the saved product record.	System displays the complete product profile with current status and attributes.

UC-002 — Configure Inventory Locations

Primary Actor: Inventory Controller

Step	Actor Action	System Response
1	Inventory Controller opens the location management function.	System displays the inventory location network and hierarchy view.
2	Inventory Controller selects to create a new location or update an existing location.	System presents the location configuration form (name, type, parent location, status).
3	Inventory Controller defines location hierarchy and location attributes.	System validates location structure and confirms hierarchy assignment.
4	Inventory Controller establishes opening stock balances for location go-live where required.	System accepts opening balance entries by product and records initial on-hand quantities.
5	Inventory Controller activates the location for inventory operations.	System confirms location is active and available for stock assignment and transactions.

UC-003 — Define Stock Policies and Thresholds

Primary Actor: Inventory Controller

Step	Actor Action	System Response
1	Inventory Controller opens stock policy configuration for a product or product group.	System displays current reorder level, minimum threshold, and safety stock settings.
2	Inventory Controller sets or updates reorder level and minimum stock threshold.	System accepts threshold values and validates against business policy limits.
3	Inventory Controller defines safety stock buffer quantity where applicable.	System records safety stock policy and associates it with the product and location scope.
4	Inventory Controller saves the stock policy configuration.	System confirms policy is saved and applies thresholds to availability monitoring.
5	Inventory Controller reviews the effective policy summary.	System displays the active policy with effective date and scope (product, location).

UC-004 — Classify Products by ABC Priority

Primary Actor: Inventory Controller

Step	Actor Action	System Response
1	Inventory Controller opens ABC classification management.	System displays products with current classification status and classification criteria.
2	Inventory Controller reviews product velocity, revenue contribution, or criticality data.	System presents supporting product performance data for classification decisions.
3	Inventory Controller assigns ABC class (A, B, or C) to each product.	System accepts classification assignment per product.
4	Inventory Controller saves the classification update.	System confirms classification is saved and applies class-based monitoring and count frequency rules.
5	Inventory Controller reviews the updated ABC classification report.	System displays products grouped by class with classification effective date.

UC-005 — Maintain Governance Policies and Reason Codes

Primary Actor: Inventory Controller

Step	Actor Action	System Response
1	Inventory Controller opens governance policy and reason code administration.	System displays the current reason code catalog and active governance policies.
2	Inventory Controller creates or updates a reason code category (e.g., adjustment, shrinkage, exception).	System presents reason code entry form with category and description fields.
3	Inventory Controller defines or updates inventory governance policy parameters.	System accepts policy definitions and associates them with applicable transaction types.
4	Inventory Controller activates the reason code or policy for operational use.	System validates completeness and confirms activation.
5	Inventory Controller reviews the published reason code and policy catalog.	System displays the active catalog available for use in inventory transactions.

Domain 2: Inbound & Receiving

UC-006 — Place and Manage Purchase Orders

Primary Actor: Buyer

Step	Actor Action	System Response
1	Buyer opens purchase order management.	System displays open, pending, and historical purchase orders.
2	Buyer creates a new purchase order and selects supplier.	System presents purchase order entry form linked to supplier and authorized products.
3	Buyer enters product lines, quantities, expected delivery date, and delivery location.	System validates product eligibility (active status, not discontinued) and calculates order totals.
4	Buyer submits the purchase order for approval.	System routes the purchase order to Purchasing Manager and records pending approval status.
5	Purchasing Manager approves the purchase order.	System confirms purchase order as authorized and records expected inbound supply.
6	Buyer reviews the confirmed purchase order.	System displays approved purchase order with status, line details, and expected delivery information.

UC-007 — Plan Expected Inbound Deliveries

Primary Actor: Buyer

Step	Actor Action	System Response
1	Buyer opens inbound delivery planning for approved purchase orders.	System displays purchase orders with expected delivery dates and shipment status.
2	Buyer records or updates expected shipment date and carrier information from supplier confirmation.	System accepts delivery schedule and associates it with the purchase order.
3	Buyer communicates planned arrival to receiving location.	System updates inbound supply visibility for the destination warehouse.
4	Buyer confirms the inbound delivery plan.	System records planned inbound delivery and makes it visible to receiving and replenishment functions.
5	Buyer reviews the inbound delivery schedule.	System displays expected arrivals by date, location, and purchase order reference.

UC-008 — Receive and Verify Supplier Goods

Primary Actor: Receiving Clerk

Step	Actor Action	System Response
1	Receiving Clerk opens goods receipt for an arriving supplier shipment.	System displays expected purchase order lines and quantities for the delivery location.
2	Receiving Clerk identifies the shipment against the purchase order reference.	System retrieves matching purchase order and presents expected products and quantities.
3	Receiving Clerk scans or enters product identifier and verifies quantity and condition of each line.	System validates product identity against purchase order and records received quantities.
4	Receiving Clerk captures lot or batch number and shelf-life data for traceable products where required.	System records lot/batch attribution and validates minimum remaining shelf life.
5	Receiving Clerk confirms receipt verification is complete.	System records verified goods receipt, updates receipt status, and stages stock for put-away (not yet available for sale).

UC-009 — Document and Resolve Receipt Discrepancies

Primary Actor: Receiving Clerk

Step	Actor Action	System Response
1	Receiving Clerk identifies a discrepancy during goods verification (short, over, wrong item, or damage).	System presents discrepancy documentation form linked to the receipt line.
2	Receiving Clerk records discrepancy type, quantity variance, and supporting notes or evidence.	System captures discrepancy details with mandatory reason code and timestamp.
3	Receiving Clerk accepts verified quantities and documents unresolved variance separately.	System records accepted quantity for inventory and holds discrepancy record for resolution.
4	Warehouse Manager reviews and approves discrepancy resolution.	System confirms resolution action and updates receipt and discrepancy status.
5	Receiving Clerk completes the receipt with documented discrepancy.	System finalizes receipt record with accepted quantities and linked discrepancy documentation.

UC-010 — Perform Stock Put-Away

Primary Actor: Warehouse Staff

Step	Actor Action	System Response
1	Warehouse Staff opens put-away tasks for verified received stock.	System displays received stock awaiting put-away with product, quantity, and suggested storage location.
2	Warehouse Staff retrieves stock from receiving area and moves it to designated storage location.	System records put-away in progress against the receipt reference.
3	Warehouse Staff confirms product, quantity, and destination storage location.	System validates storage location assignment and rotation rules (FIFO/FEFO) where applicable.
4	Warehouse Staff completes put-away confirmation.	System records put-away completion, updates on-hand quantity at storage location, and increases available-to-promise.
5	Warehouse Staff reviews completed put-away record.	System displays confirmed put-away with location, quantity, user, and timestamp.

UC-011 — Receive Return and Non-PO Inbound Stock

Primary Actor: Receiving Clerk

Step	Actor Action	System Response
1	Receiving Clerk opens non-PO or return receipt function.	System presents receipt type options (authorized return, ad-hoc receipt).
2	Receiving Clerk selects receipt type and enters or scans return authorization or ad-hoc receipt approval reference.	System validates authorization reference and retrieves linked order or approval details.
3	Receiving Clerk verifies product identity, quantity, and condition of inbound goods.	System records verified receipt lines against the authorization reference.
4	Receiving Clerk confirms receipt.	System records inbound receipt, stages stock for inspection or put-away per receipt type, and logs transaction with user and timestamp.
5	Receiving Clerk reviews the completed non-PO receipt record.	System displays receipt confirmation with reference, products, quantities, and next disposition status.

Domain 3: Inventory Visibility & Status

UC-012 — Monitor Inventory Status

Primary Actor: Inventory Controller

Step	Actor Action	System Response
1	Inventory Controller opens inventory status monitoring view.	System displays network-wide stock status summary by location, category, and status indicator.
2	Inventory Controller applies filters for location, product category, or status (in stock, low stock, out of stock).	System refreshes view with filtered inventory status results.
3	Inventory Controller reviews products at or below reorder level and out-of-stock conditions.	System highlights low-stock and out-of-stock items with threshold and ATP details.
4	Inventory Controller drills into a specific product or location for detail.	System displays on-hand, reserved, ATP, and quarantined quantities for the selection.
5	Inventory Controller completes status review.	System retains current monitoring view and last-reviewed timestamp for reference.

UC-013 — Inquire Inventory Availability

Primary Actor: Sales Representative

Step	Actor Action	System Response
1	Sales Representative opens inventory availability inquiry.	System presents product search by SKU, name, barcode, or category.
2	Sales Representative searches for the requested product.	System returns matching products with availability summary.
3	Sales Representative selects product and requests availability detail by location or network-wide.	System displays on-hand, reserved, and available-to-promise quantities per location and network total.
4	Sales Representative reviews stock status indicator (in stock, low stock, out of stock).	System presents current status and expected inbound quantity where applicable.
5	Sales Representative uses availability information to respond to customer inquiry.	System retains inquiry view for reference during the customer conversation.

Domain 4: Inventory Control & Accuracy

UC-014 — Perform Cycle Count

Primary Actor: Warehouse Staff

Step	Actor Action	System Response
1	Warehouse Staff opens assigned cycle count tasks.	System displays scheduled count assignments by product and location without revealing system expected quantity (blind count).
2	Warehouse Staff navigates to the assigned storage location and product.	System presents count entry form for the assigned SKU and location.
3	Warehouse Staff physically counts stock and enters the counted quantity.	System records the physical count result with counter identity and timestamp.
4	Warehouse Staff submits the completed count entry.	System saves count result, compares to system quantity internally, and flags variance if present.
5	Warehouse Staff reviews count submission confirmation.	System displays count submission status (recorded, variance pending review if applicable).

UC-015 — Conduct Full Physical Inventory Count

Primary Actor: Inventory Controller

Step	Actor Action	System Response
1	Inventory Controller initiates a full physical inventory count for a location.	System activates physical inventory freeze for the location (suspends movements) and generates count sheets.
2	Inventory Controller assigns count teams and distributes count assignments.	System records count team assignments by zone, aisle, or product category.
3	Warehouse Staff perform wall-to-wall counts and enter physical quantities.	System records all count entries with counter identity and timestamp under freeze conditions.
4	Inventory Controller reviews count completion status across the location.	System displays count progress and identifies any uncounted locations or products.
5	Inventory Controller closes the physical count session.	System finalizes count snapshot, releases inventory freeze, and generates variance report for investigation.

UC-016 — Investigate and Approve Count Variances

Primary Actor: Warehouse Manager

Step	Actor Action	System Response
1	Warehouse Manager opens pending count variance queue.	System displays variances with product, location, system quantity, counted quantity, and variance amount.
2	Warehouse Manager selects a variance record for investigation.	System presents variance detail and investigation notes field.
3	Warehouse Manager documents investigation findings and root cause analysis.	System records investigation notes with investigator identity and timestamp.
4	Warehouse Manager approves the variance adjustment based on investigation.	System routes approval to independent approver if required by threshold; records approval decision.
5	Inventory Controller or authorized approver confirms variance approval.	System marks variance as approved and queues approved quantity adjustment for processing.

UC-017 — Process Stock Adjustments

Primary Actor: Warehouse Manager

Step	Actor Action	System Response
1	Warehouse Manager opens stock adjustment function.	System presents adjustment entry form with product, location, and current on-hand quantity.
2	Warehouse Manager selects product, location, adjustment direction (increase or decrease), and quantity.	System displays resulting quantity preview and requires reason code selection.
3	Warehouse Manager selects mandatory reason code and enters supporting notes.	System validates reason code and checks adjustment against approval threshold.
4	Warehouse Manager submits adjustment for approval where threshold requires independent authorization.	System routes to authorized approver; approver confirms adjustment.
5	Warehouse Manager confirms approved adjustment.	System applies quantity change, updates on-hand and ATP, and records adjustment in movement history with user, timestamp, and reason code.

UC-018 — Authorize and Process Inventory Write-Offs

Primary Actor: Operations Director

Step	Actor Action	System Response
1	Operations Director opens pending write-off requests.	System displays write-off requests with product, quantity, location, reason, and supporting documentation.
2	Operations Director reviews write-off evidence and business justification.	System presents complete write-off request detail and linked adjustment history.
3	Operations Director approves the write-off authorization.	System records authorization decision with approver identity and timestamp.
4	Finance Manager acknowledges financial impact where required.	System records finance acknowledgment and marks write-off as authorized.
5	Operations Director confirms write-off execution.	System removes inventory quantity from on-hand, records write-off transaction with reason code, and updates movement and audit history.

UC-019 — Manage Quarantine and Hold Stock

Primary Actor: Warehouse Manager

Step	Actor Action	System Response
1	Warehouse Manager opens quarantine and hold stock management.	System displays current quarantined and held inventory by location.

Step	Actor Action	System Response
2	Warehouse Manager selects stock to place on quarantine or hold status.	System presents stock detail with current on-hand and ATP quantities.
3	Warehouse Manager assigns quarantine or hold reason and expected disposition.	System accepts hold reason and records hold initiation.
4	Warehouse Manager confirms quarantine placement.	System moves quantity from sellable ATP to quarantined status and excludes it from availability.
5	Warehouse Manager reviews updated stock status.	System displays product with quarantined quantity, hold reason, and date placed on hold.

UC-020 — Analyze Shrinkage, Loss, and Obsolete Stock

Primary Actor: Inventory Controller

Step	Actor Action	System Response
1	Inventory Controller opens shrinkage and obsolescence analysis reports.	System displays shrinkage trends by reason code, location, and time period.
2	Inventory Controller reviews adjustment and write-off history grouped by loss category.	System presents summarized loss values and frequency by reason code.
3	Inventory Controller identifies slow-moving and obsolete product candidates.	System displays aging inventory report with days since last movement and excess quantity indicators.
4	Inventory Controller documents recommended actions for obsolete or slow-moving stock.	System records analysis notes and recommended disposition actions.
5	Inventory Controller shares analysis output with operations and finance stakeholders.	System generates exportable shrinkage and obsolescence analysis report.

Domain 5: Order Fulfillment

UC-021 — Confirm Order Availability

Primary Actor: Sales Representative

Step	Actor Action	System Response
1	Sales Representative opens order availability confirmation for a customer order.	System presents order lines with requested products and quantities.
2	Sales Representative requests availability check for each order line.	System evaluates available-to-promise quantity by fulfillment location and network-wide.
3	Sales Representative reviews ATP result for each line (fully available, partially available, or unavailable).	System displays ATP quantity, fulfillment location options, and stock status per line.
4	Sales Representative confirms availability assessment with customer.	System records availability check result and timestamp against the order.
5	Sales Representative marks order as ready for confirmation or reservation.	System updates order availability status and indicates lines cleared for commitment.

UC-022 — Reserve Inventory for Confirmed Orders

Primary Actor: Customer Service Representative

Step	Actor Action	System Response
1	Customer Service Representative opens confirmed order for reservation.	System displays order lines with availability check results.
2	Customer Service Representative initiates inventory reservation for the order.	System validates ATP sufficiency for each line at selected fulfillment location.
3	Customer Service Representative confirms reservation quantities and fulfillment location per line.	System reserves on-hand quantity against the order and reduces ATP.
4	Customer Service Representative completes reservation.	System confirms reservation with reserved quantity, location, and reservation reference per line.
5	Customer Service Representative reviews reservation summary.	System displays order with reserved status and releases order to warehouse fulfillment queue.

UC-023 — Release Inventory Reservations

Primary Actor: Customer Service Representative

Step	Actor Action	System Response
1	Customer Service Representative opens order with active inventory reservation.	System displays reserved lines with reserved quantity and fulfillment location.
2	Customer Service Representative selects reservation to release (full order or specific lines).	System presents release confirmation with quantity to be returned to ATP.
3	Customer Service Representative confirms reservation release.	System removes reservation hold and restores quantity to available-to-promise.
4	Customer Service Representative records release reason where required.	System logs release reason, user, and timestamp in transaction history.
5	Customer Service Representative reviews updated order status.	System displays order with released reservation status and restored ATP.

UC-024 — Pick, Pack, and Ship Orders

Primary Actor: Warehouse Picker / Packer

Step	Actor Action	System Response
1	Warehouse Picker / Packer opens fulfillment queue for assigned location.	System displays orders ready for picking with product, quantity, and storage location.
2	Warehouse Picker / Packer retrieves items from storage following rotation rules (FIFO/FEFO).	System presents pick list with location sequence and lot/batch guidance for traceable products.
3	Warehouse Picker / Packer scans or confirms each picked item and quantity.	System records pick verification per line with picker identity and timestamp.
4	Warehouse Picker / Packer packs verified items and prepares shipment with carrier documentation.	System updates order status to packed and presents shipment confirmation form.
5	Warehouse Picker / Packer confirms shipment handoff to carrier.	System deducts inventory from on-hand, clears reservation, records shipment, and updates order status to shipped.

UC-025 — Handle Partial Fulfillment and Backorders

Primary Actor: Customer Service Representative

Step	Actor Action	System Response
1	Customer Service Representative identifies order line with insufficient ATP for full quantity.	System displays available quantity versus ordered quantity per affected line.

Step	Actor Action	System Response
2	Customer Service Representative presents partial fulfillment and backorder options to customer.	System presents fulfill-now quantity and backorder quantity with expected availability date if known.
3	Customer Service Representative records customer acceptance of partial shipment and backorder terms.	System captures customer acknowledgment for backordered quantity.
4	Customer Service Representative confirms partial fulfillment plan.	System reserves available quantity for immediate fulfillment and creates backorder record for remainder.
5	Customer Service Representative reviews updated order with split fulfillment status.	System displays order with shipped/pick-ready lines and backorder lines with expected fulfillment priority.

UC-026 — Handle Outbound Fulfillment Exceptions

Primary Actor: Warehouse Manager

Step	Actor Action	System Response
1	Warehouse Manager opens outbound exception queue.	System displays fulfillment exceptions (pick failure, quantity mismatch, damaged pick) with order and product detail.
2	Warehouse Manager reviews exception detail and physical investigation findings.	System presents pick record, system quantity, and reported exception reason.
3	Warehouse Manager determines resolution action (re-pick, substitute, partial ship, or quarantine).	System presents resolution options based on exception type.
4	Warehouse Manager authorizes resolution and documents reason.	System records resolution decision with manager authorization.
5	Warehouse Manager confirms exception resolution.	System updates order fulfillment status, adjusts inventory or quarantine as needed, and returns order to fulfillment flow or customer service.

UC-027 — View Order Fulfillment Status

Primary Actor: Customer Service Representative

Step	Actor Action	System Response
1	Customer Service Representative opens order fulfillment status inquiry.	System presents order search by order number, customer reference, or date.
2	Customer Service Representative selects the customer order.	System displays order header with current fulfillment status.
3	Customer Service Representative reviews line-level status (reserved, picking, packed, shipped, backordered).	System shows status, quantity, and fulfillment location per order line.
4	Customer Service Representative views shipment and carrier information for shipped lines.	System displays shipment date, carrier, and tracking reference where recorded.
5	Customer Service Representative uses status information to respond to customer inquiry.	System retains inquiry view with current real-time fulfillment status.

Domain 6: Replenishment & Inbound Supply

UC-028 — Review Low-Stock and Out-of-Stock Alerts

Primary Actor: Purchasing Manager

Step	Actor Action	System Response
1	Purchasing Manager opens low-stock and out-of-stock alert dashboard.	System displays active alerts by product, location, current ATP, and reorder threshold.
2	Purchasing Manager reviews alert priority based on ABC classification and severity.	System sorts and highlights priority alerts (Class A, out-of-stock conditions).
3	Purchasing Manager selects an alert for review.	System presents alert detail with stock history, open purchase orders, and network availability.
4	Purchasing Manager acknowledges the alert and assigns follow-up action.	System records alert acknowledgment with user and timestamp.
5	Purchasing Manager reviews remaining unacknowledged alerts.	System updates alert queue to reflect acknowledged items and pending actions.

UC-029 — Plan Replenishment and Create Purchase Requisitions

Primary Actor: Buyer

Step	Actor Action	System Response
1	Buyer opens replenishment planning from acknowledged low-stock alerts or suggestions.	System displays replenishment suggestions with recommended quantity, supplier, and network availability check results.
2	Buyer reviews network stock and confirms transfer is not preferred before purchase.	System shows excess stock at other locations and transfer alternatives.
3	Buyer creates purchase requisition with product, quantity, required date, and delivery location.	System validates product reorder eligibility and safety stock rules.
4	Buyer submits requisition for Purchasing Manager approval.	System routes requisition for approval and records pending status.
5	Purchasing Manager approves requisition.	System confirms approved requisition and makes it available for purchase order creation.

UC-030 — Track Inbound Supply Status

Primary Actor: Buyer

Step	Actor Action	System Response
1	Buyer opens inbound supply tracking view.	System displays open purchase orders with expected delivery dates and receipt status.
2	Buyer selects a purchase order to review supply status.	System presents PO lines with ordered, received, and outstanding quantities.
3	Buyer updates delivery status based on supplier communication.	System records revised expected delivery date and status.
4	Buyer reviews inbound supply position for replenishment planning.	System displays expected inbound quantity by product and arrival date.
5	Buyer confirms supply tracking review is complete.	System retains current inbound supply status for purchasing and receiving visibility.

Domain 7: Network Balance & Transfers

UC-031 — Request Inter-Location Stock Transfer

Primary Actor: Regional / Store Manager

Step	Actor Action	System Response
1	Regional / Store Manager opens inter-location transfer request.	System presents transfer request form with source and destination location options.
2	Regional / Store Manager selects source location, destination location, product, and requested quantity.	System validates source location ATP and checks source minimum threshold compliance.
3	Regional / Store Manager enters business justification for transfer.	System accepts justification and presents transfer impact preview.
4	Regional / Store Manager submits transfer request.	System records transfer request with pending approval status and notifies approver.
5	Regional / Store Manager reviews submitted transfer request.	System displays transfer request reference, status, and estimated source and destination impact.

UC-032 — Approve or Reject Transfer Requests

Primary Actor: Operations Director

Step	Actor Action	System Response
1	Operations Director opens pending transfer approval queue.	System displays transfer requests with source, destination, product, quantity, and requester.
2	Operations Director reviews transfer impact on source and destination service levels.	System presents source ATP after transfer, minimum threshold check, and destination need.
3	Operations Director approves the transfer request.	System records approval decision with approver identity and timestamp.
4	Operations Director confirms approval notification to source and destination locations.	System notifies Warehouse Manager at source and destination and updates transfer status to approved.
5	Operations Director reviews updated transfer pipeline.	System displays approved transfer ready for execution at source location.

UC-033 — Execute and Record Inter-Location Transfer

Primary Actor: Warehouse Staff

Step	Actor Action	System Response
1	Warehouse Staff opens approved transfer dispatch at source location.	System displays approved transfer with product, quantity, and destination.
2	Warehouse Staff picks and verifies transfer quantity from source storage.	System records dispatch verification with lot/batch attribution where required.
3	Warehouse Staff confirms dispatch from source location.	System deducts quantity from source on-hand and records in-transit transfer status.
4	Warehouse Staff at destination receives and verifies transferred quantity.	System presents expected in-transit receipt for confirmation.
5	Warehouse Staff confirms receipt at destination.	System adds quantity to destination on-hand, clears in-transit status, and records complete transfer in movement history.

UC-034 — Analyze and Recommend Network Rebalancing

Primary Actor: Operations Director

Step	Actor Action	System Response
1	Operations Director opens network rebalancing analysis.	System displays location-level stock imbalances (stockout vs. excess) across the network.
2	Operations Director reviews rebalancing recommendations by product and location pair.	System presents suggested transfer quantities based on demand and excess positions.

Step	Actor Action	System Response
3	Operations Director evaluates recommendations against service-level and transfer cost considerations.	System shows projected ATP impact at source and destination for each recommendation.
4	Operations Director approves selected rebalancing recommendations for transfer initiation.	System generates transfer requests from approved recommendations.
5	Operations Director reviews initiated rebalancing transfer pipeline.	System displays created transfer requests with status and network balance improvement summary.

Domain 8: Returns Management

UC-035 — Authorize and Record Customer Returns

Primary Actor: Customer Service Representative

Step	Actor Action	System Response
1	Customer Service Representative opens customer return authorization.	System presents return request form with customer and order lookup.
2	Customer Service Representative locates original order and selects lines for return.	System displays original order lines with eligible return quantities per return policy.
3	Customer Service Representative validates return reason and return eligibility.	System confirms return policy compliance and calculates authorized return quantity.
4	Customer Service Representative issues return authorization to customer.	System generates return authorization reference with authorized products, quantities, and return destination.
5	Customer Service Representative reviews issued return authorization.	System displays active return authorization with status pending receipt.

UC-036 — Receive, Inspect, and Disposition Returned Goods

Primary Actor: Receiving Clerk

Step	Actor Action	System Response
1	Receiving Clerk opens return receipt against return authorization.	System retrieves return authorization with expected products and quantities.
2	Receiving Clerk receives returned goods and verifies quantity against authorization.	System records received return quantity per line.
3	Receiving Clerk inspects returned product condition.	System presents inspection form with condition assessment options (resalable, damaged, defective).
4	Receiving Clerk records inspection outcome and disposition decision (restock, quarantine, or write-off).	System accepts disposition with reason code and routes to appropriate next action.
5	Receiving Clerk confirms return processing completion.	System updates return status, applies restock to on-hand or routes to quarantine/write-off, and closes return authorization.

Domain 9: Reporting, Performance & Financial Alignment

UC-037 — Review Inventory Dashboards and Operational Reports

Primary Actor: Operations Director

Step	Actor Action	System Response
1	Operations Director opens inventory management dashboard.	System displays inventory summary with on-hand value, stock status distribution, and alert counts.
2	Operations Director selects operational report (movement, stockout, overstock, accuracy, or location performance).	System generates report for selected type and default period.
3	Operations Director applies filters for location, category, or date range.	System refreshes report with filtered data.
4	Operations Director reviews report findings and identifies areas requiring action.	System presents report with drill-down capability to product and location detail.
5	Operations Director exports or saves report for management review.	System produces exportable report output with generation timestamp.

UC-038 — Review Inventory Performance KPIs

Primary Actor: Chief Executive Officer

Step	Actor Action	System Response
1	Chief Executive Officer opens executive inventory KPI dashboard.	System displays Tier 1 KPIs (accuracy, fill rate, turnover, stockout risk, on-time delivery).
2	Chief Executive Officer selects reporting period for KPI review.	System refreshes KPI values with period comparison to target and prior period.
3	Chief Executive Officer reviews KPI trend indicators and RAG status.	System presents KPI trends, targets, and red/amber/green status per metric.
4	Chief Executive Officer drills into underperforming KPI for contributing factors.	System displays breakdown by location, category, or product segment.
5	Chief Executive Officer completes executive KPI review.	System retains dashboard state and supports export for steering committee reporting.

UC-039 — Reconcile Inventory Valuation with Finance

Primary Actor: Finance Manager

Step	Actor Action	System Response
1	Finance Manager opens inventory valuation reconciliation view.	System displays operational inventory valuation summary by location and category.
2	Finance Manager selects reconciliation period (e.g., month-end close).	System generates valuation report for the selected period with movement and adjustment detail.
3	Finance Manager compares operational valuation to general ledger inventory balance.	System presents side-by-side comparison with variance amount and percentage.
4	Finance Manager documents reconciliation adjustments or confirms alignment within tolerance.	System records reconciliation status and finance notes.
5	Finance Manager finalizes period reconciliation.	System marks reconciliation as complete and archives reconciliation record for audit.

Domain 10: Commercial Alignment & Governance

UC-040 — Confirm Promotional and Launch Stock Readiness

Primary Actor: Marketing Manager

Step	Actor Action	System Response
1	Marketing Manager opens promotional stock readiness review for upcoming campaign.	System displays campaign or launch definition with required products and quantities.
2	Marketing Manager reviews ring-fenced promotional stock allocation.	System shows allocated promotional quantity, consumed quantity, and remaining ring-fenced stock.
3	Marketing Manager verifies ATP and inbound supply against campaign requirement.	System presents readiness status per product (confirmed, at risk, short) against campaign start date.
4	Marketing Manager confirms stock readiness for campaign launch.	System records readiness confirmation with manager identity and confirmation date.
5	Marketing Manager reviews campaign readiness summary.	System displays overall launch readiness status and any remaining gaps.

UC-041 — Conduct Inventory Audit and Control Review

Primary Actor: Internal Auditor

Step	Actor Action	System Response
1	Internal Auditor opens inventory audit and control review workspace.	System presents audit query tools for transactions, approvals, and policy compliance.
2	Internal Auditor defines audit scope (period, location, transaction type).	System retrieves scoped transaction population for review.
3	Internal Auditor samples transactions and verifies segregation of duties compliance.	System displays transaction detail with initiator, approver, timestamp, and reason codes.
4	Internal Auditor tests control policy adherence (reason codes, approvals, blind counts).	System presents compliance indicators and flags potential control breaches.
5	Internal Auditor documents audit findings and completes control review.	System records audit review session and exports audit workpaper summary.

Domain 11: Cross-Cutting System Interactions

UC-042 — View Inventory Movement and Transaction History

Primary Actor: Inventory Controller

Step	Actor Action	System Response
1	Inventory Controller opens inventory movement history inquiry.	System presents search criteria (product, location, date range, transaction type, user).
2	Inventory Controller enters search criteria for the investigation scope.	System retrieves matching movement transactions.
3	Inventory Controller reviews transaction list summary.	System displays chronologically sorted movements with type, quantity, user, and timestamp.
4	Inventory Controller selects a transaction for detail.	System presents full transaction detail including before/after quantities, reason code, and reference document.
5	Inventory Controller exports movement history for analysis.	System generates exportable movement history report for the search scope.

UC-043 — Query and Review Audit Trail

Primary Actor: Internal Auditor

Step	Actor Action	System Response
1	Internal Auditor opens audit trail query function.	System presents audit search criteria (entity type, date range, user, action type).
2	Internal Auditor defines audit query parameters.	System retrieves immutable audit log entries matching criteria.
3	Internal Auditor reviews audit log summary results.	System displays audit entries with user, timestamp, action, and entity reference.
4	Internal Auditor selects entry to view before-and-after state detail.	System presents complete audit record with prior value, new value, and reason where applicable.
5	Internal Auditor exports audit trail results.	System generates exportable audit report suitable for external audit evidence.

UC-044 — Manage User Accounts and Role Assignments

Primary Actor: System Administrator

Step	Actor Action	System Response
1	System Administrator opens user account management.	System displays user directory with account status and assigned roles.
2	System Administrator creates a new user account or selects existing account to modify.	System presents user profile form with identity, contact, and role assignment fields.
3	System Administrator assigns business role(s) and location scope to the user.	System validates role assignment against available roles and location definitions.
4	System Administrator activates the user account.	System creates or updates account, applies role permissions, and confirms activation.
5	System Administrator reviews user account summary.	System displays active account with assigned roles, location scope, and effective permissions summary.

UC-045 — Enforce Role-Based and Location-Scoped Access

Primary Actor: System Administrator

Step	Actor Action	System Response
1	System Administrator configures role permission matrix and location scoping rules.	System displays role-to-permission mapping and location scope assignments.
2	System Administrator defines or updates permissions for a business role.	System accepts permission changes and validates against segregation of duties rules.
3	System Administrator saves access control configuration.	System applies updated permissions to all users assigned the affected role.
4	Authorized user signs in and accesses a permitted function within assigned location.	System authenticates user, evaluates role and location scope, and grants access to the function.
5	System Administrator reviews access enforcement log.	System displays successful access grants with user, function, location, and timestamp.

UC-046 — Import Initial Product and Stock Master Data

Primary Actor: Inventory Controller

Step	Actor Action	System Response
1	Inventory Controller opens initial data import function.	System presents import templates for product master and opening stock balances.
2	Inventory Controller prepares and uploads import file with product and stock data.	System validates file format and required field completeness.
3	Inventory Controller reviews import validation results.	System displays validation summary with record counts, errors, and warnings.

Step	Actor Action	System Response
4	Inventory Controller confirms import execution for validated records.	System loads product master records and opening stock balances into the system.
5	Inventory Controller reviews import completion report.	System displays import results with created/updated record counts and confirms data availability for operations.

UC-047 — Receive and Respond to Operational Notifications

Primary Actor: Warehouse Manager

Step	Actor Action	System Response
1	Warehouse Manager opens notification inbox.	System displays pending notifications (low stock, approvals, exceptions, inbound delays, campaign alerts).
2	Warehouse Manager selects a notification to review.	System presents notification detail with linked business context (product, order, transfer, or campaign).
3	Warehouse Manager navigates to the related function to take action.	System opens the linked use case context (alert review, approval queue, exception record).
4	Warehouse Manager completes required action for the notification.	System records action completion against the notification.
5	Warehouse Manager marks notification as handled.	System archives notification and removes it from active pending queue.

UC-048 — Support Multi-Location Split Fulfillment

Primary Actor: Warehouse Manager

Step	Actor Action	System Response
1	Warehouse Manager opens order requiring split fulfillment across locations.	System displays order lines with single-location ATP shortfall and network-wide availability.
2	Warehouse Manager requests multi-location fulfillment plan.	System proposes fulfillment split by location with quantity per source location.
3	Warehouse Manager reviews and confirms split fulfillment allocation.	System creates location-specific reservations for each contributing location.
4	Warehouse Manager releases split fulfillment to respective warehouse queues.	System routes pick tasks to each source location with linked parent order reference.
5	Warehouse Manager monitors completion of all split shipment components.	System displays consolidated order status showing shipped/pending lines across all fulfillment locations.

Traceability

Source Document	Reference
Use Case Catalog	SRS-UC-001 (UC-001 through UC-048)
Business Requirements Document	BRD-IMS-001
Business Rule Catalog	SRS-BR-001

Document Approval

Role	Name	Signature	Date
Business Analyst			

Role	Name	Signature	Date
Product Owner			
Operations Director			

End of Use Case Main Flows

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Use Case Alternate Flows

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-UC-003
Document Title	Use Case Alternate Flows — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Business Analyst
Source Documents	SRS-UC-001 (Use Case Catalog), SRS-UC-002 (Use Case Main Flows)
Date	July 5, 2026
Scope	Valid business alternatives only — no error or exception paths

Purpose

This document identifies alternate flows for each of the 48 use cases in the Inventory Management System. An alternate flow branches from the main flow at a defined step when a valid business condition applies, leading to a different but equally legitimate business outcome. Error conditions, system failures, and validation rejections are excluded.

Alternate Flow Legend

Field	Description
Alternate Flow ID	Unique identifier in format UC-XXX-AF-NN
Starting Step	Main flow step number where the alternate path diverges
Business Condition	Valid business circumstance triggering the alternate path
Alternative Actions	Actor and system actions taken along the alternate path
Business Outcome	Resulting business state upon successful completion of the alternate path

Domain 1: Master Data & Governance

UC-001 — Manage Product Master Data

UC-001-AF-01 — Update Existing Product Record

Field	Detail
Alternate Flow ID	UC-001-AF-01
Starting Step	2
Business Condition	An existing product record requires attribute correction, identifier update, or description change rather than creation of a new SKU.
Alternative Actions	Inventory Controller searches and selects the existing product from the catalog. System loads the current product profile into the edit form. Inventory Controller updates the required attributes and submits the changes. System saves the revised record while preserving SKU identity and transaction history.
Business Outcome	Existing product master is updated with corrected attributes; all historical inventory transactions remain linked to the same SKU.

UC-001-AF-02 — Change Product Lifecycle Status

Field	Detail
Alternate Flow ID	UC-001-AF-02
Starting Step	3
Business Condition	Business decision requires changing product status to inactive or discontinued rather than modifying descriptive attributes.
Alternative Actions	Inventory Controller selects the product and chooses status change action. System presents status options (active, inactive, discontinued) with business impact summary. Inventory Controller sets new status and confirms. System applies status change, enforces discontinued reorder prohibition, and retains product record for historical reference.
Business Outcome	Product status reflects current commercial position; discontinued products are blocked from new replenishment while remaining visible in history and reporting.

UC-001-AF-03 — Resolve Duplicate Product Record

Field	Detail
Alternate Flow ID	UC-001-AF-03
Starting Step	3
Business Condition	System detects a potential duplicate SKU or identifier during create or update, and Inventory Controller confirms a merge or deduplication is required.
Alternative Actions	System presents duplicate match candidates. Inventory Controller designates the authoritative master record and selects the duplicate for resolution. Inventory Controller confirms merge or deactivation of the duplicate record. System consolidates references to the authoritative SKU and retires the duplicate.
Business Outcome	Single authoritative product identity is maintained; duplicate records are eliminated without loss of transaction traceability.

UC-002 — Configure Inventory Locations

UC-002-AF-01 — Update Existing Location Configuration

Field	Detail
Alternate Flow ID	UC-002-AF-01
Starting Step	2
Business Condition	An existing location requires hierarchy adjustment, attribute update, or status change rather than creation of a new location.
Alternative Actions	Inventory Controller selects the existing location from the network view. System loads current location configuration. Inventory Controller updates hierarchy, attributes, or operational status and saves. System applies changes and confirms updated location definition.
Business Outcome	Location configuration reflects current operational structure; stock assignments and transactions remain associated with the updated location.

UC-002-AF-02 — Plan Location Closure and Stock Disposition

Field	Detail
Alternate Flow ID	UC-002-AF-02
Starting Step	4
Business Condition	A location is scheduled for closure and requires stock disposition planning before deactivation rather than go-live activation.
Alternative Actions	Inventory Controller initiates location closure planning. System displays on-hand stock by product at the location. Inventory Controller defines disposition plan (transfer to other locations, sell-down, or write-off). System records closure plan and schedules stock movement or depletion actions. Inventory Controller confirms closure readiness when stock is dispositioned.
Business Outcome	Location closure is planned with defined stock disposition; location is deactivated without stranded inventory.

UC-003 — Define Stock Policies and Thresholds

UC-003-AF-01 — Configure Policy at Product Group Level

Field	Detail
Alternate Flow ID	UC-003-AF-01
Starting Step	1
Business Condition	Stock policy applies uniformly to a product category or group rather than individual SKUs.
Alternative Actions	Inventory Controller selects product group or category for policy configuration. System displays group members and current policy defaults. Inventory Controller sets reorder level, minimum threshold, and safety stock for the group. System applies policy to all group members and confirms group-level policy activation.
Business Outcome	Consistent stock policy is applied across the product group; monitoring and alerts operate at group-defined thresholds.

UC-003-AF-02 — Apply Location-Specific Threshold Override

Field	Detail
Alternate Flow ID	UC-003-AF-02

Field	Detail
Starting Step	2
Business Condition	A specific location requires different thresholds than the network default due to local demand, lead time, or service-level requirements.
Alternative Actions	Inventory Controller selects product and overrides thresholds for a specific location. System displays network default and location-specific fields. Inventory Controller enters location-specific reorder level and minimum threshold. System saves location override while retaining network default for other locations.
Business Outcome	Location-specific thresholds reflect local operating conditions; replenishment alerts trigger based on location-appropriate levels.

UC-004 — Classify Products by ABC Priority

UC-004-AF-01 — Bulk Reclassification from Performance Analysis

Field	Detail
Alternate Flow ID	UC-004-AF-01
Starting Step	2
Business Condition	Periodic review cycle produces system-recommended ABC assignments based on velocity, revenue, or criticality analysis.
Alternative Actions	Inventory Controller requests ABC analysis report. System generates recommended classifications based on defined criteria. Inventory Controller reviews recommendations and accepts bulk classification update. System applies new classifications and updates count frequency and monitoring rules.
Business Outcome	Product ABC classifications are refreshed based on current performance data; operational priorities align with updated segmentation.

UC-004-AF-02 — Manual Override of Single Product Classification

Field	Detail
Alternate Flow ID	UC-004-AF-02
Starting Step	3
Business Condition	Business judgment overrides system-recommended classification for a strategically critical or newly launched product.
Alternative Actions	Inventory Controller selects a product and assigns a different class than the system recommendation. System records override justification. Inventory Controller saves the manual classification. System applies override and adjusts monitoring intensity for the product.
Business Outcome	Strategically important product receives appropriate operational priority regardless of current velocity metrics.

UC-005 — Maintain Governance Policies and Reason Codes

UC-005-AF-01 — Retire or Deactivate Reason Code

Field	Detail
Alternate Flow ID	UC-005-AF-01
Starting Step	2

Field	Detail
Business Condition	An existing reason code is no longer used in operations and must be retired while preserving historical transaction references.
Alternative Actions	Inventory Controller selects existing reason code and initiates deactivation. System confirms no mandatory dependency on active policies. Inventory Controller deactivates the reason code. System marks code as inactive for new transactions while retaining it in historical records.
Business Outcome	Reason code catalog remains current; retired codes are unavailable for new transactions but visible in historical audit data.

UC-005-AF-02 — Update Governance Policy Without Reason Code Change

Field	Detail
Alternate Flow ID	UC-005-AF-02
Starting Step	3
Business Condition	Governance policy parameters (approval thresholds, count rules, safety stock override rules) require update without changes to the reason code catalog.
Alternative Actions	Inventory Controller opens governance policy configuration. System displays current policy parameters by transaction type. Inventory Controller updates policy thresholds or rules and saves. System applies updated governance policy to future transactions.
Business Outcome	Inventory governance rules reflect current business policy; reason code catalog remains unchanged.

Domain 2: Inbound & Receiving

UC-006 — Place and Manage Purchase Orders

UC-006-AF-01 — Amend Existing Open Purchase Order

Field	Detail
Alternate Flow ID	UC-006-AF-01
Starting Step	2
Business Condition	An open or approved purchase order requires quantity, delivery date, or line adjustment rather than creation of a new order.
Alternative Actions	Buyer selects existing open purchase order. System displays current order lines and status. Buyer modifies quantities, dates, or lines and submits amendment. System routes amendment for approval if required and updates expected inbound supply.
Business Outcome	Purchase order reflects revised commercial commitment; inbound supply visibility is updated accordingly.

UC-006-AF-02 — Cancel Purchase Order Before Receipt

Field	Detail
Alternate Flow ID	UC-006-AF-02
Starting Step	4
Business Condition	Business need for the purchase order no longer exists or supplier confirms order cancellation before any goods are received.
Alternative Actions	Buyer selects purchase order and initiates cancellation. System verifies no goods have been received against the order. Purchasing Manager approves cancellation. System closes purchase order, removes expected inbound supply, and records cancellation reason.

Field	Detail
Business Outcome	Purchase order is formally cancelled; no further receipt is expected and replenishment planning reflects removed inbound supply.

UC-007 — Plan Expected Inbound Deliveries

UC-007-AF-01 — Schedule Split Deliveries for Single Purchase Order

Field	Detail
Alternate Flow ID	UC-007-AF-01
Starting Step	2
Business Condition	Supplier confirms that a single purchase order will arrive in multiple shipments on different dates.
Alternative Actions	Buyer selects purchase order and defines multiple expected delivery dates by line or shipment group. System records split delivery schedule with quantities per shipment. Buyer confirms split plan. System updates inbound visibility with multiple expected arrival events.
Business Outcome	Receiving and replenishment teams have accurate visibility into phased inbound supply for a single purchase order.

UC-007-AF-02 — Reschedule Expected Delivery Date

Field	Detail
Alternate Flow ID	UC-007-AF-02
Starting Step	2
Business Condition	Supplier communicates a revised delivery date that differs from the originally planned arrival.
Alternative Actions	Buyer updates expected shipment or arrival date based on supplier confirmation. System revises inbound delivery schedule. Buyer notifies receiving location of date change. System updates replenishment and alert calculations based on revised date.
Business Outcome	Inbound delivery plan reflects current supplier commitment; downstream planning uses updated arrival expectations.

UC-008 — Receive and Verify Supplier Goods

UC-008-AF-01 — Receive Partial Shipment Against Purchase Order

Field	Detail
Alternate Flow ID	UC-008-AF-01
Starting Step	3
Business Condition	Supplier delivers only a portion of the ordered quantity with remaining quantity expected on a subsequent shipment.
Alternative Actions	Receiving Clerk verifies and records received quantities for delivered lines only. System records partial receipt against purchase order and retains outstanding quantity as open. Receiving Clerk confirms partial receipt completion. System stages received quantity for put-away and maintains open PO balance for future deliveries.
Business Outcome	Partial delivery is accurately recorded; purchase order remains open for remaining expected quantity.

UC-008-AF-02 — Receive Against Authorized Ad-Hoc Receipt

Field	Detail
Alternate Flow ID	UC-008-AF-02
Starting Step	2
Business Condition	Supplier goods arrive under an explicitly approved ad-hoc receipt authorization rather than a standard purchase order reference.
Alternative Actions	Receiving Clerk selects ad-hoc receipt option and enters approval reference. System validates authorization and presents expected receipt details. Receiving Clerk verifies and records goods per authorization. System records receipt linked to ad-hoc approval and stages for put-away.
Business Outcome	Non-standard inbound supply is received under documented authorization with full traceability.

UC-009 — Document and Resolve Receipt Discrepancies

UC-009-AF-01 — Resolve Discrepancy via Supplier Credit

Field	Detail
Alternate Flow ID	UC-009-AF-01
Starting Step	4
Business Condition	Short shipment or pricing discrepancy is resolved through supplier credit or invoice adjustment rather than physical receipt of missing quantity.
Alternative Actions	Warehouse Manager selects supplier credit resolution path. System records credit resolution against discrepancy record. Buyer confirms supplier credit agreement reference. System closes discrepancy with financial resolution and retains accepted physical receipt quantities.
Business Outcome	Receipt discrepancy is resolved commercially; inventory records reflect only physically received and verified quantities.

UC-009-AF-02 — Accept Over-Shipment with Supplier Agreement

Field	Detail
Alternate Flow ID	UC-009-AF-02
Starting Step	3
Business Condition	Supplier delivers quantity exceeding purchase order and buyer confirms acceptance of the excess under agreed commercial terms.
Alternative Actions	Receiving Clerk documents over-shipment discrepancy. Buyer confirms acceptance of excess quantity. Warehouse Manager approves receipt of accepted excess. System records full accepted quantity including over-shipment and documents supplier agreement reference.
Business Outcome	Over-shipment is formally accepted and recorded; supplier accountability and three-way match data reflect agreed quantities.

UC-010 — Perform Stock Put-Away

UC-010-AF-01 — Put-Away to Alternate Storage Location

Field	Detail
Alternate Flow ID	UC-010-AF-01
Starting Step	3
Business Condition	Suggested storage location is at capacity or operationally unsuitable; warehouse staff places stock in an approved alternate location.
Alternative Actions	Warehouse Staff selects alternate storage location within the same warehouse. System validates alternate location eligibility. Warehouse Staff confirms product, quantity, and alternate destination. System records put-away to alternate location and updates on-hand and ATP.
Business Outcome	Stock is available for fulfillment from the alternate storage location with correct rotation and location attribution.

UC-010-AF-02 — Batch Put-Away for Multiple Receipt Lines

Field	Detail
Alternate Flow ID	UC-010-AF-02
Starting Step	1
Business Condition	Multiple receipt lines from the same delivery are put away together in a single warehouse operation.
Alternative Actions	Warehouse Staff selects batch put-away for grouped receipt lines. System presents consolidated put-away task list. Warehouse Staff completes put-away confirmations for each line in sequence. System records all put-away completions and updates availability for all affected products.
Business Outcome	Efficient put-away of full delivery is completed; all received lines become available for sale and fulfillment.

UC-011 — Receive Return and Non-PO Inbound Stock

UC-011-AF-01 — Receive Authorized Customer Return

Field	Detail
Alternate Flow ID	UC-011-AF-01
Starting Step	2
Business Condition	Inbound goods are a customer return covered by a valid return authorization rather than an ad-hoc receipt.
Alternative Actions	Receiving Clerk selects authorized return receipt type and enters return authorization reference. System retrieves linked order and authorized return lines. Receiving Clerk verifies returned products and quantities. System records return receipt and routes stock to inspection per return policy.
Business Outcome	Customer return is received against valid authorization and queued for inspection and disposition.

UC-011-AF-02 — Receive Approved Ad-Hoc Inbound Stock

Field	Detail
Alternate Flow ID	UC-011-AF-02

Field	Detail
Starting Step	2
Business Condition	Inbound stock arrives under a pre-approved ad-hoc receipt (samples, emergency supply, or non-catalog inbound) without purchase order or return authorization.
Alternative Actions	Receiving Clerk selects ad-hoc receipt type and enters approval reference. System validates ad-hoc approval authority and scope. Receiving Clerk verifies and records inbound goods. System records ad-hoc receipt and stages stock per approval terms.
Business Outcome	Ad-hoc inbound stock is recorded under documented approval with defined next disposition steps.

Domain 3: Inventory Visibility & Status

UC-012 — Monitor Inventory Status

UC-012-AF-01 — Monitor Single Location Inventory Health

Field	Detail
Alternate Flow ID	UC-012-AF-01
Starting Step	2
Business Condition	Review is scoped to a single warehouse or store rather than network-wide status.
Alternative Actions	Inventory Controller filters monitoring view to a specific location. System displays stock status, alerts, and threshold breaches for the selected location only. Inventory Controller reviews location-specific low-stock and out-of-stock conditions. System retains location-scoped view for ongoing monitoring.
Business Outcome	Location-level inventory health is assessed; local replenishment and operational actions can be prioritized.

UC-012-AF-02 — Export Status Snapshot for Management Review

Field	Detail
Alternate Flow ID	UC-012-AF-02
Starting Step	5
Business Condition	Management briefing or steering review requires exported inventory status snapshot rather than live screen review only.
Alternative Actions	Inventory Controller requests export of current status view with applied filters. System generates exportable status report with timestamp. Inventory Controller distributes report to stakeholders. System archives export reference for audit.
Business Outcome	Point-in-time inventory status snapshot is available for management decision-making and meeting documentation.

UC-013 — Inquire Inventory Availability

UC-013-AF-01 — Network-Wide Availability Inquiry

Field	Detail
Alternate Flow ID	UC-013-AF-01
Starting Step	3

Field	Detail
Business Condition	Customer or sales inquiry requires total network availability rather than availability at a single fulfillment location.
Alternative Actions	Sales Representative requests network-wide availability view. System aggregates ATP across all fulfillment locations and displays network total with per-location breakdown. Sales Representative reviews network position and expected inbound supply. System presents consolidated availability for customer commitment discussion.
Business Outcome	Sales team has complete network visibility to support fulfillment options including multi-location or transfer scenarios.

UC-013-AF-02 — Product Lookup by Barcode Scan

Field	Detail
Alternate Flow ID	UC-013-AF-02
Starting Step	2
Business Condition	Product is identified by physical barcode scan rather than text search during customer-facing inquiry.
Alternative Actions	Sales Representative scans product barcode. System resolves barcode to SKU and retrieves product availability. Sales Representative reviews ATP and status for the scanned product. System displays availability result for immediate customer response.
Business Outcome	Rapid product identification and availability response during in-person or phone sales interaction.

Domain 4: Inventory Control & Accuracy

UC-014 — Perform Cycle Count

UC-014-AF-01 — Count Matches System Quantity

Field	Detail
Alternate Flow ID	UC-014-AF-01
Starting Step	4
Business Condition	Physical count result matches system quantity; no variance investigation is required.
Alternative Actions	Warehouse Staff submits count result. System compares count to system quantity and confirms match. System records count as verified accurate with no variance flag. Warehouse Staff reviews count confirmation showing no adjustment required.
Business Outcome	SKU-location accuracy is confirmed; count cycle completes without adjustment workflow.

UC-014-AF-02 — Perform Unscheduled Ad-Hoc Cycle Count

Field	Detail
Alternate Flow ID	UC-014-AF-02
Starting Step	1
Business Condition	Operations initiates an unscheduled count for a specific product or location outside the regular count schedule.
Alternative Actions	Warehouse Staff initiates ad-hoc cycle count for selected product and location. System presents blind count entry form. Warehouse Staff performs and submits count. System records ad-hoc count result and flags variance if present.

Field	Detail
Business Outcome	Targeted accuracy verification is completed outside scheduled cycle; results feed variance workflow if needed.

UC-015 — Conduct Full Physical Inventory Count

UC-015-AF-01 — Phased Zone-Based Physical Count

Field	Detail
Alternate Flow ID	UC-015-AF-01
Starting Step	2
Business Condition	Full count is executed in sequential zones rather than simultaneous wall-to-wall count across the entire location.
Alternative Actions	Inventory Controller defines zone sequence and assigns teams per zone. System generates zone-specific count sheets and applies freeze per zone as counted. Warehouse Staff complete each zone before proceeding to next. System tracks zone completion and compiles full-location count upon final zone close.
Business Outcome	Complete location count is achieved through controlled phased execution with reduced operational disruption.

UC-015-AF-02 — Recount Selected Zones Before Count Close

Field	Detail
Alternate Flow ID	UC-015-AF-02
Starting Step	4
Business Condition	Initial count results for specific zones require verification recount before final count session close.
Alternative Actions	Inventory Controller selects zones for recount. System generates recount assignments with blind count. Warehouse Staff perform recount and submit results. System compares recount to initial count and uses recount result for variance determination.
Business Outcome	Count accuracy is validated through targeted recount; final variance report reflects verified quantities.

UC-016 — Investigate and Approve Count Variances

UC-016-AF-01 — Variance Within Auto-Approval Tolerance

Field	Detail
Alternate Flow ID	UC-016-AF-01
Starting Step	4
Business Condition	Count variance falls within defined auto-approval tolerance and does not require independent approver escalation.
Alternative Actions	Warehouse Manager documents brief investigation notes. System evaluates variance against auto-approval threshold. Warehouse Manager confirms variance approval within authority. System auto-approves adjustment and queues for processing without independent approver routing.
Business Outcome	Low-materiality variance is resolved efficiently; inventory record is corrected with appropriate documentation.

UC-016-AF-02 — Investigation Concludes Count Is Accurate

Field	Detail
Alternate Flow ID	UC-016-AF-02
Starting Step	3
Business Condition	Investigation determines the physical count was correct and the system record was in error due to an unrecorded prior movement since identified and explained.
Alternative Actions	Warehouse Manager documents investigation finding that supports count accuracy. System records investigation conclusion. Warehouse Manager approves adjustment to align system to counted quantity with documented root cause. System queues adjustment reflecting corrected system record.
Business Outcome	System inventory is aligned to verified physical quantity with documented investigation supporting the correction.

UC-017 — Process Stock Adjustments

UC-017-AF-01 — Process Inventory Increase Adjustment

Field	Detail
Alternate Flow ID	UC-017-AF-01
Starting Step	2
Business Condition	Adjustment direction is an increase (found stock, receipt correction) rather than a decrease.
Alternative Actions	Warehouse Manager selects increase adjustment direction and enters quantity. System displays resulting on-hand increase preview. Warehouse Manager selects reason code and submits. System applies increase upon approval and updates ATP.
Business Outcome	On-hand and ATP are increased with documented reason; movement history records the positive adjustment.

UC-017-AF-02 — Adjustment Below Approval Threshold

Field	Detail
Alternate Flow ID	UC-017-AF-02
Starting Step	4
Business Condition	Adjustment quantity and value fall below the threshold requiring independent approver authorization.
Alternative Actions	Warehouse Manager submits adjustment within delegated authority. System confirms adjustment is below escalation threshold. Warehouse Manager confirms adjustment directly. System applies adjustment without independent approver routing.
Business Outcome	Routine low-materiality adjustment is completed within delegated authority with full audit trail.

UC-018 — Authorize and Process Inventory Write-Offs

UC-018-AF-01 — Partial Quantity Write-Off

Field	Detail
Alternate Flow ID	UC-018-AF-01

Field	Detail
Starting Step	2
Business Condition	Only a portion of on-hand quantity is damaged or obsolete; remaining quantity remains sellable.
Alternative Actions	Operations Director reviews write-off request for partial quantity. System displays on-hand total and proposed write-off quantity. Operations Director approves partial write-off. System removes only the written-off quantity from on-hand and ATP; remaining quantity stays sellable.
Business Outcome	Damaged or obsolete portion is removed from inventory; sellable remainder continues in active stock.

UC-018-AF-02 — Write-Off Initiated by Warehouse Manager

Field	Detail
Alternate Flow ID	UC-018-AF-02
Starting Step	1
Business Condition	Warehouse Manager identifies write-off need at location level and submits request for Operations Director authorization.
Alternative Actions	Warehouse Manager creates write-off request with evidence and reason code. System routes request to Operations Director approval queue. Operations Director reviews and approves. Finance Manager acknowledges where required. System executes write-off upon final authorization.
Business Outcome	Location-initiated write-off is processed through proper authorization chain with full documentation.

UC-019 — Manage Quarantine and Hold Stock

UC-019-AF-01 — Release Stock from Quarantine to Sellable

Field	Detail
Alternate Flow ID	UC-019-AF-01
Starting Step	1
Business Condition	Previously quarantined stock has passed quality review or customs clearance and is approved for return to sellable inventory.
Alternative Actions	Warehouse Manager selects quarantined stock for release review. System displays hold reason and quarantined quantity. Warehouse Manager confirms release authorization. System moves quantity from quarantined to sellable ATP and records release in movement history.
Business Outcome	Previously held stock is restored to sellable availability with documented release authorization.

UC-019-AF-02 — Place Bonded Inventory on Customs Hold

Field	Detail
Alternate Flow ID	UC-019-AF-02
Starting Step	3
Business Condition	Inbound bonded goods require customs clearance hold rather than quality quarantine.
Alternative Actions	Warehouse Manager assigns bonded customs hold status to received stock. System excludes quantity from sellable ATP under bonded hold classification. Warehouse Manager records customs reference. System maintains hold until clearance is confirmed.

Field	Detail
Business Outcome	Bonded inventory remains under regulatory hold and excluded from fulfillment until customs release is documented.

UC-020 — Analyze Shrinkage, Loss, and Obsolete Stock

UC-020-AF-01 — Shrinkage Analysis by Location

Field	Detail
Alternate Flow ID	UC-020-AF-01
Starting Step	2
Business Condition	Analysis is scoped to compare shrinkage performance across locations rather than enterprise-wide aggregation.
Alternative Actions	Inventory Controller selects location-comparative shrinkage report. System presents shrinkage by reason code and value per location. Inventory Controller identifies locations with elevated loss patterns. System supports drill-down to location-specific adjustment history.
Business Outcome	Location-level shrinkage accountability is established; targeted corrective actions can be assigned to underperforming sites.

UC-020-AF-02 — Recommend Transfer to Clearance Channel

Field	Detail
Alternate Flow ID	UC-020-AF-02
Starting Step	4
Business Condition	Slow-moving or obsolete stock is recommended for redistribution to outlet or clearance channel rather than write-off.
Alternative Actions	Inventory Controller documents transfer-to-clearance recommendation for identified SKUs. System records recommended disposition action. Inventory Controller initiates transfer request or commercial handoff to outlet channel. System links analysis recommendation to subsequent transfer or status change.
Business Outcome	Excess inventory is directed toward revenue recovery through alternate channel rather than immediate write-off.

Domain 5: Order Fulfillment

UC-021 — Confirm Order Availability

UC-021-AF-01 — Fulfill from Alternate Location

Field	Detail
Alternate Flow ID	UC-021-AF-01
Starting Step	2
Business Condition	Primary fulfillment location has insufficient ATP but another network location has adequate stock to fulfill the order.
Alternative Actions	Sales Representative requests availability check across all fulfillment locations. System identifies alternate location with sufficient ATP. Sales Representative confirms alternate fulfillment location with customer. System records availability confirmation with designated fulfillment location.

Field	Detail
Business Outcome	Order availability is confirmed using alternate location stock; customer commitment reflects achievable fulfillment source.

UC-021-AF-02 — Confirm Partial Availability for Customer Decision

Field	Detail
Alternate Flow ID	UC-021-AF-02
Starting Step	3
Business Condition	Order line has partial ATP; customer must decide on partial fulfillment before proceeding to reservation.
Alternative Actions	Sales Representative presents partial availability to customer. System displays available quantity versus ordered quantity. Customer accepts partial quantity commitment. System records partial availability confirmation and routes to partial fulfillment or reservation workflow.
Business Outcome	Customer makes informed commitment based on partial availability; order proceeds with agreed fulfillable quantity.

UC-022 — Reserve Inventory for Confirmed Orders

UC-022-AF-01 — Reserve from Alternate Fulfillment Location

Field	Detail
Alternate Flow ID	UC-022-AF-01
Starting Step	3
Business Condition	Reservation is made from a non-default fulfillment location selected during availability confirmation.
Alternative Actions	Customer Service Representative selects alternate fulfillment location for reservation. System validates ATP at selected location. Customer Service Representative confirms reservation at alternate location. System reserves stock at designated location and updates ATP.
Business Outcome	Order reservation is held at the location that will execute fulfillment; ATP is reduced at the correct source.

UC-022-AF-02 — Reserve Partial Quantity with Backorder Remainder

Field	Detail
Alternate Flow ID	UC-022-AF-02
Starting Step	3
Business Condition	Customer accepts partial reservation now with remaining quantity to be fulfilled when stock becomes available.
Alternative Actions	Customer Service Representative reserves available quantity for immediate fulfillment. System creates reservation for partial quantity and backorder record for remainder. Customer Service Representative confirms split reservation. System releases partial order to warehouse and holds backorder for replenishment priority.
Business Outcome	Available stock is reserved and queued for fulfillment; backordered remainder is tracked for future supply.

UC-023 — Release Inventory Reservations

UC-023-AF-01 — Release Single Order Line Reservation

Field	Detail
Alternate Flow ID	UC-023-AF-01
Starting Step	2
Business Condition	Only specific order lines require reservation release while other lines remain reserved for fulfillment.
Alternative Actions	Customer Service Representative selects individual lines for release. System presents per-line release quantities and ATP restoration. Customer Service Representative confirms line-level release. System releases selected lines only and maintains reservation on remaining lines.
Business Outcome	Released line quantities return to ATP; remaining reserved lines continue toward fulfillment.

UC-023-AF-02 — Release Reservation Due to Order Amendment

Field	Detail
Alternate Flow ID	UC-023-AF-02
Starting Step	4
Business Condition	Customer amends order quantity downward; excess reserved quantity must be released while order remains active.
Alternative Actions	Customer Service Representative records order amendment and initiates partial reservation release. System calculates excess reserved quantity to release. Customer Service Representative confirms release of excess only. System restores excess quantity to ATP and updates reservation to amended quantity.
Business Outcome	Reservation aligns with amended order; released stock is available for other customer commitments.

UC-024 — Pick, Pack, and Ship Orders

UC-024-AF-01 — Customer Pickup Instead of Carrier Shipment

Field	Detail
Alternate Flow ID	UC-024-AF-01
Starting Step	4
Business Condition	Customer collects order from warehouse or store rather than carrier delivery.
Alternative Actions	Warehouse Picker / Packer selects customer pickup dispatch method. System updates order to pickup-ready status. Warehouse Picker / Packer confirms customer collection upon handover. System deducts inventory and records pickup completion without carrier tracking.
Business Outcome	Order is fulfilled via customer pickup; inventory is deducted and order marked complete.

UC-024-AF-02 — Batch Pick Multiple Orders

Field	Detail
Alternate Flow ID	UC-024-AF-02

Field	Detail
Starting Step	2
Business Condition	Warehouse efficiency practice combines picking for multiple orders in a single pass through storage areas.
Alternative Actions	Warehouse Picker / Packer selects batch pick for grouped orders. System generates consolidated pick list organized by storage location. Warehouse Picker / Packer picks and verifies items per order within the batch. System records pick verification per order line across the batch.
Business Outcome	Multiple orders are picked efficiently; each order proceeds independently to pack and ship steps.

UC-025 — Handle Partial Fulfillment and Backorders

UC-025-AF-01 — Partial Shipment Only, No Backorder

Field	Detail
Alternate Flow ID	UC-025-AF-01
Starting Step	3
Business Condition	Customer accepts immediate partial shipment and declines backorder for the remaining quantity.
Alternative Actions	Customer Service Representative records customer acceptance of partial quantity only. System cancels unfulfilled remainder from order commitment. Customer Service Representative confirms partial fulfillment plan. System reserves and releases available quantity for fulfillment; remaining quantity is removed from order.
Business Outcome	Order is fulfilled for agreed partial quantity; no backorder obligation remains for declined remainder.

UC-025-AF-02 — Backorder with Replenishment Priority

Field	Detail
Alternate Flow ID	UC-025-AF-02
Starting Step	4
Business Condition	Customer accepts backorder and business assigns replenishment priority based on customer tier or product class.
Alternative Actions	Customer Service Representative confirms backorder with customer acknowledgment. System creates backorder with assigned priority level. Customer Service Representative links backorder to replenishment alert. System queues backorder for fulfillment upon next available supply at priority rank.
Business Outcome	Backorder is recorded with customer agreement and priority-based fulfillment sequencing.

UC-026 — Handle Outbound Fulfillment Exceptions

UC-026-AF-01 — Authorized Product Substitution

Field	Detail
Alternate Flow ID	UC-026-AF-01
Starting Step	3

Field	Detail
Business Condition	Original ordered product is unavailable but an authorized substitute SKU is available and approved for the customer order.
Alternative Actions	Warehouse Manager selects authorized substitution resolution. System presents approved substitute products with ATP. Warehouse Manager confirms substitute SKU and quantity. System updates order line to substitute product, adjusts reservation, and returns order to pick workflow.
Business Outcome	Order is fulfilled with approved substitute; customer receives equivalent product without order cancellation.

UC-026-AF-02 — Re-Pick from Alternate Storage Location

Field	Detail
Alternate Flow ID	UC-026-AF-02
Starting Step	3
Business Condition	Pick failure is resolved by retrieving stock from alternate storage location within the same warehouse.
Alternative Actions	Warehouse Manager authorizes re-pick from alternate location. System presents alternate location with available quantity. Warehouse Picker / Packer completes re-pick and verification. System records successful pick from alternate location and clears exception.
Business Outcome	Order fulfillment continues without substitution or cancellation; pick is completed from verified alternate location.

UC-027 — View Order Fulfillment Status

UC-027-AF-01 — Search by Customer Reference

Field	Detail
Alternate Flow ID	UC-027-AF-01
Starting Step	2
Business Condition	Customer or service inquiry provides customer reference or purchase order number rather than internal order number.
Alternative Actions	Customer Service Representative searches by customer reference. System locates matching order(s) and presents search results. Customer Service Representative selects correct order. System displays fulfillment status for the matched order.
Business Outcome	Order is located and status communicated using customer-provided reference.

UC-027-AF-02 — View Backorder and Expected Supply for Pending Lines

Field	Detail
Alternate Flow ID	UC-027-AF-02
Starting Step	3
Business Condition	Order has backordered lines and customer inquiry requires expected availability information.
Alternative Actions	Customer Service Representative selects backordered lines for detail. System displays backorder quantity, priority, and expected supply date from inbound or replenishment data. Customer Service Representative communicates expected timing to customer. System retains inquiry view with backorder supply visibility.

Field	Detail
Business Outcome	Customer receives informed expectation for backordered line fulfillment.

Domain 6: Replenishment & Inbound Supply

UC-028 — Review Low-Stock and Out-of-Stock Alerts

UC-028-AF-01 — Delegate Alert Follow-Up to Buyer

Field	Detail
Alternate Flow ID	UC-028-AF-01
Starting Step	4
Business Condition	Alert follow-up action is purchase replenishment and Purchasing Manager delegates execution to Buyer.
Alternative Actions	Purchasing Manager acknowledges alert and assigns follow-up to Buyer. System records assignment and routes alert to Buyer work queue. Buyer accepts assignment. System tracks alert as in-progress under Buyer ownership.
Business Outcome	Alert follow-up is assigned with clear ownership; replenishment action proceeds under Buyer responsibility.

UC-028-AF-02 — Initiate Transfer Instead of Purchase Response

Field	Detail
Alternate Flow ID	UC-028-AF-02
Starting Step	3
Business Condition	Network availability check reveals excess stock at another location sufficient to resolve the alert without new procurement.
Alternative Actions	Purchasing Manager reviews network availability on alert detail. System shows transfer alternative with source location excess. Purchasing Manager initiates transfer request from alert context. System creates transfer request and links to originating alert.
Business Outcome	Low-stock condition is addressed through network transfer; procurement is avoided where network stock exists.

UC-029 — Plan Replenishment and Create Purchase Requisitions

UC-029-AF-01 — Create Requisition from System Suggestion

Field	Detail
Alternate Flow ID	UC-029-AF-01
Starting Step	3
Business Condition	Buyer accepts system-generated replenishment suggestion without manual quantity modification.
Alternative Actions	Buyer selects replenishment suggestion and accepts recommended quantity and supplier. System pre-populates requisition from suggestion. Buyer submits requisition for approval. System routes for Purchasing Manager approval.
Business Outcome	Replenishment requisition is created efficiently from system recommendation with full audit trail.

UC-029-AF-02 — Initiate Transfer Request Instead of Purchase Requisition

Field	Detail
Alternate Flow ID	UC-029-AF-02
Starting Step	2
Business Condition	Network check confirms excess stock at another location; transfer is preferred over new purchase.
Alternative Actions	Buyer selects transfer alternative from replenishment planning view. System presents transfer request form pre-populated from alert context. Buyer submits transfer request. System routes transfer for approval per network balance policy.
Business Outcome	Replenishment need is addressed through inter-location transfer rather than new procurement.

UC-030 — Track Inbound Supply Status

UC-030-AF-01 — Advance Delivery Date for Early Shipment

Field	Detail
Alternate Flow ID	UC-030-AF-01
Starting Step	3
Business Condition	Supplier confirms shipment will arrive earlier than originally scheduled.
Alternative Actions	Buyer updates expected delivery to earlier date based on supplier confirmation. System revises inbound supply schedule. Buyer notifies receiving location of accelerated arrival. System updates replenishment and alert projections with earlier supply date.
Business Outcome	Inbound supply visibility reflects early delivery; receiving and replenishment planning adjust accordingly.

UC-030-AF-02 — Review Aggregate Inbound Supply Position

Field	Detail
Alternate Flow ID	UC-030-AF-02
Starting Step	2
Business Condition	Buyer reviews total expected inbound supply across all open purchase orders for a product or category rather than a single PO.
Alternative Actions	Buyer selects product or category aggregate inbound view. System displays total ordered, received, and outstanding across all open POs. Buyer reviews consolidated supply timeline. System presents aggregate inbound position for replenishment planning.
Business Outcome	Buyer has consolidated view of total inbound supply for informed replenishment and allocation decisions.

Domain 7: Network Balance & Transfers

UC-031 — Request Inter-Location Stock Transfer

UC-031-AF-01 — Request Reduced Quantity Within Source Limits

Field	Detail
Alternate Flow ID	UC-031-AF-01
Starting Step	2
Business Condition	Requested transfer quantity would breach source minimum threshold; requester reduces quantity to compliant level.
Alternative Actions	Regional / Store Manager reviews source minimum threshold warning. System displays maximum transferable quantity. Regional / Store Manager revises request to compliant quantity. System accepts revised transfer request and proceeds to submission.
Business Outcome	Transfer request complies with source protection rules; source location service level is preserved.

UC-031-AF-02 — Warehouse Manager Initiates Transfer on Behalf of Store

Field	Detail
Alternate Flow ID	UC-031-AF-02
Starting Step	1
Business Condition	Warehouse Manager identifies rebalancing need and initiates transfer request on behalf of a requesting store or regional manager.
Alternative Actions	Warehouse Manager creates transfer request specifying requesting destination. System records Warehouse Manager as initiator with destination beneficiary. Warehouse Manager completes request with justification. System submits for approval with initiator and beneficiary attribution.
Business Outcome	Transfer request is initiated by operations staff on behalf of destination; approval workflow proceeds normally.

UC-032 — Approve or Reject Transfer Requests

UC-032-AF-01 — Reject Transfer Request

Field	Detail
Alternate Flow ID	UC-032-AF-01
Starting Step	3
Business Condition	Transfer would negatively impact source service level, cost exceeds benefit, or destination need does not justify movement.
Alternative Actions	Operations Director reviews transfer impact and selects rejection. System prompts for rejection reason. Operations Director documents rejection justification and confirms. System notifies requester, records rejection, and closes transfer request.
Business Outcome	Transfer request is formally rejected with documented reason; requester can pursue alternate replenishment options.

UC-032-AF-02 — Approve Transfer with Modified Quantity

Field	Detail
Alternate Flow ID	UC-032-AF-02
Starting Step	3
Business Condition	Transfer is justified but approved quantity differs from requested quantity to protect source minimum threshold.
Alternative Actions	Operations Director modifies approved transfer quantity to compliant level. System recalculates source and destination impact. Operations Director approves modified transfer. System notifies locations of approved quantity and updates transfer record.
Business Outcome	Transfer proceeds at approved quantity that balances destination need with source protection.

UC-033 — Execute and Record Inter-Location Transfer

UC-033-AF-01 — Direct Store-to-Store Transfer

Field	Detail
Alternate Flow ID	UC-033-AF-01
Starting Step	1
Business Condition	Transfer moves directly between retail stores without routing through a central distribution warehouse.
Alternative Actions	Warehouse Staff at source store dispatches transfer to destination store. System records direct store-to-store dispatch and in-transit status. Destination store staff receive and confirm. System completes transfer between store locations without hub involvement.
Business Outcome	Stock is rebalanced directly between stores with full movement traceability.

UC-033-AF-02 — Receive Partial Transfer with Remainder In-Transit

Field	Detail
Alternate Flow ID	UC-033-AF-02
Starting Step	4
Business Condition	Destination receives partial transfer quantity in first delivery with balance to follow in subsequent shipment.
Alternative Actions	Warehouse Staff at destination confirms partial receipt quantity. System records partial receipt and maintains in-transit balance. Source or carrier delivers remainder. Destination staff confirm final receipt. System closes transfer upon full quantity receipt.
Business Outcome	Partial transfer receipt is accurately tracked; in-transit quantity remains visible until transfer is complete.

UC-034 — Analyze and Recommend Network Rebalancing

UC-034-AF-01 — Phased Rebalancing Across Multiple Periods

Field	Detail
Alternate Flow ID	UC-034-AF-01
Starting Step	4

Field	Detail
Business Condition	Operations Director approves subset of rebalancing recommendations for immediate execution and defers remainder to subsequent planning cycle.
Alternative Actions	Operations Director selects priority recommendations for immediate transfer initiation. System generates transfer requests for selected items. Operations Director defers remaining recommendations with scheduled review date. System tracks deferred recommendations for future rebalancing cycle.
Business Outcome	Network rebalancing proceeds in manageable phases; deferred items remain in planning queue.

UC-034-AF-02 — Manual Rebalancing Override

Field	Detail
Alternate Flow ID	UC-034-AF-02
Starting Step	2
Business Condition	Operations judgment identifies rebalancing need not captured by system recommendations.
Alternative Actions	Operations Director creates manual rebalancing transfer proposal. System accepts manual source, destination, product, and quantity. Operations Director submits manual proposal for execution. System generates transfer request from manual override with documented justification.
Business Outcome	Business-directed rebalancing action is initiated outside automated recommendations with full audit trail.

Domain 8: Returns Management

UC-035 — Authorize and Record Customer Returns

UC-035-AF-01 — Partial Return Authorization

Field	Detail
Alternate Flow ID	UC-035-AF-01
Starting Step	2
Business Condition	Customer returns only a subset of order lines or partial quantity within eligible return policy.
Alternative Actions	Customer Service Representative selects specific lines and quantities for return. System calculates authorized return quantity per line. Customer Service Representative validates eligibility and issues partial return authorization. System generates authorization for selected lines only.
Business Outcome	Return authorization covers only the returned portion; remaining order lines are unaffected.

UC-035-AF-02 — Exchange Authorization Instead of Refund Return

Field	Detail
Alternate Flow ID	UC-035-AF-02
Starting Step	3
Business Condition	Customer requests product exchange rather than refund return per return policy.

Field	Detail
Alternative Actions	Customer Service Representative selects exchange authorization type. System links return authorization to replacement order or product. Customer Service Representative issues exchange authorization with return and replacement details. System generates authorization referencing both return inbound and replacement fulfillment.
Business Outcome	Exchange is authorized with linked return receipt and replacement order workflow.

UC-036 — Receive, Inspect, and Disposition Returned Goods

UC-036-AF-01 — Restock to Sellable Inventory

Field	Detail
Alternate Flow ID	UC-036-AF-01
Starting Step	4
Business Condition	Inspection confirms returned product is in resalable condition meeting quality standards.
Alternative Actions	Receiving Clerk records inspection outcome as resalable. System accepts restock disposition with reason code. Receiving Clerk confirms restock to sellable inventory. System increases on-hand and ATP at designated location.
Business Outcome	Returned product re-enters sellable inventory and is available for future fulfillment.

UC-036-AF-02 — Quarantine Return for Further Inspection

Field	Detail
Alternate Flow ID	UC-036-AF-02
Starting Step	4
Business Condition	Initial inspection is inconclusive; returned product requires extended quality review before disposition.
Alternative Actions	Receiving Clerk records inspection outcome as requires further review. System routes disposition to quarantine status. Receiving Clerk confirms quarantine placement. System moves quantity to quarantined hold excluded from ATP.
Business Outcome	Returned product is held pending definitive disposition; sellable inventory is not affected until review completes.

UC-036-AF-03 — Write-Off Non-Resalable Return

Field	Detail
Alternate Flow ID	UC-036-AF-03
Starting Step	4
Business Condition	Inspection confirms returned product is damaged or defective and cannot be restocked.
Alternative Actions	Receiving Clerk records inspection outcome as non-resalable. System routes disposition to write-off with mandatory reason code. Warehouse Manager or Operations Director authorizes write-off per policy. System removes quantity from inventory and closes return line.
Business Outcome	Non-resalable return is removed from inventory with documented inspection and authorization.

Domain 9: Reporting, Performance & Financial Alignment

UC-037 — Review Inventory Dashboards and Operational Reports

UC-037-AF-01 — Dashboard Review Without Detailed Report

Field	Detail
Alternate Flow ID	UC-037-AF-01
Starting Step	2
Business Condition	Management needs summary dashboard visibility only without generating a detailed operational report.
Alternative Actions	Operations Director reviews inventory summary dashboard metrics. System displays real-time KPIs, alert counts, and status distribution. Operations Director identifies areas of concern from dashboard indicators. System supports drill-down from dashboard widgets if further detail is needed.
Business Outcome	Rapid operational health assessment completed from dashboard without formal report generation.

UC-037-AF-02 — Compare Location Performance Across Periods

Field	Detail
Alternate Flow ID	UC-037-AF-02
Starting Step	3
Business Condition	Analysis requires comparison of location performance between two time periods rather than single-period snapshot.
Alternative Actions	Operations Director selects location performance report with comparative period filter. System generates side-by-side or variance comparison between periods. Operations Director reviews period-over-period changes. System highlights significant shifts in accuracy, stockouts, or movement volume.
Business Outcome	Location performance trend is understood; improvement or deterioration is identified for management action.

UC-038 — Review Inventory Performance KPIs

UC-038-AF-01 — Executive Summary Review Without Drill-Down

Field	Detail
Alternate Flow ID	UC-038-AF-01
Starting Step	3
Business Condition	Executive review session requires Tier 1 KPI summary and RAG status only without detailed contributing factor analysis.
Alternative Actions	Chief Executive Officer reviews KPI dashboard summary view. System presents Tier 1 metrics with target comparison and RAG indicators. Chief Executive Officer assesses overall inventory program health. System retains summary view for steering committee reference.
Business Outcome	Executive has concise program health assessment suitable for governance reporting.

UC-038-AF-02 — Year-over-Year KPI Comparison

Field	Detail
Alternate Flow ID	UC-038-AF-02
Starting Step	2
Business Condition	Strategic review requires comparison of current period KPIs to same period prior year.
Alternative Actions	Chief Executive Officer selects year-over-year comparison period. System displays current and prior-year KPI values with variance. Chief Executive Officer reviews long-term trend direction. System presents trend charts for key executive metrics.
Business Outcome	Strategic progress against inventory objectives is assessed on annual comparison basis.

UC-039 — Reconcile Inventory Valuation with Finance

UC-039-AF-01 — Confirm Alignment Within Tolerance

Field	Detail
Alternate Flow ID	UC-039-AF-01
Starting Step	4
Business Condition	Operational valuation and general ledger balance are within defined reconciliation tolerance requiring no adjustment.
Alternative Actions	Finance Manager compares operational and GL balances. System displays variance within tolerance threshold. Finance Manager documents confirmation of alignment. System records reconciliation as complete with no adjustment required.
Business Outcome	Period reconciliation is closed confirming operational and financial inventory alignment.

UC-039-AF-02 — Document Reconciling Adjustment

Field	Detail
Alternate Flow ID	UC-039-AF-02
Starting Step	4
Business Condition	Variance between operational and GL valuation requires documented reconciling adjustment for finance posting.
Alternative Actions	Finance Manager identifies reconciling items causing variance. System supports drill-down to contributing transactions. Finance Manager documents reconciling adjustment with finance notes. System records reconciliation with adjustment reference for GL posting.
Business Outcome	Valuation variance is explained and reconciled; finance has documented basis for period-end adjustment.

Domain 10: Commercial Alignment & Governance

UC-040 — Confirm Promotional and Launch Stock Readiness

UC-040-AF-01 — Conditional Readiness Pending Inbound Supply

Field	Detail
Alternate Flow ID	UC-040-AF-01
Starting Step	4
Business Condition	Campaign readiness is confirmed contingent on expected inbound supply arriving before campaign start date.
Alternative Actions	Marketing Manager reviews readiness with inbound supply dependency. System displays current ring-fenced stock plus expected inbound before campaign date. Marketing Manager records conditional readiness confirmation with inbound dependency noted. System tracks conditional status and monitors inbound supply against campaign deadline.
Business Outcome	Campaign readiness is documented with explicit inbound supply dependency; monitoring continues until unconditional confirmation.

UC-040-AF-02 — Request Additional Ring-Fenced Stock Allocation

Field	Detail
Alternate Flow ID	UC-040-AF-02
Starting Step	2
Business Condition	Current ring-fenced promotional allocation is insufficient; additional stock must be reserved for the campaign.
Alternative Actions	Marketing Manager requests increase to ring-fenced promotional allocation. System presents available network stock and current allocations. Operations Director or Purchasing Manager approves additional ring-fence. System increases promotional reservation and updates readiness calculation.
Business Outcome	Promotional stock ring-fence is increased; campaign readiness reflects expanded committed inventory.

UC-041 — Conduct Inventory Audit and Control Review

UC-041-AF-01 — Statistical Sample Audit

Field	Detail
Alternate Flow ID	UC-041-AF-01
Starting Step	3
Business Condition	Audit scope is large; Internal Auditor applies statistical sampling methodology rather than full population review.
Alternative Actions	Internal Auditor defines sample size and selection criteria. System generates random or stratified sample from scoped population. Internal Auditor reviews sample transactions for control compliance. System supports extrapolation of sample results to population conclusion.
Business Outcome	Control effectiveness is assessed through valid statistical sample with documented methodology.

UC-041-AF-02 — Location-Scoped Audit Review

Field	Detail
Alternate Flow ID	UC-041-AF-02
Starting Step	2
Business Condition	Audit engagement is scoped to specific location rather than enterprise-wide transaction population.
Alternative Actions	Internal Auditor defines audit scope limited to selected location(s). System retrieves location-scoped transaction population. Internal Auditor performs control testing on location transactions. System generates location-specific audit findings summary.
Business Outcome	Location-level control compliance is assessed; findings support location-specific corrective action.

Domain 11: Cross-Cutting System Interactions

UC-042 — View Inventory Movement and Transaction History

UC-042-AF-01 — Search by Product Across All Locations

Field	Detail
Alternate Flow ID	UC-042-AF-01
Starting Step	2
Business Condition	Investigation requires all movement history for a product across the entire network without location filter.
Alternative Actions	Inventory Controller enters product identifier without location constraint. System retrieves all movements for the product across all locations. Inventory Controller reviews network-wide movement chronology. System displays consolidated history with location attribution per transaction.
Business Outcome	Complete product movement history across network is available for investigation or analysis.

UC-042-AF-02 — Navigate to Audit Trail from Movement Detail

Field	Detail
Alternate Flow ID	UC-042-AF-02
Starting Step	4
Business Condition	Movement investigation requires detailed audit trail including before-and-after state beyond standard movement summary.
Alternative Actions	Inventory Controller selects movement transaction and requests linked audit trail. System presents immutable audit record with before-and-after values. Inventory Controller reviews complete change history for the transaction. System displays audit detail suitable for governance review.
Business Outcome	Full audit evidence is accessed from movement inquiry for accountability verification.

UC-043 — Query and Review Audit Trail

UC-043-AF-01 — Query Audit Trail by User Activity

Field	Detail
Alternate Flow ID	UC-043-AF-01
Starting Step	2
Business Condition	Audit review is focused on all actions performed by a specific user during a defined period.
Alternative Actions	Internal Auditor defines query by user identity and date range. System retrieves all audit entries for the user in the period. Internal Auditor reviews user activity chronology. System displays complete user action log with entity references.
Business Outcome	User accountability is assessed through complete activity audit for the review period.

UC-043-AF-02 — Provide Audit Evidence to External Auditor

Field	Detail
Alternate Flow ID	UC-043-AF-02
Starting Step	5
Business Condition	External audit engagement requires export of audit trail evidence for independent review.
Alternative Actions	Internal Auditor prepares scoped audit trail export for external auditor. System generates export package with query parameters documented. External Auditor receives read-only access or exported evidence. System logs external auditor access for compliance record.
Business Outcome	External auditor receives complete audit evidence package with documented scope and access record.

UC-044 — Manage User Accounts and Role Assignments

UC-044-AF-01 — Modify Existing User Role Assignment

Field	Detail
Alternate Flow ID	UC-044-AF-01
Starting Step	2
Business Condition	Existing user requires role change, location scope update, or profile correction rather than new account creation.
Alternative Actions	System Administrator selects existing user account. System displays current roles and location scope. System Administrator updates role assignment or location scope. System applies changes and confirms updated permissions.
Business Outcome	User access reflects current job responsibilities without creating duplicate accounts.

UC-044-AF-02 — Deactivate User Account

Field	Detail
Alternate Flow ID	UC-044-AF-02
Starting Step	4

Field	Detail
Business Condition	User departs organization or changes role no longer requiring system access.
Alternative Actions	System Administrator selects user account for deactivation. System confirms no pending approvals or sole-authority dependencies. System Administrator deactivates account. System revokes access while preserving user attribution in historical audit records.
Business Outcome	Departed user can no longer access system; historical transactions retain correct user attribution.

UC-045 — Enforce Role-Based and Location-Scoped Access

UC-045-AF-01 — Update Location Scope for Existing Role

Field	Detail
Alternate Flow ID	UC-045-AF-01
Starting Step	2
Business Condition	User role permissions remain unchanged but location scope must be expanded or restricted.
Alternative Actions	System Administrator selects role and modifies location scope assignment. System validates location definitions and updates scope mapping. System Administrator saves location scope changes. System applies updated scope to all users with the affected role.
Business Outcome	Users retain role permissions with corrected location access boundaries.

UC-045-AF-02 — Grant Network-Wide Read Access to Executive Role

Field	Detail
Alternate Flow ID	UC-045-AF-02
Starting Step	4
Business Condition	Executive or oversight role requires read-only visibility across all locations without transactional authority.
Alternative Actions	System Administrator configures executive role with network-wide read scope. System applies read-only permission set without transactional grants. Executive user accesses inventory visibility across all locations. System grants read access without location restriction for visibility functions only.
Business Outcome	Executive stakeholders have network-wide inventory visibility without transactional or approval authority.

UC-046 — Import Initial Product and Stock Master Data

UC-046-AF-01 — Import Product Master Only

Field	Detail
Alternate Flow ID	UC-046-AF-01
Starting Step	2
Business Condition	Product catalog is imported first; opening stock balances will be entered separately after physical verification.

Field	Detail
Alternative Actions	Inventory Controller uploads product master import file without stock balances. System validates and imports product records. Inventory Controller reviews product import completion. System confirms product catalog availability; stock balances remain at zero pending separate entry.
Business Outcome	Product master is loaded; opening balances are deferred for verified physical count entry.

UC-046-AF-02 — Import Updates to Existing Catalog

Field	Detail
Alternate Flow ID	UC-046-AF-02
Starting Step	4
Business Condition	Import file updates existing product attributes or stock balances rather than initial go-live load.
Alternative Actions	Inventory Controller confirms import mode as update to existing records. System matches import records to existing SKUs and applies updates. Inventory Controller reviews update summary showing changed versus unchanged records. System confirms updated catalog and balances.
Business Outcome	Existing product and stock data is bulk-updated with import audit trail of changes applied.

UC-047 — Receive and Respond to Operational Notifications

UC-047-AF-01 — Delegate Notification to Another Role

Field	Detail
Alternate Flow ID	UC-047-AF-01
Starting Step	3
Business Condition	Notification action is outside Warehouse Manager responsibility and must be delegated to Purchasing Manager or Buyer.
Alternative Actions	Warehouse Manager reassigns notification to appropriate role owner. System routes notification to delegated recipient work queue. Delegated actor accepts and acts on notification. System records delegation chain and action completion.
Business Outcome	Notification is handled by correct role owner with documented delegation.

UC-047-AF-02 — Defer Notification for Scheduled Follow-Up

Field	Detail
Alternate Flow ID	UC-047-AF-02
Starting Step	5
Business Condition	Notification requires action at a future time; immediate handling is not operationally appropriate.
Alternative Actions	Warehouse Manager schedules notification follow-up for future date or time. System retains notification in deferred queue with reminder. System re-surfaces notification at scheduled time. Warehouse Manager completes action when deferral period ends.
Business Outcome	Notification is handled at operationally appropriate time without loss of tracking accountability.

UC-048 — Support Multi-Location Split Fulfillment

UC-048-AF-01 — Manual Adjustment of Split Allocation

Field	Detail
Alternate Flow ID	UC-048-AF-01
Starting Step	3
Business Condition	Operations judgment requires different location split than system-proposed allocation.
Alternative Actions	Warehouse Manager modifies proposed quantities per source location. System recalculates ATP impact for revised split. Warehouse Manager confirms adjusted split allocation. System creates reservations per revised location allocation.
Business Outcome	Multi-location fulfillment reflects operations-approved allocation across source locations.

UC-048-AF-02 — Consolidated Customer Delivery After Multi-Location Ship

Field	Detail
Alternate Flow ID	UC-048-AF-02
Starting Step	5
Business Condition	Multiple location shipments are consolidated at hub before single customer delivery rather than direct multi-shipment to customer.
Alternative Actions	Warehouse Manager designates consolidation hub location. System tracks in-transit shipments from source locations to hub. Hub receives and consolidates split shipment components. System records consolidated shipment to customer from hub with linked parent order reference.
Business Outcome	Customer receives single consolidated delivery while inventory deductions reflect multi-location sourcing.

Summary

Metric	Count
Use Cases	48
Alternate Flows	96
Average Alternate Flows per Use Case	2.0

Traceability

Source Document	Reference
Use Case Catalog	SRS-UC-001
Use Case Main Flows	SRS-UC-002
Business Requirements Document	BRD-IMS-001

Document Approval

Role	Name	Signature	Date
Business Analyst			

Role	Name	Signature	Date
Product Owner			
Operations Director			

End of Use Case Alternate Flows

Document ID: SRS-UC-003 | Version 1.0 | July 5, 2026

Use Case Exception Flows

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-UC-004
Document Title	Use Case Exception Flows — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Business Analyst
Source Documents	SRS-UC-001 (Use Case Catalog), SRS-UC-002 (Use Case Main Flows), SRS-BR-001 (Business Rule Catalog)
Date	July 5, 2026
Scope	Business exceptions only — technical infrastructure failures are out of scope

Purpose

This document identifies exception flows for each of the 48 use cases in the Inventory Management System. A business exception occurs when a valid business rule, authorization constraint, or operational condition prevents the main flow from completing as intended. Each exception describes how the system must respond and what business outcome results.

Exception Legend

Field	Description
Exception ID	Unique identifier in format UC-XXX-EX-NN
Trigger	Business event or condition that causes the exception
Failed Step	Main flow step where processing cannot complete successfully
Expected System Behavior	Required system response to the business exception
User Notification	Message or alert presented to the affected user or role
Business Outcome	Resulting business state; transaction is blocked, deferred, or routed for resolution

Domain 1: Master Data & Governance

UC-001 — Manage Product Master Data

UC-001-EX-01 — Duplicate SKU or Identifier Detected

Field	Detail
Exception ID	UC-001-EX-01
Trigger	Inventory Controller submits a new or updated product record with a SKU or barcode that already exists on another active product record.
Failed Step	3
Expected System Behavior	System blocks save, displays the conflicting existing product record, and prevents duplicate identity creation until resolved.
User Notification	"Duplicate SKU [or identifier] detected. Product [existing SKU/name] already uses this identifier. Resolve duplicate before saving."
Business Outcome	Product master integrity is preserved; no duplicate SKU enters the catalog. Inventory Controller must merge, correct, or choose a unique identifier.

UC-001-EX-02 — Required Product Attributes Missing

Field	Detail
Exception ID	UC-001-EX-02
Trigger	Inventory Controller submits product record without mandatory attributes (SKU, name, unit of measure, or category).
Failed Step	4
Expected System Behavior	System rejects submission, highlights missing required fields, and retains entered data for correction.
User Notification	"Required fields missing: [field list]. Complete all mandatory product attributes before saving."
Business Outcome	Incomplete product records are not saved; catalog quality standards are enforced.

UC-001-EX-03 — Discontinued Product Reactivation Without Authorization

Field	Detail
Exception ID	UC-001-EX-03
Trigger	Inventory Controller attempts to reactivate a discontinued product to active status without Operations Director authorization.
Failed Step	4
Expected System Behavior	System blocks status change and routes reactivation request for management approval per governance policy.
User Notification	"Discontinued product reactivation requires Operations Director approval. Request submitted for authorization."
Business Outcome	Discontinued product status is not changed without authorized review; obsolete inventory risk is controlled.

UC-002 — Configure Inventory Locations

UC-002-EX-01 — Invalid Location Hierarchy

Field	Detail
Exception ID	UC-002-EX-01
Trigger	Inventory Controller defines a location hierarchy that creates circular parent-child reference or assigns invalid parent location type.

Field	Detail
Failed Step	3
Expected System Behavior	System rejects hierarchy assignment and displays hierarchy validation failure detail.
User Notification	"Location hierarchy is invalid: [reason]. Correct parent assignment before saving."
Business Outcome	Location network structure remains valid; stock assignment and reporting hierarchy are not corrupted.

UC-002-EX-02 — Location Activation with Unverified Opening Balances

Field	Detail
Exception ID	UC-002-EX-02
Trigger	Inventory Controller attempts to activate a new location for operations without verified opening stock balances where go-live policy requires them.
Failed Step	5
Expected System Behavior	System blocks location activation and lists products missing opening balance entry or verification sign-off.
User Notification	"Location cannot be activated. Opening stock balances are unverified for [N] products. Complete verification before go-live."
Business Outcome	Location does not go live with embedded inventory errors; cutover integrity is protected.

UC-002-EX-03 — Location Closure with Remaining On-Hand Stock

Field	Detail
Exception ID	UC-002-EX-03
Trigger	Inventory Controller attempts to deactivate or close a location that still holds on-hand inventory without a completed disposition plan.
Failed Step	5
Expected System Behavior	System blocks deactivation and displays remaining on-hand stock by product requiring disposition.
User Notification	"Location closure blocked. [N] products with remaining on-hand stock require disposition (transfer, sell-down, or write-off) before deactivation."
Business Outcome	No location is closed with stranded inventory; stock must be dispositioned per closure policy.

UC-003 — Define Stock Policies and Thresholds

UC-003-EX-01 — Reorder Level Below Minimum Threshold

Field	Detail
Exception ID	UC-003-EX-01
Trigger	Inventory Controller sets reorder level lower than minimum stock threshold or safety stock buffer, violating stock policy consistency rules.
Failed Step	2
Expected System Behavior	System rejects threshold configuration and displays policy consistency violation.
User Notification	"Reorder level cannot be below minimum threshold or safety stock. Adjust values to maintain policy consistency."
Business Outcome	Stock policy remains logically consistent; replenishment alerts will trigger before minimum breach.

UC-003-EX-02 — Safety Stock Override Without Authorization

Field	Detail
Exception ID	UC-003-EX-02
Trigger	Inventory Controller attempts to reduce or remove safety stock below governance minimum without authorized exception.
Failed Step	3
Expected System Behavior	System blocks safety stock reduction and routes override request to Operations Director per BR-010.
User Notification	"Safety stock reduction requires authorized exception. Override request submitted to Operations Director."
Business Outcome	Safety stock protection is maintained; unauthorized buffer removal is prevented.

UC-004 — Classify Products by ABC Priority

UC-004-EX-01 — Classification Save Without Required Review Period

Field	Detail
Exception ID	UC-004-EX-01
Trigger	Inventory Controller attempts bulk reclassification outside the approved periodic review window without Operations Director override.
Failed Step	4
Expected System Behavior	System blocks bulk classification change and requires review period confirmation or management override.
User Notification	"Bulk ABC reclassification outside review cycle requires Operations Director approval. Submit override request to proceed."
Business Outcome	Classification changes follow governance cadence; ad-hoc bulk changes require explicit authorization.

UC-004-EX-02 — Inactive Product Classification Attempt

Field	Detail
Exception ID	UC-004-EX-02
Trigger	Inventory Controller assigns ABC classification to an inactive or discontinued product.
Failed Step	3
Expected System Behavior	System blocks classification assignment for non-active products.
User Notification	"ABC classification cannot be assigned to inactive or discontinued products. Activate product or exclude from classification."
Business Outcome	Operational monitoring effort is not applied to products outside active assortment.

UC-005 — Maintain Governance Policies and Reason Codes

UC-005-EX-01 — Deactivate Reason Code in Active Use

Field	Detail
Exception ID	UC-005-EX-01
Trigger	Inventory Controller attempts to deactivate a reason code that is mandated by an active governance policy or referenced in pending transactions.

Field	Detail
Failed Step	4
Expected System Behavior	System blocks deactivation and displays dependent policies or pending transaction references.
User Notification	"Reason code cannot be deactivated. Active policy [policy name] or [N] pending transactions reference this code. Reassign dependencies first."
Business Outcome	Reason code catalog integrity is maintained; in-flight transactions retain valid reason code references.

UC-005-EX-02 — Approval Threshold Policy Conflict

Field	Detail
Exception ID	UC-005-EX-02
Trigger	Inventory Controller defines governance policy with approval thresholds that conflict with segregation of duties rules (e.g., same role as initiator and approver).
Failed Step	3
Expected System Behavior	System rejects policy configuration and identifies SoD conflict per FR-084.
User Notification	"Governance policy conflicts with segregation of duties rule [SOD-XXX]. Initiator and approver roles cannot be the same."
Business Outcome	Governance policies comply with control framework; conflicting approval rules are not activated.

Domain 2: Inbound & Receiving

UC-006 — Place and Manage Purchase Orders

UC-006-EX-01 — Purchase Order for Discontinued Product

Field	Detail
Exception ID	UC-006-EX-01
Trigger	Buyer adds a discontinued product line to a purchase order.
Failed Step	3
Expected System Behavior	System blocks line entry and prevents purchase order submission for discontinued SKU per BR-027.
User Notification	"Product [SKU/name] is discontinued. New purchase orders for this SKU are not permitted."
Business Outcome	No new procurement is created for discontinued products; obsolete inventory risk is avoided.

UC-006-EX-02 — Purchase Order Submitted Without Required Approval

Field	Detail
Exception ID	UC-006-EX-02
Trigger	Buyer attempts to mark purchase order as confirmed or send to supplier before Purchasing Manager approval is recorded.
Failed Step	5
Expected System Behavior	System maintains purchase order in pending approval status and blocks supplier commitment until approved.
User Notification	"Purchase order requires Purchasing Manager approval before confirmation. Status remains Pending Approval."
Business Outcome	Unauthorized commercial commitments are prevented; procurement controls are enforced.

UC-006-EX-03 — Cancel Purchase Order with Received Goods

Field	Detail
Exception ID	UC-006-EX-03
Trigger	Buyer attempts to cancel a purchase order that has partial or full goods receipt recorded.
Failed Step	4
Expected System Behavior	System blocks full cancellation and displays received quantities requiring receipt reversal or closure processing.
User Notification	"Purchase order cannot be cancelled. [Quantity] already received. Process receipt reversal or close remaining open lines separately."
Business Outcome	Received inventory and financial records remain consistent; cancellation does not orphan receipt data.

UC-007 — Plan Expected Inbound Deliveries

UC-007-EX-01 — Delivery Plan for Unapproved Purchase Order

Field	Detail
Exception ID	UC-007-EX-01
Trigger	Buyer attempts to schedule inbound delivery for a purchase order not yet approved.
Failed Step	2
Expected System Behavior	System blocks delivery planning and indicates purchase order approval is prerequisite.
User Notification	"Inbound delivery cannot be planned. Purchase order [PO number] is not approved. Complete approval before scheduling."
Business Outcome	Receiving locations are not notified of unconfirmed supply; inbound visibility reflects authorized orders only.

UC-007-EX-02 — Delivery Date Precedes Current Date

Field	Detail
Exception ID	UC-007-EX-02
Trigger	Buyer enters expected delivery date in the past without documented receipt or correction context.
Failed Step	2
Expected System Behavior	System rejects past-date entry for future delivery planning and prompts correction or receipt action instead.
User Notification	"Expected delivery date cannot be in the past for open orders. Update to future date or process goods receipt if already delivered."
Business Outcome	Inbound schedule data remains accurate for replenishment and receiving planning.

UC-008 — Receive and Verify Supplier Goods

UC-008-EX-01 — Receipt Without Valid Purchase Order Reference

Field	Detail
Exception ID	UC-008-EX-01
Trigger	Receiving Clerk attempts to receive supplier goods without matching purchase order reference and without approved ad-hoc receipt authorization per BR-007.
Failed Step	2

Field	Detail
Expected System Behavior	System blocks receipt processing and requires valid PO reference or ad-hoc authorization before proceeding.
User Notification	"Supplier receipt requires valid purchase order reference or approved ad-hoc receipt authorization. Cannot accept goods without authorization."
Business Outcome	Unauthorized inbound supply is rejected; financial and procurement controls are preserved.

UC-008-EX-02 — Minimum Remaining Shelf Life Not Met

Field	Detail
Exception ID	UC-008-EX-02
Trigger	Receiving Clerk verifies shelf-life product with remaining shelf life below minimum policy threshold at receipt per BR-016.
Failed Step	4
Expected System Behavior	System blocks acceptance into sellable inventory and routes to discrepancy disposition workflow.
User Notification	"Product [SKU] does not meet minimum remaining shelf life requirement. Receipt blocked. Document discrepancy for supplier resolution."
Business Outcome	Short-dated product is not accepted into sellable stock; supplier accountability is preserved.

UC-008-EX-03 — Lot or Batch Not Captured for Traceable Product

Field	Detail
Exception ID	UC-008-EX-03
Trigger	Receiving Clerk attempts to confirm receipt for a traceable product without entering required lot or batch identity per BR-017.
Failed Step	4
Expected System Behavior	System blocks receipt confirmation until lot or batch is recorded.
User Notification	"Lot/batch number is required for traceable product [SKU]. Enter lot identity before completing receipt."
Business Outcome	Traceability data is captured at point of entry; recall and quality investigation capability is maintained.

UC-009 — Document and Resolve Receipt Discrepancies

UC-009-EX-01 — Discrepancy Accepted Without Documentation

Field	Detail
Exception ID	UC-009-EX-01
Trigger	Receiving Clerk attempts to complete receipt with quantity variance but without mandatory discrepancy documentation per BR-006.
Failed Step	2
Expected System Behavior	System blocks receipt completion and requires discrepancy type, variance quantity, and reason code.
User Notification	"Receipt discrepancy must be documented before acceptance. Enter discrepancy type, variance, and reason code."
Business Outcome	Supplier accountability evidence is captured at receipt; undocumented variances are prevented.

UC-009-EX-02 — Discrepancy Resolution Exceeds Approval Authority

Field	Detail
Exception ID	UC-009-EX-02
Trigger	Warehouse Manager attempts to approve receipt discrepancy resolution exceeding defined tolerance or value threshold without independent approver per SOD-007.
Failed Step	4
Expected System Behavior	System routes discrepancy resolution to authorized approver and blocks completion pending escalation.
User Notification	"Discrepancy resolution exceeds your approval authority. Escalated to [approver role] for independent approval."
Business Outcome	Material receipt variances receive proportional oversight; receiving clerk cannot self-approve beyond-tolerance discrepancies.

UC-010 — Perform Stock Put-Away

UC-010-EX-01 — Put-Away Before Receipt Verification Complete

Field	Detail
Exception ID	UC-010-EX-01
Trigger	Warehouse Staff attempts put-away for stock that has not completed goods receipt verification per BR-004.
Failed Step	1
Expected System Behavior	System excludes unverified receipts from put-away queue and displays verification prerequisite status.
User Notification	"Put-away not available. Goods receipt [reference] requires verification completion before put-away."
Business Outcome	Unverified stock does not enter storage or availability; inventory accuracy is protected at point of entry.

UC-010-EX-02 — Put-Away to Unauthorized Storage Location

Field	Detail
Exception ID	UC-010-EX-02
Trigger	Warehouse Staff confirms put-away to a storage location not authorized for the product category or location type.
Failed Step	3
Expected System Behavior	System rejects location assignment and presents list of authorized storage locations.
User Notification	"Storage location [location] is not authorized for product category [category]. Select an authorized location."
Business Outcome	Stock is stored only in appropriate locations; retrieval efficiency and product handling rules are maintained.

UC-011 — Receive Return and Non-PO Inbound Stock

UC-011-EX-01 — Return Received Without Authorization

Field	Detail
Exception ID	UC-011-EX-01
Trigger	Receiving Clerk attempts to receive customer return without valid return authorization reference per BR-025.
Failed Step	2
Expected System Behavior	System blocks return receipt and requires valid return authorization before acceptance.
User Notification	"Return receipt blocked. Valid return authorization is required. Contact Customer Service to issue authorization."

Field	Detail
Business Outcome	Unauthorized returns are not accepted; warehouse has no unlinked inbound goods.

UC-011-EX-02 — Ad-Hoc Receipt Without Pre-Approval

Field	Detail
Exception ID	UC-011-EX-02
Trigger	Receiving Clerk attempts ad-hoc non-PO receipt without valid pre-approved authorization reference.
Failed Step	2
Expected System Behavior	System blocks ad-hoc receipt and requires authorization reference from authorized approver.
User Notification	"Ad-hoc receipt requires pre-approved authorization reference. Obtain approval before receiving goods."
Business Outcome	Non-standard inbound is controlled; unauthorized ad-hoc supply entry is prevented.

Domain 3: Inventory Visibility & Status

UC-012 — Monitor Inventory Status

UC-012-EX-01 — Insufficient Access to Location Data

Field	Detail
Exception ID	UC-012-EX-01
Trigger	Inventory Controller attempts to view inventory status for a location outside assigned access scope per FR-082.
Failed Step	2
Expected System Behavior	System restricts view to authorized locations only and does not display restricted location data.
User Notification	"Access denied for location [location name]. You do not have visibility permission for this site."
Business Outcome	Location-scoped access is enforced; unauthorized network visibility is prevented.

UC-013 — Inquire Inventory Availability

UC-013-EX-01 — Product Not Found

Field	Detail
Exception ID	UC-013-EX-01
Trigger	Sales Representative searches for a product SKU, name, or barcode that does not exist in the product master.
Failed Step	2
Expected System Behavior	System returns no-match result and suggests similar products if available.
User Notification	"No product found matching '[search term]'. Verify SKU, name, or barcode. Similar products: [list if any]."
Business Outcome	Sales representative is informed that product is not in catalog; customer cannot be promised unavailable SKU.

UC-013-EX-02 — Inactive Product Availability Inquiry

Field	Detail
Exception ID	UC-013-EX-02
Trigger	Sales Representative requests availability for an inactive or discontinued product.
Failed Step	3

Field	Detail
Expected System Behavior	System displays product status as inactive or discontinued and shows remaining ATP if any with sell restriction indicator.
User Notification	"Product [SKU] is [inactive/discontinued]. Not available for new sales commitment. Remaining ATP: [quantity] (sell-through only)."
Business Outcome	Sales team cannot commit new orders against discontinued assortment without explicit visibility of status.

Domain 4: Inventory Control & Accuracy

UC-014 — Perform Cycle Count

UC-014-EX-01 — Count Entry for Unassigned Product-Location

Field	Detail
Exception ID	UC-014-EX-01
Trigger	Warehouse Staff attempts to enter cycle count for a product-location not on the assigned count schedule.
Failed Step	2
Expected System Behavior	System blocks unscheduled count entry unless ad-hoc count authorization is granted by Warehouse Manager.
User Notification	"Product-location not on your assigned count schedule. Request ad-hoc count authorization from Warehouse Manager."
Business Outcome	Count program integrity is maintained; uncontrolled counts do not disrupt scheduled accuracy program.

UC-014-EX-02 — Count Submitted with Zero Quantity for Active SKU

Field	Detail
Exception ID	UC-014-EX-02
Trigger	Warehouse Staff submits zero count for a product-location with recent movement history or open reservations.
Failed Step	4
Expected System Behavior	System flags zero-count confirmation prompt and requires explicit confirmation before accepting.
User Notification	"Zero count submitted for active SKU with recent activity. Confirm zero count is correct or re-count before submitting."
Business Outcome	Potentially erroneous zero counts receive additional verification before variance processing.

UC-015 — Conduct Full Physical Inventory Count

UC-015-EX-01 — Movement Attempted During Inventory Freeze

Field	Detail
Exception ID	UC-015-EX-01
Trigger	Any user attempts receiving, shipping, transfer, or adjustment at a location under physical inventory freeze per BR-024.
Failed Step	3
Expected System Behavior	System blocks inventory movement transactions for frozen location and displays freeze status.
User Notification	"Location [location] is under physical inventory freeze. [Transaction type] is suspended until count is completed."
Business Outcome	Count snapshot integrity is preserved; movements do not invalidate wall-to-wall count.

UC-015-EX-02 — Count Close with Uncounted Zones

Field	Detail
Exception ID	UC-015-EX-02
Trigger	Inventory Controller attempts to close full physical count session with uncounted zones or products remaining.
Failed Step	5
Expected System Behavior	System blocks count close and displays list of uncounted assignments.
User Notification	"Physical count cannot be closed. [N] zones/products remain uncounted. Complete all assignments or document approved exclusions."
Business Outcome	Incomplete count is not finalized; accuracy baseline is not established with gaps.

UC-016 — Investigate and Approve Count Variances

UC-016-EX-01 — Counter Attempts to Approve Own Count Variance

Field	Detail
Exception ID	UC-016-EX-01
Trigger	Warehouse Staff who performed the count attempts to approve the resulting variance adjustment per BR-019 and SOD-003.
Failed Step	4
Expected System Behavior	System blocks self-approval and routes variance to independent approver (Inventory Controller or Warehouse Manager if not counter).
User Notification	"You cannot approve adjustments from your own count. Variance routed to [approver role] for independent review."
Business Outcome	Segregation of duties is enforced; count accuracy oversight remains independent.

UC-016-EX-02 — Variance Approved Without Investigation Notes

Field	Detail
Exception ID	UC-016-EX-02
Trigger	Warehouse Manager attempts to approve count variance without documented investigation per BR-022.
Failed Step	3
Expected System Behavior	System blocks approval until mandatory investigation notes are entered.
User Notification	"Investigation notes are required before variance approval. Document root cause analysis or investigation effort."
Business Outcome	Adjustments are not applied without investigation; recurring variance root causes can be identified.

UC-017 — Process Stock Adjustments

UC-017-EX-01 — Adjustment Without Mandatory Reason Code

Field	Detail
Exception ID	UC-017-EX-01
Trigger	Warehouse Manager submits stock adjustment without selecting a reason code per BR-020.
Failed Step	3
Expected System Behavior	System blocks adjustment submission until reason code is selected.

Field	Detail
User Notification	"Reason code is mandatory for all adjustments. Select a reason code before submitting."
Business Outcome	All adjustments are analytically categorized; shrinkage and loss analysis integrity is maintained.

UC-017-EX-02 — Adjustment Exceeds Approval Threshold

Field	Detail
Exception ID	UC-017-EX-02
Trigger	Warehouse Manager submits adjustment exceeding defined quantity or value threshold without independent approver per BR-021.
Failed Step	4
Expected System Behavior	System routes adjustment to Operations Director or Inventory Controller and blocks application pending approval.
User Notification	"Adjustment exceeds approval threshold. Submitted to [approver role] for independent authorization."
Business Outcome	Material adjustments receive proportional oversight; high-impact changes are not self-approved.

UC-017-EX-03 — Adjustment Would Breach Safety Stock Floor

Field	Detail
Exception ID	UC-017-EX-03
Trigger	Decrease adjustment would reduce on-hand below safety stock buffer without authorized exception per BR-010.
Failed Step	2
Expected System Behavior	System warns of safety stock breach and requires authorized override or reduced adjustment quantity.
User Notification	"Adjustment would reduce stock below safety stock level. Reduce adjustment quantity or obtain authorized safety stock override."
Business Outcome	Safety stock protection is maintained; unauthorized buffer consumption is prevented.

UC-018 — Authorize and Process Inventory Write-Offs

UC-018-EX-01 — Write-Off Without Supporting Documentation

Field	Detail
Exception ID	UC-018-EX-01
Trigger	Operations Director attempts to authorize write-off without attached evidence or documentation per BR-023.
Failed Step	3
Expected System Behavior	System blocks write-off authorization until supporting documentation is attached.
User Notification	"Write-off authorization requires supporting documentation. Attach evidence before approving."
Business Outcome	Write-offs are auditable; undocumented value removal is prevented.

UC-018-EX-02 — Write-Off Exceeds Authorized Quantity

Field	Detail
Exception ID	UC-018-EX-02
Trigger	Write-off request quantity exceeds verified on-hand or quarantined quantity at location.
Failed Step	5

Field	Detail
Expected System Behavior	System blocks write-off execution and displays available quantity for write-off.
User Notification	"Write-off quantity [requested] exceeds available on-hand [available] at location [location]. Adjust write-off quantity."
Business Outcome	Inventory cannot be written off below zero; record accuracy is maintained.

UC-019 — Manage Quarantine and Hold Stock

UC-019-EX-01 — Quarantined Stock Included in Reservation Attempt

Field	Detail
Exception ID	UC-019-EX-01
Trigger	System or user workflow attempts to reserve quarantined stock for customer order per BR-013.
Failed Step	4
Expected System Behavior	System excludes quarantined quantity from reservation eligibility and displays quarantine hold reason.
User Notification	"Quantity [N] is quarantined and not available for reservation. ATP excludes quarantined stock."
Business Outcome	Defective or held stock is not committed to customers; sellable availability remains accurate.

UC-019-EX-02 — Bonded Stock Released Without Customs Clearance

Field	Detail
Exception ID	UC-019-EX-02
Trigger	Warehouse Manager attempts to release bonded inventory to sellable status without documented customs clearance per BR-030.
Failed Step	1
Expected System Behavior	System blocks release and maintains bonded hold status until clearance reference is recorded.
User Notification	"Bonded inventory cannot be released without customs clearance. Record clearance reference before release."
Business Outcome	Regulatory compliance is maintained; bonded goods are not released into domestic commerce prematurely.

UC-020 — Analyze Shrinkage, Loss, and Obsolete Stock

UC-020-EX-01 — Analysis Period Exceeds Data Retention for Reason Codes

Field	Detail
Exception ID	UC-020-EX-01
Trigger	Inventory Controller requests shrinkage analysis for a period where historical reason codes were not consistently recorded (pre-go-live or legacy data gap).
Failed Step	2
Expected System Behavior	System displays analysis with data completeness warning and excludes periods below minimum reason code compliance threshold.
User Notification	"Shrinkage analysis for [period] has incomplete reason code data ([X]% compliance). Results may not be reliable. Adjust period or scope."
Business Outcome	User is informed of data quality limitations; decisions are not made on unreliable historical analysis.

Domain 5: Order Fulfillment

UC-021 — Confirm Order Availability

UC-021-EX-01 — Insufficient ATP for Order Line

Field	Detail
Exception ID	UC-021-EX-01
Trigger	Available-to-promise quantity is less than ordered quantity at all fulfillment locations per BR-001.
Failed Step	2
Expected System Behavior	System displays ATP shortfall by line and blocks full availability confirmation.
User Notification	"Insufficient ATP for line [SKU]: ordered [N], available [M]. Confirm partial availability or revise order quantity."
Business Outcome	Customer cannot be promised unavailable quantity; overselling is prevented at confirmation stage.

UC-021-EX-02 — Promotional Stock Ring-Fence Blocks General Order

Field	Detail
Exception ID	UC-021-EX-02
Trigger	Order line draws from promotional ring-fenced stock reserved for campaign without commercial leadership approval per BR-029.
Failed Step	2
Expected System Behavior	System excludes ring-fenced promotional quantity from ATP available for non-promotional orders.
User Notification	"Available ATP excludes [N] units ring-fenced for campaign [campaign name]. Marketing Manager approval required to consume promotional stock."
Business Outcome	Campaign inventory is protected; promotional commitments are not consumed by general orders.

UC-022 — Reserve Inventory for Confirmed Orders

UC-022-EX-01 — ATP Changed Since Availability Confirmation

Field	Detail
Exception ID	UC-022-EX-01
Trigger	ATP at fulfillment location has decreased since availability confirmation due to concurrent reservation or adjustment.
Failed Step	2
Expected System Behavior	System blocks full reservation and displays current ATP versus previously confirmed quantity.
User Notification	"ATP has changed since availability check. Confirmed: [N], current ATP: [M]. Re-confirm availability or reserve partial quantity."
Business Outcome	Reservation reflects current stock position; double-allocation from stale availability data is prevented.

UC-022-EX-02 — Reservation Exceeds Location ATP

Field	Detail
Exception ID	UC-022-EX-02
Trigger	Customer Service Representative attempts to reserve quantity exceeding ATP at selected fulfillment location.
Failed Step	3

Field	Detail
Expected System Behavior	System blocks over-reservation and displays maximum reservable quantity.
User Notification	"Reservation quantity [N] exceeds available ATP [M] at [location]. Reduce reservation or select alternate location."
Business Outcome	Reserved quantity never exceeds available stock; ATP integrity is maintained.

UC-023 — Release Inventory Reservations

UC-023-EX-01 — Release Attempt on Shipped or Picked Order

Field	Detail
Exception ID	UC-023-EX-01
Trigger	Customer Service Representative attempts to release reservation for order lines already picked, packed, or shipped.
Failed Step	2
Expected System Behavior	System blocks reservation release for fulfilled lines and displays current fulfillment status.
User Notification	"Reservation cannot be released. Order line [SKU] is [picked/shipped]. Process return or cancellation workflow instead."
Business Outcome	Physical fulfillment state and reservation state remain consistent; improper ATP restoration is prevented.

UC-024 — Pick, Pack, and Ship Orders

UC-024-EX-01 — Shipment Confirmed Without Pick Verification

Field	Detail
Exception ID	UC-024-EX-01
Trigger	Warehouse Picker / Packer attempts to confirm shipment before all order lines are pick-verified per BR-003.
Failed Step	5
Expected System Behavior	System blocks shipment confirmation and lists unverified pick lines.
User Notification	"Shipment cannot be confirmed. [N] lines require pick verification. Complete verification before shipping."
Business Outcome	Inventory is not deducted and orders are not marked shipped until picks are verified.

UC-024-EX-02 — Pick Quantity Exceeds Reserved Quantity

Field	Detail
Exception ID	UC-024-EX-02
Trigger	Warehouse Picker / Packer attempts to pick quantity greater than reserved quantity for order line.
Failed Step	3
Expected System Behavior	System blocks over-pick and displays reserved quantity limit.
User Notification	"Pick quantity [N] exceeds reserved quantity [M] for order line [SKU]. Adjust pick to reserved amount."
Business Outcome	Fulfillment does not exceed committed reservation; unreserved stock is not shipped.

UC-024-EX-03 — FEFO/FIFO Rotation Violation

Field	Detail
Exception ID	UC-024-EX-03
Trigger	Warehouse Picker / Packer scans lot or batch that violates FEFO or FIFO rotation rules per BR-014 and BR-015.

Field	Detail
Failed Step	3
Expected System Behavior	System warns of rotation violation and directs picker to correct lot or batch with earlier expiry or receipt date.
User Notification	"Rotation violation: earlier [expiry/receipt date] stock exists at [location]. Pick lot [correct lot] first."
Business Outcome	Product rotation policy is enforced; expired or aged stock write-off risk is reduced.

UC-025 — Handle Partial Fulfillment and Backorders

UC-025-EX-01 — Backorder Recorded Without Customer Acknowledgment

Field	Detail
Exception ID	UC-025-EX-01
Trigger	Customer Service Representative attempts to create backorder without recorded customer acknowledgment per BR-028.
Failed Step	3
Expected System Behavior	System blocks backorder creation until customer acknowledgment is captured.
User Notification	"Backorder requires customer acknowledgment. Record customer acceptance of delay terms before proceeding."
Business Outcome	Backorders are mutual agreements; silent backorders that damage customer trust are prevented.

UC-025-EX-02 — Zero ATP for Immediate Partial Fulfillment

Field	Detail
Exception ID	UC-025-EX-02
Trigger	Customer accepts partial fulfillment but zero ATP is available for any immediate shipment quantity.
Failed Step	4
Expected System Behavior	System blocks partial fulfillment reservation and offers full backorder or order hold options only.
User Notification	"No ATP available for immediate partial fulfillment. Create full backorder or hold order until supply arrives."
Business Outcome	Customer is not promised immediate shipment when no stock exists; commitment aligns with availability.

UC-026 — Handle Outbound Fulfillment Exceptions

UC-026-EX-01 — Unauthorized Product Substitution Attempted

Field	Detail
Exception ID	UC-026-EX-01
Trigger	Warehouse Manager attempts substitution with a product not on the authorized substitute list per FR-050.
Failed Step	3
Expected System Behavior	System blocks unauthorized substitution and displays approved substitute products if any.
User Notification	"Product [SKU] is not an authorized substitute for [original SKU]. Select approved substitute or escalate to Customer Service."
Business Outcome	Customers receive only approved equivalent products; unauthorized substitutions are prevented.

UC-027 — View Order Fulfillment Status

UC-027-EX-01 — Order Not Found

Field	Detail
Exception ID	UC-027-EX-01
Trigger	Customer Service Representative searches for order number or customer reference that does not exist in the system.
Failed Step	2
Expected System Behavior	System returns no-match result with search criteria confirmation.
User Notification	"No order found for '[search term]'. Verify order number or customer reference."
Business Outcome	User is informed no matching order exists; incorrect status information is not presented.

Domain 6: Replenishment & Inbound Supply

UC-028 — Review Low-Stock and Out-of-Stock Alerts

UC-028-EX-01 — Alert for Product with Open Purchase Order

Field	Detail
Exception ID	UC-028-EX-01
Trigger	Purchasing Manager reviews low-stock alert for product with sufficient inbound supply on open approved purchase order arriving within lead time.
Failed Step	3
Expected System Behavior	System displays alert with inbound supply context and suggests alert acknowledgment with inbound coverage note.
User Notification	"Low-stock alert: [SKU]. Open PO [number] with [quantity] expected [date] may resolve condition. Review before initiating new procurement."
Business Outcome	Duplicate procurement is avoided; alert response considers existing inbound supply.

UC-029 — Plan Replenishment and Create Purchase Requisitions

UC-029-EX-01 — Requisition for Discontinued Product

Field	Detail
Exception ID	UC-029-EX-01
Trigger	Buyer attempts to create purchase requisition for discontinued product per BR-027.
Failed Step	3
Expected System Behavior	System blocks requisition creation for discontinued SKU.
User Notification	"Purchase requisition blocked. Product [SKU] is discontinued and cannot be reordered."
Business Outcome	Replenishment does not recreate obsolete inventory.

UC-029-EX-02 — Network Stock Available, Purchase Not Required

Field	Detail
Exception ID	UC-029-EX-02
Trigger	Buyer attempts to create purchase requisition when network check confirms excess stock at another location per BR-011.

Field	Detail
Failed Step	2
Expected System Behavior	System warns of network excess and recommends transfer before purchase; requires documented override to proceed with purchase.
User Notification	"Network check: [N] units excess at [source location]. Transfer recommended before new purchase. Override required to proceed with requisition."
Business Outcome	Transfer-before-buy discipline is enforced; unnecessary procurement and network imbalance are prevented.

UC-030 — Track Inbound Supply Status

UC-030-EX-01 — Purchase Order Fully Received, Status Update Attempted

Field	Detail
Exception ID	UC-030-EX-01
Trigger	Buyer attempts to update delivery status on a purchase order already fully received and closed.
Failed Step	3
Expected System Behavior	System displays closed status and prevents delivery schedule modification.
User Notification	"Purchase order [number] is fully received and closed. Delivery status cannot be modified."
Business Outcome	Historical inbound records are not altered after completion; data integrity is maintained.

Domain 7: Network Balance & Transfers

UC-031 — Request Inter-Location Stock Transfer

UC-031-EX-01 — Transfer Would Breach Source Minimum Threshold

Field	Detail
Exception ID	UC-031-EX-01
Trigger	Requested transfer quantity would reduce source location below its minimum threshold per BR-012.
Failed Step	2
Expected System Behavior	System blocks transfer request submission and displays maximum transferable quantity.
User Notification	"Transfer blocked. Requested quantity would breach source minimum threshold at [location]. Maximum transferable: [N] units."
Business Outcome	Source location service level is protected; transfer does not create shortage at sending site.

UC-031-EX-02 — Insufficient ATP at Source Location

Field	Detail
Exception ID	UC-031-EX-02
Trigger	Requested transfer quantity exceeds available ATP at source location (stock reserved or quarantined).
Failed Step	2
Expected System Behavior	System blocks transfer request and displays ATP breakdown (on-hand, reserved, quarantined, available).
User Notification	"Insufficient ATP at source [location]. Requested: [N], available: [M]. Reduce quantity or select alternate source."
Business Outcome	Transfer request reflects actually available stock; committed or held stock is not double-allocated.

UC-032 — Approve or Reject Transfer Requests

UC-032-EX-01 — Approver Is Transfer Requester

Field	Detail
Exception ID	UC-032-EX-01
Trigger	Operations Director who submitted the transfer request attempts to approve the same request per SOD-006.
Failed Step	3
Expected System Behavior	System blocks self-approval and routes to alternate authorized approver (COO or designated delegate).
User Notification	"You cannot approve your own transfer request. Routed to [alternate approver] per segregation of duties."
Business Outcome	Transfer authorization remains independent; self-approved transfers are prevented.

UC-033 — Execute and Record Inter-Location Transfer

UC-033-EX-01 — Dispatch Exceeds Approved Transfer Quantity

Field	Detail
Exception ID	UC-033-EX-01
Trigger	Warehouse Staff attempts to dispatch quantity greater than approved transfer quantity.
Failed Step	3
Expected System Behavior	System blocks over-dispatch and displays approved transfer quantity.
User Notification	"Dispatch quantity [N] exceeds approved transfer [M]. Adjust dispatch to approved amount."
Business Outcome	Physical movement matches authorized transfer; unauthorized stock movement is prevented.

UC-033-EX-02 — Receipt Quantity Exceeds Dispatched Quantity

Field	Detail
Exception ID	UC-033-EX-02
Trigger	Destination Warehouse Staff attempts to receive quantity greater than dispatched in-transit quantity.
Failed Step	4
Expected System Behavior	System blocks over-receipt and displays expected in-transit quantity.
User Notification	"Receipt quantity [N] exceeds in-transit quantity [M]. Confirm dispatch record or document variance."
Business Outcome	Transfer receipt matches dispatch; inventory inflation at destination is prevented.

UC-034 — Analyze and Recommend Network Rebalancing

UC-034-EX-01 — No Actionable Imbalance Identified

Field	Detail
Exception ID	UC-034-EX-01
Trigger	Operations Director runs network rebalancing analysis but no location pairs meet minimum imbalance threshold for recommendation.
Failed Step	2
Expected System Behavior	System displays analysis complete with no recommendations and current network balance summary.

Field	Detail
User Notification	"Network rebalancing analysis complete. No transfer recommendations — all locations within balance tolerance."
Business Outcome	Operations confirms no rebalancing action is needed; unnecessary transfers are avoided.

Domain 8: Returns Management

UC-035 — Authorize and Record Customer Returns

UC-035-EX-01 — Return Outside Policy Window

Field	Detail
Exception ID	UC-035-EX-01
Trigger	Customer Service Representative attempts return authorization for order line outside defined return policy time window.
Failed Step	3
Expected System Behavior	System blocks standard return authorization and routes to management override workflow if applicable.
User Notification	"Return outside policy window ([N] days). Standard authorization blocked. Submit management override request if applicable."
Business Outcome	Return policy is enforced; unauthorized returns outside policy require explicit management exception.

UC-035-EX-02 — Return Quantity Exceeds Original Order Quantity

Field	Detail
Exception ID	UC-035-EX-02
Trigger	Customer Service Representative requests return quantity greater than originally ordered or previously returned quantity.
Failed Step	2
Expected System Behavior	System blocks excess return authorization and displays maximum returnable quantity.
User Notification	"Return quantity [N] exceeds returnable amount [M] for order line [SKU]. Adjust to eligible quantity."
Business Outcome	Return authorization does not exceed commercial transaction scope; fraudulent or erroneous returns are limited.

UC-036 — Receive, Inspect, and Disposition Returned Goods

UC-036-EX-01 — Return Quantity Exceeds Authorization

Field	Detail
Exception ID	UC-036-EX-01
Trigger	Receiving Clerk attempts to receive return quantity exceeding return authorization per line.
Failed Step	2
Expected System Behavior	System blocks excess receipt and displays authorized return quantity.
User Notification	"Received quantity [N] exceeds authorized return [M]. Accept only authorized quantity or obtain authorization amendment."
Business Outcome	Physical return receipt matches authorization; unauthorized excess goods are not accepted.

UC-036-EX-02 — Restock Attempt on Failed Inspection

Field	Detail
Exception ID	UC-036-EX-02
Trigger	Receiving Clerk attempts to disposition returned goods as restock when inspection outcome is damaged or defective per BR-026.
Failed Step	4
Expected System Behavior	System blocks restock disposition and requires quarantine or write-off disposition for failed inspection.
User Notification	"Restock not permitted. Inspection outcome: [damaged/defective]. Select quarantine or write-off disposition."
Business Outcome	Non-resalable returns do not re-enter sellable inventory; customer quality standards are protected.

Domain 9: Reporting, Performance & Financial Alignment

UC-037 — Review Inventory Dashboards and Operational Reports

UC-037-EX-01 — Report Generation Exceeds Authorized Data Scope

Field	Detail
Exception ID	UC-037-EX-01
Trigger	Operations Director attempts to generate location performance report including locations outside authorized visibility scope.
Failed Step	3
Expected System Behavior	System generates report for authorized locations only and notes excluded locations.
User Notification	"Report generated for authorized locations only. [N] locations excluded due to access restrictions."
Business Outcome	Reporting respects location-scoped access; unauthorized data is not disclosed in reports.

UC-038 — Review Inventory Performance KPIs

UC-038-EX-01 — KPI Data Incomplete for Selected Period

Field	Detail
Exception ID	UC-038-EX-01
Trigger	Chief Executive Officer selects KPI period where baseline data or location rollout was incomplete (e.g., pre-go-live or partial network adoption).
Failed Step	2
Expected System Behavior	System displays KPI values with data completeness indicator and flags metrics with insufficient coverage.
User Notification	"KPI data for [period] is incomplete ([X]% network coverage). Metrics marked with data quality warning should be interpreted with caution."
Business Outcome	Executive is not misled by KPIs based on incomplete data; governance decisions account for data maturity.

UC-039 — Reconcile Inventory Valuation with Finance

UC-039-EX-01 — Valuation Variance Exceeds Tolerance

Field	Detail
Exception ID	UC-039-EX-01
Trigger	Operational inventory valuation differs from general ledger balance by more than defined reconciliation tolerance (target ≤2% per BO-18).
Failed Step	3
Expected System Behavior	System flags reconciliation as out of tolerance and requires documented variance investigation before close.
User Notification	"Reconciliation variance [X]% exceeds tolerance [Y]%. Investigation required before period close. Drill down to contributing transactions."
Business Outcome	Material valuation variances are investigated; month-end close does not proceed without resolution or documented exception.

UC-039-EX-02 — Consignment Stock Included in Owned Valuation

Field	Detail
Exception ID	UC-039-EX-02
Trigger	Finance Manager reconciliation includes consignment inventory in owned valuation calculation per BR-031.
Failed Step	3
Expected System Behavior	System separates consignment stock from owned valuation and displays consignment quantity excluded from GL comparison.
User Notification	"Consignment stock [N] units ([value]) excluded from owned valuation per ownership policy. Adjust reconciliation scope."
Business Outcome	Financial reconciliation reflects correct ownership; consignment stock does not inflate owned asset valuation.

Domain 10: Commercial Alignment & Governance

UC-040 — Confirm Promotional and Launch Stock Readiness

UC-040-EX-01 — Insufficient Ring-Fenced Stock for Campaign

Field	Detail
Exception ID	UC-040-EX-01
Trigger	Ring-fenced promotional stock plus expected inbound supply is insufficient to meet campaign requirement by start date.
Failed Step	3
Expected System Behavior	System displays readiness status as Not Ready and quantifies shortfall by product.
User Notification	"Campaign readiness: NOT READY. Shortfall of [N] units for [SKU] before [campaign date]. Escalate to Purchasing and Operations."
Business Outcome	Marketing cannot confirm launch readiness; campaign commitment is blocked until supply gap is resolved.

UC-040-EX-02 — Campaign Readiness Confirmation Without Inbound Coverage

Field	Detail
Exception ID	UC-040-EX-02

Field	Detail
Trigger	Marketing Manager attempts unconditional readiness confirmation when current ring-fenced stock is insufficient and inbound supply is not confirmed before campaign date.
Failed Step	4
Expected System Behavior	System blocks unconditional confirmation and requires conditional confirmation with documented inbound dependency or management risk acceptance.
User Notification	"Unconditional readiness blocked. Current stock insufficient. Confirm conditionally with inbound dependency or obtain executive risk acceptance."
Business Outcome	Promotional launch is not confirmed without adequate stock coverage; campaign failure risk is explicitly acknowledged if proceeding conditionally.

UC-041 — Conduct Inventory Audit and Control Review

UC-041-EX-01 — Segregation of Duties Violation Detected

Field	Detail
Exception ID	UC-041-EX-01
Trigger	Internal Auditor identifies transaction where initiator and approver are the same user, violating SoD rules.
Failed Step	3
Expected System Behavior	System flags transaction as SoD violation in audit review and records control breach per FR-088.
User Notification	"Control breach detected: [transaction type] [reference]. Initiator and approver are the same user [user]. Document finding for remediation."
Business Outcome	Control weakness is identified and documented; remediation action can be assigned before external audit.

UC-041-EX-02 — Adjustment Without Reason Code in Audit Sample

Field	Detail
Exception ID	UC-041-EX-02
Trigger	Internal Auditor samples adjustment transaction missing mandatory reason code per BR-020 (legacy or policy breach).
Failed Step	4
Expected System Behavior	System flags transaction as policy non-compliance and includes in audit findings report.
User Notification	"Policy non-compliance: Adjustment [reference] missing reason code. Include in audit findings and assign corrective action."
Business Outcome	Governance gap is documented; reason code compliance enforcement is reinforced.

Domain 11: Cross-Cutting System Interactions

UC-042 — View Inventory Movement and Transaction History

UC-042-EX-01 — No Transactions Found for Search Criteria

Field	Detail
Exception ID	UC-042-EX-01
Trigger	Inventory Controller searches movement history with criteria that match no transactions in the selected period.
Failed Step	2

Field	Detail
Expected System Behavior	System returns empty result set with search criteria summary and suggests broadening filters.
User Notification	"No movement transactions found for selected criteria. Broaden date range, location, or product filters."
Business Outcome	User is informed no matching history exists; investigation scope can be adjusted.

UC-043 — Query and Review Audit Trail

UC-043-EX-01 — Audit Query Exceeds Authorized Scope

Field	Detail
Exception ID	UC-043-EX-01
Trigger	Internal Auditor attempts audit trail query for entity types or locations outside authorized audit scope.
Failed Step	2
Expected System Behavior	System restricts query results to authorized audit scope and logs access attempt.
User Notification	"Audit query scope restricted. Results limited to authorized entities. Contact System Administrator for expanded audit access."
Business Outcome	Audit access is controlled; sensitive data beyond audit mandate is not exposed.

UC-044 — Manage User Accounts and Role Assignments

UC-044-EX-01 — Segregation of Duties Conflict in Role Assignment

Field	Detail
Exception ID	UC-044-EX-01
Trigger	System Administrator assigns role combination that creates segregation of duties conflict (e.g., Buyer and Purchasing Manager on same user).
Failed Step	3
Expected System Behavior	System blocks role assignment and displays conflicting SoD rule.
User Notification	"Role assignment blocked. Conflicts with segregation of duties rule [SOD-XXX]: [conflict description]. Assign non-conflicting roles."
Business Outcome	User accounts do not combine conflicting duties; control framework integrity is maintained.

UC-044-EX-02 — Deactivate User with Pending Approvals

Field	Detail
Exception ID	UC-044-EX-02
Trigger	System Administrator attempts to deactivate user account with pending approval items in user's queue.
Failed Step	4
Expected System Behavior	System blocks deactivation and displays pending approval count requiring reassignment.
User Notification	"User cannot be deactivated. [N] pending approvals in queue. Reassign approvals before deactivation."
Business Outcome	Approval workflow is not orphaned; pending business decisions are reassigned before access removal.

UC-045 — Enforce Role-Based and Location-Scoped Access

UC-045-EX-01 — Unauthorized Function Access Attempt

Field	Detail
Exception ID	UC-045-EX-01
Trigger	User attempts to access function or transaction type not granted by assigned role per FR-081.
Failed Step	4
Expected System Behavior	System denies access, does not display restricted function, and logs access denial per FR-083.
User Notification	"Access denied. Your role [role] does not have permission for [function]. Contact System Administrator if access is required."
Business Outcome	Least-privilege access is enforced; unauthorized business actions are prevented.

UC-045-EX-02 — Cross-Location Transaction Without Authorization

Field	Detail
Exception ID	UC-045-EX-02
Trigger	Transactional user attempts inventory transaction at location outside assigned location scope per FR-082 and NFR-SEC-006.
Failed Step	4
Expected System Behavior	System blocks transaction and displays user's authorized location scope.
User Notification	"Transaction blocked. Location [location] is outside your authorized scope. Authorized locations: [list]."
Business Outcome	Location-scoped access is enforced; cross-location transactions by unauthorized users are prevented.

UC-045-EX-03 — SoD Violation on Transaction Attempt

Field	Detail
Exception ID	UC-045-EX-03
Trigger	User attempts action that combines initiator and approver roles prohibited by segregation of duties (e.g., count and approve same variance).
Failed Step	4
Expected System Behavior	System blocks action and identifies conflicting SoD rule; routes to independent actor if applicable.
User Notification	"Action blocked per segregation of duties [SOD-XXX]. This action requires a different authorized role."
Business Outcome	Conflicting duties cannot be performed by same user; fraud and error risk controls are enforced.

UC-046 — Import Initial Product and Stock Master Data

UC-046-EX-01 — Import File Contains Duplicate SKUs

Field	Detail
Exception ID	UC-046-EX-01
Trigger	Import validation detects duplicate SKU entries within the import file or against existing catalog.
Failed Step	3

Field	Detail
Expected System Behavior	System rejects duplicate records, displays duplicate list, and processes only unique valid records if partial import is selected.
User Notification	"Import validation failed: [N] duplicate SKUs detected. [list]. Correct file and re-import or proceed with non-duplicate records only."
Business Outcome	Catalog integrity is protected at go-live; duplicate identities are not bulk-loaded.

UC-046-EX-02 — Import References Unknown Location

Field	Detail
Exception ID	UC-046-EX-02
Trigger	Stock balance import references location code not defined in location master.
Failed Step	3
Expected System Behavior	System rejects affected stock balance records and lists unknown location references.
User Notification	"Import blocked for [N] records. Unknown location codes: [list]. Configure locations before importing stock balances."
Business Outcome	Opening balances are not assigned to undefined locations; location master precedes stock data load.

UC-047 — Receive and Respond to Operational Notifications

UC-047-EX-01 — Notification Action Deadline Passed

Field	Detail
Exception ID	UC-047-EX-01
Trigger	Warehouse Manager opens notification after business-defined response deadline (e.g., Class A low-stock alert exceeding replenishment SLA).
Failed Step	1
Expected System Behavior	System displays notification as overdue, escalates to Operations Director or Purchasing Manager, and retains overdue status.
User Notification	"OVERDUE: [notification type] for [SKU/location]. Response deadline [date/time] passed. Escalated to [escalation role]."
Business Outcome	Overdue operational conditions receive management escalation; SLA breaches are visible and tracked.

UC-048 — Support Multi-Location Split Fulfillment

UC-048-EX-01 — Network ATP Insufficient for Full Order Despite Multi-Location Option

Field	Detail
Exception ID	UC-048-EX-01
Trigger	Combined ATP across all locations is still less than total order quantity after multi-location split evaluation.
Failed Step	2
Expected System Behavior	System displays network ATP shortfall and blocks full split fulfillment plan; routes to partial fulfillment or backorder workflow.
User Notification	"Network ATP insufficient for full order. Total available: [N], ordered: [M]. Proceed with partial fulfillment or backorder for shortfall."
Business Outcome	Full order commitment is not made when network cannot supply; customer commitment aligns with total available stock.

UC-048-EX-02 — Split Fulfillment Blocked by Source Minimum on Contributing Location

Field	Detail
Exception ID	UC-048-EX-02
Trigger	Proposed split allocation from a contributing location would breach that location's minimum threshold per BR-012.
Failed Step	3
Expected System Behavior	System reduces or removes allocation from affected location and recalculates split plan with source protection applied.
User Notification	"Split allocation adjusted. Location [location] contribution reduced to [N] to protect minimum threshold. Revised plan displayed."
Business Outcome	Multi-location fulfillment does not deplete contributing locations below service levels; split plan respects network balance rules.

Summary

Metric	Count
Use Cases	48
Business Exceptions	97
Average Exceptions per Use Case	2.0
Related Business Rules Referenced	BR-001 through BR-031, SOD-003, SOD-006, SOD-007

Exceptions by Category

Category	Representative Exceptions
Authorization & SoD	Self-approval blocked, role conflict, threshold escalation
Stock availability	Insufficient ATP, safety stock breach, quarantine exclusion
Procurement controls	Discontinued product, PO approval, network check before buy
Receiving & quality	MRSL failure, missing lot/batch, undocumented discrepancy
Fulfillment commitment	Oversell prevention, backorder acknowledgment, pick verification
Governance & audit	Missing reason code, SoD violation, incomplete data warnings

Traceability

Source Document	Reference
Use Case Catalog	SRS-UC-001
Use Case Main Flows	SRS-UC-002
Use Case Alternate Flows	SRS-UC-003
Business Rule Catalog	SRS-BR-001
Business Requirements Document	BRD-IMS-001

Document Approval

Role	Name	Signature	Date
Business Analyst			

Role	Name	Signature	Date
Product Owner			
Operations Director			

End of Use Case Exception Flows

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Use Case Preconditions and Postconditions

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-UC-005
Document Title	Use Case Preconditions and Postconditions — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Business Analyst
Source Documents	SRS-UC-001 (Use Case Catalog), SRS-UC-002 (Main Flows), SRS-BR-001 (Business Rule Catalog)
Date	July 5, 2026
Scope	Successful completion scenario — conditions for main flow outcome

Purpose

This document defines preconditions and postconditions for each of the 48 use cases in the Inventory Management System. Preconditions describe what must be true before a use case can execute successfully. Postconditions describe what will be true after successful completion. Business rule references use the identifiers defined in SRS-BR-001 (BR-001 through BR-032).

Condition Legend

Symbol	Meaning
[BR-xxx]	Condition directly enforced by or derived from business rule BR-xxx
(FR-xxx)	Condition supported by functional requirement (referenced when no direct business rule applies)

Domain 1: Master Data & Governance

UC-001 — Manage Product Master Data

Primary Actor: Inventory Controller

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with product master management permission (PG-01).	(FR-081)

#	Condition	Business Rule
P2	Product category and unit of measure reference data are configured in the system.	—
P3	For product updates, the target SKU exists as an active or inactive record in the product catalog.	—
P4	For new products, the proposed SKU and identifier are not already assigned to another active product.	—

Postconditions

#	Condition	Business Rule
PO1	Product master record exists with unique SKU, required attributes, and current lifecycle status.	—
PO2	Product identifier (barcode) is registered and available for operational lookup.	(FR-003)
PO3	If status is discontinued, the product is blocked from new purchase orders and replenishment requisitions.	[BR-027]
PO4	Product creation or update is recorded in the audit trail with user, timestamp, and changed attributes.	(FR-086)

UC-002 — Configure Inventory Locations

Primary Actor: Inventory Controller

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with location management permission (PG-02).	(FR-081)
P2	For new locations, parent location (if applicable) exists and is active in the location hierarchy.	—
P3	For go-live activation, opening stock balances are verified where location cutover policy requires them.	—
P4	For location closure, a stock disposition plan is defined if on-hand inventory remains.	(FR-007)

Postconditions

#	Condition	Business Rule
PO1	Location record exists with valid hierarchy, type, and operational status (active or planned closure).	—
PO2	For activated locations, stock can be assigned, received, and transacted against the location.	—
PO3	For go-live locations, opening balances are recorded as initial on-hand quantities.	(FR-010)
PO4	Location configuration change is recorded in the audit trail.	(FR-086)

UC-003 — Define Stock Policies and Thresholds

Primary Actor: Inventory Controller

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with stock policy configuration permission (PG-04, PG-13).	(FR-081)
P2	Target product or product group exists and is in active status.	—
P3	Target location scope (product-location or network default) is defined and active.	—
P4	Proposed reorder level is not below minimum threshold or safety stock buffer without authorized exception.	[BR-010]

Postconditions

#	Condition	Business Rule
PO1	Reorder level, minimum threshold, and safety stock policy are saved for the defined scope.	[BR-009]
PO2	Low-stock and out-of-stock alert monitoring applies the new thresholds.	[BR-009]
PO3	Safety stock buffer is protected from routine consumption without authorized override.	[BR-010]
PO4	Policy effective date and scope are available for replenishment and monitoring functions.	—

UC-004 — Classify Products by ABC Priority

Primary Actor: Inventory Controller

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with ABC classification permission (PG-01).	(FR-009)
P2	Target products exist in active status in the product catalog.	—
P3	Product performance data (velocity, revenue, or criticality) is available for classification review.	—

Postconditions

#	Condition	Business Rule
PO1	Each classified product has an assigned ABC class (A, B, or C) with effective date.	[BR-032]
PO2	Class A products are scheduled for highest cycle count frequency and monitoring intensity.	[BR-032]
PO3	Low-stock alert prioritization reflects ABC classification.	[BR-032]
PO4	Classification change is recorded in the audit trail.	(FR-086)

UC-005 — Maintain Governance Policies and Reason Codes

Primary Actor: Inventory Controller

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with governance policy administration permission (PG-13).	(FR-012)
P2	For reason code deactivation, no active governance policy or pending transaction mandates the code.	[BR-020]
P3	Proposed approval thresholds do not create segregation of duties conflicts.	(FR-084)

Postconditions

#	Condition	Business Rule
PO1	Active reason code catalog is available for adjustments, discrepancies, and exceptions.	[BR-020]
PO2	Governance policies (approval thresholds, count rules, override rules) are active for applicable transaction types.	[BR-021]
PO3	New or updated policies apply to subsequent inventory transactions.	(FR-012)
PO4	Policy and reason code changes are recorded in the audit trail.	(FR-086)

Domain 2: Inbound & Receiving

UC-006 — Place and Manage Purchase Orders

Primary Actor: Buyer

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with purchase order creation permission (PG-06).	(FR-081)
P2	Supplier is registered and authorized for procurement.	—
P3	All purchase order lines reference active, non-discontinued products.	[BR-027]
P4	Delivery location is active and configured for receiving.	—

Postconditions

#	Condition	Business Rule
PO1	Approved purchase order exists as authorized commercial commitment to supplier.	[BR-007]
PO2	Expected inbound supply is visible for replenishment and receiving planning.	(FR-055)
PO3	Purchase order is available for goods receipt matching and three-way match processing.	[BR-007], [BR-008]
PO4	Purchase order creation, approval, amendment, or cancellation is recorded in the audit trail.	(FR-086)

UC-007 — Plan Expected Inbound Deliveries

Primary Actor: Buyer

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with inbound planning permission (PG-06).	(FR-014)
P2	Target purchase order is approved and in open status.	[BR-007]
P3	Supplier delivery schedule information is available from supplier communication.	—

Postconditions

#	Condition	Business Rule
PO1	Expected delivery date and shipment details are recorded against the purchase order.	—
PO2	Receiving location has visibility of planned inbound arrival.	(FR-055)
PO3	Replenishment and alert functions reflect updated inbound supply timing.	[BR-009]
PO4	Delivery schedule changes are recorded in transaction history.	(FR-029)

UC-008 — Receive and Verify Supplier Goods

Primary Actor: Receiving Clerk

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with goods receipt permission (PG-07) for the receiving location.	(FR-082)
P2	Valid purchase order reference or approved ad-hoc receipt authorization exists for the shipment.	[BR-007]
P3	Receiving location is active and not under physical inventory freeze.	[BR-024]
P4	Supplier shipment is physically present at the receiving dock with documentation.	[BR-004]

Postconditions

#	Condition	Business Rule
PO1	Verified receipt is recorded with quantity, product identity, and condition confirmed against authorization.	[BR-004]
PO2	Lot or batch identity is captured for traceable products.	[BR-017]
PO3	Shelf-life products meeting MRSL are accepted; non-compliant product is routed to discrepancy workflow.	[BR-016]
PO4	Received stock is staged for put-away and is not yet included in available-to-promise.	[BR-005]
PO5	Receipt transaction is recorded in movement history and audit trail.	(FR-029), (FR-086)

UC-009 — Document and Resolve Receipt Discrepancies

Primary Actor: Receiving Clerk

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with receipt and discrepancy permission (PG-07).	(FR-081)
P2	Goods receipt verification is in progress or complete with identified variance.	[BR-006]
P3	Mandatory reason codes for discrepancy types are active in the system.	[BR-020]
P4	Warehouse Manager or authorized approver is available for beyond-tolerance resolution.	(SOD-007)

Postconditions

#	Condition	Business Rule
PO1	Discrepancy is documented with type, variance quantity, reason code, and timestamp.	[BR-006]
PO2	Accepted quantities are recorded; unresolved variance is held for supplier or finance resolution.	[BR-006]
PO3	Discrepancy resolution supports three-way match and supplier accountability.	[BR-008]
PO4	Discrepancy documentation is available in receipt record and audit trail.	(FR-086)

UC-010 — Perform Stock Put-Away

Primary Actor: Warehouse Staff

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with put-away permission (PG-07) for the receiving location.	(FR-082)
P2	Goods receipt verification is complete for the stock to be put away.	[BR-004], [BR-005]

#	Condition	Business Rule
P3	Destination storage location is authorized for the product category.	—
P4	Bonded inventory, if applicable, has customs clearance before sellable put-away.	[BR-030]

Postconditions

#	Condition	Business Rule
PO1	Stock is recorded at designated storage location with confirmed quantity.	[BR-005]
PO2	On-hand quantity is updated at the storage location.	—
PO3	Available-to-promise is increased by the put-away quantity (sellable stock).	[BR-005]
PO4	Put-away is recorded in movement history with user, timestamp, and location.	(FR-029)
PO5	FIFO/FEFO placement follows rotation policy for subsequent picking.	[BR-014], [BR-015]

UC-011 — Receive Return and Non-PO Inbound Stock

Primary Actor: Receiving Clerk

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with non-PO receipt permission (PG-07, PG-09).	(FR-019)
P2	For customer returns, valid return authorization exists linked to original order.	[BR-025]
P3	For ad-hoc receipts, pre-approved authorization reference exists.	[BR-007]
P4	Receiving location is active and not under physical inventory freeze.	[BR-024]

Postconditions

#	Condition	Business Rule
PO1	Non-PO inbound receipt is recorded against authorization reference.	[BR-025]
PO2	Stock is staged for inspection, put-away, or disposition per receipt type — not immediately sellable unless policy allows.	[BR-026]
PO3	Receipt is recorded in movement history and audit trail.	(FR-029), (FR-086)
PO4	Return authorization status is updated to received pending inspection.	[BR-025]

Domain 3: Inventory Visibility & Status

UC-012 — Monitor Inventory Status

Primary Actor: Inventory Controller

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with inventory visibility permission (PG-03) for authorized locations.	(FR-082)
P2	Product master and location master data exist for monitored scope.	—
P3	Stock policies and thresholds are defined for products under monitoring.	[BR-009]

Postconditions

#	Condition	Business Rule
PO1	Current on-hand, reserved, ATP, and status indicators are displayed for monitored scope.	[BR-001]
PO2	Low-stock and out-of-stock conditions are visible against defined thresholds.	[BR-009]
PO3	Quarantined stock is displayed separately and excluded from sellable ATP view.	[BR-013]
PO4	Monitoring session timestamp is recorded for accountability.	(FR-083)

UC-013 — Inquire Inventory Availability

Primary Actor: Sales Representative

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with ATP inquiry permission (PG-03, PG-10).	(FR-081)
P2	Target product exists in the product catalog.	—
P3	Inventory records exist for at least one fulfillment location (ATP may be zero).	—

Postconditions

#	Condition	Business Rule
PO1	On-hand, reserved, and ATP quantities are displayed per location and network-wide as authorized.	[BR-001]
PO2	Stock status indicator (in stock, low stock, out of stock) is presented.	—
PO3	Quarantined and ring-fenced promotional stock are excluded from quotable ATP.	[BR-013], [BR-029]
PO4	Expected inbound supply is visible where applicable for customer commitment discussion.	(FR-028)
PO5	No inventory reservation or commitment is created by inquiry alone.	[BR-001]

Domain 4: Inventory Control & Accuracy

UC-014 — Perform Cycle Count

Primary Actor: Warehouse Staff

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with cycle count permission (PG-04) for the assigned location.	(FR-082)
P2	Cycle count is scheduled and assigned to the counter for product-location combination.	[BR-018]
P3	Counter is not shown system expected quantity (blind count enforced).	[BR-018]
P4	Target location is not under full physical inventory freeze unless count is part of freeze program.	[BR-024]

Postconditions

#	Condition	Business Rule
PO1	Physical count result is recorded with counter identity and timestamp.	[BR-018]
PO2	System compares count to recorded quantity and flags variance if present.	—

#	Condition	Business Rule
PO3	Count submission status is available (verified match or variance pending review).	[BR-022]
PO4	Count transaction is recorded in audit trail.	(FR-086)

UC-015 — Conduct Full Physical Inventory Count

Primary Actor: Inventory Controller

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with physical count initiation permission (PG-04).	(FR-032)
P2	Target location is active and count schedule is approved.	—
P3	Count teams are assigned and available for wall-to-wall count execution.	—
P4	Stakeholders are notified of impending inventory freeze at the location.	[BR-024]

Postconditions

#	Condition	Business Rule
PO1	Physical inventory freeze was active during count execution at the location.	[BR-024]
PO2	Complete count snapshot exists for all assigned zones and products.	—
PO3	Inventory freeze is released upon count session close.	[BR-024]
PO4	Variance report is generated for investigation and approval workflow.	[BR-022]
PO5	Count session close is recorded in audit trail.	(FR-086)

UC-016 — Investigate and Approve Count Variances

Primary Actor: Warehouse Manager

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with variance investigation and approval permission (PG-04).	(FR-034)
P2	Count variance record exists from cycle count or physical count.	[BR-022]
P3	Approver is not the same person who performed the count.	[BR-019]
P4	Investigation notes capability is available before approval.	[BR-022]

Postconditions

#	Condition	Business Rule
PO1	Variance investigation findings are documented with investigator identity and timestamp.	[BR-022]
PO2	Variance is approved or rejected by independent authorized approver.	[BR-019]
PO3	Approved variance is queued for stock adjustment processing.	[BR-022]
PO4	Approval decision is recorded in audit trail with approver identity.	[BR-021], (FR-086)

UC-017 — Process Stock Adjustments

Primary Actor: Warehouse Manager

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with stock adjustment permission (PG-04) for the location.	(FR-082)
P2	Product exists at the target location with known current on-hand quantity.	—
P3	Mandatory reason code is selected for the adjustment.	[BR-020]
P4	For decrease adjustments, quantity does not exceed available on-hand (excluding already reserved).	—
P5	Independent approver is available if adjustment exceeds threshold.	[BR-021]

Postconditions

#	Condition	Business Rule
PO1	On-hand and ATP quantities reflect the approved adjustment.	—
PO2	Adjustment is recorded with reason code, user, timestamp, and before/after quantities.	[BR-020]
PO3	Material adjustments have independent approval recorded where threshold required.	[BR-021]
PO4	Adjustment appears in movement history and audit trail.	(FR-029), (FR-086)
PO5	Safety stock floor is not breached without documented authorized override.	[BR-010]

UC-018 — Authorize and Process Inventory Write-Offs

Primary Actor: Operations Director

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with write-off authorization permission (PG-04).	(FR-036)
P2	Write-off request exists with product, quantity, location, and reason.	[BR-023]
P3	Supporting documentation and evidence are attached to the write-off request.	[BR-023]
P4	Write-off quantity does not exceed verified on-hand at location.	—
P5	Finance acknowledgment is available where financial policy requires it.	[BR-023]

Postconditions

#	Condition	Business Rule
PO1	Written-off quantity is removed from on-hand and ATP.	[BR-023]
PO2	Write-off is recorded with reason code, authorizer, and supporting documentation reference.	[BR-020], [BR-023]
PO3	Inventory valuation reflects reduced quantity for finance reconciliation.	(FR-075)
PO4	Write-off transaction is recorded in movement history and audit trail.	(FR-029), (FR-086)

UC-019 — Manage Quarantine and Hold Stock

Primary Actor: Warehouse Manager

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with quarantine management permission (PG-04).	(FR-038)
P2	Target stock exists as on-hand at the location.	—
P3	Hold reason (quality, investigation, recall, bonded, customs) is defined.	[BR-013], [BR-030]
P4	For release, quality review or customs clearance is documented where applicable.	[BR-030]

Postconditions

#	Condition	Business Rule
PO1	Quarantined quantity is excluded from available-to-promise.	[BR-013]
PO2	On-hand reflects total physical stock; sellable ATP excludes quarantined portion.	[BR-013]
PO3	Hold reason, date, and responsible party are recorded.	—
PO4	For release, stock is restored to sellable ATP with release authorization recorded.	[BR-013], [BR-030]
PO5	Quarantine placement or release is recorded in movement history and audit trail.	(FR-029), (FR-086)

UC-020 — Analyze Shrinkage, Loss, and Obsolete Stock

Primary Actor: Inventory Controller

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with reporting and analysis permission (PG-11).	(FR-039)
P2	Adjustment and write-off history with reason codes exists for the analysis period.	[BR-020]
P3	Product movement and aging data is available for obsolescence analysis.	(FR-040)

Postconditions

#	Condition	Business Rule
PO1	Shrinkage trends by reason code, location, and period are available for review.	[BR-020]
PO2	Slow-moving and obsolete product candidates are identified with aging metrics.	(FR-040)
PO3	Recommended disposition actions are documented where analysis conclusions are recorded.	—
PO4	Analysis report is exportable for operations and finance stakeholders.	(FR-039)

Domain 5: Order Fulfillment

UC-021 — Confirm Order Availability

Primary Actor: Sales Representative

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with availability confirmation permission (PG-10).	(FR-041)
P2	Customer order exists with defined product lines and quantities.	—

#	Condition	Business Rule
P3	Product master and inventory records exist for all order lines.	—
P4	Fulfillment location(s) are identified or default fulfillment rules apply.	—

Postconditions

#	Condition	Business Rule
PO1	Availability check result is recorded per order line against ATP (not total on-hand).	[BR-001]
PO2	Ring-fenced promotional stock is excluded from quotable ATP for non-promotional orders.	[BR-029]
PO3	Order availability status indicates lines cleared, partial, or unavailable for commitment.	[BR-001]
PO4	No reservation is created until separate reservation use case is executed.	[BR-002]
PO5	Availability check timestamp and result are recorded for audit.	(FR-083)

UC-022 — Reserve Inventory for Confirmed Orders

Primary Actor: Customer Service Representative

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with reservation permission (PG-08).	(FR-042)
P2	Customer order is confirmed for fulfillment.	[BR-002]
P3	Availability confirmation is complete or current ATP is sufficient at fulfillment location.	[BR-001]
P4	ATP at fulfillment location covers reservation quantity (or partial per customer agreement).	[BR-002]
P5	Promotional ring-fence rules are satisfied if consuming campaign stock.	[BR-029]

Postconditions

#	Condition	Business Rule
PO1	Reserved quantity is committed against on-hand at fulfillment location.	[BR-002]
PO2	ATP is reduced by reserved quantity.	[BR-001], [BR-002]
PO3	Order is released to warehouse fulfillment queue with reservation reference per line.	—
PO4	Reservation is recorded in movement history with user, timestamp, and order reference.	(FR-029)
PO5	Safety stock is not consumed below buffer without authorized override.	[BR-010]

UC-023 — Release Inventory Reservations

Primary Actor: Customer Service Representative

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with reservation management permission (PG-08).	(FR-043)
P2	Active reservation exists for the order or order line.	[BR-002]
P3	Reserved lines are not already picked, packed, or shipped.	[BR-003]
P4	Release reason is available where business policy requires it.	—

Postconditions

#	Condition	Business Rule
PO1	Reservation hold is removed from specified order lines.	[BR-002]
PO2	Released quantity is restored to available-to-promise.	[BR-001]
PO3	Order status reflects released reservation.	—
PO4	Release is recorded in transaction history with reason, user, and timestamp.	(FR-029), (FR-083)

UC-024 — Pick, Pack, and Ship Orders

Primary Actor: Warehouse Picker / Packer

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with pick, pack, and ship permission (PG-08) for the fulfillment location.	(FR-082)
P2	Order has active reservation at the fulfillment location.	[BR-002]
P3	Reserved stock is in sellable status (not quarantined).	[BR-013]
P4	Pick list and storage locations are available for order lines.	—

Postconditions

#	Condition	Business Rule
PO1	All shipped lines have pick verification recorded before shipment confirmation.	[BR-003]
PO2	On-hand is deducted by shipped quantity; reservation is cleared.	[BR-003]
PO3	Lot or batch attribution is recorded for traceable products on shipment.	[BR-017]
PO4	FIFO/FEFO rotation is applied in pick execution.	[BR-014], [BR-015]
PO5	Order status is updated to shipped with shipment date and carrier reference.	—
PO6	Shipment is recorded in movement history and audit trail.	(FR-029), (FR-086)

UC-025 — Handle Partial Fulfillment and Backorders

Primary Actor: Customer Service Representative

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with partial fulfillment and backorder permission (PG-08, PG-10).	(FR-045)
P2	Order line has insufficient ATP for full quantity.	[BR-001]
P3	Partial ATP is available for immediate fulfillment or customer accepts full backorder.	—
P4	For backorder, customer acknowledgment mechanism is available.	[BR-028]

Postconditions

#	Condition	Business Rule
PO1	Available quantity is reserved for immediate fulfillment.	[BR-002]

#	Condition	Business Rule
PO2	Backorder record exists with customer acknowledgment for remaining quantity.	[BR-028]
PO3	Order reflects split status: pick-ready lines and backordered lines.	—
PO4	Backorder is queued with replenishment priority where assigned.	(FR-046)
PO5	Partial fulfillment and backorder decisions are recorded in audit trail.	(FR-086)

UC-026 — Handle Outbound Fulfillment Exceptions

Primary Actor: Warehouse Manager

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with fulfillment exception management permission (PG-08).	(FR-047)
P2	Outbound exception exists (pick failure, quantity mismatch, damage).	—
P3	Order has active reservation or is in pick/pack workflow.	[BR-002]
P4	Authorized substitute products are defined if substitution is a resolution option.	(FR-050)

Postconditions

#	Condition	Business Rule
PO1	Exception resolution is documented with manager authorization and reason.	—
PO2	Order is returned to fulfillment flow, routed to substitution, partial ship, or quarantine per resolution.	[BR-003], [BR-013]
PO3	Inventory and reservation states are adjusted to reflect resolution.	[BR-001]
PO4	Exception and resolution are recorded in audit trail.	(FR-086)

UC-027 — View Order Fulfillment Status

Primary Actor: Customer Service Representative

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with order status visibility permission (PG-08, PG-10).	(FR-049)
P2	Target order exists in the system.	—

Postconditions

#	Condition	Business Rule
PO1	Current order fulfillment status is displayed (reserved, picking, packed, shipped, backordered).	—
PO2	Line-level quantity, location, and status are visible.	—
PO3	Shipment and carrier information is displayed for shipped lines.	—
PO4	No inventory or order state is modified by status inquiry.	—

Domain 6: Replenishment & Inbound Supply

UC-028 — Review Low-Stock and Out-of-Stock Alerts

Primary Actor: Purchasing Manager

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with alert review permission (PG-06, PG-11).	(FR-052)
P2	Stock policies and thresholds are defined for alerted products.	[BR-009]
P3	Active low-stock or out-of-stock alert exists for product-location combination.	[BR-009]
P4	ABC classification is assigned for priority sorting where classification is in use.	[BR-032]

Postconditions

#	Condition	Business Rule
PO1	Alert is acknowledged with user and timestamp.	(FR-052)
PO2	Follow-up action is assigned (purchase, transfer, or deferral with reason).	[BR-009]
PO3	Alert queue reflects updated acknowledgment status.	—
PO4	Class A alert acknowledgment is recorded within monitoring SLA where defined.	[BR-032]

UC-029 — Plan Replenishment and Create Purchase Requisitions

Primary Actor: Buyer

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with requisition creation permission (PG-06).	(FR-054)
P2	Replenishment need is identified from acknowledged alert or planning cycle.	[BR-009]
P3	Network availability check is complete; transfer alternative evaluated.	[BR-011]
P4	Target product is active and not discontinued.	[BR-027]
P5	Safety stock rules are considered in requisition quantity.	[BR-010]

Postconditions

#	Condition	Business Rule
PO1	Purchase requisition exists with product, quantity, required date, and delivery location.	—
PO2	Requisition is approved and available for purchase order creation.	—
PO3	Network check outcome is documented (purchase proceeded or transfer override recorded).	[BR-011]
PO4	Requisition is recorded in audit trail.	(FR-086)

UC-030 — Track Inbound Supply Status

Primary Actor: Buyer

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with inbound supply tracking permission (PG-06).	(FR-055)
P2	Open purchase order(s) exist with expected delivery information.	[BR-007]

Postconditions

#	Condition	Business Rule
PO1	Inbound supply status reflects current ordered, received, and outstanding quantities.	(FR-055)
PO2	Expected delivery dates are current based on latest supplier communication.	—
PO3	Replenishment and alert functions have access to updated inbound supply position.	[BR-009]
PO4	Status updates are recorded in transaction history.	(FR-029)

Domain 7: Network Balance & Transfers

UC-031 — Request Inter-Location Stock Transfer

Primary Actor: Regional / Store Manager

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with transfer request permission (PG-05) for destination location.	(FR-058)
P2	Source and destination locations are active.	—
P3	Source location has sufficient ATP above minimum threshold for requested quantity.	[BR-012]
P4	Product exists at source location with available (non-quarantined) stock.	[BR-013]
P5	Network availability check supports transfer over new purchase where applicable.	[BR-011]

Postconditions

#	Condition	Business Rule
PO1	Transfer request exists with pending approval status.	—
PO2	Source and destination impact preview is recorded.	[BR-012]
PO3	Approver is notified of pending transfer request.	(FR-077)
PO4	Transfer request is recorded in audit trail.	(FR-086)

UC-032 — Approve or Reject Transfer Requests

Primary Actor: Operations Director

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with transfer approval permission (PG-05).	(FR-059)
P2	Pending transfer request exists.	—
P3	Approver is not the same person who submitted the transfer request.	(SOD-006)

#	Condition	Business Rule
P4	Source location ATP and minimum threshold impact can be evaluated.	[BR-012]

Postconditions

#	Condition	Business Rule
PO1	Transfer request status is approved or rejected with documented decision.	—
PO2	For approval, source and destination locations are notified.	(FR-077)
PO3	Approved transfer is available for execution at source location.	[BR-012]
PO4	Approval decision is recorded with approver identity and timestamp.	(FR-086)

UC-033 — Execute and Record Inter-Location Transfer

Primary Actor: Warehouse Staff

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with transfer execution permission (PG-05) at source and destination.	(FR-060)
P2	Transfer request is approved.	—
P3	Source location has approved quantity available for dispatch.	[BR-012]
P4	Source location is not under physical inventory freeze.	[BR-024]
P5	Lot or batch capture is required for traceable products.	[BR-017]

Postconditions

#	Condition	Business Rule
PO1	Source on-hand is reduced by dispatched quantity.	—
PO2	In-transit status exists from dispatch until destination receipt confirmation.	(FR-061)
PO3	Destination on-hand is increased upon receipt confirmation.	—
PO4	Transfer is complete with full movement history at source, in-transit, and destination.	(FR-029)
PO5	Lot or batch attribution is maintained through transfer for traceable products.	[BR-017]
PO6	Source location remains above minimum threshold after transfer.	[BR-012]

UC-034 — Analyze and Recommend Network Rebalancing

Primary Actor: Operations Director

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with network rebalancing analysis permission (PG-05, PG-11).	(FR-062)
P2	Inventory records exist across multiple active locations.	—
P3	Network-wide visibility is available for imbalance identification.	(FR-028)

Postconditions

#	Condition	Business Rule
PO1	Location-level imbalances (stockout vs. excess) are identified and documented.	[BR-011]
PO2	Transfer recommendations exist with source, destination, product, and quantity.	[BR-012]
PO3	Approved recommendations generate transfer requests for execution.	—
PO4	Network balance improvement summary is available for management review.	—

Domain 8: Returns Management

UC-035 — Authorize and Record Customer Returns

Primary Actor: Customer Service Representative

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with return authorization permission (PG-09).	(FR-064)
P2	Original customer order exists and is locatable in the system.	—
P3	Return request is within return policy eligibility (quantity, timeframe, product condition policy).	[BR-025]
P4	Return destination location is active and configured for return receipt.	—

Postconditions

#	Condition	Business Rule
PO1	Return authorization exists with reference, authorized products, quantities, and destination.	[BR-025]
PO2	Return authorization is linked to original order and return policy.	[BR-025]
PO3	Return status is pending receipt at warehouse.	—
PO4	Authorization is recorded in audit trail.	(FR-086)

UC-036 — Receive, Inspect, and Disposition Returned Goods

Primary Actor: Receiving Clerk

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with return receipt and inspection permission (PG-09).	(FR-065)
P2	Valid return authorization exists for the inbound return shipment.	[BR-025]
P3	Returned goods are physically present at receiving location.	—
P4	Inspection criteria and disposition options (restock, quarantine, write-off) are defined.	[BR-026]

Postconditions

#	Condition	Business Rule
PO1	Return receipt quantity is recorded against authorization.	[BR-025]
PO2	Inspection outcome is documented with condition assessment.	[BR-026]

#	Condition	Business Rule
PO3	For restock disposition, sellable on-hand and ATP are increased at designated location.	[BR-026]
PO4	For quarantine or write-off disposition, stock is routed per disposition workflow.	[BR-026], [BR-013]
PO5	Return authorization is closed or updated with final disposition status.	(FR-067)
PO6	Return processing is recorded in movement history and audit trail.	(FR-029), (FR-086)

Domain 9: Reporting, Performance & Financial Alignment

UC-037 — Review Inventory Dashboards and Operational Reports

Primary Actor: Operations Director

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with dashboard and reporting permission (PG-11).	(FR-068)
P2	Inventory transaction and balance data exists for the reporting scope.	—
P3	Reporting period and location scope are within actor's authorized visibility.	(FR-082)

Postconditions

#	Condition	Business Rule
PO1	Dashboard or report reflects current inventory data for selected scope and period.	—
PO2	Report generation timestamp is recorded.	—
PO3	Exportable report output is available where export is requested.	—
PO4	No inventory data is modified by reporting activity.	—

UC-038 — Review Inventory Performance KPIs

Primary Actor: Chief Executive Officer

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with executive KPI visibility permission (PG-11).	(FR-073)
P2	KPI baseline and target values are defined for the reporting period.	—
P3	Sufficient operational data exists for KPI calculation (data completeness indicator available).	—

Postconditions

#	Condition	Business Rule
PO1	Tier 1 inventory KPIs are displayed with target comparison and trend.	—
PO2	RAG status is indicated per KPI where thresholds are defined.	—
PO3	Drill-down to contributing factors is available for underperforming KPIs.	—
PO4	No operational data is modified by KPI review.	—

UC-039 — Reconcile Inventory Valuation with Finance

Primary Actor: Finance Manager

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with valuation and reconciliation permission (PG-12).	(FR-089)
P2	Operational inventory valuation data exists for the reconciliation period.	—
P3	General ledger inventory balance is available for comparison.	[BR-008]
P4	Consignment stock is identifiable and separable from owned inventory.	[BR-031]

Postconditions

#	Condition	Business Rule
PO1	Side-by-side comparison of operational valuation and GL balance is documented.	[BR-008]
PO2	Variance amount and percentage are calculated; tolerance status is indicated.	—
PO3	Reconciliation is marked complete with finance notes and status.	—
PO4	Consignment stock is excluded from owned valuation comparison.	[BR-031]
PO5	Reconciliation record is archived for audit.	(FR-086)

Domain 10: Commercial Alignment & Governance

UC-040 — Confirm Promotional and Launch Stock Readiness

Primary Actor: Marketing Manager

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with campaign readiness permission (PG-10).	(FR-057)
P2	Campaign or launch definition exists with required products and quantities.	[BR-029]
P3	Campaign start date is defined.	—
P4	Ring-fenced promotional stock allocation is configured or allocatable.	[BR-029]

Postconditions

#	Condition	Business Rule
PO1	Readiness status per product is recorded (confirmed, conditional, or not ready).	—
PO2	Ring-fenced promotional stock is protected from non-promotional consumption.	[BR-029]
PO3	Readiness confirmation or gap summary is documented with manager identity and date.	—
PO4	Campaign readiness alerts are updated for remaining gaps.	(FR-080)

UC-041 — Conduct Inventory Audit and Control Review

Primary Actor: Internal Auditor

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with audit query permission (PG-15).	(FR-087)
P2	Inventory transaction and audit trail data exists for the audit scope period.	(FR-086)
P3	Segregation of duties rules and governance policies are defined for compliance testing.	[BR-019], [BR-021]
P4	Audit scope (period, location, transaction type) is defined.	—

Postconditions

#	Condition	Business Rule
PO1	Audit sample or population is reviewed for control compliance.	—
PO2	SoD violations and policy non-compliance are flagged where detected.	[BR-019], [BR-020], [BR-021]
PO3	Audit findings and review session are documented.	—
PO4	Audit workpaper summary is exportable for governance records.	(FR-087)
PO5	No operational inventory data is modified by audit review (read-only).	—

Domain 11: Cross-Cutting System Interactions

UC-042 — View Inventory Movement and Transaction History

Primary Actor: Inventory Controller

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with movement history inquiry permission (PG-03, PG-11).	(FR-029)
P2	Inventory movements have been recorded for the search scope.	(FR-029)
P3	Search criteria (product, location, date range, type) are within authorized visibility.	(FR-082)

Postconditions

#	Condition	Business Rule
PO1	Matching movement transactions are displayed chronologically.	(FR-029)
PO2	Transaction detail includes quantity, type, user, timestamp, and reason code where applicable.	[BR-020]
PO3	Lot or batch attribution is visible for traceable product movements.	[BR-017]
PO4	Exportable movement history is available for the search scope.	—
PO5	No inventory data is modified by history inquiry.	—

UC-043 — Query and Review Audit Trail

Primary Actor: Internal Auditor

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with audit trail query permission (PG-15).	(FR-086)

#	Condition	Business Rule
P2	Audit log entries exist for the query scope.	(FR-086)
P3	Query parameters are within authorized audit scope.	—

Postconditions

#	Condition	Business Rule
PO1	Immutable audit log entries matching criteria are displayed.	(FR-086)
PO2	Before-and-after state is available for material changes.	(FR-086)
PO3	Exportable audit report is available for evidence purposes.	—
PO4	Audit query activity is logged for accountability.	(FR-083)
PO5	No operational data is modified by audit trail query.	—

UC-044 — Manage User Accounts and Role Assignments

Primary Actor: System Administrator

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with user administration permission (PG-14).	(FR-085)
P2	Business roles and location definitions exist in the system.	(FR-081)
P3	Proposed role assignment does not create segregation of duties conflict.	(FR-084)
P4	For deactivation, pending approvals in user's queue are reassigned or resolved.	—

Postconditions

#	Condition	Business Rule
PO1	User account exists with correct status (active or deactivated).	(FR-085)
PO2	Assigned roles and location scope reflect current job responsibilities.	(FR-081), (FR-082)
PO3	User permissions align with RBAC matrix without SoD conflicts.	(FR-084)
PO4	Account creation, modification, or deactivation is recorded in audit trail.	(FR-086)

UC-045 — Enforce Role-Based and Location-Scoped Access

Primary Actor: System Administrator

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with access control configuration permission (PG-14).	(FR-081)
P2	Role-permission matrix and location scope definitions exist.	(FR-082)
P3	Segregation of duties rules are defined for validation.	(FR-084)
P4	For access enforcement events, user account is active with assigned roles.	—

Postconditions

#	Condition	Business Rule
PO1	Updated permissions apply to all users assigned the affected role.	(FR-081)
PO2	Access attempts are granted only for permitted functions within location scope.	(FR-082)
PO3	Unauthorized access attempts are denied and logged.	(FR-083)
PO4	SoD conflicts are prevented in role permission configuration.	[BR-019], [BR-021], (FR-084)
PO5	Access control changes are recorded in audit trail.	(FR-086)

UC-046 — Import Initial Product and Stock Master Data

Primary Actor: Inventory Controller

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with data import permission and product/location management access.	(FR-090)
P2	Import file conforms to system template format.	(FR-090)
P3	Location master exists for all location codes referenced in stock balance import.	—
P4	Import file SKUs are unique within file and do not conflict with existing catalog (for initial load).	—

Postconditions

#	Condition	Business Rule
PO1	Product master records are created or updated per validated import rows.	—
PO2	Opening stock balances are recorded at specified locations.	(FR-010)
PO3	Import completion report documents created, updated, and skipped records.	—
PO4	Imported data is available for operational transactions.	—
PO5	Import execution is recorded in audit trail with user and timestamp.	(FR-086)

UC-047 — Receive and Respond to Operational Notifications

Primary Actor: Warehouse Manager

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with notification access for relevant notification types.	(FR-076)
P2	Pending notification exists (alert, approval, exception, inbound delay, campaign).	—
P3	Linked business context (product, order, transfer, campaign) is accessible to actor.	—

Postconditions

#	Condition	Business Rule
PO1	Notification action is completed or delegated with recorded outcome.	—
PO2	Linked business record reflects action taken (acknowledgment, approval, resolution).	[BR-009]

#	Condition	Business Rule
PO3	Notification is archived or removed from active pending queue.	—
PO4	Notification handling is recorded with user and timestamp.	(FR-083)

UC-048 — Support Multi-Location Split Fulfillment

Primary Actor: Warehouse Manager

Preconditions

#	Condition	Business Rule
P1	Primary actor is authenticated with multi-location fulfillment permission (PG-08).	(FR-048)
P2	Customer order is confirmed with lines requiring fulfillment.	[BR-002]
P3	Single fulfillment location has insufficient ATP; network-wide ATP can satisfy order (fully or partially).	[BR-001]
P4	Contributing source locations have available stock above minimum thresholds.	[BR-012]
P5	Availability confirmation is complete or current network ATP is validated.	[BR-001]

Postconditions

#	Condition	Business Rule
PO1	Location-specific reservations exist for each contributing fulfillment location.	[BR-002]
PO2	ATP is reduced at each contributing location by allocated quantity.	[BR-001]
PO3	Pick tasks are routed to each source location with linked parent order reference.	—
PO4	Consolidated order status reflects multi-location fulfillment progress.	—
PO5	Source locations remain above minimum threshold after allocation.	[BR-012]
PO6	Split fulfillment plan is recorded in audit trail.	(FR-086)

Business Rule Coverage Matrix

Business Rule	Use Cases with Preconditions	Use Cases with Postconditions
BR-001	UC-013, UC-021, UC-022, UC-048	UC-012, UC-013, UC-021, UC-022, UC-023, UC-026, UC-048
BR-002	UC-021, UC-022, UC-023, UC-024, UC-048	UC-021, UC-022, UC-023, UC-024, UC-025, UC-048
BR-003	UC-023, UC-024	UC-024, UC-026
BR-004	UC-008, UC-010, UC-011	UC-008
BR-005	UC-010	UC-010
BR-006	UC-009	UC-009
BR-007	UC-006, UC-007, UC-008, UC-011	UC-006, UC-030
BR-008	UC-039	UC-009, UC-039
BR-009	UC-003, UC-012, UC-028, UC-029, UC-030	UC-003, UC-028, UC-029, UC-030, UC-047
BR-010	UC-003, UC-017, UC-022, UC-029	UC-003, UC-017, UC-022
BR-011	UC-029, UC-031, UC-034	UC-028, UC-029, UC-034
BR-012	UC-031, UC-032, UC-033, UC-048	UC-031, UC-032, UC-033, UC-048
BR-013	UC-019, UC-024, UC-031	UC-012, UC-019, UC-024, UC-026, UC-031, UC-036

Business Rule	Use Cases with Preconditions	Use Cases with Postconditions
BR-014	UC-024	UC-010, UC-024
BR-015	UC-024	UC-010, UC-024
BR-016	UC-008	UC-008
BR-017	UC-008, UC-033, UC-042, UC-048	UC-008, UC-024, UC-033, UC-042
BR-018	UC-014	UC-014
BR-019	UC-016, UC-041, UC-045	UC-016, UC-041, UC-045
BR-020	UC-005, UC-009, UC-017, UC-020, UC-041	UC-005, UC-017, UC-018, UC-020, UC-041, UC-042
BR-021	UC-005, UC-016, UC-017, UC-041, UC-045	UC-005, UC-016, UC-017, UC-041, UC-045
BR-022	UC-015, UC-016	UC-015, UC-016
BR-023	UC-018	UC-018
BR-024	UC-008, UC-010, UC-011, UC-014, UC-033	UC-015
BR-025	UC-011, UC-035, UC-036	UC-011, UC-035, UC-036
BR-026	UC-036	UC-036
BR-027	UC-006, UC-029	UC-001, UC-006, UC-029
BR-028	UC-025	UC-025
BR-029	UC-021, UC-022, UC-040	UC-013, UC-021, UC-022, UC-040
BR-030	UC-010, UC-019	UC-010, UC-019
BR-031	UC-039	UC-039
BR-032	UC-004, UC-028	UC-004, UC-028

Traceability

Source Document	Reference
Use Case Catalog	SRS-UC-001
Use Case Main Flows	SRS-UC-002
Use Case Alternate Flows	SRS-UC-003
Use Case Exception Flows	SRS-UC-004
Business Rule Catalog	SRS-BR-001

Document Approval

Role	Name	Signature	Date
Business Analyst			
Product Owner			
Operations Director			

End of Use Case Preconditions and Postconditions

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Part IV — Domain Model

Ubiquitous Language

Inventory Management System

Document Information

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Scope	Business terminology only — aligned with Domain Model; no technical or implementation terms

Purpose

This document defines the **Ubiquitous Language** of the Inventory Management System — the shared, precise vocabulary used by business stakeholders and reflected in the domain model. Every term maps to a concept in the Domain Entity Model (SRS-DM-001), a Bounded Context (SRS-DM-007), or an industry business rule (SRS-BR-001).

When business and technology teams use these terms consistently, requirements, use cases, and domain model artifacts remain aligned.

How to Read This Document

Field	Meaning
Term	The canonical business word or phrase
Definition	What the term means in this organization
Context	The Bounded Context where the term is primarily defined and used
Related Terms	Other ubiquitous language terms with direct business relationship
Terms That Should Never Be Confused	Commonly misused or similar terms with different meanings

Context Reference

Context Code	Bounded Context
PC	Product Catalog
WN	Warehouse Network
IG	Inventory Governance
IB	Inventory Balance
PS	Procurement & Supplier

Context Code	Bounded Context
IR	Inbound Receiving
OF	Order Fulfillment
NT	Network Transfer
IA	Inventory Accuracy
RM	Returns Management
CC	Commercial Campaign
IAc	Identity & Access
II	Inventory Intelligence
ALL	Used across multiple contexts

Section 1: Product Catalog (PC)

SKU (Stock Keeping Unit)

Field	Detail
Definition	The unique business identifier for a distinct product tracked in inventory. SKU is the primary identity key for all stock operations across locations.
Context	Product Catalog
Related Terms	Product, Unit of Measure, Barcode, Stock Position
Terms That Should Never Be Confused	**Lot/Batch Number** — identifies a production or receipt grouping within a SKU, not the product itself; **Product Name** — descriptive label, not unique identifier

Product

Field	Detail
Definition	The master record for a SKU including name, description, unit of measure, category, lifecycle status, ABC classification, and rotation/traceability attributes.
Context	Product Catalog
Related Terms	SKU, Product Lifecycle Status, ABC Classification, Rotation Policy
Terms That Should Never Be Confused	**Stock Position** — quantity of a product at a location, not the product definition; **Product Line** — informal commercial grouping, not an IMS entity

Product Lifecycle Status

Field	Detail
Definition	The commercial availability state of a product: **Active** (orderable and fulfillable), **Inactive** (not sold but may hold stock), or **Discontinued** (exit from assortment; no reorder permitted).
Context	Product Catalog
Related Terms	Product, Purchase Order, Inventory Reservation
Terms That Should Never Be Confused	**Stock Status Indicator** — describes quantity health (in stock, low stock) at a location, not product commercial status; **Product Category** — classification for reporting, not lifecycle

ABC Classification

Field	Detail
Definition	Product segmentation by business importance — Class **A** (highest revenue/velocity/criticality), **B** (moderate), or **C** (lower) — used to prioritize count frequency, monitoring intensity, and alert priority.
Context	Product Catalog
Related Terms	Product, Cycle Count, Operational Alert
Terms That Should Never Be Confused	**Product Category** — merchandise grouping (e.g., Electronics), not operational priority; **Stock Status Indicator** — current quantity state, not importance classification

Rotation Policy

Field	Detail
Definition	The required stock rotation discipline for a product: **FIFO** (First-In, First-Out) for standard goods, or **FEFO** (First-Expired, First-Out) for shelf-life products.
Context	Product Catalog
Related Terms	Product, Lot/Batch, Put-Away, Fulfillment Execution
Terms That Should Never Be Confused	**Put-Away** — the act of placing stock in storage, not the rotation rule; **Pick Path** — warehouse navigation, not rotation policy

Unit of Measure (UOM)

Field	Detail
Definition	The standard unit in which a product is counted, ordered, stored, and sold — such as Each, Case, Pallet, Kilogram, or Liter.
Context	Product Catalog
Related Terms	Product, SKU, Quantity
Terms That Should Never Be Confused	**Quantity** — numeric amount, which always includes a UOM; bare numbers without UOM are incomplete in inventory context

Section 2: Warehouse Network (WN)

Inventory Location

Field	Detail
Definition	A physical place where stock is held, received, counted, and fulfilled — such as a warehouse, store, or distribution point. Identified by a unique Location Code.
Context	Warehouse Network
Related Terms	Storage Position, Stock Position, Location Freeze
Terms That Should Never Be Confused	**Storage Position** — subdivision within a location (bin/shelf), not the location itself; **Supplier Location** — external vendor site, not an owned inventory location

Storage Position

Field	Detail
Definition	A designated storage area within an inventory location — zone, aisle, bin, or shelf — where received goods are placed after put-away and from which picks are retrieved.
Context	Warehouse Network
Related Terms	Inventory Location, Put-Away, Fulfillment Execution
Terms That Should Never Be Confused	**Inventory Location** — the warehouse or site as a whole; **Dock** — receiving area before put-away, not a storage position

Location Freeze

Field	Detail
Definition	A temporary suspension of all receiving, shipping, transfers, and adjustments at a location during a wall-to-wall physical inventory count, ensuring a stable quantity snapshot.
Context	Warehouse Network
Related Terms	Physical Inventory Count, Inventory Location
Terms That Should Never Be Confused	**Quarantine Hold** — excludes specific stock from sellable availability, not a location-wide operational suspension; **Location Deactivation** — permanent or semi-permanent closure, not temporary count freeze

Section 3: Inventory Governance (IG)

Reason Code

Field	Detail
Definition	A standardized, categorized business explanation required for every inventory adjustment, write-off, and receipt discrepancy — enabling shrinkage analysis and audit accountability.
Context	Inventory Governance
Related Terms	Stock Adjustment, Inventory Write-Off, Receipt Discrepancy, Governance Policy
Terms That Should Never Be Confused	**Disposition Decision** — outcome for returned goods (restock/quarantine/write-off), not a general adjustment taxonomy; **Notes/Comments** — free text, not a governed categorized code

Governance Policy

Field	Detail
Definition	An organization-wide inventory control rule defining approval thresholds, tolerance limits, segregation-of-duties requirements, or exception handling — effective for a defined period.
Context	Inventory Governance
Related Terms	Reason Code, Approval Threshold, Segregation of Duties
Terms That Should Never Be Confused	**Stock Policy** — replenishment thresholds per product-location, not enterprise control rules; **Business Rule (BR-xxx)** — documented policy statement in SRS-BR-001, of which governance policy is the operational parameterization

Segregation of Duties (SoD)

Field	Detail
Definition	The separation of conflicting responsibilities — such as counting vs. approving, or initiating vs. authorizing adjustments — to prevent fraud and undetected error.
Context	Inventory Governance, Identity & Access
Related Terms	Governance Policy, Count Variance, Role Assignment
Terms That Should Never Be Confused	**Role-Based Access** — what a user may do by role, not the conflict prevention rule itself; **Dual Approval** — one instance of SoD, not the full concept

Approval Threshold

Field	Detail
Definition	A governance limit — expressed as quantity, monetary value, or both — above which an inventory action requires independent approval by an authorized person who did not initiate the underlying transaction.
Context	Inventory Governance
Related Terms	Governance Policy, Stock Adjustment, Count Variance
Terms That Should Never Be Confused	**Reorder Level** — triggers replenishment, not approval; **Safety Stock** — buffer quantity, not an approval trigger

Section 4: Inventory Balance (IB)

Stock Position

Field	Detail
Definition	The authoritative inventory balance for one product at one location — maintaining on-hand, reserved, and available-to-promise quantities, ownership classification, and optional lot/batch breakdown.
Context	Inventory Balance
Related Terms	On-Hand Stock, Available-to-Promise, Reserved Stock, Stock Policy
Terms That Should Never Be Confused	**Product** — what is being tracked; **Stock Policy** — replenishment rules, not current quantities; **Storage Position** — where physically placed within location

On-Hand Stock

Field	Detail
Definition	The physical quantity of a product present at a location at a point in time, regardless of reservation or hold status.
Context	Inventory Balance
Related Terms	Stock Position, Available-to-Promise, Reserved Stock
Terms That Should Never Be Confused	**Available-to-Promise (ATP)** — portion of on-hand available for new commitments; **On-Order** — inbound expected, not physically present; **In-Transit Stock** — left source but not yet at destination

Available-to-Promise (ATP)

Field	Detail
Definition	The quantity that may be quoted or committed to new customers — calculated as on-hand minus reserved minus quarantined/hold quantities. The only honest measure for sales promises.

Field	Detail
Context	Inventory Balance
Related Terms	On-Hand Stock, Reserved Stock, Quarantine Hold, Inventory Reservation
Terms That Should Never Be Confused	**On-Hand Stock** — total physical quantity including reserved and quarantined; **Inbound Expected** — pipeline supply not yet available; **Committed Stock** — colloquial term often meaning reserved, but ATP is the precise quotable measure

Reserved Stock

Field	Detail
Definition	On-hand quantity operationally committed to confirmed customer orders, removed from ATP until fulfilled or released.
Context	Inventory Balance, Order Fulfillment
Related Terms	Inventory Reservation, Available-to-Promise, On-Hand Stock
Terms That Should Never Be Confused	**Backorder** — customer-agreed delay for quantity not yet reserved because stock is unavailable; **Promotional Allocation** — ring-fenced for campaign, not customer order reservation; **Safety Stock** — buffer not for customer commitment

Stock Policy

Field	Detail
Definition	Replenishment thresholds defined for a specific product at a specific location: reorder level, minimum threshold, safety stock buffer, and maximum (overstock) level.
Context	Inventory Balance
Related Terms	Reorder Level, Safety Stock, Stock Position, Operational Alert
Terms That Should Never Be Confused	**Governance Policy** — enterprise approval and control rules, not replenishment thresholds; **Stock Position** — current quantities, not policy limits

Reorder Level

Field	Detail
Definition	The ATP threshold that triggers replenishment review or action, calculated to allow sufficient lead time for supply to arrive before stockout.
Context	Inventory Balance
Related Terms	Stock Policy, Safety Stock, Low Stock, Replenishment
Terms That Should Never Be Confused	**Minimum Threshold** — lowest acceptable level before critical action, may differ from reorder trigger; **Maximum Level** — overstock indicator, not reorder trigger; **Zero ATP** — stockout condition, not the proactive trigger point

Safety Stock

Field	Detail
Definition	A deliberate buffer quantity above the reorder level protecting against demand and supply uncertainty. Must not be consumed for routine fulfillment without authorized exception and replenishment plan.
Context	Inventory Balance
Related Terms	Stock Policy, Reorder Level, Inventory Reservation
Terms That Should Never Be Confused	**Reserved Stock** — committed to specific orders, not a general buffer; **Minimum Threshold** — policy floor, not the uncertainty buffer; **Reorder Level** — trigger point, which is typically above safety stock

Stock Status Indicator

Field	Detail
Definition	A derived classification of inventory health for a product at a location: **In Stock**, **Low Stock**, **Out of Stock**, or **Overstock** — based on comparison of ATP against Stock Policy thresholds.
Context	Inventory Balance
Related Terms	Stock Position, Stock Policy, Operational Alert
Terms That Should Never Be Confused	**Product Lifecycle Status** — active/inactive/discontinued, not quantity health; **Quarantine Hold** — specific hold status, not overall stock health

Lot/Batch

Field	Detail
Definition	A traceable production or receipt grouping for products requiring lot-level tracking, carrying expiration date, manufacture date, and quality attributes for FEFO rotation and recall scope.
Context	Inventory Balance
Related Terms	Product, Rotation Policy, Goods Receipt, Fulfillment Execution
Terms That Should Never Be Confused	**SKU** — product identity, not production grouping; **Serial Number** — individual item tracking, finer granularity than lot (out of MVP scope)

Quarantine Hold

Field	Detail
Definition	A business hold excluding physically present stock from sellable ATP due to quality investigation, regulatory restriction, product recall, or pending customs clearance.
Context	Inventory Balance
Related Terms	Available-to-Promise, Stock Position, Bonded Inventory
Terms That Should Never Be Confused	**Location Freeze** — suspends all location activity, not specific stock; **Reserved Stock** — committed to orders, not under investigation; **Inactive Product** — commercial status, not physical hold

Bonded Inventory

Field	Detail
Definition	Stock physically present but under customs control — duties unpaid — that must not be released to domestic sellable availability until customs clearance is confirmed.
Context	Inventory Balance, Inbound Receiving
Related Terms	Quarantine Hold, Put-Away, Ownership Classification
Terms That Should Never Be Confused	**Quarantine Hold** — broader hold category including quality and recall, not specifically customs; **Consignment Stock** — supplier-owned until consumption, not customs-controlled

Consignment Stock

Field	Detail
Definition	Inventory physically at the organization's location but owned by the supplier until a defined consumption or sale event per supplier agreement. Not treated as owned stock for valuation until ownership transfers.
Context	Inventory Balance, Procurement & Supplier

Field	Detail
Related Terms	Supplier, Ownership Classification, Inventory Valuation Snapshot
Terms That Should Never Be Confused	**On-Hand Stock (Owned)** — organization balance-sheet asset; **Bonded Inventory** — customs-controlled, not supplier-owned; **In-Transit Stock** — in movement, not ownership question

Ownership Classification

Field	Detail
Definition	The legal and financial ownership state of held inventory: **Owned**, **Consignment**, or **Bonded** — governing valuation treatment and availability rules.
Context	Inventory Balance
Related Terms	Consignment Stock, Bonded Inventory, Inventory Valuation Snapshot
Terms That Should Never Be Confused	**Product Lifecycle Status** — commercial availability, not legal ownership; **Stock Status Indicator** — quantity health, not ownership type

Section 5: Procurement & Supplier (PS)

Supplier

Field	Detail
Definition	An external vendor from whom the organization procures goods, identified by Supplier Code, carrying commercial terms including payment terms, lead time, and consignment agreement indicator.
Context	Procurement & Supplier
Related Terms	Purchase Order, Supplier Invoice, Goods Receipt
Terms That Should Never Be Confused	**Carrier/Logistics Partner** — transports goods, does not supply them; **Customer** — receives outbound goods, not inbound supplier

Purchase Requisition

Field	Detail
Definition	An internal request to procure stock — specifying product, quantity, and target location — submitted for approval before commercial Purchase Order creation.
Context	Procurement & Supplier
Related Terms	Purchase Order, Replenishment, Operational Alert
Terms That Should Never Be Confused	**Purchase Order** — authorized external commercial commitment to supplier, not internal request; **Transfer Request** — internal network redistribution, not external procurement

Purchase Order (PO)

Field	Detail
Definition	An authorized commercial commitment to a supplier specifying products, quantities, prices, delivery terms, and receiving location. Required before supplier goods receipt except approved ad-hoc exceptions.
Context	Procurement & Supplier
Related Terms	Supplier, Purchase Requisition, Goods Receipt, Expected Inbound Delivery

Field	Detail
Terms That Should Never Be Confused	**Purchase Requisition** — internal pre-approval request, not supplier-facing commitment; **Customer Order** — outbound demand, not inbound supply; **Goods Receipt** — physical arrival event, not commercial authorization

Expected Inbound Delivery

Field	Detail
Definition	A planned or confirmed supplier shipment tied to a Purchase Order, communicating anticipated arrival date and quantity for receiving preparation and pipeline visibility.
Context	Procurement & Supplier
Related Terms	Purchase Order, Goods Receipt, Inbound Expected
Terms That Should Never Be Confused	**Goods Receipt** — actual verified arrival, not plan; **In-Transit Stock** — inter-location movement within network, not supplier inbound; **Inbound Expected (ATP view)** — informational pipeline quantity, does not increase ATP

Lead Time

Field	Detail
Definition	The elapsed business time from purchase order placement until goods are available for sale or use at the receiving location, accounting for supplier processing, transit, receipt, and put-away.
Context	Procurement & Supplier, Inventory Balance
Related Terms	Purchase Order, Stock Policy, Reorder Level
Terms That Should Never Be Confused	**Transit Time** — shipping duration only, subset of lead time; **Fulfillment Cycle Time** — order-to-ship for customers, not supplier inbound

Section 6: Inbound Receiving (IR)

Goods Receipt

Field	Detail
Definition	The business event of verifying and accepting inbound goods — confirming quantity, product identity, and condition against Purchase Order or authorized exception — before stock enters inventory control.
Context	Inbound Receiving
Related Terms	Purchase Order, Receipt Discrepancy, Put-Away, Supplier
Terms That Should Never Be Confused	**Put-Away** — placement in storage after receipt, separate step; **Stock Received** — balance increase after put-away complete, not dock verification; **Customer Return** — inbound from customer, not supplier

Receipt Discrepancy

Field	Detail
Definition	A documented difference between expected and actual receipt — short shipment, over-shipment, wrong item, damage, or shelf-life non-compliance — captured with evidence at the point of receipt.
Context	Inbound Receiving
Related Terms	Goods Receipt, Reason Code, Three-Way Match
Terms That Should Never Be Confused	**Count Variance** — system vs. physical count during cycle/physical count, not receipt vs. PO; **Stock Adjustment** — correction after investigation, not receipt-time exception

Put-Away

Field	Detail
Definition	The confirmation that verified received goods have been moved from the receiving dock to a designated storage position. Stock is not available for sale or fulfillment until put-away is complete.
Context	Inbound Receiving
Related Terms	Goods Receipt, Storage Position, Stock Received, Rotation Policy
Terms That Should Never Be Confused	**Goods Receipt** — verification at dock, not storage placement; **Pick** — retrieval for outbound order, opposite direction; **Stock Received** — domain outcome after put-away increases on-hand

Minimum Remaining Shelf Life (MRSL)

Field	Detail
Definition	The minimum acceptable remaining shelf life at receipt for shelf-life products — deliveries not meeting MRSL must be rejected or dispositioned before acceptance into sellable inventory.
Context	Inbound Receiving, Product Catalog
Related Terms	Rotation Policy (FEFO), Lot/Batch, Receipt Discrepancy
Terms That Should Never Be Confused	**Expiration Date** — absolute date on lot, not the policy threshold; **FEFO** — pick rotation rule at fulfillment, not receipt acceptance criterion

Supplier Invoice

Field	Detail
Definition	The supplier's billing document for goods supplied, matched against Purchase Order and Goods Receipt in three-way match before payment authorization.
Context	Inbound Receiving
Related Terms	Three-Way Match, Purchase Order, Goods Receipt
Terms That Should Never Be Confused	**Purchase Order** — authorization to buy, not bill; **Goods Receipt** — operational receipt record, not financial document; **Inventory Valuation Snapshot** — period-end value summary, not supplier bill

Three-Way Match

Field	Detail
Definition	Financial control confirming agreement among Purchase Order (what was ordered), Goods Receipt (what was received), and Supplier Invoice (what was billed) within defined tolerance before payment is authorized.
Context	Inbound Receiving
Related Terms	Supplier Invoice, Purchase Order, Goods Receipt, Governance Policy
Terms That Should Never Be Confused	**Inventory Valuation Reconciliation** — operational stock value vs. general ledger, not PO-receipt-invoice match; **Receipt Discrepancy** — operational variance at dock, not financial match outcome

Section 7: Order Fulfillment (OF)

Customer Order

Field	Detail
Definition	A customer demand for products triggering availability confirmation, reservation, and fulfillment execution. IMS holds fulfillment-relevant order context; authoritative order data may reside in external sales systems.
Context	Order Fulfillment
Related Terms	Inventory Reservation, Fulfillment Execution, Shipment, Backorder
Terms That Should Never Be Confused	**Purchase Order** — inbound supplier commitment, not customer demand; **Shipment** — dispatch event, not the order itself; **Transfer Request** — internal stock movement, not customer sale

Inventory Reservation

Field	Detail
Definition	An operational hold on stock for a confirmed customer order line, increasing reserved quantity and decreasing ATP from the moment of order confirmation until fulfillment or release.
Context	Order Fulfillment, Inventory Balance
Related Terms	Reserved Stock, Customer Order, Available-to-Promise
Terms That Should Never Be Confused	**Backorder** — agreement to wait for unavailable quantity, not a hold on existing stock; **Promotional Allocation** — campaign ring-fence, not order-specific hold; **Quarantine Hold** — quality/regulatory exclusion, not customer commitment

Fulfillment Execution

Field	Detail
Definition	The warehouse work of picking, verifying, packing, and preparing order lines for dispatch — applying FIFO/FEFO rotation and capturing lot/batch for traceable products.
Context	Order Fulfillment
Related Terms	Pick Verification, Shipment, Customer Order, Storage Position
Terms That Should Never Be Confused	**Shipment** — carrier handoff and dispatch, after pick verification; **Goods Receipt** — inbound operation, not outbound; **Inventory Reservation** — quantity hold, not physical warehouse work

Pick Verification

Field	Detail
Definition	Confirmation that physically retrieved items match the order line in product, quantity, and lot/batch — required before shipment recording.
Context	Order Fulfillment
Related Terms	Fulfillment Execution, Shipment, Rotation Policy
Terms That Should Never Be Confused	**Goods Receipt Verification** — inbound dock check, not outbound pick check; **Cycle Count** — accuracy audit, not order fulfillment step

Shipment

Field	Detail
Definition	The confirmed dispatch of verified goods to a carrier for customer delivery, triggering on-hand deduction and reservation consumption at the ship-from location.
Context	Order Fulfillment

Field	Detail
Related Terms	Fulfillment Execution, Pick Verification, Customer Order, Stock Shipped
Terms That Should Never Be Confused	**Expected Inbound Delivery** — supplier shipment inbound, not customer outbound; **Transfer Dispatch** — inter-location movement, not customer delivery; **Fulfillment Execution** — warehouse work including pick and pack, not carrier handoff

Backorder

Field	Detail
Definition	A customer-agreed delay to fulfill remaining order quantity when full quantity is unavailable at confirmation — requiring explicit customer acknowledgment before recording. A mutual commercial commitment, not an internal note.
Context	Order Fulfillment
Related Terms	Partial Fulfillment, Customer Order, Inventory Reservation
Terms That Should Never Be Confused	**Inventory Reservation** — hold on existing available stock; **Purchase Order** — inbound supply order to replenish, not customer-facing delay; **Out of Stock** — condition, not customer agreement

Partial Fulfillment

Field	Detail
Definition	Shipping the available portion of an order line when full quantity cannot be met, with remaining quantity handled via backorder or separate fulfillment.
Context	Order Fulfillment
Related Terms	Backorder, Shipment, Customer Order
Terms That Should Never Be Confused	**Split Fulfillment** — shipping from multiple locations, not partial quantity; **Backorder** — the unfulfilled remainder agreement, not the act of shipping partial quantity

Multi-Location Split Fulfillment

Field	Detail
Definition	Fulfilling a single customer order from multiple inventory locations when no single location holds sufficient ATP for the full order quantity.
Context	Order Fulfillment
Related Terms	Customer Order, Inventory Reservation, Stock Transfer
Terms That Should Never Be Confused	**Partial Fulfillment** — less than full quantity from one location, not multiple locations; **Stock Transfer** — internal redistribution, not customer-directed fulfillment

Section 8: Network Transfer (NT)

Transfer Request

Field	Detail
Definition	A formal request to move stock from a source location to a destination location, specifying product and quantity, evaluated against network availability and source minimum thresholds before approval.
Context	Network Transfer
Related Terms	Stock Transfer, Network Rebalancing, Purchase Requisition

Field	Detail
Terms That Should Never Be Confused	**Purchase Requisition** — request to buy from supplier, not move existing stock; **Customer Order** — outbound customer demand, not internal redistribution; **Stock Transfer** — approved and executing movement, not the request

Stock Transfer

Field	Detail
Definition	The executed movement of stock between inventory locations — from dispatch at source through in-transit tracking to receipt at destination — with full transaction history.
Context	Network Transfer
Related Terms	In-Transit Stock, Transfer Request, Stock Position
Terms That Should Never Be Confused	**Shipment** — outbound to customer, not inter-location; **Goods Receipt** — inbound from supplier, not from another owned location; **Transfer Request** — pre-approval intent, not executed movement

In-Transit Stock

Field	Detail
Definition	Inventory that has left the source location but has not yet been received at the destination — excluded from both source and destination ATP until receipt is confirmed.
Context	Network Transfer
Related Terms	Stock Transfer, Expected Inbound Delivery
Terms That Should Never Be Confused	**Expected Inbound Delivery** — supplier pipeline inbound, not inter-location; **On-Hand Stock** — at a location, not in movement; **Inbound Expected (ATP view)** — informational only for supplier supply

Network Rebalancing

Field	Detail
Definition	Proactive redistribution of stock across the location network to resolve simultaneous stockout risk at one site and excess at another — preferring transfer before new procurement.
Context	Network Transfer, Inventory Balance
Related Terms	Transfer Request, Stock Transfer, Network Availability
Terms That Should Never Be Confused	**Replenishment** — restocking from external supplier, not internal redistribution; **Stock Adjustment** — quantity correction, not planned movement

Section 9: Inventory Accuracy (IA)

Cycle Count

Field	Detail
Definition	A scheduled partial physical count of selected products or locations on a rotating basis, performed blind (counter not shown system quantity) without full warehouse shutdown.
Context	Inventory Accuracy
Related Terms	Count Variance, Physical Inventory Count, ABC Classification
Terms That Should Never Be Confused	**Physical Inventory Count** — full wall-to-wall count with location freeze, not partial rotation; **Goods Receipt** — inbound verification, not accuracy audit

Physical Inventory Count

Field	Detail
Definition	A comprehensive wall-to-wall count of all products at a location, imposing location freeze until certified complete — establishing an authoritative accuracy baseline.
Context	Inventory Accuracy, Warehouse Network
Related Terms	Location Freeze, Count Variance, Cycle Count
Terms That Should Never Be Confused	**Cycle Count** — partial rotating count without full shutdown; **Opening Balance** — initial stock at go-live, not periodic audit

Count Variance

Field	Detail
Definition	An identified difference between system-recorded quantity and physically counted quantity, requiring investigation and independent approval before any Stock Adjustment is applied.
Context	Inventory Accuracy
Related Terms	Cycle Count, Physical Inventory Count, Stock Adjustment, Segregation of Duties
Terms That Should Never Be Confused	**Receipt Discrepancy** — expected vs. received at supplier inbound, not system vs. physical during count; **Stock Adjustment** — authorized correction, not the discovered discrepancy

Stock Adjustment

Field	Detail
Definition	An authorized correction to recorded inventory quantity with mandatory reason code — increasing or decreasing on-hand to align record with physical reality after investigation.
Context	Inventory Accuracy, Inventory Balance
Related Terms	Reason Code, Count Variance, Inventory Write-Off
Terms That Should Never Be Confused	**Inventory Write-Off** — permanent value removal, not reversible correction; **Stock Transfer** — movement between locations, not quantity correction; **Reservation** — customer hold, not record correction

Inventory Write-Off

Field	Detail
Definition	The formal, authorized permanent removal of inventory quantity and value for damaged, obsolete, expired, or lost goods — distinct from reversible adjustment.
Context	Inventory Accuracy, Inventory Balance
Related Terms	Reason Code, Stock Adjustment, Shrinkage, Return Disposition
Terms That Should Never Be Confused	**Stock Adjustment** — record correction that may increase or decrease, not terminal removal; **Return Disposition (Write-Off)** — write-off of returned goods specifically; **Product Discontinued** — commercial exit, not physical removal

Shrinkage

Field	Detail
Definition	Unexplained or unaccounted inventory loss from theft, damage, misplacement, or process error — analyzed through reason-coded adjustments and write-offs over time.
Context	Inventory Accuracy, Inventory Intelligence

Field	Detail
Related Terms	Stock Adjustment, Inventory Write-Off, Reason Code, Inventory Accuracy
Terms That Should Never Be Confused	**Count Variance** — specific discovered discrepancy at count time, not ongoing loss trend; **Write-Off** — individual removal event, not the aggregate loss pattern

Inventory Accuracy

Field	Detail
Definition	The degree of agreement between system-recorded quantities and physically counted stock, measured through cycle counts and physical inventory counts — expressed as a percentage or variance rate.
Context	Inventory Accuracy, Inventory Intelligence
Related Terms	Cycle Count, Count Variance, Physical Inventory Count
Terms That Should Never Be Confused	**Fill Rate** — order fulfillment performance, not record-vs-physical agreement; **Inventory Turnover** — velocity of stock movement, not accuracy

Section 10: Returns Management (RM)

Return Authorization (RMA)

Field	Detail
Definition	Approved permission for a customer to return specified products, linked to the original customer order and return policy, before the warehouse accepts physical return.
Context	Returns Management
Related Terms	Customer Return, Customer Order, Return Disposition
Terms That Should Never Be Confused	**Customer Return** — physical receipt of goods, after RMA issued; **Purchase Order** — inbound from supplier, not customer return; **Backorder** — fulfillment delay, not return permission

Customer Return

Field	Detail
Definition	The physical receipt of customer-returned goods at a warehouse, verified against a valid Return Authorization, initiating inspection and disposition workflow.
Context	Returns Management
Related Terms	Return Authorization, Return Disposition, Goods Receipt
Terms That Should Never Be Confused	**Goods Receipt** — supplier inbound, not customer return; **Return Authorization** — permission to return, not physical arrival; **Shipment** — outbound to customer, opposite direction

Return Disposition

Field	Detail
Definition	The business outcome decision for returned goods after condition inspection: **Restock** (return to sellable inventory), **Quarantine** (further investigation), or **Write-Off** (permanent removal).
Context	Returns Management
Related Terms	Customer Return, Condition Grade, Quarantine Hold, Inventory Write-Off

Field	Detail
Terms That Should Never Be Confused	**Reason Code** — categorized explanation for adjustments generally, not return-specific outcome; **Stock Adjustment** — quantity correction mechanism, not inspection decision

Condition Grade

Field	Detail
Definition	The assessed physical and resale condition of received or returned goods at inspection — determining whether restock, quarantine, or write-off disposition applies.
Context	Returns Management, Inbound Receiving
Related Terms	Return Disposition, Goods Receipt, Receipt Discrepancy
Terms That Should Never Be Confused	**Reason Code** — why an adjustment occurred, not product condition assessment; **Product Lifecycle Status** — active/discontinued, not physical condition

Section 11: Commercial Campaign (CC)

Promotional Stock Allocation

Field	Detail
Definition	A defined quantity of inventory ring-fenced for a specific promotional campaign or product launch, excluded from general ATP for non-approved orders during the campaign period.
Context	Commercial Campaign
Related Terms	Campaign Period, Channel Designation, Inventory Reservation
Terms That Should Never Be Confused	**Inventory Reservation** — hold for specific customer order, not campaign pool; **Safety Stock** — uncertainty buffer, not commercial ring-fence; **Stock Policy Maximum** — overstock limit, not campaign allocation

Campaign Period

Field	Detail
Definition	The bounded date range during which promotional stock allocation is active and protected from non-promotional consumption.
Context	Commercial Campaign
Related Terms	Promotional Stock Allocation, Promotional Readiness
Terms That Should Never Be Confused	**Lead Time** — supply timing, not marketing campaign window; **Effective Period (Governance Policy)** — policy validity dates, not campaign dates

Promotional Readiness

Field	Detail
Definition	Confirmation that ring-fenced promotional inventory is sufficient and available at fulfillment locations before campaign or launch commitment deadline.
Context	Commercial Campaign, Inventory Intelligence
Related Terms	Promotional Stock Allocation, Operational Alert
Terms That Should Never Be Confused	**Available-to-Promise** — general sellable quantity, not campaign-specific readiness; **Inbound Expected** — supplier pipeline, not campaign stock confirmation

Section 12: Identity & Access (IAc)

User Account

Field	Detail
Definition	An identified individual authorized to perform inventory operations, carrying role assignments and location scope for accountability on all transactions.
Context	Identity & Access
Related Terms	Role Assignment, Segregation of Duties, Audit Trail
Terms That Should Never Be Confused	**Supplier** — external vendor, not system user; **Customer** — external buyer, not internal operator; **Actor (BRD)** — business role concept, User Account is the system identity

Role Assignment

Field	Detail
Definition	The delegation of business permissions to a user account — specifying role (e.g., Receiving Clerk, Warehouse Manager) and authorized location scope.
Context	Identity & Access
Related Terms	User Account, Segregation of Duties, Inventory Location
Terms That Should Never Be Confused	**Job Title** — HR designation, not system permission set; **Actor** — business process participant, not access control record

Section 13: Inventory Intelligence (II)

Inventory Movement Record

Field	Detail
Definition	An immutable operational record of a single inventory quantity or status change — capturing product, location, quantity, direction, movement type, reason, user, and timestamp.
Context	Inventory Intelligence
Related Terms	Audit Trail, Stock Position, Shipment, Goods Receipt
Terms That Should Never Be Confused	**Audit Trail Record** — governance-grade before/after with approval chain, richer than operational movement; **Stock Transfer** — specific movement type, not the log concept; **Transaction** — technical term, avoid in business conversation

Audit Trail Record

Field	Detail
Definition	A governance-grade immutable record of a material inventory change including user identity, role at time of action, before/after state, and approval chain — supporting compliance and control testing.
Context	Inventory Intelligence
Related Terms	Inventory Movement Record, Segregation of Duties, Governance Policy
Terms That Should Never Be Confused	**Inventory Movement Record** — operational history of quantity changes, not necessarily full governance metadata; **System Log** — technical diagnostic output, not business audit evidence

Operational Alert

Field	Detail
Definition	A proactive notification of a business condition requiring human review — such as low stock, pending approval, transfer exception, inbound delay, or campaign readiness gap.
Context	Inventory Intelligence
Related Terms	Stock Status Indicator, Replenishment, Transfer Request
Terms That Should Never Be Confused	**Stock Status Indicator** — classification (low stock), not the notification itself; **Business Rule** — policy definition, not triggered alert instance

Inventory Valuation Snapshot

Field	Detail
Definition	A point-in-time summary of owned inventory value by product and location for finance reconciliation against the general ledger — excluding consignment, quarantined, and in-transit stock per policy.
Context	Inventory Intelligence
Related Terms	Three-Way Match, Consignment Stock, Inventory Write-Off
Terms That Should Never Be Confused	**On-Hand Stock** — operational quantity, not financial value; **Supplier Invoice** — supplier bill for specific receipt, not period-end valuation; **Inventory Turnover** — velocity metric, not point-in-time value

Replenishment

Field	Detail
Definition	The business process of restocking products to maintain target inventory levels — triggered at reorder level through purchase requisition, transfer request, or network rebalancing.
Context	Inventory Balance, Procurement & Supplier, Network Transfer
Related Terms	Reorder Level, Purchase Requisition, Transfer Request, Low Stock
Terms That Should Never Be Confused	**Purchase Order** — one replenishment mechanism (external buy), not the entire process; **Put-Away** — inbound storage step, not replenishment planning

Fill Rate

Field	Detail
Definition	The percentage of customer order lines fulfilled in full on the first shipment — a fulfillment performance KPI, not an inventory quantity measure.
Context	Inventory Intelligence, Order Fulfillment
Related Terms	Partial Fulfillment, Shipment, Customer Order
Terms That Should Never Be Confused	**Inventory Accuracy** — record vs. physical agreement, not order fulfillment performance; **Available-to-Promise** — current quotable quantity, not historical performance metric

Inventory Turnover

Field	Detail
Definition	A financial performance metric expressing how quickly inventory is sold and replaced — cost of goods sold divided by average inventory value over a period.
Context	Inventory Intelligence

Field	Detail
Related Terms	Inventory Valuation Snapshot, Carrying Cost
Terms That Should Never Be Confused	**Inventory Movement Record** — transaction history, not velocity metric; **Fill Rate** — order fulfillment completeness, not stock rotation speed

Section 14: Cross-Context Published Language (ALL)

These identifiers are shared across all bounded contexts as the **Published Language** for cross-context references.

Location Code

Field	Detail
Definition	The unique business identifier for an Inventory Location, used by all contexts to anchor stock, transactions, and authorization scope to a physical site.
Context	All contexts
Related Terms	Inventory Location, Storage Position, Stock Position
Terms That Should Never Be Confused	**Storage Position Code** — identifier within a location, not the location itself; **Supplier Code** — external vendor identifier

Product SKU Reference

Field	Detail
Definition	The cross-context reference to a Product by SKU — the standard way all contexts identify *what* product is involved without embedding full product master data.
Context	All contexts
Related Terms	SKU, Product, Stock Position
Terms That Should Never Be Confused	**Barcode** — lookup identifier that resolves to SKU, not the canonical key itself; **Lot/Batch Number** — traceability sub-identity within a SKU

Critical Distinction Reference

High-risk confusions that cause operational and financial errors if terminology is mixed:

Never Confuse	With	Key Difference
Available-to-Promise (ATP)	On-Hand Stock	ATP excludes reserved and quarantined; on-hand is total physical
Available-to-Promise (ATP)	Inbound Expected	Inbound is pipeline information; ATP is physically available now
Goods Receipt	Put-Away	Receipt verifies at dock; put-away confirms storage placement
Put-Away	Stock Received	Put-away is the action; stock received is the balance outcome
Inventory Reservation	Backorder	Reservation holds existing stock; backorder agrees to wait for unavailable quantity
Inventory Reservation	Promotional Allocation	Reservation is order-specific; allocation is campaign ring-fence
Stock Adjustment	Inventory Write-Off	Adjustment corrects record; write-off permanently removes value
Stock Adjustment	Stock Transfer	Adjustment changes quantity in place; transfer moves between locations
Count Variance	Receipt Discrepancy	Variance is system vs. physical at count; discrepancy is expected vs. received at inbound
Purchase Requisition	Purchase Order	Requisition is internal approval request; PO is supplier commitment
Purchase Order	Customer Order	PO is inbound from supplier; customer order is outbound demand

Never Confuse	With	Key Difference
Transfer Request	Stock Transfer	Request is intent pending approval; transfer is executing movement
Shipment	Stock Transfer	Shipment goes to customer; transfer goes between owned locations
Cycle Count	Physical Inventory Count	Cycle count is partial rotation; physical count is full wall-to-wall with freeze
Quarantine Hold	Location Freeze	Hold excludes specific stock from ATP; freeze suspends all location activity
Quarantine Hold	Reserved Stock	Hold is quality/regulatory exclusion; reserved is customer commitment
Bonded Inventory	Consignment Stock	Bonded is customs-controlled; consignment is supplier-owned until consumption
Reason Code	Disposition Decision	Reason code categorizes why; disposition decides return outcome
Stock Policy	Governance Policy	Stock policy is replenishment thresholds; governance is enterprise control rules
Product Lifecycle Status	Stock Status Indicator	Lifecycle is active/discontinued; status indicator is in/low/out of stock
Return Authorization	Customer Return	RMA is permission; customer return is physical receipt
Three-Way Match	Valuation Reconciliation	Three-way match is PO-receipt-invoice; valuation is operational stock vs. GL
Inventory Movement Record	Audit Trail Record	Movement is operational log; audit trail adds governance before/after and approval
Fill Rate	Inventory Accuracy	Fill rate is order fulfillment KPI; accuracy is record vs. physical agreement

Term Index (Alphabetical)

Term	Context	Section
ABC Classification	Product Catalog	1
Approval Threshold	Inventory Governance	3
Audit Trail Record	Inventory Intelligence	13
Available-to-Promise (ATP)	Inventory Balance	4
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Bonded Inventory	Inventory Balance	4
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Cycle Count	Inventory Accuracy	9
Expected Inbound Delivery	Procurement & Supplier	5
Fill Rate	Inventory Intelligence	13
Fulfillment Execution	Order Fulfillment	7
Goods Receipt	Inbound Receiving	6
Governance Policy	Inventory Governance	3
In-Transit Stock	Network Transfer	8
Inventory Accuracy	Inventory Accuracy	9
Inventory Location	Warehouse Network	2
Inventory Movement Record	Inventory Intelligence	13
Inventory Reservation	Order Fulfillment	7

Term	Context	Section
Inventory Turnover	Inventory Intelligence	13
Inventory Valuation Snapshot	Inventory Intelligence	13
Inventory Write-Off	Inventory Accuracy	9
Lead Time	Procurement & Supplier	5
Location Code	All contexts	14
Location Freeze	Warehouse Network	2
Lot/Batch	Inventory Balance	4
Minimum Remaining Shelf Life (MRSL)	Inbound Receiving	6
Multi-Location Split Fulfillment	Order Fulfillment	7
Network Rebalancing	Network Transfer	8
On-Hand Stock	Inventory Balance	4
Operational Alert	Inventory Intelligence	13
Ownership Classification	Inventory Balance	4
Partial Fulfillment	Order Fulfillment	7
Physical Inventory Count	Inventory Accuracy	9
Pick Verification	Order Fulfillment	7
Product	Product Catalog	1
Product Lifecycle Status	Product Catalog	1
Promotional Readiness	Commercial Campaign	11
Promotional Stock Allocation	Commercial Campaign	11
Purchase Order (PO)	Procurement & Supplier	5
Purchase Requisition	Procurement & Supplier	5
Put-Away	Inbound Receiving	6
Quarantine Hold	Inventory Balance	4
Reason Code	Inventory Governance	3
Receipt Discrepancy	Inbound Receiving	6
Replenishment	Multiple	13
Reorder Level	Inventory Balance	4
Reserved Stock	Inventory Balance	4
Return Authorization (RMA)	Returns Management	10
Return Disposition	Returns Management	10
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Safety Stock	Inventory Balance	4
Segregation of Duties (SoD)	Inventory Governance	3
Shipment	Order Fulfillment	7
Shrinkage	Inventory Accuracy	9
SKU	Product Catalog	1
Stock Adjustment	Inventory Accuracy	9
Stock Policy	Inventory Balance	4

Term	Context	Section
Stock Position	Inventory Balance	4
Stock Status Indicator	Inventory Balance	4
Stock Transfer	Network Transfer	8
Storage Position	Warehouse Network	2
Supplier	Procurement & Supplier	5
Supplier Invoice	Inbound Receiving	6
Three-Way Match	Inbound Receiving	6
Transfer Request	Network Transfer	8
Unit of Measure (UOM)	Product Catalog	1
User Account	Identity & Access	12

Total Terms: 72

Document Approval

Role	Name	Signature	Date
Domain Expert			
Senior Business Analyst			
Product Owner			
Inventory Controller			

End of Ubiquitous Language

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Bounded Contexts

Inventory Management System

Document Information

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Date	July 5, 2026
Scope	Strategic domain decomposition only — no microservice deployment, API design, or database partitioning

Purpose

This document defines the **Bounded Contexts** of the Inventory Management System — the explicit boundaries within which a domain model is unified, internally consistent, and expressed in a distinct ubiquitous language. Each context owns its aggregates and entities, exposes well-defined relationships to neighboring contexts, and can evolve independently without corrupting models in adjacent domains.

Bounded contexts are a **strategic design** decision. They reflect how the business organizes inventory capability — not necessarily how software is deployed in the first release.

Context Map Legend

Relationship	Meaning in IMS
Upstream → Downstream	Upstream context defines the model; downstream conforms or translates
Shared Kernel	Small shared model both contexts depend on (minimal, stable identifiers)
Customer-Supplier	Downstream (customer) negotiates needs with upstream (supplier) context
Conformist	Downstream accepts upstream model without translation
Anti-Corruption Layer (ACL)	Downstream translates foreign model to protect its own integrity
Open Host Service (OHS)	Context publishes a well-defined interface for others to consume
Published Language	Shared interchange format (e.g., SKU + Location Code identifiers)

Context Overview

#	Bounded Context	Aggregates Owned	Primary Business Processes
1	Product Catalog	Product Catalog	BP-01, BP-04
2	Warehouse Network	Inventory Location	BP-02
3	Inventory Governance	Inventory Governance	BP-01, BP-03, BP-32
4	Inventory Balance	Stock Position, Stock Policy	BP-03, BP-10, BP-21
5	Procurement & Supplier	Supplier, Purchase Requisition, Purchase Order	BP-05, BP-06, BP-22, BP-23
6	Inbound Receiving	Goods Receipt, Supplier Invoice	BP-07, BP-08, BP-09, BP-30
7	Order Fulfillment	Customer Order Fulfillment	BP-16 – BP-20
8	Network Transfer	Stock Transfer	BP-24 – BP-26
9	Inventory Accuracy	Cycle Count, Physical Inventory Count, Inventory Adjustment Case	BP-11 – BP-15
10	Returns Management	Customer Return Case	BP-27, BP-28
11	Commercial Campaign	Promotional Stock Allocation	BP-31
12	Identity & Access	User Account	Cross-cutting
13	Inventory Intelligence	*(Read models — no write aggregates)*	BP-29, BP-30, BP-32

Total Bounded Contexts: 13

Context 1: Product Catalog

Business Responsibility

Maintain the authoritative product identity catalog — SKU definitions, lifecycle status, classification (ABC), rotation and traceability attributes — that all other inventory contexts reference when identifying *what* is being bought, held, counted, or sold.

Owned Entities

Entity / Aggregate	Role in Context
Product *(aggregate root)*	SKU master, lifecycle, ABC class, rotation policy, traceability rules
Product Identifier, Product Category, Product Lifecycle Status, ABC Classification, Rotation Policy, Shelf Life Profile *(value objects)*	Embedded catalog attributes

Related Use Cases

Use Case	Name
UC-001	Manage Product Master Data
UC-004	Classify Products by ABC Priority
UC-046	Import Initial Product and Stock Master Data *(product portion)*

Relationships with Other Contexts

Related Context	Relationship	Integration
Inventory Balance	**Upstream → Downstream** (Product Catalog supplies)	Product SKU referenced by Stock Position and Stock Policy; Product Transaction Eligibility validates lifecycle before transactions
Procurement & Supplier	**Upstream → Downstream**	PO and requisition lines reference SKU; discontinued status blocks new orders
Order Fulfillment	**Upstream → Downstream**	Order lines reference SKU; inactive/discontinued products blocked from reservation
Inbound Receiving	**Upstream → Downstream**	Receipt verification validates product identity and MRSL policy
Inventory Accuracy	**Upstream → Downstream**	Cycle count scoped by product; ABC class drives count frequency
Inventory Intelligence	**Open Host Service**	Product catalog published for reporting and search

Why This Context Should Be Independent

Product master data changes **infrequently** and is owned by the **Inventory Controller** function, while stock quantities change **continuously** in warehouse operations. Merging catalog and balance into one model causes high-frequency transactional code to couple with low-frequency master data governance. Product lifecycle rules (discontinued, ABC) are assortment decisions — not warehouse execution decisions. Independent evolution allows catalog enrichment (attributes, classification) without redeploying fulfillment or receiving logic.

Context 2: Warehouse Network

Business Responsibility

Define and manage the inventory location network — warehouses, stores, distribution points, storage positions, hierarchy, operational status, and location lifecycle (go-live, closure) — that anchors all stock to physical places.

Owned Entities

Entity / Aggregate	Role in Context
Inventory Location *(aggregate root)*	Location master, hierarchy, status, freeze state
Storage Position *(entity)*	Zone, aisle, bin within location
Physical Address, Location Type, Storage Position Address *(value objects)*	Location descriptors

Related Use Cases

Use Case	Name
UC-002	Configure Inventory Locations
UC-046	Import Initial Product and Stock Master Data *(location portion)*
UC-015	Conduct Full Physical Inventory Count *(freeze enforcement)*

Relationships with Other Contexts

Related Context	Relationship	Integration
Inventory Balance	**Upstream → Downstream**	Stock Position anchored to Location Code; location must be active for transactions
Inbound Receiving	**Upstream → Downstream**	Receiving location and put-away storage positions
Order Fulfillment	**Upstream → Downstream**	Ship-from location; location freeze blocks shipment
Network Transfer	**Upstream → Downstream**	Source and destination locations
Inventory Accuracy	**Customer-Supplier**	Physical count imposes freeze on location; location context owns freeze state
Identity & Access	**Upstream → Downstream**	Role assignments scoped to location codes

Why This Context Should Be Independent

Location network structure is **organizational master data** — new warehouses, closures, and layout changes follow a different lifecycle and approval path than daily stock movements. Storage position configuration is a **facilities concern** distinct from quantity accounting. Location freeze during physical count (BR-024) is a network-wide operational control that must be authoritative without embedding freeze logic inside every transactional context. Separating network setup allows geographic expansion without modifying procurement or fulfillment domain models.

Context 3: Inventory Governance

Business Responsibility

Maintain organization-wide inventory control policies — reason codes, approval thresholds, tolerance limits, segregation-of-duties rules, and exception policies — that govern how operational contexts authorize adjustments, receipts, and financial controls.

Owned Entities

Entity / Aggregate	Role in Context
Governance Policy *(aggregate root)*	Approval thresholds, tolerances, effective periods
Reason Code *(entity)*	Standardized exception and adjustment taxonomy
Approval Threshold, Quantity Tolerance, Effective Period *(value objects)*	Policy parameters

Related Use Cases

Use Case	Name
UC-005	Maintain Governance Policies and Reason Codes
UC-041	Conduct Inventory Audit and Control Review *(policy compliance)*
UC-045	Enforce Role-Based and Location-Scoped Access *(SoD rules)*

Relationships with Other Contexts

Related Context	Relationship	Integration
Inventory Accuracy	**Upstream → Downstream**	Adjustment and count approval consult governance thresholds and SoD rules

Related Context	Relationship	Integration
Inbound Receiving	**Upstream → Downstream**	Receipt tolerance and ad-hoc exception policies
Procurement & Supplier	**Upstream → Downstream**	Three-way match tolerance limits
Identity & Access	**Shared Kernel** *(SoD rules)*	Governance defines SoD constraints; Identity & Access enforces at role assignment
All operational contexts	**Open Host Service**	Published reason code catalog and policy lookup

Why This Context Should Be Independent

Governance policies are **enterprise control standards** set by operations leadership and audit — not warehouse floor concerns. They change on **audit cycles and policy reviews**, independent of daily receipts or shipments. Centralizing reason codes and thresholds prevents each operational context from embedding its own (potentially inconsistent) approval rules. When audit findings require threshold changes, only this context updates — all consumers re-evaluate against the new policy without model changes in fulfillment or receiving.

Context 4: Inventory Balance

Business Responsibility

Own the authoritative inventory quantity state — on-hand, reserved, available-to-promise (ATP), quarantine holds, lot/batch breakdown, and replenishment thresholds — answering *what do we have, where is it, and what is available?*

Owned Entities

Entity / Aggregate	Role in Context
Stock Position *(aggregate root)*	On-hand, reserved, ATP, ownership classification
Stock Policy *(aggregate root)*	Reorder level, minimum, safety stock, maximum thresholds
Lot/Batch *(entity)*	Traceable stock grouping within Stock Position
Quarantine Hold *(entity)*	Non-sellable hold within Stock Position
Stock Availability Breakdown, Stock Status Indicator, Ownership Classification *(value objects)*	Derived quantity states

Related Use Cases

Use Case	Name
UC-003	Define Stock Policies and Thresholds
UC-012	Monitor Inventory Status
UC-013	Inquire Inventory Availability
UC-019	Manage Quarantine and Hold Stock
UC-028	Review Low-Stock and Out-of-Stock Alerts *(threshold source)*

Relationships with Other Contexts

Related Context	Relationship	Integration
Product Catalog	**Downstream (conformist)**	References Product SKU; conforms to lifecycle rules
Warehouse Network	**Downstream (conformist)**	References Location Code; respects freeze state
Order Fulfillment	**Customer-Supplier**	Fulfillment requests reservation/deduction; Balance context owns quantity
Inbound Receiving	**Customer-Supplier**	Receiving requests quantity increase after put-away
Network Transfer	**Customer-Supplier**	Transfer requests source deduction and destination increase

Related Context	Relationship	Integration
Inventory Accuracy	**Customer-Supplier**	Adjustments request quantity correction
Returns Management	**Customer-Supplier**	Restock requests quantity increase
Commercial Campaign	**Customer-Supplier**	Promotional allocation segregates ATP
Inventory Intelligence	**Upstream → Downstream**	Publishes quantity and status events for dashboards

Why This Context Should Be Independent

Inventory balance is the **system's core transactional state** — the highest-contention, highest-frequency domain. It has a distinct ubiquitous language: ATP, on-hand, reserved, quarantined — terms that mean nothing in procurement or catalog contexts. Separating balance from fulfillment prevents order lifecycle logic from corrupting quantity invariants (BR-001, BR-002, BR-013). Stock Policy belongs here because thresholds are evaluated against live quantities, even though policy *definition* is a planning act. Independent context enables the balance model to scale (per product-location aggregates) without dragging in PO or order models.

Context 5: Procurement & Supplier

Business Responsibility

Manage the inbound supply commitment lifecycle — supplier relationships, internal requisition approval, purchase order placement, and expected inbound delivery planning — authorizing *what the organization has committed to buy* before goods arrive.

Owned Entities

Entity / Aggregate	Role in Context
Supplier *(aggregate root)*	Supplier master, payment terms, consignment agreement
Purchase Requisition *(aggregate root)*	Internal replenishment request and approval
Purchase Order *(aggregate root)*	Commercial supplier commitment
Expected Inbound Delivery *(entity)*	Planned shipment schedule tied to PO
Purchase Order Line, Payment Terms, Lead Time Duration *(value objects)*	Procurement line details

Related Use Cases

Use Case	Name
UC-006	Place and Manage Purchase Orders
UC-007	Plan Expected Inbound Deliveries
UC-029	Plan Replenishment and Create Purchase Requisitions
UC-030	Track Inbound Supply Status

Relationships with Other Contexts

Related Context	Relationship	Integration
Product Catalog	**Downstream (conformist)**	PO lines reference SKU; respects discontinued status
Warehouse Network	**Downstream (conformist)**	PO specifies receiving location
Inventory Balance	**Downstream (ACL via domain service)**	Network Availability Evaluation before requisition; reads ATP network-wide
Inventory Governance	**Downstream (conformist)**	Ad-hoc receipt exception policies
Inbound Receiving	**Upstream → Downstream**	PO authorizes goods receipt
Inventory Intelligence	**Upstream → Downstream**	Open PO and inbound pipeline reporting

Why This Context Should Be Independent

Procurement speaks the language of **commercial commitment** — requisitions, POs, suppliers, lead times — distinct from warehouse language of receipt and put-away. Purchasing is owned by **Buyers and Purchasing Managers**; receiving is owned by **Receiving Clerks**. Commitment (PO placed) happens days or weeks before execution (goods arrive) — different timelines and consistency boundaries. BR-007 (PO required for receipt) establishes a clear handoff between contexts via domain events, not shared tables. Separating procurement allows PO workflow evolution (approval chains, supplier portals) without touching stock quantity logic.

Context 6: Inbound Receiving

Business Responsibility

Execute the inbound supply verification workflow — goods receipt, discrepancy documentation, put-away confirmation, and supplier invoice matching — governing *what was actually received and when it becomes available*.

Owned Entities

Entity / Aggregate	Role in Context
Goods Receipt *(aggregate root)*	Verification, receipt lines, put-away
Receipt Discrepancy *(entity)*	Documented receipt variance
Put-Away *(entity)*	Storage placement confirmation
Supplier Invoice *(aggregate root)*	Billing document and three-way match
Receipt Line Detail, Discrepancy Classification, Three-Way Match Result *(value objects)*	Receipt and match artifacts

Related Use Cases

Use Case	Name
UC-008	Receive and Verify Supplier Goods
UC-009	Document and Resolve Receipt Discrepancies
UC-010	Perform Stock Put-Away
UC-011	Receive Return and Non-PO Inbound Stock *(supplier/ad-hoc portion)*
UC-039	Reconcile Inventory Valuation with Finance *(three-way match portion)*

Relationships with Other Contexts

Related Context	Relationship	Integration
Procurement & Supplier	**Downstream (conformist)**	Receipt must match PO; conforms to authorized quantities
Product Catalog	**Downstream (conformist)**	Product identity and MRSL validation at receipt
Warehouse Network	**Downstream (conformist)**	Receiving location and storage position assignment
Inventory Governance	**Downstream (conformist)**	Tolerance limits and reason codes for discrepancies
Inventory Balance	**Upstream → Downstream**	PutAwayCompleted event triggers StockReceived in Balance context
Inventory Intelligence	**Upstream → Downstream**	Receipt and match status for supplier performance reporting

Why This Context Should Be Independent

Receiving is **operational execution at the dock** — a distinct physical workflow from commercial PO creation. Its ubiquitous language centers on verification, discrepancy, put-away, and match — not on ATP or order reservation. BR-004, BR-005, and BR-006 form a cohesive receiving rule set that should not leak into procurement or balance models. Three-way match (Supplier Invoice)

bridges to finance but originates from receipt facts — keeping it here maintains the PO→Receipt→Invoice accountability chain. Separation allows receiving workflows (mobile scanning, carrier integration) to evolve independently of PO management.

Context 7: Order Fulfillment

Business Responsibility

Manage the customer order fulfillment lifecycle — availability confirmation, reservation, pick/pack/ship, partial fulfillment, backorders, and fulfillment exceptions — governing *how customer demand consumes inventory*.

Owned Entities

Entity / Aggregate	Role in Context
Customer Order *(aggregate root)*	Order lines, fulfillment state
Inventory Reservation *(entity)*	Operational hold for order line
Fulfillment Execution *(entity)*	Pick, pack, verify workflow
Backorder *(entity)*	Customer-agreed delay for unfulfilled quantity
Shipment *(entity)*	Dispatch confirmation
Order Line Quantity, Fulfillment Quantity Breakdown, Customer Acknowledgment *(value objects)*	Fulfillment line states

Related Use Cases

Use Case	Name
UC-021	Confirm Order Availability
UC-022	Reserve Inventory for Confirmed Orders
UC-023	Release Inventory Reservations
UC-024	Pick, Pack, and Ship Orders
UC-025	Handle Partial Fulfillment and Backorders
UC-026	Handle Outbound Fulfillment Exceptions
UC-027	View Order Fulfillment Status
UC-048	Support Multi-Location Split Fulfillment

Relationships with Other Contexts

Related Context	Relationship	Integration
Inventory Balance	**Downstream (ACL via domain services)**	Reads ATP; commands reservation/deduction via events; does not own quantity
Product Catalog	**Downstream (conformist)**	Order lines reference SKU; respects product lifecycle
Warehouse Network	**Downstream (conformist)**	Ship-from location; respects location freeze
Commercial Campaign	**Downstream (conformist)**	Respects promotional ring-fence rules
Inventory Governance	**Downstream (conformist)**	Pick verification and exception policies
External Sales/CRM	**Upstream → Downstream (ACL)**	Customer order may originate externally; IMS holds fulfillment-relevant subset
Inventory Intelligence	**Upstream → Downstream**	Fulfillment status and fill rate metrics
Returns Management	**Upstream → Downstream**	Original order referenced for return authorization

Why This Context Should Be Independent

Fulfillment speaks the language of **customer commitment** — orders, reservations, picks, shipments, backorders — fundamentally different from procurement's supplier language or balance's quantity language. Fulfillment is owned by **Customer Service and Warehouse** teams with distinct peak load patterns from receiving. BR-001, BR-002, BR-003, and BR-028 form a cohesive fulfillment rule set. The BRD explicitly scopes full CRM/sales outside IMS — fulfillment context holds only what inventory needs via ACL from external order systems. Separating fulfillment prevents customer order lifecycle complexity from polluting the high-performance stock balance model.

Context 8: Network Transfer

Business Responsibility

Manage inter-location stock redistribution — transfer requests, approval, dispatch, in-transit tracking, and destination receipt — resolving network imbalance without new procurement.

Owned Entities

Entity / Aggregate	Role in Context
Stock Transfer *(aggregate root)*	Transfer lifecycle including request, approval, dispatch, receipt
In-Transit Stock *(entity)*	Pipeline quantity between locations
Transfer Line Detail, Quantity *(value objects)*	Transfer line artifacts

Note: Transfer Request is the initial state of the Stock Transfer aggregate — not a separate bounded context.

Related Use Cases

Use Case	Name
UC-031	Request Inter-Location Stock Transfer
UC-032	Approve or Reject Transfer Requests
UC-033	Execute and Record Inter-Location Transfer
UC-034	Analyze and Recommend Network Rebalancing *(initiates transfers)*

Relationships with Other Contexts

Related Context	Relationship	Integration
Inventory Balance	**Downstream (ACL via domain services)**	Transfer Capacity Evaluation reads source policy and ATP; TransferDispatched/Received events mutate Stock Positions
Warehouse Network	**Downstream (conformist)**	Source and destination location validation; freeze check
Product Catalog	**Downstream (conformist)**	Product and lot traceability on transfer lines
Procurement & Supplier	**Customer-Supplier**	Transfer alternative evaluated before new purchase (BR-011)
Inventory Intelligence	**Upstream → Downstream**	In-transit and network balance reporting

Why This Context Should Be Independent

Network transfer is a **distinct operational workflow** from both procurement (external supply) and fulfillment (customer demand). Its language — request, approve, dispatch, in-transit, receive — describes **internal redistribution**, not customer or supplier relationships. BR-011 and BR-012 are network-specific rules with no meaning in single-location fulfillment. In-transit stock is a pipeline state invisible to customer-facing ATP queries. Separating transfers allows network balancing algorithms and approval workflows to evolve without modifying PO or order models.

Context 9: Inventory Accuracy

Business Responsibility

Maintain inventory record accuracy through cycle counting, physical inventory counts, variance investigation, governed adjustments, and write-offs — ensuring *recorded stock reflects physical reality*.

Owned Entities

Entity / Aggregate	Role in Context
Cycle Count *(aggregate root)*	Scheduled partial counts
Physical Inventory Count *(aggregate root)*	Wall-to-wall count events
Inventory Adjustment Case *(aggregate root)*	Count variance, stock adjustment, write-off
Count Variance, Quantity Variance, Reason Code Attribution *(value objects)*	Variance and correction artifacts

Related Use Cases

Use Case	Name
UC-014	Perform Cycle Count
UC-015	Conduct Full Physical Inventory Count
UC-016	Investigate and Approve Count Variances
UC-017	Process Stock Adjustments
UC-018	Authorize and Process Inventory Write-Offs
UC-020	Analyze Shrinkage, Loss, and Obsolete Stock

Relationships with Other Contexts

Related Context	Relationship	Integration
Inventory Balance	**Customer-Supplier**	Approved adjustments command StockAdjusted/StockWrittenOff events; reads system quantity for variance
Warehouse Network	**Customer-Supplier**	Physical count imposes/releases location freeze
Inventory Governance	**Downstream (conformist)**	Reason codes, approval thresholds, SoD rules
Product Catalog	**Downstream (conformist)**	ABC classification drives count frequency
Identity & Access	**Downstream (conformist)**	Counter vs. approver SoD validation
Inventory Intelligence	**Upstream → Downstream**	Accuracy metrics, shrinkage trends

Why This Context Should Be Independent

Inventory accuracy is a **governance and quality discipline** — not daily execution. Its language — blind count, variance, investigation, adjustment, write-off, shrinkage — is foreign to fulfillment or procurement. Accuracy workflows enforce **segregation of duties** (BR-018, BR-019) that deliberately slow down corrections — opposite to the speed requirement of order fulfillment. Count-to-adjustment is a multi-step governed pipeline that must not be embedded in Stock Position (which should apply corrections quickly once authorized). Independent context allows audit and accuracy program maturity without disrupting transactional throughput.

Context 10: Returns Management

Business Responsibility

Manage reverse logistics — return authorization, physical return receipt, condition inspection, and disposition routing (restock, quarantine, write-off) — governing *how returned goods re-enter or exit inventory*.

Owned Entities

Entity / Aggregate	Role in Context
Return Authorization *(aggregate root)*	RMA and return eligibility
Customer Return *(entity)*	Physical return receipt
Return Disposition *(entity)*	Restock, quarantine, or write-off decision
Condition Grade, Disposition Decision *(value objects)*	Inspection outcomes

Related Use Cases

Use Case	Name
UC-035	Authorize and Record Customer Returns
UC-036	Receive, Inspect, and Disposition Returned Goods
UC-011	Receive Return and Non-PO Inbound Stock *(return portion)*

Relationships with Other Contexts

Related Context	Relationship	Integration
Order Fulfillment	**Downstream (conformist)**	RMA validates against original customer order
Inbound Receiving	**Shared Kernel** *(receipt verification pattern)*	Similar verify-before-accept discipline; may share receipt workflow concepts
Inventory Balance	**Customer-Supplier**	Restock triggers StockReceived; quarantine triggers QuarantineHoldApplied
Inventory Accuracy	**Customer-Supplier**	Write-off disposition triggers Inventory Adjustment Case
Product Catalog	**Downstream (conformist)**	Returned product identity validation
Warehouse Network	**Downstream (conformist)**	Return receiving location

Why This Context Should Be Independent

Returns are **reverse logistics** with a distinct rule chain (BR-025 → BR-026 → disposition) and customer-facing authorization requirements unlike forward fulfillment or supplier receiving. Returns involve **Customer Service** (RMA) and **Quality** (inspection) stakeholders not central to outbound pick/pack. Conflating returns with inbound supplier receiving would blur fundamentally different accountability — supplier discrepancies vs. customer policy eligibility. Independent context allows return policy, inspection grading, and disposition rules to evolve with customer experience programs without affecting PO or order models.

Context 11: Commercial Campaign

Business Responsibility

Ring-fence inventory for promotional campaigns and product launches, protect allocated stock from non-promotional consumption, and support campaign readiness confirmation.

Owned Entities

Entity / Aggregate	Role in Context
Promotional Stock Allocation *(aggregate root)*	Campaign ring-fence, channel scope, readiness

Entity / Aggregate	Role in Context
Campaign Period, Channel Designation, Quantity *(value objects)*	Campaign parameters

Related Use Cases

Use Case	Name
UC-040	Confirm Promotional and Launch Stock Readiness
UC-047	Receive and Respond to Operational Notifications *(campaign readiness alerts)*
UC-022	Reserve Inventory for Confirmed Orders *(must respect ring-fence per BR-029)*

Relationships with Other Contexts

Related Context	Relationship	Integration
Inventory Balance	**Customer-Supplier**	Allocation segregates ATP on Stock Position; released on campaign end
Product Catalog	**Downstream (conformist)**	Campaign SKUs reference product identity
Warehouse Network	**Downstream (conformist)**	Allocation scoped to fulfillment locations
Order Fulfillment	**Upstream → Downstream**	Fulfillment must check ring-fence before general reservation
Inventory Intelligence	**Upstream → Downstream**	Campaign readiness dashboards
External Marketing	**Upstream → Downstream (ACL)**	Campaign plans may originate in marketing systems

Why This Context Should Be Independent

Commercial campaign management speaks the language of **marketing commitments** — campaigns, launches, channels, readiness — not warehouse or procurement language. It is owned by **Marketing and Commercial** leadership with seasonal/event-driven cycles unrelated to daily replenishment. BR-029 (promotional ring-fencing) is a commercial policy overlay on stock availability that general fulfillment should not own. Phase 2 scope in the BRD confirms this is a distinct maturity capability. Independent context allows campaign tooling integration without modifying core balance or fulfillment aggregates.

Context 12: Identity & Access

Business Responsibility

Manage user identity, role assignments, location-scoped permissions, and segregation-of-duties enforcement for all inventory operations — governing *who may perform and approve* inventory actions.

Owned Entities

Entity / Aggregate	Role in Context
User Account *(aggregate root)*	User identity and lifecycle
Role Assignment *(entity)*	Role and location scope delegation
User Action Attribution *(value object)*	Accountability template

Related Use Cases

Use Case	Name
UC-044	Manage User Accounts and Role Assignments
UC-045	Enforce Role-Based and Location-Scoped Access
UC-041	Conduct Inventory Audit and Control Review *(access compliance)*

Relationships with Other Contexts

Related Context	Relationship	Integration
All operational contexts	**Open Host Service**	Provides user identity and authorization checks
Inventory Governance	**Shared Kernel** *(SoD rules)*	Governance defines SoD policy; Identity enforces at assignment
Inventory Accuracy	**Upstream → Downstream**	Counter/approver identity validation
Warehouse Network	**Downstream (conformist)**	Location scope references location codes

Why This Context Should Be Independent

Identity and access is **cross-cutting infrastructure capability** with its own ubiquitous language — roles, permissions, scopes, segregation of duties — unrelated to inventory quantities or orders. It changes on **HR lifecycle events** (hire, transfer, termination) independent of inventory transactions. Every operational context needs identity but none should own it — embedding user models in Stock Position or Goods Receipt would couple security to business logic. Independent context enables enterprise SSO integration, role model expansion, and audit of access changes without touching inventory domain models.

Context 13: Inventory Intelligence

Business Responsibility

Provide read-optimized views of inventory state — dashboards, KPIs, movement history, audit trails, alerts, and valuation snapshots — supporting management oversight without owning transactional write models.

Owned Entities

Concept	Role in Context
Inventory Movement Record *(read projection)*	Append-only movement history
Audit Trail Record *(read projection)*	Governance-grade change log
Operational Alert *(read projection)*	Threshold and exception notifications
Inventory Valuation Snapshot *(read projection)*	Period-end financial summary

This context owns no write aggregates — it consumes domain events from all transactional contexts.

Related Use Cases

Use Case	Name
UC-012	Monitor Inventory Status
UC-037	Review Inventory Dashboards and Operational Reports
UC-038	Review Inventory Performance KPIs
UC-042	View Inventory Movement and Transaction History
UC-043	Query and Review Audit Trail
UC-047	Receive and Respond to Operational Notifications
UC-039	Reconcile Inventory Valuation with Finance *(snapshot portion)*
UC-020	Analyze Shrinkage, Loss, and Obsolete Stock

Relationships with Other Contexts

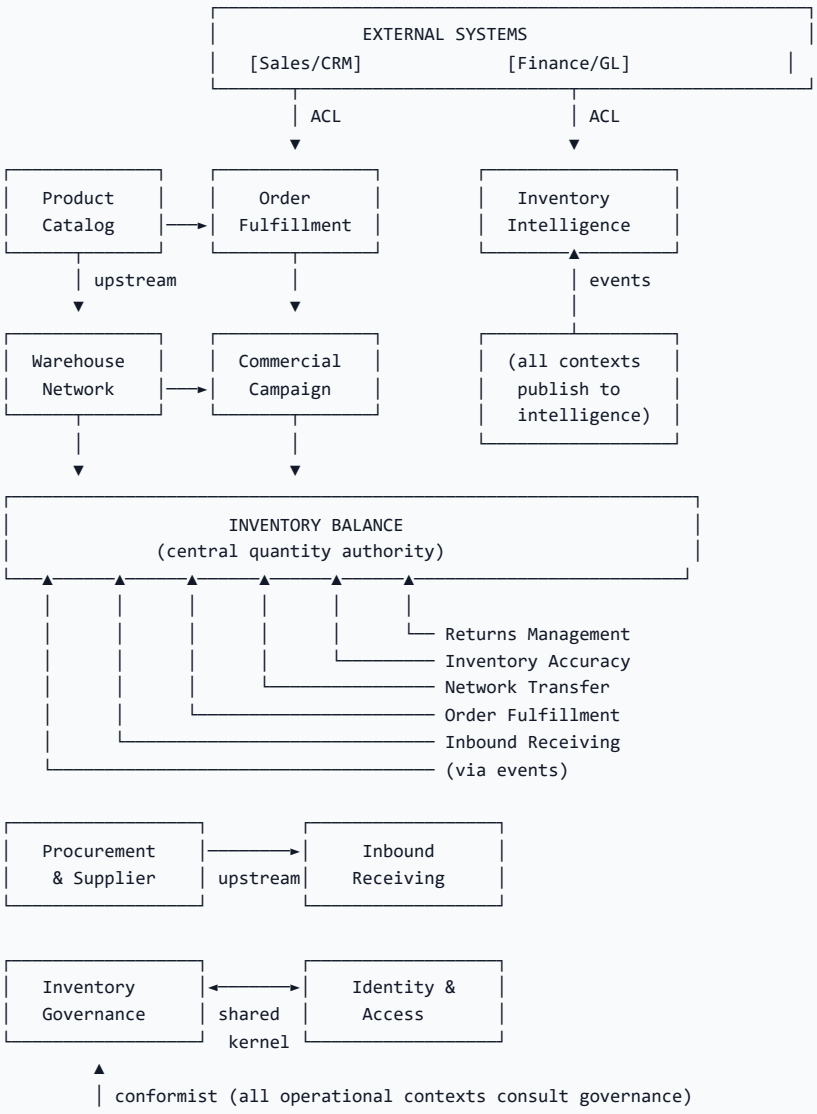
Related Context	Relationship	Integration
All transactional contexts	**Downstream (conformist)**	Consumes domain events; never writes back to source aggregates

Related Context	Relationship	Integration
Inventory Balance	**Downstream (conformist)**	Quantity and status projections
External Finance/GL	**Anti-Corruption Layer**	Valuation snapshot translated for finance reconciliation
External Executive BI	**Open Host Service**	KPI and dashboard data published

Why This Context Should Be Independent

Reporting and analytics have fundamentally different non-functional requirements — read-heavy, eventually consistent, aggregation-optimized — versus transactional contexts that require strong consistency and low-latency writes. Mixing dashboard queries into Stock Position or Order Fulfillment corrupts the write model with reporting concerns (the "reporting database inside the transaction" anti-pattern). Inventory Intelligence can rebuild projections from domain events without affecting operational correctness. Independent context allows query model optimization (materialized views, KPI caches) and finance integration without redeploying warehouse execution logic.

Context Map Diagram



Context Integration Summary

Integration Pattern	Contexts	Mechanism
Domain Events	All transactional → Inventory Balance	PutAwayCompleted, StockReserved, StockShipped, StockAdjusted, TransferReceived, ReturnRestocked
Domain Events	All transactional → Inventory Intelligence	InventoryMovementRecorded, AuditTrailRecorded, threshold alerts
Domain Services	Order Fulfillment ↔ Inventory Balance	Availability Confirmation, Reservation Coordination
Domain Services	Procurement ↔ Inventory Balance	Network Availability Evaluation before requisition
Published Language	All contexts	Product SKU, Location Code, User Identifier as shared identifiers
ACL	External Sales/CRM → Order Fulfillment	Fulfillment-relevant order subset translated
ACL	Inventory Intelligence → External Finance	Valuation snapshot mapped to GL format

Context-to-Use Case Coverage Matrix

Context	Use Cases
Product Catalog	UC-001, UC-004, UC-046
Warehouse Network	UC-002, UC-046, UC-015
Inventory Governance	UC-005, UC-041, UC-045
Inventory Balance	UC-003, UC-012, UC-013, UC-019, UC-028
Procurement & Supplier	UC-006, UC-007, UC-029, UC-030
Inbound Receiving	UC-008, UC-009, UC-010, UC-011, UC-039
Order Fulfillment	UC-021 – UC-027, UC-048
Network Transfer	UC-031 – UC-034
Inventory Accuracy	UC-014 – UC-018, UC-020
Returns Management	UC-035, UC-036, UC-011
Commercial Campaign	UC-040, UC-047
Identity & Access	UC-044, UC-045, UC-041
Inventory Intelligence	UC-012, UC-037, UC-038, UC-042, UC-043, UC-047, UC-039, UC-020

All 48 use cases are covered across contexts. Some use cases span multiple contexts as indicated.

Independence Principles Applied

Principle	How Contexts Apply It
Distinct ubiquitous language	Each context names concepts in its stakeholders' vocabulary — ATP (Balance) vs. PO (Procurement) vs. RMA (Returns)
Separate ownership	Different business functions own different contexts — Purchasing vs. Warehouse vs. Marketing vs. Finance
Independent change velocity	Catalog changes quarterly; balance changes every second; governance changes on audit cycles
Autonomous consistency	Each context enforces its own invariants internally; cross-context rules via events and domain services
Team alignment	Context boundaries align to BRD actor groups — Buyer, Receiving Clerk, Customer Service, Marketing Manager
Phased delivery	MVP contexts (Balance, Receiving, Fulfillment) deliverable without Phase 2 contexts (Campaign, full Procurement)

Document Approval

Role	Name	Signature	Date
DDD Solution Architect			
Domain Expert			
Solution Architect			
Product Owner			

End of Bounded Contexts

Document ID: SRS-DM-007 | Version 1.0 | July 5, 2026

Domain Entity Model

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-DM-001
Document Title	Domain Entity Model — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Solution Architect / Domain Expert
Source Documents	BRD-IMS-001 (v1.0), SRS-BR-001 (Business Rules, v1.0), SRS-UC-001 (Use Case Catalog, v1.0)
Date	July 5, 2026
Scope	Business domain model only — no database design, API design, or implementation detail

Purpose

This document identifies the core **domain entities** of the Inventory Management System as derived from the approved Business Requirements Document, Business Rules Catalog, and Use Case Catalog. Each entity represents a meaningful business concept with identity, responsibilities, and relationships to processes and rules that govern inventory as a strategic business asset.

Entity Domain Map

Domain	Entities
Master Data & Reference	Product, Inventory Location, Storage Position, Stock Policy, Reason Code, Governance Policy
Inventory State & Balance	Stock Position, Lot/Batch, Quarantine Hold, Promotional Stock Allocation
Procurement & Inbound	Supplier, Purchase Requisition, Purchase Order, Expected Inbound Delivery, Goods Receipt, Receipt Discrepancy, Put-Away, Supplier Invoice
Order Fulfillment	Customer Order, Inventory Reservation, Fulfillment Execution, Backorder, Shipment
Network Balance	Transfer Request, Stock Transfer, In-Transit Stock
Inventory Control	Cycle Count, Physical Inventory Count, Count Variance, Stock Adjustment, Inventory Write-Off

Domain	Entities
Returns	Return Authorization, Customer Return, Return Disposition
Accountability & Oversight	Inventory Movement Record, Audit Trail Record, Operational Alert, User Account, Role Assignment
Financial Alignment	Inventory Valuation Snapshot

Total Entities: 40

Domain 1: Master Data & Reference

1. Product

Aspect	Detail
Business Purpose	Represents a distinct stock-keeping unit (SKU) that the organization buys, holds, and sells. Product is the canonical identity for all inventory operations across locations, departments, and channels.
Identity	SKU (Stock Keeping Unit)
Responsibilities	Define what is being counted, stored, ordered, and fulfilled; carry commercial and operational attributes (name, description, unit of measure, category); govern lifecycle status (active, inactive, discontinued); support identifier lookup (SKU, barcode); enable ABC classification and rotation policy assignment (FIFO/FEFO, traceability requirements).
Important Attributes	SKU, product name, description, unit of measure (UOM), category, product status (active / inactive / discontinued), product identifiers (barcode, alternate codes), ABC classification, shelf-life indicator, traceability requirement (lot/batch), rotation policy (FIFO / FEFO), minimum remaining shelf life (MRSL) policy, supersession reference (if replaced)
Related Business Processes	BP-01 (Product Master Data Management), BP-04 (ABC Product Classification), BP-03 (Stock Policy & Threshold Definition), BP-07 (Goods Receipt & Verification), BP-18 (Pick, Pack & Ship), BP-22 (Replenishment Planning & Requisition)
Related Business Rules	BR-014 (FEFO Rotation for Shelf-Life Products), BR-015 (FIFO Rotation for Standard Goods), BR-016 (Minimum Remaining Shelf Life at Receipt), BR-017 (Lot or Batch Capture for Traceable Products), BR-027 (No Reorder of Discontinued Products), BR-032 (Class A Priority Monitoring)

2. Inventory Location

Aspect	Detail
Business Purpose	Represents a physical place in the distribution network where stock is held, received, counted, and fulfilled — such as a warehouse, store, or distribution point. Locations form the network through which inventory is balanced and reported.
Identity	Location Code
Responsibilities	Anchor all stock balances and movements to a physical site; support location hierarchy (region, site, zone); enable location-scoped visibility, authorization, and performance reporting; define where receiving, put-away, counting, and fulfillment occur; support location go-live with opening balances and closure planning.
Important Attributes	Location code, location name, location type (warehouse / store / distribution point), parent location (hierarchy), operational status (active / planned / closing), address and region, assigned responsible manager, go-live date, closure disposition plan
Related Business Processes	BP-02 (Location & Warehouse Setup), BP-10 (Inventory Status Monitoring), BP-12 (Full Physical Inventory Count), BP-24 (Inter-Location Transfer Request), BP-25 (Inter-Location Transfer Execution), BP-29 (Inventory Performance Reporting)
Related Business Rules	BR-012 (Protect Source Location Minimum), BR-024 (Physical Inventory Freeze)

3. Storage Position

Aspect	Detail
Business Purpose	Represents a designated storage area within an inventory location (aisle, zone, bin, shelf) where received goods are placed after put-away. Storage position enables efficient retrieval and rotation discipline during picking.

Aspect	Detail
Identity	Storage Position Code (unique within location)
Responsibilities	Direct put-away placement after receipt; guide pick path and rotation (FIFO/FEFO); associate physical stock with a findable address within the location; support bin-level accuracy in advanced operations.
Important Attributes	Storage position code, location reference, zone/aisle/bin designation, storage type (bulk / pick face / quarantine / bonded), capacity indicator, product assignment rules (if dedicated)
Related Business Processes	BP-09 (Stock Put-Away & Storage), BP-18 (Pick, Pack & Ship), BP-11 (Cycle Counting)
Related Business Rules	BR-005 (Put-Away Before Availability), BR-014 (FEFO Rotation for Shelf-Life Products), BR-015 (FIFO Rotation for Standard Goods)

4. Stock Policy

Aspect	Detail
Business Purpose	Defines replenishment and service-level thresholds for a product at a specific location (or network-wide default). Stock policy translates demand and lead-time assumptions into actionable reorder triggers and safety buffers.
Identity	Stock Policy Reference (Product + Location combination)
Responsibilities	Set reorder level, minimum threshold, and safety stock; trigger low-stock alerts and replenishment reviews; protect service levels during transfers and reservations; govern when procurement or network rebalancing should be initiated.
Important Attributes	Product reference, location reference, reorder level, minimum stock threshold, safety stock quantity, maximum stock level (overstock indicator), lead time assumption, replenishment review frequency, policy effective date, exception approval record
Related Business Processes	BP-03 (Stock Policy & Threshold Definition), BP-21 (Low-Stock Alert Review), BP-22 (Replenishment Planning & Requisition), BP-24 (Inter-Location Transfer Request), BP-26 (Network Stock Rebalancing)
Related Business Rules	BR-009 (Reorder at Reorder Level), BR-010 (Safety Stock Protection), BR-011 (Network Check Before Purchase), BR-012 (Protect Source Location Minimum)

5. Reason Code

Aspect	Detail
Business Purpose	Provides a standardized, categorized business explanation for inventory adjustments, exceptions, and disposition decisions. Reason codes enable shrinkage analysis, trend identification, and audit accountability.
Identity	Reason Code
Responsibilities	Categorize why inventory quantity or status changed; support mandatory attribution on adjustments, write-offs, and discrepancies; feed reporting on loss patterns and process failures; align operational actions with governance taxonomy.
Important Attributes	Reason code, description, category (adjustment / shrinkage / damage / obsolescence / receipt discrepancy / return / other), active status, approval threshold association, reporting group
Related Business Processes	BP-01 (Product Master Data Management), BP-08 (Receipt Discrepancy Resolution), BP-14 (Stock Adjustment & Write-Off), BP-15 (Shrinkage & Loss Management), BP-28 (Return-to-Stock or Disposal Decision)
Related Business Rules	BR-006 (Document Receipt Discrepancies), BR-020 (Mandatory Adjustment Reason Codes), BR-023 (Documented Write-Off Authorization)

6. Governance Policy

Aspect	Detail
Business Purpose	Captures organization-wide inventory control policies — approval thresholds, segregation-of-duties requirements, tolerance limits, and exception handling rules — that govern how operational entities may be changed.
Identity	Policy Identifier

Aspect	Detail
Responsibilities	Define financial and quantity approval thresholds for adjustments; enforce segregation of duties (count vs. approve); set receipt quantity tolerance limits; govern ad-hoc receipt exceptions; provide the policy baseline for audit and compliance review.
Important Attributes	Policy identifier, policy name, policy type (approval threshold / tolerance / access / exception), threshold value, applicable scope (location / role / transaction type), effective date, approval authority role
Related Business Processes	BP-01 (Product Master Data Management), BP-13 (Count Variance Investigation & Approval), BP-14 (Stock Adjustment & Write-Off), BP-32 (Inventory Audit & Control Review)
Related Business Rules	BR-019 (Independent Count Approval), BR-021 (Independent Approval for Material Adjustments), BR-007 (Purchase Order Required for Supplier Receipt)

Domain 2: Inventory State & Balance

7. Stock Position

Aspect	Detail
Business Purpose	Represents the current inventory balance of a product at a location — the authoritative answer to "what do we have, where is it, and what is available?" Stock position is the central entity connecting visibility, fulfillment, replenishment, and financial alignment.
Identity	Stock Position Reference (Product + Location [+ Lot/Batch when traceable])
Responsibilities	Maintain on-hand, reserved, and available-to-promise (ATP) quantities; reflect stock status indicators (in stock, low stock, out of stock, overstock); exclude non-sellable holds from ATP; support network-wide and location-level availability inquiries; change only through documented movements and adjustments.
Important Attributes	Product reference, location reference, on-hand quantity, reserved quantity, available-to-promise (ATP), inbound expected quantity (network view), stock status indicator, ownership type (owned / consignment / bonded), last movement date, quarantine flag
Related Business Processes	BP-10 (Inventory Status Monitoring), BP-16 (Order Availability Confirmation), BP-17 (Inventory Reservation), BP-21 (Low-Stock Alert Review), BP-29 (Inventory Performance Reporting), BP-30 (Inventory Valuation & Financial Reconciliation)
Related Business Rules	BR-001 (ATP-Only Customer Commitment), BR-002 (Reservation at Order Confirmation), BR-013 (Quarantine Exclusion from Sellable Stock), BR-009 (Reorder at Reorder Level), BR-031 (Consignment Ownership at Consumption)

8. Lot / Batch

Aspect	Detail
Business Purpose	Represents a traceable production or receipt grouping for products requiring lot-level tracking. Lot/batch identity enables targeted recall, quality investigation, regulatory compliance, and FEFO rotation.
Identity	Lot/Batch Number (within product scope)
Responsibilities	Capture lot identity at receipt and through all subsequent movements; support FEFO picking by expiration date; enable recall scope limitation; link stock position to quality and shelf-life attributes.
Important Attributes	Lot/batch number, product reference, expiration date, manufacture date, receipt date, remaining shelf life, quality status, supplier reference, associated stock quantity by location
Related Business Processes	BP-07 (Goods Receipt & Verification), BP-09 (Stock Put-Away & Storage), BP-18 (Pick, Pack & Ship), BP-25 (Inter-Location Transfer Execution), BP-14 (Stock Adjustment & Write-Off)
Related Business Rules	BR-017 (Lot or Batch Capture for Traceable Products), BR-014 (FEFO Rotation for Shelf-Life Products), BR-016 (Minimum Remaining Shelf Life at Receipt)

9. Quarantine Hold

Aspect	Detail
Business Purpose	Represents inventory physically present but excluded from sellable availability due to quality investigation, regulatory restriction, recall, customs hold (bonded), or other business hold.
Identity	Hold Reference
Responsibilities	Remove affected quantity from ATP; track hold reason and expected resolution; govern release back to sellable stock or disposition to write-off; support customs clearance workflow for bonded inventory.
Important Attributes	Hold reference, product reference, location reference, lot/batch reference (if applicable), held quantity, hold type (quality / regulatory / recall / customs / investigation), hold reason, hold start date, expected resolution date, release or disposition outcome, authorizing party
Related Business Processes	BP-14 (Stock Adjustment & Write-Off), BP-20 (Outbound Exception Handling), BP-07 (Goods Receipt & Verification), BP-09 (Stock Put-Away & Storage)
Related Business Rules	BR-013 (Quarantine Exclusion from Sellable Stock), BR-030 (Bonded Inventory Customs Clearance)

10. Promotional Stock Allocation

Aspect	Detail
Business Purpose	Represents inventory ring-fenced for a specific promotional campaign, product launch, or channel commitment. Promotional allocation protects campaign supply from consumption by non-promotional demand.
Identity	Allocation Reference
Responsibilities	Reserve defined quantity for campaign period and channel; exclude allocated stock from general ATP for non-approved orders; support readiness review before campaign commitment; release un consumed allocation after campaign end.
Important Attributes	Allocation reference, product reference, location(s), allocated quantity, campaign or launch name, effective start date, effective end date, channel scope, readiness status, commercial approval reference
Related Business Processes	BP-31 (Promotional & Launch Stock Readiness), BP-17 (Inventory Reservation), BP-22 (Replenishment Planning & Requisition)
Related Business Rules	BR-029 (Promotional Stock Ring-Fencing)

Domain 3: Procurement & Inbound Supply

11. Supplier

Aspect	Detail
Business Purpose	Represents an external vendor from whom the organization procures goods. Supplier is the commercial counterparty in the inbound supply chain and accountability chain for receipt and payment.
Identity	Supplier Code
Responsibilities	Anchor purchase orders and goods receipts; support supplier delivery performance tracking; enable three-way match for payment authorization; associate inbound supply with commercial terms.
Important Attributes	Supplier code, supplier name, contact information, payment terms, lead time profile, active status, consignment agreement indicator, delivery performance rating
Related Business Processes	BP-05 (Purchase Order Placement), BP-07 (Goods Receipt & Verification), BP-08 (Receipt Discrepancy Resolution), BP-23 (Inbound Supply Tracking), BP-30 (Inventory Valuation & Financial Reconciliation)
Related Business Rules	BR-007 (Purchase Order Required for Supplier Receipt), BR-008 (Three-Way Match Before Payment), BR-031 (Consignment Ownership at Consumption)

12. Purchase Requisition

Aspect	Detail
Business Purpose	Represents an internal request to procure stock, created when replenishment planning identifies a need. Requisition precedes formal purchase order creation and carries approval workflow.
Identity	Requisition Number
Responsibilities	Capture replenishment need with product, quantity, and target location; route for internal approval before PO creation; link to triggering low-stock alert or replenishment suggestion; prevent requisition for discontinued products.
Important Attributes	Requisition number, requesting location, product lines (product, quantity), requested delivery date, triggering alert reference, approval status, approver, creation date, linked purchase order reference
Related Business Processes	BP-22 (Replenishment Planning & Requisition), BP-21 (Low-Stock Alert Review), BP-05 (Purchase Order Placement)
Related Business Rules	BR-009 (Reorder at Reorder Level), BR-011 (Network Check Before Purchase), BR-027 (No Reorder of Discontinued Products), BR-010 (Safety Stock Protection)

13. Purchase Order

Aspect	Detail
Business Purpose	Represents the authorized commercial commitment to a supplier specifying products, quantities, prices, and delivery terms. The purchase order is the control document for inbound receipt and financial three-way match.
Identity	Purchase Order Number
Responsibilities	Authorize what may be received from a supplier; define expected inbound supply; enable PO-receipt-invoice matching; track open, partially received, and closed status; govern receipt without PO exceptions.
Important Attributes	Purchase order number, supplier reference, order date, expected delivery date, line items (product, ordered quantity, unit price), received quantity (per line), PO status (open / partial / closed / cancelled), delivery location, total commitment value, amendment history
Related Business Processes	BP-05 (Purchase Order Placement), BP-06 (Inbound Shipment Planning), BP-07 (Goods Receipt & Verification), BP-23 (Inbound Supply Tracking), BP-30 (Inventory Valuation & Financial Reconciliation)
Related Business Rules	BR-007 (Purchase Order Required for Supplier Receipt), BR-008 (Three-Way Match Before Payment), BR-027 (No Reorder of Discontinued Products)

14. Expected Inbound Delivery

Aspect	Detail
Business Purpose	Represents a planned or confirmed supplier shipment tied to a purchase order, communicating when and what quantity is expected to arrive. Supports receiving preparation and inbound supply visibility.
Identity	Delivery Plan Reference
Responsibilities	Communicate anticipated arrival date and quantity to receiving and replenishment teams; update when supplier changes schedule; feed inbound expected quantity into network availability views; trigger delay notifications.
Important Attributes	Delivery plan reference, purchase order reference, expected arrival date, expected quantity by product, carrier reference, shipment status (planned / in transit / arrived / delayed), receiving location
Related Business Processes	BP-06 (Inbound Shipment Planning), BP-23 (Inbound Supply Tracking), BP-10 (Inventory Status Monitoring)
Related Business Rules	—

15. Goods Receipt

Aspect	Detail
Business Purpose	Represents the business event of accepting verified inbound goods into inventory control. Goods receipt confirms that quantity, product identity, and condition have been validated before stock is recorded.
Identity	Goods Receipt Number
Responsibilities	Match inbound shipment to purchase order (or authorized exception); verify quantity, product, and condition; capture lot/batch and shelf-life data for traceable products; initiate put-away; create the receipt leg of three-way match; reject or hold non-conforming deliveries.
Important Attributes	Goods receipt number, purchase order reference (or exception authorization), supplier reference, receiving location, receipt date, received lines (product, quantity, condition), lot/batch assignments, verification status, receiving clerk, linked discrepancies
Related Business Processes	BP-07 (Goods Receipt & Verification), BP-08 (Receipt Discrepancy Resolution), BP-09 (Stock Put-Away & Storage), BP-27 (Customer Return Intake)
Related Business Rules	BR-004 (Verify Before Inventory Acceptance), BR-005 (Put-Away Before Availability), BR-006 (Document Receipt Discrepancies), BR-007 (Purchase Order Required for Supplier Receipt), BR-016 (Minimum Remaining Shelf Life at Receipt), BR-017 (Lot or Batch Capture for Traceable Products), BR-030 (Bonded Inventory Customs Clearance)

16. Receipt Discrepancy

Aspect	Detail
Business Purpose	Represents a documented difference between expected and actual receipt — short shipment, over-shipment, wrong item, damage, or shelf-life non-compliance — captured at the point of receipt with supporting evidence.
Identity	Discrepancy Reference
Responsibilities	Record variance before acceptance; preserve evidence for supplier claims; drive resolution workflow (accept partial, reject, claim); affect three-way match outcome; prevent silent acceptance of incorrect supply.
Important Attributes	Discrepancy reference, goods receipt reference, discrepancy type (short / over / wrong item / damage / shelf-life), expected vs. received quantity, product affected, evidence documentation, resolution status, resolution outcome, supplier claim reference
Related Business Processes	BP-08 (Receipt Discrepancy Resolution), BP-07 (Goods Receipt & Verification)
Related Business Rules	BR-006 (Document Receipt Discrepancies), BR-008 (Three-Way Match Before Payment), BR-016 (Minimum Remaining Shelf Life at Receipt)

17. Put-Away

Aspect	Detail
Business Purpose	Represents the business confirmation that received goods have been moved from the receiving area to their designated storage position and are now operationally available for sale and fulfillment.
Identity	Put-Away Reference
Responsibilities	Transition stock from "received but unavailable" to sellable ATP; assign storage position; apply rotation placement (FIFO/FEFO); release bonded inventory only after customs clearance; complete the inbound-to-available lifecycle.
Important Attributes	Put-away reference, goods receipt reference, product reference, quantity, from area (receiving dock), to storage position, put-away date, performing staff, availability effective timestamp
Related Business Processes	BP-09 (Stock Put-Away & Storage)
Related Business Rules	BR-005 (Put-Away Before Availability), BR-014 (FEFO Rotation for Shelf-Life Products), BR-015 (FIFO Rotation for Standard Goods), BR-030 (Bonded Inventory Customs Clearance)

18. Supplier Invoice

Aspect	Detail
Business Purpose	Represents the supplier's billing document for goods supplied. Supplier invoice is the financial leg of the three-way match against purchase order and goods receipt before payment authorization.
Identity	Supplier Invoice Number (within supplier scope)
Responsibilities	Record billed quantity and price; support three-way match evaluation against PO and receipt; flag mismatches for resolution; gate payment authorization; align operational receipt with financial obligation.
Important Attributes	Invoice number, supplier reference, invoice date, purchase order reference, line items (product, billed quantity, unit price), total amount, match status (pending / matched / exception), tolerance variance, payment authorization status
Related Business Processes	BP-30 (Inventory Valuation & Financial Reconciliation), BP-08 (Receipt Discrepancy Resolution), BP-05 (Purchase Order Placement)
Related Business Rules	BR-008 (Three-Way Match Before Payment)

Domain 4: Order Fulfillment

19. Customer Order

Aspect	Detail
Business Purpose	Represents a customer demand for products that the organization commits to fulfill. Customer order is the commercial trigger for availability confirmation, reservation, picking, and shipment. *(Authoritative order data may reside in external sales/CRM systems; IMS holds fulfillment-relevant order context.)*
Identity	Order Number
Responsibilities	Carry order lines (product, requested quantity); trigger availability confirmation against ATP; drive reservation, pick, pack, and ship; support partial fulfillment and backorder handling; link to returns via original order reference.
Important Attributes	Order number, customer reference, order date, requested delivery date, order lines (product, quantity, fulfilled quantity, backordered quantity), order status (draft / confirmed / partially fulfilled / fulfilled / cancelled), fulfillment location(s), channel
Related Business Processes	BP-16 (Order Availability Confirmation), BP-17 (Inventory Reservation), BP-18 (Pick, Pack & Ship), BP-19 (Partial Fulfillment & Backorder Management), BP-27 (Customer Return Intake)
Related Business Rules	BR-001 (ATP-Only Customer Commitment), BR-002 (Reservation at Order Confirmation), BR-028 (Backorder Customer Acknowledgment)

20. Inventory Reservation

Aspect	Detail
Business Purpose	Represents an operational hold on stock for a confirmed customer order line, reducing ATP to prevent double-allocation between confirmation and physical fulfillment.
Identity	Reservation Reference
Responsibilities	Commit on-hand stock to a specific order line; reduce ATP immediately upon order confirmation; release when order is cancelled, amended, or fulfilled; respect safety stock and promotional ring-fence rules; support multi-location split reservations.
Important Attributes	Reservation reference, order reference, order line reference, product reference, location reference, reserved quantity, reservation date, reservation status (active / released / fulfilled), expiry or priority indicator
Related Business Processes	BP-17 (Inventory Reservation), BP-16 (Order Availability Confirmation), BP-18 (Pick, Pack & Ship), BP-31 (Promotional & Launch Stock Readiness)
Related Business Rules	BR-002 (Reservation at Order Confirmation), BR-010 (Safety Stock Protection), BR-029 (Promotional Stock Ring-Fencing), BR-001 (ATP-Only Customer Commitment)

21. Fulfillment Execution

Aspect	Detail
Business Purpose	Represents the warehouse work of picking, verifying, packing, and preparing order lines for dispatch. Fulfillment execution is the operational bridge between reservation and shipment.
Identity	Fulfillment Reference
Responsibilities	Retrieve correct products from storage applying FIFO/FEFO; verify picked quantity against order; record pick confirmation before shipment; capture lot/batch on pick for traceable products; escalate exceptions (short pick, damage, shelf mismatch).
Important Attributes	Fulfillment reference, order reference, fulfillment location, pick lines (product, requested quantity, picked quantity, lot/batch), pick verification status, pack confirmation, assigned picker/packer, exception flags, pick completion timestamp
Related Business Processes	BP-18 (Pick, Pack & Ship), BP-20 (Outbound Exception Handling), BP-19 (Partial Fulfillment & Backorder Management)
Related Business Rules	BR-003 (Verified Pick Before Shipment), BR-014 (FEFO Rotation for Shelf-Life Products), BR-015 (FIFO Rotation for Standard Goods), BR-017 (Lot or Batch Capture for Traceable Products)

22. Backorder

Aspect	Detail
Business Purpose	Represents a customer-agreed delay to fulfill remaining order quantity when full quantity is unavailable at confirmation or pick. Backorder is a mutual commercial commitment, not an internal note.
Identity	Backorder Reference
Responsibilities	Record unfulfilled quantity with customer acknowledgment; communicate expected fulfillment timing; offer cancel or partial delivery options; prioritize replenishment allocation; convert to reservation when stock becomes available.
Important Attributes	Backorder reference, order reference, order line reference, product reference, backordered quantity, customer acknowledgment status, expected fulfillment date, backorder reason, priority for replenishment, resolution status
Related Business Processes	BP-19 (Partial Fulfillment & Backorder Management), BP-22 (Replenishment Planning & Requisition)
Related Business Rules	BR-028 (Backorder Customer Acknowledgment), BR-001 (ATP-Only Customer Commitment)

23. Shipment

Aspect	Detail
Business Purpose	Represents the confirmed dispatch of fulfilled goods to the customer or carrier. Shipment is the point at which inventory is deducted and the delivery promise is activated.
Identity	Shipment Number
Responsibilities	Confirm verified pick completion before dispatch; deduct on-hand and release reservation; record carrier handoff; provide shipment status for customer service; create outbound movement record.
Important Attributes	Shipment number, order reference, fulfillment reference, ship-from location, ship date, carrier reference, tracking identifier, shipped lines (product, quantity, lot/batch), shipment status (dispatched / in transit / delivered)
Related Business Processes	BP-18 (Pick, Pack & Ship)
Related Business Rules	BR-003 (Verified Pick Before Shipment)

Domain 5: Network Balance

24. Transfer Request

Aspect	Detail
Business Purpose	Represents a formal request to move stock from one location to another to resolve local shortage, rebalance network inventory, or respond to operational need — evaluated before new procurement.
Identity	Transfer Request Number
Responsibilities	Specify source location, destination location, product, and quantity; validate source can spare stock above minimum threshold; route for approval; prevent unnecessary procurement when network stock exists.
Important Attributes	Transfer request number, source location, destination location, product lines (product, requested quantity), request reason, request date, requestor, approval status, approver, rejection reason
Related Business Processes	BP-24 (Inter-Location Transfer Request), BP-26 (Network Stock Rebalancing), BP-21 (Low-Stock Alert Review)
Related Business Rules	BR-011 (Network Check Before Purchase), BR-012 (Protect Source Location Minimum), BR-010 (Safety Stock Protection)

25. Stock Transfer

Aspect	Detail
Business Purpose	Represents the executed movement of stock between inventory locations, with full transaction history from dispatch through receipt at destination.
Identity	Transfer Number
Responsibilities	Deduct stock at source upon dispatch; track in-transit state; receive and verify at destination; capture lot/batch through transfer for traceable products; update stock positions at both locations; maintain audit trail.
Important Attributes	Transfer number, transfer request reference, source location, destination location, transfer lines (product, quantity, lot/batch), dispatch date, receipt date, transfer status (requested / approved / in transit / received / cancelled), in-transit quantity
Related Business Processes	BP-25 (Inter-Location Transfer Execution), BP-24 (Inter-Location Transfer Request)
Related Business Rules	BR-012 (Protect Source Location Minimum), BR-017 (Lot or Batch Capture for Traceable Products)

26. In-Transit Stock

Aspect	Detail
Business Purpose	Represents inventory that has left the source location but has not yet been received at the destination — a distinct network availability state during inter-location transfer.
Identity	In-Transit Reference (linked to Stock Transfer)
Responsibilities	Track quantity in movement between locations; exclude from source ATP and destination ATP until receipt confirmed; support network visibility of pipeline stock; enable exception handling for delayed or lost transfers.
Important Attributes	In-transit reference, stock transfer reference, product reference, quantity, source location, destination location, dispatch date, expected receipt date, in-transit status, carrier reference (if applicable)
Related Business Processes	BP-25 (Inter-Location Transfer Execution), BP-26 (Network Stock Rebalancing), BP-10 (Inventory Status Monitoring)
Related Business Rules	BR-012 (Protect Source Location Minimum)

Domain 6: Inventory Control & Accuracy

27. Cycle Count

Aspect	Detail
Business Purpose	Represents a scheduled partial physical count of selected products or locations on a rotating basis, performed without full warehouse shutdown, to measure and maintain inventory accuracy.
Identity	Cycle Count Assignment Number
Responsibilities	Assign count scope (product, location, storage position); enforce blind counting (counter not shown system quantity); schedule by ABC priority (Class A highest frequency); record physical count result; trigger variance investigation when needed.
Important Attributes	Count assignment number, count schedule date, location reference, product scope, storage position scope, assigned counter, count status (scheduled / in progress / completed), physical count quantity, count completion date
Related Business Processes	BP-11 (Cycle Counting), BP-13 (Count Variance Investigation & Approval)
Related Business Rules	BR-018 (Blind Cycle Counting), BR-032 (Class A Priority Monitoring)

28. Physical Inventory Count

Aspect	Detail
Business Purpose	Represents a comprehensive wall-to-wall physical count at a location to establish an authoritative inventory accuracy baseline, typically on annual or audit-mandated cycle.
Identity	Physical Count Event Number
Responsibilities	Enforce location freeze (no receiving, shipping, transfers, or adjustments during count); capture full location snapshot; identify variances against system records; support audit and finance baseline reconciliation.
Important Attributes	Count event number, location reference, count start date, count completion date, freeze status, count scope (full location), assigned teams, total variances identified, count certification status
Related Business Processes	BP-12 (Full Physical Inventory Count), BP-13 (Count Variance Investigation & Approval)
Related Business Rules	BR-024 (Physical Inventory Freeze)

29. Count Variance

Aspect	Detail
Business Purpose	Represents a identified difference between system-recorded quantity and physically counted quantity, requiring investigation before any adjustment is applied.
Identity	Variance Reference
Responsibilities	Quantify discrepancy (system vs. physical); trigger root cause investigation; route for independent approval (counter cannot approve own count); gate stock adjustment creation; feed accuracy metrics.
Important Attributes	Variance reference, count reference (cycle or physical), product reference, location reference, system quantity, physical quantity, variance quantity, variance value, investigation status, investigation findings, approval status, approver (independent of counter)
Related Business Processes	BP-13 (Count Variance Investigation & Approval), BP-14 (Stock Adjustment & Write-Off)
Related Business Rules	BR-019 (Independent Count Approval), BR-021 (Independent Approval for Material Adjustments), BR-022 (Investigate Before Adjusting)

30. Stock Adjustment

Aspect	Detail
Business Purpose	Represents a correction to recorded inventory quantity when physical reality differs from system records, applied with mandatory reason code and appropriate authorization.
Identity	Adjustment Reference
Responsibilities	Increase or decrease stock position with documented reason; enforce reason code and user attribution; require independent approval above threshold; update stock position and create movement record; distinguish from write-off (recoverable correction vs. permanent removal).
Important Attributes	Adjustment reference, product reference, location reference, adjustment quantity (+/−), reason code, initiating user, approval status, approver, linked variance reference (if from count), adjustment date, before/after quantities
Related Business Processes	BP-14 (Stock Adjustment & Write-Off), BP-13 (Count Variance Investigation & Approval), BP-15 (Shrinkage & Loss Management)
Related Business Rules	BR-020 (Mandatory Adjustment Reason Codes), BR-021 (Independent Approval for Material Adjustments), BR-022 (Investigate Before Adjusting)

31. Inventory Write-Off

Aspect	Detail
Business Purpose	Represents the formal, authorized removal of inventory value for goods that are damaged beyond resale, obsolete, expired, or confirmed lost — a permanent reduction distinct from reversible adjustment.
Identity	Write-Off Reference
Responsibilities	Permanently remove quantity and value from inventory; require documented evidence and authorized approval; apply mandatory reason code; reduce stock position; support shrinkage and financial reconciliation reporting.
Important Attributes	Write-off reference, product reference, location reference, write-off quantity, reason code, supporting evidence reference, initiating user, authorizing approver, write-off date, financial value impact, disposition method (destroy / donate / scrap)
Related Business Processes	BP-14 (Stock Adjustment & Write-Off), BP-15 (Shrinkage & Loss Management), BP-28 (Return-to-Stock or Disposal Decision)
Related Business Rules	BR-023 (Documented Write-Off Authorization), BR-020 (Mandatory Adjustment Reason Codes)

Domain 7: Returns Management

32. Return Authorization

Aspect	Detail
Business Purpose	Represents approved permission for a customer to return goods, linked to the original order and return policy, before the warehouse accepts physical return.
Identity	Return Authorization Number (RMA)
Responsibilities	Validate return eligibility against original order and policy; specify expected return products and quantities; communicate return instructions to customer; provide warehouse advance notice; gate receipt of returned goods.
Important Attributes	RMA number, original order reference, customer reference, authorized return lines (product, quantity), return reason, authorization date, expiration date, return disposition expectation, authorization status
Related Business Processes	BP-27 (Customer Return Intake), BP-28 (Return-to-Stock or Disposal Decision)
Related Business Rules	BR-025 (Return Authorization Required)

33. Customer Return

Aspect	Detail
Business Purpose	Represents the physical receipt of customer-returned goods at the warehouse, linked to a valid return authorization, initiating inspection and disposition workflow.
Identity	Customer Return Number
Responsibilities	Receive returned goods only with valid RMA; verify quantity and product against authorization; initiate condition inspection; route to restock, quarantine, or write-off disposition; update return processing status.
Important Attributes	Return number, RMA reference, original order reference, receiving location, receipt date, returned lines (product, quantity, condition at receipt), inspection status, receiving clerk
Related Business Processes	BP-27 (Customer Return Intake), BP-28 (Return-to-Stock or Disposal Decision), BP-07 (Goods Receipt & Verification)
Related Business Rules	BR-025 (Return Authorization Required), BR-004 (Verify Before Inventory Acceptance)

34. Return Disposition

Aspect	Detail
Business Purpose	Represents the business decision on returned goods outcome — restock to sellable inventory, quarantine for further inspection, or write-off — based on condition assessment.
Identity	Disposition Reference
Responsibilities	Assess returned product condition against resale standards; authorize restock only for qualifying goods; route damaged goods to write-off or quarantine; update stock position upon restock; prevent used/damaged goods entering sellable inventory.
Important Attributes	Disposition reference, customer return reference, product reference, quantity, condition grade, disposition decision (restock / quarantine / write-off), inspection date, inspector, restock location (if applicable), linked write-off or hold reference
Related Business Processes	BP-28 (Return-to-Stock or Disposal Decision)
Related Business Rules	BR-026 (Inspect Returns Before Restocking), BR-013 (Quarantine Exclusion from Sellable Stock), BR-023 (Documented Write-Off Authorization)

Domain 8: Accountability & Oversight

35. Inventory Movement Record

Aspect	Detail
Business Purpose	Represents a single documented change in inventory position — the immutable business fact that stock moved, was adjusted, or changed status. Movement records form the operational history of all stock activity.
Identity	Movement Transaction Reference
Responsibilities	Record every inventory change with product, location, quantity, direction, and business event type; capture user identity and timestamp; carry reason code where applicable; support traceability by lot/batch; enable operational investigation and reporting.
Important Attributes	Movement reference, movement type (receipt / shipment / transfer / adjustment / write-off / reservation / return / put-away), product reference, location reference, quantity change, lot/batch reference, reason code, originating business document reference, user identity, movement timestamp, before/after balances
Related Business Processes	BP-10 (Inventory Status Monitoring), BP-29 (Inventory Performance Reporting), all transaction processes (BP-07 through BP-09, BP-14, BP-18, BP-25, BP-28)
Related Business Rules	BR-017 (Lot or Batch Capture for Traceable Products), BR-020 (Mandatory Adjustment Reason Codes)

36. Audit Trail Record

Aspect	Detail
Business Purpose	Represents the governance-grade record of material inventory changes including user identity, timestamp, and before/after state — supporting compliance, fraud prevention, and control testing.
Identity	Audit Event Reference
Responsibilities	Provide immutable evidence of who changed what and when; support segregation-of-duties verification; enable internal and external audit queries; complement movement records with authorization and approval context.
Important Attributes	Audit event reference, affected entity type, affected entity reference, action performed, user identity, role at time of action, timestamp, before state snapshot, after state snapshot, approval chain reference, IP or session context (business accountability)
Related Business Processes	BP-32 (Inventory Audit & Control Review), BP-14 (Stock Adjustment & Write-Off), BP-13 (Count Variance Investigation & Approval)
Related Business Rules	BR-019 (Independent Count Approval), BR-021 (Independent Approval for Material Adjustments), BR-020 (Mandatory Adjustment Reason Codes)

37. Operational Alert

Aspect	Detail
Business Purpose	Represents a proactive notification of a business condition requiring human review or action — such as low stock, pending approval, fulfillment exception, or campaign readiness gap.
Identity	Alert Reference
Responsibilities	Signal threshold breaches and exception conditions; route to responsible actor; track acknowledgment and resolution; prevent silent degradation of service levels; support replenishment and governance timeliness.
Important Attributes	Alert reference, alert type (low stock / out of stock / pending approval / transfer exception / inbound delay / campaign readiness), severity, affected product/location, triggering condition, alert date, assigned recipient role, acknowledgment status, resolution status
Related Business Processes	BP-21 (Low-Stock Alert Review), BP-24 (Inter-Location Transfer Request), BP-31 (Promotional & Launch Stock Readiness), BP-20 (Outbound Exception Handling)
Related Business Rules	BR-009 (Reorder at Reorder Level), BR-032 (Class A Priority Monitoring)

38. User Account

Aspect	Detail
Business Purpose	Represents an identified individual authorized to perform inventory operations within defined role and location scope. User account is the accountability anchor for all inventory actions.
Identity	User Identifier
Responsibilities	Authenticate identity for system access; carry role assignments and location scope; enforce segregation of duties; attribute movements, adjustments, and approvals to a named individual; support provisioning and deactivation lifecycle.
Important Attributes	User identifier, display name, assigned roles, authorized locations, account status (active / suspended / deactivated), effective date, last activity date, conflicting duty flags
Related Business Processes	All operational processes; BP-32 (Inventory Audit & Control Review)
Related Business Rules	BR-019 (Independent Count Approval), BR-021 (Independent Approval for Material Adjustments)

39. Role Assignment

Aspect	Detail
Business Purpose	Represents the delegation of business permissions to a user account — defining what functions a person may perform and within which locations, implementing segregation of duties.
Identity	Role Assignment Reference (User + Role [+ Location Scope])
Responsibilities	Grant view, transact, approve, or configure authority by role; scope permissions to assigned locations; prevent conflicting duty combinations; enable role changes on employee lifecycle events.
Important Attributes	Assignment reference, user reference, role name (e.g., Warehouse Manager, Receiving Clerk, Buyer), location scope, permission set, assignment effective date, assignment status, segregation-of-duties compliance flag
Related Business Processes	BP-32 (Inventory Audit & Control Review); cross-cutting across all processes
Related Business Rules	BR-019 (Independent Count Approval), BR-021 (Independent Approval for Material Adjustments), BR-023 (Documented Write-Off Authorization)

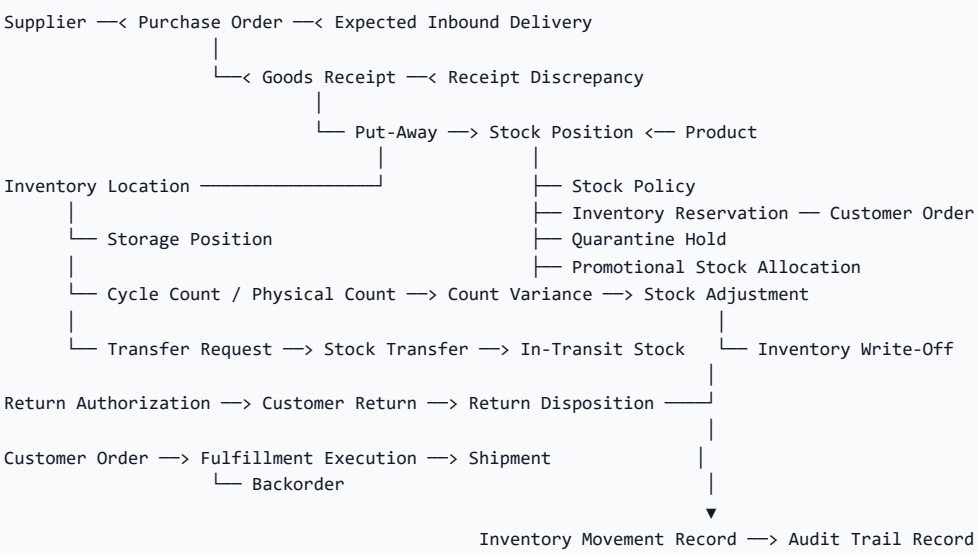
Domain 9: Financial Alignment

40. Inventory Valuation Snapshot

Aspect	Detail
Business Purpose	Represents a point-in-time summary of inventory value by product, location, and category for alignment with financial records and month-end close — bridging operational stock records to finance.
Identity	Valuation Snapshot Reference (Period + Scope)
Responsibilities	Summarize owned inventory value excluding consignment not yet consumed; support reconciliation against general ledger; identify valuation variances; inform working capital and carrying cost analysis; exclude quarantined and in-transit stock per policy.
Important Attributes	Snapshot reference, valuation date, location scope, product lines (product, quantity, unit cost, extended value), total inventory value, ownership breakdown (owned / consignment), variance vs. general ledger, reconciliation status, prepared by
Related Business Processes	BP-30 (Inventory Valuation & Financial Reconciliation), BP-29 (Inventory Performance Reporting)
Related Business Rules	BR-008 (Three-Way Match Before Payment), BR-031 (Consignment Ownership at Consumption)

Entity Relationship Overview

The following diagram summarizes primary business relationships between core entities (cardinality simplified for readability):



Entity-to-Process Traceability Matrix

Entity	Primary Processes
Product	BP-01, BP-04, BP-07, BP-18, BP-22
Inventory Location	BP-02, BP-10, BP-12, BP-24, BP-25
Storage Position	BP-09, BP-18
Stock Policy	BP-03, BP-21, BP-22
Reason Code	BP-08, BP-14, BP-15
Governance Policy	BP-13, BP-14, BP-32
Stock Position	BP-10, BP-16, BP-17, BP-21
Lot/Batch	BP-07, BP-09, BP-18, BP-25
Quarantine Hold	BP-14, BP-20
Promotional Stock Allocation	BP-31, BP-17
Supplier	BP-05, BP-07, BP-30
Purchase Requisition	BP-22
Purchase Order	BP-05, BP-06, BP-07, BP-23
Expected Inbound Delivery	BP-06, BP-23
Goods Receipt	BP-07, BP-09
Receipt Discrepancy	BP-08
Put-Away	BP-09
Supplier Invoice	BP-30
Customer Order	BP-16, BP-17, BP-18, BP-19
Inventory Reservation	BP-17
Fulfillment Execution	BP-18, BP-20
Backorder	BP-19
Shipment	BP-18
Transfer Request	BP-24
Stock Transfer	BP-25

Entity	Primary Processes
In-Transit Stock	BP-25, BP-26
Cycle Count	BP-11
Physical Inventory Count	BP-12
Count Variance	BP-13
Stock Adjustment	BP-14
Inventory Write-Off	BP-14, BP-15
Return Authorization	BP-27
Customer Return	BP-27, BP-28
Return Disposition	BP-28
Inventory Movement Record	BP-10, BP-29 (all transaction processes)
Audit Trail Record	BP-32
Operational Alert	BP-21, BP-31
User Account	Cross-cutting
Role Assignment	Cross-cutting
Inventory Valuation Snapshot	BP-30

Entity-to-Business-Rule Traceability Matrix

Business Rule	Primary Entities Governed
BR-001	Stock Position, Customer Order, Inventory Reservation
BR-002	Inventory Reservation, Stock Position
BR-003	Fulfillment Execution, Shipment
BR-004	Goods Receipt, Customer Return
BR-005	Put-Away, Stock Position
BR-006	Receipt Discrepancy, Goods Receipt
BR-007	Purchase Order, Goods Receipt
BR-008	Purchase Order, Goods Receipt, Supplier Invoice
BR-009	Stock Policy, Stock Position, Operational Alert
BR-010	Stock Policy, Inventory Reservation, Transfer Request
BR-011	Transfer Request, Purchase Requisition, Stock Position
BR-012	Transfer Request, Stock Transfer, Stock Policy
BR-013	Quarantine Hold, Stock Position
BR-014	Lot/Batch, Put-Away, Fulfillment Execution
BR-015	Stock Position, Put-Away, Fulfillment Execution
BR-016	Goods Receipt, Receipt Discrepancy, Lot/Batch
BR-017	Lot/Batch, Goods Receipt, Inventory Movement Record
BR-018	Cycle Count
BR-019	Count Variance, Cycle Count, User Account
BR-020	Stock Adjustment, Reason Code, Inventory Movement Record
BR-021	Stock Adjustment, Count Variance, Governance Policy

Business Rule	Primary Entities Governed
BR-022	Count Variance, Stock Adjustment
BR-023	Inventory Write-Off, User Account
BR-024	Physical Inventory Count, Inventory Location
BR-025	Return Authorization, Customer Return
BR-026	Return Disposition, Customer Return
BR-027	Product, Purchase Order, Purchase Requisition
BR-028	Backorder, Customer Order
BR-029	Promotional Stock Allocation, Inventory Reservation
BR-030	Quarantine Hold, Put-Away, Goods Receipt
BR-031	Stock Position, Inventory Valuation Snapshot, Supplier
BR-032	Product, Cycle Count, Operational Alert

Document Approval

Role	Name	Signature	Date
Solution Architect			
Senior Business Analyst			
Product Owner			
Inventory Controller			

End of Domain Entity Model

Document ID: SRS-DM-001 | Version 1.0 | July 5, 2026

Domain Value Objects

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-DM-002
Document Title	Domain Value Objects — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Domain Expert
Source Documents	BRD-IMS-001 (v1.0), SRS-DM-001 (Domain Entity Model, v1.0), SRS-BR-001 (Business Rules, v1.0), SRS-UC-001 (Use Case Catalog, v1.0)
Date	July 5, 2026
Scope	Immutable business concepts only — no database columns, API payloads, or implementation detail

Purpose

This document identifies **Value Objects** in the Inventory Management System domain. A value object is a business concept defined entirely by its attribute values, with no independent identity or lifecycle. Two instances with the same values are interchangeable. When business meaning changes, the value object is replaced — not mutated.

This catalog complements **SRS-DM-001 (Domain Entity Model)**, which documents entities that carry identity and lifecycle.

Value Object vs. Entity — Quick Reference

Characteristic	Entity	Value Object
Identity	Has a unique business identifier	No identity — equality by value
Mutability	Lifecycle with state changes	Immutable — replace when values change
Example	Purchase Order (PO-2026-0042)	Monetary Amount ($12.50 USD)

Value Object Domain Map

Domain	Value Objects
Measurement & Quantity	Quantity, Unit of Measure, Monetary Amount, Lead Time Duration
Product & Classification	Product Identifier, Product Category, Product Lifecycle Status, ABC Classification, Rotation Policy, Shelf Life Profile
Stock & Availability	Stock Availability Breakdown, Stock Status Indicator, Ownership Classification, Stock Threshold Set
Location & Address	Physical Address, Storage Position Address, Location Type
Time & Period	Effective Period, Campaign Period
Transaction Lines	Purchase Order Line, Order Line Quantity, Receipt Line Detail, Transfer Line Detail, Invoice Line Detail, Shipment Line Detail, Pick Line Detail, Requisition Line Detail
Variance & Inspection	Quantity Variance, Condition Grade, Discrepancy Classification, Disposition Decision
Governance & Control	Reason Code Attribution, Approval Threshold, Quantity Tolerance, Three-Way Match Result, User Action Attribution, Audit Quantity Snapshot
Fulfillment & Commercial	Fulfillment Quantity Breakdown, Customer Acknowledgment, Channel Designation, Payment Terms
Movement & Status	Movement Classification, Hold Reason Classification, Alert Priority
Financial & Traceability	Valuation Line Amount, Lot Identity

Total Value Objects: 46

Domain 1: Measurement & Quantity

1. Quantity

Aspect	Detail
Definition	A non-negative numeric amount of a product expressed in a specific unit of measure. Quantity is the fundamental measure for all inventory balances, movements, orders, and counts.
Why It Is a Value Object	A quantity of "50 Each" has no identity of its own — only meaning through its numeric amount and unit. Two quantities with the same amount and unit are identical. When count changes, a new quantity replaces the old one rather than mutating it in place.
Attributes	Amount (non-negative number), unit of measure
Related Entities	Stock Position, Goods Receipt, Shipment, Stock Adjustment, Inventory Reservation, Cycle Count, Count Variance, Purchase Order, Customer Order

Aspect	Detail
Example	`Quantity(50, Each)` — fifty units of a product counted, ordered, or moved

2. Unit of Measure

Aspect	Detail
Definition	The standard unit in which a product is counted, ordered, stored, and sold — such as Each, Case, Pallet, Kilogram, or Liter.
Why It Is a Value Object	UOM is a classification label with no lifecycle. "Case" means the same regardless of where it appears. It is defined by its code and description, not by a unique identifier.
Attributes	UOM code, display name, base-unit indicator (whether this is the product's primary counting unit)
Related Entities	Product, Quantity (embedded), Purchase Order Line, Order Line Quantity
Example	`UnitOfMeasure(CS, Case, baseUnit=false)` — one case as a ordering unit distinct from the each-level base unit

3. Monetary Amount

Aspect	Detail
Definition	A monetary value expressed as an amount in a specific currency, used for purchase pricing, invoice matching, adjustment valuation, and write-off financial impact.
Why It Is a Value Object	"$1,250.00 USD" has no independent identity — it is fully defined by amount and currency. Arithmetic and comparison operate on values; changing amount creates a new monetary amount.
Attributes	Amount (decimal), currency code (e.g., USD, EUR)
Related Entities	Purchase Order, Supplier Invoice, Inventory Write-Off, Stock Adjustment, Inventory Valuation Snapshot, Approval Threshold
Example	`MonetaryAmount(1250.00, USD)` — extended line value on a purchase order

4. Lead Time Duration

Aspect	Detail
Definition	The elapsed business time assumed between placing a replenishment order and having goods available for sale or use at a location.
Why It Is a Value Object	Lead time is a planning assumption defined by its duration value and unit — not tracked as a separate business object. Replacing "14 days" with "21 days" means replacing the whole value.
Attributes	Duration value (positive integer), duration unit (business days / calendar days), source (supplier default / product-specific / location override)
Related Entities	Stock Policy, Supplier, Purchase Order, Expected Inbound Delivery
Example	`LeadTimeDuration(14, businessDays, supplierDefault)` — two-week supplier lead time for replenishment planning

Domain 2: Product & Classification

5. Product Identifier

Aspect	Detail
Definition	A typed product lookup code — SKU, barcode, or alternate identifier — used to reference a product in operational transactions without carrying full product master data.
Why It Is a Value Object	The identifier string "SKU-8842" paired with type "SKU" is a value tuple. It identifies a product concept but has no lifecycle separate from the Product entity it references.

Aspect	Detail
Attributes	Identifier type (SKU / barcode / alternate), identifier value (non-empty string)
Related Entities	Product, Goods Receipt, Fulfillment Execution, Cycle Count, Customer Return
Example	`ProductIdentifier(SKU, SKU-8842)` — scanned or entered at receiving dock to match purchase order line

6. Product Category

Aspect	Detail
Definition	A business grouping label that classifies products for reporting, filtering, and operational segmentation.
Why It Is a Value Object	Category is defined by its code and name. "Electronics" is the same category wherever referenced; renaming creates a new category value applied going forward.
Attributes	Category code, category name, parent category code (optional, for hierarchy)
Related Entities	Product, Inventory Valuation Snapshot, Operational Alert (filtering scope)
Example	`ProductCategory(ELEC, Electronics)` — classifies a product for dashboard filtering

7. Product Lifecycle Status

Aspect	Detail
Definition	The current commercial availability state of a product: active (orderable and fulfillable), inactive (not sold but may hold stock), or discontinued (exit from assortment, no reorder).
Why It Is a Value Object	Status is an enumerated business state without its own identity. `Active` equals `Active` everywhere. Transition from active to discontinued replaces the status value on the Product entity.
Attributes	Status value (active / inactive / discontinued)
Related Entities	Product, Purchase Order, Purchase Requisition, Inventory Reservation, Customer Order
Example	`ProductLifecycleStatus(discontinued)` — blocks new purchase requisitions per BR-027

8. ABC Classification

Aspect	Detail
Definition	A priority segment (A, B, or C) assigned to a product based on business importance — revenue contribution, velocity, or criticality — to prioritize operational effort.
Why It Is a Value Object	Class "A" is a label with fixed meaning across the catalog. It has no identity beyond its letter value and the classification criteria snapshot may accompany it.
Attributes	Class value (A / B / C), classification basis note (optional: revenue / velocity / criticality)
Related Entities	Product, Cycle Count, Operational Alert, Stock Policy
Example	`ABCClassification(A, revenue)` — triggers highest cycle count frequency per BR-032

9. Rotation Policy

Aspect	Detail
Definition	The required stock rotation discipline for a product: FIFO (first-in, first-out) for standard goods or FEFO (first-expired, first-out) for shelf-life products.
Why It Is a Value Object	Rotation policy is a rule classification with no independent existence. Two products both assigned `FEFO` share the same policy value.
Attributes	Policy type (FIFO / FEFO), shelf-life applicability indicator

Aspect	Detail
Related Entities	Product, Put-Away, Fulfillment Execution, Storage Position
Example	`RotationPolicy(FEFO, shelfLifeApplicable=true)` — pick instructions prioritize earliest expiration per BR-014

10. Shelf Life Profile

Aspect	Detail
Definition	A point-in-time capture of shelf-life attributes for a received or held lot — expiration date, manufacture date, and remaining shelf life at receipt — used to enforce MRSL and FEFO decisions.
Why It Is a Value Object	The tuple of dates and remaining days at a specific moment is immutable fact. When re-measured, a new profile replaces the prior snapshot. It describes condition, not a tracked object.
Attributes	Expiration date, manufacture date (optional), remaining shelf life in days (at capture point), MRSL policy threshold in days
Related Entities	Lot/Batch, Goods Receipt, Receipt Line Detail, Receipt Discrepancy, Fulfillment Execution
Example	`ShelfLifeProfile(expiration=2026-12-01, remainingDays=180, mrslThreshold=90)` — passes MRSL check at receipt per BR-016

Domain 3: Stock & Availability

11. Stock Availability Breakdown

Aspect	Detail
Definition	The decomposed view of inventory quantity at a point in time showing on-hand, reserved, available-to-promise (ATP), and optionally inbound expected — each as distinct quantity components.
Why It Is a Value Object	The breakdown is a computed snapshot defined entirely by its component quantities. Two breakdowns with identical components represent the same availability state. ATP is derived, not independently identified.
Attributes	On-hand quantity, reserved quantity, ATP quantity, inbound expected quantity (optional), quarantined quantity (optional)
Related Entities	Stock Position, Customer Order, Inventory Reservation, Operational Alert
Example	`StockAvailabilityBreakdown(onHand=100, reserved=30, atp=65, quarantined=5)` — 65 units quotable to new customers per BR-001

12. Stock Status Indicator

Aspect	Detail
Definition	A derived classification of inventory health for a product at a location: in stock, low stock, out of stock, or overstock — based on comparison against stock policy thresholds.
Why It Is a Value Object	The indicator is a computed label with no identity. `Low Stock` at one moment is replaced by `Out of Stock` when thresholds are crossed — a new value, not a mutation.
Attributes	Status value (inStock / lowStock / outOfStock / overstock), evaluated against threshold reference (optional)
Related Entities	Stock Position, Stock Policy, Operational Alert
Example	`StockStatusIndicator(lowStock, reorderLevel)` — triggers replenishment alert per BR-009

13. Ownership Classification

Aspect	Detail
Definition	The legal and financial ownership state of held inventory: owned (organization asset), consignment (supplier-owned until consumption), or bonded (customs-controlled until clearance).
Why It Is a Value Object	Ownership type is an enumerated classification without lifecycle. It determines valuation and availability rules but is not itself a trackable business object.
Attributes	Ownership type (owned / consignment / bonded), ownership transfer trigger (for consignment: consumption / sale / issue)
Related Entities	Stock Position, Quarantine Hold, Goods Receipt, Inventory Valuation Snapshot
Example	`OwnershipClassification(consignment, transferOnConsumption)` — excluded from owned valuation until consumption per BR-031

14. Stock Threshold Set

Aspect	Detail
Definition	The replenishment policy limits for a product at a location: reorder level, minimum threshold, safety stock buffer, and maximum (overstock) level.
Why It Is a Value Object	Thresholds are a coherent set of numeric limits that define policy at a point in time. When recalibrated, the entire set is replaced with a new effective threshold set — preserving history as immutable snapshots.
Attributes	Reorder level (quantity), minimum threshold (quantity), safety stock (quantity), maximum level (quantity)
Related Entities	Stock Policy, Stock Position, Transfer Request, Operational Alert
Example	`StockThresholdSet(reorder=50, minimum=20, safety=15, maximum=200)` — reorder action triggers when ATP reaches 50

Domain 4: Location & Address

15. Physical Address

Aspect	Detail
Definition	The geographic address of an inventory location used for logistics, regional reporting, and network hierarchy grouping.
Why It Is a Value Object	An address is defined by its component lines and region. Correcting a postal code replaces the address value — it has no independent business identity.
Attributes	Street lines, city, state or region, postal code, country
Related Entities	Inventory Location, Supplier
Example	`PhysicalAddress(100 Warehouse Blvd, Dallas, TX, 75201, US)` — regional grouping for network reporting

16. Storage Position Address

Aspect	Detail
Definition	The within-location address of a storage area — zone, aisle, bin, or shelf — where stock is physically placed after put-away.
Why It Is a Value Object	"Zone A / Aisle 3 / Bin 12" is a composite location value with no lifecycle. Two pickers directed to the same address are going to the same place.
Attributes	Zone, aisle, bin/shelf, storage type (bulk / pickFace / quarantine / bonded)
Related Entities	Storage Position, Put-Away, Fulfillment Execution, Cycle Count
Example	`StoragePositionAddress(zone=A, aisle=3, bin=12, type=pickFace)` — pick path destination for order line

17. Location Type

Aspect	Detail
Definition	The operational classification of an inventory location: warehouse, store, or distribution point — determining its role in the network.
Why It Is a Value Object	Location type is a fixed enumeration. "Warehouse" carries the same business meaning across all locations of that type.
Attributes	Type value (warehouse / store / distributionPoint)
Related Entities	Inventory Location
Example	`LocationType(warehouse)` — central fulfillment site in the distribution network

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Domain 5: Time & Period

18. Effective Period

Aspect	Detail
Definition	A bounded date range during which a business policy, threshold set, or governance rule is in force.
Why It Is a Value Object	A period from July 1 to September 30 is defined by its start and end dates. Changing the end date creates a new period value; historical periods remain immutable records.
Attributes	Start date, end date (optional — open-ended if null), period label (optional)
Related Entities	Stock Policy, Governance Policy, Promotional Stock Allocation, Role Assignment
Example	`EffectivePeriod(2026-07-01, 2026-09-30, Q3-2026 Policy)` — stock threshold set valid for one quarter

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19. Campaign Period

Aspect	Detail
Definition	The bounded date range and commercial scope of a promotional campaign or product launch during which ring-fenced inventory is protected.
Why It Is a Value Object	Campaign timing is a commercial fact defined by dates and name. It has no identity separate from the promotional allocation entity that references it.
Attributes	Campaign name, start date, end date, channel scope reference
Related Entities	Promotional Stock Allocation, Operational Alert, Customer Order
Example	`CampaignPeriod(Summer Sale 2026, 2026-06-01, 2026-08-31, eCommerce)` — ring-fenced stock protected per BR-029

Domain 6: Transaction Lines

Transaction line value objects represent immutable product-quantity-price tuples embedded within document entities. They carry no independent identity — line sequence provides ordering context only.

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20. Purchase Order Line

Aspect	Detail
Definition	A single product line on a purchase order specifying ordered quantity, unit price, and expected receipt quantity tracking.
Why It Is a Value Object	PO line meaning is fully defined by product, quantity, and price. Amending a line replaces it with a new line value; the tuple "Product X, 100 units, $5.00" has no identity beyond its content.

Aspect	Detail
Attributes	Product identifier, ordered quantity, unit price (monetary amount), received quantity to date, line status (open / partial / closed)
Related Entities	Purchase Order, Goods Receipt, Supplier Invoice, Expected Inbound Delivery
Example	`PurchaseOrderLine(SKU-4420, Quantity(100, Each), $5.00 USD, received=60, partial)` — open PO line awaiting balance receipt

21. Requisition Line Detail

Aspect	Detail
Definition	A product and requested quantity on an internal purchase requisition before commercial PO creation.
Why It Is a Value Object	The requisition line is a need statement — product plus quantity — with no independent lifecycle. Approval converts it to a PO line value.
Attributes	Product identifier, requested quantity, target location, requested delivery date
Related Entities	Purchase Requisition, Purchase Order, Operational Alert
Example	`RequisitionLineDetail(SKU-4420, Quantity(200, Each), WH-DALLAS, 2026-07-15)` — replenishment need from low-stock alert

22. Receipt Line Detail

Aspect	Detail
Definition	A verified inbound line recording product, received quantity, condition assessment, and optional lot identity captured at goods receipt.
Why It Is a Value Object	What was received is an immutable fact at receipt time. Correcting a receipt creates a new receipt line value or adjustment — the original received tuple remains historical truth.
Attributes	Product identifier, received quantity, condition grade, lot identity (optional), shelf life profile (optional), verification outcome (accepted / rejected / hold)
Related Entities	Goods Receipt, Receipt Discrepancy, Put-Away, Lot/Batch
Example	`ReceiptLineDetail(SKU-4420, Quantity(98, Each), good, LotIdentity(L20260701), accepted)` — two units short triggers discrepancy

23. Order Line Quantity

Aspect	Detail
Definition	The quantity breakdown for a customer order line showing requested, reserved, fulfilled, and backordered amounts.
Why It Is a Value Object	Line quantities at a snapshot moment are a value decomposition. As fulfillment progresses, new quantity breakdown values replace prior ones.
Attributes	Product identifier, requested quantity, reserved quantity, fulfilled quantity, backordered quantity
Related Entities	Customer Order, Inventory Reservation, Fulfillment Execution, Backorder
Example	`OrderLineQuantity(SKU-4420, requested=50, reserved=50, fulfilled=30, backordered=20)` — partial fulfillment state

24. Transfer Line Detail

Aspect	Detail
Definition	A product and quantity tuple on an inter-location transfer, optionally specifying lot identity for traceable products.

Aspect	Detail
Why It Is a Value Object	Transfer line content defines what moves. The tuple has no identity beyond product, quantity, and lot — embedded in the Stock Transfer entity.
Attributes	Product identifier, transfer quantity, lot identity (optional), source storage position address (optional)
Related Entities	Transfer Request, Stock Transfer, In-Transit Stock
Example	`TransferLineDetail(SKU-4420, Quantity(25, Each), LotIdentity(L20260701))` — traceable transfer between locations per BR-017

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25. Invoice Line Detail

Aspect	Detail
Definition	A billed product line on a supplier invoice specifying billed quantity and unit price for three-way match evaluation.
Why It Is a Value Object	Invoice line is a financial fact tuple. Matching compares values across PO line and receipt line — the invoice line itself has no independent identity.
Attributes	Product identifier, billed quantity, unit price (monetary amount), extended amount (monetary amount)
Related Entities	Supplier Invoice, Purchase Order, Goods Receipt, Three-Way Match Result
Example	`InvoiceLineDetail(SKU-4420, Quantity(98, Each), $5.00 USD, $490.00 USD)` — matched against PO and receipt for payment per BR-008

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26. Pick Line Detail

Aspect	Detail
Definition	The pick execution record for an order line showing requested versus picked quantity, storage source, and lot applied during fulfillment.
Why It Is a Value Object	Pick result is an immutable operational fact at verification time. Failed picks generate new pick line values or exceptions — the verified pick tuple does not mutate.
Attributes	Product identifier, requested quantity, picked quantity, lot identity (optional), source storage position address, pick verification status (pending / verified / short)
Related Entities	Fulfillment Execution, Shipment, Lot/Batch, Storage Position
Example	`PickLineDetail(SKU-4420, requested=10, picked=10, LotIdentity(L20260701), verified)` — verified pick enables shipment per BR-003

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27. Shipment Line Detail

Aspect	Detail
Definition	A dispatched product line on a shipment recording shipped quantity and lot attribution at carrier handoff.
Why It Is a Value Object	What left the building is an immutable dispatch fact. The shipped tuple defines the outbound movement — no separate identity required.
Attributes	Product identifier, shipped quantity, lot identity (optional), order line reference
Related Entities	Shipment, Fulfillment Execution, Inventory Movement Record
Example	`ShipmentLineDetail(SKU-4420, Quantity(10, Each), LotIdentity(L20260701))` — inventory deduction at dispatch

Domain 7: Variance & Inspection

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28. Quantity Variance

Aspect	Detail
Definition	The calculated difference between system-recorded quantity and physically counted or expected quantity, expressed as absolute and signed variance.
Why It Is a Value Object	Variance is a mathematical result defined by two input quantities. System=100, Physical=95 yields Variance=−5 regardless of when or where calculated.
Attributes	System quantity, physical or expected quantity, variance quantity (signed), variance direction (short / over / match)
Related Entities	Count Variance, Cycle Count, Physical Inventory Count, Receipt Discrepancy, Stock Adjustment
Example	`QuantityVariance(system=100, physical=95, variance=-5, short)` — triggers count variance investigation per BR-022

29. Condition Grade

Aspect	Detail
Definition	The assessed physical and resale condition of received or returned goods at inspection time.
Why It Is a Value Object	Condition is an evaluated classification at a moment in time. "Damaged — not resalable" is a value judgment captured once; reassessment produces a new grade value.
Attributes	Grade value (new / good / acceptable / damaged / defective / expired), inspection notes (optional), resalable indicator (yes / no)
Related Entities	Receipt Line Detail, Customer Return, Return Disposition, Goods Receipt
Example	`ConditionGrade(damaged, resalable=false)` — returned goods routed to write-off per BR-026

30. Discrepancy Classification

Aspect	Detail
Definition	The categorized type of receipt variance: short shipment, over-shipment, wrong item, damage, or shelf-life non-compliance.
Why It Is a Value Object	Discrepancy type is an enumerated classification of a variance event. It carries business meaning for resolution workflow but has no independent identity.
Attributes	Discrepancy type (short / over / wrongItem / damage / shelfLifeNonCompliance), severity indicator (optional)
Related Entities	Receipt Discrepancy, Goods Receipt, Receipt Line Detail
Example	`DiscrepancyClassification(shortShipment)` — documented at receipt before acceptance per BR-006

31. Disposition Decision

Aspect	Detail
Definition	The outcome decision for returned or non-conforming goods: restock to sellable inventory, place in quarantine, or write off.
Why It Is a Value Object	A disposition is a final classification applied at decision time. Changing decision replaces the value — the decision moment and outcome form an immutable tuple with condition grade.
Attributes	Decision value (restock / quarantine / writeOff), decision basis (condition grade reference), decision date
Related Entities	Return Disposition, Customer Return, Quarantine Hold, Inventory Write-Off
Example	`DispositionDecision(restock, ConditionGrade(good), 2026-07-05)` — returned item returns to sellable ATP

Domain 8: Governance & Control

32. Reason Code Attribution

Aspect	Detail
Definition	An immutable snapshot of the reason code and description applied to an inventory adjustment, write-off, or exception at the time of the transaction — preserving meaning even if the master reason code is later retired.
Why It Is a Value Object	Historical transactions must retain why they occurred. Capturing `{code: SHRINK, description: "Unexplained loss"}` at transaction time creates an immutable attribution value independent of the Reason Code entity lifecycle.
Attributes	Reason code, reason description, reason category (adjustment / shrinkage / damage / obsolescence / receipt / return)
Related Entities	Stock Adjustment, Inventory Write-Off, Receipt Discrepancy, Inventory Movement Record, Reason Code (master reference entity)
Example	`ReasonCodeAttribution(SHRINK, Unexplained loss, shrinkage)` — mandatory on every adjustment per BR-020

33. Approval Threshold

Aspect	Detail
Definition	A governance limit — expressed as quantity, monetary amount, or both — above which independent approval is required before an inventory action takes effect.
Why It Is a Value Object	Threshold limits are policy numbers without identity. "$5,000 adjustment requires director approval" is a rule value applied at evaluation time.
Attributes	Threshold type (quantity / monetary / both), quantity limit (optional), monetary limit (optional), required approver role
Related Entities	Governance Policy, Stock Adjustment, Count Variance, Inventory Write-Off
Example	`ApprovalThreshold(monetary, limit=$5000, approver=OperationsDirector)` — independent approval per BR-021

34. Quantity Tolerance

Aspect	Detail
Definition	The acceptable variance range between ordered, received, and billed quantities or amounts within which automatic match approval is permitted.
Why It Is a Value Object	Tolerance of "±2% or ±5 units" is a policy parameter value. It defines acceptable bounds — not a tracked business object.
Attributes	Tolerance type (percentage / absolute quantity / absolute monetary), tolerance value, evaluation basis (quantity / amount)
Related Entities	Governance Policy, Receipt Discrepancy, Three-Way Match Result, Supplier Invoice
Example	`QuantityTolerance(percentage, 2%, quantity)` — receipt within 2% auto-accepted; beyond triggers discrepancy

35. Three-Way Match Result

Aspect	Detail
Definition	The outcome of comparing purchase order, goods receipt, and supplier invoice lines — indicating matched within tolerance or exception requiring resolution.
Why It Is a Value Object	Match result is a computed evaluation at a point in time: matched, quantity exception, price exception, or unmatched. Same inputs yield same result — no independent identity.
Attributes	Match status (matched / quantityException / priceException / unmatched), variance quantity (optional), variance amount (optional), tolerance applied
Related Entities	Supplier Invoice, Purchase Order, Goods Receipt, Supplier
Example	`ThreeWayMatchResult(matched, tolerance=QuantityTolerance(2%))` — payment authorized per BR-008

36. User Action Attribution

Aspect	Detail
Definition	An immutable record of who performed or approved a business action, including user identity and role at the time of action — supporting accountability and segregation of duties.
Why It Is a Value Object	The tuple `{user: J.Smith, role: WarehouseManager, timestamp: 2026-07-05T14:30}` at action time is historical fact. Role changes later do not alter the captured attribution value.
Attributes	User identifier, user display name, role at time of action, action timestamp
Related Entities	Inventory Movement Record, Audit Trail Record, Stock Adjustment, Count Variance, Inventory Write-Off, User Account
Example	`UserActionAttribution(WH-MGR-007, Jane Smith, WarehouseManager, 2026-07-05T14:30)` — approver distinct from counter per BR-019

37. Audit Quantity Snapshot

Aspect	Detail
Definition	A before-and-after capture of inventory quantity or status at the moment of a material change, forming the evidential basis of audit trail records.
Why It Is a Value Object	Snapshot `{before: 100, after: 95}` is immutable historical fact. It describes state transition — not an entity with its own lifecycle.
Attributes	Before quantity, after quantity, before status (optional), after status (optional), snapshot timestamp
Related Entities	Audit Trail Record, Stock Adjustment, Inventory Movement Record, Count Variance
Example	`AuditQuantitySnapshot(before=100, after=95, timestamp=2026-07-05T14:30)` — records adjustment impact for audit

Domain 9: Fulfillment & Commercial

38. Fulfillment Quantity Breakdown

Aspect	Detail
Definition	The decomposition of an order line's quantity into fulfilled, backordered, and cancelled portions after availability confirmation or pick execution.
Why It Is a Value Object	Breakdown `{requested: 50, fulfilled: 30, backordered: 20}` is a state summary defined by its components. Progress replaces the breakdown value — it does not mutate in place.
Attributes	Requested quantity, fulfilled quantity, backordered quantity, cancelled quantity
Related Entities	Customer Order, Backorder, Fulfillment Execution, Shipment
Example	`FulfillmentQuantityBreakdown(requested=50, fulfilled=30, backordered=20, cancelled=0)` — partial fulfillment with customer acknowledgment per BR-028

39. Customer Acknowledgment

Aspect	Detail
Definition	Explicit customer consent to accept a backorder or partial delivery, including acknowledgment method and timestamp — elevating backorder from internal note to mutual agreement.
Why It Is a Value Object	Acknowledgment is a captured fact: who agreed, when, and how. It has no lifecycle — either the acknowledgment value exists or it does not.
Attributes	Acknowledgment status (pending / acknowledged / declined), acknowledgment method (phone / email / portal), acknowledgment timestamp, acknowledged by (customer contact reference)

Aspect	Detail
Related Entities	Backorder, Customer Order, Fulfillment Quantity Breakdown
Example	`CustomerAcknowledgment(acknowledged, email, 2026-07-05T10:15, customerContact=ORD-Acct-442)` — required before backorder recorded per BR-028

40. Channel Designation

Aspect	Detail
Definition	The sales or fulfillment channel to which promotional stock or an order is attributed — such as eCommerce, retail store, B2B, or promotional campaign.
Why It Is a Value Object	Channel label is a commercial classification without identity. "eCommerce" means the same across all orders and allocations referencing it.
Attributes	Channel code, channel name, channel type (direct / partner / promotional)
Related Entities	Customer Order, Promotional Stock Allocation, Campaign Period
Example	`ChannelDesignation(B2B, Business Direct, direct)` — order fulfilled from B2B allocation pool

41. Payment Terms

Aspect	Detail
Definition	The agreed commercial payment conditions between the organization and a supplier, governing invoice due dates and payment authorization timing.
Why It Is a Value Object	"Net 30" is a terms classification defined by its code and due-day count. It has no lifecycle independent of the supplier relationship context.
Attributes	Terms code (e.g., NET30, NET60, COD), due days, discount terms (optional: percentage and days)
Related Entities	Supplier, Purchase Order, Supplier Invoice
Example	`PaymentTerms(NET30, dueDays=30)` — invoice due 30 days from receipt on supplier agreement

Domain 10: Movement & Status Classification

42. Movement Classification

Aspect	Detail
Definition	The categorized type of inventory movement event: receipt, shipment, transfer, adjustment, write-off, reservation, return, or put-away.
Why It Is a Value Object	Movement type is an enumerated label classifying what happened. Two receipt movements share the same classification value — identity belongs to the movement record entity.
Attributes	Movement type (receipt / shipment / transfer / adjustment / writeOff / reservation / return / putAway), direction (inbound / outbound / neutral)
Related Entities	Inventory Movement Record, Goods Receipt, Shipment, Stock Transfer, Stock Adjustment
Example	`MovementClassification(transfer, neutral)` — inter-location stock movement between source and destination

43. Hold Reason Classification

Aspect	Detail
Definition	The categorized reason inventory is placed under quarantine or hold: quality investigation, regulatory restriction, product recall, or customs/bonded clearance pending.
Why It Is a Value Object	Hold reason is a classification enum determining exclusion from ATP. The reason value defines business treatment — not a separately tracked object.
Attributes	Hold type (quality / regulatory / recall / customs / investigation), sellable exclusion indicator (always true for holds)
Related Entities	Quarantine Hold, Stock Position, Goods Receipt
Example	`HoldReasonClassification(customs, sellableExcluded=true)` — bonded stock excluded from ATP per BR-030

44. Alert Priority

Aspect	Detail
Definition	The urgency classification assigned to an operational alert based on severity of the underlying condition — such as Class A stockout versus routine low-stock warning.
Why It Is a Value Object	Priority is a severity label (critical / high / medium / low) with no identity. Same condition severity always maps to the same priority value.
Attributes	Priority level (critical / high / medium / low), escalation indicator, response time expectation (optional)
Related Entities	Operational Alert, Stock Status Indicator, ABC Classification
Example	`AlertPriority(critical, escalation=true, responseTime=sameDay)` — Class A stockout alert per BR-032

Domain 11: Financial & Traceability

45. Valuation Line Amount

Aspect	Detail
Definition	The financial value of an inventory quantity calculated as unit cost multiplied by quantity — used in valuation snapshots and write-off impact assessment.
Why It Is a Value Object	Extended value is a computed financial tuple (quantity × unit cost). Same inputs always produce the same value — no independent business identity.
Attributes	Product identifier, quantity, unit cost (monetary amount), extended value (monetary amount), ownership classification
Related Entities	Inventory Valuation Snapshot, Inventory Write-Off, Stock Adjustment
Example	`ValuationLineAmount(SKU-4420, Quantity(100, Each), $5.00 USD, $500.00 USD, owned)` — month-end valuation line

46. Lot Identity

Aspect	Detail
Definition	The identifying tuple for a traceable production or receipt batch — lot number combined with key dates — captured on movements without referencing the full Lot/Batch entity state.
Why It Is a Value Object	Lot identity `{number: L20260701, expiration: 2026-12-01}` on a pick line is an immutable identifier tuple embedded in transactions. It describes which batch was touched — the Lot/Batch entity carries ongoing stock state.
Attributes	Lot/batch number, expiration date (optional), manufacture date (optional)
Related Entities	Lot/Batch, Receipt Line Detail, Pick Line Detail, Transfer Line Detail, Shipment Line Detail, Inventory Movement Record
Example	`LotIdentity(L20260701, expiration=2026-12-01)` — captured on pick for traceability per BR-017

Value Object Composition Patterns

Several domain entities are composed of multiple value objects rather than primitive attributes:

Entity	Composed Value Objects
Stock Position	Stock Availability Breakdown, Stock Status Indicator, Ownership Classification, Quantity
Stock Policy	Stock Threshold Set, Lead Time Duration, Effective Period
Product	Product Identifier, Product Category, Product Lifecycle Status, ABC Classification, Rotation Policy, Unit of Measure
Inventory Location	Physical Address, Location Type
Storage Position	Storage Position Address
Purchase Order	Purchase Order Line (collection), Payment Terms, Lead Time Duration
Goods Receipt	Receipt Line Detail (collection), User Action Attribution
Customer Order	Order Line Quantity (collection), Fulfillment Quantity Breakdown, Channel Designation
Count Variance	Quantity Variance, Reason Code Attribution, User Action Attribution
Stock Adjustment	Quantity, Reason Code Attribution, Audit Quantity Snapshot, Approval Threshold (evaluated)
Supplier Invoice	Invoice Line Detail (collection), Three-Way Match Result
Promotional Stock Allocation	Campaign Period, Channel Designation, Quantity
Inventory Movement Record	Movement Classification, Quantity, Lot Identity, Reason Code Attribution, User Action Attribution, Audit Quantity Snapshot
Return Disposition	Condition Grade, Disposition Decision

Value Object-to-Business-Rule Traceability

Business Rule	Primary Value Objects
BR-001	Stock Availability Breakdown (ATP component)
BR-002	Order Line Quantity (reserved component)
BR-003	Pick Line Detail (verification status)
BR-004	Receipt Line Detail (verification outcome)
BR-005	Stock Availability Breakdown (ATP effective after put-away)
BR-006	Discrepancy Classification, Quantity Variance
BR-008	Three-Way Match Result, Invoice Line Detail, Purchase Order Line
BR-009	Stock Threshold Set, Stock Status Indicator
BR-010	Stock Threshold Set (safety stock component)
BR-011	Stock Availability Breakdown (network ATP view)
BR-012	Stock Threshold Set (minimum at source)
BR-013	Hold Reason Classification, Stock Availability Breakdown
BR-014	Rotation Policy, Shelf Life Profile, Lot Identity
BR-015	Rotation Policy
BR-016	Shelf Life Profile (MRSL evaluation)
BR-017	Lot Identity
BR-018	Quantity (blind count — no system quantity in count value)
BR-019	User Action Attribution (counter vs. approver)

Business Rule	Primary Value Objects
BR-020	Reason Code Attribution
BR-021	Approval Threshold, User Action Attribution
BR-022	Quantity Variance
BR-023	Reason Code Attribution, Valuation Line Amount
BR-024	Effective Period (count freeze window)
BR-025	Customer Acknowledgment (return eligibility context)
BR-026	Condition Grade, Disposition Decision
BR-027	Product Lifecycle Status
BR-028	Customer Acknowledgment, Fulfillment Quantity Breakdown
BR-029	Campaign Period, Channel Designation
BR-030	Hold Reason Classification, Ownership Classification
BR-031	Ownership Classification, Valuation Line Amount
BR-032	ABC Classification, Alert Priority

Document Approval

Role	Name	Signature	Date
Domain Expert			
Solution Architect			
Senior Business Analyst			
Inventory Controller			

End of Domain Value Objects

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Domain Aggregates

Inventory Management System

Document Information

Attribute	Detail
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Date	July 5, 2026

Attribute	Detail
Scope	Aggregate design only — no persistence, repositories, or infrastructure

Purpose

This document defines **Domain-Driven Design aggregates** for the Inventory Management System. Each aggregate is a consistency boundary: a cluster of entities and value objects mutated together in a single business transaction, accessed only through the aggregate root. External references use identity only — never direct navigation into aggregate internals.

Aggregate Design Principles Applied

Principle	Application in IMS
One transaction, one aggregate	Stock deduction and reservation never share a transaction with PO creation
Reference by identity	Fulfillment references `StockPosition` by product + location; does not hold object references
Small, business-meaningful boundaries	Stock Position scoped to one product at one location — not entire warehouse
Invariants enforced at root	All quantity changes pass through aggregate root methods that validate business rules
Eventual consistency across aggregates	Goods Receipt completion raises domain event consumed by Stock Position aggregate

Bounded Context Overview

Bounded Context	Aggregates
Catalog & Network	Product Catalog, Inventory Location, Supplier, Inventory Governance
Inventory Balance	Stock Position, Stock Policy
Procurement & Inbound	Purchase Requisition, Purchase Order, Goods Receipt, Supplier Invoice
Order Fulfillment	Customer Order Fulfillment
Network Transfer	Stock Transfer
Inventory Accuracy	Cycle Count, Physical Inventory Count, Inventory Adjustment Case
Returns	Customer Return Case
Commercial	Promotional Stock Allocation
Identity & Access	User Account

Total Aggregates: 18

Bounded Context 1: Catalog & Network

Aggregate 1: Product Catalog

Aspect	Detail
Aggregate Root	Product
Included Entities	None *(Product is a single-entity aggregate; classification and identifiers are value objects on the root)*
Included Value Objects	Product Identifier, Product Category, Product Lifecycle Status, ABC Classification, Rotation Policy, Shelf Life Profile (MRSL policy), Unit of Measure
Invariants (Business Rules)	SKU must be unique across the catalog. Product name and unit of measure are mandatory. Discontinued products must not be eligible for new replenishment or purchase orders (BR-027). Inactive or discontinued products must be blocked from new reservations unless an authorized exception exists. ABC classification must be A, B, or C when assigned (BR-032 context). Traceable products must declare lot/batch requirement and rotation policy (FIFO or FEFO). Product status transitions must record prior status, new status, and attribution.

Aspect	Detail
Responsibilities	Maintain the single authoritative product identity (SKU) used across all locations and processes. Govern product lifecycle from introduction through discontinuation. Assign operational classification (ABC) and rotation/traceability attributes that downstream aggregates enforce. Detect and resolve duplicate product identity conflicts.
Business Justification	Product master data is referenced by every inventory transaction but changes infrequently and independently of stock quantities. A dedicated aggregate prevents catalog edits from coupling to high-frequency stock mutations. Discontinued status must be enforced before any procurement aggregate accepts an order line — the Product root is the gatekeeper of assortment validity.

Aggregate 2: Inventory Location

Aspect	Detail
Aggregate Root	Inventory Location
Included Entities	Storage Position
Included Value Objects	Physical Address, Location Type, Location Hierarchy Path (parent reference by identity), Effective Period (go-live / closure)
Invariants (Business Rules)	Location name, type, and operational status are mandatory. A location cannot be its own parent or ancestor in the hierarchy. Deactivated or closing locations must not accept new receipts, reservations, or fulfillments except authorized disposition activity. Location activation for transactions requires approved opening balances when go-live policy applies. Physical inventory freeze (BR-024) must block all receiving, shipping, transfers, and adjustments at the location while a Physical Inventory Count aggregate reports active freeze for this location. Closure is blocked while on-hand stock remains unless a disposition plan is documented or waived.
Responsibilities	Define the network of places where stock is held and operated. Manage storage position structure within a location. Control location operational status (active, planned, closing). Enforce freeze compliance during wall-to-wall counts. Support location go-live and closure lifecycle.
Business Justification	Location configuration and storage layout change independently of stock balances. Storage Positions belong to exactly one location and are meaningless outside it — they share the location's lifecycle. Freeze enforcement requires the location aggregate to expose a single authoritative "frozen" state that all other aggregates consult before permitting mutations at that site.

Aggregate 3: Supplier

Aspect	Detail
Aggregate Root	Supplier
Included Entities	None
Included Value Objects	Contact Details, Payment Terms, Lead Time Duration (default)
Invariants (Business Rules)	Supplier code must be unique. Supplier must be active to appear on new purchase orders. Consignment agreement indicator governs ownership classification applied at receipt — consignment stock is not owned until consumption (BR-031). Payment terms must be defined before purchase orders are issued to the supplier.
Responsibilities	Maintain supplier commercial identity and default procurement terms. Indicate consignment relationship where ownership rules differ from standard purchase. Support supplier delivery performance tracking references.
Business Justification	Supplier is master data referenced by Purchase Order and Goods Receipt aggregates by identity. Commercial terms belong with the supplier definition and must remain consistent across all open POs — isolating suppliers prevents PO line edits from corrupting supplier master data.

Aggregate 4: Inventory Governance

Aspect	Detail
Aggregate Root	Governance Policy
Included Entities	Reason Code

Aspect	Detail
Included Value Objects	Approval Threshold, Quantity Tolerance, Effective Period
Invariants (Business Rules)	Reason codes must be unique; retired codes must not be selectable for new transactions (BR-020 context). Every adjustment, write-off, and receipt discrepancy requires an active reason code at submission time. Only one governance policy version may be effective for a given parameter at any point in time. Adjustments above defined financial or quantity thresholds require independent approval (BR-021). Receipt quantity tolerance limits govern auto-acceptance vs. discrepancy creation (BR-006, BR-008). Segregation-of-duties rules: counter must not approve adjustments from their own count (BR-019).
Responsibilities	Maintain the standardized reason code taxonomy for all inventory exceptions and corrections. Define approval thresholds, tolerance limits, and exception policies that other aggregates consult during validation. Govern policy versioning with non-overlapping effective periods.
Business Justification	Reason codes and governance thresholds are organization-wide reference data that many aggregates consult but none should own. Centralizing them in one aggregate ensures a single policy source — when a threshold changes, all aggregates evaluate against the same effective policy without embedding rules in each aggregate.

Bounded Context 2: Inventory Balance

Aggregate 5: Stock Position

Aspect	Detail
Aggregate Root	Stock Position
Included Entities	Lot/Batch *(when product is traceable)*, Quarantine Hold
Included Value Objects	Stock Availability Breakdown, Stock Status Indicator, Ownership Classification, Quantity, Lot Identity, Hold Reason Classification, Reason Code Attribution, Audit Quantity Snapshot, User Action Attribution
Invariants (Business Rules)	On-hand quantity must never be negative. Available-to-promise (ATP) = on-hand minus reserved minus quarantined/hold quantities — ATP must never exceed on-hand (BR-001, BR-013). Quarantined and bonded stock must be excluded from ATP (BR-013, BR-030). Consignment stock must not be treated as owned for fulfillment until ownership transfer event (BR-031). Safety stock buffer must not be consumed below policy minimum without authorized exception (BR-010). Lot/batch identity must be captured for all movements on traceable products (BR-017). Stock status indicator must be recalculated when quantities cross policy thresholds. All quantity mutations must produce an Inventory Movement Record as a domain event — movement history is immutable. Location must not be frozen (consult Inventory Location aggregate) when accepting quantity changes except authorized count adjustments.
Responsibilities	Maintain authoritative on-hand, reserved, and ATP quantities for one product at one location. Manage lot/batch-level quantities for traceable products. Apply and release quarantine holds. Enforce ownership classification effects on availability and valuation treatment. Accept inbound quantity increases (from receipt, transfer-in, return restock) and outbound decreases (from shipment, transfer-out, adjustment, write-off) only through root operations. Emit domain events on every balance change for audit and reporting consumers.
Business Justification	Stock Position is the system's most contended consistency boundary — reservations, picks, receipts, and adjustments all compete for the same quantities. Scoping the aggregate to one product at one location minimizes lock contention while guaranteeing ATP is always mathematically consistent within the boundary. Lot/Batch and Quarantine Hold entities belong inside because hold application immediately reduces ATP — partial updates across separate aggregates would allow overselling.

Aggregate 6: Stock Policy

Aspect	Detail
Aggregate Root	Stock Policy
Included Entities	None
Included Value Objects	Stock Threshold Set, Lead Time Duration, Effective Period

Aspect	Detail
Invariants (Business Rules)	Reorder level, minimum threshold, and safety stock must be zero or positive. Minimum threshold must not exceed reorder level unless an explicitly authorized inverted configuration exists. Threshold changes must include an effective date and retain prior values as immutable history. Replenishment action must trigger when ATP reaches reorder level (BR-009). Safety stock must not be bypassed without documented approval and replenishment plan (BR-010). Source location minimum must be enforced when this policy governs a transfer source (BR-012).
Responsibilities	Define replenishment thresholds for a specific product at a specific location. Maintain threshold change history with effective dating. Provide the policy reference that Stock Position uses to compute stock status indicators and that replenishment processes use to trigger alerts.
Business Justification	Stock policy changes are infrequent, approval-governed events distinct from high-frequency quantity mutations. Separating policy from Stock Position allows threshold recalibration without locking stock during active fulfillment. The product + location identity key aligns with how planners think about replenishment — one policy per SKU per site.

Bounded Context 3: Procurement & Inbound

Aggregate 7: Purchase Requisition

Aspect	Detail
Aggregate Root	Purchase Requisition
Included Entities	None *(requisition lines are value objects)*
Included Value Objects	Requisition Line Detail (collection), User Action Attribution
Invariants (Business Rules)	Requisition lines must reference active products — discontinued products are prohibited (BR-027). Requested quantities must be greater than zero. Network availability must be evaluated before requisition approval — excess at other locations should trigger transfer consideration before procurement (BR-011). Safety stock must not be implicitly bypassed by requisition quantity decisions (BR-010). Requisition must reach approved status before conversion to a Purchase Order.
Responsibilities	Capture internal replenishment need with product, quantity, and target location. Route through approval workflow. Convert to Purchase Order upon approval. Link to triggering low-stock alert reference when applicable.
Business Justification	Requisition is an internal intent document with its own approval lifecycle, distinct from the commercial PO commitment. Keeping it as a separate aggregate allows rejection or amendment without affecting supplier-facing PO records. Conversion to PO is a cross-aggregate operation creating a new Purchase Order aggregate from approved requisition identity.

Aggregate 8: Purchase Order

Aspect	Detail
Aggregate Root	Purchase Order
Included Entities	Expected Inbound Delivery
Included Value Objects	Purchase Order Line (collection), Payment Terms, Lead Time Duration, Monetary Amount
Invariants (Business Rules)	PO must include supplier reference, receiving location, product lines, and expected delivery date. PO lines must reference active products and active receiving locations — discontinued products prohibited (BR-027). Ordered quantity must be greater than zero. A valid PO is required before supplier goods receipt except explicitly approved ad-hoc exceptions (BR-007). Received quantity per line must not exceed ordered quantity plus defined tolerance without discrepancy workflow. PO status transitions: open → partial → closed/cancelled; cancelled POs must not accept receipts. Expected inbound delivery dates must be valid and associated with an open PO line.
Responsibilities	Record authorized commercial commitment to a supplier. Track ordered vs. received quantities per line. Manage expected inbound delivery schedules. Support PO amendment and cancellation with receipt implications. Provide the reference document for goods receipt verification and three-way match.
Business Justification	Purchase Order is the commercial control document governing inbound supply. Expected Inbound Delivery entities belong inside because delivery schedule changes are part of the same procurement commitment — splitting them risks inconsistent inbound visibility. PO enforces BR-007 as the gate for the Goods Receipt aggregate.

Aggregate 9: Goods Receipt

Aspect	Detail
Aggregate Root	Goods Receipt
Included Entities	Receipt Discrepancy, Put-Away
Included Value Objects	Receipt Line Detail (collection), Condition Grade, Discrepancy Classification, Quantity Variance, Shelf Life Profile, Lot Identity, User Action Attribution
Invariants (Business Rules)	Inbound goods must be verified before acceptance — quantity, product identity, and condition confirmed (BR-004). Valid PO reference required except approved ad-hoc exception (BR-007). Receipt discrepancies must be documented before acceptance of affected quantities (BR-006). Minimum remaining shelf life (MRSL) must be met for shelf-life products — non-compliant deliveries rejected or dispositioned (BR-016). Lot/batch identity must be captured for traceable products at receipt (BR-017). Bonded inventory must remain on hold until customs clearance — not released to sellable via put-away (BR-030). Received stock is not available for sale until put-away is confirmed (BR-005). Put-away must assign storage position and apply rotation placement (BR-014, BR-015). Received quantity must not be accepted into stock position until put-away completion triggers cross-aggregate notification.
Responsibilities	Orchestrate the inbound verification workflow from dock arrival through put-away confirmation. Document and resolve receipt discrepancies with evidence. Capture lot/batch and shelf-life data at receipt. Manage put-away as the final step before stock becomes available. Emit domain event on put-away completion for Stock Position aggregate to increase on-hand quantity.
Business Justification	Receipt, discrepancy resolution, and put-away are one operational unit of work at the receiving dock — a receiver verifies, documents exceptions, and stows goods in a continuous flow. Combining them ensures stock is never marked available (BR-005) without verified receipt and confirmed put-away within the same consistency boundary. Put-Away as an entity inside (not separate aggregate) prevents orphan "received but not stowed" states.

Aggregate 10: Supplier Invoice

Aspect	Detail
Aggregate Root	Supplier Invoice
Included Entities	None *(invoice lines are value objects)*
Included Value Objects	Invoice Line Detail (collection), Three-Way Match Result, Quantity Tolerance, Monetary Amount
Invariants (Business Rules)	Three-way match must be confirmed among PO, goods receipt, and invoice within defined tolerance before payment authorization (BR-008). Payment for unordered, unreceived, or mispriced goods is prohibited. Invoice lines must reference products and quantities that exist on the linked PO and receipt. Match exceptions must be documented and resolved or waived by authorized approver before payment status transitions to authorized.
Responsibilities	Record supplier billing document. Execute three-way match evaluation against Purchase Order and Goods Receipt aggregates (referenced by identity). Track match status and payment authorization gate. Document match exceptions for finance resolution.
Business Justification	Invoice matching is a financial control boundary that compares three independent documents. The Supplier Invoice aggregate owns the match outcome — it consults PO and GR by identity but does not mutate them. This isolates financial approval workflow from operational receipt processing while enforcing BR-008 as a single coherent decision.

Bounded Context 4: Order Fulfillment

Aggregate 11: Customer Order Fulfillment

Aspect	Detail
Aggregate Root	Customer Order
Included Entities	Inventory Reservation, Fulfillment Execution, Backorder, Shipment
Included Value Objects	Order Line Quantity (collection), Fulfillment Quantity Breakdown, Customer Acknowledgment, Pick Line Detail, Shipment Line Detail, Channel Designation, User Action Attribution

Aspect	Detail
Invariants (Business Rules)	Availability confirmation must use ATP only — not total on-hand (BR-001). Stock must be reserved at order confirmation; reserved quantity removed from ATP until fulfilled or released (BR-002). Reservation must reference active products at authorized locations. Safety stock and promotional ring-fence rules must be respected when reserving (BR-010, BR-029). Verified pick must complete before shipment recording — no ship without pick confirmation (BR-003). FEFO/FIFO rotation must be applied during pick line assignment (BR-014, BR-015). Lot/batch must be captured on pick for traceable products (BR-017). Backorder requires explicit customer acknowledgment before recording (BR-028). Partial fulfillment must decompose order line into fulfilled and backordered quantities consistently. Shipment deducts stock only after verified pick — emits domain event to Stock Position aggregate. Reservation release on cancellation must restore ATP via cross-aggregate command.
Responsibilities	Manage the complete customer order lifecycle from availability confirmation through reservation, pick/pack/ship, and partial/backorder handling. Coordinate ATP commitment and release with Stock Position aggregate. Enforce pick verification before shipment. Capture customer acknowledgment for backorders. Emit shipment events that trigger stock deduction. Provide order fulfillment status visibility.
Business Justification	Order confirmation, reservation, picking, and shipping are one coherent business transaction chain with tightly coupled state — a reservation without pick tracking or a shipment without reservation release creates overselling or phantom stock. Keeping these entities in one aggregate guarantees order line quantities reconcile: requested = reserved = picked + backordered + cancelled at every stage. Cross-aggregate calls to Stock Position handle ATP mechanics while the order aggregate owns fulfillment intent.

Bounded Context 5: Network Transfer

Aggregate 12: Stock Transfer

Aspect	Detail
Aggregate Root	Stock Transfer *(Transfer Request is the initial state; approved request promotes to transfer execution within same aggregate)*
Included Entities	In-Transit Stock
Included Value Objects	Transfer Line Detail (collection), Quantity, Lot Identity, User Action Attribution, Approval Threshold (evaluated)
Invariants (Business Rules)	Transfer must not deplete source below its minimum threshold — quantity limited to spare stock above source policy minimum (BR-012). Network availability must have been evaluated before transfer was initiated as alternative to purchase (BR-011). Safety stock at source must not be breached without authorized exception (BR-010). Transfer requires approval before dispatch when policy mandates. Lot/batch must be captured on transfer lines for traceable products (BR-017). Dispatch deducts source Stock Position; receipt at destination increases destination Stock Position — in-transit quantity tracked between. In-transit stock excluded from both source and destination ATP until receipt confirmed. Source and destination locations must not be frozen during dispatch/receipt.
Responsibilities	Manage inter-location transfer lifecycle: request, approval, dispatch, in-transit tracking, and receipt confirmation. Enforce source location minimum protection. Track in-transit inventory state. Coordinate stock deductions and additions across two Stock Position aggregates via domain events.
Business Justification	A transfer is a single business operation spanning two locations with three states (requested → in-transit → received). In-Transit Stock belongs inside because it is meaningless without the transfer that created it — splitting would orphan pipeline inventory. Source minimum enforcement (BR-012) requires evaluating source Stock Policy within the transfer decision, making the transfer aggregate the natural enforcement point before dispatch events fire.

Bounded Context 6: Inventory Accuracy

Aggregate 13: Cycle Count

Aspect	Detail
Aggregate Root	Cycle Count
Included Entities	None *(count result and variance reference are value objects; variance investigation spawns separate aggregate)*
Included Value Objects	Quantity (physical count result), Quantity Variance, User Action Attribution

Aspect	Detail
Invariants (Business Rules)	Count must be performed blind — counter must not see system expected quantity before recording result (BR-018). Class A products must receive highest count frequency (BR-032). Physical count quantity must be non-negative. Count assignment must reference valid product and location. Count completion with variance must spawn an Inventory Adjustment Case aggregate — Cycle Count does not directly mutate Stock Position. Counter identity recorded and must differ from eventual approver (BR-019).
Responsibilities	Schedule and assign cycle count work. Enforce blind counting protocol. Record physical count result. Calculate variance against system quantity (evaluated at submission, not shown to counter). Initiate variance investigation workflow when variance exists.
Business Justification	Cycle count is a bounded operational task — assign, count, record. It deliberately does not adjust stock directly because BR-022 requires investigation before adjustment and BR-019 requires independent approval. Separating count from adjustment into different aggregates enforces segregation of duties by design.

Aggregate 14: Physical Inventory Count

Aspect	Detail
Aggregate Root	Physical Inventory Count
Included Entities	None *(count teams and location scope managed on root)*
Included Value Objects	Effective Period (freeze window), Quantity Variance (per line, collected), User Action Attribution
Invariants (Business Rules)	Physical inventory freeze must apply during wall-to-wall count — no receiving, shipping, transfers, or adjustments at the location until count completed and verified (BR-024). Freeze state must be communicated to Inventory Location aggregate. Count must cover full location scope. Count certification requires all variances identified and routed to Inventory Adjustment Case aggregates. Location must return to active operational status only after count event is certified complete and freeze released.
Responsibilities	Orchestrate full wall-to-wall count at a location. Impose and release location freeze. Collect all line-level count results and variances. Certify count completion. Coordinate with Inventory Location aggregate for freeze enforcement.
Business Justification	Physical count is a location-wide event affecting every concurrent operation — BR-024 freeze is the defining invariant. This aggregate owns the count event and freeze lifecycle because only the count process knows when the location snapshot is stable. Individual variances still flow to adjustment cases rather than mutating stock during the count.

Aggregate 15: Inventory Adjustment Case

Aspect	Detail
Aggregate Root	Count Variance *(when originating from count)* or Stock Adjustment *(when direct adjustment)* — unified as **Adjustment Case** with polymorphic origin
Included Entities	Stock Adjustment, Inventory Write-Off
Included Value Objects	Quantity Variance, Reason Code Attribution, Condition Grade, Disposition Decision, Approval Threshold, Audit Quantity Snapshot, User Action Attribution, Valuation Line Amount
Invariants (Business Rules)	Variances must be investigated before adjustment is applied — root cause analysis or documented effort required (BR-022). Counter must not approve adjustments from their own count (BR-019). Every adjustment requires a mandatory reason code (BR-020). Adjustments above threshold require independent approval by someone who did not initiate the underlying transaction (BR-021). Write-off requires documented evidence and authorized approval (BR-023). Adjustment and write-off must emit domain events to Stock Position aggregate for quantity mutation. Write-off is terminal — quantity permanently removed. Adjustment quantity must not drive stock negative.
Responsibilities	Investigate count variances and document findings. Process stock adjustments with reason code and approval chain. Process inventory write-offs with evidence and authorization. Enforce segregation of duties between initiator and approver. Trigger Stock Position mutations only upon approved, finalized adjustment or write-off.
Business Justification	Investigation, approval, and execution of inventory corrections form a governance workflow that must not be bypassed. Combining Count Variance, Stock Adjustment, and Write-Off in one case aggregate ensures BR-022 (investigate before adjusting) and BR-019/BR-021 (independent approval) are enforced as a single state machine: variance → investigation → approval → stock mutation. Stock Position remains the quantity owner — this aggregate owns the decision to change it.

Bounded Context 7: Returns

Aggregate 16: Customer Return Case

Aspect	Detail
Aggregate Root	Return Authorization
Included Entities	Customer Return, Return Disposition
Included Value Objects	Condition Grade, Disposition Decision, Receipt Line Detail, Reason Code Attribution, User Action Attribution
Invariants (Business Rules)	Return authorization required before warehouse acceptance — no receipt without valid RMA linked to original order (BR-025). Returned goods must be verified on receipt — quantity and product matched to authorization (BR-004). Returned goods must be inspected before restocking — condition assessment mandatory (BR-026). Restock to sellable inventory permitted only when condition grade confirms resale quality. Damaged or defective returns must be dispositioned to quarantine or write-off — never directly to ATP. Disposition emits domain event to Stock Position (restock increase) or Inventory Adjustment Case (write-off). Original order referenced by identity for policy validation.
Responsibilities	Issue return authorization linked to original customer order. Receive and verify returned goods against authorization. Inspect condition and decide disposition (restock, quarantine, write-off). Coordinate stock position increase or write-off initiation based on disposition. Track return processing status end-to-end.
Business Justification	Return authorization, physical receipt, inspection, and disposition are one reverse-logistics workflow with strict sequencing (BR-025 → BR-004 → BR-026). Combining them prevents unauthorized returns entering stock and prevents uninspected goods reaching sellable inventory. The RMA is the natural root because it gates all subsequent steps — without authorization, receipt and disposition cannot proceed.

Bounded Context 8: Commercial

Aggregate 17: Promotional Stock Allocation

Aspect	Detail
Aggregate Root	Promotional Stock Allocation
Included Entities	None
Included Value Objects	Campaign Period, Channel Designation, Quantity, Stock Availability Breakdown (ring-fenced portion)
Invariants (Business Rules)	Promotional stock must be ring-fenced during campaign — allocated quantity must not be consumed by non-promotional orders without commercial leadership approval (BR-029). Allocation quantity must not exceed available stock at allocation time. Ring-fenced quantity reduces general ATP on Stock Position while allocation is active. Allocation must define effective campaign period — inactive outside period. Readiness status must be confirmable before campaign commitment. Release of un consumed allocation after campaign end restores quantity to general ATP.
Responsibilities	Ring-fence defined quantity for a promotional campaign or launch. Protect allocated stock from non-promotional reservation. Support readiness review before campaign start. Release un consumed allocation on campaign end. Coordinate with Stock Position aggregate to segregate ATP.
Business Justification	Promotional ring-fencing is a commercial policy overlay on stock availability with its own lifecycle (campaign start/end) independent of order fulfillment mechanics. A dedicated aggregate allows marketing and commercial teams to manage allocations without coupling to individual Customer Order Fulfillment aggregates. BR-029 enforcement requires a single authority on what quantity is promotional vs. general — this aggregate is that authority.

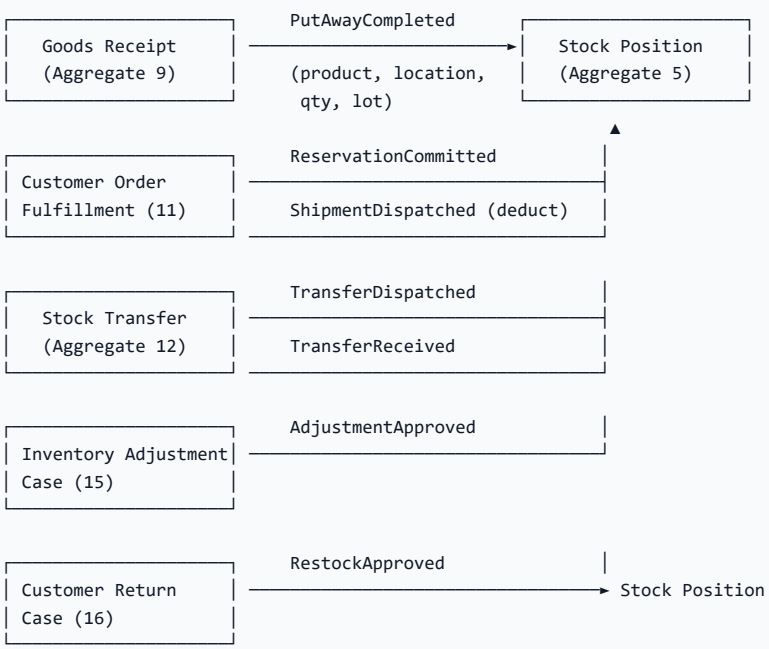
Bounded Context 9: Identity & Access

Aggregate 18: User Account

Aspect	Detail
Aggregate Root	User Account
Included Entities	Role Assignment
Included Value Objects	User Action Attribution (template), Effective Period (assignment dates)
Invariants (Business Rules)	User identifier must be unique. Account must have at least one role assignment to perform system actions. Segregation-of-duties conflicts must be blocked — e.g., user cannot hold both cycle count operator and count approval roles simultaneously (BR-019, BR-021). Location scope on role assignment must reference valid active locations. Deactivated accounts must not authenticate or perform actions. Role changes must be effective-dated; prior assignments retained as history.
Responsibilities	Provision and manage user identity for inventory operations. Assign roles with location scope and permission boundaries. Enforce segregation-of-duties constraints at assignment time. Support account lifecycle (active, suspended, deactivated). Provide user identity context consumed by all aggregates for attribution.
Business Justification	User identity and role assignments change independently of inventory transactions but govern who may initiate and approve them. Centralizing access control in one aggregate ensures SoD rules (BR-019, BR-021) are evaluated consistently at role assignment — not re-implemented in each operational aggregate. Role Assignments belong inside because they have no meaning without the user account they belong to.

Cross-Aggregate Interaction Model

Aggregates coordinate through **domain events** and **identity references** — never through shared mutable state.



Interaction Rules

Rule	Description
No cross-aggregate transaction	Goods Receipt put-away and Stock Position update are eventual consistency via domain event
Reference by identity	Customer Order Fulfillment holds `productSku + locationCode`, not Stock Position object reference
Consult, don't couple	Stock Transfer consults Stock Policy and Stock Position by identity for source minimum check before dispatch
Governance consultation	All adjustment aggregates consult Inventory Governance for reason code validity and approval thresholds
Catalog validation	Procurement and fulfillment aggregates consult Product Catalog by SKU for lifecycle status before accepting lines

Aggregate Summary

#	Aggregate	Root	Bounded Context	Primary Business Processes
1	Product Catalog	Product	Catalog & Network	BP-01, BP-04
2	Inventory Location	Inventory Location	Catalog & Network	BP-02, BP-12
3	Supplier	Supplier	Catalog & Network	BP-05, BP-07
4	Inventory Governance	Governance Policy	Catalog & Network	BP-01, BP-13, BP-14
5	Stock Position	Stock Position	Inventory Balance	BP-10, BP-16, BP-17
6	Stock Policy	Stock Policy	Inventory Balance	BP-03, BP-21
7	Purchase Requisition	Purchase Requisition	Procurement & Inbound	BP-22
8	Purchase Order	Purchase Order	Procurement & Inbound	BP-05, BP-06, BP-23
9	Goods Receipt	Goods Receipt	Procurement & Inbound	BP-07, BP-08, BP-09
10	Supplier Invoice	Supplier Invoice	Procurement & Inbound	BP-30
11	Customer Order Fulfillment	Customer Order	Order Fulfillment	BP-16, BP-17, BP-18, BP-19
12	Stock Transfer	Stock Transfer	Network Transfer	BP-24, BP-25
13	Cycle Count	Cycle Count	Inventory Accuracy	BP-11
14	Physical Inventory Count	Physical Inventory Count	Inventory Accuracy	BP-12
15	Inventory Adjustment Case	Count Variance / Stock Adjustment	Inventory Accuracy	BP-13, BP-14, BP-15
16	Customer Return Case	Return Authorization	Returns	BP-27, BP-28
17	Promotional Stock Allocation	Promotional Stock Allocation	Commercial	BP-31
18	User Account	User Account	Identity & Access	Cross-cutting

Excluded from Aggregate Design

The following domain concepts are intentionally **not** modeled as aggregates:

Concept	Treatment	Rationale
Inventory Movement Record	Domain event / append-only log	Immutable history produced by aggregates; not a consistency boundary
Audit Trail Record	Domain event / compliance log	Cross-cutting observation of aggregate mutations
Operational Alert	Read model / notification projection	Derived from threshold breaches; no transactional invariant
Inventory Valuation Snapshot	Reporting read model	Point-in-time projection for finance; rebuilt from Stock Position events

Document Approval

Role	Name	Signature	Date
DDD Solution Architect			
Domain Expert			
Senior Business Analyst			
Product Owner			

End of Domain Aggregates

Document ID: SRS-DM-003 | Version 1.0 | July 5, 2026

Domain Entity Relationships

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-DM-004
Document Title	Domain Entity Relationships — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Domain Expert
Source Documents	SRS-DM-001 (Domain Entity Model, v1.0), SRS-DM-003 (Domain Aggregates, v1.0), SRS-BR-001 (Business Rules, v1.0)
Date	July 5, 2026
Scope	Business relationships only — no database foreign keys, ORM mappings, or persistence design

Purpose

This document identifies **business relationships** between all 40 domain entities in the Inventory Management System. Each relationship expresses how entities collaborate in real operational workflows — ownership, dependency, modification, authorization, and derivation — independent of how any system stores or joins data.

Relationship Type Legend

Type	Meaning
Composition	Target is a constituent part of source; target lifecycle is bound to source
Association	Ongoing business link; both entities exist independently
Reference	Source identifies or consults target by business identity without ownership
Modification	Source operation changes quantity, status, or availability of target
Derivation	Target is computed, triggered, or produced from source state or event
Authorization	Source permits, gates, or validates an action on target
Verification	Source is validated against target before proceeding

Multiplicity Notation

Notation	Meaning
1	Exactly one
0..1	Zero or one (optional)
1..*	One or more (mandatory, many allowed)
0..*	Zero or more (optional, many allowed)
*** **	Many (zero or more)

Relationship Index by Domain

Domain	Relationship IDs	Count
Master Data & Reference	REL-001 – REL-018	18
Inventory State & Balance	REL-019 – REL-032	14
Procurement & Inbound	REL-033 – REL-058	26
Order Fulfillment	REL-059 – REL-074	16
Network Balance	REL-075 – REL-084	10
Inventory Control & Accuracy	REL-085 – REL-104	20
Returns Management	REL-105 – REL-114	10
Accountability & Oversight	REL-115 – REL-128	14
Financial Alignment	REL-129 – REL-134	6
Total	**REL-001 – REL-134**	**134**

Domain 1: Master Data & Reference

REL-001

Field	Detail
Source Entity	Product
Target Entity	Stock Position
Relationship Type	Association
Business Meaning	Every stock balance tracks exactly one product; a product may be held at zero or many locations simultaneously. Product defines *what* is counted; Stock Position defines *how much* and *where*.
Multiplicity	Product **1** → Stock Position **0..***
Business Rules	BR-027 — discontinued products must not receive new stock through procurement; existing Stock Positions may remain until depleted. BR-017 — traceable products require lot attribution on all Stock Position movements.

REL-002

Field	Detail
Source Entity	Product
Target Entity	Stock Policy
Relationship Type	Association
Business Meaning	Replenishment thresholds are defined per product (and per location on the policy). Product classification (ABC, rotation, shelf-life) determines how policy is applied and monitored.
Multiplicity	Product **1** → Stock Policy **0..***
Business Rules	BR-009 — reorder action triggers when Stock Position ATP reaches product-location reorder level defined in Stock Policy. BR-032 — Class A products require highest monitoring intensity.

REL-003

Field	Detail
Source Entity	Product
Target Entity	Lot/Batch
Relationship Type	Composition
Business Meaning	For traceable products, lot/batch records are subordinate groupings of stock identity within a product scope. Lots cannot exist without a parent product definition.
Multiplicity	Product **1** → Lot/Batch **0..***
Business Rules	BR-017 — lot identity must be captured at receipt and through all movements for traceable products. BR-014 — FEFO applies when product has shelf-life; BR-015 — FIFO for standard goods.

REL-004

Field	Detail
Source Entity	Inventory Location
Target Entity	Storage Position
Relationship Type	Composition
Business Meaning	Storage positions are physical subdivisions within a location (zones, bins, shelves). They cannot exist outside their parent location and are decommissioned when the location closes.
Multiplicity	Inventory Location **1** → Storage Position **1..***
Business Rules	BR-005 — put-away assigns received goods to a Storage Position before Stock Position ATP increases. BR-024 — location freeze suspends put-away activity during Physical Inventory Count.

REL-005

Field	Detail
Source Entity	Inventory Location
Target Entity	Stock Position
Relationship Type	Association
Business Meaning	Stock balances are anchored to a location. A location holds zero or many product balances; each balance exists at exactly one location.
Multiplicity	Inventory Location **1** → Stock Position **0..***
Business Rules	BR-012 — inter-location transfers must not deplete source location Stock Position below its Stock Policy minimum. BR-024 — location must be frozen during wall-to-wall count.

REL-006

Field	Detail
Source Entity	Inventory Location
Target Entity	Inventory Location *(parent)*
Relationship Type	Association *(hierarchy)*

Field	Detail
Business Meaning	Locations form a network hierarchy (region → site → zone) for reporting roll-up and organizational accountability. A location may optionally belong to one parent location.
Multiplicity	Inventory Location *(child)* **0..1** → Inventory Location *(parent)* **1** ; parent **1** → children **0..***
Business Rules	A location cannot be its own parent or ancestor. Deactivated parent does not automatically deactivate children — each location has independent operational status.

REL-007

Field	Detail
Source Entity	Inventory Location
Target Entity	Physical Inventory Count
Relationship Type	Association
Business Meaning	A wall-to-wall count event applies to exactly one location. The count event imposes operational freeze on that location for its duration.
Multiplicity	Inventory Location **1** → Physical Inventory Count **0..***
Business Rules	BR-024 — receiving, shipping, transfers, and adjustments suspended at location while Physical Inventory Count is active and uncertified.

REL-008

Field	Detail
Source Entity	Stock Policy
Target Entity	Product
Relationship Type	Reference
Business Meaning	Stock Policy is scoped to a specific product. The policy defines replenishment behavior for that SKU at a given location.
Multiplicity	Stock Policy **1** → Product **1**
Business Rules	BR-009 — reorder level on policy triggers replenishment when ATP breaches threshold. BR-010 — safety stock on policy must not be bypassed without authorized exception.

REL-009

Field	Detail
Source Entity	Stock Policy
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	Stock Policy is scoped to a specific location. The same product may have different thresholds at different locations reflecting local demand and lead time.
Multiplicity	Stock Policy **1** → Inventory Location **1**
Business Rules	BR-012 — minimum threshold on source location policy constrains maximum transfer quantity. Product + Location combination uniquely identifies a Stock Policy.

REL-010

Field	Detail
Source Entity	Stock Policy
Target Entity	Operational Alert
Relationship Type	Derivation
Business Meaning	When Stock Position ATP crosses policy thresholds (reorder level, minimum, zero), the system derives low-stock or out-of-stock alerts for replenishment review.
Multiplicity	Stock Policy **1** → Operational Alert **0..***
Business Rules	BR-009 — replenishment action must be initiated same business day when reorder level is reached. BR-032 — Class A products generate higher-priority alerts.

REL-011

Field	Detail
Source Entity	Reason Code
Target Entity	Stock Adjustment
Relationship Type	Authorization
Business Meaning	Every stock adjustment must cite a reason code explaining why the correction occurred. Reason code categorizes the adjustment for shrinkage analysis and audit.
Multiplicity	Reason Code **1** → Stock Adjustment **0..*** ; Stock Adjustment **1** → Reason Code **1**
Business Rules	BR-020 — no adjustment may be applied without a categorized reason code. Retired reason codes must not be used on new adjustments.

REL-012

Field	Detail
Source Entity	Reason Code
Target Entity	Inventory Write-Off
Relationship Type	Authorization
Business Meaning	Write-offs require a reason code documenting why inventory value is permanently removed (damage, obsolescence, loss).
Multiplicity	Reason Code **1** → Inventory Write-Off **0..*** ; Inventory Write-Off **1** → Reason Code **1**
Business Rules	BR-020, BR-023 — mandatory reason code plus documented evidence and authorization for write-offs.

REL-013

Field	Detail
Source Entity	Reason Code
Target Entity	Receipt Discrepancy
Relationship Type	Association
Business Meaning	Receipt discrepancies are categorized by reason code type (short shipment, damage, wrong item) to drive resolution workflow and supplier accountability.
Multiplicity	Reason Code **1** → Receipt Discrepancy **0..***

Field	Detail
Business Rules	BR-006 — discrepancies must be documented with supporting evidence before acceptance.

REL-014

Field	Detail
Source Entity	Governance Policy
Target Entity	Stock Adjustment
Relationship Type	Verification
Business Meaning	Governance policy defines approval thresholds that Stock Adjustment must satisfy before taking effect. Adjustments above threshold require independent approver per policy.
Multiplicity	Governance Policy **1** → Stock Adjustment **0..***
Business Rules	BR-021 — adjustments above financial or quantity threshold require independent approval by authorized person who did not initiate underlying transaction.

REL-015

Field	Detail
Source Entity	Governance Policy
Target Entity	Count Variance
Relationship Type	Verification
Business Meaning	Governance policy enforces segregation of duties — the counter who created the variance cannot be the approver defined by policy.
Multiplicity	Governance Policy **1** → Count Variance **0..***
Business Rules	BR-019 — person who performed count must not approve adjustments arising from their own count.

REL-016

Field	Detail
Source Entity	Governance Policy
Target Entity	Goods Receipt
Relationship Type	Verification
Business Meaning	Governance policy defines receipt quantity tolerance and ad-hoc receipt exception rules. Goods Receipt is evaluated against policy before acceptance.
Multiplicity	Governance Policy **1** → Goods Receipt **0..***
Business Rules	BR-007 — PO required except where explicitly approved ad-hoc exception applies per governance policy.

REL-017

Field	Detail
Source Entity	Storage Position
Target Entity	Put-Away
Relationship Type	Association

Field	Detail
Business Meaning	Put-away confirms placement of received goods into a designated storage position, making stock physically findable for picking.
Multiplicity	Storage Position **1** → Put-Away **0..*** ; Put-Away **1** → Storage Position **1**
Business Rules	BR-005 — stock not available until put-away to storage position is confirmed. BR-014, BR-015 — rotation placement at put-away.

REL-018

Field	Detail
Source Entity	Storage Position
Target Entity	Fulfillment Execution
Relationship Type	Reference
Business Meaning	Pick operations retrieve stock from a storage position. Fulfillment execution records which position supplied each pick line.
Multiplicity	Storage Position **1** → Fulfillment Execution **0..***
Business Rules	BR-003 — pick must be verified before shipment. BR-014, BR-015 — FEFO/FIFO guides which storage position/lot is picked first.

Domain 2: Inventory State & Balance

REL-019

Field	Detail
Source Entity	Stock Position
Target Entity	Product
Relationship Type	Reference
Business Meaning	Stock Position identifies which product's quantity is being tracked. The product must exist in the catalog and be valid for the operation context.
Multiplicity	Stock Position **1** → Product **1**
Business Rules	BR-001 — ATP on Stock Position is the only quotable quantity for customer commitment. BR-027 — discontinued product positions deplete but are not replenished.

REL-020

Field	Detail
Source Entity	Stock Position
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	Stock Position is scoped to exactly one location — the place where the physical stock resides.
Multiplicity	Stock Position **1** → Inventory Location **1**
Business Rules	BR-012 — source location minimum applies when Stock Position at source is reduced by transfer.

REL-021

Field	Detail
Source Entity	Stock Position
Target Entity	Stock Policy
Relationship Type	Reference
Business Meaning	Stock Position availability status (in stock, low stock, out of stock, overstock) is evaluated against the Stock Policy thresholds for the same product and location.
Multiplicity	Stock Position **1** → Stock Policy **0..1**
Business Rules	BR-009 — reorder triggered when ATP reaches policy reorder level. BR-010 — safety stock on policy protects against routine consumption.

REL-022

Field	Detail
Source Entity	Stock Position
Target Entity	Quarantine Hold
Relationship Type	Composition
Business Meaning	Quarantine holds are applied to specific quantities within a Stock Position, removing held quantity from ATP while remaining physically on-hand.
Multiplicity	Stock Position **1** → Quarantine Hold **0..***
Business Rules	BR-013 — quarantined quantity excluded from sellable ATP. BR-030 — bonded inventory held until customs clearance before release.

REL-023

Field	Detail
Source Entity	Stock Position
Target Entity	Lot/Batch
Relationship Type	Composition
Business Meaning	For traceable products, Stock Position quantity is subdivided by lot/batch. Each lot carries expiration and quality attributes affecting pick priority.
Multiplicity	Stock Position **1** → Lot/Batch **0..***
Business Rules	BR-017 — lot captured on all movements. BR-014 — FEFO pick order by lot expiration date.

REL-024

Field	Detail
Source Entity	Inventory Reservation
Target Entity	Stock Position
Relationship Type	Modification

Field	Detail
Business Meaning	Creating a reservation increases reserved quantity and decreases ATP on the target Stock Position. Releasing or fulfilling a reservation reverses or finalizes the hold.
Multiplicity	Inventory Reservation **0..*** → Stock Position **1**
Business Rules	BR-002 — reservation at order confirmation removes quantity from ATP. BR-010 — must not consume below safety stock without exception. BR-029 — promotional ring-fenced quantity not available for non-promotional reservations.

REL-025

Field	Detail
Source Entity	Promotional Stock Allocation
Target Entity	Stock Position
Relationship Type	Modification
Business Meaning	Promotional allocation ring-fences a portion of Stock Position quantity for campaign use, reducing general ATP available to non-promotional orders.
Multiplicity	Promotional Stock Allocation **0..*** → Stock Position **1**
Business Rules	BR-029 — promotional stock must not be consumed by non-promotional orders without commercial leadership approval.

REL-026

Field	Detail
Source Entity	Promotional Stock Allocation
Target Entity	Product
Relationship Type	Reference
Business Meaning	Allocation specifies which product's stock is ring-fenced for the campaign or launch.
Multiplicity	Promotional Stock Allocation **1** → Product **1**
Business Rules	BR-029 — allocation quantity must be confirmed before campaign commitment (BP-31).

REL-027

Field	Detail
Source Entity	Promotional Stock Allocation
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	Allocation may scope ring-fenced quantity to one or more locations supplying the promotional channel.
Multiplicity	Promotional Stock Allocation **1** → Inventory Location **1..***
Business Rules	BR-029 — readiness review confirms sufficient allocated stock at fulfillment locations before campaign start.

REL-028

Field	Detail
Source Entity	Quarantine Hold
Target Entity	Lot/Batch

Field	Detail
Relationship Type	Reference
Business Meaning	Quality holds may target a specific lot within a Stock Position for targeted recall or investigation scope limitation.
Multiplicity	Quarantine Hold **0..1** → Lot/Batch **1** ; Lot/Batch **1** → Quarantine Hold **0..***
Business Rules	BR-013 — held lot quantity excluded from ATP. BR-030 — bonded lots held pending customs clearance.

REL-029

Field	Detail
Source Entity	Goods Receipt
Target Entity	Stock Position
Relationship Type	Modification
Business Meaning	Completed put-away following goods receipt increases on-hand quantity on the Stock Position at the receiving location. Until put-away completes, Stock Position is not modified.
Multiplicity	Goods Receipt **0..*** → Stock Position **1**
Business Rules	BR-004 — verify before acceptance. BR-005 — put-away before availability increase. BR-016 — MRSL check before acceptance.

REL-030

Field	Detail
Source Entity	Shipment
Target Entity	Stock Position
Relationship Type	Modification
Business Meaning	Shipment dispatch decreases on-hand and releases reserved quantity on the ship-from Stock Position.
Multiplicity	Shipment **0..*** → Stock Position **1**
Business Rules	BR-003 — shipment only after verified pick. BR-002 — reservation released or consumed on shipment.

REL-031

Field	Detail
Source Entity	Stock Adjustment
Target Entity	Stock Position
Relationship Type	Modification
Business Meaning	Approved stock adjustment increases or decreases on-hand quantity on Stock Position to align record with physical reality.
Multiplicity	Stock Adjustment **0..*** → Stock Position **1**
Business Rules	BR-020 — reason code mandatory. BR-021 — independent approval above threshold. BR-022 — investigate before adjusting when originating from count variance.

REL-032

Field	Detail
Source Entity	Inventory Write-Off
Target Entity	Stock Position
Relationship Type	Modification
Business Meaning	Write-off permanently decreases on-hand quantity and removes inventory value from Stock Position. Distinct from adjustment — write-off is terminal.
Multiplicity	Inventory Write-Off **0..*** → Stock Position **1**
Business Rules	BR-023 — documented evidence and authorization required. BR-020 — reason code mandatory.

Domain 3: Procurement & Inbound Supply

REL-033

Field	Detail
Source Entity	Supplier
Target Entity	Purchase Order
Relationship Type	Association
Business Meaning	Purchase orders are commercial commitments to a specific supplier. A supplier may have many open and historical purchase orders.
Multiplicity	Supplier **1** → Purchase Order **0..***
Business Rules	BR-007 — PO authorizes supplier receipt. BR-031 — consignment supplier relationship affects Stock Position ownership classification at receipt.

REL-034

Field	Detail
Source Entity	Supplier
Target Entity	Supplier Invoice
Relationship Type	Association
Business Meaning	Supplier invoices are billing documents from a supplier, matched against POs and receipts for payment authorization.
Multiplicity	Supplier **1** → Supplier Invoice **0..***
Business Rules	BR-008 — three-way match before payment authorization.

REL-035

Field	Detail
Source Entity	Purchase Requisition
Target Entity	Product
Relationship Type	Reference
Business Meaning	Requisition lines specify which products the organization intends to procure and in what quantity.

Field	Detail
Multiplicity	Purchase Requisition **1** → Product **1..***
Business Rules	BR-027 — discontinued products prohibited on requisitions. BR-011 — network check before requisition approval.

REL-036

Field	Detail
Source Entity	Purchase Requisition
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	Requisition specifies the target receiving location where inbound stock is needed.
Multiplicity	Purchase Requisition **1** → Inventory Location **1**
Business Rules	BR-009 — requisition typically triggered by Stock Policy threshold breach at requesting location. BR-010 — safety stock protection evaluated before approval.

REL-037

Field	Detail
Source Entity	Purchase Requisition
Target Entity	Purchase Order
Relationship Type	Derivation
Business Meaning	An approved requisition is converted into a purchase order — the commercial document sent to the supplier. One requisition may produce zero or one PO.
Multiplicity	Purchase Requisition **1** → Purchase Order **0..1**
Business Rules	BR-027 — PO must not include discontinued products from requisition. BR-011 — network transfer alternative should be evaluated before PO creation.

REL-038

Field	Detail
Source Entity	Purchase Requisition
Target Entity	Operational Alert
Relationship Type	Reference
Business Meaning	Requisitions may be triggered by low-stock or out-of-stock alerts, maintaining traceability from alert to procurement action.
Multiplicity	Operational Alert **0..1** → Purchase Requisition **0..1**
Business Rules	BR-009 — replenishment action same business day as alert for reorder-level breach.

REL-039

Field	Detail
Source Entity	Purchase Order
Target Entity	Supplier

Field	Detail
Relationship Type	Reference
Business Meaning	Every PO identifies the supplier who will fulfill the order. PO cannot exist without a valid active supplier.
Multiplicity	Purchase Order **1** → Supplier **1**
Business Rules	BR-007 — PO is the authorization document for supplier receipt. BR-008 — PO is the first leg of three-way match.

REL-040

Field	Detail
Source Entity	Purchase Order
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	PO specifies the receiving location where goods are expected to be delivered and put away.
Multiplicity	Purchase Order **1** → Inventory Location **1**
Business Rules	Receiving location must be active. Goods Receipt must occur at PO-designated location unless authorized redirect.

REL-041

Field	Detail
Source Entity	Purchase Order
Target Entity	Product
Relationship Type	Reference
Business Meaning	PO line items reference products being ordered with quantity and price.
Multiplicity	Purchase Order **1** → Product **1..***
Business Rules	BR-027 — discontinued products prohibited on new PO lines.

REL-042

Field	Detail
Source Entity	Purchase Order
Target Entity	Expected Inbound Delivery
Relationship Type	Composition
Business Meaning	Expected inbound deliveries are shipment plans subordinate to a PO, communicating anticipated arrival dates and quantities to receiving operations.
Multiplicity	Purchase Order **1** → Expected Inbound Delivery **0..***
Business Rules	Expected delivery dates must align with PO expected delivery. Delays trigger inbound supply notifications (BP-23).

REL-043

Field	Detail
Source Entity	Purchase Order
Target Entity	Goods Receipt

Field	Detail
Relationship Type	Authorization
Business Meaning	PO authorizes what may be received from the supplier. Goods receipt must match to a valid PO except approved ad-hoc exceptions.
Multiplicity	Purchase Order **1** → Goods Receipt **0..***
Business Rules	BR-007 — valid PO required before supplier goods acceptance. BR-004 — verify received against PO quantity and product.

REL-044

Field	Detail
Source Entity	Purchase Order
Target Entity	Supplier Invoice
Relationship Type	Verification
Business Meaning	Supplier invoice is matched against the PO as the first leg of three-way match — confirming billed items were authorized for purchase.
Multiplicity	Purchase Order **1** → Supplier Invoice **0..***
Business Rules	BR-008 — PO, receipt, and invoice must agree within tolerance before payment.

REL-045

Field	Detail
Source Entity	Expected Inbound Delivery
Target Entity	Purchase Order
Relationship Type	Reference
Business Meaning	Inbound delivery plan belongs to and derives quantities from its parent purchase order.
Multiplicity	Expected Inbound Delivery **1** → Purchase Order **1**
Business Rules	Expected quantities must not exceed remaining open PO line quantities.

REL-046

Field	Detail
Source Entity	Expected Inbound Delivery
Target Entity	Stock Position
Relationship Type	Derivation
Business Meaning	Expected inbound quantity contributes to network availability views — stock "in pipeline" before physical receipt — informing replenishment and fulfillment planning.
Multiplicity	Expected Inbound Delivery **0..*** → Stock Position **1**
Business Rules	Inbound expected is informational only — does not increase ATP until put-away completes (BR-005).

REL-047

Field	Detail
Source Entity	Goods Receipt
Target Entity	Purchase Order
Relationship Type	Verification
Business Meaning	Receipt verifies inbound goods against PO — confirming product identity, quantity, and condition match the authorized order.
Multiplicity	Goods Receipt **0..*** → Purchase Order **0..1**
Business Rules	BR-007 — PO reference required except ad-hoc exception. BR-004 — verify before acceptance. BR-006 — document discrepancies before acceptance.

REL-048

Field	Detail
Source Entity	Goods Receipt
Target Entity	Supplier
Relationship Type	Reference
Business Meaning	Receipt records which supplier delivered the goods, supporting supplier performance tracking and discrepancy claims.
Multiplicity	Goods Receipt **1** → Supplier **1**
Business Rules	BR-006 — discrepancy documentation preserves supplier accountability at receipt when carrier or supplier is present.

REL-049

Field	Detail
Source Entity	Goods Receipt
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	Receipt occurs at a receiving location — the dock where goods arrive before put-away to storage positions.
Multiplicity	Goods Receipt **1** → Inventory Location **1**
Business Rules	BR-024 — receipt blocked at location under physical inventory freeze.

REL-050

Field	Detail
Source Entity	Goods Receipt
Target Entity	Receipt Discrepancy
Relationship Type	Composition
Business Meaning	Discrepancies are exception records subordinate to a goods receipt, documenting variance between expected and actual at the point of receipt.
Multiplicity	Goods Receipt **1** → Receipt Discrepancy **0..***
Business Rules	BR-006 — discrepancies documented before acceptance. BR-016 — shelf-life non-compliance recorded as discrepancy.

REL-051

Field	Detail
Source Entity	Goods Receipt
Target Entity	Put-Away
Relationship Type	Composition
Business Meaning	Put-away is the final step of the receipt workflow — moving verified goods from dock to storage. Every accepted receipt line requires put-away before availability.
Multiplicity	Goods Receipt **1** → Put-Away **1..***
Business Rules	BR-005 — ATP not increased until put-away confirmed. BR-030 — bonded goods put-away to hold storage until customs clearance.

REL-052

Field	Detail
Source Entity	Goods Receipt
Target Entity	Lot/Batch
Relationship Type	Derivation
Business Meaning	Receipt of traceable products creates or assigns lot/batch records with expiration and supplier origin captured at the point of entry.
Multiplicity	Goods Receipt **0..*** → Lot/Batch **0..***
Business Rules	BR-017 — lot/batch captured at receipt. BR-016 — MRSL validated at receipt for shelf-life products.

REL-053

Field	Detail
Source Entity	Receipt Discrepancy
Target Entity	Goods Receipt
Relationship Type	Reference
Business Meaning	Discrepancy belongs to the receipt event where the variance was discovered — evidence tied to point of arrival.
Multiplicity	Receipt Discrepancy **1** → Goods Receipt **1**
Business Rules	BR-006 — must be recorded when carrier or supplier present. BR-008 — unresolved discrepancies affect three-way match outcome.

REL-054

Field	Detail
Source Entity	Put-Away
Target Entity	Goods Receipt
Relationship Type	Reference
Business Meaning	Put-away completes a goods receipt line — without put-away the receipt is incomplete from an availability perspective.
Multiplicity	Put-Away **1** → Goods Receipt **1**
Business Rules	BR-005 — availability begins at put-away confirmation, not at dock receipt.

REL-055

Field	Detail
Source Entity	Supplier Invoice
Target Entity	Purchase Order
Relationship Type	Verification
Business Meaning	Invoice lines are matched to PO lines to confirm supplier is billing for authorized products at agreed prices.
Multiplicity	Supplier Invoice **1** → Purchase Order **1**
Business Rules	BR-008 — three-way match required before payment.

REL-056

Field	Detail
Source Entity	Supplier Invoice
Target Entity	Goods Receipt
Relationship Type	Verification
Business Meaning	Invoice lines are matched to receipt lines to confirm supplier is billing only for goods actually received and accepted.
Multiplicity	Supplier Invoice **1** → Goods Receipt **0..***
Business Rules	BR-008 — payment prohibited for unreceived or mispriced goods. Receipt discrepancies may block match.

REL-057

Field	Detail
Source Entity	Supplier Invoice
Target Entity	Supplier
Relationship Type	Reference
Business Meaning	Invoice identifies the billing supplier. Payment authorization is directed to this commercial counterparty.
Multiplicity	Supplier Invoice **1** → Supplier **1**
Business Rules	BR-008 — match must be confirmed before payment to supplier.

REL-058

Field	Detail
Source Entity	Supplier Invoice
Target Entity	Inventory Valuation Snapshot
Relationship Type	Derivation
Business Meaning	Invoice unit costs inform inventory valuation for received goods included in period-end financial reconciliation.
Multiplicity	Supplier Invoice **0..*** → Inventory Valuation Snapshot **1**
Business Rules	BR-008 — financial alignment depends on accurate PO-receipt-invoice matching. BR-031 — consignment goods excluded from owned valuation.

Domain 4: Order Fulfillment

REL-059

Field	Detail
Source Entity	Customer Order
Target Entity	Stock Position
Relationship Type	Reference
Business Meaning	Order availability confirmation consults Stock Position ATP to determine whether the order can be committed. Order does not own stock — it references availability.
Multiplicity	Customer Order **1** → Stock Position **1..***
Business Rules	BR-001 — commit only against ATP, not total on-hand. Multi-location orders may reference multiple Stock Positions.

REL-060

Field	Detail
Source Entity	Customer Order
Target Entity	Inventory Reservation
Relationship Type	Composition
Business Meaning	Reservations are operational holds created when the order is confirmed, one or more per order line, locking stock until fulfillment or cancellation.
Multiplicity	Customer Order **1** → Inventory Reservation **0..***
Business Rules	BR-002 — reservation mandatory at order confirmation. BR-029 — must respect promotional ring-fence.

REL-061

Field	Detail
Source Entity	Customer Order
Target Entity	Fulfillment Execution
Relationship Type	Composition
Business Meaning	Fulfillment execution is warehouse work (pick, pack) triggered by a confirmed order. One order may have one or more fulfillment executions (e.g., split fulfillment).
Multiplicity	Customer Order **1** → Fulfillment Execution **0..***
Business Rules	BR-003 — pick verification required. BR-014, BR-015 — rotation applied during pick.

REL-062

Field	Detail
Source Entity	Customer Order
Target Entity	Backorder
Relationship Type	Composition
Business Meaning	Backorders record customer-agreed delays for unfulfilled order line quantity when full ATP is unavailable.

Field	Detail
Multiplicity	Customer Order **1** → Backorder **0..***
Business Rules	BR-028 — explicit customer acknowledgment required before backorder is recorded.

REL-063

Field	Detail
Source Entity	Customer Order
Target Entity	Shipment
Relationship Type	Composition
Business Meaning	Shipments are dispatch events fulfilling order lines. An order may ship in one or multiple shipments (partial fulfillment).
Multiplicity	Customer Order **1** → Shipment **0..***
Business Rules	BR-003 — shipment only after verified pick. Each shipment deducts stock and consumes reservation.

REL-064

Field	Detail
Source Entity	Customer Order
Target Entity	Return Authorization
Relationship Type	Association
Business Meaning	Return authorizations reference the original customer order to validate return eligibility, products, and quantities against what was shipped.
Multiplicity	Customer Order **1** → Return Authorization **0..***
Business Rules	BR-025 — return must link to original order and return policy.

REL-065

Field	Detail
Source Entity	Inventory Reservation
Target Entity	Customer Order
Relationship Type	Reference
Business Meaning	Reservation exists to fulfill a specific order line on a specific order.
Multiplicity	Inventory Reservation **1** → Customer Order **1**
Business Rules	BR-002 — reservation created at confirmation; released on cancellation or consumed on shipment.

REL-066

Field	Detail
Source Entity	Inventory Reservation
Target Entity	Product
Relationship Type	Reference
Business Meaning	Reservation locks quantity of a specific product.

Field	Detail
Multiplicity	Inventory Reservation **1** → Product **1**
Business Rules	BR-027 — inactive/discontinued products blocked from new reservations.

REL-067

Field	Detail
Source Entity	Fulfillment Execution
Target Entity	Customer Order
Relationship Type	Reference
Business Meaning	Fulfillment execution fulfills lines on a specific customer order.
Multiplicity	Fulfillment Execution **1** → Customer Order **1**
Business Rules	BR-003 — fulfillment must complete pick verification before shipment entity is created.

REL-068

Field	Detail
Source Entity	Fulfillment Execution
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	Fulfillment executes at a ship-from location where pick and pack occur.
Multiplicity	Fulfillment Execution **1** → Inventory Location **1**
Business Rules	BR-024 — fulfillment blocked at frozen location. Split fulfillment may involve multiple locations.

REL-069

Field	Detail
Source Entity	Fulfillment Execution
Target Entity	Lot/Batch
Relationship Type	Reference
Business Meaning	Pick lines on fulfillment execution record which lot was picked for traceable products.
Multiplicity	Fulfillment Execution **0..*** → Lot/Batch **0..1** *(per pick line)*
Business Rules	BR-017 — lot captured on pick. BR-014 — earliest-expiring lot picked first (FEFO).

REL-070

Field	Detail
Source Entity	Fulfillment Execution
Target Entity	Shipment
Relationship Type	Derivation
Business Meaning	Verified fulfillment execution produces a shipment — dispatch is the outcome of successful pick and pack.
Multiplicity	Fulfillment Execution **1** → Shipment **0..1**

Field	Detail
Business Rules	BR-003 — no shipment without verified pick completion.

REL-071

Field	Detail
Source Entity	Backorder
Target Entity	Customer Order
Relationship Type	Reference
Business Meaning	Backorder records remaining unfulfilled quantity for a specific order line on a customer order.
Multiplicity	Backorder **1** → Customer Order **1**
Business Rules	BR-028 — customer acknowledgment required. Backorder may convert to reservation when stock arrives.

REL-072

Field	Detail
Source Entity	Backorder
Target Entity	Product
Relationship Type	Reference
Business Meaning	Backorder identifies which product awaits replenishment and fulfillment.
Multiplicity	Backorder **1** → Product **1**
Business Rules	Backordered products may receive replenishment priority in BP-22 planning.

REL-073

Field	Detail
Source Entity	Shipment
Target Entity	Fulfillment Execution
Relationship Type	Reference
Business Meaning	Shipment records which fulfillment execution authorized the dispatch, maintaining pick-to-ship traceability.
Multiplicity	Shipment **1** → Fulfillment Execution **1**
Business Rules	BR-003 — shipment linked to verified fulfillment only.

REL-074

Field	Detail
Source Entity	Shipment
Target Entity	Inventory Reservation
Relationship Type	Modification
Business Meaning	Shipment consumes or releases the reservation for shipped quantity — reservation status transitions to fulfilled for shipped lines.
Multiplicity	Shipment **0..*** → Inventory Reservation **1**
Business Rules	BR-002 — reservation removed from ATP at confirmation; consumed at shipment.

Domain 5: Network Balance

REL-075

Field	Detail
Source Entity	Transfer Request
Target Entity	Inventory Location *(source)*
Relationship Type	Reference
Business Meaning	Transfer request specifies the location sending stock. Source location policy minimum must be evaluated before approval.
Multiplicity	Transfer Request **1** → Inventory Location *(source)* **1**
Business Rules	BR-012 — transfer must not deplete source below minimum threshold. BR-010 — safety stock protection at source.

REL-076

Field	Detail
Source Entity	Transfer Request
Target Entity	Inventory Location *(destination)*
Relationship Type	Reference
Business Meaning	Transfer request specifies the location receiving stock to resolve local shortage or rebalance network.
Multiplicity	Transfer Request **1** → Inventory Location *(destination)* **1**
Business Rules	BR-011 — transfer evaluated before new purchase when network has excess elsewhere. Source and destination must be distinct active locations.

REL-077

Field	Detail
Source Entity	Transfer Request
Target Entity	Product
Relationship Type	Reference
Business Meaning	Transfer request specifies product and quantity to move between locations.
Multiplicity	Transfer Request **1** → Product **1..***
Business Rules	BR-017 — lot attribution required on transfer lines for traceable products.

REL-078

Field	Detail
Source Entity	Transfer Request
Target Entity	Stock Transfer
Relationship Type	Derivation
Business Meaning	Approved transfer request initiates stock transfer execution. Request may result in zero transfers if rejected.
Multiplicity	Transfer Request **1** → Stock Transfer **0..1**
Business Rules	BR-012 — approval validates source can spare quantity above minimum.

REL-079

Field	Detail
Source Entity	Transfer Request
Target Entity	Stock Policy
Relationship Type	Verification
Business Meaning	Transfer approval consults source location Stock Policy to ensure transfer quantity does not breach minimum or safety stock thresholds.
Multiplicity	Transfer Request **0..*** → Stock Policy **1** *(source product-location)*
Business Rules	BR-012, BR-010 — source minimum and safety stock protection.

REL-080

Field	Detail
Source Entity	Stock Transfer
Target Entity	Transfer Request
Relationship Type	Reference
Business Meaning	Executed transfer may originate from an approved transfer request, maintaining request-to-execution traceability.
Multiplicity	Stock Transfer **0..1** → Transfer Request **1**
Business Rules	Ad-hoc transfers without request require equivalent authorization per governance policy.

REL-081

Field	Detail
Source Entity	Stock Transfer
Target Entity	In-Transit Stock
Relationship Type	Composition
Business Meaning	In-transit stock records are pipeline state subordinate to a stock transfer — tracking quantity between dispatch and destination receipt.
Multiplicity	Stock Transfer **1** → In-Transit Stock **1..***
Business Rules	In-transit quantity excluded from both source and destination ATP until receipt confirmed.

REL-082

Field	Detail
Source Entity	Stock Transfer
Target Entity	Stock Position *(source and destination)*
Relationship Type	Modification
Business Meaning	Dispatch decreases source Stock Position; receipt increases destination Stock Position.
Multiplicity	Stock Transfer **1** → Stock Position **2** *(one source, one destination per product line)*
Business Rules	BR-012 — source not depleted below minimum. BR-017 — lot captured on transfer. BR-024 — blocked during location freeze.

REL-083

Field	Detail
Source Entity	Stock Transfer
Target Entity	Lot/Batch
Relationship Type	Reference
Business Meaning	Transfer lines for traceable products carry lot identity through the network move.
Multiplicity	Stock Transfer **0..*** → Lot/Batch **0..1** *(per line)*
Business Rules	BR-017 — lot attribution on all transfer movements.

REL-084

Field	Detail
Source Entity	In-Transit Stock
Target Entity	Stock Transfer
Relationship Type	Reference
Business Meaning	In-transit state belongs to a specific stock transfer and has no meaning outside that transfer context.
Multiplicity	In-Transit Stock **1** → Stock Transfer **1**
Business Rules	Delayed or lost in-transit stock triggers transfer exception alerts.

Domain 6: Inventory Control & Accuracy

REL-085

Field	Detail
Source Entity	Cycle Count
Target Entity	Product
Relationship Type	Reference
Business Meaning	Cycle count assignment targets a specific product (or product set) for physical verification.
Multiplicity	Cycle Count **1** → Product **1**
Business Rules	BR-032 — Class A products scheduled at highest frequency. BR-018 — blind count enforced.

REL-086

Field	Detail
Source Entity	Cycle Count
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	Cycle count occurs at a specific location where the product stock is held.
Multiplicity	Cycle Count **1** → Inventory Location **1**
Business Rules	BR-018 — counter must not see system quantity before recording physical count.

REL-087

Field	Detail
Source Entity	Cycle Count
Target Entity	Storage Position
Relationship Type	Reference
Business Meaning	Cycle count may optionally scope to a specific storage position for bin-level accuracy programs.
Multiplicity	Cycle Count **0..1** → Storage Position **1**
Business Rules	Optional bin-level scope in Phase 2 operations.

REL-088

Field	Detail
Source Entity	Cycle Count
Target Entity	Count Variance
Relationship Type	Derivation
Business Meaning	When physical count differs from system quantity, cycle count produces a count variance for investigation.
Multiplicity	Cycle Count **1** → Count Variance **0..1**
Business Rules	BR-018 — variance calculated at submission, not shown to counter during count. BR-022 — investigation required before adjustment.

REL-089

Field	Detail
Source Entity	Cycle Count
Target Entity	User Account *(counter)*
Relationship Type	Reference
Business Meaning	Cycle count records who performed the physical count for accountability and segregation-of-duties enforcement.
Multiplicity	Cycle Count **1** → User Account **1**
Business Rules	BR-019 — counter must not approve resulting adjustments.

REL-090

Field	Detail
Source Entity	Physical Inventory Count
Target Entity	Inventory Location
Relationship Type	Association
Business Meaning	Physical count applies to an entire location, imposing freeze and full scope count across all products at the site.
Multiplicity	Physical Inventory Count **1** → Inventory Location **1**
Business Rules	BR-024 — freeze all movements at location during count.

REL-091

Field	Detail
Source Entity	Physical Inventory Count
Target Entity	Count Variance
Relationship Type	Derivation
Business Meaning	Wall-to-wall count may produce multiple count variances — one per product/location combination where physical differs from system.
Multiplicity	Physical Inventory Count **1** → Count Variance **0..***
Business Rules	BR-022 — each variance investigated before adjustment. BR-024 — no movements during count that would invalidate snapshot.

REL-092

Field	Detail
Source Entity	Count Variance
Target Entity	Cycle Count *(or Physical Inventory Count)*
Relationship Type	Reference
Business Meaning	Variance originates from either a cycle count or physical inventory count event — linking discrepancy to the count that discovered it.
Multiplicity	Count Variance **1** → Cycle Count **0..1** ; Count Variance **1** → Physical Inventory Count **0..1**
Business Rules	Exactly one count origin per variance. BR-019 — counter identity on originating count must differ from approver.

REL-093

Field	Detail
Source Entity	Count Variance
Target Entity	Stock Position
Relationship Type	Reference
Business Meaning	Variance compares physical quantity to system quantity on a specific Stock Position.
Multiplicity	Count Variance **1** → Stock Position **1**
Business Rules	BR-022 — investigate root cause before adjustment applied to Stock Position.

REL-094

Field	Detail
Source Entity	Count Variance
Target Entity	Stock Adjustment
Relationship Type	Derivation
Business Meaning	Approved count variance may produce a stock adjustment to correct the Stock Position — after investigation and independent approval.
Multiplicity	Count Variance **1** → Stock Adjustment **0..1**
Business Rules	BR-019, BR-021, BR-022 — investigation, independent approval, then adjustment.

REL-095

Field	Detail
Source Entity	Count Variance
Target Entity	User Account *(approver)*
Relationship Type	Reference
Business Meaning	Variance approval records the independent approver who must differ from the original counter.
Multiplicity	Count Variance **0..1** → User Account *(approver)* **1**
Business Rules	BR-019 — approver must not be the counter who performed the originating count.

REL-096

Field	Detail
Source Entity	Stock Adjustment
Target Entity	Count Variance
Relationship Type	Reference
Business Meaning	Adjustments originating from count activities reference the variance that triggered the correction — maintaining count-to-correction traceability.
Multiplicity	Stock Adjustment **0..1** → Count Variance **1**
Business Rules	BR-022 — adjustment from count requires prior investigation on linked variance.

REL-097

Field	Detail
Source Entity	Stock Adjustment
Target Entity	Reason Code
Relationship Type	Reference
Business Meaning	Adjustment cites reason code explaining the correction category.
Multiplicity	Stock Adjustment **1** → Reason Code **1**
Business Rules	BR-020 — mandatory reason code on every adjustment.

REL-098

Field	Detail
Source Entity	Stock Adjustment
Target Entity	User Account *(initiator and approver)*
Relationship Type	Reference
Business Meaning	Adjustment records initiating user and independent approver (when threshold requires) for accountability.
Multiplicity	Stock Adjustment **1** → User Account *(initiator)* **1** ; Stock Adjustment **0..1** → User Account *(approver)* **1**
Business Rules	BR-021 — approver must differ from initiator for material adjustments.

REL-099

Field	Detail
Source Entity	Inventory Write-Off
Target Entity	Return Disposition
Relationship Type	Reference
Business Meaning	Write-off may originate from return disposition when returned goods are deemed non-resalable.
Multiplicity	Return Disposition **0..1** → Inventory Write-Off **1**
Business Rules	BR-026 — inspect before restock; damaged returns routed to write-off. BR-023 — authorization required.

REL-100

Field	Detail
Source Entity	Inventory Write-Off
Target Entity	Reason Code
Relationship Type	Reference
Business Meaning	Write-off cites reason code (obsolescence, damage, loss) for shrinkage reporting.
Multiplicity	Inventory Write-Off **1** → Reason Code **1**
Business Rules	BR-020, BR-023 — reason code and documented evidence mandatory.

REL-101

Field	Detail
Source Entity	Inventory Write-Off
Target Entity	User Account *(authorizer)*
Relationship Type	Authorization
Business Meaning	Write-off requires authorized approver distinct from initiator for high-value disposals.
Multiplicity	Inventory Write-Off **1** → User Account *(authorizer)* **1**
Business Rules	BR-023 — documented authorization by authorized person before value removal.

REL-102

Field	Detail
Source Entity	Quarantine Hold
Target Entity	Return Disposition
Relationship Type	Derivation
Business Meaning	Return disposition may place returned goods into quarantine hold pending further quality investigation rather than immediate restock or write-off.
Multiplicity	Return Disposition **0..1** → Quarantine Hold **0..1**
Business Rules	BR-026 — inspect before restock. BR-013 — quarantined quantity excluded from ATP.

REL-103

Field	Detail
Source Entity	Quarantine Hold
Target Entity	Stock Adjustment
Relationship Type	Derivation
Business Meaning	Release from quarantine or disposition to write-off may trigger stock adjustment or write-off on the held Stock Position quantity.
Multiplicity	Quarantine Hold **0..1** → Stock Adjustment **0..1** ; Quarantine Hold **0..1** → Inventory Write-Off **0..1**
Business Rules	BR-013 — hold release restores ATP only when quality/regulatory clearance confirmed. BR-030 — customs clearance required for bonded release.

REL-104

Field	Detail
Source Entity	Return Disposition
Target Entity	Stock Position
Relationship Type	Modification
Business Meaning	Restock disposition increases on-hand and ATP on Stock Position at the receiving location for resalable returned goods.
Multiplicity	Return Disposition **0..1** → Stock Position **1**
Business Rules	BR-026 — restock only after condition assessment confirms resale quality. BR-013 — quarantine path excludes from ATP until cleared.

Domain 7: Returns Management

REL-105

Field	Detail
Source Entity	Return Authorization
Target Entity	Customer Order
Relationship Type	Verification
Business Meaning	RMA validates return request against original order — confirming products, quantities, and eligibility under return policy.
Multiplicity	Return Authorization **1** → Customer Order **1**
Business Rules	BR-025 — authorization required before warehouse acceptance; must link to original order.

REL-106

Field	Detail
Source Entity	Return Authorization
Target Entity	Customer Return
Relationship Type	Authorization
Business Meaning	RMA gates physical return receipt — warehouse must not accept goods without valid authorization.

Field	Detail
Multiplicity	Return Authorization **1** → Customer Return **0..1**
Business Rules	BR-025 — no receipt without valid RMA. RMA expiration date enforced.

REL-107

Field	Detail
Source Entity	Return Authorization
Target Entity	Product
Relationship Type	Reference
Business Meaning	RMA specifies which products and quantities the customer is authorized to return.
Multiplicity	Return Authorization **1** → Product **1..***
Business Rules	Returned products must match RMA authorized lines.

REL-108

Field	Detail
Source Entity	Customer Return
Target Entity	Return Authorization
Relationship Type	Verification
Business Meaning	Physical return must match a valid RMA — verifying authorization before acceptance.
Multiplicity	Customer Return **1** → Return Authorization **1**
Business Rules	BR-025 — authorization required. BR-004 — verify quantity and product on receipt.

REL-109

Field	Detail
Source Entity	Customer Return
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	Returns are received at a designated warehouse location — same receiving infrastructure as inbound supplier goods.
Multiplicity	Customer Return **1** → Inventory Location **1**
Business Rules	BR-004 — verify before acceptance into inventory control.

REL-110

Field	Detail
Source Entity	Customer Return
Target Entity	Return Disposition
Relationship Type	Composition
Business Meaning	Disposition decisions (restock, quarantine, write-off) are subordinate outcomes of the return receipt and inspection workflow.
Multiplicity	Customer Return **1** → Return Disposition **1..***

Field	Detail
Business Rules	BR-026 — inspection mandatory before disposition. One disposition per returned line.

REL-111

Field	Detail
Source Entity	Customer Return
Target Entity	Goods Receipt
Relationship Type	Association
Business Meaning	Customer return receipt follows the same verification discipline as supplier goods receipt — parallel inbound workflow pattern.
Multiplicity	Customer Return **0..*** → Goods Receipt **0..*** *(conceptual parallel — return may use receipt workflow)*
Business Rules	BR-004 — verify before acceptance applies to return inbound same as supplier inbound.

REL-112

Field	Detail
Source Entity	Return Disposition
Target Entity	Customer Return
Relationship Type	Reference
Business Meaning	Disposition applies to goods received on a specific customer return.
Multiplicity	Return Disposition **1** → Customer Return **1**
Business Rules	BR-026 — condition assessment drives disposition outcome.

REL-113

Field	Detail
Source Entity	Return Disposition
Target Entity	Inventory Write-Off
Relationship Type	Derivation
Business Meaning	Disposition decision of "write-off" produces an inventory write-off for non-resalable returned goods.
Multiplicity	Return Disposition **0..1** → Inventory Write-Off **1**
Business Rules	BR-023 — write-off authorization required. BR-026 — damaged/defective returns not restocked.

REL-114

Field	Detail
Source Entity	Return Disposition
Target Entity	Quarantine Hold
Relationship Type	Derivation
Business Meaning	Disposition decision of "quarantine" applies hold on returned quantity pending further investigation.
Multiplicity	Return Disposition **0..1** → Quarantine Hold **1**
Business Rules	BR-013 — quarantined returns excluded from sellable ATP.

Domain 8: Accountability & Oversight

REL-115

Field	Detail
Source Entity	Inventory Movement Record
Target Entity	Stock Position
Relationship Type	Association
Business Meaning	Every movement record documents a change to a specific Stock Position — forming the immutable history of quantity and status changes.
Multiplicity	Inventory Movement Record **0..*** → Stock Position **1**
Business Rules	BR-017 — lot on movement for traceable products. BR-020 — reason code on adjustment movements. All mutations must produce movement record.

REL-116

Field	Detail
Source Entity	Inventory Movement Record
Target Entity	Goods Receipt *(and other source documents)*
Relationship Type	Reference
Business Meaning	Movement record references the originating business document (receipt, shipment, transfer, adjustment) that caused the stock change.
Multiplicity	Inventory Movement Record **1** → originating document **1**
Business Rules	Full traceability from movement back to business event. Originating document types: Goods Receipt, Shipment, Stock Transfer, Stock Adjustment, Inventory Write-Off, Put-Away, Customer Return.

REL-117

Field	Detail
Source Entity	Inventory Movement Record
Target Entity	User Account
Relationship Type	Reference
Business Meaning	Movement records who performed the action for operational accountability.
Multiplicity	Inventory Movement Record **1** → User Account **1**
Business Rules	BO-19 — 100% audit trail coverage with user identity required.

REL-118

Field	Detail
Source Entity	Audit Trail Record
Target Entity	Stock Adjustment *(and other governed entities)*
Relationship Type	Association

Field	Detail
Business Meaning	Audit trail captures governance-grade before/after state for material changes to inventory entities — complementing movement records with approval context.
Multiplicity	Audit Trail Record **0..*** → governed entity **1**
Business Rules	BR-019, BR-021 — audit trail supports segregation-of-duties verification. Immutable after write.

REL-119

Field	Detail
Source Entity	Audit Trail Record
Target Entity	User Account
Relationship Type	Reference
Business Meaning	Audit record captures user identity and role at time of action for compliance and fraud investigation.
Multiplicity	Audit Trail Record **1** → User Account **1**
Business Rules	BR-019, BR-020, BR-021 — user attribution mandatory on governed changes.

REL-120

Field	Detail
Source Entity	Operational Alert
Target Entity	Stock Position
Relationship Type	Derivation
Business Meaning	Low-stock and out-of-stock alerts are derived when Stock Position ATP crosses Stock Policy thresholds.
Multiplicity	Operational Alert **0..*** → Stock Position **1**
Business Rules	BR-009 — alert at reorder level triggers same-day replenishment review. BR-032 — Class A alerts prioritized.

REL-121

Field	Detail
Source Entity	Operational Alert
Target Entity	Transfer Request
Relationship Type	Derivation
Business Meaning	Transfer exception alerts signal pending approvals, dispatch delays, or receipt failures on inter-location transfers.
Multiplicity	Operational Alert **0..*** → Transfer Request **0..1**
Business Rules	Pending approval notifications route to Operations Director per BP-24.

REL-122

Field	Detail
Source Entity	Operational Alert
Target Entity	Promotional Stock Allocation
Relationship Type	Derivation
Business Meaning	Campaign readiness alerts signal insufficient ring-fenced stock approaching launch deadline.

Field	Detail
Multiplicity	Operational Alert **0..*** → Promotional Stock Allocation **0..1**
Business Rules	BR-029 — readiness confirmed before campaign commitment. BO-24 — 95% promotional launch readiness target.

REL-123

Field	Detail
Source Entity	User Account
Target Entity	Role Assignment
Relationship Type	Composition
Business Meaning	Role assignments delegate permissions to a user. Assignments are subordinate to the user account and meaningless without it.
Multiplicity	User Account **1** → Role Assignment **1..***
Business Rules	BR-019, BR-021 — conflicting roles blocked (counter + approver). Account must have at least one role to transact.

REL-124

Field	Detail
Source Entity	Role Assignment
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	Role assignment scopes user permissions to authorized locations — user may only operate within assigned location boundaries.
Multiplicity	Role Assignment **1** → Inventory Location **0..***
Business Rules	Location-scoped access enforced on all operational transactions. Empty scope may mean network-wide for executive roles.

REL-125

Field	Detail
Source Entity	User Account
Target Entity	Cycle Count
Relationship Type	Association
Business Meaning	User accounts are linked to cycle counts as counters or approvers — enforcing segregation of duties across the count-to-adjustment workflow.
Multiplicity	User Account **1** → Cycle Count **0..*** *(as counter)* ; User Account **1** → Count Variance **0..*** *(as approver)*
Business Rules	BR-019 — same user cannot count and approve same variance.

REL-126

Field	Detail
Source Entity	User Account
Target Entity	Goods Receipt
Relationship Type	Association
Business Meaning	Receiving clerk user account attributed on goods receipt for inbound accountability.

Field	Detail
Multiplicity	User Account **1** → Goods Receipt **0..***
Business Rules	All inbound transactions require user attribution per BO-19 audit trail objective.

REL-127

Field	Detail
Source Entity	Audit Trail Record
Target Entity	Inventory Movement Record
Relationship Type	Association
Business Meaning	Audit trail complements movement record with before/after state snapshots and approval chain — movement is operational history, audit is governance evidence.
Multiplicity	Audit Trail Record **0..1** → Inventory Movement Record **1**
Business Rules	Material changes require both movement record and audit trail per BO-19.

REL-128

Field	Detail
Source Entity	Operational Alert
Target Entity	Expected Inbound Delivery
Relationship Type	Derivation
Business Meaning	Inbound delay alerts notify when expected delivery date passes without corresponding goods receipt.
Multiplicity	Operational Alert **0..*** → Expected Inbound Delivery **0..1**
Business Rules	Delay response informs replenishment planning (BP-23).

Domain 9: Financial Alignment

REL-129

Field	Detail
Source Entity	Inventory Valuation Snapshot
Target Entity	Stock Position
Relationship Type	Derivation
Business Meaning	Valuation snapshot summarizes Stock Position quantities and values at a point in time for finance reconciliation.
Multiplicity	Inventory Valuation Snapshot **1** → Stock Position **0..***
Business Rules	BR-031 — consignment stock excluded from owned valuation. Quarantined and in-transit stock excluded per policy.

REL-130

Field	Detail
Source Entity	Inventory Valuation Snapshot
Target Entity	Product

Field	Detail
Relationship Type	Reference
Business Meaning	Valuation lines reference products with unit cost and extended value for financial reporting by SKU.
Multiplicity	Inventory Valuation Snapshot **1** → Product **0..***
Business Rules	BO-18 — valuation variance ≤2% vs. general ledger target.

REL-131

Field	Detail
Source Entity	Inventory Valuation Snapshot
Target Entity	Inventory Location
Relationship Type	Reference
Business Meaning	Snapshot may scope to specific locations or network-wide for regional and consolidated financial reporting.
Multiplicity	Inventory Valuation Snapshot **1** → Inventory Location **0..***
Business Rules	Month-end close alignment per BP-30.

REL-132

Field	Detail
Source Entity	Inventory Valuation Snapshot
Target Entity	Supplier Invoice
Relationship Type	Reference
Business Meaning	Valuation uses invoice-matched costs from supplier invoices for recently received owned inventory.
Multiplicity	Inventory Valuation Snapshot **0..*** → Supplier Invoice **0..***
Business Rules	BR-008 — three-way matched costs are authoritative for received inventory valuation.

REL-133

Field	Detail
Source Entity	Inventory Valuation Snapshot
Target Entity	Inventory Write-Off
Relationship Type	Reference
Business Meaning	Period valuation reflects write-offs that permanently removed value during the snapshot period.
Multiplicity	Inventory Valuation Snapshot **0..*** → Inventory Write-Off **0..***
Business Rules	BR-023 — authorized write-offs reduce period-end inventory value.

REL-134

Field	Detail
Source Entity	Lot/Batch
Target Entity	Supplier
Relationship Type	Reference

Field	Detail
Business Meaning	Lot/batch may record supplier origin for traceability back to source in quality investigation or recall.
Multiplicity	Lot/Batch **0..1** → Supplier **1**
Business Rules	BR-017 — lot captured at receipt with supplier context for traceable products.

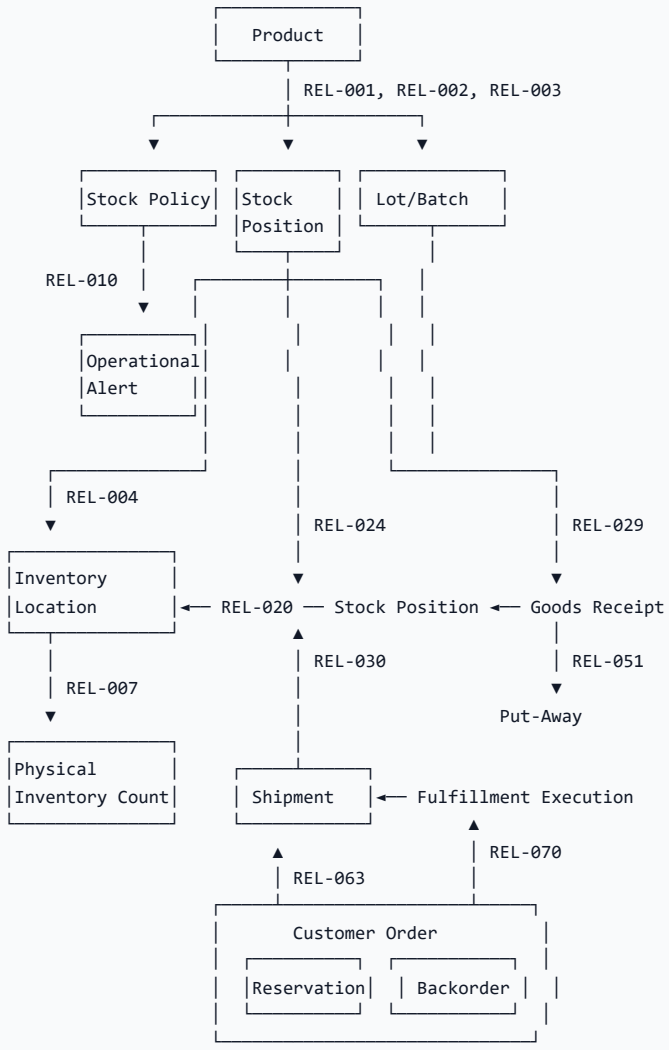
Entity Relationship Matrix

Rows = Source Entity, Columns = Target Entity. Cell shows Relationship ID(s) if related.

Source ↓ / Target →	Product	Location	Stock Pos.	PO	GR	Cust. Order	Stock Adj.
Product	—	—	REL-001	REL-041	—	—	—
Stock Position	REL-019	REL-020	—	—	REL-029	REL-059	REL-031
Purchase Order	REL-041	REL-040	—	—	REL-043	—	—
Goods Receipt	—	REL-049	REL-029	REL-047	—	—	—
Customer Order	—	—	REL-059	—	—	—	—
Count Variance	—	—	REL-093	—	—	—	REL-094
Supplier Invoice	—	—	—	REL-055	REL-056	—	—

Full relationship detail in REL-001 through REL-134 above.

Relationship Diagram — Core Inventory Flow



Document Approval

Role	Name	Signature	Date
Domain Expert			
DDD Solution Architect			
Senior Business Analyst			
Inventory Controller			

End of Domain Entity Relationships

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Domain Services

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-DM-005
Document Title	Domain Services — Inventory Management System
Version	1.0
Status	Draft
Document Owner	DDD Architect
Source Documents	SRS-DM-001 (Domain Entity Model), SRS-DM-003 (Domain Aggregates), SRS-DM-004 (Entity Relationships), SRS-BR-001 (Business Rules), SRS-UC-001 (Use Case Catalog)
Date	July 5, 2026
Scope	Domain-layer business services only — no application services, infrastructure, APIs, or persistence

Purpose

This document identifies **Domain Services** for the Inventory Management System. A domain service encapsulates business logic that is essential to the domain but does not naturally belong to a single entity or aggregate root — typically because it **coordinates**, **evaluates**, or **computes** across multiple aggregates, entities, or value objects.

Domain services are **stateless** with respect to business identity: they operate on domain objects passed to them and enforce cross-cutting business rules that no single aggregate can own without creating improper coupling.

Domain Service vs. Entity Responsibility

Criterion	Belongs on Entity/Aggregate	Belongs in Domain Service
Scope	Single consistency boundary	Spans two or more aggregates
State	Owns lifecycle and identity	Stateless operation on passed objects
Example	Stock Position maintains its own ATP formula	Network availability sums ATP across many Stock Positions

Service Catalog Overview

Domain Area	Services	IDs
Availability & Commitment	Availability Confirmation, Inventory Reservation Coordination, Multi-Location Fulfillment Allocation	DS-001 – DS-003
Network & Replenishment	Network Availability Evaluation, Replenishment Recommendation, Transfer Capacity Evaluation, Network Rebalancing Analysis	DS-004 – DS-007
Inbound & Procurement	Receipt Verification, Put-Away Activation, Three-Way Match Evaluation	DS-008 – DS-010
Fulfillment Execution	Stock Rotation Selection, Shipment Completion Coordination, Partial Fulfillment & Backorder Orchestration	DS-011 – DS-013
Inventory Control	Physical Inventory Freeze Guard, Adjustment Governance Evaluation, Count Variance Adjudication	DS-014 – DS-016
Monitoring & Classification	Stock Status Classification, Low-Stock Alert Evaluation, Promotional Readiness Evaluation	DS-017 – DS-019
Returns & Ownership	Return Eligibility & Disposition Routing, Consignment Ownership Evaluation	DS-020 – DS-021
Financial & Governance	Inventory Valuation Reconciliation, Segregation of Duties Validation, Product Transaction Eligibility	DS-022 – DS-024

Total Domain Services: 24

Domain 1: Availability & Commitment

DS-001 — Availability Confirmation Service

Aspect	Detail
Business Responsibility	Evaluate whether requested order quantities can be committed by consulting available-to-promise (ATP) across one or more Stock Positions, Promotional Stock Allocations, and Quarantine Holds. Returns a confirmation result indicating full availability, partial availability, or unavailability with per-line ATP breakdown.
Why It Should Not Belong to an Entity	Availability confirmation spans Customer Order, Stock Position (potentially multiple locations), Promotional Stock Allocation, and Product catalog validity. Customer Order does not own stock; Stock Position does not know about order context. Neither aggregate should embed the other's rules — confirmation is a cross-aggregate query and decision.
Related Business Rules	BR-001 (ATP-only customer commitment), BR-013 (quarantine exclusion from sellable stock), BR-029 (promotional ring-fencing), BR-027 (inactive/discontinued product blocking)
Related Use Cases	UC-013 (Inquire Inventory Availability), UC-021 (Confirm Order Availability), UC-048 (Support Multi-Location Split Fulfillment)

DS-002 — Inventory Reservation Coordination Service

Aspect	Detail
Business Responsibility	Orchestrate reservation of stock for confirmed customer order lines by coordinating Customer Order Fulfillment aggregate with one or more Stock Position aggregates. Validates safety stock protection, promotional ring-fence rules, and product eligibility before instructing Stock Positions to increase reserved quantity and decrease ATP. Handles reservation release on cancellation.
Why It Should Not Belong to an Entity	Reservation requires atomic business intent across order and stock boundaries — Customer Order Fulfillment owns the commercial commitment while Stock Position owns quantity state. Placing reservation logic entirely in either aggregate creates coupling: the order aggregate would mutate foreign stock, or stock would need order context. Coordination is inherently a domain service operation.
Related Business Rules	BR-002 (reservation at order confirmation), BR-010 (safety stock protection), BR-029 (promotional stock ring-fencing), BR-001 (ATP-only commitment)
Related Use Cases	UC-022 (Reserve Inventory for Confirmed Orders), UC-023 (Release Inventory Reservations), UC-025 (Handle Partial Fulfillment and Backorders)

DS-003 — Multi-Location Fulfillment Allocation Service

Aspect	Detail
Business Responsibility	When a single location cannot satisfy order quantity, determine optimal split allocation across network Stock Positions based on ATP, fulfillment cost proximity, and location operational status. Produces a fulfillment plan assigning order line quantities to one or more ship-from locations.
Why It Should Not Belong to an Entity	Split fulfillment requires simultaneous evaluation of ATP at multiple locations, transfer feasibility, and order line decomposition — no single Stock Position or Customer Order aggregate has network-wide visibility. Location selection is a policy-driven optimization across the network, not a property of any one entity.
Related Business Rules	BR-001 (commit only against ATP at each location), BR-002 (reservation per allocated location), BR-012 (source minimum if transfer needed as fallback)
Related Use Cases	UC-048 (Support Multi-Location Split Fulfillment), UC-021 (Confirm Order Availability), UC-024 (Pick, Pack, and Ship Orders)

Domain 2: Network & Replenishment

DS-004 — Network Availability Evaluation Service

Aspect	Detail
Business Responsibility	Before initiating procurement or approving a transfer request, evaluate network-wide stock availability — summing ATP and in-transit quantities across all locations for a product — to determine whether local shortage can be resolved by redistribution rather than new purchase.
Why It Should Not Belong to an Entity	Network evaluation aggregates data from Stock Positions at many locations, In-Transit Stock records, and Expected Inbound Deliveries. No single location-scoped Stock Position or Purchase Requisition aggregate should own network-wide logic — that would violate aggregate boundaries and create a god entity.
Related Business Rules	BR-011 (network check before purchase), BR-012 (protect source location minimum when evaluating transfer alternative)
Related Use Cases	UC-029 (Plan Replenishment and Create Purchase Requisitions), UC-031 (Request Inter-Location Stock Transfer), UC-034 (Analyze and Recommend Network Rebalancing), UC-028 (Review Low-Stock and Out-of-Stock Alerts)

DS-005 — Replenishment Recommendation Service

Aspect	Detail
Business Responsibility	Generate replenishment recommendations when Stock Position ATP breaches Stock Policy thresholds. Considers reorder level, safety stock, inbound expected supply, network excess at other locations, product lifecycle status, and ABC classification priority. Recommends purchase requisition, transfer request, or no action.
Why It Should Not Belong to an Entity	Replenishment decisions synthesize Stock Policy (product-location), Stock Position (current ATP), Expected Inbound Delivery (pipeline), Product (discontinued status, ABC class), and network availability. Stock Policy defines thresholds but should not scan all positions; Stock Position holds quantity but not replenishment authority. The recommendation is a derived cross-domain judgment.
Related Business Rules	BR-009 (reorder at reorder level), BR-010 (safety stock protection), BR-011 (network check before purchase), BR-027 (no reorder of discontinued products), BR-032 (Class A priority monitoring)
Related Use Cases	UC-028 (Review Low-Stock and Out-of-Stock Alerts), UC-029 (Plan Replenishment and Create Purchase Requisitions), UC-047 (Receive and Respond to Operational Notifications)

DS-006 — Transfer Capacity Evaluation Service

Aspect	Detail
Business Responsibility	Calculate the maximum transferable quantity from a source location for a given product without breaching Stock Policy minimum threshold or safety stock buffer. Used during transfer request creation and approval to enforce source location protection.
Why It Should Not Belong to an Entity	Transfer capacity requires reading source Stock Position ATP/on-hand, source Stock Policy thresholds, and existing reservations — then computing spare quantity. Stock Transfer aggregate should not own source policy rules; Stock Policy aggregate should not know about transfer requests. The calculation is a stateless evaluation spanning source stock and policy.
Related Business Rules	BR-012 (protect source location minimum), BR-010 (safety stock protection)
Related Use Cases	UC-031 (Request Inter-Location Stock Transfer), UC-032 (Approve or Reject Transfer Requests), UC-033 (Execute and Record Inter-Location Transfer)

DS-007 — Network Rebalancing Analysis Service

Aspect	Detail
Business Responsibility	Analyze network-wide stock distribution for a product or category to identify imbalance patterns — locations with stockout risk alongside locations with excess — and recommend redistribution transfers. Supports proactive rebalancing before stockouts occur.

Aspect	Detail
Why It Should Not Belong to an Entity	Rebalancing analysis compares Stock Positions and Stock Policies across the entire location network, identifying pairwise transfer opportunities. This network optimization view belongs to no single location, product, or transfer entity — it is an analytical domain operation over multiple aggregates.
Related Business Rules	BR-011 (network check before purchase), BR-012 (protect source location minimum), BR-009 (reorder level context for imbalance severity)
Related Use Cases	UC-034 (Analyze and Recommend Network Rebalancing), UC-031 (Request Inter-Location Stock Transfer), UC-012 (Monitor Inventory Status)

Domain 3: Inbound & Procurement

DS-008 — Receipt Verification Service

Aspect	Detail
Business Responsibility	Verify inbound goods against Purchase Order lines, Product catalog (identity, traceability requirements), and Governance Policy (tolerance limits). Validates minimum remaining shelf life (MRSL) for shelf-life products, captures lot/batch requirements, and determines whether receipt discrepancies must be raised before acceptance.
Why It Should Not Belong to an Entity	Verification compares Goods Receipt data against Purchase Order (separate aggregate), Product (catalog aggregate), and Governance Policy (reference aggregate). Goods Receipt should record outcome but not own PO matching rules, MRSL policy from Product, or tolerance from Governance — those are cross-aggregate validations.
Related Business Rules	BR-004 (verify before inventory acceptance), BR-006 (document receipt discrepancies), BR-007 (PO required for supplier receipt), BR-016 (minimum remaining shelf life at receipt), BR-017 (lot/batch capture for traceable products)
Related Use Cases	UC-008 (Receive and Verify Supplier Goods), UC-009 (Document and Resolve Receipt Discrepancies), UC-011 (Receive Return and Non-PO Inbound Stock)

DS-009 — Put-Away Activation Service

Aspect	Detail
Business Responsibility	Upon put-away confirmation, coordinate availability activation by validating bonded/customs clearance status (Quarantine Hold), assigning rotation-compliant storage position, and instructing the target Stock Position aggregate to increase on-hand quantity. Ensures stock does not become sellable until all preconditions are satisfied.
Why It Should Not Belong to an Entity	Put-away completion touches Goods Receipt (put-away entity), Storage Position (location aggregate), Quarantine Hold (stock aggregate), and Stock Position (quantity increase). Goods Receipt owns receipt workflow but should not directly mutate Stock Position; Stock Position should not embed put-away or customs rules. Activation is the cross-aggregate handoff point.
Related Business Rules	BR-005 (put-away before availability), BR-030 (bonded inventory customs clearance), BR-014 (FEFO rotation placement), BR-015 (FIFO rotation placement)
Related Use Cases	UC-010 (Perform Stock Put-Away), UC-008 (Receive and Verify Supplier Goods), UC-019 (Manage Quarantine and Hold Stock)

DS-010 — Three-Way Match Evaluation Service

Aspect	Detail
Business Responsibility	Compare Purchase Order, Goods Receipt, and Supplier Invoice lines within Governance Policy tolerance limits. Determine match status (matched, quantity exception, price exception, unmatched) and whether payment authorization may proceed.
Why It Should Not Belong to an Entity	Three-way match is a financial control operation spanning three independent aggregates (Purchase Order, Goods Receipt, Supplier Invoice) plus Governance Policy tolerances. None of the three document aggregates should own the comparison logic — each would create improper dependency on the other two. Match evaluation is a classic domain service.
Related Business Rules	BR-008 (three-way match before payment), BR-006 (receipt discrepancies affect match outcome)

Aspect	Detail
Related Use Cases	UC-039 (Reconcile Inventory Valuation with Finance), UC-006 (Place and Manage Purchase Orders), UC-008 (Receive and Verify Supplier Goods)

Domain 4: Fulfillment Execution

DS-011 — Stock Rotation Selection Service

Aspect	Detail
Business Responsibility	For pick and put-away operations, select the correct lot/batch or oldest stock based on Product rotation policy (FIFO for standard goods, FEFO for shelf-life products). Returns recommended pick source (storage position and lot) from available Stock Position and Lot/Batch data at a location.
Why It Should Not Belong to an Entity	Rotation selection compares multiple Lot/Batch records under a Stock Position against Product rotation policy — a decision spanning Product (policy), Lot/Batch (expiration dates), Storage Position (physical location), and Fulfillment Execution (pick context). No single entity owns the selection algorithm without duplicating policy from Product into warehouse entities.
Related Business Rules	BR-014 (FEFO rotation for shelf-life products), BR-015 (FIFO rotation for standard goods), BR-017 (lot/batch capture on selected stock)
Related Use Cases	UC-024 (Pick, Pack, and Ship Orders), UC-010 (Perform Stock Put-Away), UC-026 (Handle Outbound Fulfillment Exceptions)

DS-012 — Shipment Completion Coordination Service

Aspect	Detail
Business Responsibility	Upon verified pick confirmation, coordinate shipment recording by instructing Stock Position to deduct on-hand quantity, Customer Order Fulfillment to consume Inventory Reservation, and movement history to be recorded. Validates that pick verification preceded dispatch.
Why It Should Not Belong to an Entity	Shipment completion is the cross-aggregate moment where Fulfillment Execution, Shipment, Stock Position, and Inventory Reservation must align. Shipment entity records dispatch but should not own stock deduction; Stock Position deducts but should not know order fulfillment state. Coordination ensures BR-003 is enforced across boundaries.
Related Business Rules	BR-003 (verified pick before shipment), BR-002 (reservation consumed on shipment), BR-017 (lot captured on outbound movement)
Related Use Cases	UC-024 (Pick, Pack, and Ship Orders), UC-027 (View Order Fulfillment Status), UC-026 (Handle Outbound Fulfillment Exceptions)

DS-013 — Partial Fulfillment and Backorder Orchestration Service

Aspect	Detail
Business Responsibility	When full order quantity cannot be fulfilled, decompose order lines into fulfillable and backordered quantities, validate customer acknowledgment for backorder remainder, and coordinate partial reservation, partial shipment, and backorder recording within Customer Order Fulfillment aggregate while updating Stock Positions for fulfilled portion only.
Why It Should Not Belong to an Entity	Partial fulfillment spans availability confirmation (ATP check), reservation (partial quantity), backorder (customer acknowledgment), and shipment (partial dispatch) — each touching different aggregates with distinct lifecycle rules. Backorder acknowledgment (BR-028) is a commercial rule that neither Stock Position nor Shipment alone should own.
Related Business Rules	BR-028 (backorder customer acknowledgment), BR-001 (ATP-only for fulfillable portion), BR-002 (partial reservation)
Related Use Cases	UC-025 (Handle Partial Fulfillment and Backorders), UC-021 (Confirm Order Availability), UC-022 (Reserve Inventory for Confirmed Orders)

Domain 5: Inventory Control

DS-014 — Physical Inventory Freeze Guard Service

Aspect	Detail
Business Responsibility	Before any stock mutation operation (receipt, shipment, transfer, adjustment), verify that the target Inventory Location is not under an active Physical Inventory Count freeze. Blocks unauthorized mutations and permits only authorized count-related adjustments during freeze period.
Why It Should Not Belong to an Entity	Freeze state belongs to Physical Inventory Count and Inventory Location aggregates, but enforcement must intercept operations initiated by Goods Receipt, Stock Transfer, Customer Order Fulfillment, and Inventory Adjustment Case — four separate aggregates. No mutating aggregate should own freeze enforcement without coupling to count lifecycle.
Related Business Rules	BR-024 (physical inventory freeze during wall-to-wall count)
Related Use Cases	UC-015 (Conduct Full Physical Inventory Count), UC-008 (Receive and Verify Supplier Goods), UC-024 (Pick, Pack, and Ship Orders), UC-033 (Execute and Record Inter-Location Transfer), UC-017 (Process Stock Adjustments)

DS-015 — Adjustment Governance Evaluation Service

Aspect	Detail
Business Responsibility	Evaluate whether a proposed Stock Adjustment or Inventory Write-Off satisfies governance requirements: mandatory reason code validity, financial/quantity approval threshold from Governance Policy, and segregation of duties (initiator distinct from required approver). Returns approval requirement and authorization outcome.
Why It Should Not Belong to an Entity	Governance evaluation consults Inventory Governance (thresholds, reason codes), Inventory Adjustment Case (proposed change), and User Account (initiator/approver identities). Adjustment aggregate should not embed policy thresholds that change independently; Governance Policy should not know about specific adjustment instances.
Related Business Rules	BR-020 (mandatory adjustment reason codes), BR-021 (independent approval for material adjustments), BR-022 (investigate before adjusting), BR-023 (documented write-off authorization)
Related Use Cases	UC-017 (Process Stock Adjustments), UC-018 (Authorize and Process Inventory Write-Offs), UC-016 (Investigate and Approve Count Variances)

DS-016 — Count Variance Adjudication Service

Aspect	Detail
Business Responsibility	Adjudicate count variances by comparing Cycle Count or Physical Inventory Count results against Stock Position system quantity, enforcing blind count protocol (variance calculated at submission only), validating investigation completion, and confirming approver independence from counter before authorizing adjustment creation.
Why It Should Not Belong to an Entity	Adjudication connects Cycle Count (count result), Stock Position (system quantity), Count Variance (discrepancy), User Account (counter vs. approver), and Inventory Adjustment Case (correction). Cycle Count should not mutate Stock Position directly; Stock Position should not know count protocol. Adjudication is the governed bridge between count and correction domains.
Related Business Rules	BR-018 (blind cycle counting), BR-019 (independent count approval), BR-022 (investigate before adjusting)
Related Use Cases	UC-014 (Perform Cycle Count), UC-015 (Conduct Full Physical Inventory Count), UC-016 (Investigate and Approve Count Variances)

Domain 6: Monitoring & Classification

DS-017 — Stock Status Classification Service

Aspect	Detail
Business Responsibility	Classify a Stock Position as in stock, low stock, out of stock, or overstock by comparing current ATP and on-hand quantities against Stock Policy thresholds (reorder level, minimum, maximum). Applies ABC classification weighting for alert priority where configured.
Why It Should Not Belong to an Entity	Classification derives status from two aggregates — Stock Position (quantities) and Stock Policy (thresholds) — plus Product (ABC class). Stock Position could compute a local indicator but should not own policy thresholds; Stock Policy defines limits but not current quantities. Classification is a derived cross-aggregate evaluation refreshed on quantity or policy changes.
Related Business Rules	BR-009 (reorder at reorder level), BR-032 (Class A priority monitoring)
Related Use Cases	UC-012 (Monitor Inventory Status), UC-013 (Inquire Inventory Availability), UC-028 (Review Low-Stock and Out-of-Stock Alerts)

DS-018 — Low-Stock Alert Evaluation Service

Aspect	Detail
Business Responsibility	Scan Stock Positions against Stock Policies across the network to detect threshold breaches and generate or update Operational Alerts. Applies Class A priority escalation rules and suppresses duplicate alerts for acknowledged conditions.
Why It Should Not Belong to an Entity	Alert generation is a network-wide batch evaluation over many Stock Position and Stock Policy pairs — not a responsibility of any single position or policy. Operational Alert is a derived notification entity; the evaluation logic that creates it spans the monitoring domain without belonging to the alert itself.
Related Business Rules	BR-009 (reorder at reorder level — same business day action), BR-032 (Class A priority monitoring)
Related Use Cases	UC-028 (Review Low-Stock and Out-of-Stock Alerts), UC-047 (Receive and Respond to Operational Notifications), UC-012 (Monitor Inventory Status)

DS-019 — Promotional Readiness Evaluation Service

Aspect	Detail
Business Responsibility	Evaluate whether Promotional Stock Allocation ring-fenced quantity is sufficient for upcoming campaign requirements by comparing allocated ATP against campaign demand forecast and effective campaign period. Identifies readiness gaps and triggers readiness alerts.
Why It Should Not Belong to an Entity	Readiness evaluation crosses Promotional Stock Allocation (ring-fenced quantity), Stock Position (actual availability), Product (campaign SKU), and campaign period parameters. Marketing readiness is a commercial judgment spanning inventory and campaign entities — neither aggregate owns the full readiness question.
Related Business Rules	BR-029 (promotional stock ring-fencing), BR-001 (general ATP separate from promotional pool)
Related Use Cases	UC-040 (Confirm Promotional and Launch Stock Readiness), UC-047 (Receive and Respond to Operational Notifications), UC-022 (Reserve Inventory for Confirmed Orders)

Domain 7: Returns & Ownership

DS-020 — Return Eligibility and Disposition Routing Service

Aspect	Detail
Business Responsibility	Validate return eligibility by matching Return Authorization against original Customer Order lines and return policy. Upon inspection, route returned goods to the correct outcome — restock (Stock Position increase), quarantine (Quarantine Hold), or write-off (Inventory Write-Off) — based on condition assessment.
Why It Should Not Belong to an Entity	Return processing spans Return Authorization, Customer Order, Customer Return, Return Disposition, Stock Position, Quarantine Hold, and Inventory Write-Off. Customer Return records receipt but should not own disposition routing rules; Stock Position should not embed return policy. Routing is a cross-aggregate orchestration of reverse logistics.

Aspect	Detail
Related Business Rules	BR-025 (return authorization required), BR-026 (inspect returns before restocking), BR-004 (verify before acceptance), BR-013 (quarantine exclusion), BR-023 (write-off authorization)
Related Use Cases	UC-035 (Authorize and Record Customer Returns), UC-036 (Receive, Inspect, and Disposition Returned Goods), UC-011 (Receive Return and Non-PO Inbound Stock)

DS-021 — Consignment Ownership Evaluation Service

Aspect	Detail
Business Responsibility	Determine ownership classification (owned vs. consignment) for stock at receipt and at consumption/sale events based on Supplier consignment agreement. Governs whether Stock Position quantity counts toward owned valuation and when ownership transfer triggers financial recording.
Why It Should Not Belong to an Entity	Ownership determination consults Supplier (consignment agreement), Goods Receipt (inbound context), Stock Position (current classification), and Shipment (consumption event). Supplier owns commercial terms; Stock Position holds physical quantity — neither should fully own the ownership transfer decision that affects finance (Inventory Valuation Snapshot).
Related Business Rules	BR-031 (consignment ownership at consumption)
Related Use Cases	UC-039 (Reconcile Inventory Valuation with Finance), UC-008 (Receive and Verify Supplier Goods), UC-024 (Pick, Pack, and Ship Orders)

Domain 8: Financial & Governance

DS-022 — Inventory Valuation Reconciliation Service

Aspect	Detail
Business Responsibility	Produce Inventory Valuation Snapshot by aggregating owned Stock Position quantities (excluding consignment, quarantined, and in-transit per policy), applying unit costs from matched Supplier Invoices, and comparing result against general ledger expectations to identify reconciliation variances.
Why It Should Not Belong to an Entity	Valuation reconciliation synthesizes Stock Positions (quantities, ownership), Supplier Invoices (costs), Inventory Write-Offs (period reductions), and Quarantine/In-Transit exclusions into a point-in-time financial view. No operational entity should own finance reconciliation — it is a derived cross-domain computation.
Related Business Rules	BR-008 (three-way match — cost authority), BR-031 (consignment exclusion from owned valuation)
Related Use Cases	UC-039 (Reconcile Inventory Valuation with Finance), UC-037 (Review Inventory Dashboards and Operational Reports), UC-038 (Review Inventory Performance KPIs)

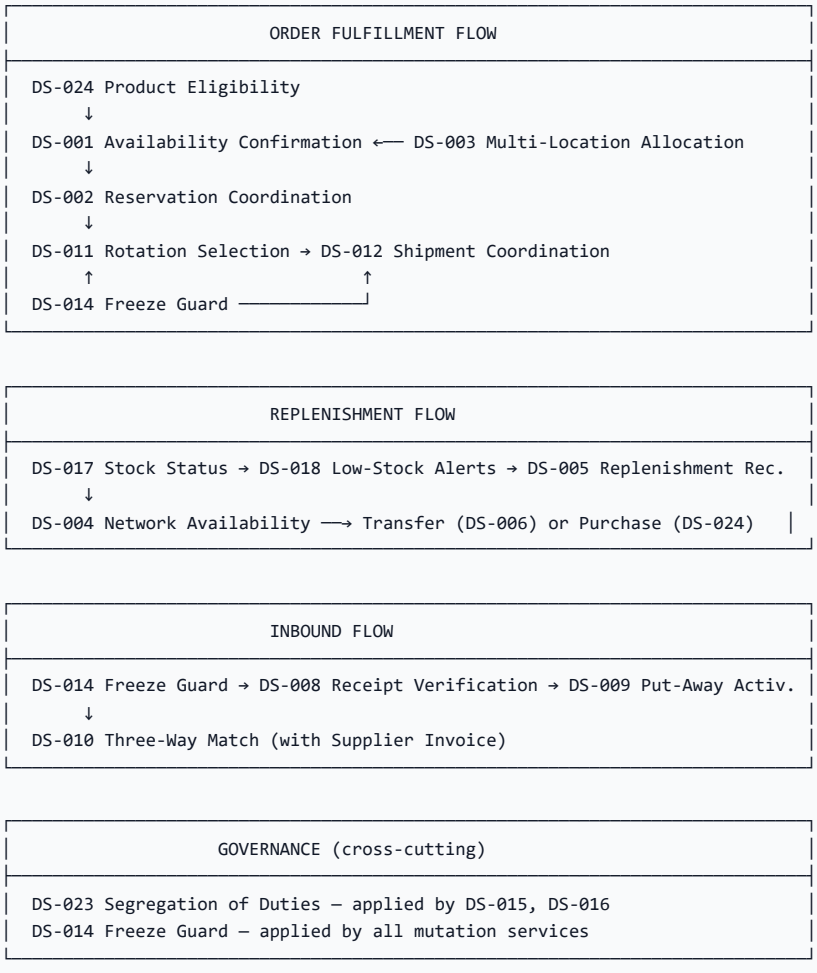
DS-023 — Segregation of Duties Validation Service

Aspect	Detail
Business Responsibility	Validate that a proposed action does not violate segregation-of-duties rules by checking User Account role assignments — e.g., counter cannot approve own count variance, initiator cannot approve own material adjustment, conflicting roles not held simultaneously.
Why It Should Not Belong to an Entity	SoD validation compares the acting user and proposed approver against User Account role assignments and Governance Policy rules across unrelated transactions (count, adjustment, write-off). Individual transaction entities should not embed user role logic — identity governance is a cross-cutting domain concern.
Related Business Rules	BR-019 (independent count approval), BR-021 (independent approval for material adjustments), BR-023 (write-off authorization by authorized person)
Related Use Cases	UC-016 (Investigate and Approve Count Variances), UC-017 (Process Stock Adjustments), UC-018 (Authorize and Process Inventory Write-Offs), UC-045 (Enforce Role-Based and Location-Scoped Access), UC-041 (Conduct Inventory Audit and Control Review)

DS-024 — Product Transaction Eligibility Service

Aspect	Detail
Business Responsibility	Validate that a product is eligible for a proposed transaction type — purchase, reservation, fulfillment, or replenishment — based on Product lifecycle status (active, inactive, discontinued) and any authorized exceptions recorded in Governance Policy.
Why It Should Not Belong to an Entity	Eligibility checks Product catalog status against transaction context from Purchase Order, Customer Order, or Purchase Requisition. Product owns its status but should not know all transaction types that consume it; transaction aggregates should not duplicate lifecycle rules. Centralized eligibility prevents inconsistent enforcement of BR-027 across procurement and fulfillment.
Related Business Rules	BR-027 (no reorder of discontinued products)
Related Use Cases	UC-006 (Place and Manage Purchase Orders), UC-022 (Reserve Inventory for Confirmed Orders), UC-029 (Plan Replenishment and Create Purchase Requisitions), UC-001 (Manage Product Master Data)

Domain Service Interaction Map



Domain Service-to-Aggregate Matrix

Domain Service	Aggregates Coordinated
DS-001 Availability Confirmation	Customer Order Fulfillment, Stock Position, Promotional Stock Allocation, Product Catalog
DS-002 Reservation Coordination	Customer Order Fulfillment, Stock Position, Stock Policy, Promotional Stock Allocation

Domain Service	Aggregates Coordinated
DS-003 Multi-Location Fulfillment Allocation	Customer Order Fulfillment, Stock Position (multiple), Inventory Location
DS-004 Network Availability Evaluation	Stock Position (network), In-Transit Stock, Expected Inbound Delivery
DS-005 Replenishment Recommendation	Stock Position, Stock Policy, Product Catalog, Expected Inbound Delivery
DS-006 Transfer Capacity Evaluation	Stock Position, Stock Policy
DS-007 Network Rebalancing Analysis	Stock Position, Stock Policy, Inventory Location (network)
DS-008 Receipt Verification	Goods Receipt, Purchase Order, Product Catalog, Inventory Governance
DS-009 Put-Away Activation	Goods Receipt, Stock Position, Inventory Location, Quarantine Hold
DS-010 Three-Way Match Evaluation	Supplier Invoice, Purchase Order, Goods Receipt, Inventory Governance
DS-011 Stock Rotation Selection	Product Catalog, Stock Position, Lot/Batch, Storage Position
DS-012 Shipment Completion Coordination	Customer Order Fulfillment, Stock Position
DS-013 Partial Fulfillment & Backorder	Customer Order Fulfillment, Stock Position
DS-014 Physical Inventory Freeze Guard	Physical Inventory Count, Inventory Location, all mutation aggregates
DS-015 Adjustment Governance Evaluation	Inventory Adjustment Case, Inventory Governance, User Account
DS-016 Count Variance Adjudication	Cycle Count, Physical Inventory Count, Stock Position, Inventory Adjustment Case, User Account
DS-017 Stock Status Classification	Stock Position, Stock Policy, Product Catalog
DS-018 Low-Stock Alert Evaluation	Stock Position, Stock Policy, Operational Alert
DS-019 Promotional Readiness Evaluation	Promotional Stock Allocation, Stock Position, Product Catalog
DS-020 Return Eligibility & Disposition	Customer Return Case, Customer Order Fulfillment, Stock Position, Inventory Adjustment Case
DS-021 Consignment Ownership Evaluation	Supplier, Stock Position, Goods Receipt, Customer Order Fulfillment
DS-022 Inventory Valuation Reconciliation	Stock Position, Supplier Invoice, Inventory Valuation Snapshot, Inventory Write-Off
DS-023 Segregation of Duties Validation	User Account, Inventory Governance
DS-024 Product Transaction Eligibility	Product Catalog, Inventory Governance

Domain Service-to-Use Case Coverage

Use Case	Domain Services Invoked
UC-008 Receive and Verify Supplier Goods	DS-014, DS-008, DS-024
UC-010 Perform Stock Put-Away	DS-009, DS-011, DS-014
UC-013 Inquire Inventory Availability	DS-001, DS-017
UC-014 Perform Cycle Count	DS-016, DS-014
UC-015 Conduct Full Physical Inventory Count	DS-014, DS-016
UC-016 Investigate and Approve Count Variances	DS-016, DS-015, DS-023
UC-017 Process Stock Adjustments	DS-015, DS-023, DS-014
UC-021 Confirm Order Availability	DS-001, DS-024, DS-003
UC-022 Reserve Inventory for Confirmed Orders	DS-002, DS-001, DS-024
UC-024 Pick, Pack, and Ship Orders	DS-011, DS-012, DS-014
UC-025 Handle Partial Fulfillment and Backorders	DS-013, DS-002, DS-001
UC-028 Review Low-Stock Alerts	DS-018, DS-017, DS-005
UC-029 Plan Replenishment	DS-005, DS-004, DS-024
UC-031 Request Inter-Location Transfer	DS-006, DS-004, DS-014

Use Case	Domain Services Invoked
UC-036 Receive, Inspect, Disposition Returns	DS-020, DS-008, DS-015
UC-039 Reconcile Inventory Valuation	DS-022, DS-010, DS-021
UC-040 Confirm Promotional Readiness	DS-019, DS-001
UC-048 Multi-Location Split Fulfillment	DS-003, DS-002, DS-001

Excluded from Domain Services

The following are **not** domain services — they belong to other layers or patterns:

Capability	Classification	Rationale
Send email/SMS notification	Application / Infrastructure Service	Delivery mechanism, not business logic
Generate PDF report	Application Service	Presentation concern
Persist Stock Position	Repository (Infrastructure)	Persistence, not domain behavior
Authenticate user session	Application / Infrastructure	Technical security, not inventory domain
Calculate inventory turnover KPI	Read Model / Analytics	Derived metric for reporting, not transactional domain logic
Import CSV product file	Application Service	File parsing and orchestration wrapping domain operations

Document Approval

Role	Name	Signature	Date
DDD Architect			
Domain Expert			
Solution Architect			
Senior Business Analyst			

End of Domain Services

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Domain Events

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-DM-006
Document Title	Domain Events — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Domain Expert

Attribute	Detail
Source Documents	SRS-DM-001 (Domain Entity Model), SRS-DM-003 (Domain Aggregates), SRS-DM-005 (Domain Services), SRS-BR-001 (Business Rules), SRS-UC-001 (Use Case Catalog)
Date	July 5, 2026
Scope	Business domain events only — no message brokers, event buses, or integration technology

Purpose

This document identifies **Domain Events** for the Inventory Management System. A domain event is an immutable record of something meaningful that happened in the business — expressed in past tense from the domain's ubiquitous language. Events announce state changes within aggregates and enable loose coupling between aggregates, read models, and downstream business reactions.

Domain events are **facts**, not commands. They describe what occurred; consumers decide how to respond.

Event Naming Convention

Events are named in **PascalCase past tense**: `StockReserved`, `PutAwayCompleted`, `CountVarianceIdentified`. Each name should be understandable to business stakeholders without technical context.

Consumer Types

Consumer Type	Role
Aggregate	Another consistency boundary that reacts by updating its own state
Domain Service	Stateless evaluator triggered to apply cross-aggregate business rules
Read Model / Projection	Derived view rebuilt from events (movement history, alerts, dashboards)
Business Process	Human workflow or notification triggered for actor action

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Event Catalog Overview

Domain Area	Events	IDs
Catalog & Network	Product and location lifecycle	DE-001 – DE-010
Inventory Balance	Stock quantity and hold state	DE-011 – DE-022
Procurement & Inbound	Requisition through receipt	DE-023 – DE-035
Order Fulfillment	Order commitment through shipment	DE-036 – DE-045
Network Transfer	Inter-location movement	DE-046 – DE-051
Inventory Control	Counting and correction	DE-052 – DE-060
Returns	Reverse logistics	DE-061 – DE-067
Commercial & Monitoring	Promotions and alerts	DE-068 – DE-073
Financial & Governance	Finance alignment and audit	DE-074 – DE-081

Total Domain Events: 81

Domain 1: Catalog & Network

DE-001 — ProductRegistered

Aspect	Detail
Trigger	Inventory Controller creates a new product record with unique SKU, name, and unit of measure in the Product Catalog aggregate.
Related Aggregate	Product Catalog
Business Meaning	The organization has introduced a new SKU into the trusted product catalog. The product is now eligible for stock policy assignment, procurement, and (when active) fulfillment referencing.
Consumers	Stock Policy aggregate *(policy may be assigned)*; Read Model — product catalog index; Business Process — master data notification to operational teams

DE-002 — ProductStatusChanged

Aspect	Detail
Trigger	Product lifecycle status transitions among active, inactive, or discontinued within the Product Catalog aggregate.
Related Aggregate	Product Catalog
Business Meaning	The commercial availability of a SKU has changed. Discontinued status prohibits new procurement and reservations; inactive status restricts new customer commitments.
Consumers	Product Transaction Eligibility domain service *(DS-024)*; Purchase Order aggregate *(blocks new lines)*; Customer Order Fulfillment aggregate *(blocks new reservations)*; Replenishment Recommendation domain service *(DS-005)*; Read Model — product status history

DE-003 — ProductABCClassificationAssigned

Aspect	Detail
Trigger	Inventory Controller assigns or updates ABC classification (A, B, or C) on a product during classification review.
Related Aggregate	Product Catalog
Business Meaning	Operational priority for the SKU has been set. Class A products receive highest count frequency, monitoring intensity, and alert priority per BR-032.
Consumers	Cycle Count aggregate *(scheduling priority)*; Low-Stock Alert Evaluation domain service *(DS-018)*; Read Model — ABC classification report

DE-004 — InventoryLocationActivated

Aspect	Detail
Trigger	Inventory Location transitions to active operational status after opening balance approval at go-live.
Related Aggregate	Inventory Location
Business Meaning	A location is authorized to receive, hold, fulfill, and transact inventory. Stock positions may be established and operational workflows enabled.
Consumers	Stock Position aggregate *(opening balances created)*; Read Model — location network index; Business Process — go-live notification to warehouse team

DE-005 — InventoryLocationDeactivated

Aspect	Detail
Trigger	Inventory Location is deactivated after stock disposition is complete or closure is authorized.
Related Aggregate	Inventory Location
Business Meaning	A location is no longer authorized for new inventory activity. Existing stock must have been cleared or dispositioned before deactivation.
Consumers	Physical Inventory Freeze Guard domain service *(DS-014)*; Purchase Order aggregate *(blocks new receiving)*; Customer Order Fulfillment aggregate *(blocks new fulfillment)*; Read Model — location status history

DE-006 — StoragePositionConfigured

Aspect	Detail
Trigger	New storage position (zone, aisle, bin) is added or updated within an Inventory Location aggregate.
Related Aggregate	Inventory Location
Business Meaning	A findable physical address within a location is available for put-away placement and pick path guidance.
Consumers	Put-Away Activation domain service *(DS-009)*; Stock Rotation Selection domain service *(DS-011)*; Read Model — warehouse layout reference

DE-007 — LocationFreezeImposed

Aspect	Detail
Trigger	Physical Inventory Count aggregate begins wall-to-wall count at a location, imposing operational freeze.
Related Aggregate	Physical Inventory Count
Business Meaning	All receiving, shipping, transfers, and adjustments at the location are suspended until the count is certified complete. Ensures stable snapshot for accurate counting per BR-024.
Consumers	Physical Inventory Freeze Guard domain service *(DS-014)*; Goods Receipt aggregate *(blocks receipt)*; Customer Order Fulfillment aggregate *(blocks shipment)*; Stock Transfer aggregate *(blocks dispatch/receipt)*; Inventory Adjustment Case aggregate *(blocks non-count adjustments)*; Business Process — freeze notification to warehouse staff

DE-008 — LocationFreezeReleased

Aspect	Detail
Trigger	Physical Inventory Count aggregate certifies count completion and releases location freeze.
Related Aggregate	Physical Inventory Count
Business Meaning	Normal inventory operations may resume at the location. Count variances have been identified and routed for investigation.
Consumers	Physical Inventory Freeze Guard domain service *(DS-014)*; Business Process — operations resume notification to warehouse staff

DE-009 — GovernancePolicyPublished

Aspect	Detail
Trigger	New or updated governance policy version becomes effective in the Inventory Governance aggregate (approval thresholds, tolerances, exception rules).

Aspect	Detail
Related Aggregate	Inventory Governance
Business Meaning	Organization-wide inventory control parameters have changed. All subsequent transactions must evaluate against the new effective policy.
Consumers	Adjustment Governance Evaluation domain service *(DS-015)*; Receipt Verification domain service *(DS-008)*; Three-Way Match Evaluation domain service *(DS-010)*; Read Model — policy version history

DE-010 — ReasonCodeRetired

Aspect	Detail
Trigger	A reason code is marked retired in the Inventory Governance aggregate and removed from active selection lists.
Related Aggregate	Inventory Governance
Business Meaning	A categorized exception reason is no longer valid for new transactions. Historical records retain prior reason code attributions unchanged.
Consumers	Adjustment Governance Evaluation domain service *(DS-015)*; Read Model — reason code catalog

Domain 2: Inventory Balance

DE-011 — StockReceived

Aspect	Detail
Trigger	Put-away is confirmed and Stock Position aggregate increases on-hand quantity for a product at a location.
Related Aggregate	Stock Position
Business Meaning	Verified inbound stock is now physically placed and recorded as on-hand inventory. ATP increases correspondingly unless quarantine or ownership rules apply.
Consumers	Stock Status Classification domain service *(DS-017)*; Inventory Movement Record read model; Audit Trail Record read model; Network Availability Evaluation domain service *(DS-004)*

DE-012 — StockReserved

Aspect	Detail
Trigger	Stock Position aggregate increases reserved quantity and decreases ATP following reservation coordination for a confirmed customer order.
Related Aggregate	Stock Position
Business Meaning	Quantity is operationally committed to a customer order and is no longer available for new sales promises. Prevents double-allocation per BR-002.
Consumers	Customer Order Fulfillment aggregate *(reservation confirmed)*; Inventory Movement Record read model; Stock Status Classification domain service *(DS-017)*

DE-013 — StockReservationReleased

Aspect	Detail
Trigger	Stock Position aggregate decreases reserved quantity and restores ATP when a reservation is cancelled or order line removed.

Aspect	Detail
Related Aggregate	Stock Position
Business Meaning	Previously committed stock is returned to available-to-promise pool and may be promised to other customers or orders.
Consumers	Customer Order Fulfillment aggregate *(reservation cleared)*; Inventory Movement Record read model; Low-Stock Alert Evaluation domain service *(DS-018)*

DE-014 — StockShipped

Aspect	Detail
Trigger	Stock Position aggregate decreases on-hand and reserved quantity upon verified shipment dispatch.
Related Aggregate	Stock Position
Business Meaning	Physical stock has left the location for customer delivery. Inventory record reflects deduction only after verified pick per BR-003.
Consumers	Customer Order Fulfillment aggregate *(shipment confirmed)*; Consignment Ownership Evaluation domain service *(DS-021)*; Inventory Movement Record read model; Stock Status Classification domain service *(DS-017)*; Read Model — fill rate metrics

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DE-015 — StockAdjusted

Aspect	Detail
Trigger	Approved Stock Adjustment is applied and Stock Position on-hand quantity is corrected with mandatory reason code.
Related Aggregate	Stock Position
Business Meaning	Recorded inventory has been aligned with physical reality following investigation and authorization. Distinct from write-off — adjustment is a correction, not permanent value removal.
Consumers	Inventory Adjustment Case aggregate *(adjustment finalized)*; Inventory Movement Record read model; Audit Trail Record read model; Stock Status Classification domain service *(DS-017)*; Read Model — shrinkage analysis

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DE-016 — StockWrittenOff

Aspect	Detail
Trigger	Authorized Inventory Write-Off permanently removes quantity and value from Stock Position.
Related Aggregate	Stock Position
Business Meaning	Inventory value has been formally removed for damaged, obsolete, expired, or lost goods. Terminal event — quantity cannot be restored without new receipt per BR-023.
Consumers	Inventory Adjustment Case aggregate *(write-off finalized)*; Inventory Valuation Reconciliation domain service *(DS-022)*; Inventory Movement Record read model; Audit Trail Record read model; Read Model — shrinkage and loss reporting

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DE-017 — QuarantineHoldApplied

Aspect	Detail
Trigger	Quarantine Hold is placed on quantity within Stock Position, excluding it from ATP.
Related Aggregate	Stock Position

Aspect	Detail
Business Meaning	Stock is physically present but not sellable due to quality investigation, regulatory restriction, recall, or customs hold. ATP reduced accordingly per BR-013.
Consumers	Availability Confirmation domain service *(DS-001)*; Inventory Movement Record read model; Business Process — quality/regulatory investigation workflow

DE-018 — QuarantineHoldReleased

Aspect	Detail
Trigger	Quarantine Hold is cleared after quality approval, customs clearance, or disposition decision, restoring quantity to ATP.
Related Aggregate	Stock Position
Business Meaning	Previously held stock is cleared for sale or fulfillment. Bonded inventory cleared for domestic commerce per BR-030.
Consumers	Put-Away Activation domain service *(DS-009)*; Availability Confirmation domain service *(DS-001)*; Inventory Movement Record read model

DE-019 — StockPolicyThresholdChanged

Aspect	Detail
Trigger	Stock Policy aggregate records new reorder level, minimum, safety stock, or maximum thresholds with effective date.
Related Aggregate	Stock Policy
Business Meaning	Replenishment and alert rules for a product at a location have been recalibrated. Future low-stock detection uses new thresholds per BR-009.
Consumers	Stock Status Classification domain service *(DS-017)*; Low-Stock Alert Evaluation domain service *(DS-018)*; Replenishment Recommendation domain service *(DS-005)*; Read Model — policy change history

DE-020 — PromotionalStockAllocated

Aspect	Detail
Trigger	Promotional Stock Allocation aggregate ring-fences quantity for a campaign or launch period.
Related Aggregate	Promotional Stock Allocation
Business Meaning	Defined inventory quantity is reserved for promotional or launch use and excluded from general ATP for non-approved orders per BR-029.
Consumers	Stock Position aggregate *(ATP segregation)*; Availability Confirmation domain service *(DS-001)*; Promotional Readiness Evaluation domain service *(DS-019)*; Business Process — marketing readiness notification

DE-021 — PromotionalAllocationReleased

Aspect	Detail
Trigger	Campaign period ends or allocation is manually released; ring-fenced quantity returns to general availability.
Related Aggregate	Promotional Stock Allocation
Business Meaning	Un consumed promotional stock is returned to general sellable pool. Campaign commitment period has concluded.
Consumers	Stock Position aggregate *(ATP restored)*; Availability Confirmation domain service *(DS-001)*; Read Model — campaign performance report

DE-022 — ConsignmentOwnershipTransferred

Aspect	Detail
Trigger	Consignment stock ownership transfers from supplier to organization upon defined consumption or sale event per supplier agreement.
Related Aggregate	Stock Position
Business Meaning	Stock previously held under consignment classification becomes owned inventory for valuation and financial recording per BR-031.
Consumers	Consignment Ownership Evaluation domain service *(DS-021)*; Inventory Valuation Reconciliation domain service *(DS-022)*; Inventory Movement Record read model

Domain 3: Procurement & Inbound

DE-023 — PurchaseRequisitionSubmitted

Aspect	Detail
Trigger	Internal replenishment need is recorded and submitted for approval in the Purchase Requisition aggregate.
Related Aggregate	Purchase Requisition
Business Meaning	The organization has identified a procurement need for specific products at a target location, pending internal approval before commercial PO creation.
Consumers	Network Availability Evaluation domain service *(DS-004)*; Business Process — purchasing manager approval workflow; Read Model — requisition pipeline

DE-024 — PurchaseRequisitionApproved

Aspect	Detail
Trigger	Purchase Requisition reaches approved status after network check and policy validation.
Related Aggregate	Purchase Requisition
Business Meaning	Internal authorization to proceed with supplier procurement is granted. Requisition is eligible for conversion to Purchase Order.
Consumers	Purchase Order aggregate *(PO creation initiated)*; Business Process — buyer notification to create PO

DE-025 — PurchaseRequisitionRejected

Aspect	Detail
Trigger	Purchase Requisition is rejected — e.g., network transfer alternative identified or policy violation.
Related Aggregate	Purchase Requisition
Business Meaning	Procurement request denied. Network redistribution or other alternative should be pursued instead of new purchase per BR-011.
Consumers	Business Process — requestor notification; Read Model — requisition status history

DE-026 — PurchaseOrderPlaced

Aspect	Detail
Trigger	Authorized commercial commitment is recorded with supplier, products, quantities, and delivery terms in the Purchase Order aggregate.
Related Aggregate	Purchase Order
Business Meaning	The organization has committed to purchase specified goods from a supplier. Inbound receipt is authorized upon delivery per BR-007.
Consumers	Expected Inbound Delivery aggregate *(delivery planning)*; Business Process — supplier notification; Read Model — open PO dashboard; Inbound supply tracking

DE-027 — PurchaseOrderCancelled

Aspect	Detail
Trigger	Open Purchase Order is cancelled before full receipt.
Related Aggregate	Purchase Order
Business Meaning	Commercial commitment to supplier is withdrawn. No further receipts authorized against cancelled PO lines.
Consumers	Expected Inbound Delivery aggregate *(schedule cancelled)*; Business Process — supplier and buyer notification; Read Model — PO status history

DE-028 — InboundDeliveryScheduled

Aspect	Detail
Trigger	Expected Inbound Delivery is planned with arrival date and quantity tied to a Purchase Order.
Related Aggregate	Purchase Order
Business Meaning	Receiving operations and replenishment planners are informed when and what quantity to expect, supporting dock preparation and inbound visibility.
Consumers	Read Model — inbound supply calendar; Network Availability Evaluation domain service *(DS-004)*; Business Process — receiving clerk preparation notification

DE-029 — InboundDeliveryDelayed

Aspect	Detail
Trigger	Expected arrival date passes without Goods Receipt, or supplier communicates schedule change.
Related Aggregate	Purchase Order
Business Meaning	Anticipated inbound supply will arrive later than planned. Replenishment timelines and fulfillment promises may be affected.
Consumers	Operational Alert read model *(inbound delay alert)*; Replenishment Recommendation domain service *(DS-005)*; Business Process — buyer and purchasing manager notification *(UC-047)*

DE-030 — GoodsReceiptVerified

Aspect	Detail
Trigger	Inbound goods pass verification against Purchase Order — quantity, product identity, and condition confirmed in Goods Receipt aggregate.
Related Aggregate	Goods Receipt
Business Meaning	The business has confirmed what actually arrived at the dock matches authorized expectation. Stock is not yet available — put-away must follow per BR-004, BR-005.
Consumers	Receipt Verification domain service *(DS-008)*; Three-Way Match Evaluation domain service *(DS-010)*; Business Process — put-away task generation

DE-031 — ReceiptDiscrepancyRecorded

Aspect	Detail
Trigger	Variance between expected and received goods is documented with evidence at receipt in Goods Receipt aggregate.
Related Aggregate	Goods Receipt
Business Meaning	Short shipment, over-shipment, wrong item, damage, or shelf-life non-compliance captured at point of receipt when evidence exists per BR-006.
Consumers	Three-Way Match Evaluation domain service *(DS-010)*; Business Process — supplier claim and buyer notification; Read Model — supplier performance tracking

DE-032 — ReceiptDiscrepancyResolved

Aspect	Detail
Trigger	Receipt discrepancy reaches resolved status — partial acceptance, rejection, or supplier claim completed.
Related Aggregate	Goods Receipt
Business Meaning	Receipt exception has been closed with documented outcome. Accepted quantities may proceed to put-away; rejected quantities will not enter inventory.
Consumers	Three-Way Match Evaluation domain service *(DS-010)*; Read Model — discrepancy resolution history

DE-033 — PutAwayCompleted

Aspect	Detail
Trigger	Received goods are confirmed placed in designated storage position within Goods Receipt aggregate.
Related Aggregate	Goods Receipt
Business Meaning	Inbound stock has transitioned from receiving dock to operational storage. Triggers availability activation — stock may now enter sellable ATP per BR-005.
Consumers	Put-Away Activation domain service *(DS-009)*; Stock Position aggregate *(StockReceived)*; Purchase Order aggregate *(received quantity updated)*; Inventory Movement Record read model

DE-034 — SupplierInvoiceReceived

Aspect	Detail
Trigger	Supplier billing document is recorded in the Supplier Invoice aggregate.

Aspect	Detail
Related Aggregate	Supplier Invoice
Business Meaning	Supplier has invoiced for goods supplied. Three-way match evaluation against PO and receipt must proceed before payment authorization per BR-008.
Consumers	Three-Way Match Evaluation domain service *(DS-010)*; Business Process — finance accounts payable workflow

DE-035 — ThreeWayMatchCompleted

Aspect	Detail
Trigger	PO, goods receipt, and supplier invoice lines agree within tolerance in Supplier Invoice aggregate.
Related Aggregate	Supplier Invoice
Business Meaning	Financial control confirmed — payment for received, authorized goods may proceed. Closes the procurement-to-payment accountability loop per BR-008.
Consumers	Inventory Valuation Reconciliation domain service *(DS-022)*; Business Process — payment authorization to finance; Read Model — match status dashboard

Domain 4: Order Fulfillment

DE-036 — OrderAvailabilityConfirmed

Aspect	Detail
Trigger	Availability Confirmation domain service validates order lines against ATP and records confirmation result on Customer Order Fulfillment aggregate.
Related Aggregate	Customer Order Fulfillment
Business Meaning	The business has verified that available-to-promise stock can support the order (fully or partially) before commercial commitment per BR-001.
Consumers	Business Process — sales/customer service confirmation to customer; Read Model — availability inquiry log

DE-037 — CustomerOrderConfirmed

Aspect	Detail
Trigger	Customer order transitions to confirmed status ready for reservation and fulfillment execution.
Related Aggregate	Customer Order Fulfillment
Business Meaning	Commercial commitment to fulfill the order is established. Reservation of stock must follow immediately per BR-002.
Consumers	Inventory Reservation Coordination domain service *(DS-002)*; Business Process — warehouse fulfillment queue

DE-038 — CustomerOrderCancelled

Aspect	Detail
Trigger	Confirmed customer order or order line is cancelled before full shipment.
Related Aggregate	Customer Order Fulfillment

Aspect	Detail
Business Meaning	Commercial commitment withdrawn. Reserved stock must be released back to ATP.
Consumers	Inventory Reservation Coordination domain service *(DS-002)* → StockReservationReleased; Business Process — customer cancellation confirmation

DE-039 — PickVerified

Aspect	Detail
Trigger	Warehouse picker confirms correct items retrieved and verified against order in Fulfillment Execution within Customer Order Fulfillment aggregate.
Related Aggregate	Customer Order Fulfillment
Business Meaning	Physical pick matches order requirements. Shipment is authorized to proceed per BR-003. Lot/batch captured for traceable products per BR-017.
Consumers	Shipment Completion Coordination domain service *(DS-012)*; Inventory Movement Record read model

DE-040 — ShipmentDispatched

Aspect	Detail
Trigger	Verified goods handed to carrier and shipment recorded in Customer Order Fulfillment aggregate.
Related Aggregate	Customer Order Fulfillment
Business Meaning	Order goods have left the warehouse. Inventory deduction and reservation consumption triggered. Delivery promise activated for customer.
Consumers	Shipment Completion Coordination domain service *(DS-012)* → StockShipped; Inventory Movement Record read model; Business Process — customer shipment notification *(UC-027)*; Read Model — on-time delivery metrics

DE-041 — PartialFulfillmentCompleted

Aspect	Detail
Trigger	Available portion of order line shipped while remainder is backordered or awaiting stock.
Related Aggregate	Customer Order Fulfillment
Business Meaning	Customer receives available quantity now; unfulfilled portion handled separately via backorder with customer acknowledgment per BR-028.
Consumers	Partial Fulfillment and Backorder Orchestration domain service *(DS-013)*; Business Process — customer partial shipment notification

DE-042 — BackorderRecorded

Aspect	Detail
Trigger	Unfulfilled order quantity recorded as backorder after customer acknowledgment in Customer Order Fulfillment aggregate.
Related Aggregate	Customer Order Fulfillment
Business Meaning	Customer has agreed to wait for remaining quantity. Mutual commercial commitment — not an internal note per BR-028.

Aspect	Detail
Consumers	Replenishment Recommendation domain service *(DS-005) (priority replenishment)*; Business Process — customer backorder confirmation; Read Model — backorder pipeline

DE-043 — BackorderFulfilled

Aspect	Detail
Trigger	Previously backordered quantity is reserved and fulfilled when stock becomes available.
Related Aggregate	Customer Order Fulfillment
Business Meaning	Delayed customer commitment is now being executed. Backorder closed for fulfilled quantity.
Consumers	Inventory Reservation Coordination domain service *(DS-002)*; Business Process — customer notification of backorder shipment

DE-044 — FulfillmentExceptionRaised

Aspect	Detail
Trigger	Pick verification fails, shelf quantity mismatch, or outbound quality issue identified during fulfillment.
Related Aggregate	Customer Order Fulfillment
Business Meaning	Warehouse cannot fulfill as planned without resolution. Inventory accuracy or product quality issue requires manager intervention per BP-20.
Consumers	Business Process — warehouse manager exception workflow *(UC-026)*; Operational Alert read model; Count Variance Adjudication domain service *(DS-016) (if shelf mismatch)*

DE-045 — MultiLocationFulfillmentPlanCreated

Aspect	Detail
Trigger	Multi-Location Fulfillment Allocation domain service assigns order lines to multiple ship-from locations.
Related Aggregate	Customer Order Fulfillment
Business Meaning	Single customer order will be fulfilled from more than one warehouse to satisfy total quantity when no single location has sufficient ATP.
Consumers	Inventory Reservation Coordination domain service *(DS-002) (reservations at each location)*; Business Process — multi-shipment customer communication *(UC-048)*

Domain 5: Network Transfer

DE-046 — TransferRequested

Aspect	Detail
Trigger	Inter-location stock transfer request submitted specifying source, destination, product, and quantity in Stock Transfer aggregate.
Related Aggregate	Stock Transfer
Business Meaning	A location requests stock from another location to resolve local shortage or rebalance network before new procurement per BR-011.
Consumers	Transfer Capacity Evaluation domain service *(DS-006)*; Business Process — operations director approval workflow *(UC-032)*; Operational Alert read model *(pending approval)*

DE-047 — TransferApproved

Aspect	Detail
Trigger	Transfer request passes approval after source minimum and safety stock validation.
Related Aggregate	Stock Transfer
Business Meaning	Network redistribution authorized. Source location can spare quantity while maintaining its own service level per BR-012.
Consumers	Business Process — source warehouse dispatch notification; Read Model — transfer pipeline

DE-048 — TransferRejected

Aspect	Detail
Trigger	Transfer request denied — source cannot spare quantity or business justification insufficient.
Related Aggregate	Stock Transfer
Business Meaning	Proposed redistribution will not proceed. Requestor must pursue alternative (procurement, different source) to resolve shortage.
Consumers	Business Process — requestor notification; Read Model — transfer request history

DE-049 — TransferDispatched

Aspect	Detail
Trigger	Stock physically leaves source location; Stock Transfer aggregate records dispatch and creates in-transit state.
Related Aggregate	Stock Transfer
Business Meaning	Source Stock Position reduced. Quantity is in transit — excluded from both source and destination ATP until receipt confirmed.
Consumers	Stock Position aggregate *(source deduction)*; In-Transit Stock tracking; Inventory Movement Record read model; Network Availability Evaluation domain service *(DS-004)*

DE-050 — TransferReceived

Aspect	Detail
Trigger	Destination location confirms receipt of transferred stock in Stock Transfer aggregate.
Related Aggregate	Stock Transfer
Business Meaning	Network redistribution complete. Destination Stock Position increased; in-transit state closed. Lot/batch preserved for traceable products per BR-017.
Consumers	Stock Position aggregate *(destination StockReceived)*; Inventory Movement Record read model; Stock Status Classification domain service *(DS-017)*

DE-051 — InTransitStockDelayed

Aspect	Detail
Trigger	Expected transfer receipt date passes without confirmation, or carrier reports delay.
Related Aggregate	Stock Transfer
Business Meaning	Pipeline stock is delayed en route. Destination location fulfillment and replenishment plans may be impacted.

Aspect	Detail
Consumers	Operational Alert read model *(transfer exception)*; Business Process — destination warehouse and requestor notification *(UC-047)*

Domain 6: Inventory Control & Accuracy

DE-052 — CycleCountScheduled

Aspect	Detail
Trigger	Cycle Count assignment created with product, location, and counter for scheduled count date.
Related Aggregate	Cycle Count
Business Meaning	Partial physical verification scheduled. Class A products scheduled at highest frequency per BR-032.
Consumers	Business Process — counter assignment notification; Read Model — count schedule calendar

DE-053 — CycleCountCompleted

Aspect	Detail
Trigger	Counter submits physical count result under blind count protocol in Cycle Count aggregate.
Related Aggregate	Cycle Count
Business Meaning	Physical quantity recorded without counter having seen system expected quantity per BR-018. Variance calculated at submission.
Consumers	Count Variance Adjudication domain service *(DS-016)*; Read Model — count completion log

DE-054 — CountVarianceIdentified

Aspect	Detail
Trigger	Physical count differs from system quantity — variance recorded in Inventory Adjustment Case aggregate.
Related Aggregate	Inventory Adjustment Case
Business Meaning	Inventory accuracy discrepancy discovered. Investigation required before any Stock Position adjustment per BR-022.
Consumers	Count Variance Adjudication domain service *(DS-016)*; Business Process — warehouse manager investigation workflow *(UC-016)*; Read Model — accuracy metrics

DE-055 — CountVarianceApproved

Aspect	Detail
Trigger	Independent approver (distinct from counter) authorizes count variance after investigation in Inventory Adjustment Case aggregate.
Related Aggregate	Inventory Adjustment Case
Business Meaning	Variance investigation complete and adjustment authorized. Segregation of duties confirmed per BR-019.
Consumers	Adjustment Governance Evaluation domain service *(DS-015)*; Stock Adjustment creation; Business Process — adjustment processing notification

DE-056 — PhysicalInventoryCountStarted

Aspect	Detail
Trigger	Wall-to-wall count begins at a location with freeze imposed in Physical Inventory Count aggregate.
Related Aggregate	Physical Inventory Count
Business Meaning	Full location snapshot count initiated. All movements suspended per BR-024. Authoritative accuracy baseline in progress.
Consumers	LocationFreezeImposed *(DE-007)*; Business Process — count team mobilization; Read Model — count event status

DE-057 — PhysicalInventoryCountCertified

Aspect	Detail
Trigger	Full count completed, all variances identified, and count event certified in Physical Inventory Count aggregate.
Related Aggregate	Physical Inventory Count
Business Meaning	Wall-to-wall count authoritative and complete. Location freeze may be released. Variances routed for investigation.
Consumers	LocationFreezeReleased *(DE-008)*; Read Model — inventory accuracy baseline; Business Process — finance notification for reconciliation

DE-058 — AdjustmentSubmitted

Aspect	Detail
Trigger	Stock adjustment or write-off proposal submitted with reason code in Inventory Adjustment Case aggregate.
Related Aggregate	Inventory Adjustment Case
Business Meaning	Correction or removal request recorded pending governance evaluation. Reason code captured per BR-020.
Consumers	Adjustment Governance Evaluation domain service *(DS-015)*; Segregation of Duties Validation domain service *(DS-023)*; Operational Alert read model *(pending approval)*

DE-059 — AdjustmentApproved

Aspect	Detail
Trigger	Independent approver authorizes adjustment or write-off after threshold and SoD validation.
Related Aggregate	Inventory Adjustment Case
Business Meaning	Governance requirements satisfied. Stock Position mutation authorized per BR-021, BR-023.
Consumers	Stock Position aggregate *(StockAdjusted or StockWrittenOff)*; Audit Trail Record read model

DE-060 — AdjustmentRejected

Aspect	Detail
Trigger	Adjustment or write-off denied due to insufficient evidence, SoD violation, or policy threshold breach.
Related Aggregate	Inventory Adjustment Case
Business Meaning	Proposed inventory change will not take effect. Initiator must revise or provide additional justification.
Consumers	Business Process — initiator notification; Read Model — adjustment case history

Domain 7: Returns Management

DE-061 — ReturnAuthorizationIssued

Aspect	Detail
Trigger	Return Authorization (RMA) created and linked to original customer order in Customer Return Case aggregate.
Related Aggregate	Customer Return Case
Business Meaning	Customer is authorized to return specified products. Warehouse notified to expect inbound return per BR-025.
Consumers	Business Process — customer return instructions; Business Process — receiving clerk advance notice

DE-062 — ReturnAuthorizationExpired

Aspect	Detail
Trigger	RMA passes expiration date without Customer Return receipt.
Related Aggregate	Customer Return Case
Business Meaning	Return window closed. Warehouse must not accept goods against expired authorization.
Consumers	Business Process — customer expiration notification; Return Eligibility domain service *(DS-020)*

DE-063 — CustomerReturnReceived

Aspect	Detail
Trigger	Returned goods arrive and pass verification against RMA in Customer Return Case aggregate.
Related Aggregate	Customer Return Case
Business Meaning	Authorized return physically received. Inspection and disposition workflow initiated per BR-004, BR-025.
Consumers	Return Eligibility and Disposition Routing domain service *(DS-020)*; Business Process — inspection task assignment

DE-064 — ReturnInspected

Aspect	Detail
Trigger	Condition assessment completed on returned goods in Customer Return Case aggregate.
Related Aggregate	Customer Return Case
Business Meaning	Resale quality of returned goods evaluated. Disposition decision (restock, quarantine, write-off) authorized to proceed per BR-026.
Consumers	Return Eligibility and Disposition Routing domain service *(DS-020)*

DE-065 — ReturnRestocked

Aspect	Detail
Trigger	Disposition decision confirms returned goods meet resale standards; restock applied to Stock Position.
Related Aggregate	Customer Return Case
Business Meaning	Qualifying returned goods re-enter sellable inventory. ATP increased at receiving location per BR-026.
Consumers	Stock Position aggregate *(StockReceived)*; Inventory Movement Record read model

DE-066 — ReturnQuarantined

Aspect	Detail
Trigger	Returned goods placed under quarantine hold pending further quality investigation.
Related Aggregate	Customer Return Case
Business Meaning	Returns requiring further evaluation excluded from sellable ATP per BR-013, BR-026.
Consumers	Stock Position aggregate *(QuarantineHoldApplied)*; Business Process — quality team investigation

DE-067 — ReturnWrittenOff

Aspect	Detail
Trigger	Returned goods deemed non-resalable; write-off authorized from Customer Return Case.
Related Aggregate	Customer Return Case
Business Meaning	Damaged or defective returns permanently removed from inventory value per BR-023, BR-026.
Consumers	Stock Position aggregate *(StockWrittenOff)*; Inventory Valuation Reconciliation domain service *(DS-022)*

Domain 8: Commercial & Monitoring

DE-068 — LowStockThresholdBreached

Aspect	Detail
Trigger	Stock Position ATP falls to or below Stock Policy reorder level; detected by Stock Status Classification domain service.
Related Aggregate	Stock Position *(state change)* / Stock Policy *(threshold reference)*
Business Meaning	Replenishment action required same business day per BR-009. Product at location needs purchasing or network transfer review.
Consumers	Low-Stock Alert Evaluation domain service *(DS-018)*; Replenishment Recommendation domain service *(DS-005)*; Operational Alert read model; Business Process — purchasing manager notification *(UC-028, UC-047)*

DE-069 — OutOfStockDetected

Aspect	Detail
Trigger	Stock Position ATP reaches zero for a product at a location.
Related Aggregate	Stock Position
Business Meaning	No quantity available to promise to new customers at this location. Existing reservations may still be fulfilled from on-hand. Stockout condition active.
Consumers	Low-Stock Alert Evaluation domain service *(DS-018)*; Operational Alert read model; Read Model — stockout report *(UC-037)*; Business Process — urgent replenishment escalation

DE-070 — OverstockDetected

Aspect	Detail
Trigger	Stock Position on-hand exceeds Stock Policy maximum level.

Aspect	Detail
Related Aggregate	Stock Position / Stock Policy
Business Meaning	Excess capital tied in inventory above policy maximum. Network rebalancing or purchasing freeze may be warranted.
Consumers	Network Rebalancing Analysis domain service *(DS-007)*; Read Model — overstock report; Business Process — inventory controller review

DE-071 — ReplenishmentRecommended

Aspect	Detail
Trigger	Replenishment Recommendation domain service produces action recommendation after threshold breach and network evaluation.
Related Aggregate	*(Cross-cutting — no single aggregate)*
Business Meaning	System recommends purchase requisition, transfer request, or no action based on policy, network availability, and product status.
Consumers	Business Process — buyer or regional manager action *(UC-029, UC-031)*; Read Model — replenishment suggestion log

DE-072 — PromotionalReadinessGapIdentified

Aspect	Detail
Trigger	Promotional Readiness Evaluation domain service detects insufficient ring-fenced stock before campaign deadline.
Related Aggregate	Promotional Stock Allocation
Business Meaning	Campaign or launch commitment at risk — allocated quantity insufficient for planned demand per BR-029.
Consumers	Operational Alert read model; Business Process — marketing and purchasing notification *(UC-040, UC-047)*

DE-073 — NetworkImbalanceDetected

Aspect	Detail
Trigger	Network Rebalancing Analysis domain service identifies simultaneous stockout risk at one location and excess at another for the same product.
Related Aggregate	*(Cross-cutting)*
Business Meaning	Network holds sufficient total stock but maldistributed. Transfer-before-buy opportunity exists per BR-011.
Consumers	Business Process — operations director rebalancing review *(UC-034)*; Replenishment Recommendation domain service *(DS-005)*

Domain 9: Financial & Governance

DE-074 — InventoryMovementRecorded

Aspect	Detail
Trigger	Any Stock Position quantity or status mutation completes (receipt, shipment, adjustment, transfer, reservation, hold).
Related Aggregate	Stock Position

Aspect	Detail
Business Meaning	Immutable operational fact recorded — what changed, where, how much, by whom, and why. Complete movement history per BO-19 audit trail objective.
Consumers	Inventory Movement Record read model *(UC-042)*; Audit Trail Record read model; Read Model — movement reports *(UC-037)*

DE-075 — AuditTrailRecorded

Aspect	Detail
Trigger	Material governed change completes with before/after state, user identity, and approval chain captured.
Related Aggregate	*(Cross-cutting — emitted by governed aggregates)*
Business Meaning	Governance-grade evidence of who changed what and when. Supports compliance, fraud investigation, and control testing per BP-32.
Consumers	Audit Trail Record read model *(UC-043)*; Business Process — internal auditor review *(UC-041)*

DE-076 — InventoryValuationSnapshotPrepared

Aspect	Detail
Trigger	Inventory Valuation Reconciliation domain service completes period-end valuation summary.
Related Aggregate	*(Read model — Inventory Valuation Snapshot)*
Business Meaning	Point-in-time owned inventory value prepared for finance reconciliation against general ledger per BO-18.
Consumers	Business Process — finance manager month-end close *(UC-039)*; Read Model — executive KPI dashboard *(UC-038)*

DE-077 — ValuationVarianceIdentified

Aspect	Detail
Trigger	Inventory Valuation Snapshot variance against general ledger exceeds tolerance threshold.
Related Aggregate	*(Read model)*
Business Meaning	Operational inventory records and financial ledger are misaligned. Investigation and reconciliation required before month-end close.
Consumers	Business Process — finance and inventory controller joint investigation *(UC-039)*; Read Model — reconciliation status dashboard

DE-078 — SegregationOfDutiesViolationDetected

Aspect	Detail
Trigger	Segregation of Duties Validation domain service blocks action due to conflicting user roles (e.g., counter attempting to approve own count).
Related Aggregate	User Account
Business Meaning	Proposed action violates accountability controls. Prevents fraud and undetected error per BR-019, BR-021.
Consumers	Business Process — security/administrator notification; Audit Trail Record read model; Business Process — internal auditor alert *(UC-041)*

DE-079 — UserAccountProvisioned

Aspect	Detail
Trigger	New User Account created with initial role assignments in User Account aggregate.
Related Aggregate	User Account
Business Meaning	Individual authorized to perform inventory operations within assigned role and location scope.
Consumers	Segregation of Duties Validation domain service *(DS-023)*; Business Process — onboarding notification *(UC-044)*

DE-080 — RoleAssignmentChanged

Aspect	Detail
Trigger	User role or location scope added, modified, or removed in User Account aggregate.
Related Aggregate	User Account
Business Meaning	User's operational authority has changed. SoD compliance re-evaluated for new role combination.
Consumers	Segregation of Duties Validation domain service *(DS-023)*; Audit Trail Record read model; Business Process — access change notification *(UC-045)*

DE-081 — ThreeWayMatchExceptionRaised

Aspect	Detail
Trigger	PO, receipt, and invoice lines fail to match within tolerance during Three-Way Match Evaluation.
Related Aggregate	Supplier Invoice
Business Meaning	Payment blocked pending resolution of quantity or price discrepancy. Financial control intervention required per BR-008.
Consumers	Business Process — finance and purchasing exception workflow; Read Model — match exception dashboard

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Event Flow — Core Business Lifecycles

Inbound Supply Lifecycle

```
PurchaseOrderPlaced → InboundDeliveryScheduled → GoodsReceiptVerified
    → [ReceiptDiscrepancyRecorded → ReceiptDiscrepancyResolved]
    → PutAwayCompleted → StockReceived → InventoryMovementRecorded
    → SupplierInvoiceReceived → ThreeWayMatchCompleted
```

Order Fulfillment Lifecycle

```
OrderAvailabilityConfirmed → CustomerOrderConfirmed → StockReserved
    → PickVerified → ShipmentDispatched → StockShipped → InventoryMovementRecorded
    → [PartialFulfillmentCompleted → BackorderRecorded → BackorderFulfilled]
```

Inventory Accuracy Lifecycle

```
CycleCountScheduled → CycleCountCompleted → CountVarianceIdentified
    → CountVarianceApproved → AdjustmentSubmitted → AdjustmentApproved
    → StockAdjusted → InventoryMovementRecorded → AuditTrailRecorded
```

Network Transfer Lifecycle

```
TransferRequested → TransferApproved → TransferDispatched → TransferReceived
    → StockReceived (destination) → InventoryMovementRecorded
```

Event-to-Aggregate Origin Matrix

Aggregate	Events Raised
Product Catalog	DE-001, DE-002, DE-003
Inventory Location	DE-004, DE-005, DE-006
Physical Inventory Count	DE-007, DE-008, DE-056, DE-057
Inventory Governance	DE-009, DE-010
Stock Position	DE-011 – DE-018, DE-022, DE-068 – DE-070, DE-074
Stock Policy	DE-019
Promotional Stock Allocation	DE-020, DE-021, DE-072
Purchase Requisition	DE-023 – DE-025
Purchase Order	DE-026 – DE-029
Goods Receipt	DE-030 – DE-033
Supplier Invoice	DE-034, DE-035, DE-081
Customer Order Fulfillment	DE-036 – DE-045, DE-042, DE-043
Stock Transfer	DE-046 – DE-051
Cycle Count	DE-052, DE-053
Inventory Adjustment Case	DE-054, DE-055, DE-058 – DE-060
Customer Return Case	DE-061 – DE-067
User Account	DE-078 – DE-080
Cross-cutting / Domain Services	DE-071, DE-073, DE-075 – DE-077

Event-to-Use Case Traceability

Use Case	Key Domain Events
UC-008 Receive and Verify Supplier Goods	DE-030, DE-031, DE-033
UC-010 Perform Stock Put-Away	DE-033, DE-011
UC-014 Perform Cycle Count	DE-052, DE-053, DE-054
UC-015 Conduct Full Physical Inventory Count	DE-056, DE-057, DE-007, DE-008
UC-016 Investigate and Approve Count Variances	DE-054, DE-055, DE-058, DE-059
UC-021 Confirm Order Availability	DE-036
UC-022 Reserve Inventory for Confirmed Orders	DE-037, DE-012
UC-024 Pick, Pack, and Ship Orders	DE-039, DE-040, DE-014
UC-025 Handle Partial Fulfillment and Backorders	DE-041, DE-042, DE-043
UC-028 Review Low-Stock Alerts	DE-068, DE-069, DE-071
UC-031 Request Inter-Location Transfer	DE-046, DE-047, DE-048
UC-033 Execute Inter-Location Transfer	DE-049, DE-050

Use Case	Key Domain Events
UC-035 Authorize Customer Returns	DE-061
UC-036 Receive and Disposition Returns	DE-063 – DE-067
UC-039 Reconcile Inventory Valuation	DE-076, DE-077, DE-035
UC-040 Confirm Promotional Readiness	DE-072, DE-020
UC-042 View Movement History	DE-074
UC-043 Query Audit Trail	DE-075

Excluded from Domain Events

Occurrence	Treatment	Rationale
User clicks "Search" on inventory screen	UI interaction	No business state change
Report query executed	Read operation	Queries do not change domain state
Validation failure (business rule rejected)	Rejection response	Nothing happened — state unchanged
Scheduled job timer fired	Infrastructure	Timer is not a business event; the resulting state change emits the event
Email delivery confirmed	Infrastructure	Notification delivery is not inventory domain concern

Document Approval

Role	Name	Signature	Date
Domain Expert			
DDD Architect			
Senior Business Analyst			
Solution Architect			

End of Domain Events

Document ID: SRS-DM-006 | Version 1.0 | July 5, 2026

Part V — Event Storming

Actors and External Systems

Inventory Management System — Event Storming

Document Information

Attribute	Detail
Document ID	SRS-ES-004
Document Title	Actors and External Systems — Event Storming Catalog

Attribute	Detail
Version	1.0
Status	Draft
Document Owner	Domain Expert
Source Documents	SRS-ES-001 (Domain Events), SRS-ES-002 (Domain Commands), SRS-ES-003 (Domain Policies), SRS-UC-001 (Use Case Catalog)
Date	July 6, 2026
Scope	Business participants only — humans and external systems outside the inventory domain core; excludes internal aggregate handlers

Purpose

This document identifies all **Actors** (human business roles) and **External Systems** (organizations and systems outside the Inventory Management System boundary) involved in event storming. For each participant it defines responsibilities, commands they initiate, and domain events they consume or react to in the course of business workflows.

Participant Types

Type	Definition
Human	Business role with accountability for inventory decisions and operations
External System	Organization or system outside the IMS bounded context that exchanges business facts or triggers workflows

Participant Summary

Category	IDs	Count
Human Actors	ACT-001 – ACT-017	17
External Systems	EXT-001 – EXT-006	6
Total		**23**

Legend — Events Consumed

"Events consumed" means the participant **acts upon**, **is notified of**, or **depends on** the business fact — via work queue, alert, dashboard, integration, or policy notification. Read-only inquiry events are included where the participant routinely monitors them for decisions.

Human Actors

ACT-001 — Inventory Controller

Field	Detail
Name	Inventory Controller
Type	Human
Responsibilities	Maintain product catalog, location network, stock policies, governance policies, and ABC classification. Schedule cycle counts and lead physical inventory counts. Monitor network inventory health, movement history, and shrinkage analysis. Establish opening balances at go-live.
Commands Initiated	`RegisterProduct`, `ChangeProductStatus`, `AssignABCClassification`, `ActivateInventoryLocation`, `DeactivateInventoryLocation`, `ConfigureStoragePosition`, `PublishGovernancePolicy`, `RetireReasonCode`, `UpdateStockPolicyThresholds`, `ScheduleCycleCount`, `StartPhysicalInventoryCount`, `CertifyPhysicalInventoryCount`, `EstablishOpeningBalances`

Field	Detail
Events Consumed	ProductRegistered, ProductStatusChanged, ProductABCClassificationAssigned, InventoryLocationActivated, InventoryLocationDeactivated, GovernancePolicyPublished, StockPolicyThresholdChanged, CycleCountScheduled, CycleCountCompleted, CountVarianceIdentified, PhysicalInventoryCountStarted, PhysicalInventoryCountCertified, LocationFreezeImposed, LocationFreezeReleased, LowStockThresholdBreached, OutOfStockDetected, OverstockDetected, InventoryMovementRecorded, StockAdjusted, StockWrittenOff

ACT-002 — Buyer

Field	Detail
Name	Buyer
Type	Human
Responsibilities	Create and manage purchase orders, plan inbound deliveries, submit purchase requisitions, and track inbound supply status. Respond to supplier delays and replenishment recommendations.
Commands Initiated	`SubmitPurchaseRequisition`, `PlacePurchaseOrder`, `CancelPurchaseOrder`, `ScheduleInboundDelivery`, `RecordInboundDelay`
Events Consumed	PurchaseRequisitionApproved, PurchaseRequisitionRejected, PurchaseOrderPlaced, PurchaseOrderCancelled, InboundDeliveryScheduled, InboundDeliveryDelayed, ReplenishmentRecommended, LowStockThresholdBreached, ReceiptDiscrepancyRecorded, GoodsReceiptVerified

ACT-003 — Purchasing Manager

Field	Detail
Name	Purchasing Manager
Type	Human
Responsibilities	Approve or reject purchase requisitions. Review low-stock and out-of-stock alerts. Oversee replenishment response and supplier accountability. Coordinate with operations on inbound delays.
Commands Initiated	`ApprovePurchaseRequisition`, `RejectPurchaseRequisition`
Events Consumed	PurchaseRequisitionSubmitted, PurchaseRequisitionApproved, PurchaseRequisitionRejected, LowStockThresholdBreached, OutOfStockDetected, ReplenishmentRecommended, InboundDeliveryDelayed, ReceiptDiscrepancyRecorded, PromotionalReadinessGapIdentified, ThreeWayMatchExceptionRaised

ACT-004 — Receiving Clerk

Field	Detail
Name	Receiving Clerk
Type	Human
Responsibilities	Verify inbound supplier shipments and non-PO receipts. Document receipt discrepancies. Receive and inspect customer returns at the dock. Capture lot/batch and shelf-life data at receipt.
Commands Initiated	`VerifyGoodsReceipt`, `RecordReceiptDiscrepancy`, `ReceiveCustomerReturn`, `InspectReturnedGoods`
Events Consumed	InboundDeliveryScheduled, InboundDeliveryDelayed, PurchaseOrderPlaced, ReturnAuthorizationIssued, ReturnAuthorizationExpired, GoodsReceiptVerified, ReceiptDiscrepancyRecorded, CustomerReturnReceived, TransferReceived

ACT-005 — Warehouse Staff

Field	Detail
Name	Warehouse Staff
Type	Human
Responsibilities	Perform put-away of verified stock. Execute cycle counts. Dispatch and receive inter-location transfers. Move stock to designated storage positions applying rotation rules.
Commands Initiated	`CompletePutAway`, `SubmitCycleCountResult`, `DispatchTransfer`, `ConfirmTransferReceipt`
Events Consumed	GoodsReceiptVerified, PutAwayCompleted, CycleCountScheduled, TransferApproved, TransferDispatched, InTransitStockDelayed, LocationFreezeImposed, LocationFreezeReleased

ACT-006 — Warehouse Picker / Packer

Field	Detail
Name	Warehouse Picker / Packer
Type	Human
Responsibilities	Physically pick, verify, pack, and dispatch customer orders. Apply FIFO/FEFO rotation during pick. Confirm lot/batch capture for traceable products.
Commands Initiated	`VerifyPick`, `DispatchShipment`
Events Consumed	CustomerOrderConfirmed, StockReserved, MultiLocationFulfillmentPlanCreated, PickVerified, FulfillmentExceptionRaised, LocationFreezeImposed

ACT-007 — Warehouse Manager

Field	Detail
Name	Warehouse Manager
Type	Human
Responsibilities	Resolve receipt discrepancies. Manage quarantine and hold stock. Investigate and approve count variances. Submit stock adjustments. Handle fulfillment exceptions. Plan multi-location fulfillment. Disposition returned goods. Respond to operational alerts and transfer delays.
Commands Initiated	`ResolveReceiptDiscrepancy`, `ApplyQuarantineHold`, `ReleaseQuarantineHold`, `ApproveCountVariance`, `SubmitAdjustment`, `RaiseFulfillmentException`, `CreateMultiLocationFulfillmentPlan`, `QuarantineReturnedGoods`, `WriteOffReturnedGoods`, `RestockReturnedGoods`, `RecordTransferDelay`
Events Consumed	ReceiptDiscrepancyRecorded, ReceiptDiscrepancyResolved, CountVarianceIdentified, CountVarianceApproved, AdjustmentSubmitted, AdjustmentApproved, AdjustmentRejected, FulfillmentExceptionRaised, TransferRequested, TransferApproved, TransferRejected, InTransitStockDelayed, LowStockThresholdBreached, OutOfStockDetected, QuarantineHoldApplied, QuarantineHoldReleased, ReturnInspected, PromotionalReadinessGapIdentified

ACT-008 — Operations Director

Field	Detail
Name	Operations Director
Type	Human
Responsibilities	Approve or reject inter-location transfer requests. Authorize material inventory adjustments and write-offs. Oversee network rebalancing. Review operational dashboards and inventory performance. Resolve cross-functional escalations.

Field	Detail
Commands Initiated	`ApproveTransferRequest`, `RejectTransferRequest`, `ApproveAdjustment`, `RejectAdjustment`, `AnalyzeNetworkBalance`
Events Consumed	TransferRequested, TransferApproved, TransferRejected, AdjustmentSubmitted, AdjustmentApproved, AdjustmentRejected, NetworkImbalanceDetected, ReplenishmentRecommended, ValuationVarianceIdentified, OutOfStockDetected, OverstockDetected, PhysicalInventoryCountCertified, SegregationOfDutiesViolationDetected, InventoryValuationSnapshotPrepared

ACT-009 — Regional / Store Manager

Field	Detail
Name	Regional / Store Manager
Type	Human
Responsibilities	Request inter-location stock transfers to resolve local shortages. Monitor local stock position and service levels. Coordinate with network locations for redistribution before procurement.
Commands Initiated	`RequestStockTransfer`
Events Consumed	TransferApproved, TransferRejected, TransferReceived, OutOfStockDetected, LowStockThresholdBreached, ReplenishmentRecommended, NetworkImbalanceDetected, StockReceived

ACT-010 — Sales Representative

Field	Detail
Name	Sales Representative
Type	Human
Responsibilities	Confirm order availability against ATP before customer commitment. Inquire network-wide stock visibility for sales conversations. Initiate availability checks on behalf of customers.
Commands Initiated	`ConfirmOrderAvailability`
Events Consumed	OrderAvailabilityConfirmed, OutOfStockDetected, LowStockThresholdBreached, PromotionalStockAllocated, QuarantineHoldApplied

ACT-011 — Customer Service Representative

Field	Detail
Name	Customer Service Representative
Type	Human
Responsibilities	Confirm customer orders and reserve stock. Release reservations on cancellation. Handle partial fulfillment and backorders with customer acknowledgment. Issue return authorizations. Communicate order and shipment status to customers.
Commands Initiated	`ConfirmCustomerOrder`, `CancelCustomerOrder`, `ReleaseReservation`, `CompletePartialFulfillment`, `RecordBackorder`, `FulfillBackorder`, `IssueReturnAuthorization`
Events Consumed	OrderAvailabilityConfirmed, CustomerOrderConfirmed, CustomerOrderCancelled, StockReserved, StockReservationReleased, ShipmentDispatched, PartialFulfillmentCompleted, BackorderRecorded, BackorderFulfilled, FulfillmentExceptionRaised, ReturnAuthorizationIssued, ReturnAuthorizationExpired, CustomerReturnReceived, ReturnRestocked

ACT-012 — Finance Manager

Field	Detail
Name	Finance Manager
Type	Human
Responsibilities	Record supplier invoices. Execute three-way match evaluation. Reconcile inventory valuation with general ledger. Authorize payment after match completion. Investigate valuation variances at month-end close.
Commands Initiated	`RecordSupplierInvoice`, `EvaluateThreeWayMatch`, `PrepareValuationSnapshot`, `ReconcileInventoryValuation`
Events Consumed	SupplierInvoiceReceived, ThreeWayMatchCompleted, ThreeWayMatchExceptionRaised, ReceiptDiscrepancyResolved, GoodsReceiptVerified, PurchaseOrderPlaced, InventoryValuationSnapshotPrepared, ValuationVarianceIdentified, StockWrittenOff, ConsignmentOwnershipTransferred, AdjustmentApproved

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ACT-013 — Marketing Manager

Field	Detail
Name	Marketing Manager
Type	Human
Responsibilities	Allocate and release promotional stock ring-fences for campaigns and product launches. Confirm promotional readiness before commercial commitment. Coordinate with purchasing on campaign stock gaps.
Commands Initiated	`AllocatePromotionalStock`, `ReleasePromotionalAllocation`, `EvaluatePromotionalReadiness`
Events Consumed	PromotionalStockAllocated, PromotionalAllocationReleased, PromotionalReadinessGapIdentified, StockReceived, LowStockThresholdBreached

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ACT-014 — Internal Auditor

Field	Detail
Name	Internal Auditor
Type	Human
Responsibilities	Review audit trails and inventory movement history. Test segregation-of-duties controls. Verify compliance with governance policies. Investigate control breaches and adjustment authorization chains.
Commands Initiated	None *(read and review workflows — no mutating inventory commands)*
Events Consumed	AuditTrailRecorded, InventoryMovementRecorded, SegregationOfDutiesViolationDetected, AdjustmentApproved, AdjustmentRejected, CountVarianceApproved, StockWrittenOff, RoleAssignmentChanged, GovernancePolicyPublished, PhysicalInventoryCountCertified

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ACT-015 — System Administrator

Field	Detail
Name	System Administrator
Type	Human
Responsibilities	Provision user accounts and manage role assignments with location scope. Enforce role-based access and segregation-of-duties constraints at assignment time. Support go-live data import orchestration.
Commands Initiated	`ProvisionUserAccount`, `ChangeRoleAssignment`
Events Consumed	UserAccountProvisioned, RoleAssignmentChanged, SegregationOfDutiesViolationDetected

ACT-016 — Chief Executive Officer

Field	Detail
Name	Chief Executive Officer
Type	Human
Responsibilities	Review executive-level inventory KPIs including turnover, fill rate, accuracy, and working capital impact. Monitor strategic inventory performance for steering decisions.
Commands Initiated	None *(executive review — no operational commands)*
Events Consumed	InventoryValuationSnapshotPrepared, OutOfStockDetected, OverstockDetected, NetworkImbalanceDetected, ValuationVarianceIdentified *(via KPI dashboards and executive reports)*

ACT-017 — Chief Financial Officer

Field	Detail
Name	Chief Financial Officer
Type	Human
Responsibilities	Oversee month-end inventory valuation alignment with financial statements. Review material valuation variances and three-way match control effectiveness.
Commands Initiated	None *(governance oversight — delegates to Finance Manager)*
Events Consumed	InventoryValuationSnapshotPrepared, ValuationVarianceIdentified, ThreeWayMatchCompleted, ThreeWayMatchExceptionRaised, StockWrittenOff *(via finance reporting)*

External Systems

EXT-001 — Supplier / Vendor

Field	Detail
Name	Supplier / Vendor
Type	External System
Responsibilities	Fulfill purchase orders by delivering goods to the receiving location. Communicate shipment schedules and delays. Submit invoices for received goods. Participate in discrepancy resolution and claims. Operate under consignment agreements where applicable.
Commands Initiated	None *(within IMS boundary — supplier actions occur outside the system)*
Events Consumed	PurchaseOrderPlaced, PurchaseOrderCancelled, InboundDeliveryScheduled, ReceiptDiscrepancyRecorded *(supplier notified of claims via buyer workflow)*
Business Triggers Into IMS	Physical shipment arrival → `VerifyGoodsReceipt`; Invoice submission → `RecordSupplierInvoice`; Schedule communication → `ScheduleInboundDelivery` / `RecordInboundDelay`

EXT-002 — Logistics / Carrier Partner

Field	Detail
Name	Logistics / Carrier Partner
Type	External System

Field	Detail
Responsibilities	Transport inbound supplier shipments, outbound customer orders, and inter-location transfers. Provide shipment tracking and delay notifications. Confirm pickup and delivery handoffs.
Commands Initiated	None
Events Consumed	ShipmentDispatched, TransferDispatched, InboundDeliveryScheduled
Business Triggers Into IMS	Delivery confirmation → supports `DispatchShipment` / `ConfirmTransferReceipt` completion; Delay notification → `RecordInboundDelay` / `RecordTransferDelay`

EXT-003 — Customer (B2B / B2C)

Field	Detail
Name	Customer (B2B / B2C)
Type	External System
Responsibilities	Place orders and request delivery commitments. Acknowledge backorders when partial fulfillment is offered. Initiate return requests per return policy. Receive shipment and return authorization communications.
Commands Initiated	None *(customer order capture may occur in sales channel — IMS receives availability and fulfillment commands from internal actors)*
Events Consumed	OrderAvailabilityConfirmed, ShipmentDispatched, PartialFulfillmentCompleted, BackorderRecorded, ReturnAuthorizationIssued, ReturnAuthorizationExpired
Business Triggers Into IMS	Order request → `ConfirmOrderAvailability` / `ConfirmCustomerOrder`; Cancellation → `CancelCustomerOrder`; Backorder acknowledgment → `RecordBackorder`; Return request → `IssueReturnAuthorization`; Return shipment → `ReceiveCustomerReturn`

EXT-004 — General Ledger / Finance ERP

Field	Detail
Name	General Ledger / Finance ERP
Type	External System
Responsibilities	Maintain authoritative financial inventory valuation in the general ledger. Receive inventory valuation snapshots for reconciliation. Record payment transactions after three-way match authorization.
Commands Initiated	None *(Finance Manager initiates reconciliation commands on behalf of finance alignment)*
Events Consumed	InventoryValuationSnapshotPrepared, ThreeWayMatchCompleted, StockWrittenOff, ConsignmentOwnershipTransferred, AdjustmentApproved
Business Triggers Into IMS	Ledger balance inquiry → `ReconcileInventoryValuation`; Period-end close cycle → `PrepareValuationSnapshot`

EXT-005 — Customs / Regulatory Authority

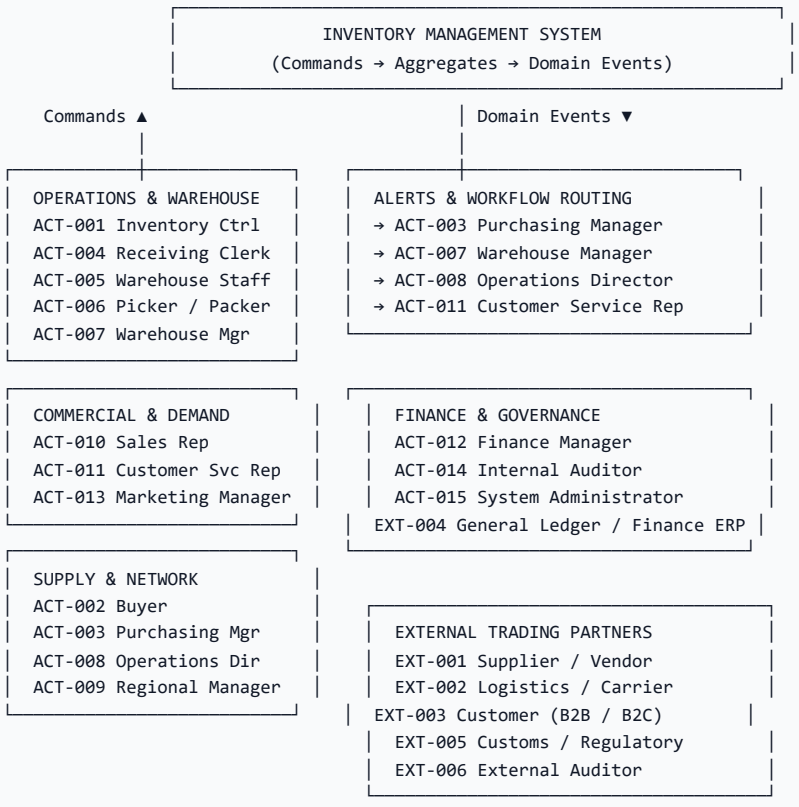
Field	Detail
Name	Customs / Regulatory Authority
Type	External System
Responsibilities	Clear bonded and imported inventory for domestic commerce. Enforce regulatory holds on recalled or restricted goods. Authorize release of quarantined stock under regulatory investigation.
Commands Initiated	None
Events Consumed	QuarantineHoldApplied, GoodsReceiptVerified *(bonded receipt)*

Field	Detail
Business Triggers Into IMS	Customs clearance granted → `ReleaseQuarantineHold`; Regulatory hold imposed → `ApplyQuarantineHold`

EXT-006 — External Auditor

Field	Detail
Name	External Auditor
Type	External System
Responsibilities	Independently review inventory controls, audit trails, and physical count certification during statutory or regulatory audit engagements. Verify three-way match and write-off authorization evidence.
Commands Initiated	None
Events Consumed	AuditTrailRecorded, PhysicalInventoryCountCertified, ThreeWayMatchCompleted, AdjustmentApproved, StockWrittenOff, SegregationOfDutiesViolationDetected, InventoryValuationSnapshotPrepared
Business Triggers Into IMS	None *(read-only access via auditor review workflows)*

Participant Interaction Map



Actor-to-Command Matrix

Actor	Commands Initiated	Count
Inventory Controller	RegisterProduct, ChangeProductStatus, AssignABCClassification, ActivateInventoryLocation, DeactivateInventoryLocation, ConfigureStoragePosition, PublishGovernancePolicy, RetireReasonCode, UpdateStockPolicyThresholds, ScheduleCycleCount, StartPhysicalInventoryCount, CertifyPhysicalInventoryCount, EstablishOpeningBalances	13
Buyer	SubmitPurchaseRequisition, PlacePurchaseOrder, CancelPurchaseOrder, ScheduleInboundDelivery, RecordInboundDelay	5
Purchasing Manager	ApprovePurchaseRequisition, RejectPurchaseRequisition	2
Receiving Clerk	VerifyGoodsReceipt, RecordReceiptDiscrepancy, ReceiveCustomerReturn, InspectReturnedGoods	4
Warehouse Staff	CompletePutAway, SubmitCycleCountResult, DispatchTransfer, ConfirmTransferReceipt	4
Warehouse Picker / Packer	VerifyPick, DispatchShipment	2
Warehouse Manager	ResolveReceiptDiscrepancy, ApplyQuarantineHold, ReleaseQuarantineHold, ApproveCountVariance, SubmitAdjustment, RaiseFulfillmentException, CreateMultiLocationFulfillmentPlan, QuarantineReturnedGoods, WriteOffReturnedGoods, RestockReturnedGoods, RecordTransferDelay	11
Operations Director	ApproveTransferRequest, RejectTransferRequest, ApproveAdjustment, RejectAdjustment, AnalyzeNetworkBalance	5
Regional / Store Manager	RequestStockTransfer	1
Sales Representative	ConfirmOrderAvailability	1
Customer Service Representative	ConfirmCustomerOrder, CancelCustomerOrder, ReleaseReservation, CompletePartialFulfillment, RecordBackorder, FulfillBackorder, IssueReturnAuthorization	7
Finance Manager	RecordSupplierInvoice, EvaluateThreeWayMatch, PrepareValuationSnapshot, ReconcileInventoryValuation	4
Marketing Manager	AllocatePromotionalStock, ReleasePromotionalAllocation, EvaluatePromotionalReadiness	3
System Administrator	ProvisionUserAccount, ChangeRoleAssignment	2
Internal Auditor, CEO, CFO	None	0

Automated domain service commands (`EvaluateStockStatus`, `RecommendReplenishmentAction`, `ReceiveStock`, `ShipStock`, etc.) are executed by IMS policies — not initiated by human actors or external systems directly.

Actor-to-Event Consumption Matrix

Actor	Primary Events Consumed (Business Reaction)
Inventory Controller	Stock policy and catalog lifecycle events; count and variance events; movement and adjustment events for investigation
Buyer	Requisition approval, PO lifecycle, inbound schedule/delay, replenishment recommendations
Purchasing Manager	Low-stock, out-of-stock, requisition submission, inbound delays, promotional readiness gaps
Receiving Clerk	Inbound delivery schedule, return authorization, transfer receipt
Warehouse Staff	Verified receipt staging, cycle count assignment, transfer approval/dispatch
Warehouse Picker / Packer	Order confirmation, reservation, pick authorization, freeze status
Warehouse Manager	Discrepancies, variances, exceptions, transfers, quarantine, operational alerts
Operations Director	Transfer requests, adjustment submissions, network imbalance, valuation variance
Regional / Store Manager	Transfer decision, local stockout, replenishment recommendations

Actor	Primary Events Consumed (Business Reaction)
Sales Representative	Availability confirmation result, stockout/low-stock for sales conversations
Customer Service Representative	Order lifecycle, shipment, backorder, return authorization and receipt
Finance Manager	Invoice, match outcome, valuation snapshot, write-off, consignment transfer
Marketing Manager	Promotional allocation, readiness gaps
Internal Auditor	Audit trail, movement history, SoD violations, governance changes
System Administrator	Account provisioning, role changes, SoD violations
CEO / CFO	Valuation, stockout, imbalance, variance *(executive reporting)*

External System Boundary Interactions

External System	Direction	Business Interaction
Supplier / Vendor	Inbound to IMS	Delivers goods → triggers goods receipt verification
Supplier / Vendor	Outbound from IMS	Notified of PO placement, cancellation, discrepancy claims
Logistics / Carrier	Bidirectional	Picks up shipments/transfers; reports delays
Customer	Inbound to IMS	Orders, cancellations, return requests, backorder acknowledgments
Customer	Outbound from IMS	Availability answers, shipment notices, RMA instructions
General Ledger / ERP	Inbound to IMS	Ledger balance for reconciliation
General Ledger / ERP	Outbound from IMS	Valuation snapshots, match completion, write-off facts
Customs / Regulatory	Inbound to IMS	Clearance decisions → quarantine release or hold
External Auditor	Inbound to IMS	Read-only review of audit trail and control evidence

Excluded Participants

Participant	Treatment	Rationale
Aggregate reaction handlers	Internal IMS mechanism	Stock Position handlers executing `ReceiveStock` are not business participants
Domain services (policy engine)	Internal IMS mechanism	Automated policies are system behavior, not external actors
Email / SMS gateway	Infrastructure	Notification delivery channel, not a business participant
Report rendering engine	Infrastructure	Presentation layer, not a business actor
Sales order capture system	Out of scope	May feed order intent; IMS receives commands from Sales/CS reps

Document Approval

Role	Name	Signature	Date
Domain Expert			
DDD Architect			
Senior Business Analyst			
Solution Architect			

End of Actors and External Systems

Domain Commands

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-ES-002
Document Title	Domain Commands — Event Storming Catalog
Version	1.0
Status	Draft
Document Owner	Domain Expert
Source Documents	SRS-ES-001 (Domain Events), SRS-DM-003 (Domain Aggregates), SRS-UC-001 (Use Case Catalog), SRS-UC-002 (Main Flows)
Date	July 6, 2026
Scope	Business commands that mutate domain state and initiate domain events — no queries, UI actions, or infrastructure signals

Purpose

This document identifies **all domain commands** that initiate domain events in the Inventory Management System. A command expresses a **business intention** — what the organization wants to happen. Commands are imperative, issued by a business actor (human or policy-driven evaluator), and target exactly one aggregate per transaction.

Commands that fail validation emit **no domain event** — the aggregate state remains unchanged. When segregation-of-duties rules block an action, `SegregationOfDutiesViolationDetected` (DE-078) is raised instead of the intended event.

Command Naming Convention

Commands use **PascalCase imperative verbs**: `RegisterProduct`, `ReserveStock`, `ApproveTransferRequest`.

Command Categories

Category	Description
Actor Command	Issued directly by a business role through a use case
Policy Command	Issued by a domain service when business rules or thresholds are evaluated
Reaction Command	Issued by an aggregate handler consuming a domain event from another aggregate

Command Catalog Summary

Domain Area	Command IDs	Count
Catalog & Network	CMD-001 – CMD-010	10
Inventory Balance	CMD-011 – CMD-020	10
Procurement & Inbound	CMD-021 – CMD-033	13
Order Fulfillment	CMD-034 – CMD-044	11
Network Transfer	CMD-045 – CMD-050	6
Inventory Control & Accuracy	CMD-051 – CMD-060	10

Domain Area	Command IDs	Count
Returns Management	CMD-061 – CMD-067	7
Commercial & Monitoring	CMD-068 – CMD-072	5
Financial & Governance	CMD-073 – CMD-078	6
Total		**78**

Domain 1: Catalog & Network

CMD-001 — RegisterProduct

Field	Detail
Command Name	RegisterProduct
Actor	Inventory Controller
Target Aggregate	Product Catalog
Business Purpose	Introduce a new SKU into the trusted product catalog so all inventory operations reference a single authoritative product identity.
Expected Domain Event(s)	ProductRegistered (DE-001), AuditTrailRecorded (DE-075)
Use Case	UC-001, UC-046

CMD-002 — ChangeProductStatus

Field	Detail
Command Name	ChangeProductStatus
Actor	Inventory Controller
Target Aggregate	Product Catalog
Business Purpose	Change commercial availability of a SKU (active, inactive, discontinued) to control eligibility for procurement, replenishment, and customer commitments.
Expected Domain Event(s)	ProductStatusChanged (DE-002), AuditTrailRecorded (DE-075)
Use Case	UC-001

CMD-003 — AssignABCClassification

Field	Detail
Command Name	AssignABCClassification
Actor	Inventory Controller
Target Aggregate	Product Catalog
Business Purpose	Set operational priority for a SKU so count frequency, monitoring intensity, and alert priority reflect business importance.
Expected Domain Event(s)	ProductABCClassificationAssigned (DE-003), AuditTrailRecorded (DE-075)
Use Case	UC-004

CMD-004 — ActivateInventoryLocation

Field	Detail
Command Name	ActivateInventoryLocation
Actor	Inventory Controller
Target Aggregate	Inventory Location
Business Purpose	Authorize a location for inventory operations after go-live readiness, enabling receipts, stock holding, fulfillment, and transfers.
Expected Domain Event(s)	InventoryLocationActivated (DE-004), AuditTrailRecorded (DE-075)
Use Case	UC-002, UC-046

CMD-005 — DeactivateInventoryLocation

Field	Detail
Command Name	DeactivateInventoryLocation
Actor	Inventory Controller
Target Aggregate	Inventory Location
Business Purpose	Close a location to new inventory activity after stock disposition is complete or authorized.
Expected Domain Event(s)	InventoryLocationDeactivated (DE-005), AuditTrailRecorded (DE-075)
Use Case	UC-002

CMD-006 — ConfigureStoragePosition

Field	Detail
Command Name	ConfigureStoragePosition
Actor	Inventory Controller
Target Aggregate	Inventory Location
Business Purpose	Define or update a findable physical address (zone, aisle, bin) within a location for put-away and pick operations.
Expected Domain Event(s)	StoragePositionConfigured (DE-006)
Use Case	UC-002

CMD-007 — PublishGovernancePolicy

Field	Detail
Command Name	PublishGovernancePolicy
Actor	Inventory Controller
Target Aggregate	Inventory Governance
Business Purpose	Make a new or updated governance policy version effective, setting approval thresholds, tolerances, and exception rules for all subsequent transactions.
Expected Domain Event(s)	GovernancePolicyPublished (DE-009), AuditTrailRecorded (DE-075)
Use Case	UC-005

CMD-008 — RetireReasonCode

Field	Detail
Command Name	RetireReasonCode
Actor	Inventory Controller
Target Aggregate	Inventory Governance
Business Purpose	Remove an exception reason from active use while preserving historical attributions on past transactions.
Expected Domain Event(s)	ReasonCodeRetired (DE-010), AuditTrailRecorded (DE-075)
Use Case	UC-005

CMD-009 — StartPhysicalInventoryCount

Field	Detail
Command Name	StartPhysicalInventoryCount
Actor	Inventory Controller
Target Aggregate	Physical Inventory Count
Business Purpose	Initiate a wall-to-wall count at a location and impose operational freeze to ensure an accurate, stable inventory snapshot.
Expected Domain Event(s)	PhysicalInventoryCountStarted (DE-056), LocationFreezeImposed (DE-007)
Use Case	UC-015

CMD-010 — CertifyPhysicalInventoryCount

Field	Detail
Command Name	CertifyPhysicalInventoryCount
Actor	Inventory Controller
Target Aggregate	Physical Inventory Count
Business Purpose	Declare the wall-to-wall count complete and authoritative, route variances for investigation, and release the location freeze.
Expected Domain Event(s)	PhysicalInventoryCountCertified (DE-057), LocationFreezeReleased (DE-008), CountVarianceIdentified (DE-054) *per variance line*
Use Case	UC-015

Domain 2: Inventory Balance

CMD-011 — UpdateStockPolicyThresholds

Field	Detail
Command Name	UpdateStockPolicyThresholds
Actor	Inventory Controller
Target Aggregate	Stock Policy
Business Purpose	Recalibrate reorder level, minimum, safety stock, and maximum thresholds to align replenishment and alerting with current service-level targets.

Field	Detail
Expected Domain Event(s)	StockPolicyThresholdChanged (DE-019), AuditTrailRecorded (DE-075)
Use Case	UC-003

CMD-012 — ReserveStock

Field	Detail
Command Name	ReserveStock
Actor	Customer Service Representative *(or system on order confirmation)*
Target Aggregate	Stock Position
Business Purpose	Commit on-hand stock to a confirmed customer order and remove reserved quantity from ATP to prevent double-allocation.
Expected Domain Event(s)	StockReserved (DE-012), InventoryMovementRecorded (DE-074)
Use Case	UC-022

CMD-013 — ReleaseReservation

Field	Detail
Command Name	ReleaseReservation
Actor	Customer Service Representative
Target Aggregate	Stock Position
Business Purpose	Return previously committed stock to the ATP pool when an order or line is cancelled or amended.
Expected Domain Event(s)	StockReservationReleased (DE-013), InventoryMovementRecorded (DE-074), LowStockThresholdBreached (DE-068) *if ATP crosses reorder level*
Use Case	UC-023

CMD-014 — ReceiveStock

Field	Detail
Command Name	ReceiveStock
Actor	Stock Position *(reaction handler — triggered by inbound completion)*
Target Aggregate	Stock Position
Business Purpose	Record verified inbound quantity as on-hand inventory after put-away, transfer receipt, return restock, or opening balance establishment.
Expected Domain Event(s)	StockReceived (DE-011), InventoryMovementRecorded (DE-074), EvaluateStockStatus → LowStockThresholdBreached / OutOfStockDetected / OverstockDetected (DE-068 – DE-070) *as applicable*
Triggered By	CompletePutAway, ConfirmTransferReceipt, RestockReturnedGoods, EstablishOpeningBalances
Use Case	UC-010, UC-033, UC-036, UC-046

CMD-015 — ShipStock

Field	Detail
Command Name	ShipStock

Field	Detail
Actor	Stock Position *(reaction handler — triggered by shipment dispatch)*
Target Aggregate	Stock Position
Business Purpose	Deduct on-hand and reserved quantities when verified goods leave the location for customer delivery.
Expected Domain Event(s)	StockShipped (DE-014), InventoryMovementRecorded (DE-074), ConsignmentOwnershipTransferred (DE-022) *if consignment*, EvaluateStockStatus → threshold events (DE-068 – DE-070) *as applicable*
Triggered By	DispatchShipment
Use Case	UC-024

CMD-016 — ApplyApprovedAdjustment

Field	Detail
Command Name	ApplyApprovedAdjustment
Actor	Stock Position *(reaction handler — triggered by adjustment approval)*
Target Aggregate	Stock Position
Business Purpose	Correct on-hand quantity to align recorded inventory with physical reality after investigation and authorization.
Expected Domain Event(s)	StockAdjusted (DE-015), InventoryMovementRecorded (DE-074), AuditTrailRecorded (DE-075)
Triggered By	ApproveAdjustment
Use Case	UC-017

CMD-017 — ApplyApprovedWriteOff

Field	Detail
Command Name	ApplyApprovedWriteOff
Actor	Stock Position *(reaction handler — triggered by write-off authorization)*
Target Aggregate	Stock Position
Business Purpose	Permanently remove damaged, obsolete, expired, or lost inventory from on-hand and valuation records.
Expected Domain Event(s)	StockWrittenOff (DE-016), InventoryMovementRecorded (DE-074), AuditTrailRecorded (DE-075)
Triggered By	ApproveAdjustment, WriteOffReturnedGoods
Use Case	UC-018, UC-036

CMD-018 — ApplyQuarantineHold

Field	Detail
Command Name	ApplyQuarantineHold
Actor	Warehouse Manager
Target Aggregate	Stock Position
Business Purpose	Exclude quantity from sellable ATP due to quality investigation, regulatory restriction, recall, or customs hold.
Expected Domain Event(s)	QuarantineHoldApplied (DE-017), InventoryMovementRecorded (DE-074)
Use Case	UC-019

CMD-019 — ReleaseQuarantineHold

Field	Detail
Command Name	ReleaseQuarantineHold
Actor	Warehouse Manager
Target Aggregate	Stock Position
Business Purpose	Clear a hold and restore quantity to sellable ATP after quality approval, customs clearance, or disposition decision.
Expected Domain Event(s)	QuarantineHoldReleased (DE-018), InventoryMovementRecorded (DE-074)
Use Case	UC-019

CMD-020 — TransferConsignmentOwnership

Field	Detail
Command Name	TransferConsignmentOwnership
Actor	Domain Service — Consignment Ownership Evaluation *(on consumption or sale)*
Target Aggregate	Stock Position
Business Purpose	Transfer ownership from supplier consignment to organization-owned inventory upon defined consumption or sale per supplier agreement.
Expected Domain Event(s)	ConsignmentOwnershipTransferred (DE-022), InventoryMovementRecorded (DE-074)
Use Case	UC-039

Domain 3: Procurement & Inbound

CMD-021 — SubmitPurchaseRequisition

Field	Detail
Command Name	SubmitPurchaseRequisition
Actor	Buyer
Target Aggregate	Purchase Requisition
Business Purpose	Record an internal procurement need for specific products at a target location and route for approval before commercial commitment.
Expected Domain Event(s)	PurchaseRequisitionSubmitted (DE-023)
Use Case	UC-029

CMD-022 — ApprovePurchaseRequisition

Field	Detail
Command Name	ApprovePurchaseRequisition
Actor	Purchasing Manager
Target Aggregate	Purchase Requisition
Business Purpose	Grant internal authorization to proceed with supplier procurement after network availability and policy validation.
Expected Domain Event(s)	PurchaseRequisitionApproved (DE-024)

Field	Detail
Use Case	UC-029

CMD-023 — RejectPurchaseRequisition

Field	Detail
Command Name	RejectPurchaseRequisition
Actor	Purchasing Manager
Target Aggregate	Purchase Requisition
Business Purpose	Deny a procurement request when network transfer is preferable or policy violations exist, directing requestor to alternatives.
Expected Domain Event(s)	PurchaseRequisitionRejected (DE-025)
Use Case	UC-029

CMD-024 — PlacePurchaseOrder

Field	Detail
Command Name	PlacePurchaseOrder
Actor	Buyer
Target Aggregate	Purchase Order
Business Purpose	Record an authorized commercial commitment to a supplier defining products, quantities, delivery terms, and receiving location.
Expected Domain Event(s)	PurchaseOrderPlaced (DE-026), AuditTrailRecorded (DE-075)
Use Case	UC-006

CMD-025 — CancelPurchaseOrder

Field	Detail
Command Name	CancelPurchaseOrder
Actor	Buyer
Target Aggregate	Purchase Order
Business Purpose	Withdraw a commercial commitment before full receipt when supply is no longer needed or terms cannot be met.
Expected Domain Event(s)	PurchaseOrderCancelled (DE-027), AuditTrailRecorded (DE-075)
Use Case	UC-006

CMD-026 — ScheduleInboundDelivery

Field	Detail
Command Name	ScheduleInboundDelivery
Actor	Buyer
Target Aggregate	Purchase Order
Business Purpose	Plan expected supplier arrival date and quantity so receiving operations and replenishment planners can prepare.

Field	Detail
Expected Domain Event(s)	InboundDeliveryScheduled (DE-028)
Use Case	UC-007

CMD-027 — RecordInboundDelay

Field	Detail
Command Name	RecordInboundDelay
Actor	Buyer
Target Aggregate	Purchase Order
Business Purpose	Document that anticipated inbound supply will arrive later than planned, enabling replenishment and fulfillment risk response.
Expected Domain Event(s)	InboundDeliveryDelayed (DE-029)
Use Case	UC-030, UC-047

CMD-028 — VerifyGoodsReceipt

Field	Detail
Command Name	VerifyGoodsReceipt
Actor	Receiving Clerk
Target Aggregate	Goods Receipt
Business Purpose	Confirm quantity, product identity, and condition of inbound goods against purchase order or authorized receipt before acceptance.
Expected Domain Event(s)	GoodsReceiptVerified (DE-030)
Use Case	UC-008, UC-011

CMD-029 — RecordReceiptDiscrepancy

Field	Detail
Command Name	RecordReceiptDiscrepancy
Actor	Receiving Clerk
Target Aggregate	Goods Receipt
Business Purpose	Document variance between expected and received goods with evidence to preserve supplier accountability.
Expected Domain Event(s)	ReceiptDiscrepancyRecorded (DE-031)
Use Case	UC-009

CMD-030 — ResolveReceiptDiscrepancy

Field	Detail
Command Name	ResolveReceiptDiscrepancy
Actor	Warehouse Manager
Target Aggregate	Goods Receipt

Field	Detail
Business Purpose	Close a receipt exception with documented outcome so accepted quantities proceed and rejected quantities are excluded from inventory.
Expected Domain Event(s)	ReceiptDiscrepancyResolved (DE-032), AuditTrailRecorded (DE-075)
Use Case	UC-009

CMD-031 — CompletePutAway

Field	Detail
Command Name	CompletePutAway
Actor	Warehouse Staff
Target Aggregate	Goods Receipt
Business Purpose	Confirm received goods are placed in designated storage position, completing the inbound workflow and enabling stock availability.
Expected Domain Event(s)	PutAwayCompleted (DE-033) → ReceiveStock → StockReceived (DE-011)
Use Case	UC-010

CMD-032 — RecordSupplierInvoice

Field	Detail
Command Name	RecordSupplierInvoice
Actor	Finance Manager
Target Aggregate	Supplier Invoice
Business Purpose	Register supplier billing document and initiate financial control evaluation against purchase order and goods receipt.
Expected Domain Event(s)	SupplierInvoiceReceived (DE-034)
Use Case	UC-039

CMD-033 — EvaluateThreeWayMatch

Field	Detail
Command Name	EvaluateThreeWayMatch
Actor	Finance Manager *(or domain service on invoice receipt)*
Target Aggregate	Supplier Invoice
Business Purpose	Compare PO, goods receipt, and invoice lines within tolerance to authorize or block payment.
Expected Domain Event(s)	ThreeWayMatchCompleted (DE-035) *on success* OR ThreeWayMatchExceptionRaised (DE-081) *on failure*
Use Case	UC-039

Domain 4: Order Fulfillment

CMD-034 — ConfirmOrderAvailability

Field	Detail
Command Name	ConfirmOrderAvailability
Actor	Sales Representative
Target Aggregate	Customer Order Fulfillment
Business Purpose	Validate that ATP can support the order (fully or partially) before the business makes a commercial commitment to the customer.
Expected Domain Event(s)	OrderAvailabilityConfirmed (DE-036)
Use Case	UC-013, UC-021

CMD-035 — ConfirmCustomerOrder

Field	Detail
Command Name	ConfirmCustomerOrder
Actor	Customer Service Representative
Target Aggregate	Customer Order Fulfillment
Business Purpose	Establish commercial commitment to fulfill the order and initiate immediate stock reservation.
Expected Domain Event(s)	CustomerOrderConfirmed (DE-037) → ReserveStock → StockReserved (DE-012)
Use Case	UC-022

CMD-036 — CancelCustomerOrder

Field	Detail
Command Name	CancelCustomerOrder
Actor	Customer Service Representative
Target Aggregate	Customer Order Fulfillment
Business Purpose	Withdraw commercial commitment and release reserved stock back to ATP when order or line is no longer required.
Expected Domain Event(s)	CustomerOrderCancelled (DE-038) → ReleaseReservation → StockReservationReleased (DE-013)
Use Case	UC-023

CMD-037 — CreateMultiLocationFulfillmentPlan

Field	Detail
Command Name	CreateMultiLocationFulfillmentPlan
Actor	Warehouse Manager
Target Aggregate	Customer Order Fulfillment
Business Purpose	Assign order lines to multiple ship-from locations when no single location holds sufficient ATP for the full order.
Expected Domain Event(s)	MultiLocationFulfillmentPlanCreated (DE-045) → ReserveStock *per location* → StockReserved (DE-012)
Use Case	UC-048

CMD-038 — VerifyPick

Field	Detail
Command Name	VerifyPick
Actor	Warehouse Picker / Packer
Target Aggregate	Customer Order Fulfillment
Business Purpose	Confirm physical pick matches order requirements and authorize shipment, capturing lot/batch for traceable products.
Expected Domain Event(s)	PickVerified (DE-039), InventoryMovementRecorded (DE-074)
Use Case	UC-024

CMD-039 — DispatchShipment

Field	Detail
Command Name	DispatchShipment
Actor	Warehouse Picker / Packer
Target Aggregate	Customer Order Fulfillment
Business Purpose	Record handoff to carrier and trigger inventory deduction for verified picked goods.
Expected Domain Event(s)	ShipmentDispatched (DE-040) → ShipStock → StockShipped (DE-014), InventoryMovementRecorded (DE-074)
Use Case	UC-024

CMD-040 — CompletePartialFulfillment

Field	Detail
Command Name	CompletePartialFulfillment
Actor	Customer Service Representative
Target Aggregate	Customer Order Fulfillment
Business Purpose	Ship available quantity now and route unfulfilled remainder to backorder workflow with customer acknowledgment.
Expected Domain Event(s)	PartialFulfillmentCompleted (DE-041) → ShipmentDispatched (DE-040) *for fulfilled portion*
Use Case	UC-025

CMD-041 — RecordBackorder

Field	Detail
Command Name	RecordBackorder
Actor	Customer Service Representative
Target Aggregate	Customer Order Fulfillment
Business Purpose	Record mutual commercial commitment for unfulfilled quantity after explicit customer acknowledgment to wait.
Expected Domain Event(s)	BackorderRecorded (DE-042)
Use Case	UC-025

CMD-042 — FulfillBackorder

Field	Detail
Command Name	FulfillBackorder
Actor	Customer Service Representative
Target Aggregate	Customer Order Fulfillment
Business Purpose	Execute a previously backordered quantity when stock becomes available, reserving and fulfilling the delayed commitment.
Expected Domain Event(s)	BackorderFulfilled (DE-043) → ReserveStock → StockReserved (DE-012) → DispatchShipment (DE-040)
Use Case	UC-025

CMD-043 — RaiseFulfillmentException

Field	Detail
Command Name	RaiseFulfillmentException
Actor	Warehouse Manager
Target Aggregate	Customer Order Fulfillment
Business Purpose	Halt fulfillment when pick verification fails or shelf quantity mismatches system record, requiring manager resolution.
Expected Domain Event(s)	FulfillmentExceptionRaised (DE-044)
Use Case	UC-026

CMD-044 — DispatchTransferStock

Field	Detail
Command Name	DispatchTransferStock
Actor	Stock Position *(reaction handler — triggered by transfer dispatch)*
Target Aggregate	Stock Position
Business Purpose	Reduce source location on-hand when stock physically leaves for inter-location transfer.
Expected Domain Event(s)	InventoryMovementRecorded (DE-074), EvaluateStockStatus → threshold events (DE-068 – DE-070) *as applicable*
Triggered By	DispatchTransfer
Use Case	UC-033

Domain 5: Network Transfer

CMD-045 — RequestStockTransfer

Field	Detail
Command Name	RequestStockTransfer
Actor	Regional / Store Manager
Target Aggregate	Stock Transfer
Business Purpose	Request stock movement from another location to resolve local shortage or rebalance network before new procurement.

Field	Detail
Expected Domain Event(s)	TransferRequested (DE-046)
Use Case	UC-031

CMD-046 — ApproveTransferRequest

Field	Detail
Command Name	ApproveTransferRequest
Actor	Operations Director
Target Aggregate	Stock Transfer
Business Purpose	Authorize network redistribution after confirming source location can spare quantity while maintaining its service level.
Expected Domain Event(s)	TransferApproved (DE-047), AuditTrailRecorded (DE-075)
Use Case	UC-032

CMD-047 — RejectTransferRequest

Field	Detail
Command Name	RejectTransferRequest
Actor	Operations Director
Target Aggregate	Stock Transfer
Business Purpose	Deny redistribution when source cannot spare quantity or business justification is insufficient.
Expected Domain Event(s)	TransferRejected (DE-048)
Use Case	UC-032

CMD-048 — DispatchTransfer

Field	Detail
Command Name	DispatchTransfer
Actor	Warehouse Staff
Target Aggregate	Stock Transfer
Business Purpose	Record physical dispatch of stock from source location and create in-transit state excluded from both locations' ATP.
Expected Domain Event(s)	TransferDispatched (DE-049) → DispatchTransferStock → InventoryMovementRecorded (DE-074)
Use Case	UC-033

CMD-049 — ConfirmTransferReceipt

Field	Detail
Command Name	ConfirmTransferReceipt
Actor	Warehouse Staff
Target Aggregate	Stock Transfer
Business Purpose	Confirm destination receipt of transferred stock, closing in-transit state and increasing destination on-hand.
Expected Domain Event(s)	TransferReceived (DE-050) → ReceiveStock → StockReceived (DE-011)

Field	Detail
Use Case	UC-033

CMD-050 — RecordTransferDelay

Field	Detail
Command Name	RecordTransferDelay
Actor	Warehouse Manager
Target Aggregate	Stock Transfer
Business Purpose	Document delayed in-transit stock so destination fulfillment and replenishment plans can be adjusted.
Expected Domain Event(s)	InTransitStockDelayed (DE-051)
Use Case	UC-030, UC-047

Domain 6: Inventory Control & Accuracy

CMD-051 — ScheduleCycleCount

Field	Detail
Command Name	ScheduleCycleCount
Actor	Inventory Controller
Target Aggregate	Cycle Count
Business Purpose	Assign partial physical verification work for a product at a location, prioritizing Class A SKUs at highest frequency.
Expected Domain Event(s)	CycleCountScheduled (DE-052)
Use Case	UC-014

CMD-052 — SubmitCycleCountResult

Field	Detail
Command Name	SubmitCycleCountResult
Actor	Warehouse Staff
Target Aggregate	Cycle Count
Business Purpose	Record physical count under blind count protocol and calculate variance against system quantity at submission.
Expected Domain Event(s)	CycleCountCompleted (DE-053) → IdentifyCountVariance → CountVarianceIdentified (DE-054) *if variance exists*
Use Case	UC-014

CMD-053 — ApproveCountVariance

Field	Detail
Command Name	ApproveCountVariance
Actor	Warehouse Manager
Target Aggregate	Inventory Adjustment Case
Business Purpose	Authorize count variance adjustment after independent investigation, enforcing segregation of duties from counter.

Field	Detail
Expected Domain Event(s)	CountVarianceApproved (DE-055), SegregationOfDutiesViolationDetected (DE-078) *if approver is counter*
Use Case	UC-016

CMD-054 — SubmitAdjustment

Field	Detail
Command Name	SubmitAdjustment
Actor	Warehouse Manager
Target Aggregate	Inventory Adjustment Case
Business Purpose	Propose inventory quantity correction with mandatory reason code, pending governance evaluation and approval.
Expected Domain Event(s)	AdjustmentSubmitted (DE-058)
Use Case	UC-017, UC-018

CMD-055 — ApproveAdjustment

Field	Detail
Command Name	ApproveAdjustment
Actor	Operations Director
Target Aggregate	Inventory Adjustment Case
Business Purpose	Authorize inventory correction or write-off after threshold validation and independent approval chain satisfaction.
Expected Domain Event(s)	AdjustmentApproved (DE-059) → ApplyApprovedAdjustment → StockAdjusted (DE-015) OR ApplyApprovedWriteOff → StockWrittenOff (DE-016); SegregationOfDutiesViolationDetected (DE-078) *if SoD violated*; AuditTrailRecorded (DE-075)
Use Case	UC-017, UC-018

CMD-056 — RejectAdjustment

Field	Detail
Command Name	RejectAdjustment
Actor	Operations Director
Target Aggregate	Inventory Adjustment Case
Business Purpose	Deny proposed inventory change due to insufficient evidence, policy threshold breach, or segregation-of-duties violation.
Expected Domain Event(s)	AdjustmentRejected (DE-060)
Use Case	UC-017, UC-018

CMD-057 — IdentifyCountVariance

Field	Detail
Command Name	IdentifyCountVariance
Actor	Inventory Adjustment Case *(reaction handler — triggered by count submission)*
Target Aggregate	Inventory Adjustment Case

Field	Detail
Business Purpose	Open an investigation case when physical count differs from system quantity, blocking direct stock mutation.
Expected Domain Event(s)	CountVarianceIdentified (DE-054)
Triggered By	SubmitCycleCountResult, CertifyPhysicalInventoryCount
Use Case	UC-016

CMD-058 — EstablishOpeningBalances

Field	Detail
Command Name	EstablishOpeningBalances
Actor	Inventory Controller
Target Aggregate	Stock Position
Business Purpose	Load verified opening on-hand quantities at location go-live to establish the authoritative inventory baseline.
Expected Domain Event(s)	StockReceived (DE-011) *per product line*, InventoryMovementRecorded (DE-074), AuditTrailRecorded (DE-075)
Use Case	UC-046

CMD-059 — ApplyQuarantineFromReturn

Field	Detail
Command Name	ApplyQuarantineFromReturn
Actor	Stock Position *(reaction handler — triggered by return quarantine disposition)*
Target Aggregate	Stock Position
Business Purpose	Place returned goods under quarantine hold pending further quality evaluation.
Expected Domain Event(s)	QuarantineHoldApplied (DE-017), InventoryMovementRecorded (DE-074)
Triggered By	QuarantineReturnedGoods
Use Case	UC-036

CMD-060 — RecordCountVarianceFromPhysicalCount

Field	Detail
Command Name	RecordCountVarianceFromPhysicalCount
Actor	Physical Inventory Count *(during certification)*
Target Aggregate	Inventory Adjustment Case
Business Purpose	Route each wall-to-wall count line variance to investigation without mutating stock during the count event.
Expected Domain Event(s)	CountVarianceIdentified (DE-054) *per variance line*
Triggered By	CertifyPhysicalInventoryCount
Use Case	UC-015, UC-016

Domain 7: Returns Management

CMD-061 — IssueReturnAuthorization

Field	Detail
Command Name	IssueReturnAuthorization
Actor	Customer Service Representative
Target Aggregate	Customer Return Case
Business Purpose	Authorize customer return linked to original order so warehouse can expect and accept inbound return goods.
Expected Domain Event(s)	ReturnAuthorizationIssued (DE-061)
Use Case	UC-035

CMD-062 — ExpireReturnAuthorization

Field	Detail
Command Name	ExpireReturnAuthorization
Actor	Domain Policy *(return window elapsed)*
Target Aggregate	Customer Return Case
Business Purpose	Close return window when authorized period passes without receipt, preventing unauthorized warehouse acceptance.
Expected Domain Event(s)	ReturnAuthorizationExpired (DE-062)
Use Case	UC-035

CMD-063 — ReceiveCustomerReturn

Field	Detail
Command Name	ReceiveCustomerReturn
Actor	Receiving Clerk
Target Aggregate	Customer Return Case
Business Purpose	Verify and accept authorized return goods at the warehouse, initiating inspection and disposition workflow.
Expected Domain Event(s)	CustomerReturnReceived (DE-063)
Use Case	UC-011, UC-036

CMD-064 — InspectReturnedGoods

Field	Detail
Command Name	InspectReturnedGoods
Actor	Receiving Clerk
Target Aggregate	Customer Return Case
Business Purpose	Assess condition of returned goods to determine whether restock, quarantine, or write-off disposition is appropriate.
Expected Domain Event(s)	ReturnInspected (DE-064)
Use Case	UC-036

CMD-065 — RestockReturnedGoods

Field	Detail
Command Name	RestockReturnedGoods
Actor	Receiving Clerk
Target Aggregate	Customer Return Case
Business Purpose	Return qualifying goods to sellable inventory when condition grade confirms resale quality.
Expected Domain Event(s)	ReturnRestocked (DE-065) → ReceiveStock → StockReceived (DE-011)
Use Case	UC-036

CMD-066 — QuarantineReturnedGoods

Field	Detail
Command Name	QuarantineReturnedGoods
Actor	Warehouse Manager
Target Aggregate	Customer Return Case
Business Purpose	Place returned goods requiring further evaluation under quarantine, excluding them from sellable ATP.
Expected Domain Event(s)	ReturnQuarantined (DE-066) → ApplyQuarantineFromReturn → QuarantineHoldApplied (DE-017)
Use Case	UC-036

CMD-067 — WriteOffReturnedGoods

Field	Detail
Command Name	WriteOffReturnedGoods
Actor	Warehouse Manager
Target Aggregate	Customer Return Case
Business Purpose	Permanently remove non-resalable returned goods from inventory value after inspection confirms damage or defect.
Expected Domain Event(s)	ReturnWrittenOff (DE-067) → ApplyApprovedWriteOff → StockWrittenOff (DE-016)
Use Case	UC-036

Domain 8: Commercial & Monitoring

CMD-068 — AllocatePromotionalStock

Field	Detail
Command Name	AllocatePromotionalStock
Actor	Marketing Manager
Target Aggregate	Promotional Stock Allocation
Business Purpose	Ring-fence defined quantity for a promotional campaign or launch, excluding it from general ATP for non-approved orders.
Expected Domain Event(s)	PromotionalStockAllocated (DE-020)
Use Case	UC-040

CMD-069 — ReleasePromotionalAllocation

Field	Detail
Command Name	ReleasePromotionalAllocation
Actor	Marketing Manager
Target Aggregate	Promotional Stock Allocation
Business Purpose	Return unconsumed promotional stock to general sellable pool when campaign period ends or allocation is manually released.
Expected Domain Event(s)	PromotionalAllocationReleased (DE-021)
Use Case	UC-040

CMD-070 — EvaluateStockStatus

Field	Detail
Command Name	EvaluateStockStatus
Actor	Domain Service — Stock Status Classification
Target Aggregate	Stock Position
Business Purpose	Reclassify stock status when quantities cross policy thresholds to trigger replenishment alerts and operational response.
Expected Domain Event(s)	LowStockThresholdBreached (DE-068) OR OutOfStockDetected (DE-069) OR OverstockDetected (DE-070) *as applicable*
Use Case	UC-012, UC-028

CMD-071 — RecommendReplenishmentAction

Field	Detail
Command Name	RecommendReplenishmentAction
Actor	Domain Service — Replenishment Recommendation
Target Aggregate	*(Cross-cutting — no single aggregate mutation)*
Business Purpose	Produce replenishment guidance (purchase, transfer, or no action) after threshold breach and network availability evaluation.
Expected Domain Event(s)	ReplenishmentRecommended (DE-071)
Use Case	UC-028, UC-029, UC-034

CMD-072 — EvaluatePromotionalReadiness

Field	Detail
Command Name	EvaluatePromotionalReadiness
Actor	Domain Service — Promotional Readiness Evaluation
Target Aggregate	Promotional Stock Allocation
Business Purpose	Assess whether ring-fenced promotional stock is sufficient before campaign or launch commitment deadline.
Expected Domain Event(s)	PromotionalReadinessGapIdentified (DE-072) *when insufficient*
Use Case	UC-040, UC-047

CMD-073 — AnalyzeNetworkBalance

Field	Detail
Command Name	AnalyzeNetworkBalance
Actor	Domain Service — Network Rebalancing Analysis
Target Aggregate	*(Cross-cutting — no single aggregate mutation)*
Business Purpose	Detect simultaneous stockout risk at one location and excess at another for the same product to recommend transfer-before-buy.
Expected Domain Event(s)	NetworkImbalanceDetected (DE-073)
Use Case	UC-034

Domain 9: Financial & Governance

CMD-074 — PrepareValuationSnapshot

Field	Detail
Command Name	PrepareValuationSnapshot
Actor	Finance Manager
Target Aggregate	Inventory Valuation Snapshot *(read model)*
Business Purpose	Compile point-in-time owned inventory value for finance reconciliation against general ledger at period end.
Expected Domain Event(s)	InventoryValuationSnapshotPrepared (DE-076)
Use Case	UC-039

CMD-075 — ReconcileInventoryValuation

Field	Detail
Command Name	ReconcileInventoryValuation
Actor	Finance Manager
Target Aggregate	Inventory Valuation Snapshot *(read model)*
Business Purpose	Compare operational inventory valuation against general ledger and identify misalignment requiring investigation.
Expected Domain Event(s)	ValuationVarianceIdentified (DE-077) *when variance exceeds tolerance*
Use Case	UC-039

CMD-076 — ProvisionUserAccount

Field	Detail
Command Name	ProvisionUserAccount
Actor	System Administrator
Target Aggregate	User Account
Business Purpose	Create user identity with initial role assignments and location scope for authorized inventory operations.
Expected Domain Event(s)	UserAccountProvisioned (DE-079), AuditTrailRecorded (DE-075)
Use Case	UC-044

CMD-077 — ChangeRoleAssignment

Field	Detail
Command Name	ChangeRoleAssignment
Actor	System Administrator
Target Aggregate	User Account
Business Purpose	Add, modify, or remove user role and location scope, re-evaluating segregation-of-duties compliance.
Expected Domain Event(s)	RoleAssignmentChanged (DE-080), SegregationOfDutiesViolationDetected (DE-078) *if conflicting roles assigned*; AuditTrailRecorded (DE-075)
Use Case	UC-044, UC-045

CMD-078 — ValidateSegregationOfDuties

Field	Detail
Command Name	ValidateSegregationOfDuties
Actor	Domain Service — Segregation of Duties Validation
Target Aggregate	User Account
Business Purpose	Block actions that violate accountability controls when the same user attempts conflicting initiate and approve roles.
Expected Domain Event(s)	SegregationOfDutiesViolationDetected (DE-078)
Triggered By	ApproveCountVariance, ApproveAdjustment, ChangeRoleAssignment, and other governed approval commands
Use Case	UC-045

Command-to-Event Quick Reference

Command	Primary Event(s)	Downstream Event Chain
RegisterProduct	DE-001	DE-075
ChangeProductStatus	DE-002	DE-075
ConfirmCustomerOrder	DE-037	DE-012, DE-074
CompletePutAway	DE-033	DE-011, DE-074, DE-068–070
DispatchShipment	DE-040	DE-014, DE-074, DE-022
ApproveAdjustment	DE-059	DE-015 or DE-016, DE-074, DE-075
DispatchTransfer	DE-049	DE-074, DE-050→DE-011
SubmitCycleCountResult	DE-053	DE-054
RestockReturnedGoods	DE-065	DE-011, DE-074
EvaluateThreeWayMatch	DE-035 or DE-081	—
EvaluateStockStatus	DE-068, DE-069, or DE-070	DE-071

Excluded from Domain Commands

Action	Treatment	Rationale
SearchProduct, ViewInventory, QueryAuditTrail	Query	Read-only — no state mutation
AcknowledgeAlert, ViewDashboard	Notification/Read	Acknowledgment does not change inventory domain state

Action	Treatment	Rationale
Failed validation (rule rejected)	Rejection response	Aggregate unchanged — no event
Login, Logout	Identity infrastructure	Outside inventory domain boundary
SendEmail, DeliverNotification	Infrastructure	Delivery channel is not a business command

Document Approval

Role	Name	Signature	Date
Domain Expert			
DDD Architect			
Senior Business Analyst			
Solution Architect			

End of Domain Commands

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Domain Events — Event Storming Catalog

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-ES-001
Document Title	Domain Events — Event Storming Catalog
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Source Documents	SRS-DM-001 (Domain Entities), SRS-DM-003 (Domain Aggregates), SRS-DM-006 (Domain Events), SRS-UC-001 (Use Case Catalog), SRS-UC-002 (Main Flows)
Date	July 6, 2026
Scope	Business domain events only — no message brokers, integration technology, or UI interactions

Purpose

This document identifies **all domain events** for the Inventory Management System as derived from the domain model and use cases. Each event records a meaningful business fact expressed in **past tense**. Events are raised by aggregates after successful command execution and describe **what happened**, not what should happen next.

Conventions

Element	Convention
Event ID	`DE-xxx` — sequential identifier

Element	Convention
Event Name	PascalCase past tense (e.g., `StockReserved`)
Aggregate	The consistency boundary that raises the event
Triggering Command	Imperative business command that caused the state change
Resulting Business State	Authoritative post-condition after the event is applied

Event Catalog Summary

Domain Area	Event IDs	Count
Catalog & Network	DE-001 – DE-010	10
Inventory Balance	DE-011 – DE-022	12
Procurement & Inbound	DE-023 – DE-035	13
Order Fulfillment	DE-036 – DE-045	10
Network Transfer	DE-046 – DE-051	6
Inventory Control & Accuracy	DE-052 – DE-060	9
Returns Management	DE-061 – DE-067	7
Commercial & Monitoring	DE-068 – DE-073	6
Financial & Governance	DE-074 – DE-081	8
Total		**81**

Domain 1: Catalog & Network

DE-001 — ProductRegistered

Field	Detail
Event Name	ProductRegistered
Business Meaning	The organization has introduced a new SKU into the trusted product catalog. The product is eligible for stock policy assignment and, when active, for procurement and fulfillment referencing.
Aggregate Responsible	Product Catalog
Triggering Command	`RegisterProduct` *(UC-001, UC-046)*
Resulting Business State	Product exists with unique SKU, mandatory attributes, and initial lifecycle status (typically active). Product is searchable across the network and eligible for downstream configuration.

DE-002 — ProductStatusChanged

Field	Detail
Event Name	ProductStatusChanged
Business Meaning	The commercial availability of a SKU has changed. Discontinued status prohibits new procurement and reservations; inactive status restricts new customer commitments.
Aggregate Responsible	Product Catalog
Triggering Command	`ChangeProductStatus` *(UC-001)*

Field	Detail
Resulting Business State	Product lifecycle status is updated with prior status, new status, and attribution recorded. Downstream eligibility rules (procurement, reservation, replenishment) reflect the new status.

DE-003 — ProductABCClassificationAssigned

Field	Detail
Event Name	ProductABCClassificationAssigned
Business Meaning	Operational priority for the SKU has been set. Class A products receive highest count frequency, monitoring intensity, and alert priority.
Aggregate Responsible	Product Catalog
Triggering Command	`AssignABCClassification` *(UC-004)*
Resulting Business State	Product carries an ABC class (A, B, or C) with effective date. Cycle count scheduling and low-stock alert prioritization use the new classification.

DE-004 — InventoryLocationActivated

Field	Detail
Event Name	InventoryLocationActivated
Business Meaning	A location is authorized to receive, hold, fulfill, and transact inventory. Stock positions may be established and operational workflows enabled.
Aggregate Responsible	Inventory Location
Triggering Command	`ActivateInventoryLocation` *(UC-002, UC-046)*
Resulting Business State	Location operational status is active. Opening balances (if required) are recorded. Location accepts receipts, reservations, fulfillments, and transfers.

DE-005 — InventoryLocationDeactivated

Field	Detail
Event Name	InventoryLocationDeactivated
Business Meaning	A location is no longer authorized for new inventory activity. Existing stock must have been cleared or dispositioned before deactivation.
Aggregate Responsible	Inventory Location
Triggering Command	`DeactivateInventoryLocation` *(UC-002)*
Resulting Business State	Location operational status is deactivated or closing. New receipts, reservations, and fulfillments are blocked. Historical records remain intact.

DE-006 — StoragePositionConfigured

Field	Detail
Event Name	StoragePositionConfigured
Business Meaning	A findable physical address within a location is available for put-away placement and pick path guidance.
Aggregate Responsible	Inventory Location

Field	Detail
Triggering Command	`ConfigureStoragePosition` *(UC-002)*
Resulting Business State	Storage position (zone, aisle, bin) exists within the location hierarchy and is available for put-away and pick operations.

DE-007 — LocationFreezeImposed

Field	Detail
Event Name	LocationFreezeImposed
Business Meaning	All receiving, shipping, transfers, and adjustments at the location are suspended until the wall-to-wall count is certified complete. Ensures a stable snapshot for accurate counting.
Aggregate Responsible	Physical Inventory Count
Triggering Command	`StartPhysicalInventoryCount` *(UC-015)*
Resulting Business State	Location is frozen. Receiving, shipping, transfers, and non-count adjustments are blocked. Count teams operate against a stable inventory snapshot.

DE-008 — LocationFreezeReleased

Field	Detail
Event Name	LocationFreezeReleased
Business Meaning	Normal inventory operations may resume at the location. Count variances have been identified and routed for investigation.
Aggregate Responsible	Physical Inventory Count
Triggering Command	`CertifyPhysicalInventoryCount` *(UC-015)*
Resulting Business State	Location freeze is lifted. Normal receiving, shipping, transfers, and adjustments may resume. Count event is certified with variances routed to adjustment cases.

DE-009 — GovernancePolicyPublished

Field	Detail
Event Name	GovernancePolicyPublished
Business Meaning	Organization-wide inventory control parameters have changed. All subsequent transactions must evaluate against the new effective policy.
Aggregate Responsible	Inventory Governance
Triggering Command	`PublishGovernancePolicy` *(UC-005)*
Resulting Business State	New governance policy version is effective with approval thresholds, tolerances, and exception rules. Subsequent adjustments, receipts, and matches evaluate against this policy.

DE-010 — ReasonCodeRetired

Field	Detail
Event Name	ReasonCodeRetired

Field	Detail
Business Meaning	A categorized exception reason is no longer valid for new transactions. Historical records retain prior reason code attributions unchanged.
Aggregate Responsible	Inventory Governance
Triggering Command	`RetireReasonCode` *(UC-005)*
Resulting Business State	Reason code status is retired and removed from active selection lists. Historical transactions retain original reason code references.

Domain 2: Inventory Balance

DE-011 — StockReceived

Field	Detail
Event Name	StockReceived
Business Meaning	Verified inbound stock is now physically placed and recorded as on-hand inventory. ATP increases correspondingly unless quarantine or ownership rules apply.
Aggregate Responsible	Stock Position
Triggering Command	`IncreaseOnHandQuantity` *(reacts to `CompletePutAway`, `ConfirmTransferReceipt`, `RestockReturnedGoods`)*
Resulting Business State	On-hand quantity increased at product-location. ATP recalculated. Stock status indicator updated. Movement history appended.

DE-012 — StockReserved

Field	Detail
Event Name	StockReserved
Business Meaning	Quantity is operationally committed to a customer order and is no longer available for new sales promises. Prevents double-allocation.
Aggregate Responsible	Stock Position
Triggering Command	`ReserveStock` *(UC-022)*
Resulting Business State	Reserved quantity increased; ATP decreased by reserved amount. Reservation linked to customer order line. On-hand unchanged.

DE-013 — StockReservationReleased

Field	Detail
Event Name	StockReservationReleased
Business Meaning	Previously committed stock is returned to the available-to-promise pool and may be promised to other customers or orders.
Aggregate Responsible	Stock Position
Triggering Command	`ReleaseReservation` *(UC-023)*
Resulting Business State	Reserved quantity decreased; ATP restored. Order reservation cleared or reduced. On-hand unchanged.

DE-014 — StockShipped

Field	Detail
Event Name	StockShipped
Business Meaning	Physical stock has left the location for customer delivery. Inventory record reflects deduction only after verified pick.
Aggregate Responsible	Stock Position
Triggering Command	`DeductShippedQuantity` *(reacts to `DispatchShipment` — UC-024)*
Resulting Business State	On-hand and reserved quantities decreased. ATP unchanged (already reserved). Lot/batch attribution recorded for traceable products.

DE-015 — StockAdjusted

Field	Detail
Event Name	StockAdjusted
Business Meaning	Recorded inventory has been aligned with physical reality following investigation and authorization. Distinct from write-off — adjustment is a correction, not permanent value removal.
Aggregate Responsible	Stock Position
Triggering Command	`ApplyApprovedAdjustment` *(reacts to `ApproveAdjustment` — UC-017)*
Resulting Business State	On-hand quantity corrected with mandatory reason code. ATP recalculated. Adjustment case finalized. Movement and audit records created.

DE-016 — StockWrittenOff

Field	Detail
Event Name	StockWrittenOff
Business Meaning	Inventory value has been formally removed for damaged, obsolete, expired, or lost goods. Terminal event — quantity cannot be restored without new receipt.
Aggregate Responsible	Stock Position
Triggering Command	`ApplyApprovedWriteOff` *(reacts to `AuthorizeWriteOff` — UC-018)*
Resulting Business State	On-hand quantity permanently reduced. ATP decreased. Write-off case closed. Valuation impact recorded for finance reconciliation.

DE-017 — QuarantineHoldApplied

Field	Detail
Event Name	QuarantineHoldApplied
Business Meaning	Stock is physically present but not sellable due to quality investigation, regulatory restriction, recall, or customs hold. ATP reduced accordingly.
Aggregate Responsible	Stock Position
Triggering Command	`ApplyQuarantineHold` *(UC-019)*

Field	Detail
Resulting Business State	Quarantine hold active on specified quantity. Held quantity excluded from ATP. Hold reason and attribution recorded.

DE-018 — QuarantineHoldReleased

Field	Detail
Event Name	QuarantineHoldReleased
Business Meaning	Previously held stock is cleared for sale or fulfillment. Bonded inventory cleared for domestic commerce.
Aggregate Responsible	Stock Position
Triggering Command	`ReleaseQuarantineHold` *(UC-019)*
Resulting Business State	Quarantine hold cleared. Quantity restored to sellable ATP (or routed to disposition). Hold closure reason recorded.

DE-019 — StockPolicyThresholdChanged

Field	Detail
Event Name	StockPolicyThresholdChanged
Business Meaning	Replenishment and alert rules for a product at a location have been recalibrated. Future low-stock detection uses new thresholds.
Aggregate Responsible	Stock Policy
Triggering Command	`UpdateStockPolicyThresholds` *(UC-003)*
Resulting Business State	Reorder level, minimum, safety stock, and/or maximum thresholds saved with effective date. Prior values retained as immutable history. Monitoring rules updated.

DE-020 — PromotionalStockAllocated

Field	Detail
Event Name	PromotionalStockAllocated
Business Meaning	Defined inventory quantity is reserved for promotional or launch use and excluded from general ATP for non-approved orders.
Aggregate Responsible	Promotional Stock Allocation
Triggering Command	`AllocatePromotionalStock` *(UC-040)*
Resulting Business State	Campaign allocation active with ring-fenced quantity. General ATP reduced by allocated amount. Campaign period and channel designation recorded.

DE-021 — PromotionalAllocationReleased

Field	Detail
Event Name	PromotionalAllocationReleased
Business Meaning	Unconsumed promotional stock is returned to the general sellable pool. Campaign commitment period has concluded.
Aggregate Responsible	Promotional Stock Allocation
Triggering Command	`ReleasePromotionalAllocation` *(UC-040)*

Field	Detail
Resulting Business State	Allocation inactive or closed. Ring-fenced quantity returned to general ATP. Campaign performance data available for review.

DE-022 — ConsignmentOwnershipTransferred

Field	Detail
Event Name	ConsignmentOwnershipTransferred
Business Meaning	Stock previously held under consignment classification becomes owned inventory for valuation and financial recording upon defined consumption or sale event.
Aggregate Responsible	Stock Position
Triggering Command	`TransferConsignmentOwnership` *(reacts to consignment consumption/sale — UC-039)*
Resulting Business State	Ownership classification changed from consignment to owned. Valuation treatment updated. Financial reconciliation triggered.

Domain 3: Procurement & Inbound

DE-023 — PurchaseRequisitionSubmitted

Field	Detail
Event Name	PurchaseRequisitionSubmitted
Business Meaning	The organization has identified a procurement need for specific products at a target location, pending internal approval before commercial PO creation.
Aggregate Responsible	Purchase Requisition
Triggering Command	`SubmitPurchaseRequisition` *(UC-029)*
Resulting Business State	Requisition status is submitted and pending approval. Lines reference active products, quantities, and target location. Linked to triggering low-stock alert when applicable.

DE-024 — PurchaseRequisitionApproved

Field	Detail
Event Name	PurchaseRequisitionApproved
Business Meaning	Internal authorization to proceed with supplier procurement is granted. Requisition is eligible for conversion to Purchase Order.
Aggregate Responsible	Purchase Requisition
Triggering Command	`ApprovePurchaseRequisition` *(UC-029)*
Resulting Business State	Requisition status is approved. Network availability and policy validation completed. Eligible for PO conversion by buyer.

DE-025 — PurchaseRequisitionRejected

Field	Detail
Event Name	PurchaseRequisitionRejected
Business Meaning	Procurement request denied. Network redistribution or other alternative should be pursued instead of new purchase.
Aggregate Responsible	Purchase Requisition
Triggering Command	`RejectPurchaseRequisition` *(UC-029)*
Resulting Business State	Requisition status is rejected with documented reason. No PO may be created from this requisition. Requestor notified to pursue alternatives.

DE-026 — PurchaseOrderPlaced

Field	Detail
Event Name	PurchaseOrderPlaced
Business Meaning	The organization has committed to purchase specified goods from a supplier. Inbound receipt is authorized upon delivery.
Aggregate Responsible	Purchase Order
Triggering Command	`PlacePurchaseOrder` *(UC-006)*
Resulting Business State	PO status is open/authorized with supplier, product lines, quantities, delivery terms, and receiving location. Inbound receipt permitted against PO lines.

DE-027 — PurchaseOrderCancelled

Field	Detail
Event Name	PurchaseOrderCancelled
Business Meaning	Commercial commitment to supplier is withdrawn. No further receipts authorized against cancelled PO lines.
Aggregate Responsible	Purchase Order
Triggering Command	`CancelPurchaseOrder` *(UC-006)*
Resulting Business State	PO status is cancelled. Expected inbound deliveries cancelled. Open receipt lines blocked. Supplier and buyer notified.

DE-028 — InboundDeliveryScheduled

Field	Detail
Event Name	InboundDeliveryScheduled
Business Meaning	Receiving operations and replenishment planners are informed when and what quantity to expect, supporting dock preparation and inbound visibility.
Aggregate Responsible	Purchase Order
Triggering Command	`ScheduleInboundDelivery` *(UC-007)*
Resulting Business State	Expected Inbound Delivery recorded with arrival date, quantity, and PO line association. Visible on inbound supply calendar.

DE-029 — InboundDeliveryDelayed

Field	Detail
Event Name	InboundDeliveryDelayed
Business Meaning	Anticipated inbound supply will arrive later than planned. Replenishment timelines and fulfillment promises may be affected.
Aggregate Responsible	Purchase Order
Triggering Command	`RecordInboundDelay` *(UC-030, UC-047)*
Resulting Business State	Expected arrival date updated or delay flag set. Inbound supply status reflects late delivery. Operational alert raised for responsible actors.

DE-030 — GoodsReceiptVerified

Field	Detail
Event Name	GoodsReceiptVerified
Business Meaning	The business has confirmed what actually arrived at the dock matches authorized expectation. Stock is not yet available — put-away must follow.
Aggregate Responsible	Goods Receipt
Triggering Command	`VerifyGoodsReceipt` *(UC-008, UC-011)*
Resulting Business State	Receipt status is verified with accepted quantities, product identity, and condition confirmed. Lot/batch and shelf-life captured where required. Stock staged for put-away — not yet in ATP.

DE-031 — ReceiptDiscrepancyRecorded

Field	Detail
Event Name	ReceiptDiscrepancyRecorded
Business Meaning	Short shipment, over-shipment, wrong item, damage, or shelf-life non-compliance captured at point of receipt when evidence exists.
Aggregate Responsible	Goods Receipt
Triggering Command	`RecordReceiptDiscrepancy` *(UC-009)*
Resulting Business State	Discrepancy documented with type, variance quantity, reason code, and evidence. Accepted quantities separated from unresolved variance. Supplier claim workflow initiated.

DE-032 — ReceiptDiscrepancyResolved

Field	Detail
Event Name	ReceiptDiscrepancyResolved
Business Meaning	Receipt exception has been closed with documented outcome. Accepted quantities may proceed to put-away; rejected quantities will not enter inventory.
Aggregate Responsible	Goods Receipt
Triggering Command	`ResolveReceiptDiscrepancy` *(UC-009)*

Field	Detail
Resulting Business State	Discrepancy status is resolved (partial acceptance, rejection, or supplier claim completed). Receipt record finalized. Three-way match evaluation updated.

DE-033 — PutAwayCompleted

Field	Detail
Event Name	PutAwayCompleted
Business Meaning	Inbound stock has transitioned from receiving dock to operational storage. Triggers availability activation — stock may now enter sellable ATP.
Aggregate Responsible	Goods Receipt
Triggering Command	`CompletePutAway` *(UC-010)*
Resulting Business State	Put-away confirmed with storage position assignment. Receipt workflow complete. Cross-aggregate notification to increase Stock Position on-hand and ATP.

DE-034 — SupplierInvoiceReceived

Field	Detail
Event Name	SupplierInvoiceReceived
Business Meaning	Supplier has invoiced for goods supplied. Three-way match evaluation against PO and receipt must proceed before payment authorization.
Aggregate Responsible	Supplier Invoice
Triggering Command	`RecordSupplierInvoice` *(UC-039)*
Resulting Business State	Invoice recorded with line details linked to PO and receipt references. Match status pending. Payment blocked until match completes.

DE-035 — ThreeWayMatchCompleted

Field	Detail
Event Name	ThreeWayMatchCompleted
Business Meaning	Financial control confirmed — payment for received, authorized goods may proceed. Closes the procurement-to-payment accountability loop.
Aggregate Responsible	Supplier Invoice
Triggering Command	`CompleteThreeWayMatch` *(UC-039)*
Resulting Business State	PO, receipt, and invoice lines agree within tolerance. Match status is complete. Payment authorization enabled for finance.

Domain 4: Order Fulfillment

DE-036 — OrderAvailabilityConfirmed

Field	Detail
Event Name	OrderAvailabilityConfirmed

Field	Detail
Business Meaning	The business has verified that available-to-promise stock can support the order (fully or partially) before commercial commitment.
Aggregate Responsible	Customer Order Fulfillment
Triggering Command	`ConfirmOrderAvailability` *(UC-013, UC-021)*
Resulting Business State	Order availability check recorded with ATP validation result per line. Order eligible for confirmation if availability sufficient. No reservation yet applied.

DE-037 — CustomerOrderConfirmed

Field	Detail
Event Name	CustomerOrderConfirmed
Business Meaning	Commercial commitment to fulfill the order is established. Reservation of stock must follow immediately.
Aggregate Responsible	Customer Order Fulfillment
Triggering Command	`ConfirmCustomerOrder` *(UC-022)*
Resulting Business State	Order status is confirmed. Reservation workflow initiated. Order released to fulfillment queue.

DE-038 — CustomerOrderCancelled

Field	Detail
Event Name	CustomerOrderCancelled
Business Meaning	Commercial commitment withdrawn. Reserved stock must be released back to ATP.
Aggregate Responsible	Customer Order Fulfillment
Triggering Command	`CancelCustomerOrder` *(UC-023)*
Resulting Business State	Order or order line status is cancelled. Reservation release triggered. Fulfillment workflow terminated for cancelled lines.

DE-039 — PickVerified

Field	Detail
Event Name	PickVerified
Business Meaning	Physical pick matches order requirements. Shipment is authorized to proceed. Lot/batch captured for traceable products.
Aggregate Responsible	Customer Order Fulfillment
Triggering Command	`VerifyPick` *(UC-024)*
Resulting Business State	Pick lines verified against order. Fulfillment execution status updated. Shipment authorized. Rotation policy (FIFO/FEFO) applied.

DE-040 — ShipmentDispatched

Field	Detail
Event Name	ShipmentDispatched
Business Meaning	Order goods have left the warehouse. Inventory deduction and reservation consumption triggered. Delivery promise activated for customer.

Field	Detail
Aggregate Responsible	Customer Order Fulfillment
Triggering Command	`DispatchShipment` *(UC-024)*
Resulting Business State	Shipment recorded with carrier handoff. Stock deduction triggered on Stock Position. Order fulfillment status updated. Customer notification eligible.

DE-041 — PartialFulfillmentCompleted

Field	Detail
Event Name	PartialFulfillmentCompleted
Business Meaning	Customer receives available quantity now; unfulfilled portion handled separately via backorder with customer acknowledgment.
Aggregate Responsible	Customer Order Fulfillment
Triggering Command	`CompletePartialFulfillment` *(UC-025)*
Resulting Business State	Order line decomposed into fulfilled and backordered quantities. Shipment dispatched for available portion. Remainder routed to backorder workflow.

DE-042 — BackorderRecorded

Field	Detail
Event Name	BackorderRecorded
Business Meaning	Customer has agreed to wait for remaining quantity. Mutual commercial commitment — not an internal note.
Aggregate Responsible	Customer Order Fulfillment
Triggering Command	`RecordBackorder` *(UC-025)*
Resulting Business State	Backorder active with customer acknowledgment. Unfulfilled quantity tracked. Priority replenishment eligible. Customer confirmation recorded.

DE-043 — BackorderFulfilled

Field	Detail
Event Name	BackorderFulfilled
Business Meaning	Delayed customer commitment is now being executed. Backorder closed for fulfilled quantity.
Aggregate Responsible	Customer Order Fulfillment
Triggering Command	`FulfillBackorder` *(UC-025)*
Resulting Business State	Backorder quantity reserved and fulfilled. Backorder reduced or closed. Customer notification of backorder shipment eligible.

DE-044 — FulfillmentExceptionRaised

Field	Detail
Event Name	FulfillmentExceptionRaised
Business Meaning	Warehouse cannot fulfill as planned without resolution. Inventory accuracy or product quality issue requires manager intervention.

Field	Detail
Aggregate Responsible	Customer Order Fulfillment
Triggering Command	`RaiseFulfillmentException` *(UC-026)*
Resulting Business State	Fulfillment execution blocked with exception reason. Operational alert raised. Manager investigation workflow initiated. Shipment prevented until resolved.

DE-045 — MultiLocationFulfillmentPlanCreated

Field	Detail
Event Name	MultiLocationFulfillmentPlanCreated
Business Meaning	Single customer order will be fulfilled from more than one warehouse to satisfy total quantity when no single location has sufficient ATP.
Aggregate Responsible	Customer Order Fulfillment
Triggering Command	`CreateMultiLocationFulfillmentPlan` *(UC-048)*
Resulting Business State	Order lines assigned to multiple ship-from locations. Reservation workflow initiated at each location. Multi-shipment customer communication eligible.

Domain 5: Network Transfer

DE-046 — TransferRequested

Field	Detail
Event Name	TransferRequested
Business Meaning	A location requests stock from another location to resolve local shortage or rebalance network before new procurement.
Aggregate Responsible	Stock Transfer
Triggering Command	`RequestStockTransfer` *(UC-031)*
Resulting Business State	Transfer request status is pending approval. Source, destination, product, and quantity documented. Network availability evaluation completed.

DE-047 — TransferApproved

Field	Detail
Event Name	TransferApproved
Business Meaning	Network redistribution authorized. Source location can spare quantity while maintaining its own service level.
Aggregate Responsible	Stock Transfer
Triggering Command	`ApproveTransferRequest` *(UC-032)*
Resulting Business State	Transfer status is approved. Source minimum and safety stock validation passed. Source warehouse dispatch notification eligible.

DE-048 — TransferRejected

Field	Detail
Event Name	TransferRejected
Business Meaning	Proposed redistribution will not proceed. Requestor must pursue alternative (procurement, different source) to resolve shortage.
Aggregate Responsible	Stock Transfer
Triggering Command	`RejectTransferRequest` *(UC-032)*
Resulting Business State	Transfer status is rejected with documented reason. No stock movement authorized. Requestor notified.

DE-049 — TransferDispatched

Field	Detail
Event Name	TransferDispatched
Business Meaning	Source Stock Position reduced. Quantity is in transit — excluded from both source and destination ATP until receipt confirmed.
Aggregate Responsible	Stock Transfer
Triggering Command	`DispatchTransfer` *(UC-033)*
Resulting Business State	Transfer status is in-transit. Source on-hand reduced. In-Transit Stock entity active. Lot/batch preserved for traceable products.

DE-050 — TransferReceived

Field	Detail
Event Name	TransferReceived
Business Meaning	Network redistribution complete. Destination Stock Position increased; in-transit state closed. Lot/batch preserved for traceable products.
Aggregate Responsible	Stock Transfer
Triggering Command	`ConfirmTransferReceipt` *(UC-033)*
Resulting Business State	Transfer status is received/complete. In-transit state closed. Destination Stock Position increase triggered. Transfer history complete.

DE-051 — InTransitStockDelayed

Field	Detail
Event Name	InTransitStockDelayed
Business Meaning	Pipeline stock is delayed en route. Destination location fulfillment and replenishment plans may be impacted.
Aggregate Responsible	Stock Transfer
Triggering Command	`RecordTransferDelay` *(UC-030, UC-047)*
Resulting Business State	In-transit delay flag set. Expected receipt date updated. Operational alert raised. Destination and requestor notification eligible.

Domain 6: Inventory Control & Accuracy

DE-052 — CycleCountScheduled

Field	Detail
Event Name	CycleCountScheduled
Business Meaning	Partial physical verification scheduled. Class A products scheduled at highest frequency.
Aggregate Responsible	Cycle Count
Triggering Command	`ScheduleCycleCount` *(UC-014)*
Resulting Business State	Cycle count assignment created with product, location, counter, and scheduled date. Counter notification eligible.

DE-053 — CycleCountCompleted

Field	Detail
Event Name	CycleCountCompleted
Business Meaning	Physical quantity recorded without counter having seen system expected quantity. Variance calculated at submission.
Aggregate Responsible	Cycle Count
Triggering Command	`SubmitCycleCountResult` *(UC-014)*
Resulting Business State	Count result recorded under blind count protocol. Variance calculated against system quantity. Count status complete. Variance investigation initiated if discrepancy exists.

DE-054 — CountVarianceIdentified

Field	Detail
Event Name	CountVarianceIdentified
Business Meaning	Inventory accuracy discrepancy discovered. Investigation required before any Stock Position adjustment.
Aggregate Responsible	Inventory Adjustment Case
Triggering Command	`IdentifyCountVariance` *(reacts to `SubmitCycleCountResult` or physical count — UC-016)*
Resulting Business State	Variance case opened with quantity difference documented. Investigation workflow pending. Stock Position unchanged pending approval.

DE-055 — CountVarianceApproved

Field	Detail
Event Name	CountVarianceApproved
Business Meaning	Variance investigation complete and adjustment authorized. Segregation of duties confirmed.
Aggregate Responsible	Inventory Adjustment Case
Triggering Command	`ApproveCountVariance` *(UC-016)*
Resulting Business State	Variance approved by independent approver (distinct from counter). Adjustment submission authorized. Segregation of duties validated.

DE-056 — PhysicalInventoryCountStarted

Field	Detail
Event Name	PhysicalInventoryCountStarted
Business Meaning	Full location snapshot count initiated. All movements suspended. Authoritative accuracy baseline in progress.
Aggregate Responsible	Physical Inventory Count
Triggering Command	`StartPhysicalInventoryCount` *(UC-015)*
Resulting Business State	Wall-to-wall count in progress. Location freeze imposed. Count teams mobilized. All concurrent movements suspended.

DE-057 — PhysicalInventoryCountCertified

Field	Detail
Event Name	PhysicalInventoryCountCertified
Business Meaning	Wall-to-wall count authoritative and complete. Location freeze may be released. Variances routed for investigation.
Aggregate Responsible	Physical Inventory Count
Triggering Command	`CertifyPhysicalInventoryCount` *(UC-015)*
Resulting Business State	Count event certified complete. All variances identified and routed. Location freeze release eligible. Finance reconciliation notification eligible.

DE-058 — AdjustmentSubmitted

Field	Detail
Event Name	AdjustmentSubmitted
Business Meaning	Correction or removal request recorded pending governance evaluation. Reason code captured.
Aggregate Responsible	Inventory Adjustment Case
Triggering Command	`SubmitAdjustment` *(UC-017, UC-018)*
Resulting Business State	Adjustment case status is pending approval. Reason code and quantity documented. Governance evaluation and SoD validation initiated. Stock Position unchanged.

DE-059 — AdjustmentApproved

Field	Detail
Event Name	AdjustmentApproved
Business Meaning	Governance requirements satisfied. Stock Position mutation authorized.
Aggregate Responsible	Inventory Adjustment Case
Triggering Command	`ApproveAdjustment` *(UC-017, UC-018)*
Resulting Business State	Adjustment or write-off approved. Stock Position mutation triggered (`StockAdjusted` or `StockWrittenOff`). Audit trail recorded. Case finalized.

DE-060 — AdjustmentRejected

Field	Detail
Event Name	AdjustmentRejected
Business Meaning	Proposed inventory change will not take effect. Initiator must revise or provide additional justification.
Aggregate Responsible	Inventory Adjustment Case
Triggering Command	`RejectAdjustment` *(UC-017, UC-018)*
Resulting Business State	Adjustment case status is rejected. Stock Position unchanged. Initiator notified with rejection reason.

Domain 7: Returns Management

DE-061 — ReturnAuthorizationIssued

Field	Detail
Event Name	ReturnAuthorizationIssued
Business Meaning	Customer is authorized to return specified products. Warehouse notified to expect inbound return.
Aggregate Responsible	Customer Return Case
Triggering Command	`IssueReturnAuthorization` *(UC-035)*
Resulting Business State	RMA active and linked to original customer order. Return window and product lines documented. Warehouse advance notice eligible.

DE-062 — ReturnAuthorizationExpired

Field	Detail
Event Name	ReturnAuthorizationExpired
Business Meaning	Return window closed. Warehouse must not accept goods against expired authorization.
Aggregate Responsible	Customer Return Case
Triggering Command	`ExpireReturnAuthorization` *(system policy — UC-035)*
Resulting Business State	RMA status is expired. Warehouse receipt blocked against this authorization. Customer expiration notification eligible.

DE-063 — CustomerReturnReceived

Field	Detail
Event Name	CustomerReturnReceived
Business Meaning	Authorized return physically received. Inspection and disposition workflow initiated.
Aggregate Responsible	Customer Return Case
Triggering Command	`ReceiveCustomerReturn` *(UC-011, UC-036)*
Resulting Business State	Return receipt verified against RMA. Inspection task assigned. Stock not yet in sellable ATP.

DE-064 — ReturnInspected

Field	Detail
Event Name	ReturnInspected
Business Meaning	Resale quality of returned goods evaluated. Disposition decision (restock, quarantine, write-off) authorized to proceed.
Aggregate Responsible	Customer Return Case
Triggering Command	`InspectReturnedGoods` *(UC-036)*
Resulting Business State	Condition grade recorded. Disposition decision pending or ready for execution. Inspection attribution documented.

DE-065 — ReturnRestocked

Field	Detail
Event Name	ReturnRestocked
Business Meaning	Qualifying returned goods re-enter sellable inventory. ATP increased at receiving location.
Aggregate Responsible	Customer Return Case
Triggering Command	`RestockReturnedGoods` *(UC-036)*
Resulting Business State	Disposition is restock. Stock Position increase triggered. Return case progressing toward closure. ATP increased at receiving location.

DE-066 — ReturnQuarantined

Field	Detail
Event Name	ReturnQuarantined
Business Meaning	Returns requiring further evaluation excluded from sellable ATP.
Aggregate Responsible	Customer Return Case
Triggering Command	`QuarantineReturnedGoods` *(UC-036)*
Resulting Business State	Disposition is quarantine. Quarantine hold applied on Stock Position. Quality investigation workflow initiated.

DE-067 — ReturnWrittenOff

Field	Detail
Event Name	ReturnWrittenOff
Business Meaning	Damaged or defective returns permanently removed from inventory value.
Aggregate Responsible	Customer Return Case
Triggering Command	`WriteOffReturnedGoods` *(UC-036)*
Resulting Business State	Disposition is write-off. Stock Position write-off triggered. Return case closed. Valuation impact recorded.

Domain 8: Commercial & Monitoring

DE-068 — LowStockThresholdBreached

Field	Detail
Event Name	LowStockThresholdBreached

Field	Detail
Business Meaning	Replenishment action required same business day. Product at location needs purchasing or network transfer review.
Aggregate Responsible	Stock Position *(state change)* / Stock Policy *(threshold reference)*
Triggering Command	`EvaluateStockStatus` *(reacts to quantity change — UC-012, UC-028)*
Resulting Business State	ATP at or below reorder level. Stock status indicator is low-stock. Replenishment recommendation and alert workflows triggered.

DE-069 — OutOfStockDetected

Field	Detail
Event Name	OutOfStockDetected
Business Meaning	No quantity available to promise to new customers at this location. Existing reservations may still be fulfilled from on-hand. Stockout condition active.
Aggregate Responsible	Stock Position
Triggering Command	`EvaluateStockStatus` *(reacts to ATP reaching zero — UC-012, UC-028)*
Resulting Business State	ATP is zero. Stock status indicator is out-of-stock. Urgent replenishment escalation and operational alert triggered.

DE-070 — OverstockDetected

Field	Detail
Event Name	OverstockDetected
Business Meaning	Excess capital tied in inventory above policy maximum. Network rebalancing or purchasing freeze may be warranted.
Aggregate Responsible	Stock Position / Stock Policy
Triggering Command	`EvaluateStockStatus` *(reacts to on-hand exceeding maximum — UC-012, UC-020)*
Resulting Business State	On-hand exceeds policy maximum. Stock status indicator is overstock. Network rebalancing analysis and inventory controller review eligible.

DE-071 — ReplenishmentRecommended

Field	Detail
Event Name	ReplenishmentRecommended
Business Meaning	System recommends purchase requisition, transfer request, or no action based on policy, network availability, and product status.
Aggregate Responsible	*(Cross-cutting — domain service outcome)*
Triggering Command	`RecommendReplenishmentAction` *(UC-028, UC-029, UC-034)*
Resulting Business State	Replenishment recommendation recorded with suggested action (purchase, transfer, or hold). Buyer or regional manager action eligible.

DE-072 — PromotionalReadinessGapIdentified

Field	Detail
Event Name	PromotionalReadinessGapIdentified

Field	Detail
Business Meaning	Campaign or launch commitment at risk — allocated quantity insufficient for planned demand.
Aggregate Responsible	Promotional Stock Allocation
Triggering Command	`EvaluatePromotionalReadiness` *(UC-040, UC-047)*
Resulting Business State	Readiness gap documented. Ring-fenced quantity below campaign requirement. Marketing and purchasing notification eligible.

DE-073 — NetworkImbalanceDetected

Field	Detail
Event Name	NetworkImbalanceDetected
Business Meaning	Network holds sufficient total stock but maldistributed. Transfer-before-buy opportunity exists.
Aggregate Responsible	*(Cross-cutting — domain service outcome)*
Triggering Command	`AnalyzeNetworkBalance` *(UC-034)*
Resulting Business State	Imbalance documented with source excess and destination shortage for same product. Transfer recommendation eligible. Operations director review triggered.

Domain 9: Financial & Governance

DE-074 — InventoryMovementRecorded

Field	Detail
Event Name	InventoryMovementRecorded
Business Meaning	Immutable operational fact recorded — what changed, where, how much, by whom, and why. Complete movement history maintained.
Aggregate Responsible	Stock Position
Triggering Command	*(Automatic — raised after any quantity mutation command)*
Resulting Business State	Movement record appended with product, location, quantity delta, reason, user, and timestamp. Movement history queryable for investigation.

DE-075 — AuditTrailRecorded

Field	Detail
Event Name	AuditTrailRecorded
Business Meaning	Governance-grade evidence of who changed what and when. Supports compliance, fraud investigation, and control testing.
Aggregate Responsible	*(Cross-cutting — emitted by governed aggregates)*
Triggering Command	*(Automatic — raised after governed state change)*
Resulting Business State	Audit record with before/after state, user identity, approval chain, and timestamp. Available for auditor query.

DE-076 — InventoryValuationSnapshotPrepared

Field	Detail
Event Name	InventoryValuationSnapshotPrepared
Business Meaning	Point-in-time owned inventory value prepared for finance reconciliation against general ledger.
Aggregate Responsible	*(Read model — Inventory Valuation Snapshot)*
Triggering Command	`PrepareValuationSnapshot` *(UC-039)*
Resulting Business State	Period-end valuation summary available. Owned inventory value calculated. Finance month-end close workflow eligible.

DE-077 — ValuationVarianceIdentified

Field	Detail
Event Name	ValuationVarianceIdentified
Business Meaning	Operational inventory records and financial ledger are misaligned. Investigation and reconciliation required before month-end close.
Aggregate Responsible	*(Read model)*
Triggering Command	`ReconcileInventoryValuation` *(UC-039)*
Resulting Business State	Variance against general ledger documented and exceeds tolerance. Joint finance and inventory investigation triggered. Reconciliation status dashboard updated.

DE-078 — SegregationOfDutiesViolationDetected

Field	Detail
Event Name	SegregationOfDutiesViolationDetected
Business Meaning	Proposed action violates accountability controls. Prevents fraud and undetected error.
Aggregate Responsible	User Account
Triggering Command	*(Blocked command — e.g., `ApproveCountVariance`, `ApproveAdjustment`, `AssignRole` — UC-045)*
Resulting Business State	Command rejected. No domain state change on target aggregate. Security violation logged. Administrator and auditor alert eligible.

DE-079 — UserAccountProvisioned

Field	Detail
Event Name	UserAccountProvisioned
Business Meaning	Individual authorized to perform inventory operations within assigned role and location scope.
Aggregate Responsible	User Account
Triggering Command	`ProvisionUserAccount` *(UC-044)*
Resulting Business State	User account active with initial role assignments and location scope. SoD rules evaluated at assignment. Onboarding notification eligible.

DE-080 — RoleAssignmentChanged

Field	Detail
Event Name	RoleAssignmentChanged
Business Meaning	User's operational authority has changed. SoD compliance re-evaluated for new role combination.
Aggregate Responsible	User Account
Triggering Command	`ChangeRoleAssignment` *(UC-044, UC-045)*
Resulting Business State	Role or location scope updated with effective dating. Prior assignments retained as history. SoD re-evaluation complete. Access change notification eligible.

DE-081 — ThreeWayMatchExceptionRaised

Field	Detail
Event Name	ThreeWayMatchExceptionRaised
Business Meaning	Payment blocked pending resolution of quantity or price discrepancy. Financial control intervention required.
Aggregate Responsible	Supplier Invoice
Triggering Command	`EvaluateThreeWayMatch` *(UC-039)*
Resulting Business State	Match status is exception. Payment authorization blocked. Discrepancy documented for finance and purchasing resolution.

Event Flow — Core Business Lifecycles

Inbound Supply

```
PlacePurchaseOrder → PurchaseOrderPlaced
ScheduleInboundDelivery → InboundDeliveryScheduled
VerifyGoodsReceipt → GoodsReceiptVerified
[RecordReceiptDiscrepancy → ReceiptDiscrepancyRecorded → ResolveReceiptDiscrepancy → ReceiptDiscrepancyResolved]
CompletePutAway → PutAwayCompleted → StockReceived → InventoryMovementRecorded
RecordSupplierInvoice → SupplierInvoiceReceived → CompleteThreeWayMatch → ThreeWayMatchCompleted
```

Order Fulfillment

```
ConfirmOrderAvailability → OrderAvailabilityConfirmed
ConfirmCustomerOrder → CustomerOrderConfirmed → ReserveStock → StockReserved
VerifyPick → PickVerified
DispatchShipment → ShipmentDispatched → StockShipped → InventoryMovementRecorded
[CompletePartialFulfillment → PartialFulfillmentCompleted → RecordBackorder → BackorderRecorded → FulfillBackorder →
BackorderFulfilled]
```

Inventory Accuracy

```
ScheduleCycleCount → CycleCountScheduled
SubmitCycleCountResult → CycleCountCompleted → IdentifyCountVariance → CountVarianceIdentified
ApproveCountVariance → CountVarianceApproved
SubmitAdjustment → AdjustmentSubmitted → ApproveAdjustment → AdjustmentApproved
→ StockAdjusted → InventoryMovementRecorded → AuditTrailRecorded
```

Network Transfer

```
RequestStockTransfer → TransferRequested
ApproveTransferRequest → TransferApproved
DispatchTransfer → TransferDispatched
ConfirmTransferReceipt → TransferReceived → StockReceived → InventoryMovementRecorded
```

Returns

```
IssueReturnAuthorization → ReturnAuthorizationIssued
ReceiveCustomerReturn → CustomerReturnReceived
InspectReturnedGoods → ReturnInspected
[RestockReturnedGoods → ReturnRestocked | QuarantineReturnedGoods → ReturnQuarantined | WriteOffReturnedGoods →
ReturnWrittenOff]
```

Command-to-Event Traceability Matrix

Command	Domain Event(s) Raised	Use Case
`RegisterProduct`	ProductRegistered	UC-001
`ChangeProductStatus`	ProductStatusChanged	UC-001
`AssignABCClassification`	ProductABCClassificationAssigned	UC-004
`ActivateInventoryLocation`	InventoryLocationActivated	UC-002
`DeactivateInventoryLocation`	InventoryLocationDeactivated	UC-002
`ConfigureStoragePosition`	StoragePositionConfigured	UC-002
`StartPhysicalInventoryCount`	PhysicalInventoryCountStarted, LocationFreezeImposed	UC-015
`CertifyPhysicalInventoryCount`	PhysicalInventoryCountCertified, LocationFreezeReleased	UC-015
`PublishGovernancePolicy`	GovernancePolicyPublished	UC-005
`RetireReasonCode`	ReasonCodeRetired	UC-005
`UpdateStockPolicyThresholds`	StockPolicyThresholdChanged	UC-003
`ReserveStock`	StockReserved	UC-022
`ReleaseReservation`	StockReservationReleased	UC-023
`ApplyQuarantineHold`	QuarantineHoldApplied	UC-019
`ReleaseQuarantineHold`	QuarantineHoldReleased	UC-019
`AllocatePromotionalStock`	PromotionalStockAllocated	UC-040
`ReleasePromotionalAllocation`	PromotionalAllocationReleased	UC-040
`SubmitPurchaseRequisition`	PurchaseRequisitionSubmitted	UC-029
`ApprovePurchaseRequisition`	PurchaseRequisitionApproved	UC-029
`RejectPurchaseRequisition`	PurchaseRequisitionRejected	UC-029
`PlacePurchaseOrder`	PurchaseOrderPlaced	UC-006
`CancelPurchaseOrder`	PurchaseOrderCancelled	UC-006
`ScheduleInboundDelivery`	InboundDeliveryScheduled	UC-007
`VerifyGoodsReceipt`	GoodsReceiptVerified	UC-008
`RecordReceiptDiscrepancy`	ReceiptDiscrepancyRecorded	UC-009
`ResolveReceiptDiscrepancy`	ReceiptDiscrepancyResolved	UC-009

Command	Domain Event(s) Raised	Use Case
`CompletePutAway`	PutAwayCompleted → StockReceived	UC-010
`ConfirmOrderAvailability`	OrderAvailabilityConfirmed	UC-021
`ConfirmCustomerOrder`	CustomerOrderConfirmed	UC-022
`CancelCustomerOrder`	CustomerOrderCancelled	UC-023
`VerifyPick`	PickVerified	UC-024
`DispatchShipment`	ShipmentDispatched → StockShipped	UC-024
`RecordBackorder`	BackorderRecorded	UC-025
`RequestStockTransfer`	TransferRequested	UC-031
`ApproveTransferRequest`	TransferApproved	UC-032
`DispatchTransfer`	TransferDispatched	UC-033
`ConfirmTransferReceipt`	TransferReceived → StockReceived	UC-033
`SubmitCycleCountResult`	CycleCountCompleted → CountVarianceIdentified	UC-014
`ApproveCountVariance`	CountVarianceApproved	UC-016
`ApproveAdjustment`	AdjustmentApproved → StockAdjusted / StockWrittenOff	UC-017, UC-018
`IssueReturnAuthorization`	ReturnAuthorizationIssued	UC-035
`ReceiveCustomerReturn`	CustomerReturnReceived	UC-036
`InspectReturnedGoods`	ReturnInspected	UC-036
`RestockReturnedGoods`	ReturnRestocked → StockReceived	UC-036
`ProvisionUserAccount`	UserAccountProvisioned	UC-044
`ChangeRoleAssignment`	RoleAssignmentChanged	UC-044

Excluded from Domain Events

Occurrence	Treatment	Rationale
User searches or views inventory	Read operation	No business state change
Report query executed	Read model projection	Queries do not mutate domain state
Validation failure / business rule rejection	Rejection response	Nothing happened — state unchanged
Scheduled timer fired	Infrastructure signal	Timer is not a business event; resulting evaluation emits the business event
Email or notification delivered	Infrastructure	Delivery channel is not an inventory domain concern

Document Approval

Role	Name	Signature	Date
Domain Expert			
DDD Architect			
Senior Business Analyst			
Solution Architect			

End of Domain Events — Event Storming Catalog

Domain Policies

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-ES-003
Document Title	Domain Policies — Event Storming Catalog
Version	1.0
Status	Draft
Document Owner	Domain Expert
Source Documents	SRS-ES-001 (Domain Events), SRS-ES-002 (Domain Commands), SRS-DM-005 (Domain Services), SRS-BR-001 (Business Rules)
Date	July 6, 2026
Scope	Business reaction policies only — no message handlers, sagas, or infrastructure orchestration

Purpose

This document identifies **Domain Policies** for the Inventory Management System. A policy is a business reaction rule expressed as:

> **Whenever** [Domain Event] **then** [Business Action]

Policies connect domain events to subsequent business intentions. They represent organizational workflow discipline — what must happen next when a meaningful business fact is recorded. Policies may issue a command, block an action, route work to a human actor, or raise an alert.

Policy Naming Convention

Policies use descriptive **PascalCase** names ending in `Policy` where helpful: `ReserveStockOnOrderConfirmationPolicy`, or concise workflow names: `Activate Stock on Put-Away`.

Policy Catalog Summary

Workflow Area	Policy IDs	Count
Order Fulfillment	POL-001 – POL-010	10
Inbound & Procurement	POL-011 – POL-020	10
Inventory Balance & Replenishment	POL-021 – POL-028	8
Network Transfer	POL-029 – POL-033	5
Inventory Accuracy & Control	POL-034 – POL-042	9
Returns Management	POL-043 – POL-048	6
Commercial & Promotional	POL-049 – POL-052	4
Governance & Access Control	POL-053 – POL-058	6
Financial Alignment	POL-059 – POL-062	4
Master Data & Reference	POL-063 – POL-066	4

Workflow Area	Policy IDs	Count
Total		**64**

Workflow 1: Order Fulfillment

POL-001 — Reserve Stock on Order Confirmation

Field	Detail
Policy Name	Reserve Stock on Order Confirmation
Triggering Domain Event	CustomerOrderConfirmed (DE-037)
Business Rule Applied	BR-002 (Reservation at Order Confirmation), BR-010 (Safety Stock Protection), BR-029 (Promotional Stock Ring-Fencing)
Action Taken	Coordinate reservation across order lines and stock positions, validating ATP, safety stock, and promotional ring-fence rules before committing quantity.
Resulting Command	`ReserveStock` (CMD-012)

POL-002 — Release Reservation on Order Cancellation

Field	Detail
Policy Name	Release Reservation on Order Cancellation
Triggering Domain Event	CustomerOrderCancelled (DE-038)
Business Rule Applied	BR-002 (Reservation at Order Confirmation)
Action Taken	Return all reserved quantity for cancelled order lines to the available-to-promise pool.
Resulting Command	`ReleaseReservation` (CMD-013)

POL-003 — Allocate Multi-Location Reservations

Field	Detail
Policy Name	Allocate Multi-Location Reservations
Triggering Domain Event	MultiLocationFulfillmentPlanCreated (DE-045)
Business Rule Applied	BR-001 (ATP-Only Customer Commitment), BR-002 (Reservation at Order Confirmation)
Action Taken	Reserve stock at each assigned ship-from location according to the fulfillment plan decomposition.
Resulting Command	`ReserveStock` (CMD-012) *per allocated location*

POL-004 — Require Verified Pick Before Shipment

Field	Detail
Policy Name	Require Verified Pick Before Shipment
Triggering Domain Event	PickVerified (DE-039)
Business Rule Applied	BR-003 (Verified Pick Before Shipment), BR-014 (FEFO), BR-015 (FIFO), BR-017 (Lot/Batch Capture)
Action Taken	Authorize shipment dispatch only after pick verification; apply rotation policy and capture lot/batch for traceable products.

Field	Detail
Resulting Command	`DispatchShipment` (CMD-039) *when warehouse proceeds to ship*

POL-005 — Deduct Stock on Shipment Dispatch

Field	Detail
Policy Name	Deduct Stock on Shipment Dispatch
Triggering Domain Event	ShipmentDispatched (DE-040)
Business Rule Applied	BR-003 (Verified Pick Before Shipment), BR-002 (Reservation consumed on shipment)
Action Taken	Deduct on-hand and reserved quantities from the source stock position and record outbound movement.
Resulting Command	`ShipStock` (CMD-015)

POL-006 — Record Partial Fulfillment and Backorder

Field	Detail
Policy Name	Record Partial Fulfillment and Backorder
Triggering Domain Event	PartialFulfillmentCompleted (DE-041)
Business Rule Applied	BR-028 (Backorder Customer Acknowledgment), BR-001 (ATP-Only Customer Commitment)
Action Taken	Ship available quantity immediately; route unfulfilled remainder to backorder workflow pending customer acknowledgment.
Resulting Command	`DispatchShipment` (CMD-039) *fulfilled portion*; `RecordBackorder` (CMD-041) *remainder*

POL-007 — Prioritize Backorder Replenishment

Field	Detail
Policy Name	Prioritize Backorder Replenishment
Triggering Domain Event	BackorderRecorded (DE-042)
Business Rule Applied	BR-028 (Backorder Customer Acknowledgment), BR-009 (Reorder at Reorder Level)
Action Taken	Flag backordered products for priority replenishment review and notify purchasing and customer service teams.
Resulting Command	`RecommendReplenishmentAction` (CMD-071)

POL-008 — Fulfill Backorder When Stock Arrives

Field	Detail
Policy Name	Fulfill Backorder When Stock Arrives
Triggering Domain Event	StockReceived (DE-011)
Business Rule Applied	BR-002 (Reservation at Order Confirmation), BR-028 (Backorder Customer Acknowledgment)
Action Taken	When inbound stock satisfies a pending backorder line, initiate reservation and fulfillment for the oldest acknowledged backorder first.
Resulting Command	`FulfillBackorder` (CMD-042) *when matching backorder exists*

POL-009 — Escalate Fulfillment Exception

Field	Detail
Policy Name	Escalate Fulfillment Exception
Triggering Domain Event	FulfillmentExceptionRaised (DE-044)
Business Rule Applied	BR-003 (Verified Pick Before Shipment), BR-022 (Investigate Before Adjusting)
Action Taken	Halt shipment, notify warehouse manager, and route shelf mismatch to count variance investigation or quality hold as appropriate.
Resulting Command	`RaiseFulfillmentException` *(already raised)*; `IdentifyCountVariance` (CMD-057) or `ApplyQuarantineHold` (CMD-018) *as applicable*

POL-010 — Block Ineligible Product Reservation

Field	Detail
Policy Name	Block Ineligible Product Reservation
Triggering Domain Event	ProductStatusChanged (DE-002) — *to inactive or discontinued*
Business Rule Applied	BR-027 (No Reorder of Discontinued Products)
Action Taken	Prevent new reservations and purchase order lines for products that are inactive or discontinued; existing reservations may fulfill but no new commitments permitted.
Resulting Command	None *(block `ConfirmCustomerOrder` and `PlacePurchaseOrder`)*

Workflow 2: Inbound & Procurement

POL-011 — Stage Stock for Put-Away After Verification

Field	Detail
Policy Name	Stage Stock for Put-Away After Verification
Triggering Domain Event	GoodsReceiptVerified (DE-030)
Business Rule Applied	BR-005 (Put-Away Before Availability), BR-004 (Verify Before Inventory Acceptance)
Action Taken	Keep received stock off sellable ATP; generate put-away work for warehouse staff to move goods to storage positions.
Resulting Command	`CompletePutAway` (CMD-031) *issued by warehouse when ready*

POL-012 — Activate Stock on Put-Away Completion

Field	Detail
Policy Name	Activate Stock on Put-Away Completion
Triggering Domain Event	PutAwayCompleted (DE-033)
Business Rule Applied	BR-005 (Put-Away Before Availability), BR-030 (Bonded Inventory Customs Clearance)
Action Taken	Validate customs/bonded clearance status; if cleared, increase on-hand quantity and ATP at the destination stock position.
Resulting Command	`ReceiveStock` (CMD-014)

POL-013 — Hold Bonded Stock Until Customs Clearance

Field	Detail
Policy Name	Hold Bonded Stock Until Customs Clearance
Triggering Domain Event	GoodsReceiptVerified (DE-030) — *bonded inventory*
Business Rule Applied	BR-030 (Bonded Inventory Customs Clearance), BR-013 (Quarantine Exclusion from Sellable Stock)
Action Taken	Place verified bonded goods under quarantine hold; exclude from put-away activation and sellable ATP until customs clearance is confirmed.
Resulting Command	`ApplyQuarantineHold` (CMD-018)

POL-014 — Document Discrepancy Before Acceptance

Field	Detail
Policy Name	Document Discrepancy Before Acceptance
Triggering Domain Event	ReceiptDiscrepancyRecorded (DE-031)
Business Rule Applied	BR-006 (Document Receipt Discrepancies), BR-008 (Three-Way Match Before Payment)
Action Taken	Separate accepted quantities from unresolved variance; notify buyer and supplier for claim resolution before financial match proceeds.
Resulting Command	`ResolveReceiptDiscrepancy` (CMD-030) *when manager resolves*

POL-015 — Evaluate Three-Way Match on Invoice Receipt

Field	Detail
Policy Name	Evaluate Three-Way Match on Invoice Receipt
Triggering Domain Event	SupplierInvoiceReceived (DE-034)
Business Rule Applied	BR-008 (Three-Way Match Before Payment)
Action Taken	Compare invoice lines against linked purchase order and goods receipt within governance tolerance limits.
Resulting Command	`EvaluateThreeWayMatch` (CMD-033)

POL-016 — Re-Evaluate Match After Receipt or Discrepancy Resolution

Field	Detail
Policy Name	Re-Evaluate Match After Receipt or Discrepancy Resolution
Triggering Domain Event	ReceiptDiscrepancyResolved (DE-032) OR GoodsReceiptVerified (DE-030) *when invoice already recorded*
Business Rule Applied	BR-008 (Three-Way Match Before Payment), BR-006 (Document Receipt Discrepancies)
Action Taken	Re-run three-way match evaluation when receipt data changes and a supplier invoice is awaiting match.
Resulting Command	`EvaluateThreeWayMatch` (CMD-033)

POL-017 — Authorize Payment on Successful Match

Field	Detail
Policy Name	Authorize Payment on Successful Match

Field	Detail
Triggering Domain Event	ThreeWayMatchCompleted (DE-035)
Business Rule Applied	BR-008 (Three-Way Match Before Payment)
Action Taken	Release payment authorization to finance for matched goods; update procurement accountability status.
Resulting Command	None *(finance payment workflow — outside inventory aggregate boundary)*

POL-018 — Block Payment on Match Exception

Field	Detail
Policy Name	Block Payment on Match Exception
Triggering Domain Event	ThreeWayMatchExceptionRaised (DE-081)
Business Rule Applied	BR-008 (Three-Way Match Before Payment)
Action Taken	Block payment authorization; route quantity or price discrepancy to finance and purchasing for joint resolution.
Resulting Command	None *(human resolution workflow)*

POL-019 — Convert Approved Requisition to Purchase Order

Field	Detail
Policy Name	Convert Approved Requisition to Purchase Order
Triggering Domain Event	PurchaseRequisitionApproved (DE-024)
Business Rule Applied	BR-027 (No Reorder of Discontinued Products), BR-011 (Network Check Before Purchase)
Action Taken	Notify buyer that internal authorization is granted; buyer creates commercial purchase order from approved requisition.
Resulting Command	`PlacePurchaseOrder` (CMD-024) *issued by buyer*

POL-020 — Plan Inbound Delivery on Purchase Order Placement

Field	Detail
Policy Name	Plan Inbound Delivery on Purchase Order Placement
Triggering Domain Event	PurchaseOrderPlaced (DE-026)
Business Rule Applied	BR-007 (Purchase Order Required for Supplier Receipt)
Action Taken	Create expected inbound delivery schedule and communicate anticipated arrival to receiving location and replenishment planners.
Resulting Command	`ScheduleInboundDelivery` (CMD-026)

Workflow 3: Inventory Balance & Replenishment

POL-021 — Reclassify Stock Status on Quantity Change

Field	Detail
Policy Name	Reclassify Stock Status on Quantity Change
Triggering Domain Event	StockReceived (DE-011), StockShipped (DE-014), StockAdjusted (DE-015), StockWrittenOff (DE-016), StockReserved (DE-012), StockReservationReleased (DE-013)

Field	Detail
Business Rule Applied	BR-009 (Reorder at Reorder Level)
Action Taken	Compare current ATP and on-hand against stock policy thresholds and reclassify status (in stock, low stock, out of stock, overstock).
Resulting Command	`EvaluateStockStatus` (CMD-070)

POL-022 — Trigger Replenishment on Low Stock

Field	Detail
Policy Name	Trigger Replenishment on Low Stock
Triggering Domain Event	LowStockThresholdBreached (DE-068)
Business Rule Applied	BR-009 (Reorder at Reorder Level — same business day action), BR-011 (Network Check Before Purchase)
Action Taken	Evaluate network availability and recommend purchase requisition, transfer request, or no action; notify purchasing manager.
Resulting Command	`RecommendReplenishmentAction` (CMD-071)

POL-023 — Escalate Out-of-Stock Urgently

Field	Detail
Policy Name	Escalate Out-of-Stock Urgently
Triggering Domain Event	OutOfStockDetected (DE-069)
Business Rule Applied	BR-009 (Reorder at Reorder Level), BR-032 (Class A Priority Monitoring)
Action Taken	Raise urgent operational alert with elevated priority for Class A products; escalate to purchasing and operations leadership.
Resulting Command	`RecommendReplenishmentAction` (CMD-071)

POL-024 — Flag Overstock for Network Rebalancing

Field	Detail
Policy Name	Flag Overstock for Network Rebalancing
Triggering Domain Event	OverstockDetected (DE-070)
Business Rule Applied	BR-011 (Network Check Before Purchase), BR-012 (Protect Source Location Minimum)
Action Taken	Identify whether excess can be redistributed to locations with shortage before new procurement is considered.
Resulting Command	`AnalyzeNetworkBalance` (CMD-073)

POL-025 — Apply Adjustment on Governance Approval

Field	Detail
Policy Name	Apply Adjustment on Governance Approval
Triggering Domain Event	AdjustmentApproved (DE-059)
Business Rule Applied	BR-021 (Independent Approval for Material Adjustments), BR-020 (Mandatory Adjustment Reason Codes)

Field	Detail
Action Taken	Mutate stock position quantity per approved adjustment case — correction or permanent write-off.
Resulting Command	`ApplyApprovedAdjustment` (CMD-016) OR `ApplyApprovedWriteOff` (CMD-017)

POL-026 — Segregate ATP on Promotional Allocation

Field	Detail
Policy Name	Segregate ATP on Promotional Allocation
Triggering Domain Event	PromotionalStockAllocated (DE-020)
Business Rule Applied	BR-029 (Promotional Stock Ring-Fencing), BR-001 (ATP-Only Customer Commitment)
Action Taken	Reduce general ATP by ring-fenced quantity; protect allocated stock from non-promotional order reservations.
Resulting Command	None *(ATP segregation applied within Stock Position coordination)*

POL-027 — Restore General ATP on Campaign End

Field	Detail
Policy Name	Restore General ATP on Campaign End
Triggering Domain Event	PromotionalAllocationReleased (DE-021)
Business Rule Applied	BR-029 (Promotional Stock Ring-Fencing)
Action Taken	Return unconsumed promotional quantity to general sellable pool for standard order fulfillment.
Resulting Command	`ReleasePromotionalAllocation` *(already executed)*; ATP restoration within Stock Position

POL-028 — Transfer Consignment Ownership on Consumption

Field	Detail
Policy Name	Transfer Consignment Ownership on Consumption
Triggering Domain Event	StockShipped (DE-014) — *consignment stock*
Business Rule Applied	BR-031 (Consignment Ownership at Consumption)
Action Taken	Transfer ownership from supplier consignment to organization-owned inventory upon sale or consumption per supplier agreement.
Resulting Command	`TransferConsignmentOwnership` (CMD-020)

Workflow 4: Network Transfer

POL-029 — Evaluate Network Before Procurement

Field	Detail
Policy Name	Evaluate Network Before Procurement
Triggering Domain Event	PurchaseRequisitionSubmitted (DE-023) OR ReplenishmentRecommended (DE-071)
Business Rule Applied	BR-011 (Network Check Before Purchase)

Field	Detail
Action Taken	Scan network-wide ATP and in-transit supply; recommend transfer instead of purchase when excess exists at another location.
Resulting Command	`RequestStockTransfer` (CMD-045) *recommended* OR `RejectPurchaseRequisition` (CMD-023) *if transfer preferred*

POL-030 — Validate Source Capacity on Transfer Request

Field	Detail
Policy Name	Validate Source Capacity on Transfer Request
Triggering Domain Event	TransferRequested (DE-046)
Business Rule Applied	BR-012 (Protect Source Location Minimum), BR-010 (Safety Stock Protection)
Action Taken	Calculate maximum transferable quantity from source without breaching minimum threshold or safety stock buffer; route for operations director approval.
Resulting Command	`ApproveTransferRequest` (CMD-046) OR `RejectTransferRequest` (CMD-047) *by operations director*

POL-031 — Deduct Source Stock on Transfer Dispatch

Field	Detail
Policy Name	Deduct Source Stock on Transfer Dispatch
Triggering Domain Event	TransferDispatched (DE-049)
Business Rule Applied	BR-012 (Protect Source Location Minimum), BR-017 (Lot/Batch Capture)
Action Taken	Reduce source location on-hand; create in-transit state excluded from both source and destination ATP.
Resulting Command	`DispatchTransferStock` (CMD-044)

POL-032 — Increase Destination Stock on Transfer Receipt

Field	Detail
Policy Name	Increase Destination Stock on Transfer Receipt
Triggering Domain Event	TransferReceived (DE-050)
Business Rule Applied	BR-017 (Lot/Batch Capture for Traceable Products)
Action Taken	Close in-transit state and increase destination stock position on-hand and ATP.
Resulting Command	`ReceiveStock` (CMD-014)

POL-033 — Detect Network Imbalance Proactively

Field	Detail
Policy Name	Detect Network Imbalance Proactively
Triggering Domain Event	NetworkImbalanceDetected (DE-073)
Business Rule Applied	BR-011 (Network Check Before Purchase), BR-012 (Protect Source Location Minimum)
Action Taken	Recommend redistribution transfers from excess locations to shortage locations before stockouts occur or procurement is initiated.

Field	Detail
Resulting Command	`RequestStockTransfer` (CMD-045) *recommended to operations director*

Workflow 5: Inventory Accuracy & Control

POL-034 — Impose Location Freeze on Wall-to-Wall Count

Field	Detail
Policy Name	Impose Location Freeze on Wall-to-Wall Count
Triggering Domain Event	PhysicalInventoryCountStarted (DE-056) / LocationFreezeImposed (DE-007)
Business Rule Applied	BR-024 (Physical Inventory Freeze)
Action Taken	Block all receiving, shipping, transfers, and non-count adjustments at the location; notify warehouse staff of freeze.
Resulting Command	None *(block mutation commands at frozen location)*

POL-035 — Release Location Freeze on Count Certification

Field	Detail
Policy Name	Release Location Freeze on Count Certification
Triggering Domain Event	PhysicalInventoryCountCertified (DE-057) / LocationFreezeReleased (DE-008)
Business Rule Applied	BR-024 (Physical Inventory Freeze)
Action Taken	Lift operational freeze and notify warehouse that normal receiving, shipping, and transfers may resume.
Resulting Command	None *(mutation commands permitted again)*

POL-036 — Open Variance Case on Count Discrepancy

Field	Detail
Policy Name	Open Variance Case on Count Discrepancy
Triggering Domain Event	CycleCountCompleted (DE-053) OR PhysicalInventoryCountCertified (DE-057) *with variance*
Business Rule Applied	BR-018 (Blind Cycle Counting), BR-022 (Investigate Before Adjusting)
Action Taken	Calculate variance at submission without counter having seen expected quantity; open investigation case without directly mutating stock.
Resulting Command	`IdentifyCountVariance` (CMD-057)

POL-037 — Require Independent Approval for Count Variance

Field	Detail
Policy Name	Require Independent Approval for Count Variance
Triggering Domain Event	CountVarianceIdentified (DE-054)
Business Rule Applied	BR-019 (Independent Count Approval), BR-022 (Investigate Before Adjusting)
Action Taken	Route variance to warehouse manager for root cause investigation; block stock adjustment until independent approver authorizes.

Field	Detail
Resulting Command	`ApproveCountVariance` (CMD-053) *after investigation*

POL-038 — Submit Adjustment After Variance Approval

Field	Detail
Policy Name	Submit Adjustment After Variance Approval
Triggering Domain Event	CountVarianceApproved (DE-055)
Business Rule Applied	BR-020 (Mandatory Adjustment Reason Codes), BR-021 (Independent Approval for Material Adjustments)
Action Taken	Create stock adjustment proposal with mandatory reason code and route through governance evaluation for final approval.
Resulting Command	`SubmitAdjustment` (CMD-054)

POL-039 — Evaluate Adjustment Governance on Submission

Field	Detail
Policy Name	Evaluate Adjustment Governance on Submission
Triggering Domain Event	AdjustmentSubmitted (DE-058)
Business Rule Applied	BR-020 (Mandatory Adjustment Reason Codes), BR-021 (Independent Approval for Material Adjustments), BR-023 (Documented Write-Off Authorization)
Action Taken	Validate reason code, check approval threshold, and verify initiator is distinct from required approver before routing for authorization.
Resulting Command	`ApproveAdjustment` (CMD-055) OR `RejectAdjustment` (CMD-056) *by authorized approver*

POL-040 — Prioritize Class A Cycle Count Scheduling

Field	Detail
Policy Name	Prioritize Class A Cycle Count Scheduling
Triggering Domain Event	ProductABCClassificationAssigned (DE-003)
Business Rule Applied	BR-032 (Class A Priority Monitoring)
Action Taken	Increase cycle count frequency and monitoring intensity for Class A products in the count schedule.
Resulting Command	`ScheduleCycleCount` (CMD-051) *at elevated frequency*

POL-041 — Block Mutations at Deactivated Location

Field	Detail
Policy Name	Block Mutations at Deactivated Location
Triggering Domain Event	InventoryLocationDeactivated (DE-005)
Business Rule Applied	BR-024 (Physical Inventory Freeze) — *closure equivalent*
Action Taken	Prevent new receipts, reservations, fulfillments, and transfers at deactivated location; allow only authorized disposition activity.
Resulting Command	None *(block mutation commands)*

POL-042 — Record Movement History on Stock Mutation

Field	Detail
Policy Name	Record Movement History on Stock Mutation
Triggering Domain Event	StockReceived (DE-011), StockShipped (DE-014), StockAdjusted (DE-015), StockWrittenOff (DE-016), StockReserved (DE-012), StockReservationReleased (DE-013)
Business Rule Applied	BR-017 (Lot/Batch Capture for Traceable Products)
Action Taken	Append immutable movement record documenting what changed, where, how much, by whom, and why for operational traceability.
Resulting Command	None *(InventoryMovementRecorded DE-074 raised automatically)*

Workflow 6: Returns Management

POL-043 — Validate Return Authorization on Receipt

Field	Detail
Policy Name	Validate Return Authorization on Receipt
Triggering Domain Event	CustomerReturnReceived (DE-063)
Business Rule Applied	BR-025 (Return Authorization Required), BR-004 (Verify Before Inventory Acceptance)
Action Taken	Match received goods to active RMA and original order; reject receipt if authorization is missing or expired.
Resulting Command	`InspectReturnedGoods` (CMD-064) *if validation passes*

POL-044 — Block Receipt on Expired Authorization

Field	Detail
Policy Name	Block Receipt on Expired Authorization
Triggering Domain Event	ReturnAuthorizationExpired (DE-062)
Business Rule Applied	BR-025 (Return Authorization Required)
Action Taken	Prevent warehouse acceptance of goods against expired RMA; notify customer that return window has closed.
Resulting Command	None *(block `ReceiveCustomerReturn`)*

POL-045 — Restock Return on Passing Inspection

Field	Detail
Policy Name	Restock Return on Passing Inspection
Triggering Domain Event	ReturnInspected (DE-064) — *resale quality confirmed*
Business Rule Applied	BR-026 (Inspect Returns Before Restocking)
Action Taken	Return qualifying goods to sellable inventory at receiving location.
Resulting Command	`RestockReturnedGoods` (CMD-065) → `ReceiveStock` (CMD-014)

POL-046 — Quarantine Return Pending Further Evaluation

Field	Detail
Policy Name	Quarantine Return Pending Further Evaluation
Triggering Domain Event	ReturnInspected (DE-064) — *condition uncertain*
Business Rule Applied	BR-026 (Inspect Returns Before Restocking), BR-013 (Quarantine Exclusion from Sellable Stock)
Action Taken	Place returned goods under quarantine hold; exclude from sellable ATP pending quality team investigation.
Resulting Command	`QuarantineReturnedGoods` (CMD-066)

POL-047 — Write Off Non-Resalable Returns

Field	Detail
Policy Name	Write Off Non-Resalable Returns
Triggering Domain Event	ReturnInspected (DE-064) — *damaged or defective*
Business Rule Applied	BR-026 (Inspect Returns Before Restocking), BR-023 (Documented Write-Off Authorization)
Action Taken	Permanently remove non-resalable returned goods from inventory value after inspection confirms disposition.
Resulting Command	`WriteOffReturnedGoods` (CMD-067) → `ApplyApprovedWriteOff` (CMD-017)

POL-048 — Notify Warehouse on Return Authorization

Field	Detail
Policy Name	Notify Warehouse on Return Authorization
Triggering Domain Event	ReturnAuthorizationIssued (DE-061)
Business Rule Applied	BR-025 (Return Authorization Required)
Action Taken	Send return instructions to customer and advance notice to receiving clerk to expect inbound return shipment.
Resulting Command	None *(notification workflow — receiving awaits `ReceiveCustomerReturn`)*

Workflow 7: Commercial & Promotional

POL-049 — Evaluate Promotional Readiness Before Campaign

Field	Detail
Policy Name	Evaluate Promotional Readiness Before Campaign
Triggering Domain Event	PromotionalStockAllocated (DE-020) OR campaign deadline approaching
Business Rule Applied	BR-029 (Promotional Stock Ring-Fencing)
Action Taken	Compare ring-fenced quantity against campaign demand forecast; alert marketing and purchasing if readiness gap exists.
Resulting Command	`EvaluatePromotionalReadiness` (CMD-072)

POL-050 — Alert on Promotional Readiness Gap

Field	Detail
Policy Name	Alert on Promotional Readiness Gap
Triggering Domain Event	PromotionalReadinessGapIdentified (DE-072)
Business Rule Applied	BR-029 (Promotional Stock Ring-Fencing)
Action Taken	Notify marketing manager and purchasing manager that campaign commitment is at risk due to insufficient ring-fenced stock.
Resulting Command	`AllocatePromotionalStock` (CMD-068) *increase allocation* OR `RecommendReplenishmentAction` (CMD-071)

POL-051 — Apply Updated Governance Thresholds

Field	Detail
Policy Name	Apply Updated Governance Thresholds
Triggering Domain Event	GovernancePolicyPublished (DE-009)
Business Rule Applied	BR-021 (Independent Approval for Material Adjustments), BR-008 (Three-Way Match Before Payment), BR-006 (Document Receipt Discrepancies)
Action Taken	All subsequent adjustments, receipt verifications, and three-way match evaluations use the newly effective policy version.
Resulting Command	None *(policy consultation on next governed command)*

POL-052 — Retire Obsolete Reason Codes from Selection

Field	Detail
Policy Name	Retire Obsolete Reason Codes from Selection
Triggering Domain Event	ReasonCodeRetired (DE-010)
Business Rule Applied	BR-020 (Mandatory Adjustment Reason Codes)
Action Taken	Remove retired reason code from active selection lists; block new adjustments or discrepancies from using retired codes.
Resulting Command	None *(validation on next `SubmitAdjustment` or `RecordReceiptDiscrepancy`)*

Workflow 8: Governance & Access Control

POL-053 — Enforce Segregation of Duties on Approval

Field	Detail
Policy Name	Enforce Segregation of Duties on Approval
Triggering Domain Event	*(Intercepts)* `ApproveCountVariance`, `ApproveAdjustment`, `ChangeRoleAssignment`
Business Rule Applied	BR-019 (Independent Count Approval), BR-021 (Independent Approval for Material Adjustments)
Action Taken	Block action when approver is the same person as counter or initiator; log accountability violation for audit review.
Resulting Command	`ValidateSegregationOfDuties` (CMD-078) → SegregationOfDutiesViolationDetected (DE-078) *on violation*

POL-054 — Record Audit Trail on Governed Change

Field	Detail
Policy Name	Record Audit Trail on Governed Change
Triggering Domain Event	AdjustmentApproved (DE-059), ProductStatusChanged (DE-002), RoleAssignmentChanged (DE-080), GovernancePolicyPublished (DE-009), and other material mutations
Business Rule Applied	BR-019, BR-020, BR-021, BR-023
Action Taken	Capture immutable before/after state, user identity, approval chain, and timestamp for compliance and control testing.
Resulting Command	None *(AuditTrailRecorded DE-075 raised automatically)*

POL-055 — Re-Evaluate SoD on Role Change

Field	Detail
Policy Name	Re-Evaluate SoD on Role Change
Triggering Domain Event	RoleAssignmentChanged (DE-080)
Business Rule Applied	BR-019 (Independent Count Approval), BR-021 (Independent Approval for Material Adjustments)
Action Taken	Re-evaluate user's role combination for segregation-of-duties conflicts; block conflicting role assignments.
Resulting Command	`ValidateSegregationOfDuties` (CMD-078)

POL-056 — Block Mutations During Location Freeze

Field	Detail
Policy Name	Block Mutations During Location Freeze
Triggering Domain Event	LocationFreezeImposed (DE-007)
Business Rule Applied	BR-024 (Physical Inventory Freeze)
Action Taken	Intercept receipt, shipment, transfer, and adjustment commands at frozen location; permit only authorized count-related activity.
Resulting Command	None *(block `VerifyGoodsReceipt`, `DispatchShipment`, `DispatchTransfer`, `SubmitAdjustment` at location)*

POL-057 — Notify Responsible Actor on Operational Alert

Field	Detail
Policy Name	Notify Responsible Actor on Operational Alert
Triggering Domain Event	LowStockThresholdBreached (DE-068), InboundDeliveryDelayed (DE-029), InTransitStockDelayed (DE-051), FulfillmentExceptionRaised (DE-044), PromotionalReadinessGapIdentified (DE-072)
Business Rule Applied	BR-009 (Reorder at Reorder Level — same business day action)
Action Taken	Deliver timely notification to the responsible business actor (purchasing manager, buyer, warehouse manager, marketing manager) for action.
Resulting Command	None *(human responds with appropriate command)*

POL-058 — Investigate SoD Violation for Audit

Field	Detail
Policy Name	Investigate SoD Violation for Audit
Triggering Domain Event	SegregationOfDutiesViolationDetected (DE-078)
Business Rule Applied	BR-019, BR-021
Action Taken	Alert system administrator and internal auditor; record violation in audit trail for control testing and investigation.
Resulting Command	None *(audit review workflow)*

Workflow 9: Financial Alignment

POL-059 — Prepare Valuation Snapshot at Period End

Field	Detail
Policy Name	Prepare Valuation Snapshot at Period End
Triggering Domain Event	ThreeWayMatchCompleted (DE-035) OR month-end close cycle
Business Rule Applied	BR-008 (Three-Way Match Before Payment), BR-031 (Consignment Ownership at Consumption)
Action Taken	Aggregate owned stock quantities (excluding consignment, quarantined, and in-transit) with matched unit costs for finance reconciliation.
Resulting Command	`PrepareValuationSnapshot` (CMD-074)

POL-060 — Investigate Valuation Variance with Finance

Field	Detail
Policy Name	Investigate Valuation Variance with Finance
Triggering Domain Event	ValuationVarianceIdentified (DE-077)
Business Rule Applied	BR-008 (Three-Way Match Before Payment), BR-031 (Consignment Ownership at Consumption)
Action Taken	Route operational-to-ledger misalignment to finance manager and inventory controller for joint investigation before month-end close.
Resulting Command	`ReconcileInventoryValuation` (CMD-075)

POL-061 — Update Valuation on Write-Off

Field	Detail
Policy Name	Update Valuation on Write-Off
Triggering Domain Event	StockWrittenOff (DE-016)
Business Rule Applied	BR-023 (Documented Write-Off Authorization)
Action Taken	Reduce owned inventory value in valuation projections; include write-off in shrinkage and loss reporting for finance.
Resulting Command	None *(valuation read model projection updated)*

POL-062 — Exclude Quarantined Stock from Valuation

Field	Detail
Policy Name	Exclude Quarantined Stock from Valuation
Triggering Domain Event	QuarantineHoldApplied (DE-017)
Business Rule Applied	BR-013 (Quarantine Exclusion from Sellable Stock), BR-031 (Consignment Ownership at Consumption)
Action Taken	Exclude quarantined quantity from sellable ATP and owned valuation snapshot until hold is released or dispositioned.
Resulting Command	None *(valuation projection rule applied)*

Workflow 10: Master Data & Reference

POL-063 — Enable Stock Policy Assignment on Product Registration

Field	Detail
Policy Name	Enable Stock Policy Assignment on Product Registration
Triggering Domain Event	ProductRegistered (DE-001)
Business Rule Applied	—
Action Taken	Make new SKU eligible for stock policy configuration, replenishment planning, and (when active) procurement and fulfillment.
Resulting Command	`UpdateStockPolicyThresholds` (CMD-011) *by inventory controller*

POL-064 — Establish Opening Balances on Location Activation

Field	Detail
Policy Name	Establish Opening Balances on Location Activation
Triggering Domain Event	InventoryLocationActivated (DE-004)
Business Rule Applied	—
Action Taken	Load verified opening on-hand quantities for go-live location to establish authoritative inventory baseline.
Resulting Command	`EstablishOpeningBalances` (CMD-058)

POL-065 — Enable Put-Away on Storage Position Configuration

Field	Detail
Policy Name	Enable Put-Away on Storage Position Configuration
Triggering Domain Event	StoragePositionConfigured (DE-006)
Business Rule Applied	BR-014 (FEFO), BR-015 (FIFO)
Action Taken	Make new storage address available for put-away placement and rotation-compliant pick path guidance.
Resulting Command	None *(next `CompletePutAway` may assign to position)*

POL-066 — Reject Procurement for Discontinued Products

Field	Detail
Policy Name	Reject Procurement for Discontinued Products
Triggering Domain Event	PurchaseRequisitionSubmitted (DE-023) OR `PlacePurchaseOrder` attempted
Business Rule Applied	BR-027 (No Reorder of Discontinued Products)
Action Taken	Validate product eligibility before procurement proceeds; reject requisition or PO lines referencing discontinued SKUs.
Resulting Command	`RejectPurchaseRequisition` (CMD-023) OR block `PlacePurchaseOrder`

Policy Flow — Core Business Lifecycles

Order-to-Cash Reaction Chain

```
CustomerOrderConfirmed
  → [POL-001] ReserveStock
    → StockReserved
      → [POL-021] EvaluateStockStatus

PickVerified
  → [POL-004] authorize DispatchShipment
    → ShipmentDispatched
      → [POL-005] ShipStock
        → StockShipped
          → [POL-028] TransferConsignmentOwnership (if consignment)
          → [POL-042] InventoryMovementRecorded
```

Procure-to-Pay Reaction Chain

```
PurchaseOrderPlaced
  → [POL-020] ScheduleInboundDelivery

GoodsReceiptVerified
  → [POL-011] stage for CompletePutAway
  → [POL-013] ApplyQuarantineHold (if bonded)

PutAwayCompleted
  → [POL-012] ReceiveStock
    → [POL-021] EvaluateStockStatus
      → [POL-022] RecommendReplenishmentAction (if low)

SupplierInvoiceReceived
  → [POL-015] EvaluateThreeWayMatch
    → ThreeWayMatchCompleted
      → [POL-017] authorize payment
      → [POL-059] PrepareValuationSnapshot
```

Count-to-Adjust Reaction Chain

```
CycleCountCompleted
  → [POL-036] IdentifyCountVariance
    → CountVarianceIdentified
      → [POL-037] route for ApproveCountVariance
        → CountVarianceApproved
          → [POL-038] SubmitAdjustment
            → [POL-039] governance evaluation
              → AdjustmentApproved
                → [POL-025] ApplyApprovedAdjustment
```

Policy-to-Event-to-Command Matrix

Policy	Trigger Event	Resulting Command(s)
POL-001	DE-037 CustomerOrderConfirmed	ReserveStock
POL-002	DE-038 CustomerOrderCancelled	ReleaseReservation
POL-005	DE-040 ShipmentDispatched	ShipStock
POL-012	DE-033 PutAwayCompleted	ReceiveStock
POL-022	DE-068 LowStockThresholdBreached	RecommendReplenishmentAction
POL-025	DE-059 AdjustmentApproved	ApplyApprovedAdjustment / ApplyApprovedWriteOff
POL-029	DE-023 PurchaseRequisitionSubmitted	RequestStockTransfer / RejectPurchaseRequisition
POL-031	DE-049 TransferDispatched	DispatchTransferStock
POL-032	DE-050 TransferReceived	ReceiveStock
POL-036	DE-053 CycleCountCompleted	IdentifyCountVariance
POL-045	DE-064 ReturnInspected	RestockReturnedGoods
POL-053	*(approval intercept)*	ValidateSegregationOfDuties

Excluded from Domain Policies

Occurrence	Treatment	Rationale
User views dashboard or report	Read model query	No reactive business workflow
User acknowledges alert without action	UI state only	Acknowledgment does not change inventory domain state
Command validation failure	Rejection response	No event raised — nothing to react to
Email/SMS delivery	Infrastructure	Delivery mechanism is not a business policy
Scheduled timer firing	Infrastructure signal	Policy reacts to the resulting domain event, not the timer

Document Approval

Role	Name	Signature	Date
Domain Expert			
DDD Architect			
Senior Business Analyst			
Solution Architect			

End of Domain Policies

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Read Models

Inventory Management System — Event Storming

Document Information

Attribute	Detail
Document ID	SRS-ES-005
Document Title	Read Models — Event Storming Catalog
Version	1.0
Status	Draft
Document Owner	Domain Expert
Source Documents	SRS-ES-001 (Domain Events), SRS-DM-003 (Domain Aggregates), SRS-DM-006 (Domain Events), SRS-UC-001 (Use Case Catalog), SRS-DM-004 (Bounded Contexts — Inventory Intelligence)
Date	July 6, 2026
Scope	Business information projections only — no database schemas, SQL, or UI wireframes

Purpose

This document identifies **Read Models** required by the Inventory Management System. A read model is a purpose-built projection that answers a specific business question by consuming domain events. Read models support inquiry, monitoring, decision-making, and compliance — they never mutate transactional aggregate state.

Read models belong to the **Inventory Intelligence** bounded context. They are eventually consistent with operational writes and may be rebuilt from the event stream.

Read Model Catalog Summary

Domain Area	Read Model IDs	Count
Inventory Visibility & Availability	RM-001 – RM-004	4
Alerts & Notifications	RM-005 – RM-006	2
Procurement & Inbound Supply	RM-007 – RM-012	6
Order Fulfillment	RM-013 – RM-016	4
Network Transfer & Balance	RM-017 – RM-019	3
Inventory Accuracy & Control	RM-020 – RM-025	6
Commercial & Promotional	RM-026 – RM-027	2
Financial & Valuation	RM-028 – RM-031	4
Governance & Compliance	RM-032 – RM-034	3
Master Data & Reference	RM-035 – RM-039	5
Total		**39**

Domain 1: Inventory Visibility & Availability

RM-001 — Network Stock Status Board

Field	Detail
Name	Network Stock Status Board
Purpose	Answer "*What do we have, where is it, and what is its health?*" across the location network for daily operational monitoring.
Information Displayed	On-hand, reserved, quarantined, and ATP quantities by product and location; stock status indicator (in stock, low stock, out of stock, overstock); ABC classification; last movement date; ownership classification (owned vs. consignment)

Field	Detail
Primary Users	Inventory Controller, Operations Director, Warehouse Manager, Purchasing Manager
Source Events	StockReceived, StockShipped, StockReserved, StockReservationReleased, StockAdjusted, StockWrittenOff, QuarantineHoldApplied, QuarantineHoldReleased, PromotionalStockAllocated, PromotionalAllocationReleased, ConsignmentOwnershipTransferred, StockPolicyThresholdChanged, ProductABCClassificationAssigned, LowStockThresholdBreached, OutOfStockDetected, OverstockDetected
Use Cases	UC-012, UC-037

RM-002 — Availability Inquiry View

Field	Detail
Name	Availability Inquiry View
Purpose	Answer "*Can we promise this quantity to a customer at this location or across the network?*" before commercial commitment.
Information Displayed	ATP by product and location; network-wide ATP rollup; reserved and quarantined exclusions; promotional ring-fence impact; product lifecycle eligibility; partial availability breakdown per order line
Primary Users	Sales Representative, Customer Service Representative, Sales Manager
Source Events	OrderAvailabilityConfirmed, StockReserved, StockReservationReleased, StockReceived, StockShipped, QuarantineHoldApplied, QuarantineHoldReleased, PromotionalStockAllocated, ProductStatusChanged, LowStockThresholdBreached, OutOfStockDetected
Use Cases	UC-013, UC-021, UC-048

RM-003 — Quarantine and Hold Register

Field	Detail
Name	Quarantine and Hold Register
Purpose	Answer "*What stock is physically present but not sellable, and why?*" for quality, regulatory, and customs oversight.
Information Displayed	Held quantity by product, location, and lot/batch; hold reason (quality, recall, customs, bonded); hold applied date and attribution; expected disposition; ATP exclusion impact
Primary Users	Warehouse Manager, Inventory Controller, Operations Director
Source Events	QuarantineHoldApplied, QuarantineHoldReleased, ReturnQuarantined, GoodsReceiptVerified *(bonded)*
Use Cases	UC-019, UC-037

RM-004 — Availability Inquiry Log

Field	Detail
Name	Availability Inquiry Log
Purpose	Answer "*What availability checks were performed and what was promised?*" for sales accountability and dispute resolution.
Information Displayed	Inquiry timestamp, product, requested quantity, location scope, ATP result, confirming user, linked order reference (if confirmed)
Primary Users	Sales Representative, Customer Service Representative, Inventory Controller
Source Events	OrderAvailabilityConfirmed
Use Cases	UC-013, UC-021

Domain 2: Alerts & Notifications

RM-005 — Operational Alert Board

Field	Detail
Name	Operational Alert Board
Purpose	Answer "*What requires my attention right now?*" by surfacing threshold breaches, delays, exceptions, and pending approvals.
Information Displayed	Alert type (low stock, out of stock, inbound delay, transfer delay, fulfillment exception, promotional readiness gap, pending approval); severity and ABC priority; affected product and location; alert age; acknowledgment status; recommended action
Primary Users	Warehouse Manager, Purchasing Manager, Buyer, Operations Director, Marketing Manager, Customer Service Representative
Source Events	LowStockThresholdBreached, OutOfStockDetected, OverstockDetected, InboundDeliveryDelayed, InTransitStockDelayed, FulfillmentExceptionRaised, PromotionalReadinessGapIdentified, TransferRequested, AdjustmentSubmitted, ThreeWayMatchExceptionRaised, ReplenishmentRecommended, NetworkImbalanceDetected
Use Cases	UC-028, UC-047

RM-006 — Replenishment Recommendation Log

Field	Detail
Name	Replenishment Recommendation Log
Purpose	Answer "*What replenishment action did the business recommend and when?*" for purchasing and network balance decisions.
Information Displayed	Recommendation timestamp, product, location, triggering event, recommended action (purchase, transfer, hold), network availability context, ABC priority, acknowledgment status
Primary Users	Purchasing Manager, Buyer, Regional / Store Manager, Operations Director
Source Events	ReplenishmentRecommended, LowStockThresholdBreached, OutOfStockDetected, BackorderRecorded, NetworkImbalanceDetected
Use Cases	UC-028, UC-029, UC-034

Domain 3: Procurement & Inbound Supply

RM-007 — Open Purchase Order Board

Field	Detail
Name	Open Purchase Order Board
Purpose	Answer "*What have we committed to buy and what is still outstanding?*" for procurement control and inbound planning.
Information Displayed	PO number, supplier, status (open, partial, closed, cancelled); line quantities ordered vs. received; receiving location; expected delivery date; linked requisition reference
Primary Users	Buyer, Purchasing Manager, Receiving Clerk, Finance Manager
Source Events	PurchaseOrderPlaced, PurchaseOrderCancelled, GoodsReceiptVerified, PutAwayCompleted, ReceiptDiscrepancyRecorded, ReceiptDiscrepancyResolved
Use Cases	UC-006, UC-030, UC-008

RM-008 — Inbound Supply Calendar

Field	Detail
Name	Inbound Supply Calendar
Purpose	Answer "*What inbound supply is expected and when?*" for dock preparation and replenishment timeline planning.

Field	Detail
Information Displayed	Expected arrival date, supplier, PO reference, product lines and quantities, destination location, carrier information, delay status, receipt completion status
Primary Users	Buyer, Receiving Clerk, Warehouse Manager, Purchasing Manager
Source Events	InboundDeliveryScheduled, InboundDeliveryDelayed, PurchaseOrderPlaced, GoodsReceiptVerified, PutAwayCompleted
Use Cases	UC-007, UC-030

RM-009 — Purchase Requisition Pipeline

Field	Detail
Name	Purchase Requisition Pipeline
Purpose	Answer "*What internal procurement needs are pending, approved, or rejected?*" before commercial PO creation.
Information Displayed	Requisition status, requestor, product lines, target location, submission date, approval outcome, linked low-stock alert, conversion to PO status
Primary Users	Buyer, Purchasing Manager, Inventory Controller
Source Events	PurchaseRequisitionSubmitted, PurchaseRequisitionApproved, PurchaseRequisitionRejected, PurchaseOrderPlaced
Use Cases	UC-029

RM-010 — Supplier Performance Scorecard

Field	Detail
Name	Supplier Performance Scorecard
Purpose	Answer "*How reliably does each supplier deliver as ordered?*" for supplier accountability and sourcing decisions.
Information Displayed	Supplier, receipt discrepancy count and types, short-shipment frequency, on-time delivery rate, shelf-life rejection rate, unresolved claim status
Primary Users	Purchasing Manager, Buyer, Finance Manager, Operations Director
Source Events	ReceiptDiscrepancyRecorded, ReceiptDiscrepancyResolved, InboundDeliveryDelayed, GoodsReceiptVerified, ThreeWayMatchExceptionRaised
Use Cases	UC-009, UC-030, UC-037

RM-011 — Receipt Discrepancy Register

Field	Detail
Name	Receipt Discrepancy Register
Purpose	Answer "*What receipt variances are open, resolved, or pending supplier action?*" for inbound exception management.
Information Displayed	Discrepancy type, PO and receipt reference, product, variance quantity, reason code, evidence notes, resolution status, accepted vs. rejected quantities
Primary Users	Receiving Clerk, Warehouse Manager, Buyer, Finance Manager
Source Events	ReceiptDiscrepancyRecorded, ReceiptDiscrepancyResolved, GoodsReceiptVerified
Use Cases	UC-009

RM-012 — Put-Away Work Queue

Field	Detail
Name	Put-Away Work Queue
Purpose	Answer "*What verified stock is waiting to be stowed before it becomes sellable?*" for warehouse task prioritization.
Information Displayed	Receipt reference, product, quantity, receiving dock location, suggested storage position, rotation policy (FIFO/FEFO), bonded hold status, put-away completion status
Primary Users	Warehouse Staff, Warehouse Manager, Receiving Clerk
Source Events	GoodsReceiptVerified, PutAwayCompleted, QuarantineHoldApplied
Use Cases	UC-010

Domain 4: Order Fulfillment

RM-013 — Order Fulfillment Status Tracker

Field	Detail
Name	Order Fulfillment Status Tracker
Purpose	Answer "*Where is each customer order in the fulfillment lifecycle?*" for customer service and operations visibility.
Information Displayed	Order reference, line status (confirmed, reserved, picking, shipped, backordered, cancelled); reservation quantity; pick verification status; shipment and carrier details; fulfillment location; exception flags
Primary Users	Customer Service Representative, Warehouse Manager, Sales Representative, Operations Director
Source Events	CustomerOrderConfirmed, CustomerOrderCancelled, StockReserved, StockReservationReleased, PickVerified, ShipmentDispatched, PartialFulfillmentCompleted, BackorderRecorded, BackorderFulfilled, FulfillmentExceptionRaised, MultiLocationFulfillmentPlanCreated
Use Cases	UC-027, UC-024, UC-025

RM-014 — Backorder Pipeline

Field	Detail
Name	Backorder Pipeline
Purpose	Answer "*What customer commitments are waiting for stock and how long?*" for priority replenishment and customer communication.
Information Displayed	Order reference, product, backordered quantity, customer acknowledgment date, age, priority, linked replenishment status, fulfillment progress
Primary Users	Customer Service Representative, Purchasing Manager, Buyer, Warehouse Manager
Source Events	BackorderRecorded, BackorderFulfilled, StockReceived, ReplenishmentRecommended
Use Cases	UC-025, UC-028

RM-015 — Fill Rate Performance Report

Field	Detail
Name	Fill Rate Performance Report
Purpose	Answer "*How completely and on time are we fulfilling customer demand?*" for service-level management.
Information Displayed	Fill rate by product, location, and period; partial fulfillment frequency; backorder rate; on-time shipment rate; order line completion time

Field	Detail
Primary Users	Operations Director, Sales Manager, Chief Executive Officer, Warehouse Manager
Source Events	ShipmentDispatched, PartialFulfillmentCompleted, BackorderRecorded, BackorderFulfilled, FulfillmentExceptionRaised, CustomerOrderConfirmed
Use Cases	UC-037, UC-038

RM-016 — Fulfillment Work Queue

Field	Detail
Name	Fulfillment Work Queue
Purpose	Answer "*What orders are ready to pick, pack, and ship?*" for warehouse execution prioritization.
Information Displayed	Order priority, ship-from location, product lines and quantities, reservation status, pick path hints, rotation policy, lot/batch requirements, freeze status at location
Primary Users	Warehouse Picker / Packer, Warehouse Manager
Source Events	CustomerOrderConfirmed, StockReserved, MultiLocationFulfillmentPlanCreated, PickVerified, LocationFreezeImposed, LocationFreezeReleased
Use Cases	UC-024

Domain 5: Network Transfer & Balance

RM-017 — Transfer Pipeline and In-Transit Tracker

Field	Detail
Name	Transfer Pipeline and In-Transit Tracker
Purpose	Answer "*What stock is moving between locations and where is it in the transfer lifecycle?*" for network visibility.
Information Displayed	Transfer reference, source and destination locations, product and quantity, status (requested, approved, in-transit, received, rejected); in-transit quantity; expected receipt date; delay flags; lot/batch attribution
Primary Users	Regional / Store Manager, Warehouse Staff, Warehouse Manager, Operations Director
Source Events	TransferRequested, TransferApproved, TransferRejected, TransferDispatched, TransferReceived, InTransitStockDelayed
Use Cases	UC-031, UC-033, UC-030

RM-018 — Network Rebalancing Opportunity View

Field	Detail
Name	Network Rebalancing Opportunity View
Purpose	Answer "*Where is stock maldistributed and what transfers would resolve imbalance?*" for proactive network optimization.
Information Displayed	Product, shortage location, excess location, imbalance severity, recommended transfer quantity, source spare capacity, transfer-before-buy opportunity flag
Primary Users	Operations Director, Inventory Controller, Purchasing Manager, Regional / Store Manager
Source Events	NetworkImbalanceDetected, OutOfStockDetected, OverstockDetected, LowStockThresholdBreached, ReplenishmentRecommended
Use Cases	UC-034, UC-031

RM-019 — Network Availability Summary

Field	Detail
Name	Network Availability Summary
Purpose	Answer "*Does the network as a whole have enough stock before we purchase?*" for replenishment and transfer decisions.
Information Displayed	Network-wide ATP by product; in-transit pipeline quantity; expected inbound quantity; location-level ATP breakdown; excess location identification
Primary Users	Buyer, Purchasing Manager, Operations Director, Regional / Store Manager
Source Events	StockReceived, StockShipped, TransferDispatched, TransferReceived, InboundDeliveryScheduled, PurchaseOrderPlaced, LowStockThresholdBreached
Use Cases	UC-029, UC-031, UC-034

Domain 6: Inventory Accuracy & Control

RM-020 — Inventory Movement History

Field	Detail
Name	Inventory Movement History
Purpose	Answer "*What changed, where, when, and why?*" for operational investigation, traceability, and accountability.
Information Displayed	Movement type (receipt, shipment, adjustment, transfer, reservation, hold); product, location, quantity delta; lot/batch; reason code; user attribution; timestamp; linked transaction reference
Primary Users	Inventory Controller, Warehouse Manager, Finance Manager, Internal Auditor, Operations Director
Source Events	InventoryMovementRecorded *(and all quantity mutation events: StockReceived, StockShipped, StockAdjusted, StockWrittenOff, StockReserved, StockReservationReleased, QuarantineHoldApplied, QuarantineHoldReleased, ConsignmentOwnershipTransferred)*
Use Cases	UC-042, UC-037

RM-021 — Cycle Count Schedule and Completion Log

Field	Detail
Name	Cycle Count Schedule and Completion Log
Purpose	Answer "*What counts are scheduled, completed, or overdue?*" for inventory accuracy program management.
Information Displayed	Scheduled count date, product, location, assigned counter, ABC class, completion status, physical count result, variance flag, count frequency compliance
Primary Users	Inventory Controller, Warehouse Staff, Warehouse Manager
Source Events	CycleCountScheduled, CycleCountCompleted, CountVarianceIdentified, ProductABCClassificationAssigned
Use Cases	UC-014

RM-022 — Physical Inventory Count Status Board

Field	Detail
Name	Physical Inventory Count Status Board
Purpose	Answer "*What is the status of the wall-to-wall count and location freeze?*" during full physical inventory events.
Information Displayed	Location, count event status, freeze window, team assignments, lines counted vs. total, variance count, certification status, freeze active indicator
Primary Users	Inventory Controller, Warehouse Manager, Finance Manager, Operations Director

Field	Detail
Source Events	PhysicalInventoryCountStarted, PhysicalInventoryCountCertified, LocationFreezeImposed, LocationFreezeReleased, CountVarianceIdentified
Use Cases	UC-015

RM-023 — Inventory Accuracy Metrics Report

Field	Detail
Name	Inventory Accuracy Metrics Report
Purpose	Answer "*How accurate is our inventory by location, product class, and period?*" for continuous improvement and audit evidence.
Information Displayed	Accuracy percentage by location and ABC class; variance count and value; count completion rate; adjustment frequency; accuracy trend over time; post-physical-count baseline
Primary Users	Inventory Controller, Operations Director, Internal Auditor, Chief Executive Officer, Finance Manager
Source Events	CycleCountCompleted, CountVarianceIdentified, CountVarianceApproved, PhysicalInventoryCountCertified, StockAdjusted, AdjustmentApproved
Use Cases	UC-016, UC-037, UC-038, UC-041

RM-024 — Count Variance Investigation Board

Field	Detail
Name	Count Variance Investigation Board
Purpose	Answer "*What count discrepancies are under investigation and awaiting approval?*" for variance workflow management.
Information Displayed	Variance reference, product, location, system vs. physical quantity, variance amount, counter identity, investigation status, approver, approval outcome, linked adjustment case
Primary Users	Warehouse Manager, Inventory Controller, Operations Director
Source Events	CountVarianceIdentified, CountVarianceApproved, AdjustmentSubmitted, AdjustmentApproved, AdjustmentRejected
Use Cases	UC-016

RM-025 — Shrinkage and Loss Analysis Report

Field	Detail
Name	Shrinkage and Loss Analysis Report
Purpose	Answer "*Where is inventory loss occurring and what are the patterns?*" for corrective action and working capital optimization.
Information Displayed	Adjustment and write-off volume by reason code, location, and product; shrinkage rate trend; obsolete and slow-moving stock; loss value by category; repeat variance hotspots
Primary Users	Inventory Controller, Finance Manager, Operations Director, Purchasing Manager
Source Events	StockAdjusted, StockWrittenOff, AdjustmentApproved, CountVarianceIdentified, ReturnWrittenOff, ProductStatusChanged
Use Cases	UC-020, UC-037

Domain 7: Commercial & Promotional

RM-026 — Promotional Allocation and Readiness View

Field	Detail
Name	Promotional Allocation and Readiness View
Purpose	Answer "*Is ring-fenced campaign stock sufficient and protected?*" before commercial launch commitment.
Information Displayed	Campaign name and period, product, allocated quantity, consumed quantity, remaining ring-fence, general ATP impact, readiness status, gap severity, channel designation
Primary Users	Marketing Manager, Purchasing Manager, Warehouse Manager, Sales Manager
Source Events	PromotionalStockAllocated, PromotionalAllocationReleased, PromotionalReadinessGapIdentified, StockReserved, StockShipped, StockReceived
Use Cases	UC-040

RM-027 — Campaign Performance Report

Field	Detail
Name	Campaign Performance Report
Purpose	Answer "*How did promotional inventory perform against campaign commitment?*" after or during campaign execution.
Information Displayed	Campaign period, allocated vs. consumed quantity, sell-through rate, unallocated remainder returned to general pool, stockout during campaign, fulfillment from promotional pool vs. general ATP
Primary Users	Marketing Manager, Operations Director, Sales Manager
Source Events	PromotionalStockAllocated, PromotionalAllocationReleased, StockShipped, StockReserved, PromotionalReadinessGapIdentified
Use Cases	UC-040, UC-037

Domain 8: Financial & Valuation

RM-028 — Inventory Valuation Snapshot

Field	Detail
Name	Inventory Valuation Snapshot
Purpose	Answer "*What is the owned inventory value at a point in time?*" for finance reconciliation and executive reporting.
Information Displayed	Period-end date, owned quantity by product and location, unit cost, extended value, consignment exclusion, quarantine exclusion, in-transit exclusion, total valuation
Primary Users	Finance Manager, Chief Financial Officer, Inventory Controller, Chief Executive Officer
Source Events	InventoryValuationSnapshotPrepared, StockReceived, StockShipped, StockWrittenOff, ConsignmentOwnershipTransferred, AdjustmentApproved, ThreeWayMatchCompleted
Use Cases	UC-039, UC-038

RM-029 — Valuation Reconciliation Status Board

Field	Detail
Name	Valuation Reconciliation Status Board
Purpose	Answer "*Are operational inventory records aligned with the general ledger?*" for month-end close management.
Information Displayed	Snapshot reference, operational valuation total, general ledger balance, variance amount, tolerance threshold, reconciliation status, investigation notes, close readiness indicator
Primary Users	Finance Manager, Chief Financial Officer, Inventory Controller, Internal Auditor

Field	Detail
Source Events	InventoryValuationSnapshotPrepared, ValuationVarianceIdentified, StockAdjusted, StockWrittenOff, ConsignmentOwnershipTransferred
Use Cases	UC-039

RM-030 — Three-Way Match Status Board

Field	Detail
Name	Three-Way Match Status Board
Purpose	Answer "*Which supplier invoices are matched, blocked, or awaiting resolution?*" for payment control.
Information Displayed	Invoice reference, supplier, PO and receipt linkage, match status (matched, exception, pending), quantity and price variance, payment authorization status
Primary Users	Finance Manager, Buyer, Purchasing Manager, Chief Financial Officer
Source Events	SupplierInvoiceReceived, ThreeWayMatchCompleted, ThreeWayMatchExceptionRaised, ReceiptDiscrepancyResolved, GoodsReceiptVerified
Use Cases	UC-039, UC-006

RM-031 — Executive KPI Dashboard

Field	Detail
Name	Executive KPI Dashboard
Purpose	Answer "*How is inventory performing as a strategic business asset?*" for leadership decision-making.
Information Displayed	Inventory turnover, fill rate, accuracy rate, stockout frequency, days of supply, working capital tied in inventory, shrinkage rate, network imbalance index, valuation trend
Primary Users	Chief Executive Officer, Chief Financial Officer, Chief Operating Officer, Operations Director
Source Events	InventoryValuationSnapshotPrepared, OutOfStockDetected, OverstockDetected, ShipmentDispatched, PartialFulfillmentCompleted, CycleCountCompleted, CountVarianceIdentified, StockWrittenOff, NetworkImbalanceDetected, LowStockThresholdBreached
Use Cases	UC-038, UC-037

Domain 9: Governance & Compliance

RM-032 — Audit Trail Register

Field	Detail
Name	Audit Trail Register
Purpose	Answer "*Who changed what, when, and with whose approval?*" for governance, compliance, and fraud investigation.
Information Displayed	Transaction type, entity affected, before/after state, user identity, timestamp, approval chain, reason code, location scope
Primary Users	Internal Auditor, External Auditor, Finance Manager, Operations Director, System Administrator
Source Events	AuditTrailRecorded *(and governed events: AdjustmentApproved, ProductStatusChanged, RoleAssignmentChanged, GovernancePolicyPublished, CountVarianceApproved, StockWrittenOff, SegregationOfDutiesViolationDetected)*
Use Cases	UC-043, UC-041

RM-033 — Adjustment Case History

Field	Detail
Name	Adjustment Case History
Purpose	Answer "*What inventory corrections and write-offs were proposed, approved, or rejected?*" for governance review.
Information Displayed	Case reference, type (variance, adjustment, write-off), product, location, quantity, reason code, initiator, approver, status, investigation notes, threshold breach flag
Primary Users	Warehouse Manager, Operations Director, Inventory Controller, Internal Auditor, Finance Manager
Source Events	AdjustmentSubmitted, AdjustmentApproved, AdjustmentRejected, CountVarianceIdentified, CountVarianceApproved, StockAdjusted, StockWrittenOff
Use Cases	UC-017, UC-018, UC-016, UC-041

RM-034 — Segregation of Duties Violation Log

Field	Detail
Name	Segregation of Duties Violation Log
Purpose	Answer "*Were accountability controls breached?*" for control testing and security review.
Information Displayed	Violation timestamp, attempted action, user identity, conflicting roles, blocked command, affected transaction reference
Primary Users	Internal Auditor, System Administrator, Operations Director
Source Events	SegregationOfDutiesViolationDetected
Use Cases	UC-041, UC-045

Domain 10: Master Data & Reference

RM-035 — Product Catalog Directory

Field	Detail
Name	Product Catalog Directory
Purpose	Answer "*What is the authoritative product identity and definition?*" for all operational lookups and references.
Information Displayed	SKU, name, description, unit of measure, category, barcode, lifecycle status, ABC class, rotation policy, traceability requirements, shelf-life profile
Primary Users	Inventory Controller, Sales Representative, Buyer, Receiving Clerk, Warehouse Staff
Source Events	ProductRegistered, ProductStatusChanged, ProductABCClassificationAssigned
Use Cases	UC-001, UC-013, UC-046

RM-036 — Location Network Directory

Field	Detail
Name	Location Network Directory
Purpose	Answer "*What locations exist, how are they organized, and are they operational?*" for network-scoped operations.
Information Displayed	Location code, name, type, hierarchy path, operational status, freeze status, go-live date, parent location, storage position count
Primary Users	Inventory Controller, Warehouse Manager, Regional / Store Manager, System Administrator

Field	Detail
Source Events	InventoryLocationActivated, InventoryLocationDeactivated, StoragePositionConfigured, LocationFreezeImposed, LocationFreezeReleased
Use Cases	UC-002, UC-046

RM-037 — Warehouse Layout Reference

Field	Detail
Name	Warehouse Layout Reference
Purpose	Answer "*Where should stock be placed or picked from?*" for put-away and pick path operations.
Information Displayed	Storage position hierarchy (zone, aisle, bin), location assignment, capacity indicators, rotation zone designation, active/inactive status
Primary Users	Warehouse Staff, Warehouse Manager, Inventory Controller
Source Events	StoragePositionConfigured, InventoryLocationActivated
Use Cases	UC-002, UC-010

RM-038 — Governance Policy and Reason Code Catalog

Field	Detail
Name	Governance Policy and Reason Code Catalog
Purpose	Answer "*What rules and reason codes govern inventory exceptions and corrections?*" for consistent accountability.
Information Displayed	Active governance policy version, effective period, approval thresholds, tolerance limits, reason code list with category and status (active/retired), applicable transaction types
Primary Users	Inventory Controller, Warehouse Manager, Internal Auditor, Finance Manager
Source Events	GovernancePolicyPublished, ReasonCodeRetired
Use Cases	UC-005, UC-017, UC-041

RM-039 — Stock Policy Registry

Field	Detail
Name	Stock Policy Registry
Purpose	Answer "*What replenishment thresholds apply to each product at each location?*" for monitoring and planning.
Information Displayed	Product, location, reorder level, minimum threshold, safety stock, maximum level, effective date, prior threshold history, policy change attribution
Primary Users	Inventory Controller, Purchasing Manager, Operations Director
Source Events	StockPolicyThresholdChanged, ProductRegistered, InventoryLocationActivated
Use Cases	UC-003, UC-012

Read Model-to-Event Source Matrix

Read Model	Primary Source Events	Event Count
RM-001 Network Stock Status Board	Stock mutation + status events	16
RM-005 Operational Alert Board	Threshold, delay, exception events	12
RM-007 Open Purchase Order Board	PO and receipt lifecycle	6

Read Model	Primary Source Events	Event Count
RM-013 Order Fulfillment Status Tracker	Order fulfillment lifecycle	11
RM-020 Inventory Movement History	InventoryMovementRecorded + mutations	10+
RM-028 Inventory Valuation Snapshot	Valuation and quantity events	7
RM-032 Audit Trail Register	AuditTrailRecorded + governed events	8+
RM-035 Product Catalog Directory	Product lifecycle events	3

Read Model-to-Use Case Coverage

Use Case	Read Models Served
UC-012 Monitor Inventory Status	RM-001, RM-039
UC-013 Inquire Availability	RM-002, RM-004, RM-035
UC-020 Analyze Shrinkage	RM-025, RM-033
UC-027 View Order Fulfillment Status	RM-013, RM-014
UC-028 Review Low-Stock Alerts	RM-005, RM-006
UC-030 Track Inbound Supply	RM-007, RM-008
UC-037 Operational Dashboards	RM-001, RM-015, RM-023, RM-025, RM-031
UC-038 Executive KPIs	RM-031, RM-028, RM-023
UC-039 Reconcile Valuation	RM-028, RM-029, RM-030
UC-040 Promotional Readiness	RM-026, RM-027
UC-041 Audit Review	RM-032, RM-033, RM-034, RM-023
UC-042 Movement History	RM-020
UC-043 Audit Trail	RM-032
UC-047 Operational Notifications	RM-005

Excluded from Read Models

Capability	Treatment	Rationale
Aggregate current state query	Write model	ATP and on-hand authority lives in Stock Position aggregate — availability views project from events but operational commands read aggregate state for consistency
Ad-hoc SQL analytics	External BI	Exploratory analysis outside IMS bounded context
Real-time command validation	Domain service	Eligibility and freeze checks are synchronous evaluations, not read models
User session and permission cache	Infrastructure	Identity layer concern, not inventory intelligence

Document Approval

Role	Name	Signature	Date
Domain Expert			
DDD Architect			
Senior Business Analyst			

Role	Name	Signature	Date
Solution Architect			

End of Read Models

Document ID: SRS-ES-005 | Version 1.0 | July 6, 2026

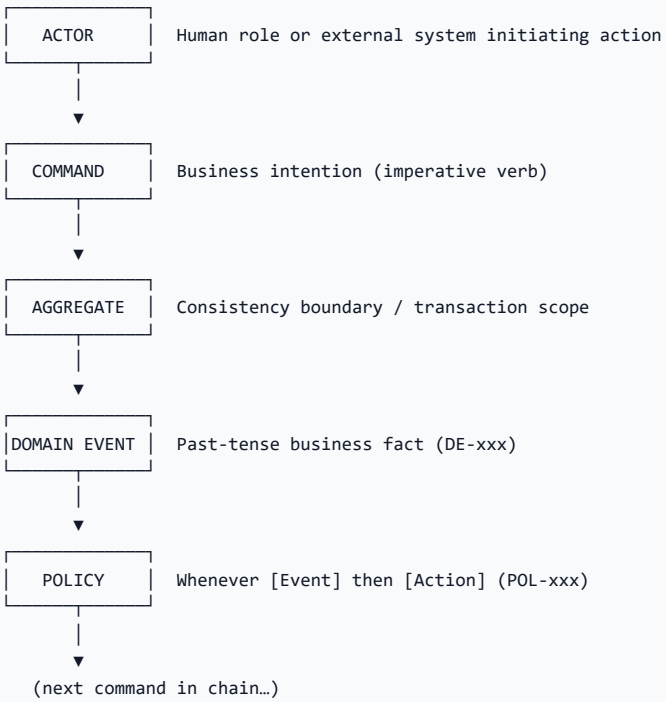
Event Storming Board

Inventory Management System — Complete Business Lifecycle

Document Information

Attribute	Detail
Document ID	SRS-ES-008
Document Title	Event Storming Board — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Domain Expert
Source Documents	SRS-ES-001 (Events), SRS-ES-002 (Commands), SRS-ES-003 (Policies), SRS-ES-004 (Actors), SRS-DM-003 (Aggregates)
Date	July 6, 2026

Board Legend



Symbol	Meaning
`[Actor]`	Business participant

Symbol	Meaning
→ CMD	Command issued
⇒ AGG	Target aggregate
⇒ EVT	Domain event raised
⇒ POL	Policy reaction
┄┄┄	Optional / conditional branch
═══	Cross-aggregate reaction (eventual consistency)

Lifecycle Index

Board	Lifecycle	Primary Actors
1	Foundation & Go-Live	Inventory Controller
2	Replenishment Detection	System / Purchasing Manager
3	Procurement & Inbound Supply	Buyer → Receiving Clerk → Warehouse Staff → Finance Manager
4	Order Fulfillment	Sales Rep → CS Rep → Picker
5	Partial Fulfillment & Backorder	Customer Service Representative
6	Network Transfer	Regional Manager → Operations Director → Warehouse Staff
7	Cycle Count & Adjustment	Warehouse Staff → Warehouse Manager → Operations Director
8	Physical Inventory Count	Inventory Controller
9	Customer Returns	CS Rep → Receiving Clerk → Warehouse Manager
10	Promotional Campaign	Marketing Manager
11	Quarantine & Hold	Warehouse Manager
12	Finance & Valuation Close	Finance Manager

Master Board — End-to-End Business Lifecycle

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PHASE 1: FOUNDATION
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Inventory Controller
  → RegisterProduct              ⇒ Product Catalog        ⇒ ProductRegistered
  → ActivateInventoryLocation    ⇒ Inventory Location     ⇒ InventoryLocationActivated
    ⇒ POL-064: Establish Opening Balances
  → EstablishOpeningBalances     ⇒ Stock Position         ⇒ StockReceived
  → UpdateStockPolicyThresholds  ⇒ Stock Policy           ⇒ StockPolicyThresholdChanged
  → PublishGovernancePolicy      ⇒ Inventory Governance   ⇒ GovernancePolicyPublished

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PHASE 2: REPLENISHMENT & INBOUND SUPPLY
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[Low stock detected]
  ⇒ Stock Position               ⇒ LowStockThresholdBreached
    ⇒ POL-022: Trigger Replenishment
  → RecommendReplenishmentAction ⇒ (Domain Service)       ⇒ ReplenishmentRecommended

Purchasing Manager / Buyer
  → SubmitPurchaseRequisition    ⇒ Purchase Requisition   ⇒ PurchaseRequisitionSubmitted
    ⇒ POL-029: Evaluate Network Before Procurement
  → ApprovePurchaseRequisition   ⇒ Purchase Requisition   ⇒ PurchaseRequisitionApproved
    ⇒ POL-019: Convert to PO
  → PlacePurchaseOrder           ⇒ Purchase Order         ⇒ PurchaseOrderPlaced
    ⇒ POL-020: Schedule Inbound
  → ScheduleInboundDelivery      ⇒ Purchase Order         ⇒ InboundDeliveryScheduled

Supplier (external) delivers goods
Receiving Clerk
  → VerifyGoodsReceipt           ⇒ Goods Receipt          ⇒ GoodsReceiptVerified
    ⇒ POL-011: Stage for Put-Away
Warehouse Staff
  → CompletePutAway              ⇒ Goods Receipt          ⇒ PutAwayCompleted
    ⇒ POL-012: Activate Stock
  → ReceiveStock                 ⇒ Stock Position         ⇒ StockReceived
    ⇒ POL-021: Reclassify Stock Status

Finance Manager
  → RecordSupplierInvoice        ⇒ Supplier Invoice       ⇒ SupplierInvoiceReceived
    ⇒ POL-015: Evaluate Match
  → EvaluateThreeWayMatch        ⇒ Supplier Invoice       ⇒ ThreeWayMatchCompleted
    ⇒ POL-017: Authorize Payment

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PHASE 3: ORDER FULFILLMENT
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Sales Representative
  → ConfirmOrderAvailability     ⇒ Customer Order         ⇒ OrderAvailabilityConfirmed
Customer Service Representative
  → ConfirmCustomerOrder           ⇒ Customer Order         ⇒ CustomerOrderConfirmed
    ⇒ POL-001: Reserve Stock
  → ReserveStock                   ⇒ Stock Position         ⇒ StockReserved
Warehouse Picker
  → VerifyPick                     ⇒ Customer Order         ⇒ PickVerified
    ⇒ POL-004: Authorize Shipment
  → DispatchShipment               ⇒ Customer Order         ⇒ ShipmentDispatched
    ⇒ POL-005: Deduct Stock
  → ShipStock                      ⇒ Stock Position         ⇒ StockShipped
Logistics / Carrier (external) picks up

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PHASE 4: INVENTORY CONTROL & RETURNS (parallel ongoing operations)
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Cycle Count path ──► CountVarianceIdentified ──► AdjustmentApproved ──► StockAdjusted
Transfer path    ──► TransferRequested ──► TransferApproved ──► TransferReceived ──► StockReceived
Returns path     ──► ReturnAuthorizationIssued ──► ReturnInspected ──► ReturnRestocked ──► StockReceived
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PHASE 5: FINANCIAL CLOSE
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Finance Manager
  → PrepareValuationSnapshot     ⇒ Valuation Snapshot     ⇒ InventoryValuationSnapshotPrepared
  → ReconcileInventoryValuation  ⇒ Valuation Snapshot     ⇒ ValuationVarianceIdentified (if mismatch)
```
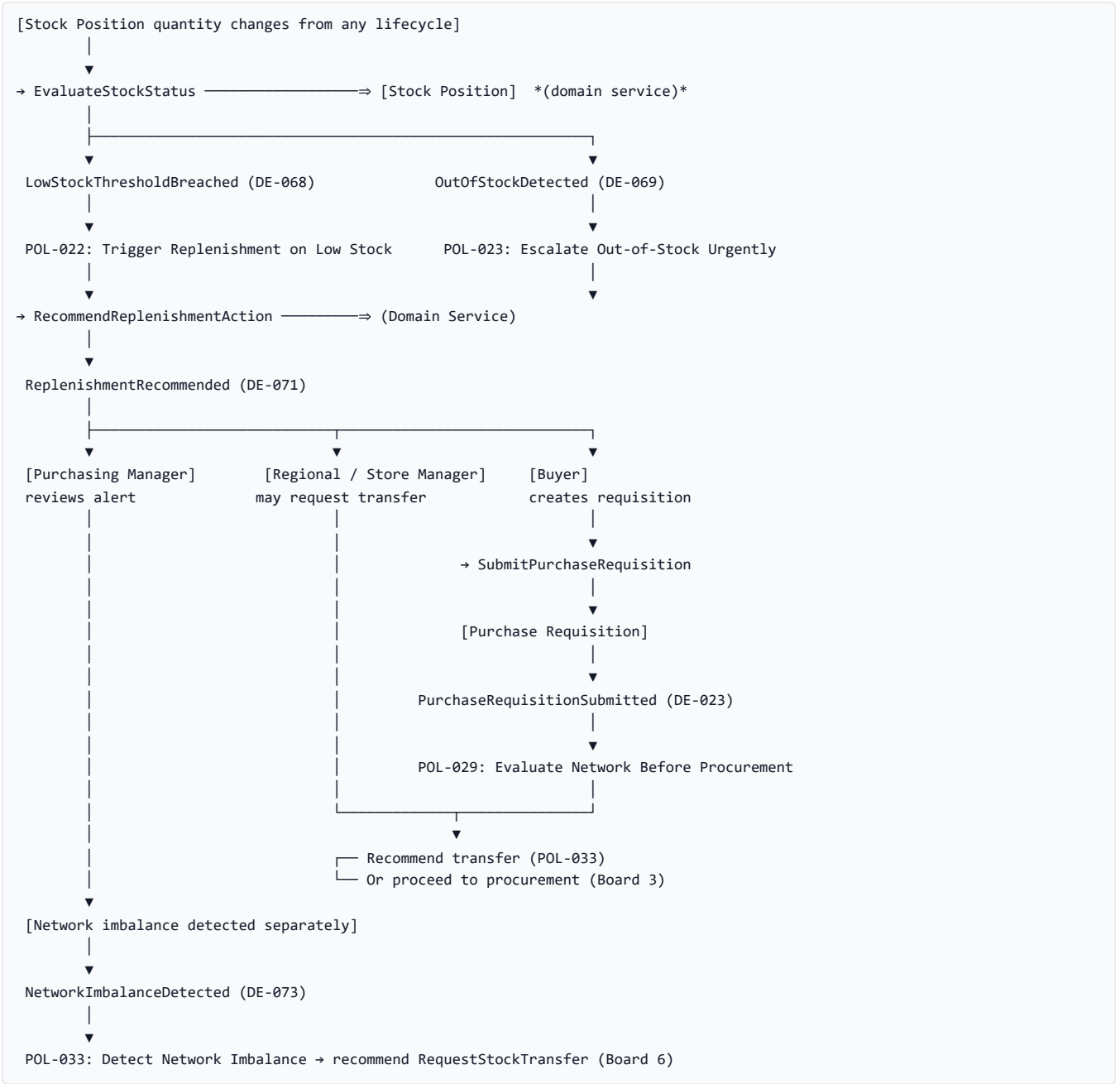

Board 1 — Foundation & Go-Live

Business Process: BP-01, BP-02, BP-03 | **Use Cases:** UC-001, UC-002, UC-003, UC-005, UC-046

Board 2 — Replenishment Detection & Planning

Business Process: BP-21, BP-22 | **Use Cases:** UC-012, UC-028, UC-029



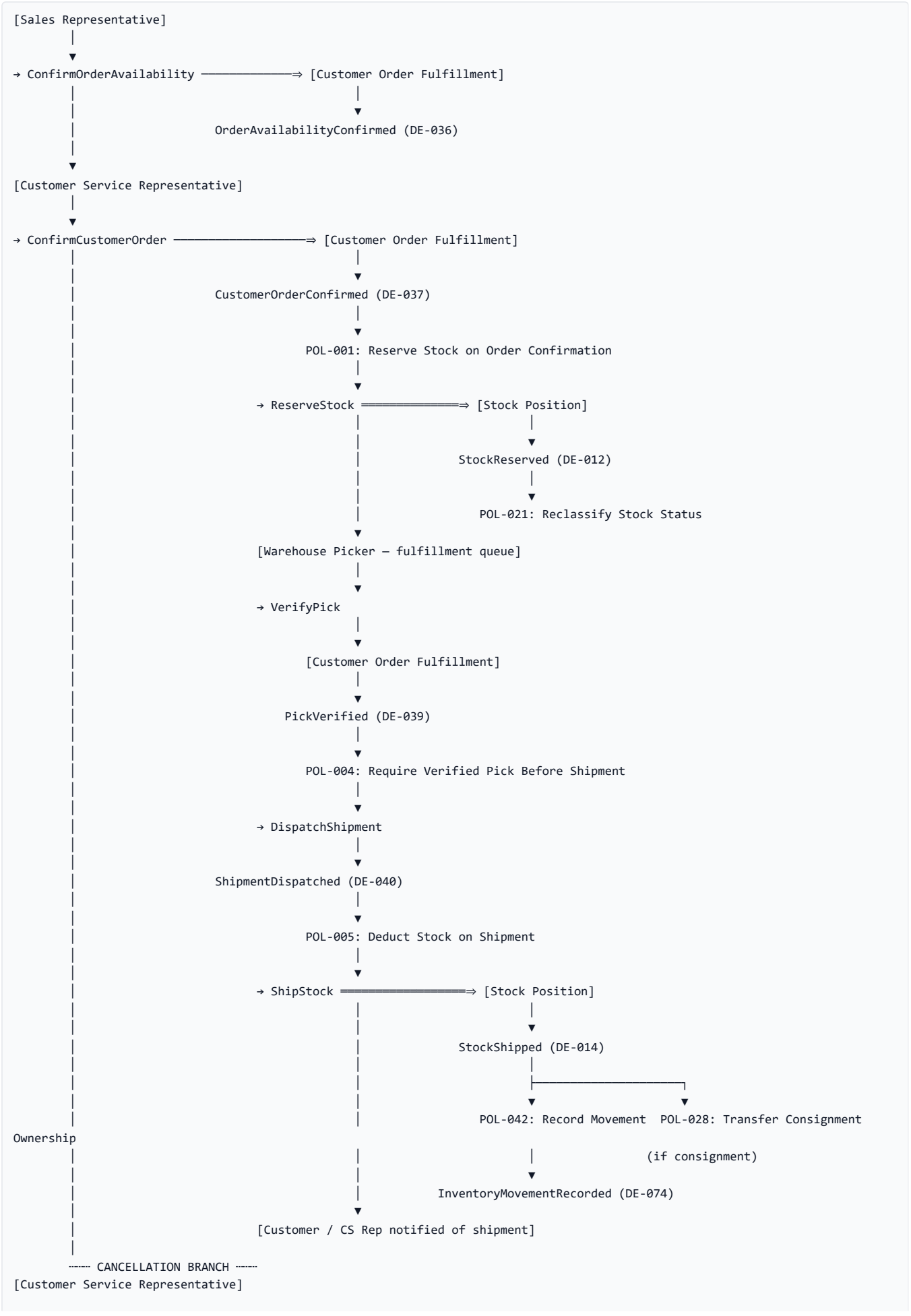
Board 3 — Procurement & Inbound Supply (Complete)

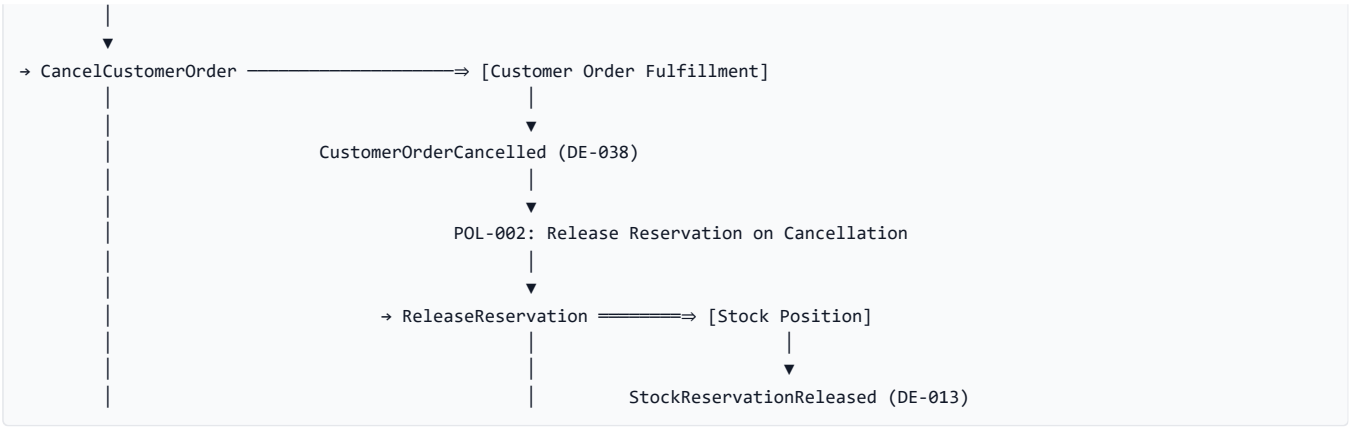
Business Process: BP-05 → BP-09, BP-30 | **Use Cases:** UC-006 – UC-010, UC-039

[Purchasing Manager]
→ ApprovePurchaseRequisition
[Purchase Requisition]
PurchaseRequisitionApproved (DE-024)
POL-019: Convert Approved Requisition to PO
[Buyer]
→ PlacePurchaseOrder
[Purchase Order]
PurchaseOrderPlaced (DE-026)
POL-020: Plan Inbound Delivery
→ ScheduleInboundDelivery
InboundDeliveryScheduled (DE-028)
[Receiving Clerk notified]
[Buyer tracks supply]
On-time delivery
Delay reported
InboundDeliveryDelayed (DE-029)
POL-057: Notify Responsible Actor
→ RecordInboundDelay
(if delayed)
[Receiving Clerk]
→ VerifyGoodsReceipt
[Goods Receipt]
(discrepancy found)
GoodsReceiptVerified (DE-030)
→ RecordReceiptDiscrepancy
POL-011: Stage for Put-Away
POL-013: Hold Bonded Stock
(if bonded)
ReceiptDiscrepancyRecorded (DE-031)
→ ApplyQuarantineHold
[Warehouse Manager]
QuarantineHoldApplied (DE-017)
→ ResolveReceiptDiscrepancy
[Warehouse Staff]
ReceiptDiscrepancyResolved (DE-032)
→ CompletePutAway
[Goods Receipt]

Board 4 — Order Fulfillment (Complete)

Business Process: BP-16 → BP-18 | **Use Cases:** UC-013, UC-021 – UC-024


[Sales Representative]
→ ConfirmOrderAvailability
[Customer Order Fulfillment]
OrderAvailabilityConfirmed (DE-036)
[Customer Service Representative]
→ ConfirmCustomerOrder
[Customer Order Fulfillment]
CustomerOrderConfirmed (DE-037)
POL-001: Reserve Stock on Order Confirmation
→ ReserveStock
[Stock Position]
StockReserved (DE-012)
POL-021: Reclassify Stock Status
[Warehouse Picker — fulfillment queue]
→ VerifyPick
[Customer Order Fulfillment]
PickVerified (DE-039)
POL-004: Require Verified Pick Before Shipment
→ DispatchShipment
ShipmentDispatched (DE-040)
POL-005: Deduct Stock on Shipment
→ ShipStock
[Stock Position]
StockShipped (DE-014)
POL-042: Record Movement
POL-028: Transfer Consignment Ownership
(if consignment)
InventoryMovementRecorded (DE-074)
[Customer / CS Rep notified of shipment]
CANCELLATION BRANCH
[Customer Service Representative]



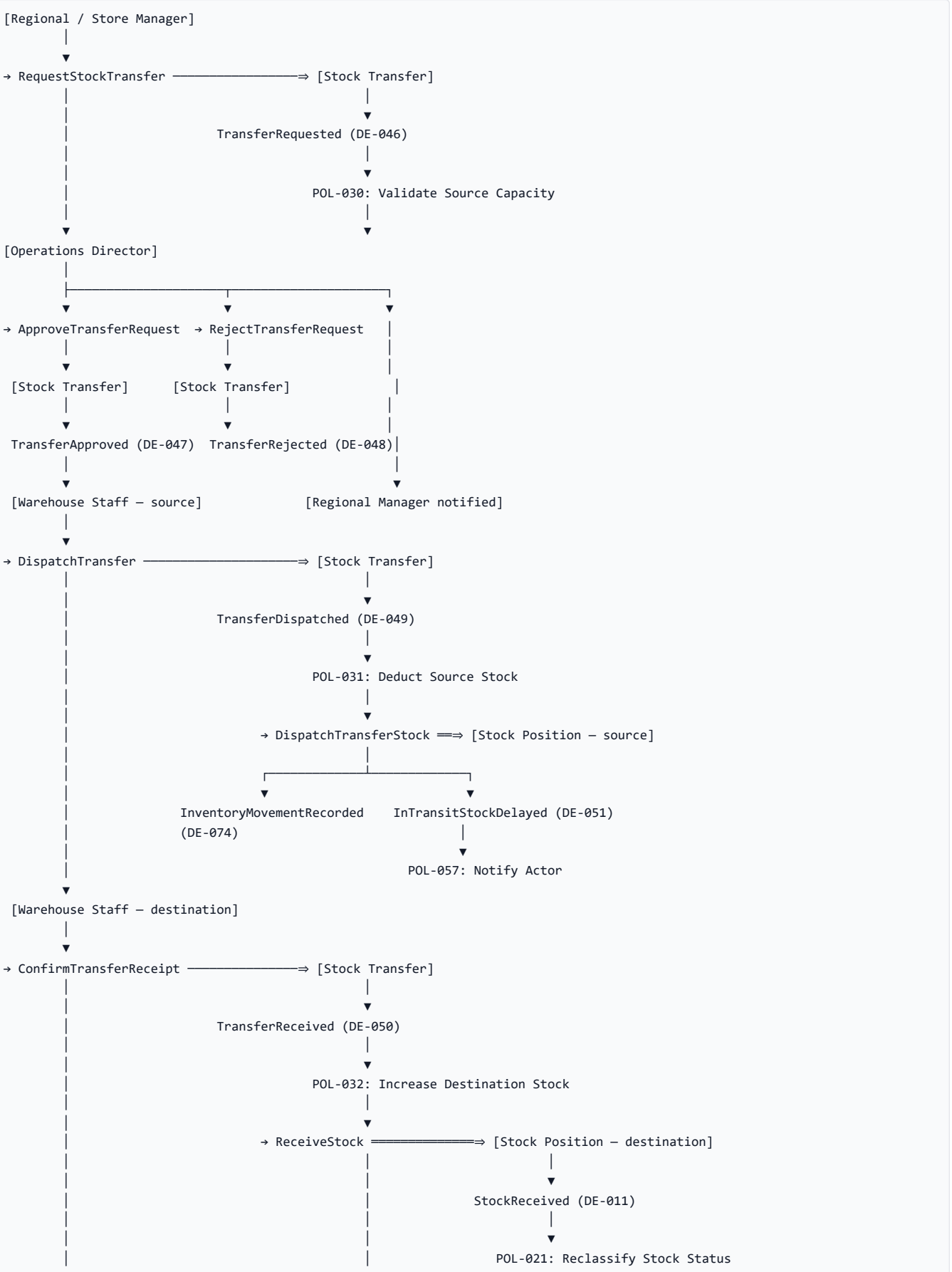
Board 5 — Partial Fulfillment, Backorder & Multi-Location

Business Process: BP-19 | **Use Cases:** UC-025, UC-048

[Customer Service Representative]
→ ConfirmCustomerOrder ⇒ [Customer Order Fulfillment]
(insufficient ATP for full quantity)
CustomerOrderConfirmed (DE-037)
→ CompletePartialFulfillment
POL-001: Reserve Stock (partial qty)
[Customer Order Fulfillment]
→ ReserveStock (partial) ⇒ StockReserved
PartialFulfillmentCompleted (DE-041)
POL-006: Record Partial Fulfillment
Ship available portion
→ RecordBackorder
→ DispatchShipment
BackorderRecorded (DE-042)
ShipmentDispatched (DE-040) → … → StockShipped
POL-007: Prioritize Backorder Replenishment
→ RecommendReplenishmentAction ⇒ ReplenishmentRecommended (DE-071)
WHEN STOCK ARRIVES (links to Board 3 / POL-008)
StockReceived (DE-011)
POL-008: Fulfill Backorder When Stock Arrives
→ FulfillBackorder
BackorderFulfilled (DE-043)
POL-001 → ReserveStock → … → DispatchShipment → StockShipped
MULTI-LOCATION BRANCH
[Warehouse Manager]
→ CreateMultiLocationFulfillmentPlan ⇒ [Customer Order Fulfillment]
MultiLocationFulfillmentPlanCreated (DE-045)
POL-003: Allocate Multi-Location Reservations
→ ReserveStock (per location) ⇒ StockReserved × N
(continues pick/ship per location on Board 4)

Board 6 — Network Transfer (Complete)

Business Process: BP-24, BP-25 | **Use Cases:** UC-031 – UC-033



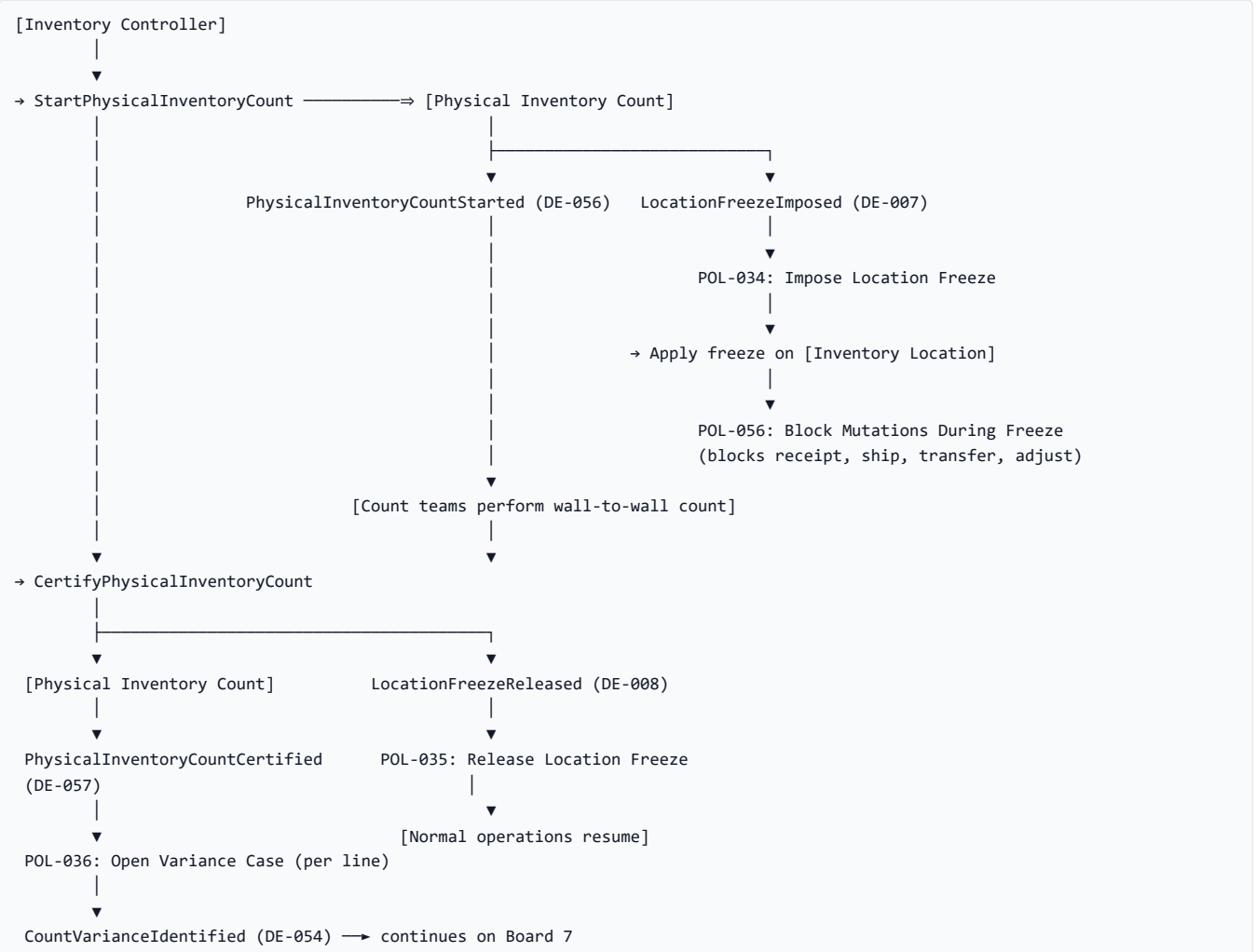
Board 7 — Cycle Count & Adjustment (Complete)

Business Process: BP-11, BP-13, BP-14 | **Use Cases:** UC-014, UC-016, UC-017, UC-018

[Inventory Controller]
→ ScheduleCycleCount
[Cycle Count]
CycleCountScheduled (DE-052)
[Warehouse Staff]
→ SubmitCycleCountResult
[Cycle Count]
CycleCountCompleted (DE-053)
(no variance)
count closed
(variance exists)
POL-036: Open Variance Case
→ IdentifyCountVariance
[Inventory Adjustment Case]
CountVarianceIdentified (DE-054)
POL-037: Require Independent Approval
[Warehouse Manager]
→ ApproveCountVariance
CountVarianceApproved (DE-055)
POL-038: Submit Adjustment After Approval
→ SubmitAdjustment
AdjustmentSubmitted (DE-058)
POL-039: Evaluate Adjustment Governance
POL-053: Enforce Segregation of Duties
[Operations Director]
→ ApproveAdjustment
→ RejectAdjustment
[Adjustment Case]
[Adjustment Case]
AdjustmentApproved
(DE-059)
AdjustmentRejected
(DE-060)

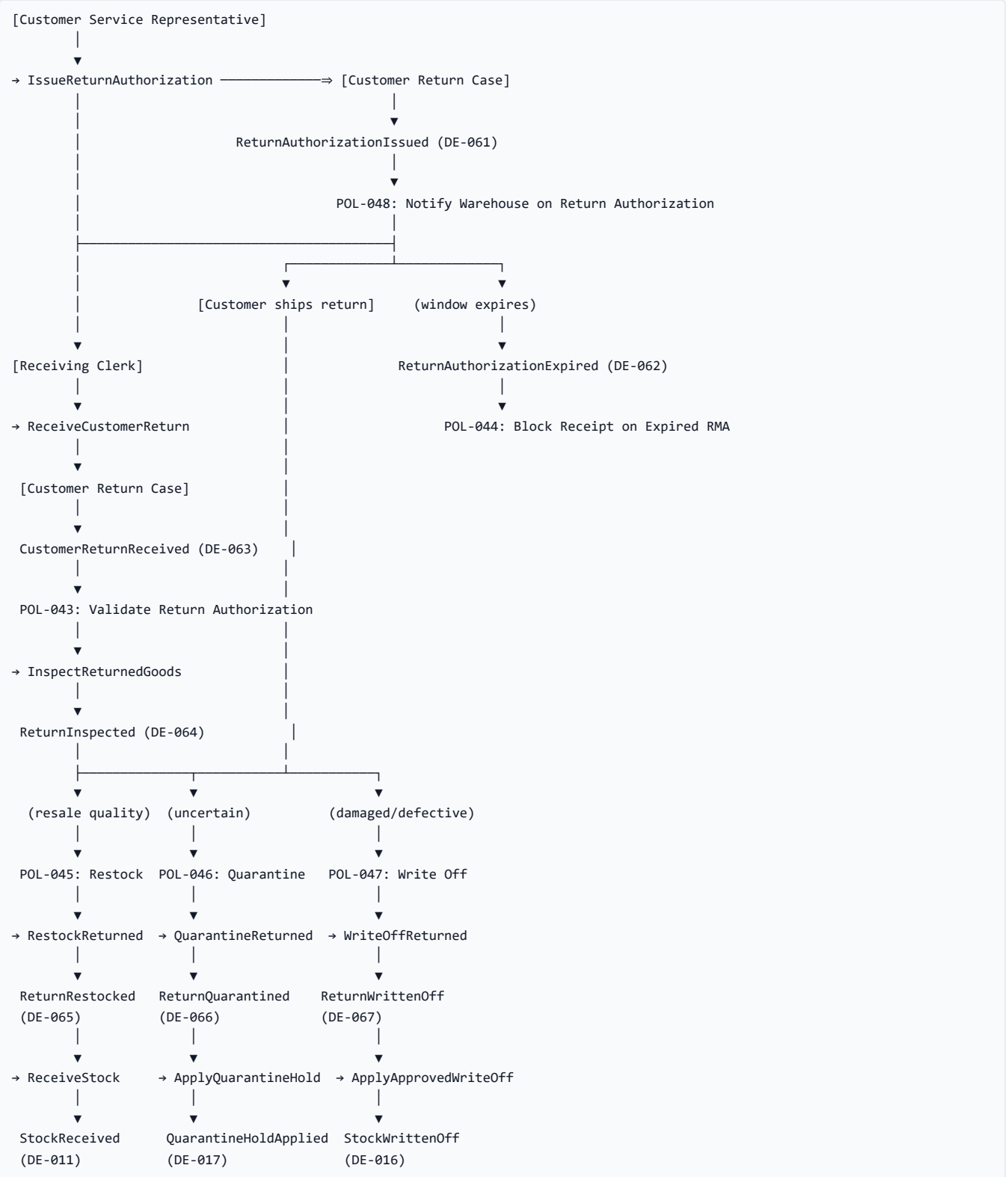
Board 8 — Physical Inventory Count & Location Freeze

Business Process: BP-12 | **Use Cases:** UC-015



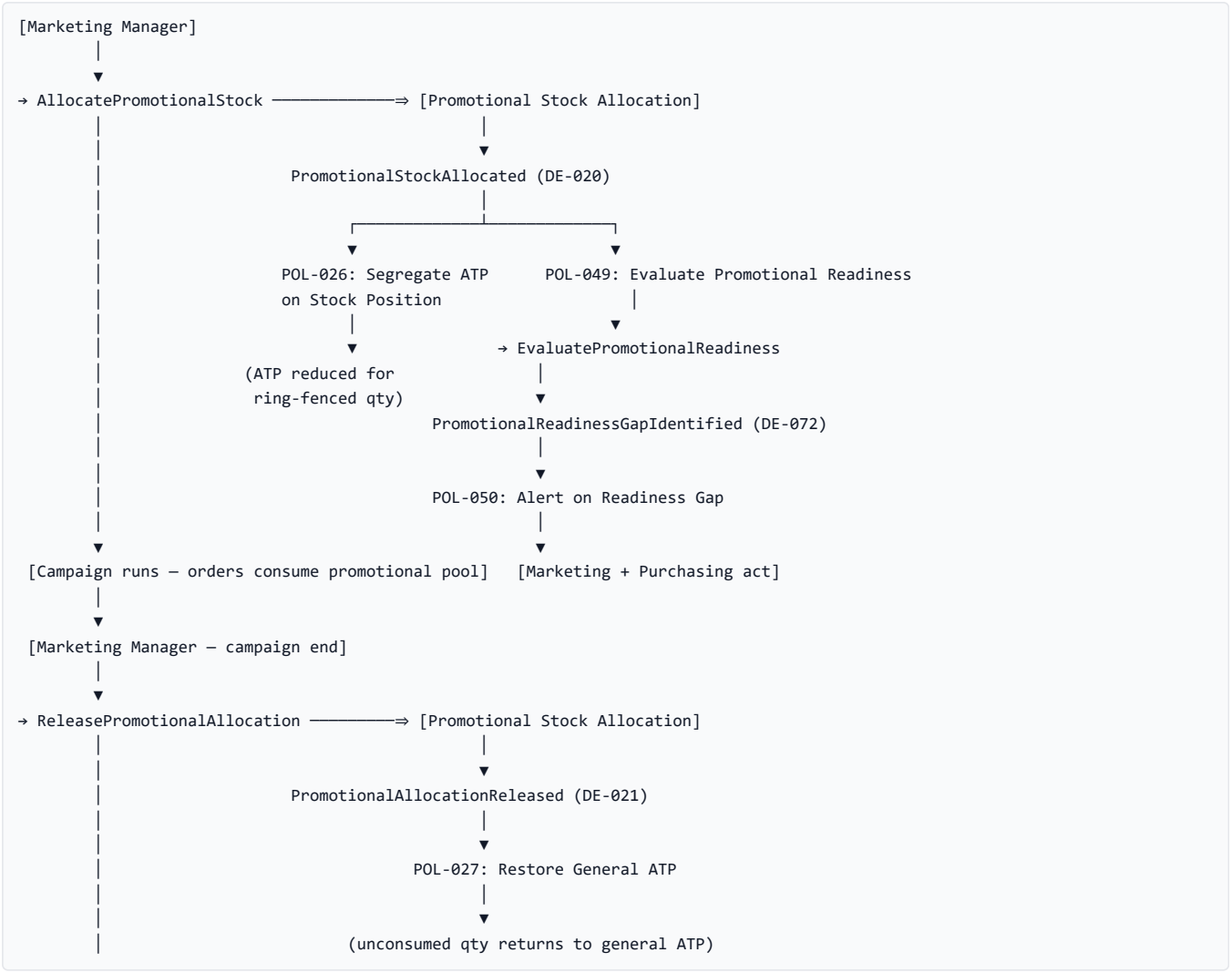
Board 9 — Customer Returns (Complete)

Business Process: BP-27, BP-28 | **Use Cases:** UC-035, UC-036



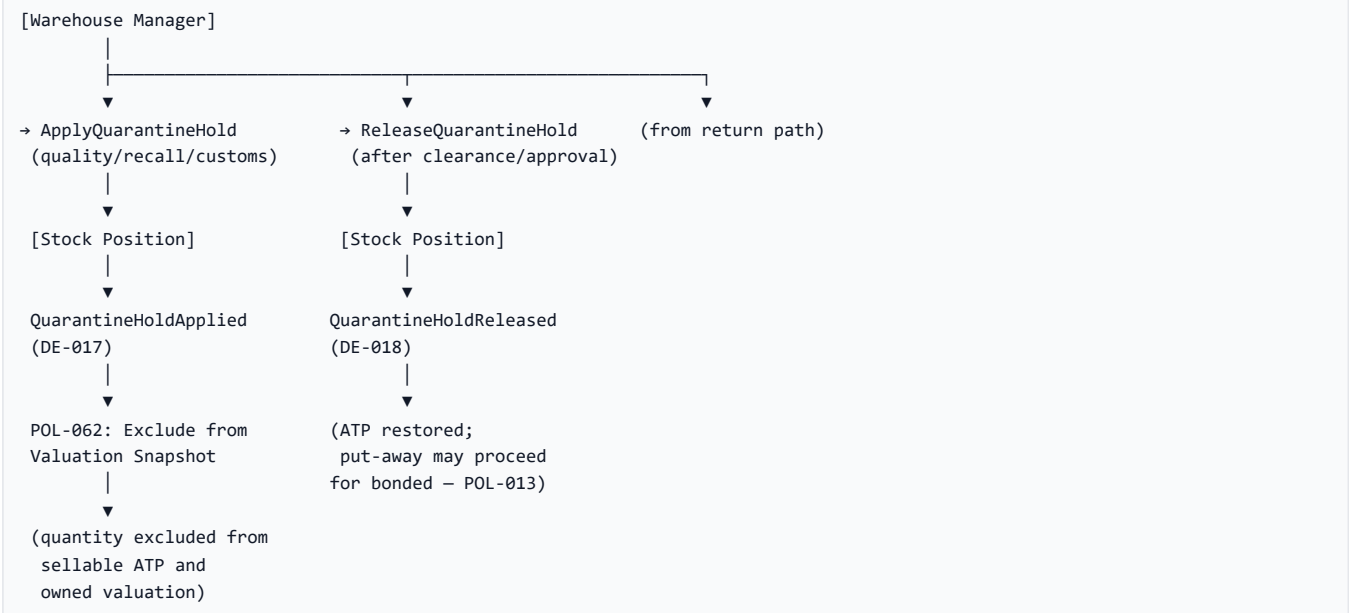
Board 10 — Promotional Campaign

Business Process: BP-31 | **Use Cases:** UC-040



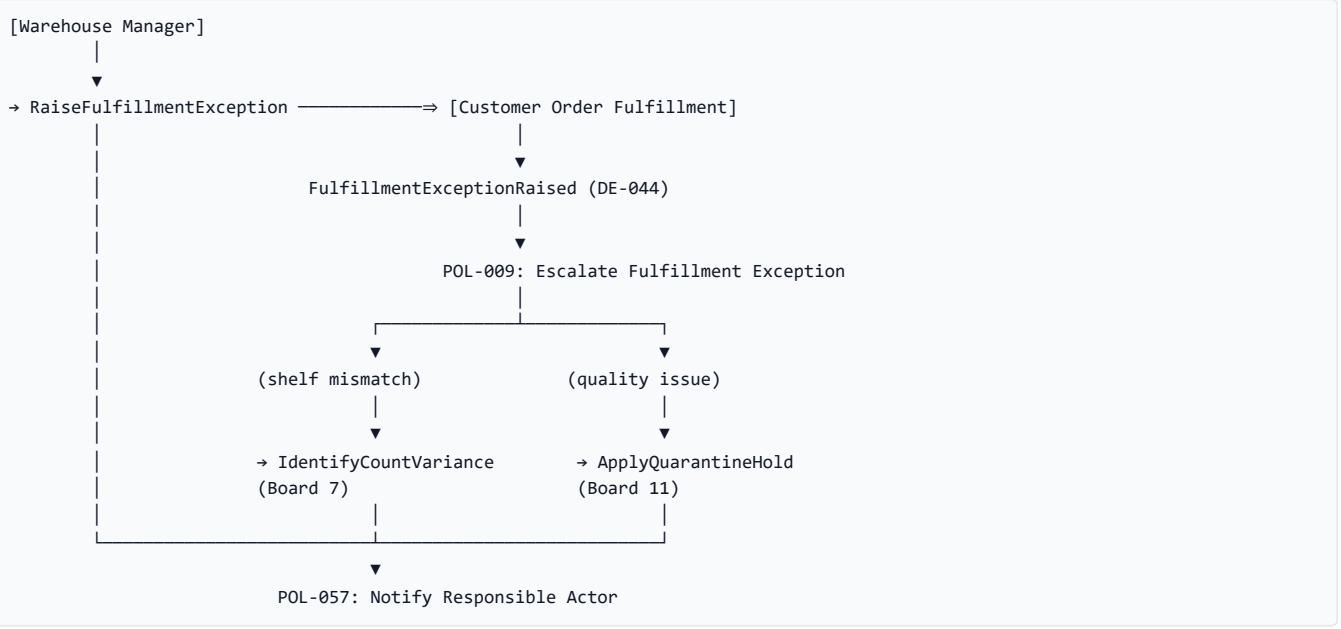
Board 11 — Quarantine & Hold Management

Business Process: BP-14, BP-20 | **Use Cases:** UC-019



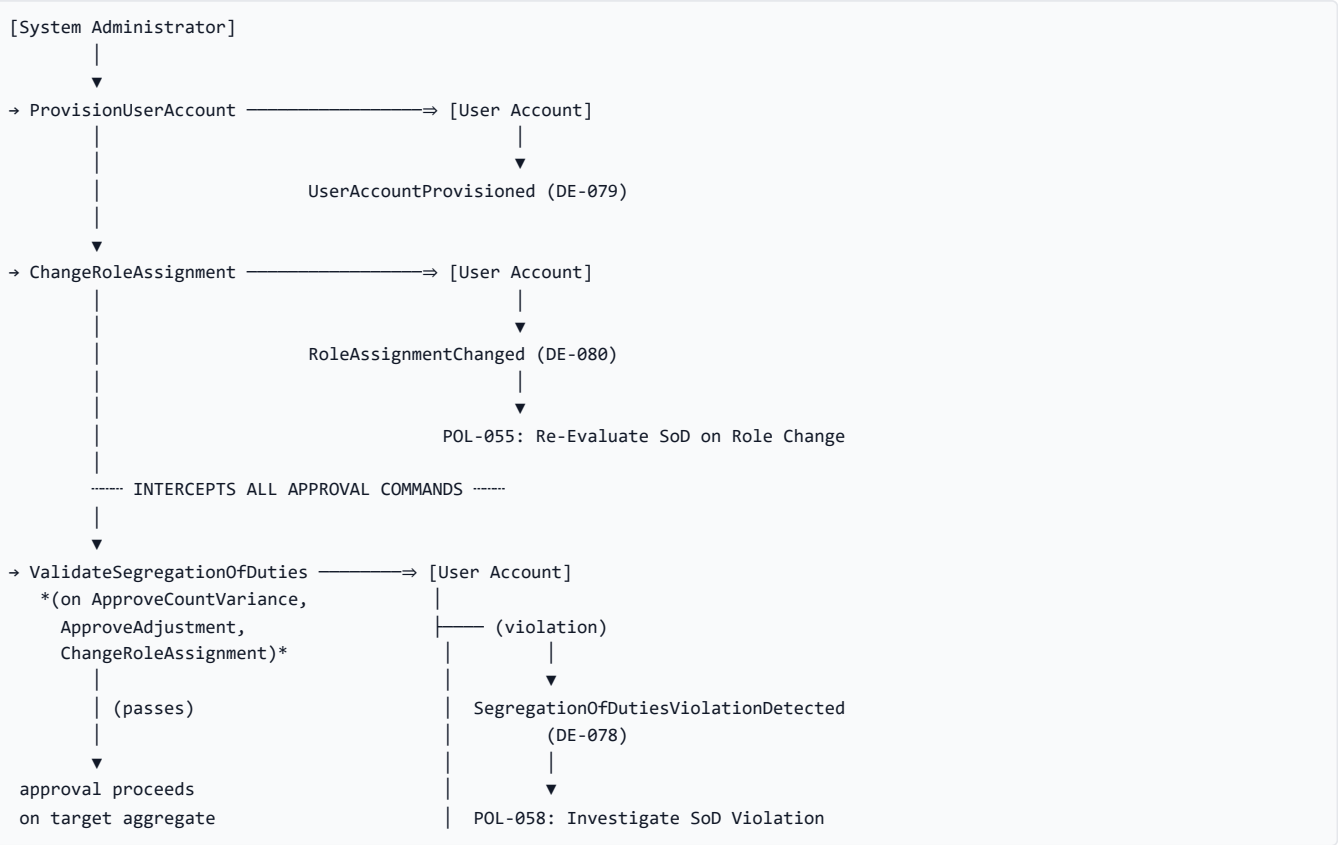
Board 12 — Fulfillment Exception Path

Business Process: BP-20 | **Use Cases:** UC-026

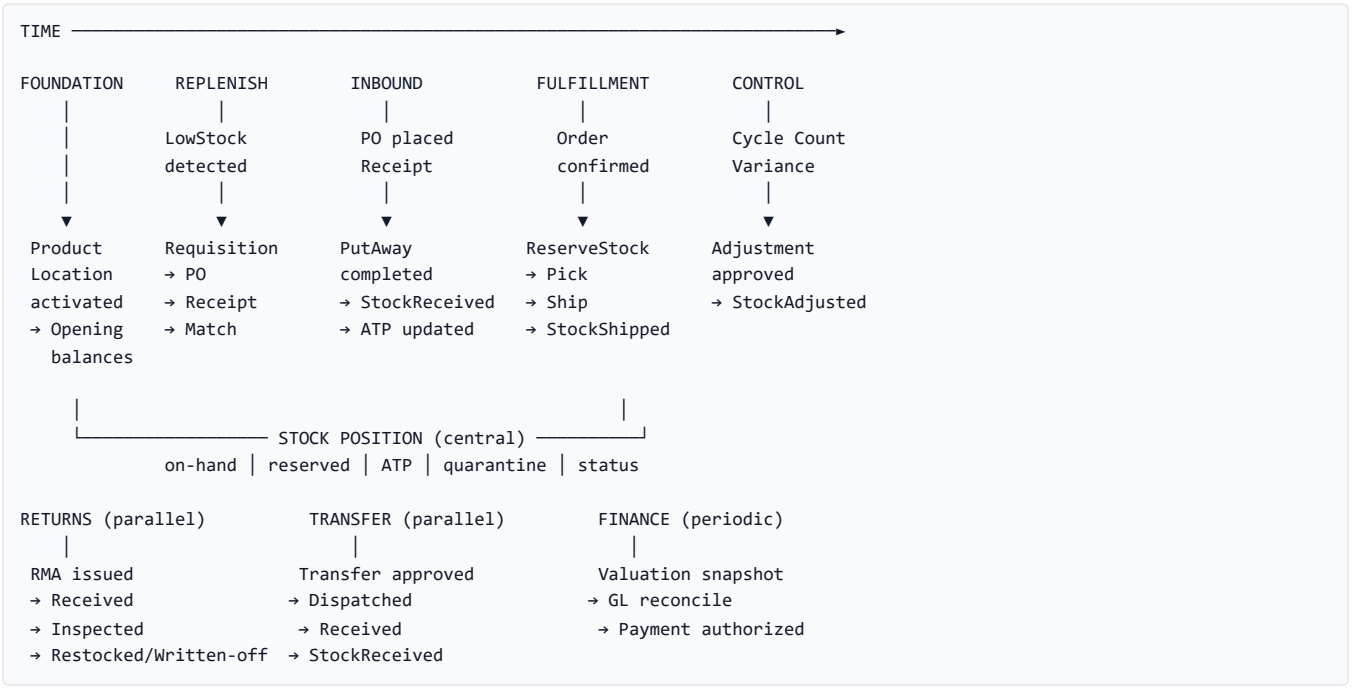


Board 13 — Governance & Access Control

Business Process: Cross-cutting | **Use Cases:** UC-044, UC-045



Complete Lifecycle Swimlane — Chronological Summary



Aggregate Participation Matrix

Aggregate	Boards	Commands Handled	Events Produced
Product Catalog	1	3	DE-001–003
Inventory Location	1, 8	3	DE-004–006
Inventory Governance	1, 7	2	DE-009–010
Stock Policy	1, 2	1	DE-019
Stock Position	1–12	11+	DE-011–018, 022, 068–070, 074
Purchase Requisition	2, 3	3	DE-023–025
Purchase Order	3	4	DE-026–029
Goods Receipt	3	4	DE-030–033
Supplier Invoice	3	2	DE-034–035, 081
Customer Order Fulfillment	4, 5, 12	10	DE-036–045
Stock Transfer	6	6	DE-046–051
Cycle Count	7	2	DE-052–053
Physical Inventory Count	8	2	DE-056–057, 007–008
Inventory Adjustment Case	7	5	DE-054–055, 058–060
Customer Return Case	9	7	DE-061–067
Promotional Stock Allocation	10	3	DE-020–021, 072
User Account	13	3	DE-078–080

External System Touchpoints on Board

External System	Board	Trigger	Event / Command
Supplier / Vendor	3	Delivers shipment	→ VerifyGoodsReceipt
Supplier / Vendor	3	Sends invoice	→ RecordSupplierInvoice
Logistics / Carrier	4, 6	Picks up goods	After ShipmentDispatched / TransferDispatched
Customer (B2B/B2C)	4, 5, 9	Places order / accepts backorder / returns	→ ConfirmCustomerOrder / RecordBackorder / IssueReturnAuthorization
General Ledger / ERP	3, 12	Month-end close	InventoryValuationSnapshotPrepared
Customs / Regulatory	3, 11	Clearance granted	→ ReleaseQuarantineHold

Document Approval

Role	Name	Signature	Date
Domain Expert			
DDD Architect			
Product Owner			

End of Event Storming Board

Document ID: SRS-ES-008 | Version 1.0 | July 6, 2026

Event Storming Hotspots

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-ES-007
Document Title	Event Storming Hotspots — Unresolved Business Issues
Version	1.0
Status	Draft
Document Owner	Domain Expert
Source Documents	SRS-ES-001 through SRS-ES-006, SRS-DM-009 (Domain Model Review), SRS-BR-001, SRS-UC-001
Date	July 6, 2026
Scope	Unresolved business uncertainties discovered during event storming — not technical design decisions

Purpose

In Event Storming, a **hotspot** marks an area where the business model is uncertain, contested, or incomplete. Hotspots block confident aggregate design, policy definition, or rule enforcement until domain experts resolve them. This document captures hotspots surfaced by cross-checking commands, events, policies, aggregates, and business rules.

Hotspot Severity

Severity	Meaning
Critical	Blocks architecture or MVP rule enforcement
High	Affects core workflows; workaround risky
Medium	Phase 2 scope or edge case; document assumption for MVP
Low	Clarification improves model; deferrable

Hotspot Summary

ID	Hotspot	Severity	Status
HS-001	Ad-Hoc Receipt Without Purchase Order	Critical	Open
HS-002	Opening Balance and Location Go-Live	Critical	Open
HS-003	Location Freeze Authority During Physical Count	High	Open
HS-004	Promotional Ring-Fence vs. General ATP	High	Open
HS-005	Lot/Batch Quantity Model for Traceable Products	High	Open
HS-006	Consignment Ownership Transfer Trigger	High	Open
HS-007	Order Reservation and Stock Position Coordination	High	Open
HS-008	Return Write-Off Governance Path	High	Open
HS-009	Bonded Inventory vs. General Quarantine	Medium	Open
HS-010	Transfer-Before-Buy vs. Purchase Decision	Medium	Open
HS-011	Backorder Priority and Auto-Fulfillment	Medium	Open
HS-012	Safety Stock Override Authorization	Medium	Open
HS-013	Supplier Master Data Ownership and Lifecycle	Medium	Open
HS-014	Finance System Boundary (Invoice and Valuation)	Medium	Open
HS-015	Product Supersession and SKU Runout	Low	Open
HS-016	Reservation Expiry and Stale Commitments	Low	Open

Total Hotspots: 16 | **Open:** 16

HS-001 — Ad-Hoc Receipt Without Purchase Order

Field	Detail
Area of Uncertainty	How supplier goods may be received when no purchase order exists, and how financial controls apply afterward
Why It Is Complex	BR-007 requires a PO for supplier receipt *except explicitly approved ad-hoc exceptions*, but the exception matrix is undefined (BRD ISS-020). Event storming has `VerifyGoodsReceipt` and POL-011 for PO-backed flow only. No command, event, authorization artifact, or policy exists for the exception path. Business Rule Dependencies flag **CF-007**: ad-hoc receipt conflicts with BR-008 three-way match assumptions. Validation rules VR-040 and VR-042 reference ad-hoc approval but no workflow owns it.
Related Business Processes	BP-07 (Goods Receipt & Verification), BP-05 (Purchase Order Placement), BP-30 (Inventory Valuation & Financial Reconciliation)
Questions Requiring Clarification	1. Who may authorize an ad-hoc receipt (role and approval level)? 2. Is retroactive PO creation required, and within what time limit? 3. What quantity/value caps apply per governance policy? 4. How does three-way match work when no PO existed at receipt — alternate match rules or retroactive PO? 5. Can ad-hoc receipt apply to consignment or bonded goods? 6. What reason codes and audit evidence are mandatory?

Field	Detail
Recommended Next Steps	1. Workshop with Purchasing, Receiving, and Finance to define ad-hoc exception matrix. 2. Add **Ad-Hoc Receipt Authorization** concept with `AdHocReceiptAuthorized` event. 3. Formalize **BR-033** (or extend BR-007) and resolve CF-007 with Finance. 4. Update Goods Receipt aggregate and POL-011 with exception branch. 5. Document in ubiquitous language.

HS-002 — Opening Balance and Location Go-Live

Field	Detail
Area of Uncertainty	How opening stock balances are established, verified, and approved before a location becomes transactionally active
Why It Is Complex	`ActivateInventoryLocation` emits DE-004 and `EstablishOpeningBalances` creates Stock Position quantities, but no aggregate governs the verification workflow. UC-046 covers bulk import; VR-025 requires approver sign-off — yet no commands or events model certification. Exception EX-004 (opening balance dispute) exists without resolution workflow. Constraint **CF-009** tensions go-live continuity (operations cannot shut down) vs. FR-010 (no transactions until balance approved). Event storming gap: first Stock Position creation lacks governing business process.
Related Business Processes	BP-02 (Location & Warehouse Setup), BP-01 (Product Master Data Management), BP-12 (Full Physical Inventory Count)
Questions Requiring Clarification	1. Is opening balance established by physical count, system import, or either? 2. Who approves opening balances (Inventory Controller, Finance, Operations Director)? 3. Can a location be *active* in read-only/staging mode before balance certification? 4. What happens when opening balance dispute arises at go-live (EX-004)? 5. Is a wall-to-wall count required for new locations or only data import with sample verification? 6. Relationship between FR-010 and FR-090 import — same workflow or alternatives?
Recommended Next Steps	1. Business decision: count-validated vs. import-only go-live. 2. Model **Opening Balance Establishment** aggregate or subprocess with `SubmitOpeningBalance`, `CertifyOpeningBalance`, `OpeningBalanceCertified` event. 3. Add **BR-043** Opening Balance Certification (per Business Rule Catalog Review). 4. Define precondition chain: certify → activate location → establish balances on Stock Position. 5. Resolve CF-009 with staging-mode business rule if needed.

HS-003 — Location Freeze Authority During Physical Count

Field	Detail
Area of Uncertainty	Which business concept owns location operational freeze during wall-to-wall physical inventory count
Why It Is Complex	BR-024 requires freeze during physical count. Events DE-007 (`LocationFreezeImposed`) and DE-008 (`LocationFreezeReleased`) originate from **Physical Inventory Count**, but **Inventory Location** aggregate also lists freeze enforcement. DS-014 (Freeze Guard) consults both. Two aggregates could mutate freeze state — risk of race conditions at count certification vs. location status changes. POL-034 and POL-056 block mutations but do not resolve ownership.
Related Business Processes	BP-12 (Full Physical Inventory Count), BP-02 (Location & Warehouse Setup), BP-07–BP-09 (Inbound), BP-18 (Pick, Pack & Ship), BP-25 (Transfer Execution)
Questions Requiring Clarification	1. Does freeze apply to the entire location hierarchy or only the counted site? 2. Can emergency receipts or shipments override freeze with director authorization? 3. Who can release freeze early if count is aborted? 4. Are count-related adjustments the *only* mutations permitted during freeze? 5. What is the business consequence if freeze is violated — reject transaction or flag exception?
Recommended Next Steps	1. Confirm pattern: Physical Count **initiates** freeze; Inventory Location **holds** authoritative frozen flag. 2. Document integration contract in bounded context map. 3. DS-014 reads Inventory Location state only. 4. Add **MR-09** formal rule: non-count adjustments blocked during freeze. 5. Workshop edge cases (emergency override, count abort).

HS-004 — Promotional Ring-Fence vs. General ATP

Field	Detail
Area of Uncertainty	How promotional stock ring-fencing interacts with available-to-promise and who may consume ring-fenced quantity

Field	Detail
Why It Is Complex	BR-029 requires ring-fencing during campaigns, but consumption rules for non-promotional orders are only partially stated (MR-10). Promotional Stock Allocation aggregate owns campaign intent; Stock Position owns ATP mechanics — dual context risks inconsistent state (P1-02). Event storming shows DE-020/DE-021 from Commercial aggregate and POL-026 affecting Stock Position, but invariant ownership is undocumented. Entity model still groups allocation under Inventory Balance domain.
Related Business Processes	BP-31 (Promotional & Launch Stock Readiness), BP-17 (Inventory Reservation), BP-16 (Order Availability Confirmation)
Questions Requiring Clarification	1. Can non-promotional orders ever consume ring-fenced stock, and who approves override (commercial leadership)? 2. Is ring-fence at product-location or network level? 3. What happens to un consumed allocation at campaign end — auto-release or manual review? 4. How is promotional ATP presented in availability inquiry (RM-002)? 5. If campaign stock is insufficient (DE-072), may general ATP backfill the campaign commitment? 6. Priority between existing reservations and new promotional allocation on same SKU?
Recommended Next Steps	1. Confirm: Commercial owns intent; Stock Position reacts to events only. 2. Formalize **MR-10** Promotional Consumption Approval. 3. Workshop Marketing, Sales, and Operations on override rules. 4. Update POL-026 with explicit ATP segregation mechanics. 5. Align SRS-DM-001 entity grouping with Commercial context.

HS-005 — Lot/Batch Quantity Model for Traceable Products

Field	Detail
Area of Uncertainty	How traceable inventory quantities are scoped, reserved, picked, and counted when lot/batch identity is mandatory
Why It Is Complex	BR-017 requires lot/batch capture on receipt, transfer, and shipment. Stock Position identity is documented ambiguously as *product + location [+ lot]* vs. product-location with embedded Lot/Batch child entities (P1-05). This fork affects reservation (`ReserveStock`), cycle count variance scope, FEFO pick selection (DS-011), and transfer dispatch. Event storming commands do not specify lot-level vs. position-level reservation.
Related Business Processes	BP-07 (Goods Receipt), BP-09 (Put-Away), BP-11 (Cycle Counting), BP-18 (Pick, Pack & Ship), BP-25 (Transfer Execution), BP-28 (Return Disposition)
Questions Requiring Clarification	1. One Stock Position per product-location with lot sub-quantities, or one position per lot? 2. Can ATP be promised without specifying lot, with lot assigned at pick? 3. How are partial lot quantities handled on count variance? 4. Mandatory FEFO — can picker override lot selection with manager approval? 5. Recall scenario: how is affected lot isolated across locations? 6. Mixed-lot storage position — allowed or prohibited?
Recommended Next Steps	1. Business workshop with Quality and Warehouse on traceability operating model. 2. Adopt **single Stock Position per product-location** with Lot/Batch child entities (recommended in domain review). 3. Document invariant: lot quantities sum to on-hand; traceable products require lot on all movements. 4. Update reservation and pick commands with lot assignment rules. 5. Add recall workflow hotspot follow-up if in scope.

HS-006 — Consignment Ownership Transfer Trigger

Field	Detail
Area of Uncertainty	When consignment stock becomes organization-owned for valuation, fulfillment, and payment purposes
Why It Is Complex	BR-031 states ownership transfers at *defined consumption or sale event per supplier agreement* — but the triggering events are not enumerated. DE-022 (`ConsignmentOwnershipTransferred`) is raised on `StockShipped` via POL-028, yet consignment agreements may trigger on issue, consumption, invoice, or other events. DEP-028 notes three-way match timing differs for consignment. VR/consignment ATP exclusion (CF-014) is unresolved — consignment may appear available for sale while not owned. Supplier aggregate has no lifecycle events to record agreement terms.
Related Business Processes	BP-07 (Goods Receipt), BP-18 (Pick, Pack & Ship), BP-30 (Valuation & Financial Reconciliation), BP-05 (Purchase Order Placement)
Questions Requiring Clarification	1. What are the standard consumption triggers (shipment, internal issue, invoice, time-based)? 2. Is consignment stock included in ATP for customer commitment (BR-001 interaction)? 3. When does three-way match and payment occur for consignment — at ownership transfer? 4. How are consignment terms captured per supplier vs. per PO line? 5. Can consignment and owned stock coexist for same SKU at same location? 6. Financial valuation treatment before and after transfer?

Field	Detail
Recommended Next Steps	1. Catalog consignment agreement types with Finance and Procurement. 2. Formalize **MR-08** with explicit trigger event list per agreement type. 3. Add Supplier terms commands/events. 4. Clarify ATP formula for consignment in BR-001 context. 5. Map consignment path in three-way match (resolve DEP-028).

HS-007 — Order Reservation and Stock Position Coordination

Field	Detail
Area of Uncertainty	How customer order commitment stays consistent with stock position quantity when operations span two aggregates without a single transaction
Why It Is Complex	Customer Order Fulfillment holds `InventoryReservation` entity; Stock Position holds `reserved` quantity — coordinated by DS-002 and POL-001/002. Partial failure scenario: order confirmed and reservation fails on Stock Position leaves inconsistent state. High-concurrency hot SKUs create contention. `CompletePartialFulfillment` decomposes lines across fulfilled and backordered — quantity reconciliation invariant must hold. Multi-location fulfillment (UC-048, DE-045) multiplies coordination points. Aggregate review flags Customer Order as split candidate post-MVP.
Related Business Processes	BP-16 (Order Availability Confirmation), BP-17 (Inventory Reservation), BP-18 (Pick, Pack & Ship), BP-19 (Partial Fulfillment & Backorder)
Questions Requiring Clarification	1. What is the business response when order is confirmed but reservation fails (retry, cancel, partial)? 2. Is reservation synchronous with confirmation or eventual within SLA? 3. How long may reservation remain open before mandatory release (see HS-016)? 4. Who wins when two orders compete for last ATP unit? 5. Multi-location split — single customer communication or per-shipment? 6. Compensation policy if shipment deduct fails after pick verified?
Recommended Next Steps	1. Define reservation SLA and failure compensation with Customer Service and Operations. 2. Document saga: ConfirmOrder → ReserveStock → on failure → CancelOrder or PartialConfirm. 3. Formalize order line invariant as published business rule. 4. Prototype contention scenario for Class A high-velocity SKUs. 5. Defer Shipment aggregate split decision until volume metrics available.

HS-008 — Return Write-Off Governance Path

Field	Detail
Area of Uncertainty	Whether returned goods written off must follow the same adjustment governance workflow as operational write-offs
Why It Is Complex	BR-023 requires documented write-off authorization. Customer Return Case command `WriteOffReturnedGoods` currently triggers `ApplyApprovedWriteOff` on Stock Position directly (POL-047). Domain review REL-ISSUE-07 states return write-off should **create Adjustment Case** for governance. Two paths to write-off create SoD and audit inconsistency. Return inspection (BR-026) and write-off authorization (BR-023) may involve different approvers.
Related Business Processes	BP-27 (Customer Return Intake), BP-28 (Return-to-Stock or Disposal), BP-14 (Stock Adjustment & Write-Off), BP-15 (Shrinkage & Loss Management)
Questions Requiring Clarification	1. Can warehouse manager authorize return write-off at inspection, or must Operations Director approve via Adjustment Case? 2. Is return write-off threshold governed by same approval limits as operational shrinkage? 3. Should return write-off reason codes differ from general adjustment codes? 4. Finance notification timing — at inspection or at governance approval? 5. Partial return line disposition — mixed restock and write-off on same RMA?
Recommended Next Steps	1. Confirm with Finance and Operations: single governance path via Adjustment Case. 2. Update POL-047: `ReturnWrittenOff` → create Adjustment Case → `ApproveAdjustment` → `ApplyApprovedWriteOff`. 3. Define return-specific reason code subset. 4. Align event storming commands with BR-019/BR-021 SoD.

HS-009 — Bonded Inventory vs. General Quarantine

Field	Detail
Area of Uncertainty	Whether customs-bonded inventory follows the same hold workflow as quality quarantine

Field	Detail
Why It Is Complex	BR-030 requires bonded goods remain on hold until customs clearance — modeled today as `QuarantineHoldApplied` (POL-013). Business may require distinct **Customs Clearance Record** (ME-05) with different release authority (Regulatory Authority vs. Warehouse Manager). Bonded release triggers put-away activation; quality quarantine may route to write-off. External participant EXT-005 (Customs) triggers `ReleaseQuarantineHold` but clearance documentation differs from quality release.
Related Business Processes	BP-07 (Goods Receipt), BP-09 (Put-Away), BP-14 (Stock Adjustment & Write-Off)
Questions Requiring Clarification	1. Is bonded hold a quarantine subtype or separate business concept? 2. Who records customs clearance — receiving clerk or finance/compliance role? 3. Can bonded goods move within bonded zone without clearance? 4. Partial clearance — some quantity cleared, remainder held? 5. Audit evidence required for clearance (document reference)?
Recommended Next Steps	1. Clarify with Compliance whether bonded = quarantine subtype for MVP. 2. If distinct, add Customs Clearance Record and `CustomsClearanceGranted` event. 3. Update POL-013 and POL-012 bonded branch. 4. Defer to Phase 2 if bonded inventory not in MVP scope.

HS-010 — Transfer-Before-Buy vs. Purchase Decision

Field	Detail
Area of Uncertainty	Who decides between network transfer and new procurement when local shortage is detected, and how mandatory the network check is
Why It Is Complex	BR-011 requires network availability evaluation before purchase. POL-029 may recommend `RequestStockTransfer` or `RejectPurchaseRequisition`, but replenishment recommendation (DE-071) is advisory — not a binding business decision. Regional manager, buyer, and operations director may disagree. Source location minimum (BR-012) may block transfer that purchasing prefers. No persistent **Replenishment Suggestion** entity for audit of overridden recommendations (ME-06).
Related Business Processes	BP-21 (Low-Stock Alert Review), BP-22 (Replenishment Planning), BP-24 (Transfer Request), BP-26 (Network Rebalancing)
Questions Requiring Clarification	1. Is network check mandatory gate before PO approval, or advisory? 2. Who has final authority when transfer is recommended but buyer prefers purchase? 3. Must rejection of transfer recommendation be documented? 4. Cost comparison (transfer cost vs. purchase) — in scope for decision? 5. Auto-generated transfer request from rebalancing (UC-034) — approval required?
Recommended Next Steps	1. Define decision authority matrix: Purchasing vs. Operations vs. Regional Manager. 2. Clarify BR-011 enforcement level (hard block vs. warning). 3. Add acknowledgment workflow when recommendation overridden. 4. Workshop BP-21/BP-22/BP-24 integrated replenishment policy.

HS-011 — Backorder Priority and Auto-Fulfillment

Field	Detail
Area of Uncertainty	How backordered customer commitments are prioritized and fulfilled when stock becomes available
Why It Is Complex	BR-028 requires customer acknowledgment for backorders. POL-008 triggers `FulfillBackorder` on `StockReceived`, but priority rules are undefined — FIFO by acknowledgment date, order value, customer tier, or product class? MR-06 states backorder must convert to reservation before fulfillment — sequence not fully modeled. Competition between backorder fulfillment and new order reservation on inbound stock. `BackorderFulfilled` (DE-043) chain spans multiple commands without documented priority policy.
Related Business Processes	BP-19 (Partial Fulfillment & Backorder), BP-17 (Inventory Reservation), BP-09 (Put-Away), BP-22 (Replenishment Planning)
Questions Requiring Clarification	1. Backorder fulfillment priority rule when multiple backorders and new orders await same SKU? 2. Auto-fulfill on stock arrival or manual customer service trigger? 3. Customer notification required before backorder shipment? 4. Partial backorder fulfillment over multiple receipts — allowed? 5. Backorder expiry or cancellation policy?
Recommended Next Steps	1. Define backorder priority policy with Customer Service. 2. Formalize MR-05 and MR-06 as business rules. 3. Update POL-008 with explicit priority algorithm. 4. Document in RM-014 (Backorder Pipeline) sort order.

HS-012 — Safety Stock Override Authorization

Field	Detail
Area of Uncertainty	When and how safety stock buffer may be consumed below policy minimum for reservations, transfers, or fulfillment
Why It Is Complex	BR-010 prohibits bypassing safety stock without documented approval and replenishment plan — but override authority, documentation, and enforcement point are unclear. DS-002 checks safety stock on reservation; DS-006 on transfer — inconsistent override path. Emergency fulfillment during stockout may pressure safety stock consumption. MR-04 proposed but not formalized. Exception scenario EX-ad-hoc policy override exists generically.
Related Business Processes	BP-17 (Inventory Reservation), BP-24 (Transfer Request), BP-21 (Low-Stock Alert Review), BP-20 (Outbound Exception Handling)
Questions Requiring Clarification	1. Who may authorize safety stock override (Warehouse Manager, Operations Director)? 2. Is override per transaction or blanket for period? 3. Must replenishment plan be created before or after override? 4. Does override apply to source location on transfer (BR-012 interaction)? 5. Audit evidence and expiry of override authorization?
Recommended Next Steps	1. Formalize MR-04 as BR-033+ with approver and evidence requirements. 2. Add `SafetyStockOverrideAuthorized` event or governance consultation pattern. 3. Align DS-002 and DS-006 override handling. 4. Include in governance policy parameters.

HS-013 — Supplier Master Data Ownership and Lifecycle

Field	Detail
Area of Uncertainty	Organizational ownership of supplier master data and how supplier lifecycle is managed in the domain model
Why It Is Complex	Supplier aggregate exists but event storming reveals **no commands or events** — gap HS from aggregate review. Bounded context placement differs: Catalog & Network (SRS-DM-003) vs. Procurement & Supplier (SRS-DM-007). Consignment terms (BR-031) live on Supplier but affect Stock Position and Finance. Supplier performance scorecard (RM-010) consumes events Supplier never produces.
Related Business Processes	BP-05 (Purchase Order Placement), BP-07 (Goods Receipt), BP-30 (Valuation & Financial Reconciliation)
Questions Requiring Clarification	1. Who owns supplier master data — Procurement or Finance? 2. Supplier onboarding and deactivation workflow? 3. How are consignment agreement terms versioned and effective-dated? 4. Can inactive supplier appear on in-flight PO? 5. Supplier performance feedback — formal review cycle or event-driven only?
Recommended Next Steps	1. Align bounded context to Procurement & Supplier. 2. Add supplier lifecycle commands and events. 3. Link consignment terms to BR-031 trigger catalog (HS-006). 4. Assign product owner for supplier master data.

HS-014 — Finance System Boundary (Invoice and Valuation)

Field	Detail
Area of Uncertainty	Whether IMS owns supplier invoicing and valuation reconciliation or synchronizes with external Finance ERP
Why It Is Complex	Supplier Invoice aggregate performs three-way match (BR-008) — typically ERP-owned in mature organizations. `InventoryValuationSnapshotPrepared` (DE-076) must align with general ledger — EXT-004. Architecture review AG-03 flags finance coupling as Phase 3 evaluation. Event storming models full match workflow; business may intend IMS as operational record with ERP as financial system of record. Valuation variance (DE-077) investigation spans both systems.
Related Business Processes	BP-30 (Inventory Valuation & Financial Reconciliation), BP-05 (Purchase Order), BP-07 (Goods Receipt)
Questions Requiring Clarification	1. Does IMS record invoices or receive match status from ERP? 2. Which system authorizes payment? 3. Valuation method (FIFO, weighted average) — owned by Finance or IMS? 4. Month-end close — IMS pushes snapshot or ERP pulls? 5. Consignment and write-off impact — which system is authoritative?
Recommended Next Steps	1. Integration workshop with CFO and Finance systems team. 2. Define anti-corruption layer for ERP boundary. 3. MVP decision: IMS owns match for accountability chain vs. ERP read-only. 4. Document in bounded context map as explicit ACL. 5. Defer if finance integration is Phase 2.

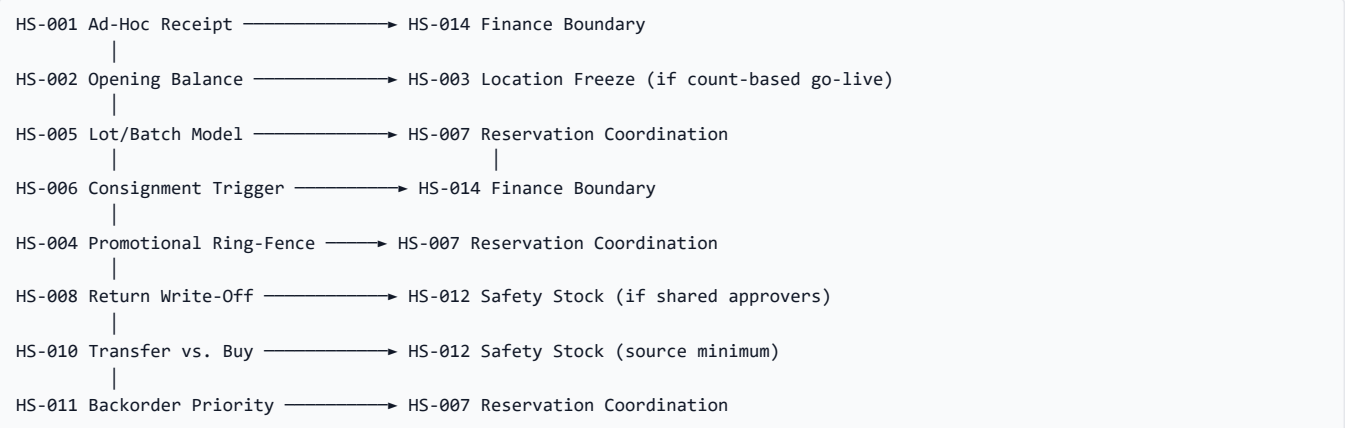
HS-015 — Product Supersession and SKU Runout

Field	Detail
Area of Uncertainty	How product supersession affects procurement, fulfillment, and inventory runout of replaced SKUs
Why It Is Complex	BR-027 blocks discontinued product reorder; BRD references supersession (FR-019→FR-065) but no **Product Supersession Link** entity (ME-03). Discontinuing SKU while stock remains creates fulfillment and transfer policy questions. Event storming POL-010 blocks new reservations on discontinued status but not migration to successor SKU.
Related Business Processes	BP-01 (Product Master Data), BP-04 (ABC Classification), BP-15 (Shrinkage & Obsolete Stock)
Questions Requiring Clarification	1. Is supersession a product catalog relationship (old SKU → new SKU)? 2. May discontinued SKU still fulfill existing backorders and reservations? 3. Auto-suggest successor on availability inquiry? 4. Runout policy — markdown, transfer, or write-off remaining stock? 5. Timeline between supersession announcement and discontinue status?
Recommended Next Steps	1. Confirm supersession in MVP scope (Phase 2 per UC-004 context). 2. If in scope, add supersession link to Product Catalog. 3. Define POL for discontinued + superseded SKU handling. 4. Link to shrinkage analysis (RM-025).

HS-016 — Reservation Expiry and Stale Commitments

Field	Detail
Area of Uncertainty	Whether inventory reservations expire and how stale commitments are released
Why It Is Complex	Reservations reduce ATP indefinitely until fulfillment or cancellation (BR-002). MR-05 proposes expiry/escalation policy — not formalized. Long-lived reservations on slow-moving orders block ATP for other customers. No command `ExpireReservation` or policy for auto-release. Customer Service may not cancel abandoned orders promptly.
Related Business Processes	BP-17 (Inventory Reservation), BP-16 (Order Availability Confirmation), BP-10 (Inventory Status Monitoring)
Questions Requiring Clarification	1. Reservation time limit — days, weeks, configurable by product class? 2. Auto-release vs. escalation notification before release? 3. Customer notification before reservation release? 4. Does expiry apply to backorders differently than confirmed reservations? 5. Re-reservation priority if customer reorders after expiry?
Recommended Next Steps	1. Define reservation TTL policy with Customer Service and Sales. 2. Formalize MR-05 as business rule if approved. 3. Add `ReservationExpired` event and POL if auto-release adopted. 4. MVP: manual release only with dashboard alert for aged reservations.

Hotspot Dependency Map



Workshop Prioritization

Session 1 — Critical Path (MVP Blockers)

Hotspot	Participants
HS-001 Ad-Hoc Receipt	Purchasing, Receiving, Finance
HS-002 Opening Balance	Inventory Controller, Operations, Finance
HS-003 Location Freeze	Inventory Controller, Warehouse Manager
HS-005 Lot/Batch Model	Warehouse, Quality, Inventory Controller

Session 2 — Core Operations

Hotspot	Participants
HS-004 Promotional Ring-Fence	Marketing, Sales, Operations
HS-007 Reservation Coordination	Customer Service, Warehouse, Operations
HS-008 Return Write-Off	Customer Service, Finance, Warehouse
HS-010 Transfer vs. Buy	Purchasing, Operations, Regional Managers

Session 3 — Financial & Advanced

Hotspot	Participants
HS-006 Consignment	Procurement, Finance
HS-014 Finance ERP Boundary	CFO, Finance systems, IMS product owner
HS-009 Bonded Inventory	Compliance, Receiving
HS-011–HS-016	As needed per phase plan

Resolution Tracking

Hotspot	Target Resolution	Owner	Resolution Document
HS-001	Before architecture	Senior BA	BR-033, Ad-Hoc Receipt Authorization
HS-002	Before architecture	Inventory Controller	Opening Balance workflow
HS-003	Before architecture	DDD Architect	Context integration contract
HS-004	Before MVP cut	Marketing / Operations	MR-10, POL-026 update
HS-005	Before architecture	Solution Architect	Stock Position identity spec
HS-006	Phase 2 or MVP if consignment in scope	Finance	MR-08, Supplier terms
HS-007	Before MVP test	Operations Director	Saga compensation spec
HS-008	Before MVP	Finance / Operations	POL-047 realignment
HS-009	Phase 2 unless bonded in MVP	Compliance	ME-05 decision
HS-010	MVP policy document	Purchasing Manager	BR-011 enforcement level
HS-011	MVP or Phase 2	Customer Service	Backorder priority policy
HS-012	MVP governance update	Operations Director	MR-04 formalization
HS-013	Before architecture	Procurement	Supplier lifecycle events
HS-014	Architecture phase	CFO / IT	ERP integration decision
HS-015	Phase 2	Product / Inventory	Supersession entity

Hotspot	Target Resolution	Owner	Resolution Document
HS-016	Phase 2	Customer Service	MR-05 decision

Document Approval

Role	Name	Signature	Date
Domain Expert			
DDD Architect			
Product Owner			
Operations Director			

End of Event Storming Hotspots

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Part VI — System Architecture

System Modules Design

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-SA-003
Document Title	System Modules Design — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Solution Architect
Prepared For	Delivery leads, technical architects, and development squads
Date	July 6, 2026
Classification	Internal — Architecture Confidential
Scope	Logical module decomposition within the modular monolith — no class design, database schema, or API specification

Related Documents

Document	ID	Relationship
Architecture Vision	SRS-SA-001	Strategic direction
Architecture Style Recommendation	SRS-SA-002	Modular monolith + Clean/Hexagonal
Bounded Contexts	SRS-DM-007	1:1 module alignment source
Domain Aggregates	SRS-DM-003	Owned aggregate definitions

Document	ID	Relationship
System Context	SRS-SA-002 (System Context)	External actors and systems

Document Purpose

This document defines the **logical system modules** of the Inventory Management System. Each module maps to a bounded context, owns specific aggregates, exposes a minimal public interface, and declares explicit dependencies. Modules are designed for **loose coupling** — integration occurs through commands, queries, domain events, and shared identifiers only.

Module Design Principles

Principle	Rule
One module per bounded context	Strategic domain boundary = compile-time module boundary
Aggregate ownership is exclusive	Each aggregate belongs to exactly one module
No cross-module aggregate access	Other modules reference aggregates by identity (SKU, Location Code, Order ID) only
No shared mutable state	Inventory Balance is the sole ATP mutator; Intelligence is read-only
Events over direct calls	Cross-module state propagation prefers domain events; synchronous queries for read-only consultation
ACL at external boundaries	Sales/CRM and Finance/ERP integrate through dedicated adapter ports — not module internals

Published Language (Cross-Module Identifiers)

All modules share these **value-only identifiers** — not domain object references:

Identifier	Owner Module	Consumers
`ProductSku`	Product Catalog	All operational modules
`LocationCode`	Warehouse Network	All operational modules
`UserId`	Identity & Access	All modules
`SupplierCode`	Procurement & Supplier	Inbound Receiving, Intelligence
`PurchaseOrderId`	Procurement & Supplier	Inbound Receiving, Intelligence
`GoodsReceiptId`	Inbound Receiving	Procurement, Intelligence, Finance ACL
`CustomerOrderId`	Order Fulfillment	Returns, Intelligence, Sales ACL
`TransferId`	Network Transfer	Inventory Balance, Intelligence
`AdjustmentCaseId`	Inventory Accuracy	Inventory Balance, Intelligence
`ReturnAuthorizationId`	Returns Management	Order Fulfillment, Inbound Receiving
`CampaignId`	Commercial Campaign	Order Fulfillment, Inventory Balance
`ReasonCode`	Inventory Governance	All modules performing adjustments or exceptions

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Module Overview

#	Module	Bounded Context	Aggregates	Phase
M-01	Product Catalog	Product Catalog	1	MVP
M-02	Warehouse Network	Warehouse Network	1	MVP
M-03	Inventory Governance	Inventory Governance	1	MVP
M-04	Inventory Balance	Inventory Balance	2	MVP
M-05	Procurement & Supplier	Procurement & Supplier	3	Phase 2*

#	Module	Bounded Context	Aggregates	Phase
M-06	Inbound Receiving	Inbound Receiving	2	MVP
M-07	Order Fulfillment	Order Fulfillment	1	MVP
M-08	Network Transfer	Network Transfer	1	Phase 2
M-09	Inventory Accuracy	Inventory Accuracy	3	MVP
M-10	Returns Management	Returns Management	1	Phase 2
M-11	Commercial Campaign	Commercial Campaign	1	Phase 2
M-12	Identity & Access	Identity & Access	1	MVP
M-13	Inventory Intelligence	Inventory Intelligence	0 (read projections)	MVP

**Procurement MVP subset: PO placement and receipt authorization without full requisition workflow may be enabled in Phase 1 per FR scope — module boundary exists from day one.*

M-01 — Product Catalog

1. Module Name

Product Catalog (`ims.catalog`)

2. Business Responsibility

Maintain the authoritative product identity catalog — SKU definitions, lifecycle status, ABC classification, rotation policy, and traceability attributes — that all other modules reference when identifying *what* is being bought, held, counted, or sold.

Process Coverage	**BP-01, BP-04**

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Product Catalog	Product	Single SKU master record and embedded classification attributes

4. Public Interfaces

Commands (Write)

Command	Description
`CreateProduct`	Register new SKU with mandatory attributes
`UpdateProductAttributes`	Amend name, category, UoM, traceability settings
`ChangeProductLifecycleStatus`	Transition active → inactive → discontinued
`AssignAbcClassification`	Set or update ABC class for a SKU
`ResolveDuplicateProductConflict`	Merge or reject duplicate identity

Queries (Read)

Query	Description
`GetProductBySku`	Retrieve product master by SKU
`SearchProducts`	Filter by name, category, status, ABC class
`ValidateProductTransactionEligibility`	Check lifecycle allows procurement/reservation

Query	Description
`ListProductsByCategory`	Catalog browse for operational screens

Published Events

Event	Consumers
`ProductCreated`	Intelligence
`ProductUpdated`	Intelligence
`ProductLifecycleStatusChanged`	Procurement, Order Fulfillment, Intelligence
`AbcClassificationAssigned`	Inventory Accuracy, Intelligence

External Adapter Ports (Inbound)

Port	Source
`BulkProductImportPort`	UC-046 go-live data import

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Identity & Access	M-12	Sync query	User attribution on mutations
Inventory Intelligence	M-13	Event publish	Catalog change projections

No dependencies on operational or quantity modules. Product Catalog is an upstream supplier context.

M-02 — Warehouse Network

1. Module Name

Warehouse Network (`ims.network`)

2. Business Responsibility

Define and manage the inventory location network — warehouses, stores, distribution points, storage positions, hierarchy, operational status, and location lifecycle (go-live, closure, freeze state) — that anchors all stock to physical places.

Process Coverage	**BP-02, BP-12 (freeze state)**

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Inventory Location	Inventory Location	Location master, hierarchy, status, **authoritative freeze flag**, storage positions

4. Public Interfaces

Commands (Write)

Command	Description
`CreateInventoryLocation`	Register new warehouse, store, or DC
`UpdateLocationHierarchy`	Assign parent, zones, layout structure
`ConfigureStoragePosition`	Add or amend bin/aisle positions within location

Command	Description
`ActivateInventoryLocation`	Enable location after opening balance certification
`DeactivateInventoryLocation`	Begin closure workflow
`ApplyLocationFreeze`	Set frozen flag (driven by Physical Count event)
`ReleaseLocationFreeze`	Clear frozen flag (driven by Physical Count event)
`CloseInventoryLocation`	Finalize closure with disposition confirmation

Queries (Read)

Query	Description
`GetLocationByCode`	Retrieve location master and status
`ListActiveLocations`	Network directory for dropdowns and filters
`GetStoragePositions`	Positions available for put-away assignment
`IsLocationFrozen`	Freeze guard check for operational modules
`IsLocationTransactionallyActive`	Go-live and status gate for mutations

Published Events

Event	Consumers
`InventoryLocationCreated`	Intelligence
`InventoryLocationActivated`	Inventory Balance, Intelligence
`InventoryLocationDeactivated`	All operational modules, Intelligence
`LocationFreezeApplied`	Inventory Balance, Inbound, Fulfillment, Transfer, Intelligence
`LocationFreezeReleased`	All operational modules, Intelligence
`StoragePositionConfigured`	Inbound Receiving, Intelligence

Subscribed Events

Event	Source Module	Reaction
`LocationFreezeImposed`	Inventory Accuracy	`ApplyLocationFreeze`
`LocationFreezeReleased`	Inventory Accuracy	`ReleaseLocationFreeze`

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Identity & Access	M-12	Sync query	User attribution; location-scoped role validation
Inventory Accuracy	M-09	Event subscribe	Freeze initiation from physical count
Inventory Intelligence	M-13	Event publish	Location network projections

Warehouse Network does not depend on Inventory Balance or any quantity module.

M-03 — Inventory Governance

1. Module Name

Inventory Governance (`ims.governance`)

2. Business Responsibility

Maintain organization-wide inventory control policies — reason codes, approval thresholds, tolerance limits, segregation-of-duties rules, and exception policies — that govern how operational modules authorize adjustments, receipts, and financial controls.

Process Coverage	**BP-01 (policy portion), BP-03 (policy portion), BP-32**

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Inventory Governance	Governance Policy	Policy versions, reason codes, thresholds, tolerances

4. Public Interfaces

Commands (Write)

Command	Description
`CreateReasonCode`	Add standardized exception/adjustment code
`RetireReasonCode`	Deactivate code for new transactions
`PublishGovernancePolicy`	Activate new policy version with effective date
`UpdateApprovalThreshold`	Amend financial/quantity approval limits
`UpdateReceiptTolerance`	Amend auto-accept vs. discrepancy thresholds

Queries (Read)

Command	Description
`GetActiveReasonCodes`	List selectable codes for transaction forms
`ValidateReasonCode`	Confirm code active for transaction type
`GetEffectiveApprovalThreshold`	Threshold lookup for adjustment value
`GetReceiptTolerance`	Tolerance for receipt discrepancy auto-accept
`EvaluateSegregationOfDuties`	Check initiator vs. approver SoD compliance

Published Events

Event	Consumers
`GovernancePolicyPublished`	All operational modules, Intelligence
`ReasonCodeCreated`	Intelligence
`ReasonCodeRetired`	All operational modules, Intelligence
`ApprovalThresholdUpdated`	Inventory Accuracy, Inbound, Intelligence

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Identity & Access	M-12	Sync query	SoD evaluation uses role assignments
Inventory Intelligence	M-13	Event publish	Policy compliance projections

Governance is consulted by operational modules; it does not call them.

M-04 — Inventory Balance

1. Module Name

Inventory Balance (`ims.balance`)

2. Business Responsibility

Own the authoritative inventory quantity state — on-hand, reserved, available-to-promise (ATP), quarantine holds, lot/batch breakdown, and replenishment thresholds — answering *what do we have, where is it, and what is available?*

This is the **core domain module** and the **sole mutator of stock quantities**.

Process Coverage	BP-03, BP-10, BP-21

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Stock Position	Stock Position	On-hand, reserved, ATP, lots, quarantine holds — per product per location
Stock Policy	Stock Policy	Reorder level, minimum, safety stock, maximum — per product per location

4. Public Interfaces

Commands (Write)

Command	Description
`EstablishOpeningBalance`	Create initial Stock Position at location go-live
`ReceiveStock`	Increase on-hand after put-away (event-driven)
`ReserveStock`	Commit ATP for order line
`ReleaseReservation`	Restore ATP on cancellation
`DeductStockForShipment`	Reduce on-hand and reserved after verified pick
`DeductStockForTransfer`	Reduce source on dispatch
`ReceiveStockFromTransfer`	Increase destination on transfer receipt
`ApplyApprovedAdjustment`	Apply governed quantity correction
`ApplyApprovedWriteOff`	Permanently remove quantity
`ApplyQuarantineHold`	Reduce ATP for held quantity
`ReleaseQuarantineHold`	Restore ATP after clearance
`SegregatePromotionalStock`	Ring-fence ATP per campaign event
`ReleasePromotionalAllocation`	Restore general ATP after campaign
`TransferConsignmentOwnership`	Reclassify ownership on consumption trigger
`DefineStockPolicy`	Set replenishment thresholds
`UpdateStockPolicy`	Amend thresholds with effective dating

Queries (Read)

Query	Description
`GetStockPosition`	On-hand, reserved, ATP for product at location
`GetAvailabilityBySku`	ATP for single SKU (hot path — NFR-PERF-002)

Query	Description
`GetNetworkAvailability`	ATP across locations for a SKU
`GetStockStatusIndicator`	In-stock / low-stock / out-of-stock derived state
`GetStockPolicy`	Thresholds for product at location
`GetLotBreakdown`	Lot-level quantities for traceable products
`EvaluateSourceTransferCapacity`	Spare quantity above minimum for transfer

Published Events

Event	Consumers
`OpeningBalanceEstablished`	Intelligence
`StockReceived`	Order Fulfillment (backorder), Intelligence
`StockReserved`	Order Fulfillment, Intelligence
`ReservationReleased`	Intelligence
`StockShipped`	Intelligence, Finance ACL
`StockAdjusted`	Intelligence, Finance ACL
`StockWrittenOff`	Intelligence, Finance ACL
`QuarantineHoldApplied`	Intelligence
`QuarantineHoldReleased`	Intelligence
`LowStockThresholdBreached`	Procurement, Intelligence
`OutOfStockDetected`	Procurement, Intelligence
`PromotionalStockSegregated`	Commercial Campaign, Intelligence
`PromotionalAllocationReleased`	Commercial Campaign, Intelligence
`ConsignmentOwnershipTransferred`	Intelligence, Finance ACL

Subscribed Events

Event	Source Module	Reaction
`PutAwayCompleted`	Inbound Receiving	`ReceiveStock`
`TransferDispatched`	Network Transfer	`DeductStockForTransfer`
`TransferReceived`	Network Transfer	`ReceiveStockFromTransfer`
`AdjustmentApproved`	Inventory Accuracy	`ApplyApprovedAdjustment` / `ApplyApprovedWriteOff`
`ReturnRestocked`	Returns Management	`ReceiveStock`
`ReturnQuarantined`	Returns Management	`ApplyQuarantineHold`
`PromotionalStockAllocated`	Commercial Campaign	`SegregatePromotionalStock`
`PromotionalAllocationReleased`	Commercial Campaign	`ReleasePromotionalAllocation`
`ShipmentDispatched`	Order Fulfillment	`DeductStockForShipment`

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Product Catalog	M-01	Sync query	SKU validation and lifecycle eligibility
Warehouse Network	M-02	Sync query	Location status and freeze guard

Dependency	Module	Coupling Type	Purpose
Inventory Governance	M-03	Sync query	Safety stock override, tolerance consultation
Identity & Access	M-12	Sync query	User attribution
Inbound Receiving	M-06	Event subscribe	Stock increase on put-away
Order Fulfillment	M-07	Command/Event	Reservation and deduction coordination
Network Transfer	M-08	Event subscribe	Transfer quantity mutations
Inventory Accuracy	M-09	Event subscribe	Approved adjustments and write-offs
Returns Management	M-10	Event subscribe	Restock and quarantine
Commercial Campaign	M-11	Event subscribe	Promotional ATP segregation
Inventory Intelligence	M-13	Event publish	All balance change projections

Inventory Balance never imports domain objects from consumer modules — only identities and events.

M-05 — Procurement & Supplier

1. Module Name

Procurement & Supplier (`ims.procurement`)

2. Business Responsibility

Manage the inbound supply commitment lifecycle — supplier relationships, internal requisition approval, purchase order placement, and expected inbound delivery planning — authorizing *what the organization has committed to buy* before goods arrive.

Process Coverage	BP-05, BP-06, BP-22, BP-23

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Supplier	Supplier	Supplier master, payment terms, consignment agreement
Purchase Requisition	Purchase Requisition	Internal replenishment request and approval
Purchase Order	Purchase Order	Commercial supplier commitment and expected deliveries

4. Public Interfaces

Commands (Write)

Command	Description
`RegisterSupplier`	Create supplier master record
`UpdateSupplierTerms`	Amend payment terms, lead time, consignment flag
`DeactivateSupplier`	Block new PO creation
`CreatePurchaseRequisition`	Internal replenishment request
`ApprovePurchaseRequisition`	Authorize requisition for PO conversion
`RejectPurchaseRequisition`	Decline with reason
`ConvertRequisitionToPurchaseOrder`	Create PO from approved requisition
`PlacePurchaseOrder`	Issue commercial commitment to supplier
`AmendPurchaseOrder`	Modify lines or delivery schedule

Command	Description
`CancelPurchaseOrder`	Cancel with receipt implications
`RecordExpectedInboundDelivery`	Update shipment schedule

Queries (Read)

Query	Description
`GetSupplierByCode`	Supplier master and terms
`GetPurchaseOrder`	PO header, lines, received quantities
`ListOpenPurchaseOrders`	Inbound pipeline for receiving and planning
`ValidatePoForReceipt`	Confirm PO authorizes receipt (BR-007)
`GetInboundPipelineStatus`	Expected deliveries by location

Published Events

Event	Consumers
`SupplierRegistered`	Intelligence
`PurchaseRequisitionCreated`	Intelligence
`PurchaseRequisitionApproved`	Intelligence
`PurchaseOrderPlaced`	Inbound Receiving, Intelligence
`PurchaseOrderAmended`	Inbound Receiving, Intelligence
`PurchaseOrderCancelled`	Inbound Receiving, Intelligence
`ExpectedInboundDeliveryUpdated`	Intelligence

Domain Service Ports (Outbound Consultation)

Port	Target Module	Purpose
`NetworkAvailabilityPort`	M-04 Inventory Balance	Evaluate network ATP before requisition (BR-011)

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Product Catalog	M-01	Sync query	SKU lifecycle validation on lines
Warehouse Network	M-02	Sync query	Receiving location validation
Inventory Balance	M-04	Sync query	Network availability evaluation
Inventory Governance	M-03	Sync query	Ad-hoc receipt policy, tolerances
Identity & Access	M-12	Sync query	User attribution and approval authority
Inbound Receiving	M-06	Event publish	PO authorizes goods receipt
Inventory Intelligence	M-13	Event publish	Procurement pipeline projections

Procurement does not mutate stock quantities.

M-06 — Inbound Receiving

1. Module Name

Inbound Receiving (`ims.inbound`)

2. Business Responsibility

Execute the inbound supply verification workflow — goods receipt, discrepancy documentation, put-away confirmation, and supplier invoice matching — governing *what was actually received and when it becomes available*.

Process Coverage	BP-07, BP-08, BP-09, BP-30 (match portion)

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Goods Receipt	Goods Receipt	Verification, receipt lines, discrepancies, put-away
Supplier Invoice	Supplier Invoice	Billing document and three-way match outcome

4. Public Interfaces

Commands (Write)

Command	Description
`CreateGoodsReceipt`	Initiate receipt against PO or ad-hoc authorization
`VerifyGoodsReceipt`	Confirm quantity, product, condition at dock
`DocumentReceiptDiscrepancy`	Record variance with classification
`ResolveReceiptDiscrepancy`	Close discrepancy with resolution
`ConfirmPutAway`	Assign storage position and complete put-away
`ReceiveAdHocInbound`	Non-PO receipt with governance authorization
`ReceiveReturnInbound`	Accept physical return (shared workflow with Returns)
`RegisterSupplierInvoice`	Record supplier billing document
`PerformThreeWayMatch`	Evaluate PO, receipt, invoice alignment
`AuthorizeInvoicePayment`	Approve match for payment gate

Queries (Read)

Query	Description
`GetGoodsReceipt`	Receipt status, lines, put-away state
`ListPendingPutAway`	Receipts awaiting storage placement
`GetReceiptDiscrepancies`	Open discrepancies by location
`GetSupplierInvoice`	Invoice and match status
`GetThreeWayMatchResult`	Match outcome for finance review

Published Events

Event	Consumers
`GoodsReceiptVerified`	Intelligence

Event	Consumers
`ReceiptDiscrepancyDocumented`	Intelligence
`PutAwayCompleted`	**Inventory Balance**, Intelligence
`SupplierInvoiceRegistered`	Intelligence
`ThreeWayMatchCompleted`	Intelligence, Finance ACL
`InvoicePaymentAuthorized`	Finance ACL

Subscribed Events

Event	Source Module	Reaction
`PurchaseOrderPlaced`	Procurement	Enable receipt against PO
`PurchaseOrderCancelled`	Procurement	Block new receipts against PO

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Product Catalog	M-01	Sync query	Product identity and MRSL validation
Warehouse Network	M-02	Sync query	Receiving location, positions, freeze check
Inventory Governance	M-03	Sync query	Tolerances, reason codes, ad-hoc policy
Procurement & Supplier	M-05	Sync query + event	PO authorization for receipt
Identity & Access	M-12	Sync query	User attribution
Inventory Balance	M-06	Event publish	Put-away triggers stock increase
Returns Management	M-10	Sync query	RMA validation for return receipt
Inventory Intelligence	M-13	Event publish	Receipt and match projections

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M-07 — Order Fulfillment

1. Module Name

Order Fulfillment (`ims.fulfillment`)

2. Business Responsibility

Manage the customer order fulfillment lifecycle — availability confirmation, reservation, pick/pack/ship, partial fulfillment, backorders, and fulfillment exceptions — governing *how customer demand consumes inventory*.

Process Coverage	BP-16, BP-17, BP-18, BP-19, BP-20

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Customer Order Fulfillment	Customer Order	Order lines, reservations, pick/pack, backorders, shipments

4. Public Interfaces

Commands (Write)

Command	Description
`ImportCustomerOrder`	Accept order from Sales/CRM ACL
`ConfirmOrderAvailability`	Validate ATP for all lines
`ConfirmCustomerOrder`	Commit order and initiate reservation saga
`ReserveInventoryForOrder`	Request stock reservation (saga step)
`ReleaseOrderReservations`	Cancel reservations on order cancel
`AssignPickList`	Allocate pick work to warehouse
`VerifyPick`	Confirm picked quantity and lot
`PackOrder`	Complete packing step
`DispatchShipment`	Record shipment and trigger stock deduction
`RecordPartialFulfillment`	Split line into fulfilled and backordered
`CreateBackorder`	Record customer-agreed delay
`FulfillBackorder`	Convert backorder to reservation on stock arrival
`HandleFulfillmentException`	Document and resolve outbound exception

Queries (Read)

Query	Description
`GetCustomerOrder`	Order status and line breakdown
`GetOrderFulfillmentStatus`	Reservation, pick, ship state
`ListOrdersAwaitingPick`	Warehouse work queue
`GetBackorderPipeline`	Open backorders by SKU

Published Events

Event	Consumers
`CustomerOrderImported`	Intelligence
`OrderAvailabilityConfirmed`	Intelligence
`CustomerOrderConfirmed`	Intelligence
`OrderReservationRequested`	Inventory Balance (via saga)
`OrderReservationFailed`	Intelligence (saga compensation)
`PickVerified`	Intelligence
`ShipmentDispatched`	**Inventory Balance**, Intelligence
`BackorderCreated`	Intelligence
`BackorderFulfilled`	Intelligence
`OrderCancelled`	Inventory Balance, Intelligence

Subscribed Events

Event	Source Module	Reaction
`StockReserved`	Inventory Balance	Advance reservation saga
`ReservationReleased`	Inventory Balance	Confirm cancellation
`StockReceived`	Inventory Balance	Trigger backorder fulfillment policy
`LocationFreezeApplied`	Warehouse Network	Block shipment at frozen location

External Adapter Ports

Port	External System	Purpose
`SalesOrderInboundPort`	EXT-01 Sales/CRM	Translate external order to fulfillment model (ACL)
`FulfillmentStatusOutboundPort`	EXT-01 Sales/CRM	Publish ship status back to sales system

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Product Catalog	M-01	Sync query	SKU lifecycle on order lines
Warehouse Network	M-02	Sync query	Ship-from location, freeze check
Inventory Balance	M-04	Sync query + command + event	ATP read, reservation, deduction
Inventory Governance	M-03	Sync query	Pick verification policy
Commercial Campaign	M-11	Sync query	Ring-fence check before reservation
Identity & Access	M-12	Sync query	User attribution
Returns Management	M-10	Sync query	Original order reference for RMA
Inventory Intelligence	M-13	Event publish	Fulfillment status projections

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M-08 — Network Transfer

1. Module Name

Network Transfer (`ims.transfer`)

2. Business Responsibility

Manage inter-location stock redistribution — transfer requests, approval, dispatch, in-transit tracking, and destination receipt — resolving network imbalance without new procurement.

Process Coverage	BP-24, BP-25, BP-26

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Stock Transfer	Stock Transfer	Request, approval, dispatch, in-transit, receipt — full transfer lifecycle

4. Public Interfaces

Commands (Write)

Command	Description
`RequestStockTransfer`	Initiate transfer from source to destination
`ApproveTransferRequest`	Authorize transfer for dispatch
`RejectTransferRequest`	Decline with reason
`DispatchTransfer`	Ship from source; create in-transit state
`ReceiveTransfer`	Confirm receipt at destination
`CancelInTransitTransfer`	Abort in-transit with reversal workflow

Queries (Read)

Query	Description
`GetStockTransfer`	Transfer status and line detail
`ListPendingTransferApprovals`	Approval queue for managers
`ListInTransitStock`	Pipeline inventory by location
`GetTransferHistory`	Completed transfers for location

Published Events

Event	Consumers
`TransferRequested`	Intelligence
`TransferApproved`	Intelligence
`TransferDispatched`	**Inventory Balance**, Intelligence
`TransferInTransit`	Intelligence
`TransferReceived`	**Inventory Balance**, Intelligence
`TransferCancelled`	Intelligence

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Product Catalog	M-01	Sync query	Product and lot traceability on lines
Warehouse Network	M-02	Sync query	Source/destination validation, freeze check
Inventory Balance	M-04	Sync query + event	Source capacity check; quantity mutations
Inventory Governance	M-03	Sync query	Approval thresholds
Identity & Access	M-12	Sync query	User attribution
Procurement & Supplier	M-05	Event/consult	Transfer recommended before purchase (BR-011)
Inventory Intelligence	M-13	Event publish	In-transit and network balance projections

M-09 — Inventory Accuracy

1. Module Name

Inventory Accuracy (`ims.accuracy`)

2. Business Responsibility

Maintain inventory record accuracy through cycle counting, physical inventory counts, variance investigation, governed adjustments, and write-offs — ensuring *recorded stock reflects physical reality*.

Process Coverage	**BP-11, BP-12, BP-13, BP-14, BP-15**

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Cycle Count	Cycle Count	Scheduled partial counts, blind count protocol
Physical Inventory Count	Physical Inventory Count	Wall-to-wall count, freeze initiation
Inventory Adjustment Case	Adjustment Case	Variance investigation, adjustment, write-off

4. Public Interfaces

Commands (Write)

Command	Description
`ScheduleCycleCount`	Assign count work by product/location
`RecordCycleCountResult`	Submit blind count result
`InitiatePhysicalInventoryCount`	Start wall-to-wall count at location
`RecordPhysicalCountResult`	Submit line-level count during freeze
`CertifyPhysicalInventoryCount`	Complete count and release freeze
`OpenAdjustmentCase`	Create case from variance or direct adjustment
`InvestigateVariance`	Document root cause analysis
`SubmitAdjustmentForApproval`	Route adjustment through governance
`ApproveAdjustment`	Independent approver authorization
`RejectAdjustment`	Decline adjustment with reason
`FinalizeWriteOff`	Approve permanent quantity removal

Queries (Read)

Query	Description
`GetCycleCountAssignment`	Count task without revealing system quantity (blind)
`GetPhysicalCountStatus`	Count progress and freeze window
`GetAdjustmentCase`	Investigation and approval state
`ListPendingApprovals`	Adjustment queue for managers

Published Events

Event	Consumers
`CycleCountCompleted`	Intelligence
`CycleCountVarianceDetected`	Intelligence
`LocationFreezeImposed`	**Warehouse Network**, Intelligence
`LocationFreezeReleased`	**Warehouse Network**, Intelligence
`PhysicalInventoryCountCertified`	Intelligence
`AdjustmentCaseOpened`	Intelligence
`AdjustmentApproved`	**Inventory Balance**, Intelligence
`WriteOffApproved`	**Inventory Balance**, Intelligence

Domain Service Ports (Outbound Consultation)

Port	Target Module	Purpose
`SystemQuantityPort`	M-04 Inventory Balance	Read system quantity at variance calculation (not shown to counter)

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Product Catalog	M-01	Sync query	ABC-driven count frequency
Warehouse Network	M-02	Sync query + event	Freeze apply/release
Inventory Balance	M-04	Sync query + event	System quantity read; approved mutations
Inventory Governance	M-03	Sync query	Reason codes, thresholds, SoD
Identity & Access	M-12	Sync query	Counter vs. approver SoD
Returns Management	M-10	Command	Return write-off may open Adjustment Case
Inventory Intelligence	M-13	Event publish	Accuracy and shrinkage projections

Inventory Accuracy never directly mutates Stock Position — only via approved events.

M-10 — Returns Management

1. Module Name

Returns Management (`ims.returns`)

2. Business Responsibility

Manage reverse logistics — return authorization, physical return receipt, condition inspection, and disposition routing (restock, quarantine, write-off) — governing *how returned goods re-enter or exit inventory*.

Process Coverage	BP-27, BP-28

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Customer Return Case	Return Authorization	RMA, return receipt, inspection, disposition

4. Public Interfaces

Commands (Write)

Command	Description
`AuthorizeCustomerReturn`	Issue RMA linked to original order
`ReceiveReturnedGoods`	Record physical return at warehouse
`InspectReturnedGoods`	Assess condition grade
`DispositionReturn`	Route to restock, quarantine, or write-off
`CancelReturnAuthorization`	Void unused RMA

Queries (Read)

Query	Description
`GetReturnAuthorization`	RMA status and eligibility
`ListPendingReturnReceipts`	Returns awaiting warehouse processing
`GetReturnDisposition`	Inspection outcome and routing

Published Events

Event	Consumers
`ReturnAuthorized`	Intelligence
`ReturnReceived`	Intelligence
`ReturnInspected`	Intelligence
`ReturnRestocked`	**Inventory Balance**, Intelligence
`ReturnQuarantined`	**Inventory Balance**, Intelligence
`ReturnWrittenOff`	**Inventory Accuracy** (Adjustment Case), Intelligence

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Product Catalog	M-01	Sync query	Returned product validation
Warehouse Network	M-02	Sync query	Return receiving location
Order Fulfillment	M-07	Sync query	Original order reference (BR-025)
Inbound Receiving	M-06	Shared workflow	Physical receipt verification pattern
Inventory Balance	M-04	Event publish	Restock and quarantine
Inventory Accuracy	M-09	Event publish	Write-off via Adjustment Case
Inventory Governance	M-03	Sync query	Disposition reason codes
Identity & Access	M-12	Sync query	User attribution
Inventory Intelligence	M-13	Event publish	Returns pipeline projections

M-11 — Commercial Campaign

1. Module Name

Commercial Campaign (`ims.campaign`)

2. Business Responsibility

Ring-fence inventory for promotional campaigns and product launches, protect allocated stock from non-promotional consumption, and support campaign readiness confirmation.

Process Coverage	BP-31

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
Promotional Stock Allocation	Promotional Stock Allocation	Campaign ring-fence, channel scope, readiness

4. Public Interfaces

Commands (Write)

Command	Description
`CreatePromotionalAllocation`	Ring-fence quantity for campaign
`AmendPromotionalAllocation`	Adjust allocated quantity or period
`ConfirmCampaignReadiness`	Certify stock sufficient for launch
`ReleasePromotionalAllocation`	End campaign and release unspent stock
`AuthorizePromotionalOverride`	Allow non-promotional consumption of ring-fenced stock

Queries (Read)

Query	Description
`GetPromotionalAllocation`	Allocation status and quantity
`CheckRingFenceForSku`	Whether SKU has active ring-fence at location
`GetCampaignReadinessStatus`	Launch readiness assessment
`ListActiveCampaigns`	Active allocations by period

Published Events

Event	Consumers
`PromotionalStockAllocated`	**Inventory Balance**, Intelligence
`CampaignReadinessConfirmed`	Intelligence
`CampaignReadinessFailed`	Intelligence
`PromotionalAllocationReleased`	**Inventory Balance**, Intelligence
`PromotionalOverrideAuthorized`	Order Fulfillment, Intelligence

External Adapter Ports

Port	External System	Purpose
`CampaignPlanInboundPort`	EXT-07 Marketing	Import campaign plan (ACL)

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Product Catalog	M-01	Sync query	Campaign SKU validation
Warehouse Network	M-02	Sync query	Allocation location scope
Inventory Balance	M-04	Sync query + event	Available quantity check; ATP segregation
Order Fulfillment	M-07	Event publish	Override authorization
Identity & Access	M-12	Sync query	User attribution
Inventory Intelligence	M-13	Event publish	Campaign readiness projections

Commercial Campaign owns allocation intent; Inventory Balance owns ATP mechanics.

M-12 — Identity & Access

1. Module Name

Identity & Access (`ims.identity`)

2. Business Responsibility

Manage user identity, role assignments, location-scoped permissions, and segregation-of-duties enforcement for all inventory operations — governing *who may perform and approve* inventory actions.

Process Coverage	Cross-cutting (UC-044, UC-045)

3. Owned Aggregates

Aggregate	Root Entity	Consistency Scope
User Account	User Account	User identity, role assignments, location scope

4. Public Interfaces

Commands (Write)

Command	Description
`ProvisionUserAccount`	Create user identity
`AssignRole`	Delegate role with location scope
`RevokeRole`	Remove role assignment
`SuspendUserAccount`	Temporarily block access
`DeactivateUserAccount`	Permanent account closure
`SyncUserFromDirectory`	Import identity from HR/IdP

Queries (Read)

Query	Description
`GetUserAccount`	User profile and role assignments
`AuthorizeAction`	Permission check for command execution
`ValidateLocationScope`	Confirm user authorized for location
`DetectSegregationOfDutiesConflict`	Block conflicting role combinations
`GetUserAttribution`	Identity template for audit records

Published Events

Event	Consumers
`UserAccountProvisioned`	Intelligence
`RoleAssigned`	Intelligence
`RoleRevoked`	Intelligence
`UserAccountDeactivated`	All modules (session invalidation), Intelligence
`SegregationOfDutiesViolationDetected`	Intelligence

External Adapter Ports

Port	External System	Purpose
`IdentityProviderPort`	SVC-01 Enterprise IdP	SSO authentication
`DirectorySyncPort`	EXT-08 HR Directory	User provisioning sync

5. Dependencies

Dependency	Module	Coupling Type	Purpose
Warehouse Network	M-02	Sync query	Location codes for scope validation
Inventory Governance	M-03	Shared policy	SoD rules defined in governance, enforced here
Inventory Intelligence	M-13	Event publish	Access audit projections

Identity & Access is consumed by all modules; it does not depend on operational modules except Network (for location scope).

M-13 — Inventory Intelligence

1. Module Name

Inventory Intelligence (`ims.intelligence`)

2. Business Responsibility

Provide read-optimized views of inventory state — dashboards, KPIs, movement history, audit trails, alerts, and valuation snapshots — supporting management oversight **without owning transactional write models**.

Process Coverage	BP-29, BP-30 (reporting), BP-32

3. Owned Aggregates

None. This module owns **read projections** only:

Projection	Source Events	Purpose
Inventory Movement Record	All balance change events	Append-only movement history
Audit Trail Record	All material command events	Governance-grade change log
Operational Alert	Threshold and exception events	Low-stock, approval pending, freeze alerts
Inventory Valuation Snapshot	Period-end aggregation	Finance reconciliation
KPI Dashboard Views	Multiple event streams	Fill rate, accuracy, turnover metrics
Backorder Pipeline View	Fulfillment events	CS and operations visibility
Inbound Pipeline View	Procurement and receipt events	Supply chain visibility

4. Public Interfaces

Queries (Read Only — No Commands)

Query	Description
`GetInventoryDashboard`	Location or network summary dashboard
`GetStockStatusSummary`	On-hand, ATP, low-stock counts
`SearchMovementHistory`	Filter movements by SKU, location, period
`QueryAuditTrail`	Compliance audit search and export
`GetOperationalAlerts`	Active alerts for user/location scope
`GetKpiReport`	Performance KPIs (fill rate, accuracy, shrinkage)
`GetValuationSnapshot`	Period-end inventory valuation
`GetBackorderPipeline`	Open backorders sorted by policy
`GetInboundPipelineReport`	Open POs and expected deliveries
`ExportAuditEvidencePackage`	Auditor export (NFR-PERF-026, NFR-COM-012)

Published Events

Event	Consumers
`OperationalAlertRaised`	Notification gateway (SVC-02)
`KpiThresholdBreached`	Notification gateway
`ValuationSnapshotPrepared`	Finance ACL (EXT-02)

Subscribed Events

Event	Source
All domain events from M-01 through M-12	Event bus — projection handlers update read models

External Adapter Ports (Outbound)

Port	External System	Purpose
`FinanceValuationExportPort`	EXT-02 Finance ERP	Valuation snapshot ACL
`ExecutiveBiExportPort`	External BI	KPI data feed

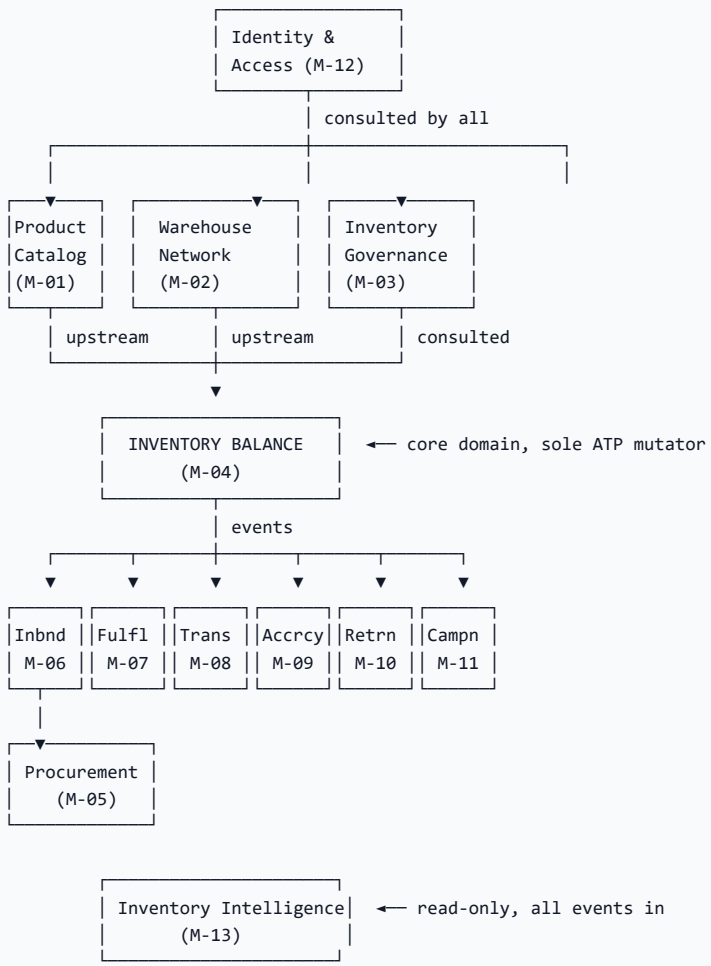
5. Dependencies

Dependency	Module	Coupling Type	Purpose
All transactional modules	M-01 – M-12	Event subscribe	Build and maintain read projections
Identity & Access	M-12	Sync query	Scope queries to user permissions

Inventory Intelligence has no write dependencies and never calls back into transactional modules.

Cross-Module Integration Map

Dependency Direction (Upstream → Downstream)



Coupling Mechanism Summary

From Module	To Module	Mechanism	Mutable State?
Any operational	Product Catalog	Sync query	No
Any operational	Warehouse Network	Sync query	No
Any operational	Inventory Governance	Sync query	No
Any operational	Identity & Access	Sync query	No
Fulfillment, Transfer, Accuracy, Returns, Campaign	Inventory Balance	Command + event	Yes (Balance only)
Inbound Receiving	Inventory Balance	Event (`PutAwayCompleted`)	Yes (Balance only)
Inventory Accuracy	Warehouse Network	Event (freeze)	Yes (Network only)

From Module	To Module	Mechanism	Mutable State?
Commercial Campaign	Inventory Balance	Event (ring-fence)	Yes (Balance only)
All transactional	Intelligence	Event publish	No (projections only)
Fulfillment	Sales/CRM	ACL adapter	External
Intelligence	Finance ERP	ACL adapter	External

Saga Ownership (Cross-Module Workflows)

Saga	Orchestrator Module	Participating Modules
Order Confirm → Reserve	Order Fulfillment (M-07)	Inventory Balance (M-04)
Put-Away → Stock Available	Inbound Receiving (M-06)	Inventory Balance (M-04)
Transfer Dispatch → Receive	Network Transfer (M-08)	Inventory Balance (M-04)
Count Variance → Adjust	Inventory Accuracy (M-09)	Inventory Balance (M-04), Warehouse Network (M-02)
Return Disposition → Restock/Write-off	Returns Management (M-10)	Inventory Balance (M-04) or Accuracy (M-09)
Opening Balance → Location Activate	Warehouse Network (M-02)	Inventory Balance (M-04)
Campaign Allocate → ATP Segregate	Commercial Campaign (M-11)	Inventory Balance (M-04)

Module Boundary Enforcement Rules

Rule ID	Rule
MB-01	No module imports another module's domain entities or aggregate internals
MB-02	Cross-module communication uses only published commands, queries, events, and shared identifiers
MB-03	Only M-04 (Inventory Balance) mutates on-hand, reserved, or quarantined quantities
MB-04	Only M-02 (Warehouse Network) mutates the location freeze flag — initiated by M-09 events
MB-05	M-13 (Intelligence) subscribes to events only — no write-back to source modules
MB-06	External systems integrate through ACL adapter ports — never directly into domain cores
MB-07	Each module owns its persistence — no shared tables across module boundaries
MB-08	Governance and Identity are consulted synchronously; state changes propagate asynchronously via events

Phased Module Activation

Phase	Active Modules	Notes
Phase 1 — MVP (0–6 mo)	M-01, M-02, M-03, M-04, M-06, M-07, M-09, M-12, M-13	Core receive, fulfill, count, adjust; basic procurement receipt against PO
Phase 2 (6–12 mo)	+ M-05 (full), M-08, M-10, M-11	Full procurement, transfers, returns, campaigns
Phase 3 (12–18 mo)	All modules at full maturity	Finance ACL, advanced Intelligence, valuation snapshots

Document Approval

Role	Name	Signature	Date
Solution Architect			
DDD Solution Architect			
Delivery Lead			

Role	Name	Signature	Date
Product Owner			

End of System Modules Design

Document ID: SRS-SA-003 | Version 1.0 | July 6, 2026

Layered Architecture Design

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-SA-004
Document Title	Layered Architecture Design — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Solution Architect
Prepared For	Development squads, technical architects, and QA leads
Date	July 6, 2026
Classification	Internal — Architecture Confidential
Scope	Layer structure and dependency rules within each bounded-context module — no framework selection, class diagrams, or code

Related Documents

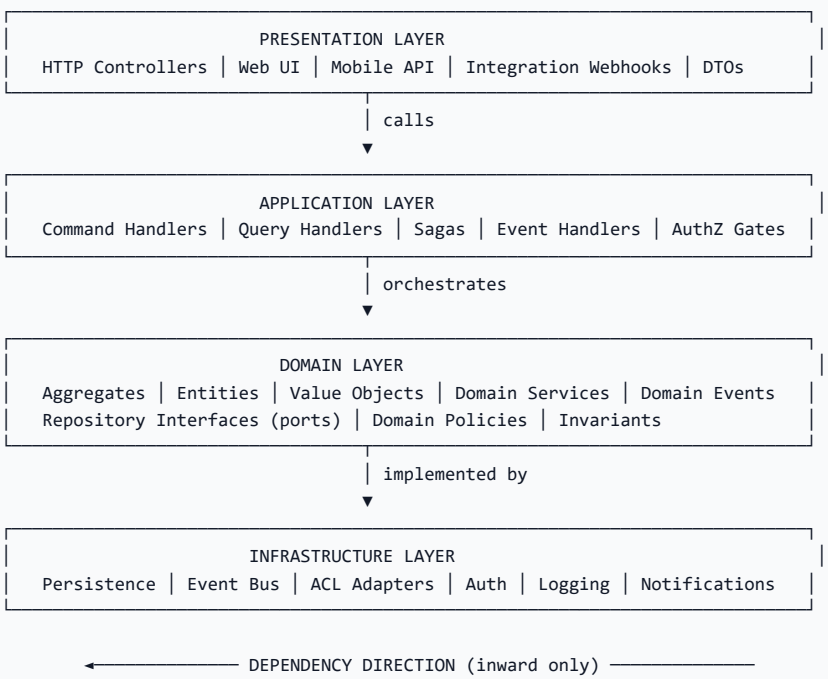
Document	ID	Relationship
Architecture Vision	SRS-SA-001	Strategic direction
Architecture Style Recommendation	SRS-SA-002	Clean/Hexagonal + modular monolith
System Modules Design	SRS-SA-003	13 module boundaries
Domain Aggregates	SRS-DM-003	Domain layer content
Domain Services	SRS-DM-005	Domain and application coordination

Document Purpose

This document defines the **four-layer internal architecture** applied within each bounded-context module of the Inventory Management System modular monolith. It follows **Clean Architecture** (dependency rule: source code dependencies point inward) combined with **Hexagonal ports and adapters** at module boundaries.

1. Clean Architecture Overview

1.1 Layer Model



1.2 The Dependency Rule

Rule	Statement
DR-01	Source code dependencies must point **inward** only — from outer layers toward the domain
DR-02	Inner layers **never** reference outer layers — no domain import of controllers, ORM, or HTTP clients
DR-03	Cross-layer collaboration uses **interfaces (ports)** defined in the domain or application layer, implemented in infrastructure
DR-04	Data crosses boundaries as **plain data structures** — DTOs, commands, queries, events — never as framework types
DR-05	Frameworks and drivers are **details** relegated to the infrastructure layer

1.3 Layer-to-Module Relationship

Each of the **13 bounded-context modules** (SRS-SA-003) contains its own complete four-layer stack. Layers are **replicated per module**, not shared as a single horizontal slice across the entire application.

```
ims/
├── catalog/          (M-01)  → presentation / application / domain / infrastructure
├── network/          (M-02)
├── governance/       (M-03)
├── balance/          (M-04)  → core domain — richest layer
├── procurement/      (M-05)
├── inbound/          (M-06)
├── fulfillment/      (M-07)
├── transfer/         (M-08)
├── accuracy/         (M-09)
├── returns/          (M-10)
├── campaign/         (M-11)
├── identity/         (M-12)
├── intelligence/     (M-13)  → query-heavy; thin domain, rich application + infra read models
└── shared-kernel/    → published identifiers only (ProductSku, LocationCode) — no layers
```

1.4 CQRS Layer Split

Command and query paths share the domain layer but diverge in application and infrastructure:

Path	Application Layer	Infrastructure Layer
Commands (writes)	Command handlers, sagas, domain event dispatch	Write repositories, transactional outbox, unit of work
Queries (reads)	Query handlers	Read repositories, projections, materialized views

Inventory Intelligence (M-13) is predominantly **query path** — minimal domain, no write command handlers.

---

2. Presentation Layer

2.1 Responsibilities

The Presentation Layer is the **outermost adapter** that translates external actor input into application-layer commands and queries, and translates application results into responses consumable by users and external systems.

Responsibility	Description
Request reception	Accept HTTP requests, UI actions, mobile scanner input, and integration webhook payloads
Input validation (syntactic)	Validate request shape, required fields, data types, and format — not business rules
Authentication presentation	Extract identity tokens or session credentials and pass to application layer
DTO mapping	Map HTTP/JSON/UI models to application commands and queries; map results to response DTOs
Authorization presentation	Invoke application-layer authorization gates before dispatching commands
Error translation	Map application/domain exceptions to HTTP status codes and user-safe error messages
API versioning and routing	Route requests to correct module endpoints
Content negotiation	JSON for APIs; HTML/view models for web UI; optimized payloads for mobile warehouse clients
Correlation ID attachment	Attach trace/correlation identifiers to requests for observability

The Presentation Layer does not:

- Enforce business rules (BR-001 – BR-032)
- Access databases or repositories directly
- Mutate domain entities
- Publish domain events
- Contain transaction boundaries

2.2 Allowed Dependencies

May Depend On	Must Not Depend On
Application Layer (command/query dispatch)	Domain Layer (direct aggregate access)
Shared presentation utilities (validation attributes, base controller)	Infrastructure Layer (repositories, ORM, message bus)
Cross-cutting presentation middleware (auth middleware invokes application)	Other modules' domain internals
Published DTO / request/response contracts	Framework-specific types leaking into application

```
Presentation ──► Application     ✅
Presentation ──► Domain          ❌
Presentation ──► Infrastructure  ❌
```

2.3 Typical Components

Component	Purpose	IMS Examples
REST API Controllers	HTTP endpoint per use case group	`StockController`, `ReceiptController`, `OrderController`
Web UI Controllers / Views	Server-rendered or SPA shell pages	Dashboard views, master data forms
Mobile Warehouse API Controllers	Optimized endpoints for scanner workflows	`PickController`, `PutAwayController`, `CycleCountController`
Integration Webhook Controllers	Inbound endpoints for external systems	`SalesOrderWebhookController` (CRM ACL entry)
Request DTOs	Inbound data contracts	`ReserveStockRequest`, `VerifyReceiptRequest`
Response DTOs	Outbound data contracts	`AvailabilityResponse`, `OrderStatusResponse`
Input Validators	Syntactic validation	Required SKU, positive quantity, valid date format
Exception Filters / Handlers	Uniform error responses	`BusinessRuleViolationException` → 422; `NotFoundException` → 404
Authentication Middleware	Token/session extraction	JWT bearer, SSO redirect
API Documentation	OpenAPI/Swagger metadata	Endpoint catalog per module
Model Binders / Mappers	DTO ↔ Command/Query translation	`ReserveStockRequest` → `ReserveStockCommand`

2.4 Presentation Layer by Actor

Actor	Presentation Entry Points
Warehouse Staff	Mobile API — put-away, cycle count, transfer
Receiving Clerk	Mobile/Web API — goods receipt, discrepancy
Picker/Packer	Mobile API — pick verification, shipment
Customer Service	Web API — availability inquiry, order confirm, RMA
Purchasing	Web API — PO, requisition, inbound pipeline
Management	Web UI — dashboards (delegates to Intelligence queries)
Finance	Web API — invoice match, valuation export
System Administrator	Web UI — user and role management
Sales/CRM (external)	Integration webhook — order import

---

3. Application Layer

3.1 Responsibilities

The Application Layer **orchestrates use cases** — it coordinates domain objects, enforces application-level workflow, manages transactions, and dispatches cross-module integration. It contains **no business invariants**; those belong in the domain layer.

Responsibility	Description
Use case execution	Receive commands/queries from presentation and execute the corresponding workflow
Transaction management	Define unit-of-work boundaries — one aggregate mutation per transaction (DR-DDD)
Domain orchestration	Load aggregates via repository ports, invoke domain methods, persist changes
Authorization gates	Verify user permission and location scope before command execution (delegates to Identity module)
Saga coordination	Manage multi-step cross-module workflows with compensation (order confirm → reserve)
Domain event dispatch	Collect domain events from aggregates and publish via event bus port
Cross-module consultation	Invoke other modules' query ports synchronously (e.g., ATP lookup, freeze check)

Responsibility	Description
Application services	Coordinate operations spanning domain services or multiple ports within one use case
Idempotency	Ensure duplicate command submission does not double-apply (receipt, reservation)
Audit context enrichment	Attach user identity and correlation ID to commands before domain execution
Query orchestration	Route read requests to appropriate read repositories or cross-module query ports
DTO assembly (reads)	Compose query results from read models into response shapes

The Application Layer does not:

- Implement business invariants (ATP formula, SoD, freeze rules)
- Contain SQL or ORM code
- Reference HTTP, JSON, or UI frameworks
- Own long-lived state beyond saga/process manager instances

3.2 Allowed Dependencies

May Depend On	Must Not Depend On
Domain Layer (aggregates, domain services, repository interfaces)	Infrastructure Layer (concrete implementations)
Application contracts (commands, queries, handler interfaces)	Presentation Layer
Cross-module **query ports** (interfaces only)	Other modules' aggregate internals
Shared kernel identifiers (`ProductSku`, `LocationCode`)	ORM entities, HTTP clients, framework types

```
Application ──► Domain          ✅
Application ──► Infrastructure  ❌  (depends on ports defined in domain/application)
Application ──► Presentation    ❌
```

Dependency inversion: Application layer defines or uses **interfaces (ports)** that infrastructure implements:

Port Type	Defined In	Implemented In
`IStockPositionRepository`	Domain	Infrastructure
`IEventBus`	Application	Infrastructure
`IUnitOfWork`	Application	Infrastructure
`IAvailabilityQueryPort`	Application (fulfillment module)	Infrastructure or Balance module adapter
`IIdentityService`	Application	Infrastructure (Identity module adapter)

3.3 Typical Components

Component	Purpose	IMS Examples
Commands	Immutable write intent records	`ReserveStockCommand`, `ConfirmPutAwayCommand`, `ApproveAdjustmentCommand`
Queries	Immutable read intent records	`GetAvailabilityQuery`, `GetStockPositionQuery`, `SearchAuditTrailQuery`
Command Handlers	One handler per command — executes use case	`ReserveStockCommandHandler`, `VerifyGoodsReceiptCommandHandler`
Query Handlers	One handler per query — returns read data	`GetAvailabilityQueryHandler`, `GetDashboardQueryHandler`
Saga / Process Managers	Multi-step cross-module workflows	`OrderConfirmationSaga`, `OpeningBalanceGoLiveSaga`
Saga Steps	Individual saga activities with compensation	`ReserveStockStep`, `CompensateOrderConfirmationStep`
Domain Event Handlers	React to events from other modules	`OnPutAwayCompletedHandler` (Balance module)

Component	Purpose	IMS Examples
Application Services	Coordinate when logic is not natural on one aggregate	`AvailabilityConfirmationService`, `TransferCapacityEvaluationService`
Authorization Behaviors	Pre-handler permission checks	`RequirePermissionBehavior`, `LocationScopeBehavior`
Validation Behaviors	Application-level validation (not domain)	Command field presence, saga preconditions
Unit of Work	Transaction boundary abstraction	`IUnitOfWork.Begin()`, `Commit()`, `Rollback()`
Command Bus / Dispatcher	Route commands to handlers	Mediator-style dispatch
Query Bus / Dispatcher	Route queries to handlers	CQRS query dispatch
Integration Command Adapters	Translate external commands to internal	`ImportSalesOrderCommandHandler`
Read Model Assemblers	Compose complex query responses	`DashboardAssembler`, `KpiReportAssembler`

3.4 Application Layer Patterns by Module

Module	Notable Application Components
M-04 Inventory Balance	`ReserveStockCommandHandler`, `ReceiveStockCommandHandler`; contention-sensitive handlers
M-07 Order Fulfillment	`OrderConfirmationSaga`, `FulfillBackorderCommandHandler`
M-06 Inbound Receiving	`ConfirmPutAwayCommandHandler` → publishes `PutAwayCompleted`
M-09 Inventory Accuracy	`ApproveAdjustmentCommandHandler` → publishes `AdjustmentApproved`
M-13 Intelligence	Query handlers only — `GetDashboardQueryHandler`, `ExportAuditEvidenceQueryHandler`

3.5 Command Handler Flow (Reference)

```
Controller
    │
    ▼
CommandBus.Dispatch(ReserveStockCommand)
    │
    ▼
AuthorizationBehavior          ──► IIdentityService.AuthorizeAction()
    │
    ▼
ReserveStockCommandHandler
    │
    ├──► IStockPositionRepository.GetBySkuAndLocation()
    ├──► StockPosition.Reserve(quantity)     [Domain Layer]
    ├──► IStockPositionRepository.Save()
    ├──► IUnitOfWork.Commit()
    └──► IEventBus.Publish(StockReserved)    [Infrastructure port]
```

---

4. Domain Layer

4.1 Responsibilities

The Domain Layer is the **center of the architecture** — it contains the business logic, rules, and models that represent the inventory management problem space. It is **framework-agnostic** and **persistence-ignorant**.

Responsibility	Description
Aggregate roots	Enforce consistency boundaries and transactional invariants
Entities	Objects with identity and lifecycle within aggregates
Value objects	Immutable, identity-less concepts (Quantity, Money, ReasonCode)

Responsibility	Description
Domain invariants	Enforce business rules at aggregate boundaries (BR-001 – BR-032)
Domain events	Record meaningful business facts in past tense (`StockReserved`, `PutAwayCompleted`)
Domain services	Stateless operations spanning concepts not natural on one entity
Domain policies	Encapsulate decision rules (FEFO lot selection, freeze guard evaluation)
Repository interfaces	Define persistence contracts — no implementation
Factory methods	Controlled aggregate creation with invariant validation
Ubiquitous language	Naming aligned to SRS-DM-008 terms

The Domain Layer does not:

- Know about databases, HTTP, UI, or message brokers
- Depend on application or infrastructure layers
- Perform I/O of any kind
- Reference other modules' aggregates directly — only shared identifiers

4.2 Allowed Dependencies

May Depend On	Must Not Depend On
Other domain types within the **same module**	Application Layer
Shared kernel value types (`ProductSku`, `LocationCode`, `UserId`)	Infrastructure Layer
Domain interfaces (repository ports) defined in same module	Presentation Layer
Standard language primitives and approved domain libraries (none required)	ORM attributes, HTTP attributes, logging frameworks
	Other modules' domain aggregates or entities

```
Domain ──► Shared Kernel identifiers   ✅
Domain ──► Application                 ❌
Domain ──► Infrastructure              ❌
Domain ──► Presentation                ❌
Domain ──► Other module domain         ❌
```

Cross-module domain interaction uses only:

- **Identity references** (`ProductSku`, not `Product` entity)
- **Domain events** consumed by application-layer handlers in other modules
- **Query ports** (interfaces) for read-only consultation — interface may be defined in calling module's domain or a shared contracts package; implementation in infrastructure

4.3 Typical Components

Component	Purpose	IMS Examples
Aggregate Roots	Consistency and mutation entry points	`StockPosition`, `GoodsReceipt`, `CustomerOrder`, `GovernancePolicy`
Entities	Identified objects within aggregates	`LotBatch`, `QuarantineHold`, `InventoryReservation`, `PutAway`
Value Objects	Immutable domain concepts	`Quantity`, `ProductSku`, `LocationCode`, `AtpBreakdown`, `ReasonCode`, `ApprovalThreshold`
Domain Events	Past-tense business facts	`StockReserved`, `PutAwayCompleted`, `AdjustmentApproved`, `LocationFreezeApplied`
Domain Services	Cross-aggregate logic within module	`FefoLotSelectionService`, `AtpCalculationService`
Domain Policies	Reusable rule evaluators	`SafetyStockPolicy`, `ProductEligibilityPolicy`, `SegregationOfDutiesPolicy`
Repository Interfaces	Persistence ports	`IStockPositionRepository`, `IGoodsReceiptRepository`, `IProductRepository`

Component	Purpose	IMS Examples
Specification Objects	Composable query predicates (domain language)	`ActiveProductSpecification`, `FrozenLocationSpecification`
Domain Exceptions	Business rule violations	`InsufficientAtpException`, `LocationFrozenException`, `SegregationOfDutiesViolationException`
Enumerations / Status Types	Lifecycle states as domain concepts	`ProductLifecycleStatus`, `TransferStatus`, `AdjustmentCaseStatus`
Aggregate Factories	Controlled creation	`StockPosition.CreateOpeningBalance()`, `GoodsReceipt.CreateFromPurchaseOrder()`

4.4 Domain Layer Richness by Module

Module	Domain Complexity	Key Aggregates / Concepts
M-04 Inventory Balance	**Highest (core domain)**	`StockPosition` invariants: ATP = on-hand − reserved − quarantined
M-07 Order Fulfillment	High	`CustomerOrder` lifecycle: confirm → reserve → pick → ship
M-06 Inbound Receiving	High	`GoodsReceipt` — verify → discrepancy → put-away chain
M-09 Inventory Accuracy	High	`AdjustmentCase` state machine: variance → investigate → approve
M-03 Inventory Governance	Medium	`GovernancePolicy` versioning, reason code lifecycle
M-01 Product Catalog	Medium	`Product` lifecycle and eligibility
M-13 Intelligence	**Minimal**	No write aggregates; optional read value types only

4.5 Aggregate Design Rules in Domain Layer

Rule	Description
DL-01	All state changes go through aggregate root public methods
DL-02	One transaction modifies one aggregate
DL-03	External references hold identity only — `ProductSku`, not `Product`
DL-04	Domain events raised inside aggregate before persistence
DL-05	Repository interfaces return aggregate roots — not ORM entities
DL-06	Value objects are immutable — replacement, not mutation
DL-07	Domain layer never queries other modules — consultation via ports at application layer

---

5. Infrastructure Layer

5.1 Responsibilities

The Infrastructure Layer provides **technical capabilities** — the concrete implementations of ports defined by inner layers. It is the **outermost implementation detail**, swappable without changing business logic.

Responsibility	Description
Persistence	Implement repository interfaces — ORM mappings, SQL, aggregate hydration and persistence
Event bus	Publish and subscribe to domain events — in-process bus (MVP) or message broker (future)
Transactional outbox	Ensure reliable event delivery with database commits
Unit of work	Manage database transactions and connection lifecycle
ACL adapters	Translate external system models to/from application commands (Sales/CRM, Finance ERP)
Authentication adapter	Integrate enterprise IdP (SSO), validate tokens, resolve `UserId`

Responsibility	Description
Authorization adapter	Implement Identity module query ports for other modules
Cross-module adapters	Implement query/command ports between modules within the monolith
Read model projections	Maintain CQRS read stores for Intelligence module
Caching	ATP hot-path cache with invalidation on balance events
Logging and audit	Structured logging, audit record persistence
Monitoring	Metrics, health checks, distributed tracing hooks
Notification delivery	Email/SMS gateway for operational alerts
Background jobs	Scheduled tasks — reservation expiry, KPI rollups, backup verification
File import/export	Bulk product/stock import (FR-090), audit evidence export
Clock / ID generation	Injectable time source and unique ID generation for testability

The Infrastructure Layer does not:

- Contain business rules or invariants
- Be referenced directly by presentation layer
- Leak ORM entities into domain or application layers

5.2 Allowed Dependencies

May Depend On	Must Not Depend On
Domain Layer (implement repository interfaces)	Presentation Layer
Application Layer (implement ports, event handlers wiring)	—
Infrastructure frameworks (ORM, HTTP clients, message brokers)	Domain aggregate internals (only through repository interfaces)
Other modules' **public query ports** (adapter implementations)	Other modules' private infrastructure
Shared kernel types	—

```
Infrastructure ──► Domain          ✅ (implements interfaces)
Infrastructure ──► Application     ✅ (implements ports, wires handlers)
Infrastructure ──► Presentation    ❌
```

5.3 Typical Components

Component	Purpose	IMS Examples
Repository Implementations	Aggregate persistence	`StockPositionRepository`, `GoodsReceiptRepository`, `ProductRepository`
ORM Entity Mappings	Database row ↔ aggregate mapping	`StockPositionConfiguration`, `GoodsReceiptConfiguration`
Unit of Work Implementation	Transaction scope	`EfUnitOfWork`, `DbContext` wrapper
In-Process Event Bus	MVP domain event dispatch	`InProcessEventBus`, `DomainEventDispatcher`
Transactional Outbox	Reliable event publish	`OutboxMessage`, `OutboxProcessor`
Event Handlers (wiring)	Register handlers with bus	`EventHandlerRegistry`, module startup configuration
Cross-Module Adapters	Inter-module port implementations	`BalanceAvailabilityQueryAdapter` (fulfillment → balance)
ACL Adapters (external)	External system translation	`SalesCrmOrderAdapter`, `FinanceErpValuationAdapter`
Identity Provider Adapter	SSO integration	`EntraIdAuthenticationAdapter`, `JwtTokenValidator`
Read Model Repositories	CQRS query stores	`MovementHistoryReadRepository`, `DashboardReadRepository`
Projection Handlers	Update read models on events	`OnStockReservedProjectionHandler`
Cache Providers	Performance optimization	`AtpCacheService` with event-driven invalidation

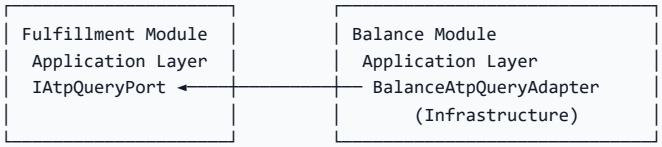
Component	Purpose	IMS Examples
Logging Adapter	Structured audit logging	`SerilogAuditLogger`, `AuditTrailWriter`
Notification Adapter	Alert delivery	`EmailNotificationGateway` (SVC-02)
Import/Export Adapters	Bulk data operations	`ProductStockBulkImporter`, `AuditEvidenceExporter`
Background Job Scheduler	Async processing	`ReservationExpiryJob`, `KpiRollupJob`
Health Check Implementations	Ops monitoring	`DatabaseHealthCheck`, `EventBusHealthCheck`
Migration Scripts	Schema versioning	Per-module database migrations

5.4 Infrastructure by Concern

Concern	Components	NFR Alignment
Write persistence	Repositories, UoW, outbox	NFR-REL-006 (integrity), NFR-BDR-001 (RPO)
Read persistence	Read repositories, projections	NFR-PERF-002 (ATP), NFR-PERF-004 (dashboards)
Security	Auth adapter, encryption	NFR-SEC-001 – NFR-SEC-032
Observability	Logging, metrics, tracing	NFR-LOG, NFR-MON
Integration	ACL adapters	TEC-04, TEC-05, HS-014
Resilience	Outbox, retry, idempotency store	NFR-REL-011 – NFR-REL-014

5.5 Cross-Module Infrastructure Pattern

Cross-module integration within the monolith uses **infrastructure adapters** implementing application-layer ports:



The port interface (`IAtpQueryPort`) is defined in the **calling module's application layer**. The adapter is implemented in **infrastructure** and may live in either module's infrastructure package, registered at application composition root.

---

6. Cross-Cutting Concerns Placement

Concern	Primary Layer	Notes
Business rules (BR-xxx)	**Domain**	Never in presentation or infrastructure
Use case orchestration	**Application**	Handlers and sagas
Syntactic input validation	**Presentation**	Format, required fields
Permission checks	**Application**	Via `IIdentityService` port before handler
Authentication	**Infrastructure** (adapter) + **Presentation** (middleware)	Token validation in infra; extraction in presentation
Transaction boundaries	**Application**	`IUnitOfWork` port
Audit trail recording	**Infrastructure** (persistence)	Triggered by domain events → Intelligence projections
API rate limiting	**Presentation** or **Infrastructure**	Edge concern
Correlation / tracing	**Presentation** (attach) + **Infrastructure** (propagate)	
Idempotency	**Application** (logic) + **Infrastructure** (store)	
Caching	**Infrastructure**	ATP cache with domain-event invalidation
Exception mapping	**Presentation**	HTTP status from domain exception types

7. Dependency Matrix (Complete)

Layer	Presentation	Application	Domain	Infrastructure
Presentation	—	✅	❌	❌
Application	❌	internal	✅	❌ (ports only)
Domain	❌	❌	internal	❌ (interfaces only)
Infrastructure	❌	✅	✅	internal

8. Request Lifecycle (End-to-End)

8.1 Write Path — Confirm Customer Order

```
1. [Presentation]  OrderController.ConfirmOrder(ConfirmOrderRequest dto)
                     → maps to ConfirmCustomerOrderCommand
2. [Application]     AuthorizationBehavior checks permission
3. [Application]     ConfirmCustomerOrderCommandHandler
                     → starts OrderConfirmationSaga
4. [Application]     Saga step: CustomerOrder.Confirm()          [Domain]
5. [Infrastructure]  CustomerOrderRepository.Save()
6. [Application]     Saga step: ReserveStockCommand → Balance module port
7. [Application]     ReserveStockCommandHandler (Balance module)
                     → StockPosition.Reserve()                   [Domain]
8. [Infrastructure]  StockPositionRepository.Save()
                     → UnitOfWork.Commit()
                     → Outbox.Publish(StockReserved)
9. [Infrastructure]  EventBus dispatches StockReserved
10.[Application]     Fulfillment saga resumes on StockReserved event
11.[Presentation]    OrderController returns ConfirmOrderResponse
```

8.2 Read Path — ATP Lookup

```
1. [Presentation]  StockController.GetAvailability(sku, location)
                     → maps to GetAvailabilityQuery
2. [Application]     GetAvailabilityQueryHandler
3. [Infrastructure]  AtpReadRepository (or cache) returns projection
4. [Presentation]    Maps to AvailabilityResponse DTO
```

Read path bypasses domain aggregate load for performance (CQRS) — NFR-PERF-002.

9. Layer Implementation Standards

Standard ID	Rule
LS-01	Every module packages layers as `ModuleName.Presentation`, `.Application`, `.Domain`, `.Infrastructure`
LS-02	Domain project has **zero** references to outer layer projects
LS-03	Application project references Domain only — infrastructure wired at composition root
LS-04	Repository interfaces live in Domain; implementations live in Infrastructure
LS-05	Commands and queries are immutable records in Application layer
LS-06	One command handler per command; handlers are stateless
LS-07	Domain events are raised inside aggregates, collected by application layer after save
LS-08	Cross-module calls use ports — never `using OtherModule.Domain`
LS-09	Intelligence module has no command handlers for transactional writes

Standard ID	Rule
LS-10	Integration tests target Application + Infrastructure with in-memory or test database adapters

10. Document Approval

Role	Name	Signature	Date
Solution Architect			
DDD Solution Architect			
Delivery Lead			
Tech Lead			

End of Layered Architecture Design

Document ID: SRS-SA-004 | Version 1.0 | July 6, 2026

Integration Architecture Design

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-SA-005
Document Title	Integration Architecture Design — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Solution Architect
Prepared For	Delivery leads, integration architects, security, and development squads
Date	July 6, 2026
Classification	Internal — Architecture Confidential
Scope	Integration patterns, boundaries, and communication styles — no message broker selection, API specification, or protocol implementation

Related Documents

Document	ID	Relationship
Architecture Vision	SRS-SA-001	Event-driven direction, ACL principles
System Modules Design	SRS-SA-003	Module public interfaces and dependencies
Layered Architecture Design	SRS-SA-004	Port/adapter placement
System Context	System Context doc	External systems and data flows
Bounded Contexts	SRS-DM-007	Context map and integration patterns
Event Storming Board	SRS-ES-008	End-to-end lifecycle chains

Document Purpose

This document defines **how the Inventory Management System integrates internally across its 13 modules and externally with enterprise systems and partners**. It establishes communication styles, synchronous versus asynchronous rules, and hard integration boundaries — so that quantity integrity, auditability, and phased delivery are preserved as integrations grow in complexity.

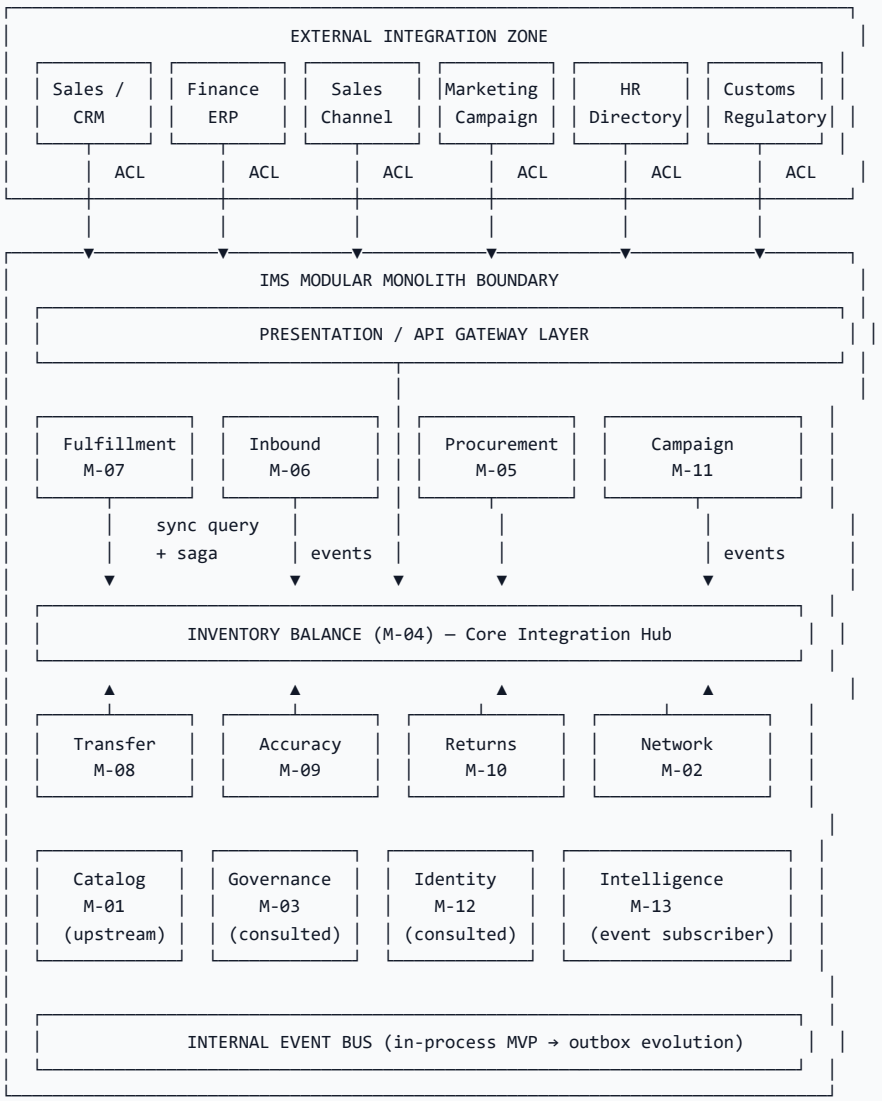
---

1. Integration Architecture Overview

1.1 Design Principles

Principle	Statement
IB-01 — Inventory Balance is the quantity authority	No module other than Inventory Balance (M-04) mutates on-hand, reserved, or quarantined quantities
IB-02 — Events for state propagation	Cross-module state changes propagate through domain events, not shared mutable data
IB-03 — Queries for read-only consultation	Synchronous cross-module calls are permitted only for read-only lookups and pre-condition checks
IB-04 — ACL at system boundary	External models are translated at the boundary; foreign concepts never enter the domain layer
IB-05 — Published language for identifiers	Cross-boundary references use shared value identifiers (`ProductSku`, `LocationCode`) — not entity references
IB-06 — Fail closed on integration uncertainty	When ATP, freeze, or authorization cannot be confirmed, the operation is rejected — not guessed
IB-07 — Idempotent integration handlers	Duplicate event or command delivery must not double-apply stock mutations
IB-08 — Observable integration flows	Every cross-module and external exchange is traceable via correlation ID in audit and movement history

1.2 Integration Topology



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2. Internal Module Communication

2.1 Communication Mechanisms

Internal integration uses **three mechanisms**, applied according to the nature of the interaction:

Mechanism	Purpose	Consistency	Coupling
Published Commands	Request a state change in another module	Strong within target aggregate; eventual across modules	Low — via port interface
Published Queries	Read-only consultation	Read-your-writes within module; eventually consistent for projections	Low — via port interface
Domain Events	Notify that a business fact occurred	Eventual consistency	Lowest — subscriber unaware of publisher internals

2.2 Module Integration Roles

Role	Modules	Integration Behaviour
Upstream reference data	M-01 Catalog, M-02 Network, M-03 Governance, M-12 Identity	Consulted synchronously; emit change events for projections
Core state authority	M-04 Inventory Balance	Receives commands and events; sole ATP mutator; emits all balance change events
Operational consumers	M-05–M-11	Consult upstream; command/event to Balance for quantity changes
Read-only subscriber	M-13 Intelligence	Subscribes to all material events; never initiates writes to transactional modules

2.3 Internal Integration Catalogue

2.3.1 Upstream Consultation (Synchronous Query)

Consumer Module	Provider Module	Query Port	Business Purpose
M-04 Balance	M-01 Catalog	`ValidateProductEligibility`	Block mutations on discontinued SKU
M-04 Balance	M-02 Network	`IsLocationTransactionallyActive`, `IsLocationFrozen`	Gate quantity changes
M-04 Balance	M-03 Governance	`EvaluateSafetyStockOverride`	BR-010 enforcement
M-05 Procurement	M-01, M-02, M-04	Product/location validation, `GetNetworkAvailability`	BR-011 before requisition
M-06 Inbound	M-01, M-02, M-03, M-05	Product, location, tolerance, PO validation	Receipt authorization
M-07 Fulfillment	M-01, M-02, M-04, M-11	Eligibility, freeze, `GetAvailability`, ring-fence check	BR-001, BR-029
M-08 Transfer	M-01, M-02, M-04	Product, freeze, `EvaluateSourceTransferCapacity`	BR-012
M-09 Accuracy	M-01, M-03, M-04, M-12	ABC class, thresholds, system quantity, SoD	Blind count variance
M-10 Returns	M-01, M-02, M-07	Product, location, original order validation	BR-025
M-11 Campaign	M-01, M-02, M-04	Product, location, available quantity	Ring-fence feasibility
All operational	M-12 Identity	`AuthorizeAction`, `GetUserAttribution`	RBAC and audit
All operational	M-03 Governance	`ValidateReasonCode`, `GetApprovalThreshold`	Governance consultation

2.3.2 Cross-Module Commands (Synchronous Write Request)

Used when one module must **request** a state change in another within a coordinated workflow (typically inside a saga):

Caller Module	Target Module	Command	Saga Context
M-07 Fulfillment	M-04 Balance	`ReserveStock`	Order confirmation saga
M-07 Fulfillment	M-04 Balance	`ReleaseReservation`	Order cancellation
M-07 Fulfillment	M-04 Balance	`DeductStockForShipment`	Post-pick shipment (or via event)
M-08 Transfer	M-04 Balance	`DeductStockForTransfer`, `ReceiveStockFromTransfer`	Transfer dispatch/receive
M-04 Balance	M-04 Balance	Internal only	Opening balance establishment

Preferred pattern: caller module publishes intent; Balance module executes atomically and publishes result event.

2.3.3 Domain Event Choreography (Asynchronous)

Publisher Module	Event	Subscriber Module(s)	Subscriber Reaction
M-06 Inbound	`PutAwayCompleted`	M-04 Balance	`ReceiveStock`

Publisher Module	Event	Subscriber Module(s)	Subscriber Reaction
M-07 Fulfillment	`ShipmentDispatched`	M-04 Balance	`DeductStockForShipment`
M-08 Transfer	`TransferDispatched`	M-04 Balance	`DeductStockForTransfer`
M-08 Transfer	`TransferReceived`	M-04 Balance	`ReceiveStockFromTransfer`
M-09 Accuracy	`AdjustmentApproved`	M-04 Balance	`ApplyApprovedAdjustment`
M-09 Accuracy	`WriteOffApproved`	M-04 Balance	`ApplyApprovedWriteOff`
M-09 Accuracy	`LocationFreezeImposed`	M-02 Network	`ApplyLocationFreeze`
M-09 Accuracy	`LocationFreezeReleased`	M-02 Network	`ReleaseLocationFreeze`
M-10 Returns	`ReturnRestocked`	M-04 Balance	`ReceiveStock`
M-10 Returns	`ReturnQuarantined`	M-04 Balance	`ApplyQuarantineHold`
M-10 Returns	`ReturnWrittenOff`	M-09 Accuracy	Open adjustment case
M-11 Campaign	`PromotionalStockAllocated`	M-04 Balance	`SegregatePromotionalStock`
M-11 Campaign	`PromotionalAllocationReleased`	M-04 Balance	`ReleasePromotionalAllocation`
M-05 Procurement	`PurchaseOrderPlaced`	M-06 Inbound	Enable receipt against PO
M-04 Balance	`StockReceived`	M-07 Fulfillment	Backorder fulfillment policy
M-04 Balance	`LowStockThresholdBreached`	M-05 Procurement	Replenishment alert
All transactional	*material events*	M-13 Intelligence	Update read projections
M-13 Intelligence	`OperationalAlertRaised`	Notification gateway	Deliver alert

2.4 Saga-Orchestrated Workflows (Internal)

Cross-module workflows that require **compensation** on partial failure are orchestrated by sagas in the **calling module's application layer**:

Saga ID	Orchestrator	Participants	Compensation on Failure
SAG-01	M-07 Fulfillment	M-04 Balance	Cancel order confirmation; release partial reservation
SAG-02	M-06 Inbound	M-04 Balance	Reverse put-away intent; flag receipt incomplete
SAG-03	M-08 Transfer	M-04 Balance (×2)	Restore source if dispatch fails; cancel in-transit
SAG-04	M-09 Accuracy	M-02 Network, M-04 Balance	Release freeze if certification fails
SAG-05	M-02 Network	M-04 Balance	Block activation if opening balance fails
SAG-06	M-11 Campaign	M-04 Balance	Release partial allocation on campaign abort
SAG-07	M-10 Returns	M-04 or M-09	Roll back disposition if downstream rejects

```
SAG-01: Order Confirmation Saga (M-07 → M-04)

  ConfirmCustomerOrder          ReserveStock              StockReserved
  ──────────────────►  ──────────────────►  ──────────────────►
        │                      │                      │
        │                      │ fail                 │ success
        │                      ▼                      ▼
        │              CompensateOrder          CompleteConfirmation
        │              (release / cancel)
        ▼
   OrderConfirmationFailed
```

2.5 Internal Event Bus Architecture

Aspect	MVP (Phase 1)	Evolution (Phase 2–3)
Transport	In-process event dispatcher within modular monolith	Transactional outbox with optional external broker for Intelligence extraction
Delivery	At-least-once within process; handler idempotency required	At-least-once across processes; deduplication by event ID
Ordering	Per-aggregate ordering guaranteed	Per-aggregate partition key preserved
Failure handling	Retry with backoff; dead-letter queue for poison events	Same + ops alerting
Transactional integrity	Domain events collected in same DB transaction as aggregate save (outbox pattern)	Outbox processor publishes after commit

2.6 Internal Integration Anti-Patterns (Prohibited)

Anti-Pattern	Why Prohibited
Shared database tables across modules	Bypasses aggregate invariants; creates hidden coupling
Direct import of another module's domain entities	Violates bounded context boundary
Synchronous cross-module call chain > 2 hops on hot path	Threatens NFR-PERF-002 (2s ATP lookup)
Balance module calling Fulfillment	Inverts dependency; Balance must not know consumers
Intelligence writing back to transactional modules	Corrupts write model; violates CQRS
Polling another module's database	Breaks encapsulation and module ownership

---

3. External Integrations

3.1 External Integration Zones

Zone	Systems	IMS Role	Integration Maturity
Commercial demand	EXT-01 Sales/CRM, EXT-03 Sales Channel	Fulfillment consumer; ATP publisher	MVP: manual/indirect → Phase 2+: API
Financial control	EXT-02 Finance ERP/GL	Operational facts publisher; match participant	MVP: export → Phase 3: ACL
Supply chain partners	EXT-04 Supplier, EXT-05 Carrier	Human-mediated operational recording	MVP: no direct system link
Commercial planning	EXT-07 Marketing	Campaign intent consumer	Phase 2
Identity	EXT-08 HR Directory, SVC-01 IdP	Identity consumer	MVP: manual/sync
Regulatory	EXT-09 Customs	Clearance fact consumer	Phase 2 (if in scope)
Notifications	SVC-02 Email/SMS	Alert publisher	MVP
Audit	External Auditor	Read-only evidence consumer	MVP

3.2 External System Integration Catalogue

EXT-01 — Sales / Order Management (CRM)

Attribute	Detail
IMS modules	M-07 Order Fulfillment (primary), M-13 Intelligence (status feed)
Direction	Inbound: orders; Outbound: ATP, reservation, fulfillment status
ACL owner	Fulfillment module — `SalesOrderInboundPort`, `FulfillmentStatusOutboundPort`

Attribute	Detail
Published language	`CustomerOrderId`, `ProductSku`, `LocationCode`, `ReservationId`, `ShipmentId`
Phase	MVP: CS enters orders manually; Phase 2+: automated order import API
Risk	Channel overselling until integration live (BRD risk register) — manual sync interim

Interaction	Style	Data Exchanged
Order intake	Sync (Phase 2+) / Manual (MVP)	Order lines, ship-from preference, customer reference
Availability check	Sync	SKU, location → ATP quantity
Reservation confirmation	Async event → outbound notification	Reservation ID, expiry
Shipment notice	Async event → outbound notification	Shipment ID, carrier ref, line quantities

EXT-02 — Finance ERP / General Ledger

Attribute	Detail
IMS modules	M-06 Inbound (three-way match), M-13 Intelligence (valuation), M-04 Balance (movement facts)
Direction	Inbound: GL balance, cost parameters; Outbound: valuation snapshot, adjustments, match results
ACL owner	Intelligence module — `FinanceValuationExportPort`; Inbound — `FinanceMatchStatusPort`
Phase	MVP: scheduled file export (TEC-05); Phase 3: automated reconciliation ACL
Open decision	HS-014 — IMS owns match vs. ERP owns match; architecture assumes IMS operational SoR, Finance payment authority

Interaction	Style	Data Exchanged
Valuation snapshot export	Async (scheduled)	Period-end inventory value by SKU/location
Adjustment/write-off facts	Async (event-driven export)	Reason code, quantity, value, approver
Three-way match result	Async	PO, receipt, invoice match status
Consignment ownership transfer	Async	Ownership change event for GL posting
GL balance import	Async (scheduled)	Reconciliation comparison input

EXT-03 — Sales Channel / E-Commerce / Marketplace

Attribute	Detail
IMS modules	M-13 Intelligence (ATP feed), M-04 Balance (source of truth)
Direction	Outbound: available quantity feed
ACL owner	Intelligence or dedicated integration adapter
Phase	Phase 2+ (TEC-04 — no real-time MVP)
Constraint	Feed must reflect ATP, not on-hand — prevent oversell (BR-001)

Interaction	Style	Data Exchanged
ATP / availability feed	Async (scheduled or event-triggered)	SKU, channel, available quantity
Stockout notification	Async	SKU, channel, alert

EXT-04 — Supplier / Vendor

Attribute	Detail
IMS modules	M-05 Procurement, M-06 Inbound
Direction	Human-mediated — buyer sends PO externally; clerk records receipt in IMS

Attribute	Detail
Phase	MVP: no EDI/API; future: ASN inbound adapter
Boundary	Supplier never executes commands in IMS

Interaction	Style	Data Exchanged
PO issuance	Outbound (human/process)	PO document
ASN / delivery notice	Inbound (human entry MVP)	Expected delivery, carrier
Supplier invoice	Inbound (human or Phase 2 scan)	Invoice lines for three-way match

EXT-05 — Logistics / Carrier Partner

Attribute	Detail
IMS modules	M-07 Fulfillment, M-08 Transfer
Direction	Human-mediated tracking entry
Phase	MVP: operational workflow; future: carrier tracking API

Interaction	Style	Data Exchanged
Dispatch notice	Outbound (human)	Shipment/transfer details
Delivery confirmation	Inbound (human)	Proof of delivery, delay reason

EXT-07 — Marketing / Campaign Planning

Attribute	Detail
IMS modules	M-11 Commercial Campaign
Direction	Inbound: campaign plan; Outbound: readiness status
ACL owner	Campaign module — `CampaignPlanInboundPort`
Phase	Phase 2

Interaction	Style	Data Exchanged
Campaign plan import	Sync or batch	SKU list, quantity, channel, period
Readiness status	Async outbound	Allocated vs. required, gap alert

EXT-08 — HR / Enterprise Directory

Attribute	Detail
IMS modules	M-12 Identity & Access
Direction	Inbound: employee identity feed
ACL owner	Identity module — `DirectorySyncPort`
Phase	MVP: manual provisioning or periodic sync

Interaction	Style	Data Exchanged
User provisioning	Async (scheduled sync)	Employee ID, name, department, status
Deprovisioning	Async	Termination status → account deactivation

EXT-09 — Customs / Regulatory Authority

Attribute	Detail
IMS modules	M-04 Balance (quarantine), M-06 Inbound
Direction	Inbound: clearance decisions via compliance role
Phase	Phase 2 if bonded inventory in scope (HS-009)

Interaction	Style	Data Exchanged
Clearance granted	Sync (human entry)	Document reference, cleared quantity
Hold / recall notice	Sync (human entry)	Affected lots, locations

SVC-01 — Enterprise Identity Provider (SSO)

Attribute	Detail
IMS modules	M-12 Identity
Direction	Inbound: authentication tokens and claims
Style	Synchronous per request (auth handshake)

SVC-02 — Email / Notification Gateway

Attribute	Detail
IMS modules	M-13 Intelligence
Direction	Outbound: alert payloads
Style	Asynchronous — fire-and-forget with delivery status callback
SLA	NFR-PERF-014 — alerts within 5 minutes of trigger

3.3 External Integration Phasing

Phase	External Integrations Activated
Phase 1 — MVP	SVC-01 SSO, SVC-02 notifications, EXT-08 manual/sync, EXT-02 file export, EXT-01 manual order entry
Phase 2	EXT-01 API order feed, EXT-03 ATP channel feed, EXT-07 campaign import, EXT-04 ASN (optional)
Phase 3	EXT-02 automated Finance ACL, EXT-03 real-time feeds, EXT-05 carrier API

4. Communication Style

4.1 Communication Style Catalogue

Style	Description	When Used
Request-Reply (Sync)	Caller waits for response before proceeding	Read-only queries, pre-condition checks, user-facing ATP lookup
Command Dispatch (Sync)	Caller sends command; target commits atomically; returns success/failure	Saga steps requiring immediate confirmation (reserve stock)
Event Notification (Async)	Publisher emits fact; subscribers react independently	Post-commit state propagation, projections, alerts
Saga Choreography (Mixed)	Sync steps coordinated with async event continuations	Multi-module workflows with compensation
Batch Export / Import	Scheduled file or bulk data transfer	Finance export, go-live import (FR-090), directory sync

Style	Description	When Used
Fire-and-Forget	Outbound notification with no response required	Email alerts, channel feed push
Human-Mediated	External party communicates outside system; internal user records outcome	Supplier PO, carrier tracking, MVP order entry

4.2 Communication Style by Integration Type

```
┌──────────────────────┬──────────────────────────────────────────────────┐
│ Integration Type     │ Primary Communication Style                      │
├──────────────────────┼──────────────────────────────────────────────────┤
│ Master data lookup   │ Request-Reply (Sync)                             │
│ Authorization        │ Request-Reply (Sync)                             │
│ User-facing ATP      │ Request-Reply (Sync) — CQRS read path            │
│ Stock mutation       │ Command Dispatch (Sync) OR Event Notification (Async)  │
│ Cross-module flow    │ Saga Choreography (Mixed)                        │
│ Read projections     │ Event Notification (Async)                       │
│ External inbound     │ ACL Translate → Command Dispatch                 │
│ External outbound    │ Event Notification → ACL Translate → Batch/API   │
│ Alerts               │ Fire-and-Forget                                  │
│ Partner (MVP)        │ Human-Mediated                                   │
└──────────────────────┴──────────────────────────────────────────────────┘
```

4.3 Published Language (Integration Contract)

All integrations — internal and external — exchange data through the **published language** of immutable identifiers and value objects:

Concept	Format	Example
Product identity	`ProductSku`	`SKU-12345`
Location identity	`LocationCode`	`WH-EAST-01`
Quantity	`Quantity` + `UnitOfMeasure`	`100 EA`
User identity	`UserId`	`usr-abc-001`
Temporal	`Instant` (UTC) + location business calendar	ISO-8601
Money	`Amount` + `Currency`	`1500.00 USD`
Event envelope	`EventId`, `CorrelationId`, `CausationId`, `OccurredAt`	Traceable audit chain

Foreign concepts are translated at the ACL — e.g., CRM `orderLineId` maps to IMS `CustomerOrderId` + line reference; never stored as CRM's native structure in domain layer.

4.4 API Exposure Style (IMS as Provider)

When IMS exposes interfaces to external consumers:

Exposure	Consumers	Content
Operational API	CRM, mobile warehouse, web UI	Commands and queries for active workflows
Read API	CRM, channels, executive BI	ATP, status, dashboards (read-only)
Integration API	Finance, marketing	Scheduled exports, webhook callbacks
Audit API	Internal/external auditor	Read-only evidence packages

Detailed API specification is out of scope for this document — style and boundary only.

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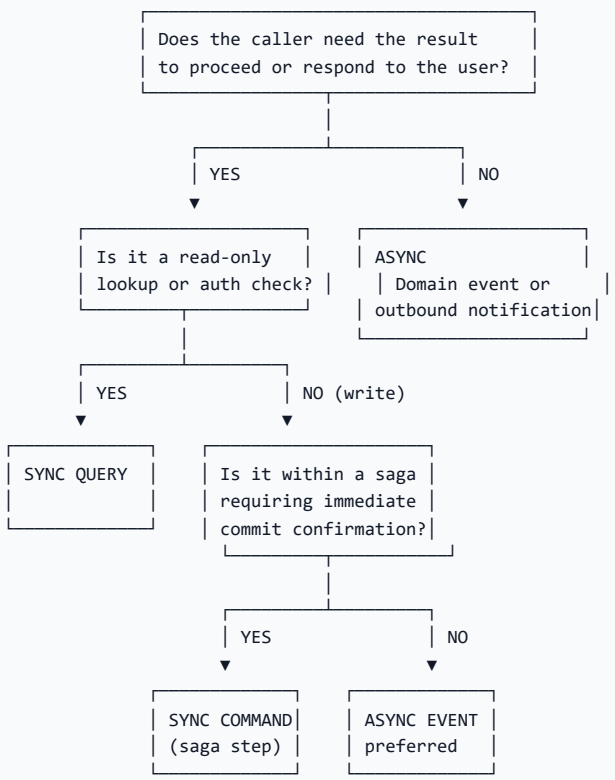
5. Synchronous vs Asynchronous Interactions

5.1 Decision Framework



5.2 Synchronous Interactions

Use when: immediate answer required; user or saga is waiting; read-only consultation; pre-condition gate.

Interaction	Modules / External	Rationale
ATP availability lookup	M-04 → caller / EXT-01	NFR-PERF-002 — 2s response; user on phone
Product lifecycle check	M-01 → consumer	Block invalid transaction before attempt
Location freeze check	M-02 → consumer	BR-024 — reject before mutation
Permission authorization	M-12 → all	NFR-SEC — fail closed before command
Governance threshold lookup	M-03 → consumer	Approval routing decision
Reserve stock (saga step)	M-07 → M-04	Order confirmation cannot complete without reservation result
SSO authentication	SVC-01 → M-12	Per-request identity validation
Campaign readiness check	M-11 → M-04	Sync quantity check before allocation command

Constraints on synchronous calls:

Constraint	Limit
Hot-path chain depth	≤ 2 cross-module sync calls (e.g., Fulfillment → Balance)
Timeout	Fail and compensate — do not block indefinitely
Retry	Idempotent queries only; commands via saga retry with deduplication
Circuit breaking	Degraded mode: reject writes; serve cached ATP with staleness indicator for reads

5.3 Asynchronous Interactions

Use when: state propagation after commit; projections; notifications; external system feeds; no caller waiting.

Interaction	Modules / External	Rationale
Put-away → stock available	M-06 → M-04	Decouple receiving from balance; eventual consistency acceptable (seconds)
Shipment → stock deduction	M-07 → M-04	Post-commit propagation
Adjustment approved → apply	M-09 → M-04	Governance pipeline complete before mutation
All events → Intelligence	M-01–M-12 → M-13	CQRS projection update
Low-stock alert → Procurement	M-04 → M-05	Replenishment is reaction, not blocking
Backorder fulfill on receipt	M-04 → M-07	Policy reaction to `StockReceived`
Valuation export → Finance	M-13 → EXT-02	Scheduled batch; no real-time GL need
ATP feed → Sales Channel	M-13 → EXT-03	Channel tolerance for minutes-level lag
Operational alert → Email	M-13 → SVC-02	NFR-PERF-014 — 5 min SLA
Directory sync → Identity	EXT-08 → M-12	HR changes not real-time critical

Asynchronous delivery guarantees:

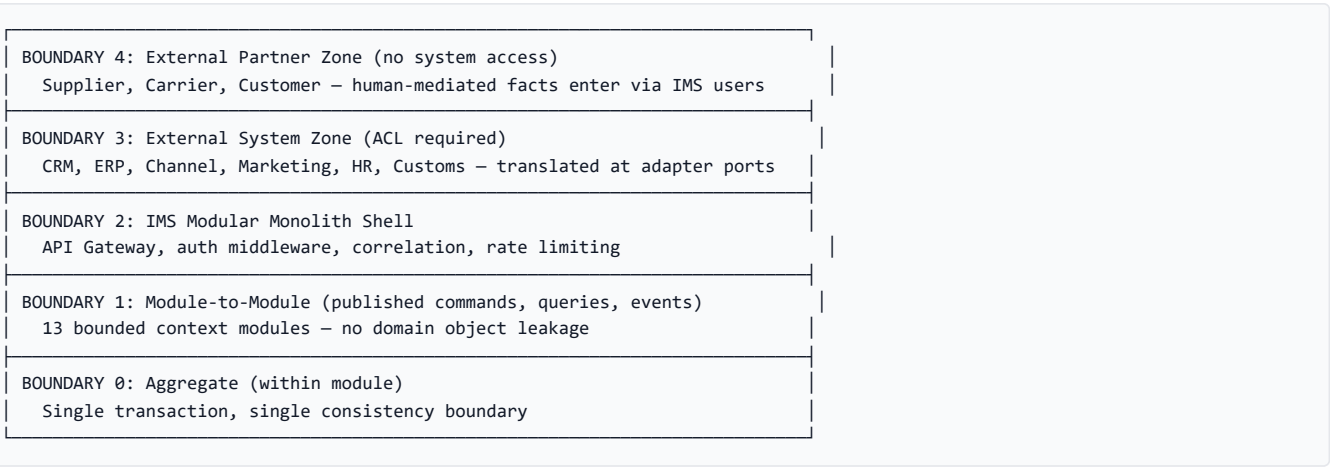
Guarantee	Requirement
Durability	Events persisted in outbox before commit acknowledgment
At-least-once delivery	Handlers must be idempotent
Ordering	Per-aggregate event sequence preserved
Latency target	Internal event propagation < 5 seconds under normal load
Visibility	Correlation ID links saga steps across async boundaries

5.4 Synchronous vs Asynchronous Summary Matrix

Flow	Sync	Async	Notes
ATP inquiry (user)	✅		CQRS read path
Product/location validation	✅		Pre-condition
Order confirm → reserve	✅		Saga step 1–2
Reserve → stock event		✅	Post-commit notification to Intelligence
Put-away → stock increase		✅	Preferred over sync command
Pick → ship → deduct	✅ confirm	✅ propagate	Pick sync; deduct via event
Count → freeze → adjust	Mixed	✅	Freeze sync apply; adjust async
Dashboard refresh	✅		Query projection
Finance month-end export		✅	Scheduled batch
CRM order import (Phase 2)	✅		Translate + command at boundary
Email alert		✅	Fire-and-forget

6. Integration Boundaries

6.1 Boundary Model



6.2 Boundary Rules

Boundary	Rule	Enforcement
B0 — Aggregate	One transaction modifies one aggregate	Application layer unit of work
B1 — Module	Modules communicate only via published interfaces	Compile-time module references; code review
B2 — Monolith shell	All external traffic enters through presentation/API layer	No infrastructure bypass
B3 — External system	ACL translates foreign models; domain never sees external schema	Adapter in infrastructure layer
B4 — Partner	Partners never hold IMS credentials or execute commands	Process design; no API exposure

6.3 Anti-Corruption Layer (ACL) Boundaries

ACL	Location	Translates	Protects
Sales Order ACL	M-07 Fulfillment infrastructure	CRM order model → `ImportCustomerOrder` command	Fulfillment domain from CRM lifecycle complexity
Finance Export ACL	M-13 Intelligence infrastructure	Valuation snapshot → GL posting format	Finance chart of accounts from IMS movement model
Finance Match ACL	M-06 Inbound infrastructure	ERP payment status → match workflow state	Three-way match from ERP payment authority (HS-014)
Channel Feed ACL	M-13 Intelligence infrastructure	ATP projection → channel-specific availability format	Channel semantics (safety buffer) from raw ATP
Campaign Plan ACL	M-11 Campaign infrastructure	Marketing plan → `CreatePromotionalAllocation`	Campaign domain from marketing tooling model
Directory Sync ACL	M-12 Identity infrastructure	HR employee record → `ProvisionUserAccount`	Identity model from HR org structure

6.4 Data Ownership Boundaries

Data Domain	System of Record	Readers	Writers
Stock quantities (on-hand, reserved, ATP)	IMS M-04	All modules, CRM (ATP), Channel	**M-04 only**
Product master	IMS M-01	All modules	M-01 only
Location network	IMS M-02	All modules	M-02 only
Customer order (fulfillment view)	IMS M-07	Returns, Intelligence	M-07 only; intake from CRM ACL
Customer order (commercial view)	EXT-01 CRM	IMS (subset via ACL)	CRM

Data Domain	System of Record	Readers	Writers
Purchase order	IMS M-05	Inbound, Intelligence	M-05 only
Goods receipt facts	IMS M-06	Balance, Finance	M-06 only
Supplier invoice / match	IMS M-06 (operational)	Finance	M-06; payment authority EXT-02
GL valuation	EXT-02 Finance	IMS (reconciliation)	Finance
User identity	IMS M-12 (operational)	All modules	M-12; source EXT-08
Audit trail	IMS M-13 (projection)	Auditors	Event-driven append only
Campaign allocation	IMS M-11	Fulfillment, Balance	M-11 only

6.5 Trust Boundaries and Security

Zone	Trust Level	Controls
Internal module calls	High — same process	Identity context propagation; no anonymous calls
External API (CRM, channel)	Medium	Mutual TLS, API keys, scoped permissions, rate limits
Finance export/import	Medium	Encrypted transfer, checksum, approval workflow
Auditor access	Read-only, time-bound	Dedicated role (NFR-SEC-016); no write endpoints
Partner (supplier/carrier)	Untrusted	No direct access; human verification
Notification gateway	Trusted service	Authenticated outbound; no inventory mutation

6.6 Integration Failure Boundaries

Failure Scenario	Boundary Behaviour
CRM unavailable (Phase 2)	CS manual order entry fallback; no auto-confirm without ATP check
Finance export fails	Retry with alert; operational IMS continues; reconciliation deferred
Event handler fails	Retry; dead-letter; no partial stock state — outbox ensures consistency
Balance reservation fails	Saga compensates order; CRM notified of failure (Phase 2)
IdP unavailable	Existing sessions continue; new auth blocked — fail closed
Channel feed stale	Feed includes `asOf` timestamp; channel applies safety buffer

---

7. Integration Patterns Reference

Pattern	Internal / External	Application in IMS
Anti-Corruption Layer	External	CRM, Finance, Marketing, HR adapters
Published Language	Both	`ProductSku`, `LocationCode` shared identifiers
Domain Events	Internal	Cross-module state propagation
CQRS	Internal	Intelligence read side; ATP query path
Saga / Process Manager	Internal	Order confirm-reserve, go-live, transfer
Transactional Outbox	Internal	Reliable event publish with DB commit
Open Host Service	Internal	Catalog, Governance, Identity query ports
Conformist	Internal	Consumers accept Catalog/Network reference models
Customer-Supplier	Internal	Fulfillment ↔ Balance; Accuracy ↔ Balance
Batch Export	External	Finance valuation, audit evidence
Human-Mediated	External	Supplier, carrier, MVP CRM

8. Integration Governance

Activity	Owner	Frequency
Integration contract review (ports, events)	Solution Architect	Per module release
ACL mapping review	Integration architect + domain expert	Per external system onboarding
Saga compensation testing	QA + Operations	Per sprint with cross-module stories
Event schema versioning	Tech lead	Breaking change requires versioned event types
External API consumer onboarding	Security + Product Owner	Per new integration
Integration failure runbook	DevOps	Per external system

9. Document Approval

Role	Name	Signature	Date
Solution Architect			
Integration Architect			
DDD Solution Architect			
Security Engineer			
Product Owner			

End of Integration Architecture Design

Document ID: SRS-SA-005 | Version 1.0 | July 6, 2026

Cross-Cutting Concerns

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-SA-006
Document Title	Cross-Cutting Concerns — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Solution Architect
Prepared For	Development squads, security, DevOps, and QA leads
Date	July 6, 2026
Classification	Internal — Architecture Confidential
Scope	Placement and responsibility of cross-cutting concerns across layers — no framework or library selection

Related Documents

Document	ID	Relationship
Layered Architecture Design	SRS-SA-004	Four-layer model and dependency rule
System Modules Design	SRS-SA-003	Identity & Access module (M-12)
Integration Architecture Design	SRS-SA-005	Security zones and correlation IDs
Security Requirements	NFR-SEC	Authentication and authorization NFRs
Logging Requirements	NFR-LOG	Operational and audit logging NFRs
Compliance Requirements	NFR-COM	Audit trail and retention NFRs

Document Purpose

Cross-cutting concerns affect every module and layer of the Inventory Management System. This document identifies **where each concern belongs** in the Clean Architecture stack, what it must **not** do, and how concerns interact — so that security, auditability, and operational quality are implemented consistently without polluting domain logic.

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1. Cross-Cutting Concerns Overview

1.1 Layer Placement Summary

```
┌──────────────────────────────────────────────────────────────────────────┐
│  PRESENTATION     Auth middleware  │ Input validators │ Exception filters  │
│                   Locale/format    │ Correlation ID attach                 │
├──────────────────────────────────────────────────────────────────────────┤
│  APPLICATION      Authorization    │ Command/query    │ Transaction (UoW)  │
│                   behaviors        │ validation       │ Saga boundaries    │
│                   Exception map    │ Audit context    │ Cache policy (use) │
├──────────────────────────────────────────────────────────────────────────┤
│  DOMAIN           Business         │ Domain           │ Domain exceptions  │
│                   invariants       │ validation       │ (rule violations)  │
├──────────────────────────────────────────────────────────────────────────┤
│  INFRASTRUCTURE   Auth adapter     │ Logger           │ Cache implementation │
│                   (IdP/SSO)        │ Audit writer     │ Config provider    │
│                   AuthZ adapter    │ Exception log    │ Localization resources │
│                   (M-12 port)      │ UoW impl         │ Transaction manager │
└──────────────────────────────────────────────────────────────────────────┘
```

1.2 Concern Placement Matrix

Concern	Primary Layer	Supporting Layers	Owning Module
Authentication	Infrastructure	Presentation, Application	M-12 Identity & Access
Authorization	Application	Infrastructure (adapter), Domain (SoD rules)	M-12 + M-03 Governance
Logging	Infrastructure	Presentation (correlation), Application (context)	Platform (all modules)
Exception handling	Presentation + Application	Domain (throw), Infrastructure (log)	Per module
Validation	Domain + Presentation	Application (orchestration)	Per module
Auditing	Infrastructure + Application	Domain (events), Intelligence (projection)	M-13 Intelligence
Caching	Infrastructure	Application (policy)	M-04 Balance (ATP), M-13 Intelligence
Transactions	Application	Infrastructure (implementation)	Per module
Configuration	Infrastructure	Application (inject)	Platform
Localization	Presentation + Infrastructure	Application (formatting ports)	Platform

---

2. Authentication

2.1 Definition in IMS

Authentication establishes *who* is performing an action — verifying that a user identity is genuine before any inventory command, query, or configuration change is accepted. Every transaction must be attributable to an individually authenticated user (NFR-SEC-001, NFR-SEC-002).

2.2 Architectural Placement

Layer	Responsibility	Components
Presentation	Extract credentials from incoming requests; attach authenticated principal to request context; reject unauthenticated requests before reaching handlers	Auth middleware, session/token extractor, `[Authorize]` guards on controllers
Application	Consume resolved `UserId` from context; never parse raw tokens; require identity for command attribution	`IUserContext` port (read-only), command enrichment with `UserId`
Domain	Accept `UserId` / `UserActionAttribution` as value objects on mutations; **no authentication logic**	`UserActionAttribution` value object on aggregates
Infrastructure	Integrate enterprise IdP (SSO); validate JWT/session; map claims to `UserId`; session store if required	`IdentityProviderAdapter`, `JwtTokenValidator`, `SessionStore`

2.3 Module Ownership

Module	Role
M-12 Identity & Access	Owns user account lifecycle; defines authentication contract
All other modules	Consume `IUserContext` / `IIdentityService` port — never implement auth

2.4 Key Rules

Rule	Detail
AUTH-01	Zero shared accounts in production (NFR-SEC-001)
AUTH-02	Authentication occurs before any handler or controller business logic
AUTH-03	Re-authentication required for high-risk actions (NFR-SEC-004): bulk import, write-offs, role changes
AUTH-04	Failed login handling: lockout after 5 attempts / 15 minutes (NFR-SEC-003)
AUTH-05	Domain layer receives resolved identity only — never raw tokens or passwords

2.5 Does Not Belong In

- **Domain** — authentication is an infrastructure concern; domain knows *who* acted, not *how* they proved identity
- **Other operational modules** — no module except M-12 implements token validation

---

3. Authorization

3.1 Definition in IMS

Authorization determines *what* an authenticated user may do — role-based and location-scoped permissions across 100 system permissions and 19 business roles, plus segregation-of-duties enforcement (FR-084, FR-085, NFR-SEC-005 – NFR-SEC-009).

3.2 Architectural Placement

Layer	Responsibility	Components
Presentation	Optional coarse endpoint-level guards; delegate fine-grained checks to application layer	Route-level role attributes (supplementary only)

Layer	Responsibility	Components
Application	**Primary authorization gate** — verify permission and location scope before every command handler; invoke SoD checks	`AuthorizationBehavior`, `RequirePermissionAttribute` handler, `LocationScopeBehavior`
Domain	Express SoD and eligibility as domain policies where business-meaningful; e.g., "counter cannot approve own count"	`SegregationOfDutiesPolicy`, governance consultation
Infrastructure	Implement `IIdentityService` / `IAuthorizationService` ports by delegating to M-12 Identity module adapter	`IdentityAuthorizationAdapter`, permission cache (short TTL)

3.3 Policy Sources

Policy Type	Defined In	Enforced In
Role → permission matrix	M-12 Identity	Application layer `AuthorizeAction`
Location scope	M-12 Identity + M-02 Network codes	Application layer before location-scoped commands
Segregation of duties	M-03 Governance + M-12 Identity	Application layer at command; M-12 at role assignment
Business eligibility (e.g., discontinued SKU)	Domain of owning module	Domain layer invariants

3.4 Key Rules

Rule	Detail
AUTHZ-01	Fail closed — if authorization cannot be determined, reject the command
AUTHZ-02	Authorization is checked in application layer, not in controllers alone
AUTHZ-03	Location-scoped roles validated against `LocationCode` on every warehouse command
AUTHZ-04	SoD violations blocked at role assignment (M-12) and at command execution (application)
AUTHZ-05	Read queries scoped to user's authorized locations and permissions

3.5 Does Not Belong In

- **Infrastructure (business rules)** — permission matrix is not hard-coded in adapters
- **Domain (role checks)** — domain enforces business SoD invariants, not RBAC matrix lookups

---

4. Logging

4.1 Definition in IMS

Logging captures operational diagnostic and security information — request traces, errors, performance markers, and security events — to support troubleshooting, monitoring (NFR-MON), and security investigation (NFR-LOG, NFR-SEC-025 – NFR-SEC-028).

Logging is distinct from **auditing** (Section 6) — logs are for operators; audit records are for compliance and governance.

4.2 Architectural Placement

Layer	Responsibility	Components
Presentation	Attach `CorrelationId`, `RequestId`, HTTP metadata to logging scope; log request start/end at info level	Request logging middleware, correlation ID middleware
Application	Enrich log scope with command name, aggregate ID, `UserId`; log handler entry/exit and saga steps	Structured log context in behaviors, saga step logging
Domain	**No logging** — domain remains pure; exceptions carry message codes only	—
Infrastructure	Logger implementation, log sinks, log format standard, retention, immutability for security logs	`StructuredLogger`, log appenders, NFR-LOG-001 standard fields

4.3 Standard Log Attributes (NFR-LOG-001)

Every log entry includes: timestamp (UTC), severity, `CorrelationId`, `UserId`, module name, operation name, outcome, and duration where applicable.

Log Category	Examples	Retention
Operational	Handler execution, query performance, cache hit/miss	Per NFR-LOG retention schedule
Security	Login success/failure, permission denied, SoD violation	≥ 7 years (NFR-COM-020)
Error	Unhandled exceptions, saga compensation, integration failure	Per operational schedule
Integration	Cross-module port calls, ACL adapter requests, event publish	Correlation-linked

4.4 Key Rules

Rule	Detail
LOG-01	Domain layer does not write logs — keeps business logic testable and pure
LOG-02	Never log passwords, tokens, or full PII — mask sensitive fields (NFR-SEC-017)
LOG-03	Correlation ID propagates from presentation through application to infrastructure and cross-module calls
LOG-04	Security events logged at warn or higher; never suppressed
LOG-05	Log structure is consistent across all 13 modules

4.5 Does Not Belong In

- **Domain** — no `ILogger` in aggregates
- **Audit trail** — operational logs are not the compliance audit record (see Auditing)

---

5. Exception Handling

5.1 Definition in IMS

Exception handling manages the translation of failures — business rule violations, validation errors, authorization denials, and infrastructure faults — into controlled responses without exposing internal state or corrupting inventory data.

5.2 Architectural Placement

Layer	Responsibility	Components
Presentation	Catch application/domain exceptions; map to HTTP status and user-safe response DTOs; never expose stack traces to clients	`ExceptionFilter`, `ProblemDetails` mapper, global exception handler
Application	Catch infrastructure exceptions; wrap in application exceptions; trigger saga compensation; decide retry vs. fail	Saga compensation handlers, `CommandExceptionBehavior`
Domain	Throw **domain exceptions** for business rule violations — typed, meaningful, no HTTP concepts	`InsufficientAtpException`, `LocationFrozenException`, `SegregationOfDutiesViolationException`
Infrastructure	Log unhandled technical exceptions; translate persistence/network errors to infrastructure exceptions	ORM exception wrapper, retry policies for transient DB errors

5.3 Exception Taxonomy

Exception Type	Origin Layer	HTTP Mapping (Presentation)	User Message
`ValidationException`	Presentation / Application	400 Bad Request	Field-level errors
`UnauthorizedException`	Application	401 Unauthorized	Generic auth failure
`ForbiddenException`	Application	403 Forbidden	Insufficient permission

Exception Type	Origin Layer	HTTP Mapping (Presentation)	User Message
`NotFoundException`	Application	404 Not Found	Resource not found
`BusinessRuleViolationException`	Domain	422 Unprocessable Entity	Business-safe rule message
`ConflictException`	Application / Domain	409 Conflict	Concurrent modification, duplicate
`InfrastructureException`	Infrastructure	503 Service Unavailable	Generic system error

5.4 Saga and Exception Interaction

Scenario	Handling
Reservation fails after order confirmed	Saga compensates — `CompensateOrderConfirmation`
Put-away fails after receipt verified	Receipt remains verified; put-away retried; no stock increase
Authorization denied mid-saga	Saga aborts; no partial commits
Transient DB error	Infrastructure retry; application remains unaware of retry count

5.5 Key Rules

Rule	Detail
EXC-01	Domain throws domain exceptions — never returns error codes as return values
EXC-02	Presentation is the only layer that maps exceptions to HTTP
EXC-03	Business rule messages use ubiquitous language terms — not technical codes in user text
EXC-04	Unhandled exceptions logged at error with full correlation context
EXC-05	Exception handling never swallows stock mutation failures silently

5.6 Does Not Belong In

- **Domain** — no HTTP status codes, no logging in catch blocks
- **Infrastructure** — no business rule interpretation; wrap and rethrow

---

6. Validation

6.1 Definition in IMS

Validation ensures inputs and state transitions are well-formed and business-legal before execution. IMS distinguishes **three validation levels** — each belongs in a different layer.

6.2 Three Levels of Validation

Level	Layer	Scope	Examples
Syntactic	Presentation	Request shape, types, required fields, format	SKU not empty, quantity is positive integer, date parseable
Application	Application	Command preconditions, cross-field rules, saga guards	"Approve adjustment" requires pending case ID; import batch size limit
Domain / Business	Domain	Invariants, business rules (BR-001 – BR-032)	ATP ≥ 0, cannot reserve on frozen location, BR-019 SoD

6.3 Architectural Placement

Layer	Responsibility	Components
Presentation	Syntactic validation on request DTOs; return 400 before dispatching command	Fluent validators, data annotations, request DTO rules

Layer	Responsibility	Components
Application	Application-level validation behaviors in command pipeline; consult governance/catalog/network ports for preconditions	`ValidationBehavior`, command validators
Domain	**Authoritative business validation** — invariants enforced in aggregate methods; cannot be bypassed	Aggregate root methods, domain policies, specifications
Infrastructure	Database constraints as last-resort safety net only — not primary validation	Unique indexes, FK constraints, check constraints

6.4 Validation Pipeline (Command Path)

```
Request DTO
    │
    ▼
[Presentation]  Syntactic validation ──fail──► 400 Bad Request
    │
    ▼
[Application]     Authorization check
    │
    ▼
[Application]     Application precondition validation ──fail──► ValidationException
    │
    ▼
[Domain]          Aggregate method enforces invariants ──fail──► BusinessRuleViolationException
    │
    ▼
[Infrastructure]  Persist (DB constraints as safety net)
```

6.5 Key Rules

Rule	Detail
VAL-01	Business rules live in domain — never only in UI or API validators
VAL-02	Presentation validation is for user experience — domain validation is for correctness
VAL-03	Validation rules from SRS-BR-001 and VR-xxx trace to domain invariants or application preconditions
VAL-04	Blind cycle count: application must not pass system quantity to presentation for counter view (BR-018)
VAL-05	Duplicate syntactic validation in domain is acceptable for defense in depth on critical invariants

6.6 Does Not Belong In

- **Infrastructure as primary validator** — DB constraints supplement, not replace, domain rules
- **Presentation for business rules** — UI cannot be the only enforcement of BR-001 (ATP)

---

7. Auditing

7.1 Definition in IMS

Auditing produces an **immutable, governance-grade record** of material inventory actions — who did what, when, where, why, and what changed — supporting BO-19 (100% audit trail), internal control (COSO), and external audit (NFR-COM-010, NFR-REL-009).

Auditing is **not** the same as operational logging. Audit records are structured business evidence; logs are diagnostic.

7.2 Architectural Placement

Layer	Responsibility	Components
Presentation	No audit logic — passes through authenticated context	—

Layer	Responsibility	Components
Application	Enrich commands with audit context (`UserId`, reason code, correlation ID); collect domain events after successful commit	`AuditContextEnricher`, post-handler event collection
Domain	Raise **domain events** that constitute auditable business facts; embed `UserActionAttribution` on mutations	`StockReserved`, `AdjustmentApproved`, attribution value objects
Infrastructure	Persist audit records; implement append-only audit store; export evidence packages	`AuditTrailWriter`, immutable log store
M-13 Intelligence	**Audit trail read projection** — query, search, export for auditors (NFR-COM-012, NFR-PERF-026)	`AuditTrailProjection`, `ExportAuditEvidenceQueryHandler`

7.3 Audit Record Sources

Source	Mechanism	Content
Domain events	Event → Intelligence projection	Business fact, before/after quantities, reason code
Command metadata	Application layer audit context	Initiator, approver, timestamp, location scope
Security events	Infrastructure security log → audit projection	Login, permission denied, role change
Governance decisions	M-03 events	Policy change, threshold amendment

7.4 Material Actions Requiring Audit (Non-Exhaustive)

Category	Actions
Stock mutations	Receipt, reservation, shipment, transfer, adjustment, write-off, quarantine
Governance	Adjustment approval, write-off approval, policy change, reason code change
Master data	Product lifecycle change, location activation/closure, supplier deactivation
Access	Role assignment, SoD violation attempt, account deactivation
Count	Cycle count result, physical count certification, variance approval
Financial	Three-way match outcome, invoice authorization

7.5 Key Rules

Rule	Detail
AUD-01	Audit records are append-only — no update or delete (NFR-LOG-026, NFR-COM-010)
AUD-02	Retained ≥ 7 years aligned with financial records (NFR-COM-019, NFR-COM-020)
AUD-03	Single authoritative audit projection in M-13 Intelligence — not duplicated per module
AUD-04	Domain events are the primary audit source — audit is a consequence of correct domain design
AUD-05	Auditor export within 15 minutes (NFR-PERF-026, NFR-COM-012)

7.6 Does Not Belong In

- **Presentation** — no audit persistence in controllers
- **Operational logs as audit substitute** — logs lack business structure and immutability guarantees
- **Per-module audit tables** — centralized projection in Intelligence

---

8. Caching

8.1 Definition in IMS

Caching stores frequently accessed read data to meet performance NFRs — especially ATP lookup (NFR-PERF-002, 2 seconds) and dashboard load (NFR-PERF-004) — without serving stale data that could cause overselling.

8.2 Architectural Placement

Layer	Responsibility	Components
Presentation	HTTP cache headers for safe static resources only — **not** for ATP or stock data	Static asset cache headers
Application	Define cache use policy — when to read from cache vs. authoritative source; never cache write paths	`ICachePolicy` consultation in query handlers
Domain	**No caching** — domain always works with authoritative state	—
Infrastructure	Cache implementation, storage, TTL, invalidation on domain events	`AtpCacheService`, `DashboardCacheService`, event-driven invalidation handlers

8.3 Cache Domains

Cache Target	Owner	Invalidation Trigger	Staleness Tolerance
ATP by SKU/location	M-04 Balance infrastructure	`StockReserved`, `StockReceived`, `StockShipped`, `StockAdjusted`, quarantine events	Near-zero — invalidate immediately on balance event
Product master	M-01 Catalog infrastructure	`ProductUpdated`, `ProductLifecycleStatusChanged`	Minutes — catalog changes infrequently
Location status / freeze	M-02 Network infrastructure	`LocationFreezeApplied`, `LocationFreezeReleased`, activation events	Near-zero for freeze; minutes for status
Dashboard KPIs	M-13 Intelligence infrastructure	Scheduled refresh + event-triggered for critical alerts	Minutes (NFR-PERF-004)
Permission lookup	M-12 Identity infrastructure	`RoleAssigned`, `RoleRevoked`	Short TTL (seconds) as safety net

8.4 Key Rules

Rule	Detail
CACHE-01	**Never cache write command paths** — cache is read-only optimization
CACHE-02	ATP cache must invalidate on any balance change event — oversell risk
CACHE-03	Cache miss falls through to authoritative read repository — no fail-to-zero ATP
CACHE-04	Cached responses include `asOf` timestamp for channel feeds (EXT-03)
CACHE-05	Domain layer is cache-unaware — testability and correctness

8.5 Does Not Belong In

- **Domain** — aggregates never read from cache
- **Application (implementation)** — application defines policy; infrastructure implements store
- **Presentation** — no in-memory ATP caching in controllers

---

9. Transactions

9.1 Definition in IMS

Transactions define atomicity boundaries — ensuring that one aggregate mutation completes fully or not at all, and that domain events are reliably published with the commit (outbox pattern).

9.2 Architectural Placement

Layer	Responsibility	Components
Presentation	No transaction management	—

Layer	Responsibility	Components
Application	**Own transaction boundaries** — `IUnitOfWork` per command handler; one aggregate per transaction; saga spans multiple transactions	`IUnitOfWork`, command handlers, saga orchestrators
Domain	Define consistency within aggregate; no transaction awareness	Aggregate atomic mutations
Infrastructure	Implement `IUnitOfWork`, database connection, transaction manager, transactional outbox	`DbUnitOfWork`, `OutboxMessage` persistence

9.3 Transaction Rules (DDD)

Rule	Scope	Detail
TX-01	Single command handler	One `UnitOfWork.Commit()` per command — one aggregate mutation
TX-02	Cross-aggregate workflow	Saga — each step is its own transaction; eventual consistency between steps
TX-03	Cross-module workflow	Same as saga — no distributed transaction across modules
TX-04	Event publish	Domain events written to outbox in same DB transaction as aggregate save
TX-05	Read queries	No transaction required for CQRS read path; read-uncommitted not used for ATP
TX-06	Intelligence projections	Separate transaction per projection handler — idempotent upsert

9.4 Transaction Flow

```
CommandHandler:
    UnitOfWork.Begin()
        aggregate = Repository.Get()
        aggregate.ExecuteCommand()     // domain invariants
        Repository.Save(aggregate)
        Outbox.Add(domainEvents)
    UnitOfWork.Commit()                // atomic: aggregate + outbox
    EventBus.Dispatch(outboxEvents)    // after commit
```

9.5 Key Rules

Rule	Detail
TX-07	Application layer opens and commits transactions — not controllers, not domain
TX-08	Stock Position reservation and order confirmation are separate transactions linked by saga
TX-09	Retry of transient failures at infrastructure layer must not double-apply — idempotent handlers
TX-10	Long-running transactions prohibited — no transaction spans user think time or external API wait

9.6 Does Not Belong In

- **Domain** — no `BeginTransaction()` in aggregates
- **Presentation** — no request-scoped transactions across multiple commands
- **Distributed two-phase commit** — prohibited across modules

---

10. Configuration

10.1 Definition in IMS

Configuration supplies environment-specific and tunable parameters — connection strings, feature flags, integration endpoints, threshold defaults, retention periods — without hard-coding them in domain or application logic.

10.2 Architectural Placement

Layer	Responsibility	Components
Presentation	UI display configuration only (e.g., page size defaults) — not system config	View settings
Application	Consume configuration via injected options interfaces — no direct file/env access	`IOptions<InventorySettings>`, `IFeatureManager`
Domain	**No configuration** — business thresholds that are governance policy live in M-03 Governance aggregate, not config files	—
Infrastructure	Load configuration from environment, secrets manager, config files; bind to options classes	`ConfigurationProvider`, secrets adapter (SVC-07)

10.3 Configuration Categories

Category	Examples	Layer	Mutable At Runtime
Infrastructure	DB connection, IdP endpoint, email gateway	Infrastructure	Deploy time
Secrets	API keys, certificates	Infrastructure (secrets manager)	Rotation via SVC-07
Feature flags	Enable campaign module, bonded inventory	Application (via infra provider)	Yes — phased rollout
Operational tuning	Cache TTL, saga timeout, retry count	Application options	Yes — ops tuning
Business policy	Approval thresholds, reason codes	**M-03 Governance aggregate**	Via governance commands — not config file
Location-specific	Business hours per location (NFR-AVL-005)	M-02 Network master data	Via location commands

10.4 Key Rules

Rule	Detail
CFG-01	Business rules (BR-xxx) are domain/governance — not `appsettings.json`
CFG-02	Secrets never in source control or domain layer
CFG-03	Configuration injected via interfaces — application depends on `IOptions<T>`, not raw config
CFG-04	Feature flags control module activation per phased delivery — not compile-time branching
CFG-05	Location business hours are master data (M-02), not environment config

10.5 Does Not Belong In

- **Domain** — no `IConfiguration` in aggregates
- **Governance thresholds in config files** — use M-03 Governance Policy aggregate for auditability

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11. Localization

11.1 Definition in IMS

Localization adapts user-facing text, number/date formats, and currency display for multi-region operations — without altering domain logic, business rules, or stored data formats.

11.2 Architectural Placement

Layer	Responsibility	Components
Presentation	**Primary localization** — UI labels, messages, error text, report headers; culture selection from user preference or browser	Resource files (`.resx` / i18n bundles), `IStringLocalizer`, culture middleware
Application	Format dates and currency for query responses via abstraction — stored values remain UTC and canonical	`ILocalizationService`, `IDateTimeFormatter` ports

Layer	Responsibility	Components
Domain	**Culture-neutral** — quantities, dates stored in canonical form (UTC, invariant decimal); reason codes stored as codes with localized display in presentation	Canonical value objects only
Infrastructure	Load resource bundles; provide localization service implementation	`LocalizationService`, resource providers

11.3 Localization Scope

Localizable	Not Localizable
UI labels and menus	`ProductSku`, `LocationCode` identifiers
Validation error messages (presentation/application)	Domain event names
Report column headers	Audit trail stored values
User notification email templates	Database-stored reason code keys (display localized separately)
Currency **display** format	Monetary amounts stored in canonical decimal
Date **display** in user's timezone	Timestamps stored UTC (NFR-LOG-001)

11.4 Key Rules

Rule	Detail
L10N-01	Domain and persistence use invariant culture for all stored values
L10N-02	Business rule exception messages may carry message keys — presentation localizes for user display
L10N-03	Ubiquitous language terms (SRS-DM-008) are the source for translation keys — ATP, not "available stock" inconsistently
L10N-04	Audit and finance exports use locale-neutral formats for machine consumption; human reports may be localized
L10N-05	MVP may ship with single locale (English); architecture supports multi-locale without domain changes

11.5 Does Not Belong In

- **Domain** — no culture-specific logic or translated strings in aggregates
- **Database** — no per-locale duplicate master data rows for products

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12. Cross-Cutting Pipeline (Command Request)

The following shows how all concerns interact on a single write request:

13. Concern Interaction Summary

Concern	Depends On	Feeds Into
Authentication	IdP (Infrastructure)	Authorization, Auditing, Logging
Authorization	Authentication, M-12, M-03	Command execution gate
Validation	—	Domain execution or early rejection
Transactions	—	Event publish, Audit record
Auditing	Authentication, Domain events	Intelligence projection, Finance export
Logging	Correlation ID	Monitoring (NFR-MON)
Exception handling	All layers	Logging, Client response
Caching	Domain events (invalidation)	Query performance
Configuration	Infrastructure	Application behavior, Feature flags
Localization	User culture (Presentation)	UI display only

14. Implementation Standards

ID	Standard
CC-01	Cross-cutting behaviors implemented as ordered pipeline behaviors in application layer — not copy-pasted per handler
CC-02	All modules use shared cross-cutting libraries for logging format, exception types, correlation, and auth ports
CC-03	Domain layer projects have zero package references to logging, config, cache, or auth libraries
CC-04	Cross-cutting concerns tested independently — auth denial, validation failure, audit record creation
CC-05	NFR traceability maintained — each concern maps to specific NFR-SEC, NFR-LOG, NFR-COM requirements

15. Document Approval

Role	Name	Signature	Date
Solution Architect			
Security Engineer			
Tech Lead			
QA Lead			

End of Cross-Cutting Concerns

Document ID: SRS-SA-006 | Version 1.0 | July 6, 2026

Technology Stack Recommendation

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-SA-007
Document Title	Technology Stack Recommendation — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Senior Solution Architect
Prepared For	Steering committee, delivery leads, development squads, DevOps, and security
Date	July 6, 2026
Classification	Internal — Architecture Confidential
Scope	Technology selection aligned to approved architecture — version pinning deferred to implementation standards

Related Documents

Document	ID	Relationship
Architecture Vision	SRS-SA-001	Modular monolith, event-driven, CQRS
Architecture Style Recommendation	SRS-SA-002	Clean/Hexagonal per module
System Modules Design	SRS-SA-003	13 bounded-context modules
Layered Architecture Design	SRS-SA-004	Layer placement
Integration Architecture Design	SRS-SA-005	Sync/async, outbox pattern
Cross-Cutting Concerns	SRS-SA-006	Security, caching, transactions
Non-Functional Requirements	204 NFRs	Performance, security, availability constraints

Stakeholder Technology Mandates

The following technologies are **organizationally mandated** for this initiative:

Mandate	Technology
Backend framework	**Java + Spring**
Frontend framework	**Angular**
Database	**PostgreSQL**

All other selections complement these mandates and align with the approved modular monolith architecture.

Document Purpose

This document recommends a complete technology stack for the Inventory Management System. For each technology it provides purpose, selection rationale, alternatives considered, and trade-offs — enabling informed approval before project bootstrap.

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1. Stack Overview

```
┌──────────────────────────────────────────────────────────────────────┐
│  FRONTEND          Angular 19 | Angular Material | TypeScript | RxJS  │
├──────────────────────────────────────────────────────────────────────┤
│  API               REST (JSON) | OpenAPI 3 | Spring MVC Controllers  │
├──────────────────────────────────────────────────────────────────────┤
│  BACKEND           Java 21 LTS | Spring Boot 3.x | Spring Modulith    │
│                    Spring Security | Spring Data JPA | MapStruct      │
├──────────────────────────────────────────────────────────────────────┤
│  MESSAGING (MVP)   Spring Application Events | Transactional Outbox (PG) │
│  MESSAGING (Ph.2+) RabbitMQ or Apache Kafka (deferred decision)      │
├──────────────────────────────────────────────────────────────────────┤
│  CACHE             Redis (ATP hot path, session optional)            │
├──────────────────────────────────────────────────────────────────────┤
│  DATABASE          PostgreSQL 16 | Flyway migrations | schema-per-module │
├──────────────────────────────────────────────────────────────────────┤
│  SECURITY          Spring Security 6 | OAuth 2.0 / OIDC | JWT Resource Srv │
├──────────────────────────────────────────────────────────────────────┤
│  OBSERVABILITY     Micrometer | Prometheus | Grafana | Structured logging │
├──────────────────────────────────────────────────────────────────────┤
│  BUILD             Maven (multi-module) | GitHub Actions / Azure DevOps CI │
├──────────────────────────────────────────────────────────────────────┤
│  TESTING           JUnit 5 | Mockito | Testcontainers | ArchUnit | Cypress │
├──────────────────────────────────────────────────────────────────────┤
│  CONTAINERIZATION  Docker | Docker Compose (dev) | Kubernetes (prod Ph.2+) │
└──────────────────────────────────────────────────────────────────────┘
```

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2. Backend

2.1 Java 21 (LTS)

Aspect	Detail
Purpose	Primary programming language for all server-side domain, application, and infrastructure code
Why selected	Mandated Java ecosystem; Java 21 is the current long-term support release with virtual threads (Project Loom) beneficial for I/O-bound warehouse API workloads; mature tooling and hiring pool for enterprise distribution/retail sector
Alternatives considered	Java 17 LTS (still supported but misses virtual threads maturity); Kotlin on JVM (concise but splits team skill set)
Trade-offs	(+) LTS stability, records/pattern matching, virtual threads for concurrency; (−) Slightly newer runtime than Java 17 — verify corporate JDK approval

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2.2 Spring Boot 3.x

Aspect	Detail
Purpose	Application framework — dependency injection, auto-configuration, REST API, transaction management, actuator health endpoints, integration with entire Spring ecosystem
Why selected	Mandated Spring stack; de facto enterprise standard; native support for security, data access, events, and testing; aligns with Clean Architecture via clear bean wiring of ports and adapters
Alternatives considered	Quarkus (faster startup, smaller footprint — less mature Modulith support); Micronaut (compile-time DI — smaller community for DDD patterns); Plain Spring Framework without Boot (more boilerplate, slower bootstrap)
Trade-offs	(+) Rapid development, extensive documentation, team familiarity; (−) Heavier runtime than Quarkus; requires discipline to avoid anemic Spring MVC + JPA anti-pattern — mitigated by Modulith and layered architecture

2.3 Spring Modulith

Aspect	Detail
Purpose	Enforces **modular monolith** boundaries — 13 bounded-context modules as separate Maven modules with compile-time dependency verification and module-scoped application events
Why selected	Directly implements SRS-SA-002 (modular monolith) and SRS-SA-003 (13 modules); provides `ApplicationModuleTest` to verify acyclic module dependencies; supports event publication APIs and JDBC event registry for outbox; avoids premature microservices while preserving decomposition path
Alternatives considered	Manual package-by-feature with ArchUnit only (works but no first-class event/outbox support); Microservices with Spring Cloud (rejected — ORC-04, BC-01); OSGi (obsolete complexity)
Trade-offs	(+) Architecture-as-code module boundaries; event infrastructure built-in; (−) Relatively newer Spring project — smaller reference base than core Spring Boot; team training required

2.4 Spring Web (Spring MVC) + REST

Aspect	Detail
Purpose	Presentation layer — expose REST JSON APIs for Angular SPA, mobile warehouse clients, and Phase 2 external integrations
Why selected	Standard Spring Boot stack; synchronous request-reply for ATP lookup (NFR-PERF-002, 2s target); well-understood by enterprise teams; integrates with Spring Security filter chain
Alternatives considered	Spring WebFlux (reactive — adds complexity without clear benefit for CRUD + domain-heavy workloads); gRPC (efficient but poor browser compatibility for Angular); GraphQL (flexible queries but overkill for defined use cases)
Trade-offs	(+) Simple mental model, excellent tooling, OpenAPI integration; (−) Thread-per-request model — mitigated by Java 21 virtual threads and connection pool tuning

2.5 MapStruct

Aspect	Detail
Purpose	Compile-time mapping between presentation DTOs, application commands/queries, and domain objects — keeps mapping out of controllers and handlers
Why selected	Type-safe, zero-reflection overhead; aligns with Clean Architecture boundary mapping (SRS-SA-004); reduces boilerplate across 48 use cases
Alternatives considered	ModelMapper (runtime reflection — harder to debug); Manual mapping (verbose, error-prone at scale); Dozer (legacy, maintenance declining)
Trade-offs	(+) Fast, compile-time errors, readable generated code; (−) Annotation processor setup; learning curve for complex mappings

2.6 Springdoc OpenAPI

Aspect	Detail
Purpose	Auto-generate OpenAPI 3 specification from Spring MVC controllers — API contract documentation and Angular client generation
Why selected	Standard for Spring Boot 3; supports contract-first collaboration between frontend and backend squads; enables Phase 2 CRM integration specification
Alternatives considered	Springfox (unmaintained, incompatible with Spring Boot 3); Manual OpenAPI YAML (high maintenance)
Trade-offs	(+) Always in sync with code; Swagger UI for QA; (−) Generated spec may need annotation tuning for clean contracts

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3. Frontend

3.1 Angular 19

Aspect	Detail
Purpose	Single-page application framework for office users (purchasing, CS, management), warehouse web views, and system administration — all role-based IMS screens
Why selected	**Mandated** frontend framework; enterprise-grade structure (modules, dependency injection, routing); strong TypeScript integration; long-term Google backing; suits large multi-team IMS with 90 functional requirements and role-based UI
Alternatives considered	React (larger ecosystem but not mandated); Vue (lighter but less enterprise governance tooling); Blazor (requires .NET stack divergence)
Trade-offs	(+) Opinionated structure aids consistency across squads; built-in routing, forms, HTTP; (−) Steeper learning curve than React; larger bundle — mitigated by lazy-loaded feature modules per bounded context

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3.2 TypeScript 5.x

Aspect	Detail
Purpose	Typed language for Angular — compile-time safety for API contracts, form models, and state
Why selected	Default and recommended language for Angular; reduces runtime errors in complex inventory forms (receipt lines, count entries, order lines)
Alternatives considered	JavaScript (no type safety — unacceptable for 90 FRs scale)
Trade-offs	(+) IDE support, refactoring safety; (−) Compilation step, type maintenance

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3.3 Angular Material

Aspect	Detail
Purpose	UI component library — tables, forms, dialogs, navigation, data grids for inventory screens, dashboards, and approval workflows
Why selected	Official Angular component library; accessible (WCAG); consistent enterprise look; data table and paginator suit stock inquiry and movement history screens
Alternatives considered	PrimeNG (richer data components but third-party); NG-ZORRO (Ant Design — less common in Western enterprise); Custom CSS (slow, inconsistent)
Trade-offs	(+) Maintained by Angular team, theming support; (−) Material Design aesthetic may need customization for corporate branding

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3.4 RxJS + Angular Signals (Hybrid)

Aspect	Detail
Purpose	Reactive state and async data flow — HTTP API calls, real-time dashboard refresh, form debounce on SKU search (NFR-PERF-002)
Why selected	RxJS is Angular's native async model; Angular 19 signals for local component state reduce boilerplate; hybrid approach suits ATP lookup streams and alert subscriptions
Alternatives considered	NgRx (full Redux — overkill for MVP, adds ceremony); Plain promises (poor composition for complex UI flows)
Trade-offs	(+) Powerful composition for search-as-you-type ATP; (−) RxJS learning curve — mitigate with shared service patterns and limited operator set

3.5 Angular PWA (Progressive Web App) — Warehouse Mobile

Aspect	Detail
Purpose	Installable web app for warehouse scanners and mobile devices — put-away, pick, cycle count without native app store deployment
Why selected	BRD constraint TEC-03 (online MVP, no offline); PWA provides home-screen install, responsive layout, and camera/barcode API access on modern devices; single Angular codebase serves office and warehouse
Alternatives considered	Native Android/iOS (higher cost, two codebases — BC-01); Ionic/Capacitor (adds layer — consider Phase 2 if scanner hardware integration deepens)
Trade-offs	(+) One codebase, faster MVP, no app store; (−) Limited offline (acceptable per TEC-03); scanner hardware integration may need Capacitor plugin in Phase 2

4. Database

4.1 PostgreSQL 16

Aspect	Detail
Purpose	Primary relational database — transactional write store for all 13 modules, transactional outbox table, CQRS read projections, audit trail storage
Why selected	**Mandated** database; ACID compliance critical for Stock Position invariants; excellent JSONB support for event payloads and flexible audit metadata; row-level locking for reservation contention; mature replication for read scaling; open source aligns with BC-01 fixed budget
Alternatives considered	Microsoft SQL Server (licensing cost under BC-01); MySQL (weaker JSON and constraint tooling); Oracle (cost, overkill for mid-sized distribution); MongoDB (document model conflicts with aggregate transactional integrity)
Trade-offs	(+) ACID, JSONB, partial indexes, logical replication, strong ecosystem; (−) Horizontal write scaling limited — acceptable for modular monolith MVP scale; team must manage vacuum/autovacuum tuning

4.2 Schema-per-Module (PostgreSQL Schemas)

Aspect	Detail
Purpose	Logical data isolation — each bounded-context module owns tables in a dedicated PostgreSQL schema (`catalog`, `balance`, `fulfillment`, etc.) enforcing SRS-SA-003 module boundary rule MB-07
Why selected	Single PostgreSQL instance (operational simplicity ORC-04) with module ownership boundaries; prevents cross-module JOIN temptation; supports per-module Flyway migration scripts
Alternatives considered	Single flat schema (no boundary enforcement); Database-per-module (microservices pattern — rejected); Separate databases per module (ops overhead)

Aspect	Detail
Trade-offs	(+) Module ownership clear, single backup target, ACID within instance; (−) Discipline required — no cross-schema FKs except shared kernel reference tables

4.3 Flyway

Aspect	Detail
Purpose	Version-controlled database migration — schema evolution per module, repeatable across environments and location rollouts
Why selected	SQL-first migrations align with DBA review practices; Spring Boot native integration; per-module migration locations map to Spring Modulith structure
Alternatives considered	Liquibase (XML/YAML abstraction — more flexible but heavier); Hibernate `ddl-auto` (prohibited for production — no governance)
Trade-offs	(+) Simple, auditable SQL scripts, CI integration; (−) No automatic rollback — requires forward-fix migration discipline

4.4 Redis 7 (Supplementary Cache Store)

Aspect	Detail
Purpose	In-memory cache for ATP hot-path queries (NFR-PERF-002), optional session store, rate limiting counters — **not** system of record
Why selected	Sub-millisecond read for high-frequency SKU availability lookups; event-driven invalidation from Balance module domain events (SRS-SA-006 caching rules); offloads PostgreSQL read pressure during peak season (NFR-AVL-003)
Alternatives considered	Caffeine (in-process only — invalid in multi-instance deployment); PostgreSQL materialized views alone (insufficient for 2s SLA under concurrent load); Hazelcast (heavier operational footprint)
Trade-offs	(+) Proven, fast, TTL and pub/sub for invalidation; (−) Additional infrastructure component; cache staleness risk — mitigated by event invalidation and `asOf` timestamp

5. ORM

5.1 Hibernate ORM 6 (via Spring Data JPA)

Aspect	Detail
Purpose	Object-relational mapping — persist and hydrate domain aggregates via repository implementations in infrastructure layer; **not** exposed to domain layer
Why selected	Default JPA provider in Spring Boot 3; mature PostgreSQL dialect; supports optimistic locking (`@Version`) for Stock Position contention; Spring Data JPA reduces repository boilerplate; team familiarity in Java enterprise
Alternatives considered	**MyBatis** (SQL-centric — more control but mapping drifts from domain model); **jOOQ** (excellent for complex queries and CQRS read side — consider supplement for Intelligence module); **EclipseLink** (less Spring community momentum); **Raw JDBC** (too low-level for 18 aggregates)
Trade-offs	(+) Productivity, Spring integration, optimistic locking, auditing (`@CreatedBy`); (−) Risk of anemic domain model and lazy-loading leaks — **mitigated by**: aggregates mapped explicitly, repository interfaces in domain, no entity exposure past infrastructure, ArchUnit rules

5.2 Spring Data JPA (Repository Abstraction)

Aspect	Detail
Purpose	Implement `IStockPositionRepository` and other domain repository ports — query methods, custom `@Query` for complex reads, pagination for movement history

Aspect	Detail
Why selected	Standard Spring stack; reduces CRUD boilerplate for master data modules (Catalog, Network, Governance); integrates with `@Transactional` unit of work
Alternatives considered	Plain Hibernate Session API (more verbose); Spring Data JDBC (no lazy loading — simpler but less aggregate mapping support)
Trade-offs	(+) Convention over configuration; (−) Derived query methods can hide expensive queries — require query review and EXPLAIN analysis for ATP paths

5.3 jOOQ (Supplementary — Intelligence Read Side)

Aspect	Detail
Purpose	Type-safe SQL for **CQRS read projections** in M-13 Intelligence — complex KPI queries, audit trail search, dashboard aggregations without loading JPA entities
Why selected	Reporting queries must not burden JPA entity model (AR-10); jOOQ generates type-safe SQL from PostgreSQL schema; complements JPA on write side
Alternatives considered	Native SQL in `@Query` (untyped, error-prone); PostgreSQL views only (limited composability in application)
Trade-offs	(+) Performance, type safety for read models; (−) Second data access pattern — limit to Intelligence module only

6. Security

6.1 Spring Security 6

Aspect	Detail
Purpose	Application security framework — authentication filter chain, authorization rules, CSRF protection, method-level security, session management
Why selected	Industry standard for Spring Boot; implements SRS-SA-006 authentication and authorization placement; integrates with OAuth2/OIDC resource server; supports location-scoped permission expressions
Alternatives considered	Apache Shiro (less Spring-native); Custom JWT filter (reinventing maintained library); Keycloak adapters deprecated in favor of OAuth2 resource server pattern
Trade-offs	(+) Comprehensive, well-documented, NFR-SEC alignment; (−) Configuration complexity — mitigate with shared security module and tested config

6.2 OAuth 2.0 / OpenID Connect (OIDC)

Aspect	Detail
Purpose	Enterprise SSO integration with identity provider (SVC-01) — Entra ID, Okta, or LDAP-federated OIDC; JWT bearer tokens for Angular SPA and API
Why selected	NFR-SEC-001 (unique user authentication); no shared accounts; standard protocol supported by all enterprise IdPs; Angular `angular-oauth2-oidc` library mature
Alternatives considered	SAML 2.0 (legacy, XML-heavy — still supported but OIDC preferred for SPA); Session-only form login (no SSO, poor for enterprise); API keys (insufficient for user attribution)
Trade-offs	(+) Enterprise SSO, token-based stateless API, refresh token flow; (−) Token management complexity in Angular; requires IdP provisioning process for 19 roles

6.3 JWT Resource Server

Aspect	Detail
Purpose	Validate JWT access tokens on every API request; extract claims for `UserId`, roles, and location scope; stateless API authentication
Why selected	Aligns with Angular SPA architecture; scales horizontally without server-side session affinity; supports NFR-SEC-021 session management policies
Alternatives considered	Server-side sessions with Redis (stateful — works but adds session management overhead); Basic auth (prohibited — NFR-SEC-001)
Trade-offs	(+) Stateless, cloud-friendly; (−) Token revocation requires short TTL + refresh or token blocklist for immediate deprovision

6.4 BCrypt / DelegatingPasswordEncoder

Aspect	Detail
Purpose	Password hashing for locally provisioned accounts (fallback when SSO unavailable) and service account secrets
Why selected	Spring Security default; NFR-SEC-010 password policy compliance; adaptive cost factor
Alternatives considered	Argon2 (stronger — supported via DelegatingPasswordEncoder if policy requires); Plain hash (prohibited)
Trade-offs	(+) Built-in, proven; (−) BCrypt 72-byte limit — irrelevant for generated passwords

7. Messaging

7.1 Spring Application Events + Spring Modulith Event API (MVP)

Aspect	Detail
Purpose	**In-process domain event bus** for MVP — cross-module choreography within modular monolith (SRS-SA-005); `PutAwayCompleted` → Balance, all events → Intelligence
Why selected	No external broker ops for MVP (ORC-04); Spring Modulith provides module-scoped event publication and subscription; transactional integrity with JDBC event registry; meets internal event latency < 5 seconds
Alternatives considered	RabbitMQ/Kafka from day one (ops overhead, BC-01); Direct method calls (tight coupling — violates module boundaries); Database polling only (higher latency)
Trade-offs	(+) Zero additional infrastructure, ACID with outbox, simple debugging; (−) Not suitable for external system integration — Phase 2 broker required for CRM webhooks and channel feeds

7.2 Transactional Outbox (PostgreSQL + Spring Modulith JDBC Registry)

Aspect	Detail
Purpose	Reliable domain event delivery — events written to outbox table in same transaction as aggregate save; async dispatcher publishes after commit (SRS-SA-005, NFR-REL-007)
Why selected	Prevents lost events on crash between DB commit and publish; idempotent event handlers; supports at-least-once delivery with deduplication by `EventId`
Alternatives considered	Dual-write (publish then save — inconsistent); XA two-phase commit (complex, performance penalty); Change Data Capture / Debezium (ops heavy for MVP)
Trade-offs	(+) Proven pattern, uses existing Postgre PostgreSQL; (−) Outbox table growth — requires archival job; slight publish latency (acceptable)

7.3 RabbitMQ (Phase 2 — Recommended External Broker)

Aspect	Detail
Purpose	External async integration — CRM order webhooks, channel ATP feed, finance export triggers, notification dispatch; decouple IMS from external consumer pace
Why selected	Mature, operable by limited IT team; sufficient throughput for IMS scale; Spring AMQP integration; easier ops than Kafka for mid-sized message volumes
Alternatives considered	**Apache Kafka** (higher throughput, event replay — overkill for MVP scale, steeper ops ORC-04); **AWS SQS** (cloud lock-in); **Redis Streams** (less durable guarantee)
Trade-offs	(+) Proven, Spring integration, manageable ops; (−) Not ideal for event sourcing replay at scale — acceptable as IMS uses PostgreSQL as source of truth, not broker log

Final broker selection deferred to Phase 2 integration workshop (HS-014, EXT-01).

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8. Build Tools

8.1 Apache Maven (Multi-Module)

Aspect	Detail
Purpose	Build orchestration — multi-module project structure mapping to 13 bounded contexts + shared kernel + application shell; dependency management, packaging, CI integration
Why selected	Enterprise standard for Java/Spring; explicit XML configuration aids governance review; Spring Modulith documentation uses Maven; predictable CI pipeline; wide IDE support
Alternatives considered	**Gradle** (faster builds, Kotlin DSL — preferred by some teams but less universal in enterprise procurement environments); **Bazel** (overkill); Single-module build (no boundary enforcement)
Trade-offs	(+) Universal, stable, audit-friendly `pom.xml`; (−) Slower than Gradle; verbose XML — mitigate with parent POM and BOM

8.2 Spring Boot Maven BOM

Aspect	Detail
Purpose	Centralized dependency version management — aligned Spring Boot, Spring Security, Hibernate, Jackson versions across all modules
Why selected	Prevents version conflicts across 13 modules; Spring-tested compatibility matrix
Alternatives considered	Manual version pinning per module (drift risk)
Trade-offs	(+) Consistency, security patch alignment; (−) Tied to Spring Boot release cadence

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8.3 GitHub Actions / Azure DevOps Pipelines (CI/CD)

Aspect	Detail
Purpose	Continuous integration — build, test, ArchUnit boundary checks, Flyway validation, Docker image publish on merge
Why selected	Automated quality gate for 13-module codebase; supports parallel module test execution; organization likely has existing pipeline platform
Alternatives considered	Jenkins (self-hosted ops overhead); GitLab CI (depends on repo hosting)
Trade-offs	(+) Automated NFR regression via test suite; (−) Pipeline maintenance — shared templates reduce per-module duplication

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9. Testing

9.1 JUnit 5

Aspect	Detail
Purpose	Unit and integration test framework — domain invariant tests, command handler tests, repository tests
Why selected	Java standard; Spring Boot Test native support; parameterized tests for business rule matrices (BR-001 – BR-032)
Alternatives considered	TestNG (less Spring ecosystem momentum); Spock/Groovy (different language)
Trade-offs	(+) Universal, IDE support; (−) Verbose compared to Spock — acceptable

9.2 Mockito

Aspect	Detail
Purpose	Mock domain repository ports and cross-module query ports in application layer unit tests — isolate handlers from infrastructure
Why selected	Standard Java mocking; aligns with Clean Architecture testing (SRS-SA-004 LS-10)
Alternatives considered	WireMock (for HTTP — use for ACL adapter tests); Manual stubs (verbose)
Trade-offs	(+) Fast unit tests; (−) Over-mocking risk — balance with integration tests

9.3 Spring Boot Test + Testcontainers

Aspect	Detail
Purpose	Integration tests with real PostgreSQL and Redis in Docker containers — verify repository mappings, Flyway migrations, saga flows, and outbox delivery
Why selected	Eliminates H2/embedded DB divergence from PostgreSQL production behaviour; Testcontainers is industry standard for Spring integration tests
Alternatives considered	H2 in-memory (PostgreSQL incompatibility — JSONB, locking); Dedicated test database (CI hygiene issues)
Trade-offs	(+) Production-parity testing; (−) Slower CI — mitigate with test slicing and parallel execution

9.4 ArchUnit

Aspect	Detail
Purpose	Architecture fitness functions — enforce module boundary rules (SRS-SA-003 MB-01 – MB-08), layer dependency rule (SRS-SA-004), domain purity (no JPA in domain package)
Why selected	Prevents modular monolith erosion; automated gate in CI; complements Spring Modulith module tests
Alternatives considered	Manual code review only (insufficient at scale); jQAssistant (heavier graph analysis)
Trade-offs	(+) Automated architecture governance; (−) Rules require maintenance as packages evolve

9.5 Cypress (E2E — Angular)

Aspect	Detail
Purpose	End-to-end browser tests for critical user journeys — ATP lookup, goods receipt, order reservation, cycle count blind entry
Why selected	Strong Angular integration; readable tests for QA collaboration; supports warehouse responsive viewport testing

Aspect	Detail
Alternatives considered	Playwright (excellent, growing — either acceptable); Selenium (verbose, flaky); Protractor (deprecated)
Trade-offs	(+) Developer-friendly, good debugging; (−) Slower than unit tests — limit to MVP critical path suite (~20 journeys)

9.6 k6 / Gatling (Performance — Phase 2)

Aspect	Detail
Purpose	Load testing for NFR-PERF validation — concurrent ATP lookups, reservation contention on Class A SKUs, peak season simulation
Why selected	Validates NFR-PERF-001 – NFR-PERF-003 before go-live; k6 scriptable in CI
Alternatives considered	JMeter (GUI-heavy, less CI-friendly)
Trade-offs	(+) Evidence for capacity model; (−) Deferred to pre-go-live phase per NFR performance baseline program

10. Containerization

10.1 Docker

Aspect	Detail
Purpose	Package IMS application, Angular static assets (via nginx sidecar or embedded), and local development dependencies (PostgreSQL, Redis) as portable containers
Why selected	Industry standard; consistent dev/prod parity; required for Testcontainers; supports ORC-04 limited ops via immutable deployments
Alternatives considered	Bare JAR deployment (environment drift); VM images (heavier, slower iteration)
Trade-offs	(+) Reproducible, portable, CI-friendly; (−) Team Docker literacy required; image security scanning needed

10.2 Docker Compose (Development & Integration Test)

Aspect	Detail
Purpose	Local developer environment — Spring Boot app + PostgreSQL + Redis + mail catcher (notifications) with single `docker compose up`
Why selected	Fast onboarding for mixed-skill delivery team; mirrors integration test environment
Alternatives considered	Local native PostgreSQL install (inconsistent versions); Devcontainers (optional supplement)
Trade-offs	(+) One-command environment; (−) Not for production — production uses orchestrator

10.3 Kubernetes (Production — Phase 2+)

Aspect	Detail
Purpose	Production orchestration — horizontal pod autoscaling for API tier, rolling deployments for zero-downtime location rollouts (OC-01), health checks (NFR-AVL-001)
Why selected	Cloud platform standard (SVC-04); supports 99.5% uptime with replica sets; secrets via SVC-07 integration
Alternatives considered	**Managed PaaS** (Azure App Service, AWS Elastic Beanstalk — simpler ops, recommended for MVP if ORC-04 is critical); **Docker Swarm** (declining adoption)

Aspect	Detail
Trade-offs	(+) Scalable, self-healing, industry standard; (−) Operational complexity — **recommend managed Kubernetes (AKS/EKS) or PaaS for MVP**, defer full K8s ops to Phase 2

10.4 Nginx (Static Asset Serving)

Aspect	Detail
Purpose	Serve Angular production build; reverse proxy to Spring Boot API; TLS termination at edge
Why selected	Lightweight, proven; separates static frontend from API scaling
Alternatives considered	Spring Boot serving static files (couples frontend deploy to backend); CDN only (premature for internal IMS)
Trade-offs	(+) Clean separation, cache headers for static assets; (−) Additional container in deployment

11. Observability Stack (Supplementary)

Technology	Purpose	Why Selected
Micrometer	Application metrics facade	Spring Boot Actuator native; NFR-MON KPI metrics
Prometheus	Metrics collection and storage	Industry standard; scrape Spring Actuator `/actuator/prometheus`
Grafana	Dashboards and alerting	Ops visibility; NFR-MON-001 5-minute outage detection
Logback + JSON encoder	Structured logging	NFR-LOG-001 standard fields; ELK/Loki ingestion ready
OpenTelemetry	Distributed tracing	Correlation ID propagation across modules (SRS-SA-006)

12. Recommended Module Structure (Maven)

```
ims-parent/                        (BOM, dependency management)
├── ims-shared-kernel/             (ProductSku, LocationCode, shared events envelope)
├── ims-catalog/                   (M-01)
│   ├── catalog-domain/
│   ├── catalog-application/
│   ├── catalog-infrastructure/
│   └── catalog-api/               (REST controllers — presentation)
├── ims-balance/                   (M-04 — core domain)
├── ims-fulfillment/               (M-07)
├── ...                            (remaining modules)
├── ims-intelligence/              (M-13 — jOOQ read side)
├── ims-identity/                  (M-12)
└── ims-application/               (Spring Boot main, Modulith assembly, security config)
```

13. Stack Decision Summary

Category	Selected	MVP / Phase
Language	Java 21 LTS	MVP
Backend framework	Spring Boot 3.x + Spring Modulith	MVP
API	REST + OpenAPI	MVP
Frontend	Angular 19 + Material + TypeScript	MVP
Database	PostgreSQL 16	MVP
Cache	Redis 7	MVP
ORM (write)	Hibernate 6 / Spring Data JPA	MVP

Category	Selected	MVP / Phase
ORM (read)	jOOQ (Intelligence module)	MVP
Migrations	Flyway	MVP
Security	Spring Security 6 + OIDC + JWT	MVP
Messaging (internal)	Spring Events + Outbox	MVP
Messaging (external)	RabbitMQ	Phase 2
Build	Maven multi-module	MVP
CI/CD	GitHub Actions / Azure DevOps	MVP
Unit/Integration test	JUnit 5 + Mockito + Testcontainers	MVP
Architecture test	ArchUnit + Spring Modulith tests	MVP
E2E test	Cypress	MVP
Containers	Docker + Compose	MVP
Orchestration	Managed PaaS or Kubernetes	PaaS MVP → K8s Phase 2

14. Risk and Mitigation

Risk	Mitigation
JPA anemic domain model	Repository ports in domain; ArchUnit rules; aggregate-focused mapping
Angular + Spring team silos	OpenAPI contract; shared API review; full-stack pairing on critical flows
PostgreSQL write contention on Stock Position	Optimistic locking; product-location aggregate scope; Redis ATP cache
Spring Modulith learning curve	Reference architecture spike in Sprint 0; module template
Redis cache staleness	Event-driven invalidation; `as0f` on ATP responses
MVP without external broker	In-process events sufficient; outbox ready for broker migration

15. Document Approval

Role	Name	Signature	Date
Senior Solution Architect			
Tech Lead			
Security Engineer			
DevOps Engineer			
Product Owner			

End of Technology Stack Recommendation

Document ID: SRS-SA-007 | Version 1.0 | July 6, 2026

Architecture Decision Records (ADR)

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-SA-008
Document Title	Architecture Decision Records — Inventory Management System
Version	1.0
Status	Active
Document Owner	Senior Solution Architect
Date	July 6, 2026
Classification	Internal — Architecture Confidential

Purpose

This document records **significant architectural decisions** for the Inventory Management System. Each ADR captures the context, decision, alternatives, consequences, and status to provide a durable rationale for future teams, auditors, and steering governance.

Status Definitions

Status	Meaning
Accepted	Decision approved and in force
Proposed	Decision direction agreed; pending workshop or gate confirmation
Deprecated	No longer recommended — see superseding ADR
Superseded	Replaced by another ADR

ADR Index

ID	Title	Status
ADR-001	Modular Monolith as Deployment Topology	Accepted
ADR-002	Domain-Driven Design with Bounded Context Modules	Accepted
ADR-003	Clean/Hexagonal Internal Layer Structure	Accepted
ADR-004	Inventory Balance as Sole Quantity Authority	Accepted
ADR-005	Event-Driven Cross-Module Integration	Accepted
ADR-006	Transactional Outbox for Reliable Event Delivery	Accepted
ADR-007	CQRS for Inventory Intelligence	Accepted
ADR-008	Saga Pattern for Cross-Module Workflows	Accepted
ADR-009	Stock Position Identity per Product-Location	Proposed
ADR-010	Location Freeze Authority Split	Proposed
ADR-011	Promotional Ring-Fence via Domain Events	Accepted
ADR-012	Anti-Corruption Layers for External Systems	Accepted
ADR-013	IMS as Operational System of Record	Accepted
ADR-014	Java Spring Boot Backend Stack	Accepted
ADR-015	Spring Modulith for Module Boundary Enforcement	Accepted
ADR-016	Angular SPA and PWA Frontend	Accepted
ADR-017	PostgreSQL as Primary Database	Accepted
ADR-018	Schema-per-Module Database Organization	Accepted

ID	Title	Status
ADR-019	JPA for Writes, jOOQ for Intelligence Reads	Accepted
ADR-020	OAuth 2.0 / OIDC with JWT Resource Server	Accepted
ADR-021	Redis Cache for ATP Hot Path	Accepted
ADR-022	In-Process Messaging for MVP	Accepted
ADR-023	Published Language for Cross-Boundary References	Accepted
ADR-024	Centralized Audit Trail Projection	Accepted
ADR-025	Architecture Fitness Functions with ArchUnit	Accepted

Source Documents

SRS-SA-001 through SRS-SA-007, SRS-DM-007, SRS-DM-009, SRS-ES-007, BRD-IMS-001, NFR suite.

ADR-001: Modular Monolith as Deployment Topology

Field	Detail
Decision ID	ADR-001
Title	Modular Monolith as Deployment Topology
Status	Accepted

Context

The IMS must support 13 bounded contexts, 18 aggregates, and 81 domain events while operating under a **fixed initiative budget (BC-01)** and **limited internal IT capacity (ORC-04)**. The organization requires phased delivery (MVP Months 0–6) without operational shutdown at go-live (OC-01). Strategic domain design (SRS-DM-007) defines clear context boundaries, but bounded contexts are explicitly **not** microservice boundaries.

The team must choose a deployment topology that balances domain alignment, operational simplicity, Stock Position consistency, and a credible path to future scale.

Decision

Deploy the Inventory Management System as a **single modular monolith** — one deployable application composed of 13 loosely coupled modules aligned to bounded contexts, with compile-time boundary enforcement and a decomposition-ready interface surface.

Microservices are **deferred** until volume metrics and operational maturity justify selective extraction (Inventory Intelligence, integration gateway).

Alternatives Considered

Alternative	Reason Not Selected
Microservices (13 services)	Distributed consistency risk on Stock Position; ops overhead exceeds ORC-04; premature with 16 open domain hotspots
Traditional layered monolith	No bounded-context boundaries; high coupling; rules scatter across layers
Service-oriented (few coarse services)	Arbitrary split; still distributed ops without clear domain mapping

Consequences

Positive

- Single deployment pipeline and runtime — lower ops cost

- ACID transactions within aggregate and in-process saga steps
- Faster MVP delivery and cross-module refactoring
- Module interfaces preserved for future extraction

Negative

- Shared runtime blast radius — defect in one module can affect availability
- Cannot independently scale Stock Position writes vs. reporting without read-side separation (mitigated by ADR-007)
- Requires discipline to prevent module boundary erosion

Neutral

- Team structure should align to core modules but shares one codebase

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ADR-002: Domain-Driven Design with Bounded Context Modules

Field	Detail
Decision ID	ADR-002
Title	Domain-Driven Design with Bounded Context Modules
Status	Accepted

Context

Domain analysis produced 13 bounded contexts, 18 aggregates, 24 domain services, 81 domain events, and 72 ubiquitous language terms. Business rules (BR-001 – BR-032) and segregation-of-duties requirements span multiple stakeholder functions (warehouse, purchasing, finance, marketing). A purely technical decomposition (UI / service / data layers only) cannot preserve domain integrity.

Decision

Adopt **Domain-Driven Design (DDD)** as the strategic and tactical design approach. Each **bounded context maps 1:1 to a system module** (M-01 through M-13). Aggregates define consistency boundaries. Cross-context integration uses published language identifiers, domain events, and application-layer ports — never shared mutable domain objects.

Alternatives Considered

Alternative	Reason Not Selected
CRUD-centric data model	Cannot express 32 business rules, SoD, and event choreography
Transaction script pattern	Business logic scatters; untestable rule enforcement
Anemic domain model with service layer only	Invariants not protected at aggregate boundaries

Consequences

Positive

- Direct traceability from BRD → domain model → modules → code
- Independent change velocity per context (catalog vs. balance vs. governance)
- Ubiquitous language (SRS-DM-008) drives API and code naming

Negative

- Higher initial design investment; DDD skill required across squads
- Cross-context workflows require explicit integration design (events, sagas)

Neutral

- Generic subdomains (Identity) may use simpler patterns within DDD module structure

ADR-003: Clean/Hexagonal Internal Layer Structure

Field	Detail
Decision ID	ADR-003
Title	Clean/Hexagonal Internal Layer Structure
Status	Accepted

Context

Each of the 13 modules requires internal organization that protects domain rules from infrastructure concerns (ORM, HTTP, SSO, messaging) while supporting testability and ACL adapters for external systems. Multiple layering models (Layered, Clean, Hexagonal, Onion) were evaluated (SRS-SA-002).

Decision

Apply **Clean Architecture with Hexagonal ports and adapters** inside every module:

- **Domain** (center): aggregates, value objects, domain events, repository interfaces
- **Application**: command/query handlers, sagas, authorization behaviors
- **Presentation** (per module API package): REST controllers, DTOs
- **Infrastructure**: JPA repositories, ACL adapters, event bus, cache

Dependency rule: source dependencies point inward only. Frameworks are details in infrastructure.

Alternatives Considered

Alternative	Reason Not Selected
Layered only (UI → Service → DAO)	Domain logic leaks; insufficient port/adapter isolation
Clean without hexagonal ports	Weaker external integration boundary protection
No internal layers (flat packages)	Unmaintainable at 13 modules × 48 use cases

Consequences

Positive

- Domain testable without database or HTTP
- ACL adapters cleanly isolated for CRM and Finance
- Consistent structure across all squads

Negative

- Mapping boilerplate (DTO ↔ command ↔ domain) — mitigated by MapStruct
- Learning curve for teams accustomed to annotation-driven MVC only

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ADR-004: Inventory Balance as Sole Quantity Authority

Field	Detail
Decision ID	ADR-004
Title	Inventory Balance as Sole Quantity Authority
Status	Accepted

Context

The system's core value is answering *what do we have, where, and what is available?* Multiple modules mutate stock-related state (receiving, fulfillment, transfers, adjustments, returns, campaigns). Without a single authority, ATP can diverge — causing overselling (BR-001), audit failures (BO-19), and customer trust loss.

Domain model review (SRS-DM-009) confirms Stock Position as the core domain center of gravity.

Decision

Module M-04 (Inventory Balance) is the sole mutator of on-hand, reserved, quarantined, and ATP quantities. All other modules request quantity changes via **commands, domain events, or sagas** — they never write to balance tables or hold mutable quantity state.

Alternatives Considered

Alternative	Reason Not Selected
Distributed quantity ownership (fulfillment owns reservation table, balance owns on-hand)	Split-brain ATP; HS-007 reservation coordination risk
Shared stock tables accessed by all modules	No aggregate protection; invariant violations
Event sourcing as sole write model	Higher complexity; team maturity and BC-01 constraints

Consequences

Positive

- Single mathematical invariant: ATP = on-hand − reserved − quarantined
- Clear audit chain for all quantity changes
- Contention localized to Stock Position aggregate (product-location scope)

Negative

- M-04 becomes highest-contention module — requires optimistic locking and caching
- All quantity workflows have dependency on Balance module availability

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ADR-005: Event-Driven Cross-Module Integration

Field	Detail
Decision ID	ADR-005
Title	Event-Driven Cross-Module Integration
Status	Accepted

Context

Event storming identified 81 domain events and 12 end-to-end lifecycles with policy chains (e.g., put-away → stock available → backorder fulfill). Synchronous coupling of all 13 modules would create fragile call chains, threaten NFR-PERF-002 (2s ATP), and embed reporting inside transactional paths.

Decision

Use **domain event choreography** as the primary cross-module state propagation mechanism. Modules publish domain events after successful aggregate commits. Subscribing modules react via application-layer event handlers. Synchronous cross-module calls are limited to **read-only queries** and **saga command steps** requiring immediate confirmation.

Alternatives Considered

Alternative	Reason Not Selected
Pure synchronous orchestration	Tight coupling; hot-path latency; cascading failures
Shared database triggers	Hidden logic; untestable; violates module ownership
Central orchestration engine (ESB)	Over-engineering; BC-01; ORC-04

Consequences

Positive

- Loose coupling; new subscribers (Intelligence, alerts) without publisher changes
- Natural audit trail from event stream
- Aligns with ADR-007 CQRS projections

Negative

- Eventual consistency between modules — requires saga compensation (ADR-008)
- Event schema versioning discipline required
- Debugging distributed flows harder than single call stack — mitigated by correlation IDs

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ADR-006: Transactional Outbox for Reliable Event Delivery

Field	Detail
Decision ID	ADR-006
Title	Transactional Outbox for Reliable Event Delivery
Status	Accepted

Context

NFR-REL-007 requires ≤15-minute recoverable data loss. NFR-BDR-001 requires continuous transaction protection. Publishing domain events **after** database commit creates a dual-write risk — crash between commit and publish loses events, causing Stock Position and Intelligence projections to diverge from operational truth.

Decision

Implement the **transactional outbox pattern** — domain events are persisted to an outbox table in the **same database transaction** as the aggregate save. A separate dispatcher publishes events after commit. Handlers must be **idempotent** (deduplicate by `EventId`).

Alternatives Considered

Alternative	Reason Not Selected
Publish-then-save	Inconsistent on failure
XA two-phase commit	Performance penalty; operational complexity
Change Data Capture (Debezium)	Additional ops infrastructure for MVP

Consequences

Positive

- At-least-once delivery with atomic write + event record
- Outbox table in PostgreSQL — no additional broker required for MVP
- Ready for external broker relay in Phase 2

Negative

- Outbox table growth — requires archival job
- Slight latency between commit and handler execution (acceptable per SRS-SA-005)

ADR-007: CQRS for Inventory Intelligence

Field	Detail
Decision ID	ADR-007
Title	CQRS for Inventory Intelligence
Status	Accepted

Context

M-13 Inventory Intelligence owns dashboards, KPIs, movement history, audit trail, and valuation snapshots — with **no write aggregates**. Running reporting queries against transactional Stock Position tables risks dock performance degradation (AR-10) and violates NFR-PERF-002/PERF-004. Intelligence has fundamentally different read-heavy, eventually consistent NFR profile.

Decision

Apply **Command Query Responsibility Segregation (CQRS)** for the Intelligence concern:

- **Commands** mutate aggregates in transactional modules (M-01 – M-12)
- **Queries** for dashboards, audit, KPIs, and ATP display read from **dedicated read projections** maintained by event handlers in M-13
- Read models are rebuildable from domain event history

Alternatives Considered

Alternative	Reason Not Selected
Single model (JPA entities for everything)	Reporting load on write tables
Database views only	Limited application-level composition; coupling to write schema
Separate reporting database via ETL batch	Stale data; complex pipeline for MVP

Consequences

Positive

- Write path protected for fulfillment and receiving performance
- Intelligence module extractable to separate service in Phase 3
- Audit trail append-only and query-optimized

Negative

- Eventual consistency on dashboards (seconds lag — acceptable)
- Projection rebuild complexity on schema evolution
- Second data access pattern (jOOQ) in Intelligence module

ADR-008: Saga Pattern for Cross-Module Workflows

Field	Detail
Decision ID	ADR-008
Title	Saga Pattern for Cross-Module Workflows

Field	Detail
Status	Accepted

Context

Business workflows span multiple aggregates and modules: order confirm → reserve stock, put-away → stock increase, transfer dispatch → source deduct + destination increase, physical count → freeze → adjust. DDD rule "one transaction, one aggregate" prohibits single ACID transaction across modules. Hotspot HS-007 highlights partial failure risk when order confirms but reservation fails.

Decision

Implement **orchestrated sagas** in the calling module's application layer for cross-module workflows. Each saga step is its own transaction. Define **explicit compensation actions** for each step failure. Critical sagas: SAG-01 (order confirm-reserve), SAG-04 (physical count-freeze), SAG-05 (opening balance go-live).

Alternatives Considered

Alternative	Reason Not Selected
Distributed two-phase commit	Impractical across modules; performance and availability impact
Choreography only (no orchestrator)	Hard to trace and compensate multi-step failures
Manual reconciliation	Unacceptable for ATP integrity and customer orders

Consequences

Positive

- Clear failure handling and compensation paths
- Each step remains single-aggregate ACID
- Testable with integration tests per saga

Negative

- Saga state management adds application complexity
- Compensation logic must be carefully designed to avoid orphaned state
- Requires HS-007 workshop for reservation failure policy

ADR-009: Stock Position Identity per Product-Location

Field	Detail
Decision ID	ADR-009
Title	Stock Position Identity per Product-Location
Status	Proposed

Context

Domain model review (P1-05) and hotspot HS-005 identify a fundamental fork: Stock Position identity as *product + location* with embedded Lot/Batch child entities, versus *product + location + lot* as separate positions. This decision affects reservation, pick, cycle count, transfer, and persistence design across all operational modules.

Decision

Adopt **one Stock Position aggregate per product-location**. Lot/Batch entities are **child entities** within the aggregate holding sub-quantities. ATP at root = sum of sellable lot quantities minus holds. Traceable products require lot capture on all movements (BR-017).

Pending confirmation at architecture gate workshop with Quality and Warehouse stakeholders.

Alternatives Considered

Alternative	Reason Not Selected
One Stock Position per product-location-lot	Explosion of aggregates; complex network ATP aggregation
Lot-less Stock Position for all products	Violates BR-017 for traceable products

Consequences

Positive

- Single ATP query point per SKU per location
- Lot quantities enforced to sum to on-hand invariant
- Aligns with FEFO pick selection within aggregate

Negative

- Larger aggregate for high-lot-count products — lock contention risk
- Pending business confirmation — **blocks detailed Balance module design until Accepted**

ADR-010: Location Freeze Authority Split

Field	Detail
Decision ID	ADR-010
Title	Location Freeze Authority Split
Status	Proposed

Context

BR-024 requires location freeze during wall-to-wall physical inventory count. Hotspot HS-003 and finding P1-03 identify dual ownership risk: Physical Inventory Count aggregate initiates freeze while Inventory Location aggregate also lists freeze enforcement. Two writers could cause race conditions at count certification.

Decision

Physical Inventory Count (M-09) initiates freeze via `LocationFreezeImposed` event. **Inventory Location (M-02) holds the authoritative frozen flag** and applies/releases state. All operational modules consult M-02 `IsLocationFrozen` query only — never M-09 directly.

Alternatives Considered

Alternative	Reason Not Selected
Freeze owned only by Physical Count module	Network module cannot enforce freeze for non-count scenarios
Freeze duplicated in both aggregates	Race conditions; inconsistent state

Consequences

Positive

- Single authoritative freeze read point (Freeze Guard DS-014)
- Clear event-driven handoff between accuracy and network modules

Negative

- Eventual consistency window between impose event and flag apply — must be sub-second

- Emergency override policy still requires business workshop

ADR-011: Promotional Ring-Fence via Domain Events

Field	Detail
Decision ID	ADR-011
Title	Promotional Ring-Fence via Domain Events
Status	Accepted

Context

Finding P1-02 and hotspot HS-004 identify dual ownership risk for promotional ATP segregation. Commercial Campaign and Inventory Balance both reference ring-fenced quantity. BR-029 requires ring-fencing during campaigns without corrupting general ATP.

Decision

M-11 Commercial Campaign owns allocation intent and lifecycle. M-04 Inventory Balance owns ATP mechanics. Campaign publishes `PromotionalStockAllocated` / `PromotionalAllocationReleased` events. Balance reacts by segregating/restoring ATP — **never self-initiates ring-fence**.

Alternatives Considered

Alternative	Reason Not Selected
Ring-fence logic only in Stock Position	Marketing lifecycle coupled to core domain
Shared mutable segregation table	Dual writers; inconsistent ATP

Consequences

Positive

- Clear ownership aligned to stakeholder functions (Marketing vs. Operations)
- Phase 2 campaign module independently evolvable

Negative

- Brief eventual consistency between allocation command and ATP segregation
- Override rules (MR-10) require business policy workshop

ADR-012: Anti-Corruption Layers for External Systems

Field	Detail
Decision ID	ADR-012
Title	Anti-Corruption Layers for External Systems
Status	Accepted

Context

IMS integrates with Sales/CRM (EXT-01), Finance ERP (EXT-02), Sales Channel (EXT-03), Marketing (EXT-07), and HR Directory (EXT-08). External systems have foreign domain models, lifecycles, and authority boundaries. BRD scope excludes full CRM and GL ownership from IMS. Absorbing foreign models corrupts fulfillment and balance domains.

Decision

Implement **Anti-Corruption Layers (ACL)** as infrastructure adapters at module boundaries:

External System	ACL Location	Translates To
Sales/CRM	M-07 Fulfillment	`ImportCustomerOrder` command
Finance ERP	M-13 Intelligence	Valuation snapshot export format
Finance ERP	M-06 Inbound	Match status (HS-014)
Marketing	M-11 Campaign	`CreatePromotionalAllocation`
HR Directory	M-12 Identity	`ProvisionUserAccount`

Foreign models **never enter the domain layer**.

Alternatives Considered

Alternative	Reason Not Selected
Shared canonical enterprise data model	Multi-year integration program; out of BC-01 scope
Direct CRM entity mapping to JPA	Couples fulfillment to CRM schema changes
Point-to-point API calls without ACL	Foreign concepts leak into application layer

Consequences

Positive

- IMS domain remains stable when CRM or ERP changes
- Phased integration depth (manual MVP → API Phase 2) without domain rewrite

Negative

- ACL maintenance overhead per external system
- Mapping logic requires tests and versioning

ADR-013: IMS as Operational System of Record

Field	Detail
Decision ID	ADR-013
Title	IMS as Operational System of Record
Status	Accepted

Context

Hotspot HS-014 questions whether IMS or Finance ERP owns supplier invoicing, three-way match, and valuation. BRD positions IMS as inventory system of record (TEC-01). Finance owns GL and payment authority. ~8% current valuation variance must reduce to ≤2% (BO-18).

Decision

IMS is the **operational system of record** for stock quantities, movements, reservations, receipt facts, and operational match outcomes. Finance ERP remains **financial system of record** for GL postings and payment authorization. IMS publishes valuation snapshots and movement facts via ACL; Finance publishes GL balance for reconciliation.

Alternatives Considered

Alternative	Reason Not Selected
ERP owns all financial inventory	IMS cannot enforce BR-004 – BR-008 at dock speed
IMS owns payment authorization	Outside BRD scope; duplicates ERP

Consequences

Positive

- Clear data ownership boundaries
- Warehouse operations not blocked by ERP availability

Negative

- Reconciliation workflow required (Phase 3 automation)
- Temporary manual finance export in MVP (TEC-05)

ADR-014: Java Spring Boot Backend Stack

Field	Detail
Decision ID	ADR-014
Title	Java Spring Boot Backend Stack
Status	Accepted

Context

Organization mandates **Java and Spring**. IMS requires enterprise-grade security, transaction management, REST APIs, and integration with PostgreSQL. Team hiring and supportability favor mainstream stack. Virtual threads in Java 21 benefit I/O-bound warehouse workloads.

Decision

Adopt **Java 21 LTS** with **Spring Boot 3.x** as the backend runtime — including Spring Web, Spring Security, Spring Data JPA, and Spring Modulith (ADR-015).

Alternatives Considered

Alternative	Reason Not Selected
Quarkus	Less Modulith maturity; not mandated
.NET	Diverges from Java mandate
Java EE/Jakarta EE standalone	More ceremony than Spring Boot

Consequences

Positive

- Mandate compliance; large hiring pool; mature ecosystem
- Spring Boot Test, Actuator, and security integration

Negative

- Heavier runtime than Quarkus — acceptable for enterprise deployment
- Requires discipline to avoid Spring MVC + JPA anemic pattern

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ADR-015: Spring Modulith for Module Boundary Enforcement

Field	Detail
Decision ID	ADR-015
Title	Spring Modulith for Module Boundary Enforcement
Status	Accepted

Context

ADR-001 modular monolith requires enforceable boundaries between 13 bounded contexts. Package naming alone is insufficient — ArchUnit helps but Spring Modulith provides first-class module APIs, event publication, JDBC event registry, and `ApplicationModuleTest` verification.

Decision

Use **Spring Modulith** to structure the modular monolith — module definitions, allowed dependencies, module-scoped events, and integration with transactional outbox.

Alternatives Considered

Alternative	Reason Not Selected
ArchUnit only	No built-in event/outbox support
OSGi	Obsolete complexity
Manual package conventions	Not compile-time verifiable

Consequences

Positive

- Architecture enforced in CI via module tests
- Event infrastructure aligned to DDD modules

Negative

- Newer Spring project — smaller community reference base
- Team training in Sprint 0 required

ADR-016: Angular SPA and PWA Frontend

Field	Detail
Decision ID	ADR-016
Title	Angular SPA and PWA Frontend
Status	Accepted

Context

Organization mandates **Angular**. IMS serves 19 user roles across office and warehouse with 90 functional requirements. TEC-03 excludes offline MVP. Warehouse staff need mobile-friendly interfaces without maintaining separate native apps under BC-01.

Decision

Build a **single Angular 19 SPA** with Angular Material, lazy-loaded feature modules per bounded context, and **PWA capabilities** for warehouse mobile (installable, responsive, barcode-ready).

Alternatives Considered

Alternative	Reason Not Selected
React	Not mandated
Separate native mobile apps	Cost and dual codebase
Server-rendered only (Thymeleaf)	Poor mobile UX for warehouse

Consequences

Positive

- One frontend codebase; mandate compliance; enterprise structure
- OpenAPI-generated TypeScript clients possible

Negative

- Large bundle — mitigated by lazy loading
- PWA limitations on specialized scanner hardware — Capacitor in Phase 2 if needed

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ADR-017: PostgreSQL as Primary Database

Field	Detail
Decision ID	ADR-017
Title	PostgreSQL as Primary Database
Status	Accepted

Context

Organization mandates **PostgreSQL**. IMS requires ACID transactions for Stock Position, JSONB for event payloads, row-level locking for reservation contention, 7-year retention (NFR-COM-019), and ≤15-minute RPO (NFR-BDR-001). Fixed budget favors open-source database.

Decision

Use **PostgreSQL 16** as the sole primary relational database for transactional writes, outbox, read projections, and audit storage.

Alternatives Considered

Alternative	Reason Not Selected
SQL Server / Oracle	Licensing cost (BC-01)
MySQL	Weaker JSON and constraint ecosystem
MongoDB	Poor fit for ATP transactional integrity

Consequences

Positive

- Mandate compliance; ACID; JSONB; logical replication for read scaling
- Testcontainers parity in CI

Negative

- Write scaling vertical — acceptable for MVP–Phase 2 scale
- DBA tuning (vacuum, indexes) required for ATP hot paths

ADR-018: Schema-per-Module Database Organization

Field	Detail
Decision ID	ADR-018
Title	Schema-per-Module Database Organization
Status	Accepted

Context

ADR-002 defines 13 modules with exclusive aggregate ownership. Module boundary rule MB-07 prohibits shared tables. Single PostgreSQL instance required for operational simplicity (ORC-04) and ACID within monolith.

Decision

Each module owns tables in a **dedicated PostgreSQL schema** (`catalog`, `balance`, `fulfillment`, etc.). Flyway migrations are per-module. **No cross-schema foreign keys** except shared kernel reference tables (`product_sku`, `location_code`).

Alternatives Considered

Alternative	Reason Not Selected
Single flat schema	No ownership enforcement
Database per module	Microservices ops overhead

Consequences

Positive

- Clear data ownership; extraction-ready
- Single backup and instance management

Negative

- Cross-schema queries discouraged — use module ports
- Migration coordination on shared kernel changes

ADR-019: JPA for Writes, jOOQ for Intelligence Reads

Field	Detail
Decision ID	ADR-019
Title	JPA for Writes, jOOQ for Intelligence Reads
Status	Accepted

Context

ADR-007 CQRS separates write and read paths. Write side requires aggregate mapping with optimistic locking. Intelligence module requires complex KPI aggregations, audit search, and dashboard queries that should not load JPA entity graphs (AR-10).

Decision

Use **Hibernate 6 / Spring Data JPA** for transactional module write repositories only. Use **jOOQ** exclusively in **M-13 Intelligence** for read projections and reports.

Alternatives Considered

Alternative	Reason Not Selected
JPA for all queries	Performance and coupling risk on reporting
jOOQ for all modules	Verbose aggregate persistence; loses optimistic locking conventions
MyBatis	Less Spring Boot native integration

Consequences

Positive

- Right tool per path; type-safe SQL for reports
- Domain layer remains ORM-free

Negative

- Two data access patterns — confined to clear module boundary
- jOOQ schema codegen pipeline required

ADR-020: OAuth 2.0 / OIDC with JWT Resource Server

Field	Detail
Decision ID	ADR-020
Title	OAuth 2.0 / OIDC with JWT Resource Server
Status	Accepted

Context

NFR-SEC-001 requires unique user authentication. NFR-SEC-005 – SEC-009 require RBAC and location-scoped permissions. Enterprise IdP (SVC-01) expected. Angular SPA requires token-based API auth. 19 roles and 20 SoD rules must be enforced.

Decision

Integrate **OAuth 2.0 / OpenID Connect** with enterprise IdP. Spring Security **JWT resource server** validates bearer tokens on every API request. Angular uses `angular-oauth2-oidc`. Authorization enforced in application layer via M-12 Identity ports.

Alternatives Considered

Alternative	Reason Not Selected
Session-only form login	No SSO; poor SPA scaling
SAML only	Heavier SPA integration than OIDC
API keys	No user attribution

Consequences

Positive

- Enterprise SSO; stateless API; horizontal scaling
- Mandate-aligned security posture

Negative

- Token revocation requires short TTL or blocklist on deprovision
- IdP role mapping to 100 IMS permissions requires governance

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ADR-021: Redis Cache for ATP Hot Path

Field	Detail
Decision ID	ADR-021
Title	Redis Cache for ATP Hot Path
Status	Accepted

Context

NFR-PERF-002 requires ≥98% of ATP lookups within 2 seconds. Stock Position is high-read, high-write contention aggregate. CS and sales conduct live availability inquiries during customer calls (BO-16). PostgreSQL alone may insufficient under peak concurrent reads.

Decision

Deploy **Redis** as distributed cache for ATP by SKU/location. Cache invalidated on **every Balance domain event** affecting quantities. Cached responses include `asOf` timestamp. Cache miss falls through to authoritative read repository — never default to zero.

Alternatives Considered

Alternative	Reason Not Selected
In-process Caffeine only	Invalid with multiple app instances
PostgreSQL materialized views only	Refresh lag; write coupling
No cache	Risk of NFR-PERF-002 breach at scale

Consequences

Positive

- Sub-millisecond reads for hot SKUs
- Event invalidation preserves BR-001 integrity

Negative

- Additional infrastructure component
- Staleness risk if invalidation fails — monitor and alert

ADR-022: In-Process Messaging for MVP

Field	Detail
Decision ID	ADR-022
Title	In-Process Messaging for MVP
Status	Accepted

Context

ADR-005 event-driven integration requires message transport. ORC-04 limits ops capacity. MVP modular monolith does not require cross-process messaging. External CRM and channel integration deferred to Phase 2 (TEC-04).

Decision

Use **Spring Application Events / Spring Modulith event API** with **transactional outbox** for MVP internal messaging. **Defer external message broker** (RabbitMQ recommended) to Phase 2 when CRM API and channel feeds require cross-process async integration.

Alternatives Considered

Alternative	Reason Not Selected
Kafka from day one	Ops overhead; overkill for monolith MVP
RabbitMQ from day one	Additional infra without external consumers
Synchronous only	Violates ADR-005; couples modules

Consequences

Positive

- Zero broker ops for MVP; simpler debugging
- Outbox schema ready for broker relay

Negative

- Cannot horizontally scale event consumers independently in MVP
- Phase 2 migration requires broker introduction and handler compatibility testing

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ADR-023: Published Language for Cross-Boundary References

Field	Detail
Decision ID	ADR-023
Title	Published Language for Cross-Boundary References
Status	Accepted

Context

13 modules and external systems must reference products, locations, users, and orders without sharing domain entity types. DDD strategic design requires identity-only references between aggregates and contexts.

Decision

Define a **shared kernel** of immutable identifier value types: `ProductSku`, `LocationCode`, `UserId`, `SupplierCode`, and standard event envelope (`EventId`, `CorrelationId`, `OccurredAt`). All cross-module and ACL integrations use published language only — **no foreign aggregate imports**.

Alternatives Considered

Alternative	Reason Not Selected
Shared entity library	Couples module lifecycles; lazy-loading leaks
Primitive strings only	No type safety; conflation of identifiers

Consequences

Positive

- Compile-time type safety for identifiers
- Modules evolve independently

Negative

- Shared kernel must remain minimal — resist feature creep
- Identifier format changes are organization-wide breaking changes

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ADR-024: Centralized Audit Trail Projection

Field	Detail
Decision ID	ADR-024
Title	Centralized Audit Trail Projection
Status	Accepted

Context

BO-19 requires 100% audit trail coverage. NFR-COM-010 requires immutable audit records. NFR review identified duplicate audit requirements across NFR-LOG, NFR-SEC, and NFR-COM. Per-module audit tables would fragment evidence and complicate 7-year retention.

Decision

Maintain a **single authoritative audit trail projection** in **M-13 Intelligence**, built from domain events and command audit context. Operational logs (NFR-LOG) remain separate diagnostic records. Auditor export via `ExportAuditEvidencePackage` query (NFR-PERF-026, NFR-COM-012).

Alternatives Considered

Alternative	Reason Not Selected
Per-module audit tables	Fragmented; inconsistent schema
Operational logs as audit	Insufficient structure and immutability
External SIEM only	Business fact context missing

Consequences

Positive

- Single query point for auditors
- Event-sourced rebuild capability

Negative

- Seconds-level lag on audit visibility — acceptable for compliance
- Intelligence module becomes critical for audit access

ADR-025: Architecture Fitness Functions with ArchUnit

Field	Detail
Decision ID	ADR-025
Title	Architecture Fitness Functions with ArchUnit
Status	Accepted

Context

Modular monolith erosion is the primary long-term risk (AR-09). ADR-003 layer rules and ADR-002 module boundaries must survive team turnover and delivery pressure across 18 months. Manual architecture review alone is insufficient.

Decision

Implement **ArchUnit** fitness functions in CI enforcing:

- Domain packages do not depend on infrastructure, application, or presentation

- Modules do not import other modules' domain packages
- JPA annotations absent from domain layer
- Controllers do not access repositories directly

Complement with **Spring Modulith** `ApplicationModuleTest`.

Alternatives Considered

Alternative	Reason Not Selected
Code review only	Inconsistent enforcement
jQAssistant	Heavier setup than needed

Consequences

Positive

- Automated regression on architecture rules
- Fast feedback on PRs

Negative

- Rules require maintenance as packages evolve
- False positives possible — tune carefully

ADR Governance

Process

Step	Action	Owner
1	Propose ADR via PR with context and alternatives	Any architect or tech lead
2	Review with DDD architect and affected squad leads	Solution Architect
3	Accept, reject, or mark Proposed pending workshop	Steering committee / ARB
4	Update status when superseded	Solution Architect
5	Link ADR in module README and relevant SRS docs	Author

Review Schedule

ADR	Next Review Trigger
ADR-009	HS-005 workshop completion
ADR-010	HS-003 workshop completion
ADR-022	Phase 2 external integration kickoff
ADR-001	30% transaction growth or multi-region deployment

Document Approval

Role	Name	Signature	Date
Senior Solution Architect			
DDD Solution Architect			
Product Owner			
Steering Committee Sponsor			

End of Architecture Decision Records

Document ID: SRS-SA-008 | Version 1.0 | July 6, 2026

Part VII — Database Design

Logical Data Model (ERD)

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-DB-001
Document Title	Logical Data Model — Entity Relationship Diagram
Version	1.0
Status	Draft
Document Owner	Senior Database Architect
Source Documents	SRS-DM-001 (Domain Entities), SRS-DM-003 (Domain Aggregates), SRS-DM-004 (Entity Relationships), SRS-BR-001 (Business Rules), SRS-ARCH (System Architecture)
Date	July 6, 2026
Scope	Logical business data model only — no SQL, physical storage, indexes, or database-vendor specifics

Purpose

This document defines the **Logical Data Model** for the Inventory Management System. It translates the approved domain model, business rules, and aggregate boundaries into a conceptual entity-relationship structure suitable for data architecture review and physical design planning.

The model contains **40 transactional and reference entities** aligned to domain entities, plus **4 derived/observability entities** modeled as logical projections. Relationships express business collaboration — not implementation join strategies.

Design Principles

Principle	Application
Domain fidelity	Entity names and semantics follow SRS-DM-008 ubiquitous language
Aggregate awareness	Each entity maps to a DDD aggregate root or child entity
Identity over navigation	Cross-aggregate links are logical references by business identity (SKU, location code, order number)
Invariant visibility	Business constraints from BR-001 – BR-032 are captured per entity
Logical only	No data types, keys, partitions, or persistence technology

Cardinality Notation

Notation	Meaning
`1`	Exactly one
`0..1`	Zero or one (optional)
`1..*`	One or more (mandatory, many allowed)
`0..*`	Zero or more
`*`	Many (zero or more)

Relationship Type Legend

Type	Meaning
Composition	Child lifecycle bound to parent
Association	Independent entities with ongoing business link
Reference	Identity lookup without ownership
Modification	Source operation changes target quantity or status
Derivation	Target produced from or computed from source
Authorization	Source gates or permits action on target
Verification	Source validated against target before proceeding

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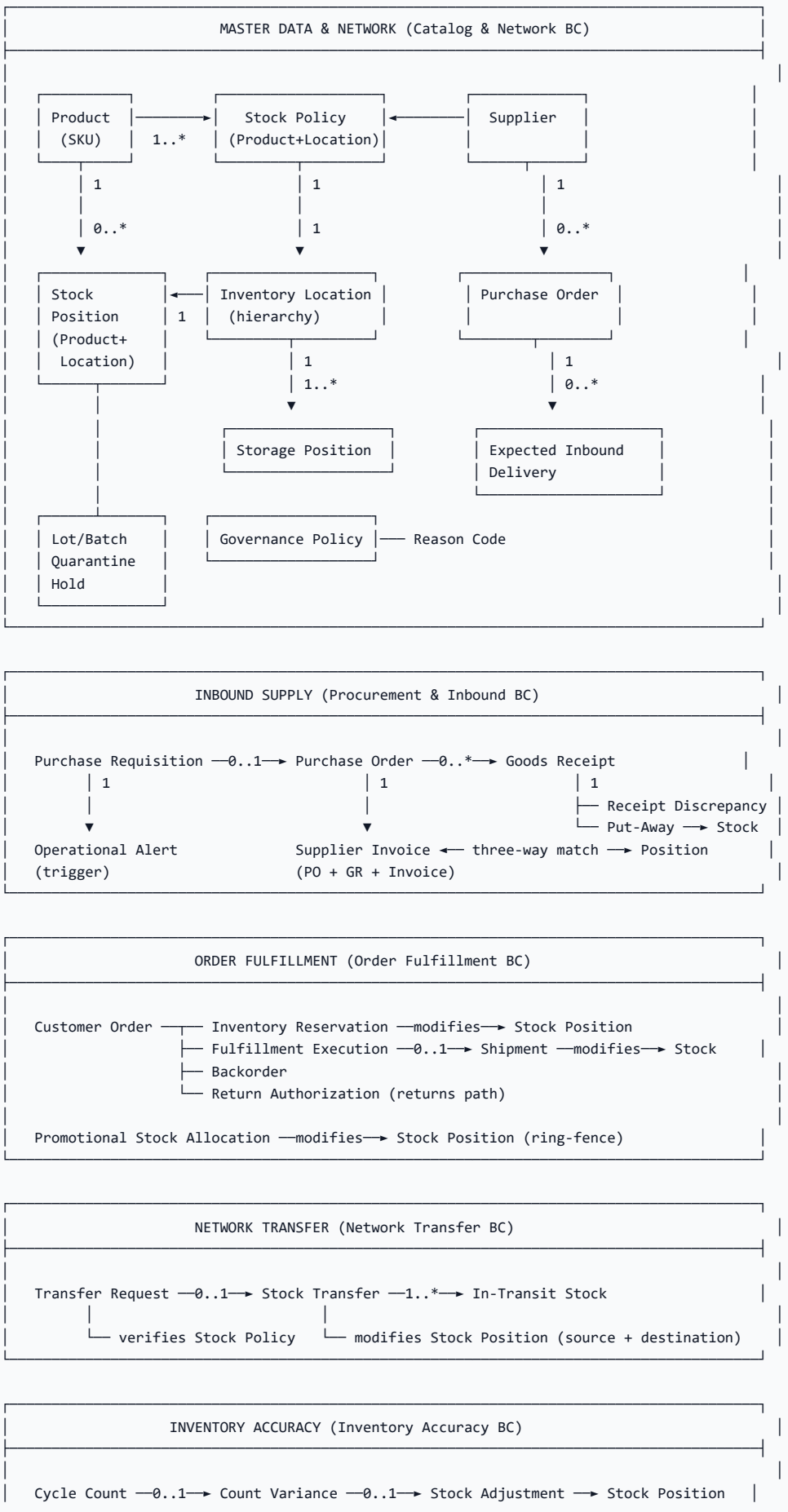
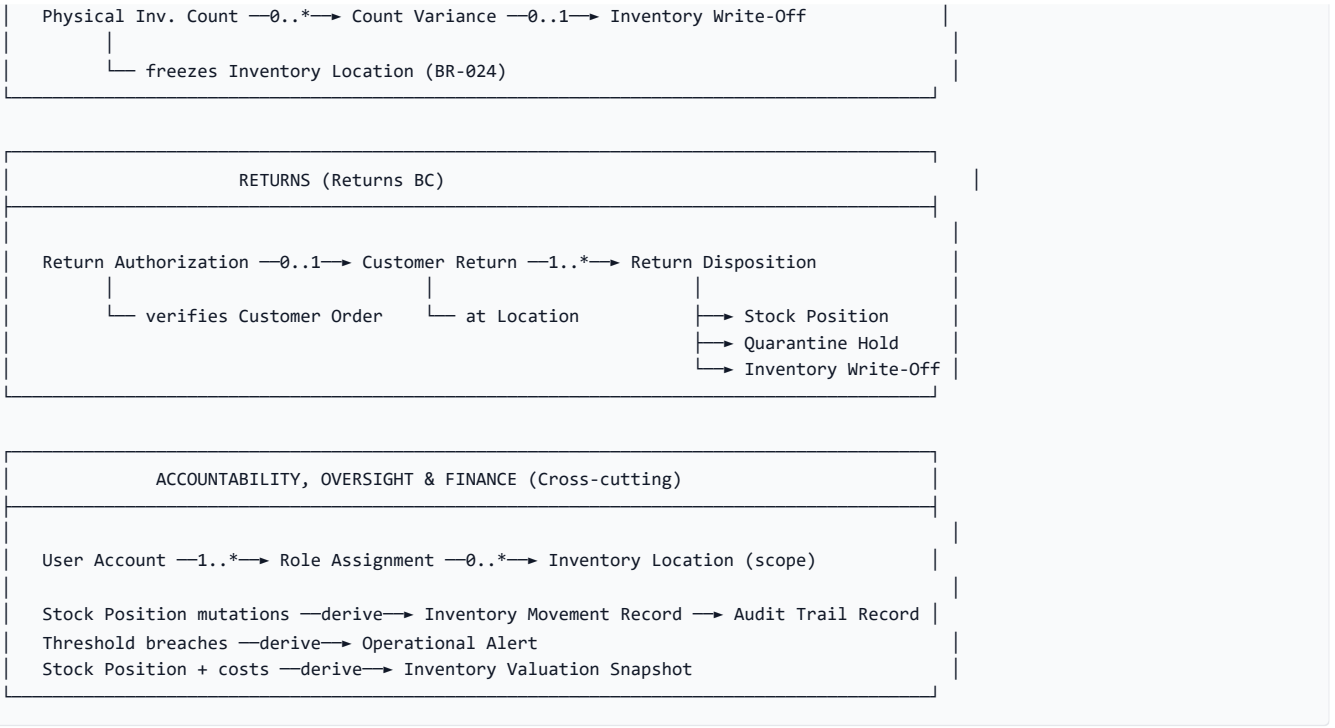
Aggregate-to-Entity Map

Aggregate	Aggregate Root	Child / Member Entities
Product Catalog	Product	—
Inventory Location	Inventory Location	Storage Position
Supplier	Supplier	—
Inventory Governance	Governance Policy	Reason Code
Stock Position	Stock Position	Lot/Batch, Quarantine Hold
Stock Policy	Stock Policy	—
Purchase Requisition	Purchase Requisition	—
Purchase Order	Purchase Order	Expected Inbound Delivery
Goods Receipt	Goods Receipt	Receipt Discrepancy, Put-Away
Supplier Invoice	Supplier Invoice	—
Customer Order Fulfillment	Customer Order	Inventory Reservation, Fulfillment Execution, Backorder, Shipment
Stock Transfer	Stock Transfer	In-Transit Stock *(Transfer Request is initial lifecycle state)*
Cycle Count	Cycle Count	—
Physical Inventory Count	Physical Inventory Count	—
Inventory Adjustment Case	Count Variance / Stock Adjustment	Stock Adjustment, Inventory Write-Off
Customer Return Case	Return Authorization	Customer Return, Return Disposition
Promotional Stock Allocation	Promotional Stock Allocation	—
User Account	User Account	Role Assignment

Aggregate	Aggregate Root	Child / Member Entities
(Derived — no aggregate)	—	Inventory Movement Record, Audit Trail Record, Operational Alert, Inventory Valuation Snapshot

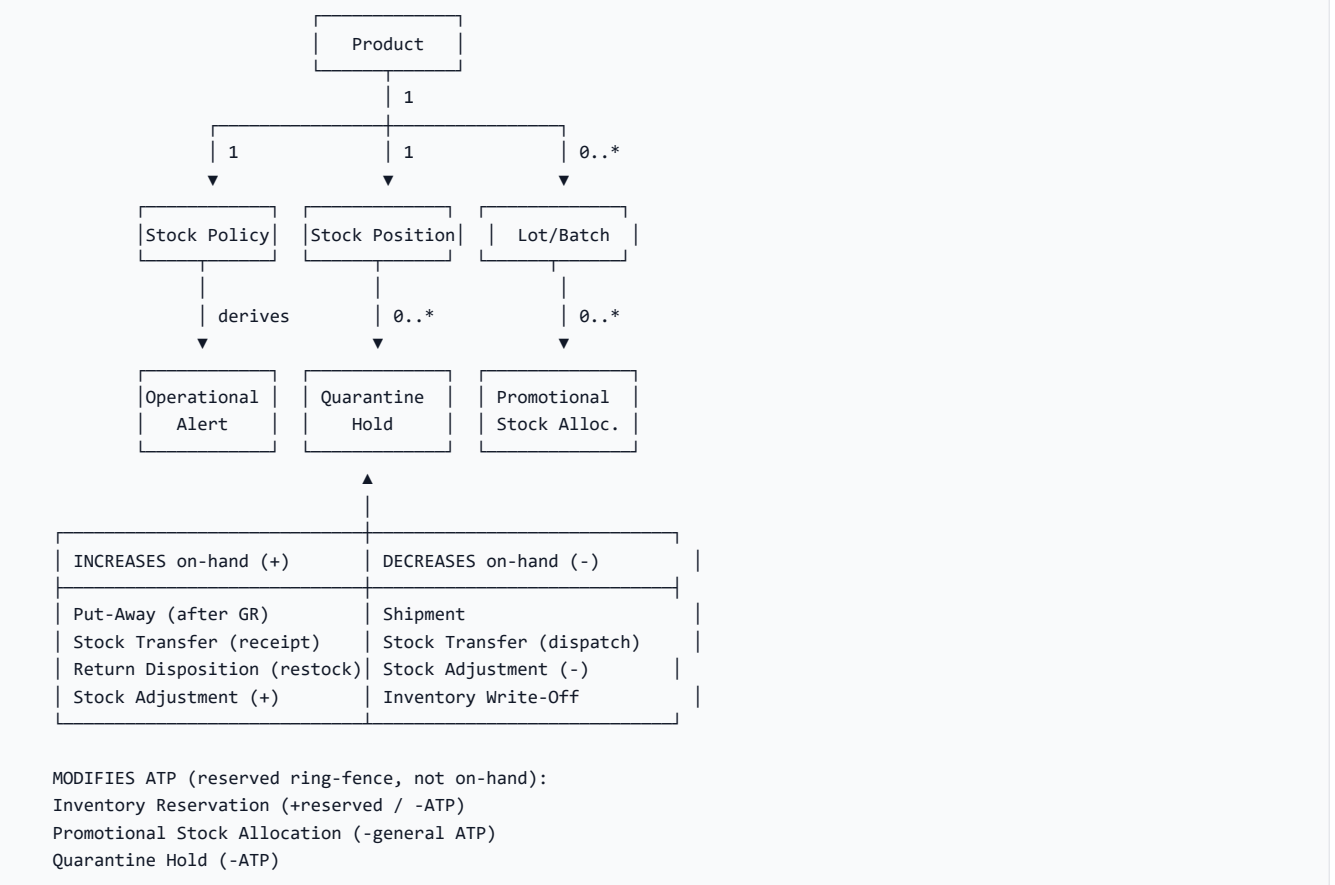
Text-Based ERD — System Overview





Text-Based ERD — Core Inventory Hub

Stock Position is the central quantity authority. This diagram emphasizes modification paths.



Entity Catalog

Entities are grouped by business domain. Each entry includes business purpose, relationships, cardinality, owning aggregate, and governing constraints.

Domain 1: Master Data & Reference

1. Product

Aspect	Detail
Business Purpose	Canonical stock-keeping unit (SKU) identity for all inventory operations — what the organization buys, holds, and sells across locations and channels.
Related Aggregate	Product Catalog *(root)*
Relationships	→ Stock Position (Association); → Stock Policy (Association); → Lot/Batch (Composition); referenced by procurement, fulfillment, transfer, count, and return line items
Cardinality	Product `1` → Stock Position `0..*`; Product `1` → Stock Policy `0..*`; Product `1` → Lot/Batch `0..*`
Important Business Constraints	SKU must be unique. Name and unit of measure are mandatory. Discontinued products must not appear on new POs or requisitions (BR-027). Traceable products must declare lot requirement and rotation policy (FIFO/FEFO). ABC classification must be A, B, or C when assigned (BR-032).

2. Inventory Location

Aspect	Detail
Business Purpose	Physical site in the distribution network where stock is received, stored, counted, and fulfilled.
Related Aggregate	Inventory Location *(root)*
Relationships	→ Storage Position (Composition); → Stock Position (Association); → Inventory Location parent (Hierarchy Association); → Physical Inventory Count (Association); referenced as receiving, fulfillment, and transfer endpoint
Cardinality	Location `1` → Storage Position `1..*`; Location `1` → Stock Position `0..*`; Child Location `0..1` → Parent Location `1`
Important Business Constraints	Location cannot be its own ancestor. Deactivated/closing locations must not accept new receipts, reservations, or fulfillments. Physical inventory freeze (BR-024) blocks movements while wall-to-wall count is active. Closure blocked while on-hand stock remains without disposition plan.

3. Storage Position

Aspect	Detail
Business Purpose	Designated storage area (zone, aisle, bin) within a location for put-away placement and pick path guidance.
Related Aggregate	Inventory Location *(child of Inventory Location aggregate)*
Relationships	← Inventory Location (Composition); → Put-Away (Association); → Fulfillment Execution (Reference); → Cycle Count (optional Reference)
Cardinality	Storage Position `1` → Inventory Location `1`; Storage Position `1` → Put-Away `0..*`; Put-Away `1` → Storage Position `1`
Important Business Constraints	Cannot exist outside parent location. Put-away must assign storage position before stock becomes available (BR-005). Rotation placement (FIFO/FEFO) applied at put-away (BR-014, BR-015). Put-away suspended during location freeze (BR-024).

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4. Stock Policy

Aspect	Detail
Business Purpose	Replenishment thresholds (reorder level, minimum, safety stock) for a product at a specific location.
Related Aggregate	Stock Policy *(root)*
Relationships	→ Product (Reference); → Inventory Location (Reference); → Operational Alert (Derivation); verified by Transfer Request
Cardinality	Stock Policy `1` → Product `1`; Stock Policy `1` → Inventory Location `1`; Product + Location uniquely identifies policy; Stock Policy `1` → Operational Alert `0..*`
Important Business Constraints	Reorder, minimum, and safety stock must be non-negative. Minimum must not exceed reorder level unless explicitly authorized. Replenishment triggers when ATP reaches reorder level (BR-009). Safety stock must not be bypassed without approval (BR-010). Source minimum constrains transfer quantity (BR-012).

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5. Reason Code

Aspect	Detail
Business Purpose	Standardized categorized explanation for adjustments, write-offs, and receipt exceptions.
Related Aggregate	Inventory Governance *(child of Governance Policy aggregate)*
Relationships	→ Stock Adjustment (Authorization); → Inventory Write-Off (Authorization); → Receipt Discrepancy (Association)
Cardinality	Reason Code `1` → Stock Adjustment `0..*`; Stock Adjustment `1` → Reason Code `1`; Write-Off `1` → Reason Code `1`
Important Business Constraints	Reason codes must be unique. Retired codes not selectable for new transactions. Mandatory on every adjustment and write-off (BR-020, BR-023).

---

6. Governance Policy

Aspect	Detail
Business Purpose	Organization-wide control policies — approval thresholds, tolerance limits, segregation-of-duties rules, and exception handling.
Related Aggregate	Inventory Governance *(root)*
Relationships	→ Stock Adjustment (Verification); → Count Variance (Verification); → Goods Receipt (Verification); contains Reason Code
Cardinality	Governance Policy `1` → Stock Adjustment `0..*`; Governance Policy `1` → Count Variance `0..*`; Governance Policy `1` → Goods Receipt `0..*`
Important Business Constraints	Only one effective policy version per parameter at any time. Adjustments above threshold require independent approval (BR-021). Counter must not approve own count variances (BR-019). PO required for supplier receipt except approved ad-hoc exception (BR-007).

Domain 2: Inventory State & Balance

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7. Stock Position

Aspect	Detail
Business Purpose	Authoritative inventory balance for a product at a location — on-hand, reserved, and available-to-promise (ATP).
Related Aggregate	Stock Position *(root)*
Relationships	→ Product, Inventory Location, Stock Policy (Reference); → Lot/Batch, Quarantine Hold (Composition); modified by Reservation, Shipment, GR/Put-Away, Transfer, Adjustment, Write-Off, Return Disposition, Promotional Allocation
Cardinality	Stock Position `1` → Product `1`; Stock Position `1` → Inventory Location `1`; Stock Position `1` → Stock Policy `0..1`; Stock Position `1` → Lot/Batch `0..*`; Stock Position `1` → Quarantine Hold `0..*`

Aspect	Detail
Important Business Constraints	On-hand never negative. ATP = on-hand − reserved − quarantined (BR-001, BR-013). ATP never exceeds on-hand. Consignment stock not owned until consumption (BR-031). All mutations produce Inventory Movement Record. Location freeze blocks changes except authorized count adjustments (BR-024).

8. Lot / Batch

Aspect	Detail
Business Purpose	Traceable production or receipt grouping enabling recall, FEFO rotation, and regulatory compliance.
Related Aggregate	Stock Position *(child when product is traceable)*
Relationships	← Product (Composition); ← Stock Position (Composition); → Supplier (Reference); referenced on GR, Put-Away, Fulfillment, Transfer, Quarantine Hold
Cardinality	Product `1` → Lot/Batch `0..*`; Stock Position `1` → Lot/Batch `0..*`; Lot/Batch `0..1` → Supplier `1`
Important Business Constraints	Lot identity captured at receipt and all movements for traceable products (BR-017). FEFO pick by expiration (BR-014). MRSL validated at receipt (BR-016).

9. Quarantine Hold

Aspect	Detail
Business Purpose	Inventory physically on-hand but excluded from sellable ATP due to quality, regulatory, recall, or customs hold.
Related Aggregate	Stock Position *(child)*
Relationships	← Stock Position (Composition); → Lot/Batch (Reference); derived from Return Disposition; may derive Adjustment or Write-Off on release
Cardinality	Stock Position `1` → Quarantine Hold `0..*`; Quarantine Hold `0..1` → Lot/Batch `1`
Important Business Constraints	Held quantity excluded from ATP (BR-013). Bonded inventory held until customs clearance (BR-030). Release restores ATP only after clearance confirmed.

10. Promotional Stock Allocation

Aspect	Detail
Business Purpose	Ring-fenced inventory for promotional campaigns or product launches, protected from general demand.
Related Aggregate	Promotional Stock Allocation *(root)*
Relationships	→ Product (Reference); → Inventory Location `1..*` (Reference); → Stock Position (Modification); → Operational Alert (Derivation)
Cardinality	Allocation `1` → Product `1`; Allocation `1` → Location `1..*`; Allocation `0..*` → Stock Position `1`
Important Business Constraints	Ring-fenced quantity must not serve non-promotional orders without commercial approval (BR-029). Allocation cannot exceed available stock at allocation time. Inactive outside campaign period. Unconsumed allocation released after campaign end.

Domain 3: Procurement & Inbound Supply

11. Supplier

Aspect	Detail
Business Purpose	External vendor — commercial counterparty for procurement, receipt, and payment.

Aspect	Detail
Related Aggregate	Supplier *(root)*
Relationships	→ Purchase Order (Association); → Supplier Invoice (Association); referenced by Goods Receipt and Lot/Batch
Cardinality	Supplier `1` → Purchase Order `0..*`; Supplier `1` → Supplier Invoice `0..*`
Important Business Constraints	Supplier code unique. Must be active for new POs. Consignment indicator governs ownership at receipt (BR-031). Payment terms required before PO issuance.

---

12. Purchase Requisition

Aspect	Detail
Business Purpose	Internal request to procure stock, preceding formal PO creation with approval workflow.
Related Aggregate	Purchase Requisition *(root)*
Relationships	→ Product `1..*` (Reference); → Inventory Location (Reference); → Purchase Order (Derivation); → Operational Alert (Reference)
Cardinality	Requisition `1` → Product `1..*`; Requisition `1` → Location `1`; Requisition `1` → Purchase Order `0..1`; Alert `0..1` → Requisition `0..1`
Important Business Constraints	Lines must reference active products — discontinued prohibited (BR-027). Network availability evaluated before approval (BR-011). Safety stock not implicitly bypassed (BR-010). Must be approved before PO conversion.

---

13. Purchase Order

Aspect	Detail
Business Purpose	Authorized commercial commitment to a supplier specifying products, quantities, prices, and delivery terms.
Related Aggregate	Purchase Order *(root)*
Relationships	→ Supplier, Inventory Location, Product (Reference); → Expected Inbound Delivery (Composition); → Goods Receipt (Authorization); → Supplier Invoice (Verification)
Cardinality	PO `1` → Supplier `1`; PO `1` → Location `1`; PO `1` → Product `1..*`; PO `1` → Expected Delivery `0..*`; PO `1` → Goods Receipt `0..*`
Important Business Constraints	Valid PO required before supplier receipt except approved exception (BR-007). Discontinued products prohibited (BR-027). Received qty per line within tolerance or discrepancy workflow. Status: open → partial → closed/cancelled.

---

14. Expected Inbound Delivery

Aspect	Detail
Business Purpose	Planned or confirmed supplier shipment communicating anticipated arrival for receiving preparation.
Related Aggregate	Purchase Order *(child)*
Relationships	→ Purchase Order (Reference); → Stock Position (Derivation — informational inbound expected); → Operational Alert (Derivation)
Cardinality	Expected Delivery `1` → Purchase Order `1`; Expected Delivery `0..*` → Stock Position `1`
Important Business Constraints	Expected quantities must not exceed remaining open PO line quantities. Inbound expected is informational only — does not increase ATP until put-away (BR-005). Delays trigger replenishment notifications.

---

15. Goods Receipt

Aspect	Detail
Business Purpose	Business event of accepting verified inbound goods into inventory control after dock verification.

Aspect	Detail
Related Aggregate	Goods Receipt *(root)*
Relationships	→ Purchase Order (Verification); → Supplier, Inventory Location (Reference); → Receipt Discrepancy, Put-Away (Composition); → Stock Position (Modification on put-away completion); → Lot/Batch (Derivation)
Cardinality	Goods Receipt `0..*` → Purchase Order `0..1`; GR `1` → Supplier `1`; GR `1` → Location `1`; GR `1` → Discrepancy `0..*`; GR `1` → Put-Away `1..*`
Important Business Constraints	Verify quantity, product, condition before acceptance (BR-004). PO required except exception (BR-007). Discrepancies documented before acceptance (BR-006). MRSL met for shelf-life products (BR-016). Lot captured for traceable products (BR-017). Receipt blocked during location freeze (BR-024).

16. Receipt Discrepancy

Aspect	Detail
Business Purpose	Documented difference between expected and actual receipt (short, over, wrong item, damage, shelf-life failure).
Related Aggregate	Goods Receipt *(child)*
Relationships	→ Goods Receipt (Composition); → Reason Code (Association)
Cardinality	Discrepancy `1` → Goods Receipt `1`; Goods Receipt `1` → Discrepancy `0..*`
Important Business Constraints	Must be recorded before acceptance of affected quantities (BR-006). Unresolved discrepancies affect three-way match (BR-008). Shelf-life non-compliance recorded as discrepancy (BR-016).

17. Put-Away

Aspect	Detail
Business Purpose	Confirmation that received goods are moved from receiving area to storage position and are operationally available.
Related Aggregate	Goods Receipt *(child)*
Relationships	→ Goods Receipt (Reference); → Storage Position (Association); → Stock Position (Modification — triggers on-hand/ATP increase)
Cardinality	Put-Away `1` → Goods Receipt `1`; Put-Away `1` → Storage Position `1`
Important Business Constraints	ATP not increased until put-away confirmed (BR-005). Rotation placement applied (BR-014, BR-015). Bonded goods to hold storage until customs clearance (BR-030).

18. Supplier Invoice

Aspect	Detail
Business Purpose	Supplier billing document — financial leg of three-way match against PO and goods receipt.
Related Aggregate	Supplier Invoice *(root)*
Relationships	→ Supplier, Purchase Order (Reference); → Goods Receipt (Verification); → Inventory Valuation Snapshot (Derivation)
Cardinality	Invoice `1` → Supplier `1`; Invoice `1` → Purchase Order `1`; Invoice `1` → Goods Receipt `0..*`
Important Business Constraints	Three-way match within tolerance before payment (BR-008). Payment prohibited for unordered, unreceived, or mispriced goods. Match exceptions require resolution or waiver.

Domain 4: Order Fulfillment

19. Customer Order

Aspect	Detail
Business Purpose	Customer demand triggering availability confirmation, reservation, picking, and shipment.
Related Aggregate	Customer Order Fulfillment *(root)*
Relationships	→ Stock Position (Reference for ATP); → Inventory Reservation, Fulfillment Execution, Backorder, Shipment (Composition); → Return Authorization (Association)
Cardinality	Order `1` → Reservation `0..*`; Order `1` → Fulfillment `0..*`; Order `1` → Backorder `0..*`; Order `1` → Shipment `0..*`; Order `1` → Stock Position `1..*`
Important Business Constraints	Commit only against ATP, not total on-hand (BR-001). Reserve at confirmation (BR-002). Respect promotional ring-fence (BR-029). Order line reconciliation: requested = reserved + fulfilled + backordered + cancelled.

---

20. Inventory Reservation

Aspect	Detail
Business Purpose	Operational hold on stock for a confirmed order line, reducing ATP until fulfillment or release.
Related Aggregate	Customer Order Fulfillment *(child)*
Relationships	→ Customer Order (Reference); → Product, Inventory Location (Reference); → Stock Position (Modification); consumed by Shipment
Cardinality	Reservation `1` → Customer Order `1`; Reservation `1` → Product `1`; Reservation `0..*` → Stock Position `1`
Important Business Constraints	Created at order confirmation (BR-002). Must not consume below safety stock without exception (BR-010). Promotional ring-fence respected (BR-029). Released on cancellation; consumed on shipment.

---

21. Fulfillment Execution

Aspect	Detail
Business Purpose	Warehouse pick, verify, and pack work bridging reservation to shipment.
Related Aggregate	Customer Order Fulfillment *(child)*
Relationships	→ Customer Order (Reference); → Inventory Location, Storage Position, Lot/Batch (Reference); → Shipment (Derivation)
Cardinality	Fulfillment `1` → Customer Order `1`; Fulfillment `1` → Location `1`; Fulfillment `1` → Shipment `0..1`
Important Business Constraints	Verified pick required before shipment (BR-003). FEFO/FIFO applied during pick (BR-014, BR-015). Lot captured on pick for traceable products (BR-017). Blocked at frozen location (BR-024).

---

22. Backorder

Aspect	Detail
Business Purpose	Customer-agreed delay for unfulfilled order line quantity when full ATP unavailable.
Related Aggregate	Customer Order Fulfillment *(child)*
Relationships	→ Customer Order (Reference); → Product (Reference)
Cardinality	Backorder `1` → Customer Order `1`; Backorder `1` → Product `1`
Important Business Constraints	Explicit customer acknowledgment required before recording (BR-028). May convert to reservation when stock arrives. Replenishment priority may apply.

---

23. Shipment

Aspect	Detail
Business Purpose	Confirmed dispatch of fulfilled goods — inventory deduction and delivery promise activation.
Related Aggregate	Customer Order Fulfillment *(child)*
Relationships	→ Customer Order, Fulfillment Execution (Reference); → Inventory Reservation (Modification); → Stock Position (Modification)
Cardinality	Shipment `1` → Fulfillment Execution `1`; Shipment `0..*` → Reservation `1`; Shipment `0..*` → Stock Position `1`
Important Business Constraints	Only after verified pick (BR-003). Deducts on-hand and consumes reservation (BR-002). Creates outbound movement record.

Domain 5: Network Balance

24. Transfer Request

Aspect	Detail
Business Purpose	Formal request to move stock between locations to resolve shortage or rebalance network before new procurement.
Related Aggregate	Stock Transfer *(initial lifecycle state within Stock Transfer aggregate)*
Relationships	→ Inventory Location source and destination (Reference); → Product `1..*` (Reference); → Stock Policy source (Verification); → Stock Transfer (Derivation)
Cardinality	Request `1` → Source Location `1`; Request `1` → Destination Location `1`; Request `1` → Product `1..*`; Request `1` → Stock Transfer `0..1`
Important Business Constraints	Must not deplete source below minimum (BR-012). Network check before purchase alternative (BR-011). Safety stock at source protected (BR-010). Source and destination must be distinct active locations.

25. Stock Transfer

Aspect	Detail
Business Purpose	Executed inter-location stock movement from dispatch through destination receipt.
Related Aggregate	Stock Transfer *(root)*
Relationships	→ Transfer Request (Reference); → In-Transit Stock (Composition); → Stock Position source and destination (Modification); → Lot/Batch (Reference)
Cardinality	Transfer `0..1` → Transfer Request `1`; Transfer `1` → In-Transit Stock `1..*`; Transfer `1` → Stock Position `2` per product line
Important Business Constraints	Source not depleted below minimum (BR-012). Lot captured for traceable products (BR-017). Blocked during location freeze (BR-024). In-transit excluded from both endpoints' ATP until receipt.

26. In-Transit Stock

Aspect	Detail
Business Purpose	Pipeline inventory that left source but is not yet received at destination.
Related Aggregate	Stock Transfer *(child)*
Relationships	→ Stock Transfer (Composition); references Product, source and destination locations
Cardinality	In-Transit `1` → Stock Transfer `1`; Transfer `1` → In-Transit `1..*`
Important Business Constraints	Excluded from source and destination ATP until receipt confirmed. Delayed/lost in-transit triggers exception alerts.

Domain 6: Inventory Control & Accuracy

27. Cycle Count

Aspect	Detail
Business Purpose	Scheduled partial physical count on rotating basis without full warehouse shutdown.
Related Aggregate	Cycle Count *(root)*
Relationships	→ Product, Inventory Location (Reference); → Storage Position (optional Reference); → Count Variance (Derivation); → User Account counter (Reference)
Cardinality	Cycle Count `1` → Product `1`; Cycle Count `1` → Location `1`; Cycle Count `0..1` → Storage Position `1`; Cycle Count `1` → Count Variance `0..1`
Important Business Constraints	Blind count — counter must not see system quantity (BR-018). Class A highest frequency (BR-032). Does not directly mutate Stock Position — spawns Adjustment Case on variance. Counter ≠ approver (BR-019).

28. Physical Inventory Count

Aspect	Detail
Business Purpose	Comprehensive wall-to-wall count at a location for audit-mandated accuracy baseline.
Related Aggregate	Physical Inventory Count *(root)*
Relationships	→ Inventory Location (Association); → Count Variance `0..*` (Derivation)
Cardinality	Physical Count `1` → Location `1`; Location `1` → Physical Count `0..*`; Physical Count `1` → Count Variance `0..*`
Important Business Constraints	Location freeze during count — no receiving, shipping, transfers, or adjustments (BR-024). Full location scope required. Certification requires variances routed to adjustment cases.

29. Count Variance

Aspect	Detail
Business Purpose	Identified difference between system and physical quantity requiring investigation before correction.
Related Aggregate	Inventory Adjustment Case *(root when originating from count)*
Relationships	→ Cycle Count or Physical Inventory Count (Reference); → Stock Position (Reference); → Stock Adjustment (Derivation); → User Account approver (Reference); verified by Governance Policy
Cardinality	Variance `1` → exactly one count origin; Variance `1` → Stock Position `1`; Variance `1` → Stock Adjustment `0..1`
Important Business Constraints	Investigate before adjusting (BR-022). Counter must not approve own variance (BR-019). Independent approval above threshold (BR-021).

30. Stock Adjustment

Aspect	Detail
Business Purpose	Corrective change to recorded inventory quantity with mandatory reason and authorization.
Related Aggregate	Inventory Adjustment Case *(child)*
Relationships	→ Stock Position (Modification); → Count Variance (optional Reference); → Reason Code (Reference); → User Account initiator and approver (Reference); verified by Governance Policy
Cardinality	Adjustment `0..1` → Count Variance `1`; Adjustment `1` → Reason Code `1`; Adjustment `0..*` → Stock Position `1`
Important Business Constraints	Mandatory reason code (BR-020). Independent approval above threshold (BR-021). Investigate when from count (BR-022). Must not drive stock negative.

31. Inventory Write-Off

Aspect	Detail
Business Purpose	Formal authorized permanent removal of inventory value for damaged, obsolete, expired, or lost goods.
Related Aggregate	Inventory Adjustment Case *(child)*
Relationships	→ Stock Position (Modification); → Reason Code (Reference); → Return Disposition (Reference); → User Account authorizer (Authorization)
Cardinality	Write-Off `0..*` → Stock Position `1`; Write-Off `1` → Reason Code `1`; Disposition `0..1` → Write-Off `1`
Important Business Constraints	Documented evidence and authorization required (BR-023). Mandatory reason code (BR-020). Terminal — quantity permanently removed.

Domain 7: Returns Management

32. Return Authorization

Aspect	Detail
Business Purpose	Approved permission (RMA) for customer return linked to original order before warehouse acceptance.
Related Aggregate	Customer Return Case *(root)*
Relationships	→ Customer Order (Verification); → Product `1..*` (Reference); → Customer Return (Authorization)
Cardinality	RMA `1` → Customer Order `1`; RMA `1` → Product `1..*`; RMA `1` → Customer Return `0..1`
Important Business Constraints	Required before warehouse acceptance (BR-025). Must link to original order. RMA expiration enforced. Returned products must match authorized lines.

33. Customer Return

Aspect	Detail
Business Purpose	Physical receipt of customer-returned goods at warehouse, initiating inspection workflow.
Related Aggregate	Customer Return Case *(child)*
Relationships	→ Return Authorization (Verification); → Inventory Location (Reference); → Return Disposition (Composition)
Cardinality	Return `1` → RMA `1`; Return `1` → Location `1`; Return `1` → Disposition `1..*`
Important Business Constraints	No receipt without valid RMA (BR-025). Verify quantity and product on receipt (BR-004). Inspection mandatory before disposition (BR-026).

34. Return Disposition

Aspect	Detail
Business Purpose	Business decision on returned goods — restock, quarantine, or write-off based on condition.
Related Aggregate	Customer Return Case *(child)*
Relationships	→ Customer Return (Reference); → Stock Position (Modification on restock); → Quarantine Hold, Inventory Write-Off (Derivation)
Cardinality	Disposition `1` → Customer Return `1`; Disposition `0..1` → Stock Position `1`; Disposition `0..1` → Quarantine Hold `0..1`; Disposition `0..1` → Write-Off `1`
Important Business Constraints	Inspect before restock (BR-026). Restock only when condition confirms resale quality. Damaged goods to quarantine or write-off — never direct to ATP.

Domain 8: Accountability & Oversight

35. Inventory Movement Record

Aspect	Detail
Business Purpose	Immutable documented change in inventory position — operational history of every stock activity.
Related Aggregate	*(Derived — append-only event log; not an aggregate consistency boundary)*
Relationships	→ Stock Position (Association); → originating business document (Reference); → User Account (Reference); → Lot/Batch, Reason Code (optional Reference)
Cardinality	Movement `0..*` → Stock Position `1`; Movement `1` → originating document `1`; Movement `1` → User Account `1`
Important Business Constraints	Every stock mutation must produce a movement record. Lot on movement for traceable products (BR-017). Reason code on adjustment movements (BR-020). Immutable after creation.

36. Audit Trail Record

Aspect	Detail
Business Purpose	Governance-grade record of material changes with user identity, timestamp, and before/after state.
Related Aggregate	*(Derived — compliance log; not an aggregate consistency boundary)*
Relationships	→ governed entity (Association); → Inventory Movement Record (Association); → User Account (Reference)
Cardinality	Audit `0..*` → governed entity `1`; Audit `0..1` → Movement Record `1`; Audit `1` → User Account `1`
Important Business Constraints	Supports segregation-of-duties verification (BR-019, BR-021). Immutable after write. Material changes require both movement and audit records.

37. Operational Alert

Aspect	Detail
Business Purpose	Proactive notification of conditions requiring human review — low stock, pending approval, exceptions.
Related Aggregate	*(Derived — read model / notification projection)*
Relationships	derived from Stock Position + Stock Policy; Transfer Request; Promotional Allocation; Expected Inbound Delivery
Cardinality	Alert `0..*` → Stock Position `1`; Alert `0..*` → Transfer Request `0..1`; Alert `0..*` → Promotional Allocation `0..1`
Important Business Constraints	Reorder-level breach triggers same-day replenishment review (BR-009). Class A products generate higher-priority alerts (BR-032). Campaign readiness alerts before launch commitment (BR-029).

38. User Account

Aspect	Detail
Business Purpose	Identified individual authorized to perform inventory operations within role and location scope.
Related Aggregate	User Account *(root)*
Relationships	→ Role Assignment (Composition); associated with Cycle Count, Goods Receipt, Adjustments, Movement and Audit records
Cardinality	User Account `1` → Role Assignment `1..*`
Important Business Constraints	User identifier unique. At least one role required to transact. Conflicting duty combinations blocked — counter + approver (BR-019, BR-021). Deactivated accounts cannot perform actions.

39. Role Assignment

Aspect	Detail
Business Purpose	Delegation of business permissions to a user with location scope and segregation-of-duties enforcement.
Related Aggregate	User Account *(child)*
Relationships	→ User Account (Composition); → Inventory Location `0..*` (Reference for scope)
Cardinality	Role Assignment `1` → User Account `1`; Role Assignment `1` → Location `0..*`
Important Business Constraints	SoD conflicts blocked at assignment. Location scope enforced on transactions. Effective-dated with history retained. Empty scope may mean network-wide for executive roles.

Domain 9: Financial Alignment

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40. Inventory Valuation Snapshot

Aspect	Detail
Business Purpose	Point-in-time inventory value summary by product and location for finance reconciliation and month-end close.
Related Aggregate	*(Derived — reporting read model; rebuilt from Stock Position events and matched costs)*
Relationships	derived from Stock Position, Product, Inventory Location; references Supplier Invoice costs, Inventory Write-Off
Cardinality	Snapshot `1` → Stock Position `0..*`; Snapshot `1` → Product `0..*`; Snapshot `1` → Location `0..*`
Important Business Constraints	Consignment stock excluded from owned valuation (BR-031). Quarantined and in-transit stock excluded per policy. Three-way matched invoice costs authoritative for received inventory (BR-008).

Relationship Summary Matrix

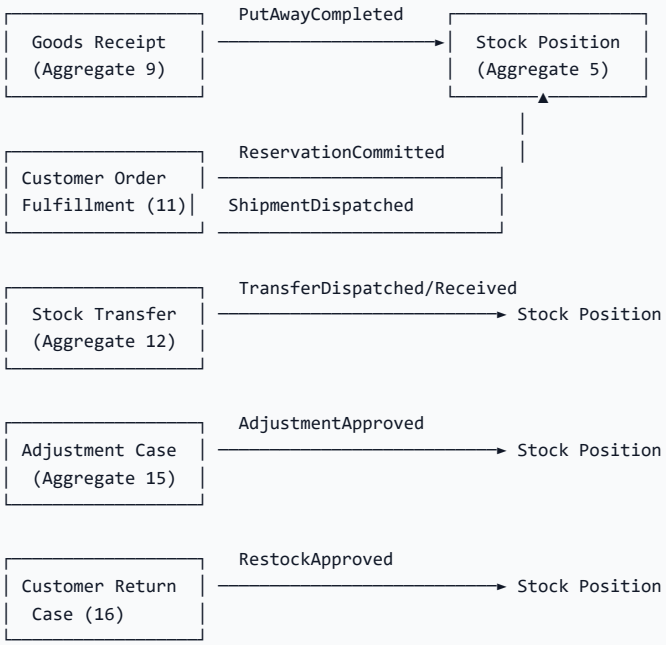
From Entity	To Entity	Type	Cardinality	Governing Rules
Product	Stock Position	Association	1 : 0..*	BR-001, BR-027, BR-017
Product	Stock Policy	Association	1 : 0..*	BR-009, BR-032
Inventory Location	Storage Position	Composition	1 : 1..*	BR-005, BR-024
Inventory Location	Stock Position	Association	1 : 0..*	BR-012, BR-024
Stock Position	Quarantine Hold	Composition	1 : 0..*	BR-013, BR-030
Stock Position	Lot/Batch	Composition	1 : 0..*	BR-017, BR-014
Inventory Reservation	Stock Position	Modification	0..* : 1	BR-002, BR-010, BR-029
Goods Receipt	Put-Away	Composition	1 : 1..*	BR-005
Put-Away	Stock Position	Modification	0..* : 1	BR-005
Purchase Order	Goods Receipt	Authorization	1 : 0..*	BR-007, BR-004
Customer Order	Inventory Reservation	Composition	1 : 0..*	BR-002
Fulfillment Execution	Shipment	Derivation	1 : 0..1	BR-003
Shipment	Stock Position	Modification	0..* : 1	BR-003
Transfer Request	Stock Transfer	Derivation	1 : 0..1	BR-012
Stock Transfer	In-Transit Stock	Composition	1 : 1..*	—
Cycle Count	Count Variance	Derivation	1 : 0..1	BR-018, BR-022
Count Variance	Stock Adjustment	Derivation	1 : 0..1	BR-019, BR-021, BR-022

From Entity	To Entity	Type	Cardinality	Governing Rules
Return Authorization	Customer Return	Authorization	1 : 0..1	BR-025
Return Disposition	Stock Position	Modification	0..1 : 1	BR-026
Supplier Invoice	Purchase Order + Goods Receipt	Verification	1 : 1 + 0..*	BR-008
User Account	Role Assignment	Composition	1 : 1..*	BR-019, BR-021

Full relationship catalog: SRS-DM-004 (REL-001 – REL-134).

Cross-Aggregate Reference Model

Aggregates coordinate through **identity references** and **domain events** — not shared mutable structures.



Interaction Rule	Description
One transaction, one aggregate	Stock deduction and PO creation never share a consistency boundary
Reference by identity	Cross-aggregate links use SKU, location code, order number — not object graphs
Consult, don't couple	Transfer consults Stock Policy and Stock Position by identity for source minimum
Governance consultation	Adjustment aggregates consult Inventory Governance for thresholds and reason codes
Catalog validation	Procurement and fulfillment consult Product Catalog for lifecycle status

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Entity Integrity Rules (Logical Level)

Rule ID	Rule	Entities Affected
LDM-01	ATP = on-hand − reserved − quarantined; ATP ≤ on-hand	Stock Position, Quarantine Hold, Inventory Reservation
LDM-02	Stock availability increases only after Put-Away completion	Goods Receipt, Put-Away, Stock Position
LDM-03	Shipment decreases on-hand only after verified Fulfillment Execution	Fulfillment Execution, Shipment, Stock Position

Rule ID	Rule	Entities Affected
LDM-04	Product + Location uniquely identifies Stock Position and Stock Policy	Stock Position, Stock Policy
LDM-05	Discontinued Product prohibited on new PO, Requisition, and Reservation lines	Product, Purchase Order, Purchase Requisition, Inventory Reservation
LDM-06	Counter User Account ≠ Approver User Account on same Count Variance	Cycle Count, Count Variance, User Account
LDM-07	Customer Return requires valid Return Authorization	Return Authorization, Customer Return
LDM-08	In-transit quantity excluded from both endpoint Stock Positions' ATP	In-Transit Stock, Stock Transfer, Stock Position
LDM-09	Physical Inventory Count imposes freeze on all Inventory Location movements	Physical Inventory Count, Inventory Location
LDM-10	Every Stock Position mutation produces Inventory Movement Record	Stock Position, Inventory Movement Record

Bounded Context Data Ownership

Bounded Context	Owning Entities	Consuming Contexts (read by identity)
Catalog & Network	Product, Inventory Location, Storage Position, Supplier, Governance Policy, Reason Code	All operational contexts
Inventory Balance	Stock Position, Lot/Batch, Quarantine Hold, Stock Policy	Fulfillment, Inbound, Transfer, Accuracy, Returns, Commercial
Procurement & Inbound	Purchase Requisition, Purchase Order, Expected Inbound Delivery, Goods Receipt, Receipt Discrepancy, Put-Away, Supplier Invoice	Inventory Balance, Financial
Order Fulfillment	Customer Order, Inventory Reservation, Fulfillment Execution, Backorder, Shipment	Inventory Balance, Returns
Network Transfer	Transfer Request, Stock Transfer, In-Transit Stock	Inventory Balance
Inventory Accuracy	Cycle Count, Physical Inventory Count, Count Variance, Stock Adjustment, Inventory Write-Off	Inventory Balance, Catalog & Network
Returns	Return Authorization, Customer Return, Return Disposition	Order Fulfillment, Inventory Balance, Inventory Accuracy
Commercial	Promotional Stock Allocation	Inventory Balance, Order Fulfillment
Identity & Access	User Account, Role Assignment	All contexts
Intelligence & Reporting	Inventory Movement Record, Audit Trail Record, Operational Alert, Inventory Valuation Snapshot	All contexts (derived)

Document Approval

Role	Name	Signature	Date
Senior Database Architect			
DDD Solution Architect			
Solution Architect			
Inventory Controller			
Product Owner			

End of Logical Data Model

Physical Data Model (PDM)

Inventory Management System — PostgreSQL

Document Information

Attribute	Detail
Document ID	SRS-DB-002
Document Title	Physical Data Model — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Database Architect
Source Document	SRS-DB-001 (Logical Data Model, v1.0)
Target Platform	PostgreSQL 15+
Date	July 6, 2026
Scope	Physical table design — no DDL/SQL statements

Purpose

This document converts the approved Logical Data Model (SRS-DB-001) into a **Physical Data Model** for PostgreSQL. It defines tables, columns, data types, nullability, defaults, and growth characteristics following PostgreSQL best practices.

Physical Design Conventions

Convention	Standard
Naming	`snake_case` for schemas, tables, and columns
Primary keys	`BIGINT` surrogate `id`, `GENERATED ALWAYS AS IDENTITY`
Business keys	Natural identifiers (`sku`, `location_code`, `order_number`) as `UNIQUE` alternate keys
Quantities	`NUMERIC(18,4)` — never `FLOAT/REAL`
Monetary amounts	`NUMERIC(18,2)`
Timestamps	`TIMESTAMPTZ` (UTC storage)
Dates	`DATE` for calendar dates without time
Status / type codes	`VARCHAR(32)` with application-enforced domain values
Descriptions / notes	`TEXT`
Flexible snapshots	`JSONB` for audit before/after state only
Optimistic locking	`version INTEGER NOT NULL DEFAULT 1` on mutable transactional headers
Audit columns	`created_at`, `updated_at` on all mutable tables; `created_by` where attribution required
Soft delete	`is_active BOOLEAN` on master data — no hard delete of reference rows
Referential integrity	Foreign keys within same bounded context; cross-context references by business key or loose FK with deferred validation
High-volume tables	Time-based range partitioning (monthly) on `movement_timestamp` / `event_timestamp`
Indexing strategy	B-tree on FKs and status filters; composite indexes on `(product_id, location_id)` hot paths

Schema Layout

Schema	Purpose
`catalog`	Product master data
`network`	Locations and storage positions
`governance`	Policies and reason codes
`inventory`	Stock balances, lots, holds, policies
`procurement`	Suppliers, requisitions, purchase orders
`inbound`	Goods receipt, discrepancies, put-away
`finance`	Supplier invoices and three-way match
`fulfillment`	Customer orders, reservations, picks, shipments
`transfer`	Inter-location transfers
`accuracy`	Cycle counts, physical counts, adjustments, write-offs
`returns`	RMA, customer returns, dispositions
`commercial`	Promotional stock allocations
`identity`	Users and role assignments
`audit`	Movement log, audit trail, alerts, valuation snapshots

Growth Tier Legend

Tier	Annual Row Estimate (mid-size deployment)	Strategy
Low	< 50K	Standard B-tree indexes
Medium	50K – 5M	Composite indexes; archive closed documents > 3 years
High	5M – 50M	Partitioning; read replicas for reporting
Very High	> 50M	Monthly partitions; cold storage archive; BRIN on timestamps

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Domain 1: Catalog (`catalog` schema)

Table: `catalog.products`

Attribute	Detail
Business Purpose	Canonical SKU master — identity for all inventory operations across locations and channels.
Estimated Growth	**Low** — ~10K–100K SKUs; stable with slow churn.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`sku`	`VARCHAR(64)`	NO	—	Business key; stock keeping unit
`product_name`	`VARCHAR(256)`	NO	—	Display name
`description`	`TEXT`	YES	NULL	Extended description
`unit_of_measure`	`VARCHAR(16)`	NO	—	EA, CS, KG, etc.
`category_code`	`VARCHAR(64)`	YES	NULL	Product category reference
`lifecycle_status`	`VARCHAR(32)`	NO	`'ACTIVE'`	ACTIVE, INACTIVE, DISCONTINUED

Column	Data Type	Nullable	Default	Description
abc_classification	VARCHAR(1)	YES	NULL	A, B, or C
is_shelf_life	BOOLEAN	NO	FALSE	Shelf-life product indicator
is_traceable	BOOLEAN	NO	FALSE	Lot/batch tracking required
rotation_policy	VARCHAR(8)	YES	NULL	FIFO or FEFO
mrsl_days	INTEGER	YES	NULL	Minimum remaining shelf life policy (days)
superseded_by_sku	VARCHAR(64)	YES	NULL	Replacement SKU if discontinued
is_active	BOOLEAN	NO	TRUE	Soft-delete flag
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification
created_by	BIGINT	YES	NULL	FK → identity.user_accounts.id

Table: catalog.product_identifiers

Attribute	Detail
Business Purpose	Alternate product identifiers (barcodes, vendor codes) for lookup at receipt and pick.
Estimated Growth	**Low** — ~1–3 identifiers per product.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
product_id	BIGINT	NO	—	FK → catalog.products.id
identifier_type	VARCHAR(32)	NO	—	BARCODE, VENDOR_CODE, GTIN, etc.
identifier_value	VARCHAR(128)	NO	—	Identifier string
is_primary	BOOLEAN	NO	FALSE	Primary scan identifier
is_active	BOOLEAN	NO	TRUE	Soft-delete flag
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Domain 2: Network (network schema)

Table: network.inventory_locations

Attribute	Detail
Business Purpose	Physical sites (warehouses, stores, DCs) anchoring all stock balances and operations.
Estimated Growth	**Low** — tens to low hundreds of locations.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
location_code	VARCHAR(32)	NO	—	Business key
location_name	VARCHAR(256)	NO	—	Display name
location_type	VARCHAR(32)	NO	—	WAREHOUSE, STORE, DISTRIBUTION_POINT
parent_location_id	BIGINT	YES	NULL	Self-FK for hierarchy

Column	Data Type	Nullable	Default	Description
operational_status	VARCHAR(32)	NO	'ACTIVE'	ACTIVE, PLANNED, CLOSING, INACTIVE
is_frozen	BOOLEAN	NO	FALSE	Physical count freeze flag (BR-024)
address_line1	VARCHAR(256)	YES	NULL	Street address
address_line2	VARCHAR(256)	YES	NULL	Address continuation
city	VARCHAR(128)	YES	NULL	City
region_code	VARCHAR(32)	YES	NULL	State/province/region
postal_code	VARCHAR(16)	YES	NULL	Postal/ZIP code
country_code	CHAR(2)	YES	NULL	ISO 3166-1 alpha-2
responsible_manager_id	BIGINT	YES	NULL	FK → identity.user_accounts.id
go_live_date	DATE	YES	NULL	Operational start date
closure_plan	TEXT	YES	NULL	Closure disposition plan
is_active	BOOLEAN	NO	TRUE	Soft-delete flag
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: network.storage_positions

Attribute	Detail
Business Purpose	Bin/zone/shelf addresses within a location for put-away and pick path.
Estimated Growth	**Low–Medium** — thousands per large warehouse.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
location_id	BIGINT	NO	—	FK → network.inventory_locations.id
position_code	VARCHAR(64)	NO	—	Unique within location
zone	VARCHAR(32)	YES	NULL	Zone designation
aisle	VARCHAR(32)	YES	NULL	Aisle designation
bin	VARCHAR(32)	YES	NULL	Bin designation
storage_type	VARCHAR(32)	NO	'BULK'	BULK, PICK_FACE, QUARANTINE, BONDED
capacity_units	NUMERIC(18,4)	YES	NULL	Optional capacity limit
is_dedicated	BOOLEAN	NO	FALSE	Product-dedicated position
dedicated_product_id	BIGINT	YES	NULL	FK → catalog.products.id
is_active	BOOLEAN	NO	TRUE	Soft-delete flag
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Domain 3: Governance (governance schema)

Table: `governance.governance_policies`

Attribute	Detail
Business Purpose	Organization-wide approval thresholds, tolerances, and exception rules.
Estimated Growth	**Low** — versioned policy set; < 1K active versions.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`policy_code`	`VARCHAR(64)`	NO	—	Business identifier
`policy_name`	`VARCHAR(256)`	NO	—	Display name
`policy_type`	`VARCHAR(32)`	NO	—	APPROVAL_THRESHOLD, TOLERANCE, ACCESS, EXCEPTION
`threshold_value`	`NUMERIC(18,4)`	YES	NULL	Numeric threshold
`threshold_currency`	`CHAR(3)`	YES	NULL	ISO 4217 for financial thresholds
`applicable_scope`	`VARCHAR(64)`	YES	NULL	Location, role, or transaction scope
`approval_authority_role`	`VARCHAR(64)`	YES	NULL	Required approver role
`effective_from`	`TIMESTAMPTZ`	NO	—	Policy effective start
`effective_to`	`TIMESTAMPTZ`	YES	NULL	Policy effective end (NULL = open)
`is_active`	`BOOLEAN`	NO	`TRUE`	Soft-delete flag
`version`	`INTEGER`	NO	`1`	Optimistic lock
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `governance.reason_codes`

Attribute	Detail
Business Purpose	Standardized categorized explanations for adjustments, write-offs, and discrepancies.
Estimated Growth	**Low** — ~50–200 codes.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`reason_code`	`VARCHAR(32)`	NO	—	Business key
`description`	`VARCHAR(512)`	NO	—	Human-readable label
`category`	`VARCHAR(32)`	NO	—	ADJUSTMENT, SHRINKAGE, DAMAGE, OBSOLESCENCE, RECEIPT_DISCREPANCY, RETURN, OTHER
`reporting_group`	`VARCHAR(64)`	YES	NULL	Reporting rollup group
`approval_threshold_id`	`BIGINT`	YES	NULL	FK → `governance.governance_policies.id`
`is_active`	`BOOLEAN`	NO	`TRUE`	Retired codes set FALSE
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Domain 4: Inventory Balance (`inventory` schema)

Table: `inventory.stock_positions`

Attribute	Detail
Business Purpose	Authoritative on-hand, reserved, and ATP quantities per product per location — central quantity hub.
Estimated Growth	**Medium** — product × location combinations; ~100K–2M rows. Hot contention table.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`product_id`	`BIGINT`	NO	—	FK → `catalog.products.id`
`location_id`	`BIGINT`	NO	—	FK → `network.inventory_locations.id`
`on_hand_qty`	`NUMERIC(18,4)`	NO	`0`	Physical on-hand quantity
`reserved_qty`	`NUMERIC(18,4)`	NO	`0`	Committed to orders
`quarantined_qty`	`NUMERIC(18,4)`	NO	`0`	Held / non-sellable quantity
`atp_qty`	`NUMERIC(18,4)`	NO	`0`	Available-to-promise (derived: on_hand − reserved − quarantined)
`inbound_expected_qty`	`NUMERIC(18,4)`	NO	`0`	Informational pipeline quantity
`stock_status`	`VARCHAR(32)`	NO	`'IN_STOCK'`	IN_STOCK, LOW_STOCK, OUT_OF_STOCK, OVERSTOCK
`ownership_type`	`VARCHAR(32)`	NO	`'OWNED'`	OWNED, CONSIGNMENT, BONDED
`last_movement_at`	`TIMESTAMPTZ`	YES	NULL	Last quantity change timestamp
`version`	`INTEGER`	NO	`1`	Optimistic lock — critical for concurrent reservations
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `inventory.lot_batches`

Attribute	Detail
Business Purpose	Traceable lot/batch identity with shelf-life attributes for FEFO and recall.
Estimated Growth	**Medium–High** — new lot per receipt batch; 500K–10M rows over time.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`stock_position_id`	`BIGINT`	NO	—	FK → `inventory.stock_positions.id`
`product_id`	`BIGINT`	NO	—	FK → `catalog.products.id` (denormalized for query)
`lot_number`	`VARCHAR(64)`	NO	—	Lot/batch business identifier
`quantity`	`NUMERIC(18,4)`	NO	`0`	Quantity in this lot at position
`expiration_date`	`DATE`	YES	NULL	Product expiration
`manufacture_date`	`DATE`	YES	NULL	Manufacture date
`receipt_date`	`DATE`	YES	NULL	First receipt date
`quality_status`	`VARCHAR(32)`	NO	`'ACCEPTED'`	ACCEPTED, HOLD, REJECTED
`supplier_id`	`BIGINT`	YES	NULL	FK → `procurement.suppliers.id`
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `inventory.quarantine_holds`

Attribute	Detail
Business Purpose	Holds removing quantity from ATP for quality, regulatory, recall, or customs reasons.
Estimated Growth	**Medium** — active holds low; historical ~100K–1M.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`hold_reference`	`VARCHAR(64)`	NO	—	Business reference number
`stock_position_id`	`BIGINT`	NO	—	FK → `inventory.stock_positions.id`
`lot_batch_id`	`BIGINT`	YES	NULL	FK → `inventory.lot_batches.id`
`held_qty`	`NUMERIC(18,4)`	NO	—	Quantity under hold
`hold_type`	`VARCHAR(32)`	NO	—	QUALITY, REGULATORY, RECALL, CUSTOMS, INVESTIGATION
`hold_reason`	`TEXT`	NO	—	Hold justification
`hold_status`	`VARCHAR(32)`	NO	`'ACTIVE'`	ACTIVE, RELEASED, DISPOSED
`hold_start_at`	`TIMESTAMPTZ`	NO	`NOW()`	Hold effective start
`expected_resolution_at`	`TIMESTAMPTZ`	YES	NULL	Expected release date
`released_at`	`TIMESTAMPTZ`	YES	NULL	Actual release timestamp
`disposition_outcome`	`VARCHAR(32)`	YES	NULL	RESTOCK, WRITE_OFF, DESTROY
`authorized_by`	`BIGINT`	YES	NULL	FK → `identity.user_accounts.id`
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `inventory.stock_policies`

Attribute	Detail
Business Purpose	Replenishment thresholds per product per location.
Estimated Growth	**Medium** — mirrors stock position cardinality; ~100K–2M.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`product_id`	`BIGINT`	NO	—	FK → `catalog.products.id`
`location_id`	`BIGINT`	NO	—	FK → `network.inventory_locations.id`
`reorder_level`	`NUMERIC(18,4)`	NO	`0`	ATP trigger for replenishment
`minimum_threshold`	`NUMERIC(18,4)`	NO	`0`	Minimum stock floor
`safety_stock_qty`	`NUMERIC(18,4)`	NO	`0`	Safety buffer
`maximum_stock_level`	`NUMERIC(18,4)`	YES	NULL	Overstock indicator
`lead_time_days`	`INTEGER`	YES	NULL	Assumed lead time
`review_frequency_days`	`INTEGER`	YES	NULL	Replenishment review cadence
`effective_from`	`TIMESTAMPTZ`	NO	`NOW()`	Current threshold effective date
`exception_approval_ref`	`VARCHAR(64)`	YES	NULL	Authorized exception reference
`version`	`INTEGER`	NO	`1`	Optimistic lock

Column	Data Type	Nullable	Default	Description
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: inventory.stock_policy_history

Attribute	Detail
Business Purpose	Immutable history of threshold changes for audit and replenishment analysis.
Estimated Growth	**Medium** — append-only; grows with policy changes.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
stock_policy_id	BIGINT	NO	—	FK → inventory.stock_policies.id
reorder_level	NUMERIC(18,4)	NO	—	Historical reorder level
minimum_threshold	NUMERIC(18,4)	NO	—	Historical minimum
safety_stock_qty	NUMERIC(18,4)	NO	—	Historical safety stock
effective_from	TIMESTAMPTZ	NO	—	Period start
effective_to	TIMESTAMPTZ	NO	—	Period end
changed_by	BIGINT	YES	NULL	FK → identity.user_accounts.id
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Domain 5: Procurement (procurement schema)

Table: procurement.suppliers

Attribute	Detail
Business Purpose	External vendor master for procurement, receipt, and payment.
Estimated Growth	**Low** — hundreds to low thousands.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
supplier_code	VARCHAR(32)	NO	—	Business key
supplier_name	VARCHAR(256)	NO	—	Legal/display name
contact_name	VARCHAR(128)	YES	NULL	Primary contact
contact_email	VARCHAR(256)	YES	NULL	Contact email
contact_phone	VARCHAR(32)	YES	NULL	Contact phone
payment_terms_days	INTEGER	NO	30	Net payment days
default_lead_time_days	INTEGER	YES	NULL	Default procurement lead time
is_consignment	BOOLEAN	NO	FALSE	Consignment agreement flag (BR-031)
delivery_rating	NUMERIC(3,2)	YES	NULL	Performance score 0.00–5.00
is_active	BOOLEAN	NO	TRUE	Soft-delete flag
version	INTEGER	NO	1	Optimistic lock

Column	Data Type	Nullable	Default	Description
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: procurement.purchase_requisitions

Attribute	Detail
Business Purpose	Internal replenishment request with approval workflow before PO creation.
Estimated Growth	**Medium** — ~50K–500K/year.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
requisition_number	VARCHAR(64)	NO	—	Business key
requesting_location_id	BIGINT	NO	—	FK → network.inventory_locations.id
requested_delivery_date	DATE	YES	NULL	Target delivery date
triggering_alert_id	BIGINT	YES	NULL	FK → audit.operational_alerts.id
approval_status	VARCHAR(32)	NO	'DRAFT'	DRAFT, SUBMITTED, APPROVED, REJECTED, CONVERTED
approved_by	BIGINT	YES	NULL	FK → identity.user_accounts.id
approved_at	TIMESTAMPTZ	YES	NULL	Approval timestamp
purchase_order_id	BIGINT	YES	NULL	FK → procurement.purchase_orders.id
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification
created_by	BIGINT	NO	—	FK → identity.user_accounts.id

Table: procurement.purchase_requisition_lines

Attribute	Detail
Business Purpose	Line-level product and quantity on a purchase requisition.
Estimated Growth	**Medium** — 1–20 lines per requisition.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
requisition_id	BIGINT	NO	—	FK → procurement.purchase_requisitions.id
line_number	INTEGER	NO	—	Sequence within requisition
product_id	BIGINT	NO	—	FK → catalog.products.id
requested_qty	NUMERIC(18,4)	NO	—	Requested quantity (> 0)
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Table: `procurement.purchase_orders`

Attribute	Detail
Business Purpose	Authorized commercial commitment to a supplier.
Estimated Growth	**Medium** — ~20K–200K/year.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`po_number`	`VARCHAR(64)`	NO	—	Business key
`supplier_id`	`BIGINT`	NO	—	FK → `procurement.suppliers.id`
`receiving_location_id`	`BIGINT`	NO	—	FK → `network.inventory_locations.id`
`order_date`	`DATE`	NO	—	PO creation date
`expected_delivery_date`	`DATE`	YES	NULL	Expected arrival
`po_status`	`VARCHAR(32)`	NO	`'OPEN'`	OPEN, PARTIAL, CLOSED, CANCELLED
`total_commitment_amount`	`NUMERIC(18,2)`	NO	`0`	Header total value
`currency_code`	`CHAR(3)`	NO	`'USD'`	ISO 4217
`payment_terms_days`	`INTEGER`	NO	`30`	Copied from supplier at creation
`version`	`INTEGER`	NO	`1`	Optimistic lock
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification
`created_by`	`BIGINT`	NO	—	FK → `identity.user_accounts.id`

Table: `procurement.purchase_order_lines`

Attribute	Detail
Business Purpose	Product, quantity, and price lines on a purchase order.
Estimated Growth	**Medium** — 1–50 lines per PO.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`purchase_order_id`	`BIGINT`	NO	—	FK → `procurement.purchase_orders.id`
`line_number`	`INTEGER`	NO	—	Sequence within PO
`product_id`	`BIGINT`	NO	—	FK → `catalog.products.id`
`ordered_qty`	`NUMERIC(18,4)`	NO	—	Ordered quantity
`received_qty`	`NUMERIC(18,4)`	NO	`0`	Cumulative received
`unit_price`	`NUMERIC(18,4)`	NO	—	Price per UOM
`line_amount`	`NUMERIC(18,2)`	NO	—	Extended line value
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `procurement.expected_inbound_deliveries`

Attribute	Detail
Business Purpose	Planned supplier shipment schedule tied to a purchase order.
Estimated Growth	**Medium** — 0–N deliveries per PO.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`delivery_reference`	`VARCHAR(64)`	NO	—	Business key
`purchase_order_id`	`BIGINT`	NO	—	FK → `procurement.purchase_orders.id`
`expected_arrival_date`	`DATE`	NO	—	Planned arrival
`carrier_reference`	`VARCHAR(128)`	YES	NULL	Carrier/shipment ID
`shipment_status`	`VARCHAR(32)`	NO	`'PLANNED'`	PLANNED, IN_TRANSIT, ARRIVED, DELAYED
`receiving_location_id`	`BIGINT`	NO	—	FK → `network.inventory_locations.id`
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `procurement.expected_inbound_delivery_lines`

Attribute	Detail
Business Purpose	Expected quantity by product on an inbound delivery plan.
Estimated Growth	**Medium** — lines per delivery.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`delivery_id`	`BIGINT`	NO	—	FK → `procurement.expected_inbound_deliveries.id`
`purchase_order_line_id`	`BIGINT`	NO	—	FK → `procurement.purchase_order_lines.id`
`product_id`	`BIGINT`	NO	—	FK → `catalog.products.id`
`expected_qty`	`NUMERIC(18,4)`	NO	—	Expected quantity
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation

Domain 6: Inbound (`inbound` schema)

Table: `inbound.goods_receipts`

Attribute	Detail
Business Purpose	Verified acceptance of inbound supplier goods at receiving dock.
Estimated Growth	**Medium–High** — ~100K–2M/year depending on receipt volume.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`gr_number`	`VARCHAR(64)`	NO	—	Business key
`purchase_order_id`	`BIGINT`	YES	NULL	FK → `procurement.purchase_orders.id`

Column	Data Type	Nullable	Default	Description
exception_authorization_ref	VARCHAR(64)	YES	NULL	Ad-hoc receipt exception (BR-007)
supplier_id	BIGINT	NO	—	FK → procurement.suppliers.id
receiving_location_id	BIGINT	NO	—	FK → network.inventory_locations.id
receipt_date	TIMESTAMPTZ	NO	NOW()	Receipt timestamp
verification_status	VARCHAR(32)	NO	'PENDING'	PENDING, VERIFIED, REJECTED, PARTIAL
receiving_clerk_id	BIGINT	NO	—	FK → identity.user_accounts.id
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: inbound.goods_receipt_lines

Attribute	Detail
Business Purpose	Line-level received product, quantity, and condition.
Estimated Growth	**Medium–High** — multiple lines per receipt.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
goods_receipt_id	BIGINT	NO	—	FK → inbound.goods_receipts.id
line_number	INTEGER	NO	—	Sequence within receipt
purchase_order_line_id	BIGINT	YES	NULL	FK → procurement.purchase_order_lines.id
product_id	BIGINT	NO	—	FK → catalog.products.id
received_qty	NUMERIC(18,4)	NO	—	Received quantity
accepted_qty	NUMERIC(18,4)	NO	0	Accepted after verification
condition_grade	VARCHAR(32)	NO	'GOOD'	GOOD, DAMAGED, REJECTED
lot_batch_id	BIGINT	YES	NULL	FK → inventory.lot_batches.id
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: inbound.receipt_discrepancies

Attribute	Detail
Business Purpose	Documented variance between expected and actual receipt.
Estimated Growth	**Medium** — ~5–15% of receipt lines.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
discrepancy_reference	VARCHAR(64)	NO	—	Business key
goods_receipt_id	BIGINT	NO	—	FK → inbound.goods_receipts.id
goods_receipt_line_id	BIGINT	YES	NULL	FK → inbound.goods_receipt_lines.id
discrepancy_type	VARCHAR(32)	NO	—	SHORT, OVER, WRONG_ITEM, DAMAGE, SHELF_LIFE

Column	Data Type	Nullable	Default	Description
expected_qty	NUMERIC(18,4)	YES	NULL	Expected quantity
actual_qty	NUMERIC(18,4)	YES	NULL	Actual quantity
product_id	BIGINT	NO	—	FK → catalog.products.id
reason_code_id	BIGINT	YES	NULL	FK → governance.reason_codes.id
evidence_notes	TEXT	YES	NULL	Supporting evidence
resolution_status	VARCHAR(32)	NO	'OPEN'	OPEN, RESOLVED, CLAIMED
resolution_outcome	VARCHAR(64)	YES	NULL	Resolution description
supplier_claim_ref	VARCHAR(64)	YES	NULL	Supplier claim reference
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: inbound.put_aways

Attribute	Detail
Business Purpose	Confirmation of stock placement from dock to storage — triggers ATP availability.
Estimated Growth	**Medium–High** — one or more per accepted receipt line.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
put_away_reference	VARCHAR(64)	NO	—	Business key
goods_receipt_id	BIGINT	NO	—	FK → inbound.goods_receipts.id
goods_receipt_line_id	BIGINT	NO	—	FK → inbound.goods_receipt_lines.id
product_id	BIGINT	NO	—	FK → catalog.products.id
quantity	NUMERIC(18,4)	NO	—	Put-away quantity
from_area	VARCHAR(64)	NO	'RECEIVING_DOCK'	Source area
storage_position_id	BIGINT	NO	—	FK → network.storage_positions.id
lot_batch_id	BIGINT	YES	NULL	FK → inventory.lot_batches.id
put_away_at	TIMESTAMPTZ	NO	NOW()	Put-away completion time
availability_effective_at	TIMESTAMPTZ	NO	NOW()	ATP effective timestamp (BR-005)
performed_by	BIGINT	NO	—	FK → identity.user_accounts.id
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Domain 7: Finance (finance schema)

Table: finance.supplier_invoices

Attribute	Detail
Business Purpose	Supplier billing document for three-way match and payment authorization.
Estimated Growth	**Medium** — ~20K–200K/year aligned with PO volume.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
invoice_number	VARCHAR(64)	NO	—	Supplier invoice number
supplier_id	BIGINT	NO	—	FK → procurement.suppliers.id
purchase_order_id	BIGINT	NO	—	FK → procurement.purchase_orders.id
invoice_date	DATE	NO	—	Invoice date
total_amount	NUMERIC(18,2)	NO	—	Invoice total
currency_code	CHAR(3)	NO	'USD'	ISO 4217
match_status	VARCHAR(32)	NO	'PENDING'	PENDING, MATCHED, EXCEPTION
tolerance_variance	NUMERIC(18,2)	YES	NULL	Match tolerance variance
payment_status	VARCHAR(32)	NO	'UNAUTHORIZED'	UNAUTHORIZED, AUTHORIZED, PAID
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: finance.supplier_invoice_lines

Attribute	Detail
Business Purpose	Billed product, quantity, and price for three-way match.
Estimated Growth	**Medium** — lines per invoice.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
supplier_invoice_id	BIGINT	NO	—	FK → finance.supplier_invoices.id
line_number	INTEGER	NO	—	Sequence within invoice
purchase_order_line_id	BIGINT	YES	NULL	FK → procurement.purchase_order_lines.id
product_id	BIGINT	NO	—	FK → catalog.products.id
billed_qty	NUMERIC(18,4)	NO	—	Billed quantity
unit_price	NUMERIC(18,4)	NO	—	Billed unit price
line_amount	NUMERIC(18,2)	NO	—	Extended amount
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Table: finance.invoice_goods_receipt_match

Attribute	Detail
Business Purpose	Associative link between supplier invoices and goods receipts for three-way match (BR-008).
Estimated Growth	**Medium** — M:N between invoices and receipts.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
supplier_invoice_id	BIGINT	NO	—	FK → finance.supplier_invoices.id
goods_receipt_id	BIGINT	NO	—	FK → inbound.goods_receipts.id

Column	Data Type	Nullable	Default	Description
match_status	VARCHAR(32)	NO	'PENDING'	PENDING, MATCHED, EXCEPTION
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Domain 8: Fulfillment (fulfillment schema)

Table: fulfillment.customer_orders

Attribute	Detail
Business Purpose	Customer demand document driving reservation, pick, and ship.
Estimated Growth	**High** — ~500K–10M/year for active retailers.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
order_number	VARCHAR(64)	NO	—	Business key
customer_reference	VARCHAR(128)	YES	NULL	External customer ID
order_date	TIMESTAMPTZ	NO	NOW()	Order placement date
requested_delivery_date	DATE	YES	NULL	Customer requested delivery
order_status	VARCHAR(32)	NO	'DRAFT'	DRAFT, CONFIRMED, PARTIAL, FULFILLED, CANCELLED
channel	VARCHAR(32)	YES	NULL	WEB, STORE, B2B, EDI
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: fulfillment.customer_order_lines

Attribute	Detail
Business Purpose	Order line product, requested, fulfilled, and backordered quantities.
Estimated Growth	**High** — 1–20 lines per order.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
customer_order_id	BIGINT	NO	—	FK → fulfillment.customer_orders.id
line_number	INTEGER	NO	—	Sequence within order
product_id	BIGINT	NO	—	FK → catalog.products.id
requested_qty	NUMERIC(18,4)	NO	—	Ordered quantity
fulfilled_qty	NUMERIC(18,4)	NO	0	Shipped quantity
backordered_qty	NUMERIC(18,4)	NO	0	Backordered quantity
fulfillment_location_id	BIGINT	YES	NULL	FK → network.inventory_locations.id
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: `fulfillment.inventory_reservations`

Attribute	Detail
Business Purpose	ATP hold for confirmed order lines until ship or release.
Estimated Growth	**High** — active reservations proportional to open orders.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`reservation_reference`	`VARCHAR(64)`	NO	—	Business key
`customer_order_id`	`BIGINT`	NO	—	FK → `fulfillment.customer_orders.id`
`order_line_id`	`BIGINT`	NO	—	FK → `fulfillment.customer_order_lines.id`
`product_id`	`BIGINT`	NO	—	FK → `catalog.products.id`
`location_id`	`BIGINT`	NO	—	FK → `network.inventory_locations.id`
`stock_position_id`	`BIGINT`	NO	—	FK → `inventory.stock_positions.id`
`reserved_qty`	`NUMERIC(18,4)`	NO	—	Reserved quantity
`reservation_status`	`VARCHAR(32)`	NO	`'ACTIVE'`	ACTIVE, RELEASED, FULFILLED
`reserved_at`	`TIMESTAMPTZ`	NO	`NOW()`	Reservation timestamp
`expires_at`	`TIMESTAMPTZ`	YES	NULL	Optional expiry
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `fulfillment.fulfillment_executions`

Attribute	Detail
Business Purpose	Pick/pack/verify warehouse work unit for an order.
Estimated Growth	**High** — 1+ per order; ~500K–10M/year.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`fulfillment_reference`	`VARCHAR(64)`	NO	—	Business key
`customer_order_id`	`BIGINT`	NO	—	FK → `fulfillment.customer_orders.id`
`fulfillment_location_id`	`BIGINT`	NO	—	FK → `network.inventory_locations.id`
`pick_verification_status`	`VARCHAR(32)`	NO	`'PENDING'`	PENDING, VERIFIED, EXCEPTION
`pack_confirmed`	`BOOLEAN`	NO	`FALSE`	Pack confirmation flag
`assigned_picker_id`	`BIGINT`	YES	NULL	FK → `identity.user_accounts.id`
`has_exception`	`BOOLEAN`	NO	`FALSE`	Exception flag
`pick_completed_at`	`TIMESTAMPTZ`	YES	NULL	Pick completion timestamp
`version`	`INTEGER`	NO	`1`	Optimistic lock
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `fulfillment.fulfillment_pick_lines`

Attribute	Detail
Business Purpose	Individual pick line with lot attribution and verification.
Estimated Growth	**High** — multiple per fulfillment execution.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`fulfillment_execution_id`	`BIGINT`	NO	—	FK → `fulfillment.fulfillment_executions.id`
`order_line_id`	`BIGINT`	NO	—	FK → `fulfillment.customer_order_lines.id`
`product_id`	`BIGINT`	NO	—	FK → `catalog.products.id`
`requested_qty`	`NUMERIC(18,4)`	NO	—	Quantity to pick
`picked_qty`	`NUMERIC(18,4)`	NO	`0`	Actual picked quantity
`storage_position_id`	`BIGINT`	YES	NULL	FK → `network.storage_positions.id`
`lot_batch_id`	`BIGINT`	YES	NULL	FK → `inventory.lot_batches.id`
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `fulfillment.backorders`

Attribute	Detail
Business Purpose	Customer-agreed delay for unfulfilled order line quantity.
Estimated Growth	**Medium** — subset of order lines during stockouts.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`backorder_reference`	`VARCHAR(64)`	NO	—	Business key
`customer_order_id`	`BIGINT`	NO	—	FK → `fulfillment.customer_orders.id`
`order_line_id`	`BIGINT`	NO	—	FK → `fulfillment.customer_order_lines.id`
`product_id`	`BIGINT`	NO	—	FK → `catalog.products.id`
`backordered_qty`	`NUMERIC(18,4)`	NO	—	Backordered quantity
`customer_acknowledged`	`BOOLEAN`	NO	`FALSE`	BR-028 acknowledgment flag
`acknowledged_at`	`TIMESTAMPTZ`	YES	NULL	Acknowledgment timestamp
`expected_fulfillment_date`	`DATE`	YES	NULL	Expected ship date
`backorder_reason`	`VARCHAR(64)`	YES	NULL	Reason code
`resolution_status`	`VARCHAR(32)`	NO	`'OPEN'`	OPEN, FULFILLED, CANCELLED
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `fulfillment.shipments`

Attribute	Detail
Business Purpose	Confirmed dispatch deducting on-hand and consuming reservations.

Attribute	Detail
Estimated Growth	**High** — ~500K–10M/year.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`shipment_number`	`VARCHAR(64)`	NO	—	Business key
`customer_order_id`	`BIGINT`	NO	—	FK → `fulfillment.customer_orders.id`
`fulfillment_execution_id`	`BIGINT`	NO	—	FK → `fulfillment.fulfillment_executions.id`
`ship_from_location_id`	`BIGINT`	NO	—	FK → `network.inventory_locations.id`
`ship_date`	`TIMESTAMPTZ`	NO	`NOW()`	Dispatch timestamp
`carrier_reference`	`VARCHAR(128)`	YES	NULL	Carrier name/code
`tracking_identifier`	`VARCHAR(128)`	YES	NULL	Tracking number
`shipment_status`	`VARCHAR(32)`	NO	`'DISPATCHED'`	DISPATCHED, IN_TRANSIT, DELIVERED
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `fulfillment.shipment_lines`

Attribute	Detail
Business Purpose	Shipped product, quantity, and lot per shipment.
Estimated Growth	**High** — multiple lines per shipment.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`shipment_id`	`BIGINT`	NO	—	FK → `fulfillment.shipments.id`
`order_line_id`	`BIGINT`	NO	—	FK → `fulfillment.customer_order_lines.id`
`product_id`	`BIGINT`	NO	—	FK → `catalog.products.id`
`shipped_qty`	`NUMERIC(18,4)`	NO	—	Shipped quantity
`lot_batch_id`	`BIGINT`	YES	NULL	FK → `inventory.lot_batches.id`
`reservation_id`	`BIGINT`	YES	NULL	FK → `fulfillment.inventory_reservations.id`
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation

Domain 9: Network Transfer (`transfer` schema)

Table: `transfer.transfer_requests`

Attribute	Detail
Business Purpose	Formal request to move stock between locations; initial lifecycle state of Stock Transfer aggregate.
Estimated Growth	**Medium** — ~10K–100K/year.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key

Column	Data Type	Nullable	Default	Description
request_number	VARCHAR(64)	NO	—	Business key
source_location_id	BIGINT	NO	—	FK → network.inventory_locations.id
destination_location_id	BIGINT	NO	—	FK → network.inventory_locations.id
request_reason	VARCHAR(128)	YES	NULL	Rebalance, shortage, operational
request_status	VARCHAR(32)	NO	'SUBMITTED'	SUBMITTED, APPROVED, REJECTED, CONVERTED
requested_by	BIGINT	NO	—	FK → identity.user_accounts.id
approved_by	BIGINT	YES	NULL	FK → identity.user_accounts.id
approved_at	TIMESTAMPTZ	YES	NULL	Approval timestamp
rejection_reason	TEXT	YES	NULL	Rejection justification
stock_transfer_id	BIGINT	YES	NULL	FK → transfer.stock_transfers.id
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: transfer.transfer_request_lines

Attribute	Detail
Business Purpose	Product and quantity requested on a transfer request.
Estimated Growth	**Medium** — lines per request.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
transfer_request_id	BIGINT	NO	—	FK → transfer.transfer_requests.id
line_number	INTEGER	NO	—	Sequence within request
product_id	BIGINT	NO	—	FK → catalog.products.id
requested_qty	NUMERIC(18,4)	NO	—	Requested quantity
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Table: transfer.stock_transfers

Attribute	Detail
Business Purpose	Executed inter-location stock movement from dispatch to receipt.
Estimated Growth	**Medium** — ~10K–100K/year.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
transfer_number	VARCHAR(64)	NO	—	Business key
transfer_request_id	BIGINT	YES	NULL	FK → transfer.transfer_requests.id
source_location_id	BIGINT	NO	—	FK → network.inventory_locations.id
destination_location_id	BIGINT	NO	—	FK → network.inventory_locations.id
transfer_status	VARCHAR(32)	NO	'APPROVED'	REQUESTED, APPROVED, IN_TRANSIT, RECEIVED, CANCELLED

Column	Data Type	Nullable	Default	Description
dispatch_date	TIMESTAMPTZ	YES	NULL	Source dispatch timestamp
receipt_date	TIMESTAMPTZ	YES	NULL	Destination receipt timestamp
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: transfer.stock_transfer_lines

Attribute	Detail
Business Purpose	Product, quantity, and lot on a stock transfer.
Estimated Growth	**Medium** — lines per transfer.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
stock_transfer_id	BIGINT	NO	—	FK → transfer.stock_transfers.id
line_number	INTEGER	NO	—	Sequence within transfer
product_id	BIGINT	NO	—	FK → catalog.products.id
quantity	NUMERIC(18,4)	NO	—	Transfer quantity
lot_batch_id	BIGINT	YES	NULL	FK → inventory.lot_batches.id
dispatched_qty	NUMERIC(18,4)	NO	0	Quantity dispatched
received_qty	NUMERIC(18,4)	NO	0	Quantity received
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: transfer.in_transit_stock

Attribute	Detail
Business Purpose	Pipeline inventory between dispatch and destination receipt.
Estimated Growth	**Medium** — active in-transit rows; short lifecycle.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
in_transit_reference	VARCHAR(64)	NO	—	Business key
stock_transfer_id	BIGINT	NO	—	FK → transfer.stock_transfers.id
stock_transfer_line_id	BIGINT	NO	—	FK → transfer.stock_transfer_lines.id
product_id	BIGINT	NO	—	FK → catalog.products.id
quantity	NUMERIC(18,4)	NO	—	In-transit quantity
source_location_id	BIGINT	NO	—	FK → network.inventory_locations.id
destination_location_id	BIGINT	NO	—	FK → network.inventory_locations.id
dispatch_date	TIMESTAMPTZ	NO	—	Dispatch timestamp
expected_receipt_date	DATE	YES	NULL	Expected arrival

Column	Data Type	Nullable	Default	Description
in_transit_status	VARCHAR(32)	NO	'IN_TRANSIT'	IN_TRANSIT, RECEIVED, LOST, DELAYED
carrier_reference	VARCHAR(128)	YES	NULL	Carrier reference
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Domain 10: Inventory Accuracy (accuracy schema)

Table: accuracy.cycle_counts

Attribute	Detail
Business Purpose	Scheduled partial physical count assignment with blind count protocol.
Estimated Growth	**Medium–High** — continuous rotating counts; ~50K–500K/year.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
count_assignment_number	VARCHAR(64)	NO	—	Business key
schedule_date	DATE	NO	—	Scheduled count date
location_id	BIGINT	NO	—	FK → network.inventory_locations.id
product_id	BIGINT	NO	—	FK → catalog.products.id
storage_position_id	BIGINT	YES	NULL	FK → network.storage_positions.id
counter_id	BIGINT	NO	—	FK → identity.user_accounts.id
count_status	VARCHAR(32)	NO	'SCHEDULED'	SCHEDULED, IN_PROGRESS, COMPLETED
physical_count_qty	NUMERIC(18,4)	YES	NULL	Recorded physical quantity
system_count_qty	NUMERIC(18,4)	YES	NULL	System qty (hidden until submission)
completed_at	TIMESTAMPTZ	YES	NULL	Completion timestamp
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: accuracy.physical_inventory_counts

Attribute	Detail
Business Purpose	Wall-to-wall count event imposing location freeze (BR-024).
Estimated Growth	**Low** — 1–4 per location per year.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
count_event_number	VARCHAR(64)	NO	—	Business key
location_id	BIGINT	NO	—	FK → network.inventory_locations.id
count_start_date	TIMESTAMPTZ	NO	—	Count start
count_completion_date	TIMESTAMPTZ	YES	NULL	Count end
freeze_status	VARCHAR(32)	NO	'ACTIVE'	ACTIVE, RELEASED

Column	Data Type	Nullable	Default	Description
total_variances	INTEGER	NO	0	Variance count
certification_status	VARCHAR(32)	NO	'IN_PROGRESS'	IN_PROGRESS, CERTIFIED
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: accuracy.count_variances

Attribute	Detail
Business Purpose	System vs. physical quantity difference requiring investigation.
Estimated Growth	**Medium** — subset of count events.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
variance_reference	VARCHAR(64)	NO	—	Business key
cycle_count_id	BIGINT	YES	NULL	FK → accuracy.cycle_counts.id
physical_count_id	BIGINT	YES	NULL	FK → accuracy.physical_inventory_counts.id
stock_position_id	BIGINT	NO	—	FK → inventory.stock_positions.id
product_id	BIGINT	NO	—	FK → catalog.products.id
location_id	BIGINT	NO	—	FK → network.inventory_locations.id
system_qty	NUMERIC(18,4)	NO	—	System quantity at count
physical_qty	NUMERIC(18,4)	NO	—	Physical quantity counted
variance_qty	NUMERIC(18,4)	NO	—	physical − system
variance_value	NUMERIC(18,2)	YES	NULL	Financial impact
investigation_status	VARCHAR(32)	NO	'OPEN'	OPEN, INVESTIGATED
investigation_findings	TEXT	YES	NULL	Root cause notes
approval_status	VARCHAR(32)	NO	'PENDING'	PENDING, APPROVED, REJECTED
approver_id	BIGINT	YES	NULL	FK → identity.user_accounts.id
approved_at	TIMESTAMPTZ	YES	NULL	Approval timestamp
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: accuracy.stock_adjustments

Attribute	Detail
Business Purpose	Authorized correction to stock position with mandatory reason code.
Estimated Growth	**Medium** — adjustments from counts and ad-hoc corrections.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
adjustment_reference	VARCHAR(64)	NO	—	Business key

Column	Data Type	Nullable	Default	Description
stock_position_id	BIGINT	NO	—	FK → inventory.stock_positions.id
product_id	BIGINT	NO	—	FK → catalog.products.id
location_id	BIGINT	NO	—	FK → network.inventory_locations.id
adjustment_qty	NUMERIC(18,4)	NO	—	Signed quantity (+/−)
qty_before	NUMERIC(18,4)	NO	—	On-hand before adjustment
qty_after	NUMERIC(18,4)	NO	—	On-hand after adjustment
reason_code_id	BIGINT	NO	—	FK → governance.reason_codes.id
count_variance_id	BIGINT	YES	NULL	FK → accuracy.count_variances.id
approval_status	VARCHAR(32)	NO	'PENDING'	PENDING, APPROVED, REJECTED
initiated_by	BIGINT	NO	—	FK → identity.user_accounts.id
approved_by	BIGINT	YES	NULL	FK → identity.user_accounts.id
adjustment_date	TIMESTAMPTZ	NO	NOW()	Effective date
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: accuracy.inventory_write_offs

Attribute	Detail
Business Purpose	Permanent authorized removal of inventory value.
Estimated Growth	**Low–Medium** — smaller volume than adjustments.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
write_off_reference	VARCHAR(64)	NO	—	Business key
stock_position_id	BIGINT	NO	—	FK → inventory.stock_positions.id
product_id	BIGINT	NO	—	FK → catalog.products.id
location_id	BIGINT	NO	—	FK → network.inventory_locations.id
write_off_qty	NUMERIC(18,4)	NO	—	Quantity written off
financial_value	NUMERIC(18,2)	YES	NULL	Value impact
reason_code_id	BIGINT	NO	—	FK → governance.reason_codes.id
evidence_reference	VARCHAR(128)	YES	NULL	Supporting evidence ref
disposition_method	VARCHAR(32)	YES	NULL	DESTROY, DONATE, SCRAP
return_disposition_id	BIGINT	YES	NULL	FK → returns.return_dispositions.id
initiated_by	BIGINT	NO	—	FK → identity.user_accounts.id
authorized_by	BIGINT	NO	—	FK → identity.user_accounts.id
write_off_date	TIMESTAMPTZ	NO	NOW()	Effective date
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Domain 11: Returns (`returns` schema)

Table: `returns.return_authorizations`

Attribute	Detail
Business Purpose	RMA gating customer return before warehouse acceptance (BR-025).
Estimated Growth	**Medium** — ~5–15% of order volume.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`rma_number`	`VARCHAR(64)`	NO	—	Business key
`customer_order_id`	`BIGINT`	NO	—	FK → `fulfillment.customer_orders.id`
`customer_reference`	`VARCHAR(128)`	YES	NULL	Customer identifier
`return_reason`	`VARCHAR(128)`	YES	NULL	Return reason
`authorization_date`	`TIMESTAMPTZ`	NO	`NOW()`	RMA issue date
`expiration_date`	`DATE`	YES	NULL	RMA expiry
`disposition_expectation`	`VARCHAR(64)`	YES	NULL	Expected disposition
`authorization_status`	`VARCHAR(32)`	NO	`'ACTIVE'`	ACTIVE, EXPIRED, USED, CANCELLED
`version`	`INTEGER`	NO	`1`	Optimistic lock
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation
`updated_at`	`TIMESTAMPTZ`	NO	`NOW()`	Last modification

Table: `returns.return_authorization_lines`

Attribute	Detail
Business Purpose	Authorized return products and quantities on an RMA.
Estimated Growth	**Medium** — lines per RMA.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`return_authorization_id`	`BIGINT`	NO	—	FK → `returns.return_authorizations.id`
`line_number`	`INTEGER`	NO	—	Sequence within RMA
`product_id`	`BIGINT`	NO	—	FK → `catalog.products.id`
`authorized_qty`	`NUMERIC(18,4)`	NO	—	Authorized return quantity
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Record creation

Table: `returns.customer_returns`

Attribute	Detail
Business Purpose	Physical receipt of customer-returned goods.
Estimated Growth	**Medium** — aligned with RMA volume.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
return_number	VARCHAR(64)	NO	—	Business key
return_authorization_id	BIGINT	NO	—	FK → returns.return_authorizations.id
customer_order_id	BIGINT	NO	—	FK → fulfillment.customer_orders.id
receiving_location_id	BIGINT	NO	—	FK → network.inventory_locations.id
receipt_date	TIMESTAMPTZ	NO	NOW()	Receipt timestamp
inspection_status	VARCHAR(32)	NO	'PENDING'	PENDING, INSPECTED
receiving_clerk_id	BIGINT	NO	—	FK → identity.user_accounts.id
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: returns.customer_return_lines

Attribute	Detail
Business Purpose	Returned product, quantity, and condition at receipt.
Estimated Growth	**Medium** — lines per return.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
customer_return_id	BIGINT	NO	—	FK → returns.customer_returns.id
line_number	INTEGER	NO	—	Sequence within return
product_id	BIGINT	NO	—	FK → catalog.products.id
returned_qty	NUMERIC(18,4)	NO	—	Returned quantity
condition_at_receipt	VARCHAR(32)	NO	—	GOOD, DAMAGED, DEFECTIVE, OPENED
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Table: returns.return_dispositions

Attribute	Detail
Business Purpose	Inspection outcome — restock, quarantine, or write-off.
Estimated Growth	**Medium** — one per return line.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
disposition_reference	VARCHAR(64)	NO	—	Business key
customer_return_id	BIGINT	NO	—	FK → returns.customer_returns.id
return_line_id	BIGINT	NO	—	FK → returns.customer_return_lines.id
product_id	BIGINT	NO	—	FK → catalog.products.id
quantity	NUMERIC(18,4)	NO	—	Disposition quantity
condition_grade	VARCHAR(32)	NO	—	Resale quality grade

Column	Data Type	Nullable	Default	Description
disposition_decision	VARCHAR(32)	NO	—	RESTOCK, QUARANTINE, WRITE_OFF
restock_location_id	BIGINT	YES	NULL	FK → network.inventory_locations.id
quarantine_hold_id	BIGINT	YES	NULL	FK → inventory.quarantine_holds.id
write_off_id	BIGINT	YES	NULL	FK → accuracy.inventory_write_offs.id
inspector_id	BIGINT	NO	—	FK → identity.user_accounts.id
inspection_date	TIMESTAMPTZ	NO	NOW()	Inspection timestamp
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Domain 12: Commercial (commercial schema)

Table: commercial.promotional_stock_allocations

Attribute	Detail
Business Purpose	Ring-fenced stock for promotional campaigns or product launches (BR-029).
Estimated Growth	**Low** — hundreds to low thousands active campaigns.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
allocation_reference	VARCHAR(64)	NO	—	Business key
product_id	BIGINT	NO	—	FK → catalog.products.id
allocated_qty	NUMERIC(18,4)	NO	—	Ring-fenced quantity
campaign_name	VARCHAR(256)	NO	—	Campaign or launch name
channel	VARCHAR(32)	YES	NULL	Channel scope
effective_start_date	DATE	NO	—	Campaign start
effective_end_date	DATE	NO	—	Campaign end
readiness_status	VARCHAR(32)	NO	'PLANNED'	PLANNED, READY, ACTIVE, CLOSED
commercial_approval_ref	VARCHAR(64)	YES	NULL	Commercial approval reference
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: commercial.promotional_allocation_locations

Attribute	Detail
Business Purpose	Locations supplying ring-fenced promotional stock.
Estimated Growth	**Low** — 1–N locations per allocation.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
allocation_id	BIGINT	NO	—	FK → commercial.promotional_stock_allocations.id
location_id	BIGINT	NO	—	FK → network.inventory_locations.id

Column	Data Type	Nullable	Default	Description
stock_position_id	BIGINT	YES	NULL	FK → inventory.stock_positions.id
allocated_qty_at_location	NUMERIC(18,4)	NO	—	Quantity at this location
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Domain 13: Identity & Access (identity schema)

Table: identity.user_accounts

Attribute	Detail
Business Purpose	Authenticated user identity and accountability anchor for all operations.
Estimated Growth	**Low** — hundreds to low thousands of users.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
user_identifier	VARCHAR(64)	NO	—	Login / employee ID
display_name	VARCHAR(256)	NO	—	Full display name
email	VARCHAR(256)	YES	NULL	Contact email
account_status	VARCHAR(32)	NO	'ACTIVE'	ACTIVE, SUSPENDED, DEACTIVATED
effective_from	DATE	NO	—	Account effective start
last_activity_at	TIMESTAMPTZ	YES	NULL	Last system activity
has_sod_conflict	BOOLEAN	NO	FALSE	Segregation-of-duties conflict flag
version	INTEGER	NO	1	Optimistic lock
created_at	TIMESTAMPTZ	NO	NOW()	Record creation
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: identity.role_assignments

Attribute	Detail
Business Purpose	Role delegation with permission boundaries per user.
Estimated Growth	**Low** — 1–5 roles per user.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
user_account_id	BIGINT	NO	—	FK → identity.user_accounts.id
role_name	VARCHAR(64)	NO	—	WAREHOUSE_MANAGER, RECEIVING_CLERK, BUYER, etc.
permission_set	VARCHAR(128)	YES	NULL	Permission bundle reference
assignment_status	VARCHAR(32)	NO	'ACTIVE'	ACTIVE, INACTIVE
effective_from	DATE	NO	—	Assignment start
effective_to	DATE	YES	NULL	Assignment end
sod_compliant	BOOLEAN	NO	TRUE	SoD compliance at assignment
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Column	Data Type	Nullable	Default	Description
updated_at	TIMESTAMPTZ	NO	NOW()	Last modification

Table: identity.role_assignment_locations

Attribute	Detail
Business Purpose	Location scope restricting role permissions to authorized sites.
Estimated Growth	**Low** — 0–N locations per assignment.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
role_assignment_id	BIGINT	NO	—	FK → identity.role_assignments.id
location_id	BIGINT	NO	—	FK → network.inventory_locations.id
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Domain 14: Audit & Reporting (audit schema)

Table: audit.inventory_movement_records

Attribute	Detail
Business Purpose	Immutable append-only log of every stock quantity change.
Estimated Growth	**Very High** — 10M–100M+/year. **Partition monthly** on movement_timestamp.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key (per partition)
movement_reference	VARCHAR(64)	NO	—	Business transaction reference
movement_type	VARCHAR(32)	NO	—	RECEIPT, SHIPMENT, TRANSFER, ADJUSTMENT, WRITE_OFF, RESERVATION, RETURN, PUT_AWAY
stock_position_id	BIGINT	NO	—	FK → inventory.stock_positions.id
product_id	BIGINT	NO	—	FK → catalog.products.id (denormalized)
location_id	BIGINT	NO	—	FK → network.inventory_locations.id (denormalized)
quantity_change	NUMERIC(18,4)	NO	—	Signed quantity delta
qty_before	NUMERIC(18,4)	NO	—	On-hand before
qty_after	NUMERIC(18,4)	NO	—	On-hand after
lot_batch_id	BIGINT	YES	NULL	FK → inventory.lot_batches.id
reason_code_id	BIGINT	YES	NULL	FK → governance.reason_codes.id
source_document_type	VARCHAR(32)	NO	—	Originating document type
source_document_id	BIGINT	NO	—	Originating document PK
performed_by	BIGINT	NO	—	FK → identity.user_accounts.id
movement_timestamp	TIMESTAMPTZ	NO	NOW()	Event timestamp (partition key)
created_at	TIMESTAMPTZ	NO	NOW()	Insert timestamp

Partitioning: Range partition by movement_timestamp (monthly). Retain 24 months online; archive older partitions to cold storage.

Table: `audit.audit_trail_records`

Attribute	Detail
Business Purpose	Governance-grade before/after snapshots for material entity changes.
Estimated Growth	**Very High** — proportional to governed mutations. **Partition monthly** on `event_timestamp`.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key (per partition)
`audit_event_reference`	`VARCHAR(64)`	NO	—	Business reference
`affected_entity_type`	`VARCHAR(64)`	NO	—	Entity/table name
`affected_entity_id`	`BIGINT`	NO	—	Affected record PK
`action_performed`	`VARCHAR(32)`	NO	—	CREATE, UPDATE, APPROVE, CANCEL
`user_id`	`BIGINT`	NO	—	FK → `identity.user_accounts.id`
`role_at_action`	`VARCHAR(64)`	YES	NULL	Role at time of action
`before_state`	`JSONB`	YES	NULL	Snapshot before change
`after_state`	`JSONB`	YES	NULL	Snapshot after change
`approval_chain_ref`	`VARCHAR(128)`	YES	NULL	Approval workflow reference
`movement_record_id`	`BIGINT`	YES	NULL	FK → `audit.inventory_movement_records.id`
`event_timestamp`	`TIMESTAMPTZ`	NO	`NOW()`	Event timestamp (partition key)
`created_at`	`TIMESTAMPTZ`	NO	`NOW()`	Insert timestamp

Partitioning: Range partition by `event_timestamp` (monthly). Immutable — no UPDATE/DELETE.

Table: `audit.operational_alerts`

Attribute	Detail
Business Purpose	Derived notifications for threshold breaches, pending approvals, and exceptions.
Estimated Growth	**High** — alerts generated and resolved continuously; purge resolved > 90 days.

Column	Data Type	Nullable	Default	Description
`id`	`BIGINT`	NO	IDENTITY	Surrogate primary key
`alert_reference`	`VARCHAR(64)`	NO	—	Business key
`alert_type`	`VARCHAR(32)`	NO	—	LOW_STOCK, OUT_OF_STOCK, PENDING_APPROVAL, TRANSFER_EXCEPTION, INBOUND_DELAY, CAMPAIGN_READINESS
`severity`	`VARCHAR(16)`	NO	`'MEDIUM'`	LOW, MEDIUM, HIGH, CRITICAL
`product_id`	`BIGINT`	YES	NULL	FK → `catalog.products.id`
`location_id`	`BIGINT`	YES	NULL	FK → `network.inventory_locations.id`
`stock_position_id`	`BIGINT`	YES	NULL	FK → `inventory.stock_positions.id`
`source_entity_type`	`VARCHAR(64)`	YES	NULL	Triggering entity type
`source_entity_id`	`BIGINT`	YES	NULL	Triggering entity PK
`triggering_condition`	`TEXT`	YES	NULL	Condition description
`assigned_role`	`VARCHAR(64)`	YES	NULL	Recipient role
`acknowledgment_status`	`VARCHAR(32)`	NO	`'OPEN'`	OPEN, ACKNOWLEDGED, RESOLVED

Column	Data Type	Nullable	Default	Description
resolution_status	VARCHAR(32)	NO	'PENDING'	PENDING, RESOLVED, DISMISSED
alert_date	TIMESTAMPTZ	NO	NOW()	Alert creation
resolved_at	TIMESTAMPTZ	YES	NULL	Resolution timestamp
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Table: audit.inventory_valuation_snapshots

Attribute	Detail
Business Purpose	Point-in-time inventory value summary for finance reconciliation (BP-30).
Estimated Growth	**Low** — monthly/periodic snapshots; ~12–52 per year plus ad-hoc.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
snapshot_reference	VARCHAR(64)	NO	—	Business key
valuation_date	DATE	NO	—	Snapshot as-of date
location_scope	VARCHAR(32)	NO	—	SINGLE, REGION, NETWORK
total_inventory_value	NUMERIC(18,2)	NO	—	Total owned value
owned_value	NUMERIC(18,2)	NO	—	Owned stock value
consignment_value	NUMERIC(18,2)	NO	0	Consignment (excluded from owned)
gl_variance	NUMERIC(18,2)	YES	NULL	Variance vs. general ledger
reconciliation_status	VARCHAR(32)	NO	'DRAFT'	DRAFT, RECONCILED, EXCEPTION
prepared_by	BIGINT	NO	—	FK → identity.user_accounts.id
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Table: audit.inventory_valuation_lines

Attribute	Detail
Business Purpose	Product-level quantity and value detail within a valuation snapshot.
Estimated Growth	**Medium per snapshot** — product × location lines; snapshot-bound.

Column	Data Type	Nullable	Default	Description
id	BIGINT	NO	IDENTITY	Surrogate primary key
snapshot_id	BIGINT	NO	—	FK → audit.inventory_valuation_snapshots.id
product_id	BIGINT	NO	—	FK → catalog.products.id
location_id	BIGINT	NO	—	FK → network.inventory_locations.id
quantity	NUMERIC(18,4)	NO	—	Snapshot quantity
unit_cost	NUMERIC(18,4)	NO	—	Unit valuation cost
extended_value	NUMERIC(18,2)	NO	—	quantity × unit_cost
ownership_type	VARCHAR(32)	NO	'OWNED'	OWNED, CONSIGNMENT
created_at	TIMESTAMPTZ	NO	NOW()	Record creation

Physical Design Appendices

Appendix A: Table Inventory Summary

Schema	Tables	Growth Tier
`catalog`	2	Low
`network`	2	Low
`governance`	2	Low
`inventory`	5	Medium
`procurement`	8	Medium
`inbound`	4	Medium–High
`finance`	3	Medium
`fulfillment`	8	High
`transfer`	5	Medium
`accuracy`	5	Medium
`returns`	5	Medium
`commercial`	2	Low
`identity`	3	Low
`audit`	5	Low–Very High
Total	**59 tables**	

Appendix B: Critical Unique Constraints (Logical)

Table	Unique Key Columns
`catalog.products`	`sku`
`network.inventory_locations`	`location_code`
`network.storage_positions`	`(location_id, position_code)`
`inventory.stock_positions`	`(product_id, location_id)`
`inventory.stock_policies`	`(product_id, location_id)`
`procurement.suppliers`	`supplier_code`
`procurement.purchase_orders`	`po_number`
`fulfillment.customer_orders`	`order_number`
`identity.user_accounts`	`user_identifier`
`governance.reason_codes`	`reason_code`

Appendix C: Partitioning & Archival Strategy

Table	Strategy	Retention
`audit.inventory_movement_records`	Monthly range on `movement_timestamp`	24 months online; archive thereafter
`audit.audit_trail_records`	Monthly range on `event_timestamp`	7 years (compliance); tiered storage
`audit.operational_alerts`	No partition; index on `acknowledgment_status`	Purge resolved > 90 days
`fulfillment.customer_orders`	Optional yearly archive of closed orders	3 years online
`inbound.goods_receipts`	Archive closed receipts > 3 years	Cold storage

Appendix D: High-Priority Index Candidates

Table	Index Columns	Rationale
`inventory.stock_positions`	`(product_id, location_id)`	Unique lookup; hottest path
`inventory.stock_positions`	`(location_id, stock_status)`	Low-stock monitoring
`fulfillment.inventory_reservations`	`(stock_position_id, reservation_status)`	Active reservation queries
`fulfillment.customer_orders`	`(order_status, order_date)`	Open order dashboards
`audit.inventory_movement_records`	`(stock_position_id, movement_timestamp DESC)`	Position history
`inbound.goods_receipts`	`(purchase_order_id)`	PO receipt lookup
`procurement.purchase_order_lines`	`(purchase_order_id, product_id)`	Open PO line queries
`inventory.lot_batches`	`(product_id, expiration_date)`	FEFO pick ordering

Appendix E: Logical-to-Physical Entity Mapping

Logical Entity (SRS-DB-001)	Physical Table(s)
Product	`catalog.products`, `catalog.product_identifiers`
Inventory Location	`network.inventory_locations`
Storage Position	`network.storage_positions`
Governance Policy	`governance.governance_policies`
Reason Code	`governance.reason_codes`
Stock Position	`inventory.stock_positions`
Lot/Batch	`inventory.lot_batches`
Quarantine Hold	`inventory.quarantine_holds`
Stock Policy	`inventory.stock_policies`, `inventory.stock_policy_history`
Supplier	`procurement.suppliers`
Purchase Requisition	`procurement.purchase_requisitions`, `_lines`
Purchase Order	`procurement.purchase_orders`, `_lines`
Expected Inbound Delivery	`procurement.expected_inbound_deliveries`, `_lines`
Goods Receipt	`inbound.goods_receipts`, `_lines`
Receipt Discrepancy	`inbound.receipt_discrepancies`
Put-Away	`inbound.put_aways`
Supplier Invoice	`finance.supplier_invoices`, `_lines`, `invoice_goods_receipt_match`
Customer Order	`fulfillment.customer_orders`, `_lines`
Inventory Reservation	`fulfillment.inventory_reservations`
Fulfillment Execution	`fulfillment.fulfillment_executions`, `fulfillment_pick_lines`
Backorder	`fulfillment.backorders`
Shipment	`fulfillment.shipments`, `shipment_lines`
Transfer Request	`transfer.transfer_requests`, `_lines`
Stock Transfer	`transfer.stock_transfers`, `_lines`
In-Transit Stock	`transfer.in_transit_stock`
Cycle Count	`accuracy.cycle_counts`

Logical Entity (SRS-DB-001)	Physical Table(s)
Physical Inventory Count	`accuracy.physical_inventory_counts`
Count Variance	`accuracy.count_variances`
Stock Adjustment	`accuracy.stock_adjustments`
Inventory Write-Off	`accuracy.inventory_write_offs`
Return Authorization	`returns.return_authorizations`, `_lines`
Customer Return	`returns.customer_returns`, `_lines`
Return Disposition	`returns.return_dispositions`
Promotional Stock Allocation	`commercial.promotional_stock_allocations`, `promotional_allocation_locations`
User Account	`identity.user_accounts`
Role Assignment	`identity.role_assignments`, `role_assignment_locations`
Inventory Movement Record	`audit.inventory_movement_records`
Audit Trail Record	`audit.audit_trail_records`
Operational Alert	`audit.operational_alerts`
Inventory Valuation Snapshot	`audit.inventory_valuation_snapshots`, `inventory_valuation_lines`

Document Approval

Role	Name	Signature	Date
Database Architect			
DDD Solution Architect			
Solution Architect			
Inventory Controller			

End of Physical Data Model

Document ID: SRS-DB-002 | Version 1.0 | July 6, 2026

Data Dictionary

Inventory Management System — PostgreSQL

Document Information

Attribute	Detail
Document ID	SRS-DB-007
Document Title	Data Dictionary — Inventory Management System
Version	1.0
Status	Draft
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Attribute	Detail
Target Platform	PostgreSQL 15+
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Purpose

This document provides a **developer- and business-analyst-friendly reference** for every table and column in the Inventory Management System database. It translates the Physical Data Model (SRS-DB-002) into plain-language business definitions, relationship notes, and data-type conventions — without DDL or SQL.

Conventions Used in This Document

Convention	Meaning
Nullable	`YES` = column may be empty; `NO` = column is required
Default	Value assigned automatically when a row is inserted without an explicit value; `–` means no default
Related Entity	`→ schema.table (column)` for foreign keys; `– (primary key)` for surrogate PKs; business-key uniqueness noted where applicable
Business Meaning	One-sentence plain-language explanation suitable for business analysts
Naming	All identifiers use `snake_case` per PDM standards

Schema Index — All Tables (58)

#	Schema.Table	One-Line Description
1	`catalog.products`	Canonical SKU master — identity for all inventory operations across locations and channels.
2	`catalog.product_identifiers`	Alternate product identifiers (barcodes, vendor codes) for lookup at receipt and pick.
3	`network.inventory_locations`	Physical sites (warehouses, stores, DCs) anchoring all stock balances and operations.
4	`network.storage_positions`	Bin/zone/shelf addresses within a location for put-away and pick path.
5	`governance.governance_policies`	Organization-wide approval thresholds, tolerances, and exception rules.
6	`governance.reason_codes`	Standardized categorized explanations for adjustments, write-offs, and discrepancies.
7	`inventory.stock_positions`	Authoritative on-hand, reserved, and ATP quantities per product per location — central quantity hub.
8	`inventory.lot_batches`	Traceable lot/batch identity with shelf-life attributes for FEFO and recall.
9	`inventory.quarantine_holds`	Holds removing quantity from ATP for quality, regulatory, recall, or customs reasons.
10	`inventory.stock_policies`	Replenishment thresholds per product per location.
11	`inventory.stock_policy_history`	Immutable history of threshold changes for audit and replenishment analysis.
12	`procurement.suppliers`	External vendor master for procurement, receipt, and payment.
13	`procurement.purchase_requisitions`	Internal replenishment request with approval workflow before PO creation.
14	`procurement.purchase_requisition_lines`	Line-level product and quantity on a purchase requisition.
15	`procurement.purchase_orders`	Authorized commercial commitment to a supplier.
16	`procurement.purchase_order_lines`	Product, quantity, and price lines on a purchase order.
17	`procurement.expected_inbound_deliveries`	Planned supplier shipment schedule tied to a purchase order.
18	`procurement.expected_inbound_delivery_lines`	Expected quantity by product on an inbound delivery plan.
19	`inbound.goods_receipts`	Verified acceptance of inbound supplier goods at receiving dock.

#	Schema.Table	One-Line Description
20	inbound.goods_receipt_lines	Line-level received product, quantity, and condition.
21	inbound.receipt_discrepancies	Documented variance between expected and actual receipt.
22	inbound.put_aways	Confirmation of stock placement from dock to storage — triggers ATP availability.
23	finance.supplier_invoices	Supplier billing document for three-way match and payment authorization.
24	finance.supplier_invoice_lines	Billed product, quantity, and price for three-way match.
25	finance.invoice_goods_receipt_match	Associative link between supplier invoices and goods receipts for three-way match (BR-008).
26	fulfillment.customer_orders	Customer demand document driving reservation, pick, and ship.
27	fulfillment.customer_order_lines	Order line product, requested, fulfilled, and backordered quantities.
28	fulfillment.inventory_reservations	ATP hold for confirmed order lines until ship or release.
29	fulfillment.fulfillment_executions	Pick/pack/verify warehouse work unit for an order.
30	fulfillment.fulfillment_pick_lines	Individual pick line with lot attribution and verification.
31	fulfillment.backorders	Customer-agreed delay for unfulfilled order line quantity.
32	fulfillment.shipments	Confirmed dispatch deducting on-hand and consuming reservations.
33	fulfillment.shipment_lines	Shipped product, quantity, and lot per shipment.
34	transfer.transfer_requests	Formal request to move stock between locations; initial lifecycle state of Stock Transfer aggregate.
35	transfer.transfer_request_lines	Product and quantity requested on a transfer request.
36	transfer.stock_transfers	Executed inter-location stock movement from dispatch to receipt.
37	transfer.stock_transfer_lines	Product, quantity, and lot on a stock transfer.
38	transfer.in_transit_stock	Pipeline inventory between dispatch and destination receipt.
39	accuracy.cycle_counts	Scheduled partial physical count assignment with blind count protocol.
40	accuracy.physical_inventory_counts	Wall-to-wall count event imposing location freeze (BR-024).
41	accuracy.count_variances	System vs. physical quantity difference requiring investigation.
42	accuracy.stock_adjustments	Authorized correction to stock position with mandatory reason code.
43	accuracy.inventory_write_offs	Permanent authorized removal of inventory value.
44	returns.return_authorizations	RMA gating customer return before warehouse acceptance (BR-025).
45	returns.return_authorization_lines	Authorized return products and quantities on an RMA.
46	returns.customer_returns	Physical receipt of customer-returned goods.
47	returns.customer_return_lines	Returned product, quantity, and condition at receipt.
48	returns.return_dispositions	Inspection outcome — restock, quarantine, or write-off.
49	commercial.promotional_stock_allocations	Ring-fenced stock for promotional campaigns or product launches (BR-029).
50	commercial.promotional_allocation_locations	Locations supplying ring-fenced promotional stock.
51	identity.user_accounts	Authenticated user identity and accountability anchor for all operations.
52	identity.role_assignments	Role delegation with permission boundaries per user.
53	identity.role_assignment_locations	Location scope restricting role permissions to authorized sites.
54	audit.inventory_movement_records	Immutable append-only log of every stock quantity change.
55	audit.audit_trail_records	Governance-grade before/after snapshots for material entity changes.
56	audit.operational_alerts	Derived notifications for threshold breaches, pending approvals, and exceptions.

#	Schema.Table	One-Line Description
57	audit.inventory_valuation_snapshots	Point-in-time inventory value summary for finance reconciliation (BP-30).
58	audit.inventory_valuation_lines	Product-level quantity and value detail within a valuation snapshot.

All table definitions present in SRS-DB-002 are documented below (58 tables). PDM Appendix A inventory summary cites 59 tables across 14 schemas.

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catalog.products

Table description: Canonical SKU master — identity for all inventory operations across locations and channels.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
sku	VARCHAR(64)	NO	—	Unique business identifier for this record (stock keeping unit).	Unique business key
product_name	VARCHAR(256)	NO	—	The human-readable name shown to users in screens and reports.	—
description	TEXT	YES	—	Optional longer text describing additional details.	—
unit_of_measure	VARCHAR(16)	NO	—	Unit of measure for counting and transacting this product (e.g., each, case, kilogram).	—
category_code	VARCHAR(64)	YES	—	Code referencing the product's merchandise or category grouping.	—
lifecycle_status	VARCHAR(32)	NO	'ACTIVE'	Catalog lifecycle status indicating whether the product is sellable.	—
abc_classification	VARCHAR(1)	YES	—	ABC classification ranking inventory importance or turnover velocity.	—
is_shelf_life	BOOLEAN	NO	FALSE	Indicates whether this product requires expiration-date tracking.	—
is_traceable	BOOLEAN	NO	FALSE	Indicates whether this product must be tracked by lot or batch number.	—
rotation_policy	VARCHAR(8)	YES	—	Stock rotation policy: first-in-first-out or first-expired-first-out.	—
mrsl_days	INTEGER	YES	—	Minimum days of shelf life required before the product may be sold or shipped.	—
superseded_by_sku	VARCHAR(64)	YES	—	SKU of the successor product when this item is discontinued.	→ catalog.products (sku)
is_active	BOOLEAN	NO	TRUE	Indicates whether this master-data record is active; deactivated records are retained for history.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—
created_by	BIGINT	YES	—	The user who created this record.	→ identity.user_accounts (id)

---

catalog.product_identifiers

Table description: Alternate product identifiers (barcodes, vendor codes) for lookup at receipt and pick.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
identifier_type	VARCHAR(32)	NO	—	Type of alternate identifier (barcode, vendor code, GTIN, etc.).	—
identifier_value	VARCHAR(128)	NO	—	The scanned or keyed identifier value.	—
is_primary	BOOLEAN	NO	FALSE	Indicates whether this is the default identifier used when scanning.	—
is_active	BOOLEAN	NO	TRUE	Indicates whether this master-data record is active; deactivated records are retained for history.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

network.inventory_locations

Table description: Physical sites (warehouses, stores, DCs) anchoring all stock balances and operations.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
location_code	VARCHAR(32)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
location_name	VARCHAR(256)	NO	—	The human-readable name shown to users in screens and reports.	—
location_type	VARCHAR(32)	NO	—	Type of physical site (warehouse, store, or distribution point).	—
parent_location_id	BIGINT	YES	—	Optional reference to a parent location in the site hierarchy.	→ network.inventory_locations (id)
operational_status	VARCHAR(32)	NO	'ACTIVE'	Operational status of the location.	—
is_frozen	BOOLEAN	NO	FALSE	Indicates the location is frozen and inventory movements are blocked during a physical count.	—
address_line1	VARCHAR(256)	YES	—	Primary street address line.	—
address_line2	VARCHAR(256)	YES	—	Secondary address line (suite, building, etc.).	—
city	VARCHAR(128)	YES	—	City name for the address.	—
region_code	VARCHAR(32)	YES	—	State, province, or regional code.	—
postal_code	VARCHAR(16)	YES	—	Postal or ZIP code for the address.	—
country_code	CHAR(2)	YES	—	Two-letter ISO country code.	—
responsible_manager_id	BIGINT	YES	—	The manager responsible for this location.	→ identity.user_accounts (id)
go_live_date	DATE	YES	—	Date when the location became or will become operational.	—
closure_plan	TEXT	YES	—	Notes describing how inventory and operations will be handled when closing.	—
is_active	BOOLEAN	NO	TRUE	Indicates whether this master-data record is active; deactivated records are retained for history.	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

network.storage_positions

Table description: Bin/zone/shelf addresses within a location for put-away and pick path.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
position_code	VARCHAR(64)	NO	—	Storage position code that is unique within its parent location.	—
zone	VARCHAR(32)	YES	—	Warehouse zone code for routing and organization.	—
aisle	VARCHAR(32)	YES	—	Aisle identifier within the warehouse.	—
bin	VARCHAR(32)	YES	—	Bin or shelf identifier completing the storage address.	—
storage_type	VARCHAR(32)	NO	'BULK'	Storage position type (bulk, pick face, quarantine, or bonded).	—
capacity_units	NUMERIC(18,4)	YES	—	Maximum quantity or capacity this position can hold, if defined.	—
is_dedicated	BOOLEAN	NO	FALSE	Indicates whether this position is reserved for a single product.	—
dedicated_product_id	BIGINT	YES	—	The product exclusively assigned to this storage position.	→ catalog.products (id)
is_active	BOOLEAN	NO	TRUE	Indicates whether this master-data record is active; deactivated records are retained for history.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

governance.governance_policies

Table description: Organization-wide approval thresholds, tolerances, and exception rules.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
policy_code	VARCHAR(64)	NO	—	Business identifier.	Unique business key

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
policy_name	VARCHAR(256)	NO	—	The human-readable name shown to users in screens and reports.	—
policy_type	VARCHAR(32)	NO	—	Category of governance policy.	—
threshold_value	NUMERIC(18,4)	YES	—	Numeric limit that triggers the policy rule.	—
threshold_currency	CHAR(3)	YES	—	ISO currency code when the threshold is a monetary amount.	—
applicable_scope	VARCHAR(64)	YES	—	Scope to which this policy applies.	—
approval_authority_role	VARCHAR(64)	YES	—	Role authorized to approve actions under this policy.	—
effective_from	TIMESTAMPTZ	NO	—	Date and time when this policy version becomes effective.	—
effective_to	TIMESTAMPTZ	YES	—	Date and time when this policy expires; null means no end date.	—
is_active	BOOLEAN	NO	TRUE	Indicates whether this master-data record is active; deactivated records are retained for history.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

governance.reason_codes

Table description: Standardized categorized explanations for adjustments, write-offs, and discrepancies.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
reason_code	VARCHAR(32)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
description	VARCHAR(512)	NO	—	Plain-language label describing this code or reason.	—
category	VARCHAR(32)	NO	—	Business category classifying the reason for the transaction.	—
reporting_group	VARCHAR(64)	YES	—	Group used to roll up this reason in management reports.	—
approval_threshold_id	BIGINT	YES	—	The governance policy defining the approval threshold for this reason.	→ governance.governance_policies (id)
is_active	BOOLEAN	NO	TRUE	Retired codes set FALSE.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

inventory.stock_positions

Table description: Authoritative on-hand, reserved, and ATP quantities per product per location — central quantity hub.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
on_hand_qty	NUMERIC(18,4)	NO	0	Total physical quantity currently in stock at this location.	—
reserved_qty	NUMERIC(18,4)	NO	0	Quantity reserved and committed to customer orders.	—
quarantined_qty	NUMERIC(18,4)	NO	0	Quantity held in quarantine and not available for sale.	—
atp_qty	NUMERIC(18,4)	NO	0	Quantity available to promise to customers after reservations and holds.	—
inbound_expected_qty	NUMERIC(18,4)	NO	0	Expected inbound quantity not yet received (informational only).	—
stock_status	VARCHAR(32)	NO	'IN_STOCK'	Derived stock health status for monitoring and alerts.	—
ownership_type	VARCHAR(32)	NO	'OWNED'	Ownership classification of the stock at this position.	—
last_movement_at	TIMESTAMPTZ	YES	—	Timestamp of the most recent quantity change at this position.	—
version	INTEGER	NO	1	Version counter critical for preventing lost updates during concurrent stock reservations.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

inventory.lot_batches

Table description: Traceable lot/batch identity with shelf-life attributes for FEFO and recall.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
stock_position_id	BIGINT	NO	—	The stock position (product at location) this record affects.	→ inventory.stock_positions (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id) (denormalized)
lot_number	VARCHAR(64)	NO	—	Business lot or batch number for traceability.	—
quantity	NUMERIC(18,4)	NO	0	Quantity of this specific lot held at the stock position.	—
expiration_date	DATE	YES	—	Expiration date for shelf-life products in this lot.	—
manufacture_date	DATE	YES	—	Date the product in this lot was manufactured.	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
receipt_date	DATE	YES	—	Date this lot was first received into inventory.	—
quality_status	VARCHAR(32)	NO	'ACCEPTED'	Quality status of this lot (accepted, on hold, or rejected).	—
supplier_id	BIGINT	YES	—	The supplier associated with this record.	→ procurement.suppliers (id)
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

inventory.quarantine_holds

Table description: Holds removing quantity from ATP for quality, regulatory, recall, or customs reasons.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
hold_reference	VARCHAR(64)	NO	—	Human-readable reference number assigned to this transaction.	—
stock_position_id	BIGINT	NO	—	The stock position (product at location) this record affects.	→ inventory.stock_positions (id)
lot_batch_id	BIGINT	YES	—	The lot or batch this record references, when applicable.	→ inventory.lot_batches (id)
held_qty	NUMERIC(18,4)	NO	—	Quantity currently blocked from available-to-promise by this hold.	—
hold_type	VARCHAR(32)	NO	—	Category of quarantine hold.	—
hold_reason	TEXT	NO	—	Business justification explaining why stock is held.	—
hold_status	VARCHAR(32)	NO	'ACTIVE'	Current status of the quarantine hold.	—
hold_start_at	TIMESTAMPTZ	NO	NOW()	Date and time when the hold became effective.	—
expected_resolution_at	TIMESTAMPTZ	YES	—	Expected date when the hold will be resolved.	—
released_at	TIMESTAMPTZ	YES	—	Date and time when the hold was actually released.	—
disposition_outcome	VARCHAR(32)	YES	—	Final disposition outcome when the hold is resolved.	—
authorized_by	BIGINT	YES	—	The user who authorized this action.	→ identity.user_accounts (id)
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

inventory.stock_policies

Table description: Replenishment thresholds per product per location.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
reorder_level	NUMERIC(18,4)	NO	0	Available-to-promise level that triggers a replenishment request.	—
minimum_threshold	NUMERIC(18,4)	NO	0	Minimum acceptable stock level before urgent action is required.	—
safety_stock_qty	NUMERIC(18,4)	NO	0	Extra buffer quantity maintained above minimum to absorb demand variability.	—
maximum_stock_level	NUMERIC(18,4)	YES	—	Maximum stock level above which the position is considered overstocked.	—
lead_time_days	INTEGER	YES	—	Expected number of days from order to receipt for replenishment planning.	—
review_frequency_days	INTEGER	YES	—	How often replenishment parameters are reviewed, in days.	—
effective_from	TIMESTAMPTZ	NO	NOW()	Date from which the current threshold values are effective.	—
exception_approval_ref	VARCHAR(64)	YES	—	Reference documenting an approved exception to standard thresholds.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

inventory.stock_policy_history

Table description: Immutable history of threshold changes for audit and replenishment analysis.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
stock_policy_id	BIGINT	NO	—	The stock policy whose historical values are captured here.	→ inventory.stock_policies (id)
reorder_level	NUMERIC(18,4)	NO	—	Reorder level that was in effect during this historical period.	—
minimum_threshold	NUMERIC(18,4)	NO	—	Minimum threshold in effect during this historical period.	—
safety_stock_qty	NUMERIC(18,4)	NO	—	Safety stock quantity in effect during this historical period.	—
effective_from	TIMESTAMPTZ	NO	—	Start of the period when these threshold values were effective.	—
effective_to	TIMESTAMPTZ	NO	—	End of the period when these threshold values were effective.	—
changed_by	BIGINT	YES	—	The user who made this policy change.	→ identity.user_accounts (id)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

procurement.suppliers

Table description: External vendor master for procurement, receipt, and payment.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
supplier_code	VARCHAR(32)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
supplier_name	VARCHAR(256)	NO	—	Legal or commonly used display name of the supplier.	—
contact_name	VARCHAR(128)	YES	—	Name of the primary contact person.	—
contact_email	VARCHAR(256)	YES	—	Email address for contacting this person or organization.	—
contact_phone	VARCHAR(32)	YES	—	Telephone number for contacting this person or organization.	—
payment_terms_days	INTEGER	NO	30	Number of days until payment is due (net terms).	—
default_lead_time_days	INTEGER	YES	—	Supplier's typical lead time in days for purchase orders.	—
is_consignment	BOOLEAN	NO	FALSE	Indicates whether stock from this supplier is held on consignment.	—
delivery_rating	NUMERIC(3,2)	YES	—	Supplier delivery performance rating on a zero-to-five scale.	—
is_active	BOOLEAN	NO	TRUE	Indicates whether this master-data record is active; deactivated records are retained for history.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

procurement.purchase_requisitions

Table description: Internal replenishment request with approval workflow before PO creation.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
requisition_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
requesting_location_id	BIGINT	NO	—	The location requesting replenishment on this requisition.	→ network.inventory_locations (id)
requested_delivery_date	DATE	YES	—	Date by which delivery is requested.	—
triggering_alert_id	BIGINT	YES	—	The operational alert that triggered this requisition, if applicable.	→ audit.operational_alerts (id)
approval_status	VARCHAR(32)	NO	'DRAFT'	Approval workflow status of the requisition.	—
approved_by	BIGINT	YES	—	The user who approved this record or transaction.	→ identity.user_accounts (id)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
approved_at	TIMESTAMPTZ	YES	—	Date and time when approval was granted.	—
purchase_order_id	BIGINT	YES	—	The purchase order linked to this invoice or receipt.	→ procurement.purchase_orders (id)
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—
created_by	BIGINT	NO	—	The user who created this record.	→ identity.user_accounts (id)

procurement.purchase_requisition_lines

Table description: Line-level product and quantity on a purchase requisition.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
requisition_id	BIGINT	NO	—	The purchase requisition this line belongs to.	→ procurement.purchase_requisitions (id)
line_number	INTEGER	NO	—	Line sequence number within the parent purchase requisition.	—
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
requested_qty	NUMERIC(18,4)	NO	—	Quantity requested on this requisition line; must be greater than zero.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

procurement.purchase_orders

Table description: Authorized commercial commitment to a supplier.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
po_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
supplier_id	BIGINT	NO	—	The supplier associated with this record.	→ procurement.suppliers (id)
receiving_location_id	BIGINT	NO	—	The location where goods will be or were received.	→ network.inventory_locations (id)
order_date	DATE	NO	—	Calendar date the purchase order was created.	—
expected_delivery_date	DATE	YES	—	Expected date when in-transit stock will arrive.	—
po_status	VARCHAR(32)	NO	'OPEN'	Fulfillment status of the purchase order.	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
total_commitment_amount	NUMERIC(18,2)	NO	0	Total committed monetary value of the purchase order.	—
currency_code	CHAR(3)	NO	'USD'	ISO 4217 three-letter currency code.	—
payment_terms_days	INTEGER	NO	30	Payment terms copied from the supplier master at PO creation.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—
created_by	BIGINT	NO	—	The user who created this record.	→ identity.user_accounts (id)

procurement.purchase_order_lines

Table description: Product, quantity, and price lines on a purchase order.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
purchase_order_id	BIGINT	NO	—	The purchase order linked to this invoice or receipt.	→ procurement.purchase_orders (id)
line_number	INTEGER	NO	—	Line sequence number within the parent purchase order.	—
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
ordered_qty	NUMERIC(18,4)	NO	—	Quantity the customer ordered on this line.	—
received_qty	NUMERIC(18,4)	NO	0	Total quantity received to date against this PO line.	—
unit_price	NUMERIC(18,4)	NO	—	Unit price per unit of measure on this line.	—
line_amount	NUMERIC(18,2)	NO	—	Line total calculated as quantity times unit price.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

procurement.expected_inbound_deliveries

Table description: Planned supplier shipment schedule tied to a purchase order.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
delivery_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
purchase_order_id	BIGINT	NO	—	The purchase order linked to this invoice or receipt.	→ procurement.purchase_orders (id)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
expected_arrival_date	DATE	NO	—	Planned arrival date for this inbound delivery.	—
carrier_reference	VARCHAR(128)	YES	—	Carrier tracking or shipment reference number.	—
shipment_status	VARCHAR(32)	NO	'PLANNED'	Status of the planned inbound shipment.	—
receiving_location_id	BIGINT	NO	—	The location where goods will be or were received.	→ network.inventory_locations (id)
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

procurement.expected_inbound_delivery_lines

Table description: Expected quantity by product on an inbound delivery plan.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
delivery_id	BIGINT	NO	—	The expected inbound delivery this line belongs to.	→ procurement.expected_inbound_deliveries (id)
purchase_order_line_id	BIGINT	NO	—	The purchase order line this record is linked to.	→ procurement.purchase_order_lines (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
expected_qty	NUMERIC(18,4)	NO	—	Quantity that was expected to be received.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

inbound.goods_receipts

Table description: Verified acceptance of inbound supplier goods at receiving dock.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
gr_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
purchase_order_id	BIGINT	YES	—	The purchase order linked to this invoice or receipt.	→ procurement.purchase_orders (id)
exception_authorization_ref	VARCHAR(64)	YES	—	Authorization reference for receiving without a matching purchase order.	—
supplier_id	BIGINT	NO	—	The supplier associated with this record.	→ procurement.suppliers (id)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
receiving_location_id	BIGINT	NO	—	The location where goods will be or were received.	→ network.inventory_locations (id)
receipt_date	TIMESTAMPTZ	NO	NOW()	Date and time when goods were received at the dock.	—
verification_status	VARCHAR(32)	NO	'PENDING'	Verification status of the goods receipt.	—
receiving_clerk_id	BIGINT	NO	—	The receiving clerk who processed this receipt.	→ identity.user_accounts (id)
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

inbound.goods_receipt_lines

Table description: Line-level received product, quantity, and condition.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
goods_receipt_id	BIGINT	NO	—	The goods receipt this record belongs to.	→ inbound.goods_receipts (id)
line_number	INTEGER	NO	—	Line sequence number within the parent goods receipt.	—
purchase_order_line_id	BIGINT	YES	—	The purchase order line this record is linked to.	→ procurement.purchase_order_lines (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
received_qty	NUMERIC(18,4)	NO	—	Quantity physically received on this line.	—
accepted_qty	NUMERIC(18,4)	NO	0	Quantity accepted into inventory after quality verification.	—
condition_grade	VARCHAR(32)	NO	'GOOD'	Physical condition grade of received goods.	—
lot_batch_id	BIGINT	YES	—	The lot or batch this record references, when applicable.	→ inventory.lot_batches (id)
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

inbound.receipt_discrepancies

Table description: Documented variance between expected and actual receipt.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
discrepancy_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
goods_receipt_id	BIGINT	NO	—	The goods receipt this record belongs to.	→ inbound.goods_receipts (id)
goods_receipt_line_id	BIGINT	YES	—	The goods receipt line this record belongs to.	→ inbound.goods_receipt_lines (id)
discrepancy_type	VARCHAR(32)	NO	—	Type of discrepancy between expected and actual receipt.	—
expected_qty	NUMERIC(18,4)	YES	—	Quantity that was expected to be received.	—
actual_qty	NUMERIC(18,4)	YES	—	Quantity that was actually received.	—
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
reason_code_id	BIGINT	YES	—	The standardized reason code explaining this transaction.	→ governance.reason_codes (id)
evidence_notes	TEXT	YES	—	Notes or references supporting the discrepancy investigation.	—
resolution_status	VARCHAR(32)	NO	'OPEN'	Resolution status of the receipt discrepancy.	—
resolution_outcome	VARCHAR(64)	YES	—	Description of how the discrepancy was resolved.	—
supplier_claim_ref	VARCHAR(64)	YES	—	Reference number for a claim filed with the supplier.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

inbound.put_aways

Table description: Confirmation of stock placement from dock to storage — triggers ATP availability.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
put_away_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
goods_receipt_id	BIGINT	NO	—	The goods receipt this record belongs to.	→ inbound.goods_receipts (id)
goods_receipt_line_id	BIGINT	NO	—	The goods receipt line this record belongs to.	→ inbound.goods_receipt_lines (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
quantity	NUMERIC(18,4)	NO	—	Quantity moved from receiving dock to storage.	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
from_area	VARCHAR(64)	NO	'RECEIVING_DOCK'	Area from which stock was moved (typically receiving dock).	—
storage_position_id	BIGINT	NO	—	The storage position (bin/shelf) where stock is or was located.	→ network.storage_positions (id)
lot_batch_id	BIGINT	YES	—	The lot or batch this record references, when applicable.	→ inventory.lot_batches (id)
put_away_at	TIMESTAMPTZ	NO	NOW()	Date and time when put-away was completed.	—
availability_effective_at	TIMESTAMPTZ	NO	NOW()	Date and time from which stock becomes available to promise after put-away.	—
performed_by	BIGINT	NO	—	The user who performed this inventory movement.	→ identity.user_accounts (id)
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

finance.supplier_invoices

Table description: Supplier billing document for three-way match and payment authorization.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
invoice_number	VARCHAR(64)	NO	—	Invoice number as printed on the supplier's billing document.	—
supplier_id	BIGINT	NO	—	The supplier associated with this record.	→ procurement.suppliers (id)
purchase_order_id	BIGINT	NO	—	The purchase order linked to this invoice or receipt.	→ procurement.purchase_orders (id)
invoice_date	DATE	NO	—	Date printed on the supplier invoice.	—
total_amount	NUMERIC(18,2)	NO	—	Total amount billed on the supplier invoice.	—
currency_code	CHAR(3)	NO	'USD'	ISO 4217 three-letter currency code.	—
match_status	VARCHAR(32)	NO	'PENDING'	Three-way match status for the invoice.	—
tolerance_variance	NUMERIC(18,2)	YES	—	Monetary variance within approved tolerance during invoice matching.	—
payment_status	VARCHAR(32)	NO	'UNAUTHORIZED'	Payment authorization and settlement status.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

finance.supplier_invoice_lines

Table description: Billed product, quantity, and price for three-way match.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
supplier_invoice_id	BIGINT	NO	—	The supplier invoice this record belongs to.	→ finance.supplier_invoices (id)
line_number	INTEGER	NO	—	Line sequence number within the parent supplier invoice.	—
purchase_order_line_id	BIGINT	YES	—	The purchase order line this record is linked to.	→ procurement.purchase_order_lines (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
billed_qty	NUMERIC(18,4)	NO	—	Quantity billed by the supplier on this invoice line.	—
unit_price	NUMERIC(18,4)	NO	—	Unit price billed by the supplier on this line.	—
line_amount	NUMERIC(18,2)	NO	—	Extended monetary amount for this invoice line.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

finance.invoice_goods_receipt_match

Table description: Associative link between supplier invoices and goods receipts for three-way match (BR-008).

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
supplier_invoice_id	BIGINT	NO	—	The supplier invoice this record belongs to.	→ finance.supplier_invoices (id)
goods_receipt_id	BIGINT	NO	—	The goods receipt this record belongs to.	→ inbound.goods_receipts (id)
match_status	VARCHAR(32)	NO	'PENDING'	Three-way match status for the invoice.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

fulfillment.customer_orders

Table description: Customer demand document driving reservation, pick, and ship.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
order_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
customer_reference	VARCHAR(128)	YES	—	Customer identifier from an external sales or CRM system.	—
order_date	TIMESTAMPTZ	NO	NOW()	Date and time when the customer placed the order.	—
requested_delivery_date	DATE	YES	—	Delivery date requested by the customer.	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
order_status	VARCHAR(32)	NO	'DRAFT'	Overall fulfillment status of the customer order.	—
channel	VARCHAR(32)	YES	—	Sales channel through which the order was placed.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

fulfillment.customer_order_lines

Table description: Order line product, requested, fulfilled, and backordered quantities.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
customer_order_id	BIGINT	NO	—	The customer order this record belongs to.	→ fulfillment.customer_orders (id)
line_number	INTEGER	NO	—	Line sequence number within the parent customer order.	—
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
requested_qty	NUMERIC(18,4)	NO	—	Quantity the customer ordered on this line.	—
fulfilled_qty	NUMERIC(18,4)	NO	0	Quantity shipped to the customer on this line to date.	—
backordered_qty	NUMERIC(18,4)	NO	0	Quantity not yet fulfilled and held as a backorder.	—
fulfillment_location_id	BIGINT	YES	—	The warehouse location fulfilling this order line or execution.	→ network.inventory_locations (id)
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

fulfillment.inventory_reservations

Table description: ATP hold for confirmed order lines until ship or release.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
reservation_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
customer_order_id	BIGINT	NO	—	The customer order this record belongs to.	→ fulfillment.customer_orders (id)
order_line_id	BIGINT	NO	—	The customer order line this record belongs to.	→ fulfillment.customer_order_lines (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
stock_position_id	BIGINT	NO	—	The stock position (product at location) this record affects.	→ inventory.stock_positions (id)
reserved_qty	NUMERIC(18,4)	NO	—	Quantity currently reserved against available stock.	—
reservation_status	VARCHAR(32)	NO	'ACTIVE'	Status of the inventory reservation.	—
reserved_at	TIMESTAMPTZ	NO	NOW()	Date and time when stock was reserved.	—
expires_at	TIMESTAMPTZ	YES	—	Optional date and time when the reservation expires if not fulfilled.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

fulfillment.fulfillment_executions

Table description: Pick/pack/verify warehouse work unit for an order.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
fulfillment_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
customer_order_id	BIGINT	NO	—	The customer order this record belongs to.	→ fulfillment.customer_orders (id)
fulfillment_location_id	BIGINT	NO	—	The warehouse location fulfilling this order line or execution.	→ network.inventory_locations (id)
pick_verification_status	VARCHAR(32)	NO	'PENDING'	Pick verification status for this fulfillment execution.	—
pack_confirmed	BOOLEAN	NO	FALSE	Indicates whether packing has been confirmed.	—
assigned_picker_id	BIGINT	YES	—	The warehouse picker assigned to this fulfillment task.	→ identity.user_accounts (id)
has_exception	BOOLEAN	NO	FALSE	Indicates whether an exception occurred during fulfillment.	—
pick_completed_at	TIMESTAMPTZ	YES	—	Date and time when picking was completed.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

fulfillment.fulfillment_pick_lines

Table description: Individual pick line with lot attribution and verification.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
fulfillment_execution_id	BIGINT	NO	—	The fulfillment execution (pick/pack unit) this record belongs to.	→ fulfillment.fulfillment_executions (id)
order_line_id	BIGINT	NO	—	The customer order line this record belongs to.	→ fulfillment.customer_order_lines (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
requested_qty	NUMERIC(18,4)	NO	—	Quantity the picker is instructed to retrieve.	—
picked_qty	NUMERIC(18,4)	NO	0	Quantity actually picked by the warehouse operator.	—
storage_position_id	BIGINT	YES	—	The storage position (bin/shelf) where stock is or was located.	→ network.storage_positions (id)
lot_batch_id	BIGINT	YES	—	The lot or batch this record references, when applicable.	→ inventory.lot_batches (id)
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

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fulfillment.backorders

Table description: Customer-agreed delay for unfulfilled order line quantity.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
backorder_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
customer_order_id	BIGINT	NO	—	The customer order this record belongs to.	→ fulfillment.customer_orders (id)
order_line_id	BIGINT	NO	—	The customer order line this record belongs to.	→ fulfillment.customer_order_lines (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
backordered_qty	NUMERIC(18,4)	NO	—	Quantity not yet fulfilled and held as a backorder.	—
customer_acknowledged	BOOLEAN	NO	FALSE	Indicates the customer has acknowledged the backorder delay.	—
acknowledged_at	TIMESTAMPTZ	YES	—	Date and time when the customer acknowledged the backorder.	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
expected_fulfillment_date	DATE	YES	—	Expected date when the backordered quantity will ship.	—
backorder_reason	VARCHAR(64)	YES	—	Short code explaining why the line is backordered.	—
resolution_status	VARCHAR(32)	NO	'OPEN'	Resolution status of the backorder.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

fulfillment.shipments

Table description: Confirmed dispatch deducting on-hand and consuming reservations.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
shipment_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
customer_order_id	BIGINT	NO	—	The customer order this record belongs to.	→ fulfillment.customer_orders (id)
fulfillment_execution_id	BIGINT	NO	—	The fulfillment execution (pick/pack unit) this record belongs to.	→ fulfillment.fulfillment_executions (id)
ship_from_location_id	BIGINT	NO	—	The location from which this shipment was dispatched.	→ network.inventory_locations (id)
ship_date	TIMESTAMPTZ	NO	NOW()	Date and time when goods were dispatched for transfer.	—
carrier_reference	VARCHAR(128)	YES	—	Name or code of the shipping carrier.	—
tracking_identifier	VARCHAR(128)	YES	—	Carrier tracking number for shipment visibility.	—
shipment_status	VARCHAR(32)	NO	'DISPATCHED'	Shipment delivery status.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

fulfillment.shipment_lines

Table description: Shipped product, quantity, and lot per shipment.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
shipment_id	BIGINT	NO	—	The shipment this line belongs to.	→ fulfillment.shipments (id)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
order_line_id	BIGINT	NO	—	The customer order line this record belongs to.	→ fulfillment.customer_order_lines (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
shipped_qty	NUMERIC(18,4)	NO	—	Quantity shipped to the customer on this line to date.	—
lot_batch_id	BIGINT	YES	—	The lot or batch this record references, when applicable.	→ inventory.lot_batches (id)
reservation_id	BIGINT	YES	—	The inventory reservation consumed by this shipment line, if applicable.	→ fulfillment.inventory_reservations (id)
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

transfer.transfer_requests

Table description: Formal request to move stock between locations; initial lifecycle state of Stock Transfer aggregate.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
request_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
source_location_id	BIGINT	NO	—	The location stock is being transferred from.	→ network.inventory_locations (id)
destination_location_id	BIGINT	NO	—	The location stock is being transferred to.	→ network.inventory_locations (id)
request_reason	VARCHAR(128)	YES	—	Business reason for requesting the inter-location transfer.	—
request_status	VARCHAR(32)	NO	'SUBMITTED'	Approval status of the transfer request.	—
requested_by	BIGINT	NO	—	The user who submitted this request.	→ identity.user_accounts (id)
approved_by	BIGINT	YES	—	The user who approved this record or transaction.	→ identity.user_accounts (id)
approved_at	TIMESTAMPTZ	YES	—	Date and time when approval was granted.	—
rejection_reason	TEXT	YES	—	Explanation provided when the transfer request was rejected.	—
stock_transfer_id	BIGINT	YES	—	The stock transfer this record belongs to.	→ transfer.stock_transfers (id)
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

transfer.transfer_request_lines

Table description: Product and quantity requested on a transfer request.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
transfer_request_id	BIGINT	NO	—	The transfer request this record belongs to.	→ transfer.transfer_requests (id)
line_number	INTEGER	NO	—	Line sequence number within the parent transfer request.	—
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
requested_qty	NUMERIC(18,4)	NO	—	Requested quantity.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

transfer.stock_transfers

Table description: Executed inter-location stock movement from dispatch to receipt.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
transfer_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
transfer_request_id	BIGINT	YES	—	The transfer request this record belongs to.	→ transfer.transfer_requests (id)
source_location_id	BIGINT	NO	—	The location stock is being transferred from.	→ network.inventory_locations (id)
destination_location_id	BIGINT	NO	—	The location stock is being transferred to.	→ network.inventory_locations (id)
transfer_status	VARCHAR(32)	NO	'APPROVED'	Lifecycle status of the stock transfer execution.	—
dispatch_date	TIMESTAMPTZ	YES	—	Date and time when stock was dispatched from the source location.	—
receipt_date	TIMESTAMPTZ	YES	—	Date and time when stock was received at the destination.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

transfer.stock_transfer_lines

Table description: Product, quantity, and lot on a stock transfer.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
stock_transfer_id	BIGINT	NO	—	The stock transfer this record belongs to.	→ transfer.stock_transfers (id)
line_number	INTEGER	NO	—	Line sequence number within the parent stock transfer.	—
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
quantity	NUMERIC(18,4)	NO	—	Quantity being transferred on this line.	—
lot_batch_id	BIGINT	YES	—	The lot or batch this record references, when applicable.	→ inventory.lot_batches (id)
dispatched_qty	NUMERIC(18,4)	NO	0	Quantity actually dispatched from the source.	—
received_qty	NUMERIC(18,4)	NO	0	Quantity actually received at the destination.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

transfer.in_transit_stock

Table description: Pipeline inventory between dispatch and destination receipt.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
in_transit_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
stock_transfer_id	BIGINT	NO	—	The stock transfer this record belongs to.	→ transfer.stock_transfers (id)
stock_transfer_line_id	BIGINT	NO	—	The stock transfer line this in-transit record belongs to.	→ transfer.stock_transfer_lines (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
quantity	NUMERIC(18,4)	NO	—	Quantity currently in transit between locations.	—
source_location_id	BIGINT	NO	—	The location stock is being transferred from.	→ network.inventory_locations (id)
destination_location_id	BIGINT	NO	—	The location stock is being transferred to.	→ network.inventory_locations (id)
dispatch_date	TIMESTAMPTZ	NO	—	Date and time when goods were dispatched for transfer.	—
expected_receipt_date	DATE	YES	—	Expected date when in-transit stock will arrive.	—
in_transit_status	VARCHAR(32)	NO	'IN_TRANSIT'	Status of stock while in transit.	—
carrier_reference	VARCHAR(128)	YES	—	Carrier reference.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

accuracy.cycle_counts

Table description: Scheduled partial physical count assignment with blind count protocol.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
count_assignment_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
schedule_date	DATE	NO	—	Calendar date when the cycle count is scheduled.	—
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
storage_position_id	BIGINT	YES	—	The storage position (bin/shelf) where stock is or was located.	→ network.storage_positions (id)
counter_id	BIGINT	NO	—	The user assigned to perform this count.	→ identity.user_accounts (id)
count_status	VARCHAR(32)	NO	'SCHEDULED'	Status of the cycle count assignment.	—
physical_count_qty	NUMERIC(18,4)	YES	—	Quantity physically counted by the operator.	—
system_count_qty	NUMERIC(18,4)	YES	—	System on-hand quantity revealed only after count submission (blind count).	—
completed_at	TIMESTAMPTZ	YES	—	Date and time when the count was completed.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

accuracy.physical_inventory_counts

Table description: Wall-to-wall count event imposing location freeze (BR-024).

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
count_event_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
count_start_date	TIMESTAMPTZ	NO	—	Date and time when the wall-to-wall physical count began.	—
count_completion_date	TIMESTAMPTZ	YES	—	Date and time when the wall-to-wall physical count ended.	—
freeze_status	VARCHAR(32)	NO	'ACTIVE'	Freeze status during the physical inventory count.	—
total_variances	INTEGER	NO	0	Number of variances discovered during the count event.	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
certification_status	VARCHAR(32)	NO	'IN_PROGRESS'	Certification status of the physical count event.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

accuracy.count_variances

Table description: System vs. physical quantity difference requiring investigation.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
variance_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
cycle_count_id	BIGINT	YES	—	The cycle count assignment that produced this variance, if applicable.	→ accuracy.cycle_counts (id)
physical_count_id	BIGINT	YES	—	The physical inventory count event that produced this variance, if applicable.	→ accuracy.physical_inventory_counts (id)
stock_position_id	BIGINT	NO	—	The stock position (product at location) this record affects.	→ inventory.stock_positions (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
system_qty	NUMERIC(18,4)	NO	—	System on-hand quantity at the time of the count.	—
physical_qty	NUMERIC(18,4)	NO	—	Quantity physically counted during the inventory event.	—
variance_qty	NUMERIC(18,4)	NO	—	Variance quantity calculated as physical minus system quantity.	—
variance_value	NUMERIC(18,2)	YES	—	Estimated monetary value of the quantity variance.	—
investigation_status	VARCHAR(32)	NO	'OPEN'	Investigation status of the count variance.	—
investigation_findings	TEXT	YES	—	Notes documenting the root cause found during investigation.	—
approval_status	VARCHAR(32)	NO	'PENDING'	Approval status for the variance or adjustment.	—
approver_id	BIGINT	YES	—	The user who approved or rejected this variance.	→ identity.user_accounts (id)
approved_at	TIMESTAMPTZ	YES	—	Date and time when approval was granted.	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

accuracy.stock_adjustments

Table description: Authorized correction to stock position with mandatory reason code.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
adjustment_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
stock_position_id	BIGINT	NO	—	The stock position (product at location) this record affects.	→ inventory.stock_positions (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
adjustment_qty	NUMERIC(18,4)	NO	—	Signed adjustment quantity (positive increases, negative decreases stock).	—
qty_before	NUMERIC(18,4)	NO	—	On-hand quantity before the adjustment was applied.	—
qty_after	NUMERIC(18,4)	NO	—	On-hand quantity after the adjustment was applied.	—
reason_code_id	BIGINT	NO	—	The standardized reason code explaining this transaction.	→ governance.reason_codes (id)
count_variance_id	BIGINT	YES	—	The count variance that triggered this stock adjustment, if applicable.	→ accuracy.count_variances (id)
approval_status	VARCHAR(32)	NO	'PENDING'	Approval status for the variance or adjustment.	—
initiated_by	BIGINT	NO	—	The user who initiated this transaction.	→ identity.user_accounts (id)
approved_by	BIGINT	YES	—	The user who approved this record or transaction.	→ identity.user_accounts (id)
adjustment_date	TIMESTAMPTZ	NO	NOW()	Date and time when the adjustment or write-off takes effect.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

accuracy.inventory_write_offs

Table description: Permanent authorized removal of inventory value.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
write_off_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
stock_position_id	BIGINT	NO	—	The stock position (product at location) this record affects.	→ inventory.stock_positions (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
write_off_qty	NUMERIC(18,4)	NO	—	Quantity permanently removed from inventory.	—
financial_value	NUMERIC(18,2)	YES	—	Monetary value impact of the write-off.	—
reason_code_id	BIGINT	NO	—	The standardized reason code explaining this transaction.	→ governance.reason_codes (id)
evidence_reference	VARCHAR(128)	YES	—	Reference to supporting documentation for the write-off.	—
disposition_method	VARCHAR(32)	YES	—	Physical disposition method for written-off goods.	—
return_disposition_id	BIGINT	YES	—	The return disposition linked to this write-off, if applicable.	→ returns.return_dispositions (id)
initiated_by	BIGINT	NO	—	The user who initiated this transaction.	→ identity.user_accounts (id)
authorized_by	BIGINT	NO	—	The user who authorized this action.	→ identity.user_accounts (id)
write_off_date	TIMESTAMPTZ	NO	NOW()	Date and time when the adjustment or write-off takes effect.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

returns.return_authorizations

Table description: RMA gating customer return before warehouse acceptance (BR-025).

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
rma_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
customer_order_id	BIGINT	NO	—	The customer order this record belongs to.	→ fulfillment.customer_orders (id)
customer_reference	VARCHAR(128)	YES	—	Customer identifier.	—
return_reason	VARCHAR(128)	YES	—	Customer-stated reason for returning the goods.	—
authorization_date	TIMESTAMPTZ	NO	NOW()	Date and time when the return authorization was issued.	—
expiration_date	DATE	YES	—	Date after which the return authorization is no longer valid.	—
disposition_expectation	VARCHAR(64)	YES	—	Expected outcome for returned goods (restock, scrap, etc.).	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
authorization_status	VARCHAR(32)	NO	'ACTIVE'	Status of the return authorization.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

returns.return_authorization_lines

Table description: Authorized return products and quantities on an RMA.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
return_authorization_id	BIGINT	NO	—	The return authorization (RMA) governing this return.	→ returns.return_authorizations (id)
line_number	INTEGER	NO	—	Line sequence number within the parent return authorization.	—
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
authorized_qty	NUMERIC(18,4)	NO	—	Quantity the customer is authorized to return.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

returns.customer_returns

Table description: Physical receipt of customer-returned goods.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
return_number	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
return_authorization_id	BIGINT	NO	—	The return authorization (RMA) governing this return.	→ returns.return_authorizations (id)
customer_order_id	BIGINT	NO	—	The customer order this record belongs to.	→ fulfillment.customer_orders (id)
receiving_location_id	BIGINT	NO	—	The location where goods will be or were received.	→ network.inventory_locations (id)
receipt_date	TIMESTAMPTZ	NO	NOW()	Date and time when goods were received at the dock.	—
inspection_status	VARCHAR(32)	NO	'PENDING'	Inspection status of the customer return receipt.	—
receiving_clerk_id	BIGINT	NO	—	The receiving clerk who processed this receipt.	→ identity.user_accounts (id)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

returns.customer_return_lines

Table description: Returned product, quantity, and condition at receipt.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
customer_return_id	BIGINT	NO	—	The customer return receipt this record belongs to.	→ returns.customer_returns (id)
line_number	INTEGER	NO	—	Line sequence number within the parent customer return.	—
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
returned_qty	NUMERIC(18,4)	NO	—	Returned quantity.	—
condition_at_receipt	VARCHAR(32)	NO	—	Condition of returned goods at receipt.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

returns.return_dispositions

Table description: Inspection outcome — restock, quarantine, or write-off.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
disposition_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
customer_return_id	BIGINT	NO	—	The customer return receipt this record belongs to.	→ returns.customer_returns (id)
return_line_id	BIGINT	NO	—	The customer return line this disposition applies to.	→ returns.customer_return_lines (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
quantity	NUMERIC(18,4)	NO	—	Disposition quantity.	—
condition_grade	VARCHAR(32)	NO	—	Quality grade assigned during return inspection.	—
disposition_decision	VARCHAR(32)	NO	—	Disposition decision for returned goods.	—
restock_location_id	BIGINT	YES	—	The location where returned goods will be restocked, if applicable.	→ network.inventory_locations (id)
quarantine_hold_id	BIGINT	YES	—	The quarantine hold created when goods are quarantined on return.	→ inventory.quarantine_holds (id)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
write_off_id	BIGINT	YES	—	The inventory write-off created when returned goods are written off.	→ accuracy.inventory_write_offs (id)
inspector_id	BIGINT	NO	—	The user who inspected and determined the disposition.	→ identity.user_accounts (id)
inspection_date	TIMESTAMPTZ	NO	NOW()	Date and time when the return inspection was completed.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

commercial.promotional_stock_allocations

Table description: Ring-fenced stock for promotional campaigns or product launches (BR-029).

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
allocation_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
allocated_qty	NUMERIC(18,4)	NO	—	Quantity reserved exclusively for the promotional campaign.	—
campaign_name	VARCHAR(256)	NO	—	Name of the promotional campaign or product launch.	—
channel	VARCHAR(32)	YES	—	Sales channel to which the promotional allocation applies.	—
effective_start_date	DATE	NO	—	First date the promotional allocation is effective.	—
effective_end_date	DATE	NO	—	Last date the promotional allocation is effective.	—
readiness_status	VARCHAR(32)	NO	'PLANNED'	Readiness lifecycle status of the promotional allocation.	—
commercial_approval_ref	VARCHAR(64)	YES	—	Reference documenting commercial team approval.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

commercial.promotional_allocation_locations

Table description: Locations supplying ring-fenced promotional stock.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
allocation_id	BIGINT	NO	—	The promotional stock allocation this location assignment belongs to.	→ commercial.promotional_stock_allocations (id)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
stock_position_id	BIGINT	YES	—	The stock position (product at location) this record affects.	→ inventory.stock_positions (id)
allocated_qty_at_location	NUMERIC(18,4)	NO	—	Portion of the promotional allocation ring-fenced at this location.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

identity.user_accounts

Table description: Authenticated user identity and accountability anchor for all operations.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
user_identifier	VARCHAR(64)	NO	—	Unique login or employee identifier used for authentication.	Unique business key
display_name	VARCHAR(256)	NO	—	The user's full name as displayed in the application.	—
email	VARCHAR(256)	YES	—	Email address for contacting this person or organization.	—
account_status	VARCHAR(32)	NO	'ACTIVE'	Account status controlling user access.	—
effective_from	DATE	NO	—	Date from which the user account is effective.	—
last_activity_at	TIMESTAMPTZ	YES	—	Timestamp of the user's most recent system activity.	—
has_sod_conflict	BOOLEAN	NO	FALSE	Indicates a segregation-of-duties conflict exists for this user.	—
version	INTEGER	NO	1	Version counter incremented on each update to prevent conflicting concurrent edits.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

identity.role_assignments

Table description: Role delegation with permission boundaries per user.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
user_account_id	BIGINT	NO	—	The user account this role assignment belongs to.	→ identity.user_accounts (id)
role_name	VARCHAR(64)	NO	—	Business role assigned to the user.	—
permission_set	VARCHAR(128)	YES	—	Reference to the permission set granted by this role.	—
assignment_status	VARCHAR(32)	NO	'ACTIVE'	Whether this role assignment is currently active.	—
effective_from	DATE	NO	—	Date from which this role assignment is effective.	—
effective_to	DATE	YES	—	Date when this role assignment expires; null means open-ended.	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
sod_compliant	BOOLEAN	NO	TRUE	Indicates whether this assignment complies with segregation-of-duties rules.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—
updated_at	TIMESTAMPTZ	NO	NOW()	Timestamp of the most recent update to this record.	—

identity.role_assignment_locations

Table description: Location scope restricting role permissions to authorized sites.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
role_assignment_id	BIGINT	NO	—	The role assignment this location scope restriction belongs to.	→ identity.role_assignments (id)
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

audit.inventory_movement_records

Table description: Immutable append-only log of every stock quantity change.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record within its monthly partition.	— (primary key)
movement_reference	VARCHAR(64)	NO	—	Reference number linking this movement to its originating business transaction.	—
movement_type	VARCHAR(32)	NO	—	Category of inventory movement recorded.	—
stock_position_id	BIGINT	NO	—	The stock position (product at location) this record affects.	→ inventory.stock_positions (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id) (denormalized)
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id) (denormalized)
quantity_change	NUMERIC(18,4)	NO	—	Signed change in quantity (positive for increases, negative for decreases).	—
qty_before	NUMERIC(18,4)	NO	—	On-hand quantity immediately before this movement.	—
qty_after	NUMERIC(18,4)	NO	—	On-hand quantity immediately after this movement.	—
lot_batch_id	BIGINT	YES	—	The lot or batch this record references, when applicable.	→ inventory.lot_batches (id)
reason_code_id	BIGINT	YES	—	The standardized reason code explaining this transaction.	→ governance.reason_codes (id)

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
source_document_type	VARCHAR(32)	NO	—	Type of business document that caused this movement.	Polymorphic document reference
source_document_id	BIGINT	NO	—	Primary key of the business document that caused this movement.	Polymorphic document reference
performed_by	BIGINT	NO	—	The user who performed this inventory movement.	→ identity.user_accounts (id)
movement_timestamp	TIMESTAMPTZ	NO	NOW()	Business event timestamp used as the monthly partition key.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this row was physically inserted into the database.	—

audit.audit_trail_records

Table description: Governance-grade before/after snapshots for material entity changes.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record within its monthly partition.	— (primary key)
audit_event_reference	VARCHAR(64)	NO	—	Human-readable reference number for this business transaction.	—
affected_entity_type	VARCHAR(64)	NO	—	Name of the entity type that was changed.	—
affected_entity_id	BIGINT	NO	—	Primary key of the record that was changed.	—
action_performed	VARCHAR(32)	NO	—	Type of action performed on the affected entity.	—
user_id	BIGINT	NO	—	The user who performed the audited action.	→ identity.user_accounts (id)
role_at_action	VARCHAR(64)	YES	—	User's role at the moment the action was performed.	—
before_state	JSONB	YES	—	JSON snapshot of the entity state before the change.	—
after_state	JSONB	YES	—	JSON snapshot of the entity state after the change.	—
approval_chain_ref	VARCHAR(128)	YES	—	Reference to the approval workflow chain for this action.	—
movement_record_id	BIGINT	YES	—	The inventory movement record associated with this audit event, if applicable.	→ audit.inventory_movement_records (id)
event_timestamp	TIMESTAMPTZ	NO	NOW()	Business event timestamp used as the monthly partition key.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this row was physically inserted into the database.	—

audit.operational_alerts

Table description: Derived notifications for threshold breaches, pending approvals, and exceptions.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
alert_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
alert_type	VARCHAR(32)	NO	—	Category of operational alert.	—
severity	VARCHAR(16)	NO	'MEDIUM'	Severity level of the alert.	—
product_id	BIGINT	YES	—	The product this record pertains to.	→ catalog.products (id)
location_id	BIGINT	YES	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
stock_position_id	BIGINT	YES	—	The stock position (product at location) this record affects.	→ inventory.stock_positions (id)
source_entity_type	VARCHAR(64)	YES	—	Type of business entity that triggered the alert.	Polymorphic source reference
source_entity_id	BIGINT	YES	—	Primary key of the entity that triggered the alert.	Polymorphic source reference
triggering_condition	TEXT	YES	—	Plain-language description of the condition that triggered the alert.	—
assigned_role	VARCHAR(64)	YES	—	Role responsible for acknowledging and resolving the alert.	—
acknowledgment_status	VARCHAR(32)	NO	'OPEN'	Whether the alert has been acknowledged.	—
resolution_status	VARCHAR(32)	NO	'PENDING'	Whether the underlying issue has been resolved.	—
alert_date	TIMESTAMPTZ	NO	NOW()	Date and time when the alert was generated.	—
resolved_at	TIMESTAMPTZ	YES	—	Date and time when the alert was resolved.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

audit.inventory_valuation_snapshots

Table description: Point-in-time inventory value summary for finance reconciliation (BP-30).

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
snapshot_reference	VARCHAR(64)	NO	—	Unique business identifier used in day-to-day operations.	Unique business key
valuation_date	DATE	NO	—	Calendar date representing the point in time for this valuation.	—
location_scope	VARCHAR(32)	NO	—	Scope of locations included in the valuation snapshot.	—
total_inventory_value	NUMERIC(18,2)	NO	—	Total monetary value of owned inventory in the snapshot.	—
owned_value	NUMERIC(18,2)	NO	—	Monetary value of inventory owned by the organization.	—
consignment_value	NUMERIC(18,2)	NO	0	Monetary value of consignment stock excluded from owned totals.	—

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
gl_variance	NUMERIC(18,2)	YES	—	Difference between inventory valuation and the general ledger balance.	—
reconciliation_status	VARCHAR(32)	NO	'DRAFT'	Reconciliation status against the general ledger.	—
prepared_by	BIGINT	NO	—	The user who prepared this valuation snapshot.	→ identity.user_accounts (id)
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

audit.inventory_valuation_lines

Table description: Product-level quantity and value detail within a valuation snapshot.

Column	Data Type	Nullable	Default	Business Meaning	Related Entity
id	BIGINT	NO	IDENTITY	System-generated unique identifier for this record.	— (primary key)
snapshot_id	BIGINT	NO	—	The valuation snapshot this detail line belongs to.	→ audit.inventory_valuation_snapshots (id)
product_id	BIGINT	NO	—	The product this record pertains to.	→ catalog.products (id)
location_id	BIGINT	NO	—	The inventory location this record pertains to.	→ network.inventory_locations (id)
quantity	NUMERIC(18,4)	NO	—	On-hand quantity captured in the valuation snapshot.	—
unit_cost	NUMERIC(18,4)	NO	—	Cost per unit used to value inventory in the snapshot.	—
extended_value	NUMERIC(18,2)	NO	—	Extended value calculated as quantity multiplied by unit cost.	—
ownership_type	VARCHAR(32)	NO	'OWNED'	OWNED, CONSIGNMENT.	—
created_at	TIMESTAMPTZ	NO	NOW()	Timestamp when this record was first created in the system.	—

Appendix A: Data Type Legend

Data Type	Business Meaning
BIGINT	Large whole number; used for primary keys and foreign key references
INTEGER	Whole number for counts, days, and version counters
NUMERIC(18,4)	Precise decimal for quantities — never floating-point
NUMERIC(18,2)	Precise decimal for monetary amounts
NUMERIC(3,2)	Decimal with limited precision (e.g., ratings 0.00–5.00)
VARCHAR(n)	Variable-length text up to n characters; used for codes, names, and status values
CHAR(n)	Fixed-length text; used for ISO codes (country, currency)
TEXT	Unlimited-length text for descriptions, notes, and evidence
BOOLEAN	True/false flag
DATE	Calendar date without time component
TIMESTAMPTZ	Date and time stored in UTC with timezone awareness

Data Type	Business Meaning
`JSONB`	Structured JSON snapshot for audit before/after state

Appendix B: Standard Audit Columns

Many tables share common audit and control columns. Their business purpose is consistent across the model:

Column	Typical Presence	Business Meaning
`id`	All tables	Surrogate primary key generated automatically by the database
`version`	Mutable transactional headers	Optimistic lock counter preventing lost updates during concurrent edits
`is_active`	Master/reference data	Soft-delete flag; inactive rows are retained but excluded from operational use
`created_at`	Most mutable tables	When the record was first created
`updated_at`	Most mutable tables	When the record was last modified
`created_by`	Documents requiring attribution	User who created the record
`line_number`	Document line tables	Sequence position of the line within its parent document

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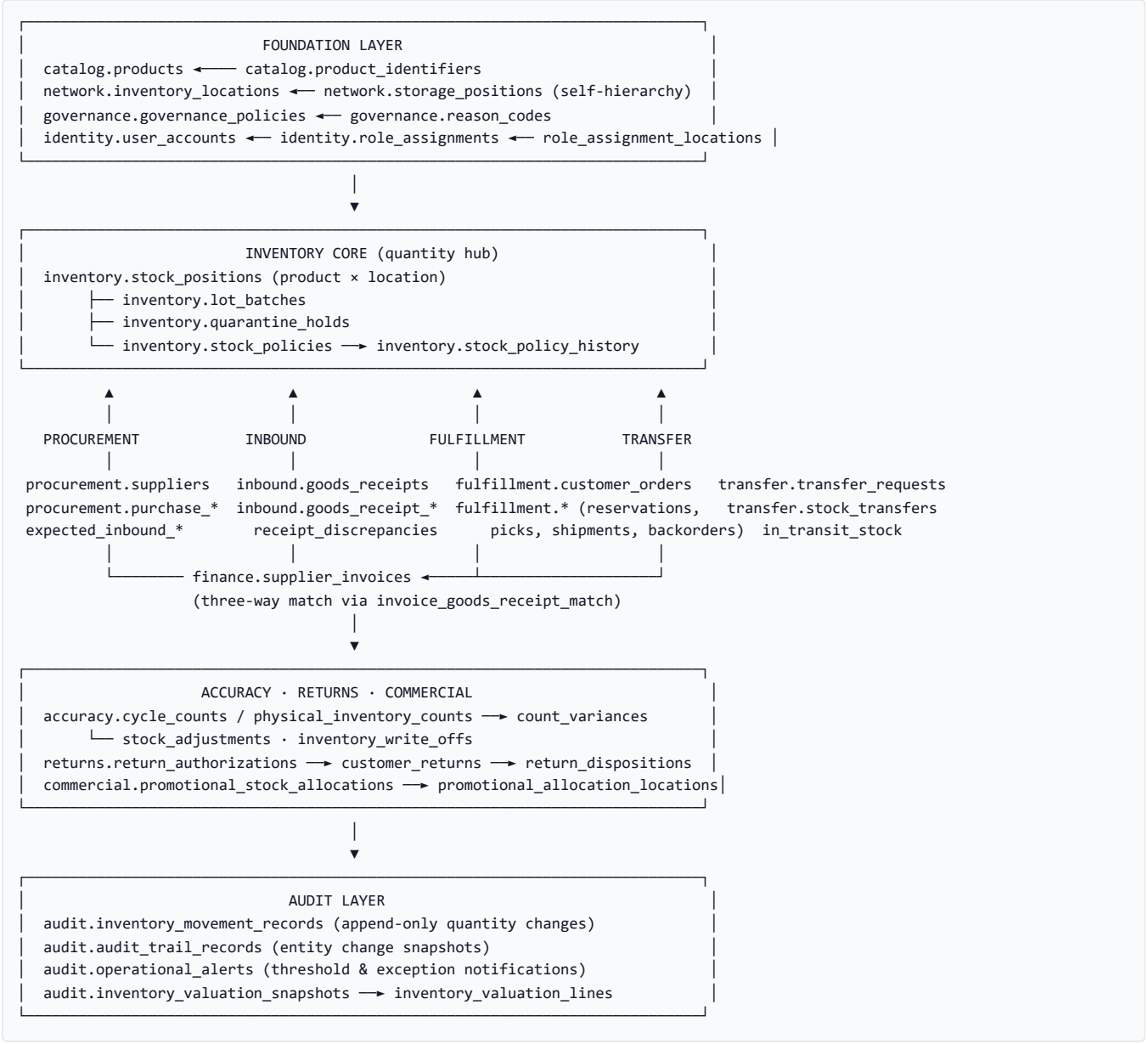
Appendix C: Entity Relationship Overview

High-level data flow and relationships across bounded contexts:



Key relationship patterns:

- **Product × Location hub:** `inventory.stock_positions` is the central quantity record; most operational tables reference it directly or via `product_id` + `location_id`.
- **Document header / line pattern:** Requisitions, POs, orders, receipts, transfers, and returns follow a header table with child line tables linked by parent FK.
- **Procure-to-receive:** `procurement.purchase_orders` → `inbound.goods_receipts` → `inbound.put_aways` → stock position update.
- **Order-to-ship:** `fulfillment.customer_orders` → `inventory_reservations` → `fulfillment_executions` → `shipments`.
- **Three-way match:** `finance.supplier_invoices` linked to `inbound.goods_receipts` via `invoice_goods_receipt_match`, matched against `purchase_order_lines`.
- **Audit trail:** Every material quantity change posts to `audit.inventory_movement_records`; governed entity changes post to `audit.audit_trail_records`.

End of Data Dictionary

Document ID: SRS-DB-007 | Version 1.0 | July 6, 2026

Database Keys Specification

Inventory Management System — PostgreSQL

Document Information

Attribute	Detail
Document ID	SRS-DB-003
Document Title	Database Keys — Primary, Foreign, Unique, and Candidate Keys
Version	1.0
Status	Draft
Document Owner	Database Architect
Source Documents	SRS-DB-002 (Physical Data Model, v1.0), SRS-DB-001 (Logical Data Model, v1.0)
Target Platform	PostgreSQL 15+
Date	July 6, 2026
Scope	Key definitions and business justification — no DDL/SQL statements

Purpose

This document defines all **database keys** for the 59 tables in the Inventory Management System physical model. For each table it identifies primary keys, foreign keys, unique keys, and candidate keys, with business justification aligned to domain rules (BR-001 – BR-032) and aggregate boundaries.

Key Design Principles

Principle	Rationale
Surrogate primary key	Every table uses `id BIGINT GENERATED ALWAYS AS IDENTITY` as PK — stable, narrow, immune to business identifier format changes
Business key as unique constraint	Natural identifiers (`sku`, `po_number`, `order_number`) enforced as `UNIQUE` alternate keys for human lookup and external integration
Composite uniqueness for documents	Header + `line_number` uniqueness on all document line tables prevents duplicate lines and supports ordered business documents
Dimensional uniqueness	Product + Location composite UK on stock positions and policies enforces "one balance per SKU per site" invariant
Referential integrity via FK	Parent-child and cross-schema references use foreign keys; optional deferred FKs on cross-bounded-context links where eventual consistency applies
Candidate keys documented	All business-viable unique identifiers recorded even when not enforced as UK — guides future constraints and integration contracts

Key Notation

Symbol	Meaning
PK	Primary Key — cluster identity, join anchor
FK	Foreign Key — referential integrity to parent table
UK	Unique Key — enforced uniqueness constraint
CK	Candidate Key — business identifier capable of unique identification; may or may not be enforced as UK

Domain 1: Catalog

`catalog.products`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_products`	Stable internal reference for all inventory transactions; SKU format may evolve without breaking FKs across millions of movement records
UK	`sku`	`uk_products_sku`	SKU is the canonical business identity (BR-027 gatekeeper); must be globally unique — two products cannot share a stock keeping unit
FK	`created_by → identity.user_accounts.id`	`fk_products_created_by`	Attributes product master creation to an accountable user for audit (BO-19)
CK	`sku`	—	Primary natural key; selected as UK because it is the ubiquitous language term used in all operational documents
CK	`(sku, lifecycle_status)`	—	Not unique globally but identifies product state context; not enforced — status changes would violate UK

`catalog.product_identifiers`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_product_identifiers`	Surrogate PK for alternate identifiers that may be added/retired without affecting product PK
FK	`product_id → catalog.products.id`	`fk_product_identifiers_product`	Identifier belongs to exactly one product; composition within Product Catalog aggregate
UK	`(identifier_type, identifier_value)`	`uk_product_identifiers_type_value`	A barcode/GTIN must map to one product globally — duplicate scan codes cause wrong picks and receipts (BR-004)
UK	`(product_id, identifier_type, identifier_value)`	`uk_product_identifiers_product_type_value`	Prevents duplicate identifier rows on same product
CK	`identifier_value` (when `is_primary = TRUE`)	—	Primary scan code per product; enforced via partial unique index concept: one primary per product

Domain 2: Network

`network.inventory_locations`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_inventory_locations`	Stable anchor for all stock balances, receipts, and fulfillments at a site
UK	`location_code`	`uk_inventory_locations_code`	Location code is the business identity used on POs, transfers, and reports; must be unique across the network
FK	`parent_location_id → network.inventory_locations.id`	`fk_inventory_locations_parent`	Optional hierarchy (region → site); self-reference enables roll-up reporting without duplicating location records
FK	`responsible_manager_id → identity.user_accounts.id`	`fk_inventory_locations_manager`	Assigns operational accountability for location performance

Key Type	Column(s)	Constraint Name (logical)	Business Justification
CK	`location_code`	—	Primary natural key; used in cross-aggregate identity references per DDD architecture
CK	`(location_code, operational_status)`	—	Identifies location in a given state; not UK because status transitions are frequent

`network.storage_positions`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_storage_positions`	Surrogate PK for bin-level addresses that may be reconfigured
FK	`location_id → network.inventory_locations.id`	`fk_storage_positions_location`	Position cannot exist outside parent location — composition within Inventory Location aggregate (REL-004)
FK	`dedicated_product_id → catalog.products.id`	`fk_storage_positions_product`	Optional dedicated-product bin assignment
UK	`(location_id, position_code)`	`uk_storage_positions_location_code`	Position code unique within location only — "A-01-03" may repeat at different warehouses
CK	`(location_id, zone, aisle, bin)`	—	Physical address composite; not enforced as UK because addressing scheme may have gaps

Domain 3: Governance

`governance.governance_policies`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_governance_policies`	Supports policy versioning without recycling business codes
UK	`(policy_code, effective_from)`	`uk_governance_policies_code_effective`	One version per policy code per effective period — prevents overlapping policy versions (BR-021)
CK	`policy_code`	—	Human-readable policy reference; not alone unique due to versioning
CK	`(policy_type, applicable_scope, effective_from)`	—	Identifies active policy for a scope at a point in time; used by application lookup

`governance.reason_codes`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_reason_codes`	Stable reference for historical adjustments even after code retirement
UK	`reason_code`	`uk_reason_codes_code`	Reason code is the business taxonomy key — mandatory on adjustments (BR-020); must not duplicate
FK	`approval_threshold_id → governance.governance_policies.id`	`fk_reason_codes_policy`	Links reason category to approval threshold when material
CK	`reason_code`	—	Primary natural key; selected as UK

Domain 4: Inventory Balance

inventory.stock_positions

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_stock_positions	Hot-path join key for reservations, movements, and adjustments — narrow BIGINT for lock efficiency
UK	(product_id, location_id)	uk_stock_positions_product_location	**Core domain invariant** — exactly one balance per SKU per location; prevents split-brain ATP (BR-001, REL-019, REL-020)
FK	product_id → catalog.products.id	fk_stock_positions_product	Balance tracks a valid catalog product
FK	location_id → network.inventory_locations.id	fk_stock_positions_location	Balance anchored to a valid location
CK	(product_id, location_id)	—	Primary candidate key; enforced as UK
CK	(product_id, location_id, ownership_type)	—	Not unique — ownership is attribute not identity

inventory.lot_batches

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_lot_batches	Surrogate PK; lot numbers may repeat across products
FK	stock_position_id → inventory.stock_positions.id	fk_lot_batches_stock_position	Lot quantity belongs to parent stock position — composition (REL-023)
FK	product_id → catalog.products.id	fk_lot_batches_product	Denormalized for FEFO queries without joining through stock position
FK	supplier_id → procurement.suppliers.id	fk_lot_batches_supplier	Traceability to supplier origin for recall (BR-017, REL-134)
UK	(product_id, lot_number, stock_position_id)	uk_lot_batches_product_lot_position	Same lot may exist at one position once; supports lot-level quantity tracking
CK	(product_id, lot_number)	—	Lot identity within product scope per domain model; may span positions via separate rows

inventory.quarantine_holds

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_quarantine_holds	Surrogate PK for hold lifecycle management
UK	hold_reference	uk_quarantine_holds_reference	Business document number for hold tracking and customs clearance (BR-030)
FK	stock_position_id → inventory.stock_positions.id	fk_quarantine_holds_stock_position	Hold applied to specific balance — composition (REL-022)
FK	lot_batch_id → inventory.lot_batches.id	fk_quarantine_holds_lot	Optional lot-scoped hold for targeted recall (REL-028)
FK	authorized_by → identity.user_accounts.id	fk_quarantine_holds_authorized_by	Hold authorization accountability

Key Type	Column(s)	Constraint Name (logical)	Business Justification
CK	`hold_reference`	—	Primary business document key

`inventory.stock_policies`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_stock_policies`	Surrogate PK; thresholds versioned via history table
UK	`(product_id, location_id)`	`uk_stock_policies_product_location`	One replenishment policy per SKU per location — mirrors stock position identity (REL-008, REL-009)
FK	`product_id` → `catalog.products.id`	`fk_stock_policies_product`	Policy scoped to product
FK	`location_id` → `network.inventory_locations.id`	`fk_stock_policies_location`	Policy scoped to location
CK	`(product_id, location_id)`	—	Primary candidate key; same composite as stock position

`inventory.stock_policy_history`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_stock_policy_history`	Append-only history rows
FK	`stock_policy_id` → `inventory.stock_policies.id`	`fk_stock_policy_history_policy`	History belongs to parent policy
FK	`changed_by` → `identity.user_accounts.id`	`fk_stock_policy_history_changed_by`	Threshold change attribution
UK	`(stock_policy_id, effective_from)`	`uk_stock_policy_history_policy_from`	Non-overlapping history periods per policy
CK	`(stock_policy_id, effective_from, effective_to)`	—	Identifies a specific threshold version window

Domain 5: Procurement

`procurement.suppliers`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_suppliers`	Stable vendor reference across PO, receipt, and invoice chains
UK	`supplier_code`	`uk_suppliers_code`	Supplier code is commercial identity — unique counterparty for procurement (BR-007, BR-008)
CK	`supplier_code`	—	Primary natural key

`procurement.purchase_requisitions`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_purchase_requisitions`	Surrogate PK for requisition lifecycle

Key Type	Column(s)	Constraint Name (logical)	Business Justification
UK	`requisition_number`	`uk_purchase_requisitions_number`	Internal document number for approval workflow traceability
FK	`requesting_location_id` → `network.inventory_locations.id`	`fk_purchase_requisitions_location`	Replenishment need anchored to destination site (REL-036)
FK	`triggering_alert_id` → `audit.operational_alerts.id`	`fk_purchase_requisitions_alert`	Links requisition to low-stock alert that triggered it (REL-038, BR-009)
FK	`approved_by` → `identity.user_accounts.id`	`fk_purchase_requisitions_approved_by`	Approval accountability
FK	`purchase_order_id` → `procurement.purchase_orders.id`	`fk_purchase_requisitions_po`	Conversion traceability — requisition to PO (REL-037)
FK	`created_by` → `identity.user_accounts.id`	`fk_purchase_requisitions_created_by`	Initiator attribution
CK	`requisition_number`	—	Primary business document key

`procurement.purchase_requisition_lines`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_purchase_requisition_lines`	Line-level surrogate PK
FK	`requisition_id` → `procurement.purchase_requisitions.id`	`fk_purchase_requisition_lines_req`	Line belongs to parent requisition
FK	`product_id` → `catalog.products.id`	`fk_purchase_requisition_lines_product`	Line references active product — discontinued blocked (BR-027)
UK	`(requisition_id, line_number)`	`uk_purchase_requisition_lines_req_line`	Ordered line sequence unique within requisition
CK	`(requisition_id, product_id)`	—	Not UK — same product could theoretically appear on amended lines; line_number disambiguates

`procurement.purchase_orders`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_purchase_orders`	Stable PO reference for receipt and three-way match
UK	`po_number`	`uk_purchase_orders_number`	PO number is authorization document for inbound receipt (BR-007)
FK	`supplier_id` → `procurement.suppliers.id`	`fk_purchase_orders_supplier`	Commercial commitment to vendor (REL-039)
FK	`receiving_location_id` → `network.inventory_locations.id`	`fk_purchase_orders_location`	Designated receipt site (REL-040)
FK	`created_by` → `identity.user_accounts.id`	`fk_purchase_orders_created_by`	Buyer attribution
CK	`po_number`	—	Primary natural key
CK	`(supplier_id, po_number)`	—	Redundant when po_number is globally unique; supplier-scoped numbering alternative if numbering scheme changes

`procurement.purchase_order_lines`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_purchase_order_lines`	Line surrogate PK
FK	`purchase_order_id` → `procurement.purchase_orders.id`	`fk_purchase_order_lines_po`	Line belongs to PO
FK	`product_id` → `catalog.products.id`	`fk_purchase_order_lines_product`	Ordered product — discontinued prohibited (BR-027, REL-041)
UK	`(purchase_order_id, line_number)`	`uk_purchase_order_lines_po_line`	Unique line sequence within PO
CK	`(purchase_order_id, product_id)`	—	Identifies product on PO; not UK if same SKU ordered on multiple lines intentionally

`procurement.expected_inbound_deliveries`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_expected_inbound_deliveries`	Surrogate PK
UK	`delivery_reference`	`uk_expected_inbound_deliveries_ref`	Shipment plan business identifier
FK	`purchase_order_id` → `procurement.purchase_orders.id`	`fk_expected_inbound_deliveries_po`	Delivery subordinate to PO — composition (REL-042)
FK	`receiving_location_id` → `network.inventory_locations.id`	`fk_expected_inbound_deliveries_location`	Expected arrival site
CK	`delivery_reference`	—	Primary business key

`procurement.expected_inbound_delivery_lines`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_expected_inbound_delivery_lines`	Line surrogate PK
FK	`delivery_id` → `procurement.expected_inbound_deliveries.id`	`fk_expected_inbound_delivery_lines_delivery`	Line belongs to delivery plan
FK	`purchase_order_line_id` → `procurement.purchase_order_lines.id`	`fk_expected_inbound_delivery_lines_pol`	Links expected qty to authorized PO line (REL-045)
FK	`product_id` → `catalog.products.id`	`fk_expected_inbound_delivery_lines_product`	Product denormalized for visibility queries
UK	`(delivery_id, purchase_order_line_id)`	`uk_expected_inbound_delivery_lines`	One expected quantity entry per PO line per delivery

Domain 6: Inbound

inbound.goods_receipts

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_goods_receipts	Receipt document anchor for put-away and three-way match
UK	gr_number	uk_goods_receipts_number	Goods receipt number is operational document identity (BR-004)
FK	purchase_order_id → procurement.purchase_orders.id	fk_goods_receipts_po	PO authorization for receipt (BR-007, REL-043); nullable for ad-hoc exception
FK	supplier_id → procurement.suppliers.id	fk_goods_receipts_supplier	Supplier accountability (REL-048)
FK	receiving_location_id → network.inventory_locations.id	fk_goods_receipts_location	Receipt site — blocked when frozen (BR-024, REL-049)
FK	receiving_clerk_id → identity.user_accounts.id	fk_goods_receipts_clerk	Receiving clerk attribution (REL-126)
CK	gr_number	—	Primary business document key

inbound.goods_receipt_lines

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_goods_receipt_lines	Line surrogate PK
FK	goods_receipt_id → inbound.goods_receipts.id	fk_goods_receipt_lines_gr	Line belongs to receipt
FK	purchase_order_line_id → procurement.purchase_order_lines.id	fk_goods_receipt_lines_pol	Verification against PO line
FK	product_id → catalog.products.id	fk_goods_receipt_lines_product	Received product identity
FK	lot_batch_id → inventory.lot_batches.id	fk_goods_receipt_lines_lot	Lot captured at receipt (BR-017, REL-052)
UK	(goods_receipt_id, line_number)	uk_goods_receipt_lines_gr_line	Unique line sequence within receipt

inbound.receipt_discrepancies

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_receipt_discrepancies	Surrogate PK
UK	discrepancy_reference	uk_receipt_discrepancies_ref	Exception document for supplier claims (BR-006)
FK	goods_receipt_id → inbound.goods_receipts.id	fk_receipt_discrepancies_gr	Discrepancy subordinate to receipt — composition (REL-050)
FK	goods_receipt_line_id → inbound.goods_receipt_lines.id	fk_receipt_discrepancies_gr_line	Optional line-level discrepancy
FK	product_id → catalog.products.id	fk_receipt_discrepancies_product	Affected product
FK	reason_code_id → governance.reason_codes.id	fk_receipt_discrepancies_reason	Categorized discrepancy (REL-013)
CK	discrepancy_reference	—	Primary business key

`inbound.put_aways`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_put_aways`	Surrogate PK
UK	`put_away_reference`	`uk_put_aways_reference`	Put-away confirmation document (BR-005)
FK	`goods_receipt_id` → `inbound.goods_receipts.id`	`fk_put_aways_gr`	Put-away completes receipt workflow (REL-051)
FK	`goods_receipt_line_id` → `inbound.goods_receipt_lines.id`	`fk_put_aways_gr_line`	Line-level put-away
FK	`product_id` → `catalog.products.id`	`fk_put_aways_product`	Product placed
FK	`storage_position_id` → `network.storage_positions.id`	`fk_put_aways_position`	Destination bin (REL-017)
FK	`lot_batch_id` → `inventory.lot_batches.id`	`fk_put_aways_lot`	Lot placement for traceable products
FK	`performed_by` → `identity.user_accounts.id`	`fk_put_aways_performed_by`	Warehouse staff attribution
CK	`put_away_reference`	—	Primary business key

Domain 7: Finance

`finance.supplier_invoices`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_supplier_invoices`	Invoice anchor for payment authorization
UK	`(supplier_id, invoice_number)`	`uk_supplier_invoices_supplier_number`	Invoice numbers unique within supplier scope — standard AP practice (REL-057)
FK	`supplier_id` → `procurement.suppliers.id`	`fk_supplier_invoices_supplier`	Billing counterparty
FK	`purchase_order_id` → `procurement.purchase_orders.id`	`fk_supplier_invoices_po`	PO leg of three-way match (BR-008, REL-055)
CK	`invoice_number`	—	Not globally unique — supplier-scoped UK enforced

`finance.supplier_invoice_lines`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_supplier_invoice_lines`	Line surrogate PK
FK	`supplier_invoice_id` → `finance.supplier_invoices.id`	`fk_supplier_invoice_lines_invoice`	Line belongs to invoice
FK	`purchase_order_line_id` → `procurement.purchase_order_lines.id`	`fk_supplier_invoice_lines_pol`	Match to authorized PO line
FK	`product_id` → `catalog.products.id`	`fk_supplier_invoice_lines_product`	Billed product
UK	`(supplier_invoice_id, line_number)`	`uk_supplier_invoice_lines_inv_line`	Unique line sequence

`finance.invoice_goods_receipt_match`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_invoice_goods_receipt_match`	Associative table surrogate PK
FK	`supplier_invoice_id` → `finance.supplier_invoices.id`	`fk_inv_gr_match_invoice`	Invoice in match set
FK	`goods_receipt_id` → `inbound.goods_receipts.id`	`fk_inv_gr_match_gr`	Receipt in match set (BR-008, REL-056)
UK	`(supplier_invoice_id, goods_receipt_id)`	`uk_inv_gr_match_pair`	One match record per invoice-receipt pair

Domain 8: Fulfillment

`fulfillment.customer_orders`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_customer_orders`	Order anchor for reservations, picks, and shipments
UK	`order_number`	`uk_customer_orders_number`	Order number is customer-facing and integration identity
CK	`order_number`	—	Primary natural key
CK	`(customer_reference, order_number)`	—	External CRM correlation when order numbers are channel-scoped

`fulfillment.customer_order_lines`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_customer_order_lines`	Line surrogate PK
FK	`customer_order_id` → `fulfillment.customer_orders.id`	`fk_customer_order_lines_order`	Line belongs to order — composition
FK	`product_id` → `catalog.products.id`	`fk_customer_order_lines_product`	Ordered SKU
FK	`fulfillment_location_id` → `network.inventory_locations.id`	`fk_customer_order_lines_location`	Ship-from site for line
UK	`(customer_order_id, line_number)`	`uk_customer_order_lines_order_line`	Unique ordered line sequence

`fulfillment.inventory_reservations`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_inventory_reservations`	Reservation surrogate PK
UK	`reservation_reference`	`uk_inventory_reservations_ref`	Operational hold document (BR-002)
FK	`customer_order_id` → `fulfillment.customer_orders.id`	`fk_inventory_reservations_order`	Reservation serves order (REL-065)
FK	`order_line_id` → `fulfillment.customer_order_lines.id`	`fk_inventory_reservations_line`	Line-level reservation
FK	`product_id` → `catalog.products.id`	`fk_inventory_reservations_product`	Reserved SKU (REL-066)

Key Type	Column(s)	Constraint Name (logical)	Business Justification
FK	`location_id` → `network.inventory_locations.id`	`fk_inventory_reservations_location`	Reservation location
FK	`stock_position_id` → `inventory.stock_positions.id`	`fk_inventory_reservations_position`	Modifies specific balance — ATP reduction (REL-024)
CK	`reservation_reference`	—	Primary business key

`fulfillment.fulfillment_executions`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_fulfillment_executions`	Pick/pack work unit PK
UK	`fulfillment_reference`	`uk_fulfillment_executions_ref`	Warehouse work document
FK	`customer_order_id` → `fulfillment.customer_orders.id`	`fk_fulfillment_executions_order`	Execution fulfills order (REL-067)
FK	`fulfillment_location_id` → `network.inventory_locations.id`	`fk_fulfillment_executions_location`	Pick site (REL-068)
FK	`assigned_picker_id` → `identity.user_accounts.id`	`fk_fulfillment_executions_picker`	Picker attribution
CK	`fulfillment_reference`	—	Primary business key

`fulfillment.fulfillment_pick_lines`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_fulfillment_pick_lines`	Pick line surrogate PK
FK	`fulfillment_execution_id` → `fulfillment.fulfillment_executions.id`	`fk_fulfillment_pick_lines_fe`	Pick belongs to execution
FK	`order_line_id` → `fulfillment.customer_order_lines.id`	`fk_fulfillment_pick_lines_ol`	Fulfills order line
FK	`product_id` → `catalog.products.id`	`fk_fulfillment_pick_lines_product`	Picked product
FK	`storage_position_id` → `network.storage_positions.id`	`fk_fulfillment_pick_lines_position`	Source bin (REL-018)
FK	`lot_batch_id` → `inventory.lot_batches.id`	`fk_fulfillment_pick_lines_lot`	Lot on pick (BR-017, REL-069)
UK	`(fulfillment_execution_id, order_line_id)`	`uk_fulfillment_pick_lines_fe_ol`	One pick line per order line per execution

`fulfillment.backorders`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_backorders`	Backorder surrogate PK
UK	`backorder_reference`	`uk_backorders_ref`	Customer-agreed delay document (BR-028)
FK	`customer_order_id` → `fulfillment.customer_orders.id`	`fk_backorders_order`	Backorder on order (REL-071)
FK	`order_line_id` → `fulfillment.customer_order_lines.id`	`fk_backorders_line`	Line-level backorder
FK	`product_id` → `catalog.products.id`	`fk_backorders_product`	Awaiting SKU (REL-072)
CK	`backorder_reference`	—	Primary business key

`fulfillment.shipments`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_shipments`	Shipment surrogate PK
UK	`shipment_number`	`uk_shipments_number`	Dispatch document identity (BR-003)
FK	`customer_order_id` → `fulfillment.customer_orders.id`	`fk_shipments_order`	Shipment fulfills order (REL-063)
FK	`fulfillment_execution_id` → `fulfillment.fulfillment_executions.id`	`fk_shipments_fe`	Verified pick required (BR-003, REL-073)
FK	`ship_from_location_id` → `network.inventory_locations.id`	`fk_shipments_location`	Ship-from site
CK	`shipment_number`	—	Primary business key
CK	`tracking_identifier`	—	Carrier tracking when assigned; not UK — carriers may reuse formats

`fulfillment.shipment_lines`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_shipment_lines`	Shipment line surrogate PK
FK	`shipment_id` → `fulfillment.shipments.id`	`fk_shipment_lines_shipment`	Line belongs to shipment
FK	`order_line_id` → `fulfillment.customer_order_lines.id`	`fk_shipment_lines_ol`	Shipped order line
FK	`product_id` → `catalog.products.id`	`fk_shipment_lines_product`	Shipped product
FK	`lot_batch_id` → `inventory.lot_batches.id`	`fk_shipment_lines_lot`	Lot traceability on dispatch
FK	`reservation_id` → `fulfillment.inventory_reservations.id`	`fk_shipment_lines_reservation`	Consumes reservation (REL-074)
UK	`(shipment_id, order_line_id)`	`uk_shipment_lines_shipment_ol`	One shipment line per order line per shipment

Domain 9: Network Transfer

`transfer.transfer_requests`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_transfer_requests`	Request surrogate PK
UK	`request_number`	`uk_transfer_requests_number`	Transfer request document
FK	`source_location_id` → `network.inventory_locations.id`	`fk_transfer_requests_source`	Source site (REL-075, BR-012)
FK	`destination_location_id` → `network.inventory_locations.id`	`fk_transfer_requests_dest`	Destination site (REL-076)
FK	`requested_by` → `identity.user_accounts.id`	`fk_transfer_requests_requester`	Requestor attribution
FK	`approved_by` → `identity.user_accounts.id`	`fk_transfer_requests_approver`	Approver attribution
FK	`stock_transfer_id` → `transfer.stock_transfers.id`	`fk_transfer_requests_transfer`	Request-to-execution link (REL-078)

Key Type	Column(s)	Constraint Name (logical)	Business Justification
CK	`request_number`	—	Primary business key

`transfer.transfer_request_lines`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_transfer_request_lines`	Line surrogate PK
FK	`transfer_request_id` → `transfer.transfer_requests.id`	`fk_transfer_request_lines_req`	Line belongs to request
FK	`product_id` → `catalog.products.id`	`fk_transfer_request_lines_product`	Product to transfer (REL-077)
UK	`(transfer_request_id, line_number)`	`uk_transfer_request_lines_req_line`	Unique line sequence

`transfer.stock_transfers`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_stock_transfers`	Transfer execution PK
UK	`transfer_number`	`uk_stock_transfers_number`	Transfer document identity
FK	`transfer_request_id` → `transfer.transfer_requests.id`	`fk_stock_transfers_request`	Optional request origin (REL-080)
FK	`source_location_id` → `network.inventory_locations.id`	`fk_stock_transfers_source`	Source deduction site
FK	`destination_location_id` → `network.inventory_locations.id`	`fk_stock_transfers_dest`	Destination receipt site
CK	`transfer_number`	—	Primary business key

`transfer.stock_transfer_lines`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_stock_transfer_lines`	Line surrogate PK
FK	`stock_transfer_id` → `transfer.stock_transfers.id`	`fk_stock_transfer_lines_transfer`	Line belongs to transfer
FK	`product_id` → `catalog.products.id`	`fk_stock_transfer_lines_product`	Transferred SKU
FK	`lot_batch_id` → `inventory.lot_batches.id`	`fk_stock_transfer_lines_lot`	Lot on transfer (BR-017, REL-083)
UK	`(stock_transfer_id, line_number)`	`uk_stock_transfer_lines_transfer_line`	Unique line sequence

`transfer.in_transit_stock`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_in_transit_stock`	In-transit surrogate PK
UK	`in_transit_reference`	`uk_in_transit_stock_ref`	Pipeline stock document
FK	`stock_transfer_id` → `transfer.stock_transfers.id`	`fk_in_transit_stock_transfer`	Subordinate to transfer — composition (REL-081)
FK	`stock_transfer_line_id` → `transfer.stock_transfer_lines.id`	`fk_in_transit_stock_line`	Line-level in-transit
FK	`product_id` → `catalog.products.id`	`fk_in_transit_stock_product`	Product in pipeline
FK	`source_location_id` → `network.inventory_locations.id`	`fk_in_transit_stock_source`	Origin site

Key Type	Column(s)	Constraint Name (logical)	Business Justification
FK	`destination_location_id` → `network.inventory_locations.id`	`fk_in_transit_stock_dest`	Destination site
CK	`in_transit_reference`	—	Primary business key

Domain 10: Inventory Accuracy

`accuracy.cycle_counts`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_cycle_counts`	Count assignment PK
UK	`count_assignment_number`	`uk_cycle_counts_number`	Count work document (BR-018)
FK	`location_id` → `network.inventory_locations.id`	`fk_cycle_counts_location`	Count site (REL-086)
FK	`product_id` → `catalog.products.id`	`fk_cycle_counts_product`	Counted SKU (REL-085)
FK	`storage_position_id` → `network.storage_positions.id`	`fk_cycle_counts_position`	Optional bin scope (REL-087)
FK	`counter_id` → `identity.user_accounts.id`	`fk_cycle_counts_counter`	Counter for SoD — must differ from approver (BR-019, REL-089)
CK	`count_assignment_number`	—	Primary business key

`accuracy.physical_inventory_counts`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_physical_inventory_counts`	Wall-to-wall count event PK
UK	`count_event_number`	`uk_physical_inventory_counts_number`	Count event document (BR-024)
FK	`location_id` → `network.inventory_locations.id`	`fk_physical_inventory_counts_location`	Frozen location (REL-007, REL-090)
CK	`count_event_number`	—	Primary business key

`accuracy.count_variances`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_count_variances`	Variance surrogate PK
UK	`variance_reference`	`uk_count_variances_ref`	Variance investigation document
FK	`cycle_count_id` → `accuracy.cycle_counts.id`	`fk_count_variances_cycle`	Cycle count origin (REL-092) — mutually exclusive with physical count
FK	`physical_count_id` → `accuracy.physical_inventory_counts.id`	`fk_count_variances_physical`	Physical count origin (REL-092)
FK	`stock_position_id` → `inventory.stock_positions.id`	`fk_count_variances_position`	Affected balance (REL-093)
FK	`product_id` → `catalog.products.id`	`fk_count_variances_product`	Denormalized product
FK	`location_id` → `network.inventory_locations.id`	`fk_count_variances_location`	Denormalized location

Key Type	Column(s)	Constraint Name (logical)	Business Justification
FK	`approver_id` → `identity.user_accounts.id`	`fk_count_variances_approver`	Independent approver ≠ counter (BR-019, REL-095)
CK	`variance_reference`	—	Primary business key
CK	`(cycle_count_id)` or `(physical_count_id)`	—	Exactly one count origin per variance — application CHECK, not UK

`accuracy.stock_adjustments`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_stock_adjustments`	Adjustment surrogate PK
UK	`adjustment_reference`	`uk_stock_adjustments_ref`	Adjustment document (BR-020)
FK	`stock_position_id` → `inventory.stock_positions.id`	`fk_stock_adjustments_position`	Modified balance (REL-031)
FK	`product_id` → `catalog.products.id`	`fk_stock_adjustments_product`	Denormalized product
FK	`location_id` → `network.inventory_locations.id`	`fk_stock_adjustments_location`	Denormalized location
FK	`reason_code_id` → `governance.reason_codes.id`	`fk_stock_adjustments_reason`	Mandatory reason (BR-020, REL-097)
FK	`count_variance_id` → `accuracy.count_variances.id`	`fk_stock_adjustments_variance`	Count-to-correction traceability (REL-096, BR-022)
FK	`initiated_by` → `identity.user_accounts.id`	`fk_stock_adjustments_initiator`	Initiator (REL-098)
FK	`approved_by` → `identity.user_accounts.id`	`fk_stock_adjustments_approver`	Independent approver (BR-021)
CK	`adjustment_reference`	—	Primary business key

`accuracy.inventory_write_offs`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_inventory_write_offs`	Write-off surrogate PK
UK	`write_off_reference`	`uk_inventory_write_offs_ref`	Write-off document (BR-023)
FK	`stock_position_id` → `inventory.stock_positions.id`	`fk_inventory_write_offs_position`	Permanent quantity removal (REL-032)
FK	`product_id` → `catalog.products.id`	`fk_inventory_write_offs_product`	Denormalized product
FK	`location_id` → `network.inventory_locations.id`	`fk_inventory_write_offs_location`	Denormalized location
FK	`reason_code_id` → `governance.reason_codes.id`	`fk_inventory_write_offs_reason`	Mandatory reason (BR-020)
FK	`return_disposition_id` → `returns.return_dispositions.id`	`fk_inventory_write_offs_disposition`	Return-originated write-off (REL-099)
FK	`initiated_by` → `identity.user_accounts.id`	`fk_inventory_write_offs_initiator`	Initiator
FK	`authorized_by` → `identity.user_accounts.id`	`fk_inventory_write_offs_authorizer`	Authorizer (BR-023, REL-101)
CK	`write_off_reference`	—	Primary business key

Domain 11: Returns

returns.return_authorizations

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_return_authorizations	RMA surrogate PK
UK	rma_number	uk_return_authorizations_rma	RMA is gate document for returns (BR-025)
FK	customer_order_id → fulfillment.customer_orders.id	fk_return_authorizations_order	Validates against original order (REL-105)
CK	rma_number	—	Primary business key

returns.return_authorization_lines

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_return_authorization_lines	Line surrogate PK
FK	return_authorization_id → returns.return_authorizations.id	fk_return_authorization_lines_rma	Line belongs to RMA
FK	product_id → catalog.products.id	fk_return_authorization_lines_product	Authorized return SKU (REL-107)
UK	(return_authorization_id, line_number)	uk_return_authorization_lines_rma_line	Unique line sequence

returns.customer_returns

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_customer_returns	Return receipt surrogate PK
UK	return_number	uk_customer_returns_number	Return receipt document
FK	return_authorization_id → returns.return_authorizations.id	fk_customer_returns_rma	Valid RMA required (BR-025, REL-108)
FK	customer_order_id → fulfillment.customer_orders.id	fk_customer_returns_order	Original order reference
FK	receiving_location_id → network.inventory_locations.id	fk_customer_returns_location	Receipt site (REL-109)
FK	receiving_clerk_id → identity.user_accounts.id	fk_customer_returns_clerk	Clerk attribution
CK	return_number	—	Primary business key

returns.customer_return_lines

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_customer_return_lines	Line surrogate PK
FK	customer_return_id → returns.customer_returns.id	fk_customer_return_lines_return	Line belongs to return
FK	product_id → catalog.products.id	fk_customer_return_lines_product	Returned product
UK	(customer_return_id, line_number)	uk_customer_return_lines_return_line	Unique line sequence

`returns.return_dispositions`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_return_dispositions`	Disposition surrogate PK
UK	`disposition_reference`	`uk_return_dispositions_ref`	Disposition document (BR-026)
FK	`customer_return_id` → `returns.customer_returns.id`	`fk_return_dispositions_return`	Disposition of return (REL-112)
FK	`return_line_id` → `returns.customer_return_lines.id`	`fk_return_dispositions_line`	Line-level disposition
FK	`product_id` → `catalog.products.id`	`fk_return_dispositions_product`	Dispositioned product
FK	`restock_location_id` → `network.inventory_locations.id`	`fk_return_dispositions_location`	Restock target
FK	`quarantine_hold_id` → `inventory.quarantine_holds.id`	`fk_return_dispositions_hold`	Quarantine path (REL-114)
FK	`write_off_id` → `accuracy.inventory_write_offs.id`	`fk_return_dispositions_writeoff`	Write-off path (REL-113)
FK	`inspector_id` → `identity.user_accounts.id`	`fk_return_dispositions_inspector`	Inspector attribution
CK	`disposition_reference`	—	Primary business key

Domain 12: Commercial

`commercial.promotional_stock_allocations`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_promotional_stock_allocations`	Allocation surrogate PK
UK	`allocation_reference`	`uk_promotional_allocations_ref`	Campaign ring-fence document (BR-029)
FK	`product_id` → `catalog.products.id`	`fk_promotional_allocations_product`	Ring-fenced SKU (REL-026)
CK	`allocation_reference`	—	Primary business key
CK	`(product_id, campaign_name, effective_start_date)`	—	Identifies campaign allocation window; not UK if campaigns rerun annually

`commercial.promotional_allocation_locations`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_promotional_allocation_locations`	Junction surrogate PK
FK	`allocation_id` → `commercial.promotional_stock_allocations.id`	`fk_promo_alloc_locations_allocation`	Location scope of allocation (REL-027)
FK	`location_id` → `network.inventory_locations.id`	`fk_promo_alloc_locations_location`	Supplying location
FK	`stock_position_id` → `inventory.stock_positions.id`	`fk_promo_alloc_locations_position`	Specific balance ring-fenced (REL-025)
UK	`(allocation_id, location_id)`	`uk_promo_alloc_locations_alloc_loc`	One allocation entry per location per campaign

Domain 13: Identity & Access

identity.user_accounts

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_user_accounts	User surrogate PK — referenced across all transactional tables for attribution
UK	user_identifier	uk_user_accounts_identifier	Login/employee ID must be unique — accountability anchor (BO-19)
UK	email	uk_user_accounts_email	Optional; unique when present for authentication and notification
CK	user_identifier	—	Primary natural key
CK	email	—	Alternate login identifier when used

identity.role_assignments

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_role_assignments	Assignment surrogate PK
FK	user_account_id → identity.user_accounts.id	fk_role_assignments_user	Assignment belongs to user — composition (REL-123)
UK	(user_account_id, role_name, effective_from)	uk_role_assignments_user_role_from	Same role not double-assigned at same effective date; supports history
CK	(user_account_id, role_name)	—	Active role lookup; not UK due to effective dating

identity.role_assignment_locations

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_role_assignment_locations	Junction surrogate PK
FK	role_assignment_id → identity.role_assignments.id	fk_role_assignment_locations_assignment	Scope belongs to assignment
FK	location_id → network.inventory_locations.id	fk_role_assignment_locations_location	Authorized site (REL-124)
UK	(role_assignment_id, location_id)	uk_role_assignment_locations	One scope entry per location per assignment

Domain 14: Audit & Reporting

audit.inventory_movement_records

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	id	pk_inventory_movement_records	Partition-local surrogate PK; append-only
UK	(movement_reference, movement_timestamp)	uk_movement_records_ref_ts	Reference unique within partition window; supports idempotent event ingestion
FK	stock_position_id → inventory.stock_positions.id	fk_movement_records_position	Affected balance (REL-115)
FK	product_id → catalog.products.id	fk_movement_records_product	Denormalized for partition-local queries
FK	location_id → network.inventory_locations.id	fk_movement_records_location	Denormalized for reporting

Key Type	Column(s)	Constraint Name (logical)	Business Justification
FK	`lot_batch_id → inventory.lot_batches.id`	`fk_movement_records_lot`	Lot on movement (BR-017)
FK	`reason_code_id → governance.reason_codes.id`	`fk_movement_records_reason`	Reason on adjustment movements (BR-020)
FK	`performed_by → identity.user_accounts.id`	`fk_movement_records_user`	User attribution (REL-117)
CK	`movement_reference`	—	Business transaction reference
CK	`(source_document_type, source_document_id)`	—	Polymorphic origin link (REL-116) — not FK due to multiple source tables

---

`audit.audit_trail_records`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_audit_trail_records`	Partition-local surrogate PK; immutable
UK	`(audit_event_reference, event_timestamp)`	`uk_audit_trail_ref_ts`	Unique event within partition
FK	`user_id → identity.user_accounts.id`	`fk_audit_trail_user`	Actor identity (REL-119)
FK	`movement_record_id → audit.inventory_movement_records.id`	`fk_audit_trail_movement`	Links governance snapshot to operational movement (REL-127)
CK	`(affected_entity_type, affected_entity_id)`	—	Polymorphic governed entity (REL-118)
CK	`audit_event_reference`	—	Primary business event key

`audit.operational_alerts`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_operational_alerts`	Alert surrogate PK
UK	`alert_reference`	`uk_operational_alerts_ref`	Alert document for workflow routing
FK	`product_id → catalog.products.id`	`fk_operational_alerts_product`	Alert context SKU
FK	`location_id → network.inventory_locations.id`	`fk_operational_alerts_location`	Alert context site
FK	`stock_position_id → inventory.stock_positions.id`	`fk_operational_alerts_position`	Threshold breach source (REL-120)
CK	`(source_entity_type, source_entity_id)`	—	Polymorphic trigger (transfer, delivery, allocation)
CK	`alert_reference`	—	Primary business key

`audit.inventory_valuation_snapshots`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_inventory_valuation_snapshots`	Snapshot surrogate PK
UK	`snapshot_reference`	`uk_valuation_snapshots_ref`	Finance reconciliation document
UK	`(valuation_date, location_scope)`	`uk_valuation_snapshots_date_scope`	One network/regional snapshot per period — prevents duplicate month-end close

Key Type	Column(s)	Constraint Name (logical)	Business Justification
FK	`prepared_by` → `identity.user_accounts.id`	`fk_valuation_snapshots_prepared_by`	Preparer attribution
CK	`snapshot_reference`	—	Primary business key

`audit.inventory_valuation_lines`

Key Type	Column(s)	Constraint Name (logical)	Business Justification
PK	`id`	`pk_inventory_valuation_lines`	Line surrogate PK
FK	`snapshot_id` → `audit.inventory_valuation_snapshots.id`	`fk_valuation_lines_snapshot`	Line belongs to snapshot
FK	`product_id` → `catalog.products.id`	`fk_valuation_lines_product`	Valued SKU (REL-130)
FK	`location_id` → `network.inventory_locations.id`	`fk_valuation_lines_location`	Valued site (REL-131)
UK	`(snapshot_id, product_id, location_id)`	`uk_valuation_lines_snapshot_prod_loc`	One valuation line per SKU per location per snapshot

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Appendix A: Primary Key Summary

All 59 tables use `id BIGINT GENERATED ALWAYS AS IDENTITY` as the sole primary key column.

Design Decision	Business Justification
Surrogate over natural PK	Business identifiers (SKU, PO number) can change format, merge, or require soft-retirement without cascading PK updates across high-volume child tables
Single-column PK	Simplifies ORM mapping, partition management, and FK definitions
BIGINT not INT	Supports very high-volume tables (movement records, order lines) without ID exhaustion
No composite PKs	Composite business uniqueness enforced via UK constraints — separates physical row identity from business identity

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Appendix B: Foreign Key Relationship Map

Child Table	Parent Table	Cardinality	ON DELETE Policy (recommended)	Business Rule
`inventory.stock_positions`	`catalog.products`	N:1	RESTRICT	Cannot delete product with balances
`inventory.stock_positions`	`network.inventory_locations`	N:1	RESTRICT	Cannot delete location with stock
`fulfillment.inventory_reservations`	`inventory.stock_positions`	N:1	RESTRICT	Reservation requires valid balance
`inbound.goods_receipts`	`procurement.purchase_orders`	N:1	RESTRICT	PO required for receipt (BR-007)
`inbound.put_aways`	`inbound.goods_receipts`	N:1	RESTRICT	Put-away requires receipt
`accuracy.stock_adjustments`	`governance.reason_codes`	N:1	RESTRICT	Reason mandatory (BR-020)
`accuracy.count_variances`	`accuracy.cycle_counts`	N:1	RESTRICT	Variance traceable to count
`returns.customer_returns`	`returns.return_authorizations`	N:1	RESTRICT	RMA required (BR-025)

Child Table	Parent Table	Cardinality	ON DELETE Policy (recommended)	Business Rule
finance.invoice_goods_receipt_match	finance.supplier_invoices	N:1	CASCADE	Match rows die with invoice draft
identity.role_assignments	identity.user_accounts	N:1	CASCADE	Assignments removed with user deprovisioning
audit.inventory_movement_records	inventory.stock_positions	N:1	RESTRICT	Movement history preserved

Cross-schema FK note: Foreign keys spanning bounded contexts (e.g., `fulfillment` → `inventory`) enforce referential integrity at persistence time. Cross-aggregate coordination still uses domain events per architecture — FKs prevent orphan rows, not distributed transactions.

Appendix C: Unique Key Catalog

Table	Unique Key Column(s)	Enforces Business Rule
catalog.products	sku	Global SKU uniqueness
catalog.product_identifiers	(identifier_type, identifier_value)	No duplicate scan codes
network.inventory_locations	location_code	Network location identity
network.storage_positions	(location_id, position_code)	Bin unique within site
inventory.stock_positions	(product_id, location_id)	One balance per SKU per site
inventory.stock_policies	(product_id, location_id)	One policy per SKU per site
inventory.lot_batches	(product_id, lot_number, stock_position_id)	Lot quantity uniqueness at position
procurement.suppliers	supplier_code	Vendor identity
procurement.purchase_orders	po_number	PO document identity
finance.supplier_invoices	(supplier_id, invoice_number)	AP invoice uniqueness
fulfillment.customer_orders	order_number	Order identity
governance.reason_codes	reason_code	Taxonomy uniqueness
identity.user_accounts	user_identifier	Login uniqueness
governance.governance_policies	(policy_code, effective_from)	Policy version uniqueness
finance.invoice_goods_receipt_match	(supplier_invoice_id, goods_receipt_id)	Single match pair
commercial.promotional_allocation_locations	(allocation_id, location_id)	One location entry per campaign
audit.inventory_valuation_lines	(snapshot_id, product_id, location_id)	Snapshot line uniqueness
All *_lines tables	(parent_id, line_number)	Document line ordering

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Appendix D: Candidate Key Reference

Candidate keys not enforced as UK are retained for integration contracts and future constraint hardening.

Table	Candidate Key	Why Not Enforced as UK
fulfillment.shipments	tracking_identifier	Carrier may assign tracking late or reuse across carriers
procurement.purchase_order_lines	(purchase_order_id, product_id)	Same SKU may appear on multiple lines (different prices/dates)

Table	Candidate Key	Why Not Enforced as UK
`audit.inventory_movement_records`	`(source_document_type, source_document_id, stock_position_id)`	Multiple movements per source document (partial shipments)
`accuracy.count_variances`	`(cycle_count_id)` XOR `(physical_count_id)`	Enforced via CHECK constraint, not UK
`commercial.promotional_stock_allocations`	`(product_id, campaign_name, effective_start_date)`	Campaign names may repeat across years

Appendix E: Keys Enforcing Critical Business Rules

Business Rule	Key Enforcement
BR-001 ATP-only commitment	`uk_stock_positions_product_location` ensures single authoritative balance per SKU/site
BR-002 Reservation at confirmation	FK `inventory_reservations.stock_position_id` binds hold to specific balance
BR-005 Put-away before availability	FK chain `put_aways` → `goods_receipts` → `purchase_orders` enforces receipt workflow
BR-007 PO required for receipt	FK `goods_receipts.purchase_order_id` (nullable only for documented exception)
BR-008 Three-way match	UK `invoice_goods_receipt_match (supplier_invoice_id, goods_receipt_id)`
BR-019 Independent count approval	FK `count_variances.approver_id` ≠ `cycle_counts.counter_id` — application-enforced SoD
BR-020 Mandatory reason code	FK `stock_adjustments.reason_code_id` NOT NULL + RESTRICT
BR-025 RMA required	FK `customer_returns.return_authorization_id` NOT NULL
BR-027 No discontinued reorder	FK to `products` + application lifecycle check on insert
BR-029 Promotional ring-fence	UK `promotional_allocation_locations (allocation_id, location_id)`

Document Approval

Role	Name	Signature	Date
Database Architect			
DDD Solution Architect			
Solution Architect			
Inventory Controller			

End of Database Keys Specification

Document ID: SRS-DB-003 | Version 1.0 | July 6, 2026

Database Constraints Specification

Inventory Management System — PostgreSQL

Document Information

Attribute	Detail
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Scope	Constraint definitions and business justification — no DDL/SQL statements

Purpose

This document defines all **database constraints** for the Inventory Management System. Each constraint enforces a business rule or data integrity invariant at the persistence layer. Constraints are named using the pattern `{type}_{table}_{column_or_scope}`.

Constraint Type Legend

Type	Prefix	Description
NOT NULL	`nn_`	Column must contain a value
UNIQUE	`uk_`	Column or combination must be unique
CHECK	`chk_`	Value must satisfy a business rule expression
FOREIGN KEY	`fk_`	Referential integrity to parent table
DEFAULT	`def_`	Default value when not supplied on INSERT

Standard Platform Constraints

Applied uniformly unless a table explicitly omits a column.

Table Pattern	Column	Type	Constraint	Business Rule	Justification
All tables	`id`	NOT NULL	`nn_*_id`	—	Primary key identity must always exist
All tables	`id`	DEFAULT	`def_*_id` IDENTITY	—	System-generated stable surrogate key
Mutable tables	`created_at`	NOT NULL	`nn_*_created_at`	BO-19	Every record has auditable creation timestamp
Mutable tables	`created_at`	DEFAULT	`def_*_created_at` = NOW()	BO-19	Auto-stamp creation time in UTC
Mutable transactional headers	`version`	NOT NULL	`nn_*_version`	—	Optimistic locking counter required
Mutable transactional headers	`version`	DEFAULT	`def_*_version` = 1	—	Initial version on insert
Master data tables	`is_active`	NOT NULL	`nn_*_is_active`	—	Soft-delete flag must be explicit
Master data tables	`is_active`	DEFAULT	`def_*_is_active` = TRUE	—	New master records active by default

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Shared Domain CHECK Registry

Reusable enumerated value constraints referenced by multiple tables.

Constraint Name	Allowed Values	Business Rule	Justification
chk_domain_lifecycle_status	ACTIVE, INACTIVE, DISCONTINUED	BR-027	Controls product eligibility for procurement and reservation
chk_domain_abc_classification	A, B, C	BR-032	ABC class drives count frequency and alert priority
chk_domain_rotation_policy	FIFO, FEFO	BR-014, BR-015	Rotation discipline for picking and put-away
chk_domain_ownership_type	OWNED, CONSIGNMENT, BONDED	BR-031, BR-030	Ownership affects ATP, valuation, and customs hold
chk_domain_quantity_non_negative	value >= 0	BR-001, LDM-01	Inventory quantities cannot be negative
chk_domain_quantity_positive	value > 0	—	Order/receipt line quantities must be meaningful
chk_domain_order_status	DRAFT, CONFIRMED, PARTIAL, FULFILLED, CANCELLED	BR-001, BR-002	Order lifecycle states
chk_domain_po_status	OPEN, PARTIAL, CLOSED, CANCELLED	BR-007	PO receipt authorization lifecycle
chk_domain_approval_status	PENDING, APPROVED, REJECTED (+ variants)	BR-019, BR-021	Governance workflow states
chk_domain_hold_type	QUALITY, REGULATORY, RECALL, CUSTOMS, INVESTIGATION	BR-013, BR-030	Quarantine hold classification
chk_domain_disposition_decision	RESTOCK, QUARANTINE, WRITE_OFF	BR-026	Return outcome paths
chk_domain_movement_type	RECEIPT, SHIPMENT, TRANSFER, ADJUSTMENT, WRITE_OFF, RESERVATION, RETURN, PUT_AWAY	BR-017, BR-020	Movement log taxonomy

Domain 1: Catalog

catalog.products

Column	Type	Constraint	Business Rule	Justification
sku	NOT NULL	nn_products_sku	—	SKU is mandatory product identity
sku	UNIQUE	uk_products_sku	BR-027	Globally unique stock keeping unit
product_name	NOT NULL	nn_products_product_name	—	Product must be identifiable by name in operations
unit_of_measure	NOT NULL	nn_products_uom	—	All inventory quantities require a UOM
lifecycle_status	NOT NULL	nn_products_lifecycle_status	BR-027	Status governs eligibility for new transactions
lifecycle_status	DEFAULT	def_products_lifecycle_status = 'ACTIVE'	—	New products enter as active
lifecycle_status	CHECK	chk_products_lifecycle_status	BR-027	Only valid lifecycle states permitted
abc_classification	CHECK	chk_products_abc_classification	BR-032	ABC must be A, B, or C when assigned
is_shelf_life	NOT NULL	nn_products_is_shelf_life	BR-014, BR-016	Shelf-life flag required for MRSL enforcement
is_shelf_life	DEFAULT	def_products_is_shelf_life = FALSE	—	Most products are non-shelf-life
is_traceable	NOT NULL	nn_products_is_traceable	BR-017	Traceability flag required
is_traceable	DEFAULT	def_products_is_traceable = FALSE	—	Traceability opt-in per product

Column	Type	Constraint	Business Rule	Justification
rotation_policy	CHECK	chk_products_rotation_policy	BR-014, BR-015	FIFO/FEFO only when specified
(is_traceable, rotation_policy)	CHECK	chk_products_traceable_rotation	BR-017	Traceable products must declare rotation policy
(is_shelf_life, mrsl_days)	CHECK	chk_products_shelf_life_mrsl	BR-016	Shelf-life products must have MRSL days > 0
is_active	DEFAULT	def_products_is_active = TRUE	—	Soft-delete default
version	DEFAULT	def_products_version = 1	—	Optimistic lock initial value
created_by	FOREIGN KEY	fk_products_created_by → identity.user_accounts	BO-19	Product creation attribution

catalog.product_identifiers

Column	Type	Constraint	Business Rule	Justification
product_id	NOT NULL	nn_product_identifiers_product_id	—	Identifier must belong to a product
product_id	FOREIGN KEY	fk_product_identifiers_product → catalog.products	BR-004	Parent product must exist for scan lookup
identifier_type	NOT NULL	nn_product_identifiers_type	BR-004	Type distinguishes barcode from vendor code
identifier_value	NOT NULL	nn_product_identifiers_value	BR-004	Empty identifier unusable at scan
(identifier_type, identifier_value)	UNIQUE	uk_product_identifiers_type_value	BR-004	One product per scan code globally
(product_id, identifier_type, identifier_value)	UNIQUE	uk_product_identifiers_composite	—	No duplicate identifiers on same product
is_primary	NOT NULL	nn_product_identifiers_is_primary	—	Primary flag must be explicit
is_primary	DEFAULT	def_product_identifiers_is_primary = FALSE	—	Non-primary by default
is_active	DEFAULT	def_product_identifiers_is_active = TRUE	—	Active identifier by default

Domain 2: Network

network.inventory_locations

Column	Type	Constraint	Business Rule	Justification
location_code	NOT NULL	nn_inventory_locations_code	—	Location code is mandatory network identity
location_code	UNIQUE	uk_inventory_locations_code	—	Unique site identifier across network
location_name	NOT NULL	nn_inventory_locations_name	—	Human-readable site name required
location_type	NOT NULL	nn_inventory_locations_type	—	Type drives operational capabilities
location_type	CHECK	chk_inventory_locations_type IN (WAREHOUSE, STORE, DISTRIBUTION_POINT)	—	Valid location types only

Column	Type	Constraint	Business Rule	Justification
parent_location_id	FOREIGN KEY	fk_inventory_locations_parent → self	—	Hierarchy reference must be valid location
parent_location_id	CHECK	chk_inventory_locations_not_self_parent id ≠ parent_location_id	—	Location cannot be its own parent
operational_status	NOT NULL	nn_inventory_locations_status	BR-024	Status controls transaction eligibility
operational_status	DEFAULT	def_inventory_locations_status = 'ACTIVE'	—	New locations active by default
operational_status	CHECK	chk_inventory_locations_status	—	ACTIVE, PLANNED, CLOSING, INACTIVE only
is_frozen	NOT NULL	nn_inventory_locations_is_frozen	BR-024	Freeze flag must be explicit during wall count
is_frozen	DEFAULT	def_inventory_locations_is_frozen = FALSE	BR-024	Locations not frozen by default
responsible_manager_id	FOREIGN KEY	fk_inventory_locations_manager → identity.user_accounts	—	Optional manager accountability
country_code	CHECK	chk_inventory_locations_country length = 2	—	ISO 3166-1 alpha-2 when provided

network.storage_positions

Column	Type	Constraint	Business Rule	Justification
location_id	NOT NULL	nn_storage_positions_location_id	BR-005	Position must belong to a location
location_id	FOREIGN KEY	fk_storage_positions_location → network.inventory_locations	REL-004	Composition within location aggregate
position_code	NOT NULL	nn_storage_positions_code	BR-005	Bin address required for put-away
(location_id, position_code)	UNIQUE	uk_storage_positions_location_code	—	Code unique within location only
storage_type	NOT NULL	nn_storage_positions_type	BR-030	Storage type drives bonded/quarantine routing
storage_type	DEFAULT	def_storage_positions_type = 'BULK'	—	Default bulk storage
storage_type	CHECK	chk_storage_positions_type IN (BULK, PICK_FACE, QUARANTINE, BONDED)	BR-030	Valid storage classifications
capacity_units	CHECK	chk_storage_positions_capacity >= 0 OR NULL	—	Capacity cannot be negative
dedicated_product_id	FOREIGN KEY	fk_storage_positions_product → catalog.products	—	Dedicated bin references valid product
(is_dedicated, dedicated_product_id)	CHECK	chk_storage_positions_dedicated	—	Dedicated product required when is_dedicated = TRUE

Domain 3: Governance

`governance.governance_policies`

Column	Type	Constraint	Business Rule	Justification
`policy_code`	NOT NULL	`nn_governance_policies_code`	BR-021	Policy must be identifiable
`policy_name`	NOT NULL	`nn_governance_policies_name`	—	Human-readable policy label
`policy_type`	NOT NULL	`nn_governance_policies_type`	BR-021	Type determines enforcement context
`policy_type`	CHECK	`chk_governance_policies_type`	—	APPROVAL_THRESHOLD, TOLERANCE, ACCESS, EXCEPTION
`effective_from`	NOT NULL	`nn_governance_policies_effective_from`	BR-021	Policy must have effective start
`(effective_from, effective_to)`	CHECK	`chk_governance_policies_dates` effective_to IS NULL OR effective_to > effective_from	—	No inverted effective periods
`(policy_code, effective_from)`	UNIQUE	`uk_governance_policies_code_from`	BR-021	One version per code per effective date
`threshold_value`	CHECK	`chk_governance_policies_threshold` >= 0 OR NULL	BR-021	Thresholds are non-negative
`is_active`	DEFAULT	`def_governance_policies_is_active` = TRUE	—	Active policy by default

`governance.reason_codes`

Column	Type	Constraint	Business Rule	Justification
`reason_code`	NOT NULL	`nn_reason_codes_code`	BR-020	Reason code mandatory on adjustments
`reason_code`	UNIQUE	`uk_reason_codes_code`	BR-020	Unique taxonomy key
`description`	NOT NULL	`nn_reason_codes_description`	BR-020	Operators must understand reason meaning
`category`	NOT NULL	`nn_reason_codes_category`	BR-020	Category drives reporting and analysis
`category`	CHECK	`chk_reason_codes_category`	BR-020	Valid adjustment/discrepancy categories
`approval_threshold_id`	FOREIGN KEY	`fk_reason_codes_policy` → `governance.governance_policies`	BR-021	Links to approval threshold when applicable
`is_active`	NOT NULL	`nn_reason_codes_is_active`	BR-020	Retired codes must be flagged, not deleted
`is_active`	DEFAULT	`def_reason_codes_is_active` = TRUE	BR-020	New codes active by default

Domain 4: Inventory Balance

`inventory.stock_positions`

Column	Type	Constraint	Business Rule	Justification
`product_id`	NOT NULL	`nn_stock_positions_product_id`	—	Balance must reference a product

Column	Type	Constraint	Business Rule	Justification
product_id	FOREIGN KEY	fk_stock_positions_product → catalog.products	BR-001	Valid catalog product required
location_id	NOT NULL	nn_stock_positions_location_id	—	Balance must reference a location
location_id	FOREIGN KEY	fk_stock_positions_location → network.inventory_locations	BR-001	Valid location required
(product_id, location_id)	UNIQUE	uk_stock_positions_product_location	BR-001, LDM-04	One authoritative balance per SKU per site
on_hand_qty	NOT NULL	nn_stock_positions_on_hand	LDM-01	On-hand is mandatory quantity
on_hand_qty	DEFAULT	def_stock_positions_on_hand = 0	—	Zero balance at creation
on_hand_qty	CHECK	chk_stock_positions_on_hand >= 0	LDM-01	On-hand never negative
reserved_qty	NOT NULL	nn_stock_positions_reserved	BR-002	Reserved quantity always tracked
reserved_qty	DEFAULT	def_stock_positions_reserved = 0	BR-002	No reservation at creation
reserved_qty	CHECK	chk_stock_positions_reserved >= 0	BR-002	Reserved cannot be negative
reserved_qty	CHECK	chk_stock_positions_reserved_le_on_hand reserved_qty <= on_hand_qty	BR-002	Cannot reserve more than on-hand
quarantined_qty	NOT NULL	nn_stock_positions_quarantined	BR-013	Quarantined quantity always tracked
quarantined_qty	DEFAULT	def_stock_positions_quarantined = 0	BR-013	No quarantine at creation
quarantined_qty	CHECK	chk_stock_positions_quarantined >= 0	BR-013	Quarantine qty non-negative
quarantined_qty	CHECK	chk_stock_positions_quarantine_le_on_hand quarantined_qty <= on_hand_qty	BR-013	Hold cannot exceed on-hand
atp_qty	NOT NULL	nn_stock_positions_atp	BR-001	ATP is mandatory availability measure
atp_qty	DEFAULT	def_stock_positions_atp = 0	BR-001	Zero ATP at creation
atp_qty	CHECK	chk_stock_positions_atp >= 0	BR-001	ATP never negative
atp_qty	CHECK	chk_stock_positions_atp_le_on_hand atp_qty <= on_hand_qty	BR-001	ATP cannot exceed on-hand
(on_hand_qty, reserved_qty, quarantined_qty, atp_qty)	CHECK	chk_stock_positions_atp_formula atp_qty = on_hand_qty - reserved_qty - quarantined_qty	BR-001, LDM-01	ATP formula integrity at persistence
inbound_expected_qty	NOT NULL	nn_stock_positions_inbound_expected	BR-005	Pipeline qty tracked separately from ATP
inbound_expected_qty	DEFAULT	def_stock_positions_inbound_expected = 0	BR-005	Informational only — not ATP
inbound_expected_qty	CHECK	chk_stock_positions_inbound_expected >= 0	BR-005	Expected inbound non-negative
stock_status	NOT NULL	nn_stock_positions_status	BR-009	Status indicator mandatory
stock_status	DEFAULT	def_stock_positions_status = 'IN_STOCK'	—	Default healthy status

Column	Type	Constraint	Business Rule	Justification
stock_status	CHECK	chk_stock_positions_status IN (IN_STOCK, LOW_STOCK, OUT_OF_STOCK, OVERSTOCK)	BR-009	Valid status values
ownership_type	NOT NULL	nn_stock_positions_ownership	BR-031	Ownership classification required
ownership_type	DEFAULT	def_stock_positions_ownership = 'OWNED'	BR-031	Owned stock by default
ownership_type	CHECK	chk_stock_positions_ownership	BR-031, BR-030	OWNED, CONSIGNMENT, BONDED only
version	DEFAULT	def_stock_positions_version = 1	—	Optimistic lock for concurrent reservations

inventory.lot_batches

Column	Type	Constraint	Business Rule	Justification
stock_position_id	NOT NULL	nn_lot_batches_position_id	BR-017	Lot belongs to a stock position
stock_position_id	FOREIGN KEY	fk_lot_batches_position → inventory.stock_positions	BR-017	Parent balance must exist
product_id	NOT NULL	nn_lot_batches_product_id	BR-017	Lot scoped to product
product_id	FOREIGN KEY	fk_lot_batches_product → catalog.products	BR-017	Denormalized product for FEFO queries
lot_number	NOT NULL	nn_lot_batches_lot_number	BR-017	Lot identity mandatory for traceable products
(product_id, lot_number, stock_position_id)	UNIQUE	uk_lot_batches_composite	BR-017	One lot row per position
quantity	NOT NULL	nn_lot_batches_quantity	BR-017	Lot quantity required
quantity	DEFAULT	def_lot_batches_quantity = 0	—	Zero at lot creation
quantity	CHECK	chk_lot_batches_quantity >= 0	LDM-01	Lot quantity non-negative
(expiration_date, manufacture_date)	CHECK	chk_lot_batches_dates expiration IS NULL OR manufacture IS NULL OR expiration >= manufacture	BR-014	Expiration after manufacture
quality_status	NOT NULL	nn_lot_batches_quality	BR-013	Quality status required
quality_status	DEFAULT	def_lot_batches_quality = 'ACCEPTED'	—	Accepted at receipt default
quality_status	CHECK	chk_lot_batches_quality IN (ACCEPTED, HOLD, REJECTED)	BR-013	Valid quality states
supplier_id	FOREIGN KEY	fk_lot_batches_supplier → procurement.suppliers	BR-017	Supplier origin for recall traceability

inventory.quarantine_holds

Column	Type	Constraint	Business Rule	Justification
hold_reference	NOT NULL	nn_quarantine_holds_reference	BR-013	Hold document identity
hold_reference	UNIQUE	uk_quarantine_holds_reference	BR-013	Unique hold reference
stock_position_id	NOT NULL	nn_quarantine_holds_position_id	BR-013	Hold applied to specific balance

Column	Type	Constraint	Business Rule	Justification
stock_position_id	FOREIGN KEY	fk_quarantine_holds_position → inventory.stock_positions	BR-013	Valid balance required
lot_batch_id	FOREIGN KEY	fk_quarantine_holds_lot → inventory.lot_batches	BR-013	Optional lot-scoped hold
held_qty	NOT NULL	nn_quarantine_holds_held_qty	BR-013	Held quantity mandatory
held_qty	CHECK	chk_quarantine_holds_held_qty > 0	BR-013	Hold must affect positive quantity
hold_type	NOT NULL	nn_quarantine_holds_type	BR-013, BR-030	Hold classification required
hold_type	CHECK	chk_quarantine_holds_type	BR-013, BR-030	Valid hold types
hold_reason	NOT NULL	nn_quarantine_holds_reason	BR-013	Hold must be justified
hold_status	NOT NULL	nn_quarantine_holds_status	BR-013	Lifecycle state required
hold_status	DEFAULT	def_quarantine_holds_status = 'ACTIVE'	BR-013	Active hold by default
hold_status	CHECK	chk_quarantine_holds_status IN (ACTIVE, RELEASED, DISPOSED)	BR-013	Valid hold lifecycle
hold_start_at	NOT NULL	nn_quarantine_holds_start	BR-013	Hold effective time required
hold_start_at	DEFAULT	def_quarantine_holds_start = NOW()	BR-013	Auto-stamp hold start
(hold_status, released_at)	CHECK	chk_quarantine_holds_release	BR-013	RELEASED status requires released_at
authorized_by	FOREIGN KEY	fk_quarantine_holds_authorized_by → identity.user_accounts	BR-013	Hold authorization attribution

inventory.stock_policies

Column	Type	Constraint	Business Rule	Justification
product_id	NOT NULL	nn_stock_policies_product_id	BR-009	Policy scoped to product
product_id	FOREIGN KEY	fk_stock_policies_product → catalog.products	BR-009	Valid product required
location_id	NOT NULL	nn_stock_policies_location_id	BR-009	Policy scoped to location
location_id	FOREIGN KEY	fk_stock_policies_location → network.inventory_locations	BR-009	Valid location required
(product_id, location_id)	UNIQUE	uk_stock_policies_product_location	BR-009, LDM-04	One policy per SKU per site
reorder_level	NOT NULL	nn_stock_policies_reorder	BR-009	Reorder trigger required
reorder_level	DEFAULT	def_stock_policies_reorder = 0	—	Zero default
reorder_level	CHECK	chk_stock_policies_reorder >= 0	BR-009	Non-negative threshold
minimum_threshold	NOT NULL	nn_stock_policies_minimum	BR-010, BR-012	Minimum floor required
minimum_threshold	DEFAULT	def_stock_policies_minimum = 0	—	Zero default
minimum_threshold	CHECK	chk_stock_policies_minimum >= 0	BR-010	Non-negative minimum
safety_stock_qty	NOT NULL	nn_stock_policies_safety	BR-010	Safety stock required
safety_stock_qty	DEFAULT	def_stock_policies_safety = 0	BR-010	Zero default

Column	Type	Constraint	Business Rule	Justification
safety_stock_qty	CHECK	chk_stock_policies_safety >= 0	BR-010	Safety stock non-negative
(minimum_threshold, reorder_level)	CHECK	chk_stock_policies_threshold_order minimum_threshold <= reorder_level	BR-009, BR-010	Minimum ≤ reorder unless authorized exception
maximum_stock_level	CHECK	chk_stock_policies_maximum > reorder_level OR NULL	—	Overstock level above reorder when set
lead_time_days	CHECK	chk_stock_policies_lead_time > 0 OR NULL	—	Lead time must be positive
effective_from	NOT NULL	nn_stock_policies_effective_from	BR-009	Effective dating required
effective_from	DEFAULT	def_stock_policies_effective_from = NOW()	—	Current timestamp default

inventory.stock_policy_history

Column	Type	Constraint	Business Rule	Justification
stock_policy_id	NOT NULL	nn_stock_policy_history_policy_id	BR-009	History belongs to policy
stock_policy_id	FOREIGN KEY	fk_stock_policy_history_policy → inventory.stock_policies	BR-009	Parent policy must exist
reorder_level	NOT NULL	nn_stock_policy_history_reorder	BR-009	Historical value captured
minimum_threshold	NOT NULL	nn_stock_policy_history_minimum	BR-010	Historical value captured
safety_stock_qty	NOT NULL	nn_stock_policy_history_safety	BR-010	Historical value captured
effective_from	NOT NULL	nn_stock_policy_history_from	BR-009	Period start required
effective_to	NOT NULL	nn_stock_policy_history_to	BR-009	Period end required
(effective_from, effective_to)	CHECK	chk_stock_policy_history_dates effective_to > effective_from	BR-009	Valid history window
(stock_policy_id, effective_from)	UNIQUE	uk_stock_policy_history_policy_from	BR-009	Non-overlapping history periods
changed_by	FOREIGN KEY	fk_stock_policy_history_changed_by → identity.user_accounts	BO-19	Change attribution

Domain 5: Procurement

procurement.suppliers

Column	Type	Constraint	Business Rule	Justification
supplier_code	NOT NULL	nn_suppliers_code	BR-007	Vendor code mandatory
supplier_code	UNIQUE	uk_suppliers_code	BR-007	Unique commercial counterparty
supplier_name	NOT NULL	nn_suppliers_name	BR-007	Vendor must be identifiable
payment_terms_days	NOT NULL	nn_suppliers_payment_terms	BR-008	Payment terms required before PO
payment_terms_days	DEFAULT	def_suppliers_payment_terms = 30	—	Standard net-30 default
payment_terms_days	CHECK	chk_suppliers_payment_terms > 0	BR-008	Positive payment terms
is_consignment	NOT NULL	nn_suppliers_is_consignment	BR-031	Consignment flag explicit
is_consignment	DEFAULT	def_suppliers_is_consignment = FALSE	BR-031	Standard purchase by default
delivery_rating	CHECK	chk_suppliers_rating BETWEEN 0 AND 5 OR NULL	—	Valid rating scale
default_lead_time_days	CHECK	chk_suppliers_lead_time > 0 OR NULL	—	Positive lead time when set

procurement.purchase_requisitions

Column	Type	Constraint	Business Rule	Justification
requisition_number	NOT NULL	nn_purchase_requisitions_number	BR-009	Internal document identity
requisition_number	UNIQUE	uk_purchase_requisitions_number	—	Unique requisition reference
requesting_location_id	NOT NULL	nn_purchase_requisitions_location	BR-009	Replenishment destination required
requesting_location_id	FOREIGN KEY	fk_purchase_requisitions_location → network.inventory_locations	BR-009	Valid receiving site
approval_status	NOT NULL	nn_purchase_requisitions_status	BR-009	Workflow state required
approval_status	DEFAULT	def_purchase_requisitions_status = 'DRAFT'	—	Starts as draft
approval_status	CHECK	chk_purchase_requisitions_status IN (DRAFT, SUBMITTED, APPROVED, REJECTED, CONVERTED)	BR-009	Valid approval states
(approval_status, approved_by)	CHECK	chk_purchase_requisitions_approval	BR-021	APPROVED requires approver
triggering_alert_id	FOREIGN KEY	fk_purchase_requisitions_alert → audit.operational_alerts	BR-009	Alert-to-requisition traceability
approved_by	FOREIGN KEY	fk_purchase_requisitions_approved_by → identity.user_accounts	BR-021	Approver attribution
purchase_order_id	FOREIGN KEY	fk_purchase_requisitions_po → procurement.purchase_orders	BR-009	PO conversion link
created_by	NOT NULL	nn_purchase_requisitions_created_by	BO-19	Initiator required
created_by	FOREIGN KEY	fk_purchase_requisitions_created_by → identity.user_accounts	BO-19	Initiator attribution

procurement.purchase_requisition_lines

Column	Type	Constraint	Business Rule	Justification
requisition_id	NOT NULL	nn_purchase_requisition_lines_req_id	—	Line belongs to requisition
requisition_id	FOREIGN KEY	fk_purchase_requisition_lines_req → procurement.purchase_requisitions	—	Parent document exists
line_number	NOT NULL	nn_purchase_requisition_lines_line	—	Line sequence required
line_number	CHECK	chk_purchase_requisition_lines_line > 0	—	Positive line numbers
(requisition_id, line_number)	UNIQUE	uk_purchase_requisition_lines_req_line	—	Unique line per requisition
product_id	NOT NULL	nn_purchase_requisition_lines_product	BR-027	Product required
product_id	FOREIGN KEY	fk_purchase_requisition_lines_product → catalog.products	BR-027	Active product validation at application layer
requested_qty	NOT NULL	nn_purchase_requisition_lines_qty	BR-009	Quantity required
requested_qty	CHECK	chk_purchase_requisition_lines_qty > 0	BR-009	Positive requested quantity

procurement.purchase_orders

Column	Type	Constraint	Business Rule	Justification
po_number	NOT NULL	nn_purchase_orders_number	BR-007	PO document identity
po_number	UNIQUE	uk_purchase_orders_number	BR-007	Unique authorization document
supplier_id	NOT NULL	nn_purchase_orders_supplier	BR-007	PO requires supplier
supplier_id	FOREIGN KEY	fk_purchase_orders_supplier → procurement.suppliers	BR-007	Valid active supplier
receiving_location_id	NOT NULL	nn_purchase_orders_location	BR-007	Receiving site required
receiving_location_id	FOREIGN KEY	fk_purchase_orders_location → network.inventory_locations	BR-007	Valid receiving location
order_date	NOT NULL	nn_purchase_orders_order_date	BR-007	PO date mandatory
po_status	NOT NULL	nn_purchase_orders_status	BR-007	PO lifecycle state
po_status	DEFAULT	def_purchase_orders_status = 'OPEN'	BR-007	New POs open
po_status	CHECK	chk_purchase_orders_status	BR-007	OPEN, PARTIAL, CLOSED, CANCELLED
total_commitment_amount	NOT NULL	nn_purchase_orders_total	BR-008	Financial commitment tracked
total_commitment_amount	DEFAULT	def_purchase_orders_total = 0	—	Zero until lines added
total_commitment_amount	CHECK	chk_purchase_orders_total >= 0	BR-008	Non-negative commitment
currency_code	NOT NULL	nn_purchase_orders_currency	BR-008	Currency required
currency_code	DEFAULT	def_purchase_orders_currency = 'USD'	—	Default currency
currency_code	CHECK	chk_purchase_orders_currency length = 3	—	ISO 4217 format
payment_terms_days	NOT NULL	nn_purchase_orders_terms	BR-008	Terms on PO
payment_terms_days	DEFAULT	def_purchase_orders_terms = 30	—	Default net-30
created_by	NOT NULL	nn_purchase_orders_created_by	BO-19	Buyer attribution
created_by	FOREIGN KEY	fk_purchase_orders_created_by → identity.user_accounts	BO-19	Buyer identity

procurement.purchase_order_lines

Column	Type	Constraint	Business Rule	Justification
purchase_order_id	NOT NULL	nn_purchase_order_lines_po_id	BR-007	Line belongs to PO
purchase_order_id	FOREIGN KEY	fk_purchase_order_lines_po → procurement.purchase_orders	BR-007	Parent PO exists
line_number	NOT NULL	nn_purchase_order_lines_line	—	Line sequence required
line_number	CHECK	chk_purchase_order_lines_line > 0	—	Positive line number
(purchase_order_id, line_number)	UNIQUE	uk_purchase_order_lines_po_line	—	Unique line per PO
product_id	NOT NULL	nn_purchase_order_lines_product	BR-027	Product required
product_id	FOREIGN KEY	fk_purchase_order_lines_product → catalog.products	BR-027	Discontinued blocked at app layer
ordered_qty	NOT NULL	nn_purchase_order_lines_ordered	BR-007	Ordered quantity required

Column	Type	Constraint	Business Rule	Justification
ordered_qty	CHECK	chk_purchase_order_lines_ordered > 0	BR-007	Positive order quantity
received_qty	NOT NULL	nn_purchase_order_lines_received	BR-007	Received cumulative tracked
received_qty	DEFAULT	def_purchase_order_lines_received = 0	—	Zero at line creation
received_qty	CHECK	chk_purchase_order_lines_received >= 0	BR-007	Non-negative received
received_qty	CHECK	chk_purchase_order_lines_received_le_ordered received_qty <= ordered_qty + tolerance	BR-006, BR-007	Over-receipt requires discrepancy workflow
unit_price	NOT NULL	nn_purchase_order_lines_price	BR-008	Price required for match
unit_price	CHECK	chk_purchase_order_lines_price >= 0	BR-008	Non-negative price
line_amount	NOT NULL	nn_purchase_order_lines_amount	BR-008	Extended amount required
line_amount	CHECK	chk_purchase_order_lines_amount >= 0	BR-008	Non-negative line value

procurement.expected_inbound_deliveries

Column	Type	Constraint	Business Rule	Justification
delivery_reference	NOT NULL	nn_expected_inbound_deliveries_ref	—	Shipment plan identity
delivery_reference	UNIQUE	uk_expected_inbound_deliveries_ref	—	Unique delivery reference
purchase_order_id	NOT NULL	nn_expected_inbound_deliveries_po	BR-007	Delivery tied to PO
purchase_order_id	FOREIGN KEY	fk_expected_inbound_deliveries_po → procurement.purchase_orders	BR-007	Valid open PO
expected_arrival_date	NOT NULL	nn_expected_inbound_deliveries_date	—	Arrival date required
shipment_status	NOT NULL	nn_expected_inbound_deliveries_status	—	Shipment state required
shipment_status	DEFAULT	def_expected_inbound_deliveries_status = 'PLANNED'	—	Planned by default
shipment_status	CHECK	chk_expected_inbound_deliveries_status IN (PLANNED, IN_TRANSIT, ARRIVED, DELAYED)	—	Valid shipment states
receiving_location_id	NOT NULL	nn_expected_inbound_deliveries_location	—	Destination required
receiving_location_id	FOREIGN KEY	fk_expected_inbound_deliveries_location → network.inventory_locations	—	Valid location

procurement.expected_inbound_delivery_lines

Column	Type	Constraint	Business Rule	Justification
delivery_id	NOT NULL	nn_expected_inbound_delivery_lines_delivery	—	Line belongs to delivery
delivery_id	FOREIGN KEY	fk_expected_inbound_delivery_lines_delivery → procurement.expected_inbound_deliveries	—	Parent delivery exists
purchase_order_line_id	NOT NULL	nn_expected_inbound_delivery_lines_pol	BR-007	Links to PO line

Column	Type	Constraint	Business Rule	Justification
purchase_order_line_id	FOREIGN KEY	fk_expected_inbound_delivery_lines_pol → procurement.purchase_order_lines	BR-007	Valid PO line
product_id	NOT NULL	nn_expected_inbound_delivery_lines_product	—	Product required
product_id	FOREIGN KEY	fk_expected_inbound_delivery_lines_product → catalog.products	—	Valid product
expected_qty	NOT NULL	nn_expected_inbound_delivery_lines_qty	—	Expected quantity required
expected_qty	CHECK	chk_expected_inbound_delivery_lines_qty > 0	—	Positive expected qty
(delivery_id, purchase_order_line_id)	UNIQUE	uk_expected_inbound_delivery_lines	—	One entry per PO line per delivery

Domain 6: Inbound

inbound.goods_receipts

Column	Type	Constraint	Business Rule	Justification
gr_number	NOT NULL	nn_goods_receipts_number	BR-004	Receipt document identity
gr_number	UNIQUE	uk_goods_receipts_number	BR-004	Unique receipt reference
purchase_order_id	FOREIGN KEY	fk_goods_receipts_po → procurement.purchase_orders	BR-007	PO authorization when not ad-hoc
(purchase_order_id, exception_authorization_ref)	CHECK	chk_goods_receipts_po_or_exception	BR-007	PO or documented exception required
supplier_id	NOT NULL	nn_goods_receipts_supplier	BR-004	Supplier accountability
supplier_id	FOREIGN KEY	fk_goods_receipts_supplier → procurement.suppliers	BR-004	Valid supplier
receiving_location_id	NOT NULL	nn_goods_receipts_location	BR-004, BR-024	Receipt site required
receiving_location_id	FOREIGN KEY	fk_goods_receipts_location → network.inventory_locations	BR-024	Location not frozen (app layer)
receipt_date	NOT NULL	nn_goods_receipts_date	BR-004	Receipt timestamp required
receipt_date	DEFAULT	def_goods_receipts_date = NOW()	BR-004	Auto-stamp receipt
verification_status	NOT NULL	nn_goods_receipts_verification	BR-004	Verification state required
verification_status	DEFAULT	def_goods_receipts_verification = 'PENDING'	BR-004	Pending until verified
verification_status	CHECK	chk_goods_receipts_verification IN (PENDING, VERIFIED, REJECTED, PARTIAL)	BR-004	Valid verification states
receiving_clerk_id	NOT NULL	nn_goods_receipts_clerk	BO-19	Receiving clerk required
receiving_clerk_id	FOREIGN KEY	fk_goods_receipts_clerk → identity.user_accounts	BO-19	Clerk attribution

inbound.goods_receipt_lines

Column	Type	Constraint	Business Rule	Justification
goods_receipt_id	NOT NULL	nn_goods_receipt_lines_gr_id	BR-004	Line belongs to receipt
goods_receipt_id	FOREIGN KEY	fk_goods_receipt_lines_gr → inbound.goods_receipts	BR-004	Parent receipt exists
line_number	NOT NULL	nn_goods_receipt_lines_line	—	Line sequence required
(goods_receipt_id, line_number)	UNIQUE	uk_goods_receipt_lines_gr_line	—	Unique line per receipt
product_id	NOT NULL	nn_goods_receipt_lines_product	BR-004	Product identity required
product_id	FOREIGN KEY	fk_goods_receipt_lines_product → catalog.products	BR-004	Valid product
purchase_order_line_id	FOREIGN KEY	fk_goods_receipt_lines_pol → procurement.purchase_order_lines	BR-007	PO line match when applicable
received_qty	NOT NULL	nn_goods_receipt_lines_received	BR-004	Received qty required
received_qty	CHECK	chk_goods_receipt_lines_received >= 0	BR-004	Non-negative received
accepted_qty	NOT NULL	nn_goods_receipt_lines_accepted	BR-004	Accepted qty tracked
accepted_qty	DEFAULT	def_goods_receipt_lines_accepted = 0	BR-004	Zero until verified
accepted_qty	CHECK	chk_goods_receipt_lines_accepted >= 0	BR-004	Non-negative accepted
accepted_qty	CHECK	chk_goods_receipt_lines_accepted_le_received accepted_qty <= received_qty	BR-004	Cannot accept more than received
condition_grade	NOT NULL	nn_goods_receipt_lines_condition	BR-004	Condition assessment required
condition_grade	DEFAULT	def_goods_receipt_lines_condition = 'GOOD'	—	Good condition default
condition_grade	CHECK	chk_goods_receipt_lines_condition IN (GOOD, DAMAGED, REJECTED)	BR-004	Valid condition grades
lot_batch_id	FOREIGN KEY	fk_goods_receipt_lines_lot → inventory.lot_batches	BR-017	Lot when traceable

inbound.receipt_discrepancies

Column	Type	Constraint	Business Rule	Justification
discrepancy_reference	NOT NULL	nn_receipt_discrepancies_ref	BR-006	Discrepancy document identity
discrepancy_reference	UNIQUE	uk_receipt_discrepancies_ref	BR-006	Unique discrepancy reference
goods_receipt_id	NOT NULL	nn_receipt_discrepancies_gr	BR-006	Discrepancy at receipt point
goods_receipt_id	FOREIGN KEY	fk_receipt_discrepancies_gr → inbound.goods_receipts	BR-006	Parent receipt exists
goods_receipt_line_id	FOREIGN KEY	fk_receipt_discrepancies_gr_line → inbound.goods_receipt_lines	BR-006	Optional line-level discrepancy

Column	Type	Constraint	Business Rule	Justification
discrepancy_type	NOT NULL	nn_receipt_discrepancies_type	BR-006	Discrepancy classification
discrepancy_type	CHECK	chk_receipt_discrepancies_type IN (SHORT, OVER, WRONG_ITEM, DAMAGE, SHELF_LIFE)	BR-006, BR-016	Valid discrepancy types
product_id	NOT NULL	nn_receipt_discrepancies_product	BR-006	Affected product required
product_id	FOREIGN KEY	fk_receipt_discrepancies_product → catalog.products	BR-006	Valid product
reason_code_id	FOREIGN KEY	fk_receipt_discrepancies_reason → governance.reason_codes	BR-006	Categorized reason
resolution_status	NOT NULL	nn_receipt_discrepancies_resolution	BR-006	Resolution tracking
resolution_status	DEFAULT	def_receipt_discrepancies_resolution = 'OPEN'	BR-006	Open until resolved
resolution_status	CHECK	chk_receipt_discrepancies_resolution IN (OPEN, RESOLVED, CLAIMED)	BR-006	Valid resolution states

inbound.put_aways

Column	Type	Constraint	Business Rule	Justification
put_away_reference	NOT NULL	nn_put_aways_reference	BR-005	Put-away document identity
put_away_reference	UNIQUE	uk_put_aways_reference	BR-005	Unique put-away reference
goods_receipt_id	NOT NULL	nn_put_aways_gr	BR-005	Put-away completes receipt
goods_receipt_id	FOREIGN KEY	fk_put_aways_gr → inbound.goods_receipts	BR-005	Parent receipt exists
goods_receipt_line_id	NOT NULL	nn_put_aways_gr_line	BR-005	Line-level put-away
goods_receipt_line_id	FOREIGN KEY	fk_put_aways_gr_line → inbound.goods_receipt_lines	BR-005	Valid receipt line
product_id	NOT NULL	nn_put_aways_product	BR-005	Product placed
product_id	FOREIGN KEY	fk_put_aways_product → catalog.products	BR-005	Valid product
quantity	NOT NULL	nn_put_aways_quantity	BR-005	Put-away quantity required
quantity	CHECK	chk_put_aways_quantity > 0	BR-005	Positive placement quantity
storage_position_id	NOT NULL	nn_put_aways_position	BR-005	Destination bin required
storage_position_id	FOREIGN KEY	fk_put_aways_position → network.storage_positions	BR-005, BR-014	Valid storage for rotation
lot_batch_id	FOREIGN KEY	fk_put_aways_lot → inventory.lot_batches	BR-017	Lot placement when traceable
put_away_at	NOT NULL	nn_put_aways_at	BR-005	Completion time required
put_away_at	DEFAULT	def_put_aways_at = NOW()	BR-005	Auto-stamp completion
availability_effective_at	NOT NULL	nn_put_aways_availability	BR-005	ATP effective time required
availability_effective_at	DEFAULT	def_put_aways_availability = NOW()	BR-005	ATP starts at put-away
performed_by	NOT NULL	nn_put_aways_performed_by	BO-19	Staff attribution required
performed_by	FOREIGN KEY	fk_put_aways_performed_by → identity.user_accounts	BO-19	Performer identity

Domain 7: Finance

finance.supplier_invoices

Column	Type	Constraint	Business Rule	Justification
invoice_number	NOT NULL	nn_supplier_invoices_number	BR-008	Invoice identity required
(supplier_id, invoice_number)	UNIQUE	uk_supplier_invoices_supplier_number	BR-008	Unique within supplier
supplier_id	NOT NULL	nn_supplier_invoices_supplier	BR-008	Billing party required
supplier_id	FOREIGN KEY	fk_supplier_invoices_supplier → procurement.suppliers	BR-008	Valid supplier
purchase_order_id	NOT NULL	nn_supplier_invoices_po	BR-008	PO leg of three-way match
purchase_order_id	FOREIGN KEY	fk_supplier_invoices_po → procurement.purchase_orders	BR-008	Valid PO reference
invoice_date	NOT NULL	nn_supplier_invoices_date	BR-008	Invoice date required
total_amount	NOT NULL	nn_supplier_invoices_total	BR-008	Invoice total required
total_amount	CHECK	chk_supplier_invoices_total >= 0	BR-008	Non-negative invoice
currency_code	NOT NULL	nn_supplier_invoices_currency	BR-008	Currency required
currency_code	DEFAULT	def_supplier_invoices_currency = 'USD'	—	Default currency
match_status	NOT NULL	nn_supplier_invoices_match	BR-008	Match state required
match_status	DEFAULT	def_supplier_invoices_match = 'PENDING'	BR-008	Pending until matched
match_status	CHECK	chk_supplier_invoices_match IN (PENDING, MATCHED, EXCEPTION)	BR-008	Valid match states
payment_status	NOT NULL	nn_supplier_invoices_payment	BR-008	Payment gate required
payment_status	DEFAULT	def_supplier_invoices_payment = 'UNAUTHORIZED'	BR-008	Unauthorized until matched
payment_status	CHECK	chk_supplier_invoices_payment IN (UNAUTHORIZED, AUTHORIZED, PAID)	BR-008	Valid payment states
(match_status, payment_status)	CHECK	chk_supplier_invoices_payment_gate	BR-008	AUTHORIZED/PAID only when MATCHED

finance.supplier_invoice_lines

Column	Type	Constraint	Business Rule	Justification
supplier_invoice_id	NOT NULL	nn_supplier_invoice_lines_invoice	BR-008	Line belongs to invoice
supplier_invoice_id	FOREIGN KEY	fk_supplier_invoice_lines_invoice → finance.supplier_invoices	BR-008	Parent invoice exists
line_number	NOT NULL	nn_supplier_invoice_lines_line	—	Line sequence required
(supplier_invoice_id, line_number)	UNIQUE	uk_supplier_invoice_lines_inv_line	—	Unique line per invoice
product_id	NOT NULL	nn_supplier_invoice_lines_product	BR-008	Billed product required
product_id	FOREIGN KEY	fk_supplier_invoice_lines_product → catalog.products	BR-008	Valid product

Column	Type	Constraint	Business Rule	Justification
purchase_order_line_id	FOREIGN KEY	fk_supplier_invoice_lines_pol → procurement.purchase_order_lines	BR-008	PO line match
billed_qty	NOT NULL	nn_supplier_invoice_lines_qty	BR-008	Billed quantity required
billed_qty	CHECK	chk_supplier_invoice_lines_qty > 0	BR-008	Positive billed qty
unit_price	NOT NULL	nn_supplier_invoice_lines_price	BR-008	Unit price required
unit_price	CHECK	chk_supplier_invoice_lines_price >= 0	BR-008	Non-negative price
line_amount	NOT NULL	nn_supplier_invoice_lines_amount	BR-008	Extended amount required

finance.invoice_goods_receipt_match

Column	Type	Constraint	Business Rule	Justification
supplier_invoice_id	NOT NULL	nn_inv_gr_match_invoice	BR-008	Invoice in match set
supplier_invoice_id	FOREIGN KEY	fk_inv_gr_match_invoice → finance.supplier_invoices	BR-008	Valid invoice
goods_receipt_id	NOT NULL	nn_inv_gr_match_gr	BR-008	Receipt in match set
goods_receipt_id	FOREIGN KEY	fk_inv_gr_match_gr → inbound.goods_receipts	BR-008	Valid receipt
(supplier_invoice_id, goods_receipt_id)	UNIQUE	uk_inv_gr_match_pair	BR-008	One match record per pair
match_status	NOT NULL	nn_inv_gr_match_status	BR-008	Match outcome tracked
match_status	DEFAULT	def_inv_gr_match_status = 'PENDING'	BR-008	Pending by default
match_status	CHECK	chk_inv_gr_match_status IN (PENDING, MATCHED, EXCEPTION)	BR-008	Valid match states

Domain 8: Fulfillment

fulfillment.customer_orders

Column	Type	Constraint	Business Rule	Justification
order_number	NOT NULL	nn_customer_orders_number	BR-001	Order document identity
order_number	UNIQUE	uk_customer_orders_number	—	Unique order reference
order_date	NOT NULL	nn_customer_orders_date	—	Order date required
order_date	DEFAULT	def_customer_orders_date = NOW()	—	Auto-stamp order date
order_status	NOT NULL	nn_customer_orders_status	BR-001, BR-002	Lifecycle state required
order_status	DEFAULT	def_customer_orders_status = 'DRAFT'	—	Draft until confirmed
order_status	CHECK	chk_customer_orders_status	BR-001	DRAFT, CONFIRMED, PARTIAL, FULFILLED, CANCELLED

fulfillment.customer_order_lines

Column	Type	Constraint	Business Rule	Justification
customer_order_id	NOT NULL	nn_customer_order_lines_order	—	Line belongs to order
customer_order_id	FOREIGN KEY	fk_customer_order_lines_order → fulfillment.customer_orders	—	Parent order exists
line_number	NOT NULL	nn_customer_order_lines_line	—	Line sequence required
(customer_order_id, line_number)	UNIQUE	uk_customer_order_lines_order_line	—	Unique line per order
product_id	NOT NULL	nn_customer_order_lines_product	BR-027	Product required
product_id	FOREIGN KEY	fk_customer_order_lines_product → catalog.products	BR-027	Valid product
requested_qty	NOT NULL	nn_customer_order_lines_requested	BR-001	Requested qty required
requested_qty	CHECK	chk_customer_order_lines_requested > 0	BR-001	Positive order quantity
fulfilled_qty	NOT NULL	nn_customer_order_lines_fulfilled	BR-003	Fulfilled tracked
fulfilled_qty	DEFAULT	def_customer_order_lines_fulfilled = 0	—	Zero at creation
fulfilled_qty	CHECK	chk_customer_order_lines_fulfilled >= 0	BR-003	Non-negative fulfilled
backordered_qty	NOT NULL	nn_customer_order_lines_backordered	BR-028	Backorder tracked
backordered_qty	DEFAULT	def_customer_order_lines_backordered = 0	BR-028	Zero at creation
backordered_qty	CHECK	chk_customer_order_lines_backordered >= 0	BR-028	Non-negative backorder
(requested_qty, fulfilled_qty, backordered_qty)	CHECK	chk_customer_order_lines_qty_balance fulfilled + backordered <= requested	BR-001, BR-028	Line quantity reconciliation
fulfillment_location_id	FOREIGN KEY	fk_customer_order_lines_location → network.inventory_locations	BR-001	Ship-from site

fulfillment.inventory_reservations

Column	Type	Constraint	Business Rule	Justification
reservation_reference	NOT NULL	nn_inventory_reservations_ref	BR-002	Reservation document identity
reservation_reference	UNIQUE	uk_inventory_reservations_ref	BR-002	Unique reservation reference
customer_order_id	NOT NULL	nn_inventory_reservations_order	BR-002	Reservation serves order
customer_order_id	FOREIGN KEY	fk_inventory_reservations_order → fulfillment.customer_orders	BR-002	Valid order
order_line_id	NOT NULL	nn_inventory_reservations_line	BR-002	Line-level reservation
order_line_id	FOREIGN KEY	fk_inventory_reservations_line → fulfillment.customer_order_lines	BR-002	Valid order line
product_id	NOT NULL	nn_inventory_reservations_product	BR-002	Reserved SKU
product_id	FOREIGN KEY	fk_inventory_reservations_product → catalog.products	BR-027	Valid product

Column	Type	Constraint	Business Rule	Justification
location_id	NOT NULL	nn_inventory_reservations_location	BR-002	Reservation location
location_id	FOREIGN KEY	fk_inventory_reservations_location → network.inventory_locations	BR-002	Valid location
stock_position_id	NOT NULL	nn_inventory_reservations_position	BR-002	Target balance
stock_position_id	FOREIGN KEY	fk_inventory_reservations_position → inventory.stock_positions	BR-002	Valid stock position
reserved_qty	NOT NULL	nn_inventory_reservations_qty	BR-002	Reserved quantity required
reserved_qty	CHECK	chk_inventory_reservations_qty > 0	BR-002	Positive reservation
reservation_status	NOT NULL	nn_inventory_reservations_status	BR-002	Lifecycle state
reservation_status	DEFAULT	def_inventory_reservations_status = 'ACTIVE'	BR-002	Active on creation
reservation_status	CHECK	chk_inventory_reservations_status IN (ACTIVE, RELEASED, FULFILLED)	BR-002	Valid states
reserved_at	NOT NULL	nn_inventory_reservations_at	BR-002	Reservation timestamp
reserved_at	DEFAULT	def_inventory_reservations_at = NOW()	BR-002	Auto-stamp at confirmation

fulfillment.fulfillment_executions

Column	Type	Constraint	Business Rule	Justification
fulfillment_reference	NOT NULL	nn_fulfillment_executions_ref	BR-003	Pick work document
fulfillment_reference	UNIQUE	uk_fulfillment_executions_ref	BR-003	Unique fulfillment reference
customer_order_id	NOT NULL	nn_fulfillment_executions_order	BR-003	Execution serves order
customer_order_id	FOREIGN KEY	fk_fulfillment_executions_order → fulfillment.customer_orders	BR-003	Valid order
fulfillment_location_id	NOT NULL	nn_fulfillment_executions_location	BR-003, BR-024	Pick site required
fulfillment_location_id	FOREIGN KEY	fk_fulfillment_executions_location → network.inventory_locations	BR-024	Valid non-frozen location
pick_verification_status	NOT NULL	nn_fulfillment_executions_pick_status	BR-003	Pick verification state
pick_verification_status	DEFAULT	def_fulfillment_executions_pick_status = 'PENDING'	BR-003	Pending until verified
pick_verification_status	CHECK	chk_fulfillment_executions_pick_status IN (PENDING, VERIFIED, EXCEPTION)	BR-003	Valid pick states
pack_confirmed	NOT NULL	nn_fulfillment_executions_pack	BR-003	Pack flag explicit
pack_confirmed	DEFAULT	def_fulfillment_executions_pack = FALSE	BR-003	Not packed by default
has_exception	NOT NULL	nn_fulfillment_executions_exception	BR-003	Exception flag explicit
has_exception	DEFAULT	def_fulfillment_executions_exception = FALSE	—	No exception by default
assigned_picker_id	FOREIGN KEY	fk_fulfillment_executions_picker → identity.user_accounts	BO-19	Picker attribution

fulfillment.fulfillment_pick_lines

Column	Type	Constraint	Business Rule	Justification
fulfillment_execution_id	NOT NULL	nn_fulfillment_pick_lines_fe	BR-003	Pick belongs to execution
fulfillment_execution_id	FOREIGN KEY	fk_fulfillment_pick_lines_fe → fulfillment.fulfillment_executions	BR-003	Parent execution exists
order_line_id	NOT NULL	nn_fulfillment_pick_lines_ol	BR-003	Fulfills order line
order_line_id	FOREIGN KEY	fk_fulfillment_pick_lines_ol → fulfillment.customer_order_lines	BR-003	Valid order line
product_id	NOT NULL	nn_fulfillment_pick_lines_product	BR-003	Picked product
product_id	FOREIGN KEY	fk_fulfillment_pick_lines_product → catalog.products	BR-003	Valid product
(fulfillment_execution_id, order_line_id)	UNIQUE	uk_fulfillment_pick_lines_fe_ol	BR-003	One pick per line per execution
requested_qty	NOT NULL	nn_fulfillment_pick_lines_requested	BR-003	Pick quantity required
requested_qty	CHECK	chk_fulfillment_pick_lines_requested > 0	BR-003	Positive pick qty
picked_qty	NOT NULL	nn_fulfillment_pick_lines_picked	BR-003	Actual picked tracked
picked_qty	DEFAULT	def_fulfillment_pick_lines_picked = 0	BR-003	Zero until picked
picked_qty	CHECK	chk_fulfillment_pick_lines_picked >= 0	BR-003	Non-negative picked
picked_qty	CHECK	chk_fulfillment_pick_lines_picked_le_requested picked_qty <= requested_qty	BR-003	Cannot over-pick without exception
storage_position_id	FOREIGN KEY	fk_fulfillment_pick_lines_position → network.storage_positions	BR-014	Source bin
lot_batch_id	FOREIGN KEY	fk_fulfillment_pick_lines_lot → inventory.lot_batches	BR-017	Lot on pick when traceable

fulfillment.backorders

Column	Type	Constraint	Business Rule	Justification
backorder_reference	NOT NULL	nn_backorders_ref	BR-028	Backorder document identity
backorder_reference	UNIQUE	uk_backorders_ref	BR-028	Unique backorder reference
customer_order_id	NOT NULL	nn_backorders_order	BR-028	Backorder on order
customer_order_id	FOREIGN KEY	fk_backorders_order → fulfillment.customer_orders	BR-028	Valid order
order_line_id	NOT NULL	nn_backorders_line	BR-028	Line-level backorder
order_line_id	FOREIGN KEY	fk_backorders_line → fulfillment.customer_order_lines	BR-028	Valid order line
product_id	NOT NULL	nn_backorders_product	BR-028	Awaiting SKU
product_id	FOREIGN KEY	fk_backorders_product → catalog.products	BR-028	Valid product

Column	Type	Constraint	Business Rule	Justification
backordered_qty	NOT NULL	nn_backorders_qty	BR-028	Backordered quantity
backordered_qty	CHECK	chk_backorders_qty > 0	BR-028	Positive backorder qty
customer_acknowledged	NOT NULL	nn_backorders_acknowledged	BR-028	Acknowledgment flag required
customer_acknowledged	DEFAULT	def_backorders_acknowledged = FALSE	BR-028	Not acknowledged by default
(customer_acknowledged, acknowledged_at)	CHECK	chk_backorders_ack_timestamp	BR-028	Acknowledged requires timestamp
resolution_status	NOT NULL	nn_backorders_resolution	BR-028	Resolution tracking
resolution_status	DEFAULT	def_backorders_resolution = 'OPEN'	BR-028	Open until resolved
resolution_status	CHECK	chk_backorders_resolution IN (OPEN, FULFILLED, CANCELLED)	BR-028	Valid resolution states

fulfillment.shipments

Column	Type	Constraint	Business Rule	Justification
shipment_number	NOT NULL	nn_shipments_number	BR-003	Shipment document identity
shipment_number	UNIQUE	uk_shipments_number	BR-003	Unique shipment reference
customer_order_id	NOT NULL	nn_shipments_order	BR-003	Shipment fulfills order
customer_order_id	FOREIGN KEY	fk_shipments_order → fulfillment.customer_orders	BR-003	Valid order
fulfillment_execution_id	NOT NULL	nn_shipments_fe	BR-003	Verified pick required
fulfillment_execution_id	FOREIGN KEY	fk_shipments_fe → fulfillment.fulfillment_executions	BR-003	Valid verified execution
ship_from_location_id	NOT NULL	nn_shipments_location	BR-003	Ship-from site
ship_from_location_id	FOREIGN KEY	fk_shipments_location → network.inventory_locations	BR-003	Valid location
ship_date	NOT NULL	nn_shipments_date	BR-003	Dispatch timestamp
ship_date	DEFAULT	def_shipments_date = NOW()	BR-003	Auto-stamp dispatch
shipment_status	NOT NULL	nn_shipments_status	BR-003	Shipment lifecycle
shipment_status	DEFAULT	def_shipments_status = 'DISPATCHED'	BR-003	Dispatched on creation
shipment_status	CHECK	chk_shipments_status IN (DISPATCHED, IN_TRANSIT, DELIVERED)	BR-003	Valid shipment states

fulfillment.shipment_lines

Column	Type	Constraint	Business Rule	Justification
shipment_id	NOT NULL	nn_shipment_lines_shipment	BR-003	Line belongs to shipment
shipment_id	FOREIGN KEY	fk_shipment_lines_shipment → fulfillment.shipments	BR-003	Parent shipment exists
order_line_id	NOT NULL	nn_shipment_lines_ol	BR-003	Shipped order line

Column	Type	Constraint	Business Rule	Justification
order_line_id	FOREIGN KEY	fk_shipment_lines_ol → fulfillment.customer_order_lines	BR-003	Valid order line
(shipment_id, order_line_id)	UNIQUE	uk_shipment_lines_shipment_ol	BR-003	One line per shipment
product_id	NOT NULL	nn_shipment_lines_product	BR-003	Shipped product
product_id	FOREIGN KEY	fk_shipment_lines_product → catalog.products	BR-003	Valid product
shipped_qty	NOT NULL	nn_shipment_lines_qty	BR-003	Shipped quantity
shipped_qty	CHECK	chk_shipment_lines_qty > 0	BR-003	Positive shipped qty
lot_batch_id	FOREIGN KEY	fk_shipment_lines_lot → inventory.lot_batches	BR-017	Lot traceability on dispatch
reservation_id	FOREIGN KEY	fk_shipment_lines_reservation → fulfillment.inventory_reservations	BR-002	Consumes reservation

Domain 9: Network Transfer

transfer.transfer_requests

Column	Type	Constraint	Business Rule	Justification
request_number	NOT NULL	nn_transfer_requests_number	BR-011	Transfer request identity
request_number	UNIQUE	uk_transfer_requests_number	—	Unique request reference
source_location_id	NOT NULL	nn_transfer_requests_source	BR-012	Source site required
source_location_id	FOREIGN KEY	fk_transfer_requests_source → network.inventory_locations	BR-012	Valid source
destination_location_id	NOT NULL	nn_transfer_requests_dest	BR-011	Destination required
destination_location_id	FOREIGN KEY	fk_transfer_requests_dest → network.inventory_locations	BR-011	Valid destination
(source_location_id, destination_location_id)	CHECK	chk_transfer_requests_distinct_locations source ≠ destination	BR-011	Source and destination must differ
request_status	NOT NULL	nn_transfer_requests_status	BR-012	Approval workflow state
request_status	DEFAULT	def_transfer_requests_status = 'SUBMITTED'	—	Submitted on creation
request_status	CHECK	chk_transfer_requests_status IN (SUBMITTED, APPROVED, REJECTED, CONVERTED)	BR-012	Valid states
requested_by	NOT NULL	nn_transfer_requests_requester	BO-19	Requestor required
requested_by	FOREIGN KEY	fk_transfer_requests_requester → identity.user_accounts	BO-19	Requestor attribution
approved_by	FOREIGN KEY	fk_transfer_requests_approver → identity.user_accounts	BR-012	Approver attribution
stock_transfer_id	FOREIGN KEY	fk_transfer_requests_transfer → transfer.stock_transfers	BR-012	Execution link

transfer.transfer_request_lines

Column	Type	Constraint	Business Rule	Justification
transfer_request_id	NOT NULL	nn_transfer_request_lines_req	BR-012	Line belongs to request
transfer_request_id	FOREIGN KEY	fk_transfer_request_lines_req → transfer.transfer_requests	BR-012	Parent request exists
(transfer_request_id, line_number)	UNIQUE	uk_transfer_request_lines_req_line	—	Unique line per request
product_id	NOT NULL	nn_transfer_request_lines_product	BR-017	Product to transfer
product_id	FOREIGN KEY	fk_transfer_request_lines_product → catalog.products	BR-017	Valid product
requested_qty	NOT NULL	nn_transfer_request_lines_qty	BR-012	Requested quantity
requested_qty	CHECK	chk_transfer_request_lines_qty > 0	BR-012	Positive transfer qty

transfer.stock_transfers

Column	Type	Constraint	Business Rule	Justification
transfer_number	NOT NULL	nn_stock_transfers_number	BR-012	Transfer document identity
transfer_number	UNIQUE	uk_stock_transfers_number	—	Unique transfer reference
transfer_request_id	FOREIGN KEY	fk_stock_transfers_request → transfer.transfer_requests	BR-012	Optional request origin
source_location_id	NOT NULL	nn_stock_transfers_source	BR-012	Source site
source_location_id	FOREIGN KEY	fk_stock_transfers_source → network.inventory_locations	BR-012	Valid source
destination_location_id	NOT NULL	nn_stock_transfers_dest	BR-012	Destination site
destination_location_id	FOREIGN KEY	fk_stock_transfers_dest → network.inventory_locations	BR-012	Valid destination
(source_location_id, destination_location_id)	CHECK	chk_stock_transfers_distinct_locations	BR-011	Distinct endpoints
transfer_status	NOT NULL	nn_stock_transfers_status	BR-012	Transfer lifecycle
transfer_status	DEFAULT	def_stock_transfers_status = 'APPROVED'	—	Approved when created
transfer_status	CHECK	chk_stock_transfers_status IN (REQUESTED, APPROVED, IN_TRANSIT, RECEIVED, CANCELLED)	BR-012	Valid states
(transfer_status, dispatch_date)	CHECK	chk_stock_transfers_dispatch	BR-012	IN_TRANSIT/RECEIVED requires dispatch_date
(transfer_status, receipt_date)	CHECK	chk_stock_transfers_receipt	BR-012	RECEIVED requires receipt_date

transfer.stock_transfer_lines

Column	Type	Constraint	Business Rule	Justification
stock_transfer_id	NOT NULL	nn_stock_transfer_lines_transfer	BR-012	Line belongs to transfer
stock_transfer_id	FOREIGN KEY	fk_stock_transfer_lines_transfer → transfer.stock_transfers	BR-012	Parent transfer exists
(stock_transfer_id, line_number)	UNIQUE	uk_stock_transfer_lines_transfer_line	—	Unique line per transfer
product_id	NOT NULL	nn_stock_transfer_lines_product	BR-017	Transferred product
product_id	FOREIGN KEY	fk_stock_transfer_lines_product → catalog.products	BR-017	Valid product
quantity	NOT NULL	nn_stock_transfer_lines_qty	BR-012	Transfer quantity
quantity	CHECK	chk_stock_transfer_lines_qty > 0	BR-012	Positive transfer qty
dispatched_qty	NOT NULL	nn_stock_transfer_lines_dispatched	BR-012	Dispatched tracked
dispatched_qty	DEFAULT	def_stock_transfer_lines_dispatched = 0	—	Zero until dispatch
dispatched_qty	CHECK	chk_stock_transfer_lines_dispatched >= 0	BR-012	Non-negative
received_qty	NOT NULL	nn_stock_transfer_lines_received	BR-012	Received tracked
received_qty	DEFAULT	def_stock_transfer_lines_received = 0	—	Zero until receipt
received_qty	CHECK	chk_stock_transfer_lines_received >= 0	BR-012	Non-negative
received_qty	CHECK	chk_stock_transfer_lines_received_le_dispatched received_qty <= dispatched_qty	BR-012	Cannot receive more than dispatched
lot_batch_id	FOREIGN KEY	fk_stock_transfer_lines_lot → inventory.lot_batches	BR-017	Lot on transfer

transfer.in_transit_stock

Column	Type	Constraint	Business Rule	Justification
in_transit_reference	NOT NULL	nn_in_transit_stock_ref	BR-012	Pipeline stock identity
in_transit_reference	UNIQUE	uk_in_transit_stock_ref	—	Unique in-transit reference
stock_transfer_id	NOT NULL	nn_in_transit_stock_transfer	BR-012	Belongs to transfer
stock_transfer_id	FOREIGN KEY	fk_in_transit_stock_transfer → transfer.stock_transfers	BR-012	Parent transfer exists
stock_transfer_line_id	NOT NULL	nn_in_transit_stock_line	BR-012	Line-level tracking
stock_transfer_line_id	FOREIGN KEY	fk_in_transit_stock_line → transfer.stock_transfer_lines	BR-012	Valid transfer line
product_id	NOT NULL	nn_in_transit_stock_product	—	Product in pipeline
product_id	FOREIGN KEY	fk_in_transit_stock_product → catalog.products	—	Valid product
quantity	NOT NULL	nn_in_transit_stock_qty	BR-012	In-transit quantity
quantity	CHECK	chk_in_transit_stock_qty > 0	BR-012	Positive pipeline qty
source_location_id	NOT NULL	nn_in_transit_stock_source	BR-012	Origin site
source_location_id	FOREIGN KEY	fk_in_transit_stock_source → network.inventory_locations	BR-012	Valid source

Column	Type	Constraint	Business Rule	Justification
destination_location_id	NOT NULL	nn_in_transit_stock_dest	BR-012	Destination site
destination_location_id	FOREIGN KEY	fk_in_transit_stock_dest → network.inventory_locations	BR-012	Valid destination
dispatch_date	NOT NULL	nn_in_transit_stock_dispatch	BR-012	Dispatch time required
in_transit_status	NOT NULL	nn_in_transit_stock_status	BR-012	Pipeline state
in_transit_status	DEFAULT	def_in_transit_stock_status = 'IN_TRANSIT'	BR-012	In-transit by default
in_transit_status	CHECK	chk_in_transit_stock_status IN (IN_TRANSIT, RECEIVED, LOST, DELAYED)	BR-012	Valid pipeline states

Domain 10: Inventory Accuracy

accuracy.cycle_counts

Column	Type	Constraint	Business Rule	Justification
count_assignment_number	NOT NULL	nn_cycle_counts_number	BR-018	Count assignment identity
count_assignment_number	UNIQUE	uk_cycle_counts_number	BR-018	Unique count reference
schedule_date	NOT NULL	nn_cycle_counts_schedule	BR-018, BR-032	Scheduled date required
location_id	NOT NULL	nn_cycle_counts_location	BR-018	Count site required
location_id	FOREIGN KEY	fk_cycle_counts_location → network.inventory_locations	BR-018	Valid location
product_id	NOT NULL	nn_cycle_counts_product	BR-018	Counted SKU required
product_id	FOREIGN KEY	fk_cycle_counts_product → catalog.products	BR-032	Valid product
storage_position_id	FOREIGN KEY	fk_cycle_counts_position → network.storage_positions	—	Optional bin scope
counter_id	NOT NULL	nn_cycle_counts_counter	BR-018, BR-019	Counter required
counter_id	FOREIGN KEY	fk_cycle_counts_counter → identity.user_accounts	BR-019	Counter attribution
count_status	NOT NULL	nn_cycle_counts_status	BR-018	Count workflow state
count_status	DEFAULT	def_cycle_counts_status = 'SCHEDULED'	BR-018	Scheduled by default
count_status	CHECK	chk_cycle_counts_status IN (SCHEDULED, IN_PROGRESS, COMPLETED)	BR-018	Valid count states
physical_count_qty	CHECK	chk_cycle_counts_physical >= 0 OR NULL	BR-018	Physical qty non-negative when recorded
(count_status, physical_count_qty)	CHECK	chk_cycle_counts_completed	BR-018	COMPLETED requires physical_count_qty

accuracy.physical_inventory_counts

Column	Type	Constraint	Business Rule	Justification
count_event_number	NOT NULL	nn_physical_inventory_counts_number	BR-024	Count event identity

Column	Type	Constraint	Business Rule	Justification
count_event_number	UNIQUE	uk_physical_inventory_counts_number	BR-024	Unique event reference
location_id	NOT NULL	nn_physical_inventory_counts_location	BR-024	Wall-to-wall site
location_id	FOREIGN KEY	fk_physical_inventory_counts_location → network.inventory_locations	BR-024	Valid location
count_start_date	NOT NULL	nn_physical_inventory_counts_start	BR-024	Count start required
freeze_status	NOT NULL	nn_physical_inventory_counts_freeze	BR-024	Freeze state tracked
freeze_status	DEFAULT	def_physical_inventory_counts_freeze = 'ACTIVE'	BR-024	Active freeze on start
freeze_status	CHECK	chk_physical_inventory_counts_freeze IN (ACTIVE, RELEASED)	BR-024	Valid freeze states
total_variances	NOT NULL	nn_physical_inventory_counts_variances	BR-024	Variance count tracked
total_variances	DEFAULT	def_physical_inventory_counts_variances = 0	—	Zero at start
total_variances	CHECK	chk_physical_inventory_counts_variances >= 0	—	Non-negative count
certification_status	NOT NULL	nn_physical_inventory_counts_cert	BR-024	Certification tracking
certification_status	DEFAULT	def_physical_inventory_counts_cert = 'IN_PROGRESS'	BR-024	In progress on start
certification_status	CHECK	chk_physical_inventory_counts_cert IN (IN_PROGRESS, CERTIFIED)	BR-024	Valid certification states
(freeze_status, certification_status)	CHECK	chk_physical_inventory_counts_freeze_release	BR-024	CERTIFIED requires freeze RELEASED

accuracy.count_variances

Column	Type	Constraint	Business Rule	Justification
variance_reference	NOT NULL	nn_count_variances_ref	BR-022	Variance document identity
variance_reference	UNIQUE	uk_count_variances_ref	BR-022	Unique variance reference
cycle_count_id	FOREIGN KEY	fk_count_variances_cycle → accuracy.cycle_counts	BR-022	Cycle count origin
physical_count_id	FOREIGN KEY	fk_count_variances_physical → accuracy.physical_inventory_counts	BR-022	Physical count origin
(cycle_count_id, physical_count_id)	CHECK	chk_count_variances_single_origin	BR-022	Exactly one count origin — XOR not both null, not both set
stock_position_id	NOT NULL	nn_count_variances_position	BR-022	Affected balance
stock_position_id	FOREIGN KEY	fk_count_variances_position → inventory.stock_positions	BR-022	Valid stock position
product_id	NOT NULL	nn_count_variances_product	—	Denormalized product
product_id	FOREIGN KEY	fk_count_variances_product → catalog.products	—	Valid product
location_id	NOT NULL	nn_count_variances_location	—	Denormalized location
location_id	FOREIGN KEY	fk_count_variances_location → network.inventory_locations	—	Valid location
system_qty	NOT NULL	nn_count_variances_system	BR-018	System quantity at count
system_qty	CHECK	chk_count_variances_system >= 0	LDM-01	Non-negative system qty

Column	Type	Constraint	Business Rule	Justification
physical_qty	NOT NULL	nn_count_variances_physical	BR-018	Physical quantity recorded
physical_qty	CHECK	chk_count_variances_physical >= 0	BR-018	Non-negative physical qty
variance_qty	NOT NULL	nn_count_variances_variance	BR-022	Variance quantity
investigation_status	NOT NULL	nn_count_variances_investigation	BR-022	Investigation required
investigation_status	DEFAULT	def_count_variances_investigation = 'OPEN'	BR-022	Open until investigated
investigation_status	CHECK	chk_count_variances_investigation IN (OPEN, INVESTIGATED)	BR-022	Valid investigation states
approval_status	NOT NULL	nn_count_variances_approval	BR-019, BR-021	Approval workflow
approval_status	DEFAULT	def_count_variances_approval = 'PENDING'	BR-019	Pending approval
approval_status	CHECK	chk_count_variances_approval IN (PENDING, APPROVED, REJECTED)	BR-019	Valid approval states
approver_id	FOREIGN KEY	fk_count_variances_approver → identity.user_accounts	BR-019	Independent approver
(approval_status, approver_id)	CHECK	chk_count_variances_approver_required	BR-019	APPROVED requires approver

accuracy.stock_adjustments

Column	Type	Constraint	Business Rule	Justification
adjustment_reference	NOT NULL	nn_stock_adjustments_ref	BR-020	Adjustment document identity
adjustment_reference	UNIQUE	uk_stock_adjustments_ref	BR-020	Unique adjustment reference
stock_position_id	NOT NULL	nn_stock_adjustments_position	BR-020	Target balance
stock_position_id	FOREIGN KEY	fk_stock_adjustments_position → inventory.stock_positions	BR-020	Valid stock position
product_id	NOT NULL	nn_stock_adjustments_product	BR-020	Denormalized product
product_id	FOREIGN KEY	fk_stock_adjustments_product → catalog.products	BR-020	Valid product
location_id	NOT NULL	nn_stock_adjustments_location	BR-020	Denormalized location
location_id	FOREIGN KEY	fk_stock_adjustments_location → network.inventory_locations	BR-020	Valid location
adjustment_qty	NOT NULL	nn_stock_adjustments_qty	BR-020	Signed adjustment required
adjustment_qty	CHECK	chk_stock_adjustments_qty <> 0	BR-020	Zero adjustment meaningless
qty_before	NOT NULL	nn_stock_adjustments_before	BR-020	Before snapshot required
qty_before	CHECK	chk_stock_adjustments_before >= 0	LDM-01	Non-negative before
qty_after	NOT NULL	nn_stock_adjustments_after	BR-020	After snapshot required
qty_after	CHECK	chk_stock_adjustments_after >= 0	LDM-01	Adjustment must not drive negative (LDM-10)
(qty_before, qty_after, adjustment_qty)	CHECK	chk_stock_adjustments_balance qty_after = qty_before + adjustment_qty	BR-020	Quantity arithmetic integrity
reason_code_id	NOT NULL	nn_stock_adjustments_reason	BR-020	Reason mandatory

Column	Type	Constraint	Business Rule	Justification
reason_code_id	FOREIGN KEY	fk_stock_adjustments_reason → governance.reason_codes	BR-020	Valid active reason code
count_variance_id	FOREIGN KEY	fk_stock_adjustments_variance → accuracy.count_variances	BR-022	Count-to-correction link
approval_status	NOT NULL	nn_stock_adjustments_approval	BR-021	Approval workflow
approval_status	DEFAULT	def_stock_adjustments_approval = 'PENDING'	BR-021	Pending by default
approval_status	CHECK	chk_stock_adjustments_approval IN (PENDING, APPROVED, REJECTED)	BR-021	Valid states
initiated_by	NOT NULL	nn_stock_adjustments_initiator	BR-021	Initiator required
initiated_by	FOREIGN KEY	fk_stock_adjustments_initiator → identity.user_accounts	BR-021	Initiator attribution
approved_by	FOREIGN KEY	fk_stock_adjustments_approver → identity.user_accounts	BR-021	Approver attribution
(initiated_by, approved_by)	CHECK	chk_stock_adjustments_sod	BR-021	Approver ≠ initiator when both set
adjustment_date	NOT NULL	nn_stock_adjustments_date	BR-020	Effective date required
adjustment_date	DEFAULT	def_stock_adjustments_date = NOW()	BR-020	Auto-stamp effective date

accuracy.inventory_write_offs

Column	Type	Constraint	Business Rule	Justification
write_off_reference	NOT NULL	nn_inventory_write_offs_ref	BR-023	Write-off document identity
write_off_reference	UNIQUE	uk_inventory_write_offs_ref	BR-023	Unique write-off reference
stock_position_id	NOT NULL	nn_inventory_write_offs_position	BR-023	Target balance
stock_position_id	FOREIGN KEY	fk_inventory_write_offs_position → inventory.stock_positions	BR-023	Valid stock position
product_id	NOT NULL	nn_inventory_write_offs_product	BR-023	Denormalized product
product_id	FOREIGN KEY	fk_inventory_write_offs_product → catalog.products	BR-023	Valid product
location_id	NOT NULL	nn_inventory_write_offs_location	BR-023	Denormalized location
location_id	FOREIGN KEY	fk_inventory_write_offs_location → network.inventory_locations	BR-023	Valid location
write_off_qty	NOT NULL	nn_inventory_write_offs_qty	BR-023	Write-off quantity
write_off_qty	CHECK	chk_inventory_write_offs_qty > 0	BR-023	Positive removal quantity
reason_code_id	NOT NULL	nn_inventory_write_offs_reason	BR-020, BR-023	Reason mandatory
reason_code_id	FOREIGN KEY	fk_inventory_write_offs_reason → governance.reason_codes	BR-020	Valid reason code
return_disposition_id	FOREIGN KEY	fk_inventory_write_offs_disposition → returns.return_dispositions	BR-026	Return-originated write-off

Column	Type	Constraint	Business Rule	Justification
initiated_by	NOT NULL	nn_inventory_write_offs_initiator	BR-023	Initiator required
initiated_by	FOREIGN KEY	fk_inventory_write_offs_initiator → identity.user_accounts	BR-023	Initiator attribution
authorized_by	NOT NULL	nn_inventory_write_offs_authorizer	BR-023	Authorization required
authorized_by	FOREIGN KEY	fk_inventory_write_offs_authorizer → identity.user_accounts	BR-023	Authorizer attribution
(initiated_by, authorized_by)	CHECK	chk_inventory_write_offs_sod	BR-023	Authorizer ≠ initiator
write_off_date	NOT NULL	nn_inventory_write_offs_date	BR-023	Effective date
write_off_date	DEFAULT	def_inventory_write_offs_date = NOW()	BR-023	Auto-stamp effective date
disposition_method	CHECK	chk_inventory_write_offs_disposition IN (DESTROY, DONATE, SCRAP) OR NULL	BR-023	Valid disposition methods

Domain 11: Returns

returns.return_authorizations

Column	Type	Constraint	Business Rule	Justification
rma_number	NOT NULL	nn_return_authorizations_rma	BR-025	RMA document identity
rma_number	UNIQUE	uk_return_authorizations_rma	BR-025	Unique RMA reference
customer_order_id	NOT NULL	nn_return_authorizations_order	BR-025	Original order required
customer_order_id	FOREIGN KEY	fk_return_authorizations_order → fulfillment.customer_orders	BR-025	Valid original order
authorization_date	NOT NULL	nn_return_authorizations_date	BR-025	Issue date required
authorization_date	DEFAULT	def_return_authorizations_date = NOW()	BR-025	Auto-stamp issue
(authorization_date, expiration_date)	CHECK	chk_return_authorizations_expiry expiration IS NULL OR expiration >= authorization_date::date	BR-025	Valid expiry window
authorization_status	NOT NULL	nn_return_authorizations_status	BR-025	RMA lifecycle
authorization_status	DEFAULT	def_return_authorizations_status = 'ACTIVE'	BR-025	Active on issue
authorization_status	CHECK	chk_return_authorizations_status IN (ACTIVE, EXPIRED, USED, CANCELLED)	BR-025	Valid RMA states

returns.return_authorization_lines

Column	Type	Constraint	Business Rule	Justification
return_authorization_id	NOT NULL	nn_return_authorization_lines_rma	BR-025	Line belongs to RMA
return_authorization_id	FOREIGN KEY	fk_return_authorization_lines_rma → returns.return_authorizations	BR-025	Parent RMA exists

Column	Type	Constraint	Business Rule	Justification
(return_authorization_id, line_number)	UNIQUE	uk_return_authorization_lines_rma_line	BR-025	Unique line per RMA
product_id	NOT NULL	nn_return_authorization_lines_product	BR-025	Authorized product
product_id	FOREIGN KEY	fk_return_authorization_lines_product → catalog.products	BR-025	Valid product
authorized_qty	NOT NULL	nn_return_authorization_lines_qty	BR-025	Authorized quantity
authorized_qty	CHECK	chk_return_authorization_lines_qty > 0	BR-025	Positive authorized qty

returns.customer_returns

Column	Type	Constraint	Business Rule	Justification
return_number	NOT NULL	nn_customer_returns_number	BR-025	Return receipt identity
return_number	UNIQUE	uk_customer_returns_number	BR-025	Unique return reference
return_authorization_id	NOT NULL	nn_customer_returns_rma	BR-025	Valid RMA required
return_authorization_id	FOREIGN KEY	fk_customer_returns_rma → returns.return_authorizations	BR-025	RMA must exist and be active
customer_order_id	NOT NULL	nn_customer_returns_order	BR-025	Original order link
customer_order_id	FOREIGN KEY	fk_customer_returns_order → fulfillment.customer_orders	BR-025	Valid order
receiving_location_id	NOT NULL	nn_customer_returns_location	BR-004	Receipt site
receiving_location_id	FOREIGN KEY	fk_customer_returns_location → network.inventory_locations	BR-004	Valid location
receipt_date	NOT NULL	nn_customer_returns_date	BR-004	Receipt timestamp
receipt_date	DEFAULT	def_customer_returns_date = NOW()	BR-004	Auto-stamp receipt
inspection_status	NOT NULL	nn_customer_returns_inspection	BR-026	Inspection workflow
inspection_status	DEFAULT	def_customer_returns_inspection = 'PENDING'	BR-026	Pending inspection
inspection_status	CHECK	chk_customer_returns_inspection IN (PENDING, INSPECTED)	BR-026	Valid inspection states
receiving_clerk_id	NOT NULL	nn_customer_returns_clerk	BO-19	Clerk attribution
receiving_clerk_id	FOREIGN KEY	fk_customer_returns_clerk → identity.user_accounts	BO-19	Clerk identity

returns.customer_return_lines

Column	Type	Constraint	Business Rule	Justification
customer_return_id	NOT NULL	nn_customer_return_lines_return	BR-004	Line belongs to return
customer_return_id	FOREIGN KEY	fk_customer_return_lines_return → returns.customer_returns	BR-004	Parent return exists
(customer_return_id, line_number)	UNIQUE	uk_customer_return_lines_return_line	—	Unique line per return

Column	Type	Constraint	Business Rule	Justification
product_id	NOT NULL	nn_customer_return_lines_product	BR-004	Returned product
product_id	FOREIGN KEY	fk_customer_return_lines_product → catalog.products	BR-004	Valid product
returned_qty	NOT NULL	nn_customer_return_lines_qty	BR-004	Returned quantity
returned_qty	CHECK	chk_customer_return_lines_qty > 0	BR-004	Positive returned qty
condition_at_receipt	NOT NULL	nn_customer_return_lines_condition	BR-004	Condition at receipt
condition_at_receipt	CHECK	chk_customer_return_lines_condition IN (GOOD, DAMAGED, DEFECTIVE, OPENED)	BR-004	Valid condition codes

returns.return_dispositions

Column	Type	Constraint	Business Rule	Justification
disposition_reference	NOT NULL	nn_return_dispositions_ref	BR-026	Disposition document
disposition_reference	UNIQUE	uk_return_dispositions_ref	BR-026	Unique disposition reference
customer_return_id	NOT NULL	nn_return_dispositions_return	BR-026	Parent return
customer_return_id	FOREIGN KEY	fk_return_dispositions_return → returns.customer_returns	BR-026	Valid return
return_line_id	NOT NULL	nn_return_dispositions_line	BR-026	Return line
return_line_id	FOREIGN KEY	fk_return_dispositions_line → returns.customer_return_lines	BR-026	Valid return line
product_id	NOT NULL	nn_return_dispositions_product	BR-026	Dispositioned product
product_id	FOREIGN KEY	fk_return_dispositions_product → catalog.products	BR-026	Valid product
quantity	NOT NULL	nn_return_dispositions_qty	BR-026	Disposition quantity
quantity	CHECK	chk_return_dispositions_qty > 0	BR-026	Positive quantity
condition_grade	NOT NULL	nn_return_dispositions_grade	BR-026	Quality grade required
disposition_decision	NOT NULL	nn_return_dispositions_decision	BR-026	Outcome required
disposition_decision	CHECK	chk_return_dispositions_decision	BR-026	RESTOCK, QUARANTINE, WRITE_OFF
restock_location_id	FOREIGN KEY	fk_return_dispositions_location → network.inventory_locations	BR-026	Restock target when RESTOCK
quarantine_hold_id	FOREIGN KEY	fk_return_dispositions_hold → inventory.quarantine_holds	BR-013	Quarantine path
write_off_id	FOREIGN KEY	fk_return_dispositions_writeoff → accuracy.inventory_write_offs	BR-023	Write-off path
inspector_id	NOT NULL	nn_return_dispositions_inspector	BR-026	Inspector required
inspector_id	FOREIGN KEY	fk_return_dispositions_inspector → identity.user_accounts	BR-026	Inspector attribution
inspection_date	NOT NULL	nn_return_dispositions_date	BR-026	Inspection timestamp
inspection_date	DEFAULT	def_return_dispositions_date = NOW()	BR-026	Auto-stamp inspection

Column	Type	Constraint	Business Rule	Justification
(disposition_decision, restock_location_id)	CHECK	chk_return_dispositions_restock	BR-026	RESTOCK requires restock_location_id
(disposition_decision, quarantine_hold_id)	CHECK	chk_return_dispositions_quarantine	BR-013	QUARANTINE requires hold reference
(disposition_decision, write_off_id)	CHECK	chk_return_dispositions_writeoff	BR-023	WRITE_OFF requires write_off_id

Domain 12: Commercial

commercial.promotional_stock_allocations

Column	Type	Constraint	Business Rule	Justification
allocation_reference	NOT NULL	nn_promotional_allocations_ref	BR-029	Allocation document identity
allocation_reference	UNIQUE	uk_promotional_allocations_ref	BR-029	Unique allocation reference
product_id	NOT NULL	nn_promotional_allocations_product	BR-029	Ring-fenced SKU
product_id	FOREIGN KEY	fk_promotional_allocations_product → catalog.products	BR-029	Valid product
allocated_qty	NOT NULL	nn_promotional_allocations_qty	BR-029	Ring-fenced quantity
allocated_qty	CHECK	chk_promotional_allocations_qty > 0	BR-029	Positive allocation
campaign_name	NOT NULL	nn_promotional_allocations_campaign	BR-029	Campaign identity required
effective_start_date	NOT NULL	nn_promotional_allocations_start	BR-029	Campaign start
effective_end_date	NOT NULL	nn_promotional_allocations_end	BR-029	Campaign end
(effective_start_date, effective_end_date)	CHECK	chk_promotional_allocations_dates end_date >= start_date	BR-029	Valid campaign window
readiness_status	NOT NULL	nn_promotional_allocations_readiness	BR-029	Readiness tracking
readiness_status	DEFAULT	def_promotional_allocations_readiness = 'PLANNED'	BR-029	Planned until confirmed
readiness_status	CHECK	chk_promotional_allocations_readiness IN (PLANNED, READY, ACTIVE, CLOSED)	BR-029	Valid readiness states

commercial.promotional_allocation_locations

Column	Type	Constraint	Business Rule	Justification
allocation_id	NOT NULL	nn_promo_alloc_locations_allocation	BR-029	Belongs to allocation
allocation_id	FOREIGN KEY	fk_promo_alloc_locations_allocation → commercial.promotional_stock_allocations	BR-029	Parent allocation exists
location_id	NOT NULL	nn_promo_alloc_locations_location	BR-029	Supplying location
location_id	FOREIGN KEY	fk_promo_alloc_locations_location → network.inventory_locations	BR-029	Valid location

Column	Type	Constraint	Business Rule	Justification
stock_position_id	FOREIGN KEY	fk_promo_alloc_locations_position → inventory.stock_positions	BR-029	Specific balance ring-fenced
(allocation_id, location_id)	UNIQUE	uk_promo_alloc_locations_alloc_loc	BR-029	One entry per location per campaign
allocated_qty_at_location	NOT NULL	nn_promo_alloc_locations_qty	BR-029	Location-level quantity
allocated_qty_at_location	CHECK	chk_promo_alloc_locations_qty > 0	BR-029	Positive location allocation

Domain 13: Identity & Access

identity.user_accounts

Column	Type	Constraint	Business Rule	Justification
user_identifier	NOT NULL	nn_user_accounts_identifier	BO-19	Login identity required
user_identifier	UNIQUE	uk_user_accounts_identifier	BO-19	Unique user identity
display_name	NOT NULL	nn_user_accounts_name	BO-19	Display name for attribution
email	UNIQUE	uk_user_accounts_email	—	Unique email when provided
account_status	NOT NULL	nn_user_accounts_status	BR-019	Account state required
account_status	DEFAULT	def_user_accounts_status = 'ACTIVE'	—	Active by default
account_status	CHECK	chk_user_accounts_status IN (ACTIVE, SUSPENDED, DEACTIVATED)	BR-019	Valid account states
effective_from	NOT NULL	nn_user_accounts_effective_from	—	Account effective date
has_sod_conflict	NOT NULL	nn_user_accounts_sod	BR-019, BR-021	SoD flag explicit
has_sod_conflict	DEFAULT	def_user_accounts_sod = FALSE	BR-019	No conflict by default

identity.role_assignments

Column	Type	Constraint	Business Rule	Justification
user_account_id	NOT NULL	nn_role_assignments_user	BR-019	Assignment belongs to user
user_account_id	FOREIGN KEY	fk_role_assignments_user → identity.user_accounts	BR-019	Valid user account
role_name	NOT NULL	nn_role_assignments_role	BR-019	Role name required
(user_account_id, role_name, effective_from)	UNIQUE	uk_role_assignments_user_role_from	BR-019	No duplicate role at same effective date
assignment_status	NOT NULL	nn_role_assignments_status	BR-019	Assignment state
assignment_status	DEFAULT	def_role_assignments_status = 'ACTIVE'	—	Active by default
assignment_status	CHECK	chk_role_assignments_status IN (ACTIVE, INACTIVE)	BR-019	Valid assignment states

Column	Type	Constraint	Business Rule	Justification
effective_from	NOT NULL	nn_role_assignments_from	BR-019	Effective start required
(effective_from, effective_to)	CHECK	chk_role_assignments_dates effective_to IS NULL OR effective_to >= effective_from	—	Valid assignment period
sod_compliant	NOT NULL	nn_role_assignments_sod	BR-019, BR-021	SoD compliance flag
sod_compliant	DEFAULT	def_role_assignments_sod = TRUE	BR-019	Compliant by default

identity.role_assignment_locations

Column	Type	Constraint	Business Rule	Justification
role_assignment_id	NOT NULL	nn_role_assignment_locations_assignment	BR-019	Scope belongs to assignment
role_assignment_id	FOREIGN KEY	fk_role_assignment_locations_assignment → identity.role_assignments	BR-019	Valid assignment
location_id	NOT NULL	nn_role_assignment_locations_location	BR-019	Scoped location
location_id	FOREIGN KEY	fk_role_assignment_locations_location → network.inventory_locations	BR-019	Valid location
(role_assignment_id, location_id)	UNIQUE	uk_role_assignment_locations	BR-019	One scope entry per location

Domain 14: Audit & Reporting

audit.inventory_movement_records

Column	Type	Constraint	Business Rule	Justification
movement_reference	NOT NULL	nn_movement_records_ref	BO-19, BR-017	Movement document identity
(movement_reference, movement_timestamp)	UNIQUE	uk_movement_records_ref_ts	BO-19	Idempotent event ingestion per partition
movement_type	NOT NULL	nn_movement_records_type	BR-017, BR-020	Movement classification
movement_type	CHECK	chk_movement_records_type	—	Valid movement taxonomy
stock_position_id	NOT NULL	nn_movement_records_position	BO-19	Affected balance
stock_position_id	FOREIGN KEY	fk_movement_records_position → inventory.stock_positions	BO-19	Valid stock position
product_id	NOT NULL	nn_movement_records_product	BR-017	Denormalized product
product_id	FOREIGN KEY	fk_movement_records_product → catalog.products	BR-017	Valid product
location_id	NOT NULL	nn_movement_records_location	BO-19	Denormalized location
location_id	FOREIGN KEY	fk_movement_records_location → network.inventory_locations	BO-19	Valid location
quantity_change	NOT NULL	nn_movement_records_change	BO-19	Signed delta required
quantity_change	CHECK	chk_movement_records_change <> 0	BO-19	Zero change not logged

Column	Type	Constraint	Business Rule	Justification
qty_before	NOT NULL	nn_movement_records_before	BO-19	Before snapshot
qty_before	CHECK	chk_movement_records_before >= 0	LDM-01	Non-negative before
qty_after	NOT NULL	nn_movement_records_after	BO-19	After snapshot
qty_after	CHECK	chk_movement_records_after >= 0	LDM-01	Non-negative after
(qty_before, qty_after, quantity_change)	CHECK	chk_movement_records_balance qty_after = qty_before + quantity_change	BO-19	Movement arithmetic integrity
lot_batch_id	FOREIGN KEY	fk_movement_records_lot → inventory.lot_batches	BR-017	Lot on traceable movements
reason_code_id	FOREIGN KEY	fk_movement_records_reason → governance.reason_codes	BR-020	Reason on adjustment movements
source_document_type	NOT NULL	nn_movement_records_source_type	BO-19	Origin document type
source_document_id	NOT NULL	nn_movement_records_source_id	BO-19	Origin document PK
performed_by	NOT NULL	nn_movement_records_user	BO-19	User attribution
performed_by	FOREIGN KEY	fk_movement_records_user → identity.user_accounts	BO-19	Valid user
movement_timestamp	NOT NULL	nn_movement_records_timestamp	BO-19	Event time — partition key
movement_timestamp	DEFAULT	def_movement_records_timestamp = NOW()	BO-19	Auto-stamp event time

audit.audit_trail_records

Column	Type	Constraint	Business Rule	Justification
audit_event_reference	NOT NULL	nn_audit_trail_ref	BO-19	Audit event identity
(audit_event_reference, event_timestamp)	UNIQUE	uk_audit_trail_ref_ts	BO-19	Unique event per partition
affected_entity_type	NOT NULL	nn_audit_trail_entity_type	BO-19	Entity type required
affected_entity_id	NOT NULL	nn_audit_trail_entity_id	BO-19	Entity PK required
action_performed	NOT NULL	nn_audit_trail_action	BO-19	Action classification
action_performed	CHECK	chk_audit_trail_action IN (CREATE, UPDATE, APPROVE, CANCEL)	BO-19	Valid audit actions
user_id	NOT NULL	nn_audit_trail_user	BO-19	Actor required
user_id	FOREIGN KEY	fk_audit_trail_user → identity.user_accounts	BO-19	Valid user
movement_record_id	FOREIGN KEY	fk_audit_trail_movement → audit.inventory_movement_records	BO-19	Links to operational movement
event_timestamp	NOT NULL	nn_audit_trail_timestamp	BO-19	Event time — partition key
event_timestamp	DEFAULT	def_audit_trail_timestamp = NOW()	BO-19	Auto-stamp event time

audit.operational_alerts

Column	Type	Constraint	Business Rule	Justification
alert_reference	NOT NULL	nn_operational_alerts_ref	BR-009, BR-032	Alert document identity
alert_reference	UNIQUE	uk_operational_alerts_ref	—	Unique alert reference
alert_type	NOT NULL	nn_operational_alerts_type	BR-009	Alert classification
alert_type	CHECK	chk_operational_alerts_type IN (LOW_STOCK, OUT_OF_STOCK, PENDING_APPROVAL, TRANSFER_EXCEPTION, INBOUND_DELAY, CAMPAIGN_READINESS)	BR-009, BR-032	Valid alert types
severity	NOT NULL	nn_operational_alerts_severity	BR-032	Priority required
severity	DEFAULT	def_operational_alerts_severity = 'MEDIUM'	BR-032	Medium default
severity	CHECK	chk_operational_alerts_severity IN (LOW, MEDIUM, HIGH, CRITICAL)	BR-032	Valid severity levels
product_id	FOREIGN KEY	fk_operational_alerts_product → catalog.products	BR-009	Alert context SKU
location_id	FOREIGN KEY	fk_operational_alerts_location → network.inventory_locations	BR-009	Alert context site
stock_position_id	FOREIGN KEY	fk_operational_alerts_position → inventory.stock_positions	BR-009	Threshold breach source
acknowledgment_status	NOT NULL	nn_operational_alerts_ack	BR-009	Acknowledgment tracking
acknowledgment_status	DEFAULT	def_operational_alerts_ack = 'OPEN'	BR-009	Open until acknowledged
acknowledgment_status	CHECK	chk_operational_alerts_ack IN (OPEN, ACKNOWLEDGED, RESOLVED)	BR-009	Valid ack states
resolution_status	NOT NULL	nn_operational_alerts_resolution	BR-009	Resolution tracking
resolution_status	DEFAULT	def_operational_alerts_resolution = 'PENDING'	BR-009	Pending by default
resolution_status	CHECK	chk_operational_alerts_resolution IN (PENDING, RESOLVED, DISMISSED)	BR-009	Valid resolution states
alert_date	NOT NULL	nn_operational_alerts_date	BR-009	Alert creation time
alert_date	DEFAULT	def_operational_alerts_date = NOW()	BR-009	Auto-stamp alert
(resolution_status, resolved_at)	CHECK	chk_operational_alerts_resolved	BR-009	RESOLVED requires resolved_at

audit.inventory_valuation_snapshots

Column	Type	Constraint	Business Rule	Justification
snapshot_reference	NOT NULL	nn_valuation_snapshots_ref	BR-008, BR-031	Snapshot document identity
snapshot_reference	UNIQUE	uk_valuation_snapshots_ref	—	Unique snapshot reference
valuation_date	NOT NULL	nn_valuation_snapshots_date	BR-031	As-of date required

Column	Type	Constraint	Business Rule	Justification
location_scope	NOT NULL	nn_valuation_snapshots_scope	BR-031	Scope classification
location_scope	CHECK	chk_valuation_snapshots_scope IN (SINGLE, REGION, NETWORK)	BR-031	Valid scope values
(valuation_date, location_scope)	UNIQUE	uk_valuation_snapshots_date_scope	BR-031	One snapshot per period per scope
total_inventory_value	NOT NULL	nn_valuation_snapshots_total	BR-031	Total value required
total_inventory_value	CHECK	chk_valuation_snapshots_total >= 0	BR-031	Non-negative total
owned_value	NOT NULL	nn_valuation_snapshots_owned	BR-031	Owned value required
owned_value	CHECK	chk_valuation_snapshots_owned >= 0	BR-031	Consignment excluded from owned
consignment_value	NOT NULL	nn_valuation_snapshots_consignment	BR-031	Consignment tracked separately
consignment_value	DEFAULT	def_valuation_snapshots_consignment = 0	BR-031	Zero consignment default
consignment_value	CHECK	chk_valuation_snapshots_consignment >= 0	BR-031	Non-negative consignment
(owned_value, consignment_value, total_inventory_value)	CHECK	chk_valuation_snapshots_sum total >= owned	BR-031	Total includes owned component
reconciliation_status	NOT NULL	nn_valuation_snapshots_recon	BR-008	Reconciliation state
reconciliation_status	DEFAULT	def_valuation_snapshots_recon = 'DRAFT'	—	Draft until reconciled
reconciliation_status	CHECK	chk_valuation_snapshots_recon IN (DRAFT, RECONCILED, EXCEPTION)	BR-008	Valid recon states
prepared_by	NOT NULL	nn_valuation_snapshots_prepared_by	BO-19	Preparer attribution
prepared_by	FOREIGN KEY	fk_valuation_snapshots_prepared_by → identity.user_accounts	BO-19	Valid user

audit.inventory_valuation_lines

Column	Type	Constraint	Business Rule	Justification
snapshot_id	NOT NULL	nn_valuation_lines_snapshot	BR-031	Line belongs to snapshot
snapshot_id	FOREIGN KEY	fk_valuation_lines_snapshot → audit.inventory_valuation_snapshots	BR-031	Parent snapshot exists
product_id	NOT NULL	nn_valuation_lines_product	BR-031	Valued SKU
product_id	FOREIGN KEY	fk_valuation_lines_product → catalog.products	BR-031	Valid product
location_id	NOT NULL	nn_valuation_lines_location	BR-031	Valued site
location_id	FOREIGN KEY	fk_valuation_lines_location → network.inventory_locations	BR-031	Valid location
(snapshot_id, product_id, location_id)	UNIQUE	uk_valuation_lines_snapshot_prod_loc	BR-031	One line per SKU per site per snapshot
quantity	NOT NULL	nn_valuation_lines_qty	BR-031	Snapshot quantity
quantity	CHECK	chk_valuation_lines_qty >= 0	BR-031	Non-negative snapshot qty
unit_cost	NOT NULL	nn_valuation_lines_cost	BR-008	Unit cost from matched invoices

Column	Type	Constraint	Business Rule	Justification
unit_cost	CHECK	chk_valuation_lines_cost >= 0	BR-008	Non-negative unit cost
extended_value	NOT NULL	nn_valuation_lines_extended	BR-031	Extended value required
(quantity, unit_cost, extended_value)	CHECK	chk_valuation_lines_calc extended_value = quantity * unit_cost	BR-031	Valuation arithmetic integrity
ownership_type	NOT NULL	nn_valuation_lines_ownership	BR-031	Ownership classification
ownership_type	DEFAULT	def_valuation_lines_ownership = 'OWNED'	BR-031	Owned by default
ownership_type	CHECK	chk_valuation_lines_ownership IN (OWNED, CONSIGNMENT)	BR-031	Consignment excluded from owned total

Appendix A: Business Rule to Constraint Index

Business Rule	Key Constraints Enforcing Rule
BR-001 ATP-only commitment	chk_stock_positions_atp_formula, chk_stock_positions_atp_le_on_hand, chk_customer_order_lines_qty_balance
BR-002 Reservation at confirmation	nn_inventory_reservations_*, fk_inventory_reservations_position, chk_stock_positions_reserved_le_on_hand
BR-003 Verified pick before shipment	nn_shipments_fe, fk_shipments_fe, chk_fulfillment_executions_pick_status
BR-004 Verify before acceptance	nn_goods_receipts_verification, chk_goods_receipt_lines_accepted_le_received
BR-005 Put-away before availability	nn_put_aways_*, fk_put_aways_position, def_put_aways_availability
BR-006 Document discrepancies	nn_receipt_discrepancies_*, uk_receipt_discrepancies_ref
BR-007 PO required for receipt	chk_goods_receipts_po_or_exception, fk_goods_receipts_po
BR-008 Three-way match	chk_supplier_invoices_payment_gate, uk_inv_gr_match_pair, chk_valuation_lines_cost
BR-009 Reorder at reorder level	nn_stock_policies_*, uk_stock_policies_product_location, chk_operational_alerts_type
BR-010 Safety stock protection	chk_stock_policies_safety, chk_stock_policies_threshold_order
BR-012 Protect source minimum	chk_transfer_requests_distinct_locations, chk_stock_transfer_lines_qty
BR-013 Quarantine exclusion	chk_stock_positions_atp_formula, chk_quarantine_holds_held_qty
BR-014 FEFO rotation	chk_products_rotation_policy, chk_lot_batches_dates
BR-016 MRSL at receipt	chk_products_shelf_life_mrsl, chk_receipt_discrepancies_type
BR-017 Lot capture	chk_products_traceable_rotation, fk_*_lot across receipt/pick/transfer
BR-019 Independent count approval	chk_count_variances_single_origin, chk_stock_adjustments_sod
BR-020 Mandatory reason codes	nn_stock_adjustments_reason, fk_stock_adjustments_reason
BR-021 Independent approval	chk_stock_adjustments_sod, chk_count_variances_approver_required
BR-022 Investigate before adjusting	chk_count_variances_investigation, fk_stock_adjustments_variance
BR-023 Write-off authorization	nn_inventory_write_offs_authorizer, chk_inventory_write_offs_sod
BR-024 Physical inventory freeze	nn_physical_inventory_counts_freeze, chk_physical_inventory_counts_freeze_release
BR-025 RMA required	nn_customer_returns_rma, fk_customer_returns_rma
BR-026 Inspect returns	chk_return_dispositions_decision, chk_return_dispositions_restock

Business Rule	Key Constraints Enforcing Rule
BR-027 No discontinued reorder	FK to `products` + application lifecycle check on insert
BR-028 Backorder acknowledgment	`nn_backorders_acknowledged`, `chk_backorders_ack_timestamp`
BR-029 Promotional ring-fence	`chk_promotional_allocations_qty`, `uk_promo_alloc_locations_alloc_loc`
BR-030 Bonded customs	`chk_stock_positions_ownership`, `chk_quarantine_holds_type`, `chk_storage_positions_type`
BR-031 Consignment ownership	`chk_stock_positions_ownership`, `chk_valuation_lines_ownership`
BR-032 Class A monitoring	`chk_products_abc_classification`, `chk_operational_alerts_severity`

Appendix B: Constraint Count Summary

Constraint Type	Approximate Count	Primary Purpose
NOT NULL	~420	Mandatory business fields and audit columns
UNIQUE	~75	Business keys, document lines, dimensional identity
CHECK	~130	Quantity invariants, status enums, SoD, date logic, ATP formula
FOREIGN KEY	~145	Referential integrity across 14 schemas
DEFAULT	~95	Sensible initial states and auto-timestamps

Appendix C: Application-Layer Constraints (Not Database-Enforced)

These business rules require runtime validation beyond static CHECK constraints:

Business Rule	Reason Not DB-Enforced	Application Enforcement
BR-027 Discontinued product block	Requires join to `products.lifecycle_status` on child insert	Trigger or application service before PO/requisition/reservation insert
BR-019 Counter ≠ approver	Requires cross-row comparison on `cycle_counts` and `count_variances`	Application SoD check or deferred constraint trigger
BR-024 Location freeze	Requires checking `inventory_locations.is_frozen` on transaction insert	Application guard before receipt/shipment/transfer
BR-010 Safety stock bypass	Requires policy evaluation against live ATP	Domain service on reservation insert
BR-012 Source minimum on transfer	Requires `stock_policies` and `stock_positions` evaluation	Transfer approval service
BR-029 Promotional ring-fence	Requires live ATP segregation calculation	Stock Position aggregate coordination

Document Approval

Role	Name	Signature	Date
Database Architect			
DDD Solution Architect			
Solution Architect			
Inventory Controller			

End of Database Constraints Specification

Database Indexing Strategy

Inventory Management System — PostgreSQL

Document Information

Attribute	Detail
Document ID	SRS-DB-005
Document Title	Database Indexing Strategy
Version	1.0
Status	Draft
Document Owner	Database Performance Specialist
Source Documents	SRS-DB-002 (Physical Data Model), SRS-DB-003 (Database Keys), SRS-DB-004 (Database Constraints)
Target Platform	PostgreSQL 15+
Date	July 6, 2026
Scope	Index design and justification — no DDL/SQL statements

Purpose

This document defines the **indexing strategy** for the Inventory Management System. Indexes are chosen only where they support real query patterns from business operations — not as a default on every column. Each index includes junior-friendly performance explanation.

Part 1: Indexing Principles (Junior Guide)

What is an index?

An index is a **sorted lookup structure** (like a book index) that helps PostgreSQL find rows quickly without scanning the entire table.

Without index	With index
Full table scan — reads every row	Index seek — jumps to matching rows
Fast on tiny tables	Essential on large tables (millions of rows)
OK for rare admin queries	Required for high-frequency operational queries

Why indexes matter in IMS

The hottest path is: **"How much of product X is available at location Y?"** — asked on every order confirmation, reservation, and replenishment check. That query runs thousands of times per hour. A missing index here blocks the whole warehouse.

How PostgreSQL uses an index

1. Query arrives: `WHERE product_id = 42 AND location_id = 7`
2. Planner checks: is there an index on `(product_id, location_id)`?
3. If yes → **Index Scan** — reads only matching pages (milliseconds)
4. If no → **Seq Scan** — reads all stock position rows (seconds at scale)

Column order matters in composite indexes: `(product_id, location_id)` helps `WHERE product_id = ?` and `WHERE product_id = ? AND location_id = ?`, but is weaker for `WHERE location_id = ?` alone. Design indexes to match real `WHERE` clauses.

Costs of too many indexes

Benefit	Cost
Faster SELECT	Slower INSERT/UPDATE/DELETE (every index must be updated)
Faster JOIN	More disk space
Faster ORDER BY	More memory during maintenance (VACUUM, REINDEX)

Rule: Index for queries that run **often** and **filter/join** on the indexed columns. Do not index "just in case."

Index types used in this system

Type	When to use	IMS example
B-tree (default)	Equality, range, sorting, most FK lookups	`stock_positions (product_id, location_id)`
Partial B-tree	Frequent queries on a **subset** of rows	Active reservations only: `WHERE reservation_status = 'ACTIVE'`
BRIN	Very large time-ordered tables (append-only logs)	`movement_timestamp` on movement records
GIN	JSONB containment queries (rare)	`audit_trail_records.before_state` — only if JSON search needed

Not used: Hash indexes (no advantage over B-tree in modern PostgreSQL), GiST (no spatial data in IMS).

Part 2: Index Tiers

Tier	Purpose	Count	Maintenance
Tier 1 — Critical	Hot-path operational queries; system fails SLA without these	18	Monitor weekly
Tier 2 — Operational	High-volume workflows; noticeable slowdown if missing	22	Monitor monthly
Tier 3 — Reporting & audit	Dashboards, history, compliance; tolerates slightly slower reads	12	Review quarterly
Implicit (UNIQUE/PK)	Created automatically by constraints — do not duplicate	~75	Maintained with constraints

Total explicit indexes defined: 52 (excluding PK/UK implicit indexes)

Part 3: Tier 1 — Critical Indexes

These support the highest-frequency queries. Deploy in the initial migration.

IDX-001

Attribute	Detail
Table	`inventory.stock_positions`
Indexed columns	`(product_id, location_id)`
Index type	B-tree **UNIQUE** (implicit via `uk_stock_positions_product_location`)
Expected query pattern	`SELECT * FROM stock_positions WHERE product_id = ? AND location_id = ?`
Business justification	Central ATP lookup for order confirmation (BR-001), reservation (BR-002), and replenishment (BR-009). One row per SKU per site.

Attribute	Detail
Performance clarity (junior)	**Why:** Every order check asks "do we have stock here?" **How:** UNIQUE constraint automatically creates this B-tree index — PostgreSQL finds the single matching row in O(log n) time instead of scanning all balances. **Do not create a duplicate non-unique index** on the same columns.

IDX-002

Attribute	Detail
Table	`catalog.products`
Indexed columns	`sku`
Index type	B-tree **UNIQUE** (implicit via `uk_products_sku`)
Expected query pattern	`SELECT id FROM products WHERE sku = ?`
Business justification	SKU is the ubiquitous lookup key at receipt, pick, and catalog integration.
Performance clarity (junior)	**Why:** External systems and scanners send SKU strings, not internal `id`. **How:** UNIQUE index on `sku` converts string lookup to fast seek before any stock operation.

IDX-003

Attribute	Detail
Table	`catalog.product_identifiers`
Indexed columns	`(identifier_type, identifier_value)`
Index type	B-tree **UNIQUE** (implicit via UK)
Expected query pattern	`SELECT product_id FROM product_identifiers WHERE identifier_type = 'BARCODE' AND identifier_value = ?`
Business justification	Barcode scan at receiving dock and pick face (BR-004). Wrong index = slow scan response.
Performance clarity (junior)	**Why:** Warehouse staff scan barcodes, not SKUs. **How:** Composite UK lets PostgreSQL resolve barcode → product_id in one index lookup.

IDX-004

Attribute	Detail
Table	`fulfillment.inventory_reservations`
Indexed columns	`(stock_position_id)` WHERE `reservation_status = 'ACTIVE'`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM inventory_reservations WHERE stock_position_id = ? AND reservation_status = 'ACTIVE'`
Business justification	Only active reservations affect ATP; released/fulfilled rows are historical noise (BR-002).
Performance clarity (junior)	**Why:** Table grows with all historical reservations but queries care about ACTIVE rows only. **How:** Partial index stores only active rows — smaller, faster, cheaper to maintain than a full composite index on `(stock_position_id, reservation_status)`. **Note:** Do not also create a non-partial index on the same columns — one index only.

IDX-005

Attribute	Detail
Table	`inventory.lot_batches`

Attribute	Detail
Indexed columns	`(stock_position_id, expiration_date)`
Index type	B-tree
Expected query pattern	`SELECT * FROM lot_batches WHERE stock_position_id = ? ORDER BY expiration_date ASC LIMIT 1`
Business justification	FEFO pick — earliest expiring lot first at a specific bin (BR-014).
Performance clarity (junior)	**Why:** Picker needs "which lot expires first at this bin?" **How:** Index sorted by `expiration_date` within each `stock_position_id` avoids sorting thousands of lots at pick time.

IDX-006

Attribute	Detail
Table	`inventory.lot_batches`
Indexed columns	`(product_id, expiration_date)`
Index type	B-tree
Expected query pattern	`SELECT * FROM lot_batches WHERE product_id = ? AND expiration_date > CURRENT_DATE ORDER BY expiration_date`
Business justification	Network-wide FEFO and recall scope queries across locations.
Performance clarity (junior)	**Why:** Quality team asks "all lots of product X expiring soon network-wide." **How:** Product_id narrows scope; expiration_date supports range scan and ORDER BY.

IDX-008

Attribute	Detail
Table	`fulfillment.customer_orders`
Indexed columns	`(order_status, order_date DESC)`
Index type	B-tree
Expected query pattern	`SELECT * FROM customer_orders WHERE order_status IN ('CONFIRMED','PARTIAL') ORDER BY order_date DESC LIMIT 50`
Business justification	Open order queue for warehouse dashboard and fulfillment planning.
Performance clarity (junior)	**Why:** Warehouse managers view "orders needing action today." **How:** `order_status` filters open orders; `order_date DESC` returns newest first without separate sort step.

IDX-009

Attribute	Detail
Table	`fulfillment.customer_orders`
Indexed columns	`order_number`
Index type	B-tree **UNIQUE** (implicit via UK)
Expected query pattern	`SELECT * FROM customer_orders WHERE order_number = ?`
Business justification	Customer service and integration lookup by order number.
Performance clarity (junior)	**Why:** Call center searches by order number constantly. **How:** UK provides index automatically — no extra index needed.

IDX-010

Attribute	Detail
Table	`network.inventory_locations`
Indexed columns	`location_code`
Index type	B-tree **UNIQUE** (implicit via UK)
Expected query pattern	`SELECT id FROM inventory_locations WHERE location_code = ?`
Business justification	Location code used in UI, reports, and cross-system references.
Performance clarity (junior)	**Why:** Users think in "WH-01" not internal id 7. **How:** UK on location_code — implicit index, do not duplicate.

IDX-011

Attribute	Detail
Table	`inventory.stock_positions`
Indexed columns	`(location_id, stock_status)` WHERE `stock_status IN ('LOW_STOCK','OUT_OF_STOCK')`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM stock_positions WHERE location_id = ? AND stock_status IN ('LOW_STOCK','OUT_OF_STOCK')`
Business justification	Low-stock alert generation and replenishment review (BR-009, BR-032).
Performance clarity (junior)	**Why:** Replenishment team monitors exceptions, not every IN_STOCK row. **How:** Partial index includes only alert-worthy rows — tiny index, fast scan per location. Most rows are IN_STOCK and excluded.

IDX-012

Attribute	Detail
Table	`procurement.purchase_orders`
Indexed columns	`po_number`
Index type	B-tree **UNIQUE** (implicit via UK)
Expected query pattern	`SELECT * FROM purchase_orders WHERE po_number = ?`
Business justification	PO lookup at receiving dock and three-way match (BR-007, BR-008).
Performance clarity (junior)	**Why:** Receiver scans/types PO number on every inbound shipment. **How:** UK provides index — no duplicate needed.

IDX-013

Attribute	Detail
Table	`inbound.goods_receipts`
Indexed columns	`purchase_order_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM goods_receipts WHERE purchase_order_id = ?`
Business justification	"What has been received against this PO?" — receiving and match workflows.
Performance clarity (junior)	**Why:** Three-way match joins invoice → PO → all receipts for that PO. **How:** FK column is not auto-indexed in PostgreSQL — explicit index required for JOIN performance.

IDX-014

Attribute	Detail
Table	`procurement.purchase_order_lines`
Indexed columns	`(purchase_order_id, product_id)`
Index type	B-tree
Expected query pattern	`SELECT * FROM purchase_order_lines WHERE purchase_order_id = ? AND product_id = ?`
Business justification	Receipt verification against specific PO line (BR-004, BR-007).
Performance clarity (junior)	**Why:** Receiver confirms "PO line for SKU X." **How:** Composite index matches both filter columns — one seek instead of fetching all PO lines.

IDX-015

Attribute	Detail
Table	`fulfillment.customer_order_lines`
Indexed columns	`customer_order_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM customer_order_lines WHERE customer_order_id = ?`
Business justification	Load all lines when opening an order for reservation and fulfillment.
Performance clarity (junior)	**Why:** Every order detail screen loads header + lines. **How:** Index on FK parent id — classic parent-child pattern. Without it, full scan of all order lines in the system.

IDX-016

Attribute	Detail
Table	`fulfillment.inventory_reservations`
Indexed columns	`order_line_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM inventory_reservations WHERE order_line_id = ?`
Business justification	Find reservation for order line during ship and cancellation.
Performance clarity (junior)	**Why:** Shipment consumes reservation per order line. **How:** Direct lookup by line id when processing fulfillment.

IDX-017

Attribute	Detail
Table	`inventory.stock_policies`
Indexed columns	`(product_id, location_id)`
Index type	B-tree **UNIQUE** (implicit via UK)
Expected query pattern	`SELECT * FROM stock_policies WHERE product_id = ? AND location_id = ?`
Business justification	Threshold lookup paired with stock position for reorder alerts (BR-009).
Performance clarity (junior)	**Why:** ATP status compared to reorder level on same product+location key. **How:** UK mirrors stock_positions key — often joined together; both have implicit indexes.

IDX-018

Attribute	Detail
Table	`identity.user_accounts`
Indexed columns	`user_identifier`
Index type	B-tree **UNIQUE** (implicit via UK)
Expected query pattern	`SELECT * FROM user_accounts WHERE user_identifier = ?`
Business justification	Login and attribution on every transactional row.
Performance clarity (junior)	**Why:** Every operation records "who did it." **How:** UK on login id — implicit index at authentication time.

Part 4: Tier 2 — Operational Indexes

High-volume workflows beyond the absolute hot path.

IDX-019

Attribute	Detail
Table	`inbound.goods_receipt_lines`
Indexed columns	`goods_receipt_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM goods_receipt_lines WHERE goods_receipt_id = ?`
Business justification	Load receipt lines for verification and put-away.
Performance clarity (junior)	**Why:** Parent-child document pattern. **How:** Index on receipt header FK — same pattern as order lines.

IDX-020

Attribute	Detail
Table	`inbound.put_aways`
Indexed columns	`goods_receipt_line_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM put_aways WHERE goods_receipt_line_id = ?`
Business justification	Check put-away completion before ATP increase (BR-005).
Performance clarity (junior)	**Why:** System asks "is this receipt line put away yet?" **How:** Direct lookup on receipt line FK.

IDX-021

Attribute	Detail
Table	`procurement.purchase_orders`
Indexed columns	`(supplier_id, po_status)` WHERE `po_status IN ('OPEN','PARTIAL')`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM purchase_orders WHERE supplier_id = ? AND po_status IN ('OPEN','PARTIAL')`
Business justification	Open PO list per supplier for receiving and AP.

Attribute	Detail
Performance clarity (junior)	**Why:** Buyers care about open POs, not 5 years of closed history. **How:** Partial index shrinks size by excluding CLOSED/CANCELLED rows.

IDX-022

Attribute	Detail
Table	`finance.supplier_invoices`
Indexed columns	`purchase_order_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM supplier_invoices WHERE purchase_order_id = ?`
Business justification	Three-way match PO leg (BR-008).
Performance clarity (junior)	**Why:** AP clerk finds invoices for a PO. **How:** FK index for JOIN in match workflow.

IDX-023

Attribute	Detail
Table	`finance.supplier_invoices`
Indexed columns	`(supplier_id, invoice_number)`
Index type	B-tree **UNIQUE** (implicit via UK)
Expected query pattern	`SELECT * FROM supplier_invoices WHERE supplier_id = ? AND invoice_number = ?`
Business justification	Duplicate invoice detection.
Performance clarity (junior)	**Why:** UK enforces uniqueness and serves lookup — no extra index.

IDX-024

Attribute	Detail
Table	`finance.invoice_goods_receipt_match`
Indexed columns	`goods_receipt_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM invoice_goods_receipt_match WHERE goods_receipt_id = ?`
Business justification	Reverse lookup: which invoices matched this receipt.
Performance clarity (junior)	**Why:** UK covers `(supplier_invoice_id, goods_receipt_id)` but queries from receipt side need `goods_receipt_id` leading column. **How:** Second index justified — different access path than UK.

IDX-025

Attribute	Detail
Table	`fulfillment.fulfillment_executions`
Indexed columns	`customer_order_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM fulfillment_executions WHERE customer_order_id = ?`

Attribute	Detail
Business justification	Pick queue per order.
Performance clarity (junior)	**Why:** Warehouse loads pick work for one order. **How:** Standard FK parent index.

IDX-026

Attribute	Detail
Table	`fulfillment.fulfillment_executions`
Indexed columns	`(fulfillment_location_id, pick_verification_status)` WHERE `pick_verification_status = 'PENDING'`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM fulfillment_executions WHERE fulfillment_location_id = ? AND pick_verification_status = 'PENDING'`
Business justification	Warehouse pick queue by location.
Performance clarity (junior)	**Why:** Pickers see only pending work at their site. **How:** Partial index on PENDING rows only — verified picks are dead for queue purposes.

IDX-027

Attribute	Detail
Table	`fulfillment.shipments`
Indexed columns	`customer_order_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM shipments WHERE customer_order_id = ?`
Business justification	Order tracking and shipment history.
Performance clarity (junior)	**Why:** Customer service views shipments for an order. **How:** FK index on order.

IDX-028

Attribute	Detail
Table	`fulfillment.backorders`
Indexed columns	`(resolution_status, product_id)` WHERE `resolution_status = 'OPEN'`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM backorders WHERE resolution_status = 'OPEN' AND product_id = ?`
Business justification	Replenishment priority for open backorders (BR-028).
Performance clarity (junior)	**Why:** Planners ask "what SKUs have open backorders?" **How:** Partial index on OPEN only — fulfilled backorders excluded.

IDX-029

Attribute	Detail
Table	`transfer.stock_transfers`
Indexed columns	`(destination_location_id, transfer_status)` WHERE `transfer_status = 'IN_TRANSIT'`
Index type	**Partial B-tree**

Attribute	Detail
Expected query pattern	`SELECT * FROM stock_transfers WHERE destination_location_id = ? AND transfer_status = 'IN_TRANSIT'`
Business justification	Expected inbound transfers at destination (BR-012).
Performance clarity (junior)	**Why:** Destination warehouse prepares for incoming transfers. **How:** Small partial index — few IN_TRANSIT rows at any time.

IDX-030

Attribute	Detail
Table	`transfer.in_transit_stock`
Indexed columns	`stock_transfer_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM in_transit_stock WHERE stock_transfer_id = ?`
Business justification	Pipeline detail for active transfer.
Performance clarity (junior)	**Why:** Child rows of transfer header. **How:** FK index — composition pattern.

IDX-031

Attribute	Detail
Table	`inventory.quarantine_holds`
Indexed columns	`(stock_position_id)` WHERE `hold_status = 'ACTIVE'`
Index type	**Partial B-tree**
Expected query pattern	`SELECT SUM(held_qty) FROM quarantine_holds WHERE stock_position_id = ? AND hold_status = 'ACTIVE'`
Business justification	ATP calculation excludes active holds (BR-013).
Performance clarity (junior)	**Why:** Same pattern as reservations — only ACTIVE holds reduce ATP. **How:** Partial index for sum aggregation per stock position.

IDX-032

Attribute	Detail
Table	`network.storage_positions`
Indexed columns	`(location_id, position_code)`
Index type	B-tree **UNIQUE** (implicit via UK)
Expected query pattern	`SELECT id FROM storage_positions WHERE location_id = ? AND position_code = ?`
Business justification	Put-away and pick bin lookup.
Performance clarity (junior)	**Why:** "Put this in bin A-01-03 at WH-01." **How:** UK provides composite index — no duplicate.

IDX-033

Attribute	Detail
Table	`network.inventory_locations`
Indexed columns	`parent_location_id`
Index type	B-tree

Attribute	Detail
Expected query pattern	`SELECT * FROM inventory_locations WHERE parent_location_id = ?`
Business justification	Hierarchy roll-up — list child sites under region.
Performance clarity (junior)	**Why:** Reports aggregate by region. **How:** Index on self-referencing FK — tree traversal from parent to children.

IDX-034

Attribute	Detail
Table	`accuracy.cycle_counts`
Indexed columns	`(location_id, count_status, schedule_date)` WHERE `count_status IN ('SCHEDULED','IN_PROGRESS')`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM cycle_counts WHERE location_id = ? AND count_status IN ('SCHEDULED','IN_PROGRESS') ORDER BY schedule_date`
Business justification	Counter work queue (BR-018).
Performance clarity (junior)	**Why:** Counters see today's assignments at their warehouse. **How:** Partial index excludes completed historical counts.

IDX-035

Attribute	Detail
Table	`accuracy.count_variances`
Indexed columns	`(approval_status, investigation_status)` WHERE `approval_status = 'PENDING'`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM count_variances WHERE approval_status = 'PENDING' ORDER BY created_at`
Business justification	Approver work queue (BR-019, BR-021).
Performance clarity (junior)	**Why:** Managers approve pending variances — not thousands of old approved rows. **How:** Partial index on PENDING only.

IDX-036

Attribute	Detail
Table	`accuracy.stock_adjustments`
Indexed columns	`(approval_status)` WHERE `approval_status = 'PENDING'`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM stock_adjustments WHERE approval_status = 'PENDING'`
Business justification	Adjustment approval queue (BR-021).
Performance clarity (junior)	**Why:** Same approval-queue pattern as variances. **How:** Tiny partial index of pending items.

IDX-037

Attribute	Detail
Table	`returns.return_authorizations`
Indexed columns	`customer_order_id`

Attribute	Detail
Index type	B-tree
Expected query pattern	`SELECT * FROM return_authorizations WHERE customer_order_id = ?`
Business justification	RMA history per order (BR-025).
Performance clarity (junior)	**Why:** Customer service checks return eligibility on original order. **How:** FK index.

IDX-038

Attribute	Detail
Table	`returns.return_authorizations`
Indexed columns	`rma_number`
Index type	B-tree **UNIQUE** (implicit via UK)
Expected query pattern	`SELECT * FROM return_authorizations WHERE rma_number = ?`
Business justification	Warehouse scans RMA at return dock.
Performance clarity (junior)	**Why:** UK on rma_number — implicit index, no duplicate.

IDX-039

Attribute	Detail
Table	`procurement.purchase_requisitions`
Indexed columns	`(approval_status) WHERE approval_status IN ('SUBMITTED','APPROVED')`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM purchase_requisitions WHERE approval_status = 'SUBMITTED'`
Business justification	Buyer approval queue.
Performance clarity (junior)	**Why:** Buyers approve submitted requisitions — not drafts or converted. **How:** Partial index on workflow-active states.

IDX-040

Attribute	Detail
Table	`transfer.transfer_requests`
Indexed columns	`(request_status) WHERE request_status = 'SUBMITTED'`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM transfer_requests WHERE request_status = 'SUBMITTED'`
Business justification	Transfer approval queue (BR-012).
Performance clarity (junior)	**Why:** Operations director approves pending transfers. **How:** Partial index — small pending set.

Part 5: Tier 3 — Reporting & Audit Indexes

IDX-041

Attribute	Detail
Table	`audit.inventory_movement_records`

Attribute	Detail
Indexed columns	`(stock_position_id, movement_timestamp DESC)`
Index type	B-tree (per monthly partition)
Expected query pattern	`SELECT * FROM inventory_movement_records WHERE stock_position_id = ? ORDER BY movement_timestamp DESC LIMIT 100`
Business justification	Stock position history for investigation and audit (BO-19).
Performance clarity (junior)	**Why:** "Show me last 100 movements for this SKU at this site." **How:** Composite index with DESC timestamp — PostgreSQL reads index backward for recent-first without sort. **Partition note:** Create this index on each monthly partition, not only parent table.

IDX-042

Attribute	Detail
Table	`audit.inventory_movement_records`
Indexed columns	`movement_timestamp`
Index type	**BRIN**
Expected query pattern	`SELECT COUNT(*) FROM inventory_movement_records WHERE movement_timestamp BETWEEN ? AND ?`
Business justification	Time-range reporting on very large append-only log (10M+ rows/year).
Performance clarity (junior)	**Why:** BRIN is tiny compared to B-tree on timestamp — stores min/max per block range. **How:** Perfect for append-only data inserted in time order. **When NOT to use:** Point lookups by stock_position_id — use IDX-041 instead. BRIN complements, does not replace IDX-041.

IDX-043

Attribute	Detail
Table	`audit.audit_trail_records`
Indexed columns	`(affected_entity_type, affected_entity_id, event_timestamp DESC)`
Index type	B-tree (per monthly partition)
Expected query pattern	`SELECT * FROM audit_trail_records WHERE affected_entity_type = 'stock_adjustments' AND affected_entity_id = ? ORDER BY event_timestamp DESC`
Business justification	Compliance audit trail per entity (BR-019, BO-19).
Performance clarity (junior)	**Why:** Auditor asks "full history of changes to adjustment #12345." **How:** Polymorphic lookup by type + id, sorted by time.

IDX-044

Attribute	Detail
Table	`audit.operational_alerts`
Indexed columns	`(acknowledgment_status, alert_type, severity)` WHERE `acknowledgment_status = 'OPEN'`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM operational_alerts WHERE acknowledgment_status = 'OPEN' ORDER BY severity DESC, alert_date`
Business justification	Alert dashboard and replenishment review (BR-009, BR-032).

Attribute	Detail
Performance clarity (junior)	**Why:** Users see open alerts only — resolved alerts purged after 90 days. **How:** Partial index keeps dashboard query fast as alert table grows.

IDX-045

Attribute	Detail
Table	`audit.inventory_valuation_snapshots`
Indexed columns	`(valuation_date DESC, location_scope)`
Index type	B-tree
Expected query pattern	`SELECT * FROM inventory_valuation_snapshots WHERE location_scope = 'NETWORK' ORDER BY valuation_date DESC LIMIT 12`
Business justification	Month-end valuation history (BR-031, BP-30).
Performance clarity (junior)	**Why:** Finance views last 12 monthly snapshots — low volume table, simple index sufficient.

IDX-046

Attribute	Detail
Table	`audit.inventory_valuation_lines`
Indexed columns	`snapshot_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM inventory_valuation_lines WHERE snapshot_id = ?`
Business justification	Load snapshot detail lines for reconciliation report.
Performance clarity (junior)	**Why:** Parent-child report pattern. **How:** FK index on snapshot header.

IDX-047

Attribute	Detail
Table	`inventory.stock_policy_history`
Indexed columns	`(stock_policy_id, effective_from DESC)`
Index type	B-tree
Expected query pattern	`SELECT * FROM stock_policy_history WHERE stock_policy_id = ? ORDER BY effective_from DESC`
Business justification	Threshold change audit for replenishment analysis.
Performance clarity (junior)	**Why:** Planner reviews "when did reorder level change?" **How:** History lookup by policy id, recent first.

IDX-048

Attribute	Detail
Table	`identity.role_assignments`
Indexed columns	`(user_account_id)` WHERE `assignment_status = 'ACTIVE'`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM role_assignments WHERE user_account_id = ? AND assignment_status = 'ACTIVE'`

Attribute	Detail
Business justification	Authorization check on every API request.
Performance clarity (junior)	**Why:** Login loads active roles only — not historical inactive assignments. **How:** Partial index for permission resolution.

IDX-049

Attribute	Detail
Table	`identity.role_assignment_locations`
Indexed columns	`role_assignment_id`
Index type	B-tree
Expected query pattern	`SELECT location_id FROM role_assignment_locations WHERE role_assignment_id = ?`
Business justification	Location scope check for warehouse operations.
Performance clarity (junior)	**Why:** After loading roles, load scoped locations. **How:** FK child index.

IDX-050

Attribute	Detail
Table	`commercial.promotional_stock_allocations`
Indexed columns	`(product_id, readiness_status)` WHERE `readiness_status IN ('PLANNED','READY','ACTIVE')`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM promotional_stock_allocations WHERE product_id = ? AND readiness_status = 'ACTIVE'`
Business justification	Campaign readiness review (BR-029).
Performance clarity (junior)	**Why:** Small table, few active campaigns — partial index for active/planned only.

IDX-051

Attribute	Detail
Table	`commercial.promotional_allocation_locations`
Indexed columns	`stock_position_id`
Index type	B-tree
Expected query pattern	`SELECT * FROM promotional_allocation_locations WHERE stock_position_id = ?`
Business justification	Check ring-fence when reserving stock (BR-029).
Performance clarity (junior)	**Why:** Reservation service asks "is this balance ring-fenced for a campaign?" **How:** Reverse lookup from stock position to active allocation.

IDX-052

Attribute	Detail
Table	`governance.governance_policies`
Indexed columns	`(policy_type, effective_from DESC)` WHERE `is_active = TRUE AND effective_to IS NULL`
Index type	**Partial B-tree**
Expected query pattern	`SELECT * FROM governance_policies WHERE policy_type = 'APPROVAL_THRESHOLD' AND is_active = TRUE AND effective_to IS NULL`

Attribute	Detail
Business justification	Load current effective policy for approval threshold check (BR-021).
Performance clarity (junior)	**Why:** Only current active policies matter at runtime — not retired versions. **How:** Partial index on "currently effective" rows.

Part 6: Indexes Explicitly NOT Created

Avoiding unnecessary indexes is as important as creating the right ones.

Table / Column	Why NOT indexed	Junior explanation
`products.description`	Never in WHERE/JOIN; TEXT search not required in IMS	Full table scan on description search would need full-text (GIN) only if search feature added later
`products.category_code`	Low-selectivity filter alone; network reports use stock_positions join	Index on low-cardinality column alone rarely helps unless combined in composite with high-selectivity column
Every FK automatically	PostgreSQL does **not** auto-index FKs	Only index FKs that appear in JOIN/WHERE on the **child** side of frequent queries
`stock_positions.product_id` alone	Covered by UK (`product_id, location_id`) leftmost prefix	Composite UK already helps `WHERE product_id = ?` — separate single-column index is redundant
`customer_orders.customer_reference`	Optional, low-cardinality, rare lookup	Add only if CRM integration requires frequent search by customer_reference
`shipments.tracking_identifier`	Lookup only after ship; nullable; not hot path	Index when carrier tracking portal integration is confirmed
`goods_receipts.supplier_id`	Receipts usually found by PO or GR number, not supplier	Supplier performance reports are batch — acceptable sequential scan with date filter
`audit_trail_records.before_state` (JSONB)	No JSON containment queries in scope	GIN index is large — add only if "search audit JSON" feature is built
Duplicate index on `reservation_status` alone	Low cardinality (3 values) without leading selective column	Always combine with `stock_position_id` in partial index
`inventory_write_offs`, `stock_adjustments` by `stock_position_id`	Low volume; approval queues use status partial indexes	Historical adjustment lookup is rare — add if investigation tool requires it
All `created_at` columns	Rarely filtered alone; use composite with status	`WHERE created_at > ?` alone causes large index — prefer (`status, created_at`) if date filtering needed

Part 7: Index Maintenance Guide (Junior)

Monitoring

Check	Tool	Frequency
Unused indexes	`pg_stat_user_indexes` where `idx_scan = 0`	Monthly
Bloated indexes	`pgstatindex` or pg_repack	Quarterly
Slow queries missing indexes	`pg_stat_statements`, EXPLAIN ANALYZE	Weekly on Tier 1

When INSERT/UPDATE feels slow

Ask: **did we add an index we don't use?** Each extra index adds write cost. Check `idx_scan` — if zero after 30 days in production, drop the index.

Partial index reminder

```
-- Conceptual example (not DDL for deployment)
Index: reservations WHERE status = 'ACTIVE'
Stores: only active rows
Query must include: reservation_status = 'ACTIVE' (or equivalent) for index to be used
```

If the query omits the `WHERE status = 'ACTIVE'` predicate, PostgreSQL cannot use the partial index.

Partitioned tables

`inventory_movement_records` and `audit_trail_records` are partitioned by month. **Indexes must exist on each partition** (PostgreSQL 11+ can create on parent and propagate). BRIN on `movement_timestamp` works well **inside** each partition.

Part 8: Index Summary Matrix

Schema	Tier 1	Tier 2	Tier 3	Implicit UK/PK
`catalog`	3	0	0	3
`network`	1	2	0	2
`inventory`	4	1	1	3
`procurement`	2	2	0	4
`inbound`	1	2	0	1
`finance`	0	3	0	1
`fulfillment`	4	5	0	1
`transfer`	0	2	0	1
`accuracy`	0	3	0	4
`returns`	0	1	0	1
`commercial`	0	0	2	1
`identity`	1	0	2	2
`governance`	0	0	1	2
`audit`	0	0	6	2
Total explicit	**16**	**21**	**12**	**~28 used**

*Note: Several Tier 1 entries document implicit UK/PK indexes only. Total explicit indexes to CREATE: **49**.*

Part 9: Query Pattern → Index Quick Reference

Business operation	Query pattern	Index
Confirm order availability	`stock_positions BY product_id + location_id`	UK implicit (IDX-001)
Scan barcode at receipt	`product_identifiers BY type + value`	UK implicit (IDX-003)
Reserve stock	`reservations ACTIVE BY stock_position_id`	IDX-004 partial
FEFO pick	`lot_batches BY stock_position_id ORDER BY expiration`	IDX-005
Open order dashboard	`customer_orders BY status + date`	IDX-008
Low stock alert	`stock_positions LOW/OUT at location`	IDX-011 partial
Receive against PO	`goods_receipts BY purchase_order_id`	IDX-013

Business operation	Query pattern	Index
Three-way match	`invoices + receipts BY purchase_order_id`	IDX-013, IDX-022
Pick queue	`fulfillment_executions PENDING at location`	IDX-026 partial
Approval queues	`* WHERE approval_status = 'PENDING'`	IDX-035, IDX-036 partial
Movement history	`movement_records BY position + timestamp`	IDX-041
Open alerts	`operational_alerts OPEN`	IDX-044 partial

Document Approval

Role	Name	Signature	Date
Database Performance Specialist			
Database Architect			
DDD Solution Architect			

End of Database Indexing Strategy

Document ID: SRS-DB-005 | Version 1.0 | July 6, 2026

Database Audit Strategy

Inventory Management System — PostgreSQL

Document Information

Attribute	Detail
Document ID	SRS-DB-006
Document Title	Database Audit Strategy
Version	1.0
Status	Draft
Document Owner	Database Architect
Source Documents	SRS-DB-002 (Physical Data Model), SRS-DB-004 (Database Constraints), SRS-DM-006 (Domain Events), SRS-BR-001 (Business Rules), ADR-024 (Centralized Audit Trail Projection), Cross-Cutting Concerns §7
Target Platform	PostgreSQL 15+
Date	July 6, 2026
Scope	Audit design, field standards, table classification, event catalog, retention — no DDL/SQL statements

Purpose

This document defines the **database audit strategy** for the Inventory Management System. Auditing supports **traceability** (who changed what, when, where, and why) and **compliance** (COSO internal control, financial record alignment, lot recall, segregation of duties). The strategy avoids audit sprawl: not every table receives the same treatment — audit depth matches business materiality.

Part 1: Audit Architecture Overview

1.1 Three-Layer Audit Model

Layer	Mechanism	Purpose	Immutability
L1 — Row attribution	Standard and entity-specific columns on transactional tables	Current-state accountability; lightweight trace on every mutable row	Mutable table rows can UPDATE; attribution columns capture last actor where applicable
L2 — Movement log	`audit.inventory_movement_records`	Append-only record of **every stock quantity change** with before/after quantities	**Append-only** — no UPDATE/DELETE
L3 — Governance trail	`audit.audit_trail_records`	Before/after snapshots for **material** entity and policy changes	**Append-only** — no UPDATE/DELETE

Supporting projections (not primary audit evidence):

Artifact	Table	Role
Threshold history	`inventory.stock_policy_history`	Dedicated history for replenishment policy changes
Operational alerts	`audit.operational_alerts`	Derived notifications — trace to source entity, not standalone evidence
Financial snapshots	`audit.inventory_valuation_snapshots` + `inventory_valuation_lines`	Point-in-time valuation evidence for reconciliation (BP-30)

1.2 Architectural Rules (from ADR-024, AUD-01–AUD-05)

Rule	Requirement
AUD-01	Audit store records are append-only — application role has INSERT only on L2/L3 tables
AUD-02	Governance trail retained ≥ 7 years (NFR-COM-019, NFR-COM-020)
AUD-03	Single authoritative audit projection — no per-module duplicate audit tables
AUD-04	Domain events are the primary audit source; database audit tables are the persistence projection
AUD-05	Auditor evidence export within 15 minutes (NFR-PERF-026, NFR-COM-012)

1.3 What Auditing Is Not

Not audit	Instead
Application debug logs	Operational logging (NFR-LOG) — diagnostic, shorter retention
`updated_at` alone	Insufficient for compliance — no actor, no before state, no reason
Duplicating movement + trail for every SELECT	Read operations are authorized via security logs, not business audit trail
Indexing JSONB audit fields for ad-hoc search	Structured query dimensions (`affected_entity_type`, `user_id`, `event_timestamp`)

Part 2: Standard Audit Fields

2.1 Platform-Wide Columns (L1)

Applied per SRS-DB-002 conventions on **mutable** tables.

Field	Data Type	Tables	Business Purpose
`created_at`	`TIMESTAMPTZ` NOT NULL DEFAULT NOW()	All mutable tables	When the record was first persisted (UTC)

Field	Data Type	Tables	Business Purpose
`updated_at`	`TIMESTAMPTZ` NOT NULL DEFAULT NOW()	Mutable tables with UPDATE lifecycle	Last modification timestamp — maintained by application/trigger
`version`	`INTEGER` NOT NULL DEFAULT 1	Transactional headers with concurrent updates	Optimistic locking — detects lost updates; audit-relevant as conflict signal

2.2 Attribution Columns (L1 — actor tracking)

Field	Data Type	When Required	Business Purpose
`created_by`	`BIGINT` FK → `identity.user_accounts`	Document creation (requisitions, receipts, orders)	Who initiated the business document
`initiated_by`	`BIGINT` FK	Adjustments, write-offs, count workflows	Who started a correction requiring approval
`approved_by`	`BIGINT` FK	Any approval-gated action (BR-019, BR-021, BR-023)	Who authorized — **SoD verification** (must ≠ initiator/counter)
`authorized_by`	`BIGINT` FK	Quarantine release, write-off authorization	Formal authorization for irreversible or high-risk actions
`performed_by`	`BIGINT` FK	Put-away, movement source events	Who executed the physical operation
`prepared_by`	`BIGINT` FK	Valuation snapshots	Who certified the financial snapshot

2.3 Workflow and Context Columns (L1 — traceability enrichment)

Field	Data Type	Tables	Business Purpose
`reason_code_id`	`BIGINT` FK → `governance.reason_codes`	Adjustments, write-offs, some movements	**Why** the change occurred (BR-020)
`adjustment_reference` / `*_reference`	`VARCHAR(64)`	Headers across domains	Human-readable business key for external audit correspondence
`qty_before` / `qty_after`	`NUMERIC(18,4)`	Adjustments, movement records	Quantity state transition evidence
`approval_status`	`VARCHAR(32)`	Gated documents	Workflow state at time of audit
`approval_chain_ref`	`VARCHAR(128)`	Audit trail records	Link to workflow engine instance
`evidence_reference`	`VARCHAR(128)`	Write-offs, variances	External evidence (photo, inspector report)

2.4 L2 Movement Record Fields

Every stock mutation produces one row in `audit.inventory_movement_records`:

Field	Purpose
`movement_reference`	Correlates to source document line or transaction ID
`movement_type`	RECEIPT, SHIPMENT, TRANSFER, ADJUSTMENT, WRITE_OFF, RESERVATION, RETURN, PUT_AWAY
`stock_position_id`, `product_id`, `location_id`	**Where** and **what** changed (denormalized for query performance)
`quantity_change`, `qty_before`, `qty_after`	**What** changed numerically
`lot_batch_id`	Lot traceability (BR-017)
`reason_code_id`	**Why** for adjustments and write-offs
`source_document_type`, `source_document_id`	**Originating document** — receipt, shipment, adjustment, etc.
`performed_by`	**Who** executed
`movement_timestamp`	**When** (partition key)

2.5 L3 Audit Trail Record Fields

Material non-movement or co-material changes produce rows in `audit.audit_trail_records`:

Field	Purpose
`audit_event_reference`	Unique business event ID (correlates to domain event / outbox message)
`affected_entity_type`	Logical table or entity name (e.g. `stock_adjustments`, `products`)
`affected_entity_id`	PK of governed record
`action_performed`	CREATE, UPDATE, APPROVE, CANCEL
`user_id`	Actor
`role_at_action`	Role at time of action — supports SoD retrospective review
`before_state` / `after_state`	JSONB snapshots of changed attributes
`approval_chain_ref`	Workflow reference when approval involved
`movement_record_id`	Optional FK linking governance trail to L2 movement when both apply
`event_timestamp`	**When** (partition key)

2.6 Recommended Application Audit Context (not stored on every row)

Passed from application layer to audit writer on each command — may be embedded in `before_state`/`after_state` or logged to security audit:

Context Field	Purpose
`correlation_id`	Traces one user action across multiple tables/events
`session_id`	Links to authentication session
`client_ip` / `user_agent`	Security investigation (security log primary; optional in trail)
`location_scope`	Location context of the acting user
`command_name`	Application command that triggered the change

Part 3: Tables Requiring Auditing

Tables are classified by **audit tier**. Higher tiers require more layers.

3.1 Tier Definitions

Tier	L1 Columns	L2 Movement	L3 Trail	Rationale
T0 — Audit store	N/A (is audit)	—	Self	Append-only audit tables
T1 — Stock mutation	Full attribution	**Required** on every qty change	**Required** on approval/status transitions	Financial and operational materiality
T2 — Governance & master	`created_at/updated_at` + actor on change	If qty affected	**Required** on CREATE/UPDATE/DEACTIVATE	Master data and policy integrity
T3 — Transactional document	Full attribution on header	Via lines/execution	On status transitions (submit, approve, cancel, ship)	Document lifecycle traceability
T4 — Line / child	`created_at` only	Via parent workflow	Only if line is approval point	Inherits header audit chain via FK
T5 — Reference / junction	`created_at` only	No	No — parent audited	Low independent materiality

Tier	L1 Columns	L2 Movement	L3 Trail	Rationale
T6 — Derived / snapshot	`created_at` + `prepared_by` where applicable	No	No — immutable snapshot IS the record	Point-in-time evidence

3.2 Complete Table Classification (59 tables)

T0 — Audit Store (5 tables)

Table	Audit Treatment
`audit.inventory_movement_records`	Append-only; DB role INSERT-only; monthly partition
`audit.audit_trail_records`	Append-only; DB role INSERT-only; monthly partition
`audit.operational_alerts`	Mutable status (ACKNOWLEDGED/RESOLVED); derived — not primary evidence
`audit.inventory_valuation_snapshots`	Immutable after publish; financial evidence
`audit.inventory_valuation_lines`	Immutable child of snapshot

T1 — Stock Mutation (11 tables)

Table	L1 Fields	L2	L3 Events
`inventory.stock_positions`	`created_at`, `updated_at`, `version`, `last_movement_at`	**Every** qty change	Status/ownership changes
`inventory.lot_batches`	`created_at`, `updated_at`	Qty changes per lot	Quality status, expiration corrections
`inventory.quarantine_holds`	`authorized_by`, hold/release timestamps	Hold apply/release qty	CREATE, RELEASE, DISPOSE
`fulfillment.inventory_reservations`	`created_at`, `updated_at`	Reserve/release/fulfill	Status transitions
`inbound.put_aways`	`performed_by`	ATP increase on completion	Completion event
`fulfillment.shipments`	`created_at`, `updated_at`	Ship qty deduction	Ship confirm, cancel
`transfer.stock_transfers`	`created_at`, `updated_at`	Transfer in/out	Status transitions
`transfer.in_transit_stock`	`created_at`, `updated_at`	In-transit qty tracking	Receive, loss, exception
`accuracy.stock_adjustments`	`initiated_by`, `approved_by`, `qty_before/after`, `reason_code_id`	On APPROVED	PENDING→APPROVED/REJECTED
`accuracy.inventory_write_offs`	`initiated_by`, `authorized_by`, `reason_code_id`	On authorization	CREATE, AUTHORIZE
`returns.return_dispositions`	`created_at`, `updated_at`	Restock/scrap qty	Disposition decision

T2 — Governance & Master Data (11 tables)

Table	L1 Fields	L3 Events
`catalog.products`	`created_at`, `updated_at`, `created_by`	ProductRegistered, ProductStatusChanged, ABC classification
`catalog.product_identifiers`	`created_at`, `updated_at`	Identifier add/change (barcode integrity)
`network.inventory_locations`	`created_at`, `updated_at`	LocationActivated, LocationDeactivated
`network.storage_positions`	`created_at`, `updated_at`	StoragePositionConfigured
`governance.governance_policies`	`created_at`, `updated_at`	Policy threshold changes
`governance.reason_codes`	`created_at`, `updated_at`	Reason code add/deactivate
`procurement.suppliers`	`created_at`, `updated_at`, `is_active`	Supplier activation/deactivation

Table	L1 Fields	L3 Events
inventory.stock_policies	created_at, updated_at	Threshold create/update — also writes stock_policy_history
inventory.stock_policy_history	Dedicated history table	**Is** the audit artifact for policy changes
identity.user_accounts	created_at, updated_at, account_status	Account create, suspend, deactivate
identity.role_assignments	created_at, updated_at, effective_from/to	Role grant/revoke — **security-critical**

T3 — Transactional Documents (16 tables)

Table	L1 Fields	L3 Events
procurement.purchase_requisitions	created_by, approved_by, version	Submit, approve, reject, convert
procurement.purchase_orders	created_at, updated_at, version	Place, amend, cancel, close
inbound.goods_receipts	created_at, updated_at, version	Receive, close, discrepancy
inbound.receipt_discrepancies	created_at, updated_at	Discrepancy resolution
finance.supplier_invoices	created_at, updated_at	Register, match, authorize payment
finance.invoice_goods_receipt_match	created_at	Three-way match outcome (BR-008)
fulfillment.customer_orders	created_at, updated_at, version	Confirm, cancel, complete
fulfillment.fulfillment_executions	created_at, updated_at	Pick start, complete, exception
fulfillment.backorders	created_at, updated_at	Backorder create, fulfill, cancel
transfer.transfer_requests	created_at, updated_at, approved_by	Request, approve, reject
accuracy.cycle_counts	created_at, updated_at	Schedule, start, complete
accuracy.physical_inventory_counts	created_at, updated_at	Count certification (BR-024 freeze)
accuracy.count_variances	initiated_by, approved_by	Variance identification, approval (BR-019)
returns.return_authorizations	created_at, updated_at, version	RMA issue, cancel, expire
returns.customer_returns	created_at, updated_at	Return receipt
commercial.promotional_stock_allocations	created_at, updated_at	Campaign allocation create/amend (BR-029)

T4 — Line / Child Tables (15 tables)

Inherit audit chain from parent header via FK. L3 trail written at header level unless line is the approval point.

Table	Parent	Line-Level L3 When
catalog.product_identifiers	products	Barcode change (also T2)
procurement.purchase_requisition_lines	purchase_requisitions	—
procurement.purchase_order_lines	purchase_orders	—
procurement.expected_inbound_deliveries	—	Status changes
procurement.expected_inbound_delivery_lines	expected_inbound_deliveries	—
inbound.goods_receipt_lines	goods_receipts	Lot capture (BR-017)
finance.supplier_invoice_lines	supplier_invoices	—
fulfillment.customer_order_lines	customer_orders	—
fulfillment.fulfillment_pick_lines	fulfillment_executions	Pick confirmation
fulfillment.shipment_lines	shipments	—
transfer.transfer_request_lines	transfer_requests	—

Table	Parent	Line-Level L3 When
`transfer.stock_transfer_lines`	`stock_transfers`	—
`returns.return_authorization_lines`	`return_authorizations`	—
`returns.customer_return_lines`	`customer_returns`	—
`commercial.promotional_allocation_locations`	`promotional_stock_allocations`	—

T5 — Junction / Scope (1 table)

Table	Treatment
`identity.role_assignment_locations`	`created_at` only; scope changes audited via parent `role_assignments` L3 entry

T6 — Derived Snapshots (2 tables — also in T0)

Table	Treatment
`audit.inventory_valuation_snapshots`	Immutable financial point-in-time; `prepared_by` required
`audit.inventory_valuation_lines`	Immutable detail; no UPDATE after snapshot publish

3.3 Tables Explicitly NOT Requiring L3 Trail

Table	Reason
`identity.role_assignment_locations`	Junction — parent role assignment carries audit
All T4 line tables (unless noted)	Parent document audit + movement log captures qty impact
`audit.operational_alerts`	Derived notification — auditors trace via `source_entity_type/id` to source

---

Part 4: Audit Tables

4.1 Primary Audit Tables

`audit.inventory_movement_records` — Operational Quantity Audit

Aspect	Detail
Purpose	Immutable log of every stock quantity change
Populated by	Domain event handlers after successful stock mutation commit
Cardinality	1+ per stock mutation; links to source document
Growth	Very high (10M–100M+/year)
Partitioning	Monthly range on `movement_timestamp`
Permissions	Application role: INSERT only
Linked from	`audit.audit_trail_records.movement_record_id` when co-material

`audit.audit_trail_records` — Governance Compliance Audit

Aspect	Detail
Purpose	Before/after snapshots for material entity and policy changes
Populated by	M-13 Intelligence projection from domain events + command audit context
Cardinality	0–1 per material command; may pair with movement record

Aspect	Detail
Growth	Very high (proportional to governed mutations)
Partitioning	Monthly range on `event_timestamp`
Permissions	Application role: INSERT only
JSONB scope	Changed attributes only — not full row dump unless material

`inventory.stock_policy_history` — Dedicated Policy History

Aspect	Detail
Purpose	Threshold change history without polluting operational stock tables
Populated by	Trigger or application on `stock_policies` UPDATE
Retention	Align with governance trail (7 years)

4.2 Supporting Audit Artifacts

Table	Audit Role	Retention
`audit.operational_alerts`	Exception and threshold **notification** trail	Purge resolved > 90 days
`audit.inventory_valuation_snapshots`	Financial reconciliation evidence (BP-30)	7 years
`audit.inventory_valuation_lines`	Snapshot detail	7 years (with parent)

4.3 Audit Write Flow

```
User Command
    │
    ├─► [Application] Enrich AuditContext (user, role, correlation, reason)
    │
    ├─► [Domain] Mutate aggregate + raise domain event
    │
    ├─► [DB Transaction] UPDATE transactional table (L1 columns)
    │                    INSERT movement record (L2) if qty changed
    │
    └─► [Post-commit Handler] INSERT audit_trail_record (L3) from event payload
```

Rule: L2 and L3 writes occur in the **same database transaction** as the stock mutation when possible. L3 from async projection is acceptable only for non-qty master data changes with eventual consistency ≤ 5 seconds.

4.4 Before/After Snapshot Guidelines (JSONB)

Entity Type	Include in `before_state` / `after_state`
`stock_positions`	`on_hand_qty`, `reserved_qty`, `quarantined_qty`, `atp_qty`, `stock_status`, `ownership_type`
`stock_adjustments`	`adjustment_qty`, `approval_status`, `reason_code_id`, `initiated_by`, `approved_by`
`products`	`sku`, `product_status`, `abc_classification`, `is_traceable`, `unit_of_measure`
`inventory_locations`	`location_code`, `operational_status`, `location_type`
`role_assignments`	`role_name`, `assignment_status`, `effective_from`, `effective_to`, `sod_compliant`
`supplier_invoices`	`invoice_status`, `match_status`, `total_amount`, `authorization_status`
`governance_policies`	Policy type, threshold values, effective dates

Exclude: Large TEXT blobs, redundant denormalized fields already on movement record, PII beyond `user_id` reference.

Part 5: Business Events to Audit

Events map to domain events (SRS-DM-006). Each **material** event produces at minimum L1 update; stock events also produce L2; governance events produce L3.

5.1 Stock & Balance Events (L2 + L3)

Event ID	Domain Event	Source Table(s)	Movement Type	L3 Action
AE-001	StockIncreased (receipt)	`goods_receipts`, `put_aways`, `stock_positions`	RECEIPT / PUT_AWAY	APPROVE on put-away complete
AE-002	StockReserved	`inventory_reservations`, `stock_positions`	RESERVATION	UPDATE
AE-003	ReservationReleased	`inventory_reservations`	RESERVATION (negative)	UPDATE
AE-004	StockShipped	`shipments`, `shipment_lines`	SHIPMENT	APPROVE
AE-005	StockTransferred	`stock_transfers`, `in_transit_stock`	TRANSFER	UPDATE
AE-006	TransferReceived	`stock_transfers`, `in_transit_stock`	TRANSFER	APPROVE
AE-007	StockAdjusted	`stock_adjustments`	ADJUSTMENT	APPROVE
AE-008	StockWrittenOff	`inventory_write_offs`	WRITE_OFF	APPROVE
AE-009	StockQuarantined	`quarantine_holds`	ADJUSTMENT (quarantine bucket)	CREATE
AE-010	QuarantineReleased	`quarantine_holds`	ADJUSTMENT	UPDATE
AE-011	ReturnRestocked	`return_dispositions`, `customer_returns`	RETURN	APPROVE
AE-012	LotQuantityChanged	`lot_batches`	Per parent movement	UPDATE if lot attrs change

5.2 Inbound & Procurement Events (L3; L2 on receipt)

Event ID	Domain Event	Source Table(s)	L3 Action
AE-020	PurchaseRequisitionSubmitted	`purchase_requisitions`	UPDATE
AE-021	PurchaseRequisitionApproved	`purchase_requisitions`	APPROVE
AE-022	PurchaseOrderPlaced	`purchase_orders`	CREATE
AE-023	PurchaseOrderCancelled	`purchase_orders`	CANCEL
AE-024	GoodsReceived	`goods_receipts`	CREATE
AE-025	ReceiptDiscrepancyIdentified	`receipt_discrepancies`	CREATE
AE-026	PutAwayCompleted	`put_aways`	APPROVE

5.3 Fulfillment Events (L2 + L3)

Event ID	Domain Event	Source Table(s)	L3 Action
AE-030	CustomerOrderConfirmed	`customer_orders`	UPDATE
AE-031	CustomerOrderCancelled	`customer_orders`	CANCEL
AE-032	FulfillmentStarted	`fulfillment_executions`	UPDATE
AE-033	PickConfirmed	`fulfillment_pick_lines`	UPDATE
AE-034	ShipmentDispatched	`shipments`	APPROVE
AE-035	BackorderCreated	`backorders`	CREATE

5.4 Inventory Control Events (L2 + L3 — SoD critical)

Event ID	Domain Event	Source Table(s)	L3 Action	Compliance Note
AE-040	CycleCountStarted	`cycle_counts`	UPDATE	—
AE-041	PhysicalInventoryCountCertified	`physical_inventory_counts`	APPROVE	BR-024 freeze evidence
AE-042	CountVarianceIdentified	`count_variances`	CREATE	Counter recorded
AE-043	CountVarianceApproved	`count_variances`	APPROVE	**Approver ≠ counter** (BR-019)
AE-044	AdjustmentApproved	`stock_adjustments`	APPROVE	**Approver ≠ initiator** (BR-021)
AE-045	WriteOffAuthorized	`inventory_write_offs`	APPROVE	Dual authorization (BR-023)

5.5 Financial & Governance Events (L3)

Event ID	Domain Event	Source Table(s)	L3 Action	Compliance Note
AE-050	ThreeWayMatchCompleted	`invoice_goods_receipt_match`	APPROVE	BR-008 payment gate
AE-051	InvoicePaymentAuthorized	`supplier_invoices`	APPROVE	Financial control
AE-052	GovernancePolicyChanged	`governance_policies`	UPDATE	Threshold audit
AE-053	ReasonCodeDeactivated	`reason_codes`	UPDATE	Prevents orphan reasons
AE-054	ValuationSnapshotPublished	`inventory_valuation_snapshots`	CREATE	BP-30 reconciliation

5.6 Master Data & Security Events (L3)

Event ID	Domain Event	Source Table(s)	L3 Action	Compliance Note
AE-060	ProductStatusChanged	`products`	UPDATE	BR-027 discontinued
AE-061	InventoryLocationDeactivated	`inventory_locations`	UPDATE	Closure authorization
AE-062	SupplierDeactivated	`suppliers`	UPDATE	Procurement block
AE-063	RoleAssigned	`role_assignments`	CREATE	Access control
AE-064	RoleRevoked	`role_assignments`	UPDATE	Access control
AE-065	UserAccountDeactivated	`user_accounts`	UPDATE	Security
AE-066	SoDViolationAttemptBlocked	— (security log + L3)	CREATE	BR-019, BR-021

5.7 Traceability-Specific Events (Lot Recall)

Event ID	Domain Event	Tables	Audit Requirement
AE-070	LotCapturedAtReceipt	`goods_receipt_lines`, `lot_batches`	L2 with `lot_batch_id`; L3 on lot create
AE-071	LotMovedWithTransfer	`stock_transfer_lines`, `lot_batches`	L2 must carry `lot_batch_id`
AE-072	LotShippedToCustomer	`shipment_lines`, `lot_batches`	L2 for recall forward trace
AE-073	LotRecalled	`quarantine_holds`, `lot_batches`	L3 + L2; `hold_type = RECALL`

Recall query pattern: Forward trace from `lot_batches` → `inventory_movement_records` by `lot_batch_id`; backward from shipment to receipt via movement chain.

5.8 Events NOT Written to L3 Trail

Event / Operation	Reason
Read-only queries (ATP lookup, dashboards)	Not a state change — security log only if required
Alert generation	Derived — source condition already audited

Event / Operation	Reason
Cache invalidation	Infrastructure — not business audit
Failed validation (pre-commit)	No state change — operational log
Optimistic lock retry (success)	Only final successful commit audited

---

Part 6: Retention Policy Considerations

6.1 Retention Tiers

Tier	Storage	Access Pattern	Technology
Hot	0–24 months	Sub-second operational queries	PostgreSQL primary — partitioned
Warm	25 months–7 years	Auditor search, compliance export	PostgreSQL archive schema or read replica
Cold	7+ years (if mandated beyond)	Legal hold, rare retrieval	Object storage (Parquet/CSV), compressed
Purge	After retention expiry	None	Secure deletion per legal approval

6.2 Table-Specific Retention

Table / Data Class	Hot (online)	Total Retention	Purge Rule	Regulatory Driver
`audit.inventory_movement_records`	24 months	**7 years** total	Archive partition to warm/cold; drop after 7 years	Financial records alignment (NFR-COM-020); lot traceability (BR-017)
`audit.audit_trail_records`	24 months	**7 years** minimum	Tiered archive; legal hold overrides purge	COSO, external audit (BO-19, NFR-COM-010)
`inventory.stock_policy_history`	Full period	**7 years**	Archive with governance trail	Replenishment decision audit
`audit.inventory_valuation_snapshots` + lines	36 months	**7 years**	Archive with finance records	Month-end close, BP-30
Security audit events (login, SoD)	24 months	**7 years**	Separate security log store	NFR-COM-020
`audit.operational_alerts`	90 days resolved	90 days after RESOLVED	Scheduled purge job	Operational — not compliance primary
Transactional documents (closed PO, orders)	36 months	7 years	Archive closed documents	Commercial dispute resolution
`identity.user_accounts` (deactivated)	Active + 36 months	7 years after deactivation	Pseudonymize PII per legal matrix	GDPR vs audit (C-05) — legal sign-off required
Operational application logs	90 days	1 year	Standard log rotation	NFR-LOG — not compliance audit

6.3 Partitioning and Archival Mechanics

Table	Partition Key	Archival Action
`inventory_movement_records`	`movement_timestamp` (monthly)	`DETACH PARTITION` → export to cold → `DROP` from hot after 24 months
`audit_trail_records`	`event_timestamp` (monthly)	Same; retain detached partitions in warm store until year 7
Others	Not partitioned	Row-level archive job by `created_at` / status

6.4 Legal Hold and Compliance Overrides

Scenario	Action
Active litigation / regulatory investigation	Suspend purge for affected `lot_batch_id`, `product_id`, date range, or `user_id`
Recall event (AE-073)	Extend movement retention for affected lots — **no purge** until recall closed

Scenario	Action
GDPR erasure request (C-05)	Pseudonymize `user_accounts.display_name/email`; retain `user_id` in audit trail for compliance
External auditor export (AUD-05)	Hot+warm data exportable within 15 minutes — pre-built evidence package views

6.5 Immutability Enforcement

Control	Implementation
Database permissions	`audit_writer` role: INSERT on L2/L3 only; no UPDATE/DELETE
Application	No ORM update paths to audit tables
Trigger guard (optional)	`BEFORE UPDATE OR DELETE` → RAISE EXCEPTION on audit tables
Monitoring	Alert on any DDL against `audit` schema
Hash chain (future)	Optional `record_hash` + `previous_hash` on L3 for tamper evidence — not in v1.0

6.6 Storage Growth Estimates (Planning)

Table	Annual Rows (est.)	Row Size (est.)	7-Year Storage (est.)
`inventory_movement_records`	50M	~500 bytes	~175 GB + indexes
`audit_trail_records`	10M	~2 KB (JSONB)	~140 GB + indexes
`operational_alerts`	2M (rolling)	~300 bytes	< 1 GB (purged)

Implication: Hot PostgreSQL cannot hold 7 years of movement data at scale — **tiered architecture is mandatory** (ADR-026, Architecture Review G-05).

Part 7: Segregation of Duties (SoD) Audit Requirements

Business Rule	Audit Evidence Required	Tables
BR-019	Counter ≠ variance approver	`count_variances.initiated_by` vs `approved_by` + L3 trail
BR-021	Initiator ≠ adjustment approver	`stock_adjustments.initiated_by` vs `approved_by` + L3 trail
BR-023	Write-off requires authorization	`inventory_write_offs.initiated_by` vs `authorized_by`
BR-008	Three-way match before payment	`invoice_goods_receipt_match` + L3 APPROVE
Identity SoD	Role conflict blocked	`role_assignments.sod_compliant`, AE-066 on violation attempt

Auditor query pattern: Join L3 `audit_trail_records` where `action_performed = 'APPROVE'` and compare `user_id` to `before_state.initiated_by`.

Part 8: Implementation Checklist

#	Task	Owner
1	Apply L1 audit columns per table tier in migrations	DBA
2	Create `audit_writer` DB role with INSERT-only on L2/L3	DBA
3	Implement movement record writer in stock mutation transaction	Backend
4	Implement audit trail projection handler (M-13)	Backend
5	Map domain events AE-001–AE-073 to audit writers	Backend
6	Configure monthly partitions on movement and trail tables	DBA
7	Build partition detach/archive job (hot → warm)	DevOps

#	Task	Owner
8	Define legal hold flag and purge suspension	Legal + DBA
9	Privacy matrix for GDPR vs 7-year retention (G-09)	Legal
10	Auditor export views / evidence package (AUD-05)	Intelligence module

Part 9: Summary Matrix

Audit Layer	Tables	Write Trigger	Retention
L1 Row attribution	54 mutable tables	Every INSERT/UPDATE	Life of row
L2 Movement log	Stock mutations (11 source tables)	Every qty change	7 years (tiered)
L3 Governance trail	35 material tables/events	Material commands	7 years (tiered)
Policy history	`stock_policy_history`	Policy UPDATE	7 years
Financial snapshot	Valuation tables	Period close	7 years
Alerts (supporting)	`operational_alerts`	Threshold/exception	90 days resolved

Document Approval

Role	Name	Signature	Date
Database Architect			
Database Performance Specialist			
Internal Audit / Compliance			
Solution Architect			

End of Database Audit Strategy

Document ID: SRS-DB-006 | Version 1.0 | July 6, 2026

Database Naming Conventions

Inventory Management System — PostgreSQL

Document Information

Attribute	Detail
Document ID	SRS-DB-008
Document Title	Database Naming Conventions
Version	1.0
Status	Draft
Document Owner	Database Architect
Source Documents	SRS-DB-002 (PDM), SRS-DB-003 (Keys), SRS-DB-004 (Constraints), SRS-DB-005 (Indexing), SRS-DB-006 (Audit)
Target Platform	PostgreSQL 15+

Attribute	Detail
Date	July 6, 2026
Scope	Naming standards for all database objects — no DDL/SQL statements

Purpose

This document defines **consistent, scalable naming conventions** for the Inventory Management System database. All new objects — tables, columns, keys, constraints, indexes, sequences, and views — must follow these standards. Conventions align with existing SRS-DB artifacts and PostgreSQL identifier limits (63 characters).

Global Rules

Rule	Standard
Case	Lowercase only — `snake_case` for all identifiers
Characters	Letters, digits, underscores; start with a letter
Length	Target ≤ 40 characters; hard limit 63 (PostgreSQL)
Reserved words	Avoid PostgreSQL reserved words (`user`, `order`, `position`) — qualify with schema or use plural/descriptive names (`user_accounts`, `customer_orders`, `storage_positions`)
Abbreviations	Use only approved abbreviations (see Appendix A); prefer full words for clarity
Pluralization	**Tables:** plural nouns (`products`, `stock_positions`). **Columns:** singular attribute names (`product_id`, `order_date`)
Language	American English spelling in identifiers (`canceled` → use domain value `CANCELLED` in data, not in object names)
Bounded context	Schema name reflects domain module — not `dbo` or `public` for application tables

---

1. Schemas

1.1 Naming Pattern

```
{bounded_context_noun}
```

Rule	Detail
Format	Single lowercase noun or compound noun — no prefix/suffix
Count	One schema per bounded context (14 schemas in IMS)
`public`	Reserved for extensions only — application tables do not use `public`

1.2 Approved Schemas

Schema	Purpose
`catalog`	Product master data
`network`	Locations and storage positions
`governance`	Policies and reason codes
`inventory`	Stock balances, lots, holds, policies
`procurement`	Suppliers, requisitions, purchase orders
`inbound`	Goods receipt, discrepancies, put-away
`finance`	Supplier invoices and three-way match
`fulfillment`	Customer orders, reservations, picks, shipments

Schema	Purpose
`transfer`	Inter-location transfers
`accuracy`	Cycle counts, adjustments, write-offs
`returns`	RMA and customer returns
`commercial`	Promotional allocations
`identity`	Users and role assignments
`audit`	Movement log, audit trail, alerts, valuation

1.3 Fully Qualified References

Always use **schema-qualified** names in documentation, migrations, and cross-schema FK definitions:

```
inventory.stock_positions
fulfillment.customer_orders
```

---

2. Tables

2.1 Naming Pattern

```
{entity_noun_plural}
```

or for associative / junction tables:

```
{parent_a}_{parent_b}          -- when relationship is the entity
{parent_entity}_{role_noun}      -- e.g. stock_policy_history, invoice_goods_receipt_match
```

Rule	Detail
Plural	Tables hold collections — `products`, not `product`
Prefix	No `tbl_`, `t_`, or module prefix — schema provides context
Suffix	Avoid `_table`; use `_history`, `_lines`, `_match` only when semantically required
Line tables	Parent name + `_lines` — `purchase_order_lines`, `goods_receipt_lines`
Header documents	Domain noun — `purchase_orders`, `goods_receipts`, `customer_orders`
Log / audit tables	`_records` suffix for append-only logs — `inventory_movement_records`, `audit_trail_records`
Max length	≤ 40 characters recommended

2.2 Examples

Good	Bad	Reason
`stock_positions`	`StockPosition`	Must be snake_case lowercase
`inventory_reservations`	`inv_res`	No unapproved abbreviations
`invoice_goods_receipt_match`	`inv_gr_match_tbl`	No `tbl_` prefix; spell out
`user_accounts`	`users`	Avoid reserved word `user`
`put_aways`	`putaway`	Compound operations use underscore

2.3 Table Partition Naming (High-Volume Tables)

```
{parent_table}_{partition_bound}
```

Partition Type	Pattern	Example
Monthly range	`{table}_y{YYYY}m{MM}`	`inventory_movement_records_y2026m07`
Yearly archive	`{table}_y{YYYY}`	`customer_orders_y2023`

Partition child tables inherit parent naming root. Partition key column: `movement_timestamp`, `event_timestamp`.

---

3. Columns

3.1 General Rules

Rule	Detail
Format	`snake_case`, singular attribute name
Boolean	Prefix `is_` or `has_` — `is_active`, `has_sod_conflict`, `pack_confirmed`
Timestamps	Suffix `_at` for point-in-time — `created_at`, `reserved_at`, `released_at`
Dates	Suffix `_date` for calendar date — `order_date`, `expiration_date`
Quantities	Suffix `_qty` — `on_hand_qty`, `reserved_qty`, `adjustment_qty`
Amounts	Suffix `_amount` or `_value` — `total_amount`, `financial_value`, `line_amount`
Status	Suffix `_status` — `order_status`, `approval_status`, `match_status`
Type / category	Suffix `_type` — `movement_type`, `hold_type`, `ownership_type`
Codes	Suffix `_code` — `reason_code`, `category_code`, `currency_code`
References	Suffix `_reference` or `_ref` — `movement_reference`, `approval_chain_ref`
Counts	Suffix `_count` — `total_variances` (exception: use `_number` for business document numbers)

3.2 Standard Platform Columns

Apply consistently per table type (see SRS-DB-002):

Column	Type	Tables	Meaning
`id`	`BIGINT`	All	Surrogate primary key
`version`	`INTEGER`	Mutable transactional headers	Optimistic lock counter
`is_active`	`BOOLEAN`	Master / reference data	Soft-delete flag
`created_at`	`TIMESTAMPTZ`	Mutable tables	Row creation timestamp (UTC)
`updated_at`	`TIMESTAMPTZ`	Mutable tables with UPDATE	Last modification timestamp
`created_by`	`BIGINT`	Document headers	Creating user FK
`line_number`	`INTEGER`	Document line tables	Sequence within parent document

3.3 Foreign Key Column Naming

```
{referenced_table_singular}_id
```

Referenced Table	FK Column	Notes
`products`	`product_id`	Singular entity + `_id`
`inventory_locations`	`location_id`	Drop redundant prefix when unambiguous in context
`stock_positions`	`stock_position_id`	Full entity name when abbreviation is ambiguous
`user_accounts`	`user_id` or role-specific — see below	

Role-specific user FKs (when multiple users on one row):

Column	Meaning
`created_by`	Record creator
`initiated_by`	Workflow initiator
`approved_by`	Approver
`authorized_by`	Formal authorizer
`performed_by`	Physical executor
`prepared_by`	Snapshot preparer
`counter_id`	Cycle count performer
`inspector_id`	Return inspector
`receiving_clerk_id`	Receipt operator
`assigned_picker_id`	Pick assignee
`approver_id`	Variance approver (when `approved_by` not used)

Do **not** use `user_id` when a role-specific name is clearer.

3.4 Business Key Columns

Natural identifiers use domain terms — not `business_key` or `code` alone:

Entity	Column
Product	`sku`
Location	`location_code`
Purchase order	`po_number`
Customer order	`order_number`
Goods receipt	`gr_number`
Shipment	`shipment_number`
RMA	`rma_number`

3.5 Denormalized Columns

When FK target is duplicated for query performance, use same name as canonical FK:

```
product_id    -- denormalized on lot_batches, movement_records
location_id   -- denormalized on movement_records
```

Comment in data dictionary: "(denormalized)" — do not use `product_id_dup` or `product_id_fk2`.

3.6 Anti-Patterns

Avoid	Use Instead
`desc`	`description`
`amt`	`amount`
`qty1`, `qty2`	`qty_before`, `qty_after`
`status1`	`approval_status`, `order_status`
`data`, `info`, `misc`	Domain-specific name

Avoid	Use Instead
Hungarian notation (`strName`, `iCount`)	`product_name`, `line_count`

---

4. Primary Keys

4.1 Column Standard

Attribute	Standard
Column name	`id` — always, on every table
Data type	`BIGINT`
Generation	`GENERATED ALWAYS AS IDENTITY` (preferred over sequences)
Nullability	`NOT NULL`
Uniqueness	Single-column primary key only — no composite PKs

4.2 Constraint Naming

```
pk_{table_name}
```

`{table_name}` is the table name **without schema prefix**.

Table	PK Constraint Name
`catalog.products`	`pk_products`
`inventory.stock_positions`	`pk_stock_positions`
`audit.inventory_movement_records`	`pk_inventory_movement_records`

4.3 Rationale

- Surrogate `id` is stable across business key changes (`sku` format evolution).
- Narrow `BIGINT` PK optimizes FK indexes and joins.
- Single-column PK simplifies ORM mapping and partition key design.
- Business uniqueness enforced via **unique constraints**, not composite PKs.

---

5. Foreign Keys

5.1 Column Standard

See §3.3. Every FK column references parent `id` unless documented otherwise.

5.2 Constraint Naming

```
fk_{child_table}_{parent_entity_or_role}
```

Rule	Detail
Prefix	`fk_`
Child	Child table name (without schema)
Parent	Parent table singular noun or role qualifier
Length	Truncate parent qualifier if > 63 chars — keep child table name intact

5.3 Examples

Child Table	Column	FK Constraint Name
product_identifiers	product_id	fk_product_identifiers_product
stock_positions	product_id	fk_stock_positions_product
stock_positions	location_id	fk_stock_positions_location
products	created_by	fk_products_created_by
purchase_requisitions	triggering_alert_id	fk_purchase_requisitions_alert
audit_trail_records	movement_record_id	fk_audit_trail_movement

5.4 Self-Referencing FKs

```
fk_{table}_{relationship_role}
```

Table	Column	Constraint Name
inventory_locations	parent_location_id	fk_inventory_locations_parent

5.5 Cross-Schema FKs

Permitted and named identically. Schema qualification appears in DDL target, not constraint name:

```
fk_goods_receipts_purchase_order   → procurement.purchase_orders(id)
fk_stock_adjustments_reason_code   → governance.reason_codes(id)
```

5.6 Index Requirement

PostgreSQL does **not** auto-index FK columns. Create supporting index per SRS-DB-005 when column appears in JOIN/WHERE. Index name follows §7 — not the FK constraint name.

---

6. Constraints

6.1 Naming Pattern

```
{type_prefix}_{table_name}_{column_or_scope}
```

6.2 Type Prefixes

Constraint Type	Prefix	Example
Primary key	pk_	pk_products
Foreign key	fk_	fk_products_created_by
Unique	uk_	uk_products_sku
Check	chk_	chk_products_lifecycle_status
Not null	nn_	nn_products_sku
Default	def_	def_products_created_at

Note: PostgreSQL merges NOT NULL and DEFAULT into column definitions — `nn_` and `def_` names are logical/documentation identifiers used in SRS-DB-004; use in migration comments or extended property metadata.

6.3 Unique Constraint Naming

```
uk_{table}_{column}
uk_{table}_{col1}_{col2}        -- composite; omit redundant table prefix on columns
```

Table	Column(s)	Constraint Name
products	sku	uk_products_sku
stock_positions	(product_id, location_id)	uk_stock_positions_product_location
product_identifiers	(identifier_type, identifier_value)	uk_product_identifiers_type_value
supplier_invoices	(supplier_id, invoice_number)	uk_supplier_invoices_supplier_number
audit_trail_records	(audit_event_reference, event_timestamp)	uk_audit_trail_ref_ts

6.4 Check Constraint Naming

```
chk_{table}_{column_or_rule}
```

Scope	Pattern	Example
Single column domain	chk_{table}_{column}	chk_products_lifecycle_status
Cross-column rule	chk_{table}_{rule_short_name}	chk_stock_positions_atp_formula
Shared domain (reusable)	chk_domain_{domain_name}	chk_domain_approval_status

Shared domain checks — when identical value sets apply across tables, use `chk_domain_*` prefix and document in SRS-DB-004 Appendix. Table-level checks may alias the same rule.

6.5 Composite Constraint Scope Abbreviations

When column list exceeds name length, use semantic scope:

Scope Token	Meaning
product_location	(product_id, location_id)
type_value	(identifier_type, identifier_value)
ref_ts	(reference, timestamp) partition uniqueness
atp_formula	atp_qty = on_hand_qty - reserved_qty - quarantined_qty

---

7. Indexes

7.1 Naming Pattern

```
idx_{table}_{column_or_scope}[_{suffix}]
```

Rule	Detail
Prefix	idx_ — distinguishes from unique constraints (uk_)
Table	Child table name without schema
Columns	Leading column names joined by underscore; omit _id suffix on single FK when clear
Suffix	Optional qualifier for special index types

7.2 Index Type Suffixes

Index Type	Suffix	Example
B-tree (default)	*(none)*	`idx_reservations_stock_position`
Partial index	`_partial_{predicate_short}`	`idx_reservations_stock_position_partial_active`
BRIN	`_brin`	`idx_movement_records_brin`
GIN (JSONB)	`_gin`	`idx_audit_trail_before_state_gin`
DESC sort	`_desc` on column	`idx_movement_records_position_timestamp_desc`

7.3 Examples

Table	Index Columns	Index Name
`stock_positions`	`(location_id, stock_status)` partial LOW/OUT	`idx_stock_positions_location_status_partial_low`
`inventory_reservations`	`(stock_position_id)` WHERE ACTIVE	`idx_reservations_stock_position_partial_active`
`inventory_movement_records`	`(stock_position_id, movement_timestamp DESC)`	`idx_movement_records_position_timestamp_desc`
`inventory_movement_records`	`movement_timestamp` BRIN	`idx_movement_records_brin`
`goods_receipts`	`purchase_order_id`	`idx_goods_receipts_purchase_order`
`customer_orders`	`(order_status, order_date DESC)`	`idx_customer_orders_status_date_desc`

7.4 Rules

Rule	Detail
UK/PK indexes	**Do not** create duplicate `idx_` when `uk_` or `pk_` already provides the index
FK indexes	Name after FK column, not FK constraint name
Max length	Abbreviate column scope tokens (§6.5) before truncating table name
Partitioned tables	Index on parent propagates to partitions — name on parent only
One partial per predicate	Do not create both full and partial index on same columns

7.5 Strategy Document Cross-Reference

Logical index IDs in SRS-DB-005 (`IDX-001` – `IDX-052`) map to physical names during DDL generation:

```
IDX-004 → idx_reservations_stock_position_partial_active
IDX-005 → idx_lot_batches_position_expiration
```

---

8. Sequences

8.1 Preferred Approach — Identity Columns

IMS **does not use standalone sequences** for primary keys. All tables use:

```
id BIGINT GENERATED ALWAYS AS IDENTITY
```

Implicit sequence names are PostgreSQL-generated (`{table}_id_seq`) — **do not rename** unless required for replication tooling.

8.2 When Standalone Sequences Are Permitted

Use only for **business reference number generation** outside PK identity:

Use Case	Example
Document number pools	PO number series per location
External-facing reference ranges	Movement reference batch

8.3 Sequence Naming Pattern

```
seq_{purpose}_{scope}
```

Rule	Detail
Prefix	`seq_`
Purpose	Document or reference type
Scope	Optional location or year qualifier

8.4 Examples

Sequence Name	Purpose
`seq_movement_reference`	Global movement reference counter
`seq_po_number_wh01`	PO numbers scoped to location WH-01
`seq_audit_event_y2026`	Annual audit event reference partition

8.5 Business Reference Column Naming

Generated business references use `_reference` or domain-specific `_number` — not the sequence name:

Sequence	Populates Column
`seq_movement_reference`	`movement_reference`
`seq_po_number_wh01`	`po_number`

---

9. Views

9.1 Naming Pattern

```
vw_{purpose}_{entity_or_scope}
```

Rule	Detail
Prefix	`vw_` — distinguishes from tables
Purpose	`rpt_` segment optional for reporting — `vw_rpt_open_orders`
Materialized	Prefix `mvw_` — `mvw_dashboard_atp_summary`
No `v_`	Use `vw_` consistently (two-letter prefix improves readability)

9.2 Schema Placement

View Type	Schema	Example
Operational read models	Same schema as source or `audit`	`audit.vw_rpt_open_alerts`
Cross-domain reporting	`audit` (M-13 Intelligence)	`audit.vw_rpt_atp_by_location`
Auditor export (AUD-05)	`audit`	`audit.vw_export_audit_evidence`
Application convenience	Avoid — prefer explicit queries or ORM projections	

9.3 Examples

View Name	Purpose
`vw_rpt_stock_position_atp`	ATP summary with product and location names
`vw_rpt_open_purchase_orders`	Open PO dashboard
`vw_rpt_pending_approvals`	Cross-table approval queue
`vw_rpt_movement_history`	Movement log with denormalized product SKU
`mvw_rpt_low_stock_alerts`	Materialized low-stock dashboard (refreshed on schedule)
`vw_export_audit_trail`	Auditor evidence package source

9.4 Column Naming in Views

Rule	Detail
Inherit	Use source column names unchanged when exposing base columns
Computed	`snake_case` descriptive — `days_until_expiry`, `is_below_reorder`
Alias prefix	Do not prefix with `vw_` — `vw_atp_qty` is wrong; use `atp_qty`
Consistency	Same business term as base table — `sku`, not `product_sku_code`

9.5 View Object Comments

Every view must have a `COMMENT ON VIEW` describing purpose, refresh policy (if materialized), and source tables.

---

10. Additional Database Objects

10.1 Triggers

```
trg_{table}_{timing}_{event}_{purpose}
```

Example	Event
`trg_stock_policies_after_update_history`	Append `stock_policy_history` on policy change
`trg_audit_trail_records_before_update_block`	Prevent UPDATE on audit tables
`trg_stock_positions_before_update_timestamp`	Maintain `updated_at`

Timing: `before` / `after`. Event: `insert` / `update` / `delete`.

10.2 Functions and Procedures

```
fn_{verb}_{entity}          -- returns scalar or set
sp_{verb}_{entity}          -- procedure (side effects)
```

Example	Purpose
`fn_calculate_atp`	ATP derivation
`sp_archive_movement_partition`	Partition detach and archive

10.3 Database Roles

```
ims_{module}_{access_level}
```

Example	Purpose
`ims_app_readwrite`	Application DML role
`ims_audit_writer`	INSERT-only on audit tables
`ims_reporting_read`	SELECT-only for BI
`ims_migration_admin`	DDL migration role

10.4 Enum Types (If Used)

Prefer `VARCHAR(32)` + CHECK constraints in IMS. If native PostgreSQL enums are introduced:

```
{schema}.{domain}_enum
```

Example: `inventory.ownership_type_enum` — use only when value set is truly immutable.

---

11. Scalability Guidelines

11.1 Adding New Tables

1. Assign to correct **schema** (bounded context).
2. Use plural **table name** ≤ 40 characters.
3. Include `id BIGINT GENERATED ALWAYS AS IDENTITY` as PK.
4. Apply standard audit columns per table tier (SRS-DB-006).
5. Name constraints and indexes before first deployment — renaming in production is costly.

11.2 Adding New Columns

1. Follow suffix conventions (§3.1).
2. FK columns: `{entity}_id` referencing parent `id`.
3. Add `nn_`, `chk_`, `fk_` constraints with names per §6.
4. Add index per §7 if column is filtered or joined frequently.
5. Update SRS-DB-007 Data Dictionary.

11.3 Name Collision Resolution

Situation	Resolution
Same entity name in two schemas	Schema qualifies — `catalog.products` vs external `erp.products` (avoid — use `erp_product_id` FK column instead)
Long composite UK name	Abbreviate scope tokens (§6.5), not table name
Index name > 63 chars	Drop `_id` suffixes from column tokens first, then abbreviate scope

11.4 Prohibited Practices

Practice	Reason
Hungarian notation	Reduces readability for BAs and developers
Single-letter column names	Not self-documenting
Duplicate indexes (UK + idx)	Wastes write performance and storage
Renaming `id` to `{table}_id` on PK	Breaks ORM and convention consistency
Mixed case identifiers	PostgreSQL folds unquoted identifiers to lowercase — use snake_case explicitly
Quotes in identifiers	Requires quoted SQL everywhere — avoid

Appendix A: Approved Abbreviations

Use only when necessary for length or established domain usage:

Abbreviation	Full Term	Context
`qty`	quantity	Column suffix
`po`	purchase order	`po_number`, `po_status`
`gr`	goods receipt	`gr_number`
`rma`	return merchandise authorization	`rma_number`
`atp`	available to promise	`atp_qty`
`mrsl`	minimum remaining shelf life	`mrsl_days`
`fefo`	first expired first out	Domain values, not object names
`fifo`	first in first out	Domain values
`sod`	segregation of duties	`sod_compliant`, `has_sod_conflict`
`ref`	reference	Suffix only — `approval_chain_ref`
`desc`	**Never** — use `description`	—

Appendix B: Quick Reference Card

Object	Pattern	Example
Schema	`{context}`	`inventory`
Table	`{entities_plural}`	`stock_positions`
Column	`{attribute}` / `{entity}_id`	`on_hand_qty`, `product_id`
Primary key	`id` / `pk_{table}`	`pk_stock_positions`
Foreign key	`fk_{child}_{parent}`	`fk_stock_positions_product`
Unique	`uk_{table}_{cols}`	`uk_products_sku`
Check	`chk_{table}_{col}`	`chk_products_lifecycle_status`
Index	`idx_{table}_{cols}`	`idx_goods_receipts_purchase_order`
Partial index	`idx_{table}_{cols}_partial_{pred}`	`idx_reservations_stock_position_partial_active`
Sequence	`seq_{purpose}`	`seq_movement_reference`
View	`vw_{purpose}_{entity}`	`vw_rpt_open_orders`
Materialized view	`mvw_{purpose}_{entity}`	`mvw_rpt_low_stock_alerts`
Partition	`{table}_y{YYYY}m{MM}`	`inventory_movement_records_y2026m07`
Trigger	`trg_{table}_{timing}_{event}_{purpose}`	`trg_audit_trail_records_before_update_block`
Role	`ims_{scope}_{access}`	`ims_audit_writer`

Appendix C: Cross-Document Alignment

SRS Document	Naming Content
SRS-DB-002 (PDM)	Table and column names — canonical examples
SRS-DB-003 (Keys)	`pk_`, `uk_`, `fk_` logical constraint names

SRS Document	Naming Content
SRS-DB-004 (Constraints)	`nn_`, `uk_`, `chk_`, `fk_`, `def_` prefixes
SRS-DB-005 (Indexing)	`idx_` patterns; IDX-NNN logical IDs
SRS-DB-006 (Audit)	Audit table and column standards
SRS-DB-007 (Data Dictionary)	Business meanings — updated when names change
SRS-DB-008 (this document)	Authoritative naming standard

Document Approval

Role	Name	Signature	Date
Database Architect			
Database Performance Specialist			
Solution Architect			
Development Lead			

End of Database Naming Conventions

Document ID: SRS-DB-008 | Version 1.0 | July 6, 2026

Part VIII — API Design

REST API Catalog

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-API-001
Document Title	REST API Catalog — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Senior Backend Architect
Source Documents	BRD-IMS-001 (v1.0), SRS-UC-001 (Use Case Catalog), SRS-DM-003 (Domain Aggregates), SRS-ES-002 (Domain Commands), SRS-ES-005 (Read Models), SRS-SA-003 (System Modules Design), SRS-SA-005 (Integration Architecture Design), SRS-DB (Database Design)
Date	July 6, 2026
Scope	REST API identification only — endpoints, methods, purpose, and traceability. Request/response structures are out of scope.

Purpose

This catalog identifies **all REST APIs** required to support the Inventory Management System. Each API is traced to a business module, use case, and domain aggregate derived from approved business and architecture artifacts. APIs align with the 13 logical system modules (M-01–M-13) and expose the published commands and queries defined in the System Modules Design.

API Conventions

Convention	Rule
Base path	`/api/v1`
Resource naming	Plural nouns, kebab-case path segments
Identifiers	`ProductSku`, `LocationCode`, and other published-language identifiers in path or query
Command actions	`POST` to sub-resource action paths (e.g., `/orders/{id}/confirm`)
Queries	`GET` with filter query parameters
Idempotency	Safe retries on `PUT` and idempotent `POST` where saga compensation applies
Authorization	All endpoints require authenticated session (SVC-01 IdP); location-scoped RBAC enforced per UC-045

Traceability Legend

Reference	Meaning
UC-xxx	Use Case (SRS-UC-001)
M-xx	Business Module (SRS-SA-003)
CMD-xxx	Domain Command (SRS-ES-002) — write API origin
RM-xxx	Read Model (SRS-ES-005) — query API origin

API Catalog Summary

Business Module	API ID Range	Count
M-01 Product Catalog	API-001 – API-010	10
M-02 Warehouse Network	API-011 – API-021	11
M-03 Inventory Governance	API-022 – API-031	10
M-04 Inventory Balance	API-032 – API-048	17
M-05 Procurement & Supplier	API-049 – API-065	17
M-06 Inbound Receiving	API-066 – API-080	15
M-07 Order Fulfillment	API-081 – API-097	17
M-08 Network Transfer	API-098 – API-107	10
M-09 Inventory Accuracy	API-108 – API-123	16
M-10 Returns Management	API-124 – API-131	8
M-11 Commercial Campaign	API-132 – API-139	8
M-12 Identity & Access	API-140 – API-149	10
M-13 Inventory Intelligence	API-150 – API-172	23
Platform & Integration	API-173 – API-178	6
Total		**178**

M-01 — Product Catalog

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-001	Product Catalog	`/api/v1/products`	POST	Register a new product SKU with mandatory catalog attributes	UC-001, UC-046	Product Catalog
API-002	Product Catalog	`/api/v1/products`	GET	Search and filter products by name, category, status, and ABC class	UC-001, UC-013	Product Catalog
API-003	Product Catalog	`/api/v1/products/{sku}`	GET	Retrieve product master record by SKU	UC-001, UC-013	Product Catalog
API-004	Product Catalog	`/api/v1/products/{sku}`	PATCH	Update product attributes (name, category, UoM, traceability settings)	UC-001	Product Catalog
API-005	Product Catalog	`/api/v1/products/{sku}/lifecycle-status`	POST	Change product lifecycle status (active, inactive, discontinued)	UC-001	Product Catalog
API-006	Product Catalog	`/api/v1/products/{sku}/abc-classification`	PUT	Assign or update ABC operational priority classification	UC-004	Product Catalog
API-007	Product Catalog	`/api/v1/products/{sku}/eligibility`	GET	Validate whether product is eligible for procurement, reservation, or fulfillment	UC-006, UC-021, UC-022	Product Catalog
API-008	Product Catalog	`/api/v1/products/categories/{categoryCode}/products`	GET	List products belonging to a category for catalog browse	UC-001	Product Catalog
API-009	Product Catalog	`/api/v1/products/duplicate-resolutions`	POST	Resolve duplicate product identity conflict (merge or reject)	UC-001	Product Catalog
API-010	Product Catalog	`/api/v1/products/import`	POST	Bulk import product master data at location or system go-live	UC-046	Product Catalog

M-02 — Warehouse Network

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-011	Warehouse Network	`/api/v1/locations`	POST	Register a new inventory location (warehouse, store, distribution point)	UC-002	Inventory Location
API-012	Warehouse Network	`/api/v1/locations`	GET	List active locations in the inventory network	UC-002, UC-012, UC-013	Inventory Location
API-013	Warehouse Network	`/api/v1/locations/{locationCode}`	GET	Retrieve location master, hierarchy, and operational status	UC-002	Inventory Location
API-014	Warehouse Network	`/api/v1/locations/{locationCode}/hierarchy`	PATCH	Update location hierarchy (parent assignment, zones, layout)	UC-002	Inventory Location

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-015	Warehouse Network	`/api/v1/locations/{locationCode}/storage-positions`	POST	Configure or update a storage position (zone, aisle, bin) within a location	UC-002, UC-010	Inventory Location
API-016	Warehouse Network	`/api/v1/locations/{locationCode}/storage-positions`	GET	List storage positions available for put-away and pick assignment	UC-010	Inventory Location
API-017	Warehouse Network	`/api/v1/locations/{locationCode}/activate`	POST	Activate location for inventory operations after go-live readiness	UC-002, UC-046	Inventory Location
API-018	Warehouse Network	`/api/v1/locations/{locationCode}/deactivate`	POST	Deactivate location and begin closure workflow	UC-002	Inventory Location
API-019	Warehouse Network	`/api/v1/locations/{locationCode}/close`	POST	Finalize location closure after stock disposition confirmation	UC-002	Inventory Location
API-020	Warehouse Network	`/api/v1/locations/{locationCode}/freeze-status`	GET	Check whether location is under physical inventory freeze	UC-015	Inventory Location
API-021	Warehouse Network	`/api/v1/locations/{locationCode}/operational-status`	GET	Check whether location accepts transactional inventory activity	UC-002	Inventory Location

M-03 — Inventory Governance

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-022	Inventory Governance	`/api/v1/governance/reason-codes`	POST	Create a standardized reason code for adjustments and exceptions	UC-005	Inventory Governance
API-023	Inventory Governance	`/api/v1/governance/reason-codes`	GET	List active reason codes selectable for transactions	UC-005, UC-017, UC-009	Inventory Governance
API-024	Inventory Governance	`/api/v1/governance/reason-codes/{code}/retire`	POST	Retire a reason code from active use while preserving history	UC-005	Inventory Governance
API-025	Inventory Governance	`/api/v1/governance/reason-codes/{code}/validate`	GET	Validate reason code is active and applicable to a transaction type	UC-017, UC-009	Inventory Governance
API-026	Inventory Governance	`/api/v1/governance/policies`	POST	Publish a new governance policy version with effective date	UC-005	Inventory Governance
API-027	Inventory Governance	`/api/v1/governance/policies/{policyId}/approval-thresholds`	PATCH	Update financial or quantity approval thresholds	UC-005	Inventory Governance
API-028	Inventory Governance	`/api/v1/governance/policies/{policyId}/receipt-tolerances`	PATCH	Update receipt quantity tolerance limits for auto-accept vs. discrepancy	UC-005, UC-009	Inventory Governance
API-029	Inventory Governance	`/api/v1/governance/approval-thresholds`	GET	Retrieve effective approval threshold for a transaction context	UC-017, UC-018	Inventory Governance

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-030	Inventory Governance	`/api/v1/governance/receipt-tolerances`	GET	Retrieve effective receipt tolerance for discrepancy evaluation	UC-008, UC-009	Inventory Governance
API-031	Inventory Governance	`/api/v1/governance/segregation-of-duties/evaluate`	POST	Evaluate segregation-of-duties compliance between initiator and approver	UC-045	Inventory Governance

M-04 — Inventory Balance

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-032	Inventory Balance	`/api/v1/inventory/opening-balances`	POST	Establish verified opening on-hand balances at location go-live	UC-046	Stock Position
API-033	Inventory Balance	`/api/v1/inventory/stock-positions`	GET	Retrieve on-hand, reserved, and ATP for product at location	UC-012, UC-013	Stock Position
API-034	Inventory Balance	`/api/v1/inventory/stock-positions/{locationCode}/{sku}`	GET	Retrieve stock position detail for a specific product-location pair	UC-012, UC-013	Stock Position
API-035	Inventory Balance	`/api/v1/inventory/availability/{sku}`	GET	Retrieve ATP for a SKU at a location (hot-path availability inquiry)	UC-013, UC-021	Stock Position
API-036	Inventory Balance	`/api/v1/inventory/availability/{sku}/network`	GET	Retrieve network-wide ATP rollup across all locations	UC-013, UC-021, UC-034	Stock Position
API-037	Inventory Balance	`/api/v1/inventory/stock-status/{locationCode}/{sku}`	GET	Retrieve derived stock status indicator (in stock, low, out, overstock)	UC-012, UC-028	Stock Position
API-038	Inventory Balance	`/api/v1/inventory/stock-policies`	POST	Define replenishment thresholds for a product at a location	UC-003	Stock Policy

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-039	Inventory Balance	`/api/v1/inventory/stock-policies/{locationCode}/{sku}`	GET	Retrieve stock policy thresholds for a product-location pair	UC-003, UC-028	Stock Policy
API-040	Inventory Balance	`/api/v1/inventory/stock-policies/{locationCode}/{sku}`	PATCH	Update stock policy thresholds with effective dating	UC-003	Stock Policy
API-041	Inventory Balance	`/api/v1/inventory/stock-positions/{locationCode}/{sku}/lots`	GET	Retrieve lot/batch-level quantity breakdown for traceable products	UC-024, UC-033	Stock Position
API-042	Inventory Balance	`/api/v1/inventory/stock-positions/{locationCode}/{sku}/transfer-capacity`	GET	Evaluate spare quantity above source minimum available for transfer	UC-031, UC-032	Stock Position, Stock Policy
API-043	Inventory Balance	`/api/v1/inventory/stock-positions/{locationCode}/{sku}/reservations`	POST	Reserve stock and reduce ATP for a confirmed order line	UC-022	Stock Position
API-044	Inventory Balance	`/api/v1/inventory/stock-positions/{locationCode}/{sku}/reservations/{reservationId}/release`	POST	Release reserved stock and restore ATP on order cancellation	UC-023	Stock Position
API-045	Inventory Balance	`/api/v1/inventory/stock-positions/{locationCode}/{sku}/quarantine-holds`	POST	Apply quarantine hold and exclude quantity from sellable ATP	UC-019	Stock Position
API-046	Inventory Balance	`/api/v1/inventory/stock-positions/{locationCode}/{sku}/quarantine-holds/{holdId}/release`	POST	Release quarantine hold and restore quantity to sellable ATP	UC-019	Stock Position
API-047	Inventory Balance	`/api/v1/inventory/stock-positions/{locationCode}/{sku}/consignment-ownership`	POST	Transfer consignment ownership to organization-owned upon consumption	UC-039	Stock Position

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-048	Inventory Balance	`/api/v1/inventory/stock-positions/{locationCode}/{sku}/promotional-segregation`	POST	Segregate ATP for promotional ring-fence (event-driven from campaign)	UC-040	Stock Position

M-05 — Procurement & Supplier

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-049	Procurement & Supplier	`/api/v1/suppliers`	POST	Register a new supplier master record	UC-006	Supplier
API-050	Procurement & Supplier	`/api/v1/suppliers`	GET	List suppliers with optional status filter	UC-006, UC-029	Supplier
API-051	Procurement & Supplier	`/api/v1/suppliers/{supplierCode}`	GET	Retrieve supplier master, terms, and consignment indicator	UC-006	Supplier
API-052	Procurement & Supplier	`/api/v1/suppliers/{supplierCode}`	PATCH	Update supplier payment terms, lead time, and consignment flag	UC-006	Supplier
API-053	Procurement & Supplier	`/api/v1/suppliers/{supplierCode}/deactivate`	POST	Deactivate supplier and block new purchase order creation	UC-006	Supplier
API-054	Procurement & Supplier	`/api/v1/procurement/requisitions`	POST	Create internal purchase requisition for replenishment need	UC-029	Purchase Requisition
API-055	Procurement & Supplier	`/api/v1/procurement/requisitions/{requisitionId}`	GET	Retrieve purchase requisition status and line detail	UC-029	Purchase Requisition
API-056	Procurement & Supplier	`/api/v1/procurement/requisitions/{requisitionId}/approve`	POST	Approve purchase requisition after network availability validation	UC-029	Purchase Requisition
API-057	Procurement & Supplier	`/api/v1/procurement/requisitions/{requisitionId}/reject`	POST	Reject purchase requisition with documented reason	UC-029	Purchase Requisition

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-058	Procurement & Supplier	`/api/v1/procurement/requisitions/{requisitionId}/convert-to-po`	POST	Convert approved requisition into a purchase order	UC-029, UC-006	Purchase Requisition, Purchase Order
API-059	Procurement & Supplier	`/api/v1/procurement/purchase-orders`	POST	Place purchase order as authorized commercial commitment to supplier	UC-006	Purchase Order
API-060	Procurement & Supplier	`/api/v1/procurement/purchase-orders`	GET	List open purchase orders for inbound pipeline visibility	UC-006, UC-030	Purchase Order
API-061	Procurement & Supplier	`/api/v1/procurement/purchase-orders/{purchaseOrderId}`	GET	Retrieve purchase order header, lines, and received quantities	UC-006, UC-008, UC-030	Purchase Order
API-062	Procurement & Supplier	`/api/v1/procurement/purchase-orders/{purchaseOrderId}`	PATCH	Amend purchase order lines or delivery schedule	UC-006	Purchase Order
API-063	Procurement & Supplier	`/api/v1/procurement/purchase-orders/{purchaseOrderId}/cancel`	POST	Cancel purchase order with receipt implications	UC-006	Purchase Order
API-064	Procurement & Supplier	`/api/v1/procurement/purchase-orders/{purchaseOrderId}/expected-deliveries`	POST	Schedule or update expected inbound delivery dates	UC-007, UC-030	Purchase Order
API-065	Procurement & Supplier	`/api/v1/procurement/purchase-orders/{purchaseOrderId}/validate-for-receipt`	GET	Validate purchase order authorizes goods receipt (BR-007)	UC-008	Purchase Order

M-06 — Inbound Receiving

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-066	Inbound Receiving	`/api/v1/inbound/goods-receipts`	POST	Initiate goods receipt against purchase order or ad-hoc authorization	UC-008, UC-011	Goods Receipt
API-067	Inbound Receiving	`/api/v1/inbound/goods-receipts`	GET	List goods receipts with status and location filters	UC-008, UC-010	Goods Receipt
API-068	Inbound Receiving	`/api/v1/inbound/goods-receipts/{goodsReceiptId}`	GET	Retrieve goods receipt status, lines, and put-away state	UC-008, UC-010	Goods Receipt

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-069	Inbound Receiving	`/api/v1/inbound/goods-receipts/{goodsReceiptId}/verify`	POST	Verify inbound quantity, product identity, and condition at dock	UC-008, UC-011	Goods Receipt
API-070	Inbound Receiving	`/api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies`	POST	Document receipt discrepancy with classification and evidence	UC-009	Goods Receipt
API-071	Inbound Receiving	`/api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies`	GET	List open discrepancies for a goods receipt	UC-009	Goods Receipt
API-072	Inbound Receiving	`/api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies/{discrepancyId}/resolve`	POST	Resolve receipt discrepancy with documented outcome	UC-009	Goods Receipt
API-073	Inbound Receiving	`/api/v1/inbound/goods-receipts/{goodsReceiptId}/put-away`	POST	Confirm put-away to storage position, completing inbound workflow	UC-010	Goods Receipt
API-074	Inbound Receiving	`/api/v1/inbound/pending-put-away`	GET	List receipts awaiting storage placement at a location	UC-010	Goods Receipt
API-075	Inbound Receiving	`/api/v1/inbound/ad-hoc-receipts`	POST	Record non-PO inbound receipt with governance authorization	UC-011	Goods Receipt
API-076	Inbound Receiving	`/api/v1/inbound/return-receipts`	POST	Accept physical customer return shipment at receiving dock	UC-011, UC-036	Goods Receipt, Customer Return Case
API-077	Inbound Receiving	`/api/v1/inbound/supplier-invoices`	POST	Register supplier billing document for financial control	UC-039	Supplier Invoice
API-078	Inbound Receiving	`/api/v1/inbound/supplier-invoices/{invoiceId}`	GET	Retrieve supplier invoice and three-way match status	UC-039	Supplier Invoice

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-079	Inbound Receiving	`/api/v1/inbound/supplier-invoices/{invoiceId}/three-way-match`	POST	Execute three-way match evaluation against PO and goods receipt	UC-039	Supplier Invoice
API-080	Inbound Receiving	`/api/v1/inbound/supplier-invoices/{invoiceId}/authorize-payment`	POST	Authorize invoice payment after successful three-way match	UC-039	Supplier Invoice

M-07 — Order Fulfillment

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-081	Order Fulfillment	`/api/v1/fulfillment/orders`	POST	Import or create customer order for fulfillment processing	UC-021, UC-022	Customer Order Fulfillment
API-082	Order Fulfillment	`/api/v1/fulfillment/orders`	GET	List customer orders with status and location filters	UC-027	Customer Order Fulfillment
API-083	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}`	GET	Retrieve customer order detail and line breakdown	UC-027	Customer Order Fulfillment
API-084	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/availability`	POST	Confirm order availability using ATP before commercial commitment	UC-013, UC-021	Customer Order Fulfillment
API-085	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/confirm`	POST	Confirm customer order and initiate stock reservation saga	UC-022	Customer Order Fulfillment
API-086	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/cancel`	POST	Cancel customer order and release reserved stock	UC-023	Customer Order Fulfillment
API-087	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/multi-location-plan`	POST	Create multi-location fulfillment plan when single location ATP is insufficient	UC-048	Customer Order Fulfillment
API-088	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/pick-lists`	POST	Assign pick work to warehouse for order fulfillment	UC-024	Customer Order Fulfillment
API-089	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/pick-lists/awaiting`	GET	List orders awaiting pick at a fulfillment location	UC-024	Customer Order Fulfillment
API-090	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/picks/{pickId}/verify`	POST	Verify physical pick matches order requirements and capture lot/batch	UC-024	Customer Order Fulfillment

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-091	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/pack`	POST	Complete packing step for verified pick	UC-024	Customer Order Fulfillment
API-092	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/shipments`	POST	Dispatch shipment and trigger inventory deduction for picked goods	UC-024	Customer Order Fulfillment
API-093	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/partial-fulfillment`	POST	Fulfill available quantity and route remainder to backorder workflow	UC-025	Customer Order Fulfillment
API-094	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/backorders`	POST	Record backorder with customer acknowledgment for unfulfilled quantity	UC-025	Customer Order Fulfillment
API-095	Order Fulfillment	`/api/v1/fulfillment/backorders/{backorderId}/fulfill`	POST	Fulfill previously backordered quantity when stock becomes available	UC-025	Customer Order Fulfillment
API-096	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/exceptions`	POST	Raise and document outbound fulfillment exception for manager resolution	UC-026	Customer Order Fulfillment
API-097	Order Fulfillment	`/api/v1/fulfillment/orders/{orderId}/status`	GET	Retrieve order fulfillment status (reservation, pick, ship, exception)	UC-027	Customer Order Fulfillment

M-08 — Network Transfer

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-098	Network Transfer	`/api/v1/transfers`	POST	Request inter-location stock transfer to resolve local shortage	UC-031	Stock Transfer
API-099	Network Transfer	`/api/v1/transfers`	GET	List stock transfers with status and location filters	UC-031, UC-033	Stock Transfer
API-100	Network Transfer	`/api/v1/transfers/{transferId}`	GET	Retrieve transfer request status, lines, and in-transit state	UC-031, UC-033	Stock Transfer
API-101	Network Transfer	`/api/v1/transfers/{transferId}/approve`	POST	Approve transfer request after source capacity validation	UC-032	Stock Transfer
API-102	Network Transfer	`/api/v1/transfers/{transferId}/reject`	POST	Reject transfer request with documented reason	UC-032	Stock Transfer
API-103	Network Transfer	`/api/v1/transfers/{transferId}/dispatch`	POST	Dispatch stock from source location and create in-transit state	UC-033	Stock Transfer
API-104	Network Transfer	`/api/v1/transfers/{transferId}/receive`	POST	Confirm receipt of transferred stock at destination location	UC-033	Stock Transfer
API-105	Network Transfer	`/api/v1/transfers/{transferId}/cancel`	POST	Cancel in-transit transfer with reversal workflow	UC-033	Stock Transfer
API-106	Network Transfer	`/api/v1/transfers/pending-approvals`	GET	List transfer requests awaiting management approval	UC-032, UC-047	Stock Transfer

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-107	Network Transfer	`/api/v1/transfers/in-transit`	GET	List in-transit stock pipeline by location	UC-030, UC-033	Stock Transfer

M-09 — Inventory Accuracy

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-108	Inventory Accuracy	`/api/v1/accuracy/cycle-counts`	POST	Schedule cycle count assignment for product at location	UC-014	Cycle Count
API-109	Inventory Accuracy	`/api/v1/accuracy/cycle-counts`	GET	List cycle count assignments by location and status	UC-014	Cycle Count
API-110	Inventory Accuracy	`/api/v1/accuracy/cycle-counts/{cycleCountId}`	GET	Retrieve cycle count assignment without revealing system quantity (blind count)	UC-014	Cycle Count
API-111	Inventory Accuracy	`/api/v1/accuracy/cycle-counts/{cycleCountId}/result`	POST	Submit blind cycle count physical result	UC-014	Cycle Count
API-112	Inventory Accuracy	`/api/v1/accuracy/physical-counts`	POST	Initiate wall-to-wall physical inventory count and impose location freeze	UC-015	Physical Inventory Count
API-113	Inventory Accuracy	`/api/v1/accuracy/physical-counts/{physicalCountId}`	GET	Retrieve physical count progress and freeze window status	UC-015	Physical Inventory Count
API-114	Inventory Accuracy	`/api/v1/accuracy/physical-counts/{physicalCountId}/results`	POST	Submit line-level count result during freeze period	UC-015	Physical Inventory Count
API-115	Inventory Accuracy	`/api/v1/accuracy/physical-counts/{physicalCountId}/certify`	POST	Certify physical count complete and release location freeze	UC-015	Physical Inventory Count
API-116	Inventory Accuracy	`/api/v1/accuracy/adjustment-cases`	POST	Open adjustment case from count variance or direct adjustment request	UC-016, UC-017	Inventory Adjustment Case
API-117	Inventory Accuracy	`/api/v1/accuracy/adjustment-cases/{caseId}`	GET	Retrieve adjustment case investigation and approval state	UC-016, UC-017	Inventory Adjustment Case
API-118	Inventory Accuracy	`/api/v1/accuracy/adjustment-cases/{caseId}/investigate`	POST	Document variance root cause investigation findings	UC-016	Inventory Adjustment Case
API-119	Inventory Accuracy	`/api/v1/accuracy/adjustment-cases/{caseId}/variances/{varianceId}/approve`	POST	Approve count variance for adjustment with independent approver	UC-016	Inventory Adjustment Case
API-120	Inventory Accuracy	`/api/v1/accuracy/adjustment-cases/{caseId}/submit`	POST	Submit adjustment or write-off for governance approval	UC-017, UC-018	Inventory Adjustment Case
API-121	Inventory Accuracy	`/api/v1/accuracy/adjustment-cases/{caseId}/approve`	POST	Approve inventory adjustment or write-off after threshold validation	UC-017, UC-018	Inventory Adjustment Case

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-122	Inventory Accuracy	`/api/v1/accuracy/adjustment-cases/{caseId}/reject`	POST	Reject proposed adjustment with documented reason	UC-017, UC-018	Inventory Adjustment Case
API-123	Inventory Accuracy	`/api/v1/accuracy/pending-approvals`	GET	List adjustment cases awaiting independent approval	UC-016, UC-017, UC-047	Inventory Adjustment Case

M-10 — Returns Management

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-124	Returns Management	`/api/v1/returns/authorizations`	POST	Issue return authorization (RMA) linked to original customer order	UC-035	Customer Return Case
API-125	Returns Management	`/api/v1/returns/authorizations`	GET	List return authorizations with status filter	UC-035, UC-036	Customer Return Case
API-126	Returns Management	`/api/v1/returns/authorizations/{returnAuthorizationId}`	GET	Retrieve RMA status, eligibility, and linked order reference	UC-035	Customer Return Case
API-127	Returns Management	`/api/v1/returns/authorizations/{returnAuthorizationId}/cancel`	POST	Cancel unused return authorization	UC-035	Customer Return Case
API-128	Returns Management	`/api/v1/returns/{returnId}/receive`	POST	Record physical receipt and verification of authorized return goods	UC-036	Customer Return Case
API-129	Returns Management	`/api/v1/returns/{returnId}/inspect`	POST	Inspect returned goods and record condition grade	UC-036	Customer Return Case
API-130	Returns Management	`/api/v1/returns/{returnId}/disposition`	POST	Route return to restock, quarantine, or write-off disposition	UC-036	Customer Return Case
API-131	Returns Management	`/api/v1/returns/pending-receipts`	GET	List authorized returns awaiting warehouse receipt	UC-036	Customer Return Case

M-11 — Commercial Campaign

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-132	Commercial Campaign	`/api/v1/campaigns/allocations`	POST	Create promotional stock allocation and ring-fence quantity	UC-040	Promotional Stock Allocation
API-133	Commercial Campaign	`/api/v1/campaigns/allocations`	GET	List promotional allocations by campaign period and status	UC-040	Promotional Stock Allocation
API-134	Commercial Campaign	`/api/v1/campaigns/allocations/{allocationId}`	GET	Retrieve promotional allocation status and ring-fenced quantity	UC-040	Promotional Stock Allocation
API-135	Commercial Campaign	`/api/v1/campaigns/allocations/{allocationId}`	PATCH	Amend allocated quantity or campaign period	UC-040	Promotional Stock Allocation
API-136	Commercial Campaign	`/api/v1/campaigns/allocations/{allocationId}/readiness`	POST	Confirm or evaluate campaign stock readiness before launch	UC-040, UC-047	Promotional Stock Allocation
API-137	Commercial Campaign	`/api/v1/campaigns/allocations/{allocationId}/release`	POST	Release unconsumed promotional allocation after campaign end	UC-040	Promotional Stock Allocation
API-138	Commercial Campaign	`/api/v1/campaigns/allocations/{allocationId}/override`	POST	Authorize non-promotional consumption of ring-fenced stock	UC-040	Promotional Stock Allocation
API-139	Commercial Campaign	`/api/v1/campaigns/ring-fence/{locationCode}/{sku}`	GET	Check whether SKU has active promotional ring-fence at location	UC-022, UC-040	Promotional Stock Allocation

M-12 — Identity & Access

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-140	Identity & Access	`/api/v1/users`	POST	Provision new user account with initial role assignments	UC-044	User Account
API-141	Identity & Access	`/api/v1/users`	GET	List user accounts with role and status filters	UC-044	User Account
API-142	Identity & Access	`/api/v1/users/{userId}`	GET	Retrieve user profile and role assignments	UC-044, UC-045	User Account
API-143	Identity & Access	`/api/v1/users/{userId}/roles`	POST	Assign role with location scope to user account	UC-044, UC-045	User Account
API-144	Identity & Access	`/api/v1/users/{userId}/roles/{roleAssignmentId}`	DELETE	Revoke role assignment from user account	UC-044	User Account
API-145	Identity & Access	`/api/v1/users/{userId}/suspend`	POST	Temporarily suspend user account access	UC-044	User Account

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-146	Identity & Access	`/api/v1/users/{userId}/deactivate`	POST	Permanently deactivate user account	UC-044	User Account
API-147	Identity & Access	`/api/v1/users/sync`	POST	Synchronize user identities from enterprise HR directory	UC-044	User Account
API-148	Identity & Access	`/api/v1/authorization/actions/evaluate`	POST	Evaluate whether authenticated user may execute a command	UC-045	User Account
API-149	Identity & Access	`/api/v1/authorization/location-scope/validate`	POST	Validate user is authorized for a specific location scope	UC-045	User Account

M-13 — Inventory Intelligence

Read-only APIs serving dashboards, reports, movement history, audit trails, alerts, and valuation projections. No aggregate mutation.

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-150	Inventory Intelligence	`/api/v1/intelligence/dashboard`	GET	Retrieve inventory summary dashboard by location or network	UC-037	*(Read projection — no aggregate)*
API-151	Inventory Intelligence	`/api/v1/intelligence/stock-status/summary`	GET	Retrieve aggregated on-hand, ATP, and status counts	UC-012, UC-037	*(Read projection)*
API-152	Inventory Intelligence	`/api/v1/intelligence/stock-status/board`	GET	Retrieve network stock status board with health indicators	UC-012, UC-037	Stock Position *(projection)*
API-153	Inventory Intelligence	`/api/v1/intelligence/availability/inquiries`	GET	Retrieve availability inquiry log for sales accountability	UC-013, UC-021	Customer Order Fulfillment *(projection)*
API-154	Inventory Intelligence	`/api/v1/intelligence/quarantine-register`	GET	Retrieve quarantine and hold register across locations	UC-019, UC-037	Stock Position *(projection)*
API-155	Inventory Intelligence	`/api/v1/intelligence/movements`	GET	Search inventory movement history by SKU, location, user, and date	UC-042	*(Inventory Movement Record projection)*
API-156	Inventory Intelligence	`/api/v1/intelligence/audit-trail`	GET	Query immutable audit trail of material inventory changes	UC-043, UC-041	*(Audit Trail Record projection)*
API-157	Inventory Intelligence	`/api/v1/intelligence/audit-trail/export`	GET	Export audit evidence package for internal or external audit	UC-041, UC-043	*(Audit Trail Record projection)*
API-158	Inventory Intelligence	`/api/v1/intelligence/alerts`	GET	Retrieve active operational alerts for user and location scope	UC-028, UC-047	*(Operational Alert projection)*
API-159	Inventory Intelligence	`/api/v1/intelligence/alerts/{alertId}/acknowledge`	POST	Acknowledge operational alert without mutating inventory state	UC-028, UC-047	*(Operational Alert projection)*

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-160	Inventory Intelligence	`/api/v1/intelligence/replenishment/recommendations`	GET	Retrieve replenishment recommendation log	UC-028, UC-029, UC-034	*(Read projection)*
API-161	Inventory Intelligence	`/api/v1/intelligence/procurement/open-pos`	GET	Retrieve open purchase order board for inbound pipeline	UC-006, UC-030	Purchase Order *(projection)*
API-162	Inventory Intelligence	`/api/v1/intelligence/procurement/inbound-calendar`	GET	Retrieve expected inbound supply calendar	UC-007, UC-030	Purchase Order *(projection)*
API-163	Inventory Intelligence	`/api/v1/intelligence/procurement/inbound-delays`	GET	Retrieve inbound delivery delay log	UC-030, UC-047	Purchase Order *(projection)*
API-164	Inventory Intelligence	`/api/v1/intelligence/fulfillment/backorders`	GET	Retrieve backorder pipeline sorted by policy priority	UC-025, UC-027	Customer Order Fulfillment *(projection)*
API-165	Inventory Intelligence	`/api/v1/intelligence/fulfillment/exceptions`	GET	Retrieve open fulfillment exception log	UC-026, UC-027	Customer Order Fulfillment *(projection)*
API-166	Inventory Intelligence	`/api/v1/intelligence/network/imbalance`	GET	Retrieve network imbalance analysis and transfer recommendations	UC-034	*(Read projection)*
API-167	Inventory Intelligence	`/api/v1/intelligence/accuracy/shrinkage`	GET	Retrieve shrinkage, loss, and obsolete stock analysis	UC-020	Inventory Adjustment Case *(projection)*
API-168	Inventory Intelligence	`/api/v1/intelligence/accuracy/count-performance`	GET	Retrieve cycle count and physical count accuracy metrics	UC-014, UC-015, UC-037	Cycle Count, Physical Inventory Count *(projection)*
API-169	Inventory Intelligence	`/api/v1/intelligence/campaigns/readiness`	GET	Retrieve promotional campaign readiness status	UC-040, UC-047	Promotional Stock Allocation *(projection)*
API-170	Inventory Intelligence	`/api/v1/intelligence/kpis`	GET	Retrieve executive inventory performance KPIs	UC-038	*(KPI Dashboard projection)*
API-171	Inventory Intelligence	`/api/v1/intelligence/valuation/snapshots`	GET	Retrieve period-end inventory valuation snapshots	UC-039	*(Inventory Valuation Snapshot projection)*
API-172	Inventory Intelligence	`/api/v1/intelligence/valuation/reconciliation`	GET	Retrieve inventory valuation vs. general ledger reconciliation report	UC-039	*(Inventory Valuation Snapshot projection)*

Platform & Integration

Cross-cutting and external-integration APIs supporting authentication, health monitoring, and enterprise system boundaries.

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-173	Identity & Access	`/api/v1/auth/session`	GET	Validate authenticated session and retrieve user context	UC-045	User Account

API ID	Business Module	Endpoint	HTTP Method	Purpose	Related Use Case	Related Aggregate
API-174	Identity & Access	`/api/v1/auth/logout`	POST	Terminate authenticated session	UC-045	*(Infrastructure — no aggregate)*
API-175	Platform	`/api/v1/health`	GET	Application health check for operations monitoring	—	*(Infrastructure)*
API-176	Integration (EXT-01)	`/api/v1/integrations/sales/orders`	POST	Import customer order from Sales/CRM system (ACL inbound)	UC-021, UC-022	Customer Order Fulfillment
API-177	Integration (EXT-01)	`/api/v1/integrations/sales/fulfillment-status`	GET	Publish fulfillment status feed to Sales/CRM (ACL outbound)	UC-027	Customer Order Fulfillment *(projection)*
API-178	Integration (EXT-07)	`/api/v1/integrations/marketing/campaign-plans`	POST	Import campaign plan from marketing system (ACL inbound)	UC-040	Promotional Stock Allocation

Use Case Coverage Matrix

All 48 use cases from SRS-UC-001 are supported by at least one REST API in this catalog.

Use Case	Supporting API IDs
UC-001 Manage Product Master Data	API-001 – API-009
UC-002 Configure Inventory Locations	API-011 – API-021
UC-003 Define Stock Policies and Thresholds	API-038 – API-040
UC-004 Classify Products by ABC Priority	API-006
UC-005 Maintain Governance Policies and Reason Codes	API-022 – API-031
UC-006 Place and Manage Purchase Orders	API-049 – API-063, API-161
UC-007 Plan Expected Inbound Deliveries	API-064, API-162
UC-008 Receive and Verify Supplier Goods	API-065 – API-069
UC-009 Document and Resolve Receipt Discrepancies	API-070 – API-072, API-025, API-030
UC-010 Perform Stock Put-Away	API-073, API-074, API-015, API-016
UC-011 Receive Return and Non-PO Inbound Stock	API-075, API-076, API-128
UC-012 Monitor Inventory Status	API-033 – API-037, API-151, API-152
UC-013 Inquire Inventory Availability	API-002, API-003, API-035, API-036, API-153
UC-014 Perform Cycle Count	API-108 – API-111, API-168
UC-015 Conduct Full Physical Inventory Count	API-112 – API-115, API-020
UC-016 Investigate and Approve Count Variances	API-116 – API-119, API-123
UC-017 Process Stock Adjustments	API-120 – API-122
UC-018 Authorize and Process Inventory Write-Offs	API-120 – API-122
UC-019 Manage Quarantine and Hold Stock	API-045, API-046, API-154
UC-020 Analyze Shrinkage, Loss, and Obsolete Stock	API-167
UC-021 Confirm Order Availability	API-035, API-036, API-084, API-176
UC-022 Reserve Inventory for Confirmed Orders	API-043, API-085, API-139
UC-023 Release Inventory Reservations	API-044, API-086
UC-024 Pick, Pack, and Ship Orders	API-088 – API-092, API-041

Use Case	Supporting API IDs
UC-025 Handle Partial Fulfillment and Backorders	API-093 – API-095, API-164
UC-026 Handle Outbound Fulfillment Exceptions	API-096, API-165
UC-027 View Order Fulfillment Status	API-082, API-083, API-097, API-177
UC-028 Review Low-Stock and Out-of-Stock Alerts	API-037, API-158, API-159, API-160
UC-029 Plan Replenishment and Create Purchase Requisitions	API-054 – API-058, API-160
UC-030 Track Inbound Supply Status	API-060, API-061, API-064, API-107, API-161 – API-163
UC-031 Request Inter-Location Stock Transfer	API-098 – API-100, API-042
UC-032 Approve or Reject Transfer Requests	API-101, API-102, API-106
UC-033 Execute and Record Inter-Location Transfer	API-103 – API-105, API-107
UC-034 Analyze and Recommend Network Rebalancing	API-036, API-166, API-160
UC-035 Authorize and Record Customer Returns	API-124 – API-127
UC-036 Receive, Inspect, and Disposition Returned Goods	API-076, API-128 – API-131
UC-037 Review Inventory Dashboards and Operational Reports	API-150 – API-152, API-154, API-168
UC-038 Review Inventory Performance KPIs	API-170
UC-039 Reconcile Inventory Valuation with Finance	API-077 – API-080, API-047, API-171, API-172
UC-040 Confirm Promotional and Launch Stock Readiness	API-132 – API-139, API-169, API-178
UC-041 Conduct Inventory Audit and Control Review	API-156, API-157
UC-042 View Inventory Movement and Transaction History	API-155
UC-043 Query and Review Audit Trail	API-156, API-157
UC-044 Manage User Accounts and Role Assignments	API-140 – API-147
UC-045 Enforce Role-Based and Location-Scoped Access	API-031, API-148, API-149, API-173
UC-046 Import Initial Product and Stock Master Data	API-010, API-017, API-032
UC-047 Receive and Respond to Operational Notifications	API-158, API-159, API-106, API-123, API-163, API-169
UC-048 Support Multi-Location Split Fulfillment	API-036, API-087

Aggregate Coverage Matrix

All 18 domain aggregates from SRS-DM-003 are accessed through at least one REST API. Read projections and infrastructure endpoints are noted separately.

Aggregate	REST API IDs
Product Catalog	API-001 – API-010
Inventory Location	API-011 – API-021
Inventory Governance	API-022 – API-031
Stock Position	API-032 – API-037, API-041 – API-048
Stock Policy	API-038 – API-040, API-042
Supplier	API-049 – API-053
Purchase Requisition	API-054 – API-058
Purchase Order	API-059 – API-065, API-161 – API-163
Goods Receipt	API-066 – API-076
Supplier Invoice	API-077 – API-080

Aggregate	REST API IDs
Customer Order Fulfillment	API-081 – API-097, API-153, API-164 – API-165, API-176, API-177
Stock Transfer	API-098 – API-107
Cycle Count	API-108 – API-111, API-168
Physical Inventory Count	API-112 – API-115, API-168
Inventory Adjustment Case	API-116 – API-123, API-167
Customer Return Case	API-124 – API-131, API-076
Promotional Stock Allocation	API-132 – API-139, API-169, API-178
User Account	API-140 – API-149, API-173

Domain Command to API Mapping

Write APIs trace to domain commands (SRS-ES-002). Internal reaction commands (triggered by domain events) are not exposed as public REST endpoints.

Domain Command	REST API
CMD-001 RegisterProduct	API-001
CMD-002 ChangeProductStatus	API-005
CMD-003 AssignABCClassification	API-006
CMD-004 ActivateInventoryLocation	API-017
CMD-005 DeactivateInventoryLocation	API-018
CMD-006 ConfigureStoragePosition	API-015
CMD-007 PublishGovernancePolicy	API-026
CMD-008 RetireReasonCode	API-024
CMD-009 StartPhysicalInventoryCount	API-112
CMD-010 CertifyPhysicalInventoryCount	API-115
CMD-011 UpdateStockPolicyThresholds	API-040
CMD-012 ReserveStock	API-043, API-085
CMD-013 ReleaseReservation	API-044, API-086
CMD-014 ReceiveStock	*(Internal — API-073, API-104, API-130 trigger via events)*
CMD-015 ShipStock	*(Internal — API-092 triggers via event)*
CMD-016 ApplyApprovedAdjustment	*(Internal — API-121 triggers via event)*
CMD-017 ApplyApprovedWriteOff	*(Internal — API-121, API-130 trigger via events)*
CMD-018 ApplyQuarantineHold	API-045
CMD-019 ReleaseQuarantineHold	API-046
CMD-020 TransferConsignmentOwnership	API-047
CMD-021 SubmitPurchaseRequisition	API-054
CMD-022 ApprovePurchaseRequisition	API-056
CMD-023 RejectPurchaseRequisition	API-057
CMD-024 PlacePurchaseOrder	API-059
CMD-025 CancelPurchaseOrder	API-063
CMD-026 ScheduleInboundDelivery	API-064

Domain Command	REST API
CMD-027 RecordInboundDelay	*(Intelligence projection — API-163)*
CMD-028 VerifyGoodsReceipt	API-069
CMD-029 RecordReceiptDiscrepancy	API-070
CMD-030 ResolveReceiptDiscrepancy	API-072
CMD-031 CompletePutAway	API-073
CMD-032 RecordSupplierInvoice	API-077
CMD-033 EvaluateThreeWayMatch	API-079
CMD-034 ConfirmOrderAvailability	API-084
CMD-035 ConfirmCustomerOrder	API-085
CMD-036 CancelCustomerOrder	API-086
CMD-037 CreateMultiLocationFulfillmentPlan	API-087
CMD-038 VerifyPick	API-090
CMD-039 DispatchShipment	API-092
CMD-040 CompletePartialFulfillment	API-093
CMD-041 RecordBackorder	API-094
CMD-042 FulfillBackorder	API-095
CMD-043 RaiseFulfillmentException	API-096
CMD-044 DispatchTransferStock	*(Internal — API-103 triggers via event)*
CMD-045 RequestStockTransfer	API-098
CMD-046 ApproveTransferRequest	API-101
CMD-047 RejectTransferRequest	API-102
CMD-048 DispatchTransfer	API-103
CMD-049 ConfirmTransferReceipt	API-104
CMD-050 RecordTransferDelay	*(Intelligence projection — API-107)*
CMD-051 ScheduleCycleCount	API-108
CMD-052 SubmitCycleCountResult	API-111
CMD-053 ApproveCountVariance	API-119
CMD-054 SubmitAdjustment	API-120
CMD-055 ApproveAdjustment	API-121
CMD-056 RejectAdjustment	API-122
CMD-057 IdentifyCountVariance	*(Internal — API-111, API-115 trigger)*
CMD-058 EstablishOpeningBalances	API-032
CMD-059 ApplyQuarantineFromReturn	*(Internal — API-130 triggers via event)*
CMD-060 RecordCountVarianceFromPhysicalCount	*(Internal — API-115 triggers)*
CMD-061 IssueReturnAuthorization	API-124
CMD-062 ExpireReturnAuthorization	*(Policy-driven — no direct REST)*
CMD-063 ReceiveCustomerReturn	API-128
CMD-064 InspectReturnedGoods	API-129
CMD-065 RestockReturnedGoods	API-130 *(disposition: restock)*
CMD-066 QuarantineReturnedGoods	API-130 *(disposition: quarantine)*

Domain Command	REST API
CMD-067 WriteOffReturnedGoods	API-130 *(disposition: write-off)*
CMD-068 AllocatePromotionalStock	API-132
CMD-069 ReleasePromotionalAllocation	API-137
CMD-070 EvaluateStockStatus	*(Internal policy — surfaces via API-037, API-158)*
CMD-071 RecommendReplenishmentAction	*(Intelligence — API-160)*
CMD-072 EvaluatePromotionalReadiness	API-136
CMD-073 AnalyzeNetworkBalance	*(Intelligence — API-166)*
CMD-074 PrepareValuationSnapshot	*(Intelligence — API-171)*
CMD-075 ReconcileInventoryValuation	*(Intelligence — API-172)*
CMD-076 ProvisionUserAccount	API-140
CMD-077 ChangeRoleAssignment	API-143, API-144
CMD-078 ValidateSegregationOfDuties	API-031, API-148

Excluded from Public REST Exposure

The following capabilities are intentionally not exposed as REST APIs per architecture boundaries:

Capability	Treatment	Rationale
Internal domain event handlers (CMD-014–017, CMD-044, CMD-057–060)	Event-driven choreography	Cross-aggregate consistency via events, not direct client invocation
Notification delivery (email/SMS)	SVC-02 gateway	Infrastructure concern; Intelligence raises alerts, gateway delivers
SSO authentication handshake	SVC-01 IdP	Standard OIDC/SAML — not IMS domain API
Finance ERP file export (MVP)	Scheduled batch job	TEC-05 file export; Phase 3 moves to API-172 feed
Supplier EDI / carrier tracking (MVP)	Human-mediated	EXT-04, EXT-05 — no direct system integration in MVP
Database audit triggers	Persistence layer	SRS Database Audit Strategy — transparent to API consumers

Document Approval

Role	Name	Signature	Date
Senior Backend Architect			
Solution Architect			
DDD Solution Architect			
Product Owner			

End of REST API Catalog

Document ID: SRS-API-001 | Version 1.0 | July 6, 2026

Complete API Specification

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-API-009
Document Title	Complete REST API Specification
Version	1.0
Status	Draft
Document Owner	Senior Backend Architect
Source Documents	SRS-API-001 through SRS-API-008
Date	July 6, 2026
Base URL	`https://{host}/api/v1`
Scope	All 178 REST endpoints — full contract per endpoint

Purpose

Single authoritative specification combining HTTP contract, validation, authorization, business rules, and examples for every IMS REST API.

Specification Template (18 Fields)

Every endpoint section uses fields **1–18** in identical order. Cross-references: Request detail → SRS-API-002/004; Response shapes → SRS-API-003; Errors → SRS-API-005; Auth → SRS-API-007; Collection queries → SRS-API-006.

Global Conventions

Standard Request Headers

Header	Required	Applies To	Description
`Authorization`	Yes*	All except API-175	`Bearer {access_token}` from IdP (SVC-01)
`Content-Type`	Yes	POST/PATCH/PUT with body	`application/json`
`Accept`	Recommended	All	`application/json`
`X-Trace-Id`	Optional	All	Client correlation UUID; echoed in errors
`Idempotency-Key`	Recommended	Stock mutation, reservation, payment commands	UUID; 24h deduplication window
`X-API-Major-Version`	Optional	All	Client-declared major version (default `1`)

* API-176–178 use OAuth2 **Client Credentials** (`Authorization: Bearer {client_token}`).

Standard Success Headers

Header	Description
`X-API-Version`	Contract release e.g. `1.0.0`
`X-Trace-Id`	Server trace ID

Header	Description
`Location`	On `201 Created` — URI of created resource
`ETag`	On mutable resources — optimistic lock version

Standard Error Responses (All Endpoints)

`Content-Type: application/problem+json`

HTTP	When	Typical `errorCode`
400	Validation failure	`VAL_REQUEST_INVALID`, `VAL_REQUIRED_FIELD`
401	Missing/invalid auth	`AUTH_MISSING_CREDENTIALS`, `AUTH_TOKEN_EXPIRED`
403	Permission or location scope denied	`AUTHZ_INSUFFICIENT_PERMISSION`, `AUTHZ_LOCATION_SCOPE_DENIED`
404	Resource not found	`NF_*` per resource type
409	Duplicate, state conflict, optimistic lock	`CONF_*`
422	Business rule violation	`BIZ_*`
500	Unhandled error	`SYS_INTERNAL_ERROR`
503	Dependency unavailable	`SYS_DATABASE_UNAVAILABLE`

Standard error body (reference for all APIs):

```
{
  "type": "https://api.ims.example/problems/validation-error",
  "title": "Validation Failed",
  "status": 400,
  "detail": "The request contains validation errors.",
  "instance": "/api/v1/{path}",
  "errorCode": "VAL_REQUEST_INVALID",
  "traceId": "7f3c9a2e-4b1d-4e8a-9c0f-2a8b6d4e1f30",
  "timestamp": "2026-07-06T14:30:00.000Z",
  "errors": [{ "field": "fieldName", "code": "VAL_REQUIRED_FIELD", "message": "..." }]
}
```

Standard Pagination Query (Collection GETs)

Parameter	Type	Default	Max	Description
`page`	integer	1	—	1-based page index
`pageSize`	integer	20	100	Records per page

See SRS-API-006 for endpoint-specific filters, sort, and search parameters.

M-01 — Product Catalog

API-001: Register Product

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/products`
3	**Purpose**	Register a new product SKU with mandatory catalog attributes
4	**Authentication**	Required — Bearer JWT

#	Field	Value
5	**Required Permission**	`PERM-001` (Create Product) — Role: `INV_CTRL`
6	**Request Headers**	`Authorization`, `Content-Type: application/json`, optional `X-Trace-Id`
7	**Path Parameters**	None
8	**Query Parameters**	None
9	**Request Body**	`CreateProductRequest` — `sku`, `name`, `unitOfMeasure` (required); `description`, `categoryCode`, `barcode`, `isTraceable`, `rotationPolicy`, `shelfLifeManaged`, `mrslThresholdDays` (optional/conditional)
10	**Validation Rules**	`sku` 1–50, pattern `^[A-Z0-9][A-Z0-9_-]*$`, unique (VR-001, VR-002); `name` 1–200 non-blank; `unitOfMeasure` 1–10; if `isTraceable=true` → `rotationPolicy` required; if `shelfLifeManaged=true` → `mrslThresholdDays` ≥1 (VR-035)
11	**Success Response**	`ProductDetailResponse` — full product with `audit`
12	**Error Responses**	400 validation; 401 auth; 403 `INV_CTRL` only; 409 `CONF_DUPLICATE_SKU`; 422 discontinued rules N/A on create
13	**HTTP Status Codes**	Success: **201 Created**; Errors: 400, 401, 403, 409, 500
14	**Business Rules Applied**	BR-014, BR-015, BR-016, BR-017 (traceability attributes at definition)
15	**Related Use Case**	UC-001, UC-046

16. Example Request

```
POST /api/v1/products HTTP/1.1
Authorization: Bearer eyJhbGciOiJSUzI1NiIs...
Content-Type: application/json

{
  "sku": "BOLT-M10",
  "name": "M10 Hex Bolt",
  "unitOfMeasure": "EA",
  "categoryCode": "FAST",
  "isTraceable": false
}
```

17. Example Success Response — `201 Created`

```
{
  "sku": "BOLT-M10",
  "name": "M10 Hex Bolt",
  "lifecycleStatus": "ACTIVE",
  "abcClass": null,
  "unitOfMeasure": "EA",
  "description": null,
  "categoryCode": "FAST",
  "barcode": null,
  "isTraceable": false,
  "rotationPolicy": "FIFO",
  "shelfLifeManaged": false,
  "mrslThresholdDays": null,
  "audit": {
    "createdAt": "2026-07-06T14:00:00.000Z",
    "createdBy": "inv.controller",
    "updatedAt": "2026-07-06T14:00:00.000Z",
    "updatedBy": "inv.controller"
  }
}
```

18. Example Error Response — `409 Conflict`

```
{
  "type": "https://api.ims.example/problems/conflict",
  "title": "Resource Conflict",
  "status": 409,
  "detail": "A product with SKU 'BOLT-M10' already exists in the catalog.",
  "instance": "/api/v1/products",
  "errorCode": "CONF_DUPLICATE_SKU",
  "traceId": "f7a8b9c0-1234-5678-9012-def012345678",
  "timestamp": "2026-07-06T14:01:00.000Z",
  "validationRuleId": "VR-002"
}
```

API-002: Search Products

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/products`
3	**Purpose**	Search and filter products by name, category, status, ABC class
4	**Authentication**	Required — Bearer JWT
5	**Required Permission**	`PERM-004` (Lookup Product)
6	**Request Headers**	`Authorization`, optional `Accept`
7	**Path Parameters**	None
8	**Query Parameters**	`name` (partial), `sku` (prefix), `categoryCode`, `lifecycleStatus`, `abcClass`, `barcode`, `q` (search name/sku/barcode), `page`, `pageSize`, `sortBy` (`sku`,`name`,`categoryCode`,`lifecycleStatus`), `sortDirection` (`ASC`,`DESC`)
9	**Request Body**	None
10	**Validation Rules**	`page`≥1; `pageSize` 1–100; valid enums for status/ABC; `sortBy` whitelist (VR-054)
11	**Success Response**	`PagedResponse<ProductSummaryResponse>`
12	**Error Responses**	400 invalid query; 401; 403
13	**HTTP Status Codes**	Success: **200 OK**
14	**Business Rules Applied**	BR-027 (filter by discontinued via `lifecycleStatus`)
15	**Related Use Case**	UC-001, UC-013

16. Example Request

```
GET /api/v1/products?lifecycleStatus=ACTIVE&page=1&pageSize=20&sortBy=sku HTTP/1.1
Authorization: Bearer eyJhbGciOiJSUzI1NiIs...
```

17. Example Success Response — `200 OK`

```
{
  "data": [
    { "sku": "BOLT-M10", "name": "M10 Hex Bolt", "lifecycleStatus": "ACTIVE", "abcClass": "B", "unitOfMeasure": "EA" }
  ],
  "page": 1,
  "pageSize": 20,
  "totalElements": 1,
  "totalPages": 1
}
```

18. Example Error Response — `400 Bad Request`

```
{
  "type": "https://api.ims.example/problems/validation-error",
  "title": "Validation Failed",
  "status": 400,
  "detail": "Invalid sortBy value.",
  "instance": "/api/v1/products",
  "errorCode": "VAL_INVALID_ENUM",
  "traceId": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "timestamp": "2026-07-06T14:05:00.000Z",
  "errors": [{ "field": "sortBy", "code": "VAL_INVALID_ENUM", "message": "sortBy must be one of: sku, name, categoryCode,
lifecycleStatus" }]
}
```

API-003: Get Product by SKU

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/products/{sku}`
3	**Purpose**	Retrieve product master record by SKU
4	**Authentication**	Required
5	**Required Permission**	`PERM-004`
6	**Request Headers**	`Authorization`
7	**Path Parameters**	`sku` (string, required) — ProductSku pattern
8	**Query Parameters**	None
9	**Request Body**	None
10	**Validation Rules**	`sku` format VR-001; must exist VR-007
11	**Success Response**	`ProductDetailResponse`
12	**Error Responses**	401; 403; 404 `NF_PRODUCT_NOT_FOUND`
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	—
15	**Related Use Case**	UC-001, UC-013

16. **Example Request:** `GET /api/v1/products/BOLT-M10`
17. **Example Success Response:** Same structure as API-001 success body.
18. **Example Error Response:** `404` with `errorCode: NF_PRODUCT_NOT_FOUND`, `validationRuleId: VR-007`.

API-004: Update Product Attributes

#	Field	Value
1	**HTTP Method**	`PATCH`
2	**URL**	`/api/v1/products/{sku}`
3	**Purpose**	Update mutable product attributes
4	**Authentication**	Required
5	**Required Permission**	`PERM-002` — `INV_CTRL`
6	**Request Headers**	`Authorization`, `Content-Type`, optional `If-Match` (ETag)
7	**Path Parameters**	`sku` (required)

#	Field	Value
8	**Query Parameters**	None
9	**Request Body**	`UpdateProductAttributesRequest` — at least one of: `name`, `description`, `unitOfMeasure`, `categoryCode`, `barcode`, `isTraceable`, `rotationPolicy`, `shelfLifeManaged`, `mrslThresholdDays`
10	**Validation Rules**	Same field rules as API-001; at least one field required; traceability cross-rules (VR-003)
11	**Success Response**	`ProductDetailResponse`
12	**Error Responses**	400; 401; 403; 404; 409 `CONF_OPTIMISTIC_LOCK`
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	BR-014–BR-017
15	**Related Use Case**	UC-001

16–18: PATCH body with `name` update; 200 returns updated `ProductDetailResponse`; 400 if empty body (`VAL_PATCH_NO_FIELDS`).

API-005: Change Product Lifecycle Status

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/products/{sku}/lifecycle-status`
3	**Purpose**	Change lifecycle status (ACTIVE, INACTIVE, DISCONTINUED)
4	**Authentication**	Required
5	**Required Permission**	`PERM-003` — `INV_CTRL`
6	**Request Headers**	`Authorization`, `Content-Type`
7	**Path Parameters**	`sku`
8	**Query Parameters**	None
9	**Request Body**	`newStatus` (required), `reason` (optional), `effectiveDate` (optional)
10	**Validation Rules**	Valid status transition VR-004, VR-005; `reason` max 500
11	**Success Response**	`CommandResultResponse` + embedded `ProductDetailResponse`
12	**Error Responses**	400; 403; 404; 422 invalid transition
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	**BR-027** (DISCONTINUED blocks replenishment)
15	**Related Use Case**	UC-001

16. Example Request

```
{ "newStatus": "DISCONTINUED", "reason": "Supplier end-of-life", "effectiveDate": "2026-07-15" }
```

17. Example Success: `{ "id": "BOLT-M10", "status": "DISCONTINUED", "processedAt": "2026-07-06T14:10:00.000Z", "message": "Lifecycle status updated." }`

18. Example Error: `422 BIZ_PRODUCT_DISCONTINUED` / invalid transition per VR-005.

API-006: Assign ABC Classification

#	Field	Value
1	**HTTP Method**	`PUT`

#	Field	Value
2	**URL**	`/api/v1/products/{sku}/abc-classification`
3	**Purpose**	Assign or update ABC priority classification
4	**Authentication**	Required
5	**Required Permission**	`PERM-012` — `INV_CTRL`
6	**Request Headers**	`Authorization`, `Content-Type`
7	**Path Parameters**	`sku`
8	**Query Parameters**	None
9	**Request Body**	`abcClass` (required: A/B/C), `classificationBasis` (optional), `effectiveDate` (optional)
10	**Validation Rules**	VR-021, VR-022; product exists
11	**Success Response**	`ProductDetailResponse`
12	**Error Responses**	400; 403; 404
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	**BR-032** (Class A priority monitoring)
15	**Related Use Case**	UC-004

16–18: PUT `{ "abcClass": "A", "classificationBasis": "REVENUE" }` → 200 product with `abcClass: "A"`.

API-007: Validate Product Eligibility

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/products/{sku}/eligibility`
3	**Purpose**	Evaluate product eligibility for procurement, reservation, fulfillment, transfer
4	**Authentication**	Required
5	**Required Permission**	`PERM-004`
6	**Request Headers**	`Authorization`
7	**Path Parameters**	`sku`
8	**Query Parameters**	`transactionType` (required enum), `locationCode` (conditional for location-scoped types)
9	**Request Body**	None
10	**Validation Rules**	VR-004, VR-006, VR-008; location scope
11	**Success Response**	`EligibilityResponse`
12	**Error Responses**	400; 403 location scope
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	**BR-027**
15	**Related Use Case**	UC-006, UC-021, UC-022

17. Example Success: `{ "eligible": true, "transactionType": "RESERVATION", "reason": null, "lifecycleStatus": "ACTIVE" }`
18. Example Error: `{ "eligible": false, "reason": "Product is discontinued" }` returned in **200** body (not error HTTP) when ineligible; HTTP 422 only on invalid query.

API-008: List Products by Category

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/products/categories/{categoryCode}/products`
3	**Purpose**	List products in a category
4	**Authentication**	Required
5	**Required Permission**	`PERM-004`
6	**Request Headers**	`Authorization`
7	**Path Parameters**	`categoryCode` (required)
8	**Query Parameters**	`lifecycleStatus`, `page`, `pageSize`, `sortBy`, `sortDirection`
9	**Request Body**	None
10	**Validation Rules**	Category exists; pagination per SRS-API-006
11	**Success Response**	`PagedResponse<ProductSummaryResponse>`
12	**Error Responses**	400; 404 category
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	—
15	**Related Use Case**	UC-001

16–18: `GET .../categories/FAST/products?page=1` → paginated product list; 404 if category unknown.

API-009: Resolve Duplicate Product Conflict

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/products/duplicate-resolutions`
3	**Purpose**	Merge, reject, or keep both duplicate SKUs
4	**Authentication**	Required
5	**Required Permission**	`PERM-005` — `INV_CTRL`
6	**Request Headers**	`Authorization`, `Content-Type`
7	**Path Parameters**	None
8	**Query Parameters**	None
9	**Request Body**	`resolutionAction`, `sourceSku`, `targetSku` (if MERGE), `mergeAuthorizationReference` (conditional), `notes`
10	**Validation Rules**	VR-009, VR-010; merge auth when both have on-hand
11	**Success Response**	`CommandResultResponse`
12	**Error Responses**	400; 403; 422 merge constraints
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	Catalog identity integrity
15	**Related Use Case**	UC-001

16–18: POST merge action with `mergeAuthorizationReference` when required; 200 command result.

API-010: Bulk Product Import

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/products/import`
3	**Purpose**	Bulk import/validate product master at go-live
4	**Authentication**	Required
5	**Required Permission**	`PERM-015` — `OPS_DIR`, `INV_CTRL`
6	**Request Headers**	`Authorization`, `Content-Type`
7	**Path Parameters**	None
8	**Query Parameters**	None
9	**Request Body**	`importMode` (`VALIDATE_ONLY`,`UPSERT`,`CREATE_ONLY`), `sourceReference`, `products[]` (1–5000 lines)
10	**Validation Rules**	Each line per API-001; unique SKU in batch VR-001, VR-002
11	**Success Response**	`BulkImportResultResponse` — `importedCount`, `errorCount`, `errors[]`
12	**Error Responses**	400 batch invalid; 403; 422 blocking errors on UPSERT
13	**HTTP Status Codes**	**200 OK** (validate/partial) or **201 Created** (pure create success)
14	**Business Rules Applied**	BR-014–BR-017; PERM-016 approval for activation
15	**Related Use Case**	UC-046

17. Example Success: `{ "importedCount": 498, "errorCount": 2, "errors": [{ "rowIndex": 15, "sku": "BAD SKU", "errorCode": "VAL_INVALID_FORMAT", "message": "..." }] }`
18. Example Error: `400` when `products` empty or >5000 items.

M-02 — Warehouse Network

API-011: Register Location

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/locations`
3	**Purpose**	Register new inventory location (warehouse, store, DC)
4	**Authentication**	Required — Bearer JWT
5	**Required Permission**	`PERM-006` — `COO`, `OPS_DIR`
6	**Request Headers**	`Authorization`, `Content-Type`
7	**Path Parameters**	None
8	**Query Parameters**	None
9	**Request Body**	`locationCode`, `name`, `locationType`, `operationalStatus`; optional address fields, `parentLocationCode`, `plannedGoLiveDate`
10	**Validation Rules**	VR-011; unique `locationCode`; no circular hierarchy VR-014
11	**Success Response**	`LocationDetailResponse`
12	**Error Responses**	400; 401; 403; 409 `CONF_DUPLICATE_LOCATION`
13	**HTTP Status Codes**	**201 Created**

#	Field	Value
14	**Business Rules Applied**	Network structure governance
15	**Related Use Case**	UC-002

16. `POST /api/v1/locations` with `{ "locationCode": "WH-EAST", "name": "East Warehouse", "locationType": "WAREHOUSE", "operationalStatus": "PLANNED" }`
17. `201` — `LocationDetailResponse` with `locationCode: WH-EAST`
18. `409 CONF_DUPLICATE_LOCATION` — standard ProblemDetail envelope

API-012: List Locations

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/locations`
3	**Purpose**	List locations with filters
4	**Authentication**	Required
5	**Required Permission**	`PERM-004`
6	**Request Headers**	`Authorization`
7	**Path Parameters**	None
8	**Query Parameters**	`operationalStatus` (default ACTIVE), `locationType`, `parentLocationCode`, `name`, `q`, `page`, `pageSize`, `sortBy`, `sortDirection`
9	**Request Body**	None
10	**Validation Rules**	Pagination SRS-API-006; user location scope VR-053
11	**Success Response**	`PagedResponse<LocationSummaryResponse>`
12	**Error Responses**	400; 401; 403
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	—
15	**Related Use Case**	UC-002, UC-012, UC-013

16. `GET /api/v1/locations?operationalStatus=ACTIVE&page=1`
17. `200` paginated `LocationSummaryResponse[]`
18. `400 VAL_INVALID_ENUM` for invalid `sortBy`

API-013: Get Location

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/locations/{locationCode}`
3	**Purpose**	Retrieve location master and hierarchy
4	**Authentication**	Required
5	**Required Permission**	`PERM-004`
6	**Request Headers**	`Authorization`
7	**Path Parameters**	`locationCode`

#	Field	Value
8	**Query Parameters**	None
9	**Request Body**	None
10	**Validation Rules**	Exists; user scope VR-053
11	**Success Response**	`LocationDetailResponse`
12	**Error Responses**	401; 403; 404 `NF_LOCATION_NOT_FOUND`
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	—
15	**Related Use Case**	UC-002

16–18: `GET /api/v1/locations/WH-EAST` → 200 detail; 404 if not found or out of scope.

API-014: Update Location Hierarchy

#	Field	Value
1	**HTTP Method**	`PATCH`
2	**URL**	`/api/v1/locations/{locationCode}/hierarchy`
3	**Purpose**	Update parent, zone, layout
4	**Authentication**	Required
5	**Required Permission**	`PERM-008` — `COO`, `OPS_DIR`
6	**Request Headers**	`Authorization`, `Content-Type`
7	**Path Parameters**	`locationCode`
8	**Query Parameters**	None
9	**Request Body**	≥1 of: `parentLocationCode`, `zoneCode`, `layoutNotes`
10	**Validation Rules**	VR-014, VR-015; no self/ancestor parent
11	**Success Response**	`LocationDetailResponse`
12	**Error Responses**	400; 403; 404; 422 hierarchy violation
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	—
15	**Related Use Case**	UC-002

16–18: PATCH parent assignment → 200; 422 if circular hierarchy.

API-015: Configure Storage Position

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/locations/{locationCode}/storage-positions`
3	**Purpose**	Create/configure storage position
4	**Authentication**	Required
5	**Required Permission**	`PERM-007` — `COO`, `OPS_DIR`
6	**Request Headers**	`Authorization`, `Content-Type`
7	**Path Parameters**	`locationCode`

#	Field	Value
8	**Query Parameters**	None
9	**Request Body**	`positionCode` (required); `zone`, `aisle`, `bin`, `positionType`, `isActive`
10	**Validation Rules**	Location active VR-012; unique position per location
11	**Success Response**	`StoragePositionResponse`
12	**Error Responses**	400; 403; 404; 409 duplicate position
13	**HTTP Status Codes**	**201 Created**
14	**Business Rules Applied**	—
15	**Related Use Case**	UC-002, UC-010

16–18: POST position `RCV-STAGE-01` → 201; 404 inactive location.

API-016: List Storage Positions

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/locations/{locationCode}/storage-positions`
3	**Purpose**	List storage positions for put-away/pick
4	**Authentication**	Required
5	**Required Permission**	`PERM-004`
6	**Request Headers**	`Authorization`
7	**Path Parameters**	`locationCode`
8	**Query Parameters**	`positionType`, `zone`, `isActive`, `page`, `pageSize`, `sortBy`
9	**Request Body**	None
10	**Validation Rules**	SRS-API-006 pagination
11	**Success Response**	`PagedResponse<StoragePositionResponse>`
12	**Error Responses**	401; 403; 404
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	—
15	**Related Use Case**	UC-010

16–18: GET with `zone=RCV` → 200 paged positions.

API-017: Activate Location

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/locations/{locationCode}/activate`
3	**Purpose**	Activate location for inventory operations
4	**Authentication**	Required
5	**Required Permission**	`PERM-007` — `COO`, `OPS_DIR`
6	**Request Headers**	`Authorization`, `Content-Type`
7	**Path Parameters**	`locationCode`

#	Field	Value
8	**Query Parameters**	None
9	**Request Body**	`openingBalanceApprovalReference` (conditional), `activationDate`, `notes`
10	**Validation Rules**	VR-013, VR-025 opening balance policy
11	**Success Response**	`CommandResultResponse`
12	**Error Responses**	400; 403; 422 missing approval
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	Go-live integrity (BA-16)
15	**Related Use Case**	UC-002, UC-046

16–18: POST activate → 200 `{ "status": "ACTIVE" }`; 422 without approved opening balance.

API-018: Deactivate Location

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/locations/{locationCode}/deactivate`
3	**Purpose**	Deactivate location; block new transactions
4	**Authentication**	Required
5	**Required Permission**	`PERM-007`
6	**Request Headers**	`Authorization`, `Content-Type`
7	**Path Parameters**	`locationCode`
8	**Query Parameters**	None
9	**Request Body**	`reason` (required), `effectiveDate`
10	**Validation Rules**	VR-012, VR-017
11	**Success Response**	`CommandResultResponse`
12	**Error Responses**	400; 403; 409 active transactions
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	—
15	**Related Use Case**	UC-002

16–18: POST with reason → 200 deactivated status.

API-019: Close Location

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/locations/{locationCode}/close`
3	**Purpose**	Finalize location closure after disposition
4	**Authentication**	Required
5	**Required Permission**	`PERM-009` — `CEO`, `COO`, `OPS_DIR`
6	**Request Headers**	`Authorization`, `Content-Type`
7	**Path Parameters**	`locationCode`

#	Field	Value
8	**Query Parameters**	None
9	**Request Body**	`dispositionPlanReference` or `closureWaiverReference` (conditional), `closureDate`, `notes`
10	**Validation Rules**	VR-016; required when on-hand > 0
11	**Success Response**	`CommandResultResponse`
12	**Error Responses**	400; 403; 422 remaining stock
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	Strategic closure governance
15	**Related Use Case**	UC-002

16–18: POST close with disposition plan → 200; 422 if stock remains without plan.

API-020: Get Freeze Status

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/locations/{locationCode}/freeze-status`
3	**Purpose**	Check physical inventory freeze state
4	**Authentication**	Required
5	**Required Permission**	`PERM-040`
6	**Request Headers**	`Authorization`
7	**Path Parameters**	`locationCode`
8	**Query Parameters**	None
9	**Request Body**	None
10	**Validation Rules**	Location scope VR-053
11	**Success Response**	`LocationFreezeStatusResponse` — `frozen`, `freezeReason`, `freezeUntil`
12	**Error Responses**	401; 403; 404
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	**BR-024**
15	**Related Use Case**	UC-015

16–18: GET → `{ "frozen": true, "freezeReason": "PHYSICAL_COUNT", "freezeUntil": "2026-07-10" }`

API-021: Get Operational Status

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/locations/{locationCode}/operational-status`
3	**Purpose**	Check if location accepts transactions
4	**Authentication**	Required
5	**Required Permission**	`PERM-004`
6	**Request Headers**	`Authorization`
7	**Path Parameters**	`locationCode`

#	Field	Value
8	**Query Parameters**	None
9	**Request Body**	None
10	**Validation Rules**	—
11	**Success Response**	`LocationOperationalStatusResponse`
12	**Error Responses**	401; 403; 404
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	—
15	**Related Use Case**	UC-002

16–18: GET → `{ "operationalStatus": "ACTIVE", "acceptsTransactions": true }`

Continued — M-03 through M-07 in `Complete-API-Specification-Part2.md`; M-08 through Platform in `Complete-API-Specification-Part3.md`

Complete API Specification — Part 2

M-03 through M-07 (API-022 – API-097)

Parent document: `Complete-API-Specification.md` (SRS-API-009)
Conventions: Global headers, errors, and pagination apply from Part 1.

M-03 — Inventory Governance

API-022: Create Reason Code

| 1 | `POST` | 2 | `/api/v1/governance/reason-codes` | 3 | Create standardized reason code | 4 | Bearer JWT | 5 | `PERM-014` / `INV_CTRL` | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `code`, `description`, `transactionTypes[]`, `isActive` | 10 | VR-026, VR-027; unique code | 11 | `ReasonCodeResponse` | 12 | 400,403,409 | 13 | **201** | 14 | **BR-020** | 15 | UC-005 | **16** POST `{ "code": "DAMAGE_RCPT", "description": "Damaged at receipt", "transactionTypes": ["RECEIPT_DISCREPANCY"] }`
17 `201` reason code object
18 `409` duplicate code

API-023: List Reason Codes

| 1 | `GET` | 2 | `/api/v1/governance/reason-codes` | 3 | List active reason codes | 4 | Bearer | 5 | `PERM-014` | 6 | `Authorization` | 7 | — | 8 | `transactionType`, `isActive`, `page`, `pageSize` | 9 | — | 10 | VR-026 | 11 | `PagedResponse<ReasonCodeResponse>` | 12 | 400,401,403 | 13 | **200** | 14 | BR-020 | 15 | UC-005, UC-017 | **16–18** GET `?transactionType=ADJUSTMENT` → 200 paged list; 400 invalid enum

API-024: Retire Reason Code

| 1 | `POST` | 2 | `/api/v1/governance/reason-codes/{code}/retire` | 3 | Retire reason code | 4 | Bearer | 5 | `PERM-014` | 6 | `Authorization`, `Content-Type` | 7 | `code` | 8 | — | 9 | `retirementReason`, `effectiveDate` | 10 | VR-027 | 11 | `CommandResultResponse` | 12 | 400,403,404 | 13 | **200** | 14 | BR-020 | 15 | UC-005 | **16–18** POST retire → 200; 404 unknown code

API-025: Validate Reason Code

| 1 | `GET` | 2 | `/api/v1/governance/reason-codes/{code}/validate` | 3 | Validate code for transaction type | 4 | Bearer | 5 | `PERM-014` | 6 | `Authorization` | 7 | `code` | 8 | `transactionType` (required) | 9 | — | 10 | VR-026, VR-027 | 11 | `{ "valid": true, "applicable": true }` | 12 | 400,401 | 13 | **200** | 14 | BR-020 | 15 | UC-017 | **16–18** GET validate → 200 `{ "valid": true }`

API-026: Publish Governance Policy

| 1 | `POST` | 2 | `/api/v1/governance/policies` | 3 | Publish policy version | 4 | Bearer | 5 | `PERM-013` / `OPS_DIR,COO` | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `policyName`, `effectiveFrom`, `parameters[]` | 10 | VR-028 no overlap | 11 | `GovernancePolicyResponse` | 12 | 400,403,409 | 13 | **201** | 14 | Governance | 15 | UC-005 | **16–18** POST policy → 201; 409 overlapping effective dates

API-027: Update Approval Thresholds

| 1 | `PATCH` | 2 | `/api/v1/governance/policies/{policyId}/approval-thresholds` | 3 | Update approval thresholds | 4 | Bearer | 5 | `PERM-013` | 6 | `Authorization`, `Content-Type` | 7 | `policyId` | 8 | — | 9 | `transactionType`, `effectiveFrom`, thresholds | 10 | VR-020, VR-082 | 11 | `GovernancePolicyResponse` | 12 | 400,403,404 | 13 | **200** | 14 | **BR-021** | 15 | UC-005 | **16–18** PATCH thresholds → 200

API-028: Update Receipt Tolerances

| 1 | `PATCH` | 2 | `/api/v1/governance/policies/{policyId}/receipt-tolerances` | 3 | Update receipt tolerances | 4 | Bearer | 5 | `PERM-013` | 6 | `Authorization`, `Content-Type` | 7 | `policyId` | 8 | — | 9 | tolerance fields, `effectiveFrom` | 10 | VR-048 | 11 | `GovernancePolicyResponse` | 12 | 400,403 | 13 | **200** | 14 | **BR-006** | 15 | UC-005, UC-009 | **16–18** PATCH tolerance percent → 200

API-029: Get Approval Thresholds

| 1 | `GET` | 2 | `/api/v1/governance/approval-thresholds` | 3 | Lookup effective threshold | 4 | Bearer | 5 | `PERM-013` | 6 | `Authorization` | 7 | — | 8 | `transactionType`, `locationCode`, `asOfDate` | 9 | — | 10 | VR-082 | 11 | `ApprovalThresholdResponse` | 12 | 401,403 | 13 | **200** | 14 | BR-021 | 15 | UC-017 | **16–18** GET → 200 threshold object

API-030: Get Receipt Tolerances

| 1 | `GET` | 2 | `/api/v1/governance/receipt-tolerances` | 3 | Lookup receipt tolerance | 4 | Bearer | 5 | `PERM-032` / `WH_MGR` | 6 | `Authorization` | 7 | — | 8 | `locationCode`, `supplierCode`, `asOfDate` | 9 | — | 10 | VR-048 | 11 | `ReceiptToleranceResponse` | 12 | 401,403 | 13 | **200** | 14 | BR-006 | 15 | UC-008 | **16–18** GET → 200 tolerance

API-031: Evaluate Segregation of Duties

| 1 | `POST` | 2 | `/api/v1/governance/segregation-of-duties/evaluate` | 3 | SoD compliance check | 4 | Bearer | 5 | `PERM-013` | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `initiatorUserId`, `approverUserId`, `transactionType` | 10 | VR-075, VR-083; approver≠initiator | 11 | `{ "compliant": false, "violations": [] }` | 12 | 400,401 | 13 | **200** | 14 | **BR-019**, **BR-021** | 15 | UC-045 | **16** POST evaluate SoD
17 `{ "compliant": false, "violations": ["SOD_INITIATOR_APPROVER_SAME"] }`
18 `400` missing `approverUserId`

M-04 — Inventory Balance

API-032: Establish Opening Balances

| 1 | `POST` | 2 | `/api/v1/inventory/opening-balances` | 3 | Set go-live opening balances | 4 | Bearer | 5 | `PERM-010` | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `locationCode`, `balanceLines[]`, `approvalReference` | 10 | VR-024, VR-025; lot if traceable | 11 | `OpeningBalanceResultResponse` | 12 | 400,403,422 | 13 | **201** | 14 | BR-017 | 15 | UC-046 | **16–18** POST balances → 201; 422 location not pre-activation

API-033: List Stock Positions

| 1 | `GET` | 2 | `/api/v1/inventory/stock-positions` | 3 | Search stock positions | 4 | Bearer | 5 | `PERM-017,PERM-021` | 6 | `Authorization` | 7 | — | 8 | `sku`, `locationCode`, `stockStatus`, `abcClass`, pagination | 9 | — | 10 | VR-054 at least one filter; VR-053 scope | 11 | `PagedResponse<StockPositionSummaryResponse>` | 12 | 400,403 | 13 | **200** | 14 | — | 15 | UC-012, UC-013 | **16–18** GET `?locationCode=WH-EAST` → 200 paged positions

API-034: Get Stock Position

| 1 | `GET` | 2 | `/api/v1/inventory/stock-positions/{locationCode}/{sku}` | 3 | Position detail | 4 | Bearer | 5 | `PERM-017` | 6 | `Authorization` | 7 | `locationCode`, `sku` | 8 | — | 9 | — | 10 | VR-053, VR-054 | 11 | `StockPositionDetailResponse` | 12 | 403,404 | 13 | **200** | 14 | — | 15 | UC-012 | **16–18** GET → 200 with `StockAvailabilityResponse`

API-035: Get Availability by SKU

| 1 | `GET` | 2 | `/api/v1/inventory/availability/{sku}` | 3 | ATP inquiry | 4 | Bearer | 5 | `PERM-018` | 6 | `Authorization` | 7 | `sku` | 8 | `locationCode` (required), `requestedQuantity` | 9 | — | 10 | VR-055 | 11 | `AvailabilityResponse` | 12 | 400,403 | 13 | **200** | 14 | **BR-001** | 15 | UC-013, UC-021 | **17** `{ "sku": "BOLT-M10", "atp": { "amount": 45, "unitOfMeasure": "EA" }, "canFulfill": true }`

API-036: Get Network Availability

| 1 | `GET` | 2 | `/api/v1/inventory/availability/{sku}/network` | 3 | Network ATP rollup | 4 | Bearer | 5 | `PERM-022` | 6 | `Authorization` | 7 | `sku` | 8 | `locationCodes[]`, `excludeQuarantined`, `excludePromotionalRingFence` | 9 | — | 10 | VR-056, VR-064 | 11 | `NetworkAvailabilityResponse` | 12 | 401,403 | 13 | **200** | 14 | **BR-001**, **BR-013** | 15 | UC-013, UC-034 | **16–18** GET network ATP → 200 rollup by location

API-037: Get Stock Status Indicator

| 1 | `GET` | 2 | `/api/v1/inventory/stock-status/{locationCode}/{sku}` | 3 | Derived stock status | 4 | Bearer | 5 | `PERM-020` | 6 | `Authorization` | 7 | `locationCode`, `sku` | 8 | — | 9 | — | 10 | VR-059 policy exists | 11 | `StockStatusIndicatorResponse` | 12 | 404 | 13 | **200** | 14 | **BR-009** | 15 | UC-012, UC-028 | **17** `{ "stockStatus": "LOW_STOCK", "reorderLevel": 100, "atp": 25 }`

API-038: Define Stock Policy

| 1 | `POST` | 2 | `/api/v1/inventory/stock-policies` | 3 | Create stock policy | 4 | Bearer | 5 | `PERM-011` | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `locationCode`, `sku`, thresholds, `effectiveFrom` | 10 | VR-018–VR-020 | 11 | `StockPolicyResponse` | 12 | 400,403 | 13 | **201** | 14 | **BR-009**, **BR-010** | 15 | UC-003 | **16–18** POST policy → 201

API-039: Get Stock Policy

| 1 | `GET` | 2 | `/api/v1/inventory/stock-policies/{locationCode}/{sku}` | 3 | Get effective policy | 4 | Bearer | 5 | `PERM-011` | 6 | `Authorization` | 7 | `locationCode`, `sku` | 8 | `asOfDate` | 9 | — | 10 | VR-020 | 11 | `StockPolicyResponse` | 12 | 404 | 13 | **200** | 14 | BR-009 | 15 | UC-003 | **16–18** GET → 200 policy

API-040: Update Stock Policy

| 1 | `PATCH` | 2 | `/api/v1/inventory/stock-policies/{locationCode}/{sku}` | 3 | Update thresholds | 4 | Bearer | 5 | `PERM-011` | 6 | `Authorization`, `Content-Type` | 7 | `locationCode`, `sku` | 8 | — | 9 | threshold fields, `effectiveFrom` | 10 | VR-018–VR-020; ≥1 field | 11 | `StockPolicyResponse` | 12 | 400,403 | 13 | **200** | 14 | BR-009, BR-010 | 15 | UC-003 | **16–18** PATCH reorder level → 200

API-041: Get Lot Breakdown

| 1 | `GET` | 2 | `/api/v1/inventory/stock-positions/{locationCode}/{sku}/lots` | 3 | Lot-level quantities | 4 | Bearer | 5 | `PERM-017` | 6 | `Authorization` | 7 | `locationCode`, `sku` | 8 | `includeQuarantined` | 9 | — | 10 | Product traceable | 11 | `BoundedListResponse<LotQuantityResponse>` | 12 | 404 | 13 | **200** | 14 | **BR-017** | 15 | UC-024 | **16–18** GET lots → 200 lot array

API-042: Evaluate Transfer Capacity

| 1 | `GET` | 2 | `/api/v1/inventory/stock-positions/{locationCode}/{sku}/transfer-capacity` | 3 | Max transferable spare | 4 | Bearer | 5 | `PERM-076` | 6 | `Authorization` | 7 | `locationCode`, `sku` | 8 | `requestedQuantity` | 9 | — | 10 | VR-140 | 11 | `TransferCapacityResponse` | 12 | 403 | 13 | **200** | 14 | **BR-012** | 15 | UC-031 | **17** `{ "maxTransferable": { "amount": 30, "unitOfMeasure": "EA" } }`

API-043: Reserve Stock

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/inventory/stock-positions/{locationCode}/{sku}/reservations`

#	Field	Value
3	**Purpose**	Reserve stock; reduce ATP for order line
4	**Authentication**	Required
5	**Required Permission**	`PERM-052` — COO, OPS_DIR, WH_MGR
6	**Request Headers**	`Authorization`, `Content-Type`, **`Idempotency-Key`** (recommended)
7	**Path Parameters**	`locationCode`, `sku`
8	**Query Parameters**	None
9	**Request Body**	`quantity`, `orderId`, `orderLineId`; optional `reservationExpiry`, `safetyStockExceptionReference`
10	**Validation Rules**	VR-092, VR-093, VR-094; qty ≤ ATP; location not frozen
11	**Success Response**	`ReservationResponse` — `reservationId`, `reservedQuantity`, `remainingAtp`
12	**Error Responses**	400; 403; 422 `BIZ_INSUFFICIENT_ATP`, `BIZ_SAFETY_STOCK_BREACH`
13	**HTTP Status Codes**	**201 Created**
14	**Business Rules Applied**	**BR-001**, **BR-002**, **BR-010**
15	**Related Use Case**	UC-022

16. Example Request

```
POST /api/v1/inventory/stock-positions/WH-EAST/BOLT-M10/reservations HTTP/1.1
Authorization: Bearer eyJ...
Content-Type: application/json
Idempotency-Key: 550e8400-e29b-41d4-a716-446655440000

{
  "quantity": { "amount": 10, "unitOfMeasure": "EA" },
  "orderId": "ORD-2026-001",
  "orderLineId": "LINE-001"
}
```

17. Example Success — `201`

```
{
  "reservationId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "locationCode": "WH-EAST",
  "sku": "BOLT-M10",
  "reservedQuantity": { "amount": 10, "unitOfMeasure": "EA" },
  "remainingAtp": { "amount": 35, "unitOfMeasure": "EA" },
  "status": "ACTIVE"
}
```

18. Example Error — `422`

```
{
  "type": "https://api.ims.example/problems/business-rule-violation",
  "title": "Business Rule Violation",
  "status": 422,
  "detail": "Cannot reserve 100 units. Only 45 units are available to promise.",
  "instance": "/api/v1/inventory/stock-positions/WH-EAST/BOLT-M10/reservations",
  "errorCode": "BIZ_INSUFFICIENT_ATP",
  "businessRuleId": "BR-001",
  "validationRuleId": "VR-092",
  "traceId": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "timestamp": "2026-07-06T15:00:00.000Z"
}
```

API-044: Release Reservation

| 1 | `POST` | 2 | `.../reservations/{reservationId}/release` | 3 | Release reserved stock | 4 | Bearer | 5 | `PERM-053` | 6 | `Authorization`, `Content-Type`, `Idempotency-Key` | 7 | `locationCode`, `sku`, `reservationId` | 8 | — | 9 | `releaseReason`, optional `releaseQuantity` | 10 | VR-095, VR-096 | 11 | `CommandResultResponse` | 12 | 400,404,422 | 13 | **200** | 14 | **BR-002** | 15 | UC-023 | **16–18** POST release → 200; 422 exceeds reserved qty

API-045: Apply Quarantine Hold

| 1 | `POST` | 2 | `.../quarantine-holds` | 3 | Place stock on quarantine | 4 | Bearer | 5 | `PERM-047` | 6 | `Authorization`, `Content-Type` | 7 | `locationCode`, `sku` | 8 | — | 9 | `quantity`, `holdReason`, `reasonCode`, `lotIdentity` | 10 | VR-084, VR-026 | 11 | `QuarantineHoldResponse` | 12 | 422 | 13 | **201** | 14 | **BR-013**, **BR-030** | 15 | UC-019 | **16–18** POST hold → 201 holdId

API-046: Release Quarantine Hold

| 1 | `POST` | 2 | `.../quarantine-holds/{holdId}/release` | 3 | Release quarantine | 4 | Bearer | 5 | `PERM-048` | 6 | `Authorization`, `Content-Type` | 7 | `locationCode`, `sku`, `holdId` | 8 | — | 9 | `disposition`, `reasonCode`, `inspectionOutcomeReference` | 10 | VR-085 | 11 | `CommandResultResponse` | 12 | 422 | 13 | **200** | 14 | **BR-026** | 15 | UC-019 | **16–18** POST RESTORE_TO_SELLABLE with inspection ref → 200

API-047: Transfer Consignment Ownership

| 1 | `POST` | 2 | `.../consignment-ownership` | 3 | Transfer consignment to owned | 4 | Bearer | 5 | `PERM-017` / finance | 6 | `Authorization`, `Content-Type` | 7 | `locationCode`, `sku` | 8 | — | 9 | `quantity`, `consumptionReference` | 10 | consignment on-hand | 11 | `CommandResultResponse` | 12 | 422 | 13 | **200** | 14 | **BR-031** | 15 | UC-039 | **16–18** POST consumption → 200

API-048: Segregate Promotional Stock

| 1 | `POST` | 2 | `.../promotional-segregation` | 3 | Ring-fence ATP for campaign | 4 | Bearer / internal | 5 | `PERM-093` | 6 | `Authorization`, `Content-Type` | 7 | `locationCode`, `sku` | 8 | — | 9 | `campaignId`, `allocationId`, `quantity` | 10 | VR-127 | 11 | `CommandResultResponse` | 12 | 422 | 13 | **200** | 14 | **BR-029** | 15 | UC-040 | **16–18** Event-driven ring-fence → 200

M-05 — Procurement & Supplier (API-049 – API-065)

API	Method	URL	Permission	UC	Success	Key BR
049	POST	`/suppliers`	PERM-024 PUR_MGR	UC-006	201 SupplierResponse	—
050	GET	`/suppliers`	PERM-035	UC-006	200 Paged	—
051	GET	`/suppliers/{supplierCode}`	PERM-035	UC-006	200 Detail	—
052	PATCH	`/suppliers/{supplierCode}`	PERM-025 PUR_MGR	UC-006	200 Detail	—
053	POST	`/suppliers/{code}/deactivate`	PERM-026 PUR_MGR	UC-006	200 Command	—
054	POST	`/procurement/requisitions`	PERM-067 BUYER,PUR_MGR	UC-029	201 Requisition	BR-011, BR-027
055	GET	`/procurement/requisitions/{id}`	PERM-067	UC-029	200 Detail	—
056	POST	`.../approve`	PERM-068 PUR_MGR	UC-029	200 Command	BR-011 SoD
057	POST	`.../reject`	PERM-068	UC-029	200 Command	—
058	POST	`.../convert-to-po`	PERM-024	UC-029	201 PO	BR-007
059	POST	`/procurement/purchase-orders`	PERM-024 BUYER	UC-006	201 PO	BR-007, BR-027
060	GET	`/procurement/purchase-orders`	PERM-027	UC-006	200 Paged	—
061	GET	`/procurement/purchase-orders/{id}`	PERM-027	UC-006	200 Detail	—
062	PATCH	`/procurement/purchase-orders/{id}`	PERM-025	UC-006	200 Detail	VR-051
063	POST	`.../cancel`	PERM-026	UC-006	200 Command	—
064	POST	`.../expected-deliveries`	PERM-027	UC-007	200 Schedule	—

API	Method	URL	Permission	UC	Success	Key BR
065	GET	`.../validate-for-receipt`	PERM-028 WH_MGR	UC-008	200 Validation	**BR-007**

Each API-049–065 follows the 18-field template: Auth=Bearer; Headers per Part 1; Validation per SRS-API-004; Errors=standard set; Example request=minimal HTTP+JSON for POST commands, query string for GET.

API-059 Example (Place PO):

16 `POST /api/v1/procurement/purchase-orders` + body `{ "supplierCode": "SUP-01", "receivingLocationCode": "WH-EAST", "expectedDeliveryDate": "2026-08-01", "lines": [{ "sku": "BOLT-M10", "orderedQuantity": { "amount": 500, "unitOfMeasure": "EA" } }] }`
17 `201` `PurchaseOrderDetailResponse` with `purchaseOrderId`, `status: OPEN`
18 `422` `BIZ_PRODUCT_DISCONTINUED` if SKU discontinued (BR-027)

M-06 — Inbound Receiving (API-066 – API-080)

API	Method	URL	Permission	UC	Success	Key BR
066	POST	`/inbound/goods-receipts`	PERM-028	UC-008	201 Receipt	**BR-007**
067	GET	`/inbound/goods-receipts`	PERM-028	UC-008	200 Paged	—
068	GET	`/inbound/goods-receipts/{id}`	PERM-028	UC-008	200 Detail	—
069	POST	`.../verify`	PERM-028	UC-008	200 Verified	**BR-004**, **BR-016**, **BR-017**
070	POST	`.../discrepancies`	PERM-029	UC-009	201 Discrepancy	**BR-006**
071	GET	`.../discrepancies`	PERM-029	UC-009	200 List	—
072	POST	`.../discrepancies/{id}/resolve`	PERM-030	UC-009	200 Resolved	BR-006
073	POST	`.../put-away`	PERM-033	UC-010	200 Complete	**BR-005**
074	GET	`/inbound/pending-put-away`	PERM-033	UC-010	200 Paged	—
075	POST	`/inbound/ad-hoc-receipts`	PERM-031 WH_MGR	UC-011	201 Receipt	**BR-007** override
076	POST	`/inbound/return-receipts`	PERM-034	UC-011	201 Receipt	**BR-025**
077	POST	`/inbound/supplier-invoices`	PERM-094 FIN_MGR	UC-039	201 Invoice	—
078	GET	`/inbound/supplier-invoices/{id}`	PERM-094	UC-039	200 Detail	—
079	POST	`.../three-way-match`	PERM-094	UC-039	200 MatchResult	**BR-008**
080	POST	`.../authorize-payment`	PERM-094 CFO,FIN_MGR	UC-039	200 Authorized	**BR-008**

API-069 Verify Example:

16 POST verify lines with `acceptedQuantity`, `lotIdentity` for traceable SKU
17 `200` `{ "goodsReceiptId": "...", "status": "VERIFIED" }`
18 `422` `BIZ_MRSL_NON_COMPLIANT` (BR-016)

M-07 — Order Fulfillment (API-081 – API-097)

API	Method	URL	Permission	UC	Success	Key BR
081	POST	`/fulfillment/orders`	PERM-054	UC-021	201 Order	—
082	GET	`/fulfillment/orders`	PERM-062	UC-027	200 Paged	—
083	GET	`/fulfillment/orders/{id}`	PERM-062	UC-027	200 Detail	—
084	POST	`.../availability`	PERM-051	UC-021	200 Availability	**BR-001**

API	Method	URL	Permission	UC	Success	Key BR
085	POST	`.../confirm`	PERM-054	UC-022	200 Confirmed	**BR-002**
086	POST	`.../cancel`	PERM-054	UC-023	200 Cancelled	—
087	POST	`.../multi-location-plan`	PERM-061	UC-048	200 Plan	BR-001
088	POST	`.../pick-lists`	PERM-055 WH_MGR	UC-024	201 PickList	BR-014/015
089	GET	`.../pick-lists/awaiting`	PERM-055	UC-024	200 List	—
090	POST	`.../picks/{id}/verify`	PERM-055	UC-024	200 Verified	**BR-003**, **BR-017**
091	POST	`.../pack`	PERM-055	UC-024	200 Packed	BR-003
092	POST	`.../shipments`	PERM-056	UC-024	201 Shipment	**BR-003**
093	POST	`.../partial-fulfillment`	PERM-057	UC-025	200 Partial	**BR-028**
094	POST	`.../backorders`	PERM-058	UC-025	201 Backorder	**BR-028**
095	POST	`/fulfillment/backorders/{id}/fulfill`	PERM-058	UC-025	200 Fulfilled	BR-001
096	POST	`.../exceptions`	PERM-059	UC-026	201 Exception	—
097	GET	`.../status`	PERM-062	UC-027	200 Status	—

API-092 Dispatch Shipment (full example):

#	Field	Value
1–5	POST `/api/v1/fulfillment/orders/{orderId}/shipments` — Auth Bearer — `PERM-056` (`WH_MGR`, `PIK_PAK`)	
6–9	Headers: `Authorization`, `Content-Type`, `Idempotency-Key`; body: `shipmentLines[]`, optional `carrierCode`, `trackingNumber`	
10	VR-097, VR-098; pick verified; qty ≤ reserved	
11–13	`201` `ShipmentResponse`; errors 400,403,422	
14	**BR-003**	
15	UC-024	

16 POST shipment with `shipmentLines[{ orderLineId, shippedQuantity, lotIdentity }]`
17 `201` `{ "shipmentId": "...", "status": "SHIPPED", "shippedAt": "2026-07-06T16:00:00Z" }`
18 `422` `BIZ_*` pick not verified

Continued in `Complete-API-Specification-Part3.md` — API-098 through API-178 (M-08 Network Transfer through Platform & Integration)

Complete API Specification — Part 3

M-08 through Platform (API-098 – API-178)

Parent document: `Complete-API-Specification.md` (SRS-API-009)
Conventions: Global headers, errors, and pagination apply from Part 1. All URLs are prefixed with `/api/v1`.

M-08 — Network Transfer

API-098: Request Stock Transfer

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/transfers`
3	**Purpose**	Request inter-location stock transfer to resolve local shortage
4	**Authentication**	Required — Bearer JWT
5	**Required Permission**	`PERM-071` — `OPS_DIR`, `WH_MGR`, `REG_MGR`
6	**Request Headers**	`Authorization`, `Content-Type: application/json`, optional `Idempotency-Key`, `X-Trace-Id`
7	**Path Parameters**	None
8	**Query Parameters**	None
9	**Request Body**	`sourceLocationCode`, `destinationLocationCode`, `justification`, `lines[]` (`sku`, `requestedQuantity`); optional `priority`, `requiredByDate`
10	**Validation Rules**	VR-128–VR-130; source ≠ destination; qty ≤ source ATP; respects source minimum (VR-140)
11	**Success Response**	`StockTransferDetailResponse` — `transferId`, `status: REQUESTED`, lines, audit
12	**Error Responses**	400 validation; 401; 403 location scope; 422 `BIZ_INSUFFICIENT_ATP`, `BIZ_SOURCE_MINIMUM_BREACH`
13	**HTTP Status Codes**	Success: **201 Created**; Errors: 400, 401, 403, 422, 500
14	**Business Rules Applied**	**BR-011**, **BR-012**
15	**Related Use Case**	UC-031

16. Example Request

```
POST /api/v1/transfers HTTP/1.1
Authorization: Bearer eyJ...
Content-Type: application/json

{
  "sourceLocationCode": "WH-WEST",
  "destinationLocationCode": "WH-EAST",
  "justification": "East warehouse below safety stock for BOLT-M10",
  "lines": [{ "sku": "BOLT-M10", "requestedQuantity": { "amount": 50, "unitOfMeasure": "EA" } }]
}
```

17. Example Success — `201`

```
{
  "transferId": "TRF-2026-0042",
  "status": "REQUESTED",
  "sourceLocationCode": "WH-WEST",
  "destinationLocationCode": "WH-EAST",
  "lines": [{ "sku": "BOLT-M10", "requestedQuantity": { "amount": 50, "unitOfMeasure": "EA" }, "approvedQuantity": null }]
}
```

18. Example Error — `422`

```
{
  "status": 422,
  "errorCode": "BIZ_SOURCE_MINIMUM_BREACH",
  "businessRuleId": "BR-012",
  "detail": "Transfer would reduce WH-WEST below minimum stock level."
}
```

API-099: List Stock Transfers

| 1 | `GET` | 2 | `/api/v1/transfers` | 3 | List transfers with status and location filters | 4 | Bearer | 5 | `PERM-075` | 6 | `Authorization` | 7 | — | 8 | `status`, `sourceLocationCode`, `destinationLocationCode`, `startDate`, `endDate`, `page`, `pageSize` | 9 | — | 10 | VR-136 date range; user location scope | 11 | `PagedResponse<StockTransferSummaryResponse>` | 12 | 400,401,403 | 13 | **200** | 14 | — | 15 | UC-031, UC-033 | **16** GET `?status=IN_TRANSIT&destinationLocationCode=WH-EAST`
17 `200` paged transfer list
18 `400 VAL_INVALID_DATE_RANGE`

API-100: Get Stock Transfer

| 1 | `GET` | 2 | `/api/v1/transfers/{transferId}` | 3 | Transfer detail and in-transit state | 4 | Bearer | 5 | `PERM-075` | 6 | `Authorization` | 7 | `transferId` | 8 | — | 9 | — | 10 | Transfer exists; user scope | 11 | `StockTransferDetailResponse` | 12 | 401,403,404 | 13 | **200** | 14 | — | 15 | UC-031, UC-033 | **16–18** GET by id → 200 detail; 404 `NF_TRANSFER_NOT_FOUND`

API-101: Approve Transfer

| 1 | `POST` | 2 | `/api/v1/transfers/{transferId}/approve` | 3 | Approve after source capacity validation | 4 | Bearer | 5 | `PERM-072` (`OPS_DIR`, `COO`) | 6 | `Authorization`, `Content-Type` | 7 | `transferId` | 8 | — | 9 | optional `approvalNotes`, `approvedLines[]` | 10 | VR-132, VR-141; status `REQUESTED`; approver ≠ requester (SoD) | 11 | `CommandResultResponse` + transfer detail | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-012** | 15 | UC-032 | **16** POST approve `{}`
17 `200 { "transferId": "TRF-2026-0042", "status": "APPROVED" }`
18 `403 AUTHZ_INSUFFICIENT_PERMISSION` if requester approves own transfer

API-102: Reject Transfer

| 1 | `POST` | 2 | `/api/v1/transfers/{transferId}/reject` | 3 | Reject transfer with documented reason | 4 | Bearer | 5 | `PERM-072` | 6 | `Authorization`, `Content-Type` | 7 | `transferId` | 8 | — | 9 | `rejectionReason` (required) | 10 | VR-131; status `REQUESTED` | 11 | `CommandResultResponse` | 12 | 400,403,404 | 13 | **200** | 14 | — | 15 | UC-032 | **16–18** POST reject with reason → 200; 400 missing `rejectionReason`

API-103: Dispatch Transfer

| 1 | `POST` | 2 | `/api/v1/transfers/{transferId}/dispatch` | 3 | Dispatch from source; create in-transit state | 4 | Bearer | 5 | `PERM-073` (`WH_MGR`) | 6 | `Authorization`, `Content-Type`, `Idempotency-Key` | 7 | `transferId` | 8 | — | 9 | `dispatchLines[]` (min 1) | 10 | VR-133; status `APPROVED`; source not frozen; dispatched ≤ approved | 11 | `StockTransferDetailResponse` (`status: IN_TRANSIT`) | 12 | 400,403,404,422 | 13 | **200** | 14 | — | 15 | UC-033 | **16–18** POST dispatch lines → 200 in-transit; 422 location frozen

API-104: Confirm Transfer Receipt

| 1 | `POST` | 2 | `/api/v1/transfers/{transferId}/receive` | 3 | Confirm receipt at destination | 4 | Bearer | 5 | `PERM-074` (`WH_MGR`) | 6 | `Authorization`, `Content-Type`, `Idempotency-Key` | 7 | `transferId` | 8 | — | 9 | `receiptLines[]` (min 1); optional `varianceNotes` | 10 | VR-134, VR-135; status `IN_TRANSIT`; variance notes when qty mismatch | 11 | `StockTransferDetailResponse` (`status: RECEIVED`) | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-020** (variance) | 15 | UC-033 | **16–18** POST receipt lines → 200 received; 422 if not in-transit

API-105: Cancel In-Transit Transfer

| 1 | `POST` | 2 | `/api/v1/transfers/{transferId}/cancel` | 3 | Cancel in-transit transfer with reversal | 4 | Bearer | 5 | `PERM-073` | 6 | `Authorization`, `Content-Type` | 7 | `transferId` | 8 | — | 9 | `cancellationReason` (required) | 10 | Eligible in-transit status | 11 | `CommandResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | — | 15 | UC-033 | **16–18** POST cancel → 200; 422 if not cancellable state

API-106: List Pending Transfer Approvals

| 1 | `GET` | 2 | `/api/v1/transfers/pending-approvals` | 3 | Approval queue for transfers | 4 | Bearer | 5 | `PERM-072` | 6 | `Authorization` | 7 | — | 8 | `sourceLocationCode`, `page`, `pageSize` | 9 | — | 10 | VR-136; network scope | 11 | `PagedResponse<StockTransferSummaryResponse>` | 12 | 401,403 | 13 | **200** | 14 | — | 15 | UC-032, UC-047 | **16–18** GET pending queue → 200 paged list

API-107: List In-Transit Transfers

| 1 | `GET` | 2 | `/api/v1/transfers/in-transit` | 3 | In-transit pipeline by location | 4 | Bearer | 5 | `PERM-075` | 6 | `Authorization` | 7 | — | 8 | `locationCode`, `overdueOnly`, `page`, `pageSize` | 9 | — | 10 | VR-137 overdue filter; location scope | 11 | `PagedResponse<InTransitTransferResponse>` | 12 | 401,403 | 13 | **200** | 14 | — | 15 | UC-030, UC-033 | **16–18** GET `?locationCode=WH-EAST&overdueOnly=true` → 200 pipeline

---

M-09 — Inventory Accuracy

API-108: Schedule Cycle Count

| 1 | `POST` | 2 | `/api/v1/accuracy/cycle-counts` | 3 | Schedule cycle count assignment | 4 | Bearer | 5 | `PERM-037` | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `locationCode`, `sku`, `scheduledDate`; optional `priority` | 10 | VR-067, VR-068; active location/product | 11 | `CycleCountAssignmentResponse` | 12 | 400,403,422 | 13 | **201** | 14 | **BR-018**, **BR-032** | 15 | UC-014 | **16–18** POST schedule → 201 assignment id; 422 inactive SKU

API-109: List Cycle Counts

| 1 | `GET` | 2 | `/api/v1/accuracy/cycle-counts` | 3 | List count assignments | 4 | Bearer | 5 | `PERM-037` | 6 | `Authorization` | 7 | — | 8 | `locationCode`, `status`, `startDate`, `endDate`, `page`, `pageSize` | 9 | — | 10 | Date range cross-field; user scope | 11 | `PagedResponse<CycleCountSummaryResponse>` | 12 | 400,401,403 | 13 | **200** | 14 | BR-018 | 15 | UC-014 | **16–18** GET by location/status → 200 paged list

API-110: Get Cycle Count Assignment

| 1 | `GET` | 2 | `/api/v1/accuracy/cycle-counts/{cycleCountId}` | 3 | Blind count assignment (no system qty) | 4 | Bearer | 5 | `PERM-038` | 6 | `Authorization` | 7 | `cycleCountId` | 8 | — | 9 | — | 10 | VR-069; response must not expose expected quantity | 11 | `CycleCountAssignmentResponse` (blind) | 12 | 401,403,404 | 13 | **200** | 14 | **BR-018** | 15 | UC-014 | **17** `{ "cycleCountId": "...", "sku": "BOLT-M10", "locationCode": "WH-EAST", "status": "SCHEDULED" }` (no `expectedQuantity`)
18 404 `NF_CYCLE_COUNT_NOT_FOUND`

API-111: Submit Cycle Count Result

| 1 | `POST` | 2 | `/api/v1/accuracy/cycle-counts/{cycleCountId}/result` | 3 | Submit blind physical count result | 4 | Bearer | 5 | `PERM-038` | 6 | `Authorization`, `Content-Type` | 7 | `cycleCountId` | 8 | — | 9 | `physicalQuantity` (required) | 10 | VR-070; status scheduled/in-progress; counter ≠ approver (SoD) | 11 | `CycleCountResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-018** | 15 | UC-014 | **16** POST `{ "physicalQuantity": { "amount": 98, "unitOfMeasure": "EA" } }`
17 `200` `{ "status": "VARIANCE_DETECTED", "varianceQuantity": { "amount": -2, "unitOfMeasure": "EA" } }`
18 `422` invalid count status

API-112: Initiate Physical Inventory Count

| 1 | `POST` | 2 | `/api/v1/accuracy/physical-counts` | 3 | Wall-to-wall count; impose location freeze | 4 | Bearer | 5 | `PERM-039` | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `locationCode`, `plannedStartDate`, `plannedEndDate` | 10 | VR-071; end ≥ start | 11 | `PhysicalCountResponse` | 12 | 400,403,409,422 | 13 | **201** | 14 | **BR-024** | 15 | UC-015 | **16–18** POST physical count → 201 freeze active; 409 count already in progress

API-113: Get Physical Count Status

| 1 | `GET` | 2 | `/api/v1/accuracy/physical-counts/{physicalCountId}` | 3 | Count progress and freeze window | 4 | Bearer | 5 | `PERM-039` | 6 | `Authorization` | 7 | `physicalCountId` | 8 | — | 9 | — | 10 | Exists | 11 | `PhysicalCountStatusResponse` | 12 | 401,403,404 | 13 | **200** | 14 | **BR-024** | 15 | UC-015 | **17** `{ "physicalCountId": "...", "freezeActive": true, "completionPercent": 45 }`
18 404 not found

API-114: Record Physical Count Result

| 1 | `POST` | 2 | `/api/v1/accuracy/physical-counts/{physicalCountId}/results` | 3 | Submit line results during freeze | 4 | Bearer | 5 | `PERM-039` | 6 | `Authorization`, `Content-Type` | 7 | `physicalCountId` | 8 | — | 9 | `countLines[]` (min 1) | 10 | VR-070, VR-071; location frozen | 11 | `PhysicalCountResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-024** | 15 | UC-015 | **16–18** POST count lines → 200; 422 if freeze not active

API-115: Certify Physical Count

| 1 | `POST` | 2 | `/api/v1/accuracy/physical-counts/{physicalCountId}/certify` | 3 | Certify complete; release freeze | 4 | Bearer | 5 | `PERM-039` | 6 | `Authorization`, `Content-Type` | 7 | `physicalCountId` | 8 | — | 9 | optional `certificationNotes` | 10 | VR-072; all SKUs counted | 11 | `CommandResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-024** | 15 | UC-015 | **16–18** POST certify → 200 freeze released; 422 incomplete count

API-116: Open Adjustment Case

| 1 | `POST` | 2 | `/api/v1/accuracy/adjustment-cases` | 3 | Open case from variance or direct request | 4 | Bearer | 5 | `PERM-043` | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `originType`, `locationCode`, `sku`, `reasonCode`; conditional `varianceId`, direction/qty | 10 | VR-077–VR-079 | 11 | `AdjustmentCaseResponse` | 12 | 400,403,422 | 13 | **201** | 14 | **BR-020**, **BR-022** | 15 | UC-016, UC-017 | **16–18** POST case → 201 caseId

API-117: Get Adjustment Case

| 1 | `GET` | 2 | `/api/v1/accuracy/adjustment-cases/{caseId}` | 3 | Case investigation and approval state | 4 | Bearer | 5 | `PERM-040` | 6 | `Authorization` | 7 | `caseId` | 8 | — | 9 | — | 10 | Exists | 11 | `AdjustmentCaseDetailResponse` | 12 | 401,403,404 | 13 | **200** | 14 | — | 15 | UC-016, UC-017 | **16–18** GET → 200 case detail

API-118: Investigate Variance

| 1 | `POST` | 2 | `/api/v1/accuracy/adjustment-cases/{caseId}/investigate` | 3 | Document root cause investigation | 4 | Bearer | 5 | `PERM-041` | 6 | `Authorization`, `Content-Type` | 7 | `caseId` | 8 | — | 9 | `rootCauseCategory`, `investigationFindings` | 10 | VR-076; findings 1–5000 chars | 11 | `AdjustmentCaseResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-022** | 15 | UC-016 | **16–18** POST investigation → 200; 422 if investigation already complete

API-119: Approve Count Variance

| 1 | `POST` | 2 | `/api/v1/accuracy/adjustment-cases/{caseId}/variances/{varianceId}/approve` | 3 | Approve variance with independent approver | 4 | Bearer | 5 | `PERM-042` | 6 | `Authorization`, `Content-Type` | 7 | `caseId`, `varianceId` | 8 | — | 9 | `approvedAdjustmentQuantity` | 10 | VR-075; approver ≠ counter | 11 | `CommandResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-019** | 15 | UC-016 | **16–18** POST approve variance → 200; 422 `BIZ_SEGREGATION_OF_DUTIES`

API-120: Submit Adjustment for Approval

| 1 | `POST` | 2 | `/api/v1/accuracy/adjustment-cases/{caseId}/submit` | 3 | Submit adjustment/write-off for governance approval | 4 | Bearer | 5 | `PERM-043`, `PERM-045` | 6 | `Authorization`, `Content-Type` | 7 | `caseId` | 8 | — | 9 | `adjustmentType`, `quantity`, `reasonCode`; optional `evidenceReference` | 10 | VR-077, VR-080, VR-081; write-off evidence required | 11 | `AdjustmentCaseResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-020**, **BR-023** | 15 | UC-017, UC-018 | **16–18** POST submit → 200 pending approval

API-121: Approve Adjustment

| 1 | `POST` | 2 | `/api/v1/accuracy/adjustment-cases/{caseId}/approve` | 3 | Approve adjustment after threshold validation | 4 | Bearer | 5 | `PERM-044`, `PERM-046` | 6 | `Authorization`, `Content-Type` | 7 | `caseId` | 8 | — | 9 | optional `approvalNotes`, `approvedQuantity` | 10 | VR-083; approver ≠ initiator and counter | 11 | `CommandResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-021** | 15 | UC-017, UC-018 | **16** POST approve

17 `200` `{ "caseId": "...", "status": "APPROVED" }`

18 `422` `BIZ_SEGREGATION_OF_DUTIES`

API-122: Reject Adjustment

| 1 | `POST` | 2 | `/api/v1/accuracy/adjustment-cases/{caseId}/reject` | 3 | Reject proposed adjustment | 4 | Bearer | 5 | `PERM-044` | 6 | `Authorization, Content-Type` | 7 | `caseId` | 8 | — | 9 | `rejectionReason` (required, 1–2000) | 10 | Case in pending approval | 11 | `CommandResultResponse` | 12 | 400,403,404 | 13 | **200** | 14 | — | 15 | UC-017, UC-018 | **16–18** POST reject → 200

API-123: List Pending Approvals

| 1 | `GET` | 2 | `/api/v1/accuracy/pending-approvals` | 3 | Adjustment cases awaiting approval | 4 | Bearer | 5 | `PERM-044, PERM-046` | 6 | `Authorization` | 7 | — | 8 | `caseType, locationCode, page, pageSize` | 9 | — | 10 | User location scope | 11 | `PagedResponse<AdjustmentCaseSummaryResponse>` | 12 | 401,403 | 13 | **200** | 14 | — | 15 | UC-016, UC-017, UC-047 | **16–18** GET approval queue → 200 paged list

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M-10 — Returns Management

API-124: Issue Return Authorization

| 1 | `POST` | 2 | `/api/v1/returns/authorizations` | 3 | Issue RMA linked to original order | 4 | Bearer | 5 | `PERM-077` (`CS_REP`) | 6 | `Authorization, Content-Type` | 7 | — | 8 | — | 9 | `originalOrderId, returnReason, lines[]` | 10 | VR-142, VR-143; qty ≤ shipped; order return-eligible | 11 | `ReturnAuthorizationResponse` | 12 | 400,403,404,422 | 13 | **201** | 14 | **BR-025** | 15 | UC-035 | **16** POST RMA for order line
17 `201` `{ "returnAuthorizationId": "RMA-2026-001", "status": "AUTHORIZED" }`
18 `422` no valid order reference

API-125: List Return Authorizations

| 1 | `GET` | 2 | `/api/v1/returns/authorizations` | 3 | List RMAs with status filter | 4 | Bearer | 5 | `PERM-082` | 6 | `Authorization` | 7 | — | 8 | `status, originalOrderId, page, pageSize` | 9 | — | 10 | Query filters | 11 | `PagedResponse<ReturnAuthorizationSummaryResponse>` | 12 | 401,403 | 13 | **200** | 14 | BR-025 | 15 | UC-035, UC-036 | **16–18** GET `?status=AUTHORIZED` → 200 paged list

API-126: Get Return Authorization

| 1 | `GET` | 2 | `/api/v1/returns/authorizations/{returnAuthorizationId}` | 3 | RMA detail and order reference | 4 | Bearer | 5 | `PERM-082` | 6 | `Authorization` | 7 | `returnAuthorizationId` | 8 | — | 9 | — | 10 | VR-149; exists | 11 | `ReturnAuthorizationDetailResponse` | 12 | 401,403,404 | 13 | **200** | 14 | **BR-025** | 15 | UC-035 | **16–18** GET → 200 detail; 404 not found

API-127: Cancel Return Authorization

| 1 | `POST` | 2 | `/api/v1/returns/authorizations/{returnAuthorizationId}/cancel` | 3 | Cancel unused RMA | 4 | Bearer | 5 | `PERM-077` | 6 | `Authorization, Content-Type` | 7 | `returnAuthorizationId` | 8 | — | 9 | `cancellationReason` (required) | 10 | Open/unused RMA only | 11 | `CommandResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | — | 15 | UC-035 | **16–18** POST cancel → 200; 422 if already received

API-128: Receive Customer Return

| 1 | `POST` | 2 | `/api/v1/returns/{returnId}/receive` | 3 | Record physical receipt of return | 4 | Bearer | 5 | `PERM-078` | 6 | `Authorization, Content-Type` | 7 | `returnId` | 8 | — | 9 | `receivingLocationCode, receivedLines[]` | 10 | VR-144, VR-145; SKU matches auth; qty ≤ authorized | 11 | `CustomerReturnCaseResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-025** | 15 | UC-036 | **16–18** POST receive lines → 200; 422 qty exceeds authorized

API-129: Inspect Returned Goods

| 1 | `POST` | 2 | `/api/v1/returns/{returnId}/inspect` | 3 | Inspect and record condition grade | 4 | Bearer | 5 | `PERM-079` | 6 | `Authorization, Content-Type` | 7 | `returnId` | 8 | — | 9 | `inspectionLines[]` (min 1) | 10 | VR-146; inspect before restock | 11 | `InspectionResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-026** | 15 | UC-036 | **16–18** POST inspection grades → 200

API-130: Disposition Return

| 1 | `POST` | 2 | `/api/v1/returns/{returnId}/disposition` | 3 | Route to restock, quarantine, or write-off | 4 | Bearer | 5 | `PERM-080`, `PERM-081` | 6 | `Authorization`, `Content-Type` | 7 | `returnId` | 8 | — | 9 | `dispositionAction`, `dispositionLines[]` | 10 | VR-147, VR-148; sum = inspected qty; write-off refs | 11 | `DispositionResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-023**, **BR-026** | 15 | UC-036 | **16–18** POST RESTOCK/WRITE_OFF disposition → 200

API-131: List Pending Return Receipts

| 1 | `GET` | 2 | `/api/v1/returns/pending-receipts` | 3 | Authorized returns awaiting receipt | 4 | Bearer | 5 | `PERM-078` | 6 | `Authorization` | 7 | — | 8 | `locationCode`, `page`, `pageSize` | 9 | — | 10 | User location scope | 11 | `PagedResponse<PendingReturnReceiptResponse>` | 12 | 401,403 | 13 | **200** | 14 | — | 15 | UC-036 | **16–18** GET pending receipts → 200 paged list

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M-11 — Commercial Campaign

API-132: Create Promotional Allocation

| 1 | `POST` | 2 | `/api/v1/campaigns/allocations` | 3 | Create ring-fenced promotional allocation | 4 | Bearer | 5 | `PERM-093` | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `campaignId`, `campaignName`, `locationCode`, `sku`, `allocatedQuantity`, `campaignStartDate`, `campaignEndDate` | 10 | VR-126, VR-127; allocated ≤ ATP; end ≥ start | 11 | `PromotionalAllocationResponse` | 12 | 400,403,422 | 13 | **201** | 14 | **BR-029** | 15 | UC-040 | **16** POST allocation 500 EA ring-fence
17 `201` allocation with `allocationId`
18 `422 BIZ_PROMOTIONAL_RING_FENCE` exceeds ATP

API-133: List Promotional Allocations

| 1 | `GET` | 2 | `/api/v1/campaigns/allocations` | 3 | List allocations by campaign/status | 4 | Bearer | 5 | `PERM-092` | 6 | `Authorization` | 7 | — | 8 | `campaignId`, `status`, `locationCode`, `page`, `pageSize` | 9 | — | 10 | User scope | 11 | `PagedResponse<PromotionalAllocationSummaryResponse>` | 12 | 401,403 | 13 | **200** | 14 | BR-029 | 15 | UC-040 | **16–18** GET allocations → 200 paged list

API-134: Get Promotional Allocation

| 1 | `GET` | 2 | `/api/v1/campaigns/allocations/{allocationId}` | 3 | Allocation detail and ring-fenced qty | 4 | Bearer | 5 | `PERM-092` | 6 | `Authorization` | 7 | `allocationId` | 8 | — | 9 | — | 10 | Exists; user scope | 11 | `PromotionalAllocationDetailResponse` | 12 | 401,403,404 | 13 | **200** | 14 | **BR-029** | 15 | UC-040 | **16–18** GET → 200 detail

API-135: Amend Promotional Allocation

| 1 | `PATCH` | 2 | `/api/v1/campaigns/allocations/{allocationId}` | 3 | Amend quantity or campaign period | 4 | Bearer | 5 | `PERM-093` | 6 | `Authorization`, `Content-Type` | 7 | `allocationId` | 8 | — | 9 | `amendmentReason`; ≥1 of `allocatedQuantity`, dates | 10 | VR-127; increased qty ≤ ATP | 11 | `PromotionalAllocationResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-029** | 15 | UC-040 | **16–18** PATCH amend → 200

API-136: Evaluate Campaign Readiness

| 1 | `POST` | 2 | `/api/v1/campaigns/allocations/{allocationId}/readiness` | 3 | Evaluate stock readiness before launch | 4 | Bearer | 5 | `PERM-092` | 6 | `Authorization`, `Content-Type` | 7 | `allocationId` | 8 | — | 9 | optional `targetQuantity` | 10 | VR-126; defaults to allocated qty | 11 | `CampaignReadinessResponse` | 12 | 400,403,404 | 13 | **200** | 14 | BR-029 | 15 | UC-040, UC-047 | **17** `{ "ready": true, "shortfallQuantity": null }`
18 404 allocation not found

API-137: Release Promotional Allocation

| 1 | `POST` | 2 | `/api/v1/campaigns/allocations/{allocationId}/release` | 3 | Release unconsumed ring-fence after campaign | 4 | Bearer | 5 | `PERM-093` | 6 | `Authorization`, `Content-Type` | 7 | `allocationId` | 8 | — | 9 | `releaseReason` (enum) | 10 | Allocation releasable state | 11 | `CommandResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-029** | 15 | UC-040 | **16–18** POST release → 200 ATP restored

API-138: Authorize Promotional Override

| 1 | `POST` | 2 | `/api/v1/campaigns/allocations/{allocationId}/override` | 3 | Override ring-fence for non-promotional order | 4 | Bearer | 5 | `PERM-093` | 6 | `Authorization`, `Content-Type` | 7 | `allocationId` | 8 | — | 9 | `orderId`, `overrideQuantity`, `authorizationReference`, `justification` | 10 | Override ≤ ring-fenced remaining | 11 | `PromotionalOverrideResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | **BR-029** | 15 | UC-040 | **16–18** POST override → 200; 422 exceeds ring-fence

API-139: Check Ring-Fence Status

| 1 | `GET` | 2 | `/api/v1/campaigns/ring-fence/{locationCode}/{sku}` | 3 | Check active promotional ring-fence | 4 | Bearer | 5 | `PERM-092`, `PERM-051` | 6 | `Authorization` | 7 | `locationCode`, `sku` | 8 | `asOfDate` | 9 | — | 10 | BR-029 active allocation check | 11 | `RingFenceStatusResponse` | 12 | 401,403,404 | 13 | **200** | 14 | **BR-029** | 15 | UC-022, UC-040 | **17** `{ "ringFenced": true, "allocatedQuantity": { "amount": 500, "unitOfMeasure": "EA" } }`
18 404 unknown SKU/location

M-12 — Identity & Access

API-140: Provision User Account

| 1 | `POST` | 2 | `/api/v1/users` | 3 | Provision user with initial roles | 4 | Bearer | 5 | `PERM-095` (`SYS_ADMIN`) | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `userId`, `email`, `displayName`, `initialRoles[]` | 10 | VR-083; unique userId; no SoD in role set | 11 | `UserAccountResponse` | 12 | 400,403,409 | 13 | **201** | 14 | — | 15 | UC-044 | **16–18** POST user → 201; 409 duplicate userId

API-141: List User Accounts

| 1 | `GET` | 2 | `/api/v1/users` | 3 | List users with role/status filters | 4 | Bearer | 5 | `PERM-095` | 6 | `Authorization` | 7 | — | 8 | `status`, `roleCode`, `page`, `pageSize` | 9 | — | 10 | Pagination | 11 | `PagedResponse<UserAccountSummaryResponse>` | 12 | 401,403 | 13 | **200** | 14 | — | 15 | UC-044 | **16–18** GET users → 200 paged list

API-142: Get User Account

| 1 | `GET` | 2 | `/api/v1/users/{userId}` | 3 | User profile and role assignments | 4 | Bearer | 5 | `PERM-095` or self (`PERM-062`) | 6 | `Authorization` | 7 | `userId` | 8 | — | 9 | — | 10 | Exists; self-access allowed | 11 | `UserAccountDetailResponse` | 12 | 401,403,404 | 13 | **200** | 14 | — | 15 | UC-044, UC-045 | **16–18** GET own profile → 200; 403 if non-admin reads other user

API-143: Assign Role

| 1 | `POST` | 2 | `/api/v1/users/{userId}/roles` | 3 | Assign role with location scope | 4 | Bearer | 5 | `PERM-096` | 6 | `Authorization`, `Content-Type` | 7 | `userId` | 8 | — | 9 | `roleAssignment: { roleCode, locationCodes[], effectiveFrom }` | 10 | VR-083; account active; no SoD conflict | 11 | `RoleAssignmentResponse` | 12 | 400,403,404,422 | 13 | **201** | 14 | — | 15 | UC-044, UC-045 | **16** POST `{ "roleAssignment": { "roleCode": "WAREHOUSE_MANAGER", "locationCodes": ["WH-EAST"], "effectiveFrom": "2026-07-06" } }`
17 `201` role assignment id
18 `403 AUTHZ_ROLE_ASSIGNMENT_DENIED`

API-144: Revoke Role Assignment

| 1 | `DELETE` | 2 | `/api/v1/users/{userId}/roles/{roleAssignmentId}` | 3 | Revoke role from user | 4 | Bearer | 5 | `PERM-096` | 6 | `Authorization` | 7 | `userId`, `roleAssignmentId` | 8 | — | 9 | — | 10 | Assignment exists on user | 11 | `CommandResultResponse` | 12 | 401,403,404 | 13 | **200** | 14 | — | 15 | UC-044 | **16–18** DELETE assignment → 200

API-145: Suspend User Account

| 1 | `POST` | 2 | `/api/v1/users/{userId}/suspend` | 3 | Temporarily suspend access | 4 | Bearer | 5 | `PERM-095` | 6 | `Authorization`, `Content-Type` | 7 | `userId` | 8 | — | 9 | `suspensionReason` (required) | 10 | Account active | 11 | `CommandResultResponse` | 12 | 400,403,404,422 | 13 | **200** | 14 | — | 15 | UC-044 | **16–18** POST suspend → 200

API-146: Deactivate User Account

| 1 | `POST` | 2 | `/api/v1/users/{userId}/deactivate` | 3 | Permanently deactivate account | 4 | Bearer | 5 | `PERM-095` | 6 | `Authorization`, `Content-Type` | 7 | `userId` | 8 | — | 9 | `deactivationReason` (required) | 10 | Valid transition | 11 | `CommandResultResponse` | 12 | 400,403,404 | 13 | **200** | 14 | — | 15 | UC-044 | **16–18** POST deactivate → 200

API-147: Sync Users from Directory

| 1 | `POST` | 2 | `/api/v1/users/sync` | 3 | HR directory identity sync | 4 | Bearer | 5 | `PERM-095` | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `syncMode` (`FULL`/`INCREMENTAL`); `sinceTimestamp` when incremental | 10 | Cross-field incremental requires timestamp | 11 | `UserSyncResultResponse` | 12 | 400,403,500 | 13 | **200** | 14 | — | 15 | UC-044 | **16–18** POST `{ "syncMode": "INCREMENTAL", "sinceTimestamp": "2026-07-01T00:00:00Z" }` → 200 sync summary

API-148: Evaluate Authorization

| 1 | `POST` | 2 | `/api/v1/authorization/actions/evaluate` | 3 | Pre-check if user may execute command | 4 | Bearer | 5 | authenticated (`PERM-062` self) | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `action` (required); optional `userId`, `locationCode` | 10 | VR-008; user defaults to caller | 11 | `{ "permitted": true, "denialReason": null }` | 12 | 400,401 | 13 | **200** | 14 | — | 15 | UC-045 | **16–18** POST evaluate → 200 `{ "permitted": false, "denialReason": "AUTHZ_LOCATION_SCOPE_DENIED" }`

API-149: Validate Location Scope

| 1 | `POST` | 2 | `/api/v1/authorization/location-scope/validate` | 3 | Validate user authorized for location | 4 | Bearer | 5 | authenticated | 6 | `Authorization`, `Content-Type` | 7 | — | 8 | — | 9 | `locationCode` (required) | 10 | VR-008, VR-053; location exists | 11 | `{ "authorized": true, "locationCode": "WH-EAST" }` | 12 | 400,401,404 | 13 | **200** | 14 | — | 15 | UC-045 | **16–18** POST validate → 200 authorized flag

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M-13 — Inventory Intelligence

Read-only projections unless noted. Auth=Bearer; Headers=`Authorization`; Body=none for GET; standard errors 400,401,403,500.

API-150: Get Inventory Dashboard

| 1 | `GET` | 2 | `/api/v1/intelligence/dashboard` | 3 | Inventory summary dashboard | 4 | Bearer | 5 | `PERM-083` | 6 | `Authorization` | 7 | — | 8 | `scope` (`NETWORK`/`LOCATION`), `locationCode` (when scope=LOCATION) | 9 | — | 10 | VR-151; location required when scoped | 11 | `InventoryDashboardResponse` | 12 | 400,401,403 | 13 | **200** | 14 | — | 15 | UC-037 | **16–18** GET `?scope=LOCATION&locationCode=WH-EAST` → 200 dashboard widgets

API-151: Get Stock Status Summary

| 1 | `GET` | 2 | `/api/v1/intelligence/stock-status/summary` | 3 | Aggregated on-hand, ATP, status counts | 4 | Bearer | 5 | `PERM-083` | 6 | `Authorization` | 7 | — | 8 | `locationCode`, `categoryCode`, `abcClass` | 9 | — | 10 | VR-151 authorized locations | 11 | `StockStatusSummaryResponse` | 12 | 401,403 | 13 | **200** | 14 | — | 15 | UC-012, UC-037 | **16–18** GET summary → 200 aggregates

API-152: Get Stock Status Board

| 1 | `GET` | 2 | `/api/v1/intelligence/stock-status/board` | 3 | Network stock status health board | 4 | Bearer | 5 | `PERM-085` | 6 | `Authorization` | 7 | — | 8 | `locationCodes[]`, `stockStatus`, `page`, `pageSize` | 9 | — | 10 | VR-151 | 11 | `PagedResponse<StockStatusBoardItemResponse>` | 12 | 401,403 | 13 | **200** | 14 | **BR-009** | 15 | UC-012, UC-037 | **16–18** GET board → 200 paged health indicators

API-153: Get Availability Inquiry Log

| 1 | `GET` | 2 | `/api/v1/intelligence/availability/inquiries` | 3 | Sales availability inquiry accountability log | 4 | Bearer | 5 | `PERM-018` | 6 | `Authorization` | 7 | — | 8 | `startDate`, `endDate`, `sku`, `userId`, `page`, `pageSize` | 9 | — | 10 | VR-062 date range | 11 | `PagedResponse<AvailabilityInquiryLogEntryResponse>` | 12 | 400,401,403 | 13 | **200** | 14 | **BR-001** | 15 | UC-013, UC-021 | **16–18** GET inquiry log → 200 paged entries

API-154: Get Quarantine Register

| 1 | `GET` | 2 | `/api/v1/intelligence/quarantine-register` | 3 | Quarantine and hold register | 4 | Bearer | 5 | `PERM-047` | 6 | `Authorization` | 7 | — | 8 | `locationCode`, `holdReason`, `page`, `pageSize` | 9 | — | 10 | Hold reason enum; location scope | 11 | `PagedResponse<QuarantineRegisterEntryResponse>` | 12 | 401,403 | 13 | **200** | 14 | **BR-013** | 15 | UC-019, UC-037 | **16–18** GET register → 200 quarantine entries

API-155: Search Movement History

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/intelligence/movements`
3	**Purpose**	Search inventory movement history by SKU, location, user, date
4	**Authentication**	Required — Bearer JWT
5	**Required Permission**	`PERM-023`, `PERM-084`
6	**Request Headers**	`Authorization`
7	**Path Parameters**	None
8	**Query Parameters**	`startDate`, `endDate` (required); `sku`, `locationCode`, `userId`, `movementType`, `page`, `pageSize`
9	**Request Body**	None
10	**Validation Rules**	VR-153 date range; VR-154 at least one filter recommended; max 365-day range
11	**Success Response**	`PagedResponse<MovementHistoryEntryResponse>`
12	**Error Responses**	400 invalid date range; 401; 403 location scope
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	Audit history — immutable projection
15	**Related Use Case**	UC-042

16 `GET /api/v1/intelligence/movements?startDate=2026-07-01&endDate=2026-07-06&sku=BOLT-M10`
17 `200` paged movement entries with `movementType`, `quantity`, `performedBy`
18 `400 VAL_INVALID_DATE_RANGE`

API-156: Query Audit Trail

| 1 | `GET` | 2 | `/api/v1/intelligence/audit-trail` | 3 | Immutable audit trail of inventory changes | 4 | Bearer | 5 | `PERM-098` (audit roles only) | 6 | `Authorization` | 7 | — | 8 | `startDate`, `endDate`, `entityType`, `entityId`, `userId`, `page`, `pageSize` | 9 | — | 10 | VR-153; date range required | 11 | `PagedResponse<AuditTrailEntryResponse>` | 12 | 401,403 | 13 | **200** | 14 | S5 sensitive audit access | 15 | UC-043, UC-041 | **18** `403 AUTHZ_INSUFFICIENT_PERMISSION` for operational roles

API-157: Export Audit Evidence

| 1 | `GET` | 2 | `/api/v1/intelligence/audit-trail/export` | 3 | Export audit evidence package | 4 | Bearer | 5 | `PERM-099` | 6 | `Authorization` | 7 | — | 8 | `startDate`, `endDate`, `format` (PDF/CSV), `locationCodes[]` | 9 | — | 10 | VR-153, VR-154; max 365-day range | 11 | `ExportJobResponse` or file stream | 12 | 400,401,403 | 13 | **200** / **202** | 14 | S5 formal audit export | 15 | UC-041, UC-043 | **16–18** GET export → 202 async job id; 400 range too large

API-158: Get Operational Alerts

| 1 | `GET` | 2 | `/api/v1/intelligence/alerts` | 3 | Active operational alerts for user scope | 4 | Bearer | 5 | `PERM-064` | 6 | `Authorization` | 7 | — | 8 | `alertType`, `severity`, `locationCode`, `acknowledged`, `page`, `pageSize` | 9 | — | 10 | Enums; location scope | 11 | `PagedResponse<OperationalAlertResponse>` | 12 | 401,403 | 13 | **200** | 14 | **BR-009** | 15 | UC-028, UC-047 | **16–18** GET alerts → 200 paged alerts

API-159: Acknowledge Operational Alert

| 1 | `POST` | 2 | `/api/v1/intelligence/alerts/{alertId}/acknowledge` | 3 | Acknowledge alert (no stock mutation) | 4 | Bearer | 5 | `PERM-065` | 6 | `Authorization`, `Content-Type` | 7 | `alertId` | 8 | — | 9 | `actionTaken` (`REQUISITION`/`TRANSFER`/`DISMISS`/`DEFER`); `acknowledgmentNotes` when DISMISS/DEFER | 10 | VR-113–VR-115 | 11 | `CommandResultResponse` | 12 | 400,401,403,404 | 13 | **200** | 14 | BR-009 alert workflow | 15 | UC-028, UC-047 | **16** POST `{ "actionTaken": "TRANSFER", "acknowledgmentNotes": "Transfer TRF-0042 raised" }`
17 `200` `{ "alertId": "...", "acknowledged": true }`
18 `400` missing notes for DISMISS

API-160: Get Replenishment Recommendations

| 1 | `GET` | 2 | `/api/v1/intelligence/replenishment/recommendations` | 3 | Replenishment recommendation log | 4 | Bearer | 5 | `PERM-066` | 6 | `Authorization` | 7 | — | 8 | `startDate`, `endDate`, `locationCode`, `recommendationAction`, `page`, `pageSize` | 9 | — | 10 | Date range; location scope | 11 | `PagedResponse<ReplenishmentRecommendationResponse>` | 12 | 400,401,403 | 13 | **200** | 14 | **BR-009**, **BR-011** | 15 | UC-028, UC-029, UC-034 | **16–18** GET recommendations → 200 paged list

API-161: Get Open Purchase Order Board

| 1 | `GET` | 2 | `/api/v1/intelligence/procurement/open-pos` | 3 | Open PO board for inbound pipeline | 4 | Bearer | 5 | `PERM-035` | 6 | `Authorization` | 7 | — | 8 | `locationCode`, `supplierCode`, `overdueOnly`, `page`, `pageSize` | 9 | — | 10 | VR-034 location scope; VR-123 overdue | 11 | `PagedResponse<OpenPurchaseOrderBoardItemResponse>` | 12 | 401,403 | 13 | **200** | 14 | — | 15 | UC-006, UC-030 | **16–18** GET open POs → 200 board

API-162: Get Inbound Supply Calendar

| 1 | `GET` | 2 | `/api/v1/intelligence/procurement/inbound-calendar` | 3 | Expected inbound supply calendar | 4 | Bearer | 5 | `PERM-027` | 6 | `Authorization` | 7 | — | 8 | `startDate`, `endDate` (required), `locationCode` | 9 | — | 10 | VR-033; max 90-day range | 11 | `InboundSupplyCalendarResponse` | 12 | 400,401,403 | 13 | **200** | 14 | — | 15 | UC-007, UC-030 | **16–18** GET calendar → 200 delivery schedule

API-163: Get Inbound Delivery Delays

| 1 | `GET` | 2 | `/api/v1/intelligence/procurement/inbound-delays` | 3 | Inbound delivery delay log | 4 | Bearer | 5 | `PERM-036` | 6 | `Authorization` | 7 | — | 8 | `startDate`, `endDate`, `supplierCode`, `locationCode`, `page`, `pageSize` | 9 | — | 10 | Date range; location scope | 11 | `PagedResponse<InboundDelayLogEntryResponse>` | 12 | 400,401,403 | 13 | **200** | 14 | — | 15 | UC-030, UC-047 | **16–18** GET delays → 200 paged log

API-164: Get Backorder Pipeline

| 1 | `GET` | 2 | `/api/v1/intelligence/fulfillment/backorders` | 3 | Backorder pipeline by policy priority | 4 | Bearer | 5 | `PERM-058` | 6 | `Authorization` | 7 | — | 8 | `locationCode`, `prioritySort`, `page`, `pageSize` | 9 | — | 10 | VR-102; location scope | 11 | `PagedResponse<BackorderPipelineItemResponse>` | 12 | 401,403 | 13 | **200** | 14 | **BR-028** | 15 | UC-025, UC-027 | **16–18** GET backorders → 200 priority-sorted pipeline

API-165: Get Fulfillment Exceptions

| 1 | `GET` | 2 | `/api/v1/intelligence/fulfillment/exceptions` | 3 | Open fulfillment exception log | 4 | Bearer | 5 | `PERM-059` | 6 | `Authorization` | 7 | — | 8 | `exceptionType`, `status`, `startDate`, `endDate`, `page`, `pageSize` | 9 | — | 10 | Enums; date range | 11 | `PagedResponse<FulfillmentExceptionLogEntryResponse>` | 12 | 400,401,403 | 13 | **200** | 14 | **BR-003** | 15 | UC-026, UC-027 | **16–18** GET exceptions → 200 paged log

API-166: Get Network Imbalance Analysis

| 1 | `GET` | 2 | `/api/v1/intelligence/network/imbalance` | 3 | Network imbalance and transfer recommendations | 4 | Bearer | 5 | `PERM-076` | 6 | `Authorization` | 7 | — | 8 | `sku`, `locationCodes[]` | 9 | — | 10 | VR-139; active locations/SKU | 11 | `NetworkImbalanceAnalysisResponse` | 12 | 401,403 | 13 | **200** | 14 | **BR-011** | 15 | UC-034 | **16–18** GET imbalance → 200 surplus/deficit analysis

API-167: Get Shrinkage Analysis

| 1 | `GET` | 2 | `/api/v1/intelligence/accuracy/shrinkage` | 3 | Shrinkage, loss, obsolete stock analysis | 4 | Bearer | 5 | `PERM-049` | 6 | `Authorization` | 7 | — | 8 | `startDate`, `endDate` (required), `locationCode`, `reasonCode` | 9 | — | 10 | VR-086, VR-087; financial sensitivity (S4) | 11 | `ShrinkageAnalysisResponse` | 12 | 401,403 | 13 | **200** | 14 | **BR-020** | 15 | UC-020 | **16–18** GET shrinkage → 200 analysis; 403 for warehouse staff

API-168: Get Count Performance Metrics

| 1 | `GET` | 2 | `/api/v1/intelligence/accuracy/count-performance` | 3 | Cycle/physical count accuracy metrics | 4 | Bearer | 5 | `PERM-086` | 6 | `Authorization` | 7 | — | 8 | `startDate`, `endDate`, `locationCode`, `countType` | 9 | — | 10 | VR-157, VR-158; exclude pending variances default | 11 | `CountPerformanceMetricsResponse` | 12 | 400,401,403 | 13 | **200** | 14 | **BR-018** | 15 | UC-014, UC-015, UC-037 | **16–18** GET metrics → 200 accuracy KPIs

API-169: Get Campaign Readiness Status

| 1 | `GET` | 2 | `/api/v1/intelligence/campaigns/readiness` | 3 | Promotional campaign readiness status | 4 | Bearer | 5 | `PERM-092` | 6 | `Authorization` | 7 | — | 8 | `campaignId`, `daysUntilStart` | 9 | — | 10 | VR-126; campaignId max 50 chars | 11 | `CampaignReadinessStatusResponse` | 12 | 401,403 | 13 | **200** | 14 | **BR-029** | 15 | UC-040, UC-047 | **16–18** GET readiness → 200 campaign status rollup

API-170: Get Inventory KPIs

| 1 | `GET` | 2 | `/api/v1/intelligence/kpis` | 3 | Executive inventory performance KPIs | 4 | Bearer | 5 | `PERM-088` | 6 | `Authorization` | 7 | — | 8 | `startDate`, `endDate`, `locationCodes[]`, `kpiCodes[]` | 9 | — | 10 | VR-151, VR-161, VR-162 | 11 | `InventoryKpiDashboardResponse` | 12 | 400,401,403 | 13 | **200** | 14 | — | 15 | UC-038 | **16–18** GET KPIs → 200 executive metrics

API-171: Get Valuation Snapshots

| 1 | `GET` | 2 | `/api/v1/intelligence/valuation/snapshots` | 3 | Period-end inventory valuation snapshots | 4 | Bearer | 5 | `PERM-090` | 6 | `Authorization` | 7 | — | 8 | `startDate`, `endDate`, `locationCode`, `page`, `pageSize` | 9 | — | 10 | Date range; finance/audit only (S4) | 11 | `PagedResponse<ValuationSnapshotResponse>` | 12 | 401,403 | 13 | **200** | 14 | Unit cost exposure | 15 | UC-039 | **16–18** GET snapshots → 200 paged valuations; 403 ops roles

API-172: Get Valuation Reconciliation

| 1 | `GET` | 2 | `/api/v1/intelligence/valuation/reconciliation` | 3 | Inventory vs. GL reconciliation report | 4 | Bearer | 5 | `PERM-094` (`CFO`, `FIN_MGR`) | 6 | `Authorization` | 7 | — | 8 | `periodEndDate` (required), `varianceThresholdPercent` | 9 | — | 10 | VR period end; threshold 0–100 | 11 | `ValuationReconciliationResponse` | 12 | 400,401,403 | 13 | **200** | 14 | **BR-008** finance alignment | 15 | UC-039 | **16** GET `?periodEndDate=2026-06-30&varianceThresholdPercent=2`
17 `200` reconciliation with variances above threshold
18 `403 AUTHZ_INSUFFICIENT_PERMISSION`

Platform & Integration

API-173: Get Session

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/auth/session`
3	**Purpose**	Validate session and return user context
4	**Authentication**	Required — Bearer JWT (any authenticated user)
5	**Required Permission**	— (authenticated principal)
6	**Request Headers**	`Authorization`
7	**Path Parameters**	None

#	Field	Value
8	**Query Parameters**	`includePermissions` (bool), `includeLocationScope` (bool, default true)
9	**Request Body**	None
10	**Validation Rules**	VR-008 via token
11	**Success Response**	`SessionResponse` — `userId`, `roles[]`, `locationScope[]`, optional `permissions[]`
12	**Error Responses**	401 token expired/invalid
13	**HTTP Status Codes**	**200 OK**
14	**Business Rules Applied**	—
15	**Related Use Case**	UC-045

16 `GET /api/v1/auth/session?includePermissions=true`
17 `{ "userId": "wh.manager", "roles": ["WAREHOUSE_MANAGER"], "locationScope": ["WH-EAST"], "permissions": ["PERM-052"] }`
18 `401 AUTH_TOKEN_EXPIRED`

API-174: Logout

| 1 | `POST` | 2 | `/api/v1/auth/logout` | 3 | Terminate authenticated session | 4 | Bearer | 5 | — (authenticated) | 6 | `Authorization`, optional `Content-Type` | 7 | — | 8 | — | 9 | optional `logoutAllSessions` (bool) | 10 | Empty body accepted | 11 | `{ "message": "Session terminated" }` | 12 | 401 | 13 | **200** | 14 | — | 15 | UC-045 | **16** POST `{}` or `{ "logoutAllSessions": true }`
17 `200` session terminated
18 `401` invalid token

API-175: Health Check

#	Field	Value
1	**HTTP Method**	`GET`
2	**URL**	`/api/v1/health`
3	**Purpose**	Application health for operations monitoring
4	**Authentication**	**None** (public)
5	**Required Permission**	—
6	**Request Headers**	None required
7–9	None	
10	—	
11	`{ "status": "UP", "components": { "database": "UP" } }`	
12	503 when unhealthy	
13	**200 OK** / **503 Service Unavailable**	
14–15	—	

16 `GET /api/v1/health` (no Authorization)
17 `{ "status": "UP" }`
18 `503 { "status": "DOWN" }`

API-176: Import Sales Order (Integration)

#	Field	Value
1	**HTTP Method**	`POST`
2	**URL**	`/api/v1/integrations/sales/orders`
3	**Purpose**	Import customer order from CRM (ACL inbound)

#	Field	Value
4	**Authentication**	**Client Credentials** — `INT_SALES` service principal
5	**Required Permission**	`PERM-054`
6	**Request Headers**	`Authorization: Bearer {client_token}`, `Content-Type`, **`Idempotency-Key`** (required), `X-Correlation-Id`
7–8	None	
9	`externalOrderId`, `externalSystemCode`, `channel`, `lines[]`; optional `shipFromLocationCode`, `correlationId`	
10	VR-007; idempotent on `externalOrderId`; SKU exists; qty > 0; channel ∈ {B2B,B2C,MARKETPLACE}	
11	`CustomerOrderDetailResponse`	
12	400,401,403,409 duplicate external id,422	
13	**201 Created**	
14	**BR-001** (downstream availability)	
15	UC-021, UC-022	

16

```
POST /api/v1/integrations/sales/orders
Idempotency-Key: 7f3c9a2e-4b1d-4e8a-9c0f-2a8b6d4e1f30

{
  "externalOrderId": "CRM-99881",
  "externalSystemCode": "SALESFORCE",
  "channel": "B2B",
  "lines": [{ "externalLineId": "1", "sku": "BOLT-M10", "quantity": { "amount": 5, "unitOfMeasure": "EA" } }]
}
```

17 `201` order with internal `orderId`
18 `401` `AUTH_INTEGRATION_CREDENTIAL_INVALID`

API-177: Get Fulfillment Status Feed

| 1 | `GET` | 2 | `/api/v1/integrations/sales/fulfillment-status` | 3 | Publish fulfillment status to CRM | 4 | Client Credentials (`INT_SALES`) | 5 | `PERM-062` | 6 | `Authorization` | 7 | — | 8 | `externalSystemCode` (required), `externalOrderId`, `updatedSince`, `status`, `cursor`, `limit` | 9 | — | 10 | VR external system registered; ISO `updatedSince` | 11 | `CursorPagedResponse<FulfillmentStatusFeedItem>` | 12 | 401,403 | 13 | **200** | 14 | — | 15 | UC-027 | **16** GET `?externalSystemCode=SALESFORCE&updatedSince=2026-07-01T00:00:00Z`
17 cursor page of order fulfillment statuses
18 `403` unregistered external system

API-178: Import Campaign Plan (Integration)

| 1 | `POST` | 2 | `/api/v1/integrations/marketing/campaign-plans` | 3 | Import campaign plan from marketing system | 4 | Client Credentials (`INT_MKT`) | 5 | `PERM-093` | 6 | `Authorization`, `Content-Type`, `Idempotency-Key` | 7 | — | 8 | — | 9 | `externalCampaignId`, `externalSystemCode`, `campaignName`, `campaignStartDate`, `campaignEndDate`, `allocationLines[]` | 10 | VR-126, VR-127; active SKU/location; end ≥ start | 11 | `PromotionalAllocationBatchResponse` | 12 | 400,401,403,422 | 13 | **201** | 14 | **BR-029** | 15 | UC-040 | **16** POST campaign plan with allocation lines per location/SKU
17 `201` created allocation ids
18 `422` invalid SKU or location on allocation line

Complete API Index (178 Endpoints)

Range	Module	Document
API-001 – 010	M-01 Product Catalog	`Complete-API-Specification.md`
API-011 – 021	M-02 Warehouse Network	`Complete-API-Specification.md`
API-022 – 097	M-03 – M-07	`Complete-API-Specification-Part2.md`
API-098 – 178	M-08 – Platform	`Complete-API-Specification-Part3.md` (this document)

Document Approval

Role	Name	Signature	Date
Senior Backend Architect			
Solution Architect			
Product Owner			

SRS-API-009 Part 3 | July 6, 2026 — Completes all 178 REST API specifications.

Request DTO Specification

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-API-002
Document Title	Request DTO Specification — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Backend Architect
Source Documents	SRS-API-001 (REST API Catalog), SRS-VR-001 (Validation Rules), SRS-DM-002 (Value Objects), SRS-DB (Database Design)
Date	July 6, 2026
Scope	Request DTOs only — path parameters, query parameters, and request bodies. Response DTOs are out of scope.

Purpose

This document defines the **request data transfer object** for every REST API in SRS-API-001. Each field specifies name, data type, required/optional status, validation rules, and business description. Validation rule references use **VR-xxx** from SRS-VR-001 and **BR-xxx** from the Business Rule Catalog where applicable.

Request Composition Conventions

Layer	Convention
Path parameters	Included in every applicable API's Request DTO
Query parameters	Used for `GET` list, search, and filter operations
Request body	Used for `POST`, `PUT`, `PATCH` command operations

Layer	Convention
Headers	`Authorization: Bearer <token>` required on all APIs except API-175; not repeated per DTO
Idempotency	Optional `Idempotency-Key` header (UUID) on financial and reservation commands — documented per API where recommended

Shared Primitive Types

Type	Java/Spring Mapping	Validation
`ProductSku`	`String`	Required when present; max 50 chars; pattern `^[A-Z0-9][A-Z0-9_-]*$` (VR-001, VR-007)
`LocationCode`	`String`	Required when present; max 30 chars; pattern `^[A-Z0-9][A-Z0-9_-]*$` (VR-011)
`SupplierCode`	`String`	Required when present; max 30 chars; pattern `^[A-Z0-9][A-Z0-9_-]*$`
`ReasonCode`	`String`	Required when present; max 20 chars; must reference active reason code (VR-026, VR-027)
`Quantity`	`BigDecimal`	Scale ≤ 4; min 0 (VR-018, VR-023, VR-070)
`PositiveQuantity`	`BigDecimal`	Scale ≤ 4; min exclusive 0 (VR-032, VR-035, VR-066)
`CurrencyCode`	`String`	ISO 4217; exactly 3 uppercase letters
`LocalDate`	`LocalDate`	ISO-8601 `YYYY-MM-DD`
`Instant`	`Instant`	ISO-8601 UTC datetime
`Uuid`	`UUID`	RFC 4122 format
`AbcClass`	`String`	Enum: `A`, `B`, `C` (VR-021)
`LifecycleStatus`	`String`	Enum: `ACTIVE`, `INACTIVE`, `DISCONTINUED` (VR-004–VR-006)
`PageNumber`	`Integer`	Min 1; default 1
`PageSize`	`Integer`	Min 1; max 100; default 20

Shared Nested DTOs

Reused across multiple API request bodies.

`PaginationQuery` (query)

Field	Type	Required	Validation Rules	Description
`page`	Integer	Optional	Min 1; default 1	Page number for paginated results
`pageSize`	Integer	Optional	Min 1; max 100; default 20	Number of records per page
`sortBy`	String	Optional	Max 50 chars; allowed values per resource	Sort field name
`sortDirection`	String	Optional	Enum: `ASC`, `DESC`; default `ASC`	Sort direction

`DateRangeQuery` (query)

Field	Type	Required	Validation Rules	Description
`startDate`	LocalDate	Required	Valid date; on or before `endDate` (VR-062, VR-153)	Period start (inclusive)
`endDate`	LocalDate	Required	Valid date; on or after `startDate`	Period end (inclusive)

`QuantityDto` (body)

Field	Type	Required	Validation Rules	Description
`amount`	BigDecimal	Required	Scale ≤ 4; min 0	Numeric quantity amount
`unitOfMeasure`	String	Required	Max 10 chars; must match product UoM or approved alternate	Unit of measure code

`PositiveQuantityDto` **(body)**

Field	Type	Required	Validation Rules	Description
`amount`	BigDecimal	Required	Scale ≤ 4; min exclusive 0 (VR-032)	Numeric quantity greater than zero
`unitOfMeasure`	String	Required	Max 10 chars	Unit of measure code

`LotIdentityDto` **(body)**

Field	Type	Required	Validation Rules	Description
`lotNumber`	String	Required	Max 50 chars; non-blank (BR-017)	Lot or batch identifier
`expirationDate`	LocalDate	Optional	Future or present when product is shelf-life managed	Lot expiration date for FEFO

`ShelfLifeProfileDto` **(body)**

Field	Type	Required	Validation Rules	Description
`expirationDate`	LocalDate	Required	Valid date; remaining life must meet MRSL (BR-016, VR-035)	Product expiration date at receipt
`manufactureDate`	LocalDate	Optional	On or before `expirationDate`	Manufacture date when available
`remainingShelfLifeDays`	Integer	Optional	Min 0	Remaining shelf life in days at capture

`MonetaryAmountDto` **(body)**

Field	Type	Required	Validation Rules	Description
`amount`	BigDecimal	Required	Scale 2; min 0	Monetary value
`currencyCode`	CurrencyCode	Required	ISO 4217 3-letter code	Currency of the amount

`RoleAssignmentDto` **(body)**

Field	Type	Required	Validation Rules	Description
`roleCode`	String	Required	Max 50 chars; must be valid system role	Role to assign
`locationCodes`	Array<LocationCode>	Optional	Each valid `LocationCode`; empty means network-wide where role permits	Location scope for the role
`effectiveFrom`	LocalDate	Optional	Valid date; default today	Role assignment effective start
`effectiveTo`	LocalDate	Optional	On or after `effectiveFrom` when provided	Role assignment effective end

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M-01 — Product Catalog

API-001 — `POST /api/v1/products`

DTO: `CreateProductRequest`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Unique across catalog (VR-002); duplicate check on similar name/barcode (VR-009)	Stock keeping unit identifier
`name`	String	Required	Max 200 chars; non-blank (VR-001)	Product display name

Field	Type	Required	Validation Rules	Description
`description`	String	Optional	Max 2000 chars	Product description
`unitOfMeasure`	String	Required	Max 10 chars; non-blank (VR-001)	Primary counting and ordering unit
`categoryCode`	String	Optional	Max 30 chars; must exist when provided	Product category reference
`barcode`	String	Optional	Max 50 chars	Primary barcode for operational lookup
`isTraceable`	Boolean	Optional	Default `false`; when `true`, `rotationPolicy` required	Whether lot/batch capture is mandatory
`rotationPolicy`	String	Conditional	Required if `isTraceable=true`; enum: `FIFO`, `FEFO`	Stock rotation discipline
`shelfLifeManaged`	Boolean	Optional	Default `false`	Whether MRSL rules apply at receipt
`mrslThresholdDays`	Integer	Conditional	Required if `shelfLifeManaged=true`; min 1	Minimum remaining shelf life days at receipt

---

API-002 — `GET /api/v1/products`

DTO: `SearchProductsQuery`

Field	Type	Required	Validation Rules	Description
`name`	String	Optional	Max 200 chars; partial match	Filter by product name
`sku`	ProductSku	Optional	Exact or prefix match	Filter by SKU
`categoryCode`	String	Optional	Max 30 chars	Filter by category
`lifecycleStatus`	LifecycleStatus	Optional	Valid enum value	Filter by lifecycle status
`abcClass`	AbcClass	Optional	`A`, `B`, or `C` (VR-021)	Filter by ABC classification
`barcode`	String	Optional	Max 50 chars	Lookup by barcode (VR-007)
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size
`sortBy`	String	Optional	Enum: `sku`, `name`, `categoryCode`, `lifecycleStatus`	Sort field
`sortDirection`	String	Optional	`ASC` or `DESC`	Sort direction

API-003 — `GET /api/v1/products/{sku}`

DTO: `GetProductBySkuRequest`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Path param; must exist (VR-007)	Product SKU to retrieve

---

API-004 — `PATCH /api/v1/products/{sku}`

DTO: `UpdateProductAttributesRequest`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Path param; must exist	Product SKU to update
`name`	String	Optional	Max 200 chars; non-blank when provided	Updated product name
`description`	String	Optional	Max 2000 chars	Updated description

Field	Type	Required	Validation Rules	Description
`unitOfMeasure`	String	Optional	Max 10 chars; non-blank when provided	Updated primary UoM
`categoryCode`	String	Optional	Max 30 chars	Updated category
`barcode`	String	Optional	Max 50 chars	Updated barcode
`isTraceable`	Boolean	Optional	When set `true`, `rotationPolicy` required	Updated traceability flag
`rotationPolicy`	String	Conditional	Enum: `FIFO`, `FEFO`	Updated rotation policy
`shelfLifeManaged`	Boolean	Optional	—	Updated shelf-life management flag
`mrslThresholdDays`	Integer	Conditional	Min 1 when `shelfLifeManaged=true`	Updated MRSL threshold

At least one body field must be present.

API-005 — `POST /api/v1/products/{sku}/lifecycle-status`

DTO: `ChangeProductLifecycleStatusRequest`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Path param; must exist	Product SKU
`newStatus`	LifecycleStatus	Required	Valid transition; `DISCONTINUED` blocks replenishment (VR-004, VR-005)	Target lifecycle status
`reason`	String	Optional	Max 500 chars	Business reason for status change
`effectiveDate`	LocalDate	Optional	Default today	Date status change takes effect

---

API-006 — `PUT /api/v1/products/{sku}/abc-classification`

DTO: `AssignAbcClassificationRequest`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Path param; product must exist (VR-022)	Product SKU
`abcClass`	AbcClass	Required	`A`, `B`, or `C` (VR-021)	ABC priority class
`classificationBasis`	String	Optional	Max 100 chars; enum: `REVENUE`, `VELOCITY`, `CRITICALITY`	Basis used for classification
`effectiveDate`	LocalDate	Optional	Default today	Classification effective date

---

API-007 — `GET /api/v1/products/{sku}/eligibility`

DTO: `ValidateProductEligibilityQuery`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Path param	Product SKU
`transactionType`	String	Required	Enum: `PROCUREMENT`, `RESERVATION`, `FULFILLMENT`, `TRANSFER`	Transaction context to evaluate
`locationCode`	LocationCode	Optional	Required for location-scoped transactions; user must have scope (VR-008)	Location context

---

API-008 — `GET /api/v1/products/categories/{categoryCode}/products`

DTO: `ListProductsByCategoryQuery`

Field	Type	Required	Validation Rules	Description
`categoryCode`	String	Required	Path param; max 30 chars	Category code
`lifecycleStatus`	LifecycleStatus	Optional	Valid enum	Filter by status within category
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-009 — `POST /api/v1/products/duplicate-resolutions`

DTO: `ResolveDuplicateProductConflictRequest`

Field	Type	Required	Validation Rules	Description
`resolutionAction`	String	Required	Enum: `MERGE`, `REJECT_NEW`, `KEEP_BOTH`	Resolution decision
`sourceSku`	ProductSku	Required	Must exist	SKU proposed as duplicate or merge source
`targetSku`	ProductSku	Conditional	Required if `resolutionAction=MERGE`; must exist	Canonical SKU to merge into
`mergeAuthorizationReference`	String	Conditional	Required when both SKUs have on-hand stock (VR-010)	Inventory Controller approval reference
`notes`	String	Optional	Max 1000 chars	Resolution notes

API-010 — `POST /api/v1/products/import`

DTO: `BulkProductImportRequest`

Field	Type	Required	Validation Rules	Description
`importMode`	String	Required	Enum: `VALIDATE_ONLY`, `UPSERT`, `CREATE_ONLY`	Import execution mode
`sourceReference`	String	Optional	Max 100 chars	External file or batch reference
`products`	Array<BulkProductLineDto>	Required	Min 1 item; max 5000 items per request	Product records to import

`BulkProductLineDto`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Unique within batch; catalog rules (VR-001, VR-002)	Product SKU
`name`	String	Required	Max 200 chars	Product name
`unitOfMeasure`	String	Required	Max 10 chars	Primary UoM
`categoryCode`	String	Optional	Max 30 chars	Category code
`barcode`	String	Optional	Max 50 chars	Barcode
`lifecycleStatus`	LifecycleStatus	Optional	Default `ACTIVE`	Initial lifecycle status
`abcClass`	AbcClass	Optional	`A`, `B`, or `C`	ABC classification

---

M-02 — Warehouse Network

API-011 — `POST /api/v1/locations`

DTO: `CreateInventoryLocationRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Unique; max 30 chars (VR-011)	Location identifier
`name`	String	Required	Max 200 chars; non-blank (VR-011)	Location display name
`locationType`	String	Required	Enum: `WAREHOUSE`, `STORE`, `DISTRIBUTION_CENTER`, `TRANSIT` (VR-011)	Location type
`operationalStatus`	String	Required	Enum: `PLANNED`, `ACTIVE`, `CLOSING`, `INACTIVE`	Initial operational status
`parentLocationCode`	LocationCode	Optional	Must not create circular hierarchy (VR-014, VR-015)	Parent location in hierarchy
`addressLine1`	String	Optional	Max 200 chars	Street address line 1
`addressLine2`	String	Optional	Max 200 chars	Street address line 2
`city`	String	Optional	Max 100 chars	City
`stateProvince`	String	Optional	Max 100 chars	State or province
`postalCode`	String	Optional	Max 20 chars	Postal or ZIP code
`countryCode`	String	Optional	ISO 3166-1 alpha-2	Country code
`plannedGoLiveDate`	LocalDate	Optional	Valid date	Planned activation date

API-012 — `GET /api/v1/locations`

DTO: `ListActiveLocationsQuery`

Field	Type	Required	Validation Rules	Description
`operationalStatus`	String	Optional	Enum: `PLANNED`, `ACTIVE`, `CLOSING`, `INACTIVE`; default `ACTIVE`	Filter by status
`locationType`	String	Optional	Valid location type enum	Filter by type
`parentLocationCode`	LocationCode	Optional	Valid location code	Filter by parent
`name`	String	Optional	Max 200 chars; partial match	Filter by name
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-013 — `GET /api/v1/locations/{locationCode}`

DTO: `GetLocationByCodeRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param; must exist; user location scope (VR-053)	Location code

---

API-014 — `PATCH /api/v1/locations/{locationCode}/hierarchy`

DTO: `UpdateLocationHierarchyRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location to update
`parentLocationCode`	LocationCode	Optional	No self/ancestor assignment (VR-014); single parent (VR-015)	New parent location
`zoneCode`	String	Optional	Max 30 chars	Zone within location layout
`layoutNotes`	String	Optional	Max 500 chars	Layout change notes

At least one body field must be present.

API-015 — `POST /api/v1/locations/{locationCode}/storage-positions`

DTO: `ConfigureStoragePositionRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param; location must be active	Parent location
`positionCode`	String	Required	Max 30 chars; unique within location	Storage position identifier
`zone`	String	Optional	Max 30 chars	Zone code
`aisle`	String	Optional	Max 20 chars	Aisle identifier
`bin`	String	Optional	Max 20 chars	Bin identifier
`positionType`	String	Optional	Enum: `BULK`, `PICK_FACE`, `STAGING`, `QUARANTINE`, `BONDED`	Position purpose
`isActive`	Boolean	Optional	Default `true`	Whether position accepts put-away

---

API-016 — `GET /api/v1/locations/{locationCode}/storage-positions`

DTO: `GetStoragePositionsQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code
`positionType`	String	Optional	Valid position type enum	Filter by position type
`zone`	String	Optional	Max 30 chars	Filter by zone
`isActive`	Boolean	Optional	Default `true`	Filter active positions only
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-017 — `POST /api/v1/locations/{locationCode}/activate`

DTO: `ActivateInventoryLocationRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param; opening balances approved when policy applies (VR-013, VR-025)	Location to activate
`openingBalanceApprovalReference`	String	Conditional	Required when go-live policy mandates approved opening balance	Approval reference for opening balances
`activationDate`	LocalDate	Optional	Default today	Effective activation date
`notes`	String	Optional	Max 500 chars	Activation notes

API-018 — `POST /api/v1/locations/{locationCode}/deactivate`

DTO: `DeactivateInventoryLocationRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location to deactivate
`reason`	String	Required	Max 500 chars; non-blank	Reason for deactivation
`effectiveDate`	LocalDate	Optional	Default today	Deactivation effective date

---

API-019 — `POST /api/v1/locations/{locationCode}/close`

DTO: `CloseInventoryLocationRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param; no on-hand stock unless disposition plan (VR-016)	Location to close
`dispositionPlanReference`	String	Conditional	Required when on-hand stock remains	Documented stock disposition plan reference
`closureWaiverReference`	String	Conditional	Authorized waiver when closing with remaining stock	Executive waiver reference
`closureDate`	LocalDate	Optional	Default today	Final closure date
`notes`	String	Optional	Max 1000 chars	Closure notes

API-020 — `GET /api/v1/locations/{locationCode}/freeze-status`

DTO: `GetLocationFreezeStatusRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param; user location scope	Location code

---

API-021 — `GET /api/v1/locations/{locationCode}/operational-status`

DTO: `GetLocationOperationalStatusRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code

M-03 — Inventory Governance

API-022 — `POST /api/v1/governance/reason-codes`

DTO: `CreateReasonCodeRequest`

Field	Type	Required	Validation Rules	Description
`code`	ReasonCode	Required	Unique; max 20 chars; uppercase alphanumeric and underscore	Reason code identifier
`description`	String	Required	Max 200 chars; non-blank	Human-readable description
`transactionTypes`	Array<String>	Required	Min 1; values from: `ADJUSTMENT`, `WRITE_OFF`, `RECEIPT_DISCREPANCY`, `RETURN`, `TRANSFER_EXCEPTION`, `FULFILLMENT_EXCEPTION`	Applicable transaction types
`isActive`	Boolean	Optional	Default `true`	Initial active status

API-023 — `GET /api/v1/governance/reason-codes`

DTO: `GetActiveReasonCodesQuery`

Field	Type	Required	Validation Rules	Description
`transactionType`	String	Optional	Valid transaction type enum	Filter codes applicable to transaction type

Field	Type	Required	Validation Rules	Description
`isActive`	Boolean	Optional	Default `true`	Include only active codes
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-024 — `POST /api/v1/governance/reason-codes/{code}/retire`

DTO: `RetireReasonCodeRequest`

Field	Type	Required	Validation Rules	Description
`code`	ReasonCode	Required	Path param; must exist and be active	Reason code to retire
`retirementReason`	String	Optional	Max 500 chars	Reason for retiring the code
`effectiveDate`	LocalDate	Optional	Default today	Date code becomes inactive for new transactions

---

API-025 — `GET /api/v1/governance/reason-codes/{code}/validate`

DTO: `ValidateReasonCodeQuery`

Field	Type	Required	Validation Rules	Description
`code`	ReasonCode	Required	Path param	Reason code to validate
`transactionType`	String	Required	Valid transaction type enum (VR-026)	Transaction context

---

API-026 — `POST /api/v1/governance/policies`

DTO: `PublishGovernancePolicyRequest`

Field	Type	Required	Validation Rules	Description
`policyName`	String	Required	Max 100 chars; non-blank	Policy version name
`effectiveFrom`	Instant	Required	Must not overlap existing effective version (VR-028)	Policy effective start timestamp
`effectiveTo`	Instant	Optional	After `effectiveFrom`	Policy effective end; null means open-ended
`description`	String	Optional	Max 1000 chars	Policy description
`parameters`	Array<PolicyParameterDto>	Required	Min 1 parameter	Policy parameter definitions

`PolicyParameterDto`

Field	Type	Required	Validation Rules	Description
`parameterKey`	String	Required	Max 50 chars	Parameter identifier
`parameterValue`	String	Required	Max 500 chars	Parameter value
`parameterType`	String	Required	Enum: `QUANTITY`, `MONETARY`, `PERCENTAGE`, `BOOLEAN`, `DURATION`	Value type

---

API-027 — `PATCH /api/v1/governance/policies/{policyId}/approval-thresholds`

DTO: `UpdateApprovalThresholdRequest`

Field	Type	Required	Validation Rules	Description
`policyId`	Uuid	Required	Path param; policy must exist	Governance policy identifier
`quantityThreshold`	BigDecimal	Optional	Min 0 (VR-082)	Quantity above which independent approval required
`monetaryThreshold`	MonetaryAmountDto	Optional	Amount min 0	Financial value threshold for approval
`transactionType`	String	Required	Enum: `ADJUSTMENT`, `WRITE_OFF`, `TRANSFER`, `REQUISITION`	Transaction type threshold applies to
`effectiveFrom`	LocalDate	Required	Valid date (VR-020)	Threshold effective date

---

API-028 — `PATCH /api/v1/governance/policies/{policyId}/receipt-tolerances`

DTO: `UpdateReceiptToleranceRequest`

Field	Type	Required	Validation Rules	Description
`policyId`	Uuid	Required	Path param	Governance policy identifier
`quantityTolerancePercent`	BigDecimal	Optional	Min 0; max 100	Percentage over/under receipt tolerance
`quantityToleranceAbsolute`	BigDecimal	Optional	Min 0	Absolute quantity tolerance
`autoAcceptWithinTolerance`	Boolean	Optional	Default `true` (VR-048)	Auto-accept variances within tolerance
`effectiveFrom`	LocalDate	Required	Valid date	Tolerance effective date

API-029 — `GET /api/v1/governance/approval-thresholds`

DTO: `GetEffectiveApprovalThresholdQuery`

Field	Type	Required	Validation Rules	Description
`transactionType`	String	Required	Valid transaction type enum	Transaction context
`locationCode`	LocationCode	Optional	User location scope	Location-specific threshold lookup
`asOfDate`	LocalDate	Optional	Default today	Effective date for threshold evaluation

---

API-030 — `GET /api/v1/governance/receipt-tolerances`

DTO: `GetReceiptToleranceQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Optional	User location scope	Receiving location context
`supplierCode`	SupplierCode	Optional	Valid supplier when provided	Supplier-specific tolerance
`asOfDate`	LocalDate	Optional	Default today	Effective date for tolerance

---

API-031 — `POST /api/v1/governance/segregation-of-duties/evaluate`

DTO: `EvaluateSegregationOfDutiesRequest`

Field	Type	Required	Validation Rules	Description
`initiatorUserId`	String	Required	Max 50 chars; must exist	User who initiated the transaction
`approverUserId`	String	Required	Max 50 chars; must exist; must differ from initiator (VR-075, VR-083)	Proposed approver user

Field	Type	Required	Validation Rules	Description
`transactionType`	String	Required	Enum: `COUNT_VARIANCE_APPROVAL`, `ADJUSTMENT_APPROVAL`, `WRITE_OFF_APPROVAL`, `ROLE_ASSIGNMENT`	Transaction context for SoD check
`transactionReference`	String	Optional	Max 100 chars	Reference to underlying transaction

---

M-04 — Inventory Balance

API-032 — `POST /api/v1/inventory/opening-balances`

DTO: `EstablishOpeningBalancesRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Location must be in `PLANNED` or pre-activation state (VR-024, VR-025)	Target location for opening balances
`approvalReference`	String	Conditional	Required before location activation (VR-025)	Authorized approval reference
`balanceLines`	Array<OpeningBalanceLineDto>	Required	Min 1 line	Opening balance per SKU

`OpeningBalanceLineDto`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Must exist in catalog (VR-024)	Product SKU
`quantity`	PositiveQuantityDto	Required	Amount min 0 (VR-023)	Verified opening on-hand quantity
`lotIdentity`	LotIdentityDto	Conditional	Required for traceable products	Lot/batch at go-live
`ownershipType`	String	Optional	Enum: `OWNED`, `CONSIGNMENT`; default `OWNED`	Ownership classification

---

API-033 — `GET /api/v1/inventory/stock-positions`

DTO: `GetStockPositionsQuery`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Optional	At least one of `sku` or `locationCode` required (VR-054)	Product filter
`locationCode`	LocationCode	Optional	User location scope (VR-053)	Location filter
`stockStatus`	String	Optional	Enum: `IN_STOCK`, `LOW_STOCK`, `OUT_OF_STOCK`, `OVERSTOCK`	Status filter
`abcClass`	AbcClass	Optional	`A`, `B`, or `C`	ABC class filter
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-034 — `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}`

DTO: `GetStockPositionRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param; user location scope (VR-053)	Location code
`sku`	ProductSku	Required	Path param; must exist (VR-054)	Product SKU

API-035 — `GET /api/v1/inventory/availability/{sku}`

DTO: `GetAvailabilityBySkuQuery`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Path param	Product SKU
`locationCode`	LocationCode	Required	Query param; user location scope (VR-055)	Fulfillment location for ATP inquiry
`requestedQuantity`	BigDecimal	Optional	Min 0	Quantity to evaluate partial availability against

---

API-036 — `GET /api/v1/inventory/availability/{sku}/network`

DTO: `GetNetworkAvailabilityQuery`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Path param	Product SKU
`locationCodes`	Array<LocationCode>	Optional	Subset of user-authorized locations only (VR-064)	Locations to include; default all authorized
`excludeQuarantined`	Boolean	Optional	Default `true` (VR-056)	Exclude quarantined from ATP rollup
`excludePromotionalRingFence`	Boolean	Optional	Default `false`	Exclude promotional ring-fenced quantity

---

API-037 — `GET /api/v1/inventory/stock-status/{locationCode}/{sku}`

DTO: `GetStockStatusIndicatorRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param; stock policy must exist (VR-059)	Location code
`sku`	ProductSku	Required	Path param	Product SKU

API-038 — `POST /api/v1/inventory/stock-policies`

DTO: `DefineStockPolicyRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Active location	Policy location
`sku`	ProductSku	Required	Active product	Policy product
`reorderLevel`	BigDecimal	Required	Min 0 (VR-018)	ATP level triggering replenishment (BR-009)
`minimumThreshold`	BigDecimal	Required	Min 0; must not exceed `reorderLevel` unless inverted policy (VR-019)	Minimum stock threshold
`safetyStock`	BigDecimal	Required	Min 0 (VR-018, BR-010)	Safety stock buffer
`maximumThreshold`	BigDecimal	Optional	Min 0; must be ≥ `reorderLevel` when provided	Overstock threshold
`leadTimeDays`	Integer	Optional	Min 0	Replenishment lead time in business days
`effectiveFrom`	LocalDate	Required	Valid date (VR-020)	Policy effective date

---

API-039 — `GET /api/v1/inventory/stock-policies/{locationCode}/{sku}`

DTO: `GetStockPolicyRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code
`sku`	ProductSku	Required	Path param	Product SKU
`asOfDate`	LocalDate	Optional	Default today	Effective date for policy lookup

---

API-040 — `PATCH /api/v1/inventory/stock-policies/{locationCode}/{sku}`

DTO: `UpdateStockPolicyRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code
`sku`	ProductSku	Required	Path param	Product SKU
`reorderLevel`	BigDecimal	Optional	Min 0 (VR-018)	Updated reorder level
`minimumThreshold`	BigDecimal	Optional	Min 0 (VR-019)	Updated minimum threshold
`safetyStock`	BigDecimal	Optional	Min 0	Updated safety stock
`maximumThreshold`	BigDecimal	Optional	Min 0	Updated maximum threshold
`leadTimeDays`	Integer	Optional	Min 0	Updated lead time
`effectiveFrom`	LocalDate	Required	Valid date; retains history (VR-020)	Effective date for threshold change

At least one threshold field must be present.

API-041 — `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/lots`

DTO: `GetLotBreakdownRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code
`sku`	ProductSku	Required	Path param; product must be traceable	Product SKU
`includeQuarantined`	Boolean	Optional	Default `false`	Include quarantined lot quantities

---

API-042 — `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/transfer-capacity`

DTO: `EvaluateSourceTransferCapacityQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param; source location	Source location code
`sku`	ProductSku	Required	Path param	Product SKU
`requestedQuantity`	BigDecimal	Optional	Min 0; evaluated against max transferable (VR-140)	Quantity to validate against source spare capacity

---

API-043 — `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/reservations`

DTO: `ReserveStockRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param; location active and not frozen	Location code
`sku`	ProductSku	Required	Path param; product active (VR-006)	Product SKU
`quantity`	PositiveQuantityDto	Required	Must not exceed ATP (VR-092); safety stock rules (VR-094, BR-010)	Quantity to reserve
`orderId`	String	Required	Max 50 chars; valid order reference (VR-093)	Customer order identifier
`orderLineId`	String	Required	Max 50 chars	Order line identifier
`reservationExpiry`	Instant	Optional	Future timestamp	Reservation expiry for auto-release
`safetyStockExceptionReference`	String	Conditional	Required when reservation breaches safety stock (VR-124, VR-125)	Authorized safety stock exception

---

API-044 — `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/reservations/{reservationId}/release`

DTO: `ReleaseReservationRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code
`sku`	ProductSku	Required	Path param	Product SKU
`reservationId`	Uuid	Required	Path param; must exist	Reservation identifier
`releaseQuantity`	PositiveQuantityDto	Optional	Must not exceed reserved quantity (VR-095); default full release	Quantity to release
`releaseReason`	String	Required	Enum: `ORDER_CANCELLED`, `LINE_REMOVED`, `QUANTITY_CHANGED`, `FULFILLED`, `POLICY_EXPIRY` (VR-096)	Reason for release

---

API-045 — `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/quarantine-holds`

DTO: `ApplyQuarantineHoldRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code
`sku`	ProductSku	Required	Path param	Product SKU
`quantity`	PositiveQuantityDto	Required	Must not exceed on-hand (VR-084)	Quantity to place on hold
`holdReason`	String	Required	Enum: `QUALITY`, `RECALL`, `CUSTOMS`, `BONDED`, `INVESTIGATION` (BR-013, BR-030)	Hold reason classification
`reasonCode`	ReasonCode	Required	Active reason code (VR-026)	Governance reason code
`lotIdentity`	LotIdentityDto	Conditional	Required for traceable products	Affected lot/batch
`notes`	String	Optional	Max 1000 chars	Hold documentation

API-046 — `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/quarantine-holds/{holdId}/release`

DTO: `ReleaseQuarantineHoldRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code
`sku`	ProductSku	Required	Path param	Product SKU
`holdId`	Uuid	Required	Path param; hold must be active	Quarantine hold identifier
`releaseQuantity`	PositiveQuantityDto	Optional	Must not exceed held quantity; default full release	Quantity to release from hold
`disposition`	String	Required	Enum: `RESTORE_TO_SELLABLE`, `WRITE_OFF`, `RETURN_TO_SUPPLIER` (VR-085)	Post-release disposition
`inspectionOutcomeReference`	String	Conditional	Required for `RESTORE_TO_SELLABLE` (VR-085)	Quality inspection approval reference
`reasonCode`	ReasonCode	Required	Active reason code	Release reason code

---

API-047 — `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/consignment-ownership`

DTO: `TransferConsignmentOwnershipRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code
`sku`	ProductSku	Required	Path param; stock must be consignment (BR-031)	Product SKU
`quantity`	PositiveQuantityDto	Required	Must not exceed consignment on-hand	Quantity to transfer ownership
`consumptionReference`	String	Required	Max 100 chars	Sale or consumption event reference
`lotIdentity`	LotIdentityDto	Conditional	Required for traceable products	Affected lot

---

API-048 — `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/promotional-segregation`

DTO: `SegregatePromotionalStockRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code
`sku`	ProductSku	Required	Path param	Product SKU
`campaignId`	String	Required	Max 50 chars	Promotional campaign identifier
`allocationId`	Uuid	Required	Valid promotional allocation (BR-029)	Allocation triggering segregation
`quantity`	PositiveQuantityDto	Required	Must not exceed available ATP at allocation time (VR-127)	Quantity to ring-fence

Internal/event-driven API — typically invoked by campaign module, not end-user UI.

M-05 — Procurement & Supplier

API-049 — `POST /api/v1/suppliers`

DTO: `RegisterSupplierRequest`

Field	Type	Required	Validation Rules	Description
`supplierCode`	SupplierCode	Required	Unique; max 30 chars	Supplier identifier
`name`	String	Required	Max 200 chars; non-blank	Supplier legal or trading name

Field	Type	Required	Validation Rules	Description
`contactEmail`	String	Optional	Valid email format; max 254 chars	Primary contact email
`contactPhone`	String	Optional	Max 30 chars	Primary contact phone
`paymentTermsDays`	Integer	Required	Min 0	Standard payment terms in days
`defaultLeadTimeDays`	Integer	Optional	Min 0	Default procurement lead time
`isConsignment`	Boolean	Optional	Default `false` (BR-031)	Consignment agreement indicator
`currencyCode`	CurrencyCode	Optional	Default organization currency	Default transaction currency

---

API-050 — `GET /api/v1/suppliers`

DTO: `ListSuppliersQuery`

Field	Type	Required	Validation Rules	Description
`isActive`	Boolean	Optional	Default `true`	Filter by active status
`name`	String	Optional	Max 200 chars; partial match	Filter by supplier name
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-051 — `GET /api/v1/suppliers/{supplierCode}`

DTO: `GetSupplierByCodeRequest`

Field	Type	Required	Validation Rules	Description
`supplierCode`	SupplierCode	Required	Path param; must exist	Supplier code

---

API-052 — `PATCH /api/v1/suppliers/{supplierCode}`

DTO: `UpdateSupplierTermsRequest`

Field	Type	Required	Validation Rules	Description
`supplierCode`	SupplierCode	Required	Path param	Supplier code
`name`	String	Optional	Max 200 chars	Updated supplier name
`contactEmail`	String	Optional	Valid email format	Updated contact email
`contactPhone`	String	Optional	Max 30 chars	Updated contact phone
`paymentTermsDays`	Integer	Optional	Min 0	Updated payment terms
`defaultLeadTimeDays`	Integer	Optional	Min 0	Updated lead time
`isConsignment`	Boolean	Optional	—	Updated consignment flag
`currencyCode`	CurrencyCode	Optional	ISO 4217	Updated currency

At least one body field must be present.

API-053 — `POST /api/v1/suppliers/{supplierCode}/deactivate`

DTO: `DeactivateSupplierRequest`

Field	Type	Required	Validation Rules	Description
`supplierCode`	SupplierCode	Required	Path param; no open POs unless policy allows	Supplier to deactivate
`reason`	String	Required	Max 500 chars	Deactivation reason
`effectiveDate`	LocalDate	Optional	Default today	Effective deactivation date

---

API-054 — `POST /api/v1/procurement/requisitions`

DTO: `CreatePurchaseRequisitionRequest`

Field	Type	Required	Validation Rules	Description
`receivingLocationCode`	LocationCode	Required	Active location (VR-119)	Target receiving location
`justification`	String	Required	Max 1000 chars; non-blank (VR-119)	Business justification
`lines`	Array<RequisitionLineDto>	Required	Min 1 line	Requisition line items
`triggeringAlertId`	Uuid	Optional	Valid alert when replenishment-triggered	Low-stock alert reference

`RequisitionLineDto`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Active product; not discontinued (VR-120, BR-027)	Product to requisition
`quantity`	PositiveQuantityDto	Required	Min exclusive 0 (VR-117)	Requested quantity
`neededByDate`	LocalDate	Optional	Valid future or present date	Required delivery date

---

API-055 — `GET /api/v1/procurement/requisitions/{requisitionId}`

DTO: `GetPurchaseRequisitionRequest`

Field	Type	Required	Validation Rules	Description
`requisitionId`	Uuid	Required	Path param; must exist	Requisition identifier

---

API-056 — `POST /api/v1/procurement/requisitions/{requisitionId}/approve`

DTO: `ApprovePurchaseRequisitionRequest`

Field	Type	Required	Validation Rules	Description
`requisitionId`	Uuid	Required	Path param; status must be submitted	Requisition identifier
`approvalNotes`	String	Optional	Max 1000 chars	Approver notes
`networkTransferDeclined`	Boolean	Optional	Required true when network stock exists and transfer not pursued (VR-118)	Confirms transfer-before-buy evaluation
`networkTransferDeclineReason`	String	Conditional	Required when `networkTransferDeclined=true`	Reason transfer was not selected

---

API-057 — `POST /api/v1/procurement/requisitions/{requisitionId}/reject`

DTO: `RejectPurchaseRequisitionRequest`

Field	Type	Required	Validation Rules	Description
`requisitionId`	Uuid	Required	Path param	Requisition identifier

Field	Type	Required	Validation Rules	Description
`rejectionReason`	String	Required	Max 1000 chars; non-blank (VR-131)	Documented rejection reason

---

API-058 — `POST /api/v1/procurement/requisitions/{requisitionId}/convert-to-po`

DTO: `ConvertRequisitionToPurchaseOrderRequest`

Field	Type	Required	Validation Rules	Description
`requisitionId`	Uuid	Required	Path param; must be approved	Requisition identifier
`supplierCode`	SupplierCode	Required	Active supplier	Selected supplier
`expectedDeliveryDate`	LocalDate	Required	Valid date not in distant past (VR-033)	Expected delivery date
`paymentTermsDays`	Integer	Optional	Overrides supplier default	PO payment terms
`lineOverrides`	Array<PoLineOverrideDto>	Optional	Quantity must not exceed requisition line	Optional line-level overrides

`PoLineOverrideDto`

Field	Type	Required	Validation Rules	Description
`requisitionLineId`	Uuid	Required	Must match requisition line	Requisition line reference
`orderedQuantity`	PositiveQuantityDto	Optional	Min exclusive 0; ≤ requisition quantity	Override ordered quantity
`unitPrice`	MonetaryAmountDto	Optional	Amount min 0	Line unit price

---

API-059 — `POST /api/v1/procurement/purchase-orders`

DTO: `PlacePurchaseOrderRequest`

Field	Type	Required	Validation Rules	Description
`supplierCode`	SupplierCode	Required	Active supplier (VR-030)	Supplier code
`receivingLocationCode`	LocationCode	Required	Active location (VR-031)	Receiving location
`expectedDeliveryDate`	LocalDate	Required	Valid date (VR-033)	Expected delivery date
`currencyCode`	CurrencyCode	Optional	Default supplier currency	PO currency
`paymentTermsDays`	Integer	Optional	Min 0	Payment terms
`lines`	Array<PurchaseOrderLineDto>	Required	Min 1 line (VR-030)	PO line items
`notes`	String	Optional	Max 1000 chars	PO notes

`PurchaseOrderLineDto`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Active product; not discontinued (VR-031, BR-027)	Ordered product
`orderedQuantity`	PositiveQuantityDto	Required	Min exclusive 0 (VR-032)	Ordered quantity
`unitPrice`	MonetaryAmountDto	Optional	Amount min 0	Unit price
`lineNumber`	Integer	Optional	Min 1	Client line sequence

API-060 — `GET /api/v1/procurement/purchase-orders`

DTO: `ListOpenPurchaseOrdersQuery`

Field	Type	Required	Validation Rules	Description
`status`	String	Optional	Enum: `OPEN`, `PARTIAL`, `CLOSED`, `CANCELLED`; default `OPEN,PARTIAL`	PO status filter
`supplierCode`	SupplierCode	Optional	Valid supplier	Supplier filter
`receivingLocationCode`	LocationCode	Optional	User location scope (VR-034)	Receiving location filter
`expectedDeliveryFrom`	LocalDate	Optional	On or before `expectedDeliveryTo`	Expected delivery range start
`expectedDeliveryTo`	LocalDate	Optional	On or after `expectedDeliveryFrom`	Expected delivery range end
`isOverdue`	Boolean	Optional	—	Filter overdue PO lines (VR-123)
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-061 — `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}`

DTO: `GetPurchaseOrderRequest`

Field	Type	Required	Validation Rules	Description
`purchaseOrderId`	Uuid	Required	Path param; must exist	Purchase order identifier

---

API-062 — `PATCH /api/v1/procurement/purchase-orders/{purchaseOrderId}`

DTO: `AmendPurchaseOrderRequest`

Field	Type	Required	Validation Rules	Description
`purchaseOrderId`	Uuid	Required	Path param; authorized user (VR-051)	Purchase order identifier
`expectedDeliveryDate`	LocalDate	Optional	Valid date (VR-033)	Updated expected delivery
`amendmentReason`	String	Required	Max 1000 chars; non-blank	Reason for amendment
`lineAmendments`	Array<PoLineAmendmentDto>	Optional	Cannot reduce below received qty (VR-050)	Line-level amendments

`PoLineAmendmentDto`

Field	Type	Required	Validation Rules	Description
`lineId`	Uuid	Required	Must exist on PO	PO line identifier
`orderedQuantity`	PositiveQuantityDto	Optional	≥ received quantity (VR-050)	Amended ordered quantity
`unitPrice`	MonetaryAmountDto	Optional	Amount min 0	Amended unit price
`action`	String	Optional	Enum: `UPDATE`, `CANCEL_LINE`	Line amendment action

---

API-063 — `POST /api/v1/procurement/purchase-orders/{purchaseOrderId}/cancel`

DTO: `CancelPurchaseOrderRequest`

Field	Type	Required	Validation Rules	Description
`purchaseOrderId`	Uuid	Required	Path param; authorized user (VR-051)	Purchase order identifier
`cancellationReason`	String	Required	Max 1000 chars; non-blank	Cancellation reason
`effectiveDate`	LocalDate	Optional	Default today	Cancellation effective date

---

API-064 — `POST /api/v1/procurement/purchase-orders/{purchaseOrderId}/expected-deliveries`

DTO: `ScheduleInboundDeliveryRequest`

Field	Type	Required	Validation Rules	Description
`purchaseOrderId`	Uuid	Required	Path param; PO must be open	Purchase order identifier
`expectedArrivalDate`	LocalDate	Required	Valid date not in distant past (VR-033)	Expected arrival date
`carrierReference`	String	Optional	Max 100 chars	Carrier or shipment reference
`lineDeliveries`	Array<LineDeliveryScheduleDto>	Optional	Line must exist on PO	Per-line delivery schedule

`LineDeliveryScheduleDto`

Field	Type	Required	Validation Rules	Description
`lineId`	Uuid	Required	Valid PO line	PO line identifier
`expectedQuantity`	PositiveQuantityDto	Required	≤ remaining open quantity	Expected quantity on this delivery
`expectedArrivalDate`	LocalDate	Optional	Overrides header date for line	Line-specific arrival date

---

API-065 — `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}/validate-for-receipt`

DTO: `ValidatePoForReceiptQuery`

Field	Type	Required	Validation Rules	Description
`purchaseOrderId`	Uuid	Required	Path param	Purchase order identifier
`receivingLocationCode`	LocationCode	Optional	Must match PO receiving location when provided	Intended receiving location

---

M-06 — Inbound Receiving

API-066 — `POST /api/v1/inbound/goods-receipts`

DTO: `CreateGoodsReceiptRequest`

Field	Type	Required	Validation Rules	Description
`purchaseOrderId`	Uuid	Conditional	Required unless `adHocAuthorizationReference` provided (VR-040, BR-007)	Linked purchase order
`receivingLocationCode`	LocationCode	Required	Active location; not frozen (VR-012)	Receiving dock location
`supplierCode`	SupplierCode	Optional	Must match PO supplier when PO provided	Supplier reference
`carrierReference`	String	Optional	Max 100 chars	Carrier or ASN reference
`expectedReceiptDate`	LocalDate	Optional	Valid date	Expected receipt date

API-067 — `GET /api/v1/inbound/goods-receipts`

DTO: `ListGoodsReceiptsQuery`

Field	Type	Required	Validation Rules	Description
`receivingLocationCode`	LocationCode	Optional	User location scope (VR-034)	Filter by receiving location

Field	Type	Required	Validation Rules	Description
`status`	String	Optional	Enum: `DRAFT`, `VERIFIED`, `DISCREPANCY`, `PUT_AWAY_PENDING`, `COMPLETED`	Receipt status filter
`purchaseOrderId`	Uuid	Optional	Valid UUID	Filter by PO
`receivedFrom`	LocalDate	Optional	On or before `receivedTo`	Receipt date range start
`receivedTo`	LocalDate	Optional	On or after `receivedFrom`	Receipt date range end
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-068 — `GET /api/v1/inbound/goods-receipts/{goodsReceiptId}`

DTO: `GetGoodsReceiptRequest`

Field	Type	Required	Validation Rules	Description
`goodsReceiptId`	Uuid	Required	Path param; must exist	Goods receipt identifier

---

API-069 — `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/verify`

DTO: `VerifyGoodsReceiptRequest`

Field	Type	Required	Validation Rules	Description
`goodsReceiptId`	Uuid	Required	Path param	Goods receipt identifier
`lines`	Array<ReceiptVerificationLineDto>	Required	Min 1 line (VR-035, VR-036)	Per-line verification results

`ReceiptVerificationLineDto`

Field	Type	Required	Validation Rules	Description
`lineId`	Uuid	Required	Must exist on receipt	Receipt line identifier
`sku`	ProductSku	Required	Must match PO line product (VR-036)	Verified product SKU
`acceptedQuantity`	PositiveQuantityDto	Required	≤ physically verified qty (VR-035)	Accepted quantity
`rejectedQuantity`	QuantityDto	Optional	Min 0; default 0 (VR-037)	Rejected or damaged quantity
`conditionGrade`	String	Optional	Enum: `NEW`, `DAMAGED`, `EXPIRED`	Condition at receipt
`lotIdentity`	LotIdentityDto	Conditional	Required for traceable products (BR-017)	Lot/batch captured at receipt
`shelfLifeProfile`	ShelfLifeProfileDto	Conditional	Required for shelf-life products; MRSL check (BR-016)	Shelf-life attributes

---

API-070 — `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies`

DTO: `RecordReceiptDiscrepancyRequest`

Field	Type	Required	Validation Rules	Description
`goodsReceiptId`	Uuid	Required	Path param	Goods receipt identifier
`lineId`	Uuid	Required	Must exist on receipt	Affected receipt line
`discrepancyType`	String	Required	Enum: `SHORT_SHIPMENT`, `OVER_SHIPMENT`, `WRONG_ITEM`, `DAMAGE`, `SHELF_LIFE_NON_COMPLIANCE` (VR-038)	Discrepancy classification
`varianceQuantity`	PositiveQuantityDto	Required	Min exclusive 0	Variance quantity

Field	Type	Required	Validation Rules	Description
`reasonCode`	ReasonCode	Required	Active code (VR-026)	Reason code
`notes`	String	Required	Max 2000 chars; non-blank (VR-038)	Discrepancy documentation
`evidenceReference`	String	Optional	Max 200 chars	Photo or document reference

API-071 — `GET /api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies`

DTO: `ListReceiptDiscrepanciesQuery`

Field	Type	Required	Validation Rules	Description
`goodsReceiptId`	Uuid	Required	Path param	Goods receipt identifier
`status`	String	Optional	Enum: `OPEN`, `RESOLVED`; default `OPEN`	Discrepancy status filter

API-072 — `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies/{discrepancyId}/resolve`

DTO: `ResolveReceiptDiscrepancyRequest`

Field	Type	Required	Validation Rules	Description
`goodsReceiptId`	Uuid	Required	Path param	Goods receipt identifier
`discrepancyId`	Uuid	Required	Path param; must be open	Discrepancy identifier
`resolutionOutcome`	String	Required	Enum: `ACCEPT_VARIANCE`, `REJECT_QUANTITY`, `SUPPLIER_CLAIM`, `CREDIT_NOTE`	Resolution decision
`resolvedQuantity`	QuantityDto	Optional	Min 0	Quantity accepted after resolution
`resolutionNotes`	String	Required	Max 2000 chars; non-blank	Resolution documentation
`supplierClaimReference`	String	Optional	Max 100 chars	Supplier claim reference

API-073 — `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/put-away`

DTO: `CompletePutAwayRequest`

Field	Type	Required	Validation Rules	Description
`goodsReceiptId`	Uuid	Required	Path param; receipt must be verified	Goods receipt identifier
`putAwayLines`	Array<PutAwayLineDto>	Required	Min 1 line (VR-043, BR-005)	Put-away line assignments

`PutAwayLineDto`

Field	Type	Required	Validation Rules	Description
`receiptLineId`	Uuid	Required	Must exist on receipt	Receipt line reference
`quantity`	PositiveQuantityDto	Required	≤ unreceived put-away remaining qty (VR-043)	Quantity put away
`storagePositionCode`	String	Required	Max 30 chars; must exist at location	Target storage position
`lotIdentity`	LotIdentityDto	Conditional	Required for traceable products	Lot placed in position

API-074 — `GET /api/v1/inbound/pending-put-away`

DTO: `ListPendingPutAwayQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	User location scope	Receiving location
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-075 — `POST /api/v1/inbound/ad-hoc-receipts`

DTO: `ReceiveAdHocInboundRequest`

Field	Type	Required	Validation Rules	Description
`receivingLocationCode`	LocationCode	Required	Active location (VR-042)	Receiving location
`adHocAuthorizationReference`	String	Required	Max 100 chars; authorized approver (VR-042)	Governance authorization reference
`reasonCode`	ReasonCode	Required	Active code (VR-026)	Ad-hoc receipt reason
`sourceDescription`	String	Required	Max 500 chars	Source of inbound stock
`lines`	Array<AdHocReceiptLineDto>	Required	Min 1 line	Receipt lines

`AdHocReceiptLineDto`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Active product	Product received
`acceptedQuantity`	PositiveQuantityDto	Required	Min exclusive 0 (VR-035)	Accepted quantity
`lotIdentity`	LotIdentityDto	Conditional	Required for traceable products	Lot/batch
`storagePositionCode`	String	Optional	Valid position at location	Target storage position

---

API-076 — `POST /api/v1/inbound/return-receipts`

DTO: `ReceiveReturnInboundRequest`

Field	Type	Required	Validation Rules	Description
`returnAuthorizationId`	Uuid	Required	Valid open RMA (VR-045, BR-025)	Return authorization identifier
`receivingLocationCode`	LocationCode	Required	Active location	Receiving location
`carrierReference`	String	Optional	Max 100 chars	Return shipment carrier reference
`lines`	Array<ReturnReceiptLineDto>	Required	Min 1 line	Return receipt lines

`ReturnReceiptLineDto`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Must match RMA line (VR-145)	Returned product
`receivedQuantity`	PositiveQuantityDto	Required	≤ authorized return qty (VR-046)	Received quantity
`conditionGrade`	String	Optional	Enum: `NEW`, `OPENED`, `DAMAGED`	Initial condition observation

---

API-077 — `POST /api/v1/inbound/supplier-invoices`

DTO: `RegisterSupplierInvoiceRequest`

Field	Type	Required	Validation Rules	Description
`supplierCode`	SupplierCode	Required	Active supplier	Supplier code
`invoiceNumber`	String	Required	Max 50 chars; unique per supplier	Supplier invoice number
`invoiceDate`	LocalDate	Required	Valid date	Invoice date
`purchaseOrderId`	Uuid	Required	Valid PO linked to supplier	Purchase order reference
`goodsReceiptId`	Uuid	Optional	Must reference same PO when provided	Goods receipt reference
`currencyCode`	CurrencyCode	Required	ISO 4217	Invoice currency
`lines`	Array<InvoiceLineDto>	Required	Min 1 line	Invoice line items

`InvoiceLineDto`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Must exist on PO/receipt	Invoiced product
`quantity`	PositiveQuantityDto	Required	Min exclusive 0	Invoiced quantity
`unitPrice`	MonetaryAmountDto	Required	Amount min 0	Unit price
`poLineId`	Uuid	Optional	Must match PO line	PO line reference

API-078 — `GET /api/v1/inbound/supplier-invoices/{invoiceId}`

DTO: `GetSupplierInvoiceRequest`

Field	Type	Required	Validation Rules	Description
`invoiceId`	Uuid	Required	Path param; must exist	Supplier invoice identifier

---

API-079 — `POST /api/v1/inbound/supplier-invoices/{invoiceId}/three-way-match`

DTO: `PerformThreeWayMatchRequest`

Field	Type	Required	Validation Rules	Description
`invoiceId`	Uuid	Required	Path param	Supplier invoice identifier
`toleranceOverrideReference`	String	Optional	Max 100 chars; authorized when exceeding tolerance (VR-048)	Tolerance override authorization
`forceReevaluation`	Boolean	Optional	Default `false`	Force re-run of match evaluation

---

API-080 — `POST /api/v1/inbound/supplier-invoices/{invoiceId}/authorize-payment`

DTO: `AuthorizeInvoicePaymentRequest`

Field	Type	Required	Validation Rules	Description
`invoiceId`	Uuid	Required	Path param; three-way match must be completed (BR-008)	Supplier invoice identifier
`authorizationNotes`	String	Optional	Max 1000 chars	Payment authorization notes
`paymentDueDate`	LocalDate	Optional	On or after invoice date	Scheduled payment date

M-07 — Order Fulfillment

API-081 — `POST /api/v1/fulfillment/orders`

DTO: `CreateCustomerOrderRequest`

Field	Type	Required	Validation Rules	Description
`externalOrderReference`	String	Optional	Max 100 chars	CRM or channel order reference
`customerReference`	String	Optional	Max 100 chars	Customer account reference
`channel`	String	Required	Enum: `B2B`, `B2C`, `INTERNAL`, `MARKETPLACE`	Sales channel
`shipFromLocationCode`	LocationCode	Optional	Active location when specified	Preferred fulfillment location
`requestedShipDate`	LocalDate	Optional	Valid date	Requested ship date
`lines`	Array<OrderLineDto>	Required	Min 1 line	Order line items

`OrderLineDto`

Field	Type	Required	Validation Rules	Description
`lineNumber`	Integer	Required	Min 1	Client line sequence
`sku`	ProductSku	Required	Active product (VR-006)	Ordered product
`requestedQuantity`	PositiveQuantityDto	Required	Min exclusive 0	Requested order quantity
`substitutionAllowed`	Boolean	Optional	Default `false`	Whether substitution is permitted

API-082 — `GET /api/v1/fulfillment/orders`

DTO: `ListCustomerOrdersQuery`

Field	Type	Required	Validation Rules	Description
`status`	String	Optional	Enum: `DRAFT`, `AVAILABILITY_CHECKED`, `CONFIRMED`, `PICKING`, `SHIPPED`, `CANCELLED`, `EXCEPTION`	Order status filter
`shipFromLocationCode`	LocationCode	Optional	User location scope	Fulfillment location filter
`customerReference`	String	Optional	Max 100 chars	Customer reference filter
`externalOrderReference`	String	Optional	Max 100 chars	External order reference
`orderDateFrom`	LocalDate	Optional	On or before `orderDateTo`	Order date range start
`orderDateTo`	LocalDate	Optional	On or after `orderDateFrom`	Order date range end
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-083 — `GET /api/v1/fulfillment/orders/{orderId}`

DTO: `GetCustomerOrderRequest`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param; must exist (VR-108)	Customer order identifier

---

API-084 — `POST /api/v1/fulfillment/orders/{orderId}/availability`

DTO: `ConfirmOrderAvailabilityRequest`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param	Customer order identifier
`locationCode`	LocationCode	Optional	Default order ship-from location; ATP-based (VR-090, BR-001)	Location for availability check
`includeInboundSupply`	Boolean	Optional	Default `false` (VR-091)	Include approved inbound in commitment calculation
`lineOverrides`	Array<AvailabilityLineCheckDto>	Optional	Per-line location override	Per-line availability check parameters

`AvailabilityLineCheckDto`

Field	Type	Required	Validation Rules	Description
`lineId`	Uuid	Required	Must exist on order	Order line identifier
`locationCode`	LocationCode	Optional	User location scope	Line-specific fulfillment location
`requestedQuantity`	PositiveQuantityDto	Optional	Defaults to line quantity	Quantity to check

---

API-085 — `POST /api/v1/fulfillment/orders/{orderId}/confirm`

DTO: `ConfirmCustomerOrderRequest`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param; availability must be confirmed	Customer order identifier
`confirmationNotes`	String	Optional	Max 1000 chars	Confirmation notes
`reservationExpiry`	Instant	Optional	Future timestamp	Reservation expiry for all lines
`safetyStockExceptionReference`	String	Conditional	Required when reservation breaches safety stock (VR-124)	Safety stock override authorization

---

API-086 — `POST /api/v1/fulfillment/orders/{orderId}/cancel`

DTO: `CancelCustomerOrderRequest`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param	Customer order identifier
`cancellationReason`	String	Required	Max 1000 chars; non-blank (VR-096)	Cancellation reason
`lineIds`	Array<Uuid>	Optional	Must exist on order; empty means full order cancel	Specific lines to cancel

---

API-087 — `POST /api/v1/fulfillment/orders/{orderId}/multi-location-plan`

DTO: `CreateMultiLocationFulfillmentPlanRequest`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param	Customer order identifier
`allocations`	Array<LocationAllocationDto>	Required	Min 1; sum must equal line qty (VR-106)	Per-location line allocations

`LocationAllocationDto`

Field	Type	Required	Validation Rules	Description
`orderLineId`	Uuid	Required	Must exist on order	Order line identifier
`locationCode`	LocationCode	Required	ATP must suffice (VR-107)	Ship-from location
`quantity`	PositiveQuantityDto	Required	≤ location ATP (VR-107)	Allocated quantity

---

API-088 — `POST /api/v1/fulfillment/orders/{orderId}/pick-lists`

DTO: `AssignPickListRequest`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param; order must be confirmed	Customer order identifier
`assignedToUserId`	String	Optional	Max 50 chars; valid user	Picker user assignment
`priority`	String	Optional	Enum: `NORMAL`, `HIGH`, `URGENT`	Pick priority
`pickStrategy`	String	Optional	Enum: `FIFO`, `FEFO`; default per product policy (BR-014, BR-015)	Rotation strategy override

---

API-089 — `GET /api/v1/fulfillment/orders/{orderId}/pick-lists/awaiting`

DTO: `ListOrdersAwaitingPickQuery`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param	Customer order identifier
`locationCode`	LocationCode	Optional	User location scope	Filter by fulfillment location

---

API-090 — `POST /api/v1/fulfillment/orders/{orderId}/picks/{pickId}/verify`

DTO: `VerifyPickRequest`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param	Customer order identifier
`pickId`	Uuid	Required	Path param	Pick list identifier
`pickLines`	Array<PickVerificationLineDto>	Required	Min 1 line (VR-097, BR-003)	Verified pick lines

`PickVerificationLineDto`

Field	Type	Required	Validation Rules	Description
`orderLineId`	Uuid	Required	Must exist on order	Order line identifier
`pickedQuantity`	PositiveQuantityDto	Required	≤ reserved quantity (VR-098)	Verified picked quantity
`sku`	ProductSku	Required	Must match order line or approved substitute (VR-099, VR-110)	Picked product SKU
`lotIdentity`	LotIdentityDto	Conditional	Required for traceable products (BR-017)	Picked lot/batch

Field	Type	Required	Validation Rules	Description
substitutionReference	String	Conditional	Required when SKU differs from order line (VR-110, VR-111)	Approved substitution reference

---

API-091 — POST /api/v1/fulfillment/orders/{orderId}/pack

DTO: PackOrderRequest

Field	Type	Required	Validation Rules	Description
orderId	Uuid	Required	Path param; pick must be verified	Customer order identifier
packages	Array<PackageDto>	Required	Min 1 package	Packing details

PackageDto

Field	Type	Required	Validation Rules	Description
packageNumber	Integer	Required	Min 1	Package sequence
weight	BigDecimal	Optional	Min 0	Package weight
weightUnit	String	Optional	Enum: KG, LB	Weight unit
lineItems	Array<PackageLineDto>	Required	Min 1 item	Contents of package

PackageLineDto

Field	Type	Required	Validation Rules	Description
orderLineId	Uuid	Required	Must exist on order	Order line in package
quantity	PositiveQuantityDto	Required	≤ picked quantity	Quantity in package

---

API-092 — POST /api/v1/fulfillment/orders/{orderId}/shipments

DTO: DispatchShipmentRequest

Field	Type	Required	Validation Rules	Description
orderId	Uuid	Required	Path param; pick verified (VR-097, BR-003)	Customer order identifier
carrierCode	String	Optional	Max 30 chars	Carrier identifier
trackingNumber	String	Optional	Max 100 chars	Shipment tracking number
shippedAt	Instant	Optional	Default now	Shipment dispatch timestamp
shipmentLines	Array<ShipmentLineDto>	Required	Min 1 line; ≤ reserved qty (VR-098)	Shipped line quantities

ShipmentLineDto

Field	Type	Required	Validation Rules	Description
orderLineId	Uuid	Required	Must exist on order	Order line identifier
shippedQuantity	PositiveQuantityDto	Required	≤ picked and reserved quantity	Shipped quantity
lotIdentity	LotIdentityDto	Conditional	Required for traceable products	Shipped lot/batch

---

API-093 — POST /api/v1/fulfillment/orders/{orderId}/partial-fulfillment

DTO: CompletePartialFulfillmentRequest

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param	Customer order identifier
`fulfillmentLines`	Array<PartialFulfillmentLineDto>	Required	Min 1 line (VR-100)	Lines to partially fulfill now

`PartialFulfillmentLineDto`

Field	Type	Required	Validation Rules	Description
`orderLineId`	Uuid	Required	Must exist on order	Order line identifier
`fulfillQuantity`	PositiveQuantityDto	Required	> 0 and < full line qty (VR-100)	Quantity to ship now
`createBackorderForRemainder`	Boolean	Required	Default `true`	Whether remainder goes to backorder
`customerAcknowledgmentReference`	String	Conditional	Required when `createBackorderForRemainder=true` (VR-101, BR-028)	Customer acknowledgment reference

---

API-094 — `POST /api/v1/fulfillment/orders/{orderId}/backorders`

DTO: `RecordBackorderRequest`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param	Customer order identifier
`backorderLines`	Array<BackorderLineDto>	Required	Min 1 line (VR-103)	Backorder line items

`BackorderLineDto`

Field	Type	Required	Validation Rules	Description
`orderLineId`	Uuid	Required	Must exist on order	Order line identifier
`backorderQuantity`	PositiveQuantityDto	Required	Equals unfulfilled remainder (VR-103)	Backordered quantity
`customerAcknowledgmentReference`	String	Required	Max 100 chars (VR-101, BR-028)	Customer acknowledgment reference
`expectedFulfillmentDate`	LocalDate	Optional	Valid date	Expected fulfillment date

---

API-095 — `POST /api/v1/fulfillment/backorders/{backorderId}/fulfill`

DTO: `FulfillBackorderRequest`

Field	Type	Required	Validation Rules	Description
`backorderId`	Uuid	Required	Path param; must be open (VR-102)	Backorder identifier
`fulfillQuantity`	PositiveQuantityDto	Optional	≤ backorder quantity; default full	Quantity to fulfill now
`locationCode`	LocationCode	Optional	ATP must suffice (VR-092)	Fulfillment location
`priorityOverrideReference`	String	Conditional	Required when fulfilling out of priority sequence (VR-102)	Priority override authorization

---

API-096 — `POST /api/v1/fulfillment/orders/{orderId}/exceptions`

DTO: `RaiseFulfillmentExceptionRequest`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param	Customer order identifier
`exceptionType`	String	Required	Enum: `PICK_DISCREPANCY`, `SHELF_MISMATCH`, `DAMAGED_GOODS`, `SHORT_PICK` (VR-104)	Exception classification
`affectedLineId`	Uuid	Optional	Must exist on order when provided	Affected order line
`affectedQuantity`	PositiveQuantityDto	Optional	Min exclusive 0 when provided	Affected quantity
`description`	String	Required	Max 2000 chars; non-blank (VR-104)	Exception description
`proposedResolution`	String	Optional	Enum: `RE_PICK`, `PARTIAL_SHIP`, `CANCEL_LINE`, `ADJUST_INVENTORY`	Proposed resolution action

API-097 — `GET /api/v1/fulfillment/orders/{orderId}/status`

DTO: `GetOrderFulfillmentStatusRequest`

Field	Type	Required	Validation Rules	Description
`orderId`	Uuid	Required	Path param; must exist (VR-108)	Customer order identifier
`includeLineDetail`	Boolean	Optional	Default `true`	Include per-line status breakdown

M-08 — Network Transfer

API-098 — `POST /api/v1/transfers`

DTO: `RequestStockTransferRequest`

Field	Type	Required	Validation Rules	Description
`sourceLocationCode`	LocationCode	Required	Active; distinct from destination (VR-129)	Source location
`destinationLocationCode`	LocationCode	Required	Active; distinct from source (VR-129)	Destination location
`justification`	String	Required	Max 1000 chars; non-blank (VR-128)	Business justification
`lines`	Array<TransferLineDto>	Required	Min 1 line	Transfer line items

`TransferLineDto`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Active product	Product to transfer
`quantity`	PositiveQuantityDto	Required	≤ source ATP (VR-130); respects source minimum (VR-140, BR-012)	Transfer quantity
`lotIdentity`	LotIdentityDto	Conditional	Required for traceable products (BR-017)	Lot/batch to transfer

API-099 — `GET /api/v1/transfers`

DTO: `ListStockTransfersQuery`

Field	Type	Required	Validation Rules	Description
`status`	String	Optional	Enum: `REQUESTED`, `APPROVED`, `DISPATCHED`, `IN_TRANSIT`, `RECEIVED`, `CANCELLED`, `REJECTED`	Transfer status filter
`sourceLocationCode`	LocationCode	Optional	User location scope	Source location filter

Field	Type	Required	Validation Rules	Description
`destinationLocationCode`	LocationCode	Optional	User location scope	Destination location filter
`sku`	ProductSku	Optional	Valid SKU	Product filter
`requestedFrom`	LocalDate	Optional	On or before `requestedTo`	Request date range start
`requestedTo`	LocalDate	Optional	On or after `requestedFrom`	Request date range end
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-100 — `GET /api/v1/transfers/{transferId}`

DTO: `GetStockTransferRequest`

Field	Type	Required	Validation Rules	Description
`transferId`	Uuid	Required	Path param; must exist	Transfer identifier

---

API-101 — `POST /api/v1/transfers/{transferId}/approve`

DTO: `ApproveTransferRequest`

Field	Type	Required	Validation Rules	Description
`transferId`	Uuid	Required	Path param; status `REQUESTED`	Transfer identifier
`approvalNotes`	String	Optional	Max 1000 chars	Approval notes
`approvedLines`	Array<ApprovedTransferLineDto>	Optional	Revalidated against source ATP (VR-132, VR-141)	Approved line quantities

`ApprovedTransferLineDto`

Field	Type	Required	Validation Rules	Description
`lineId`	Uuid	Required	Must exist on transfer	Transfer line identifier
`approvedQuantity`	PositiveQuantityDto	Required	≤ requested and ≤ max transferable (VR-141)	Approved transfer quantity

---

API-102 — `POST /api/v1/transfers/{transferId}/reject`

DTO: `RejectTransferRequest`

Field	Type	Required	Validation Rules	Description
`transferId`	Uuid	Required	Path param	Transfer identifier
`rejectionReason`	String	Required	Max 1000 chars; non-blank (VR-131)	Documented rejection reason

---

API-103 — `POST /api/v1/transfers/{transferId}/dispatch`

DTO: `DispatchTransferRequest`

Field	Type	Required	Validation Rules	Description
`transferId`	Uuid	Required	Path param; status `APPROVED`; source not frozen	Transfer identifier

Field	Type	Required	Validation Rules	Description
`dispatchedAt`	Instant	Optional	Default now	Dispatch timestamp
`carrierReference`	String	Optional	Max 100 chars	Carrier or shipment reference
`dispatchLines`	Array<DispatchLineDto>	Required	Min 1 line; ≤ approved qty (VR-133)	Dispatched line quantities

`DispatchLineDto`

Field	Type	Required	Validation Rules	Description
`lineId`	Uuid	Required	Must exist on transfer	Transfer line identifier
`dispatchedQuantity`	PositiveQuantityDto	Required	≤ approved quantity (VR-133)	Dispatched quantity
`lotIdentity`	LotIdentityDto	Conditional	Required for traceable products	Dispatched lot/batch

---

API-104 — `POST /api/v1/transfers/{transferId}/receive`

DTO: `ConfirmTransferReceiptRequest`

Field	Type	Required	Validation Rules	Description
`transferId`	Uuid	Required	Path param; status `IN_TRANSIT`	Transfer identifier
`receivedAt`	Instant	Optional	Default now	Receipt timestamp
`receiptLines`	Array<TransferReceiptLineDto>	Required	Min 1 line (VR-134, VR-135)	Received line quantities

`TransferReceiptLineDto`

Field	Type	Required	Validation Rules	Description
`lineId`	Uuid	Required	Must exist on transfer	Transfer line identifier
`receivedQuantity`	QuantityDto	Required	Min 0; variance triggers investigation (VR-135)	Received quantity
`storagePositionCode`	String	Optional	Valid at destination location	Destination storage position
`varianceNotes`	String	Conditional	Required when received ≠ dispatched	Receipt variance notes

---

API-105 — `POST /api/v1/transfers/{transferId}/cancel`

DTO: `CancelInTransitTransferRequest`

Field	Type	Required	Validation Rules	Description
`transferId`	Uuid	Required	Path param; must be in-transit or approved-not-dispatched	Transfer identifier
`cancellationReason`	String	Required	Max 1000 chars; non-blank	Cancellation reason
`reversalAction`	String	Optional	Enum: `RESTORE_SOURCE`, `INVESTIGATE_LOSS`	Reversal workflow action

---

API-106 — `GET /api/v1/transfers/pending-approvals`

DTO: `ListPendingTransferApprovalsQuery`

Field	Type	Required	Validation Rules	Description
`sourceLocationCode`	LocationCode	Optional	User location scope	Source location filter
`destinationLocationCode`	LocationCode	Optional	User location scope	Destination location filter
`page`	PageNumber	Optional	Min 1	Pagination page

Field	Type	Required	Validation Rules	Description
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-107 — `GET /api/v1/transfers/in-transit`

DTO: `ListInTransitStockQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Optional	Source or destination; user scope (VR-136)	Location filter
`sku`	ProductSku	Optional	Valid SKU	Product filter
`isOverdue`	Boolean	Optional	—	Filter overdue in-transit (VR-137)
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

M-09 — Inventory Accuracy

API-108 — `POST /api/v1/accuracy/cycle-counts`

DTO: `ScheduleCycleCountRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Active location (VR-067)	Count location
`sku`	ProductSku	Required	Active product (VR-067)	Product to count
`scheduledDate`	LocalDate	Required	Valid date	Scheduled count date
`assignedToUserId`	String	Optional	Max 50 chars; valid user	Assigned counter
`priority`	String	Optional	Enum: `NORMAL`, `HIGH`; Class A default HIGH (VR-068, BR-032)	Count priority

API-109 — `GET /api/v1/accuracy/cycle-counts`

DTO: `ListCycleCountsQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Optional	User location scope	Location filter
`status`	String	Optional	Enum: `SCHEDULED`, `IN_PROGRESS`, `COMPLETED`, `VARIANCE_DETECTED`	Count status filter
`sku`	ProductSku	Optional	Valid SKU	Product filter
`scheduledFrom`	LocalDate	Optional	On or before `scheduledTo`	Scheduled date range start
`scheduledTo`	LocalDate	Optional	On or after `scheduledFrom`	Scheduled date range end
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-110 — `GET /api/v1/accuracy/cycle-counts/{cycleCountId}`

DTO: `GetCycleCountAssignmentRequest`

Field	Type	Required	Validation Rules	Description
`cycleCountId`	Uuid	Required	Path param; must exist	Cycle count identifier

Response must not include system expected quantity — blind count (VR-069, BR-018).

API-111 — `POST /api/v1/accuracy/cycle-counts/{cycleCountId}/result`

DTO: `SubmitCycleCountResultRequest`

Field	Type	Required	Validation Rules	Description
`cycleCountId`	Uuid	Required	Path param; status `SCHEDULED` or `IN_PROGRESS`	Cycle count identifier
`physicalQuantity`	QuantityDto	Required	Min 0 (VR-070)	Blind physical count result
`countNotes`	String	Optional	Max 1000 chars	Counter notes
`countedAt`	Instant	Optional	Default now	Count submission timestamp

---

API-112 — `POST /api/v1/accuracy/physical-counts`

DTO: `InitiatePhysicalInventoryCountRequest`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Active location (VR-071)	Location for wall-to-wall count
`plannedStartDate`	LocalDate	Required	Valid date	Planned count start
`plannedEndDate`	LocalDate	Required	On or after `plannedStartDate`	Planned count end
`countScope`	String	Optional	Enum: `FULL_LOCATION`; default full	Count scope
`notes`	String	Optional	Max 1000 chars	Count initiation notes

API-113 — `GET /api/v1/accuracy/physical-counts/{physicalCountId}`

DTO: `GetPhysicalCountStatusRequest`

Field	Type	Required	Validation Rules	Description
`physicalCountId`	Uuid	Required	Path param; must exist	Physical count identifier

---

API-114 — `POST /api/v1/accuracy/physical-counts/{physicalCountId}/results`

DTO: `RecordPhysicalCountResultRequest`

Field	Type	Required	Validation Rules	Description
`physicalCountId`	Uuid	Required	Path param; location frozen (VR-071)	Physical count identifier
`countLines`	Array<PhysicalCountLineDto>	Required	Min 1 line	Line-level count results

`PhysicalCountLineDto`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Must have on-hand or receiving status at location	Product counted
`physicalQuantity`	QuantityDto	Required	Min 0 (VR-070)	Physical count quantity
`storagePositionCode`	String	Optional	Valid position at location	Storage position counted

Field	Type	Required	Validation Rules	Description
`lotIdentity`	LotIdentityDto	Conditional	Required for traceable products	Lot-level count

API-115 — `POST /api/v1/accuracy/physical-counts/{physicalCountId}/certify`

DTO: `CertifyPhysicalInventoryCountRequest`

Field	Type	Required	Validation Rules	Description
`physicalCountId`	Uuid	Required	Path param; all SKUs counted (VR-072)	Physical count identifier
`certificationNotes`	String	Optional	Max 2000 chars	Certification notes
`certifiedByReference`	String	Optional	Max 100 chars	Certifying authority reference

API-116 — `POST /api/v1/accuracy/adjustment-cases`

DTO: `OpenAdjustmentCaseRequest`

Field	Type	Required	Validation Rules	Description
`originType`	String	Required	Enum: `COUNT_VARIANCE`, `DIRECT_ADJUSTMENT`, `FULFILLMENT_EXCEPTION`	Case origin type
`locationCode`	LocationCode	Required	User authorized at location (VR-079)	Affected location
`sku`	ProductSku	Required	Valid product	Affected product
`varianceId`	Uuid	Conditional	Required when `originType=COUNT_VARIANCE`	Linked count variance
`proposedAdjustmentQuantity`	BigDecimal	Conditional	Non-zero when direct adjustment; decrease ≤ on-hand (VR-078)	Proposed quantity change
`adjustmentDirection`	String	Conditional	Enum: `INCREASE`, `DECREASE`; required for direct adjustment	Adjustment direction
`reasonCode`	ReasonCode	Required	Active code (VR-077, BR-020)	Adjustment reason
`initialNotes`	String	Optional	Max 2000 chars	Initial case notes

API-117 — `GET /api/v1/accuracy/adjustment-cases/{caseId}`

DTO: `GetAdjustmentCaseRequest`

Field	Type	Required	Validation Rules	Description
`caseId`	Uuid	Required	Path param; must exist	Adjustment case identifier

API-118 — `POST /api/v1/accuracy/adjustment-cases/{caseId}/investigate`

DTO: `InvestigateVarianceRequest`

Field	Type	Required	Validation Rules	Description
`caseId`	Uuid	Required	Path param	Adjustment case identifier
`rootCauseCategory`	String	Required	Enum: `MISPLACEMENT`, `DAMAGE`, `THEFT`, `SYSTEM_ERROR`, `RECEIVING_ERROR`, `UNKNOWN`	Root cause category
`investigationFindings`	String	Required	Max 5000 chars; non-blank (VR-076, BR-022)	Investigation documentation
`correctiveAction`	String	Optional	Max 2000 chars	Proposed corrective action

API-119 — `POST /api/v1/accuracy/adjustment-cases/{caseId}/variances/{varianceId}/approve`

DTO: `ApproveCountVarianceRequest`

Field	Type	Required	Validation Rules	Description
`caseId`	Uuid	Required	Path param	Adjustment case identifier
`varianceId`	Uuid	Required	Path param	Count variance identifier
`approvalNotes`	String	Optional	Max 2000 chars	Approver notes
`approvedAdjustmentQuantity`	BigDecimal	Required	Non-zero; approver ≠ counter (VR-075, BR-019)	Approved adjustment quantity

---

API-120 — `POST /api/v1/accuracy/adjustment-cases/{caseId}/submit`

DTO: `SubmitAdjustmentForApprovalRequest`

Field	Type	Required	Validation Rules	Description
`caseId`	Uuid	Required	Path param; investigation complete (VR-076)	Adjustment case identifier
`adjustmentType`	String	Required	Enum: `ADJUSTMENT`, `WRITE_OFF`	Submission type
`quantity`	PositiveQuantityDto	Required	Decrease ≤ on-hand (VR-078, VR-081)	Adjustment or write-off quantity
`adjustmentDirection`	String	Conditional	Enum: `INCREASE`, `DECREASE`; required for `ADJUSTMENT`	Direction of adjustment
`reasonCode`	ReasonCode	Required	Active code (VR-077, BR-020)	Reason code
`evidenceReference`	String	Conditional	Required for write-off (VR-080, BR-023)	Supporting evidence reference
`financialValue`	MonetaryAmountDto	Optional	For threshold evaluation (VR-082)	Financial impact of adjustment

---

API-121 — `POST /api/v1/accuracy/adjustment-cases/{caseId}/approve`

DTO: `ApproveAdjustmentRequest`

Field	Type	Required	Validation Rules	Description
`caseId`	Uuid	Required	Path param; status submitted	Adjustment case identifier
`approvalNotes`	String	Optional	Max 2000 chars	Independent approver notes
`approvedQuantity`	PositiveQuantityDto	Optional	May differ from submitted within policy	Final approved quantity

Approver must differ from initiator and counter (VR-083, BR-021).

API-122 — `POST /api/v1/accuracy/adjustment-cases/{caseId}/reject`

DTO: `RejectAdjustmentRequest`

Field	Type	Required	Validation Rules	Description
`caseId`	Uuid	Required	Path param	Adjustment case identifier
`rejectionReason`	String	Required	Max 2000 chars; non-blank	Rejection reason

---

API-123 — `GET /api/v1/accuracy/pending-approvals`

DTO: `ListPendingApprovalsQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Optional	User location scope	Location filter
`caseType`	String	Optional	Enum: `COUNT_VARIANCE`, `ADJUSTMENT`, `WRITE_OFF`	Case type filter
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

M-10 — Returns Management

API-124 — `POST /api/v1/returns/authorizations`

DTO: `IssueReturnAuthorizationRequest`

Field	Type	Required	Validation Rules	Description
`originalOrderId`	Uuid	Required	Valid return-eligible order (VR-142, BR-025)	Original customer order
`returnReason`	String	Required	Max 500 chars; non-blank	Customer return reason
`returnWindowDays`	Integer	Optional	Min 1; default per policy	Authorization validity in days
`lines`	Array<ReturnAuthorizationLineDto>	Required	Min 1 line	Authorized return lines

`ReturnAuthorizationLineDto`

Field	Type	Required	Validation Rules	Description
`originalOrderLineId`	Uuid	Required	Must exist on order	Original order line
`sku`	ProductSku	Required	Must match order line product	Return product
`authorizedQuantity`	PositiveQuantityDto	Required	≤ originally shipped qty (VR-143)	Authorized return quantity

---

API-125 — `GET /api/v1/returns/authorizations`

DTO: `ListReturnAuthorizationsQuery`

Field	Type	Required	Validation Rules	Description
`status`	String	Optional	Enum: `AUTHORIZED`, `IN_TRANSIT`, `RECEIVED`, `INSPECTING`, `COMPLETED`, `EXPIRED`, `CANCELLED`	RMA status filter
`originalOrderId`	Uuid	Optional	Valid UUID	Filter by original order
`customerReference`	String	Optional	Max 100 chars	Customer reference filter
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-126 — `GET /api/v1/returns/authorizations/{returnAuthorizationId}`

DTO: `GetReturnAuthorizationRequest`

Field	Type	Required	Validation Rules	Description
`returnAuthorizationId`	Uuid	Required	Path param; must exist (VR-149)	Return authorization identifier

---

API-127 — `POST /api/v1/returns/authorizations/{returnAuthorizationId}/cancel`

DTO: `CancelReturnAuthorizationRequest`

Field	Type	Required	Validation Rules	Description
`returnAuthorizationId`	Uuid	Required	Path param; must be open/unused	Return authorization identifier
`cancellationReason`	String	Required	Max 1000 chars; non-blank	Cancellation reason

---

API-128 — `POST /api/v1/returns/{returnId}/receive`

DTO: `ReceiveCustomerReturnRequest`

Field	Type	Required	Validation Rules	Description
`returnId`	Uuid	Required	Path param; valid open RMA (VR-144)	Customer return case identifier
`receivingLocationCode`	LocationCode	Required	Active location	Receiving location
`receivedLines`	Array<ReturnReceiveLineDto>	Required	Min 1 line	Received return lines

`ReturnReceiveLineDto`

Field	Type	Required	Validation Rules	Description
`authorizationLineId`	Uuid	Required	Must exist on RMA	RMA line reference
`sku`	ProductSku	Required	Must match authorization (VR-145)	Received product
`receivedQuantity`	PositiveQuantityDto	Required	≤ authorized qty (VR-046)	Received quantity

---

API-129 — `POST /api/v1/returns/{returnId}/inspect`

DTO: `InspectReturnedGoodsRequest`

Field	Type	Required	Validation Rules	Description
`returnId`	Uuid	Required	Path param; must be received	Customer return case identifier
`inspectionLines`	Array<ReturnInspectionLineDto>	Required	Min 1 line (VR-146, BR-026)	Inspection results

`ReturnInspectionLineDto`

Field	Type	Required	Validation Rules	Description
`returnLineId`	Uuid	Required	Must exist on return	Return line identifier
`conditionGrade`	String	Required	Enum: `RESELLABLE`, `DAMAGED`, `DEFECTIVE`, `EXPIRED`	Condition assessment
`inspectedQuantity`	PositiveQuantityDto	Required	≤ received quantity	Inspected quantity
`inspectionNotes`	String	Optional	Max 2000 chars	Inspection notes

API-130 — `POST /api/v1/returns/{returnId}/disposition`

DTO: `DispositionReturnRequest`

Field	Type	Required	Validation Rules	Description
`returnId`	Uuid	Required	Path param; inspection complete (VR-146)	Customer return case identifier
`dispositionAction`	String	Required	Enum: `RESTOCK`, `QUARANTINE`, `WRITE_OFF`	Disposition routing

Field	Type	Required	Validation Rules	Description
`dispositionLines`	Array<ReturnDispositionLineDto>	Required	Min 1 line; sum must equal inspected qty (VR-147)	Per-line disposition

`ReturnDispositionLineDto`

Field	Type	Required	Validation Rules	Description
`returnLineId`	Uuid	Required	Must exist on return	Return line identifier
`quantity`	PositiveQuantityDto	Required	≤ inspected quantity	Disposition quantity
`restockLocationCode`	LocationCode	Conditional	Required when `dispositionAction=RESTOCK`	Restock target location
`reasonCode`	ReasonCode	Conditional	Required for `WRITE_OFF` (VR-148, BR-023)	Write-off reason code
`writeOffApprovalReference`	String	Conditional	Required for `WRITE_OFF` above threshold (VR-080)	Write-off approval reference

API-131 — `GET /api/v1/returns/pending-receipts`

DTO: `ListPendingReturnReceiptsQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Optional	User location scope	Receiving location filter
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

M-11 — Commercial Campaign

API-132 — `POST /api/v1/campaigns/allocations`

DTO: `CreatePromotionalAllocationRequest`

Field	Type	Required	Validation Rules	Description
`campaignId`	String	Required	Max 50 chars	Campaign identifier
`campaignName`	String	Required	Max 200 chars; non-blank	Campaign display name
`locationCode`	LocationCode	Required	Active location	Allocation location
`sku`	ProductSku	Required	Active product	Allocated product
`allocatedQuantity`	PositiveQuantityDto	Required	≤ available ATP at allocation (VR-127, BR-029)	Ring-fenced quantity
`channel`	String	Optional	Enum: `B2B`, `B2C`, `ALL`; default `ALL`	Channel designation
`campaignStartDate`	LocalDate	Required	Valid date (VR-126)	Campaign start date
`campaignEndDate`	LocalDate	Required	On or after `campaignStartDate`	Campaign end date

API-133 — `GET /api/v1/campaigns/allocations`

DTO: `ListPromotionalAllocationsQuery`

Field	Type	Required	Validation Rules	Description
`campaignId`	String	Optional	Max 50 chars	Campaign filter
`status`	String	Optional	Enum: `PLANNED`, `ACTIVE`, `COMPLETED`, `RELEASED`	Allocation status
`locationCode`	LocationCode	Optional	User location scope	Location filter

Field	Type	Required	Validation Rules	Description
`activeOnDate`	LocalDate	Optional	Valid date	Allocations active on date
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-134 — `GET /api/v1/campaigns/allocations/{allocationId}`

DTO: `GetPromotionalAllocationRequest`

Field	Type	Required	Validation Rules	Description
`allocationId`	Uuid	Required	Path param; must exist	Allocation identifier

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API-135 — `PATCH /api/v1/campaigns/allocations/{allocationId}`

DTO: `AmendPromotionalAllocationRequest`

Field	Type	Required	Validation Rules	Description
`allocationId`	Uuid	Required	Path param	Allocation identifier
`allocatedQuantity`	PositiveQuantityDto	Optional	≤ available ATP when increasing	Amended allocated quantity
`campaignStartDate`	LocalDate	Optional	Valid date	Amended start date
`campaignEndDate`	LocalDate	Optional	On or after start date	Amended end date
`amendmentReason`	String	Required	Max 1000 chars; non-blank	Reason for amendment

At least one amendable field must be present.

API-136 — `POST /api/v1/campaigns/allocations/{allocationId}/readiness`

DTO: `EvaluateCampaignReadinessRequest`

Field	Type	Required	Validation Rules	Description
`allocationId`	Uuid	Required	Path param	Allocation identifier
`targetQuantity`	PositiveQuantityDto	Optional	Defaults to allocated quantity (VR-126)	Target quantity for readiness check
`readinessDeadline`	LocalDate	Optional	Valid date	Campaign commitment deadline
`confirmReadiness`	Boolean	Optional	Default `false`	Whether to certify readiness if sufficient

---

API-137 — `POST /api/v1/campaigns/allocations/{allocationId}/release`

DTO: `ReleasePromotionalAllocationRequest`

Field	Type	Required	Validation Rules	Description
`allocationId`	Uuid	Required	Path param	Allocation identifier
`releaseReason`	String	Required	Enum: `CAMPAIGN_ENDED`, `MANUAL_RELEASE`, `CAMPAIGN_CANCELLED`	Release reason
`releaseNotes`	String	Optional	Max 1000 chars	Release notes

API-138 — `POST /api/v1/campaigns/allocations/{allocationId}/override`

DTO: `AuthorizePromotionalOverrideRequest`

Field	Type	Required	Validation Rules	Description
`allocationId`	Uuid	Required	Path param; allocation active	Allocation identifier
`orderId`	Uuid	Required	Valid customer order	Non-promotional order consuming ring-fence
`overrideQuantity`	PositiveQuantityDto	Required	≤ ring-fenced remaining (BR-029)	Quantity to override
`authorizationReference`	String	Required	Max 100 chars; commercial leadership approval	Override authorization reference
`justification`	String	Required	Max 1000 chars; non-blank	Business justification

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API-139 — `GET /api/v1/campaigns/ring-fence/{locationCode}/{sku}`

DTO: `CheckRingFenceQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Required	Path param	Location code
`sku`	ProductSku	Required	Path param	Product SKU
`asOfDate`	LocalDate	Optional	Default today	Date for active ring-fence check

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M-12 — Identity & Access

API-140 — `POST /api/v1/users`

DTO: `ProvisionUserAccountRequest`

Field	Type	Required	Validation Rules	Description
`userId`	String	Required	Max 50 chars; unique; alphanumeric and underscore	User identifier
`email`	String	Required	Valid email; max 254 chars	User email address
`displayName`	String	Required	Max 200 chars; non-blank	User display name
`initialRoles`	Array<RoleAssignmentDto>	Required	Min 1 role; no SoD conflicts (VR-083)	Initial role assignments
`externalDirectoryId`	String	Optional	Max 100 chars	HR/IdP directory reference

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API-141 — `GET /api/v1/users`

DTO: `ListUserAccountsQuery`

Field	Type	Required	Validation Rules	Description
`status`	String	Optional	Enum: `ACTIVE`, `SUSPENDED`, `DEACTIVATED`	Account status filter
`roleCode`	String	Optional	Max 50 chars	Filter by assigned role
`locationCode`	LocationCode	Optional	Filter users with scope at location	Location scope filter
`search`	String	Optional	Max 200 chars; matches name or email	Search term
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-142 — `GET /api/v1/users/{userId}`

DTO: `GetUserAccountRequest`

Field	Type	Required	Validation Rules	Description
`userId`	String	Required	Path param; must exist	User identifier

API-143 — `POST /api/v1/users/{userId}/roles`

DTO: `AssignRoleRequest`

Field	Type	Required	Validation Rules	Description
`userId`	String	Required	Path param; account must be active	User identifier
`roleAssignment`	RoleAssignmentDto	Required	No SoD conflict with existing roles (VR-083)	Role assignment details

---

API-144 — `DELETE /api/v1/users/{userId}/roles/{roleAssignmentId}`

DTO: `RevokeRoleAssignmentRequest`

Field	Type	Required	Validation Rules	Description
`userId`	String	Required	Path param	User identifier
`roleAssignmentId`	Uuid	Required	Path param; must exist on user	Role assignment identifier
`revocationReason`	String	Optional	Max 500 chars	Reason for revocation

API-145 — `POST /api/v1/users/{userId}/suspend`

DTO: `SuspendUserAccountRequest`

Field	Type	Required	Validation Rules	Description
`userId`	String	Required	Path param; must be active	User identifier
`suspensionReason`	String	Required	Max 1000 chars; non-blank	Suspension reason
`suspendedUntil`	LocalDate	Optional	Future date when temporary	Suspension end date

---

API-146 — `POST /api/v1/users/{userId}/deactivate`

DTO: `DeactivateUserAccountRequest`

Field	Type	Required	Validation Rules	Description
`userId`	String	Required	Path param	User identifier
`deactivationReason`	String	Required	Max 1000 chars; non-blank	Deactivation reason
`effectiveDate`	LocalDate	Optional	Default today	Deactivation effective date

---

API-147 — `POST /api/v1/users/sync`

DTO: `SyncUsersFromDirectoryRequest`

Field	Type	Required	Validation Rules	Description
`syncMode`	String	Required	Enum: `FULL`, `INCREMENTAL`	Directory sync mode
`sinceTimestamp`	Instant	Conditional	Required when `syncMode=INCREMENTAL`	Sync changes since timestamp
`dryRun`	Boolean	Optional	Default `false`	Validate without persisting changes

API-148 — `POST /api/v1/authorization/actions/evaluate`

DTO: `EvaluateAuthorizationRequest`

Field	Type	Required	Validation Rules	Description
`userId`	String	Optional	Max 50 chars; defaults to authenticated user	User to evaluate
`action`	String	Required	Max 100 chars; valid permission code	Action or command to authorize
`resourceType`	String	Optional	Max 50 chars	Resource type context
`resourceId`	String	Optional	Max 100 chars	Resource identifier
`locationCode`	LocationCode	Optional	Location scope for action	Location context

API-149 — `POST /api/v1/authorization/location-scope/validate`

DTO: `ValidateLocationScopeRequest`

Field	Type	Required	Validation Rules	Description
`userId`	String	Optional	Defaults to authenticated user	User to validate
`locationCode`	LocationCode	Required	Must exist (VR-008, VR-053)	Location to validate access
`requiredPermission`	String	Optional	Max 100 chars	Specific permission required at location

M-13 — Inventory Intelligence

All intelligence APIs are read-only. Request DTOs comprise path and query parameters only.

API-150 — `GET /api/v1/intelligence/dashboard`

DTO: `GetInventoryDashboardQuery`

Field	Type	Required	Validation Rules	Description
`scope`	String	Optional	Enum: `LOCATION`, `NETWORK`; default `LOCATION`	Dashboard scope
`locationCode`	LocationCode	Conditional	Required when `scope=LOCATION`; user scope (VR-151)	Location for dashboard
`asOfTimestamp`	Instant	Optional	Default latest projection (VR-152)	Point-in-time for dashboard data

API-151 — `GET /api/v1/intelligence/stock-status/summary`

DTO: `GetStockStatusSummaryQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Optional	User location scope (VR-151)	Location filter; omit for authorized network
`categoryCode`	String	Optional	Max 30 chars	Category filter
`abcClass`	AbcClass	Optional	`A`, `B`, or `C`	ABC class filter

API-152 — `GET /api/v1/intelligence/stock-status/board`

DTO: `GetStockStatusBoardQuery`

Field	Type	Required	Validation Rules	Description
`locationCodes`	Array<LocationCode>	Optional	Authorized locations only (VR-151)	Locations to include
`stockStatus`	String	Optional	Enum: `IN_STOCK`, `LOW_STOCK`, `OUT_OF_STOCK`, `OVERSTOCK`	Status filter
`sku`	ProductSku	Optional	Valid SKU	Product filter
`abcClass`	AbcClass	Optional	ABC filter	ABC class filter
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-153 — `GET /api/v1/intelligence/availability/inquiries`

DTO: `GetAvailabilityInquiryLogQuery`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Optional	Valid SKU	Product filter
`locationCode`	LocationCode	Optional	User location scope	Location filter
`userId`	String	Optional	Max 50 chars	Inquiring user filter
`startDate`	LocalDate	Optional	On or before `endDate` (VR-062)	Inquiry period start
`endDate`	LocalDate	Optional	On or after `startDate`	Inquiry period end
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-154 — `GET /api/v1/intelligence/quarantine-register`

DTO: `GetQuarantineRegisterQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Optional	User location scope	Location filter
`holdReason`	String	Optional	Enum: `QUALITY`, `RECALL`, `CUSTOMS`, `BONDED`, `INVESTIGATION`	Hold reason filter
`sku`	ProductSku	Optional	Valid SKU	Product filter
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-155 — `GET /api/v1/intelligence/movements`

DTO: `SearchMovementHistoryQuery`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Optional	At least one filter recommended (VR-154)	Product filter
`locationCode`	LocationCode	Optional	User location scope (VR-154)	Location filter
`userId`	String	Optional	Max 50 chars	User attribution filter

Field	Type	Required	Validation Rules	Description
`movementType`	String	Optional	Enum: `RECEIPT`, `SHIPMENT`, `ADJUSTMENT`, `TRANSFER`, `RESERVATION`, `WRITE_OFF`, `RETURN`	Movement type filter
`startDate`	LocalDate	Required	On or before `endDate` (VR-153)	Movement period start
`endDate`	LocalDate	Required	On or after `startDate`	Movement period end
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-156 — `GET /api/v1/intelligence/audit-trail`

DTO: `QueryAuditTrailQuery`

Field	Type	Required	Validation Rules	Description
`entityType`	String	Optional	Max 50 chars	Entity type filter
`entityId`	String	Optional	Max 100 chars	Entity identifier filter
`userId`	String	Optional	Max 50 chars	Acting user filter
`action`	String	Optional	Max 100 chars	Action type filter
`startDate`	LocalDate	Required	On or before `endDate` (VR-153)	Audit period start
`endDate`	LocalDate	Required	On or after `startDate`	Audit period end
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-157 — `GET /api/v1/intelligence/audit-trail/export`

DTO: `ExportAuditEvidenceQuery`

Field	Type	Required	Validation Rules	Description
`startDate`	LocalDate	Required	On or before `endDate`; max 365-day range	Export period start
`endDate`	LocalDate	Required	On or after `startDate`	Export period end
`entityTypes`	Array<String>	Optional	Max 20 types	Entity types to include
`locationCodes`	Array<LocationCode>	Optional	User authorized only (VR-154)	Location scope for export
`format`	String	Optional	Enum: `CSV`, `JSON`, `PDF`; default `CSV`	Export file format

---

API-158 — `GET /api/v1/intelligence/alerts`

DTO: `GetOperationalAlertsQuery`

Field	Type	Required	Validation Rules	Description
`alertType`	String	Optional	Enum: `LOW_STOCK`, `OUT_OF_STOCK`, `INBOUND_DELAY`, `TRANSFER_DELAY`, `FULFILLMENT_EXCEPTION`, `PROMOTIONAL_READINESS`, `PENDING_APPROVAL`	Alert type filter
`severity`	String	Optional	Enum: `INFO`, `WARNING`, `CRITICAL`	Severity filter
`locationCode`	LocationCode	Optional	User location scope	Location filter
`acknowledged`	Boolean	Optional	Default `false`	Include acknowledged alerts

Field	Type	Required	Validation Rules	Description
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-159 — `POST /api/v1/intelligence/alerts/{alertId}/acknowledge`

DTO: `AcknowledgeOperationalAlertRequest`

Field	Type	Required	Validation Rules	Description
`alertId`	Uuid	Required	Path param; must be active (VR-113)	Alert identifier
`actionTaken`	String	Required	Enum: `REQUISITION`, `TRANSFER`, `DISMISS`, `DEFER` (VR-115)	Action taken on alert
`acknowledgmentNotes`	String	Conditional	Required when `actionTaken=DISMISS` or `DEFER` (VR-114)	Acknowledgment rationale

API-160 — `GET /api/v1/intelligence/replenishment/recommendations`

DTO: `GetReplenishmentRecommendationsQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Optional	User location scope	Location filter
`sku`	ProductSku	Optional	Valid SKU	Product filter
`recommendationAction`	String	Optional	Enum: `PURCHASE`, `TRANSFER`, `HOLD`, `NO_ACTION`	Recommended action filter
`acknowledged`	Boolean	Optional	—	Filter by acknowledgment status
`startDate`	LocalDate	Optional	On or before `endDate`	Recommendation period start
`endDate`	LocalDate	Optional	On or after `startDate`	Recommendation period end
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-161 — `GET /api/v1/intelligence/procurement/open-pos`

DTO: `GetOpenPurchaseOrderBoardQuery`

Field	Type	Required	Validation Rules	Description
`receivingLocationCode`	LocationCode	Optional	User location scope (VR-034)	Receiving location filter
`supplierCode`	SupplierCode	Optional	Valid supplier	Supplier filter
`isOverdue`	Boolean	Optional	—	Overdue PO filter (VR-123)
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-162 — `GET /api/v1/intelligence/procurement/inbound-calendar`

DTO: `GetInboundSupplyCalendarQuery`

Field	Type	Required	Validation Rules	Description
`startDate`	LocalDate	Required	On or before `endDate` (VR-033)	Calendar period start
`endDate`	LocalDate	Required	On or after `startDate`; max 90-day range	Calendar period end

Field	Type	Required	Validation Rules	Description
`receivingLocationCode`	LocationCode	Optional	User location scope	Receiving location filter
`supplierCode`	SupplierCode	Optional	Valid supplier	Supplier filter

API-163 — `GET /api/v1/intelligence/procurement/inbound-delays`

DTO: `GetInboundDeliveryDelayLogQuery`

Field	Type	Required	Validation Rules	Description
`receivingLocationCode`	LocationCode	Optional	User location scope	Location filter
`supplierCode`	SupplierCode	Optional	Valid supplier	Supplier filter
`startDate`	LocalDate	Optional	On or before `endDate`	Delay log period start
`endDate`	LocalDate	Optional	On or after `startDate`	Delay log period end
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-164 — `GET /api/v1/intelligence/fulfillment/backorders`

DTO: `GetBackorderPipelineQuery`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Optional	Valid SKU	Product filter
`locationCode`	LocationCode	Optional	User location scope	Fulfillment location filter
`prioritySort`	Boolean	Optional	Default `true` (VR-102)	Sort by backorder priority policy
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-165 — `GET /api/v1/intelligence/fulfillment/exceptions`

DTO: `GetFulfillmentExceptionLogQuery`

Field	Type	Required	Validation Rules	Description
`locationCode`	LocationCode	Optional	User location scope	Location filter
`exceptionType`	String	Optional	Enum: `PICK_DISCREPANCY`, `SHELF_MISMATCH`, `DAMAGED_GOODS`, `SHORT_PICK`	Exception type filter
`status`	String	Optional	Enum: `OPEN`, `RESOLVED`; default `OPEN`	Exception status
`startDate`	LocalDate	Optional	On or before `endDate`	Exception period start
`endDate`	LocalDate	Optional	On or after `startDate`	Exception period end
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-166 — `GET /api/v1/intelligence/network/imbalance`

DTO: `GetNetworkImbalanceAnalysisQuery`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Optional	Valid SKU; active product (VR-139)	Product filter
`sourceLocationCode`	LocationCode	Optional	Active location	Excess location filter
`destinationLocationCode`	LocationCode	Optional	Active location	Shortage location filter
`includeRecommendations`	Boolean	Optional	Default `true`	Include transfer recommendations

---

API-167 — `GET /api/v1/intelligence/accuracy/shrinkage`

DTO: `GetShrinkageAnalysisQuery`

Field	Type	Required	Validation Rules	Description
`startDate`	LocalDate	Required	On or before `endDate` (VR-086)	Analysis period start
`endDate`	LocalDate	Required	On or after `startDate`	Analysis period end
`locationCode`	LocationCode	Optional	User location scope	Location filter
`categoryCode`	String	Optional	Max 30 chars	Category filter
`reasonCode`	ReasonCode	Optional	Valid reason code (VR-087)	Reason code filter
`groupBy`	String	Optional	Enum: `LOCATION`, `SKU`, `REASON_CODE`, `CATEGORY`; default `SKU`	Aggregation dimension

---

API-168 — `GET /api/v1/intelligence/accuracy/count-performance`

DTO: `GetCountPerformanceMetricsQuery`

Field	Type	Required	Validation Rules	Description
`startDate`	LocalDate	Required	On or before `endDate` (VR-157)	Reporting period start
`endDate`	LocalDate	Required	On or after `startDate`	Reporting period end
`locationCode`	LocationCode	Optional	User location scope	Location filter
`countType`	String	Optional	Enum: `CYCLE_COUNT`, `PHYSICAL_COUNT`, `ALL`; default `ALL`	Count type filter
`abcClass`	AbcClass	Optional	ABC filter	ABC class filter
`excludePendingVariances`	Boolean	Optional	Default `true` (VR-158)	Exclude unresolved variances from accuracy rate

---

API-169 — `GET /api/v1/intelligence/campaigns/readiness`

DTO: `GetCampaignReadinessStatusQuery`

Field	Type	Required	Validation Rules	Description
`campaignId`	String	Optional	Max 50 chars	Campaign filter
`locationCode`	LocationCode	Optional	User location scope	Location filter
`readinessStatus`	String	Optional	Enum: `READY`, `GAP_IDENTIFIED`, `PENDING`	Readiness status filter
`daysUntilStart`	Integer	Optional	Min 0	Campaigns starting within N days

---

API-170 — `GET /api/v1/intelligence/kpis`

DTO: `GetInventoryKpisQuery`

Field	Type	Required	Validation Rules	Description
`startDate`	LocalDate	Required	On or before `endDate` (VR-162)	KPI period start
`endDate`	LocalDate	Required	On or after `startDate`	KPI period end
`kpiCodes`	Array<String>	Optional	Valid configured KPI codes (VR-161)	Specific KPIs to retrieve
`locationCodes`	Array<LocationCode>	Optional	User authorized only (VR-151)	Location scope for KPIs
`compareToTarget`	Boolean	Optional	Default `true`	Include target comparison

---

API-171 — `GET /api/v1/intelligence/valuation/snapshots`

DTO: `GetValuationSnapshotsQuery`

Field	Type	Required	Validation Rules	Description
`periodEndDate`	LocalDate	Optional	Valid date	Specific period-end snapshot
`startDate`	LocalDate	Optional	On or before `endDate`	Snapshot period range start
`endDate`	LocalDate	Optional	On or after `startDate`	Snapshot period range end
`locationCode`	LocationCode	Optional	User location scope	Location filter
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-172 — `GET /api/v1/intelligence/valuation/reconciliation`

DTO: `GetValuationReconciliationQuery`

Field	Type	Required	Validation Rules	Description
`periodEndDate`	LocalDate	Required	Valid period-end date	Reconciliation period end
`locationCode`	LocationCode	Optional	User location scope	Location filter
`varianceThresholdPercent`	BigDecimal	Optional	Min 0; max 100; default per policy	Variance threshold for flagging
`includeGlComparison`	Boolean	Optional	Default `true`	Include general ledger comparison data

---

Platform & Integration

API-173 — `GET /api/v1/auth/session`

DTO: `GetSessionRequest`

Field	Type	Required	Validation Rules	Description
`includePermissions`	Boolean	Optional	Default `false`	Include resolved permission list in response
`includeLocationScope`	Boolean	Optional	Default `true`	Include user's location scope

Authentication via `Authorization` header — no request body.

API-174 — `POST /api/v1/auth/logout`

DTO: `LogoutRequest`

Field	Type	Required	Validation Rules	Description
`logoutAllSessions`	Boolean	Optional	Default `false`	Terminate all active sessions for user

No request body required; empty body accepted.

API-175 — `GET /api/v1/health`

DTO: `HealthCheckRequest`

No path parameters, query parameters, or request body. Public endpoint — no authentication required.

API-176 — `POST /api/v1/integrations/sales/orders`

DTO: `ImportSalesOrderRequest`

Field	Type	Required	Validation Rules	Description
`externalOrderId`	String	Required	Max 100 chars; idempotent on re-import	CRM external order identifier
`externalSystemCode`	String	Required	Max 30 chars; registered integration source	Source system code
`customerReference`	String	Optional	Max 100 chars	CRM customer reference
`channel`	String	Required	Enum: `B2B`, `B2C`, `MARKETPLACE`	Sales channel
`shipFromLocationCode`	LocationCode	Optional	Active location when specified	Preferred ship-from location
`requestedShipDate`	LocalDate	Optional	Valid date	Requested ship date
`correlationId`	String	Optional	Max 100 chars; traceability (IB-08)	Integration correlation identifier
`lines`	Array<ExternalOrderLineDto>	Required	Min 1 line	Order lines from CRM

`ExternalOrderLineDto`

Field	Type	Required	Validation Rules	Description
`externalLineId`	String	Required	Max 50 chars	CRM line identifier
`sku`	ProductSku	Required	Must exist in catalog (VR-007)	Product SKU
`quantity`	PositiveQuantityDto	Required	Min exclusive 0	Ordered quantity
`unitPrice`	MonetaryAmountDto	Optional	Amount min 0	Line unit price from CRM

API-177 — `GET /api/v1/integrations/sales/fulfillment-status`

DTO: `GetFulfillmentStatusFeedQuery`

Field	Type	Required	Validation Rules	Description
`externalSystemCode`	String	Required	Max 30 chars	Requesting CRM system code
`updatedSince`	Instant	Optional	ISO-8601 UTC	Return orders updated since timestamp
`externalOrderId`	String	Optional	Max 100 chars	Filter by external order
`status`	String	Optional	Enum: `CONFIRMED`, `PICKING`, `SHIPPED`, `CANCELLED`, `BACKORDERED`, `EXCEPTION`	Fulfillment status filter
`page`	PageNumber	Optional	Min 1	Pagination page
`pageSize`	PageSize	Optional	Min 1; max 100	Pagination size

API-178 — `POST /api/v1/integrations/marketing/campaign-plans`

DTO: `ImportCampaignPlanRequest`

Field	Type	Required	Validation Rules	Description
`externalCampaignId`	String	Required	Max 50 chars; idempotent on re-import	Marketing system campaign ID
`externalSystemCode`	String	Required	Max 30 chars	Source marketing system code
`campaignName`	String	Required	Max 200 chars	Campaign name
`campaignStartDate`	LocalDate	Required	Valid date (VR-126)	Campaign start date
`campaignEndDate`	LocalDate	Required	On or after start date	Campaign end date
`channel`	String	Optional	Enum: `B2B`, `B2C`, `ALL`	Target channel
`correlationId`	String	Optional	Max 100 chars	Integration correlation identifier
`allocationLines`	Array<CampaignAllocationImportLineDto>	Required	Min 1 line	Planned stock allocations

`CampaignAllocationImportLineDto`

Field	Type	Required	Validation Rules	Description
`sku`	ProductSku	Required	Active product	Campaign product
`locationCode`	LocationCode	Required	Active location	Allocation location
`targetQuantity`	PositiveQuantityDto	Required	Min exclusive 0 (VR-126)	Target promotional quantity

---

Request DTO Index

API ID	DTO Name	HTTP Method	Has Body
API-001	`CreateProductRequest`	POST	Yes
API-002	`SearchProductsQuery`	GET	No
API-003	`GetProductBySkuRequest`	GET	No
API-004	`UpdateProductAttributesRequest`	PATCH	Yes
API-005	`ChangeProductLifecycleStatusRequest`	POST	Yes
API-006	`AssignAbcClassificationRequest`	PUT	Yes
API-007	`ValidateProductEligibilityQuery`	GET	No
API-008	`ListProductsByCategoryQuery`	GET	No
API-009	`ResolveDuplicateProductConflictRequest`	POST	Yes
API-010	`BulkProductImportRequest`	POST	Yes
API-011 – API-178	*(see sections above)*	—	—

Total APIs documented: 178
Total shared nested DTOs: 12
Validation rules referenced: VR-001 through VR-162 (SRS-VR-001)

Document Approval

Role	Name	Signature	Date
Backend Architect			
Solution Architect			
Senior Business Analyst			
Product Owner			

End of Request DTO Specification

Document ID: SRS-API-002 | Version 1.0 | July 6, 2026

Response DTO Specification

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-API-003
Document Title	Response DTO Specification — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Senior Backend Architect
Source Documents	SRS-API-001 (REST API Catalog), SRS-API-002 (Request DTOs), SRS-VR-001, SRS-DM-002
Date	July 6, 2026
Scope	Success response DTOs only — error responses follow RFC 7807 Problem Details (separate contract)

Purpose

Defines the **success response structure** for every REST API in SRS-API-001: HTTP status code, response fields, nested objects, and example JSON. Naming aligns with SRS-API-002 request DTOs.

HTTP Status Code Conventions

Operation	Success Status	Response Body
`POST` — resource creation	`201 Created`	Created resource or creation result
`GET` — single resource	`200 OK`	Resource detail
`GET` — collection / search	`200 OK`	Paginated or array wrapper
`PATCH` / `PUT`	`200 OK`	Updated resource
`POST` — command / action	`200 OK`	Command result with updated state
`DELETE`	`200 OK`	Revocation result (or `204 No Content` when no body — IMS uses `200` with body for traceability)

Shared Response Wrappers

`PagedResponse<T>`

Field	Type	Description
`data`	Array<T>	Page of result items
`page`	Integer	Current page number (1-based)
`pageSize`	Integer	Page size
`totalElements`	Long	Total matching records
`totalPages`	Integer	Total pages

`CommandResultResponse`

Used by approve, reject, cancel, acknowledge, and similar action endpoints.

Field	Type	Description
`id`	UUID	Affected resource identifier
`status`	String	New resource status after command
`processedAt`	Instant	Command processing timestamp
`message`	String	Optional human-readable confirmation

`AuditInfoResponse`

Field	Type	Description
`createdAt`	Instant	Record creation timestamp
`createdBy`	String	Creating user ID
`updatedAt`	Instant	Last update timestamp
`updatedBy`	String	Last updating user ID

Shared Nested Response Objects

`QuantityResponse`

Field	Type	Description
`amount`	BigDecimal	Quantity amount
`unitOfMeasure`	String	UoM code

`MonetaryAmountResponse`

Field	Type	Description
`amount`	BigDecimal	Monetary value
`currencyCode`	String	ISO 4217 currency

`LotIdentityResponse`

Field	Type	Description
`lotNumber`	String	Lot/batch number

Field	Type	Description
`expirationDate`	LocalDate	Expiration date

`StockAvailabilityResponse`

Field	Type	Description
`onHand`	QuantityResponse	Physical on-hand quantity
`reserved`	QuantityResponse	Reserved quantity
`quarantined`	QuantityResponse	Quarantined quantity
`atp`	QuantityResponse	Available-to-promise
`ownershipType`	String	`OWNED` or `CONSIGNMENT`

`AddressResponse`

Field	Type	Description
`line1`	String	Address line 1
`line2`	String	Address line 2
`city`	String	City
`stateProvince`	String	State or province
`postalCode`	String	Postal code
`countryCode`	String	ISO country code

`ProductSummaryResponse`

Field	Type	Description
`sku`	String	Product SKU
`name`	String	Product name
`lifecycleStatus`	String	`ACTIVE`, `INACTIVE`, `DISCONTINUED`
`abcClass`	String	`A`, `B`, `C`, or null
`unitOfMeasure`	String	Primary UoM

`ProductDetailResponse` ***(extends summary)***

Field	Type	Description
(all ProductSummaryResponse fields)		
`description`	String	Product description
`categoryCode`	String	Category code
`barcode`	String	Primary barcode
`isTraceable`	Boolean	Lot/batch required
`rotationPolicy`	String	`FIFO` or `FEFO`
`shelfLifeManaged`	Boolean	MRSL applies
`mrslThresholdDays`	Integer	MRSL threshold days
`audit`	AuditInfoResponse	Audit metadata

`LocationSummaryResponse`

Field	Type	Description
`locationCode`	String	Location code
`name`	String	Location name
`locationType`	String	Location type enum
`operationalStatus`	String	Operational status
`parentLocationCode`	String	Parent location code

`EligibilityResponse`

Field	Type	Description
`eligible`	Boolean	Whether transaction is permitted
`transactionType`	String	Evaluated transaction type
`reason`	String	Ineligibility reason when `eligible=false`
`lifecycleStatus`	String	Current product lifecycle status

M-01 — Product Catalog

API-001 — `POST /api/v1/products`

Attribute	Value
HTTP Status	`201 Created`
Response DTO	`ProductDetailResponse`
Nested Objects	`AuditInfoResponse`

Example:

```
{
  "sku": "SKU-WIDGET-001",
  "name": "Industrial Widget 100mm",
  "lifecycleStatus": "ACTIVE",
  "abcClass": null,
  "unitOfMeasure": "EA",
  "description": "Standard 100mm widget",
  "categoryCode": "HARDW",
  "barcode": "0123456789012",
  "isTraceable": false,
  "rotationPolicy": "FIFO",
  "shelfLifeManaged": false,
  "mrslThresholdDays": null,
  "audit": {
    "createdAt": "2026-07-06T14:30:00Z",
    "createdBy": "usr-inv-ctrl-01",
    "updatedAt": "2026-07-06T14:30:00Z",
    "updatedBy": "usr-inv-ctrl-01"
  }
}
```

API-002 — `GET /api/v1/products`

Attribute	Value
HTTP Status	`200 OK`
Response DTO	`PagedResponse<ProductSummaryResponse>`
Nested Objects	`ProductSummaryResponse`

Example:

```
{
  "data": [
    {
      "sku": "SKU-WIDGET-001",
      "name": "Industrial Widget 100mm",
      "lifecycleStatus": "ACTIVE",
      "abcClass": "A",
      "unitOfMeasure": "EA"
    }
  ],
  "page": 1,
  "pageSize": 20,
  "totalElements": 1,
  "totalPages": 1
}
```

API-003 — `GET /api/v1/products/{sku}`

Attribute	Value
HTTP Status	`200 OK`
Response DTO	`ProductDetailResponse`
Nested Objects	`AuditInfoResponse`

Example: Same structure as API-001 response.

API-004 — `PATCH /api/v1/products/{sku}`

Attribute	Value
HTTP Status	`200 OK`
Response DTO	`ProductDetailResponse`
Nested Objects	`AuditInfoResponse`

Example: Same structure as API-001 with updated fields.

API-005 — `POST /api/v1/products/{sku}/lifecycle-status`

Attribute	Value
HTTP Status	`200 OK`
Response DTO	`ProductStatusChangeResponse`

Field	Type	Description
`sku`	String	Product SKU

Field	Type	Description
`priorStatus`	String	Previous lifecycle status
`newStatus`	String	New lifecycle status
`effectiveDate`	LocalDate	Effective date
`changedAt`	Instant	Change timestamp
`changedBy`	String	User who made the change

Example:

```
{
  "sku": "SKU-WIDGET-001",
  "priorStatus": "ACTIVE",
  "newStatus": "DISCONTINUED",
  "effectiveDate": "2026-07-06",
  "changedAt": "2026-07-06T14:35:00Z",
  "changedBy": "usr-inv-ctrl-01"
}
```

API-006 — `PUT /api/v1/products/{sku}/abc-classification`

Attribute	Value
HTTP Status	`200 OK`
Response DTO	`AbcClassificationResponse`

Field	Type	Description
`sku`	String	Product SKU
`abcClass`	String	Assigned class `A`, `B`, or `C`
`classificationBasis`	String	Basis for classification
`effectiveDate`	LocalDate	Effective date
`assignedAt`	Instant	Assignment timestamp

Example:

```
{
  "sku": "SKU-WIDGET-001",
  "abcClass": "A",
  "classificationBasis": "REVENUE",
  "effectiveDate": "2026-07-06",
  "assignedAt": "2026-07-06T14:36:00Z"
}
```

API-007 — `GET /api/v1/products/{sku}/eligibility`

Attribute	Value
HTTP Status	`200 OK`
Response DTO	`EligibilityResponse`

Example:

```
{
  "eligible": true,
  "transactionType": "RESERVATION",
  "reason": null,
  "lifecycleStatus": "ACTIVE"
}
```

API-008 — `GET /api/v1/products/categories/{categoryCode}/products`

Attribute	Value
HTTP Status	`200 OK`
Response DTO	`PagedResponse<ProductSummaryResponse>`

Example: Same structure as API-002.

API-009 — `POST /api/v1/products/duplicate-resolutions`

Attribute	Value
HTTP Status	`200 OK`
Response DTO	`DuplicateResolutionResponse`

Field	Type	Description
`resolutionAction`	String	Action taken
`sourceSku`	String	Source SKU
`targetSku`	String	Target SKU when merged
`resolvedAt`	Instant	Resolution timestamp
`resolvedBy`	String	Resolving user

Example:

```
{
  "resolutionAction": "MERGE",
  "sourceSku": "SKU-WIDGET-OLD",
  "targetSku": "SKU-WIDGET-001",
  "resolvedAt": "2026-07-06T14:40:00Z",
  "resolvedBy": "usr-inv-ctrl-01"
}
```

API-010 — `POST /api/v1/products/import`

Attribute	Value
HTTP Status	`200 OK`
Response DTO	`BulkImportResultResponse`

Field	Type	Description
`importMode`	String	Import mode executed
`totalRecords`	Integer	Records processed
`successCount`	Integer	Successfully imported
`failureCount`	Integer	Failed records

Field	Type	Description
`errors`	Array<ImportErrorResponse>	Per-record errors

`ImportErrorResponse`: `rowIndex` (Integer), `sku` (String), `errorCode` (String), `message` (String)

Example:

```
{
  "importMode": "UPSERT",
  "totalRecords": 100,
  "successCount": 98,
  "failureCount": 2,
  "errors": [
    {
      "rowIndex": 15,
      "sku": "SKU-DUP-001",
      "errorCode": "DUPLICATE_SKU",
      "message": "SKU already exists"
    }
  ]
}
```

M-02 — Warehouse Network

API-011 — `POST /api/v1/locations`

| HTTP Status | `201 Created` | **DTO:** `LocationDetailResponse`

Field	Type	Description
`locationCode`	String	Location code
`name`	String	Location name
`locationType`	String	Type enum
`operationalStatus`	String	Status enum
`parentLocationCode`	String	Parent code
`address`	AddressResponse	Physical address
`plannedGoLiveDate`	LocalDate	Planned go-live
`isFrozen`	Boolean	Freeze flag
`audit`	AuditInfoResponse	Audit metadata

Example:

```
{
  "locationCode": "WH-EAST-01",
  "name": "East Coast Distribution Center",
  "locationType": "WAREHOUSE",
  "operationalStatus": "PLANNED",
  "parentLocationCode": "NET-US-EAST",
  "address": {
    "line1": "100 Warehouse Blvd",
    "line2": null,
    "city": "Newark",
    "stateProvince": "NJ",
    "postalCode": "07102",
    "countryCode": "US"
  },
  "plannedGoLiveDate": "2026-08-01",
  "isFrozen": false,
  "audit": {
    "createdAt": "2026-07-06T14:45:00Z",
    "createdBy": "usr-inv-ctrl-01",
    "updatedAt": "2026-07-06T14:45:00Z",
    "updatedBy": "usr-inv-ctrl-01"
  }
}
```

API-012 — `GET /api/v1/locations`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<LocationSummaryResponse>`

Example:

```
{
  "data": [
    {
      "locationCode": "WH-EAST-01",
      "name": "East Coast Distribution Center",
      "locationType": "WAREHOUSE",
      "operationalStatus": "ACTIVE",
      "parentLocationCode": "NET-US-EAST"
    }
  ],
  "page": 1,
  "pageSize": 20,
  "totalElements": 1,
  "totalPages": 1
}
```

API-013 — `GET /api/v1/locations/{locationCode}`

| HTTP Status | `200 OK` | **DTO:** `LocationDetailResponse` — same as API-011.

API-014 — `PATCH /api/v1/locations/{locationCode}/hierarchy`

| HTTP Status | `200 OK` | **DTO:** `LocationDetailResponse` — updated hierarchy fields.

API-015 — `POST /api/v1/locations/{locationCode}/storage-positions`

| HTTP Status | `201 Created` | **DTO:** `StoragePositionResponse`

Field	Type	Description
`locationCode`	String	Parent location

Field	Type	Description
`positionCode`	String	Position code
`zone`	String	Zone
`aisle`	String	Aisle
`bin`	String	Bin
`positionType`	String	Position type
`isActive`	Boolean	Active flag
`configuredAt`	Instant	Configuration timestamp

Example:

```
{
  "locationCode": "WH-EAST-01",
  "positionCode": "A-01-001",
  "zone": "A",
  "aisle": "01",
  "bin": "001",
  "positionType": "PICK_FACE",
  "isActive": true,
  "configuredAt": "2026-07-06T14:50:00Z"
}
```

API-016 — `GET /api/v1/locations/{locationCode}/storage-positions`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<StoragePositionResponse>`

API-017 — `POST /api/v1/locations/{locationCode}/activate`

| HTTP Status | `200 OK` | **DTO:** `LocationDetailResponse` with `operationalStatus: "ACTIVE"`.

API-018 — `POST /api/v1/locations/{locationCode}/deactivate`

| HTTP Status | `200 OK` | **DTO:** `LocationDetailResponse` with `operationalStatus: "INACTIVE"`.

API-019 — `POST /api/v1/locations/{locationCode}/close`

| HTTP Status | `200 OK` | **DTO:** `LocationClosureResponse`

Field	Type	Description
`locationCode`	String	Closed location
`operationalStatus`	String	`INACTIVE`
`closureDate`	LocalDate	Closure date
`dispositionPlanReference`	String	Disposition plan ref
`closedAt`	Instant	Closure timestamp

Example:

```
{
  "locationCode": "WH-EAST-01",
  "operationalStatus": "INACTIVE",
  "closureDate": "2026-07-06",
  "dispositionPlanReference": "DISP-PLAN-2026-042",
  "closedAt": "2026-07-06T15:00:00Z"
}
```

API-020 — `GET /api/v1/locations/{locationCode}/freeze-status`

| HTTP Status | `200 OK` | **DTO:** `LocationFreezeStatusResponse`

Field	Type	Description
`locationCode`	String	Location code
`isFrozen`	Boolean	Whether location is frozen
`freezeReason`	String	Reason when frozen
`frozenSince`	Instant	Freeze start time
`physicalCountId`	UUID	Linked physical count when applicable

Example:

```
{
  "locationCode": "WH-EAST-01",
  "isFrozen": true,
  "freezeReason": "PHYSICAL_INVENTORY_COUNT",
  "frozenSince": "2026-07-06T08:00:00Z",
  "physicalCountId": "a1b2c3d4-e5f6-7890-abcd-ef1234567890"
}
```

API-021 — `GET /api/v1/locations/{locationCode}/operational-status`

| HTTP Status | `200 OK` | **DTO:** `LocationOperationalStatusResponse`

Field	Type	Description
`locationCode`	String	Location code
`operationalStatus`	String	Current status
`transactionallyActive`	Boolean	Accepts inventory transactions
`openingBalanceApproved`	Boolean	Opening balance approved

Example:

```
{
  "locationCode": "WH-EAST-01",
  "operationalStatus": "ACTIVE",
  "transactionallyActive": true,
  "openingBalanceApproved": true
}
```

M-03 — Inventory Governance

API-022 — `POST /api/v1/governance/reason-codes`

| HTTP Status | `201 Created` | **DTO:** `ReasonCodeResponse`

Field	Type	Description
`code`	String	Reason code
`description`	String	Description
`transactionTypes`	Array<String>	Applicable transaction types
`isActive`	Boolean	Active flag
`audit`	AuditInfoResponse	Audit metadata

Example:

```
{
  "code": "ADJ-DAMAGE",
  "description": "Damaged goods adjustment",
  "transactionTypes": ["ADJUSTMENT", "WRITE_OFF"],
  "isActive": true,
  "audit": {
    "createdAt": "2026-07-06T15:05:00Z",
    "createdBy": "usr-inv-ctrl-01",
    "updatedAt": "2026-07-06T15:05:00Z",
    "updatedBy": "usr-inv-ctrl-01"
  }
}
```

API-023 — `GET /api/v1/governance/reason-codes`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<ReasonCodeResponse>`

API-024 — `POST /api/v1/governance/reason-codes/{code}/retire`

| HTTP Status | `200 OK` | **DTO:** `ReasonCodeResponse` with `isActive: false`

API-025 — `GET /api/v1/governance/reason-codes/{code}/validate`

| HTTP Status | `200 OK` | **DTO:** `ReasonCodeValidationResponse`

Field	Type	Description
`code`	String	Reason code
`valid`	Boolean	Valid for transaction type
`transactionType`	String	Evaluated type
`message`	String	Invalid reason when `valid=false`

Example:

```
{
  "code": "ADJ-DAMAGE",
  "valid": true,
  "transactionType": "ADJUSTMENT",
  "message": null
}
```

API-026 — `POST /api/v1/governance/policies`

| HTTP Status | `201 Created` | **DTO:** `GovernancePolicyResponse`

Field	Type	Description
`policyId`	UUID	Policy identifier
`policyName`	String	Policy name
`effectiveFrom`	Instant	Effective start
`effectiveTo`	Instant	Effective end
`parameters`	Array<PolicyParameterResponse>	Policy parameters
`audit`	AuditInfoResponse	Audit metadata

Example:

```
{
  "policyId": "b2c3d4e5-f6a7-8901-bcde-f12345678901",
  "policyName": "FY2026 Adjustment Policy",
  "effectiveFrom": "2026-07-01T00:00:00Z",
  "effectiveTo": null,
  "parameters": [
    {
      "parameterKey": "ADJUSTMENT_QTY_THRESHOLD",
      "parameterValue": "50",
      "parameterType": "QUANTITY"
    }
  ],
  "audit": {
    "createdAt": "2026-07-06T15:10:00Z",
    "createdBy": "usr-inv-ctrl-01",
    "updatedAt": "2026-07-06T15:10:00Z",
    "updatedBy": "usr-inv-ctrl-01"
  }
}
```

API-027 — `PATCH /api/v1/governance/policies/{policyId}/approval-thresholds`

| HTTP Status | `200 OK` | **DTO:** `ApprovalThresholdResponse`

Field	Type	Description
`policyId`	UUID	Policy ID
`transactionType`	String	Transaction type
`quantityThreshold`	BigDecimal	Quantity threshold
`monetaryThreshold`	MonetaryAmountResponse	Monetary threshold
`effectiveFrom`	LocalDate	Effective date

API-028 — `PATCH /api/v1/governance/policies/{policyId}/receipt-tolerances`

| HTTP Status | `200 OK` | **DTO:** `ReceiptToleranceResponse`

Field	Type	Description
`policyId`	UUID	Policy ID
`quantityTolerancePercent`	BigDecimal	Percent tolerance
`quantityToleranceAbsolute`	BigDecimal	Absolute tolerance
`autoAcceptWithinTolerance`	Boolean	Auto-accept flag
`effectiveFrom`	LocalDate	Effective date

API-029 — `GET /api/v1/governance/approval-thresholds`

| HTTP Status | `200 OK` | **DTO:** `ApprovalThresholdResponse`

API-030 — `GET /api/v1/governance/receipt-tolerances`

| HTTP Status | `200 OK` | **DTO:** `ReceiptToleranceResponse`

API-031 — `POST /api/v1/governance/segregation-of-duties/evaluate`

| HTTP Status | `200 OK` | **DTO:** `SegregationOfDutiesEvaluationResponse`

Field	Type	Description
`compliant`	Boolean	SoD compliant
`initiatorUserId`	String	Initiator user
`approverUserId`	String	Approver user
`transactionType`	String	Transaction type
`violationCode`	String	Violation code when non-compliant
`message`	String	Explanation

Example:

```
{
  "compliant": false,
  "initiatorUserId": "usr-wh-staff-05",
  "approverUserId": "usr-wh-staff-05",
  "transactionType": "COUNT_VARIANCE_APPROVAL",
  "violationCode": "SOD_COUNTER_APPROVER",
  "message": "Counter cannot approve own count variance"
}
```

M-04 — Inventory Balance

API-032 — `POST /api/v1/inventory/opening-balances`

| HTTP Status | `201 Created` | **DTO:** `OpeningBalancesResultResponse`

Field	Type	Description
`locationCode`	String	Location
`linesProcessed`	Integer	Lines established
`totalOnHandValue`	MonetaryAmountResponse	Optional total value
`establishedAt`	Instant	Establishment timestamp
`approvalReference`	String	Approval reference

Example:

```
{
  "locationCode": "WH-EAST-01",
  "linesProcessed": 250,
  "totalOnHandValue": { "amount": 1250000.00, "currencyCode": "USD" },
  "establishedAt": "2026-07-06T15:15:00Z",
  "approvalReference": "OB-APPR-2026-001"
}
```

API-033 — `GET /api/v1/inventory/stock-positions`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<StockPositionSummaryResponse>`

Field	Type	Description
`locationCode`	String	Location
`sku`	String	Product SKU
`productName`	String	Product name
`availability`	StockAvailabilityResponse	Quantity breakdown
`stockStatus`	String	Status indicator
`abcClass`	String	ABC class
`lastMovementAt`	Instant	Last movement date

Example:

```
{
  "data": [
    {
      "locationCode": "WH-EAST-01",
      "sku": "SKU-WIDGET-001",
      "productName": "Industrial Widget 100mm",
      "availability": {
        "onHand": { "amount": 500, "unitOfMeasure": "EA" },
        "reserved": { "amount": 30, "unitOfMeasure": "EA" },
        "quarantined": { "amount": 0, "unitOfMeasure": "EA" },
        "atp": { "amount": 470, "unitOfMeasure": "EA" },
        "ownershipType": "OWNED"
      },
      "stockStatus": "IN_STOCK",
      "abcClass": "A",
      "lastMovementAt": "2026-07-06T12:00:00Z"
    }
  ],
  "page": 1,
  "pageSize": 20,
  "totalElements": 1,
  "totalPages": 1
}
```

API-034 — `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}`

| HTTP Status | `200 OK` | **DTO:** `StockPositionDetailResponse`

Extends summary with: `lots` (Array<LotQuantityResponse>), `activeHolds` (Array<QuarantineHoldSummaryResponse>), `policy` (StockPolicySummaryResponse), `audit` (AuditInfoResponse).

Example:

```
{
  "locationCode": "WH-EAST-01",
  "sku": "SKU-WIDGET-001",
  "productName": "Industrial Widget 100mm",
  "availability": {
    "onHand": { "amount": 500, "unitOfMeasure": "EA" },
    "reserved": { "amount": 30, "unitOfMeasure": "EA" },
    "quarantined": { "amount": 0, "unitOfMeasure": "EA" },
    "atp": { "amount": 470, "unitOfMeasure": "EA" },
    "ownershipType": "OWNED"
  },
  "stockStatus": "IN_STOCK",
  "lots": [],
  "activeHolds": [],
  "policy": {
    "reorderLevel": 100,
    "minimumThreshold": 50,
    "safetyStock": 25
  },
  "audit": {
    "createdAt": "2026-07-01T00:00:00Z",
    "createdBy": "system",
    "updatedAt": "2026-07-06T12:00:00Z",
    "updatedBy": "usr-rcv-01"
  }
}
```

API-035 — `GET /api/v1/inventory/availability/{sku}`

| HTTP Status | `200 OK` | **DTO:** `AvailabilityInquiryResponse`

Field	Type	Description
`sku`	String	Product SKU
`locationCode`	String	Location
`atp`	QuantityResponse	Available quantity
`canFulfillRequested`	Boolean	Meets requested qty when provided
`requestedQuantity`	QuantityResponse	Requested qty evaluated
`inquiredAt`	Instant	Inquiry timestamp

Example:

```
{
  "sku": "SKU-WIDGET-001",
  "locationCode": "WH-EAST-01",
  "atp": { "amount": 470, "unitOfMeasure": "EA" },
  "canFulfillRequested": true,
  "requestedQuantity": { "amount": 100, "unitOfMeasure": "EA" },
  "inquiredAt": "2026-07-06T15:20:00Z"
}
```

API-036 — `GET /api/v1/inventory/availability/{sku}/network`

| HTTP Status | `200 OK` | **DTO:** `NetworkAvailabilityResponse`

Field	Type	Description
`sku`	String	Product SKU
`networkAtp`	QuantityResponse	Total network ATP

Field	Type	Description
`locations`	Array<LocationAvailabilityResponse>	Per-location ATP

Example:

```
{
  "sku": "SKU-WIDGET-001",
  "networkAtp": { "amount": 1200, "unitOfMeasure": "EA" },
  "locations": [
    {
      "locationCode": "WH-EAST-01",
      "atp": { "amount": 470, "unitOfMeasure": "EA" },
      "stockStatus": "IN_STOCK"
    },
    {
      "locationCode": "WH-WEST-01",
      "atp": { "amount": 730, "unitOfMeasure": "EA" },
      "stockStatus": "IN_STOCK"
    }
  ]
}
```

API-037 — `GET /api/v1/inventory/stock-status/{locationCode}/{sku}`

| HTTP Status | `200 OK` | **DTO:** `StockStatusIndicatorResponse`

Field	Type	Description
`locationCode`	String	Location
`sku`	String	SKU
`stockStatus`	String	`IN_STOCK`, `LOW_STOCK`, `OUT_OF_STOCK`, `OVERSTOCK`
`atp`	QuantityResponse	Current ATP
`reorderLevel`	BigDecimal	Reorder threshold
`minimumThreshold`	BigDecimal	Minimum threshold

API-038 — `POST /api/v1/inventory/stock-policies`

| HTTP Status | `201 Created` | **DTO:** `StockPolicyResponse`

Field	Type	Description
`locationCode`	String	Location
`sku`	String	SKU
`reorderLevel`	BigDecimal	Reorder level
`minimumThreshold`	BigDecimal	Minimum
`safetyStock`	BigDecimal	Safety stock
`maximumThreshold`	BigDecimal	Maximum
`leadTimeDays`	Integer	Lead time
`effectiveFrom`	LocalDate	Effective date
`audit`	AuditInfoResponse	Audit metadata

API-039 — `GET /api/v1/inventory/stock-policies/{locationCode}/{sku}`

| HTTP Status | `200 OK` | **DTO:** `StockPolicyResponse`

API-040 — `PATCH /api/v1/inventory/stock-policies/{locationCode}/{sku}`

| HTTP Status | `200 OK` | **DTO:** `StockPolicyResponse` with updated thresholds.

API-041 — `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/lots`

| HTTP Status | `200 OK` | **DTO:** `LotBreakdownResponse`

Field	Type	Description
`locationCode`	String	Location
`sku`	String	SKU
`lots`	Array<LotQuantityResponse>	Lot-level quantities

`LotQuantityResponse`: `lotIdentity` (LotIdentityResponse), `onHand` (QuantityResponse), `reserved` (QuantityResponse), `atp` (QuantityResponse), `isQuarantined` (Boolean)

API-042 — `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/transfer-capacity`

| HTTP Status | `200 OK` | **DTO:** `TransferCapacityResponse`

Field	Type	Description
`locationCode`	String	Source location
`sku`	String	SKU
`onHand`	QuantityResponse	On-hand
`minimumThreshold`	BigDecimal	Source minimum
`safetyStock`	BigDecimal	Safety stock
`maxTransferable`	QuantityResponse	Max spare transferable
`canFulfillRequested`	Boolean	Meets requested qty

Example:

```
{
  "locationCode": "WH-WEST-01",
  "sku": "SKU-WIDGET-001",
  "onHand": { "amount": 730, "unitOfMeasure": "EA" },
  "minimumThreshold": 100,
  "safetyStock": 50,
  "maxTransferable": { "amount": 580, "unitOfMeasure": "EA" },
  "canFulfillRequested": true
}
```

API-043 — `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/reservations`

| HTTP Status | `201 Created` | **DTO:** `ReservationResponse`

Field	Type	Description
`reservationId`	UUID	Reservation ID

Field	Type	Description
`locationCode`	String	Location
`sku`	String	SKU
`reservedQuantity`	QuantityResponse	Reserved qty
`orderId`	String	Order reference
`orderLineId`	String	Order line reference
`reservedAt`	Instant	Reservation time
`expiresAt`	Instant	Expiry time
`remainingAtp`	QuantityResponse	ATP after reservation

Example:

```
{
  "reservationId": "c3d4e5f6-a7b8-9012-cdef-123456789012",
  "locationCode": "WH-EAST-01",
  "sku": "SKU-WIDGET-001",
  "reservedQuantity": { "amount": 50, "unitOfMeasure": "EA" },
  "orderId": "ORD-2026-0042",
  "orderLineId": "LINE-001",
  "reservedAt": "2026-07-06T15:25:00Z",
  "expiresAt": "2026-07-13T15:25:00Z",
  "remainingAtp": { "amount": 420, "unitOfMeasure": "EA" }
}
```

API-044 — `POST .../reservations/{reservationId}/release`

| HTTP Status | `200 OK` | **DTO:** `ReservationReleaseResponse`

Field	Type	Description
`reservationId`	UUID	Released reservation
`releasedQuantity`	QuantityResponse	Released qty
`releaseReason`	String	Release reason
`releasedAt`	Instant	Release timestamp
`restoredAtp`	QuantityResponse	ATP after release

API-045 — `POST .../quarantine-holds`

| HTTP Status | `201 Created` | **DTO:** `QuarantineHoldResponse`

Field	Type	Description
`holdId`	UUID	Hold ID
`locationCode`	String	Location
`sku`	String	SKU
`heldQuantity`	QuantityResponse	Held qty
`holdReason`	String	Hold reason
`reasonCode`	String	Reason code
`appliedAt`	Instant	Applied timestamp
`remainingAtp`	QuantityResponse	ATP after hold

API-046 — `POST .../quarantine-holds/{holdId}/release`

| HTTP Status | `200 OK` | **DTO:** `QuarantineHoldReleaseResponse` — `holdId`, `releasedQuantity`, `disposition`, `releasedAt`, `restoredAtp`.

API-047 — `POST .../consignment-ownership`

| HTTP Status | `200 OK` | **DTO:** `ConsignmentOwnershipTransferResponse`

Field	Type	Description
`locationCode`	String	Location
`sku`	String	SKU
`transferredQuantity`	QuantityResponse	Qty transferred
`consumptionReference`	String	Consumption ref
`transferredAt`	Instant	Transfer timestamp
`newOwnershipType`	String	`OWNED`

API-048 — `POST .../promotional-segregation`

| HTTP Status | `200 OK` | **DTO:** `PromotionalSegregationResponse`

Field	Type	Description
`locationCode`	String	Location
`sku`	String	SKU
`campaignId`	String	Campaign ID
`segregatedQuantity`	QuantityResponse	Ring-fenced qty
`generalAtp`	QuantityResponse	Remaining general ATP
`segregatedAt`	Instant	Segregation timestamp

M-05 — Procurement & Supplier

API-049 — `POST /api/v1/suppliers`

| HTTP Status | `201 Created` | **DTO:** `SupplierDetailResponse`

Field	Type	Description
`supplierCode`	String	Supplier code
`name`	String	Supplier name
`contactEmail`	String	Contact email
`contactPhone`	String	Contact phone
`paymentTermsDays`	Integer	Payment terms
`defaultLeadTimeDays`	Integer	Lead time
`isConsignment`	Boolean	Consignment flag
`isActive`	Boolean	Active flag
`currencyCode`	String	Default currency

Field	Type	Description
`audit`	AuditInfoResponse	Audit metadata

Example:

```
{
  "supplierCode": "SUP-ACME-01",
  "name": "Acme Industrial Supply",
  "contactEmail": "orders@acme.example.com",
  "contactPhone": "+1-555-0100",
  "paymentTermsDays": 30,
  "defaultLeadTimeDays": 14,
  "isConsignment": false,
  "isActive": true,
  "currencyCode": "USD",
  "audit": {
    "createdAt": "2026-07-06T15:30:00Z",
    "createdBy": "usr-buyer-01",
    "updatedAt": "2026-07-06T15:30:00Z",
    "updatedBy": "usr-buyer-01"
  }
}
```

API-050 — `GET /api/v1/suppliers`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<SupplierSummaryResponse>` — `supplierCode`, `name`, `isActive`, `isConsignment`.

API-051 — `GET /api/v1/suppliers/{supplierCode}`

| HTTP Status | `200 OK` | **DTO:** `SupplierDetailResponse`

API-052 — `PATCH /api/v1/suppliers/{supplierCode}`

| HTTP Status | `200 OK` | **DTO:** `SupplierDetailResponse` (updated).

API-053 — `POST /api/v1/suppliers/{supplierCode}/deactivate`

| HTTP Status | `200 OK` | **DTO:** `SupplierDetailResponse` with `isActive: false`.

API-054 — `POST /api/v1/procurement/requisitions`

| HTTP Status | `201 Created` | **DTO:** `PurchaseRequisitionResponse`

Field	Type	Description
`requisitionId`	UUID	Requisition ID
`status`	String	`SUBMITTED`
`receivingLocationCode`	String	Target location
`justification`	String	Justification
`lines`	Array<RequisitionLineResponse>	Line items
`submittedAt`	Instant	Submission time
`audit`	AuditInfoResponse	Audit metadata

`RequisitionLineResponse`: `lineId`, `sku`, `productName`, `requestedQuantity` (QuantityResponse), `neededByDate`

Example:

```
{
  "requisitionId": "d4e5f6a7-b8c9-0123-def0-123456789abc",
  "status": "SUBMITTED",
  "receivingLocationCode": "WH-EAST-01",
  "justification": "Class A SKU below reorder level",
  "lines": [
    {
      "lineId": "e5f6a7b8-c9d0-1234-ef01-234567890bcd",
      "sku": "SKU-WIDGET-001",
      "productName": "Industrial Widget 100mm",
      "requestedQuantity": { "amount": 200, "unitOfMeasure": "EA" },
      "neededByDate": "2026-07-20"
    }
  ],
  "submittedAt": "2026-07-06T15:35:00Z",
  "audit": {
    "createdAt": "2026-07-06T15:35:00Z",
    "createdBy": "usr-buyer-01",
    "updatedAt": "2026-07-06T15:35:00Z",
    "updatedBy": "usr-buyer-01"
  }
}
```

API-055 — `GET /api/v1/procurement/requisitions/{requisitionId}`

| HTTP Status | `200 OK` | **DTO:** `PurchaseRequisitionResponse`

API-056 — `POST .../requisitions/{requisitionId}/approve`

| HTTP Status | `200 OK` | **DTO:** `CommandResultResponse` — `id`, `status: "APPROVED"`, `processedAt`, `message`

API-057 — `POST .../requisitions/{requisitionId}/reject`

| HTTP Status | `200 OK` | **DTO:** `CommandResultResponse` — `status: "REJECTED"`

API-058 — `POST .../requisitions/{requisitionId}/convert-to-po`

| HTTP Status | `201 Created` | **DTO:** `PurchaseOrderResponse` (see API-059).

API-059 — `POST /api/v1/procurement/purchase-orders`

| HTTP Status | `201 Created` | **DTO:** `PurchaseOrderResponse`

Field	Type	Description
`purchaseOrderId`	UUID	PO ID
`poNumber`	String	Business PO number
`status`	String	`OPEN`
`supplierCode`	String	Supplier
`supplierName`	String	Supplier name
`receivingLocationCode`	String	Receiving location
`expectedDeliveryDate`	LocalDate	Expected delivery
`currencyCode`	String	Currency

Field	Type	Description
`totalAmount`	MonetaryAmountResponse	PO total
`lines`	Array<PurchaseOrderLineResponse>	PO lines
`audit`	AuditInfoResponse	Audit metadata

`PurchaseOrderLineResponse`: `lineId`, `lineNumber`, `sku`, `productName`, `orderedQuantity`, `receivedQuantity`, `remainingQuantity`, `unitPrice`

Example:

```
{
  "purchaseOrderId": "f6a7b8c9-d0e1-2345-f012-345678901cde",
  "poNumber": "PO-2026-0088",
  "status": "OPEN",
  "supplierCode": "SUP-ACME-01",
  "supplierName": "Acme Industrial Supply",
  "receivingLocationCode": "WH-EAST-01",
  "expectedDeliveryDate": "2026-07-20",
  "currencyCode": "USD",
  "totalAmount": { "amount": 5000.00, "currencyCode": "USD" },
  "lines": [
    {
      "lineId": "a7b8c9d0-e1f2-3456-0123-456789012def",
      "lineNumber": 1,
      "sku": "SKU-WIDGET-001",
      "productName": "Industrial Widget 100mm",
      "orderedQuantity": { "amount": 200, "unitOfMeasure": "EA" },
      "receivedQuantity": { "amount": 0, "unitOfMeasure": "EA" },
      "remainingQuantity": { "amount": 200, "unitOfMeasure": "EA" },
      "unitPrice": { "amount": 25.00, "currencyCode": "USD" }
    }
  ],
  "audit": {
    "createdAt": "2026-07-06T15:40:00Z",
    "createdBy": "usr-buyer-01",
    "updatedAt": "2026-07-06T15:40:00Z",
    "updatedBy": "usr-buyer-01"
  }
}
```

API-060 — `GET /api/v1/procurement/purchase-orders`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<PurchaseOrderSummaryResponse>`

API-061 — `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}`

| HTTP Status | `200 OK` | **DTO:** `PurchaseOrderResponse`

API-062 — `PATCH /api/v1/procurement/purchase-orders/{purchaseOrderId}`

| HTTP Status | `200 OK` | **DTO:** `PurchaseOrderResponse` (amended).

API-063 — `POST .../purchase-orders/{purchaseOrderId}/cancel`

| HTTP Status | `200 OK` | **DTO:** `CommandResultResponse` — `status: "CANCELLED"`

API-064 — `POST .../expected-deliveries`

| HTTP Status | `200 OK` | **DTO:** `InboundDeliveryScheduleResponse`

Field	Type	Description
`purchaseOrderId`	UUID	PO ID
`expectedArrivalDate`	LocalDate	Expected arrival
`carrierReference`	String	Carrier ref
`lineDeliveries`	Array<LineDeliveryScheduleResponse>	Per-line schedule
`scheduledAt`	Instant	Schedule timestamp

API-065 — `GET .../validate-for-receipt`

| HTTP Status | `200 OK` | **DTO:** `PoReceiptValidationResponse`

Field	Type	Description
`purchaseOrderId`	UUID	PO ID
`validForReceipt`	Boolean	Receipt authorized
`poStatus`	String	PO status
`receivingLocationCode`	String	Authorized location
`message`	String	Invalid reason when false

Example:

```
{
  "purchaseOrderId": "f6a7b8c9-d0e1-2345-f012-345678901cde",
  "validForReceipt": true,
  "poStatus": "OPEN",
  "receivingLocationCode": "WH-EAST-01",
  "message": null
}
```

M-06 — Inbound Receiving

API-066 — `POST /api/v1/inbound/goods-receipts`

| HTTP Status | `201 Created` | **DTO:** `GoodsReceiptResponse`

Field	Type	Description
`goodsReceiptId`	UUID	Receipt ID
`receiptNumber`	String	Business receipt number
`status`	String	`DRAFT`
`purchaseOrderId`	UUID	Linked PO
`receivingLocationCode`	String	Location
`supplierCode`	String	Supplier
`lines`	Array<GoodsReceiptLineResponse>	Receipt lines
`audit`	AuditInfoResponse	Audit metadata

Example:

```
{
  "goodsReceiptId": "b8c9d0e1-f2a3-4567-8901-234567890abc",
  "receiptNumber": "GR-2026-0156",
  "status": "DRAFT",
  "purchaseOrderId": "f6a7b8c9-d0e1-2345-f012-345678901cde",
  "receivingLocationCode": "WH-EAST-01",
  "supplierCode": "SUP-ACME-01",
  "lines": [
    {
      "lineId": "c9d0e1f2-a3b4-5678-9012-345678901bcd",
      "sku": "SKU-WIDGET-001",
      "expectedQuantity": { "amount": 200, "unitOfMeasure": "EA" },
      "acceptedQuantity": { "amount": 0, "unitOfMeasure": "EA" },
      "putAwayStatus": "PENDING"
    }
  ],
  "audit": {
    "createdAt": "2026-07-06T16:00:00Z",
    "createdBy": "usr-rcv-01",
    "updatedAt": "2026-07-06T16:00:00Z",
    "updatedBy": "usr-rcv-01"
  }
}
```

API-067 — `GET /api/v1/inbound/goods-receipts`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<GoodsReceiptSummaryResponse>`

API-068 — `GET /api/v1/inbound/goods-receipts/{goodsReceiptId}`

| HTTP Status | `200 OK` | **DTO:** `GoodsReceiptResponse` (full detail with discrepancies, put-away state).

API-069 — `POST .../verify`

| HTTP Status | `200 OK` | **DTO:** `GoodsReceiptResponse` with `status: "VERIFIED"`.

API-070 — `POST .../discrepancies`

| HTTP Status | `201 Created` | **DTO:** `ReceiptDiscrepancyResponse`

Field	Type	Description
`discrepancyId`	UUID	Discrepancy ID
`goodsReceiptId`	UUID	Receipt ID
`lineId`	UUID	Line ID
`discrepancyType`	String	Type enum
`varianceQuantity`	QuantityResponse	Variance qty
`status`	String	`OPEN`
`recordedAt`	Instant	Recorded timestamp

API-071 — `GET .../discrepancies`

| HTTP Status | `200 OK` | **DTO:** `Array<ReceiptDiscrepancyResponse>` or `PagedResponse`

API-072 — `POST .../discrepancies/{discrepancyId}/resolve`

| HTTP Status | `200 OK` | **DTO:** `ReceiptDiscrepancyResponse` with `status: "RESOLVED"`.

API-073 — `POST .../put-away`

| HTTP Status | `200 OK` | **DTO:** `PutAwayCompletionResponse`

Field	Type	Description
`goodsReceiptId`	UUID	Receipt ID
`status`	String	`COMPLETED`
`putAwayLines`	Array<PutAwayLineResultResponse>	Put-away results
`stockAvailableAt`	Instant	When stock became ATP-available

Example:

```
{
  "goodsReceiptId": "b8c9d0e1-f2a3-4567-8901-234567890abc",
  "status": "COMPLETED",
  "putAwayLines": [
    {
      "receiptLineId": "c9d0e1f2-a3b4-5678-9012-345678901bcd",
      "quantity": { "amount": 200, "unitOfMeasure": "EA" },
      "storagePositionCode": "A-01-001",
      "putAwayAt": "2026-07-06T16:30:00Z"
    }
  ],
  "stockAvailableAt": "2026-07-06T16:30:00Z"
}
```

API-074 — `GET /api/v1/inbound/pending-put-away`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<PendingPutAwayResponse>`

API-075 — `POST /api/v1/inbound/ad-hoc-receipts`

| HTTP Status | `201 Created` | **DTO:** `GoodsReceiptResponse` (ad-hoc type).

API-076 — `POST /api/v1/inbound/return-receipts`

| HTTP Status | `201 Created` | **DTO:** `ReturnReceiptResponse`

Field	Type	Description
`returnReceiptId`	UUID	Return receipt ID
`returnAuthorizationId`	UUID	RMA reference
`goodsReceiptId`	UUID	Linked goods receipt
`status`	String	`RECEIVED`
`lines`	Array<ReturnReceiptLineResponse>	Received lines
`receivedAt`	Instant	Receipt timestamp

API-077 — `POST /api/v1/inbound/supplier-invoices`

| HTTP Status | `201 Created` | **DTO:** `SupplierInvoiceResponse`

Field	Type	Description
`invoiceId`	UUID	Invoice ID
`invoiceNumber`	String	Supplier invoice number
`supplierCode`	String	Supplier
`purchaseOrderId`	UUID	PO reference
`status`	String	`REGISTERED`
`totalAmount`	MonetaryAmountResponse	Invoice total
`matchStatus`	String	`PENDING`
`lines`	Array<InvoiceLineResponse>	Invoice lines
`audit`	AuditInfoResponse	Audit metadata

API-078 — `GET /api/v1/inbound/supplier-invoices/{invoiceId}`

| HTTP Status | `200 OK` | **DTO:** `SupplierInvoiceResponse` with `matchStatus`, `threeWayMatchResult`.

API-079 — `POST .../three-way-match`

| HTTP Status | `200 OK` | **DTO:** `ThreeWayMatchResultResponse`

Field	Type	Description
`invoiceId`	UUID	Invoice ID
`matchStatus`	String	`MATCHED` or `EXCEPTION`
`poMatch`	Boolean	PO aligned
`receiptMatch`	Boolean	Receipt aligned
`invoiceMatch`	Boolean	Invoice aligned
`exceptions`	Array<MatchExceptionResponse>	Line-level exceptions
`evaluatedAt`	Instant	Evaluation timestamp

Example:

```
{
  "invoiceId": "d0e1f2a3-b4c5-6789-0123-456789012def",
  "matchStatus": "MATCHED",
  "poMatch": true,
  "receiptMatch": true,
  "invoiceMatch": true,
  "exceptions": [],
  "evaluatedAt": "2026-07-06T17:00:00Z"
}
```

API-080 — `POST .../authorize-payment`

| HTTP Status | `200 OK` | **DTO:** `InvoicePaymentAuthorizationResponse`

Field	Type	Description
`invoiceId`	UUID	Invoice ID
`paymentAuthorized`	Boolean	Authorization granted
`authorizedAt`	Instant	Authorization time
`authorizedBy`	String	Authorizing user
`paymentDueDate`	LocalDate	Payment due date

M-07 — Order Fulfillment

API-081 — `POST /api/v1/fulfillment/orders`

| HTTP Status | `201 Created` | **DTO:** `CustomerOrderResponse`

Field	Type	Description
`orderId`	UUID	Order ID
`orderNumber`	String	Business order number
`status`	String	`DRAFT`
`channel`	String	Sales channel
`customerReference`	String	Customer ref
`shipFromLocationCode`	String	Ship-from location
`lines`	Array<OrderLineResponse>	Order lines
`audit`	AuditInfoResponse	Audit metadata

Example:

```
{
  "orderId": "e1f2a3b4-c5d6-7890-1234-567890123def",
  "orderNumber": "ORD-2026-0042",
  "status": "DRAFT",
  "channel": "B2B",
  "customerReference": "CUST-ACME-001",
  "shipFromLocationCode": "WH-EAST-01",
  "lines": [
    {
      "lineId": "f2a3b4c5-d6e7-8901-2345-678901234ef0",
      "lineNumber": 1,
      "sku": "SKU-WIDGET-001",
      "productName": "Industrial Widget 100mm",
      "requestedQuantity": { "amount": 50, "unitOfMeasure": "EA" },
      "reservedQuantity": { "amount": 0, "unitOfMeasure": "EA" },
      "shippedQuantity": { "amount": 0, "unitOfMeasure": "EA" },
      "lineStatus": "OPEN"
    }
  ],
  "audit": {
    "createdAt": "2026-07-06T17:05:00Z",
    "createdBy": "usr-cs-01",
    "updatedAt": "2026-07-06T17:05:00Z",
    "updatedBy": "usr-cs-01"
  }
}
```

API-082 — `GET /api/v1/fulfillment/orders`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<CustomerOrderSummaryResponse>`

API-083 — `GET /api/v1/fulfillment/orders/{orderId}`

| HTTP Status | `200 OK` | **DTO:** `CustomerOrderResponse` (full detail).

API-084 — `POST .../availability`

| HTTP Status | `200 OK` | **DTO:** `OrderAvailabilityResponse`

Field	Type	Description
`orderId`	UUID	Order ID
`fullyAvailable`	Boolean	All lines fulfillable
`lines`	Array<LineAvailabilityResponse>	Per-line ATP check
`confirmedAt`	Instant	Confirmation timestamp

`LineAvailabilityResponse`: `lineId`, `sku`, `requestedQuantity`, `atp`, `canFulfill`, `shortfallQuantity`

Example:

```
{
  "orderId": "e1f2a3b4-c5d6-7890-1234-567890123def",
  "fullyAvailable": true,
  "lines": [
    {
      "lineId": "f2a3b4c5-d6e7-8901-2345-678901234ef0",
      "sku": "SKU-WIDGET-001",
      "requestedQuantity": { "amount": 50, "unitOfMeasure": "EA" },
      "atp": { "amount": 470, "unitOfMeasure": "EA" },
      "canFulfill": true,
      "shortfallQuantity": null
    }
  ],
  "confirmedAt": "2026-07-06T17:10:00Z"
}
```

API-085 — `POST .../confirm`

| HTTP Status | `200 OK` | **DTO:** `CustomerOrderResponse` with `status: "CONFIRMED"` and reservation IDs on lines.

API-086 — `POST .../cancel`

| HTTP Status | `200 OK` | **DTO:** `CommandResultResponse` — `status: "CANCELLED"`

API-087 — `POST .../multi-location-plan`

| HTTP Status | `200 OK` | **DTO:** `MultiLocationFulfillmentPlanResponse`

Field	Type	Description
`orderId`	UUID	Order ID
`planId`	UUID	Plan ID
`allocations`	Array<LocationAllocationResponse>	Location allocations

Field	Type	Description
`reservations`	Array<ReservationResponse>	Created reservations
`createdAt`	Instant	Plan creation time

API-088 — `POST .../pick-lists`

| HTTP Status | `201 Created` | **DTO:** `PickListResponse`

Field	Type	Description
`pickId`	UUID	Pick list ID
`orderId`	UUID	Order ID
`status`	String	`ASSIGNED`
`assignedToUserId`	String	Picker user
`pickLines`	Array<PickLineResponse>	Lines to pick
`assignedAt`	Instant	Assignment time

API-089 — `GET .../pick-lists/awaiting`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<PickListSummaryResponse>`

API-090 — `POST .../picks/{pickId}/verify`

| HTTP Status | `200 OK` | **DTO:** `PickVerificationResponse`

Field	Type	Description
`pickId`	UUID	Pick ID
`orderId`	UUID	Order ID
`status`	String	`VERIFIED`
`verifiedLines`	Array<VerifiedPickLineResponse>	Verified picks
`verifiedAt`	Instant	Verification time

API-091 — `POST .../pack`

| HTTP Status | `200 OK` | **DTO:** `PackCompletionResponse`

Field	Type	Description
`orderId`	UUID	Order ID
`status`	String	`PACKED`
`packages`	Array<PackageResponse>	Packed packages
`packedAt`	Instant	Pack completion time

API-092 — `POST .../shipments`

| HTTP Status | `201 Created` | **DTO:** `ShipmentResponse`

Field	Type	Description
`shipmentId`	UUID	Shipment ID
`orderId`	UUID	Order ID
`status`	String	`DISPATCHED`
`carrierCode`	String	Carrier
`trackingNumber`	String	Tracking number
`shipmentLines`	Array<ShipmentLineResponse>	Shipped lines
`shippedAt`	Instant	Dispatch time

Example:

```
{
  "shipmentId": "a3b4c5d6-e7f8-9012-3456-789012345abc",
  "orderId": "e1f2a3b4-c5d6-7890-1234-567890123def",
  "status": "DISPATCHED",
  "carrierCode": "FEDEX",
  "trackingNumber": "794612345678",
  "shipmentLines": [
    {
      "orderLineId": "f2a3b4c5-d6e7-8901-2345-678901234ef0",
      "sku": "SKU-WIDGET-001",
      "shippedQuantity": { "amount": 50, "unitOfMeasure": "EA" }
    }
  ],
  "shippedAt": "2026-07-06T18:00:00Z"
}
```

API-093 — `POST .../partial-fulfillment`

| HTTP Status | `200 OK` | **DTO:** `PartialFulfillmentResponse`

Field	Type	Description
`orderId`	UUID	Order ID
`fulfilledLines`	Array<FulfilledLineResponse>	Fulfilled portion
`backorderCreated`	Boolean	Backorder created for remainder
`backorderId`	UUID	Backorder ID when created
`shipmentId`	UUID	Partial shipment ID

API-094 — `POST .../backorders`

| HTTP Status | `201 Created` | **DTO:** `BackorderResponse`

Field	Type	Description
`backorderId`	UUID	Backorder ID
`orderId`	UUID	Order ID
`status`	String	`OPEN`
`lines`	Array<BackorderLineResponse>	Backordered lines
`customerAcknowledgmentReference`	String	Customer ack ref
`createdAt`	Instant	Creation time

API-095 — `POST /api/v1/fulfillment/backorders/{backorderId}/fulfill`

| HTTP Status | `200 OK` | **DTO:** `BackorderFulfillmentResponse` — `backorderId`, `fulfilledQuantity`, `reservationId`, `shipmentId`, `fulfilledAt`.

API-096 — `POST .../exceptions`

| HTTP Status | `201 Created` | **DTO:** `FulfillmentExceptionResponse`

Field	Type	Description
`exceptionId`	UUID	Exception ID
`orderId`	UUID	Order ID
`exceptionType`	String	Exception type
`status`	String	`OPEN`
`description`	String	Description
`raisedAt`	Instant	Raised timestamp

API-097 — `GET .../status`

| HTTP Status | `200 OK` | **DTO:** `OrderFulfillmentStatusResponse`

Field	Type	Description
`orderId`	UUID	Order ID
`orderNumber`	String	Order number
`overallStatus`	String	Aggregate status
`reservationStatus`	String	Reservation state
`pickStatus`	String	Pick state
`shipStatus`	String	Ship state
`lines`	Array<LineFulfillmentStatusResponse>	Per-line status
`lastUpdatedAt`	Instant	Last status update

Example:

```
{
  "orderId": "e1f2a3b4-c5d6-7890-1234-567890123def",
  "orderNumber": "ORD-2026-0042",
  "overallStatus": "SHIPPED",
  "reservationStatus": "FULFILLED",
  "pickStatus": "VERIFIED",
  "shipStatus": "DISPATCHED",
  "lines": [
    {
      "lineId": "f2a3b4c5-d6e7-8901-2345-678901234ef0",
      "sku": "SKU-WIDGET-001",
      "requestedQuantity": { "amount": 50, "unitOfMeasure": "EA" },
      "reservedQuantity": { "amount": 50, "unitOfMeasure": "EA" },
      "shippedQuantity": { "amount": 50, "unitOfMeasure": "EA" },
      "backorderedQuantity": { "amount": 0, "unitOfMeasure": "EA" },
      "lineStatus": "SHIPPED"
    }
  ],
  "lastUpdatedAt": "2026-07-06T18:00:00Z"
}
```

M-08 — Network Transfer

API-098 — `POST /api/v1/transfers`

| HTTP Status | `201 Created` | **DTO:** `StockTransferResponse`

Field	Type	Description
`transferId`	UUID	Transfer ID
`transferNumber`	String	Business transfer number
`status`	String	`REQUESTED`
`sourceLocationCode`	String	Source
`destinationLocationCode`	String	Destination
`justification`	String	Justification
`lines`	Array<TransferLineResponse>	Transfer lines
`requestedAt`	Instant	Request time

Example:

```
{
  "transferId": "b4c5d6e7-f8a9-0123-4567-890123456abc",
  "transferNumber": "TRF-2026-0033",
  "status": "REQUESTED",
  "sourceLocationCode": "WH-WEST-01",
  "destinationLocationCode": "WH-EAST-01",
  "justification": "East coast stockout; excess at west",
  "lines": [
    {
      "lineId": "c5d6e7f8-a9b0-1234-5678-901234567bcd",
      "sku": "SKU-WIDGET-001",
      "requestedQuantity": { "amount": 100, "unitOfMeasure": "EA" },
      "approvedQuantity": null,
      "dispatchedQuantity": null,
      "receivedQuantity": null
    }
  ],
  "requestedAt": "2026-07-06T18:10:00Z"
}
```

API-099 — `GET /api/v1/transfers`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<StockTransferSummaryResponse>`

API-100 — `GET /api/v1/transfers/{transferId}`

| HTTP Status | `200 OK` | **DTO:** `StockTransferResponse` (full with in-transit state).

API-101 — `POST .../approve`

| HTTP Status | `200 OK` | **DTO:** `StockTransferResponse` with `status: "APPROVED"`.

API-102 — `POST .../reject`

| HTTP Status | `200 OK` | **DTO:** `CommandResultResponse` — `status: "REJECTED"`

API-103 — `POST .../dispatch`

| HTTP Status | `200 OK` | **DTO:** `StockTransferResponse` with `status: "IN_TRANSIT"`.

API-104 — `POST .../receive`

| HTTP Status | `200 OK` | **DTO:** `StockTransferResponse` with `status: "RECEIVED"`.

API-105 — `POST .../cancel`

| HTTP Status | `200 OK` | **DTO:** `CommandResultResponse` — `status: "CANCELLED"`

API-106 — `GET /api/v1/transfers/pending-approvals`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<StockTransferSummaryResponse>`

API-107 — `GET /api/v1/transfers/in-transit`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<InTransitStockResponse>`

Field	Type	Description
`transferId`	UUID	Transfer ID
`sku`	String	SKU
`sourceLocationCode`	String	Source
`destinationLocationCode`	String	Destination
`inTransitQuantity`	QuantityResponse	In-transit qty
`dispatchedAt`	Instant	Dispatch time
`isOverdue`	Boolean	Overdue flag

M-09 — Inventory Accuracy

API-108 — `POST /api/v1/accuracy/cycle-counts`

| HTTP Status | `201 Created` | **DTO:** `CycleCountResponse`

Field	Type	Description
`cycleCountId`	UUID	Count ID
`locationCode`	String	Location
`sku`	String	SKU
`status`	String	`SCHEDULED`
`scheduledDate`	LocalDate	Scheduled date
`assignedToUserId`	String	Counter
`priority`	String	Priority

Example:

```
{
  "cycleCountId": "d6e7f8a9-b0c1-2345-6789-012345678cde",
  "locationCode": "WH-EAST-01",
  "sku": "SKU-WIDGET-001",
  "status": "SCHEDULED",
  "scheduledDate": "2026-07-07",
  "assignedToUserId": "usr-wh-staff-03",
  "priority": "HIGH"
}
```

API-109 — `GET /api/v1/accuracy/cycle-counts`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<CycleCountSummaryResponse>`

API-110 — `GET /api/v1/accuracy/cycle-counts/{cycleCountId}`

| HTTP Status | `200 OK` | **DTO:** `CycleCountAssignmentResponse` — **excludes system quantity** (blind count).

Field	Type	Description
`cycleCountId`	UUID	Count ID
`locationCode`	String	Location
`sku`	String	SKU
`productName`	String	Product name
`status`	String	Status
`scheduledDate`	LocalDate	Scheduled date
`storagePositionCode`	String	Position hint (optional)

API-111 — `POST .../result`

| HTTP Status | `200 OK` | **DTO:** `CycleCountResultResponse`

Field	Type	Description
`cycleCountId`	UUID	Count ID
`status`	String	`COMPLETED` or `VARIANCE_DETECTED`
`physicalQuantity`	QuantityResponse	Submitted count
`varianceQuantity`	QuantityResponse	Variance (returned after submission)
`varianceId`	UUID	Variance case ID when variance exists
`submittedAt`	Instant	Submission time

Example:

```
{
  "cycleCountId": "d6e7f8a9-b0c1-2345-6789-012345678cde",
  "status": "VARIANCE_DETECTED",
  "physicalQuantity": { "amount": 495, "unitOfMeasure": "EA" },
  "varianceQuantity": { "amount": -5, "unitOfMeasure": "EA" },
  "varianceId": "e7f8a9b0-c1d2-3456-7890-123456789def",
  "submittedAt": "2026-07-07T09:00:00Z"
}
```

API-112 — `POST /api/v1/accuracy/physical-counts`

| HTTP Status | `201 Created` | **DTO:** `PhysicalInventoryCountResponse`

Field	Type	Description
`physicalCountId`	UUID	Count ID
`locationCode`	String	Location
`status`	String	`IN_PROGRESS`
`freezeImposed`	Boolean	`true`
`plannedStartDate`	LocalDate	Start date
`plannedEndDate`	LocalDate	End date
`startedAt`	Instant	Start timestamp

API-113 — `GET /api/v1/accuracy/physical-counts/{physicalCountId}`

| HTTP Status | `200 OK` | **DTO:** `PhysicalInventoryCountResponse` with progress metrics.

API-114 — `POST .../results`

| HTTP Status | `200 OK` | **DTO:** `PhysicalCountLineResultResponse`

Field	Type	Description
`physicalCountId`	UUID	Count ID
`linesRecorded`	Integer	Lines recorded in batch
`totalLinesRecorded`	Integer	Cumulative lines
`recordedAt`	Instant	Batch timestamp

API-115 — `POST .../certify`

| HTTP Status | `200 OK` | **DTO:** `PhysicalCountCertificationResponse`

Field	Type	Description
`physicalCountId`	UUID	Count ID
`status`	String	`CERTIFIED`
`freezeReleased`	Boolean	`true`
`varianceCount`	Integer	Variances identified
`varianceIds`	Array<UUID>	Variance case IDs
`certifiedAt`	Instant	Certification time

API-116 — `POST /api/v1/accuracy/adjustment-cases`

| HTTP Status | `201 Created` | **DTO:** `AdjustmentCaseResponse`

Field	Type	Description
`caseId`	UUID	Case ID
`caseNumber`	String	Business case number

Field	Type	Description
`status`	String	`OPEN`
`originType`	String	Origin type
`locationCode`	String	Location
`sku`	String	SKU
`reasonCode`	String	Reason code
`openedAt`	Instant	Open time

API-117 — `GET /api/v1/accuracy/adjustment-cases/{caseId}`

| HTTP Status | `200 OK` | **DTO:** `AdjustmentCaseDetailResponse` — includes investigation, approval chain, proposed adjustment.

API-118 — `POST .../investigate`

| HTTP Status | `200 OK` | **DTO:** `AdjustmentCaseResponse` with `status: "UNDER_INVESTIGATION"`.

API-119 — `POST .../variances/{varianceId}/approve`

| HTTP Status | `200 OK` | **DTO:** `CommandResultResponse` — `status: "VARIANCE_APPROVED"`

API-120 — `POST .../submit`

| HTTP Status | `200 OK` | **DTO:** `AdjustmentCaseResponse` with `status: "PENDING_APPROVAL"`.

API-121 — `POST .../approve`

| HTTP Status | `200 OK` | **DTO:** `AdjustmentApprovalResponse`

Field	Type	Description
`caseId`	UUID	Case ID
`status`	String	`APPROVED`
`adjustmentType`	String	`ADJUSTMENT` or `WRITE_OFF`
`approvedQuantity`	QuantityResponse	Approved qty
`stockAdjustedAt`	Instant	When stock mutation applied

Example:

```
{
  "caseId": "f8a9b0c1-d2e3-4567-8901-234567890ef0",
  "status": "APPROVED",
  "adjustmentType": "ADJUSTMENT",
  "approvedQuantity": { "amount": 5, "unitOfMeasure": "EA" },
  "stockAdjustedAt": "2026-07-07T14:00:00Z"
}
```

API-122 — `POST .../reject`

| HTTP Status | `200 OK` | **DTO:** `CommandResultResponse` — `status: "REJECTED"`

API-123 — `GET /api/v1/accuracy/pending-approvals`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<AdjustmentCaseSummaryResponse>`

M-10 — Returns Management

API-124 — `POST /api/v1/returns/authorizations`

| HTTP Status | `201 Created` | **DTO:** `ReturnAuthorizationResponse`

Field	Type	Description
`returnAuthorizationId`	UUID	RMA ID
`rmaNumber`	String	RMA number
`status`	String	`AUTHORIZED`
`originalOrderId`	UUID	Original order
`originalOrderNumber`	String	Order number
`validUntil`	LocalDate	Authorization expiry
`lines`	Array<ReturnAuthorizationLineResponse>	Authorized lines
`issuedAt`	Instant	Issue time

Example:

```
{
  "returnAuthorizationId": "a9b0c1d2-e3f4-5678-9012-345678901abc",
  "rmaNumber": "RMA-2026-0012",
  "status": "AUTHORIZED",
  "originalOrderId": "e1f2a3b4-c5d6-7890-1234-567890123def",
  "originalOrderNumber": "ORD-2026-0042",
  "validUntil": "2026-07-20",
  "lines": [
    {
      "lineId": "b0c1d2e3-f4a5-6789-0123-456789012bcd",
      "sku": "SKU-WIDGET-001",
      "authorizedQuantity": { "amount": 5, "unitOfMeasure": "EA" }
    }
  ],
  "issuedAt": "2026-07-06T19:00:00Z"
}
```

API-125 — `GET /api/v1/returns/authorizations`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<ReturnAuthorizationSummaryResponse>`

API-126 — `GET /api/v1/returns/authorizations/{returnAuthorizationId}`

| HTTP Status | `200 OK` | **DTO:** `ReturnAuthorizationResponse`

API-127 — `POST .../cancel`

| HTTP Status | `200 OK` | **DTO:** `CommandResultResponse` — `status: "CANCELLED"`

API-128 — `POST /api/v1/returns/{returnId}/receive`

| HTTP Status | `200 OK` | **DTO:** `CustomerReturnResponse` with `status: "RECEIVED"`.

API-129 — `POST /api/v1/returns/{returnId}/inspect`

| HTTP Status | `200 OK` | **DTO:** `ReturnInspectionResponse`

Field	Type	Description
`returnId`	UUID	Return ID
`status`	String	`INSPECTED`
`inspectionLines`	Array<InspectionLineResponse>	Inspection results
`inspectedAt`	Instant	Inspection time

API-130 — `POST /api/v1/returns/{returnId}/disposition`

| HTTP Status | `200 OK` | **DTO:** `ReturnDispositionResponse`

Field	Type	Description
`returnId`	UUID	Return ID
`status`	String	`COMPLETED`
`dispositionAction`	String	`RESTOCK`, `QUARANTINE`, or `WRITE_OFF`
`dispositionLines`	Array<DispositionLineResultResponse>	Per-line results
`dispositionedAt`	Instant	Disposition time

Example:

```
{
  "returnId": "c1d2e3f4-a5b6-7890-1234-567890123cde",
  "status": "COMPLETED",
  "dispositionAction": "RESTOCK",
  "dispositionLines": [
    {
      "returnLineId": "d2e3f4a5-b6c7-8901-2345-678901234def",
      "quantity": { "amount": 5, "unitOfMeasure": "EA" },
      "outcome": "RESTOCKED",
      "restockLocationCode": "WH-EAST-01"
    }
  ],
  "dispositionedAt": "2026-07-06T19:30:00Z"
}
```

API-131 — `GET /api/v1/returns/pending-receipts`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<PendingReturnReceiptResponse>`

M-11 — Commercial Campaign

API-132 — `POST /api/v1/campaigns/allocations`

| HTTP Status | `201 Created` | **DTO:** `PromotionalAllocationResponse`

Field	Type	Description
`allocationId`	UUID	Allocation ID
`campaignId`	String	Campaign ID
`campaignName`	String	Campaign name
`locationCode`	String	Location
`sku`	String	SKU
`allocatedQuantity`	QuantityResponse	Ring-fenced qty
`status`	String	`PLANNED` or `ACTIVE`
`campaignStartDate`	LocalDate	Start date
`campaignEndDate`	LocalDate	End date
`readinessStatus`	String	Readiness status

Example:

```
{
  "allocationId": "e3f4a5b6-c7d8-9012-3456-789012345def",
  "campaignId": "CAMP-SUMMER-2026",
  "campaignName": "Summer Promotion 2026",
  "locationCode": "WH-EAST-01",
  "sku": "SKU-WIDGET-001",
  "allocatedQuantity": { "amount": 100, "unitOfMeasure": "EA" },
  "status": "PLANNED",
  "campaignStartDate": "2026-07-15",
  "campaignEndDate": "2026-08-15",
  "readinessStatus": "PENDING"
}
```

API-133 — `GET /api/v1/campaigns/allocations`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<PromotionalAllocationSummaryResponse>`

API-134 — `GET /api/v1/campaigns/allocations/{allocationId}`

| HTTP Status | `200 OK` | **DTO:** `PromotionalAllocationResponse`

API-135 — `PATCH /api/v1/campaigns/allocations/{allocationId}`

| HTTP Status | `200 OK` | **DTO:** `PromotionalAllocationResponse` (amended).

API-136 — `POST .../readiness`

| HTTP Status | `200 OK` | **DTO:** `CampaignReadinessResponse`

Field	Type	Description
`allocationId`	UUID	Allocation ID
`readinessStatus`	String	`READY` or `GAP_IDENTIFIED`
`targetQuantity`	QuantityResponse	Target qty
`availableQuantity`	QuantityResponse	Ring-fenced available
`gapQuantity`	QuantityResponse	Shortfall when gap

Field	Type	Description
`evaluatedAt`	Instant	Evaluation time

API-137 — `POST .../release`

| HTTP Status | `200 OK` | **DTO:** `CommandResultResponse` — `status: "RELEASED"`

API-138 — `POST .../override`

| HTTP Status | `200 OK` | **DTO:** `PromotionalOverrideResponse`

Field	Type	Description
`allocationId`	UUID	Allocation ID
`orderId`	UUID	Consuming order
`overrideQuantity`	QuantityResponse	Overridden qty
`authorizationReference`	String	Authorization ref
`authorizedAt`	Instant	Authorization time

API-139 — `GET /api/v1/campaigns/ring-fence/{locationCode}/{sku}`

| HTTP Status | `200 OK` | **DTO:** `RingFenceStatusResponse`

Field	Type	Description
`locationCode`	String	Location
`sku`	String	SKU
`hasActiveRingFence`	Boolean	Active ring-fence exists
`allocations`	Array<ActiveAllocationSummaryResponse>	Active allocations

Example:

```
{
  "locationCode": "WH-EAST-01",
  "sku": "SKU-WIDGET-001",
  "hasActiveRingFence": true,
  "allocations": [
    {
      "allocationId": "e3f4a5b6-c7d8-9012-3456-789012345def",
      "campaignId": "CAMP-SUMMER-2026",
      "allocatedQuantity": { "amount": 100, "unitOfMeasure": "EA" },
      "campaignEndDate": "2026-08-15"
    }
  ]
}
```

M-12 — Identity & Access

API-140 — `POST /api/v1/users`

| HTTP Status | `201 Created` | **DTO:** `UserAccountResponse`

Field	Type	Description
`userId`	String	User ID
`email`	String	Email
`displayName`	String	Display name
`status`	String	`ACTIVE`
`roles`	Array<RoleAssignmentResponse>	Role assignments
`audit`	AuditInfoResponse	Audit metadata

Example:

```
{
  "userId": "usr-wh-mgr-02",
  "email": "j.smith@company.example.com",
  "displayName": "Jane Smith",
  "status": "ACTIVE",
  "roles": [
    {
      "roleAssignmentId": "f4a5b6c7-d8e9-0123-4567-890123456ef0",
      "roleCode": "WAREHOUSE_MANAGER",
      "locationCodes": ["WH-EAST-01"],
      "effectiveFrom": "2026-07-06",
      "effectiveTo": null
    }
  ],
  "audit": {
    "createdAt": "2026-07-06T20:00:00Z",
    "createdBy": "usr-admin-01",
    "updatedAt": "2026-07-06T20:00:00Z",
    "updatedBy": "usr-admin-01"
  }
}
```

API-141 — `GET /api/v1/users`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<UserAccountSummaryResponse>`

API-142 — `GET /api/v1/users/{userId}`

| HTTP Status | `200 OK` | **DTO:** `UserAccountResponse`

API-143 — `POST /api/v1/users/{userId}/roles`

| HTTP Status | `201 Created` | **DTO:** `RoleAssignmentResponse`

API-144 — `DELETE /api/v1/users/{userId}/roles/{roleAssignmentId}`

| HTTP Status | `200 OK` | **DTO:** `RoleRevocationResponse`

Field	Type	Description
`roleAssignmentId`	UUID	Revoked assignment
`userId`	String	User ID
`revokedAt`	Instant	Revocation time

API-145 — `POST /api/v1/users/{userId}/suspend`

| HTTP Status | `200 OK` | **DTO:** `UserAccountResponse` with `status: "SUSPENDED"`.

API-146 — `POST /api/v1/users/{userId}/deactivate`

| HTTP Status | `200 OK` | **DTO:** `UserAccountResponse` with `status: "DEACTIVATED"`.

API-147 — `POST /api/v1/users/sync`

| HTTP Status | `200 OK` | **DTO:** `DirectorySyncResultResponse`

Field	Type	Description
`syncMode`	String	Sync mode
`usersCreated`	Integer	Created count
`usersUpdated`	Integer	Updated count
`usersDeactivated`	Integer	Deactivated count
`errors`	Array<SyncErrorResponse>	Sync errors
`syncedAt`	Instant	Sync completion time

API-148 — `POST /api/v1/authorization/actions/evaluate`

| HTTP Status | `200 OK` | **DTO:** `AuthorizationEvaluationResponse`

Field	Type	Description
`authorized`	Boolean	Action permitted
`userId`	String	Evaluated user
`action`	String	Action code
`denialReason`	String	Reason when denied

Example:

```
{
  "authorized": true,
  "userId": "usr-wh-mgr-02",
  "action": "ADJUSTMENT_APPROVE",
  "denialReason": null
}
```

API-149 — `POST /api/v1/authorization/location-scope/validate`

| HTTP Status | `200 OK` | **DTO:** `LocationScopeValidationResponse`

Field	Type	Description
`authorized`	Boolean	Location access granted
`userId`	String	User ID
`locationCode`	String	Location
`scopedLocations`	Array<String>	User's authorized locations

M-13 — Inventory Intelligence

API-150 — `GET /api/v1/intelligence/dashboard`

| HTTP Status | `200 OK` | **DTO:** `InventoryDashboardResponse`

Field	Type	Description
`scope`	String	`LOCATION` or `NETWORK`
`locationCode`	String	Location when scoped
`asOfTimestamp`	Instant	Data freshness
`summary`	DashboardSummaryResponse	KPI summary tiles
`lowStockCount`	Integer	Low-stock SKUs
`outOfStockCount`	Integer	Out-of-stock SKUs
`pendingApprovals`	Integer	Pending approvals
`openAlerts`	Integer	Open alerts

Example:

```
{
  "scope": "LOCATION",
  "locationCode": "WH-EAST-01",
  "asOfTimestamp": "2026-07-06T20:30:00Z",
  "summary": {
    "totalSkus": 1250,
    "totalOnHandValue": { "amount": 2500000.00, "currencyCode": "USD" },
    "averageFillRate": 0.94
  },
  "lowStockCount": 23,
  "outOfStockCount": 5,
  "pendingApprovals": 7,
  "openAlerts": 12
}
```

API-151 — `GET /api/v1/intelligence/stock-status/summary`

| HTTP Status | `200 OK` | **DTO:** `StockStatusSummaryResponse`

Field	Type	Description
`inStockCount`	Integer	In-stock positions
`lowStockCount`	Integer	Low-stock positions
`outOfStockCount`	Integer	Out-of-stock positions
`overstockCount`	Integer	Overstock positions
`totalOnHand`	QuantityResponse	Aggregated on-hand
`totalAtp`	QuantityResponse	Aggregated ATP

API-152 — `GET /api/v1/intelligence/stock-status/board`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<StockStatusBoardRowResponse>`

API-153 — `GET /api/v1/intelligence/availability/inquiries`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<AvailabilityInquiryLogEntryResponse>`

API-154 — `GET /api/v1/intelligence/quarantine-register`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<QuarantineRegisterEntryResponse>`

API-155 — `GET /api/v1/intelligence/movements`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<MovementHistoryEntryResponse>`

Field	Type	Description
`movementId`	UUID	Movement ID
`sku`	String	SKU
`locationCode`	String	Location
`movementType`	String	Movement type
`quantity`	QuantityResponse	Quantity moved
`userId`	String	Acting user
`occurredAt`	Instant	Movement timestamp
`reference`	String	Source document reference

Example:

```
{
  "data": [
    {
      "movementId": "a5b6c7d8-e9f0-1234-5678-901234567abc",
      "sku": "SKU-WIDGET-001",
      "locationCode": "WH-EAST-01",
      "movementType": "SHIPMENT",
      "quantity": { "amount": 50, "unitOfMeasure": "EA" },
      "userId": "usr-pick-01",
      "occurredAt": "2026-07-06T18:00:00Z",
      "reference": "ORD-2026-0042"
    }
  ],
  "page": 1,
  "pageSize": 20,
  "totalElements": 1,
  "totalPages": 1
}
```

API-156 — `GET /api/v1/intelligence/audit-trail`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<AuditTrailEntryResponse>`

Field	Type	Description
`auditId`	UUID	Audit entry ID
`entityType`	String	Entity type
`entityId`	String	Entity ID
`action`	String	Action performed
`userId`	String	Acting user

Field	Type	Description
`occurredAt`	Instant	Timestamp
`beforeState`	Object	Prior state snapshot
`afterState`	Object	New state snapshot

API-157 — `GET /api/v1/intelligence/audit-trail/export`

| HTTP Status | `200 OK` | **DTO:** Binary file stream **or** `ExportJobResponse` for async export.

Field	Type	Description
`exportId`	UUID	Export job ID
`status`	String	`COMPLETED`
`format`	String	`CSV`, `JSON`, or `PDF`
`downloadUrl`	String	Signed download URL
`recordCount`	Integer	Records exported
`expiresAt`	Instant	URL expiry

API-158 — `GET /api/v1/intelligence/alerts`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<OperationalAlertResponse>`

Field	Type	Description
`alertId`	UUID	Alert ID
`alertType`	String	Alert type
`severity`	String	Severity
`sku`	String	SKU
`locationCode`	String	Location
`message`	String	Alert message
`acknowledged`	Boolean	Acknowledgment flag
`raisedAt`	Instant	Raised time

API-159 — `POST /api/v1/intelligence/alerts/{alertId}/acknowledge`

| HTTP Status | `200 OK` | **DTO:** `OperationalAlertResponse` with `acknowledged: true`.

API-160 — `GET /api/v1/intelligence/replenishment/recommendations`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<ReplenishmentRecommendationResponse>`

API-161 — `GET /api/v1/intelligence/procurement/open-pos`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<OpenPurchaseOrderBoardRowResponse>`

API-162 — `GET /api/v1/intelligence/procurement/inbound-calendar`

| HTTP Status | `200 OK` | **DTO:** `InboundSupplyCalendarResponse`

Field	Type	Description
`startDate`	LocalDate	Calendar start
`endDate`	LocalDate	Calendar end
`entries`	Array<InboundCalendarEntryResponse>	Calendar entries

`InboundCalendarEntryResponse`: `date`, `purchaseOrderId`, `poNumber`, `supplierCode`, `sku`, `expectedQuantity`, `receivingLocationCode`

API-163 — `GET /api/v1/intelligence/procurement/inbound-delays`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<InboundDelayLogEntryResponse>`

API-164 — `GET /api/v1/intelligence/fulfillment/backorders`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<BackorderPipelineEntryResponse>`

API-165 — `GET /api/v1/intelligence/fulfillment/exceptions`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<FulfillmentExceptionLogEntryResponse>`

API-166 — `GET /api/v1/intelligence/network/imbalance`

| HTTP Status | `200 OK` | **DTO:** `NetworkImbalanceAnalysisResponse`

Field	Type	Description
`analyzedAt`	Instant	Analysis timestamp
`imbalances`	Array<NetworkImbalanceEntryResponse>	Imbalance entries
`recommendations`	Array<TransferRecommendationResponse>	Transfer recommendations

Example:

```
{
  "analyzedAt": "2026-07-06T21:00:00Z",
  "imbalances": [
    {
      "sku": "SKU-WIDGET-001",
      "shortageLocationCode": "WH-EAST-01",
      "shortageQuantity": { "amount": 50, "unitOfMeasure": "EA" },
      "excessLocationCode": "WH-WEST-01",
      "excessQuantity": { "amount": 200, "unitOfMeasure": "EA" }
    }
  ],
  "recommendations": [
    {
      "sku": "SKU-WIDGET-001",
      "sourceLocationCode": "WH-WEST-01",
      "destinationLocationCode": "WH-EAST-01",
      "recommendedQuantity": { "amount": 100, "unitOfMeasure": "EA" }
    }
  ]
}
```

API-167 — `GET /api/v1/intelligence/accuracy/shrinkage`

| HTTP Status | `200 OK` | **DTO:** `ShrinkageAnalysisResponse`

Field	Type	Description
`periodStart`	LocalDate	Period start
`periodEnd`	LocalDate	Period end
`totalShrinkageValue`	MonetaryAmountResponse	Total shrinkage value
`shrinkageRate`	BigDecimal	Shrinkage rate %
`breakdown`	Array<ShrinkageBreakdownRowResponse>	Grouped breakdown

API-168 — `GET /api/v1/intelligence/accuracy/count-performance`

| HTTP Status | `200 OK` | **DTO:** `CountPerformanceMetricsResponse`

Field	Type	Description
`periodStart`	LocalDate	Period start
`periodEnd`	LocalDate	Period end
`accuracyRate`	BigDecimal	Inventory accuracy %
`cycleCountsCompleted`	Integer	Cycle counts done
`physicalCountsCompleted`	Integer	Physical counts done
`varianceCount`	Integer	Variances detected

API-169 — `GET /api/v1/intelligence/campaigns/readiness`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<CampaignReadinessStatusResponse>`

API-170 — `GET /api/v1/intelligence/kpis`

| HTTP Status | `200 OK` | **DTO:** `InventoryKpiReportResponse`

Field	Type	Description
`periodStart`	LocalDate	Period start
`periodEnd`	LocalDate	Period end
`kpis`	Array<KpiMetricResponse>	KPI metrics

`KpiMetricResponse`: `kpiCode`, `name`, `value`, `target`, `unit`, `trend`, `status` (`ON_TARGET`, `BELOW_TARGET`, `ABOVE_TARGET`)

Example:

```
{
  "periodStart": "2026-06-01",
  "periodEnd": "2026-06-30",
  "kpis": [
    {
      "kpiCode": "FILL_RATE",
      "name": "Order Fill Rate",
      "value": 0.94,
      "target": 0.95,
      "unit": "PERCENT",
      "trend": "UP",
      "status": "BELOW_TARGET"
    },
    {
      "kpiCode": "INVENTORY_ACCURACY",
      "name": "Inventory Record Accuracy",
      "value": 0.97,
      "target": 0.95,
      "unit": "PERCENT",
      "trend": "STABLE",
      "status": "ON_TARGET"
    }
  ]
}
```

API-171 — `GET /api/v1/intelligence/valuation/snapshots`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<ValuationSnapshotResponse>`

Field	Type	Description
`snapshotId`	UUID	Snapshot ID
`periodEndDate`	LocalDate	Period end
`totalValue`	MonetaryAmountResponse	Total inventory value
`locationCode`	String	Location scope
`lineCount`	Integer	Valuation lines
`preparedAt`	Instant	Preparation time

API-172 — `GET /api/v1/intelligence/valuation/reconciliation`

| HTTP Status | `200 OK` | **DTO:** `ValuationReconciliationReportResponse`

Field	Type	Description
`periodEndDate`	LocalDate	Period end
`operationalValue`	MonetaryAmountResponse	IMS valuation
`generalLedgerValue`	MonetaryAmountResponse	GL valuation
`variance`	MonetaryAmountResponse	Variance amount
`variancePercent`	BigDecimal	Variance %
`withinTolerance`	Boolean	Within tolerance
`lineVariances`	Array<ValuationLineVarianceResponse>	Line-level variances

Example:

```
{
  "periodEndDate": "2026-06-30",
  "operationalValue": { "amount": 2500000.00, "currencyCode": "USD" },
  "generalLedgerValue": { "amount": 2495000.00, "currencyCode": "USD" },
  "variance": { "amount": 5000.00, "currencyCode": "USD" },
  "variancePercent": 0.20,
  "withinTolerance": true,
  "lineVariances": []
}
```

Platform & Integration

API-173 — `GET /api/v1/auth/session`

| HTTP Status | `200 OK` | **DTO:** `SessionResponse`

Field	Type	Description
`userId`	String	Authenticated user
`email`	String	Email
`displayName`	String	Display name
`roles`	Array<String>	Role codes
`locationScope`	Array<String>	Authorized locations
`permissions`	Array<String>	Resolved permissions (when requested)
`sessionExpiresAt`	Instant	Session expiry

Example:

```
{
  "userId": "usr-wh-mgr-02",
  "email": "j.smith@company.example.com",
  "displayName": "Jane Smith",
  "roles": ["WAREHOUSE_MANAGER"],
  "locationScope": ["WH-EAST-01"],
  "permissions": ["STOCK_VIEW", "ADJUSTMENT_APPROVE"],
  "sessionExpiresAt": "2026-07-06T22:00:00Z"
}
```

API-174 — `POST /api/v1/auth/logout`

| HTTP Status | `200 OK` | **DTO:** `LogoutResponse`

Field	Type	Description
`loggedOut`	Boolean	`true`
`loggedOutAt`	Instant	Logout timestamp

Example:

```
{
  "loggedOut": true,
  "loggedOutAt": "2026-07-06T21:45:00Z"
}
```

API-175 — `GET /api/v1/health`

| HTTP Status | `200 OK` | **DTO:** `HealthCheckResponse`

Field	Type	Description
`status`	String	`UP`
`timestamp`	Instant	Check timestamp
`version`	String	Application version
`components`	Array<HealthComponentResponse>	Component health

Example:

```
{
  "status": "UP",
  "timestamp": "2026-07-06T21:50:00Z",
  "version": "1.0.0",
  "components": [
    { "name": "database", "status": "UP" },
    { "name": "eventBus", "status": "UP" }
  ]
}
```

API-176 — `POST /api/v1/integrations/sales/orders`

| HTTP Status | `201 Created` | **DTO:** `SalesOrderImportResponse`

Field	Type	Description
`orderId`	UUID	Created IMS order ID
`orderNumber`	String	IMS order number
`externalOrderId`	String	CRM order ID
`status`	String	`DRAFT`
`correlationId`	String	Integration correlation ID
`importedAt`	Instant	Import timestamp

Example:

```
{
  "orderId": "f5a6b7c8-d9e0-1234-5678-901234567abc",
  "orderNumber": "ORD-2026-0043",
  "externalOrderId": "CRM-ORD-998877",
  "status": "DRAFT",
  "correlationId": "corr-20260706-001",
  "importedAt": "2026-07-06T21:55:00Z"
}
```

API-177 — `GET /api/v1/integrations/sales/fulfillment-status`

| HTTP Status | `200 OK` | **DTO:** `PagedResponse<FulfillmentStatusFeedEntryResponse>`

Field	Type	Description
`externalOrderId`	String	CRM order ID
`orderNumber`	String	IMS order number
`status`	String	Fulfillment status

Field	Type	Description
`shipmentTrackingNumber`	String	Tracking number
`shippedAt`	Instant	Ship time
`updatedAt`	Instant	Last update

API-178 — `POST /api/v1/integrations/marketing/campaign-plans`

| HTTP Status | `201 Created` | **DTO:** `CampaignPlanImportResponse`

Field	Type	Description
`campaignId`	String	Campaign ID
`allocationsCreated`	Integer	Allocations created
`allocationIds`	Array<UUID>	Created allocation IDs
`correlationId`	String	Correlation ID
`importedAt`	Instant	Import timestamp

Example:

```
{
  "campaignId": "CAMP-FALL-2026",
  "allocationsCreated": 15,
  "allocationIds": [
    "a6b7c8d9-e0f1-2345-6789-012345678abc",
    "b7c8d9e0-f1a2-3456-7890-123456789bcd"
  ],
  "correlationId": "corr-mkt-20260706-001",
  "importedAt": "2026-07-06T22:00:00Z"
}
```

Response DTO Index

Pattern	APIs	HTTP Status
`ProductDetailResponse`	API-001, 003, 004	201 / 200
`PagedResponse<T>`	API-002, 008, 012, 023, 033, 050, 060, 067, 082, 099, 109, 123, 125, 131, 133, 141, 152–165, 169, 171, 177	200
`CommandResultResponse`	API-056, 057, 063, 086, 101, 102, 105, 119, 122, 127, 137	200
`CustomerOrderResponse`	API-081, 083, 085	201 / 200
`StockTransferResponse`	API-098, 100–104	201 / 200
`GoodsReceiptResponse`	API-066–069, 075	201 / 200
`PurchaseOrderResponse`	API-058, 059, 061, 062	201 / 200

Total APIs documented: 178

Document Approval

Role	Name	Signature	Date
Senior Backend Architect			
Solution Architect			

Role	Name	Signature	Date
Product Owner			

End of Response DTO Specification

Document ID: SRS-API-003 | Version 1.0 | July 6, 2026

Request DTO Validation Specification

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-API-004
Document Title	Request DTO Validation Specification
Version	1.0
Status	Draft
Document Owner	Senior Backend Architect
Source Documents	SRS-API-002 (Request DTOs), SRS-VR-001 (Validation Rules), SRS-BR-001 (Business Rules)
Date	July 6, 2026
Scope	Input validation for all Request DTOs — success-path only; error contract is RFC 7807

Purpose

Defines **enforceable validation** for every Request DTO field across all 178 REST APIs. Each constraint is classified as required, length, format, business, or cross-field validation and traced to **BR-xxx** (business rule) and/or **VR-xxx** (validation rule).

Coverage Summary

Module	API Range	DTOs	Validation Depth
M-01 Product Catalog	API-001 – API-010	10	Field matrix + profiles
M-02 Warehouse Network	API-011 – API-021	11	Field matrix + profiles
M-03 Inventory Governance	API-022 – API-031	10	Validation profiles
M-04 Inventory Balance	API-032 – API-048	17	Validation profiles
M-05 Procurement & Supplier	API-049 – API-065	17	Validation profiles
M-06 Inbound Receiving	API-066 – API-080	15	Validation profiles
M-07 Order Fulfillment	API-081 – API-097	17	Validation profiles
M-08 Network Transfer	API-098 – API-107	10	Validation profiles
M-09 Inventory Accuracy	API-108 – API-123	16	Validation profiles
M-10 Returns Management	API-124 – API-131	8	Validation profiles
M-11 Commercial Campaign	API-132 – API-139	8	Validation profiles
M-12 Identity & Access	API-140 – API-149	10	Validation profiles
M-13 Inventory Intelligence	API-150 – API-172	23	Query validation profiles

Module	API Range	DTOs	Validation Depth
Platform & Integration	API-173 – API-178	6	Validation profiles
Total	**API-001 – API-178**	**178**	—

Validation Execution Layers

Layer	Responsibility	Examples
L1 — Structural	Required, length, format, enum	`@NotBlank`, `@Size`, `@Pattern`, `@DecimalMin`
L2 — Referential	FK existence, catalog lookup	Product exists, location active
L3 — Business	Domain invariants, BR enforcement	ATP check, SoD check, freeze guard
L4 — Cross-field	Multi-field logic	`startDate ≤ endDate`, PO **or** ad-hoc auth

L1–L2 run at API/presentation boundary; L3–L4 run in application/domain services.

Validation Matrix Legend

Column	Meaning
Req	`R`=Required, `O`=Optional, `C`=Conditional
Length	Max/min character or numeric bounds
Format	Pattern, enum, ISO type
Business Validation	L2/L3 domain checks
Cross-Field	L4 dependencies
BR	Business rule ID(s)
VR	Validation rule ID(s)

Shared Primitive Validation

Primitive	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`ProductSku`	R when present	1–50 chars	`^[A-Z0-9][A-Z0-9_-]*$`	Must exist in catalog when mutating stock	—	—	VR-001, VR-002, VR-007
`LocationCode`	R when present	1–30 chars	`^[A-Z0-9][A-Z0-9_-]*$`	Must exist; user location scope; not frozen for mutations	—	BR-024	VR-011, VR-012, VR-053
`SupplierCode`	R when present	1–30 chars	`^[A-Z0-9][A-Z0-9_-]*$`	Must exist; active for new PO	—	—	VR-030
`ReasonCode`	R when present	1–20 chars	`^[A-Z0-9_]+$`	Must be active; applicable to transaction type	—	BR-020	VR-026, VR-027
`Quantity.amount`	R	scale ≤4	decimal ≥0	Must not exceed on-hand for decreases	`unitOfMeasure` required	—	VR-018, VR-023, VR-070

Primitive	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`PositiveQuantity.amount`	R	scale ≤4	decimal >0	Must not exceed ATP/reservation/PO open qty per context	`unitOfMeasure` required	BR-001, BR-002	VR-032, VR-035, VR-066
`CurrencyCode`	R when present	exactly 3	`^[A-Z]{3}$` ISO 4217	Must match org or supplier currency policy	—	—	—
`LocalDate`	R when present	—	ISO `YYYY-MM-DD`	Not unreasonably past for delivery schedules	≤ paired end date	—	VR-033, VR-062
`Instant`	R when present	—	ISO-8601 UTC	Future when used as expiry	—	—	—
`Uuid`	R when present	36 chars	RFC 4122	Referenced entity must exist	—	—	—
`AbcClass`	R when present	1 char	enum `A,B,C`	Product must exist	—	BR-032	VR-021
`LifecycleStatus`	R when present	—	enum `ACTIVE,INACTIVE,DISCONTINUED`	Valid state transition	—	BR-027	VR-004–VR-006
`PageNumber`	O	—	integer ≥1	default 1	—	—	—
`PageSize`	O	—	integer 1–100	default 20	—	—	—

Shared Nested DTO Validation

`QuantityDto`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`amount`	R	scale ≤4	decimal ≥0	Context-specific ceiling	—	—	VR-018
`unitOfMeasure`	R	1–10	alphanumeric	Must match product primary UoM or approved alternate	Both required together	—	VR-001

`PositiveQuantityDto`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`amount`	R	scale ≤4	decimal >0	≤ ATP, reservation, PO remaining per API	—	BR-001, BR-002	VR-032, VR-092
`unitOfMeasure`	R	1–10	alphanumeric	Match product UoM	Required with amount	—	VR-001

`LotIdentityDto`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`lotNumber`	R	1–50	non-blank	Required only for traceable products	—	BR-017	—
`expirationDate`	O	—	ISO date	Required for shelf-life products at receipt	≤ product shelf policy	BR-014, BR-016	VR-035

`ShelfLifeProfileDto`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`expirationDate`	R	—	ISO date	Remaining life ≥ MRSL threshold	`manufactureDate ≤ expirationDate`	BR-016	VR-035

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`manufactureDate`	O	—	ISO date	On or before expiration	—	—	—
`remainingShelfLifeDays`	O	—	integer ≥0	Must meet MRSL when shelf-life managed	Computed from dates if omitted	BR-016	—

`MonetaryAmountDto`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`amount`	R	scale 2	decimal ≥0	Used for approval threshold evaluation	`currencyCode` required	BR-021	VR-082
`currencyCode`	R	3	ISO 4217	Match PO/invoice currency	Required with amount	BR-008	—

`RoleAssignmentDto`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`roleCode`	R	1–50	known role enum	Must not create SoD conflict	—	BR-019, BR-021	VR-083
`locationCodes[]`	O	each 1–30	LocationCode	Each location active; user admin scope	—	—	VR-008
`effectiveFrom`	O	—	ISO date	default today	`effectiveTo ≥ effectiveFrom`	—	—
`effectiveTo`	O	—	ISO date	Future end optional	`≥ effectiveFrom`	—	—

`PaginationQuery` / `DateRangeQuery`

DTO	Cross-Field	BR	VR
`PaginationQuery`	`sortDirection` required when `sortBy` set	—	—
`DateRangeQuery`	`startDate ≤ endDate`; max range 365 days for exports	—	VR-062, VR-153

M-01 — Product Catalog

API-001 — `CreateProductRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`sku`	R	1–50	ProductSku pattern	Unique across catalog	—	—	VR-001, VR-002, VR-009
`name`	R	1–200	non-blank	—	—	—	VR-001
`description`	O	0–2000	text	—	—	—	—
`unitOfMeasure`	R	1–10	non-blank	Valid UoM code in reference data	—	—	VR-001
`categoryCode`	O	1–30	alphanumeric	Category exists if provided	—	—	—
`barcode`	O	1–50	—	No duplicate barcode (VR-009)	—	—	VR-009
`isTraceable`	O	—	boolean	default false	If true → `rotationPolicy` R	BR-017	—
`rotationPolicy`	C	—	`FIFO,FEFO`	Required for traceable	Required when `isTraceable=true`	BR-014, BR-015	—
`shelfLifeManaged`	O	—	boolean	default false	If true → `mrslThresholdDays` R	BR-016	—

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`mrslThresholdDays`	C	—	integer ≥1	—	Required when `shelfLifeManaged=true`	BR-016	—

Cross-field: (`isTraceable=true` → `rotationPolicy` required); (`shelfLifeManaged=true` → `mrslThresholdDays` required).
Business rules: BR-014, BR-015, BR-016, BR-017 (traceability attributes at catalog definition).

API-002 — `SearchProductsQuery`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`name`	O	1–200	partial match	—	—	—	VR-054
`sku`	O	1–50	ProductSku	—	—	—	VR-007
`categoryCode`	O	1–30	—	—	—	—	—
`lifecycleStatus`	O	—	LifecycleStatus enum	—	—	BR-027	—
`abcClass`	O	1	AbcClass enum	—	—	BR-032	VR-021
`barcode`	O	1–50	—	—	—	—	VR-007
`page,pageSize,sortBy,sortDirection`	O	per PaginationQuery	—	—	sort pair rule	—	—

API-003 — `GetProductBySkuRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
sku (path)	R	1–50	ProductSku	Product must exist	—	—	VR-007

API-004 — `UpdateProductAttributesRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
sku (path)	R	1–50	ProductSku	Product exists	—	—	VR-007
body fields	O*	per API-001	—	At least one field present	Traceability cross-rules same as API-001	BR-014–BR-017	VR-003

Cross-field: At least one mutable field required; traceability rules mirror API-001.

API-005 — `ChangeProductLifecycleStatusRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
sku (path)	R	1–50	ProductSku	Product exists	—	—	VR-007
`newStatus`	R	—	LifecycleStatus	Valid transition; audit attribution	—	BR-027	VR-004, VR-005
`reason`	O	0–500	text	—	—	—	VR-005
`effectiveDate`	O	—	ISO date	default today	—	—	VR-005

Business validation: `DISCONTINUED` blocks new PO/requisition (BR-027). Status change records user+date (VR-005).

API-006 — `AssignAbcClassificationRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
sku (path)	R	1–50	ProductSku	Product exists, not invalid ref	—	BR-032	VR-021, VR-022

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`abcClass`	R	1	A,B,C	—	—	BR-032	VR-021
`classificationBasis`	O	1–100	enum basis	—	—	BR-032	—
`effectiveDate`	O	—	ISO date	default today	—	—	—

API-007 — `ValidateProductEligibilityQuery`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`sku` (path)	R	1–50	ProductSku	Product exists	—	—	VR-007
`transactionType`	R	—	enum	Evaluated per type	Location R for location-scoped types	BR-027	VR-004, VR-006
`locationCode`	C	1–30	LocationCode	User scope	Required for RESERVATION/FULFILLMENT	—	VR-008

API-008 — `ListProductsByCategoryQuery`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`categoryCode` (path)	R	1–30	—	Category exists	—	—	—
`lifecycleStatus`	O	—	enum	—	—	—	—
pagination	O	per PaginationQuery	—	—	—	—	—

API-009 — `ResolveDuplicateProductConflictRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`resolutionAction`	R	—	`MERGE,REJECT_NEW,KEEP_BOTH`	Authorized role	—	—	VR-009, VR-010
`sourceSku`	R	1–50	ProductSku	Exists	—	—	VR-009
`targetSku`	C	1–50	ProductSku	Exists	Required if `MERGE`	—	VR-010
`mergeAuthorizationReference`	C	1–100	—	Inventory Controller approval	Required if both SKUs have on-hand	—	VR-010
`notes`	O	0–1000	—	—	—	—	—

API-010 — `BulkProductImportRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`importMode`	R	—	`VALIDATE_ONLY,UPSERT,CREATE_ONLY`	—	—	—	—
`sourceReference`	O	0–100	—	—	—	—	—
`products[]`	R	1–5000 items	array	Each line validates as API-001	Unique SKU within batch	BR-014–BR-017	VR-001, VR-002

`BulkProductLineDto`: same rules as API-001 per line.

M-02 — Warehouse Network

API-011 — `CreateInventoryLocationRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`locationCode`	R	1–30	LocationCode	Unique	—	—	VR-011
`name`	R	1–200	non-blank	—	—	—	VR-011
`locationType`	R	—	location type enum	—	—	—	VR-011
`operationalStatus`	R	—	status enum	—	—	—	VR-011
`parentLocationCode`	O	1–30	LocationCode	Parent exists; no circular hierarchy	≠ `locationCode`	—	VR-014, VR-015
address fields	O	per AddressDto	—	—	—	—	—
`plannedGoLiveDate`	O	—	ISO date	—	—	—	—

API-012 — `ListActiveLocationsQuery`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
filters	O	per field	enums	User location scope on results	—	—	VR-053, VR-061
pagination	O	PaginationQuery	—	—	—	—	—

API-013 to API-014 — Location read/update

DTO	Key business validation	BR	VR
`GetLocationByCodeRequest`	Location exists; user scope	—	VR-053
`UpdateLocationHierarchyRequest`	No self/ancestor parent (VR-014); single parent	—	VR-014, VR-015

API-015 — `ConfigureStoragePositionRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`locationCode` (path)	R	1–30	LocationCode	Location active	—	BR-024	VR-012
`positionCode`	R	1–30	unique per location	—	—	—	—
`zone`,`aisle`,`bin`	O	1–30 each	—	—	—	—	—
`positionType`	O	—	position enum	—	—	—	—
`isActive`	O	—	boolean	default true	—	—	—

API-016 — `GetStoragePositionsQuery`

Pagination + location scope; `locationCode` path required.

API-017 — `ActivateInventoryLocationRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`locationCode` (path)	R	1–30	LocationCode	Opening balance approved when policy applies	`openingBalanceApprovalReferenceC`	—	VR-013, VR-025

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`openingBalanceApprovalReference`	C	1–100	—	Valid approval record	Required per go-live policy	—	VR-025
`activationDate`	O	—	ISO date	—	—	—	—

API-018 — `DeactivateInventoryLocationRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`locationCode` (path)	R	1–30	LocationCode	No new transactions after deactivation	—	—	VR-012, VR-017
`reason`	R	1–500	non-blank	—	—	—	—

API-019 — `CloseInventoryLocationRequest`

Field	Req	Length	Format	Business Validation	Cross-Field	BR	VR
`locationCode` (path)	R	1–30	LocationCode	No on-hand unless disposition	`dispositionPlanReference` OR `closureWaiverReference` when stock remains	—	VR-016
`dispositionPlanReference`	C	1–100	—	Valid plan	Required if on-hand > 0	—	VR-016
`closureWaiverReference`	C	1–100	—	Executive waiver	Alternative to disposition plan	—	VR-016

API-020 / API-021 — Freeze and operational status

Path-only `locationCode` (R, LocationCode, exists, user scope). No body validation.

M-03 — Inventory Governance

API-022 — `CreateReasonCodeRequest`

Category	Rule	BR	VR
Required	`code`, `description`, `transactionTypes` (min 1)	BR-020	VR-026
Length	`code` 1–20; `description` 1–200	—	—
Format	`code` uppercase alphanumeric+underscore; `transactionTypes[]` ∈ {`ADJUSTMENT`,`WRITE_OFF`,`RECEIPT_DISCREPANCY`,`RETURN`,`TRANSFER_EXCEPTION`,`FULFILLMENT_EXCEPTION`}	—	VR-027
Business	`code` unique; `isActive` default true	BR-020	VR-026
Cross-field	—	—	—

API-023 — `GetActiveReasonCodesQuery`

Category	Rule	BR	VR
Required	None (all optional)	—	—
Length	Pagination per shared	—	—
Format	`transactionType` valid enum; `isActive` boolean	—	VR-026
Business	Default `isActive=true`	—	—

Category	Rule	BR	VR
Cross-field	—	—	—

API-024 — `RetireReasonCodeRequest`

Category	Rule	BR	VR
Required	`code` (path)	—	VR-026
Length	`retirementReason` 0–500	—	—
Format	`code` ReasonCode; `effectiveDate` ISO date	—	—
Business	Code exists and active; retired codes blocked for new transactions	BR-020	VR-027
Cross-field	—	—	—

API-025 — `ValidateReasonCodeQuery`

Category	Rule	BR	VR
Required	`code` (path), `transactionType`	BR-020	VR-026
Format	`transactionType` valid enum	—	VR-026, VR-027
Business	Code active and applicable to transaction type	BR-020	VR-027

API-026 — `PublishGovernancePolicyRequest`

Category	Rule	BR	VR
Required	`policyName`, `effectiveFrom`, `parameters` (min 1)	—	VR-028
Length	`policyName` 1–100; `description` 0–1000; param key 1–50; value 1–500	—	—
Format	`effectiveFrom`/`effectiveTo` ISO instant; `parameterType` enum	—	VR-028
Business	No overlapping effective policy version	—	VR-028
Cross-field	`effectiveTo` > `effectiveFrom` when provided	—	VR-028

API-027 — `UpdateApprovalThresholdRequest`

Category	Rule	BR	VR
Required	`policyId` (path), `transactionType`, `effectiveFrom`; at least one threshold	BR-021	VR-020, VR-082
Length	—	—	—
Format	`transactionType` ∈ {`ADJUSTMENT`,`WRITE_OFF`,`TRANSFER`,`REQUISITION`}; thresholds ≥0	—	VR-082
Business	Policy exists; threshold drives independent approval	BR-021	VR-082
Cross-field	At least one of `quantityThreshold` or `monetaryThreshold`	BR-021	—

API-028 — `UpdateReceiptToleranceRequest`

Category	Rule	BR	VR
Required	`policyId` (path), `effectiveFrom`	BR-006	VR-048
Format	`quantityTolerancePercent` 0–100; absolutes ≥0	—	VR-048
Business	Tolerance governs auto-accept of receipt variances	BR-006	VR-048
Cross-field	At least one tolerance field when amending	—	—

API-029 / API-030 — Threshold/tolerance queries

DTO	Required	Business	BR	VR
`GetEffectiveApprovalThresholdQuery`	`transactionType`	User location scope	BR-021	VR-082
`GetReceiptToleranceQuery`	None	User location scope; supplier exists if provided	BR-006	VR-048

API-031 — `EvaluateSegregationOfDutiesRequest`

Category	Rule	BR	VR
Required	`initiatorUserId`, `approverUserId`, `transactionType`	BR-019, BR-021	VR-075, VR-083
Length	User IDs 1–50; `transactionReference` 0–100	—	—
Format	`transactionType` ∈ {`COUNT_VARIANCE_APPROVAL`,`ADJUSTMENT_APPROVAL`,`WRITE_OFF_APPROVAL`,`ROLE_ASSIGNMENT`}	—	VR-083
Business	Both users exist; approver ≠ initiator	BR-019, BR-021	VR-075, VR-083
Cross-field	`approverUserId` ≠ `initiatorUserId`	BR-019	VR-075

M-04 — Inventory Balance

API-032 — `EstablishOpeningBalancesRequest`

Category	Rule	BR	VR
Required	`locationCode`, `balanceLines` (min 1); `approvalReference` conditional	—	VR-024, VR-025
Business	Location `PLANNED`/pre-activation; SKU exists; lot required if traceable	BR-017	VR-024, VR-025
Cross-field	`approvalReference` required before activation policy	—	VR-025

`OpeningBalanceLineDto`: `sku` R; `quantity` R (≥0); `lotIdentity` C if traceable; `ownershipType` enum default `OWNED` (BR-031).

API-033 — `GetStockPositionsQuery`

Category	Rule	BR	VR
Required	At least one of `sku` or `locationCode`	—	VR-054
Business	User location scope on `locationCode`	—	VR-053
Cross-field	`sku` OR `locationCode` required	—	VR-054

API-034 to API-037 — Stock read APIs

API	DTO	Required	Business	BR	VR
API-034	`GetStockPositionRequest`	`locationCode`, `sku` (path)	User scope; SKU exists	—	VR-053, VR-054
API-035	`GetAvailabilityBySkuQuery`	`sku` (path), `locationCode` (query)	ATP inquiry scope	BR-001	VR-055
API-036	`GetNetworkAvailabilityQuery`	`sku` (path)	Authorized locations only; quarantine excluded default	BR-001, BR-013	VR-056, VR-064
API-037	`GetStockStatusIndicatorRequest`	`locationCode`, `sku` (path)	Stock policy must exist	BR-009, BR-010	VR-059

API-038 — `DefineStockPolicyRequest`

Category	Rule	BR	VR
Required	`locationCode`, `sku`, `reorderLevel`, `minimumThreshold`, `safetyStock`, `effectiveFrom`	BR-009, BR-010	VR-018–VR-020
Format	Thresholds ≥0; `maximumThreshold` ≥ `reorderLevel` when set	BR-009	VR-019
Business	Active location and product	BR-009, BR-010	VR-018
Cross-field	`minimumThreshold` ≤ `reorderLevel` unless inverted policy documented	BR-010	VR-019

API-039 — `GetStockPolicyRequest`

Path params R; `asOfDate` O (default today). Business: policy lookup effective date (VR-020).

API-040 — `UpdateStockPolicyRequest`

Category	Rule	BR	VR
Required	`locationCode`, `sku` (path), `effectiveFrom`; ≥1 threshold field	BR-009, BR-010	VR-018–VR-020
Cross-field	At least one threshold field; same ordering rules as API-038	BR-009, BR-010	VR-019

API-041 — `GetLotBreakdownRequest`

Path `locationCode`, `sku` R; product traceable; `includeQuarantined` default false (BR-013, BR-017).

API-042 — `EvaluateSourceTransferCapacityQuery`

Path params R; `requestedQuantity` O ≥0; evaluated against max transferable spare (BR-012, VR-140).

API-043 — `ReserveStockRequest`

Category	Rule	BR	VR
Required	`locationCode`, `sku`, `quantity`, `orderId`, `orderLineId`	BR-002	VR-092–VR-094
Length	`orderId`, `orderLineId` 1–50	—	VR-093
Business	Location active, not frozen; product active; qty ≤ ATP; safety stock protected	BR-001, BR-002, BR-010	VR-006, VR-092, VR-094
Cross-field	`safetyStockExceptionReference` required when breaching safety stock	BR-010	VR-124, VR-125

API-044 — `ReleaseReservationRequest`

Category	Rule	BR	VR
Required	path params, `releaseReason`	—	VR-095, VR-096
Business	Reservation exists; release qty ≤ reserved	—	VR-095
Format	`releaseReason` enum	—	VR-096

API-045 — `ApplyQuarantineHoldRequest`

Category	Rule	BR	VR
Required	`quantity`, `holdReason`, `reasonCode`; `lotIdentity` if traceable	BR-013, BR-030	VR-026, VR-084
Business	Qty ≤ on-hand; hold excludes from sellable ATP	BR-013	VR-084, VR-056
Format	`holdReason` ∈ {`QUALITY`,`RECALL`,`CUSTOMS`,`BONDED`,`INVESTIGATION`}	BR-030	—

API-046 — `ReleaseQuarantineHoldRequest`

Category	Rule	BR	VR
Required	path params, `disposition`, `reasonCode`	BR-013, BR-026	VR-085
Cross-field	`inspectionOutcomeReference` required when `disposition=RESTORE_TO_SELLABLE`	BR-026	VR-085

API-047 — `TransferConsignmentOwnershipRequest`

Category	Rule	BR	VR
Required	`quantity`, `consumptionReference`; `lotIdentity` if traceable	BR-031	—
Length	`consumptionReference` 1–100	—	—
Business	Stock must be consignment; qty ≤ consignment on-hand	BR-031	—

API-048 — `SegregatePromotionalStockRequest`

Category	Rule	BR	VR
Required	`campaignId`, `allocationId`, `quantity`	BR-029	VR-127
Business	Valid allocation; qty ≤ ATP at allocation time	BR-029	VR-127

M-05 — Procurement & Supplier

API-049 — `RegisterSupplierRequest`

Category	Rule	BR	VR
Required	`supplierCode`, `name`, `paymentTermsDays`	—	VR-030
Length	`supplierCode` 1–30; `name` 1–200; email 0–254; phone 0–30	—	—
Format	`supplierCode` pattern; email RFC; `currencyCode` ISO 4217	—	—
Business	`supplierCode` unique; `isConsignment` flags consignment agreement	BR-031	VR-030

API-050 to API-053 — Supplier read/update/deactivate

API	Key validation	BR	VR
API-050	Query filters; pagination	—	—
API-051	`supplierCode` exists (path)	—	VR-030
API-052	≥1 body field; email format	—	—
API-053	`reason` R 1–500; no open POs unless policy	—	VR-030

API-054 — `CreatePurchaseRequisitionRequest`

Category	Rule	BR	VR
Required	`receivingLocationCode`, `justification`, `lines` (min 1)	BR-011, BR-027	VR-117, VR-119, VR-120
Length	`justification` 1–1000	—	VR-119
Business	Active location; SKU active, not discontinued	BR-027	VR-006, VR-120
Cross-field	Each line: `quantity` > 0	—	VR-117

API-055 — `GetPurchaseRequisitionRequest`

`requisitionId` (path) R; must exist.

API-056 — `ApprovePurchaseRequisitionRequest`

Category	Rule	BR	VR
Required	`requisitionId` (path); status submitted	BR-011	VR-118
Cross-field	`networkTransferDeclineReason` required when `networkTransferDeclined=true`	BR-011	VR-118

API-057 — `RejectPurchaseRequisitionRequest`

`requisitionId` R; `rejectionReason` R 1–1000 (VR-131).

API-058 — `ConvertRequisitionToPurchaseOrderRequest`

Category	Rule	BR	VR
Required	`requisitionId`, `supplierCode`, `expectedDeliveryDate`	BR-007, BR-027	VR-030, VR-033
Business	Requisition approved; supplier active	BR-007	VR-030
Cross-field	Line override qty ≤ requisition line qty	—	VR-117

API-059 — `PlacePurchaseOrderRequest`

Category	Rule	BR	VR
Required	`supplierCode`, `receivingLocationCode`, `expectedDeliveryDate`, `lines` (min 1)	BR-007, BR-027	VR-030–VR-033
Business	Supplier/location active; SKU not discontinued	BR-027	VR-006, VR-031
Cross-field	Each line `orderedQuantity` > 0	—	VR-032

API-060 — `ListOpenPurchaseOrdersQuery`

Date range cross-field; user location scope (VR-034); overdue filter (VR-123).

API-061 — `GetPurchaseOrderRequest`

`purchaseOrderId` (path) R; exists.

API-062 — `AmendPurchaseOrderRequest`

Category	Rule	BR	VR
Required	`purchaseOrderId`, `amendmentReason`	—	VR-050, VR-051
Length	`amendmentReason` 1–1000	—	—
Business	Authorized user; amended qty ≥ received qty	—	VR-050, VR-051

API-063 — `CancelPurchaseOrderRequest`

`purchaseOrderId` R; `cancellationReason` R 1–1000; authorized user (VR-051).

API-064 — `ScheduleInboundDeliveryRequest`

Category	Rule	BR	VR
Required	`purchaseOrderId`, `expectedArrivalDate`	—	VR-033
Business	PO open	—	—

Category	Rule	BR	VR
Cross-field	Line `expectedQuantity` ≤ remaining open qty	—	VR-050

API-065 — `ValidatePoForReceiptQuery`

`purchaseOrderId` R; `receivingLocationCode` must match PO when provided (BR-007, VR-040).

M-06 — Inbound Receiving

API-066 — `CreateGoodsReceiptRequest`

Category	Rule	BR	VR
Required	`receivingLocationCode`; `purchaseOrderId` OR ad-hoc auth (via separate API)	BR-007	VR-040, VR-012
Business	Location active, not frozen; PO valid when provided	BR-007	VR-040
Cross-field	`supplierCode` must match PO supplier when PO set	BR-007	VR-036

API-067 — `ListGoodsReceiptsQuery`

Date range cross-field; user location scope (VR-034).

API-068 — `GetGoodsReceiptRequest`

`goodsReceiptId` (path) R; exists.

API-069 — `VerifyGoodsReceiptRequest`

Category	Rule	BR	VR
Required	`lines` (min 1) per line: `lineId`, `sku`, `acceptedQuantity`	BR-004, BR-016, BR-017	VR-035–VR-037
Business	SKU matches PO; MRSL check for shelf-life; lot if traceable	BR-004, BR-016, BR-017	VR-035, VR-036
Cross-field	`acceptedQuantity` + `rejectedQuantity` ≤ physically verified	BR-004	VR-037

API-070 — `RecordReceiptDiscrepancyRequest`

Category	Rule	BR	VR
Required	`lineId`, `discrepancyType`, `varianceQuantity`, `reasonCode`, `notes`	BR-006	VR-038, VR-026
Length	`notes` 1–2000; `evidenceReference` 0–200	BR-006	VR-038
Business	Document before acceptance	BR-006	VR-038

API-071 — `ListReceiptDiscrepanciesQuery`

`goodsReceiptId` (path) R; `status` enum optional.

API-072 — `ResolveReceiptDiscrepancyRequest`

Category	Rule	BR	VR
Required	path params, `resolutionOutcome`, `resolutionNotes`	BR-006	VR-038
Length	`resolutionNotes` 1–2000	—	—
Business	Discrepancy open	BR-006	—

API-073 — `CompletePutAwayRequest`

Category	Rule	BR	VR
Required	`putAwayLines` (min 1): `receiptLineId`, `quantity`, `storagePositionCode`	BR-005	VR-043
Business	Receipt verified; position exists; ATP increases only after put-away	BR-005	VR-043
Cross-field	Put-away qty ≤ remaining unplaced per line	BR-005	VR-043

API-074 — `ListPendingPutAwayQuery`

`locationCode` R; user scope.

API-075 — `ReceiveAdHocInboundRequest`

Category	Rule	BR	VR
Required	`receivingLocationCode`, `adHocAuthorizationReference`, `reasonCode`, `sourceDescription`, `lines`	BR-007	VR-042, VR-026
Length	`adHocAuthorizationReference` 1–100; `sourceDescription` 1–500	—	VR-042
Business	Governed exception to PO requirement	BR-007	VR-042

API-076 — `ReceiveReturnInboundRequest`

Category	Rule	BR	VR
Required	`returnAuthorizationId`, `receivingLocationCode`, `lines`	BR-025	VR-045, VR-046
Business	Valid open RMA; qty ≤ authorized	BR-025	VR-145, VR-046

API-077 — `RegisterSupplierInvoiceRequest`

Category	Rule	BR	VR
Required	`supplierCode`, `invoiceNumber`, `invoiceDate`, `purchaseOrderId`, `currencyCode`, `lines`	BR-008	—
Length	`invoiceNumber` 1–50 unique per supplier	—	—
Business	PO linked to supplier; lines match PO/receipt	BR-008	—

API-078 — `GetSupplierInvoiceRequest`

`invoiceId` (path) R; exists.

API-079 — `PerformThreeWayMatchRequest`

Category	Rule	BR	VR
Required	`invoiceId` (path)	BR-008	VR-048
Business	Match PO–receipt–invoice; tolerance override authorized when exceeded	BR-008	VR-048

API-080 — `AuthorizeInvoicePaymentRequest`

Category	Rule	BR	VR
Required	`invoiceId` (path)	BR-008	—
Business	Three-way match completed	BR-008	—
Cross-field	`paymentDueDate` ≥ `invoiceDate`	—	—

M-07 — Order Fulfillment

API-081 — `CreateCustomerOrderRequest`

Category	Rule	BR	VR
Required	`channel`, `lines` (min 1); per line: `lineNumber`, `sku`, `requestedQuantity`	—	VR-006
Format	`channel` ∈ {B2B,B2C,INTERNAL,MARKETPLACE}	—	—
Business	SKU active	—	VR-006

API-082 — `ListCustomerOrdersQuery`

Date range cross-field; user location scope on `shipFromLocationCode`.

API-083 — `GetCustomerOrderRequest`

`orderId` (path) R; exists (VR-108).

API-084 — `ConfirmOrderAvailabilityRequest`

Category	Rule	BR	VR
Required	`orderId` (path)	BR-001	VR-090, VR-091
Business	ATP-based availability; optional inbound supply flag	BR-001	VR-090, VR-091

API-085 — `ConfirmCustomerOrderRequest`

Category	Rule	BR	VR
Required	`orderId` (path)	BR-002	VR-092
Business	Availability confirmed; creates reservation	BR-001, BR-002	VR-092
Cross-field	`safetyStockExceptionReference` when breaching safety stock	BR-010	VR-124

API-086 — `CancelCustomerOrderRequest`

`orderId` R; `cancellationReason` R 1–1000 (VR-096); `lineIds` must exist on order.

API-087 — `CreateMultiLocationFulfillmentPlanRequest`

Category	Rule	BR	VR
Required	`allocations` (min 1)	BR-001	VR-106, VR-107
Business	Per-location ATP sufficient	BR-001	VR-107
Cross-field	Sum of allocations per line = line requested qty	—	VR-106

API-088 — `AssignPickListRequest`

`orderId` R; order confirmed; `pickStrategy` default per product FIFO/FEFO (BR-014, BR-015).

API-089 — `ListOrdersAwaitingPickQuery`

`orderId` (path) R; optional location scope.

API-090 — `VerifyPickRequest`

Category	Rule	BR	VR
Required	`pickLines` (min 1)	BR-003, BR-017	VR-097–VR-099

Category	Rule	BR	VR
Business	Picked qty ≤ reserved; SKU match or approved substitute	BR-003	VR-098, VR-110
Cross-field	`substitutionReference` when SKU differs	—	VR-110, VR-111

API-091 — `PackOrderRequest`

Category	Rule	BR	VR
Required	`packages` (min 1); each package `lineItems` (min 1)	BR-003	—
Cross-field	Package line qty ≤ picked qty per order line	BR-003	—

API-092 — `DispatchShipmentRequest`

Category	Rule	BR	VR
Required	`shipmentLines` (min 1)	BR-003	VR-097, VR-098
Business	Pick verified; shipped qty ≤ picked/reserved	BR-003	VR-097, VR-098
Cross-field	Lot required if traceable	BR-017	—

API-093 — `CompletePartialFulfillmentRequest`

Category	Rule	BR	VR
Required	`fulfillmentLines`; `createBackorderForRemainder`	BR-028	VR-100, VR-101
Cross-field	`fulfillQuantity` > 0 and < full line; `customerAcknowledgmentReference` when backorder	BR-028	VR-101

API-094 — `RecordBackorderRequest`

Category	Rule	BR	VR
Required	`backorderLines`; `customerAcknowledgmentReference` per line	BR-028	VR-101, VR-103
Business	Backorder qty = unfulfilled remainder	BR-028	VR-103

API-095 — `FulfillBackorderRequest`

Category	Rule	BR	VR
Required	`backorderId` (path)	BR-028	VR-102, VR-092
Business	Backorder open; ATP sufficient	BR-001	VR-092
Cross-field	`priorityOverrideReference` when out of sequence	—	VR-102

API-096 — `RaiseFulfillmentExceptionRequest`

`orderId` R; `exceptionType` R; `description` R 1–2000 (VR-104).

API-097 — `GetOrderFulfillmentStatusRequest`

`orderId` (path) R; exists (VR-108).

M-08 — Network Transfer

API-098 — `RequestStockTransferRequest`

Category	Rule	BR	VR
Required	`sourceLocationCode`, `destinationLocationCode`, `justification`, `lines`	BR-011, BR-012	VR-128–VR-130
Length	`justification` 1–1000	—	VR-128
Business	Qty ≤ source ATP; respects source minimum	BR-012	VR-130, VR-140
Cross-field	Source ≠ destination	—	VR-129

API-099 — `ListStockTransfersQuery`

Date range cross-field; user location scope.

API-100 — `GetStockTransferRequest`

`transferId` (path) R; exists.

API-101 — `ApproveTransferRequest`

Category	Rule	BR	VR
Required	`transferId` (path); status `REQUESTED`	BR-012	VR-132, VR-141
Business	Revalidated ATP; approved qty ≤ max transferable	BR-012	VR-141

API-102 — `RejectTransferRequest`

`transferId` R; `rejectionReason` R 1–1000 (VR-131).

API-103 — `DispatchTransferRequest`

Category	Rule	BR	VR
Required	`dispatchLines` (min 1)	—	VR-133
Business	Status `APPROVED`; source not frozen; dispatched ≤ approved	—	VR-133

API-104 — `ConfirmTransferReceiptRequest`

Category	Rule	BR	VR
Required	`receiptLines` (min 1)	—	VR-134, VR-135
Business	Status `IN_TRANSIT`	—	—
Cross-field	`varianceNotes` when received ≠ dispatched	BR-020	VR-135

API-105 — `CancelInTransitTransferRequest`

`transferId` R; `cancellationReason` R 1–1000; eligible status.

API-106 / API-107 — Transfer list queries

User location scope (VR-136); overdue in-transit (VR-137).

M-09 — Inventory Accuracy

API-108 — `ScheduleCycleCountRequest`

Category	Rule	BR	VR
Required	`locationCode`, `sku`, `scheduledDate`	BR-018, BR-032	VR-067, VR-068
Business	Active location/product; Class A default HIGH priority	BR-032	VR-068

API-109 — `ListCycleCountsQuery`

Date range cross-field; user scope.

API-110 — `GetCycleCountAssignmentRequest`

`cycleCountId` R; response must not expose expected qty (BR-018, VR-069).

API-111 — `SubmitCycleCountResultRequest`

Category	Rule	BR	VR
Required	`physicalQuantity`	BR-018	VR-070
Business	Blind count submission; status scheduled/in-progress	BR-018	VR-069

API-112 — `InitiatePhysicalInventoryCountRequest`

Category	Rule	BR	VR
Required	`locationCode`, `plannedStartDate`, `plannedEndDate`	BR-024	VR-071
Cross-field	`plannedEndDate` ≥ `plannedStartDate`	BR-024	VR-071

API-113 — `GetPhysicalCountStatusRequest`

`physicalCountId` (path) R.

API-114 — `RecordPhysicalCountResultRequest`

Category	Rule	BR	VR
Required	`countLines` (min 1)	BR-018, BR-024	VR-070, VR-071
Business	Location frozen during count	BR-024	VR-071

API-115 — `CertifyPhysicalInventoryCountRequest`

Category	Rule	BR	VR
Required	`physicalCountId` (path)	BR-024	VR-072
Business	All SKUs counted before certification	BR-024	VR-072

API-116 — `OpenAdjustmentCaseRequest`

Category	Rule	BR	VR
Required	`originType`, `locationCode`, `sku`, `reasonCode`	BR-020, BR-022	VR-077, VR-079
Cross-field	`varianceId` when `originType=COUNT_VARIANCE`; direction/qty for direct	BR-022	VR-078

API-117 — `GetAdjustmentCaseRequest`

`caseId` (path) R.

API-118 — `InvestigateVarianceRequest`

Category	Rule	BR	VR
Required	`rootCauseCategory`, `investigationFindings`	BR-022	VR-076
Length	`investigationFindings` 1–5000	BR-022	VR-076
Business	Investigation before adjustment	BR-022	VR-076

API-119 — `ApproveCountVarianceRequest`

Category	Rule	BR	VR
Required	path params, `approvedAdjustmentQuantity`	BR-019	VR-075
Business	Approver ≠ counter	BR-019	VR-075

API-120 — `SubmitAdjustmentForApprovalRequest`

Category	Rule	BR	VR
Required	`adjustmentType`, `quantity`, `reasonCode`	BR-020, BR-023	VR-077, VR-080, VR-081
Cross-field	`evidenceReference` for write-off; decrease ≤ on-hand	BR-023	VR-080, VR-078

API-121 — `ApproveAdjustmentRequest`

Category	Rule	BR	VR
Required	`caseId` (path)	BR-021	VR-083
Business	Approver ≠ initiator and counter; independent approval	BR-019, BR-021	VR-083, VR-075

API-122 — `RejectAdjustmentRequest`

`caseId` R; `rejectionReason` R 1–2000.

API-123 — `ListPendingApprovalsQuery`

User location scope; optional case type filter.

M-10 — Returns Management

API-124 — `IssueReturnAuthorizationRequest`

Category	Rule	BR	VR
Required	`originalOrderId`, `returnReason`, `lines`	BR-025	VR-142, VR-143
Length	`returnReason` 1–500	—	—
Business	Order return-eligible; authorized qty ≤ shipped	BR-025	VR-143

API-125 to API-127 — RMA list/get/cancel

API	Key validation	BR	VR
API-125	Query filters	—	—
API-126	`returnAuthorizationId` exists	BR-025	VR-149
API-127	`cancellationReason` R 1–1000; open/unused RMA	—	—

API-128 — `ReceiveCustomerReturnRequest`

Category	Rule	BR	VR
Required	`returnId`, `receivingLocationCode`, `receivedLines`	BR-025	VR-144, VR-145, VR-046
Business	SKU matches authorization; qty ≤ authorized	BR-025	VR-145

API-129 — `InspectReturnedGoodsRequest`

Category	Rule	BR	VR
Required	`inspectionLines` (min 1)	BR-026	VR-146
Business	Inspect before restocking	BR-026	VR-146

API-130 — `DispositionReturnRequest`

Category	Rule	BR	VR
Required	`dispositionAction`, `dispositionLines`	BR-023, BR-026	VR-147, VR-148
Cross-field	Sum disposition qty = inspected qty; `restockLocationCode` if RESTOCK; write-off refs if WRITE_OFF	BR-023, BR-026	VR-147, VR-080

API-131 — `ListPendingReturnReceiptsQuery`

Optional `locationCode`; user scope.

M-11 — Commercial Campaign

API-132 — `CreatePromotionalAllocationRequest`

Category	Rule	BR	VR
Required	`campaignId`, `campaignName`, `locationCode`, `sku`, `allocatedQuantity`, `campaignStartDate`, `campaignEndDate`	BR-029	VR-126, VR-127
Length	`campaignId` 1–50; `campaignName` 1–200	—	—
Business	Allocated qty ≤ ATP	BR-029	VR-127
Cross-field	`campaignEndDate` ≥ `campaignStartDate`	—	VR-126

API-133 / API-134 — Allocation list/get

Standard query/path validation; user scope (BR-029).

API-135 — `AmendPromotionalAllocationRequest`

Category	Rule	BR	VR
Required	`allocationId`, `amendmentReason`; ≥1 amendable field	BR-029	VR-127
Business	Increased qty ≤ available ATP	BR-029	VR-127

API-136 — `EvaluateCampaignReadinessRequest`

`allocationId` R; `targetQuantity` defaults to allocated (VR-126).

API-137 — `ReleasePromotionalAllocationRequest`

`allocationId` R; `releaseReason` R enum.

API-138 — `AuthorizePromotionalOverrideRequest`

Category	Rule	BR	VR
Required	`allocationId`, `orderId`, `overrideQuantity`, `authorizationReference`, `justification`	BR-029	—
Business	Override qty ≤ ring-fenced remaining; commercial leadership auth	BR-029	—

API-139 — `CheckRingFenceQuery`

Path `locationCode`, `sku` R; optional `asOfDate` (BR-029).

M-12 — Identity & Access

API-140 — `ProvisionUserAccountRequest`

Category	Rule	BR	VR
Required	`userId`, `email`, `displayName`, `initialRoles` (min 1)	—	VR-083
Length	`userId` 1–50; `email` 1–254; `displayName` 1–200	—	—
Format	`userId` alphanumeric+underscore; valid email	—	—
Business	`userId` unique; no SoD in role set	—	VR-083

API-141 to API-142 — User list/get

Standard pagination; `userId` exists for get.

API-143 — `AssignRoleRequest`

Category	Rule	BR	VR
Required	`userId` (path), `roleAssignment`	—	VR-083
Business	Account active; no SoD conflict	—	VR-083

API-144 — `RevokeRoleAssignmentRequest`

Path params R; assignment exists on user.

API-145 — `SuspendUserAccountRequest`

`userId` R; `suspensionReason` R 1–1000; account active.

API-146 — `DeactivateUserAccountRequest`

`userId` R; `deactivationReason` R 1–1000.

API-147 — `SyncUsersFromDirectoryRequest`

Category	Rule	BR	VR
Required	`syncMode`	—	—
Format	`syncMode` ∈ {`FULL`,`INCREMENTAL`}	—	—
Cross-field	`sinceTimestamp` required when `INCREMENTAL`	—	—

API-148 — `EvaluateAuthorizationRequest`

`action` R 1–100; optional user defaults to caller; location scope when provided (VR-008).

API-149 — `ValidateLocationScopeRequest`

`locationCode` R; exists; user access check (VR-008, VR-053).

M-13 — Inventory Intelligence (Read-Only)

All intelligence DTOs: path/query only. Common patterns:

Pattern	Validation	BR	VR
Location scope	User authorized locations only	—	VR-151, VR-154
Date range	`startDate` ≤ `endDate`; max 365 days for export	—	VR-062, VR-153
Pagination	Shared `PaginationQuery`	—	—
Point-in-time	`asOfTimestamp` default latest projection	—	VR-152

API-150 — `GetInventoryDashboardQuery`

| Cross-field | `locationCode` required when `scope=LOCATION` | VR-151 |

API-151 to API-152 — Stock status summary/board

Optional location/category/ABC filters; authorized locations (VR-151).

API-153 — `GetAvailabilityInquiryLogQuery`

Date range cross-field (VR-062); user/location scope.

API-154 — `GetQuarantineRegisterQuery`

Hold reason enum; location scope (BR-013).

API-155 — `SearchMovementHistoryQuery`

| Required | `startDate`, `endDate` | | Cross-field | At least one filter recommended (VR-154) |

API-156 — `QueryAuditTrailQuery`

`startDate`, `endDate` R; date range (VR-153).

API-157 — `ExportAuditEvidenceQuery`

`startDate`, `endDate` R; max 365-day range; authorized locations (VR-153, VR-154).

API-158 — `GetOperationalAlertsQuery`

Alert type/severity enums; location scope.

API-159 — `AcknowledgeOperationalAlertRequest`

Category	Rule	BR	VR
Required	`alertId` (path), `actionTaken`	BR-009	VR-113, VR-115
Format	`actionTaken` ∈ {`REQUISITION`,`TRANSFER`,`DISMISS`,`DEFER`}	—	VR-115
Cross-field	`acknowledgmentNotes` required when `actionTaken` ∈ {`DISMISS`,`DEFER`}	—	VR-114

API-160 — `GetReplenishmentRecommendationsQuery`

Date range cross-field; user location scope; `recommendationAction` enum (BR-009, BR-011).

API-161 — `GetOpenPurchaseOrderBoardQuery`

User location scope (VR-034); overdue filter (VR-123).

API-162 — `GetInboundSupplyCalendarQuery`

| Required | `startDate`, `endDate` | | Cross-field | `startDate` ≤ `endDate`; max 90-day range | VR-033 |

API-163 — `GetInboundDeliveryDelayLogQuery`

Date range cross-field; location/supplier filters; user scope.

API-164 — `GetBackorderPipelineQuery`

`prioritySort` default true (VR-102, BR-028); location scope.

API-165 — `GetFulfillmentExceptionLogQuery`

Exception type/status enums; date range cross-field (BR-003).

API-166 — `GetNetworkImbalanceAnalysisQuery`

Active locations; optional SKU active (VR-139, BR-011).

API-167 — `GetShrinkageAnalysisQuery`

| Required | `startDate`, `endDate` | | Business | Reason code filter if provided (BR-020) | VR-086, VR-087 |

API-168 — `GetCountPerformanceMetricsQuery`

| Required | `startDate`, `endDate` | | Business | Exclude pending variances default (BR-018) | VR-157, VR-158 |

API-169 — `GetCampaignReadinessStatusQuery`

`campaignId` max 50; `daysUntilStart` ≥ 0 (BR-029, VR-126).

API-170 — `GetInventoryKpisQuery`

| Required | `startDate`, `endDate` | | Business | Authorized `locationCodes[]`; valid `kpiCodes[]` | VR-151, VR-161, VR-162 |

API-171 — `GetValuationSnapshotsQuery`

Date range cross-field; user location scope.

API-172 — `GetValuationReconciliationQuery`

| Required | `periodEndDate` | | Format | `varianceThresholdPercent` 0–100 | BR-008 |

Platform & Integration

API-173 — `GetSessionRequest`

Optional query flags only; auth via `Authorization` header (VR-008).

API-174 — `LogoutRequest`

`logoutAllSessions` optional boolean; empty body accepted.

API-175 — `HealthCheckRequest`

No parameters; public endpoint — no validation.

API-176 — `ImportSalesOrderRequest`

Category	Rule	BR	VR
Required	`externalOrderId`, `externalSystemCode`, `channel`, `lines` (min 1)	—	VR-007
Length	`externalOrderId` 1–100; `externalSystemCode` 1–30	—	—
Format	`channel` ∈ {B2B,B2C,MARKETPLACE}	—	—
Business	Idempotent re-import; SKU exists; qty > 0 per line	BR-001	VR-006, VR-007

`ExternalOrderLineDto`: `externalLineId` R 1–50; `sku` R; `quantity` R > 0.

API-177 — `GetFulfillmentStatusFeedQuery`

| Required | `externalSystemCode` | | Length | `externalSystemCode` 1–30; `externalOrderId` 0–100 | | Format | `updatedSince` ISO instant; `status` enum |

API-178 — `ImportCampaignPlanRequest`

Category	Rule	BR	VR
Required	`externalCampaignId`, `externalSystemCode`, `campaignName`, `campaignStartDate`, `campaignEndDate`, `allocationLines` (min 1)	BR-029	VR-126
Length	`externalCampaignId` 1–50; `campaignName` 1–200	—	—
Cross-field	`campaignEndDate` ≥ `campaignStartDate`	—	VR-126
Business	Active SKU/location per allocation line; target qty > 0	BR-029	VR-127

Business Rule Traceability Index

Maps each **BR-xxx** to Request DTOs where it is enforced.

BR	Rule Name	Enforced In (API IDs)
BR-001	ATP-Only Customer Commitment	API-035, API-036, API-043, API-084, API-085, API-087, API-095, API-098, API-176
BR-002	Reservation at Order Confirmation	API-043, API-085, API-164
BR-003	Verified Pick Before Shipment	API-090, API-091, API-092
BR-004	Verify Before Inventory Acceptance	API-069
BR-005	Put-Away Before Availability	API-073
BR-006	Document Receipt Discrepancies	API-028, API-070, API-072
BR-007	Purchase Order Required for Supplier Receipt	API-058, API-059, API-065, API-066, API-075
BR-008	Three-Way Match Before Payment	API-077, API-079, API-080
BR-009	Reorder at Reorder Level	API-038, API-040, API-159
BR-010	Safety Stock Protection	API-038, API-040, API-043, API-085
BR-011	Network Check Before Purchase	API-054, API-056, API-098
BR-012	Protect Source Location Minimum	API-042, API-098, API-101
BR-013	Quarantine Exclusion from Sellable Stock	API-036, API-045, API-046, API-154
BR-014	FEFO Rotation for Shelf-Life Products	API-001, API-004, API-088, API-090
BR-015	FIFO Rotation for Standard Goods	API-001, API-004, API-088, API-090
BR-016	Minimum Remaining Shelf Life at Receipt	API-001, API-004, API-069

BR	Rule Name	Enforced In (API IDs)
BR-017	Lot or Batch Capture for Traceable Products	API-001, API-004, API-032, API-045, API-069, API-090, API-092, API-098, API-103
BR-018	Blind Cycle Counting	API-108, API-110, API-111, API-114, API-168
BR-019	Independent Count Approval	API-031, API-119, API-121
BR-020	Mandatory Adjustment Reason Codes	API-022, API-024, API-104, API-116, API-120, API-167
BR-021	Independent Approval for Material Adjustments	API-027, API-031, API-121
BR-022	Investigate Before Adjusting	API-116, API-118
BR-023	Documented Write-Off Authorization	API-120, API-130
BR-024	Physical Inventory Freeze	API-112, API-114, API-115
BR-025	Return Authorization Required	API-076, API-124, API-126, API-128
BR-026	Inspect Returns Before Restocking	API-046, API-129, API-130
BR-027	No Reorder of Discontinued Products	API-005, API-007, API-054, API-059
BR-028	Backorder Customer Acknowledgment	API-093, API-094, API-095
BR-029	Promotional Stock Ring-Fencing	API-048, API-132, API-135, API-138, API-139, API-169, API-178
BR-030	Bonded Inventory Customs Clearance	API-045
BR-031	Consignment Ownership at Consumption	API-032, API-047, API-049
BR-032	Class A Priority Monitoring	API-006, API-108, API-168

Validation Rule Cross-Reference Index

VR Range	Domain	Primary APIs
VR-001–VR-010	Product catalog	API-001–API-010
VR-011–VR-017	Locations	API-011–API-021
VR-018–VR-028	Governance & thresholds	API-026–API-031, API-038–API-040
VR-030–VR-051	Procurement & PO	API-049–API-065, API-062–API-063
VR-035–VR-048	Receiving & matching	API-066–API-080
VR-053–VR-056, VR-059	Stock positions & ATP	API-033–API-037
VR-062, VR-153–VR-154	Date ranges & audit queries	API-153–API-157, intelligence APIs
VR-066–VR-072	Physical accuracy	API-108–API-115
VR-075–VR-085	Adjustments & quarantine	API-045–API-046, API-116–API-122
VR-090–VR-111	Fulfillment	API-081–API-096
VR-117–VR-125	Requisition & replenishment	API-054–API-056, API-043, API-085
VR-126–VR-127	Promotional	API-048, API-132–API-138, API-178
VR-128–VR-141	Transfers	API-042, API-098–API-107
VR-142–VR-149	Returns	API-124–API-130, API-076
VR-151–VR-162	Intelligence scope & KPIs	API-150–API-172

Document Approval

Role	Name	Signature	Date
Senior Backend Architect			

Role	Name	Signature	Date
Solution Architect			
Senior Business Analyst			
Product Owner			

End of Request DTO Validation Specification

Document ID: SRS-API-004 | Version 1.0 | July 6, 2026

Error Response Specification

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-API-005
Document Title	Standard Error Response Model — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Senior Backend Architect
Source Documents	SRS-API-001, SRS-API-004, SRS-BR-001, SRS-VR-001, Cross-Cutting Concerns (Exception Handling)
Date	July 6, 2026
Scope	All REST APIs under `/api/v1`

Purpose

Defines the **standard error response contract** for every REST API. All error responses conform to RFC 7807 Problem Details for HTTP APIs with IMS extensions for machine-readable error codes, field-level validation, business rule traceability, and operational correlation.

Design Principles

Principle	Detail
RFC 7807 baseline	`type`, `title`, `status`, `detail`, `instance` on every response
Machine + human	`errorCode` for clients/automation; `detail` / `message` for operators
No leakage	`500` responses never expose stack traces, SQL, or internal paths
Traceability	`traceId` on every error; `businessRuleId` / `validationRuleId` when applicable
Layer mapping	Presentation maps domain/application exceptions — domain never emits HTTP codes

Standard Envelope

`ProblemDetailResponse` (base)

Field	Type	Required	Description
`type`	String (URI)	Yes	Problem type URI — stable identifier for error category
`title`	String	Yes	Short, human-readable summary of the problem type
`status`	Integer	Yes	HTTP status code (duplicate of response status for convenience)
`detail`	String	Yes	Human-readable explanation specific to this occurrence
`instance`	String (URI)	Yes	Request path that triggered the error (e.g. `/api/v1/products`)
`errorCode`	String	Yes	IMS machine-readable error code (see naming convention)
`traceId`	String (UUID)	Yes	Distributed trace / correlation ID for support and logs
`timestamp`	String (Instant)	Yes	ISO-8601 UTC time the error was generated
`details`	Object	No	Structured context — shape varies by error category
`errors`	Array<FieldError>	No	Field-level validation errors (400 only)

`FieldError` (validation extension)

Field	Type	Required	Description
`field`	String	Yes	JSON path to the invalid field (e.g. `lines[2].quantity.amount`)
`code`	String	Yes	Field-level validation error code
`message`	String	Yes	Human-readable field error
`rejectedValue`	Any	No	Submitted value when safe to return (omit secrets)

Problem Type URI Convention

```
https://api.{environment}.ims.example/problems/{category}
```

Category slug	HTTP Status	Used for
`validation-error`	400	Request syntactic / structural validation
`authentication-error`	401	Missing or invalid credentials
`authorization-error`	403	Authenticated but not permitted
`not-found`	404	Resource does not exist or not visible
`conflict`	409	State conflict, duplicate, concurrency
`business-rule-violation`	422	Domain business rule or invariant failure
`internal-server-error`	500	Unexpected unhandled failure
`service-unavailable`	503	Infrastructure / dependency unavailable

Error Code Naming Convention

```
{CATEGORY}_{DOMAIN}_{REASON}
```

Prefix	Category
`VAL_`	Validation (400)
`AUTH_`	Authentication (401)

Prefix	Category
`AUTHZ_`	Authorization (403)
`NF_`	Not Found (404)
`CONF_`	Conflict (409)
`BIZ_`	Business rule (422)
`SYS_`	System / internal (500, 503)

Error Category Catalog

1. Validation Errors — `400 Bad Request`

When: Request fails syntactic validation (L1–L2) before or during command dispatch. Bean Validation, FluentValidation, or request binder failures.

Attribute	Value
HTTP Status Code	`400`
Problem Type	`https://api.ims.example/problems/validation-error`
Exception Source	`ValidationException` (Presentation / Application)

Standard Error Codes

Error Code	Error Message (title/detail pattern)	Trigger
`VAL_REQUEST_INVALID`	The request contains one or more validation errors.	Generic wrapper when multiple field errors
`VAL_REQUIRED_FIELD`	Required field is missing or blank.	`@NotNull`, `@NotBlank`
`VAL_INVALID_FORMAT`	Field value does not match the required format.	Pattern, type coercion failure
`VAL_OUT_OF_RANGE`	Value is outside the allowed range.	Min/max, scale violations
`VAL_INVALID_ENUM`	Value is not a permitted enumeration member.	Unknown enum value
`VAL_INVALID_DATE_RANGE`	Start date must be on or before end date.	Cross-field date (VR-062, VR-153)
`VAL_PATCH_NO_FIELDS`	At least one field must be provided for update.	Empty PATCH body
`VAL_QUERY_FILTER_REQUIRED`	At least one filter parameter is required.	e.g. `sku` OR `locationCode` (VR-054)

Details Object Shape

```
{
  "details": {
    "requestId": "optional-client-correlation"
  }
}
```

Example JSON Response

```
{
  "type": "https://api.ims.example/problems/validation-error",
  "title": "Validation Failed",
  "status": 400,
  "detail": "The request contains 2 validation errors. Correct the highlighted fields and resubmit.",
  "instance": "/api/v1/products",
  "errorCode": "VAL_REQUEST_INVALID",
  "traceId": "7f3c9a2e-4b1d-4e8a-9c0f-2a8b6d4e1f30",
  "timestamp": "2026-07-06T14:30:00.000Z",
  "errors": [
    {
      "field": "sku",
      "code": "VAL_INVALID_FORMAT",
      "message": "SKU must match pattern ^[A-Z0-9][A-Z0-9_-]*$ and be 1–50 characters.",
      "rejectedValue": "widget-01"
    },
    {
      "field": "mrslThresholdDays",
      "code": "VAL_REQUIRED_FIELD",
      "message": "Minimum remaining shelf life days is required when shelfLifeManaged is true.",
      "rejectedValue": null
    }
  ]
}
```

Business rule link: Validation enforces VR-xxx constraints defined in SRS-API-004; field `code` may mirror VR ID (e.g. `VR-001`).

---

2. Business Errors — `422 Unprocessable Entity`

When: Request is well-formed but violates a domain invariant or business rule (L3–L4). Aggregate refuses the command.

Attribute	Value
HTTP Status Code	`422`
Problem Type	`https://api.ims.example/problems/business-rule-violation`
Exception Source	`BusinessRuleViolationException` (Domain)

Standard Error Codes

Error Code	Error Message	BR	VR
`BIZ_INSUFFICIENT_ATP`	Cannot commit quantity exceeding available-to-promise.	BR-001	VR-092
`BIZ_RESERVATION_EXCEEDS_ATP`	Reservation quantity exceeds available stock at this location.	BR-001, BR-002	VR-092
`BIZ_SAFETY_STOCK_BREACH`	Operation would breach safety stock. Provide an authorized exception reference.	BR-010	VR-124, VR-125
`BIZ_LOCATION_FROZEN`	Location is frozen for inventory transactions.	BR-024	VR-053
`BIZ_PRODUCT_DISCONTINUED`	Product is discontinued and cannot be ordered or replenished.	BR-027	VR-004, VR-120
`BIZ_PO_REQUIRED_FOR_RECEIPT`	A valid purchase order is required before receiving supplier goods.	BR-007	VR-040
`BIZ_VERIFY_BEFORE_ACCEPTANCE`	Goods must be verified before acceptance into inventory.	BR-004	VR-035
`BIZ_PUT_AWAY_REQUIRED`	Stock is not available until put-away is complete.	BR-005	VR-043
`BIZ_MRSL_NON_COMPLIANT`	Remaining shelf life does not meet minimum receiving threshold.	BR-016	VR-035
`BIZ_LOT_REQUIRED`	Lot or batch capture is required for this traceable product.	BR-017	—

Error Code	Error Message	BR	VR
`BIZ_SEGREGATION_OF_DUTIES`	Proposed approver cannot approve this transaction due to segregation of duties.	BR-019, BR-021	VR-075, VR-083
`BIZ_INVESTIGATION_REQUIRED`	Variance must be investigated before adjustment submission.	BR-022	VR-076
`BIZ_BACKORDER_ACK_REQUIRED`	Customer acknowledgment is required before recording a backorder.	BR-028	VR-101
`BIZ_SOURCE_MINIMUM_PROTECTED`	Transfer would breach minimum stock at the source location.	BR-012	VR-140
`BIZ_NETWORK_TRANSFER_DECLINED`	Network transfer evaluation must be documented before purchase approval.	BR-011	VR-118
`BIZ_THREE_WAY_MATCH_INCOMPLETE`	Payment cannot be authorized until three-way match is complete.	BR-008	—
`BIZ_RETURN_AUTH_REQUIRED`	Return authorization is required before receiving customer returns.	BR-025	VR-045
`BIZ_PROMOTIONAL_RING_FENCE`	Quantity exceeds promotional ring-fenced allocation.	BR-029	VR-127

Details Object Extensions

Field	Type	Description
`businessRuleId`	String	BR-xxx identifier when applicable
`validationRuleId`	String	VR-xxx identifier when applicable
`details`	Object	Domain-specific context (SKU, quantities, statuses)

Example JSON Response

```
{
  "type": "https://api.ims.example/problems/business-rule-violation",
  "title": "Business Rule Violation",
  "status": 422,
  "detail": "Cannot reserve 100 units. Only 45 units are available to promise at this location.",
  "instance": "/api/v1/inventory/stock-positions/WH-EAST/WIDGET-01/reservations",
  "errorCode": "BIZ_INSUFFICIENT_ATP",
  "traceId": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "timestamp": "2026-07-06T14:35:12.456Z",
  "businessRuleId": "BR-001",
  "validationRuleId": "VR-092",
  "details": {
    "sku": "WIDGET-01",
    "locationCode": "WH-EAST",
    "orderId": "ORD-2026-004521",
    "orderLineId": "LINE-003",
    "requestedQuantity": {
      "amount": 100.0000,
      "unitOfMeasure": "EA"
    },
    "availableToPromise": {
      "amount": 45.0000,
      "unitOfMeasure": "EA"
    },
    "onHand": {
      "amount": 120.0000,
      "unitOfMeasure": "EA"
    },
    "reserved": {
      "amount": 60.0000,
      "unitOfMeasure": "EA"
    },
    "quarantined": {
      "amount": 15.0000,
      "unitOfMeasure": "EA"
    }
  }
}
```

3. Authentication Errors — `401 Unauthorized`

When: Request lacks valid credentials, token is expired, or authentication fails. **No resource existence is implied.**

Attribute	Value
HTTP Status Code	`401`
Problem Type	`https://api.ims.example/problems/authentication-error`
Exception Source	`UnauthorizedException` (Application)
Response Header	`WWW-Authenticate: Bearer realm="ims-api"` when applicable

Standard Error Codes

Error Code	Error Message	Details
`AUTH_MISSING_CREDENTIALS`	Authentication credentials were not provided.	No `Authorization` header
`AUTH_INVALID_TOKEN`	The access token is invalid or malformed.	JWT parse/signature failure
`AUTH_TOKEN_EXPIRED`	The access token has expired.	`exp` claim in the past
`AUTH_SESSION_TERMINATED`	The session has been terminated.	Logout or admin revocation
`AUTH_ACCOUNT_SUSPENDED`	The user account is suspended.	Account status check
`AUTH_ACCOUNT_DEACTIVATED`	The user account is deactivated.	Account status check
`AUTH_INTEGRATION_CREDENTIAL_INVALID`	Integration client credentials are invalid.	API-176 ACL inbound

Example JSON Response

```
{
  "type": "https://api.ims.example/problems/authentication-error",
  "title": "Authentication Required",
  "status": 401,
  "detail": "The access token has expired. Obtain a new token and retry the request.",
  "instance": "/api/v1/fulfillment/orders/3fa85f64-5717-4562-b3fc-2c963f66afa6/confirm",
  "errorCode": "AUTH_TOKEN_EXPIRED",
  "traceId": "c4d5e6f7-8901-2345-6789-abcdef012345",
  "timestamp": "2026-07-06T14:40:00.000Z",
  "details": {
    "tokenExpiredAt": "2026-07-06T14:38:22.000Z"
  }
}
```

Security note: Do not reveal whether a user ID exists. Use generic messages for credential failures.

---

4. Authorization Errors — `403 Forbidden`

When: Caller is authenticated but lacks permission, location scope, or role for the action.

Attribute	Value
HTTP Status Code	`403`
Problem Type	`https://api.ims.example/problems/authorization-error`
Exception Source	`ForbiddenException` (Application)

Standard Error Codes

Error Code	Error Message	Details
`AUTHZ_INSUFFICIENT_PERMISSION`	You do not have permission to perform this action.	Missing role/permission
`AUTHZ_LOCATION_SCOPE_DENIED`	You are not authorized to access this location.	VR-008, VR-053
`AUTHZ_RESOURCE_SCOPE_DENIED`	You are not authorized to access this resource.	Row-level / ownership
`AUTHZ_APPROVAL_THRESHOLD`	This transaction requires independent approval above your authority.	BR-021, VR-082
`AUTHZ_PO_AMENDMENT_DENIED`	You are not authorized to amend or cancel this purchase order.	VR-051
`AUTHZ_ROLE_ASSIGNMENT_DENIED`	You are not authorized to assign or revoke this role.	VR-083
`AUTHZ_INTEGRATION_NOT_REGISTERED`	The integration source system is not registered or enabled.	API-176, API-177

Example JSON Response

```
{
  "type": "https://api.ims.example/problems/authorization-error",
  "title": "Access Denied",
  "status": 403,
  "detail": "You are not authorized to perform inventory transactions at location WH-WEST.",
  "instance": "/api/v1/inventory/stock-positions/WH-WEST/BOLT-M10/reservations",
  "errorCode": "AUTHZ_LOCATION_SCOPE_DENIED",
  "traceId": "d5e6f7a8-9012-3456-7890-bcdef0123456",
  "timestamp": "2026-07-06T14:42:18.789Z",
  "validationRuleId": "VR-053",
  "details": {
    "userId": "picker.jones",
    "locationCode": "WH-WEST",
    "requiredPermission": "INVENTORY_RESERVE",
    "authorizedLocations": ["WH-EAST", "WH-CENTRAL"]
  }
}
```

---

5. Resource Not Found — `404 Not Found`

When: Requested resource does not exist, or exists but is not visible to the caller (optional policy: return 404 instead of 403 to prevent enumeration).

Attribute	Value
HTTP Status Code	`404`
Problem Type	`https://api.ims.example/problems/not-found`
Exception Source	`NotFoundException` (Application)

Standard Error Codes

Error Code	Error Message	Resource
`NF_PRODUCT_NOT_FOUND`	Product not found.	SKU (VR-007)
`NF_LOCATION_NOT_FOUND`	Location not found.	LocationCode
`NF_ORDER_NOT_FOUND`	Customer order not found.	orderId (VR-108)
`NF_PURCHASE_ORDER_NOT_FOUND`	Purchase order not found.	purchaseOrderId
`NF_GOODS_RECEIPT_NOT_FOUND`	Goods receipt not found.	goodsReceiptId
`NF_RESERVATION_NOT_FOUND`	Reservation not found.	reservationId
`NF_TRANSFER_NOT_FOUND`	Stock transfer not found.	transferId

Error Code	Error Message	Resource
NF_ADJUSTMENT_CASE_NOT_FOUND	Adjustment case not found.	caseId
NF_RETURN_AUTHORIZATION_NOT_FOUND	Return authorization not found.	VR-149
NF_USER_NOT_FOUND	User account not found.	userId
NF_RESOURCE_NOT_FOUND	The requested resource was not found.	Generic fallback

Example JSON Response

```
{
  "type": "https://api.ims.example/problems/not-found",
  "title": "Resource Not Found",
  "status": 404,
  "detail": "No product exists with SKU 'UNKNOWN-SKU'.",
  "instance": "/api/v1/products/UNKNOWN-SKU",
  "errorCode": "NF_PRODUCT_NOT_FOUND",
  "traceId": "e6f7a8b9-0123-4567-8901-cdef01234567",
  "timestamp": "2026-07-06T14:45:00.000Z",
  "validationRuleId": "VR-007",
  "details": {
    "resourceType": "Product",
    "sku": "UNKNOWN-SKU"
  }
}
```

---

6. Conflict — `409 Conflict`

When: Request conflicts with current resource state — duplicates, optimistic locking, illegal state transition, or concurrent modification.

Attribute	Value
HTTP Status Code	`409`
Problem Type	`https://api.ims.example/problems/conflict`
Exception Source	`ConflictException` (Application / Domain)

Standard Error Codes

Error Code	Error Message	Scenario
CONF_DUPLICATE_SKU	A product with this SKU already exists.	API-001 (VR-002)
CONF_DUPLICATE_BARCODE	A product with this barcode already exists.	VR-009
CONF_DUPLICATE_LOCATION	A location with this code already exists.	VR-011
CONF_DUPLICATE_SUPPLIER	A supplier with this code already exists.	VR-030
CONF_DUPLICATE_INVOICE	An invoice with this number already exists for the supplier.	API-077
CONF_OPTIMISTIC_LOCK	The resource was modified by another user. Refresh and retry.	ETag / version mismatch
CONF_INVALID_STATE_TRANSITION	The operation is not valid for the current resource status.	e.g. ship unconfirmed order
CONF_IDEMPOTENCY_KEY_REUSED	Idempotency key was reused with a different request payload.	Financial/reservation commands
CONF_ORDER_ALREADY_CONFIRMED	Order is already confirmed.	API-085
CONF_RECEIPT_ALREADY_VERIFIED	Goods receipt is already verified.	API-069
CONF_TRANSFER_NOT_APPROVED	Transfer must be approved before dispatch.	API-103

Example JSON Response

```
{
  "type": "https://api.ims.example/problems/conflict",
  "title": "Resource Conflict",
  "status": 409,
  "detail": "A product with SKU 'BOLT-M10' already exists in the catalog.",
  "instance": "/api/v1/products",
  "errorCode": "CONF_DUPLICATE_SKU",
  "traceId": "f7a8b9c0-1234-5678-9012-def012345678",
  "timestamp": "2026-07-06T14:48:33.210Z",
  "validationRuleId": "VR-002",
  "details": {
    "resourceType": "Product",
    "sku": "BOLT-M10",
    "existingResourceId": "BOLT-M10",
    "conflictType": "DUPLICATE_KEY"
  }
}
```

Optimistic Locking Example

```
{
  "type": "https://api.ims.example/problems/conflict",
  "title": "Resource Conflict",
  "status": 409,
  "detail": "Purchase order was updated by another user. Retrieve the latest version and retry.",
  "instance": "/api/v1/procurement/purchase-orders/8b2e4f1a-9c3d-4e5f-a6b7-c8d9e0f1a2b3",
  "errorCode": "CONF_OPTIMISTIC_LOCK",
  "traceId": "0a1b2c3d-4e5f-6789-0abc-def123456789",
  "timestamp": "2026-07-06T14:50:00.000Z",
  "details": {
    "resourceType": "PurchaseOrder",
    "resourceId": "8b2e4f1a-9c3d-4e5f-a6b7-c8d9e0f1a2b3",
    "expectedVersion": 3,
    "actualVersion": 4
  }
}
```

---

7. Internal Server Errors — `500 Internal Server Error`

When: Unexpected unhandled exception. Programming defect, unhandled edge case, or non-transient failure not classified as 503.

Attribute	Value
HTTP Status Code	`500`
Problem Type	`https://api.ims.example/problems/internal-server-error`
Exception Source	Unhandled `Exception` (any layer)
Logging	Full stack trace logged server-side with `traceId` (EXC-04)

Standard Error Codes

Error Code	Error Message	Client exposure
`SYS_INTERNAL_ERROR`	An unexpected error occurred. Contact support with the trace ID.	Generic only
`SYS_DATA_INTEGRITY_ERROR`	An unexpected data integrity error occurred.	Generic only
`SYS_MAPPING_ERROR`	An unexpected error occurred while processing the response.	Generic only

Example JSON Response

```
{
  "type": "https://api.ims.example/problems/internal-server-error",
  "title": "Internal Server Error",
  "status": 500,
  "detail": "An unexpected error occurred while processing your request. Contact support and provide the trace ID.",
  "instance": "/api/v1/inbound/goods-receipts/1a2b3c4d-5e6f-7890-abcd-ef1234567890/verify",
  "errorCode": "SYS_INTERNAL_ERROR",
  "traceId": "1a2b3c4d-5e6f-7890-abcd-ef1234567890",
  "timestamp": "2026-07-06T14:55:00.000Z",
  "details": {
    "supportReference": "1a2b3c4d-5e6f-7890-abcd-ef1234567890"
  }
}
```

Prohibited in 500 responses: `exception`, `stackTrace`, `sql`, `cause`, file paths, connection strings.

---

8. Service Unavailable — `503 Service Unavailable`

When: Transient infrastructure failure — database unreachable, message broker down, dependency timeout after retries exhausted.

Attribute	Value
HTTP Status Code	`503`
Problem Type	`https://api.ims.example/problems/service-unavailable`
Exception Source	`InfrastructureException` (Infrastructure)
Response Header	`Retry-After: <seconds>` when retry is advisable

Standard Error Codes

Error Code	Error Message
`SYS_DATABASE_UNAVAILABLE`	The service is temporarily unavailable. Please retry later.
`SYS_DEPENDENCY_TIMEOUT`	A dependent service did not respond in time. Please retry later.
`SYS_MAINTENANCE_MODE`	The system is under maintenance. Please retry later.

Example JSON Response

```
{
  "type": "https://api.ims.example/problems/service-unavailable",
  "title": "Service Unavailable",
  "status": 503,
  "detail": "The service is temporarily unavailable due to a database connectivity issue. Please retry after 30 seconds.",
  "instance": "/api/v1/inventory/availability/WIDGET-01",
  "errorCode": "SYS_DATABASE_UNAVAILABLE",
  "traceId": "2b3c4d5e-6f70-8901-bcde-f12345678901",
  "timestamp": "2026-07-06T15:00:00.000Z",
  "details": {
    "retryAfterSeconds": 30
  }
}
```

HTTP Status Code Summary

Category	HTTP Status	Problem Type Slug	Primary `errorCode` Prefix
Validation	`400`	`validation-error`	`VAL_`

Category	HTTP Status	Problem Type Slug	Primary `errorCode` Prefix
Authentication	`401`	`authentication-error`	`AUTH_`
Authorization	`403`	`authorization-error`	`AUTHZ_`
Not Found	`404`	`not-found`	`NF_`
Conflict	`409`	`conflict`	`CONF_`
Business Rule	`422`	`business-rule-violation`	`BIZ_`
Internal Server	`500`	`internal-server-error`	`SYS_`
Service Unavailable	`503`	`service-unavailable`	`SYS_`

Exception-to-HTTP Mapping (Implementation Reference)

Aligns with Cross-Cutting Concerns §5.3.

Java Exception	HTTP	`errorCode` family
`ValidationException`	400	`VAL_*`
`UnauthorizedException`	401	`AUTH_*`
`ForbiddenException`	403	`AUTHZ_*`
`NotFoundException`	404	`NF_*`
`ConflictException`	409	`CONF_*`
`BusinessRuleViolationException`	422	`BIZ_*`
`InfrastructureException` (transient)	503	`SYS_*`
`Exception` (unhandled)	500	`SYS_INTERNAL_ERROR`

Domain Exception Examples → Error Codes

Domain Exception	Error Code	BR
`InsufficientAtpException`	`BIZ_INSUFFICIENT_ATP`	BR-001
`LocationFrozenException`	`BIZ_LOCATION_FROZEN`	BR-024
`SegregationOfDutiesViolationException`	`BIZ_SEGREGATION_OF_DUTIES`	BR-019
`MrslNonComplianceException`	`BIZ_MRSL_NON_COMPLIANT`	BR-016
`DiscontinuedProductException`	`BIZ_PRODUCT_DISCONTINUED`	BR-027

Response Headers

Header	When	Value
`Content-Type`	Always	`application/problem+json`
`WWW-Authenticate`	401	`Bearer realm="ims-api"`
`Retry-After`	503	Seconds until retry advised
`X-Trace-Id`	Always	Same value as `traceId` in body
`ETag`	409 optimistic lock	Current resource version when available

Bulk / Partial Success Errors

Import and sync endpoints (API-010, API-147) return **200 OK** with per-row errors in the success body for non-blocking validation. **Blocking** failures use standard error envelope:

Scenario	HTTP	Pattern
Entire request invalid (empty batch, bad mode)	400	Standard `ProblemDetailResponse`
Batch has blocking errors on activate	422	`BIZ_IMPORT_BLOCKED` with error count in `details`
Per-row errors in validate-only mode	200	`errors[]` in success response (SRS-API-003)

OpenAPI Schema Sketch

```
ProblemDetailResponse:
  type: object
  required: [type, title, status, detail, instance, errorCode, traceId, timestamp]
  properties:
    type: { type: string, format: uri }
    title: { type: string }
    status: { type: integer }
    detail: { type: string }
    instance: { type: string }
    errorCode: { type: string }
    traceId: { type: string, format: uuid }
    timestamp: { type: string, format: date-time }
    businessRuleId: { type: string, pattern: '^BR-\\d{3}$' }
    validationRuleId: { type: string, pattern: '^VR-\\d{3}$' }
    details: { type: object, additionalProperties: true }
    errors:
      type: array
      items:
        $ref: '#/components/schemas/FieldError'

FieldError:
  type: object
  required: [field, code, message]
  properties:
    field: { type: string }
    code: { type: string }
    message: { type: string }
    rejectedValue: {}
```

Document Approval

Role	Name	Signature	Date
Senior Backend Architect			
Solution Architect			
Tech Lead			
Product Owner			

End of Error Response Specification

Document ID: SRS-API-005 | Version 1.0 | July 6, 2026

API Authorization Specification

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-API-007
Document Title	API Authorization Specification — RBAC Mapping
Version	1.0
Status	Draft
Document Owner	Senior Backend Architect
Source Documents	SRS-API-001, `project_business/Permissions/rbac-matrix.md`, `project_business/Permissions/system-permissions.md`, SRS-API-005
Date	July 6, 2026
Scope	Authorization requirements for all 178 REST APIs

Purpose

Maps every REST API to the **RBAC model** (UR-01–UR-19, PERM-001–PERM-100). For each endpoint specifies authentication, authorized roles, required permissions, sensitive-operation classification, and business justification.

Authorization Model

```
Request → Authenticate (JWT / Client Credentials)
        → Resolve user roles + location scope
        → Check PERM-xxx (any assigned role grants permission)
        → Enforce location scope on transactional APIs
        → Evaluate SoD on approval APIs (API-031, dual-control endpoints)
        → Execute or 401/403
```

Mechanism	Detail
Authentication	OAuth 2.0 Bearer JWT (SVC-01 IdP) for users; Client Credentials for integration principals
Authorization	Permission-based (`PERM-xxx`); roles are permission bundles
Location scope	Transactional APIs require `locationCode` ∈ user's assigned locations (except network-scoped roles)
SoD	Initiator ≠ approver enforced at domain layer; pre-checked via API-031 where applicable
Deny default	No permission → `403 AUTHZ_INSUFFICIENT_PERMISSION`

Role Code Reference

Code	UR	Role
`CEO`	UR-01	Chief Executive Officer
`COO`	UR-02	Chief Operating Officer
`CFO`	UR-03	Chief Financial Officer
`OPS_DIR`	UR-04	Operations Director
`INV_CTRL`	UR-05	Inventory Controller
`WH_MGR`	UR-06	Warehouse Manager

Code	UR	Role
RCV_CLK	UR-07	Receiving Clerk
PIK_PAK	UR-08	Warehouse Picker / Packer
WH_STF	UR-09	Warehouse Staff
REG_MGR	UR-10	Regional / Store Manager
SAL_MGR	UR-11	Sales Manager
SAL_REP	UR-12	Sales Representative
CS_REP	UR-13	Customer Service Representative
MKT_MGR	UR-14	Marketing Manager
PUR_MGR	UR-15	Purchasing Manager
BUYER	UR-16	Buyer / Procurement Officer
FIN_MGR	UR-17	Finance Manager / Accountant
INT_AUD	UR-18	Internal Auditor
EXT_AUD	UR-19	External Auditor
SYS_ADMIN	—	System Administrator (technical; PERM-095/096 only)
INT_SALES	—	Integration principal — Sales/CRM (EXT-01)
INT_MKT	—	Integration principal — Marketing (EXT-07)

Sensitive Operation Categories

Code	Category	Examples
—	Not sensitive	Read-only visibility, lookups
S1	Approval	PO, transfer, adjustment, write-off approvals
S2	Control override	Ad-hoc receipt, tolerance override, safety stock override
S3	Governance / policy	Policies, thresholds, reason codes, location closure
S4	Financial	Valuation, reconciliation, invoice payment
S5	Audit / compliance	Audit trail, evidence export, breach records
S6	Master data mutation	Product, location, supplier create/update
S7	Identity / admin	User provisioning, role assignment
S8	Stock mutation	Reservation, shipment, transfer dispatch, adjustments

Location scope key: NET = network-wide; LOC = assigned location(s); NONE = not location-bound; SELF = own session/profile only.

M-01 — Product Catalog

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-001	POST /products	Yes	PERM-001	INV_CTRL	S6	NET	Product master integrity centralized with Inventory Controller (BR catalog identity)

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-002	GET `/products`	Yes	PERM-004	COO, OPS_DIR, INV_CTRL, WH_MGR, RCV_CLK, PIK_PAK, WH_STF, REG_MGR, SAL_MGR, SAL_REP, CS_REP, MKT_MGR, PUR_MGR, BUYER, FIN_MGR	—	NET	Operational lookup without master-data mutation rights
API-003	GET `/products/{sku}`	Yes	PERM-004	Same as API-002	—	NET	Single-product retrieval for transactions and inquiries
API-004	PATCH `/products/{sku}`	Yes	PERM-002	INV_CTRL	S6	NET	Attribute changes affect all locations; restricted to control owner
API-005	POST `.../lifecycle-status`	Yes	PERM-003	INV_CTRL	S6	NET	Discontinued status blocks replenishment (BR-027)
API-006	PUT `.../abc-classification`	Yes	PERM-012	INV_CTRL	S3	NET	ABC drives monitoring priority (BR-032)
API-007	GET `.../eligibility`	Yes	PERM-004	Same as API-002	—	LOC	Pre-check before procurement/reservation/fulfillment
API-008	GET `.../categories/.../products`	Yes	PERM-004	Same as API-002	—	NET	Category browse for catalog navigation
API-009	POST `/products/duplicate-resolutions`	Yes	PERM-005	INV_CTRL	S6	NET	Merge affects on-hand identity; requires control owner + merge auth when stock exists
API-010	POST `/products/import`	Yes	PERM-015	OPS_DIR, INV_CTRL	S6	NET	Bulk cutover data; OpsDir submits, InvCtrl approves via separate flow (PERM-016)

M-02 — Warehouse Network

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-011	POST `/locations`	Yes	PERM-006	COO, OPS_DIR	S6	NET	Location network structure is executive/ops leadership responsibility
API-012	GET `/locations`	Yes	PERM-004	All roles with visibility (PERM-017+)	—	NET/LOC	List locations within user scope
API-013	GET `/locations/{code}`	Yes	PERM-004	Same as API-012	—	LOC	Location detail for operational context
API-014	PATCH `.../hierarchy`	Yes	PERM-008	COO, OPS_DIR	S6	NET	Hierarchy changes affect reporting and accountability
API-015	POST `.../storage-positions`	Yes	PERM-007	COO, OPS_DIR	S6	LOC	Storage layout owned by ops; WH_MGR uses positions operationally
API-016	GET `.../storage-positions`	Yes	PERM-004	WH_MGR, RCV_CLK, WH_STF, OPS_DIR, INV_CTRL + visibility roles	—	LOC	Put-away and pick face lookup
API-017	POST `.../activate`	Yes	PERM-007	COO, OPS_DIR	S3	LOC	Go-live enables transactions; requires opening balance readiness
API-018	POST `.../deactivate`	Yes	PERM-007	COO, OPS_DIR	S3	LOC	Stops new transactions at location
API-019	POST `.../close`	Yes	PERM-009	CEO, COO, OPS_DIR	S3	LOC	Strategic closure with disposition planning

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-020	GET `.../freeze-status`	Yes	PERM-040	OPS_DIR, INV_CTRL, WH_MGR, CEO, CFO, FIN_MGR, INT_AUD, EXT_AUD, REG_MGR	—	LOC	Physical count freeze visibility (BR-024)
API-021	GET `.../operational-status`	Yes	PERM-004	Same as API-012	—	LOC	Transaction eligibility check

M-03 — Inventory Governance

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-022	POST `/governance/reason-codes`	Yes	PERM-014	INV_CTRL	S3	NET	Reason codes substantiate adjustments (BR-020)
API-023	GET `/governance/reason-codes`	Yes	PERM-014	INV_CTRL, OPS_DIR + transactional roles using codes	—	NET	Selectable codes for transactions
API-024	POST `.../reason-codes/{code}/retire`	Yes	PERM-014	INV_CTRL	S3	NET	Retire obsolete codes while preserving history
API-025	GET `.../reason-codes/{code}/validate`	Yes	PERM-014	All roles posting adjustments/receipts	—	NET	Pre-validate code applicability (VR-026)
API-026	POST `/governance/policies`	Yes	PERM-013	OPS_DIR, COO	S3	NET	Policy versions govern thresholds and tolerances network-wide
API-027	PATCH `.../approval-thresholds`	Yes	PERM-013	OPS_DIR, COO	S3	NET	Approval thresholds enforce SoD and materiality (BR-021)
API-028	PATCH `.../receipt-tolerances`	Yes	PERM-013	OPS_DIR, COO	S3	NET	Receipt tolerance affects payment and inventory acceptance
API-029	GET `/governance/approval-thresholds`	Yes	PERM-013	OPS_DIR, COO, INV_CTRL, WH_MGR, FIN_MGR	—	NET/LOC	Threshold lookup before submission
API-030	GET `/governance/receipt-tolerances`	Yes	PERM-032	WH_MGR, OPS_DIR, COO	—	LOC	Tolerance evaluation at receipt
API-031	POST `/governance/segregation-of-duties/evaluate`	Yes	PERM-013	OPS_DIR, COO, INV_CTRL, FIN_MGR, INT_AUD	S3	NET	SoD pre-check (BR-019, BR-021); no mutation

M-04 — Inventory Balance

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-032	POST `/inventory/opening-balances`	Yes	PERM-010	OPS_DIR, INV_CTRL	S3	LOC	Go-live balances embed baseline accuracy (BA-16)
API-033	GET `/inventory/stock-positions`	Yes	PERM-017, PERM-021	All visibility roles (PG-03)	—	LOC	On-hand inquiry scoped to authorized locations
API-034	GET `.../stock-positions/{loc}/{sku}`	Yes	PERM-017	PG-03 roles	—	LOC	Detail position for operational decisions
API-035	GET `/inventory/availability/{sku}`	Yes	PERM-018	PG-03 except WH_STF, PIK_PAK (no ATP for some)	—	LOC	ATP hot-path for sales/fulfillment (BR-001)

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-036	GET `.../availability/{sku}/network`	Yes	PERM-022	Network-scoped visibility roles	—	NET	Network ATP for transfer/buy decisions (BR-011)
API-037	GET `.../stock-status/{loc}/{sku}`	Yes	PERM-020	PG-03 roles	—	LOC	Derived status for replenishment alerts
API-038	POST `/inventory/stock-policies`	Yes	PERM-011	INV_CTRL, OPS_DIR	S3	LOC	Thresholds drive replenishment (BR-009, BR-010)
API-039	GET `.../stock-policies/{loc}/{sku}`	Yes	PERM-011	INV_CTRL, OPS_DIR, WH_MGR, PUR_MGR	—	LOC	Policy lookup
API-040	PATCH `.../stock-policies/{loc}/{sku}`	Yes	PERM-011	INV_CTRL, OPS_DIR	S3	LOC	Threshold changes affect network replenishment
API-041	GET `.../lots`	Yes	PERM-017	PG-03 with location scope	—	LOC	Lot traceability (BR-017)
API-042	GET `.../transfer-capacity`	Yes	PERM-076	COO, OPS_DIR, INV_CTRL, REG_MGR, PUR_MGR, FIN_MGR, INT_AUD, EXT_AUD	—	LOC	Source spare capacity before transfer (BR-012)
API-043	POST `.../reservations`	Yes	PERM-052	COO, OPS_DIR, WH_MGR	S8	LOC	ATP commitment (BR-001, BR-002); may need PERM-070 if safety stock breach
API-044	POST `.../reservations/.../release`	Yes	PERM-053	COO, OPS_DIR, WH_MGR	S8	LOC	Release on cancel/change (BR-002)
API-045	POST `.../quarantine-holds`	Yes	PERM-047	OPS_DIR, INV_CTRL, WH_MGR	S8	LOC	Exclude from sellable ATP (BR-013)
API-046	POST `.../quarantine-holds/.../release`	Yes	PERM-048	OPS_DIR, INV_CTRL, WH_MGR	S2	LOC	Restore requires inspection for resellable (BR-026)
API-047	POST `.../consignment-ownership`	Yes	PERM-017	FIN_MGR, CFO (operational trigger via finance)	S4	LOC	Consignment consumption accounting (BR-031)
API-048	POST `.../promotional-segregation`	Yes	PERM-093	COO, OPS_DIR, MKT_MGR	S2	LOC	Ring-fence ATP (BR-029); event-driven from campaign

M-05 — Procurement & Supplier

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-049	POST `/suppliers`	Yes	PERM-024	PUR_MGR	S6	NET	Supplier master enables PO placement
API-050	GET `/suppliers`	Yes	PERM-035	PG-03 + PG-06 visibility roles	—	NET	Supplier selection for procurement
API-051	GET `/suppliers/{code}`	Yes	PERM-035	Same as API-050	—	NET	Supplier terms lookup
API-052	PATCH `/suppliers/{code}`	Yes	PERM-025	PUR_MGR	S6	NET	Terms affect payment and lead time
API-053	POST `.../deactivate`	Yes	PERM-026	PUR_MGR	S3	NET	Block unauthorized new POs to inactive supplier
API-054	POST `/procurement/requisitions`	Yes	PERM-067	PUR_MGR, BUYER	—	LOC	Replenishment need documentation (BR-011)
API-055	GET `/procurement/requisitions/{id}`	Yes	PERM-067, PERM-068	PUR_MGR, BUYER, COO, OPS_DIR	—	NET	Requisition status tracking

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-056	POST `.../approve`	Yes	PERM-068	PUR_MGR	S1	NET	Spend commitment approval; Buyer cannot approve (SoD)
API-057	POST `.../reject`	Yes	PERM-068	PUR_MGR	—	NET	Documented rejection (VR-131)
API-058	POST `.../convert-to-po`	Yes	PERM-024	PUR_MGR, BUYER	—	LOC	Convert approved need to commercial commitment
API-059	POST `/procurement/purchase-orders`	Yes	PERM-024	PUR_MGR, BUYER	S1	LOC	PO creates binding supplier commitment (BR-007)
API-060	GET `/procurement/purchase-orders`	Yes	PERM-027, PERM-035	PG-06 visibility roles	—	LOC	Inbound pipeline visibility
API-061	GET `/procurement/purchase-orders/{id}`	Yes	PERM-027	PG-06 visibility roles	—	LOC	PO detail for receipt and finance
API-062	PATCH `/procurement/purchase-orders/{id}`	Yes	PERM-025	PUR_MGR, BUYER	S1	LOC	Amendment affects commitment; authorized roles only (VR-051)
API-063	POST `.../cancel`	Yes	PERM-026	PUR_MGR, BUYER	S1	LOC	Cancellation with receipt implications
API-064	POST `.../expected-deliveries`	Yes	PERM-027	PUR_MGR, BUYER, WH_MGR	—	LOC	Inbound planning (UC-007)
API-065	GET `.../validate-for-receipt`	Yes	PERM-028	WH_MGR, RCV_CLK	—	LOC	PO authorization check before receipt (BR-007)

M-06 — Inbound Receiving

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-066	POST `/inbound/goods-receipts`	Yes	PERM-028	WH_MGR, RCV_CLK	—	LOC	Initiate receipt against PO (BR-007)
API-067	GET `/inbound/goods-receipts`	Yes	PERM-028	WH_MGR, RCV_CLK, PUR_MGR, BUYER	—	LOC	Receipt status tracking
API-068	GET `/inbound/goods-receipts/{id}`	Yes	PERM-028	Same as API-067	—	LOC	Receipt detail
API-069	POST `.../verify`	Yes	PERM-028	WH_MGR, RCV_CLK	S8	LOC	Verify before acceptance (BR-004)
API-070	POST `.../discrepancies`	Yes	PERM-029	WH_MGR, RCV_CLK	—	LOC	Document discrepancies (BR-006)
API-071	GET `.../discrepancies`	Yes	PERM-029	WH_MGR, RCV_CLK, PUR_MGR, BUYER	—	LOC	Open discrepancy list
API-072	POST `.../discrepancies/.../resolve`	Yes	PERM-030	WH_MGR, PUR_MGR, BUYER	S1	LOC	Resolution affects acceptance and supplier claim
API-073	POST `.../put-away`	Yes	PERM-033	WH_MGR, WH_STF	S8	LOC	ATP increases only after put-away (BR-005)
API-074	GET `/inbound/pending-put-away`	Yes	PERM-033	WH_MGR, WH_STF, RCV_CLK	—	LOC	Put-away queue
API-075	POST `/inbound/ad-hoc-receipts`	Yes	PERM-031	WH_MGR	S2	LOC	Non-PO receipt bypasses procurement controls
API-076	POST `/inbound/return-receipts`	Yes	PERM-034	WH_MGR, RCV_CLK	—	LOC	Return dock receipt (BR-025)
API-077	POST `/inbound/supplier-invoices`	Yes	PERM-094	FIN_MGR, CFO	S4	NET	Invoice registration for three-way match

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-078	GET `/inbound/supplier-invoices/{id}`	Yes	PERM-094	FIN_MGR, CFO, INT_AUD	S4	NET	Match status visibility
API-079	POST `.../three-way-match`	Yes	PERM-094	FIN_MGR, CFO	S4	NET	PO–receipt–invoice match (BR-008)
API-080	POST `.../authorize-payment`	Yes	PERM-094	FIN_MGR, CFO	S1	NET	Payment only after successful match (BR-008)

M-07 — Order Fulfillment

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-081	POST `/fulfillment/orders`	Yes	PERM-054	COO, OPS_DIR, WH_MGR, SAL_MGR	—	LOC	Create/import order for fulfillment pipeline
API-082	GET `/fulfillment/orders`	Yes	PERM-062	PG-08 status viewers	—	LOC	Order list for operations and customer service
API-083	GET `/fulfillment/orders/{id}`	Yes	PERM-062	PG-08 status viewers	—	LOC	Order detail
API-084	POST `.../availability`	Yes	PERM-051	COO, OPS_DIR, WH_MGR, SAL_MGR, SAL_REP, CS_REP, MKT_MGR	—	LOC	ATP check before promise (BR-001)
API-085	POST `.../confirm`	Yes	PERM-054	COO, OPS_DIR, WH_MGR	S8	LOC	Confirms order and reserves stock (BR-002); PERM-070 if safety override
API-086	POST `.../cancel`	Yes	PERM-054	COO, OPS_DIR, WH_MGR	S8	LOC	Cancel releases reservation
API-087	POST `.../multi-location-plan`	Yes	PERM-061	COO, OPS_DIR	S8	NET	Multi-location allocation when single-site ATP insufficient
API-088	POST `.../pick-lists`	Yes	PERM-055	WH_MGR	—	LOC	Assign pick work
API-089	GET `.../pick-lists/awaiting`	Yes	PERM-055	WH_MGR, PIK_PAK	—	LOC	Pick queue
API-090	POST `.../picks/{id}/verify`	Yes	PERM-055	WH_MGR, PIK_PAK	S8	LOC	Verified pick before ship (BR-003); PERM-063 if substitution
API-091	POST `.../pack`	Yes	PERM-055	WH_MGR, PIK_PAK	—	LOC	Pack verified pick
API-092	POST `.../shipments`	Yes	PERM-056	WH_MGR, PIK_PAK	S8	LOC	Shipment deducts inventory (BR-003)
API-093	POST `.../partial-fulfillment`	Yes	PERM-057	COO, OPS_DIR, WH_MGR	S8	LOC	Partial ship with optional backorder (BR-028)
API-094	POST `.../backorders`	Yes	PERM-058	COO, OPS_DIR, WH_MGR, SAL_MGR, CS_REP, PUR_MGR	S1	LOC	Backorder requires customer acknowledgment (BR-028)
API-095	POST `/fulfillment/backorders/{id}/fulfill`	Yes	PERM-058	COO, OPS_DIR, WH_MGR	S8	LOC	Fulfill when ATP available
API-096	POST `.../exceptions`	Yes	PERM-059	WH_MGR, PIK_PAK	—	LOC	Report pick/ship exceptions
API-097	GET `.../status`	Yes	PERM-062	PG-08 status viewers + auditors	—	LOC	Fulfillment status for CS and audit

M-08 — Network Transfer

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-098	POST `/transfers`	Yes	PERM-071	OPS_DIR, WH_MGR, REG_MGR	—	LOC	Transfer request; source capacity validated (BR-012)
API-099	GET `/transfers`	Yes	PERM-075	PG-05 visibility roles	—	LOC	Transfer history and status
API-100	GET `/transfers/{id}`	Yes	PERM-075	PG-05 visibility roles	—	LOC	Transfer detail
API-101	POST `.../approve`	Yes	PERM-072	OPS_DIR, COO	S1	NET	Approver ≠ requester (SoD); protects source minimum
API-102	POST `.../reject`	Yes	PERM-072	OPS_DIR, COO	—	NET	Documented rejection
API-103	POST `.../dispatch`	Yes	PERM-073	WH_MGR	S8	LOC	Dispatch creates in-transit stock
API-104	POST `.../receive`	Yes	PERM-074	WH_MGR	S8	LOC	Destination receipt completes transfer
API-105	POST `.../cancel`	Yes	PERM-073	OPS_DIR, WH_MGR	S8	NET	Cancel in-transit with reversal
API-106	GET `/transfers/pending-approvals`	Yes	PERM-072	OPS_DIR, COO	—	NET	Approval queue (UC-047)
API-107	GET `/transfers/in-transit`	Yes	PERM-075	PG-05 visibility roles	—	LOC	Pipeline visibility (UC-030)

M-09 — Inventory Accuracy

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-108	POST `/accuracy/cycle-counts`	Yes	PERM-037	OPS_DIR, INV_CTRL	S3	LOC	Schedule count program (BR-018)
API-109	GET `/accuracy/cycle-counts`	Yes	PERM-037	OPS_DIR, INV_CTRL, WH_MGR	—	LOC	Count assignment list
API-110	GET `/accuracy/cycle-counts/{id}`	Yes	PERM-038	WH_STF, REG_MGR, OPS_DIR, INV_CTRL	—	LOC	Blind count — no system qty exposed (BR-018)
API-111	POST `.../result`	Yes	PERM-038	WH_STF, REG_MGR	S8	LOC	Submit blind count; counter cannot approve (SoD)
API-112	POST `/accuracy/physical-counts`	Yes	PERM-039	OPS_DIR	S3	LOC	Wall-to-wall count imposes freeze (BR-024)
API-113	GET `/accuracy/physical-counts/{id}`	Yes	PERM-039	OPS_DIR, INV_CTRL, WH_MGR	—	LOC	Count progress during freeze
API-114	POST `.../results`	Yes	PERM-039	OPS_DIR, WH_STF (during freeze)	S8	LOC	Line results during freeze window
API-115	POST `.../certify`	Yes	PERM-039	OPS_DIR	S1	LOC	Certify complete; release freeze
API-116	POST `/accuracy/adjustment-cases`	Yes	PERM-043	OPS_DIR, INV_CTRL, WH_MGR	—	LOC	Open adjustment case (BR-022)
API-117	GET `/accuracy/adjustment-cases/{id}`	Yes	PERM-040	OPS_DIR, INV_CTRL, WH_MGR, CEO, CFO, FIN_MGR, INT_AUD, EXT_AUD, REG_MGR	—	LOC	Case investigation state

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-118	POST `.../investigate`	Yes	PERM-041	OPS_DIR, INV_CTRL	S3	LOC	Investigation before adjustment (BR-022)
API-119	POST `.../variances/.../approve`	Yes	PERM-042	OPS_DIR, INV_CTRL, WH_MGR	S1	LOC	Count variance approval; counter ≠ approver (BR-019)
API-120	POST `.../submit`	Yes	PERM-043, PERM-045	OPS_DIR, INV_CTRL, WH_MGR	S8	LOC	Submit adjustment/write-off for approval
API-121	POST `.../approve`	Yes	PERM-044, PERM-046	OPS_DIR, COO, CFO	S1	LOC	Independent material adjustment/write-off approval (BR-021)
API-122	POST `.../reject`	Yes	PERM-044	OPS_DIR, COO, CFO	—	LOC	Reject proposed adjustment
API-123	GET `/accuracy/pending-approvals`	Yes	PERM-044, PERM-046	OPS_DIR, COO, CFO, INV_CTRL	—	LOC	Approval queue (UC-047)

M-10 — Returns Management

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-124	POST `/returns/authorizations`	Yes	PERM-077	CS_REP	S1	NONE	RMA required before return (BR-025)
API-125	GET `/returns/authorizations`	Yes	PERM-082	CS_REP, WH_MGR, INV_CTRL, FIN_MGR, INT_AUD	—	NONE	RMA status list
API-126	GET `/returns/authorizations/{id}`	Yes	PERM-082	Same as API-125	—	NONE	RMA detail
API-127	POST `.../cancel`	Yes	PERM-077	CS_REP	—	NONE	Cancel unused authorization
API-128	POST `/returns/{id}/receive`	Yes	PERM-078	WH_MGR, RCV_CLK	—	LOC	Physical return receipt
API-129	POST `/returns/{id}/inspect`	Yes	PERM-079	WH_MGR	S8	LOC	Inspect before restock (BR-026)
API-130	POST `/returns/{id}/disposition`	Yes	PERM-080, PERM-081	WH_MGR, INV_CTRL	S1	LOC	Restock or write-off; write-off needs approval refs (BR-023)
API-131	GET `/returns/pending-receipts`	Yes	PERM-078	WH_MGR, RCV_CLK	—	LOC	Returns awaiting dock receipt

M-11 — Commercial Campaign

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-132	POST `/campaigns/allocations`	Yes	PERM-093	COO, OPS_DIR, MKT_MGR	S2	LOC	Ring-fence removes ATP from general pool (BR-029)
API-133	GET `/campaigns/allocations`	Yes	PERM-092	COO, OPS_DIR, SAL_MGR, MKT_MGR, PUR_MGR	—	LOC	Allocation visibility
API-134	GET `/campaigns/allocations/{id}`	Yes	PERM-092	Same as API-133	—	LOC	Allocation detail
API-135	PATCH `/campaigns/allocations/{id}`	Yes	PERM-093	COO, OPS_DIR, MKT_MGR	S2	LOC	Amend ring-fence quantity/period

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-136	POST `.../readiness`	Yes	PERM-092	COO, OPS_DIR, SAL_MGR, MKT_MGR, PUR_MGR	—	LOC	Campaign readiness evaluation (UC-040)
API-137	POST `.../release`	Yes	PERM-093	COO, OPS_DIR, MKT_MGR	S2	LOC	Release unconsumed ring-fence
API-138	POST `.../override`	Yes	PERM-093	COO, OPS_DIR, MKT_MGR	S2	LOC	Override ring-fence for non-promotional order
API-139	GET `/campaigns/ring-fence/{loc}/{sku}`	Yes	PERM-092, PERM-051	SAL_MGR, SAL_REP, WH_MGR, MKT_MGR	—	LOC	Check active ring-fence before availability

M-12 — Identity & Access

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-140	POST `/users`	Yes	PERM-095	SYS_ADMIN	S7	NET	Account lifecycle outside business roles (RBAC §1)
API-141	GET `/users`	Yes	PERM-095	SYS_ADMIN	S7	NET	User directory for administration
API-142	GET `/users/{userId}`	Yes	PERM-095, PERM-062	SYS_ADMIN; users may read SELF profile	S7	SELF	Profile and role visibility
API-143	POST `/users/{userId}/roles`	Yes	PERM-096	SYS_ADMIN	S7	NET	Role assignment determines all access boundaries
API-144	DELETE `/users/.../roles/{id}`	Yes	PERM-096	SYS_ADMIN	S7	NET	Revoke access; prevents privilege retention
API-145	POST `/users/{userId}/suspend`	Yes	PERM-095	SYS_ADMIN	S7	NET	Temporary access suspension
API-146	POST `/users/{userId}/deactivate`	Yes	PERM-095	SYS_ADMIN	S7	NET	Permanent deactivation
API-147	POST `/users/sync`	Yes	PERM-095	SYS_ADMIN	S7	NET	HR directory sync
API-148	POST `/authorization/actions/evaluate`	Yes	PERM-062	Any authenticated user (evaluate self)	—	LOC	Client-side permission pre-check (UC-045)
API-149	POST `/authorization/location-scope/validate`	Yes	PERM-062	Any authenticated user	—	LOC	Location scope validation before command

M-13 — Inventory Intelligence

Read-only projections. Write API-159 is command on alert acknowledgment.

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-150	GET `/intelligence/dashboard`	Yes	PERM-083	PG-11 dashboard roles	—	NET/LOC	Executive/ops dashboard (UC-037)
API-151	GET `.../stock-status/summary`	Yes	PERM-083	PG-11 roles	—	NET/LOC	Aggregated stock health
API-152	GET `.../stock-status/board`	Yes	PERM-085	PG-11 roles	—	NET/LOC	Stock status board
API-153	GET `.../availability/inquiries`	Yes	PERM-018	PG-03 sales/accountability roles	—	LOC	Sales inquiry accountability log

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-154	GET `.../quarantine-register`	Yes	PERM-047	OPS_DIR, INV_CTRL, WH_MGR + report roles	—	NET	Quarantine visibility (BR-013)
API-155	GET `.../movements`	Yes	PERM-023, PERM-084	PG-03 + PG-11 movement report roles	—	LOC	Movement history (UC-042)
API-156	GET `.../audit-trail`	Yes	PERM-098	FIN_MGR, INT_AUD, EXT_AUD	S5	NET	Immutable change log — audit only
API-157	GET `.../audit-trail/export`	Yes	PERM-099	FIN_MGR, INT_AUD, EXT_AUD	S5	NET	Formal audit evidence package
API-158	GET `/intelligence/alerts`	Yes	PERM-064	PG-11 + replenishment alert roles	—	LOC	Operational alerts (UC-047)
API-159	POST `.../alerts/{id}/acknowledge`	Yes	PERM-065	COO, OPS_DIR, PUR_MGR, SAL_MGR, MKT_MGR	—	LOC	Alert acknowledgment without stock mutation
API-160	GET `.../replenishment/recommendations`	Yes	PERM-066	COO, OPS_DIR, INV_CTRL, SAL_MGR, MKT_MGR, PUR_MGR, BUYER	—	NET	Replenishment recommendations
API-161	GET `.../procurement/open-pos`	Yes	PERM-035	PG-06 visibility roles	—	LOC	Open PO board
API-162	GET `.../procurement/inbound-calendar`	Yes	PERM-027	PG-06 visibility roles	—	LOC	Inbound supply calendar
API-163	GET `.../procurement/inbound-delays`	Yes	PERM-036	PG-06 + alert roles	—	LOC	Supplier delay log
API-164	GET `.../fulfillment/backorders`	Yes	PERM-058	PG-08 backorder viewers	—	LOC	Backorder pipeline (BR-028)
API-165	GET `.../fulfillment/exceptions`	Yes	PERM-059	WH_MGR, COO, OPS_DIR + status viewers	—	LOC	Exception log
API-166	GET `.../network/imbalance`	Yes	PERM-076	COO, OPS_DIR, INV_CTRL, REG_MGR, PUR_MGR, FIN_MGR, INT_AUD	—	NET	Rebalancing analysis (UC-034)
API-167	GET `.../accuracy/shrinkage`	Yes	PERM-049	CEO, COO, CFO, OPS_DIR, INV_CTRL, WH_MGR, FIN_MGR, INT_AUD, EXT_AUD	S4	NET	Shrinkage analysis — financial sensitivity
API-168	GET `.../accuracy/count-performance`	Yes	PERM-086	PG-11 accuracy report roles	—	NET	Count accuracy metrics
API-169	GET `.../campaigns/readiness`	Yes	PERM-092	COO, OPS_DIR, SAL_MGR, MKT_MGR, PUR_MGR	—	NET	Campaign readiness status
API-170	GET `/intelligence/kpis`	Yes	PERM-088	CEO, COO, CFO, OPS_DIR, INV_CTRL, SAL_MGR, FIN_MGR, INT_AUD	—	NET	Executive KPIs (UC-038)
API-171	GET `.../valuation/snapshots`	Yes	PERM-090	CEO, COO, CFO, FIN_MGR, INT_AUD, EXT_AUD	S4	NET	Unit cost exposure — finance/audit only
API-172	GET `.../valuation/reconciliation`	Yes	PERM-094	CFO, FIN_MGR	S4	NET	GL reconciliation — finance independence

Platform & Integration

API	Method	Auth	Permission(s)	Authorized Roles	Sensitive	Loc	Business Justification
API-173	GET `/auth/session`	Yes	— (authenticated)	All authenticated users	—	SELF	Session validation and context (UC-045)
API-174	POST `/auth/logout`	Yes	— (authenticated)	All authenticated users	—	SELF	Terminate session
API-175	GET `/health`	**No**	—	Public (monitoring)	—	NONE	Operations health probe — no business data
API-176	POST `/integrations/sales/orders`	Client Credentials	PERM-054	INT_SALES service principal	S8	LOC	CRM order import; machine identity with order release permission
API-177	GET `/integrations/sales/fulfillment-status`	Client Credentials	PERM-062	INT_SALES service principal	—	NET	CRM status feed — read fulfillment state
API-178	POST `/integrations/marketing/campaign-plans`	Client Credentials	PERM-093	INT_MKT service principal	S2	LOC	Marketing campaign import triggers ring-fence planning

Sensitive Operations Summary

Category	Count	API Examples	SoD / Control Notes
S1 Approval	18	API-056, API-101, API-119, API-121, API-080	Initiator ≠ approver enforced
S2 Control override	8	API-075, API-046, API-048, API-132, API-138	WH_MGR / leadership only
S3 Governance	14	API-026–028, API-038, API-112, API-053	Policy drift prevention
S4 Financial	7	API-077–080, API-171, API-172, API-047	Finance independence from ops
S5 Audit	2	API-156, API-157	Auditors read-only on operations
S6 Master data	12	API-001–005, API-011, API-049	Centralized ownership
S7 Identity	8	API-140–147	SYS_ADMIN only
S8 Stock mutation	22	API-043, API-092, API-103, API-111	Location-scoped transactional

API-to-Permission Quick Index

Permission	API IDs
PERM-001	API-001
PERM-002	API-004
PERM-003	API-005
PERM-004	API-002, API-003, API-007, API-008, API-012–016, API-021
PERM-005	API-009
PERM-006–009	API-011, API-014–019
PERM-010	API-032, API-017
PERM-011	API-038–040

Permission	API IDs
PERM-012	API-006
PERM-013	API-026–031
PERM-014	API-022–025
PERM-015–016	API-010
PERM-017–023	API-033–037, API-041, API-152–155
PERM-024–027	API-049, API-058–065, API-161–162
PERM-028–034	API-066–076
PERM-035–036	API-050–051, API-060–061, API-163
PERM-037–042	API-108–111, API-119
PERM-043–046	API-116, API-118, API-120–123
PERM-047–048	API-045–046, API-154
PERM-049–050	API-167
PERM-051–063	API-084–096, API-139
PERM-064–066	API-158–160
PERM-067–069	API-054–057
PERM-070	API-043, API-085 (conditional)
PERM-071–076	API-098–107, API-042, API-166
PERM-077–082	API-124–131
PERM-083–091	API-150–152, API-157–158, API-168–170
PERM-092–093	API-132–139, API-178
PERM-094	API-077–080, API-172
PERM-095–096	API-140–147
PERM-097–100	API-156–157 (098–099); activity/breach via future APIs
Integration	API-176–178

Enforcement Implementation Notes

1. **Spring Security:** `@PreAuthorize("hasAuthority('PERM-052')")` on controller methods; custom `LocationScopeAuthorizationManager` for `LOC` APIs.
2. **Integration clients:** Map `INT_SALES` / `INT_MKT` to OAuth client IDs with scoped permissions only — no human role assignment.
3. **Self-service:** API-142, API-148, API-149 allow `userId = principal` without PERM-095 for own profile.
4. **Dual control APIs:** API-119, API-121, API-056, API-101, API-080 — domain validates initiator ≠ approver after API-031 optional pre-check.
5. **Audit logging:** All **S1–S8** operations emit audit events with `userId`, `roles`, `permission`, `locationCode`, `traceId`.

Document Approval

Role	Name	Signature	Date
Senior Backend Architect			
Security Engineer			
Solution Architect			

End of API Authorization Specification

Document ID: SRS-API-007 | Version 1.0 | July 6, 2026

API Versioning Strategy

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-API-008
Document Title	API Versioning Strategy — Inventory Management System
Version	1.0
Status	Draft
Document Owner	Senior Backend Architect
Source Documents	SRS-API-001, SRS-SA-005 (Integration Architecture), SRS-SA-004 (Layered Architecture), ADR backlog item ADR-030
Date	July 6, 2026
Scope	Public REST APIs under `/api`; internal domain events are governed separately

Purpose

Defines how the Inventory Management System **versions, evolves, deprecates, and retires** its REST API surface. Ensures warehouse operations, finance integrations, and external ACL partners (Sales/CRM, Marketing) can adopt changes predictably without disrupting stock integrity or audit trails.

Executive Recommendation

Recommended Strategy: **URI Path Versioning (`/api/v{major}`)**

Decision	Choice
Primary mechanism	Major version in URL path — `/api/v1/...`, `/api/v2/...`
Secondary signals	`Deprecation` and `Sunset` HTTP headers on deprecated endpoints
Contract artifact	OpenAPI 3 document per major version (`openapi-v1.yaml`, `openapi-v2.yaml`)
Release alignment	API major version increments only on **breaking** changes

Why this fits IMS:

1. **Already adopted** — All 178 APIs in SRS-API-001 use `/api/v1`; zero migration cost at launch.
2. **Integration-heavy** — EXT-01 (Sales) and EXT-07 (Marketing) partners configure fixed base URLs; path versioning is the most discoverable and gateway-friendly model for B2B ACL.
3. **Operational visibility** — Warehouse mobile clients, API gateways, and WAF rules route and log version explicitly.
4. **Enterprise lifecycle** — Finance and audit consumers need long deprecation windows; a visible major version supports parallel operation of v1 and v2 during migration.
5. **Team scale** — 13 modules share a presentation kernel; path versioning keeps module routing uniform (`/api/v1/inventory/...`, `/api/v1/fulfillment/...`).

Not recommended for IMS:

Strategy	Reason excluded
Header-only (`Accept-Version`, custom headers)	Harder for operations to debug; integration partners often miss header requirements
Media type (HATEOAS) (`application/vnd.ims.v2+json`)	High complexity for 178 endpoints; limited client ecosystem benefit
Query parameter (`?version=2`)	Cache-unfriendly; easy to omit accidentally on stock-mutation commands

Header-based negotiation may supplement path versioning for **minor** contract metadata but must not replace the major version in the URL.

---

1. Version Format

1.1 URL Structure

```
https://{host}/api/v{major}/{resource-path}
```

Component	Format	Example	Rules
Base path	`/api`	`/api`	Fixed prefix for all REST APIs
Major version	`v` + positive integer	`v1`, `v2`	No minor/patch in URL
Resource path	kebab-case plural nouns	`/inventory/stock-positions`	Unversioned within major
Action sub-paths	verb noun commands	`/orders/{id}/confirm`	Stable within major version

Current baseline: `v1` — covers APIs API-001 through API-178.

1.2 API Release Identifier (Documentation)

Each major version has a **contract release tag** for changelog and OpenAPI publishing:

```
IMS-API-{major}.{minor}.{patch}
```

Segment	Meaning	Example
major	Breaking change → new `/api/v{n}`	`2` → `/api/v2`
minor	Backward-compatible addition within same major	`1.3.0` — new optional fields, new endpoints
patch	Documentation or non-contract fix	`1.3.1` — typo in OpenAPI description

Example: `IMS-API-1.4.0` = fourth minor release of v1 contract (all under `/api/v1`).

1.3 Version Discovery Endpoints

Endpoint	Auth	Response
`GET /api/versions`	Public	Supported major versions, current, deprecated, sunset dates
`GET /api/v1/openapi`	Authenticated	OpenAPI 3 spec for v1
`GET /api/v1/changelog`	Authenticated	Human-readable contract changelog

`GET /api/health` (API-175) remains unversioned for load balancers.

1.4 Response Headers (All Versioned APIs)

Header	When	Example
`X-API-Version`	Always	`1.4.0` (contract release)
`X-API-Major-Version`	Always	`1`

Header	When	Example
`Deprecation`	Deprecated endpoint	`true`
`Sunset`	Deprecated endpoint	`Sat, 01 Jan 2028 00:00:00 GMT`
`Link`	Deprecated endpoint	`<https://api.ims.example/api/v2/products>; rel="successor-version"`

---

2. Backward Compatibility Approach

2.1 Compatibility Guarantee

Within a **major version**, IMS guarantees **backward compatibility** for existing clients:

> A client written against contract release `IMS-API-1.0.0` must continue to work against `IMS-API-1.x.x` without code changes.

2.2 Non-Breaking (Allowed in Same Major Version)

Change Type	Rule	Example
Add optional request field	Client may omit	New `correlationId` on command body
Add response field	Clients must ignore unknown fields	New `lastCountedAt` on stock position
Add new endpoint	No impact on existing	`GET /api/v1/intelligence/new-report`
Add optional query parameter	Default preserves prior behavior	New filter `abcClass` on product search
Add enum value	Only if clients tolerate unknown values	New `holdReason=BONDED`
Relax validation	Wider acceptance	Increase `pageSize` max from 100 to 200
Add HTTP header	Optional on request/response	`X-Trace-Id`
Bug fix	Correct behavior to match documented contract	Fix ATP calculation matching BR-001

2.3 Breaking (Requires New Major Version)

See Section 4. Same major version must **never** include breaking changes.

2.4 Client Responsibilities (Tolerant Reader Pattern)

Rule	Rationale
Ignore unknown JSON properties in responses	Forward compatibility
Do not depend on response field ordering	JSON is unordered
Treat new enum values as opaque or map to `UNKNOWN`	Enum extension
Use `errorCode` not HTTP status alone	Stable error handling (SRS-API-005)
Pin integration tests to contract release tag	Detect unintended drift

2.5 DTO and OpenAPI Versioning

Artifact	Versioning
Request DTOs (SRS-API-002)	Annotated with `since: IMS-API-1.x.0` per field
Response DTOs (SRS-API-003)	New fields marked `optional` in OpenAPI
Validation rules (SRS-API-004)	Stricter validation is **breaking** unless behind new optional field
Error codes (SRS-API-005)	New codes are non-breaking; removing/renaming codes is breaking

2.6 Database vs API Versioning

API major version is **independent** of database schema version. The application layer maps v1 and v2 DTOs to the current domain model via anti-corruption adapters in the presentation layer — domain aggregates do not fork per API version.

---

3. Deprecation Policy

3.1 Deprecation Triggers

An endpoint or field is **deprecated** when:

1. A successor exists in the same or next major version
2. A module is refactored and the endpoint is superseded
3. A business process is retired (with BRD approval)
4. Security or compliance requires replacement

Deprecation does **not** remove functionality — it signals intent to remove.

3.2 Deprecation Timeline

Consumer Type	Minimum Notice	Parallel Support
Internal UI (Angular)	1 minor release (≈ 6 weeks)	Same major version
Internal modules / BFF	1 minor release	Same major version
External integration (EXT-01, EXT-07)	**12 months** from deprecation announcement	v1 + v2 parallel
Partner/custom API consumers	**12 months**	v1 + v2 parallel
Deprecated major version (v1 entire)	**18 months** after v2 GA	Full v1 stack operational

3.3 Deprecation Process

```
1. Mark deprecated in OpenAPI (deprecated: true)
2. Add Deprecation + Sunset headers on responses
3. Log Warning on each deprecated call (metric: api.deprecated.calls)
4. Publish changelog entry and notify integration owners
5. Update SRS-API-001 with DEPRECATED badge
6. Remove only after sunset date + zero critical consumers
```

3.4 Deprecation Documentation Requirements

Each deprecation entry must include:

Field	Content
Deprecated artifact	Endpoint, field, or enum value
Since release	`IMS-API-1.x.0`
Sunset date	ISO date
Successor	Replacement endpoint or migration path
Migration guide	Request/response mapping
Business owner sign-off	Product Owner or Integration Architect

3.5 Field-Level Deprecation

Prefer **field-level** deprecation before endpoint removal:

```
{
  "sku": "WIDGET-01",
  "legacyBarcode": "12345",
  "_deprecatedFields": ["legacyBarcode"]
}
```

OpenAPI `deprecated: true` on the field. Field removed only in next **major** version.

---

4. Breaking Change Policy

4.1 Definition

A change is **breaking** if an existing conforming client would fail or behave differently without modification.

4.2 Breaking Change Catalog

#	Breaking Change	Example
B1	Remove endpoint	Delete `GET /api/v1/legacy-report`
B2	Remove request/response field	Remove `mrslThresholdDays`
B3	Rename field or endpoint	`orderedQuantity` → `quantity`
B4	Change field type	`quantity` string → number
B5	Change field semantics	`atp` includes quarantined stock
B6	Add required request field	New required `reasonCode` on POST
B7	Tighten validation	SKU max length 50 → 30
B8	Change URL path or HTTP method	`PATCH` → `PUT` for product update
B9	Change authentication model	Bearer scope requirements
B10	Change error code for same condition	`BIZ_INSUFFICIENT_ATP` renamed
B11	Change pagination defaults materially	Default `pageSize` 20 → 5 with no opt-out
B12	Remove enum value	Remove `DISCONTINUED` status
B13	Change idempotency behavior	Idempotency key scope change
B14	Change HTTP status for same outcome	`201` → `200` for create

4.3 Breaking Change Approval

Step	Owner	Gate
1. Propose RFC	Backend Architect	Impact analysis on 178 APIs + integrations
2. Classify breaking/non-breaking	API governance board	Check against §2.2 / §4.2
3. Business impact review	Product Owner	Operational/finance impact
4. Integration impact review	Integration Architect	EXT-01, EXT-07 partner notice
5. Security review	Security Engineer	Auth/RBAC changes
6. Approve major version bump	Tech Lead + PO	RFC signed
7. Implement v{n+1} parallel to v{n}	Development squads	No removal of v{n} until sunset

4.4 Emergency Breaking Changes

Security vulnerabilities (e.g., data exposure, auth bypass) may require **accelerated** breaking changes:

- Minimum notice: **30 days** for integrations (unless active exploit)

- Requires CFO/COO notification for finance-facing APIs
- Documented as `SECURITY` in changelog

---

5. Version Lifecycle

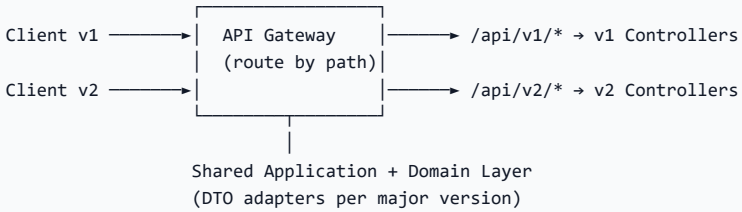
5.1 States

```
PLANNED → BETA → CURRENT → DEPRECATED → SUNSET → RETIRED
```

State	`/api/v1` active	Headers	New clients
PLANNED	No	—	Not accepted
BETA	Preview path optional	`X-API-Version: 2.0.0-beta`	Opt-in partners only
CURRENT	Yes	Standard	Default for new integrations
DEPRECATED	Yes	`Deprecation: true`	Discouraged; migrate to CURRENT
SUNSET	Yes (limited)	`Sunset` date passed warning	Existing only; no new keys
RETIRED	**No**	`410 Gone`	Not accepted

5.2 Multi-Version Operation

During migration from v1 to v2:



- **v1 and v2 share** domain services and aggregates — only presentation DTOs and controllers differ.
- Stock mutations from v1 and v2 use the **same** domain commands — no duplicate business logic.
- Integration tests run against **both** versions during overlap period.

5.3 Version Support Matrix (Illustrative)

Major Version	Status	GA Date	Deprecation	Sunset	Notes
v1	CURRENT	Jul 2026	TBD	TBD	Initial 178-API catalog
v2	PLANNED	—	—	—	First major when breaking need arises

---

6. Implementation Standards

6.1 Code Organization

```
presentation/
  api/
    v1/
      product/ProductControllerV1.java
      inventory/StockPositionControllerV1.java
    v2/
      product/ProductControllerV2.java
    common/
      VersionFilter.java          # X-API-Version headers
      DeprecationInterceptor.java # Deprecation/Sunset headers
```

6.2 Spring Boot Routing

```
@RestController
@RequestMapping("/api/v1/products")
public class ProductControllerV1 { ... }
```

Gateway rule: `/api/v1/**` and `/api/v2/**` route to same service; unknown version → `404` with available versions in body.

6.3 Contract Testing

Test Type	Scope
OpenAPI diff	CI fails on breaking change in same major (oasdiff)
Consumer-driven contracts	EXT-01, EXT-07 pact tests pinned to `IMS-API-1.x.0`
Regression suite	Golden JSON fixtures per release tag

6.4 Changelog Discipline

Every API change requires an entry in `CHANGELOG-API.md`:

```
## [IMS-API-1.5.0] - 2026-09-01
### Added
- API-179 `GET /api/v1/intelligence/reservation-utilization` (non-breaking)

### Deprecated
- `GET /api/v1/intelligence/alerts` — use API-158 filters instead; sunset 2027-09-01
```

---

7. Versioning by API Category

Category	Version Sensitivity	Guidance
Stock mutations (reserve, ship, transfer)	High	Breaking changes require v2; never alter semantics in minor
Approvals (adjustment, PO, transfer)	High	Error codes and status transitions are contract
Master data (product, location)	Medium	Additive fields OK; identifier format changes are breaking
Read/query (intelligence, lists)	Medium	New filters OK; removing filters is breaking
Integration ACL (API-176–178)	**Highest**	12-month deprecation; partner sign-off on any change
Platform (health, session)	Low	Health unversioned; session follows auth provider

---

8. Relationship to Other Versioning Concerns

Concern	Versioning mechanism	Document
REST API	URI `/api/v{major}`	This document
Domain events	Event type version + upcasters	ADR / event schema catalog
Database schema	Flyway migrations	SRS-DB
RBAC permissions	Additive PERM-xxx only in minor	rbac-matrix.md
Error codes	Additive in minor; remove in major	SRS-API-005

---

9. Decision Summary

Question	Decision
Which versioning strategy?	**URI path major versioning** (`/api/v1`, `/api/v2`)
Version format?	`v{major}` in URL; `IMS-API-{major}.{minor}.{patch}` for contract releases
Backward compatibility?	Guaranteed within major; tolerant reader; additive-only minors
Deprecation?	`Deprecation`/`Sunset` headers; 12-month external notice; documented successors
Breaking changes?	New major version only; RFC + governance approval; parallel operation period
When to create v2?	First catalogued B1–B14 breaking change that cannot be avoided

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Document Approval

Role	Name	Signature	Date
Senior Backend Architect			
Solution Architect			
Tech Lead			
Product Owner			
Integration Architect			

End of API Versioning Strategy

Document ID: SRS-API-008 | Version 1.0 | July 6, 2026

Collection API Strategy

Inventory Management System

Document Information

Attribute	Detail
Document ID	SRS-API-006
Document Title	Collection API Strategy — Pagination, Filtering, Sorting, Searching
Version	1.0
Status	Draft
Document Owner	Senior Backend Architect
Source Documents	SRS-API-001, SRS-API-002, SRS-API-003, SRS-API-004, SRS-API-005
Date	July 6, 2026
Scope	All `GET` collection, search, and list endpoints under `/api/v1`

Purpose

Defines a **consistent, REST-aligned strategy** for paginating, filtering, sorting, and searching collection responses across the IMS API surface. Every collection endpoint follows one of five collection profiles with explicit query parameters, defaults, limits, and response metadata.

Design Principles

Principle	Implementation
Resource-oriented URLs	Collections at plural nouns (`/products`, `/transfers`); filters as query params, not path segments
Predictable pagination	Offset (`page`/`pageSize`) by default; cursor for high-volume feeds
Whitelist sorting	`sortBy` accepts only documented fields per endpoint — unknown values → `400`
Explicit filter semantics	Each filter documents match type: exact, prefix, partial, range, enum
Scope by default	Results constrained to caller's authorized locations unless network-scoped
Empty is success	`200 OK` with empty `data[]` — never `404` for zero matches
Stable ordering	Tie-breaker sort on primary key (`sku`, `id`) when not explicitly sorted
No over-fetching	`pageSize` capped per profile; export endpoints use separate limits

Global Query Parameter Standards

Pagination (Offset — Profile `P`)

Parameter	Type	Default	Limits	Description
`page`	integer	`1`	min `1`	1-based page index
`pageSize`	integer	`20`	min `1`, max per profile	Records per page

Behavior: `totalPages = ceil(totalElements / pageSize)`. Requesting `page` beyond `totalPages` returns `200` with empty `data[]` and correct metadata.

Pagination (Cursor — Profile `C`)

Parameter	Type	Default	Limits	Description
`cursor`	string	—	opaque token	Position from prior response `links.next`
`limit`	integer	`100`	min `1`, max `500`	Records per page

Behavior: No `totalElements` guarantee. Response includes `hasMore` boolean and `nextCursor`.

Sorting

Parameter	Type	Default	Description
`sortBy`	string	per endpoint	Whitelisted field name
`sortDirection`	string	per endpoint	`ASC` or `DESC`

Rules:

- Invalid `sortBy` → `400 VAL_INVALID_ENUM`
- `sortDirection` without `sortBy` → ignored; default sort applies
- Compound default: primary business field ASC + `id` ASC as tie-breaker

Filtering Conventions

Match Type	Syntax	Example
Exact	param=value	`sku=WIDGET-01`
Prefix	param=value (documented)	`sku=WID` matches `WIDGET-01`

Match Type	Syntax	Example
Partial (contains)	param=value, case-insensitive	`name=bolt`
Enum	param=ENUM_VALUE	`status=OPEN`
Boolean	param=true|false	`isActive=true`
Date range	`from`/`to` or `startDate`/`endDate`	inclusive both ends
Multi-value	repeat param or comma-separated	`locationCode=WH-EAST&locationCode=WH-WEST`
Null-safe	omit param = no filter	—

Searching

Parameter	Type	Description
`q`	string	Cross-field search where supported (max 200 chars)
`search`	string	Alternate name on user directory — same semantics as `q`

Rules: `q` applies OR across documented fields. Combining `q` with field filters = AND logic. Minimum 2 characters for `q` unless exact ID lookup field is used.

Common Date Range Parameters

Parameter	Required	Max Range	Used By
`startDate` / `endDate`	varies	365 days (audit); 90 days (calendar)	Logs, movements, KPIs
`*From` / `*To`	optional pairs	same	PO delivery, transfers, counts

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Response Metadata Standards

`PagedResponse<T>` (Offset — extends SRS-API-003)

```
{
  "data": [ ],
  "page": 1,
  "pageSize": 20,
  "totalElements": 156,
  "totalPages": 8,
  "sort": {
    "field": "name",
    "direction": "ASC"
  },
  "appliedFilters": {
    "lifecycleStatus": "ACTIVE"
  },
  "links": {
    "self": "/api/v1/products?page=1&pageSize=20&lifecycleStatus=ACTIVE",
    "first": "/api/v1/products?page=1&pageSize=20&lifecycleStatus=ACTIVE",
    "prev": null,
    "next": "/api/v1/products?page=2&pageSize=20&lifecycleStatus=ACTIVE",
    "last": "/api/v1/products?page=8&pageSize=20&lifecycleStatus=ACTIVE"
  }
}
```

Field	Type	Description
`data`	array	Result items for current page
`page`	integer	Current page (echo)
`pageSize`	integer	Page size (echo)

Field	Type	Description
`totalElements`	long	Total matching records
`totalPages`	integer	Total pages
`sort`	object	Applied sort field and direction
`appliedFilters`	object	Echo of non-pagination query params
`links`	object	RFC 8288-style pagination URIs (relative)

`CursorPagedResponse<T>` (Cursor)

```
{
  "data": [ ],
  "limit": 100,
  "hasMore": true,
  "nextCursor": "eyJvcmRlcklkIjoiLi4uIn0",
  "sort": { "field": "updatedAt", "direction": "DESC" },
  "appliedFilters": { "status": "SHIPPED" }
}
```

`BoundedListResponse<T>` (Small nested collections)

```
{
  "data": [ ],
  "totalElements": 4,
  "truncated": false
}
```

Used when parent scope bounds result set (e.g. discrepancies on one receipt). Pagination optional; if `totalElements > maxPageSize`, pagination required.

Response Headers

Header	When
`X-Total-Count`	Offset pagination — duplicates `totalElements` for simple clients
`Link`	Optional RFC 5988 equivalent of `links` object

Collection Profiles

Profile	Code	Default `pageSize`	Max `pageSize`	Sort	Typical Use
Standard Collection	`P-STD`	20	100	Yes	Master data, transactional lists
Operational Queue	`P-OPS`	20	50	Yes	Pending approvals, put-away queues
Audit / Log	`P-LOG`	50	100	Yes (timestamp DESC)	Movements, audit trail, inquiries
Intelligence Board	`P-INT`	20	100	Yes	Dashboards, alerts, boards
Integration Feed	`C-FED`	100	500	Fixed	CRM fulfillment status feed
Bounded Nested	`B-NST`	all	50	Limited	Child collections under parent resource
Report / Aggregate	`R-AGG`	n/a	n/a	n/a	Summaries, calendars, KPIs — no pagination

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Endpoint Catalog

Legend: **Profile** = collection profile code. **P/S/F** = Pagination / Sort / Filter supported.

M-01 — Product Catalog

API-002 — `GET /api/v1/products`

Attribute	Specification
Profile	`P-STD`
DTO	`SearchProductsQuery`

Parameter	Type	Match	Default	Notes
`name`	string	partial	—	Display name contains
`sku`	string	prefix	—	SKU starts-with
`categoryCode`	string	exact	—	
`lifecycleStatus`	enum	exact	—	`ACTIVE`, `INACTIVE`, `DISCONTINUED`
`abcClass`	enum	exact	—	`A`, `B`, `C`
`barcode`	string	exact	—	
`q`	string	search	—	Searches `name`, `sku`, `barcode` (OR)
`page`, `pageSize`	int	—	1, 20	max 100
`sortBy`	enum	—	`sku`	`sku`, `name`, `categoryCode`, `lifecycleStatus`
`sortDirection`	enum	—	`ASC`	

Default behavior: All active products in catalog scope; sort `sku ASC`.
Response: `PagedResponse<ProductSummaryResponse>`

API-008 — `GET /api/v1/products/categories/{categoryCode}/products`

Attribute	Specification
Profile	`P-STD`
Path filter	`categoryCode` (exact, required)

Parameter	Type	Match	Default
`lifecycleStatus`	enum	exact	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`sku` (`sku`, `name`)
`sortDirection`	enum	—	`ASC`

Default behavior: Products in category; active only when `lifecycleStatus` omitted.
Response: `PagedResponse<ProductSummaryResponse>`

M-02 — Warehouse Network

API-012 — `GET /api/v1/locations`

Parameter	Type	Match	Default
`operationalStatus`	enum	exact	`ACTIVE`
`locationType`	enum	exact	—
`parentLocationCode`	string	exact	—

Parameter	Type	Match	Default
`name`	string	partial	—
`q`	string	search	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`locationCode` (`locationCode`, `name`, `operationalStatus`)
`sortDirection`	enum	—	`ASC`

Profile: `P-STD` | **Scope:** User-authorized locations
Response: `PagedResponse<LocationSummaryResponse>`

API-016 — `GET /api/v1/locations/{locationCode}/storage-positions`

Parameter	Type	Match	Default
`positionType`	enum	exact	—
`zone`	string	exact	—
`isActive`	boolean	exact	`true`
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`positionCode` (`positionCode`, `zone`, `aisle`)
`sortDirection`	enum	—	`ASC`

Profile: `P-STD` | **Path:** `locationCode` required
Response: `PagedResponse<StoragePositionResponse>`

M-03 — Inventory Governance

API-023 — `GET /api/v1/governance/reason-codes`

Parameter	Type	Match	Default
`transactionType`	enum	exact	—
`isActive`	boolean	exact	`true`
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`code` (`code`, `description`)
`sortDirection`	enum	—	`ASC`

Profile: `P-STD` (reference data, typically small)
Response: `PagedResponse<ReasonCodeResponse>`

M-04 — Inventory Balance

API-033 — `GET /api/v1/inventory/stock-positions`

Parameter	Type	Match	Default
`sku`	string	exact/prefix	—
`locationCode`	string	exact	—
`stockStatus`	enum	exact	—

Parameter	Type	Match	Default
`abcClass`	enum	exact	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`sku` (`sku`, `locationCode`, `atp`, `stockStatus`)
`sortDirection`	enum	—	`ASC`

Profile: `P-STD` | **Required:** at least one of `sku` or `locationCode` (VR-054)
Scope: User location scope on `locationCode`
Response: `PagedResponse<StockPositionSummaryResponse>`

API-041 — `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/lots`

Parameter	Type	Match	Default
`includeQuarantined`	boolean	exact	`false`
`page`, `pageSize`	int	—	1, 20 (optional)
`sortBy`	enum	—	`expirationDate` (`expirationDate`, `lotNumber`)
`sortDirection`	enum	—	`ASC` (FEFO-friendly)

Profile: `B-NST` | max 50 lots; pagination required if > 50
Response: `BoundedListResponse<LotQuantityResponse>` or `PagedResponse` when paginated

M-05 — Procurement & Supplier

API-050 — `GET /api/v1/suppliers`

Parameter	Type	Match	Default
`isActive`	boolean	exact	`true`
`name`	string	partial	—
`q`	string	search	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`supplierCode` (`supplierCode`, `name`)
`sortDirection`	enum	—	`ASC`

Profile: `P-STD`
Response: `PagedResponse<SupplierSummaryResponse>`

API-060 — `GET /api/v1/procurement/purchase-orders`

Parameter	Type	Match	Default
`status`	enum	exact	`OPEN,PARTIAL`
`supplierCode`	string	exact	—
`receivingLocationCode`	string	exact	—
`expectedDeliveryFrom`	date	range start	—
`expectedDeliveryTo`	date	range end	—
`isOverdue`	boolean	exact	—

Parameter	Type	Match	Default
page, pageSize	int	—	1, 20
sortBy	enum	—	expectedDeliveryDate (expectedDeliveryDate, purchaseOrderId, supplierCode)
sortDirection	enum	—	ASC

Profile: P-STD | **Cross-field:** delivery date range
Scope: User location scope
Response: PagedResponse<PurchaseOrderSummaryResponse>

M-06 — Inbound Receiving

API-067 — GET /api/v1/inbound/goods-receipts

Parameter	Type	Match	Default
receivingLocationCode	string	exact	—
status	enum	exact	—
purchaseOrderId	uuid	exact	—
receivedFrom / receivedTo	date	range	—
page, pageSize	int	—	1, 20
sortBy	enum	—	receivedAt (receivedAt, goodsReceiptId, status)
sortDirection	enum	—	DESC

Profile: P-STD
Response: PagedResponse<GoodsReceiptSummaryResponse>

API-071 — GET /api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies

Parameter	Type	Match	Default
status	enum	exact	OPEN
page, pageSize	int	—	1, 20 (optional)
sortBy	enum	—	recordedAt (recordedAt, discrepancyType)
sortDirection	enum	—	DESC

Profile: B-NST | max 50 without pagination
Response: BoundedListResponse<ReceiptDiscrepancyResponse>

API-074 — GET /api/v1/inbound/pending-put-away

Parameter	Type	Match	Default
locationCode	string	exact	required
page, pageSize	int	—	1, 20
sortBy	enum	—	verifiedAt (verifiedAt, goodsReceiptId)
sortDirection	enum	—	ASC (FIFO queue)

Profile: P-OPS | max pageSize 50
Response: PagedResponse<PendingPutAwayResponse>

M-07 — Order Fulfillment

API-082 — `GET /api/v1/fulfillment/orders`

Parameter	Type	Match	Default
`status`	enum	exact	—
`shipFromLocationCode`	string	exact	—
`customerReference`	string	exact	—
`externalOrderReference`	string	exact	—
`orderDateFrom` / `orderDateTo`	date	range	—
`q`	string	search	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`orderDate` (`orderDate`, `status`, `orderId`)
`sortDirection`	enum	—	`DESC`

Profile: `P-STD`
Response: `PagedResponse<CustomerOrderSummaryResponse>`

API-089 — `GET /api/v1/fulfillment/orders/{orderId}/pick-lists/awaiting`

Parameter	Type	Match	Default
`locationCode`	string	exact	—

Profile: `B-NST` | No pagination — returns pick lists for single order (typically ≤ 10)
Response: `BoundedListResponse<PickListSummaryResponse>`

M-08 — Network Transfer

API-099 — `GET /api/v1/transfers`

Parameter	Type	Match	Default
`status`	enum	exact	—
`sourceLocationCode`	string	exact	—
`destinationLocationCode`	string	exact	—
`sku`	string	exact	—
`requestedFrom` / `requestedTo`	date	range	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`requestedAt` (`requestedAt`, `status`, `transferId`)
`sortDirection`	enum	—	`DESC`

Profile: `P-STD`
Response: `PagedResponse<StockTransferSummaryResponse>`

API-106 — `GET /api/v1/transfers/pending-approvals`

Parameter	Type	Match	Default
`sourceLocationCode`	string	exact	—
`destinationLocationCode`	string	exact	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`requestedAt` (fixed whitelist)
`sortDirection`	enum	—	`ASC` (oldest first)

Profile: `P-OPS` | max pageSize 50 | implicit filter `status=REQUESTED`
Response: `PagedResponse<StockTransferSummaryResponse>`

API-107 — `GET /api/v1/transfers/in-transit`

Parameter	Type	Match	Default
`locationCode`	string	exact	—
`sku`	string	exact	—
`isOverdue`	boolean	exact	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`dispatchedAt` (`dispatchedAt`, `transferId`)
`sortDirection`	enum	—	`ASC`

Profile: `P-OPS` | implicit filter `status=IN_TRANSIT`
Response: `PagedResponse<InTransitStockResponse>`

M-09 — Inventory Accuracy

API-109 — `GET /api/v1/accuracy/cycle-counts`

Parameter	Type	Match	Default
`locationCode`	string	exact	—
`status`	enum	exact	—
`sku`	string	exact	—
`scheduledFrom` / `scheduledTo`	date	range	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`scheduledDate` (`scheduledDate`, `priority`, `status`)
`sortDirection`	enum	—	`ASC`

Profile: `P-STD`
Response: `PagedResponse<CycleCountSummaryResponse>`

API-123 — `GET /api/v1/accuracy/pending-approvals`

Parameter	Type	Match	Default
`locationCode`	string	exact	—

Parameter	Type	Match	Default
caseType	enum	exact	—
page, pageSize	int	—	1, 20
sortBy	enum	—	submittedAt (submittedAt, caseType)
sortDirection	enum	—	ASC

Profile: P-OPS | max pageSize 50
Response: PagedResponse<AdjustmentCaseSummaryResponse>

M-10 — Returns Management

API-125 — GET /api/v1/returns/authorizations

Parameter	Type	Match	Default
status	enum	exact	—
originalOrderId	uuid	exact	—
customerReference	string	exact	—
page, pageSize	int	—	1, 20
sortBy	enum	—	authorizedAt (authorizedAt, status)
sortDirection	enum	—	DESC

Profile: P-STD
Response: PagedResponse<ReturnAuthorizationSummaryResponse>

API-131 — GET /api/v1/returns/pending-receipts

Parameter	Type	Match	Default
locationCode	string	exact	—
page, pageSize	int	—	1, 20
sortBy	enum	—	authorizedAt
sortDirection	enum	—	ASC

Profile: P-OPS | implicit filter status AUTHORIZED/IN_TRANSIT
Response: PagedResponse<PendingReturnReceiptResponse>

M-11 — Commercial Campaign

API-133 — GET /api/v1/campaigns/allocations

Parameter	Type	Match	Default
campaignId	string	exact	—
status	enum	exact	—
locationCode	string	exact	—
activeOnDate	date	exact	—
page, pageSize	int	—	1, 20

Parameter	Type	Match	Default
sortBy	enum	—	campaignStartDate (campaignStartDate, campaignId)
sortDirection	enum	—	ASC

Profile: P-STD
Response: PagedResponse<PromotionalAllocationSummaryResponse>

M-12 — Identity & Access

API-141 — GET /api/v1/users

Parameter	Type	Match	Default
status	enum	exact	—
roleCode	string	exact	—
locationCode	string	exact	—
search	string	partial	—
page, pageSize	int	—	1, 20
sortBy	enum	—	displayName (displayName, userId, email)
sortDirection	enum	—	ASC

Profile: P-STD | Admin scope required
Response: PagedResponse<UserAccountSummaryResponse>

M-13 — Inventory Intelligence

API-152 — GET /api/v1/intelligence/stock-status/board

Parameter	Type	Match	Default
locationCodes	string[]	exact multi	authorized locations
stockStatus	enum	exact	—
sku	string	exact/prefix	—
abcClass	enum	exact	—
page, pageSize	int	—	1, 20
sortBy	enum	—	stockStatus (stockStatus, sku, atp)
sortDirection	enum	—	ASC

Profile: P-INT
Response: PagedResponse<StockStatusBoardItemResponse>

API-153 — GET /api/v1/intelligence/availability/inquiries

Parameter	Type	Match	Default
sku	string	exact	—
locationCode	string	exact	—
userId	string	exact	—

Parameter	Type	Match	Default
`startDate` / `endDate`	date	range	last 30 days
`page`, `pageSize`	int	—	1, 50
`sortBy`	enum	—	`inquiredAt` (fixed)
`sortDirection`	enum	—	`DESC`

Profile: `P-LOG` | max range 365 days
Response: `PagedResponse<AvailabilityInquiryLogEntryResponse>`

API-154 — `GET /api/v1/intelligence/quarantine-register`

Parameter	Type	Match	Default
`locationCode`	string	exact	—
`holdReason`	enum	exact	—
`sku`	string	exact	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`heldAt` (`heldAt`, `sku`, `holdReason`)
`sortDirection`	enum	—	`DESC`

Profile: `P-INT`
Response: `PagedResponse<QuarantineRegisterEntryResponse>`

API-155 — `GET /api/v1/intelligence/movements`

Parameter	Type	Match	Default
`sku`	string	exact	—
`locationCode`	string	exact	—
`userId`	string	exact	—
`movementType`	enum	exact	—
`startDate` / `endDate`	date	range	**required**; max 365 days
`page`, `pageSize`	int	—	1, 50
`sortBy`	enum	—	`occurredAt` (fixed)
`sortDirection`	enum	—	`DESC`

Profile: `P-LOG` | Recommend ≥1 filter (VR-154)
Response: `PagedResponse<MovementHistoryEntryResponse>`

API-156 — `GET /api/v1/intelligence/audit-trail`

Parameter	Type	Match	Default
`entityType`	string	exact	—
`entityId`	string	exact	—
`userId`	string	exact	—
`action`	string	exact	—

Parameter	Type	Match	Default
`startDate` / `endDate`	date	range	**required**; max 365 days
`page`, `pageSize`	int	—	1, 50
`sortBy`	enum	—	`occurredAt` (fixed)
`sortDirection`	enum	—	`DESC`

Profile: `P-LOG`
Response: `PagedResponse<AuditTrailEntryResponse>`

API-157 — `GET /api/v1/intelligence/audit-trail/export`

Parameter	Type	Match	Default
`startDate` / `endDate`	date	range	**required**; max 365 days
`entityTypes`	string[]	multi	max 20 types
`locationCodes`	string[]	multi	authorized only
`format`	enum	exact	`CSV`

Profile: `R-AGG` | No pagination — async export if > 100k rows; returns `202` with `exportJobId`
Response: File stream (`200`) or `ExportJobResponse` (`202`)

API-158 — `GET /api/v1/intelligence/alerts`

Parameter	Type	Match	Default
`alertType`	enum	exact	—
`severity`	enum	exact	—
`locationCode`	string	exact	—
`acknowledged`	boolean	exact	`false`
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`severity` (`severity`, `raisedAt`)
`sortDirection`	enum	—	`DESC`

Profile: `P-INT` | Severity sort: CRITICAL → WARNING → INFO
Response: `PagedResponse<OperationalAlertResponse>`

API-160 — `GET /api/v1/intelligence/replenishment/recommendations`

Parameter	Type	Match	Default
`locationCode`	string	exact	—
`sku`	string	exact	—
`recommendationAction`	enum	exact	—
`acknowledged`	boolean	exact	—
`startDate` / `endDate`	date	range	optional
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`recommendedAt`

Parameter	Type	Match	Default
`sortDirection`	enum	—	`DESC`

Profile: `P-INT`
Response: `PagedResponse<ReplenishmentRecommendationResponse>`

API-161 — `GET /api/v1/intelligence/procurement/open-pos`

Parameter	Type	Match	Default
`receivingLocationCode`	string	exact	—
`supplierCode`	string	exact	—
`isOverdue`	boolean	exact	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`expectedDeliveryDate`
`sortDirection`	enum	—	`ASC`

Profile: `P-INT` | Projection of open PO lines
Response: `PagedResponse<OpenPurchaseOrderBoardItemResponse>`

API-162 — `GET /api/v1/intelligence/procurement/inbound-calendar`

Parameter	Type	Match	Default
`startDate` / `endDate`	date	range	**required**; max **90 days**
`receivingLocationCode`	string	exact	—
`supplierCode`	string	exact	—

Profile: `R-AGG` | Calendar buckets by date — no pagination
Response: `InboundSupplyCalendarResponse` (date-keyed arrays)

API-163 — `GET /api/v1/intelligence/procurement/inbound-delays`

Parameter	Type	Match	Default
`receivingLocationCode`	string	exact	—
`supplierCode`	string	exact	—
`startDate` / `endDate`	date	range	optional
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`delayDays` (`delayDays`, `expectedArrivalDate`)
`sortDirection`	enum	—	`DESC`

Profile: `P-LOG`
Response: `PagedResponse<InboundDelayLogEntryResponse>`

API-164 — `GET /api/v1/intelligence/fulfillment/backorders`

Parameter	Type	Match	Default
`sku`	string	exact	—
`locationCode`	string	exact	—
`prioritySort`	boolean	exact	`true`
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	policy priority when `prioritySort=true`; else `backorderDate`
`sortDirection`	enum	—	`ASC`

Profile: `P-OPS` | BR-028 priority policy (VR-102)
Response: `PagedResponse<BackorderPipelineItemResponse>`

API-165 — `GET /api/v1/intelligence/fulfillment/exceptions`

Parameter	Type	Match	Default
`locationCode`	string	exact	—
`exceptionType`	enum	exact	—
`status`	enum	exact	`OPEN`
`startDate` / `endDate`	date	range	optional
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`raisedAt`
`sortDirection`	enum	—	`DESC`

Profile: `P-LOG`
Response: `PagedResponse<FulfillmentExceptionLogEntryResponse>`

API-166 — `GET /api/v1/intelligence/network/imbalance`

Parameter	Type	Match	Default
`sku`	string	exact	—
`sourceLocationCode`	string	exact	—
`destinationLocationCode`	string	exact	—
`includeRecommendations`	boolean	exact	`true`

Profile: `R-AGG` | Analysis result — no pagination; max 500 imbalance rows inline
Response: `NetworkImbalanceAnalysisResponse`

API-167 — `GET /api/v1/intelligence/accuracy/shrinkage`

Parameter	Type	Match	Default
`startDate` / `endDate`	date	range	**required**
`locationCode`	string	exact	—
`categoryCode`	string	exact	—

Parameter	Type	Match	Default
`reasonCode`	string	exact	—
`groupBy`	enum	exact	`SKU`

Profile: `R-AGG` | Aggregated analysis — no pagination
Response: `ShrinkageAnalysisResponse`

API-168 — `GET /api/v1/intelligence/accuracy/count-performance`

Parameter	Type	Match	Default
`startDate` / `endDate`	date	range	**required**
`locationCode`	string	exact	—
`countType`	enum	exact	`ALL`
`abcClass`	enum	exact	—
`excludePendingVariances`	boolean	exact	`true`

Profile: `R-AGG`
Response: `CountPerformanceMetricsResponse`

API-169 — `GET /api/v1/intelligence/campaigns/readiness`

Parameter	Type	Match	Default
`campaignId`	string	exact	—
`locationCode`	string	exact	—
`readinessStatus`	enum	exact	—
`daysUntilStart`	integer	max	—

Profile: `P-INT` | Optional pagination when result > 100: `page`, `pageSize`
Response: `PagedResponse<CampaignReadinessStatusResponse>` or array if ≤ 100

API-170 — `GET /api/v1/intelligence/kpis`

Parameter	Type	Match	Default
`startDate` / `endDate`	date	range	**required**
`kpiCodes`	string[]	multi	all configured
`locationCodes`	string[]	multi	authorized
`compareToTarget`	boolean	exact	`true`

Profile: `R-AGG`
Response: `InventoryKpisResponse`

API-171 — `GET /api/v1/intelligence/valuation/snapshots`

Parameter	Type	Match	Default
`periodEndDate`	date	exact	—

Parameter	Type	Match	Default
`startDate` / `endDate`	date	range	optional
`locationCode`	string	exact	—
`page`, `pageSize`	int	—	1, 20
`sortBy`	enum	—	`periodEndDate`
`sortDirection`	enum	—	`DESC`

Profile: `P-INT`
Response: `PagedResponse<ValuationSnapshotSummaryResponse>`

API-172 — `GET /api/v1/intelligence/valuation/reconciliation`

Parameter	Type	Match	Default
`periodEndDate`	date	exact	**required**
`locationCode`	string	exact	—
`varianceThresholdPercent`	decimal	min/max	policy default
`includeGlComparison`	boolean	exact	`true`

Profile: `R-AGG`
Response: `ValuationReconciliationResponse`

API-150 / API-151 — Dashboard & summary (non-paginated)

API	Profile	Key Parameters	Response
API-150	`R-AGG`	`scope`, `locationCode` (conditional), `asOfTimestamp`	`InventoryDashboardResponse`
API-151	`R-AGG`	`locationCode`, `categoryCode`, `abcClass`	`StockStatusSummaryResponse`

Platform & Integration

API-177 — `GET /api/v1/integrations/sales/fulfillment-status`

Parameter	Type	Match	Default
`externalSystemCode`	string	exact	**required**
`updatedSince`	instant	range start	—
`externalOrderId`	string	exact	—
`status`	enum	exact	—
`cursor`	string	—	—
`limit`	int	—	100

Profile: `C-FED` | Cursor pagination; sort fixed `updatedAt DESC`
Response: `CursorPagedResponse<FulfillmentStatusFeedItemResponse>`

Collection Endpoint Index

API	Endpoint	Profile	Pagination	Sort	Filter	Search (q)
API-002	`/products`	P-STD	✓	✓	✓	✓
API-008	`/products/categories/{code}/products`	P-STD	✓	✓	✓	—
API-012	`/locations`	P-STD	✓	✓	✓	✓
API-016	`/locations/{code}/storage-positions`	P-STD	✓	✓	✓	—
API-023	`/governance/reason-codes`	P-STD	✓	✓	✓	—
API-033	`/inventory/stock-positions`	P-STD	✓	✓	✓	—
API-041	`/inventory/.../lots`	B-NST	optional	✓	✓	—
API-050	`/suppliers`	P-STD	✓	✓	✓	✓
API-060	`/procurement/purchase-orders`	P-STD	✓	✓	✓	—
API-067	`/inbound/goods-receipts`	P-STD	✓	✓	✓	—
API-071	`/inbound/.../discrepancies`	B-NST	optional	✓	✓	—
API-074	`/inbound/pending-put-away`	P-OPS	✓	✓	✓	—
API-082	`/fulfillment/orders`	P-STD	✓	✓	✓	✓
API-089	`/fulfillment/orders/{id}/pick-lists/awaiting`	B-NST	—	—	✓	—
API-099	`/transfers`	P-STD	✓	✓	✓	—
API-106	`/transfers/pending-approvals`	P-OPS	✓	✓	✓	—
API-107	`/transfers/in-transit`	P-OPS	✓	✓	✓	—
API-109	`/accuracy/cycle-counts`	P-STD	✓	✓	✓	—
API-123	`/accuracy/pending-approvals`	P-OPS	✓	✓	✓	—
API-125	`/returns/authorizations`	P-STD	✓	✓	✓	—
API-131	`/returns/pending-receipts`	P-OPS	✓	✓	✓	—
API-133	`/campaigns/allocations`	P-STD	✓	✓	✓	—
API-141	`/users`	P-STD	✓	✓	✓	✓ (`search`)
API-152	`/intelligence/stock-status/board`	P-INT	✓	✓	✓	—
API-153	`/intelligence/availability/inquiries`	P-LOG	✓	fixed	✓	—
API-154	`/intelligence/quarantine-register`	P-INT	✓	✓	✓	—
API-155	`/intelligence/movements`	P-LOG	✓	fixed	✓	—
API-156	`/intelligence/audit-trail`	P-LOG	✓	fixed	✓	—
API-157	`/intelligence/audit-trail/export`	R-AGG	—	—	✓	—
API-158	`/intelligence/alerts`	P-INT	✓	✓	✓	—
API-160	`/intelligence/replenishment/recommendations`	P-INT	✓	✓	✓	—
API-161	`/intelligence/procurement/open-pos`	P-INT	✓	✓	✓	—
API-162	`/intelligence/procurement/inbound-calendar`	R-AGG	—	—	✓	—
API-163	`/intelligence/procurement/inbound-delays`	P-LOG	✓	✓	✓	—
API-164	`/intelligence/fulfillment/backorders`	P-OPS	✓	policy	✓	—
API-165	`/intelligence/fulfillment/exceptions`	P-LOG	✓	✓	✓	—
API-166	`/intelligence/network/imbalance`	R-AGG	—	—	✓	—

API	Endpoint	Profile	Pagination	Sort	Filter	Search (q)
API-167	`/intelligence/accuracy/shrinkage`	R-AGG	—	—	✓	—
API-168	`/intelligence/accuracy/count-performance`	R-AGG	—	—	✓	—
API-169	`/intelligence/campaigns/readiness`	P-INT	conditional	—	✓	—
API-170	`/intelligence/kpis`	R-AGG	—	—	✓	—
API-171	`/intelligence/valuation/snapshots`	P-INT	✓	✓	✓	—
API-172	`/intelligence/valuation/reconciliation`	R-AGG	—	—	✓	—
API-150	`/intelligence/dashboard`	R-AGG	—	—	✓	—
API-151	`/intelligence/stock-status/summary`	R-AGG	—	—	✓	—
API-177	`/integrations/sales/fulfillment-status`	C-FED	cursor	fixed	✓	—

Total collection/search endpoints: 44

Error Handling for Collection Queries

Condition	HTTP	Error Code
Invalid `page` or `pageSize`	400	`VAL_OUT_OF_RANGE`
Invalid `sortBy`	400	`VAL_INVALID_ENUM`
Invalid date range	400	`VAL_INVALID_DATE_RANGE`
Date range exceeds max	400	`VAL_OUT_OF_RANGE`
Missing required filter	400	`VAL_QUERY_FILTER_REQUIRED`
Unauthorized location in filter	403	`AUTHZ_LOCATION_SCOPE_DENIED`

Implementation Notes

1. **Link generation:** Preserve all query parameters except `page`/`cursor` when building `links.next` / `links.prev`.
2. **Count query:** Execute `COUNT(*)` with same WHERE clause; cache counts ≤ 30s for intelligence endpoints.
3. **Index alignment:** Every `sortBy` whitelist field must have a supporting database index or composite index leading column.
4. **OpenAPI:** Document per-endpoint `sortBy` enum and filter match types in parameter descriptions.
5. **Versioning:** New sort/filter fields are additive; removing fields requires API version bump.

Document Approval

Role	Name	Signature	Date
Senior Backend Architect			
Solution Architect			
Tech Lead			

End of Collection API Strategy

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Part IX — UI/UX Specification

Inventory Management System Screen Inventory

Document Purpose

This document identifies the screens required for the Inventory Management System based on the approved BRD, use-case package, and API design package. It defines the screen inventory only and intentionally does not propose layouts, navigation, or wireframes.

Scope Notes

- The screen list covers MVP plus approved future-phase capabilities already defined in the BRD, use cases, and API catalog.
- Screens are defined at page/workspace level. Closely related actions such as create, edit, review, and approve are grouped into one screen when they belong to the same user task.
- Where the API review identified an unresolved endpoint gap, the affected screen references the current APIs plus the noted gap.

Access and Shared Experience

UI-001

- **Screen name:** Sign In / Session Entry
- **Purpose:** Authenticate the user and establish the authorized IMS session and role/location context before any protected function is accessed.
- **Primary users:** All internal IMS users
- **Related use cases:** UC-045
- **Related APIs:** API-173 `GET /api/v1/auth/session`, API-174 `POST /api/v1/auth/logout`

UI-002

- **Screen name:** Operational Home Dashboard
- **Purpose:** Provide the day-to-day landing view for operational users with inventory summary, stock health, alert counts, and shortcuts into core workflows.
- **Primary users:** Operations Director, Inventory Controller, Warehouse Manager, Purchasing Manager, Sales Manager
- **Related use cases:** UC-012, UC-037, UC-047
- **Related APIs:** API-150 `GET /api/v1/intelligence/dashboard`, API-151 `GET /api/v1/intelligence/stock-status/summary`, API-152 `GET /api/v1/intelligence/stock-status/board`, API-158 `GET /api/v1/intelligence/alerts`

UI-003

- **Screen name:** Executive KPI Dashboard
- **Purpose:** Present strategic inventory KPIs, trends, target comparisons, and drill-downs for executive review.
- **Primary users:** Chief Executive Officer, Chief Operating Officer, Chief Financial Officer, Operations Director, Finance Manager
- **Related use cases:** UC-038
- **Related APIs:** API-170 `GET /api/v1/intelligence/kpis`

UI-004

- **Screen name:** Notification Center
- **Purpose:** Centralize low-stock alerts, pending approvals, transfer issues, inbound delays, and campaign readiness notifications so users can review and act on them.
- **Primary users:** Warehouse Manager, Purchasing Manager, Operations Director, Buyer, Marketing Manager, Customer Service Representative

- **Related use cases:** UC-028, UC-032, UC-047
- **Related APIs:** API-158 `GET /api/v1/intelligence/alerts`, API-159 `POST /api/v1/intelligence/alerts/{alertId}/acknowledge`, API-106 `GET /api/v1/transfers/pending-approvals`, API-123 `GET /api/v1/accuracy/pending-approvals`, API-163 `GET /api/v1/intelligence/procurement/inbound-delays`, API-169 `GET /api/v1/intelligence/campaigns/readiness`

Master Data and Governance

UI-005

- **Screen name:** Product Catalog Management
- **Purpose:** Search, create, update, review, and maintain product master records including lifecycle status and identifiers.
- **Primary users:** Inventory Controller
- **Related use cases:** UC-001
- **Related APIs:** API-001 `POST /api/v1/products`, API-002 `GET /api/v1/products`, API-003 `GET /api/v1/products/{sku}`, API-004 `PATCH /api/v1/products/{sku}`, API-005 `POST /api/v1/products/{sku}/lifecycle-status`, API-008 `GET /api/v1/products/categories/{categoryCode}/products`, API-009 `POST /api/v1/products/duplicate-resolutions`

UI-006

- **Screen name:** Product Import Workspace
- **Purpose:** Upload, validate, review, and complete bulk product and opening data imports during go-live or controlled data loads.
- **Primary users:** Inventory Controller, System Administrator
- **Related use cases:** UC-046
- **Related APIs:** API-010 `POST /api/v1/products/import`, API-032 `POST /api/v1/inventory/opening-balances`, API-017 `POST /api/v1/locations/{locationCode}/activate`

UI-007

- **Screen name:** Location and Warehouse Management
- **Purpose:** Configure inventory locations, location hierarchy, storage positions, and operating status for the network.
- **Primary users:** Inventory Controller, Operations Director
- **Related use cases:** UC-002
- **Related APIs:** API-011 `POST /api/v1/locations`, API-012 `GET /api/v1/locations`, API-013 `GET /api/v1/locations/{locationCode}`, API-014 `PATCH /api/v1/locations/{locationCode}/hierarchy`, API-015 `POST /api/v1/locations/{locationCode}/storage-positions`, API-016 `GET /api/v1/locations/{locationCode}/storage-positions`, API-017 `POST /api/v1/locations/{locationCode}/activate`, API-018 `POST /api/v1/locations/{locationCode}/deactivate`, API-019 `POST /api/v1/locations/{locationCode}/close`, API-021 `GET /api/v1/locations/{locationCode}/operational-status`

UI-008

- **Screen name:** Stock Policy and Threshold Management
- **Purpose:** Define and maintain reorder levels, minimum thresholds, and safety stock settings by product and location.
- **Primary users:** Inventory Controller, Purchasing Manager
- **Related use cases:** UC-003
- **Related APIs:** API-038 `POST /api/v1/inventory/stock-policies`, API-039 `GET /api/v1/inventory/stock-policies/{locationCode}/{sku}`, API-040 `PATCH /api/v1/inventory/stock-policies/{locationCode}/{sku}`

UI-009

- **Screen name:** ABC Classification Management
- **Purpose:** Assign and review ABC product classes to support differentiated monitoring and counting discipline.
- **Primary users:** Inventory Controller
- **Related use cases:** UC-004
- **Related APIs:** API-006 `PUT /api/v1/products/{sku}/abc-classification`, API-002 `GET /api/v1/products`

UI-010

- **Screen name:** Governance Policies and Reason Codes

- **Purpose:** Maintain standardized reason codes, approval thresholds, receipt tolerances, and governance policy versions.
- **Primary users:** Inventory Controller, Internal Auditor, Finance Manager, Operations Director
- **Related use cases:** UC-005
- **Related APIs:** API-022 `POST /api/v1/governance/reason-codes`, API-023 `GET /api/v1/governance/reason-codes`, API-024 `POST /api/v1/governance/reason-codes/{code}/retire`, API-025 `GET /api/v1/governance/reason-codes/{code}/validate`, API-026 `POST /api/v1/governance/policies`, API-027 `PATCH /api/v1/governance/policies/{policyId}/approval-thresholds`, API-028 `PATCH /api/v1/governance/policies/{policyId}/receipt-tolerances`, API-029 `GET /api/v1/governance/approval-thresholds`, API-030 `GET /api/v1/governance/receipt-tolerances`

Procurement and Inbound

UI-011

- **Screen name:** Supplier Management
- **Purpose:** Maintain supplier master data and supplier operational status used by procurement workflows.
- **Primary users:** Buyer, Purchasing Manager
- **Related use cases:** UC-006
- **Related APIs:** API-049 `POST /api/v1/suppliers`, API-050 `GET /api/v1/suppliers`, API-051 `GET /api/v1/suppliers/{supplierCode}`, API-052 `PATCH /api/v1/suppliers/{supplierCode}`, API-053 `POST /api/v1/suppliers/{supplierCode}/deactivate`

UI-012

- **Screen name:** Purchase Order Management
- **Purpose:** Create, review, approve-state track, amend, and cancel purchase orders.
- **Primary users:** Buyer, Purchasing Manager
- **Related use cases:** UC-006
- **Related APIs:** API-059 `POST /api/v1/procurement/purchase-orders`, API-060 `GET /api/v1/procurement/purchase-orders`, API-061 `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}`, API-062 `PATCH /api/v1/procurement/purchase-orders/{purchaseOrderId}`, API-063 `POST /api/v1/procurement/purchase-orders/{purchaseOrderId}/cancel`, API-065 `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}/validate-for-receipt`, API-161 `GET /api/v1/intelligence/procurement/open-pos`

UI-013

- **Screen name:** Purchase Requisition Management
- **Purpose:** Review replenishment-driven requisitions, submit them for approval, and convert approved requisitions into purchase orders.
- **Primary users:** Buyer, Purchasing Manager
- **Related use cases:** UC-029
- **Related APIs:** API-054 `POST /api/v1/procurement/requisitions`, API-055 `GET /api/v1/procurement/requisitions/{requisitionId}`, API-056 `POST /api/v1/procurement/requisitions/{requisitionId}/approve`, API-057 `POST /api/v1/procurement/requisitions/{requisitionId}/reject`, API-058 `POST /api/v1/procurement/requisitions/{requisitionId}/convert-to-po`
- **API note:** API review item `C-03` identifies the missing requisition list endpoint needed for full queue browsing.

UI-014

- **Screen name:** Inbound Delivery Planning
- **Purpose:** Plan, update, and communicate expected supplier deliveries to receiving locations.
- **Primary users:** Buyer, Warehouse Manager, Receiving Clerk
- **Related use cases:** UC-007, UC-030
- **Related APIs:** API-064 `POST /api/v1/procurement/purchase-orders/{purchaseOrderId}/expected-deliveries`, API-162 `GET /api/v1/intelligence/procurement/inbound-calendar`, API-163 `GET /api/v1/intelligence/procurement/inbound-delays`

UI-015

- **Screen name:** Goods Receipt Workbench
- **Purpose:** Receive supplier goods, verify products and quantities, capture lot and shelf-life data, and stage stock for put-away.
- **Primary users:** Receiving Clerk

- **Related use cases:** UC-008
- **Related APIs:** API-065 `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}/validate-for-receipt`, API-066 `POST /api/v1/inbound/goods-receipts`, API-067 `GET /api/v1/inbound/goods-receipts`, API-068 `GET /api/v1/inbound/goods-receipts/{goodsReceiptId}`, API-069 `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/verify`

UI-016

- **Screen name:** Receipt Discrepancy Resolution
- **Purpose:** Record, review, and resolve inbound discrepancies such as shortages, wrong items, damage, or tolerance breaches.
- **Primary users:** Receiving Clerk, Warehouse Manager, Buyer
- **Related use cases:** UC-009
- **Related APIs:** API-070 `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies`, API-071 `GET /api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies`, API-072 `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies/{discrepancyId}/resolve`, API-025 `GET /api/v1/governance/reason-codes/{code}/validate`, API-030 `GET /api/v1/governance/receipt-tolerances`

UI-017

- **Screen name:** Put-Away Workbench
- **Purpose:** Present pending put-away tasks, storage destination choices, and completion confirmation for received stock.
- **Primary users:** Warehouse Staff
- **Related use cases:** UC-010
- **Related APIs:** API-074 `GET /api/v1/inbound/pending-put-away`, API-073 `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/put-away`, API-015 `POST /api/v1/locations/{locationCode}/storage-positions`, API-016 `GET /api/v1/locations/{locationCode}/storage-positions`

UI-018

- **Screen name:** Non-PO and Return Receipt Intake
- **Purpose:** Record authorized non-PO inbound stock and return-related inbound receipts that arrive outside standard supplier PO flow.
- **Primary users:** Receiving Clerk
- **Related use cases:** UC-011
- **Related APIs:** API-075 `POST /api/v1/inbound/ad-hoc-receipts`, API-076 `POST /api/v1/inbound/return-receipts`
- **API note:** API review item `C-04` recommends API-128 as the canonical returns-receipt path for returns UI.

UI-019

- **Screen name:** Inbound Supply Tracking
- **Purpose:** Monitor open purchase orders, expected arrivals, delays, and outstanding inbound quantities for replenishment visibility.
- **Primary users:** Buyer, Purchasing Manager, Warehouse Manager
- **Related use cases:** UC-030
- **Related APIs:** API-060 `GET /api/v1/procurement/purchase-orders`, API-061 `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}`, API-064 `POST /api/v1/procurement/purchase-orders/{purchaseOrderId}/expected-deliveries`, API-107 `GET /api/v1/transfers/in-transit`, API-161 `GET /api/v1/intelligence/procurement/open-pos`, API-162 `GET /api/v1/intelligence/procurement/inbound-calendar`, API-163 `GET /api/v1/intelligence/procurement/inbound-delays`

Inventory Visibility and Control

UI-020

- **Screen name:** Inventory Status Monitor
- **Purpose:** Give network-wide visibility into on-hand, reserved, ATP, low-stock, out-of-stock, overstock, and quarantined conditions.
- **Primary users:** Inventory Controller, Operations Director, Warehouse Manager, Purchasing Manager
- **Related use cases:** UC-012

- **Related APIs:** API-012 `GET /api/v1/locations`, API-033 `GET /api/v1/inventory/stock-positions`, API-034 `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}`, API-037 `GET /api/v1/inventory/stock-status/{locationCode}/{sku}`, API-151 `GET /api/v1/intelligence/stock-status/summary`, API-152 `GET /api/v1/intelligence/stock-status/board`

UI-021

- **Screen name:** Inventory Availability Inquiry
- **Purpose:** Allow sales and service users to quickly search products and confirm ATP and network availability during customer conversations.
- **Primary users:** Sales Representative, Customer Service Representative, Sales Manager
- **Related use cases:** UC-013, UC-021
- **Related APIs:** API-002 `GET /api/v1/products`, API-003 `GET /api/v1/products/{sku}`, API-035 `GET /api/v1/inventory/availability/{sku}`, API-036 `GET /api/v1/inventory/availability/{sku}/network`, API-084 `POST /api/v1/fulfillment/orders/{orderId}/availability`, API-153 `GET /api/v1/intelligence/availability/inquiries`

UI-022

- **Screen name:** Inventory Movement History
- **Purpose:** Trace inventory transactions by product, location, user, type, and date for investigation and operational accountability.
- **Primary users:** Inventory Controller, Warehouse Manager, Finance Manager, Internal Auditor, Operations Director
- **Related use cases:** UC-042
- **Related APIs:** API-155 `GET /api/v1/intelligence/movements`

UI-023

- **Screen name:** Cycle Count Workbench
- **Purpose:** Present assigned blind cycle counts, collect physical count results, and track submission status.
- **Primary users:** Warehouse Staff, Inventory Controller
- **Related use cases:** UC-014
- **Related APIs:** API-108 `POST /api/v1/accuracy/cycle-counts`, API-109 `GET /api/v1/accuracy/cycle-counts`, API-110 `GET /api/v1/accuracy/cycle-counts/{cycleCountId}`, API-111 `POST /api/v1/accuracy/cycle-counts/{cycleCountId}/result`, API-168 `GET /api/v1/intelligence/accuracy/count-performance`

UI-024

- **Screen name:** Physical Inventory Count Management
- **Purpose:** Initiate, monitor, and close wall-to-wall counts including freeze status, team progress, and certification.
- **Primary users:** Inventory Controller, Warehouse Manager, Warehouse Staff, Operations Director
- **Related use cases:** UC-015
- **Related APIs:** API-020 `GET /api/v1/locations/{locationCode}/freeze-status`, API-112 `POST /api/v1/accuracy/physical-counts`, API-113 `GET /api/v1/accuracy/physical-counts/{physicalCountId}`, API-114 `POST /api/v1/accuracy/physical-counts/{physicalCountId}/results`, API-115 `POST /api/v1/accuracy/physical-counts/{physicalCountId}/certify`, API-168 `GET /api/v1/intelligence/accuracy/count-performance`

UI-025

- **Screen name:** Variance Investigation and Approval
- **Purpose:** Review count variances, capture investigation findings, and approve or reject quantity adjustments through proper independent controls.
- **Primary users:** Warehouse Manager, Inventory Controller, Operations Director
- **Related use cases:** UC-016
- **Related APIs:** API-116 `POST /api/v1/accuracy/adjustment-cases`, API-117 `GET /api/v1/accuracy/adjustment-cases/{caseId}`, API-118 `POST /api/v1/accuracy/adjustment-cases/{caseId}/investigate`, API-119 `POST /api/v1/accuracy/adjustment-cases/{caseId}/variances/{varianceId}/approve`, API-123 `GET /api/v1/accuracy/pending-approvals`

UI-026

- **Screen name:** Stock Adjustment Management
- **Purpose:** Submit, approve-route, and finalize stock adjustments with mandatory reason codes and full traceability.

- **Primary users:** Warehouse Manager, Inventory Controller, Operations Director
- **Related use cases:** UC-017
- **Related APIs:** API-023 `GET /api/v1/governance/reason-codes`, API-029 `GET /api/v1/governance/approval-thresholds`, API-116 `POST /api/v1/accuracy/adjustment-cases`, API-117 `GET /api/v1/accuracy/adjustment-cases/{caseId}`, API-120 `POST /api/v1/accuracy/adjustment-cases/{caseId}/submit`, API-121 `POST /api/v1/accuracy/adjustment-cases/{caseId}/approve`, API-122 `POST /api/v1/accuracy/adjustment-cases/{caseId}/reject`

UI-027

- **Screen name:** Write-Off Authorization
- **Purpose:** Review write-off requests, supporting evidence, and finance acknowledgment before permanent inventory removal.
- **Primary users:** Operations Director, Finance Manager, Inventory Controller, Warehouse Manager
- **Related use cases:** UC-018
- **Related APIs:** API-117 `GET /api/v1/accuracy/adjustment-cases/{caseId}`, API-120 `POST /api/v1/accuracy/adjustment-cases/{caseId}/submit`, API-121 `POST /api/v1/accuracy/adjustment-cases/{caseId}/approve`, API-122 `POST /api/v1/accuracy/adjustment-cases/{caseId}/reject`

UI-028

- **Screen name:** Quarantine and Hold Register
- **Purpose:** Apply, review, and release quarantine or hold status while showing non-sellable quantities and reasons.
- **Primary users:** Warehouse Manager, Inventory Controller, Operations Director
- **Related use cases:** UC-019
- **Related APIs:** API-045 `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/quarantine-holds`, API-046 `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/quarantine-holds/{holdId}/release`, API-154 `GET /api/v1/intelligence/quarantine-register`

UI-029

- **Screen name:** Shrinkage, Loss, and Obsolescence Analysis
- **Purpose:** Analyze loss trends, obsolete stock, and slow-moving inventory to support corrective action and working-capital decisions.
- **Primary users:** Inventory Controller, Finance Manager, Operations Director, Purchasing Manager
- **Related use cases:** UC-020
- **Related APIs:** API-167 `GET /api/v1/intelligence/accuracy/shrinkage`

Order Fulfillment

UI-030

- **Screen name:** Order Reservation Workbench
- **Purpose:** Confirm order readiness, reserve inventory, and release reservations when orders are changed or canceled.
- **Primary users:** Customer Service Representative, Sales Representative
- **Related use cases:** UC-021, UC-022, UC-023
- **Related APIs:** API-081 `POST /api/v1/fulfillment/orders`, API-084 `POST /api/v1/fulfillment/orders/{orderId}/availability`, API-085 `POST /api/v1/fulfillment/orders/{orderId}/confirm`, API-086 `POST /api/v1/fulfillment/orders/{orderId}/cancel`, API-043 `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/reservations`, API-044 `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/reservations/{reservationId}/release`, API-139 `GET /api/v1/campaigns/ring-fence/{locationCode}/{sku}`

UI-031

- **Screen name:** Pick, Pack, and Ship Workbench
- **Purpose:** Show warehouse fulfillment queues, guide pick verification and packing, and confirm shipment handoff.
- **Primary users:** Warehouse Picker / Packer, Warehouse Manager
- **Related use cases:** UC-024

- **Related APIs:** API-088 `POST /api/v1/fulfillment/orders/{orderId}/pick-lists`, API-089 `GET /api/v1/fulfillment/orders/{orderId}/pick-lists/awaiting`, API-090 `POST /api/v1/fulfillment/orders/{orderId}/picks/{pickId}/verify`, API-091 `POST /api/v1/fulfillment/orders/{orderId}/pack`, API-092 `POST /api/v1/fulfillment/orders/{orderId}/shipments`, API-041 `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/lots`
- **API note:** API review item `C-01` identifies the missing location-level pick queue endpoint required for a complete warehouse workbench.

UI-032

- **Screen name:** Partial Fulfillment and Backorder Management
- **Purpose:** Manage split shipments, customer acknowledgment, and backorder follow-up when full quantity is unavailable.
- **Primary users:** Customer Service Representative, Sales Representative, Purchasing Manager
- **Related use cases:** UC-025
- **Related APIs:** API-093 `POST /api/v1/fulfillment/orders/{orderId}/partial-fulfillment`, API-094 `POST /api/v1/fulfillment/orders/{orderId}/backorders`, API-095 `POST /api/v1/fulfillment/backorders/{backorderId}/fulfill`, API-164 `GET /api/v1/intelligence/fulfillment/backorders`

UI-033

- **Screen name:** Fulfillment Exception Management
- **Purpose:** Resolve pick failures, quantity mismatches, damaged picks, or other outbound exceptions and route the order back into the correct flow.
- **Primary users:** Warehouse Manager, Customer Service Representative
- **Related use cases:** UC-026
- **Related APIs:** API-096 `POST /api/v1/fulfillment/orders/{orderId}/exceptions`, API-165 `GET /api/v1/intelligence/fulfillment/exceptions`, API-045 `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/quarantine-holds`

UI-034

- **Screen name:** Order Fulfillment Status
- **Purpose:** Provide current reservation, picking, packing, shipment, exception, and backorder status for order inquiries.
- **Primary users:** Customer Service Representative, Sales Representative, Sales Manager, Warehouse Manager
- **Related use cases:** UC-027
- **Related APIs:** API-082 `GET /api/v1/fulfillment/orders`, API-083 `GET /api/v1/fulfillment/orders/{orderId}`, API-097 `GET /api/v1/fulfillment/orders/{orderId}/status`, API-177 `GET /api/v1/integrations/sales/fulfillment-status`

UI-035

- **Screen name:** Multi-Location Split Fulfillment Planner
- **Purpose:** Plan and monitor fulfillment of a single order from multiple locations when no single location can satisfy it alone.
- **Primary users:** Warehouse Manager, Customer Service Representative, Operations Director
- **Related use cases:** UC-048
- **Related APIs:** API-036 `GET /api/v1/inventory/availability/{sku}/network`, API-087 `POST /api/v1/fulfillment/orders/{orderId}/multi-location-plan`

Replenishment and Transfers

UI-036

- **Screen name:** Low-Stock Alert Review
- **Purpose:** Review active replenishment alerts, prioritize them, and acknowledge follow-up actions.
- **Primary users:** Purchasing Manager, Buyer, Inventory Controller, Operations Director
- **Related use cases:** UC-028
- **Related APIs:** API-037 `GET /api/v1/inventory/stock-status/{locationCode}/{sku}`, API-039 `GET /api/v1/inventory/stock-policies/{locationCode}/{sku}`, API-158 `GET /api/v1/intelligence/alerts`, API-159 `POST /api/v1/intelligence/alerts/{alertId}/acknowledge`, API-160 `GET /api/v1/intelligence/replenishment/recommendations`

UI-037

- **Screen name:** Replenishment Planning
- **Purpose:** Turn alerts and recommendations into replenishment decisions, including transfer-before-buy review and requisition creation.
- **Primary users:** Buyer, Purchasing Manager
- **Related use cases:** UC-029
- **Related APIs:** API-054 `POST /api/v1/procurement/requisitions`, API-055 `GET /api/v1/procurement/requisitions/{requisitionId}`, API-056 `POST /api/v1/procurement/requisitions/{requisitionId}/approve`, API-057 `POST /api/v1/procurement/requisitions/{requisitionId}/reject`, API-058 `POST /api/v1/procurement/requisitions/{requisitionId}/convert-to-po`, API-160 `GET /api/v1/intelligence/replenishment/recommendations`, API-166 `GET /api/v1/intelligence/network/imbalance`
- **API note:** Full list/queue browsing depends on the missing requisition list endpoint noted in API review item `C-03`.

UI-038

- **Screen name:** Transfer Request
- **Purpose:** Submit inter-location transfer requests with source, destination, product, quantity, and business justification.
- **Primary users:** Regional / Store Manager, Warehouse Manager
- **Related use cases:** UC-031
- **Related APIs:** API-098 `POST /api/v1/transfers`, API-099 `GET /api/v1/transfers`, API-100 `GET /api/v1/transfers/{transferId}`, API-042 `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/transfer-capacity`

UI-039

- **Screen name:** Transfer Approval Queue
- **Purpose:** Allow management to approve or reject transfer requests based on source capacity and network impact.
- **Primary users:** Operations Director
- **Related use cases:** UC-032
- **Related APIs:** API-101 `POST /api/v1/transfers/{transferId}/approve`, API-102 `POST /api/v1/transfers/{transferId}/reject`, API-106 `GET /api/v1/transfers/pending-approvals`, API-042 `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/transfer-capacity`

UI-040

- **Screen name:** Transfer Execution and In-Transit Tracking
- **Purpose:** Dispatch approved transfers, receive them at destination, and monitor in-transit stock movement.
- **Primary users:** Warehouse Staff, Warehouse Manager, Regional / Store Manager
- **Related use cases:** UC-033
- **Related APIs:** API-099 `GET /api/v1/transfers`, API-100 `GET /api/v1/transfers/{transferId}`, API-103 `POST /api/v1/transfers/{transferId}/dispatch`, API-104 `POST /api/v1/transfers/{transferId}/receive`, API-105 `POST /api/v1/transfers/{transferId}/cancel`, API-107 `GET /api/v1/transfers/in-transit`

UI-041

- **Screen name:** Network Rebalancing Analysis
- **Purpose:** Analyze stock imbalance across locations and generate recommended redistribution actions.
- **Primary users:** Operations Director, Inventory Controller, Purchasing Manager
- **Related use cases:** UC-034
- **Related APIs:** API-036 `GET /api/v1/inventory/availability/{sku}/network`, API-160 `GET /api/v1/intelligence/replenishment/recommendations`, API-166 `GET /api/v1/intelligence/network/imbalance`

Returns

UI-042

- **Screen name:** Return Authorization Management
- **Purpose:** Look up original orders, validate return eligibility, and issue return authorizations.
- **Primary users:** Customer Service Representative
- **Related use cases:** UC-035

- **Related APIs:** API-124 `POST /api/v1/returns/authorizations`, API-125 `GET /api/v1/returns/authorizations`, API-126 `GET /api/v1/returns/authorizations/{returnAuthorizationId}`, API-127 `POST /api/v1/returns/authorizations/{returnAuthorizationId}/cancel`

UI-043

- **Screen name:** Return Inspection and Disposition
- **Purpose:** Receive authorized returns, inspect their condition, and route them to restock, quarantine, or write-off.
- **Primary users:** Receiving Clerk, Warehouse Manager, Inventory Controller, Finance Manager
- **Related use cases:** UC-036
- **Related APIs:** API-128 `POST /api/v1/returns/{returnId}/receive`, API-129 `POST /api/v1/returns/{returnId}/inspect`, API-130 `POST /api/v1/returns/{returnId}/disposition`, API-131 `GET /api/v1/returns/pending-receipts`, API-076 `POST /api/v1/inbound/return-receipts`
- **API note:** API review item `C-04` recommends API-128 as the canonical receipt path for this screen.

Reporting, Finance, Audit, and Administration

UI-044

- **Screen name:** Operational Reports and Analytics
- **Purpose:** Access dashboard and operational reports for stock movement, stockout, overstock, accuracy, location performance, and related management analysis.
- **Primary users:** Operations Director, Chief Operating Officer, Inventory Controller, Warehouse Manager, Purchasing Manager, Sales Manager
- **Related use cases:** UC-037
- **Related APIs:** API-150 `GET /api/v1/intelligence/dashboard`, API-151 `GET /api/v1/intelligence/stock-status/summary`, API-152 `GET /api/v1/intelligence/stock-status/board`, API-154 `GET /api/v1/intelligence/quarantine-register`, API-155 `GET /api/v1/intelligence/movements`, API-167 `GET /api/v1/intelligence/accuracy/shrinkage`, API-168 `GET /api/v1/intelligence/accuracy/count-performance`

UI-045

- **Screen name:** Inventory Valuation Reconciliation
- **Purpose:** Compare operational valuation with finance records, investigate variances, and finalize reconciliation status.
- **Primary users:** Finance Manager, Chief Financial Officer, Inventory Controller, Internal Auditor
- **Related use cases:** UC-039
- **Related APIs:** API-077 `POST /api/v1/inbound/supplier-invoices`, API-078 `GET /api/v1/inbound/supplier-invoices/{invoiceId}`, API-079 `POST /api/v1/inbound/supplier-invoices/{invoiceId}/three-way-match`, API-080 `POST /api/v1/inbound/supplier-invoices/{invoiceId}/authorize-payment`, API-047 `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/consignment-ownership`, API-171 `GET /api/v1/intelligence/valuation/snapshots`, API-172 `GET /api/v1/intelligence/valuation/reconciliation`

UI-046

- **Screen name:** Promotional Stock Readiness
- **Purpose:** Allocate and ring-fence promotional stock, assess readiness, and confirm launch stock coverage.
- **Primary users:** Marketing Manager, Purchasing Manager, Warehouse Manager, Sales Manager, Operations Director
- **Related use cases:** UC-040
- **Related APIs:** API-132 `POST /api/v1/campaigns/allocations`, API-133 `GET /api/v1/campaigns/allocations`, API-134 `GET /api/v1/campaigns/allocations/{allocationId}`, API-135 `PATCH /api/v1/campaigns/allocations/{allocationId}`, API-136 `POST /api/v1/campaigns/allocations/{allocationId}/readiness`, API-137 `POST /api/v1/campaigns/allocations/{allocationId}/release`, API-138 `POST /api/v1/campaigns/allocations/{allocationId}/override`, API-139 `GET /api/v1/campaigns/ring-fence/{locationCode}/{sku}`, API-169 `GET /api/v1/intelligence/campaigns/readiness`, API-178 `POST /api/v1/integrations/marketing/campaign-plans`

UI-047

- **Screen name:** Audit and Control Review Workspace
- **Purpose:** Let auditors define review scope, inspect transaction and approval populations, test control compliance, and document findings.

- **Primary users:** Internal Auditor, External Auditor, Finance Manager
- **Related use cases:** UC-041
- **Related APIs:** API-156 `GET /api/v1/intelligence/audit-trail`, API-157 `GET /api/v1/intelligence/audit-trail/export`

UI-048

- **Screen name:** Audit Trail Query
- **Purpose:** Search immutable audit records by entity, date, user, and action and inspect before/after values.
- **Primary users:** Internal Auditor, External Auditor, Finance Manager, Operations Director
- **Related use cases:** UC-043
- **Related APIs:** API-156 `GET /api/v1/intelligence/audit-trail`, API-157 `GET /api/v1/intelligence/audit-trail/export`

UI-049

- **Screen name:** User Account Management
- **Purpose:** Provision, modify, suspend, deactivate, and review user accounts and role assignments.
- **Primary users:** System Administrator
- **Related use cases:** UC-044
- **Related APIs:** API-140 `POST /api/v1/users`, API-141 `GET /api/v1/users`, API-142 `GET /api/v1/users/{userId}`, API-143 `POST /api/v1/users/{userId}/roles`, API-144 `DELETE /api/v1/users/{userId}/roles/{roleAssignmentId}`, API-145 `POST /api/v1/users/{userId}/suspend`, API-146 `POST /api/v1/users/{userId}/deactivate`, API-147 `POST /api/v1/users/sync`

UI-050

- **Screen name:** Role and Access Control Administration
- **Purpose:** Configure role permissions, location scope rules, and access policy enforcement behavior.
- **Primary users:** System Administrator, Internal Auditor
- **Related use cases:** UC-045
- **Related APIs:** API-031 `POST /api/v1/governance/segregation-of-duties/evaluate`, API-148 `POST /api/v1/authorization/actions/evaluate`, API-149 `POST /api/v1/authorization/location-scope/validate`, API-173 `GET /api/v1/auth/session`

Summary

- **Total screens identified:** 50
- **Coverage basis:** BRD scope and functional requirements, 48 use cases, and the REST API catalog
- **Design status:** Screen inventory only; no layout, navigation, or wireframes defined in this document

Inventory Management System Component Inventory

Purpose

This document defines the reusable UI component inventory for the Inventory Management System. It focuses on shared design elements that can be used across multiple screens, modules, and user roles to support consistency, efficiency, and maintainability.

Scope

The inventory is derived from the screen catalog, navigation structure, user journeys, user flows, and low-fidelity wireframe specifications. It emphasizes reusable components rather than one-off page content.

For each component, this document provides:

1. Component name
2. Purpose
3. Reusability
4. Variants

5. Related screens

Global Layout Components

C-001 Global Header

- **Purpose:** Provide persistent access to product identity, global search, notifications, tasks, and user account controls.
- **Reusability:** Global; used across nearly all authenticated screens.
- **Variants:** Standard operational header, executive header, admin-focused header, unauthenticated header.
- **Related screens:** `UI-002` to `UI-050`, plus simplified use on `UI-001`.

C-002 Main Navigation Bar

- **Purpose:** Support movement between primary business modules.
- **Reusability:** Global; reused across all authenticated screens.
- **Variants:** Full menu, role-filtered menu, collapsed menu, highlighted-current-module state.
- **Related screens:** `UI-002` to `UI-050`.

C-003 Secondary Navigation Panel

- **Purpose:** Organize module-level sections and workflows inside the selected business domain.
- **Reusability:** High; reused by all module-driven screens.
- **Variants:** Vertical sidebar, horizontal tabs, accordion section list.
- **Related screens:** `UI-005` to `UI-050`.

C-004 Breadcrumb Trail

- **Purpose:** Show hierarchical location and allow quick return to parent levels.
- **Reusability:** Global on all non-home authenticated screens.
- **Variants:** Standard breadcrumb, breadcrumb with final record ID, breadcrumb with action state.
- **Related screens:** `UI-005` to `UI-050`.

C-005 Page Title Bar

- **Purpose:** Provide a page title, context subtitle, primary action, and optional status indicator.
- **Reusability:** Very high; present on most application screens.
- **Variants:** Simple title bar, title with primary CTA, title with status badge, title with filter controls.
- **Related screens:** `UI-002` to `UI-050`.

C-006 Footer Metadata Strip

- **Purpose:** Show system metadata such as timestamps, versioning, refresh state, or record audit information.
- **Reusability:** High; useful on dashboards, workbenches, and record detail screens.
- **Variants:** Page footer, report footer, record audit footer, lightweight timestamp-only footer.
- **Related screens:** `UI-001` to `UI-050`.

Navigation and Utility Components

C-007 Global Search Bar

- **Purpose:** Let users quickly search products, orders, transfers, users, or other indexed entities from anywhere.
- **Reusability:** High; shared utility in the header and some workbench screens.
- **Variants:** Header search, contextual scoped search, autocomplete search, advanced search modal.
- **Related screens:** `UI-002`, `UI-005`, `UI-020`, `UI-021`, `UI-030`, `UI-034`, `UI-049`.

C-008 Notification Icon and Inbox Trigger

- **Purpose:** Surface pending alerts and open the notification center quickly.

- **Reusability:** Global utility for all operational roles.
- **Variants:** Header icon with count badge, inline notification shortcut, compact task-alert widget.
- **Related screens:** `UI-002`, `UI-004`, `UI-036`, `UI-039`, `UI-047`.

C-009 User Menu

- **Purpose:** Provide session access, profile/context display, and sign-out actions.
- **Reusability:** Global.
- **Variants:** Minimal profile menu, profile with role/location context, admin profile menu.
- **Related screens:** `UI-002` to `UI-050`.

C-010 Quick Action Tile

- **Purpose:** Offer shortcut navigation from dashboards to frequently used tasks.
- **Reusability:** Medium to high; mainly used on dashboards and home pages.
- **Variants:** Icon tile, KPI-linked tile, role-priority shortcut tile.
- **Related screens:** `UI-002`, `UI-003`.

C-011 Task / Approval Shortcut Widget

- **Purpose:** Summarize work awaiting the current user and route them directly into task queues.
- **Reusability:** High for operational and managerial roles.
- **Variants:** Approvals summary, task summary, mixed workload widget.
- **Related screens:** `UI-002`, `UI-004`, `UI-025`, `UI-039`, `UI-047`.

Filter, Search, and Query Components

C-012 Filter Bar

- **Purpose:** Narrow lists, reports, dashboards, and queues by key criteria.
- **Reusability:** Very high; used across list and analytics screens.
- **Variants:** Inline filter row, expandable advanced filters, sticky filter bar, compact dashboard filters.
- **Related screens:** `UI-002`, `UI-003`, `UI-004`, `UI-005`, `UI-019`, `UI-020`, `UI-022`, `UI-029`, `UI-036`, `UI-041`, `UI-044`, `UI-048`.

C-013 Date Range Picker

- **Purpose:** Select periods for dashboards, reports, analytics, audit, and reconciliation workflows.
- **Reusability:** High.
- **Variants:** Preset range selector, custom date range, single-period selector.
- **Related screens:** `UI-003`, `UI-019`, `UI-029`, `UI-044`, `UI-045`, `UI-047`, `UI-048`.

C-014 Search Results List

- **Purpose:** Present matching records after a search action in a structured, scannable way.
- **Reusability:** High.
- **Variants:** Product search results, order results, user results, audit query results.
- **Related screens:** `UI-005`, `UI-021`, `UI-034`, `UI-042`, `UI-049`, `UI-048`.

C-015 Saved Filter / View Selector

- **Purpose:** Allow users to switch among common views such as open items, urgent items, my location, or all network.
- **Reusability:** Medium to high in operational systems.
- **Variants:** Dropdown saved views, segmented buttons, tabbed saved views.
- **Related screens:** `UI-004`, `UI-020`, `UI-036`, `UI-039`, `UI-044`.

Data Display Components

C-016 KPI Summary Card

- **Purpose:** Display a key metric with value, trend, status, and optional target comparison.
- **Reusability:** Very high across dashboards and reporting.
- **Variants:** Operational KPI card, executive KPI card, status count card, alert count card.
- **Related screens:** `UI-002`, `UI-003`, `UI-020`, `UI-024`, `UI-029`, `UI-045`, `UI-046`.

C-017 Status Badge

- **Purpose:** Communicate record state clearly using short labels and color/state semantics.
- **Reusability:** Global.
- **Variants:** Stock status, workflow status, approval status, audit status, campaign readiness status.
- **Related screens:** Used widely across `UI-005` to `UI-050`.

C-018 Summary Statistic Strip

- **Purpose:** Group several related top-line metrics in a compact horizontal summary region.
- **Reusability:** High for dashboards, workbenches, and record detail screens.
- **Variants:** Dashboard summary strip, record summary strip, queue summary strip.
- **Related screens:** `UI-002`, `UI-015`, `UI-020`, `UI-024`, `UI-030`, `UI-040`, `UI-046`.

C-019 Data Table

- **Purpose:** Display structured records with sorting, filtering, selection, and row actions.
- **Reusability:** Core reusable component across almost all modules.
- **Variants:** Simple table, selectable table, expandable row table, sticky-header table, paginated table.
- **Related screens:** Nearly all list-based screens including `UI-005`, `UI-012`, `UI-015`, `UI-020`, `UI-034`, `UI-049`.

C-020 Table Row Action Menu

- **Purpose:** Provide compact actions for a record without overcrowding the table.
- **Reusability:** Very high.
- **Variants:** Kebab menu, inline text actions, hover action set.
- **Related screens:** All screens using `C-019`, especially `UI-005`, `UI-012`, `UI-036`, `UI-039`, `UI-049`.

C-021 Detail Panel

- **Purpose:** Show detailed information for a selected row or record without leaving the current context.
- **Reusability:** High in list-detail workflows.
- **Variants:** Right-side drawer, bottom panel, inline expandable detail.
- **Related screens:** `UI-005`, `UI-007`, `UI-015`, `UI-016`, `UI-025`, `UI-039`, `UI-049`.

C-022 Comparison Panel

- **Purpose:** Help users compare two or more states, such as expected vs actual, source vs destination, or operational vs finance.
- **Reusability:** Medium but important for business-critical flows.
- **Variants:** Side-by-side columns, stacked comparison card, split-table comparison.
- **Related screens:** `UI-016`, `UI-025`, `UI-032`, `UI-039`, `UI-045`.

C-023 Record Metadata Block

- **Purpose:** Display created by, updated by, timestamps, reference IDs, and versioning.
- **Reusability:** Very high on detail pages.
- **Variants:** Inline metadata row, side metadata card, footer metadata strip.
- **Related screens:** `UI-005`, `UI-012`, `UI-015`, `UI-026`, `UI-034`, `UI-042`, `UI-049`.

Form Components

C-024 Standard Form Section

- **Purpose:** Group related input fields into a clear, reusable form structure.
- **Reusability:** Very high.
- **Variants:** Single-column form, two-column form, stepped form, grouped accordion form.
- **Related screens:** Most create/edit screens including `UI-005`, `UI-007`, `UI-010`, `UI-012`, `UI-026`, `UI-049`.

C-025 Field Group Header

- **Purpose:** Label and separate major form sections such as header details, line items, approvals, or notes.
- **Reusability:** High.
- **Variants:** Plain section heading, collapsible section heading, heading with helper text.
- **Related screens:** `UI-005`, `UI-012`, `UI-015`, `UI-017`, `UI-026`, `UI-043`, `UI-049`.

C-026 Text Input

- **Purpose:** Capture short free-text data such as names, codes, and references.
- **Reusability:** Universal.
- **Variants:** Single-line, masked input, searchable text input, read-only text field.
- **Related screens:** Used across almost all form-based screens.

C-027 Text Area / Notes Field

- **Purpose:** Capture longer comments, rationale, investigation notes, or exception details.
- **Reusability:** High.
- **Variants:** Plain notes field, expandable text area, rich note field with character count.
- **Related screens:** `UI-010`, `UI-016`, `UI-025`, `UI-026`, `UI-027`, `UI-033`, `UI-045`, `UI-047`.

C-028 Dropdown / Select Control

- **Purpose:** Let users choose from predefined options such as status, role, location, reason code, or supplier.
- **Reusability:** Universal.
- **Variants:** Single-select, multi-select, searchable dropdown, grouped dropdown.
- **Related screens:** Broad use across all forms.

C-029 Checkbox Group

- **Purpose:** Support multiple non-exclusive selections or confirmations.
- **Reusability:** Medium.
- **Variants:** Inline checkbox list, stacked checkbox list, checkbox with help text.
- **Related screens:** `UI-001`, `UI-010`, `UI-024`, `UI-046`, `UI-050`.

C-030 Radio Group / Segmented Choice

- **Purpose:** Support mutually exclusive selections where quick comparison matters.
- **Reusability:** High.
- **Variants:** Traditional radio group, segmented control, card-select radio set.
- **Related screens:** `UI-018`, `UI-026`, `UI-028`, `UI-035`, `UI-043`.

C-031 Number Input

- **Purpose:** Capture quantities, thresholds, counts, values, and numeric adjustments.
- **Reusability:** Very high in inventory workflows.
- **Variants:** Integer field, decimal field, quantity-with-unit, plus/minus stepper.
- **Related screens:** `UI-008`, `UI-012`, `UI-015`, `UI-023`, `UI-026`, `UI-032`, `UI-038`, `UI-043`.

C-032 Date / Date-Time Field

- **Purpose:** Capture effective dates, expected delivery dates, reporting periods, and session times.
- **Reusability:** High.
- **Variants:** Date picker, date-time picker, read-only timestamp field.
- **Related screens:** `UI-008`, `UI-012`, `UI-014`, `UI-024`, `UI-037`, `UI-045`, `UI-046`.

C-033 File Upload Control

- **Purpose:** Support import files, evidence attachments, and supporting documents.
- **Reusability:** Medium to high.
- **Variants:** Drag-and-drop upload, button upload, evidence attachment upload with file list.
- **Related screens:** `UI-006`, `UI-016`, `UI-025`, `UI-027`, `UI-047`.

C-034 Validation Message Block

- **Purpose:** Show field-level or form-level validation feedback before submission.
- **Reusability:** Universal.
- **Variants:** Inline field error, form summary banner, warning advisory block.
- **Related screens:** All form-based screens.

C-035 Read-Only Review Form

- **Purpose:** Present captured data in review mode before confirmation, approval, or publication.
- **Reusability:** High.
- **Variants:** Review summary card, approval review panel, final confirmation screen section.
- **Related screens:** `UI-006`, `UI-013`, `UI-015`, `UI-027`, `UI-038`, `UI-042`.

Workflow and Transaction Components

C-036 Line Item Editor

- **Purpose:** Allow users to add, edit, and review multiple structured items within one transaction.
- **Reusability:** Very high for orders, POs, receipts, and returns.
- **Variants:** Inline editable table, add-row editor, master-detail line editor.
- **Related screens:** `UI-012`, `UI-015`, `UI-030`, `UI-034`, `UI-042`, `UI-043`.

C-037 Queue List

- **Purpose:** Present task-based operational records waiting for user action.
- **Reusability:** Very high in warehouse and approval workflows.
- **Variants:** Pick queue, receipt queue, approvals queue, alerts queue, returns queue.
- **Related screens:** `UI-015`, `UI-017`, `UI-023`, `UI-025`, `UI-031`, `UI-033`, `UI-039`, `UI-043`.

C-038 Step Progress Indicator

- **Purpose:** Show where the user is in a multi-step workflow such as import, receiving, packing, or return processing.
- **Reusability:** Medium to high.
- **Variants:** Horizontal steps, vertical progress steps, checklist progress.
- **Related screens:** `UI-006`, `UI-015`, `UI-031`, `UI-043`, `UI-047`.

C-039 Approval Decision Panel

- **Purpose:** Standardize approve/reject workflows with decision note capture.
- **Reusability:** High for governance-heavy processes.
- **Variants:** Binary approve/reject, approve/reject/escalate, approve with threshold warning.
- **Related screens:** `UI-013`, `UI-025`, `UI-027`, `UI-039`.

C-040 Status Timeline / History

- **Purpose:** Display the chronological progression of a transaction or workflow.
- **Reusability:** High across operational records.
- **Variants:** Vertical timeline, status history table, compact event log.
- **Related screens:** `UI-013`, `UI-015`, `UI-030`, `UI-034`, `UI-038`, `UI-042`, `UI-043`.

C-041 Alert Detail Card

- **Purpose:** Present key alert context, severity, and recommended next action.
- **Reusability:** High in notification and replenishment workflows.
- **Variants:** Low-stock alert card, exception alert card, delay alert card, readiness alert card.
- **Related screens:** `UI-004`, `UI-036`, `UI-037`, `UI-046`.

C-042 Recommendation Card

- **Purpose:** Present system-identified next-best actions such as replenishment, transfer, or rebalancing.
- **Reusability:** Medium to high.
- **Variants:** Replenishment recommendation, transfer recommendation, network rebalance recommendation.
- **Related screens:** `UI-037`, `UI-041`.

C-043 Confirmation Dialog

- **Purpose:** Prevent accidental execution of sensitive or irreversible actions.
- **Reusability:** Universal for critical actions.
- **Variants:** Delete/deactivate confirmation, approve/reject confirmation, ship/dispatch confirmation, write-off confirmation.
- **Related screens:** `UI-007`, `UI-012`, `UI-026`, `UI-027`, `UI-031`, `UI-039`, `UI-049`.

C-044 Success / Warning / Error Banner

- **Purpose:** Provide immediate feedback after a user action or system event.
- **Reusability:** Universal.
- **Variants:** Success banner, warning banner, blocking error banner, informational banner.
- **Related screens:** Broad use across all transactional screens.

Reporting and Analytics Components

C-045 Report Viewer

- **Purpose:** Display tabular or summarized report output with filtering and export capability.
- **Reusability:** High in analytics and management review.
- **Variants:** Operational report viewer, audit evidence viewer, valuation report viewer.
- **Related screens:** `UI-044`, `UI-045`, `UI-047`, `UI-048`.

C-046 Export Action Group

- **Purpose:** Standardize export options for reports, audit results, dashboards, and movement histories.
- **Reusability:** High.
- **Variants:** Export CSV, export PDF, export evidence package, print-friendly export.
- **Related screens:** `UI-003`, `UI-022`, `UI-029`, `UI-044`, `UI-045`, `UI-047`, `UI-048`.

C-047 Trend Indicator

- **Purpose:** Show whether a metric is improving, declining, or stable relative to target or previous period.
- **Reusability:** High in dashboards and reports.
- **Variants:** Up/down arrow, percentage delta, RAG trend chip.
- **Related screens:** `UI-002`, `UI-003`, `UI-029`, `UI-044`, `UI-045`, `UI-046`.

C-048 Variance Highlight Block

- **Purpose:** Emphasize differences between actual and expected values in operational or finance contexts.
- **Reusability:** Medium to high.
- **Variants:** Count variance block, receipt discrepancy block, valuation variance block.
- **Related screens:** `UI-016`, `UI-025`, `UI-045`.

C-049 Drill-Down Link Group

- **Purpose:** Help users move from summary insights into the underlying records driving them.
- **Reusability:** High for analytical screens.
- **Variants:** KPI drill-down links, report-to-record links, alert-to-action links.
- **Related screens:** `UI-002`, `UI-003`, `UI-020`, `UI-029`, `UI-044`, `UI-046`.

Identity, Access, and Audit Components

C-050 Role Assignment Widget

- **Purpose:** Assign one or more business roles and scope them to locations or permissions.
- **Reusability:** High in admin workflows.
- **Variants:** Simple role picker, role plus location-scope matrix, effective-role summary.
- **Related screens:** `UI-049`, `UI-050`.

C-051 Permission Matrix

- **Purpose:** Show or edit the mapping between roles, permission groups, and scope.
- **Reusability:** High for access control management.
- **Variants:** Read-only matrix, editable matrix, compare-two-roles matrix.
- **Related screens:** `UI-050`.

C-052 Access Scope Selector

- **Purpose:** Specify the organizational or location boundaries for a role or action.
- **Reusability:** Medium to high.
- **Variants:** Single-location selector, multi-location selector, network-wide scope selector.
- **Related screens:** `UI-049`, `UI-050`, optionally `UI-001` and `UI-002` for context display.

C-053 Audit Record Viewer

- **Purpose:** Display before/after values, actor identity, timestamp, and reason reference for an auditable event.
- **Reusability:** High in audit, governance, and exception review.
- **Variants:** Inline audit record, modal audit record viewer, split before/after viewer.
- **Related screens:** `UI-047`, `UI-048`, and supporting detail usage in `UI-025`, `UI-026`, `UI-049`.

C-054 Evidence Attachment Panel

- **Purpose:** Collect and display uploaded evidence linked to discrepancies, write-offs, investigations, and audits.
- **Reusability:** Medium to high.
- **Variants:** File list with thumbnails, document list with uploader metadata, attachment upload panel.
- **Related screens:** `UI-016`, `UI-025`, `UI-027`, `UI-047`.

C-055 Session / Context Badge

- **Purpose:** Display the current role, location scope, or operating context to reduce user mistakes.
- **Reusability:** High for multi-role and location-scoped systems.
- **Variants:** Role badge, location badge, combined role-location badge.
- **Related screens:** `UI-002`, `UI-015`, `UI-017`, `UI-023`, `UI-031`, `UI-040`, `UI-049`.

Selection and State Components

C-056 Empty State Panel

- **Purpose:** Guide users when no records, tasks, alerts, or results are available.
- **Reusability:** Universal.
- **Variants:** No search results, no tasks, no alerts, no data for selected period.
- **Related screens:** All list, queue, and report screens.

C-057 Loading State / Skeleton

- **Purpose:** Show that data is being retrieved while preserving page structure.
- **Reusability:** Universal.
- **Variants:** Table skeleton, card skeleton, detail panel skeleton, dashboard skeleton.
- **Related screens:** All data-driven screens.

C-058 Bulk Selection Toolbar

- **Purpose:** Support actions on multiple selected rows at once.
- **Reusability:** Medium; useful for admin and operational batch tasks.
- **Variants:** Bulk acknowledge, bulk export, bulk classification update, bulk approve/reject.
- **Related screens:** `UI-004`, `UI-009`, `UI-036`, `UI-049`.

C-059 Toggle / Switch Control

- **Purpose:** Support fast binary settings such as active/inactive, include/exclude, or network/local view.
- **Reusability:** High.
- **Variants:** Status switch, view toggle, scope toggle.
- **Related screens:** `UI-005`, `UI-007`, `UI-010`, `UI-020`, `UI-049`, `UI-050`.

C-060 Tabs / Segmented Content Switcher

- **Purpose:** Let users switch between related content views without leaving the page.
- **Reusability:** High.
- **Variants:** Detail tabs, record subview tabs, mode switcher.
- **Related screens:** `UI-010`, `UI-034`, `UI-045`, `UI-047`.

Summary

- **Total reusable components identified:** 60
- **Primary focus:** Shared structural, workflow, form, data display, and governance components
- **Design intent:** Promote consistency across inventory, procurement, warehouse, fulfillment, reporting, and administration experiences

Recommended Next Use

This component inventory should be used as the baseline for:

- A design system starter library
- Component-level wireframes
- State definitions for validation, loading, empty, and error behavior
- UX acceptance criteria for reusable UI patterns

Inventory Management System Navigation Structure

Purpose

This document defines the application's navigation structure for the Inventory Management System. It is based on the approved BRD, use cases, API design, and the screen inventory. It describes how users move through the application but does not define visual layout or page wireframes.

Navigation Model

- **Primary navigation:** Persistent top-level modules used to move across major business domains.
- **Secondary navigation:** Section-level navigation within a selected module.
- **Breadcrumbs:** Hierarchical path showing where the user is and enabling quick backtracking.
- **Contextual navigation:** In-page task links, related records, status-driven actions, and work queues.
- **Role-specific menus:** Navigation visibility and priority adapted to each user role.

Main Navigation

The main navigation should present the following top-level modules:

1. `Home`
2. `Inventory`
3. `Products`
4. `Locations`
5. `Procurement`
6. `Inbound`
7. `Fulfillment`
8. `Transfers`
9. `Returns`
10. `Reports & Analytics`
11. `Governance & Audit`
12. `Administration`

Supporting persistent utilities:

- `Notifications`
- `Global Search`
- `My Tasks / Approvals`
- `User Menu / Session`

Secondary Navigation

Secondary navigation should appear after a user enters a main module.

Home

- `Operational Dashboard`
- `Executive KPI Dashboard`
- `My Notifications`
- `My Approvals`

Inventory

- `Inventory Status Monitor`
- `Availability Inquiry`

- Movement History
- Cycle Counts
- Physical Counts
- Variance Approvals
- Stock Adjustments
- Write-Offs
- Quarantine & Holds
- Shrinkage & Obsolescence

Products

- Product Catalog
- Product Import
- ABC Classification
- Stock Policies

Locations

- Location Directory
- Location Hierarchy
- Storage Positions
- Location Activation / Closure

Procurement

- Suppliers
- Purchase Requisitions
- Purchase Orders
- Inbound Delivery Planning
- Inbound Supply Tracking
- Low-Stock Alerts
- Replenishment Planning

Inbound

- Goods Receipt Workbench
- Receipt Discrepancies
- Put-Away Workbench
- Non-PO Receipt Intake

Fulfillment

- Order Reservation
- Pick / Pack / Ship
- Partial Fulfillment & Backorders
- Fulfillment Exceptions
- Order Status
- Multi-Location Fulfillment

Transfers

- Transfer Requests
- Transfer Approvals
- Transfer Execution
- In-Transit Tracking
- Network Rebalancing

Returns

- Return Authorizations

- `Return Inspection & Disposition`

Reports & Analytics

- `Operational Reports`
- `Inventory Dashboard`
- `KPI Dashboard`
- `Valuation Reconciliation`
- `Promotional Readiness`

Governance & Audit

- `Governance Policies`
- `Reason Codes`
- `Audit Workspace`
- `Audit Trail Query`
- `Approvals Inbox`

Administration

- `User Accounts`
- `Roles & Access Control`

Breadcrumbs

Breadcrumbs should reflect the navigation hierarchy in the format:

```
Home > Module > Section > Record / Action
```

Examples:

- `Home > Inventory > Inventory Status Monitor`
- `Home > Products > Product Catalog > SKU-100245`
- `Home > Procurement > Purchase Orders > PO-2026-00418`
- `Home > Inbound > Goods Receipt Workbench > GR-2026-00291`
- `Home > Fulfillment > Order Status > SO-2026-01872`
- `Home > Transfers > Transfer Requests > TR-2026-00088`
- `Home > Returns > Return Authorizations > RA-2026-00116`
- `Home > Governance & Audit > Audit Trail Query > Export Results`
- `Home > Administration > User Accounts > User Profile > Edit Role Assignments`

Breadcrumb behavior:

- The first crumb should always return to `Home`.
- Intermediate crumbs should return to the parent list or workspace.
- The final crumb should represent the current screen, record, or action state.
- Modal or side-panel actions should not replace the breadcrumb unless they open as full screens.

Contextual Navigation

Contextual navigation should be available within screens to help users complete related tasks without returning to module menus.

Inventory contexts

- From `Inventory Status Monitor`:
  - `View Product`
  - `View Location`
  - `Open Movement History`

    - `Create Adjustment`
    - `Request Transfer`
    - `Open Replenishment Planning`

Product contexts

- From `Product Catalog` or a product detail:
    - `Edit Product`
    - `View Stock by Location`
    - `Open Stock Policy`
    - `Open ABC Classification`
    - `View Movement History`
    - `Import Related Data`

Procurement contexts

- From `Low-Stock Alerts`:
    - `Acknowledge Alert`
    - `Review Network Availability`
    - `Create Requisition`
    - `Create Transfer Request`
- From `Purchase Requisition`:
    - `Approve`
    - `Reject`
    - `Convert to PO`

Inbound contexts

- From `Goods Receipt Workbench`:
    - `Verify Receipt`
    - `Log Discrepancy`
    - `Send to Put-Away`
    - `View PO`

Fulfillment contexts

- From `Order Reservation`:
    - `Check Availability`
    - `Reserve Stock`
    - `Release Reservation`
    - `Open Split Fulfillment`
- From `Pick / Pack / Ship`:
    - `Verify Pick`
    - `View Lot / Batch`
    - `Raise Exception`
    - `Confirm Shipment`

Transfer contexts

- From `Transfer Request`:
    - `Approve / Reject`
    - `View Source Capacity`
    - `Dispatch Transfer`
    - `Receive Transfer`

Returns contexts

- From `Return Authorization`:
    - `View Original Order`

    - `Receive Return`
    - `Inspect Return`
    - `Disposition Return`

Audit and admin contexts

- From `Audit Trail Query`:
    - `Open Record`
    - `Export Evidence`
- From `User Account Management`:
    - `Assign Role`
    - `Suspend User`
    - `Deactivate User`
    - `View Access Scope`

Role-Specific Menus

Role-specific menus should simplify navigation by exposing only relevant modules and prioritizing frequent tasks.

Executive roles

Roles:

- `Chief Executive Officer`
- `Chief Operating Officer`
- `Chief Financial Officer`

Priority menu:

- `Home`
- `Reports & Analytics`
- `Governance & Audit`

Common secondary focus:

- `Executive KPI Dashboard`
- `Operational Reports`
- `Valuation Reconciliation`
- `Audit Trail Query`

Inventory control roles

Roles:

- `Inventory Controller`
- `Operations Director`

Priority menu:

- `Home`
- `Inventory`
- `Products`
- `Locations`
- `Transfers`
- `Reports & Analytics`
- `Governance & Audit`

Common secondary focus:

- `Inventory Status Monitor`

- Cycle Counts
- Physical Counts
- Variance Approvals
- Stock Adjustments
- Quarantine & Holds
- Shrinkage & Obsolescence

Warehouse operations roles

Roles:

- Warehouse Manager
- Receiving Clerk
- Warehouse Staff
- Warehouse Picker / Packer

Priority menu:

- Home
- Inbound
- Inventory
- Fulfillment
- Transfers
- Returns

Common secondary focus:

- Goods Receipt Workbench
- Put-Away Workbench
- Cycle Counts
- Pick / Pack / Ship
- Fulfillment Exceptions
- Transfer Execution
- Return Inspection & Disposition

Procurement roles

Roles:

- Purchasing Manager
- Buyer / Procurement Officer

Priority menu:

- Home
- Procurement
- Inventory
- Transfers
- Reports & Analytics

Common secondary focus:

- Suppliers
- Purchase Requisitions
- Purchase Orders
- Inbound Delivery Planning
- Inbound Supply Tracking
- Low-Stock Alerts
- Replenishment Planning

Commercial roles

Roles:

- `Sales Manager`
- `Sales Representative`
- `Customer Service Representative`
- `Marketing Manager`

Priority menu:

- `Home`
- `Inventory`
- `Fulfillment`
- `Returns`
- `Reports & Analytics`

Common secondary focus:

- `Availability Inquiry`
- `Order Reservation`
- `Order Status`
- `Partial Fulfillment & Backorders`
- `Return Authorizations`
- `Promotional Readiness`

Audit and compliance roles

Roles:

- `Internal Auditor`
- `External Auditor`

Priority menu:

- `Home`
- `Governance & Audit`
- `Reports & Analytics`

Common secondary focus:

- `Audit Workspace`
- `Audit Trail Query`
- `Valuation Reconciliation`
- `Operational Reports`

System administration roles

Roles:

- `System Administrator`

Priority menu:

- `Home`
- `Administration`
- `Governance & Audit`

Common secondary focus:

- `User Accounts`
- `Roles & Access Control`
- `Reason Codes`

Text-Based Sitemap

```
Home
├── Operational Dashboard
├── Executive KPI Dashboard
├── Notifications
└── My Tasks / Approvals

Inventory
├── Inventory Status Monitor
│   ├── Product Stock Detail
│   ├── Location Stock Detail
│   ├── Request Transfer
│   ├── Create Adjustment
│   └── Open Replenishment Planning
├── Availability Inquiry
│   ├── Product Search Results
│   └── Network Availability Detail
├── Movement History
│   └── Transaction Detail
├── Cycle Counts
│   ├── Assigned Count Queue
│   └── Count Submission
├── Physical Counts
│   ├── Count Session Detail
│   └── Count Certification
├── Variance Approvals
│   └── Variance Investigation Detail
├── Stock Adjustments
│   └── Adjustment Case Detail
├── Write-Offs
│   └── Write-Off Authorization Detail
├── Quarantine & Holds
│   └── Hold Detail
└── Shrinkage & Obsolescence

Products
├── Product Catalog
│   ├── Product Detail
│   ├── Create Product
│   └── Edit Product
├── Product Import
│   ├── Upload File
│   ├── Validation Results
│   └── Import Completion
├── ABC Classification
└── Stock Policies
    └── Product-Location Policy Detail

Locations
├── Location Directory
│   └── Location Detail
├── Location Hierarchy
├── Storage Positions
└── Location Activation / Closure

Procurement
├── Suppliers
│   └── Supplier Detail
├── Purchase Requisitions
│   └── Requisition Detail
├── Purchase Orders
│   └── Purchase Order Detail
├── Inbound Delivery Planning
├── Inbound Supply Tracking
├── Low-Stock Alerts
│   └── Alert Detail
└── Replenishment Planning
    ├── Replenishment Recommendation Detail
    └── Convert to Requisition / PO
```

```
Inbound
├── Goods Receipt Workbench
│   └── Goods Receipt Detail
├── Receipt Discrepancies
│   └── Discrepancy Detail
├── Put-Away Workbench
│   └── Put-Away Task Detail
└── Non-PO Receipt Intake

Fulfillment
├── Order Reservation
│   └── Order Reservation Detail
├── Pick / Pack / Ship
│   ├── Pick Queue
│   ├── Pick Verification
│   ├── Packing
│   └── Shipment Confirmation
├── Partial Fulfillment & Backorders
│   └── Backorder Detail
├── Fulfillment Exceptions
│   └── Exception Detail
├── Order Status
│   └── Order Detail
└── Multi-Location Fulfillment
    └── Split Fulfillment Plan

Transfers
├── Transfer Requests
│   └── Transfer Request Detail
├── Transfer Approvals
│   └── Approval Detail
├── Transfer Execution
│   ├── Dispatch Detail
│   └── Receipt Detail
├── In-Transit Tracking
└── Network Rebalancing
    └── Rebalancing Recommendation Detail

Returns
├── Return Authorizations
│   └── Return Authorization Detail
└── Return Inspection & Disposition
    ├── Return Receipt Detail
    ├── Inspection Detail
    └── Disposition Detail

Reports & Analytics
├── Inventory Dashboard
├── Operational Reports
│   ├── Movement Report
│   ├── Stockout / Overstock Report
│   ├── Accuracy Report
│   └── Location Performance Report
├── Executive KPI Dashboard
├── Valuation Reconciliation
└── Promotional Readiness

Governance & Audit
├── Governance Policies
├── Reason Codes
├── Audit Workspace
│   └── Audit Review Detail
├── Audit Trail Query
│   └── Audit Record Detail
└── Approvals Inbox
    ├── Transfer Approvals
    └── Adjustment / Variance Approvals

Administration
├── User Accounts
```

```
│   └── User Account Detail
└── Roles & Access Control
    └── Role Permission Detail
```

Navigation Principles

1. `Home` should act as the universal return point for all roles.
2. Modules should reflect business domains, not technical services.
3. Work-queue screens should be reachable from both module navigation and `My Tasks / Approvals`.
4. High-frequency operational screens should be no more than two clicks from the main navigation.
5. Role-based visibility should hide irrelevant modules rather than merely disabling them.
6. Contextual links should connect adjacent workflows such as alert to requisition, receipt to discrepancy, and order to shipment.

Inventory Management System User Journeys

Purpose

This document defines user journeys for the major business processes in the Inventory Management System. Each journey is written from the user's perspective and focuses on the user's goal, decision points, friction, and successful outcome rather than internal system behavior.

Scope

The journeys below cover the major business processes defined in the approved BRD:

1. Product master data management
2. Location and warehouse setup
3. Stock policy and threshold definition
4. Goods receipt and verification
5. Receipt discrepancy resolution
6. Stock put-away
7. Inventory status monitoring
8. Inventory availability inquiry
9. Cycle counting
10. Full physical inventory count
11. Variance investigation and approval
12. Stock adjustment
13. Inventory write-off
14. Quarantine and hold management
15. Order availability confirmation
16. Inventory reservation
17. Pick, pack, and ship
18. Partial fulfillment and backorder handling
19. Low-stock alert review
20. Replenishment planning
21. Inter-location transfer request
22. Transfer approval
23. Transfer execution
24. Customer return authorization
25. Return inspection and disposition
26. Reporting and operational review
27. Inventory valuation reconciliation
28. Promotional stock readiness
29. Audit and control review
30. User account and access management

1. Product Master Data Management

- **User role:** Inventory Controller
- **Goal:** Create or maintain an accurate product record so every team works from one trusted SKU definition.
- **Steps:**
  1. Open the product catalog.
  2. Search for the product to confirm it does not already exist.
  3. Create a new product or open an existing one for update.
  4. Enter or revise SKU, name, category, unit of measure, barcode, and status.
  5. Review the record for completeness.
  6. Save the product and confirm it is available for operations.
- **Decisions:**
  - Is this a new product or an update to an existing one?
  - Is there a possible duplicate that should be resolved before saving?
  - Should the product be active, inactive, or discontinued?
- **Pain points:**
  - Duplicate products create confusion and downstream errors.
  - Missing attributes can block receiving, reservation, or reporting later.
  - Product status decisions may affect purchasing and sales unexpectedly.
- **Success outcome:** A complete, unique, and trusted product record is available for all inventory processes.

2. Location and Warehouse Setup

- **User role:** Inventory Controller
- **Goal:** Configure a location correctly so stock can be assigned, counted, received, moved, and reported there.
- **Steps:**
  1. Open location management.
  2. Create a new location or update an existing one.
  3. Define the location type, hierarchy, and operational status.
  4. Configure storage positions if needed.
  5. Load or confirm opening balances for go-live.
  6. Activate the location for daily operations.
- **Decisions:**
  - Is the location operationally ready for activation?
  - Does it belong under an existing parent location?
  - Are opening balances verified enough to trust the location at go-live?
- **Pain points:**
  - Poor setup causes inventory to be tracked in the wrong place.
  - Unclear hierarchy can affect reporting and transfer flows.
  - Inaccurate opening balances create immediate mistrust in the system.
- **Success outcome:** The location is correctly configured, active, and ready for controlled stock activity.

3. Stock Policy and Threshold Definition

- **User role:** Inventory Controller
- **Goal:** Set thresholds that help the business replenish on time without overstocking.
- **Steps:**
  1. Open stock policy management.
  2. Find the product and relevant location.
  3. Review current stock levels, demand expectations, and service needs.
  4. Set reorder level, minimum threshold, and safety stock.
  5. Save the policy.
  6. Confirm the policy is active for monitoring and alerts.
- **Decisions:**
  - Does this product need a location-specific policy?
  - Is current safety stock sufficient or too conservative?

    - Should thresholds change because demand or lead time changed?
- **Pain points:**
    - Users may lack confidence in the data behind threshold decisions.
    - Overly low thresholds increase stockout risk.
    - Overly high thresholds tie up working capital.
- **Success outcome:** The product is governed by clear thresholds that support proactive replenishment.

4. Goods Receipt and Verification

- **User role:** Receiving Clerk
- **Goal:** Receive incoming supplier goods accurately so only verified stock enters inventory.
- **Steps:**
    1. Open the goods receipt workbench.
    2. Locate the expected purchase order or shipment.
    3. Scan or enter received items.
    4. Verify item identity, quantity, and visible condition.
    5. Capture lot, batch, or shelf-life details when required.
    6. Confirm the receipt and stage stock for put-away.
- **Decisions:**
    - Does the shipment match the purchase order?
    - Is there damage, shortage, wrong product, or shelf-life issue?
    - Can the receipt be accepted as normal or does it require discrepancy handling?
- **Pain points:**
    - Matching paperwork to physical goods can be slow.
    - Traceable products add extra data-entry pressure.
    - Receiving the wrong item creates immediate process interruption.
- **Success outcome:** Verified goods are accurately received and ready for the next step without compromising stock accuracy.

5. Receipt Discrepancy Resolution

- **User role:** Receiving Clerk
- **Goal:** Record and resolve inbound differences without losing supplier accountability or inventory control.
- **Steps:**
    1. Identify a mismatch during receiving.
    2. Open discrepancy logging for the affected receipt line.
    3. Record discrepancy type, quantity impact, and notes.
    4. Attach evidence if needed.
    5. Escalate for manager review or resolution.
    6. Complete the receipt with accepted quantities and documented variance.
- **Decisions:**
    - Is the issue a shortage, overage, wrong item, damage, or tolerance breach?
    - Should the quantity be accepted, partially accepted, or held for review?
    - Does the discrepancy require manager approval or supplier follow-up?
- **Pain points:**
    - Users can feel pressured to keep receiving moving quickly.
    - Incomplete evidence makes later disputes harder to resolve.
    - Rework increases when discrepancies are documented inconsistently.
- **Success outcome:** The discrepancy is fully documented, the accepted stock is correct, and the issue can be followed through to closure.

6. Stock Put-Away

- **User role:** Warehouse Staff
- **Goal:** Move received goods into the correct storage location so they become available for fulfillment and tracking.
- **Steps:**
    1. Open the put-away task queue.
    2. Select the next receipt awaiting storage.

3. Move goods from receiving to the suggested storage position.
4. Confirm product, quantity, and storage location.
5. Complete put-away confirmation.
6. Verify the task is cleared from the queue.

- **Decisions:**
  - Is the suggested location appropriate and available?
  - Does product rotation require FIFO or FEFO handling?
  - Is there any mismatch between the staged quantity and the physical goods?
- **Pain points:**
  - Storage congestion can force manual workaround decisions.
  - Incorrect placement causes future pick failures and count problems.
  - Staff may lose time if receiving and storage labeling are unclear.
- **Success outcome:** Stock is in the right location, properly recorded, and available for downstream use.

7. Inventory Status Monitoring

- **User role:** Inventory Controller
- **Goal:** Understand overall stock health and identify issues before they disrupt operations.
- **Steps:**
  1. Open the inventory status monitor.
  2. Filter by location, product type, or status.
  3. Review low-stock, out-of-stock, overstock, and quarantined items.
  4. Drill into a product or location when something looks wrong.
  5. Decide whether follow-up action is needed.
  6. Route the issue to replenishment, transfer, adjustment, or investigation.
- **Decisions:**
  - Is this a temporary issue or something that requires immediate action?
  - Should the next action be replenishment, transfer, or investigation?
  - Is the problem local to one location or network-wide?
- **Pain points:**
  - Users may see symptoms before they see the root cause.
  - Too many alerts can make prioritization difficult.
  - Network-level visibility can be overwhelming without filtering.
- **Success outcome:** The user quickly identifies stock risks and knows what action to take next.

8. Inventory Availability Inquiry

- **User role:** Sales Representative
- **Goal:** Answer a customer stock question confidently without needing to call the warehouse.
- **Steps:**
  1. Open availability inquiry.
  2. Search by SKU, product name, or barcode.
  3. Review on-hand, reserved, and ATP figures.
  4. Check relevant locations or network-wide availability.
  5. Confirm stock status and expected inbound if needed.
  6. Respond to the customer.
- **Decisions:**
  - Is there enough ATP to support a commitment?
  - Should the answer be location-specific or network-wide?
  - Is expected inbound relevant to the customer conversation?
- **Pain points:**
  - Users need fast answers during live customer conversations.
  - Confusion between on-hand and ATP can lead to bad promises.
  - If stock is fragmented across locations, the answer may be less straightforward.
- **Success outcome:** The customer receives a fast, accurate, and credible availability answer.

9. Cycle Counting

- **User role:** Warehouse Staff
- **Goal:** Count assigned stock accurately without being biased by system quantities.
- **Steps:**
  1. Open assigned cycle counts.
  2. Go to the specified product and storage location.
  3. Count the physical stock.
  4. Enter the counted quantity.
  5. Submit the result.
  6. Move to the next assignment.
- **Decisions:**
  - Does the physical stock clearly match one count result, or should it be recounted?
  - Is the storage area clear enough to count confidently?
  - Should an unusual condition be noted for later investigation?
- **Pain points:**
  - Cluttered or mixed storage slows counting.
  - Staff may worry about making mistakes that trigger investigations.
  - Repeated counting tasks can feel disruptive to warehouse throughput.
- **Success outcome:** The physical count is submitted accurately and supports improved inventory confidence.

10. Full Physical Inventory Count

- **User role:** Inventory Controller
- **Goal:** Complete a controlled wall-to-wall count that establishes a trusted inventory baseline.
- **Steps:**
  1. Start a physical inventory count session.
  2. Confirm the location is frozen for normal movement.
  3. Assign teams or zones for counting.
  4. Monitor progress as counts are entered.
  5. Identify any incomplete or questionable areas.
  6. Close and certify the count session.
- **Decisions:**
  - Is the location ready to enter freeze mode?
  - Are all areas fully counted and verified?
  - Can the count be certified now or should unresolved issues be addressed first?
- **Pain points:**
  - Full counts are operationally disruptive.
  - Users must coordinate many people under time pressure.
  - Incomplete counts reduce trust in the final result.
- **Success outcome:** The location completes a reliable full count and generates a trustworthy variance baseline.

11. Variance Investigation and Approval

- **User role:** Warehouse Manager
- **Goal:** Understand why a count variance happened and approve only justified corrections.
- **Steps:**
  1. Open the pending variance queue.
  2. Select a variance case.
  3. Review system quantity versus counted quantity.
  4. Investigate likely causes with notes and evidence.
  5. Decide whether to approve, escalate, or reject the proposed correction.
  6. Route the case to the proper next step.
- **Decisions:**
  - Is the variance caused by process error, damage, shrinkage, or timing?
  - Does the variance exceed an approval threshold?
  - Is more investigation needed before approval?

- **Pain points:**
  - Root causes are not always obvious.
  - The user may be balancing speed against control discipline.
  - Large variances can create pressure from multiple stakeholders.
- **Success outcome:** Only justified variances move forward, preserving control and data accuracy.

12. Stock Adjustment

- **User role:** Warehouse Manager
- **Goal:** Correct recorded stock safely when reality does not match the system.
- **Steps:**
  1. Open stock adjustment management.
  2. Identify the product and location.
  3. Enter increase or decrease quantity.
  4. Select the required reason code and add notes.
  5. Submit for approval if threshold rules require it.
  6. Confirm the adjustment is applied.
- **Decisions:**
  - Is an adjustment truly needed or should the item be investigated first?
  - Which reason code best explains the change?
  - Does the value or quantity require independent approval?
- **Pain points:**
  - Users may choose vague reason codes under time pressure.
  - Incorrect adjustments can worsen the data instead of fixing it.
  - Approval delays can block urgent corrections.
- **Success outcome:** Stock is corrected with proper justification, traceability, and governance.

13. Inventory Write-Off

- **User role:** Operations Director
- **Goal:** Remove unsellable inventory in a controlled way with clear accountability.
- **Steps:**
  1. Open pending write-off requests.
  2. Review the affected inventory, quantity, and supporting evidence.
  3. Check financial and operational implications.
  4. Approve or reject the write-off.
  5. Confirm finance acknowledgment if required.
  6. Complete write-off execution.
- **Decisions:**
  - Is the inventory truly unsellable or should another disposition be used?
  - Is the evidence strong enough to support write-off?
  - Does finance need to review before final approval?
- **Pain points:**
  - Write-offs can be sensitive and highly visible.
  - Weak documentation makes decisions harder.
  - Users may want better root-cause context before approval.
- **Success outcome:** Unsellable stock is removed with defensible authorization and a full audit trail.

14. Quarantine and Hold Management

- **User role:** Warehouse Manager
- **Goal:** Prevent questionable or restricted stock from being used or sold until it is cleared.
- **Steps:**
  1. Open quarantine and hold management.
  2. Locate the affected stock.
  3. Apply hold or quarantine status with a reason.
  4. Confirm the quantity is excluded from sellable stock.

5. Monitor the hold until resolution.
6. Release or disposition the stock when authorized.

- **Decisions:**
  - Should the stock be quarantined, held temporarily, or written off?
  - How much quantity is affected?
  - Is the issue quality-related, regulatory, or operational?
- **Pain points:**
  - Users need to move fast to avoid accidental sale or shipment.
  - Unclear ownership of follow-up can leave stock stuck in hold.
  - Long-running holds reduce usable inventory and frustrate teams.
- **Success outcome:** Risky stock is contained quickly and kept out of normal operations until resolved.

15. Order Availability Confirmation

- **User role:** Sales Representative
- **Goal:** Determine whether the business can commit to a customer's requested order.
- **Steps:**
  1. Open order availability confirmation.
  2. Review the requested products and quantities.
  3. Trigger an ATP check.
  4. Review line-by-line availability results.
  5. Consider location options or partial availability.
  6. Communicate the commitment position to the customer.
- **Decisions:**
  - Is the order fully available, partially available, or unavailable?
  - Should the promise be made from one location or across the network?
  - Should the next step be reservation, split fulfillment, or backorder discussion?
- **Pain points:**
  - Customer expectations are often immediate.
  - Users need clarity on what can actually be promised.
  - Network stock can complicate what seems like a simple answer.
- **Success outcome:** The user gives a realistic and supportable commitment based on ATP, not guesswork.

16. Inventory Reservation

- **User role:** Customer Service Representative
- **Goal:** Reserve confirmed stock so it is protected for a specific customer order.
- **Steps:**
  1. Open the confirmed order.
  2. Review availability and selected fulfillment location.
  3. Reserve the needed quantity.
  4. Confirm the reservation result.
  5. Release the order to warehouse fulfillment.
  6. Revisit the reservation later if the order changes.
- **Decisions:**
  - Which location should fulfill the order?
  - Is there enough ATP for all lines?
  - Should a reservation be released if the customer changes or cancels the order?
- **Pain points:**
  - Reservation mistakes can create downstream fulfillment issues.
  - Users need confidence that ATP is current at the moment of reservation.
  - Changing orders can require tedious reservation cleanup.
- **Success outcome:** Stock is securely committed to the order and protected from double allocation.

17. Pick, Pack, and Ship

- **User role:** Warehouse Picker / Packer

- **Goal:** Fulfill the order accurately and get it shipped without causing stock errors.
- **Steps:**
  1. Open the pick queue.
  2. Select the assigned order or task.
  3. Pick items from the indicated locations.
  4. Verify the picked products and quantities.
  5. Pack the shipment.
  6. Confirm shipment handoff and complete the task.
- **Decisions:**
  - Does the shelf stock match what the task expects?
  - Is the picked quantity correct and complete?
  - If something is wrong, should the task be stopped and an exception raised?
- **Pain points:**
  - Pick failures are disruptive and time-sensitive.
  - Staff need accurate location guidance to move fast.
  - Errors discovered late in packing waste effort and delay shipment.
- **Success outcome:** The correct order is shipped accurately and inventory is deducted at the right moment.

18. Partial Fulfillment and Backorder Handling

- **User role:** Customer Service Representative
- **Goal:** Save as much of the order as possible when full quantity cannot be supplied immediately.
- **Steps:**
  1. Open the affected order.
  2. Review available quantity versus ordered quantity.
  3. Discuss options with the customer.
  4. Record the customer's acceptance of a partial shipment if applicable.
  5. Confirm immediate fulfillment for available quantity.
  6. Create or monitor the backorder for the remainder.
- **Decisions:**
  - Is partial shipment acceptable to the customer?
  - Should the remainder be backordered or canceled?
  - Is an expected future date credible enough to communicate?
- **Pain points:**
  - Customer conversations can be sensitive and time-consuming.
  - Users may not always know when the remainder will be available.
  - Splitting an order adds coordination overhead.
- **Success outcome:** The customer receives a clear, agreed plan and the available stock still moves forward.

19. Low-Stock Alert Review

- **User role:** Purchasing Manager
- **Goal:** Notice and prioritize stock risks early enough to prevent service problems.
- **Steps:**
  1. Open low-stock alerts.
  2. Review active alerts by severity and product importance.
  3. Drill into the most urgent items.
  4. Check current stock, open supply, and network alternatives.
  5. Acknowledge each alert and assign next action.
  6. Monitor remaining unresolved alerts.
- **Decisions:**
  - Which alert needs action first?
  - Should the answer be transfer, purchase, or watchful waiting?
  - Is the issue local or network-wide?
- **Pain points:**
  - Too many alerts can create noise.
  - Users need enough context to act, not just know there is a problem.

    - Prioritization is harder when many SKUs are at risk simultaneously.
- **Success outcome:** Critical alerts are acknowledged quickly and turned into concrete next actions.

20. Replenishment Planning

- **User role:** Buyer
- **Goal:** Convert stock signals into the right replenishment action at the right time.
- **Steps:**
    1. Open replenishment planning.
    2. Review recommended reorder items.
    3. Check whether another location can cover the need first.
    4. Decide to create a requisition or take another action.
    5. Submit the requisition for approval.
    6. Follow it through to purchase order conversion.
- **Decisions:**
    - Should the need be covered by transfer or purchase?
    - Is the recommended quantity appropriate?
    - Does the product remain eligible for reorder?
- **Pain points:**
    - Users must balance urgency, cost, and available alternatives.
    - Data gaps in lead times or demand can reduce confidence.
    - Approval delay can keep urgent replenishment from moving fast enough.
- **Success outcome:** A justified replenishment action is created and ready for procurement execution.

21. Inter-Location Transfer Request

- **User role:** Regional / Store Manager
- **Goal:** Solve a local shortage by using stock already available elsewhere in the network.
- **Steps:**
    1. Open transfer request.
    2. Select destination location, source location, product, and quantity.
    3. Review source availability and transfer impact.
    4. Enter the business reason.
    5. Submit the request for approval.
    6. Track the request status.
- **Decisions:**
    - Which source location can spare the stock?
    - Is transfer better than purchasing new stock?
    - Is the requested quantity appropriate and sustainable for the source?
- **Pain points:**
    - Users may not always know the best source location.
    - Transfer urgency can conflict with approval latency.
    - Poor network visibility can lead to unnecessary transfers.
- **Success outcome:** A well-justified transfer request is submitted to relieve the shortage with minimal network disruption.

22. Transfer Approval

- **User role:** Operations Director
- **Goal:** Approve only transfers that help the network without harming the source location.
- **Steps:**
    1. Open the transfer approval queue.
    2. Review source and destination need.
    3. Check source stock after transfer.
    4. Decide to approve or reject.
    5. Record the decision.
    6. Notify locations through workflow status.
- **Decisions:**

    - Will this transfer create a new risk at the source?
    - Is the shortage urgent enough to justify movement?
    - Should the request be approved as-is or rejected?
- **Pain points:**
    - The approver must think network-wide, not locally.
    - Approval queues can become bottlenecks.
    - Incomplete justification makes confident decisions harder.
- **Success outcome:** The best transfer requests move forward while risky ones are stopped before causing new problems.

23. Transfer Execution

- **User role:** Warehouse Staff
- **Goal:** Move stock from one location to another accurately and keep both locations' inventory correct.
- **Steps:**
    1. Open approved transfer tasks.
    2. Pick the transfer quantity at the source.
    3. Verify and dispatch the stock.
    4. Send or hand off the stock for transport.
    5. Receive and verify it at the destination.
    6. Confirm completion of the transfer.
- **Decisions:**
    - Does the picked quantity exactly match the transfer order?
    - Is any part of the stock damaged or mismatched during movement?
    - Can the destination accept the transfer as received?
- **Pain points:**
    - Transfers have two operational handoffs, increasing failure risk.
    - In-transit gaps can create anxiety if status is unclear.
    - Errors at either end affect both source and destination records.
- **Success outcome:** Stock leaves the source, arrives at the destination, and is fully traceable throughout the move.

24. Customer Return Authorization

- **User role:** Customer Service Representative
- **Goal:** Approve valid returns fairly while keeping control over what the warehouse should expect back.
- **Steps:**
    1. Open return authorization.
    2. Find the original order.
    3. Select the item and quantity requested for return.
    4. Validate the reason and policy eligibility.
    5. Issue the return authorization.
    6. Share return instructions with the customer.
- **Decisions:**
    - Is the return eligible under policy?
    - Is the requested quantity within allowed limits?
    - Should the authorization be issued or declined?
- **Pain points:**
    - Users often need to balance customer satisfaction with policy compliance.
    - Missing order context slows the conversation.
    - Unclear reasons can make approval harder.
- **Success outcome:** The customer gets a clear authorization path and the warehouse receives a controlled expectation of what will return.

25. Return Inspection and Disposition

- **User role:** Receiving Clerk
- **Goal:** Decide what to do with returned goods so good stock is recovered and bad stock is contained.
- **Steps:**

1. Open pending return receipts.
2. Match the incoming return to its authorization.
3. Verify quantity and product identity.
4. Inspect physical condition.
5. Choose disposition: restock, quarantine, or write-off.
6. Confirm completion of the return process.

- **Decisions:**
  - Is the return truly authorized?
  - Is the product resalable, damaged, or defective?
  - Should the stock go back into inventory, be held, or be removed?
- **Pain points:**
  - Condition assessments can be subjective.
  - Returned goods may arrive incomplete or in poor condition.
  - Users need clarity on what qualifies for restock versus quarantine.
- **Success outcome:** Returned goods are processed consistently and routed to the correct inventory state.

26. Reporting and Operational Review

- **User role:** Operations Director
- **Goal:** Understand operational performance and identify where intervention is needed.
- **Steps:**
  1. Open reports and analytics.
  2. Choose the relevant report or dashboard.
  3. Filter by date, location, or category.
  4. Review trends, exceptions, and outliers.
  5. Drill into problem areas if needed.
  6. Export or share findings for follow-up action.
- **Decisions:**
  - Which metric or report matters most right now?
  - Is this a one-off issue or a pattern?
  - Does the issue need operational action, policy change, or further investigation?
- **Pain points:**
  - Large data sets can slow understanding if not well focused.
  - Users may struggle to connect metrics to actionable next steps.
  - Different stakeholders may need different levels of detail.
- **Success outcome:** The user gains actionable insight and can direct operations based on evidence rather than instinct.

27. Inventory Valuation Reconciliation

- **User role:** Finance Manager
- **Goal:** Confirm that operational inventory records align with financial expectations for a period close.
- **Steps:**
  1. Open valuation reconciliation.
  2. Select the relevant period.
  3. Review operational valuation and comparison figures.
  4. Investigate material variances.
  5. Document reconciliation notes.
  6. Finalize the period review.
- **Decisions:**
  - Is the variance acceptable or material?
  - Does the issue stem from operations, timing, or finance treatment?
  - Can the period be closed with confidence?
- **Pain points:**
  - Finance needs confidence in data produced by operational processes.
  - Large unexplained variances can create time pressure at period end.
  - Users may need to cross-check multiple underlying events.
- **Success outcome:** Finance can complete reconciliation with a clear view of differences and documented closure confidence.

28. Promotional Stock Readiness

- **User role:** Marketing Manager
- **Goal:** Make sure enough inventory is secured before a campaign or launch commitment is made.
- **Steps:**
  1. Open promotional readiness.
  2. Review campaign products and required quantities.
  3. Check ring-fenced stock, ATP, and inbound supply.
  4. Identify products at risk.
  5. Confirm readiness or escalate gaps.
  6. Communicate launch confidence to stakeholders.
- **Decisions:**
  - Is current ring-fenced stock sufficient?
  - Can inbound supply close any shortfall in time?
  - Should the campaign proceed, be adjusted, or be escalated?
- **Pain points:**
  - Promotional demand creates pressure on shared inventory.
  - Users need certainty before public commitments are made.
  - Gaps discovered late can disrupt campaigns and trust.
- **Success outcome:** The campaign team has a credible readiness view and can commit with confidence.

29. Audit and Control Review

- **User role:** Internal Auditor
- **Goal:** Verify that inventory activity follows control rules and that the audit trail is complete.
- **Steps:**
  1. Open the audit workspace.
  2. Define the audit scope.
  3. Review transactions, approvals, and control evidence.
  4. Sample high-risk records.
  5. Note findings, breaches, or clean results.
  6. Export evidence and complete the review.
- **Decisions:**
  - Which locations, periods, or transaction types are highest risk?
  - Is a control failure isolated or systemic?
  - Does a finding require escalation or remediation tracking?
- **Pain points:**
  - Audit users need complete, reliable history without ambiguity.
  - Missing approval evidence undermines trust quickly.
  - Investigations can be time-consuming when records are fragmented.
- **Success outcome:** The auditor can confidently assess compliance and produce defensible findings.

30. User Account and Access Management

- **User role:** System Administrator
- **Goal:** Give each user the right access for their job while protecting control boundaries.
- **Steps:**
  1. Open user account management.
  2. Search for an existing user or create a new one.
  3. Assign role and location scope.
  4. Review any access-control conflicts.
  5. Activate, suspend, or update the account as needed.
  6. Confirm effective permissions.
- **Decisions:**
  - Does the user need a new account or an update to an existing one?
  - Which role best matches the user's responsibility?
  - Does the requested access create a segregation-of-duties issue?

- **Pain points:**
  - Over-permissioning creates control risk.
  - Under-permissioning blocks operations and frustrates users.
  - Role complexity can make correct assignment difficult.
- **Success outcome:** The user receives correct, scoped access that supports work without compromising governance.

Journey Design Notes

- These journeys are intentionally written from the user's point of view rather than the system's.
- The journeys can be used as input for task flows, wireframes, usability scenarios, and acceptance criteria.
- Where multiple roles collaborate in one process, the journey is centered on the primary user who owns the outcome.

Inventory Management System User Flows

Purpose

This document defines user flows for each major feature of the Inventory Management System. The flows are screen-oriented and are intended to support UX design, wireframes, interaction design, and future usability validation.

Scope

The flows are based on the major features already identified across the BRD, use cases, screen inventory, navigation structure, and user journeys. Each flow includes:

1. Starting screen
2. User actions
3. Navigation path
4. Decision points
5. Destination screen

Each flow is represented as a text-based diagram from the user's perspective.

---

1. Product Catalog Management

- **Starting screen:** `Home > Products > Product Catalog`
- **User actions:** Search product, create product, edit product, update status, save record
- **Navigation path:** `Product Catalog -> Product Detail / Create Product -> Save -> Product Detail`
- **Decision points:** New product or existing product? Duplicate detected? Active, inactive, or discontinued?
- **Destination screen:** `Product Detail`

```
Product Catalog
  -> Search by SKU / name / category
  -> [Decision: Product exists?]
     -> No -> Create Product
           -> Enter product attributes
           -> [Decision: Duplicate detected?]
              -> Yes -> Review duplicate resolution
              -> No -> Save
           -> Product Detail
     -> Yes -> Product Detail
           -> Edit Product
           -> Update attributes / status
           -> Save
           -> Product Detail
```

2. Product Import

- **Starting screen:** `Home > Products > Product Import`
- **User actions:** Download template, upload file, review validation, confirm import
- **Navigation path:** `Product Import -> Upload File -> Validation Results -> Import Completion`
- **Decision points:** File valid? Errors blocking import? Proceed with approved records only?
- **Destination screen:** `Import Completion`

```
Product Import
  -> Upload File
  -> Validation Results
  -> [Decision: Validation passed?]
     -> No -> Correct file -> Re-upload
     -> Yes -> Confirm Import
           -> Import Completion
```

3. Location and Warehouse Management

- **Starting screen:** `Home > Locations > Location Directory`
- **User actions:** Create location, edit hierarchy, configure storage positions, activate location
- **Navigation path:** `Location Directory -> Location Detail -> Storage Positions / Hierarchy -> Activate`
- **Decision points:** New or existing location? Ready for activation? Opening balances complete?
- **Destination screen:** `Location Detail`

```
Location Directory
  -> [Decision: Create new location?]
     -> Yes -> Create Location
           -> Enter location details
           -> Configure hierarchy
           -> Configure storage positions
           -> [Decision: Ready to activate?]
              -> No -> Save draft -> Location Detail
              -> Yes -> Activate Location
                     -> Location Detail
     -> No -> Location Detail
           -> Edit Hierarchy / Storage Positions / Status
           -> Save
           -> Location Detail
```

4. Stock Policy and Threshold Management

- **Starting screen:** `Home > Products > Stock Policies`
- **User actions:** Select product-location pair, review policy, update thresholds, save
- **Navigation path:** `Stock Policies -> Product-Location Policy Detail -> Save`
- **Decision points:** Use existing policy? Update reorder level? Adjust safety stock?
- **Destination screen:** `Product-Location Policy Detail`

```
Stock Policies
  -> Search product and location
  -> Product-Location Policy Detail
  -> Update reorder level / minimum / safety stock
  -> [Decision: Values acceptable?]
     -> No -> Revise values
     -> Yes -> Save
           -> Product-Location Policy Detail
```

5. ABC Classification Management

- **Starting screen:** `Home > Products > ABC Classification`
- **User actions:** Review products, assign class, save classification
- **Navigation path:** `ABC Classification -> Classification Review -> Save`

- **Decision points:** Class A, B, or C? Reclassify now or defer?
- **Destination screen:** `ABC Classification`

```
ABC Classification
  -> Review product performance indicators
  -> Select product
  -> Assign A / B / C class
  -> [Decision: Classification final?]
     -> No -> Continue review
     -> Yes -> Save
           -> ABC Classification
```

6. Governance Policies and Reason Codes

- **Starting screen:** `Home > Governance & Audit > Governance Policies`
- **User actions:** Create or update policy, manage reason code, publish changes
- **Navigation path:** `Governance Policies -> Policy Detail / Reason Code Detail -> Publish`
- **Decision points:** New policy or update? Retire reason code? Threshold change required?
- **Destination screen:** `Governance Policies`

```
Governance Policies
  -> [Decision: Manage policy or reason code?]
     -> Policy -> Policy Detail
                -> Update thresholds / tolerances
                -> Publish
                -> Governance Policies
     -> Reason Code -> Reason Code Detail
                    -> Create / edit / retire code
                    -> Save
                    -> Governance Policies
```

7. Supplier Management

- **Starting screen:** `Home > Procurement > Suppliers`
- **User actions:** Search supplier, create supplier, edit terms, deactivate supplier
- **Navigation path:** `Suppliers -> Supplier Detail -> Save / Deactivate`
- **Decision points:** New or existing supplier? Supplier still active?
- **Destination screen:** `Supplier Detail`

```
Suppliers
  -> Search supplier
  -> [Decision: Supplier exists?]
     -> No -> Create Supplier
           -> Enter supplier details
           -> Save
           -> Supplier Detail
     -> Yes -> Supplier Detail
           -> Edit terms / lead time / status
           -> [Decision: Deactivate supplier?]
              -> Yes -> Confirm deactivation -> Supplier Detail
              -> No -> Save updates -> Supplier Detail
```

8. Purchase Requisition Management

- **Starting screen:** `Home > Procurement > Purchase Requisitions`
- **User actions:** Review requisition, approve, reject, convert to PO
- **Navigation path:** `Purchase Requisitions -> Requisition Detail -> Approve / Reject / Convert`
- **Decision points:** Approve or reject? Ready to convert to PO?
- **Destination screen:** `Requisition Detail` or `Purchase Order Detail`

```
Purchase Requisitions
  -> Select requisition
  -> Requisition Detail
  -> [Decision: Approve?]
     -> No -> Reject with reason
           -> Requisition Detail
     -> Yes -> Approve
           -> [Decision: Convert to PO now?]
              -> No -> Requisition Detail
              -> Yes -> Convert to PO
                     -> Purchase Order Detail
```

9. Purchase Order Management

- **Starting screen:** `Home > Procurement > Purchase Orders`
- **User actions:** Create PO, edit PO, cancel PO, review open PO
- **Navigation path:** `Purchase Orders -> Purchase Order Detail -> Save / Cancel`
- **Decision points:** Create from requisition or manual? Amend or cancel?
- **Destination screen:** `Purchase Order Detail`

```
Purchase Orders
  -> [Decision: Create new PO?]
     -> Yes -> Create Purchase Order
           -> Enter supplier / lines / dates
           -> Save
           -> Purchase Order Detail
     -> No -> Purchase Order Detail
           -> [Decision: Amend or cancel?]
              -> Amend -> Update lines / dates -> Save -> Purchase Order Detail
              -> Cancel -> Confirm cancel -> Purchase Order Detail
```

10. Inbound Delivery Planning

- **Starting screen:** `Home > Procurement > Inbound Delivery Planning`
- **User actions:** Select PO, update expected delivery, confirm plan
- **Navigation path:** `Inbound Delivery Planning -> Purchase Order Detail -> Save expected delivery`
- **Decision points:** Delivery confirmed? Delay or change received?
- **Destination screen:** `Inbound Delivery Planning`

```
Inbound Delivery Planning
  -> Select purchase order
  -> Review expected delivery
  -> Update shipment date / carrier / destination
  -> [Decision: Delivery plan final?]
     -> No -> Continue editing
     -> Yes -> Save
           -> Inbound Delivery Planning
```

11. Goods Receipt Workbench

- **Starting screen:** `Home > Inbound > Goods Receipt Workbench`
- **User actions:** Open expected receipt, verify items, confirm receipt
- **Navigation path:** `Goods Receipt Workbench -> Goods Receipt Detail -> Verify -> Send to Put-Away`
- **Decision points:** PO match? Traceability data required? Discrepancy found?
- **Destination screen:** `Goods Receipt Detail`

```
Goods Receipt Workbench
  -> Select arriving shipment
  -> Goods Receipt Detail
  -> Scan / verify items and quantities
  -> [Decision: Discrepancy found?]
     -> Yes -> Receipt Discrepancy Detail
           -> Resolve / document
           -> Return to Goods Receipt Detail
     -> No -> Confirm Receipt
           -> Send to Put-Away
           -> Goods Receipt Detail
```

12. Receipt Discrepancy Resolution

- **Starting screen:** `Home > Inbound > Receipt Discrepancies`
- **User actions:** Select discrepancy, review, resolve, close
- **Navigation path:** `Receipt Discrepancies -> Discrepancy Detail -> Resolve`
- **Decision points:** Accept quantity? Escalate? Hold for supplier follow-up?
- **Destination screen:** `Discrepancy Detail`

```
Receipt Discrepancies
  -> Select discrepancy
  -> Discrepancy Detail
  -> Review variance and notes
  -> [Decision: Resolve now?]
     -> No -> Leave open / escalate
     -> Yes -> Record resolution
           -> Close discrepancy
           -> Discrepancy Detail
```

13. Put-Away Workbench

- **Starting screen:** `Home > Inbound > Put-Away Workbench`
- **User actions:** Select task, move stock, confirm location, complete task
- **Navigation path:** `Put-Away Workbench -> Put-Away Task Detail -> Confirm`
- **Decision points:** Suggested location valid? Rotation rule applies? Quantity mismatch?
- **Destination screen:** `Put-Away Task Detail`

```
Put-Away Workbench
  -> Select pending task
  -> Put-Away Task Detail
  -> Move goods to storage
  -> Confirm quantity and destination
  -> [Decision: Location / quantity valid?]
     -> No -> Correct issue / choose alternate location
     -> Yes -> Complete Put-Away
           -> Put-Away Task Detail
```

14. Inventory Status Monitor

- **Starting screen:** `Home > Inventory > Inventory Status Monitor`
- **User actions:** Filter results, drill into stock detail, route next action
- **Navigation path:** `Inventory Status Monitor -> Product Stock Detail / Location Stock Detail -> Related action`
- **Decision points:** Low stock? Overstock? Investigate, transfer, replenish, or adjust?
- **Destination screen:** `Product Stock Detail`, `Transfer Request`, `Replenishment Planning`, or `Stock Adjustments`

```
Inventory Status Monitor
  -> Filter by location / status / category
  -> Review stock board
  -> Select product or location
  -> Product Stock Detail
  -> [Decision: What action is needed?]
     -> Replenish -> Replenishment Planning
     -> Transfer -> Transfer Request
     -> Investigate -> Movement History
     -> Correct stock -> Stock Adjustments
```

15. Availability Inquiry

- **Starting screen:** `Home > Inventory > Availability Inquiry`
- **User actions:** Search product, view ATP, check network availability
- **Navigation path:** `Availability Inquiry -> Search Results -> Network Availability Detail`
- **Decision points:** Enough ATP? Need network view? Customer-ready answer available?
- **Destination screen:** `Network Availability Detail`

```
Availability Inquiry
  -> Search product
  -> Search Results
  -> Select product
  -> View ATP / reserved / on-hand
  -> [Decision: Need network availability?]
     -> No -> Respond to customer
     -> Yes -> Network Availability Detail
               -> Respond to customer
```

16. Cycle Count Workbench

- **Starting screen:** `Home > Inventory > Cycle Counts`
- **User actions:** Open assigned count, enter quantity, submit result
- **Navigation path:** `Cycle Counts -> Count Submission -> Submit`
- **Decision points:** Count confident? Recount needed? Variance expected?
- **Destination screen:** `Cycle Counts`

```
Cycle Counts
  -> Select assigned count
  -> Count Submission
  -> Enter physical quantity
  -> [Decision: Recount needed?]
     -> Yes -> Recount -> Enter revised quantity
     -> No -> Submit
           -> Cycle Counts
```

17. Physical Count Management

- **Starting screen:** `Home > Inventory > Physical Counts`
- **User actions:** Start count, monitor progress, certify count
- **Navigation path:** `Physical Counts -> Count Session Detail -> Count Certification`
- **Decision points:** Freeze location? All areas counted? Certify now?
- **Destination screen:** `Count Certification`

```
Physical Counts
  -> Start new count session
  -> Count Session Detail
  -> Monitor zones / teams / progress
  -> [Decision: Count complete?]
     -> No -> Continue counting
     -> Yes -> Count Certification
           -> Confirm certification
           -> Count Certification
```

18. Variance Investigation and Approval

- **Starting screen:** `Home > Inventory > Variance Approvals`
- **User actions:** Select variance, investigate, approve or reject
- **Navigation path:** `Variance Approvals -> Variance Investigation Detail -> Approval result`
- **Decision points:** Root cause clear? Approval threshold exceeded? Approve or reject?
- **Destination screen:** `Variance Investigation Detail`

```
Variance Approvals
  -> Select variance case
  -> Variance Investigation Detail
  -> Review evidence / notes / quantity gap
  -> [Decision: Approve variance?]
     -> No -> Reject / escalate
           -> Variance Investigation Detail
     -> Yes -> Approve
           -> Route to Stock Adjustment
           -> Variance Investigation Detail
```

19. Stock Adjustment Management

- **Starting screen:** `Home > Inventory > Stock Adjustments`
- **User actions:** Create adjustment, select reason code, submit, approve if needed
- **Navigation path:** `Stock Adjustments -> Adjustment Case Detail -> Submit -> Approval / Apply`
- **Decision points:** Increase or decrease? Threshold approval needed? Apply now?
- **Destination screen:** `Adjustment Case Detail`

```
Stock Adjustments
  -> Create Adjustment
  -> Adjustment Case Detail
  -> Enter quantity and reason code
  -> Submit
  -> [Decision: Approval required?]
     -> No -> Apply adjustment -> Adjustment Case Detail
     -> Yes -> Pending approval
           -> Approval result
           -> [Decision: Approved?]
              -> No -> Adjustment Case Detail
              -> Yes -> Apply adjustment -> Adjustment Case Detail
```

20. Write-Off Authorization

- **Starting screen:** `Home > Inventory > Write-Offs`
- **User actions:** Review request, inspect evidence, approve or reject
- **Navigation path:** `Write-Offs -> Write-Off Authorization Detail -> Approve / Reject`
- **Decision points:** Evidence sufficient? Finance acknowledgment needed? Approve?
- **Destination screen:** `Write-Off Authorization Detail`

```
Write-Offs
  -> Select write-off request
  -> Write-Off Authorization Detail
  -> Review evidence / quantity / reason
  -> [Decision: Approve write-off?]
     -> No -> Reject
           -> Write-Off Authorization Detail
     -> Yes -> [Decision: Finance acknowledgment required?]
               -> Yes -> Await finance acknowledgment
               -> No -> Continue
           -> Approve and execute
           -> Write-Off Authorization Detail
```

21. Quarantine and Hold Register

- **Starting screen:** `Home > Inventory > Quarantine & Holds`
- **User actions:** Select stock, apply hold, release hold, view hold detail
- **Navigation path:** `Quarantine & Holds -> Hold Detail -> Apply / Release`
- **Decision points:** Apply hold or release? Quarantine or temporary hold?
- **Destination screen:** `Hold Detail`

```
Quarantine & Holds
  -> Search product / location
  -> Hold Detail
  -> [Decision: Apply new hold or release existing hold?]
     -> Apply -> Select quantity / reason / type
             -> Confirm
             -> Hold Detail
     -> Release -> Review hold
               -> Confirm release
               -> Hold Detail
```

22. Order Reservation

- **Starting screen:** `Home > Fulfillment > Order Reservation`
- **User actions:** Open order, confirm availability, reserve stock, release if needed
- **Navigation path:** `Order Reservation -> Order Reservation Detail -> Reserve / Release`
- **Decision points:** ATP sufficient? Which location? Reservation or release?
- **Destination screen:** `Order Reservation Detail`

```
Order Reservation
  -> Search / open order
  -> Order Reservation Detail
  -> Check availability
  -> [Decision: Enough ATP?]
     -> No -> Partial Fulfillment & Backorders
     -> Yes -> Select fulfillment location
           -> Reserve stock
           -> [Decision: Order changed / canceled later?]
              -> Yes -> Release reservation
              -> No -> Continue to warehouse queue
           -> Order Reservation Detail
```

23. Pick / Pack / Ship

- **Starting screen:** `Home > Fulfillment > Pick / Pack / Ship`
- **User actions:** Open pick queue, verify pick, pack order, confirm shipment
- **Navigation path:** `Pick Queue -> Pick Verification -> Packing -> Shipment Confirmation`
- **Decision points:** Pick mismatch? Raise exception? Shipment complete?
- **Destination screen:** `Shipment Confirmation`

```
Pick / Pack / Ship
  -> Pick Queue
  -> Select order
  -> Pick Verification
  -> [Decision: Pick successful?]
     -> No -> Fulfillment Exception Detail
     -> Yes -> Packing
           -> Shipment Confirmation
           -> Confirm ship
           -> Shipment Confirmation
```

24. Partial Fulfillment and Backorders

- **Starting screen:** `Home > Fulfillment > Partial Fulfillment & Backorders`
- **User actions:** Open affected order, confirm partial plan, create backorder
- **Navigation path:** `Partial Fulfillment & Backorders -> Backorder Detail -> Confirm`
- **Decision points:** Customer accepts partial? Backorder or cancel remainder?
- **Destination screen:** `Backorder Detail`

```
Partial Fulfillment & Backorders
  -> Select affected order
  -> Review available vs ordered quantity
  -> [Decision: Customer accepts partial fulfillment?]
     -> No -> Cancel or revise order
     -> Yes -> Confirm immediate shipment quantity
           -> Create backorder for remainder
           -> Backorder Detail
```

25. Fulfillment Exception Management

- **Starting screen:** `Home > Fulfillment > Fulfillment Exceptions`
- **User actions:** Open exception, choose resolution, route order
- **Navigation path:** `Fulfillment Exceptions -> Exception Detail -> Resolve`
- **Decision points:** Re-pick, partial ship, quarantine, substitute, or return to service team?
- **Destination screen:** `Exception Detail`

```
Fulfillment Exceptions
  -> Select exception
  -> Exception Detail
  -> Review issue
  -> [Decision: Resolution path?]
     -> Re-pick -> Return to Pick Queue
     -> Partial ship -> Partial Fulfillment & Backorders
     -> Quarantine -> Quarantine & Holds
     -> Customer follow-up -> Order Status
```

26. Order Status Inquiry

- **Starting screen:** `Home > Fulfillment > Order Status`
- **User actions:** Search order, view status, drill into shipment or exception
- **Navigation path:** `Order Status -> Order Detail -> Shipment / Exception / Backorder detail`
- **Decision points:** Need shipment detail? Need backorder detail? Need customer follow-up?
- **Destination screen:** `Order Detail`

```
Order Status
  -> Search order
  -> Order Detail
  -> Review line status
  -> [Decision: Need deeper detail?]
     -> Shipment -> Shipment Confirmation
     -> Exception -> Exception Detail
     -> Backorder -> Backorder Detail
     -> No -> Order Detail
```

27. Multi-Location Fulfillment

- **Starting screen:** `Home > Fulfillment > Multi-Location Fulfillment`
- **User actions:** Open order, generate split plan, confirm allocation, release to locations
- **Navigation path:** `Multi-Location Fulfillment -> Split Fulfillment Plan -> Release`
- **Decision points:** Single location enough? Accept split plan? Adjust allocation?
- **Destination screen:** `Split Fulfillment Plan`

```
Multi-Location Fulfillment
  -> Open order with shortage
  -> Generate split plan
  -> Split Fulfillment Plan
  -> [Decision: Accept proposed allocation?]
     -> No -> Adjust location quantities
     -> Yes -> Confirm
           -> Release to warehouse queues
           -> Split Fulfillment Plan
```

28. Low-Stock Alert Review

- **Starting screen:** `Home > Procurement > Low-Stock Alerts`
- **User actions:** Review alert, acknowledge, open related action
- **Navigation path:** `Low-Stock Alerts -> Alert Detail -> Replenishment Planning / Transfer Request`
- **Decision points:** Replenish, transfer, or monitor?
- **Destination screen:** `Alert Detail` or related action screen

```
Low-Stock Alerts
  -> Select alert
  -> Alert Detail
  -> Review stock / threshold / open supply
  -> [Decision: Best next action?]
     -> Replenish -> Replenishment Planning
     -> Transfer -> Transfer Request
     -> Monitor -> Acknowledge alert -> Alert Detail
```

29. Replenishment Planning

- **Starting screen:** `Home > Procurement > Replenishment Planning`
- **User actions:** Review recommendation, create requisition, submit for approval
- **Navigation path:** `Replenishment Planning -> Replenishment Recommendation Detail -> Purchase Requisition`
- **Decision points:** Transfer-before-buy? Recommended quantity accepted? Submit now?
- **Destination screen:** `Requisition Detail`

```
Replenishment Planning
  -> Review recommendations
  -> Replenishment Recommendation Detail
  -> [Decision: Transfer can solve need?]
     -> Yes -> Transfer Request
     -> No -> Create Purchase Requisition
           -> Submit for approval
           -> Requisition Detail
```

30. Transfer Request

- **Starting screen:** `Home > Transfers > Transfer Requests`
- **User actions:** Create request, select source and destination, submit
- **Navigation path:** `Transfer Requests -> Transfer Request Detail -> Submit`
- **Decision points:** Best source location? Quantity acceptable? Submit for approval?
- **Destination screen:** `Transfer Request Detail`

```
Transfer Requests
  -> Create transfer request
  -> Transfer Request Detail
  -> Select source / destination / SKU / quantity
  -> Review transfer impact
  -> [Decision: Submit request?]
     -> No -> Save draft / cancel
     -> Yes -> Submit
           -> Transfer Request Detail
```

31. Transfer Approval

- **Starting screen:** `Home > Transfers > Transfer Approvals`
- **User actions:** Open request, review impact, approve or reject
- **Navigation path:** `Transfer Approvals -> Approval Detail -> Approve / Reject`
- **Decision points:** Protect source threshold? Approve?
- **Destination screen:** `Approval Detail`

```
Transfer Approvals
  -> Select pending transfer
  -> Approval Detail
  -> Review source and destination impact
  -> [Decision: Approve transfer?]
     -> No -> Reject
           -> Approval Detail
     -> Yes -> Approve
           -> Approval Detail
```

32. Transfer Execution

- **Starting screen:** `Home > Transfers > Transfer Execution`
- **User actions:** Dispatch stock, receive stock, confirm completion
- **Navigation path:** `Transfer Execution -> Dispatch Detail -> Receipt Detail`
- **Decision points:** Dispatch correct quantity? Destination receives full quantity?
- **Destination screen:** `Receipt Detail`

```
Transfer Execution
  -> Select approved transfer
  -> Dispatch Detail
  -> Pick and verify transfer stock
  -> Confirm dispatch
  -> Receipt Detail
  -> [Decision: Destination received full transfer?]
     -> No -> Investigate discrepancy
     -> Yes -> Confirm receipt
           -> Receipt Detail
```

33. Network Rebalancing Analysis

- **Starting screen:** `Home > Transfers > Network Rebalancing`
- **User actions:** Review imbalances, select recommendation, generate transfer request
- **Navigation path:** `Network Rebalancing -> Rebalancing Recommendation Detail -> Transfer Request`

- **Decision points:** Recommendation valid? Approve recommendation now?
- **Destination screen:** `Transfer Request Detail`

```
Network Rebalancing
  -> Review imbalance dashboard
  -> Select recommendation
  -> Rebalancing Recommendation Detail
  -> [Decision: Approve recommendation?]
     -> No -> Return to dashboard
     -> Yes -> Generate transfer request
           -> Transfer Request Detail
```

34. Return Authorization Management

- **Starting screen:** `Home > Returns > Return Authorizations`
- **User actions:** Search order, validate return, issue authorization
- **Navigation path:** `Return Authorizations -> Return Authorization Detail -> Issue authorization`
- **Decision points:** Return eligible? Quantity allowed? Approve return?
- **Destination screen:** `Return Authorization Detail`

```
Return Authorizations
  -> Search original order
  -> Return Authorization Detail
  -> Select items and quantity
  -> Validate return reason / policy
  -> [Decision: Eligible return?]
     -> No -> Decline request
     -> Yes -> Issue authorization
           -> Return Authorization Detail
```

35. Return Inspection and Disposition

- **Starting screen:** `Home > Returns > Return Inspection & Disposition`
- **User actions:** Receive return, inspect item, select disposition
- **Navigation path:** `Return Inspection & Disposition -> Return Receipt Detail -> Inspection Detail -> Disposition Detail`
- **Decision points:** Authorized return? Resalable, quarantine, or write-off?
- **Destination screen:** `Disposition Detail`

```
Return Inspection & Disposition
  -> Select pending return
  -> Return Receipt Detail
  -> Verify received quantity
  -> Inspection Detail
  -> Assess condition
  -> [Decision: Disposition?]
     -> Restock -> Confirm restock
     -> Quarantine -> Quarantine & Holds
     -> Write-off -> Write-Off Authorization Detail
  -> Disposition Detail
```

36. Operational Reports and Analytics

- **Starting screen:** `Home > Reports & Analytics > Operational Reports`
- **User actions:** Select report, filter results, drill down, export
- **Navigation path:** `Operational Reports -> Report View -> Detail / Export`
- **Decision points:** Which report? Need drill-down? Export needed?
- **Destination screen:** `Report View`

```
Operational Reports
  -> Select report type
  -> Report View
  -> Apply filters
  -> Review results
  -> [Decision: Need drill-down?]
     -> Yes -> Detail View
     -> No -> Continue
  -> [Decision: Export report?]
     -> Yes -> Export
     -> No -> Report View
```

37. Executive KPI Dashboard

- **Starting screen:** `Home > Executive KPI Dashboard`
- **User actions:** Change period, review KPIs, drill into issue
- **Navigation path:** `Executive KPI Dashboard -> KPI Detail -> Supporting report`
- **Decision points:** KPI on target? Drill into underperformance?
- **Destination screen:** `KPI Detail` or supporting analytics view

```
Executive KPI Dashboard
  -> Select reporting period
  -> Review KPIs and trends
  -> [Decision: KPI underperforming?]
     -> No -> Continue review
     -> Yes -> KPI Detail
           -> Open supporting report / analysis
```

38. Inventory Valuation Reconciliation

- **Starting screen:** `Home > Reports & Analytics > Valuation Reconciliation`
- **User actions:** Select period, compare values, document notes, finalize
- **Navigation path:** `Valuation Reconciliation -> Reconciliation Detail -> Finalize`
- **Decision points:** Variance acceptable? Need deeper investigation?
- **Destination screen:** `Reconciliation Detail`

```
Valuation Reconciliation
  -> Select period
  -> Reconciliation Detail
  -> Compare operational vs finance view
  -> [Decision: Variance within tolerance?]
     -> Yes -> Finalize reconciliation
     -> No -> Investigate underlying drivers
           -> Document notes
           -> Finalize or escalate
```

39. Promotional Stock Readiness

- **Starting screen:** `Home > Reports & Analytics > Promotional Readiness`
- **User actions:** Open campaign, review readiness, confirm or escalate
- **Navigation path:** `Promotional Readiness -> Campaign Detail -> Confirm readiness / Escalate`
- **Decision points:** Stock sufficient? Inbound closes gap? Ready to confirm?
- **Destination screen:** `Campaign Detail`

```
Promotional Readiness
  -> Select campaign
  -> Campaign Detail
  -> Review allocated / ring-fenced / inbound supply
  -> [Decision: Campaign stock ready?]
     -> No -> Escalate gap / update plan
     -> Yes -> Confirm readiness
           -> Campaign Detail
```

40. Audit Workspace

- **Starting screen:** `Home > Governance & Audit > Audit Workspace`
- **User actions:** Define scope, review records, document findings, export evidence
- **Navigation path:** `Audit Workspace -> Audit Review Detail -> Export`
- **Decision points:** Control issue found? Sample more records? Export evidence?
- **Destination screen:** `Audit Review Detail`

```
Audit Workspace
  -> Define scope
  -> Audit Review Detail
  -> Review transactions / approvals / controls
  -> [Decision: Finding identified?]
     -> No -> Continue sampling / close review
     -> Yes -> Document finding
  -> [Decision: Export evidence?]
     -> Yes -> Export package
     -> No -> Audit Review Detail
```

41. Audit Trail Query

- **Starting screen:** `Home > Governance & Audit > Audit Trail Query`
- **User actions:** Enter search criteria, review log entries, inspect record details, export
- **Navigation path:** `Audit Trail Query -> Audit Record Detail -> Export`
- **Decision points:** Results found? Need before/after detail? Export required?
- **Destination screen:** `Audit Record Detail`

```
Audit Trail Query
  -> Enter search criteria
  -> Review results
  -> [Decision: Matching records found?]
     -> No -> Refine search
     -> Yes -> Audit Record Detail
           -> Review before / after state
           -> [Decision: Export results?]
              -> Yes -> Export
              -> No -> Audit Record Detail
```

42. User Account Management

- **Starting screen:** `Home > Administration > User Accounts`
- **User actions:** Create user, edit user, assign role, suspend or deactivate account
- **Navigation path:** `User Accounts -> User Account Detail -> Save / Suspend / Deactivate`
- **Decision points:** New or existing user? Change role? Suspend or deactivate?
- **Destination screen:** `User Account Detail`

```
User Accounts
  -> Search user
  -> [Decision: Existing user?]
     -> No -> Create User
           -> Enter profile and scope
           -> Save
           -> User Account Detail
     -> Yes -> User Account Detail
           -> Edit user / assign role / change scope
           -> [Decision: Suspend or deactivate?]
              -> Suspend -> Confirm -> User Account Detail
              -> Deactivate -> Confirm -> User Account Detail
              -> Neither -> Save changes -> User Account Detail
```

43. Roles and Access Control

- **Starting screen:** `Home > Administration > Roles & Access Control`
- **User actions:** Open role, review permissions, update scope rules, save
- **Navigation path:** `Roles & Access Control -> Role Permission Detail -> Save`
- **Decision points:** Permission conflict? Location scope valid? SoD breach?
- **Destination screen:** `Role Permission Detail`

```
Roles & Access Control
  -> Select role
  -> Role Permission Detail
  -> Review permissions and location scope
  -> [Decision: SoD conflict detected?]
     -> Yes -> Revise permissions / scope
     -> No -> Save
           -> Role Permission Detail
```

Notes

- These flows focus on major features rather than every minor interaction.
- Flows are aligned to previously defined screens and navigation structure.
- The diagrams can be used directly as the basis for task flows, page transitions, and low-fidelity UX design.

Inventory Management System Low-Fidelity Wireframe Specifications

Purpose

This document defines low-fidelity wireframe specifications for every identified Inventory Management System screen. The specifications are text-based and describe structural content only. They do not define visual styling, spacing, branding, or high-fidelity interaction behavior.

Structure Standard

Each screen specification includes:

1. Header
2. Navigation
3. Main content
4. Forms
5. Tables
6. Action buttons

7. Footer (if applicable)

Shared Layout Assumptions

- The application uses a persistent top header.
- Main navigation is available globally.
- Secondary navigation appears at module level when applicable.
- Breadcrumbs appear below the header on all non-home screens.
- Most operational screens use a list-detail or dashboard-workbench structure.
- Footer is optional and used mainly for system metadata, help links, and record timestamps.

UI-001 Sign In / Session Entry

Header

- Product name
- Environment indicator
- Optional help link

Navigation

- No primary application navigation before authentication

Main Content

- Centered sign-in panel
- Brief system purpose statement
- Optional support/contact note

Forms

- Username / email
- Password
- Optional remember session checkbox

Tables

- None

Action Buttons

- `Sign In`
- `Forgot Password` or `Need Help`

Footer

- Version
- Security/privacy notice

UI-002 Operational Home Dashboard

Header

- Page title
- Global search
- Notification icon
- My tasks / approvals shortcut
- User menu

Navigation

- Main navigation
- Optional role-based quick links

Main Content

- KPI summary cards
- Stock health summary
- Alert snapshot
- Task / approval snapshot
- Shortcut tiles to core workflows

Forms

- Dashboard filter bar: location, date range, role scope

Tables

- Recent alerts table
- Pending tasks table

Action Buttons

- `View Alerts`
- `Open Approvals`
- `Go to Inventory`
- `Refresh`

Footer

- Last refreshed timestamp

UI-003 Executive KPI Dashboard

Header

- Dashboard title
- Reporting period selector
- Export option

Navigation

- Main navigation
- Secondary navigation for dashboards/reports
- Breadcrumbs

Main Content

- Executive KPI cards
- Trend summaries
- Variance/target indicators
- Drill-down summaries

Forms

- Filter controls: period, business unit, location group

Tables

- KPI summary table

Action Buttons

- `Apply Filters`
- `Export Dashboard`
- `View Detail`

Footer

- Data as-of timestamp

UI-004 Notification Center

Header

- Page title
- Unread count

Navigation

- Main navigation
- Breadcrumbs

Main Content

- Notification inbox layout
- Left filter panel or top filter bar
- Selected notification detail pane

Forms

- Filter controls: type, priority, module, status, date

Tables

- Notification list table

Action Buttons

- `Open Related Record`
- `Mark as Handled`
- `Acknowledge`
- `Refresh`

Footer

- Optional archive retention note

UI-005 Product Catalog Management

Header

- Page title
- Create product shortcut

Navigation

- Main navigation
- Products secondary navigation
- Breadcrumbs

Main Content

- Search/filter toolbar
- Product results list
- Product detail side panel or full detail region

Forms

- Search fields: SKU, name, category, status, ABC class
- Product form fields: SKU, name, description, UOM, category, barcode, lifecycle status

Tables

- Product catalog table

Action Buttons

- `Create Product`
- `Edit`
- `Save`
- `Change Status`
- `Resolve Duplicate`

Footer

- Created/updated metadata

UI-006 Product Import Workspace

Header

- Page title
- Import instructions link

Navigation

- Main navigation
- Products secondary navigation
- Breadcrumbs

Main Content

- Import steps panel
- Template download area
- Upload zone
- Validation results region
- Completion summary region

Forms

- File upload field
- Optional import type selector

Tables

- Validation errors table
- Import summary table

Action Buttons

- `Download Template`
- `Upload File`
- `Validate`
- `Confirm Import`
- `Cancel`

Footer

- Supported file format note

UI-007 Location and Warehouse Management

Header

- Page title
- Create location shortcut

Navigation

- Main navigation
- Locations secondary navigation
- Breadcrumbs

Main Content

- Location directory panel
- Hierarchy view
- Location detail view
- Storage positions subsection

Forms

- Location form: code, name, type, parent, status
- Storage position form: zone, aisle, bin, status

Tables

- Location list table
- Storage positions table

Action Buttons

- `Create Location`
- `Edit`
- `Save`
- `Activate`
- `Deactivate`
- `Close Location`

Footer

- Location lifecycle metadata

UI-008 Stock Policy and Threshold Management

Header

- Page title

Navigation

- Main navigation
- Products secondary navigation
- Breadcrumbs

Main Content

- Product-location selector
- Current stock policy summary
- Threshold editing area

Forms

- Search fields: product, location
- Policy form: reorder level, minimum threshold, safety stock, effective date

Tables

- Existing policy table by product/location

Action Buttons

- `Search`
- `Edit Policy`
- `Save`
- `Reset`

Footer

- Policy version/effective timestamp

UI-009 ABC Classification Management

Header

- Page title

Navigation

- Main navigation
- Products secondary navigation
- Breadcrumbs

Main Content

- Classification summary
- Product review list
- Classification detail panel

Forms

- Filter fields: category, current class, activity status

- Classification edit fields: ABC class, reason, effective date

Tables

- Product classification table

Action Buttons

- `Assign Class`
- `Save`
- `Bulk Update`

Footer

- Classification last-updated metadata

UI-010 Governance Policies and Reason Codes

Header

- Page title
- Tab switcher for policies/reason codes

Navigation

- Main navigation
- Governance & Audit secondary navigation
- Breadcrumbs

Main Content

- Policy list and detail area
- Reason code list and detail area

Forms

- Policy form: name, version, thresholds, tolerances, effective date, status
- Reason code form: code, category, description, active status

Tables

- Governance policies table
- Reason codes table

Action Buttons

- `Create Policy`
- `Publish`
- `Create Reason Code`
- `Retire`
- `Save`

Footer

- Version and publication metadata

UI-011 Supplier Management

Header

- Page title
- Create supplier shortcut

Navigation

- Main navigation
- Procurement secondary navigation
- Breadcrumbs

Main Content

- Supplier list
- Supplier detail panel

Forms

- Supplier form: code, name, contact, payment terms, lead time, active status

Tables

- Supplier table

Action Buttons

- `Create Supplier`
- `Edit`
- `Save`
- `Deactivate`

Footer

- Supplier update metadata

UI-012 Purchase Order Management

Header

- Page title
- Create purchase order shortcut

Navigation

- Main navigation
- Procurement secondary navigation
- Breadcrumbs

Main Content

- PO search/filter area
- PO list
- PO detail and line items section

Forms

- PO header form: supplier, destination, expected date, status
- PO lines form: product, quantity, unit, price, notes

Tables

- Purchase orders table
- PO line items table

Action Buttons

- `Create PO`
- `Save Draft`
- `Submit`
- `Amend`
- `Cancel PO`

Footer

- Approval/status metadata

UI-013 Purchase Requisition Management

Header

- Page title

Navigation

- Main navigation
- Procurement secondary navigation
- Breadcrumbs

Main Content

- Requisition queue
- Requisition detail panel
- Status history section

Forms

- Requisition form: product, quantity, required date, destination, rationale

Tables

- Requisition table

Action Buttons

- `Create Requisition`
- `Approve`
- `Reject`
- `Convert to PO`
- `Save`

Footer

- Approval trail summary

UI-014 Inbound Delivery Planning

Header

- Page title

Navigation

- Main navigation
- Procurement secondary navigation
- Breadcrumbs

Main Content

- Expected deliveries calendar/list
- Selected delivery detail panel

Forms

- Delivery planning fields: PO reference, shipment date, ETA, carrier, destination

Tables

- Inbound deliveries table

Action Buttons

- `Update Delivery`
- `Save`
- `Notify Receiving`

Footer

- Latest supplier update timestamp

UI-015 Goods Receipt Workbench

Header

- Page title
- Location context

Navigation

- Main navigation
- Inbound secondary navigation
- Breadcrumbs

Main Content

- Pending receipt queue
- Receipt detail section
- Verification area

Forms

- Receipt identification fields: PO/reference, supplier, dock, receipt type
- Verification fields: item scan, quantity received, condition, lot/batch, shelf-life

Tables

- Pending receipts table
- Receipt line items table

Action Buttons

- `Open Receipt`
- `Verify`
- `Log Discrepancy`
- `Confirm Receipt`
- `Send to Put-Away`

Footer

- Receipt confirmation metadata

UI-016 Receipt Discrepancy Resolution

Header

- Page title

Navigation

- Main navigation
- Inbound secondary navigation
- Breadcrumbs

Main Content

- Open discrepancies queue
- Discrepancy detail panel
- Evidence/notes section

Forms

- Discrepancy form: discrepancy type, quantity variance, reason code, notes, evidence reference
- Resolution form: accepted quantity, resolution note, approver note

Tables

- Discrepancies table

Action Buttons

- `Save Discrepancy`
- `Resolve`
- `Escalate`
- `Close`

Footer

- Resolution history

UI-017 Put-Away Workbench

Header

- Page title
- Current location context

Navigation

- Main navigation
- Inbound secondary navigation
- Breadcrumbs

Main Content

- Pending put-away task queue
- Task detail section
- Destination confirmation section

Forms

- Put-away fields: receipt reference, product, quantity, suggested location, actual location

Tables

- Pending put-away table

Action Buttons

- `Open Task`
- `Confirm Location`
- `Complete Put-Away`
- `Reassign Location`

Footer

- Task completion timestamp

UI-018 Non-PO and Return Receipt Intake

Header

- Page title

Navigation

- Main navigation
- Inbound secondary navigation
- Breadcrumbs

Main Content

- Receipt type selector
- Intake form area
- Verification results area

Forms

- Receipt type: non-PO / authorized return
- Reference fields: approval number / return authorization

- Verification fields: product, quantity, condition, notes

Tables

- Optional received line items table

Action Buttons

- `Start Intake`
- `Verify`
- `Confirm Receipt`
- `Cancel`

Footer

- Intake transaction metadata

UI-019 Inbound Supply Tracking

Header

- Page title

Navigation

- Main navigation
- Procurement secondary navigation
- Breadcrumbs

Main Content

- Open inbound supply overview
- Delay status indicators
- PO detail drill-in panel

Forms

- Filter controls: supplier, location, status, expected date range

Tables

- Open PO/inbound supply table

Action Buttons

- `View PO`
- `Update Delivery Status`
- `Refresh`

Footer

- Data refresh timestamp

UI-020 Inventory Status Monitor

Header

- Page title
- Location/network scope

Navigation

- Main navigation
- Inventory secondary navigation
- Breadcrumbs

Main Content

- Status summary cards
- Stock status board
- Detail drill-down area

Forms

- Filter controls: location, category, product, status, ABC class

Tables

- Inventory status table

Action Buttons

- `View Detail`
- `Request Transfer`
- `Open Replenishment`
- `Create Adjustment`
- `Refresh`

Footer

- Last reviewed/refresh timestamp

UI-021 Inventory Availability Inquiry

Header

- Page title

Navigation

- Main navigation
- Inventory secondary navigation
- Breadcrumbs

Main Content

- Search panel
- Results list
- Availability detail view

Forms

- Search fields: SKU, name, barcode, category
- Optional location scope selector

Tables

- Product search results table
- Availability by location table

Action Buttons

- `Search`
- `View Network Availability`
- `Reserve from Order`

Footer

- Inquiry timestamp

UI-022 Inventory Movement History

Header

- Page title

Navigation

- Main navigation
- Inventory secondary navigation
- Breadcrumbs

Main Content

- Search/filter region
- Movement list
- Transaction detail view

Forms

- Filter fields: SKU, location, transaction type, user, date range

Tables

- Movement history table

Action Buttons

- `Search`
- `Open Transaction`
- `Export`

Footer

- Export/query metadata

UI-023 Cycle Count Workbench

Header

- Page title
- Assigned user / location context

Navigation

- Main navigation
- Inventory secondary navigation
- Breadcrumbs

Main Content

- Assigned count queue
- Count entry panel
- Submission confirmation area

Forms

- Count entry fields: SKU, location, counted quantity, notes

Tables

- Assigned cycle counts table

Action Buttons

- `Open Count`
- `Submit Count`
- `Save Draft`

Footer

- Assignment and submission timestamps

UI-024 Physical Inventory Count Management

Header

- Page title
- Count session status badge

Navigation

- Main navigation
- Inventory secondary navigation
- Breadcrumbs

Main Content

- Count session summary
- Progress by area/team
- Zone/task breakdown
- Certification panel

Forms

- Session fields: location, freeze start, count scope
- Certification fields: unresolved issues note, certifier

Tables

- Count progress table
- Zone/team assignments table

Action Buttons

- `Start Count`
- `Assign Teams`
- `Monitor Progress`
- `Certify Count`

- `Release Freeze`

Footer

- Session audit metadata

UI-025 Variance Investigation and Approval

Header

- Page title
- Pending approvals count

Navigation

- Main navigation
- Inventory secondary navigation
- Breadcrumbs

Main Content

- Variance queue
- Variance comparison panel
- Investigation notes and evidence area

Forms

- Investigation form: root cause, notes, supporting evidence
- Approval form: decision, approver note

Tables

- Variance cases table

Action Buttons

- `Investigate`
- `Approve`
- `Reject`
- `Escalate`

Footer

- Approval history

UI-026 Stock Adjustment Management

Header

- Page title

Navigation

- Main navigation
- Inventory secondary navigation
- Breadcrumbs

Main Content

- Adjustment cases list
- Adjustment form
- Approval/status summary

Forms

- Adjustment fields: product, location, direction, quantity, reason code, note

Tables

- Adjustment cases table

Action Buttons

- `Create Adjustment`
- `Submit`
- `Approve`
- `Reject`
- `Apply`

Footer

- Adjustment audit metadata

UI-027 Write-Off Authorization

Header

- Page title

Navigation

- Main navigation
- Inventory secondary navigation
- Breadcrumbs

Main Content

- Write-off request list
- Write-off request detail
- Evidence and finance acknowledgment section

Forms

- Request fields: reason, quantity, value impact, evidence reference, finance note

Tables

- Pending write-offs table

Action Buttons

- `Review`
- `Approve`
- `Reject`
- `Request More Info`

Footer

- Authorization metadata

UI-028 Quarantine and Hold Register

Header

- Page title

Navigation

- Main navigation
- Inventory secondary navigation
- Breadcrumbs

Main Content

- Hold/quarantine register
- Selected hold detail
- Apply/release panel

Forms

- Hold form: product, location, quantity, hold type, reason, expected disposition

Tables

- Quarantine/hold table

Action Buttons

- `Apply Hold`
- `Release Hold`
- `View Product`

Footer

- Hold duration and update metadata

UI-029 Shrinkage, Loss, and Obsolescence Analysis

Header

- Page title

Navigation

- Main navigation
- Inventory secondary navigation
- Breadcrumbs

Main Content

- Trend summary cards
- Analysis charts/summary placeholders
- Recommendations section

Forms

- Filter controls: date range, location, category, reason code

Tables

- Shrinkage summary table
- Aging/obsolete items table

Action Buttons

- `Apply Filters`
- `Export`
- `Open Adjustment Cases`

Footer

- Report generation timestamp

UI-030 Order Reservation Workbench

Header

- Page title

Navigation

- Main navigation
- Fulfillment secondary navigation
- Breadcrumbs

Main Content

- Order search and queue
- Order line availability panel
- Reservation summary section

Forms

- Search fields: order number, customer reference, date
- Reservation fields: location, quantity per line, release reason if applicable

Tables

- Orders table
- Order line table

Action Buttons

- `Check Availability`
- `Reserve Stock`
- `Release Reservation`
- `Open Split Fulfillment`

Footer

- Reservation timestamps/status

UI-031 Pick, Pack, and Ship Workbench

Header

- Page title
- Location/shift context

Navigation

- Main navigation
- Fulfillment secondary navigation
- Breadcrumbs

Main Content

- Pick queue
- Pick verification panel
- Packing panel
- Shipment confirmation section

Forms

- Verification fields: item scan, lot/batch, picked quantity
- Packing fields: package reference, carrier, notes

Tables

- Pick queue table
- Pick line items table

Action Buttons

- `Open Pick`
- `Verify Pick`
- `Pack`
- `Raise Exception`
- `Confirm Shipment`

Footer

- Task/shipment metadata

UI-032 Partial Fulfillment and Backorder Management

Header

- Page title

Navigation

- Main navigation
- Fulfillment secondary navigation
- Breadcrumbs

Main Content

- Affected orders list
- Available vs ordered comparison
- Backorder plan section

Forms

- Partial fulfillment fields: fulfill-now quantity, backorder quantity, customer acknowledgment note, expected date

Tables

- Partial/backorder cases table

Action Buttons

- `Confirm Partial Fulfillment`
- `Create Backorder`
- `Cancel Remainder`

Footer

- Customer acknowledgment metadata

UI-033 Fulfillment Exception Management

Header

- Page title
- Open exceptions count

Navigation

- Main navigation
- Fulfillment secondary navigation
- Breadcrumbs

Main Content

- Exception queue
- Exception detail and context
- Resolution options panel

Forms

- Resolution fields: resolution type, notes, reassignment, customer follow-up flag

Tables

- Fulfillment exceptions table

Action Buttons

- `Resolve`
- `Re-Pick`
- `Partial Ship`
- `Quarantine`
- `Route to Customer Service`

Footer

- Resolution history

UI-034 Order Fulfillment Status

Header

- Page title

Navigation

- Main navigation
- Fulfillment secondary navigation
- Breadcrumbs

Main Content

- Order search area
- Order status summary
- Line-level progress view
- Shipment/backorder/exception detail tabs

Forms

- Search fields: order ID, customer reference, date, status

Tables

- Orders table
- Order lines status table

Action Buttons

- `Search`
- `View Shipment`
- `View Exception`
- `View Backorder`

Footer

- Live status timestamp

UI-035 Multi-Location Split Fulfillment Planner

Header

- Page title

Navigation

- Main navigation
- Fulfillment secondary navigation
- Breadcrumbs

Main Content

- Order shortage summary
- Proposed split allocation panel
- Location-specific reservation section

Forms

- Allocation fields: source locations, quantities, shipping grouping, notes

Tables

- Split allocation table

Action Buttons

- `Generate Plan`
- `Adjust Allocation`
- `Confirm Split`
- `Release to Queues`

Footer

- Plan creation metadata

UI-036 Low-Stock Alert Review

Header

- Page title
- Active alerts count

Navigation

- Main navigation
- Procurement secondary navigation
- Breadcrumbs

Main Content

- Alert priority summary
- Alert list
- Alert detail and recommended action panel

Forms

- Filter controls: priority, location, category, status

Tables

- Low-stock alerts table

Action Buttons

- `Acknowledge`
- `Open Replenishment`
- `Create Transfer Request`
- `Refresh`

Footer

- Alert generation timestamp

UI-037 Replenishment Planning

Header

- Page title

Navigation

- Main navigation
- Procurement secondary navigation
- Breadcrumbs

Main Content

- Recommendation list
- Network availability comparison
- Requisition creation panel

Forms

- Replenishment fields: product, suggested quantity, final quantity, required date, supplier, destination

Tables

- Recommendations table
- Open supply/network availability table

Action Buttons

- `Create Requisition`
- `Adjust Quantity`
- `View Open PO`
- `Open Transfer Option`

Footer

- Recommendation calculation timestamp

UI-038 Transfer Request

Header

- Page title

Navigation

- Main navigation
- Transfers secondary navigation
- Breadcrumbs

Main Content

- Transfer request form
- Source impact summary
- Request status panel

Forms

- Transfer fields: source, destination, SKU, quantity, justification

Tables

- Recent transfer requests table

Action Buttons

- `Check Capacity`
- `Submit Request`
- `Save Draft`

Footer

- Request tracking metadata

UI-039 Transfer Approval Queue

Header

- Page title
- Pending approvals count

Navigation

- Main navigation
- Transfers secondary navigation
- Breadcrumbs

Main Content

- Approval queue
- Selected request detail
- Source/destination impact comparison

Forms

- Decision fields: approve/reject, approver note

Tables

- Transfer approvals table

Action Buttons

- `Approve`
- `Reject`
- `View Source Capacity`

Footer

- Approval history

UI-040 Transfer Execution and In-Transit Tracking

Header

- Page title
- Location context

Navigation

- Main navigation
- Transfers secondary navigation

- Breadcrumbs

Main Content

- Approved dispatch queue
- In-transit tracking section
- Destination receipt confirmation section

Forms

- Dispatch fields: transfer ID, picked quantity, lot/batch if applicable
- Receipt fields: quantity received, variance note

Tables

- Transfer queue table
- In-transit transfers table

Action Buttons

- `Dispatch`
- `Confirm Receipt`
- `Cancel Transfer`

Footer

- Transit status metadata

UI-041 Network Rebalancing Analysis

Header

- Page title

Navigation

- Main navigation
- Transfers secondary navigation
- Breadcrumbs

Main Content

- Network imbalance summary
- Recommendations list
- Impact preview section

Forms

- Filter controls: location pair, product category, date range

Tables

- Rebalancing recommendations table

Action Buttons

- `Generate Recommendation`
- `Create Transfer`
- `Export`

Footer

- Analysis timestamp

UI-042 Return Authorization Management

Header

- Page title

Navigation

- Main navigation
- Returns secondary navigation
- Breadcrumbs

Main Content

- Order search panel
- Return eligibility panel
- Authorization detail section

Forms

- Search fields: order number, customer reference
- Authorization fields: item, quantity, reason, authorization notes

Tables

- Eligible return lines table
- Return authorizations table

Action Buttons

- `Validate Eligibility`
- `Issue Authorization`
- `Cancel Authorization`

Footer

- Authorization status metadata

UI-043 Return Inspection and Disposition

Header

- Page title

Navigation

- Main navigation
- Returns secondary navigation
- Breadcrumbs

Main Content

- Pending returns queue
- Receipt verification section

- Inspection section
- Disposition selection section

Forms

- Receipt fields: authorization reference, quantity received
- Inspection fields: condition, notes
- Disposition fields: restock/quarantine/write-off, reason code

Tables

- Pending returns table

Action Buttons

- `Receive Return`
- `Record Inspection`
- `Confirm Disposition`

Footer

- Return processing metadata

UI-044 Operational Reports and Analytics

Header

- Page title
- Report selector

Navigation

- Main navigation
- Reports & Analytics secondary navigation
- Breadcrumbs

Main Content

- Report filter bar
- Report result area
- Drill-down panel

Forms

- Filter controls: report type, location, category, date range, status

Tables

- Report output table

Action Buttons

- `Run Report`
- `Export`
- `Open Detail`

Footer

- Report run timestamp

UI-045 Inventory Valuation Reconciliation

Header

- Page title
- Period selector

Navigation

- Main navigation
- Reports & Analytics secondary navigation
- Breadcrumbs

Main Content

- Valuation summary cards
- Side-by-side comparison area
- Reconciliation notes section

Forms

- Reconciliation fields: period, comment, variance explanation, status

Tables

- Valuation comparison table
- Adjustment/variance detail table

Action Buttons

- `Load Period`
- `Add Note`
- `Finalize Reconciliation`
- `Export`

Footer

- Period close metadata

UI-046 Promotional Stock Readiness

Header

- Page title

Navigation

- Main navigation
- Reports & Analytics secondary navigation
- Breadcrumbs

Main Content

- Campaign selector
- Readiness summary cards
- Product readiness grid
- Gap/escalation section

Forms

- Campaign fields: campaign, launch date, location scope
- Readiness confirmation fields: status, note, escalation reason

Tables

- Campaign product readiness table

Action Buttons

- `Check Readiness`
- `Confirm Readiness`
- `Escalate Gap`
- `Release Allocation`

Footer

- Campaign readiness timestamp

UI-047 Audit and Control Review Workspace

Header

- Page title

Navigation

- Main navigation
- Governance & Audit secondary navigation
- Breadcrumbs

Main Content

- Audit scope setup area
- Sample/review workspace
- Findings panel

Forms

- Scope fields: period, location, transaction type, reviewer
- Finding fields: issue type, severity, note, evidence reference

Tables

- Reviewed transactions table
- Findings table

Action Buttons

- `Start Review`
- `Record Finding`
- `Export Evidence`
- `Complete Review`

Footer

- Audit session metadata

UI-048 Audit Trail Query

Header

- Page title

Navigation

- Main navigation
- Governance & Audit secondary navigation
- Breadcrumbs

Main Content

- Query filter panel
- Audit log result list
- Selected record before/after detail

Forms

- Query fields: entity type, entity ID, user, date range, action type

Tables

- Audit trail results table

Action Buttons

- `Run Query`
- `Open Record`
- `Export Results`

Footer

- Query/extract timestamp

UI-049 User Account Management

Header

- Page title
- Create user shortcut

Navigation

- Main navigation
- Administration secondary navigation
- Breadcrumbs

Main Content

- User directory
- User account detail panel
- Role assignment section

Forms

- User fields: name, email, status, department
- Role assignment fields: role, location scope, effective status

Tables

- Users table
- Role assignments table

Action Buttons

- `Create User`
- `Save`
- `Suspend`
- `Deactivate`
- `Sync Users`

Footer

- Account audit metadata

UI-050 Role and Access Control Administration

Header

- Page title

Navigation

- Main navigation
- Administration secondary navigation
- Breadcrumbs

Main Content

- Role list
- Permission matrix/detail panel
- Location scope rules section
- Conflict/SoD warning area

Forms

- Role fields: role name, description, active status
- Permission fields: permission group, allow/deny, scope

Tables

- Roles table
- Permissions matrix table

Action Buttons

- `Edit Role`
- `Save Permissions`
- `Validate SoD`
- `Reset Changes`

Footer

- Last updated metadata

Notes

- These specifications are intentionally structural and low-fidelity.
- They should be used to create wireframes, not as final UI designs.
- Screen-level interaction states such as empty, loading, error, and success states should be added during detailed wireframe and prototype work.

Inventory Management System Form Specifications and Validation

Purpose

This document defines the UX form specification for every major form in the Inventory Management System. It aligns user-facing validation, inline behavior, success behavior, and error behavior with the approved API validation and error specifications in:

- `SRS/API_Design/Request-DTO-Validation-Spec.md`
- `SRS/API_Design/Error-Response-Specification.md`

Scope Notes

- This document focuses on application forms, not read-only dashboards or query-only screens with no editable input.
- Where multiple screens use the same underlying form pattern, a single shared form specification is used.
- Validation messages are written for users but remain aligned to backend rules such as `VAL_*`, `BIZ_*`, `CONF_*`, and supporting `VR-xxx` / `BR-xxx` rules.
- Input types are expressed in UX terms such as text field, dropdown, date picker, numeric field, file upload, and checkbox.

Shared Validation Behavior

These rules apply to all forms unless a form-specific rule overrides them.

Inline Validation Behavior

- Validate required fields on blur and on submit.
- Validate format-sensitive fields on blur once the user leaves the field.
- Validate cross-field rules when any dependent field changes.
- Do not show an error before the user has interacted with a field, except after submit.
- Replace blocking field errors with success/neutral state immediately when corrected.
- For numeric quantity and amount fields, enforce allowed precision in the input where practical and confirm again on blur.
- For code fields such as SKU, Location Code, Supplier Code, and Reason Code, preserve uppercase formatting where the API expects uppercase patterns.

Success Behavior

- After successful submit, show a visible success banner with a concise outcome statement.
- Keep the user on the record detail screen for create/update forms unless the workflow naturally returns to a queue.
- Refresh related summary panels, tables, and status badges immediately after success.
- For multi-step or queued work, mark the completed item as processed and move the user to the next logical task.

Error Behavior

- For `400 VAL_*` errors, map field-level errors inline and show a form-level summary banner.
- For `409 CONF_*` errors, keep user-entered values intact and explain the conflicting record or state.
- For `422 BIZ_*` errors, show a form-level business-rule message and highlight any affected fields or sections.
- For `401 AUTH_*` errors, redirect to re-authentication or session recovery flow.
- For `403 AUTHZ_*` errors, disable retry-only behavior and explain that the user lacks access.

- For `500/503 SYS_*` errors, preserve draft state when possible and offer retry behavior.

Standard Form-Level Error Summary Pattern

- Title: `There is a problem with this form.`
- Body: `Review the highlighted fields and try again.`
- If the API returns `traceId`, show it in support-facing error details but not as primary user content.

Shared Field Patterns

Shared Pattern A: Product SKU

- **Input type:** Single-line text field
- **Required fields:** Required when the form creates, updates, or references a specific product
- **Validation messages:**
  - `SKU is required.`
  - `SKU must use uppercase letters, numbers, hyphens, or underscores only.`
  - `SKU cannot be longer than 50 characters.`
  - `This product could not be found.` (`NF_PRODUCT_NOT_FOUND`)
- **Inline validation behavior:** Convert to uppercase on blur where appropriate; validate pattern `^[A-Z0-9][A-Z0-9_-]*$`.

Shared Pattern B: Location Code

- **Input type:** Searchable dropdown or code text field depending on context
- **Required fields:** Required when location-scoped activity occurs
- **Validation messages:**
  - `Location is required.`
  - `Select a valid location.`
  - `You are not authorized for this location.` (`AUTHZ_LOCATION_SCOPE_DENIED`)
- **Inline validation behavior:** Validate existence and scope after selection when API lookup is available.

Shared Pattern C: Quantity

- **Input type:** Numeric decimal field with unit label
- **Required fields:** Required when stock, order, receipt, or transfer quantity is submitted
- **Validation messages:**
  - `Quantity is required.`
  - `Quantity must be greater than 0.` for positive quantity contexts
  - `Quantity cannot be negative.`
  - `Quantity has too many decimal places.`
  - `Quantity exceeds the allowed available amount.` (`BIZ_INSUFFICIENT_ATP`, `BIZ_SOURCE_MINIMUM_PROTECTED`, or context equivalent)
- **Inline validation behavior:** Validate numeric format immediately; validate business ceilings after related context fields are complete.

Shared Pattern D: Reason Code

- **Input type:** Dropdown
- **Required fields:** Required for adjustments, write-offs, discrepancies, holds, and many exception flows
- **Validation messages:**
  - `Reason code is required.`
  - `Select a valid active reason code.`
  - `This reason code cannot be used for this transaction type.`
- **Inline validation behavior:** Filter reason-code options by transaction type where possible.

Shared Pattern E: Date Range

- **Input type:** Start date + end date date pickers
- **Required fields:** Required for reporting and audit forms where specified

- **Validation messages:**
  - `Start date is required.`
  - `End date is required.`
  - `Start date must be on or before end date.` (`VAL_INVALID_DATE_RANGE`)
  - `The selected date range is too large for this request.`
- **Inline validation behavior:** Revalidate both fields whenever either date changes.

Form Specifications

F-001 Sign In Form

- **Related screens:** `UI-001`
- **Aligned APIs:** Session/auth integration, `API-173`, `API-174`

Fields

Field	Input type	Required
Username / Email	Text field	Yes
Password	Password field	Yes
Remember Session	Checkbox	No

Validation messages

- `Username or email is required.`
- `Password is required.`
- `Your session could not be started. Check your credentials and try again.` (`AUTH_INVALID_TOKEN` equivalent UX messaging)
- `Your account is suspended.` (`AUTH_ACCOUNT_SUSPENDED`)
- `Your account is deactivated.` (`AUTH_ACCOUNT_DEACTIVATED`)

Inline validation behavior

- Required validation on blur and submit.
- Password field does not show content-specific validation, only presence.

Success behavior

- Redirect to `Operational Home Dashboard` or role-specific landing page.

Error behavior

- Authentication failures show a form-level error banner.
- Preserve entered username/email after failure; clear password field.

F-002 Product Form

- **Related screens:** `UI-005`
- **Aligned APIs:** `API-001`, `API-004`, `API-005`

Fields

Field	Input type	Required
SKU	Text field	Yes on create
Name	Text field	Yes
Description	Text area	No

Field	Input type	Required
Unit of Measure	Dropdown	Yes
Category Code	Searchable dropdown	No
Barcode	Text field	No
Is Traceable	Toggle	No
Rotation Policy	Dropdown	Conditional
Shelf Life Managed	Toggle	No
MRSL Threshold Days	Numeric field	Conditional
Lifecycle Status	Dropdown	Yes on edit/status action
Status Change Reason	Text area	No
Effective Date	Date picker	No

Validation messages

- `SKU is required.`
- `SKU must use uppercase letters, numbers, hyphens, or underscores only.`
- `SKU cannot be longer than 50 characters.`
- `A product with this SKU already exists.` (`CONF_DUPLICATE_SKU`)
- `Name is required.`
- `Name cannot be longer than 200 characters.`
- `Unit of measure is required.`
- `Barcode cannot be longer than 50 characters.`
- `A product with this barcode already exists.` (`CONF_DUPLICATE_BARCODE`)
- `Rotation policy is required when the product is traceable.`
- `MRSL threshold days is required when shelf life management is enabled.`
- `MRSL threshold days must be at least 1.`
- `Select a valid lifecycle status.`
- `This lifecycle change is not allowed for the current product state.`

Inline validation behavior

- SKU: uppercase normalization, pattern validation on blur.
- Conditional fields appear and become required immediately when toggles are enabled.
- Duplicate SKU/barcode checks can run after blur if asynchronous validation is available.

Success behavior

- Show success banner: `Product saved successfully.`
- Remain on `Product Detail`.

Error behavior

- `400` field errors map inline.
- `409` duplicate conflicts show inline on SKU or barcode and in banner.
- `422` lifecycle/business-rule failures show form-level message and keep values intact.

F-003 Product Duplicate Resolution Form

- **Related screens:** `UI-005`
- **Aligned APIs:** `API-009`

Fields

Field	Input type	Required
Resolution Action	Radio group	Yes
Source SKU	Read-only/select field	Yes
Target SKU	Searchable dropdown	Conditional
Merge Authorization Reference	Text field	Conditional
Notes	Text area	No

Validation messages

- `Resolution action is required.`
- `Target product is required when merging duplicates.`
- `Merge authorization reference is required when inventory exists on both products.`
- `Select a valid target product.`

Inline validation behavior

- Conditional fields appear when `MERGE` is selected.
- Revalidate target and authorization reference immediately when resolution action changes.

Success behavior

- Show success banner and refresh the related product records.

Error behavior

- Keep all selections intact.
- Show conflict/business-rule guidance if merge is blocked.

F-004 Bulk Product Import Form

- **Related screens:** `UI-006`
- **Aligned APIs:** `API-010`

Fields

Field	Input type	Required
Import Mode	Dropdown	Yes
Source Reference	Text field	No
Import File / Product Batch	File upload	Yes

Validation messages

- `Import mode is required.`
- `Import file is required.`
- `The selected file format is not supported.`
- `The batch cannot be empty.`
- `This import contains duplicate SKUs in the same batch.`
- `Some rows contain blocking validation errors.`

Inline validation behavior

- Validate file presence immediately after selection.
- Validate file type and size before upload.

- Row-level validation is shown in the results table after validation/upload.

Success behavior

- `VALIDATE_ONLY`: show validation summary.
- `UPSERT` / `CREATE_ONLY`: show import completion summary with created/updated counts.

Error behavior

- Entire-request validation issues show form-level error.
- Per-row errors stay in the results grid with row-level messages.

F-005 Location Form

- **Related screens:** `UI-007`
- **Aligned APIs:** `API-011`, `API-013`, `API-014`, `API-017`, `API-018`, `API-019`

Fields

Field	Input type	Required
Location Code	Text field	Yes on create
Name	Text field	Yes
Location Type	Dropdown	Yes
Operational Status	Dropdown	Yes
Parent Location Code	Searchable dropdown	No
Planned Go-Live Date	Date picker	No
Address Fields	Text fields	No
Activation Date	Date picker	No
Opening Balance Approval Reference	Text field	Conditional
Deactivation Reason	Text area	Conditional
Disposition Plan Reference	Text field	Conditional
Closure Waiver Reference	Text field	Conditional

Validation messages

- `Location code is required.`
- `Location code must use uppercase letters, numbers, hyphens, or underscores only.`
- `A location with this code already exists.` (`CONF_DUPLICATE_LOCATION`)
- `Name is required.`
- `Location type is required.`
- `Operational status is required.`
- `A location cannot be its own parent.`
- `Select a valid parent location.`
- `Opening balance approval reference is required before activation.`
- `A reason is required when deactivating a location.`
- `A disposition plan or closure waiver is required when stock remains at this location.`

Inline validation behavior

- Validate hierarchy rules when parent changes.
- Activation-specific fields validate only when the user initiates activation.
- Closure fields validate only when the user initiates closure.

Success behavior

- Save/update returns the user to `Location Detail` with updated status.

Error behavior

- Hierarchy conflicts show inline on parent field and as banner.
- Business-rule closure and activation failures show form-level guidance.

F-006 Storage Position Form

- **Related screens:** `UI-007`
- **Aligned APIs:** `API-015`

Fields

Field	Input type	Required
Location Code	Context field	Yes
Position Code	Text field	Yes
Zone	Text field	No
Aisle	Text field	No
Bin	Text field	No
Position Type	Dropdown	No
Active	Toggle	No

Validation messages

- `Position code is required.`
- `Position code must be unique within this location.`
- `Select a valid location.`

Inline validation behavior

- Validate uniqueness after blur if supported.

Success behavior

- Add/update row in storage positions table and clear or close form.

Error behavior

- Preserve entered values; highlight duplicate or invalid position issues.

F-007 Stock Policy Form

- **Related screens:** `UI-008`
- **Aligned APIs:** `API-038`, `API-040`

Fields

Field	Input type	Required
Location Code	Searchable dropdown	Yes
SKU	Searchable dropdown / text field	Yes

Field	Input type	Required
Reorder Level	Numeric field	Yes
Minimum Threshold	Numeric field	Yes
Safety Stock	Numeric field	Yes
Maximum Threshold	Numeric field	No
Effective From	Date picker	Yes

Validation messages

- `Location is required.`
- `Product is required.`
- `Reorder level is required.`
- `Minimum threshold is required.`
- `Safety stock is required.`
- `Values cannot be negative.`
- `Maximum threshold must be greater than or equal to reorder level.`
- `Minimum threshold must be less than or equal to reorder level.`
- `Effective from date is required.`

Inline validation behavior

- Revalidate all threshold relationships whenever one threshold field changes.
- Numeric inputs enforce decimal/precision rules immediately.

Success behavior

- Show success banner and refresh the policy summary view.

Error behavior

- Field-level threshold errors shown inline.
- If location or product is inactive, show form-level business-rule message.

F-008 ABC Classification Form

- **Related screens:** `UI-009`
- **Aligned APIs:** `API-006`

Fields

Field	Input type	Required
SKU	Context/select field	Yes
ABC Class	Dropdown / segmented control	Yes
Classification Basis	Dropdown	No
Effective Date	Date picker	No

Validation messages

- `Product is required.`
- `ABC class is required.`
- `Select A, B, or C.`

Inline validation behavior

- Immediate validation when class is selected.

Success behavior

- Update classification in list/table and show success banner.

Error behavior

- Preserve selected class and surface any product-state conflict.

F-009 Governance Policy Form

- **Related screens:** `UI-010`
- **Aligned APIs:** `API-026`, `API-027`, `API-028`

Fields

Field	Input type	Required
Policy Name	Text field	Yes
Description	Text area	No
Effective From	Date-time picker	Yes
Effective To	Date-time picker	No
Transaction Type	Dropdown	Conditional
Quantity Threshold	Numeric field	Conditional
Monetary Threshold	Numeric amount field	Conditional
Currency Code	Dropdown	Conditional
Quantity Tolerance Percent	Numeric field	Conditional
Absolute Tolerance	Numeric field	Conditional
Parameter Key/Value Pairs	Repeating fields	Conditional

Validation messages

- `Policy name is required.`
- `Effective from date is required.`
- `Effective to date must be after effective from date.`
- `At least one policy parameter is required.`
- `Transaction type is required for threshold updates.`
- `At least one threshold value is required.`
- `Thresholds cannot be negative.`
- `Tolerance percent must be between 0 and 100.`
- `A currency code is required when a monetary threshold is entered.`
- `This policy overlaps an existing effective policy version.`

Inline validation behavior

- Cross-field validation runs when effective dates or threshold fields change.
- Monetary threshold and currency validate as a pair.

Success behavior

- Show `Policy published successfully.` or equivalent outcome based on action.

Error behavior

- Overlap and governance-rule issues surface as form-level errors with related fields highlighted.

F-010 Reason Code Form

- **Related screens:** `UI-010`
- **Aligned APIs:** `API-022`, `API-024`

Fields

Field	Input type	Required
Code	Text field	Yes
Description	Text field	Yes
Transaction Types	Multi-select	Yes
Active	Toggle	No
Retirement Reason	Text area	Conditional
Effective Date	Date picker	No

Validation messages

- `Code is required.`
- `Code must use uppercase letters, numbers, and underscores only.`
- `Code cannot be longer than 20 characters.`
- `Description is required.`
- `Select at least one transaction type.`
- `Retirement reason is required when retiring a reason code.`

Inline validation behavior

- Validate code format on blur.
- Retirement fields only validate when retire action is selected.

Success behavior

- Refresh reason-code list and show success feedback.

Error behavior

- Keep form open with code-specific errors preserved.

F-011 Supplier Form

- **Related screens:** `UI-011`
- **Aligned APIs:** `API-049`, `API-052`, `API-053`

Fields

Field	Input type	Required
Supplier Code	Text field	Yes on create
Name	Text field	Yes
Email	Email field	No
Phone	Text field	No

Field	Input type	Required
Payment Terms Days	Numeric field	Yes
Lead Time	Numeric field	No
Currency Code	Dropdown	No
Is Consignment	Checkbox / toggle	No
Active Status	Toggle	No
Deactivation Reason	Text area	Conditional

Validation messages

- `Supplier code is required.`
- `A supplier with this code already exists.`
- `Name is required.`
- `Enter a valid email address.`
- `Payment terms days is required.`
- `Payment terms days cannot be negative.`
- `A reason is required when deactivating a supplier.`

Inline validation behavior

- Validate code uniqueness and email format on blur when possible.

Success behavior

- Stay on supplier detail and refresh supplier summary.

Error behavior

- Duplicate/conflict errors stay attached to supplier code.

F-012 Purchase Requisition Form

- **Related screens:** `UI-013`, `UI-037`
- **Aligned APIs:** `API-054`, `API-056`, `API-057`, `API-058`

Fields

Field	Input type	Required
Receiving Location Code	Searchable dropdown	Yes
Justification	Text area	Yes
Requisition Lines	Repeating line-item editor	Yes
Line SKU	Searchable dropdown / text field	Yes
Line Quantity	Numeric field	Yes
Required Date	Date picker	Yes per workflow if used
Supplier Code	Searchable dropdown	Conditional on conversion
Expected Delivery Date	Date picker	Conditional on conversion
Network Transfer Declined	Checkbox	No
Network Transfer Decline Reason	Text area	Conditional
Rejection Reason	Text area	Conditional

Validation messages

- `Receiving location is required.`
- `Justification is required.`
- `Add at least one requisition line.`
- `Each line must include a product and quantity.`
- `Quantity must be greater than 0.`
- `Discontinued products cannot be requisitioned.` (`BIZ_PRODUCT_DISCONTINUED`)
- `Provide a reason if network transfer was declined.`
- `Supplier is required to convert this requisition to a purchase order.`
- `Expected delivery date is required to convert this requisition.`
- `Rejection reason is required when rejecting a requisition.`

Inline validation behavior

- Line-item validation runs per row and at whole-form level.
- Conditional conversion and approval fields appear only in those actions.

Success behavior

- On create: show requisition detail.
- On approve/reject: update requisition status in place.
- On convert: navigate to resulting `Purchase Order Detail`.

Error behavior

- Preserve all line items and entered notes.
- Surface network-transfer and discontinued-product rules as form-level business messages.

F-013 Purchase Order Form

- **Related screens:** `UI-012`
- **Aligned APIs:** `API-059`, `API-062`, `API-063`, `API-064`

Fields

Field	Input type	Required
Supplier Code	Searchable dropdown	Yes
Receiving Location Code	Searchable dropdown	Yes
Expected Delivery Date	Date picker	Yes
PO Lines	Repeating line-item editor	Yes
Line SKU	Searchable dropdown / text field	Yes
Ordered Quantity	Numeric field	Yes
Amendment Reason	Text area	Conditional
Cancellation Reason	Text area	Conditional
Shipment / Delivery Update Quantity	Numeric field	Conditional

Validation messages

- `Supplier is required.`
- `Receiving location is required.`
- `Expected delivery date is required.`
- `Add at least one purchase order line.`
- `Each line must include a valid product and quantity.`

- `Quantity must be greater than 0.`
- `Discontinued products cannot be ordered.` (`BIZ_PRODUCT_DISCONTINUED`)
- `Supplier must be active.`
- `Amendment reason is required when changing a purchase order.`
- `Cancellation reason is required when canceling a purchase order.`
- `Amended quantity cannot be less than quantity already received.`

Inline validation behavior

- Validate lines per row.
- Amendment/cancellation reason fields only validate during those actions.

Success behavior

- Show PO detail with updated status and line summary.

Error behavior

- State conflicts such as already-amended/canceled PO show form-level conflict banner.
- Preserve user changes for retry when possible.

F-014 Inbound Delivery Planning Form

- **Related screens:** `UI-014`
- **Aligned APIs:** `API-064`

Fields

Field	Input type	Required
Purchase Order ID	Context field	Yes
Expected Arrival Date	Date picker	Yes
Carrier	Text field	No
Delivery Location	Context/select field	No
Expected Quantities by Line	Numeric fields	Conditional

Validation messages

- `Expected arrival date is required.`
- `Expected quantity cannot exceed the remaining open quantity.`

Inline validation behavior

- Revalidate line quantities against outstanding PO quantity whenever edited.

Success behavior

- Delivery schedule updates on the screen and in inbound calendar context.

Error behavior

- Invalid line override quantities stay highlighted inline.

F-015 Goods Receipt Creation / Verification Form

- **Related screens:** `UI-015`
- **Aligned APIs:** `API-066`, `API-069`

Fields

Field	Input type	Required
Receiving Location Code	Searchable dropdown	Yes
Purchase Order ID	Search/select field	Conditional
Supplier Code	Search/select field	Conditional
Receipt Lines	Repeating line-item editor	Yes on verify
Line ID	Hidden/context	Yes
SKU	Read-only/context	Yes
Accepted Quantity	Numeric field	Yes
Rejected Quantity	Numeric field	No
Condition	Dropdown	No
Lot Number	Text field	Conditional
Expiration Date	Date picker	Conditional
Manufacture Date	Date picker	No

Validation messages

- `Receiving location is required.`
- `A valid purchase order is required for supplier receipts.` (`BIZ_PO_REQUIRED_FOR_RECEIPT`)
- `Add at least one verified line.`
- `Accepted quantity is required.`
- `Accepted quantity cannot be negative.`
- `Accepted and rejected quantities cannot exceed the physically verified quantity.`
- `Lot number is required for traceable products.` (`BIZ_LOT_REQUIRED`)
- `Expiration date is required for shelf-life products.`
- `Remaining shelf life does not meet the minimum threshold.` (`BIZ_MRSL_NON_COMPLIANT`)

Inline validation behavior

- Line-level quantities validate immediately on blur.
- Traceability and shelf-life fields become required based on selected/loaded product attributes.

Success behavior

- Receipt is marked verified and becomes available for put-away workflow.

Error behavior

- Business-rule errors appear at line level when specific, otherwise as form-level banner.

F-016 Receipt Discrepancy Form

- **Related screens:** `UI-016`
- **Aligned APIs:** `API-070`, `API-072`

Fields

Field	Input type	Required
Receipt Line ID	Context field	Yes
Discrepancy Type	Dropdown	Yes

Field	Input type	Required
Variance Quantity	Numeric field	Yes
Reason Code	Dropdown	Yes
Notes	Text area	Yes
Evidence Reference / Attachment	File upload or text field	No
Resolution Outcome	Dropdown	Conditional
Resolution Notes	Text area	Conditional

Validation messages

- `Discrepancy type is required.`
- `Variance quantity is required.`
- `Reason code is required.`
- `Notes are required.`
- `Resolution outcome is required to close this discrepancy.`
- `Resolution notes are required to close this discrepancy.`

Inline validation behavior

- Close-resolution fields validate only when the user chooses a resolve action.

Success behavior

- Discrepancy updates to resolved/open status and related receipt reflects accepted outcome.

Error behavior

- Keep all evidence and notes attached to the draft state.

F-017 Put-Away Form

- **Related screens:** `UI-017`
- **Aligned APIs:** `API-073`

Fields

Field	Input type	Required
Put-Away Lines	Repeating line-item editor	Yes
Receipt Line ID	Hidden/context	Yes
Quantity	Numeric field	Yes
Storage Position Code	Searchable dropdown	Yes

Validation messages

- `Add at least one put-away line.`
- `Quantity is required.`
- `Storage position is required.`
- `Put-away quantity cannot exceed the remaining unplaced quantity.`
- `Select a valid storage position.`

Inline validation behavior

- Quantity and storage position validate per row.
- Remaining unplaced quantity should be visible as helper text.

Success behavior

- Task is marked complete and quantity is moved to active stock position.

Error behavior

- Keep the task open and preserve entered row values.

F-018 Ad-Hoc Inbound Receipt Form

- **Related screens:** `UI-018`
- **Aligned APIs:** `API-075`

Fields

Field	Input type	Required
Receiving Location Code	Searchable dropdown	Yes
Ad-Hoc Authorization Reference	Text field	Yes
Reason Code	Dropdown	Yes
Source Description	Text area / text field	Yes
Lines	Repeating line-item editor	Yes
SKU	Searchable dropdown / text field	Yes
Quantity	Numeric field	Yes

Validation messages

- `Receiving location is required.`
- `Authorization reference is required.`
- `Reason code is required.`
- `Source description is required.`
- `Add at least one line.`
- `Quantity must be greater than 0.`

Inline validation behavior

- Validate reference and reason code on blur/selection.

Success behavior

- Create ad-hoc receipt and route to verification/put-away states.

Error behavior

- Show governance-rule error if authorization is not valid.

F-019 Inventory Status Filter Form

- **Related screens:** `UI-020`
- **Aligned APIs:** `API-033` to `API-037`

Fields

Field	Input type	Required
Location	Searchable dropdown	Conditional

Field	Input type	Required
SKU	Search field	Conditional
Category	Dropdown	No
Status	Dropdown	No
ABC Class	Dropdown	No

Validation messages

- `Enter at least one search filter.` when the query requires SKU or location context
- `You are not authorized for this location.`

Inline validation behavior

- Validate scope-sensitive locations after selection.

Success behavior

- Refresh the board/table results in place.

Error behavior

- Keep selected filters and show a banner if the query is invalid or unauthorized.

F-020 Availability Inquiry Form

- **Related screens:** `UI-021`
- **Aligned APIs:** `API-002`, `API-035`, `API-036`

Fields

Field	Input type	Required
SKU	Search field	Conditional
Product Name	Search field	Conditional
Barcode	Search field	Conditional
Category	Dropdown	No
Location	Searchable dropdown	Conditional

Validation messages

- `Enter a product search value.`
- `Select a valid product.`
- `You are not authorized for this location.`

Inline validation behavior

- Search inputs can validate on submit rather than blur to avoid interrupting discovery.

Success behavior

- Show matching products and availability detail.

Error behavior

- No-match state uses empty-state messaging, not error styling.

F-021 Movement History Query Form

- **Related screens:** `UI-022`
- **Aligned APIs:** `API-155`

Fields

Field	Input type	Required
Start Date	Date picker	Yes
End Date	Date picker	Yes
SKU	Search field	No
Location	Searchable dropdown	No
Transaction Type	Dropdown	No
User	Search field	No

Validation messages

- `Start date is required.`
- `End date is required.`
- `Start date must be on or before end date.`
- `Add at least one additional filter to narrow the result set.` when applicable to UX guidance

Inline validation behavior

- Date range validation runs whenever either date changes.

Success behavior

- Populate movement result list and detail panel.

Error behavior

- Date-range issues shown inline on both date fields.

F-022 Cycle Count Submission Form

- **Related screens:** `UI-023`
- **Aligned APIs:** `API-111`

Fields

Field	Input type	Required
Physical Quantity	Numeric field	Yes
Notes	Text area	No

Validation messages

- `Counted quantity is required.`
- `Counted quantity cannot be negative.`

Inline validation behavior

- Quantity validates on blur and before submit.
- Do not expose expected system quantity during this workflow.

Success behavior

- Show `Count submitted successfully.` and return to queue or next assignment.

Error behavior

- Keep entered quantity intact unless the API rejects the case status.

F-023 Physical Count Session Form

- **Related screens:** `UI-024`
- **Aligned APIs:** `API-112`, `API-114`, `API-115`

Fields

Field	Input type	Required
Location Code	Searchable dropdown	Yes
Planned Start Date	Date-time picker	Yes
Planned End Date	Date-time picker	Yes
Count Lines	Repeating line-item editor	Conditional during result entry
Physical Quantity	Numeric field	Yes per line during result entry
Certification Notes	Text area	No

Validation messages

- `Location is required.`
- `Planned start date is required.`
- `Planned end date is required.`
- `Planned end date must be on or after the start date.`
- `All required count lines must be entered before certification.`

Inline validation behavior

- Start/end date validation runs immediately when either changes.
- Certification checks all required work is complete before submit.

Success behavior

- Start action creates count session.
- Certification closes session and updates status badges.

Error behavior

- Block certification until incomplete sections are resolved.

F-024 Variance Investigation Form

- **Related screens:** `UI-025`
- **Aligned APIs:** `API-118`, `API-119`

Fields

Field	Input type	Required
Root Cause Category	Dropdown	Yes

Field	Input type	Required
Investigation Findings	Text area	Yes
Approved Adjustment Quantity	Numeric field	Conditional on approval
Approver Note	Text area	No

Validation messages

- `Root cause category is required.`
- `Investigation findings are required.`
- `Approved adjustment quantity is required to approve the variance.`
- `The approver cannot be the same person who performed the count.` (`BIZ_SEGREGATION_OF_DUTIES`)

Inline validation behavior

- Approval quantity validates only when approval action is selected.

Success behavior

- Case advances to approved/rejected state and updates pending approvals queue.

Error behavior

- SoD and approval-rule failures show a form-level governance error.

F-025 Stock Adjustment Form

- **Related screens:** `UI-026`
- **Aligned APIs:** `API-116`, `API-120`, `API-121`, `API-122`

Fields

Field	Input type	Required
Origin Type	Dropdown	Yes
Location Code	Searchable dropdown	Yes
SKU	Searchable dropdown / text field	Yes
Adjustment Type	Dropdown	Yes
Quantity	Numeric field	Yes
Reason Code	Dropdown	Yes
Notes	Text area	No
Evidence Reference	File upload / text field	Conditional for write-off
Rejection Reason	Text area	Conditional

Validation messages

- `Origin type is required.`
- `Location is required.`
- `Product is required.`
- `Adjustment type is required.`
- `Quantity is required.`
- `Reason code is required.`
- `Quantity must be greater than 0.`
- `Decrease quantity cannot exceed on-hand quantity.`
- `Evidence is required for write-offs.`

- `Rejection reason is required when rejecting an adjustment.`
- `This adjustment requires independent approval.` (`AUTHZ_APPROVAL_THRESHOLD` / `BIZ_SEGREGATION_OF_DUTIES`)

Inline validation behavior

- Quantity direction and on-hand constraints validate when type and quantity are both known.
- Write-off evidence becomes required when adjustment type indicates write-off.

Success behavior

- Case moves to pending approval or applied state depending on thresholds.

Error behavior

- Preserve full draft including attachments and notes.
- Business-rule and approval-threshold failures remain at form level.

F-026 Write-Off Authorization Form

- **Related screens:** `UI-027`
- **Aligned APIs:** `API-120`, `API-121`, `API-122`

Fields

Field	Input type	Required
Case ID	Context field	Yes
Approval Decision	Radio group	Yes
Finance Acknowledgment	Checkbox / status input	Conditional
Approver Note	Text area	No
Rejection Reason	Text area	Conditional

Validation messages

- `Approval decision is required.`
- `Finance acknowledgment is required before this write-off can be finalized.` when policy applies
- `Rejection reason is required when rejecting a write-off.`

Inline validation behavior

- Decision-sensitive fields appear immediately after decision selection.

Success behavior

- Show updated authorization status and refresh pending requests list.

Error behavior

- If blocked by threshold or SoD rules, show governance message and keep user input.

F-027 Quarantine / Hold Form

- **Related screens:** `UI-028`
- **Aligned APIs:** `API-045`, `API-046`

Fields

Field	Input type	Required
Quantity	Numeric field	Yes
Hold Reason	Dropdown	Yes
Reason Code	Dropdown	Yes
Lot Identity	Text field	Conditional
Disposition	Dropdown	Conditional on release
Inspection Outcome Reference	Text field	Conditional on restore to sellable

Validation messages

- `Quantity is required.`
- `Hold reason is required.`
- `Reason code is required.`
- `Lot number is required for traceable products.`
- `Hold quantity cannot exceed on-hand quantity.`
- `Disposition is required when releasing a hold.`
- `Inspection outcome reference is required before restoring stock to sellable status.`

Inline validation behavior

- Release-specific fields validate only in release mode.
- Traceability rules activate based on product context.

Success behavior

- Register updates instantly and affected availability metrics refresh.

Error behavior

- Business-rule conflicts such as bonded/customs restrictions show as form-level errors.

F-028 Order Creation / Import Form

- **Related screens:** `UI-030` supporting context
- **Aligned APIs:** `API-081`, `API-176`

Fields

Field	Input type	Required
Channel	Dropdown	Yes
External Order ID	Text field	Conditional for integration/import contexts
External System Code	Text field	Conditional for integration/import contexts
Order Lines	Repeating line-item editor	Yes
Line Number	Text field / auto-generated	Yes
SKU	Searchable dropdown / text field	Yes
Requested Quantity	Numeric field	Yes

Validation messages

- `Channel is required.`

- `Add at least one order line.`
- `Each order line must include a product and quantity.`
- `Quantity must be greater than 0.`
- `Product must be active.`
- `External order ID is required for imported orders.`
- `External system code is required for imported orders.`

Inline validation behavior

- Validate each row independently and show row-level errors.

Success behavior

- Order is created and available for availability/reservation flow.

Error behavior

- Idempotent integration conflicts show clear conflict banner while preserving payload if user-entered.

F-029 Order Availability / Reservation Form

- **Related screens:** `UI-030`
- **Aligned APIs:** `API-084`, `API-085`, `API-043`, `API-044`, `API-086`

Fields

Field	Input type	Required
Order ID	Context field	Yes
Selected Fulfillment Location	Searchable dropdown	Conditional
Reservation Quantity by Line	Numeric field	Conditional
Safety Stock Exception Reference	Text field	Conditional
Release Reason	Dropdown / text area	Conditional
Cancellation Reason	Text area	Conditional

Validation messages

- `Select a fulfillment location.`
- `Reservation quantity cannot exceed available-to-promise.`
- `A safety stock exception reference is required to continue.` (`BIZ_SAFETY_STOCK_BREACH`)
- `Release reason is required when releasing a reservation.`
- `Cancellation reason is required when canceling the order.`
- `This order is already confirmed.` (`CONF_ORDER_ALREADY_CONFIRMED`)

Inline validation behavior

- Quantity validates against ATP after location is selected.
- Safety-stock reference becomes required only when a breach condition is detected.

Success behavior

- Order status updates to reserved/released/canceled and warehouse queue state refreshes.

Error behavior

- ATP and safety-stock failures show form-level business messages and line-level highlights where applicable.

F-030 Pick Verification Form

- **Related screens:** `UI-031`
- **Aligned APIs:** `API-090`

Fields

Field	Input type	Required
Pick Lines	Repeating line-item editor	Yes
Picked Quantity	Numeric field	Yes
Picked SKU	Context / scanned input	Yes
Lot Number	Text field	Conditional
Substitution Reference	Text field	Conditional

Validation messages

- `Add at least one pick line.`
- `Picked quantity is required.`
- `Picked quantity cannot exceed reserved quantity.`
- `Lot number is required for traceable products.`
- `A substitution reference is required when the picked product differs from the order line.`

Inline validation behavior

- Validate line by line as scans/entries occur.
- Substitution reference appears only when SKU mismatch is detected or selected.

Success behavior

- Pick lines move to packed-ready state.

Error behavior

- Quantity or SKU mismatches keep the pick in exception-ready state without losing scanned values.

F-031 Pack Order Form

- **Related screens:** `UI-031`
- **Aligned APIs:** `API-091`

Fields

Field	Input type	Required
Packages	Repeating package editor	Yes
Package Reference	Text field	No
Package Line Items	Repeating line-item editor	Yes
Quantity per Line	Numeric field	Yes

Validation messages

- `Add at least one package.`
- `Each package must contain at least one line item.`
- `Packed quantity cannot exceed picked quantity.`

Inline validation behavior

- Validate package totals against picked quantities whenever package lines change.

Success behavior

- Order moves to packed state and enables shipment confirmation.

Error behavior

- Preserve package structure and highlight the package row with the issue.

F-032 Shipment Confirmation Form

- **Related screens:** `UI-031`
- **Aligned APIs:** `API-092`

Fields

Field	Input type	Required
Shipment Lines	Repeating line-item editor	Yes
Shipped Quantity	Numeric field	Yes
Carrier	Text field / dropdown	No
Tracking Reference	Text field	No
Lot Number	Text field	Conditional

Validation messages

- `Add at least one shipment line.`
- `Shipped quantity cannot exceed picked quantity.`
- `Lot number is required for traceable products.`
- `Pick verification must be completed before shipment.` (`BIZ_VERIFY_BEFORE_ACCEPTANCE`-style fulfillment rule / `BR-003`)

Inline validation behavior

- Shipment line validation runs per row; traceability rules appear by product context.

Success behavior

- Order status changes to shipped and reservation is cleared.

Error behavior

- Keep shipment draft values if dispatch fails or network issue occurs.

F-033 Partial Fulfillment / Backorder Form

- **Related screens:** `UI-032`
- **Aligned APIs:** `API-093`, `API-094`, `API-095`

Fields

Field	Input type	Required
Fulfillment Lines	Repeating line-item editor	Yes
Fulfill Quantity	Numeric field	Yes

Field	Input type	Required
Create Backorder for Remainder	Checkbox	Yes
Customer Acknowledgment Reference	Text field	Conditional
Priority Override Reference	Text field	Conditional for backorder fulfillment

Validation messages

- `Add at least one fulfillment line.`
- `Fulfill quantity must be greater than 0.`
- `Fulfill quantity must be less than the full ordered line for a partial fulfillment flow.`
- `Customer acknowledgment reference is required before creating a backorder.` (`BIZ_BACKORDER_ACK_REQUIRED`)
- `Backorder quantity must match the unfulfilled remainder.`
- `Priority override reference is required when fulfilling out of sequence.`

Inline validation behavior

- Backorder-required fields validate as soon as the user indicates the remainder will be backordered.
- Recalculate remainder live as fulfill quantity changes.

Success behavior

- Immediate quantity progresses to fulfillment; remainder moves into backorder state.

Error behavior

- Keep line breakdown visible and preserve customer acknowledgment entry.

F-034 Fulfillment Exception Form

- **Related screens:** `UI-033`
- **Aligned APIs:** `API-096`

Fields

Field	Input type	Required
Exception Type	Dropdown	Yes
Description	Text area	Yes
Resolution Path	Dropdown	No at creation; yes in resolution step
Notes	Text area	No

Validation messages

- `Exception type is required.`
- `Description is required.`
- `Description cannot be longer than 2000 characters.`

Inline validation behavior

- Basic validation only on create; richer resolution validation occurs in downstream resolution interactions.

Success behavior

- Exception case is logged and visible in exception queue.

Error behavior

- Keep entered description and route context if submission fails.

F-035 Multi-Location Fulfillment Plan Form

- **Related screens:** `UI-035`
- **Aligned APIs:** `API-087`

Fields

Field	Input type	Required
Allocations	Repeating line-item allocation editor	Yes
Location Code	Searchable dropdown	Yes
Quantity	Numeric field	Yes

Validation messages

- `Add at least one allocation.`
- `Each allocation must include a location and quantity.`
- `Allocated quantity cannot exceed ATP at the selected location.`
- `The total allocated quantity must equal the order line quantity.`

Inline validation behavior

- Validate totals continuously as allocations change.
- Show remaining/unallocated quantity helper text.

Success behavior

- Reservations are split by location and downstream pick queues are prepared.

Error behavior

- Keep allocations intact when any one row fails.

F-036 Alert Acknowledgment Form

- **Related screens:** `UI-004`, `UI-036`
- **Aligned APIs:** `API-159`

Fields

Field	Input type	Required
Action Taken	Dropdown / radio group	Yes
Acknowledgment Notes	Text area	Conditional

Validation messages

- `Action taken is required.`
- `Notes are required when dismissing or deferring an alert.`

Inline validation behavior

- Notes validate immediately when `Dismiss` or `Defer` is selected.

Success behavior

- Alert state updates and is removed from active queue when appropriate.

Error behavior

- Preserve selected action and notes.

F-037 Transfer Request Form

- **Related screens:** `UI-038`, `UI-041`
- **Aligned APIs:** `API-098`

Fields

Field	Input type	Required
Source Location Code	Searchable dropdown	Yes
Destination Location Code	Searchable dropdown	Yes
Justification	Text area	Yes
Lines	Repeating line-item editor	Yes
SKU	Searchable dropdown / text field	Yes
Quantity	Numeric field	Yes

Validation messages

- `Source location is required.`
- `Destination location is required.`
- `Source and destination cannot be the same.`
- `Justification is required.`
- `Add at least one transfer line.`
- `Quantity must be greater than 0.`
- `Requested quantity exceeds available transferable stock.`
- `This transfer would breach the source minimum threshold.` (`BIZ_SOURCE_MINIMUM_PROTECTED`)

Inline validation behavior

- Source vs destination validation runs immediately when either location changes.
- Quantity validates against transferable capacity when source and SKU are known.

Success behavior

- Transfer request is created and appears in approval queue.

Error behavior

- Preserve all lines and justification.

F-038 Transfer Approval Form

- **Related screens:** `UI-039`
- **Aligned APIs:** `API-101`, `API-102`

Fields

Field	Input type	Required
Decision	Radio group	Yes
Rejection Reason	Text area	Conditional

Validation messages

- `Approval decision is required.`
- `Rejection reason is required when rejecting a transfer.`
- `This transfer can no longer be approved in its current state.`

Inline validation behavior

- Rejection reason validates immediately when reject is selected.

Success behavior

- Request status updates to approved/rejected and dependent queues refresh.

Error behavior

- Revalidation failures due to changed ATP show business-rule banner.

F-039 Transfer Dispatch / Receipt Form

- **Related screens:** `UI-040`
- **Aligned APIs:** `API-103`, `API-104`, `API-105`

Fields

Field	Input type	Required
Dispatch Lines	Repeating line-item editor	Conditional on dispatch
Receipt Lines	Repeating line-item editor	Conditional on receipt
Quantity	Numeric field	Yes
Lot Number	Text field	Conditional
Variance Notes	Text area	Conditional
Cancellation Reason	Text area	Conditional

Validation messages

- `Add at least one transfer line.`
- `Dispatched quantity cannot exceed approved quantity.`
- `Received quantity is required.`
- `Variance notes are required when received quantity differs from dispatched quantity.`
- `A cancellation reason is required to cancel an in-transit transfer.`

Inline validation behavior

- Dispatch and receipt validation depend on current workflow state.
- Variance notes appear only when mismatch is detected.

Success behavior

- Dispatch moves transfer to in-transit state.

- Receipt completes transfer and updates both locations.

Error behavior

- Preserve row edits and mismatch notes.

F-040 Return Authorization Form

- **Related screens:** `UI-042`
- **Aligned APIs:** `API-124`, `API-127`

Fields

Field	Input type	Required
Original Order ID	Search/select field	Yes
Return Reason	Text area / text field	Yes
Return Lines	Repeating line-item editor	Yes
SKU / Order Line	Context/select field	Yes
Authorized Quantity	Numeric field	Yes
Cancellation Reason	Text area	Conditional

Validation messages

- `Original order is required.`
- `Return reason is required.`
- `Add at least one return line.`
- `Authorized quantity must be greater than 0.`
- `Authorized quantity cannot exceed shipped quantity.`
- `A cancellation reason is required when canceling a return authorization.`
- `This order line is not eligible for return.` (`BIZ_RETURN_AUTH_REQUIRED`-adjacent eligibility rule)

Inline validation behavior

- Quantity validates against available shipped quantity after order line selection.

Success behavior

- Return authorization is issued and visible in pending returns.

Error behavior

- Preserve selected lines and quantities.

F-041 Return Receipt / Inspection / Disposition Form

- **Related screens:** `UI-043`
- **Aligned APIs:** `API-128`, `API-129`, `API-130`

Fields

Field	Input type	Required
Return ID / Authorization Reference	Context field	Yes
Receiving Location Code	Searchable dropdown	Yes on receipt

Field	Input type	Required
Received Lines	Repeating line-item editor	Yes
Received Quantity	Numeric field	Yes
Inspection Lines	Repeating line-item editor	Yes on inspection
Condition	Dropdown	Yes
Disposition Action	Dropdown	Yes
Disposition Lines	Repeating line-item editor	Yes
Restock Location Code	Searchable dropdown	Conditional
Write-Off Reference / Evidence	Text field / upload	Conditional

Validation messages

- `Receiving location is required.`
- `Add at least one received line.`
- `Received quantity cannot exceed authorized quantity.`
- `Inspection must be completed before disposition.` (`BR-026`)
- `Disposition action is required.`
- `Disposition quantities must equal the inspected quantity.`
- `Restock location is required when returning inventory to stock.`
- `Write-off reference or evidence is required when disposition is write-off.`

Inline validation behavior

- Workflow-step validation changes as the record moves from receipt to inspection to disposition.
- Quantity balancing runs continuously in disposition step.

Success behavior

- Return progresses to the next state or completes with final inventory outcome.

Error behavior

- Preserve line breakdowns and condition assessments across retries.

F-042 Report Filter Form

- **Related screens:** `UI-044`
- **Aligned APIs:** `API-150` to `API-172` depending on selected report

Fields

Field	Input type	Required
Report Type	Dropdown	Yes
Start Date	Date picker	Conditional
End Date	Date picker	Conditional
Location	Searchable dropdown / multi-select	No
Category	Dropdown	No
Status	Dropdown	No
KPI Codes	Multi-select	Conditional
Period End Date	Date picker	Conditional for valuation reconciliation

Validation messages

- `Report type is required.`
- `Start date is required for this report.`
- `End date is required for this report.`
- `Start date must be on or before end date.`
- `The selected date range is too large for this report.`
- `Select valid KPI codes.`

Inline validation behavior

- Required fields change based on selected report type.
- Date-range validation runs immediately after either date changes.

Success behavior

- Report output refreshes in place.

Error behavior

- Keep filters intact and focus the user on invalid filter fields.

F-043 Valuation Reconciliation Form

- **Related screens:** `UI-045`
- **Aligned APIs:** `API-171`, `API-172`, `API-077` to `API-080` supporting context

Fields

Field	Input type	Required
Period End Date	Date picker	Yes
Variance Threshold Percent	Numeric field	No
Reconciliation Notes	Text area	No
Finalization Status	Dropdown / action choice	Conditional

Validation messages

- `Period end date is required.`
- `Variance threshold percent must be between 0 and 100.`

Inline validation behavior

- Threshold field validates numeric range on blur.

Success behavior

- The reconciliation view refreshes with final status and saved notes.

Error behavior

- Preserve notes and selected period on failure.

F-044 Promotional Allocation / Readiness Form

- **Related screens:** `UI-046`
- **Aligned APIs:** `API-132`, `API-135`, `API-136`, `API-137`, `API-138`

Fields

Field	Input type	Required
Campaign ID	Text field	Yes
Campaign Name	Text field	Yes
Location Code	Searchable dropdown	Yes
SKU	Searchable dropdown / text field	Yes
Allocated Quantity	Numeric field	Yes
Campaign Start Date	Date picker	Yes
Campaign End Date	Date picker	Yes
Amendment Reason	Text area	Conditional
Target Quantity	Numeric field	Conditional
Release Reason	Dropdown	Conditional
Override Quantity	Numeric field	Conditional
Authorization Reference	Text field	Conditional
Justification	Text area	Conditional

Validation messages

- `Campaign ID is required.`
- `Campaign name is required.`
- `Location is required.`
- `Product is required.`
- `Allocated quantity is required.`
- `Allocated quantity must be greater than 0.`
- `Campaign start date is required.`
- `Campaign end date is required.`
- `Campaign end date must be on or after campaign start date.`
- `Allocated quantity cannot exceed available stock.`
- `Authorization reference is required for promotional override.`
- `Justification is required for promotional override.`

Inline validation behavior

- Date range validation and quantity-vs-ATP validation run immediately when inputs change.
- Action-specific fields validate only during amend/release/override actions.

Success behavior

- Allocation/readiness status refreshes and campaign grid updates.

Error behavior

- Ring-fence and ATP errors remain as form-level business messages.

F-045 Audit Review Scope Form

- **Related screens:** `UI-047`
- **Aligned APIs:** `API-156`, `API-157`, `API-167`, `API-168`

Fields

Field	Input type	Required
Start Date	Date picker	Yes
End Date	Date picker	Yes
Location	Searchable dropdown / multi-select	No
Transaction Type	Dropdown	No
Reviewer	Text field / context field	No
Finding Type	Dropdown	Conditional when recording finding
Severity	Dropdown	Conditional when recording finding
Finding Note	Text area	Conditional when recording finding
Evidence Reference	File upload / text field	No

Validation messages

- `Start date is required.`
- `End date is required.`
- `Start date must be on or before end date.`
- `The selected date range is too large for audit export.`
- `Finding type is required when recording a finding.`
- `Severity is required when recording a finding.`
- `Finding note is required when recording a finding.`

Inline validation behavior

- Date-range validation runs immediately.
- Finding fields only validate after the user chooses to record a finding.

Success behavior

- Review scope loads results; findings save without leaving the review.

Error behavior

- Export failures preserve the audit scope and show retry guidance.

F-046 Audit Trail Query Form

- **Related screens:** `UI-048`
- **Aligned APIs:** `API-156`, `API-157`

Fields

Field	Input type	Required
Entity Type	Dropdown	No
Entity ID	Text field	No
User	Text field	No
Start Date	Date picker	Yes
End Date	Date picker	Yes
Action Type	Dropdown	No

Validation messages

- `Start date is required.`
- `End date is required.`
- `Start date must be on or before end date.`
- `The selected date range is too large for export.`

Inline validation behavior

- Date fields revalidate together.

Success behavior

- Results populate in query table; detail panel opens on row selection.

Error behavior

- Keep query filters intact and show export/range messages inline and in banner if needed.

F-047 User Account Form

- **Related screens:** `UI-049`
- **Aligned APIs:** `API-140`, `API-143`, `API-145`, `API-146`

Fields

Field	Input type	Required
User ID	Text field	Yes on create
Email	Email field	Yes
Display Name	Text field	Yes
Initial Roles / Role Assignments	Role assignment widget	Yes
Role Code	Dropdown	Yes
Location Codes	Multi-select	No
Effective From	Date picker	No
Effective To	Date picker	No
Suspension Reason	Text area	Conditional
Deactivation Reason	Text area	Conditional

Validation messages

- `User ID is required.`
- `User ID can contain letters, numbers, and underscores only.`
- `Email is required.`
- `Enter a valid email address.`
- `Display name is required.`
- `Assign at least one role.`
- `Effective to date must be on or after effective from date.`
- `This role assignment creates a segregation-of-duties conflict.` (`BIZ_SEGREGATION_OF_DUTIES`)
- `Suspension reason is required when suspending a user.`
- `Deactivation reason is required when deactivating a user.`

Inline validation behavior

- Role and location assignments validate as a group.

- SoD checks can run after role selection if asynchronous evaluation is available.

Success behavior

- User detail refreshes with active roles and status.

Error behavior

- Preserve all role assignments and notes on error.

F-048 Role and Access Control Form

- **Related screens:** `UI-050`
- **Aligned APIs:** `API-031`, `API-148`, `API-149`

Fields

Field	Input type	Required
Role Name	Text field / context field	Yes
Permission Group Assignments	Permission matrix	Yes
Location Scope Rules	Multi-select / scope selector	No
Initiator User ID	Text field	Conditional for SoD evaluation
Approver User ID	Text field	Conditional for SoD evaluation
Transaction Type	Dropdown	Conditional for SoD evaluation

Validation messages

- `Select at least one permission for this role.`
- `Approver cannot be the same as the initiator.` (`BIZ_SEGREGATION_OF_DUTIES`)
- `Select a valid transaction type for segregation-of-duties evaluation.`
- `You are not authorized to assign this role.` (`AUTHZ_ROLE_ASSIGNMENT_DENIED`)

Inline validation behavior

- SoD evaluation can run when both users and transaction type are present.
- Permission matrix should show conflict/warning state before save where possible.

Success behavior

- Permission matrix saves and confirmation banner appears.

Error behavior

- Keep all edited permission changes in memory until user chooses to discard or retry.

F-049 Session / Logout Form

- **Related screens:** Global user menu
- **Aligned APIs:** `API-174`

Fields

Field	Input type	Required
Logout All Sessions	Checkbox	No

Validation messages

- None beyond system-level failure messaging

Inline validation behavior

- No inline validation required

Success behavior

- End session and redirect to sign-in screen

Error behavior

- If logout fails, show lightweight banner and allow retry

Traceability Summary

- Product, catalog, and lifecycle forms align primarily to `VR-001` to `VR-010`.
- Location and storage forms align primarily to `VR-011` to `VR-017`.
- Stock policy, governance, and threshold forms align primarily to `VR-018` to `VR-028`.
- Procurement and supplier forms align primarily to `VR-030` to `VR-051`.
- Receiving and discrepancy forms align primarily to `VR-035` to `VR-048`.
- Fulfillment forms align primarily to `VR-090` to `VR-111`.
- Replenishment and transfer forms align primarily to `VR-117` to `VR-141`.
- Returns forms align primarily to `VR-142` to `VR-149`.
- Reporting and audit date-range forms align primarily to `VR-062`, `VR-153`, and related intelligence query rules.
- User and access forms align primarily to `VR-083` and location-scope validations such as `VR-008` and `VR-053`.

Notes

- This specification is written for UX and frontend implementation and should be read together with the backend validation and error contracts.
- Final field-level implementation should preserve API naming, value constraints, and business-rule enforcement even if labels are simplified for users.
- Empty, loading, disabled, and async-check states should be specified in the next UI interaction-state pass.

Inventory Management System Screen State Specifications

Purpose

This document defines the expected user experience for key screen states across every Inventory Management System screen. It covers:

1. Loading state
2. Empty state
3. Error state
4. Success state
5. No-permission state

The goal is to ensure every screen behaves consistently and gives users clear guidance when data is loading, absent, successful, blocked, or unavailable.

Shared UX Principles

- **Loading:** Preserve page structure with skeletons or placeholders so users understand what is coming.

- **Empty:** Explain why nothing is shown and what the user can do next.
- **Error:** State what went wrong in user language, preserve work where possible, and offer recovery.
- **Success:** Confirm completion clearly and update the affected data or workflow immediately.
- **No-permission:** Explain access limits without exposing sensitive detail and point users to the right next action.

UI-001 Sign In / Session Entry

Loading state

- Show a disabled sign-in button with inline spinner after submit.
- Keep username visible and mask password.
- Message: `Signing you in...`

Empty state

- Not applicable as a data state.
- Default first-load experience acts as the neutral state.

Error state

- Show a form-level error for invalid credentials or session problems.
- Preserve username/email; clear password.
- Offer retry and help link.

Success state

- Show brief transition feedback such as `Sign-in successful`.
- Redirect immediately to the correct landing screen.

No-permission state

- If authentication succeeds but the account has no usable access, show:
  - `Your account is active, but no application access has been assigned.`
  - Action: `Contact Administrator`

UI-002 Operational Home Dashboard

Loading state

- Show skeleton KPI cards, placeholder alert rows, and shortcut tile skeletons.
- Keep page header and navigation visible.

Empty state

- If there are no alerts or tasks, show:
  - `You are up to date.`
  - Explain that no urgent items or pending tasks exist.

Error state

- Show banner: `We could not load the dashboard right now.`
- Keep navigation available.
- Offer `Retry`.

Success state

- After refresh or task completion, show a non-blocking success banner:
  - `Dashboard updated successfully.`

No-permission state

- If user can access home but not some widgets, show available widgets and replace restricted ones with:
  - `You do not have access to this data.`

UI-003 Executive KPI Dashboard

Loading state

- Show KPI card skeletons and trend placeholders.
- Preserve selected period if already chosen.

Empty state

- Show:
  - `No KPI data is available for the selected period.`
  - Suggest changing period or scope.

Error state

- Show banner: `KPI data could not be loaded.`
- Keep filters intact.
- Offer `Retry`.

Success state

- After filters change, update KPI cards and show subtle confirmation:
  - `Dashboard updated.`

No-permission state

- Show:
  - `You do not have permission to view executive KPI data.`
  - Offer return to `Home`.

UI-004 Notification Center

Loading state

- Show placeholder notification rows and a blank detail pane with shimmer state.

Empty state

- Show:
  - `No notifications to review.`
  - Explain that alerts, approvals, and operational notices will appear here.

Error state

- Show:
  - `Notifications could not be loaded.`
  - Preserve current filters if any.

Success state

- After acknowledge/handle:
  - Remove or update the item in place.
  - Show `Notification updated.`

No-permission state

- Show:
  - `You do not have permission to view this notification set.`
  - If partial access exists, show only allowed items.

UI-005 Product Catalog Management

Loading state

- Show product table skeleton and product detail placeholder.
- Keep search/filter bar visible.

Empty state

- If catalog is empty:
  - `No products have been created yet.`
  - Primary action: `Create Product`
- If search returns no matches:
  - `No products match these filters.`

Error state

- For catalog load failure:
  - `Products could not be loaded.`
- For save failure:
  - Preserve form values and show inline plus banner errors.

Success state

- After create/update:
  - Show `Product saved successfully.`
  - Refresh detail and table row.

No-permission state

- Read-only users see product data without edit actions.
- Fully blocked users see:
  - `You do not have permission to manage products.`

UI-006 Product Import Workspace

Loading state

- Show step progress placeholder, upload area active, validation/result panels skeletonized while processing.

Empty state

- Show:
  - `No import file uploaded yet.`
  - Provide template guidance and upload prompt.

Error state

- If file-level failure:
  - `The import could not be processed.`
- If validation fails:
  - Keep file summary and display row-level issues clearly.

Success state

- Show created/updated counts and any non-blocking warnings.
- Message: `Import completed successfully.`

No-permission state

- Show:
  - `You do not have permission to import product data.`
  - Hide upload control.

UI-007 Location and Warehouse Management

Loading state

- Show location list skeleton, hierarchy tree placeholders, and empty detail shell.

Empty state

- `No locations have been configured yet.`
- Primary action: `Create Location`

Error state

- On load: `Locations could not be loaded.`
- On save/activate/close: preserve edits and show relevant field or business-rule errors.

Success state

- `Location saved successfully.`
- `Location activated successfully.` or equivalent action-specific feedback.

No-permission state

- Show read-only location directory if view-only access exists.
- Otherwise show blocked-state message with return navigation.

UI-008 Stock Policy and Threshold Management

Loading state

- Show filter shell and policy-detail skeleton.

Empty state

- `No stock policy exists for this product and location.`
- Action: `Create Policy`

Error state

- Show invalid threshold relationships inline.
- For load failures, preserve current filter values.

Success state

- `Stock policy saved successfully.`
- Refresh effective thresholds summary.

No-permission state

- `You do not have permission to manage stock policies.`

UI-009 ABC Classification Management

Loading state

- Show classification table skeleton and summary placeholders.

Empty state

- `No products are available for classification in this view.`
- Suggest adjusting filters.

Error state

- `Classifications could not be loaded.`
- Save errors preserve class selections.

Success state

- `Classification updated successfully.`
- Updated class appears immediately in the list.

No-permission state

- `You do not have permission to manage ABC classifications.`

UI-010 Governance Policies and Reason Codes

Loading state

- Show tabs and table skeletons; keep module shell visible.

Empty state

- Policies tab: `No governance policies have been published yet.`
- Reason codes tab: `No reason codes are available yet.`

Error state

- `Governance data could not be loaded.`
- Publish/update failures preserve draft values.

Success state

- `Policy published successfully.` or `Reason code saved successfully.`

No-permission state

- If read-only audit access exists, disable editing and publishing.
- Otherwise show blocked-state panel.

UI-011 Supplier Management

Loading state

- Show supplier table skeleton and detail placeholder.

Empty state

- `No suppliers have been added yet.`
- Primary action: `Create Supplier`

Error state

- `Suppliers could not be loaded.`
- Save/deactivate errors preserve the form.

Success state

- `Supplier saved successfully.`
- Deactivation uses action-specific confirmation.

No-permission state

- `You do not have permission to manage suppliers.`

UI-012 Purchase Order Management

Loading state

- Show PO table skeleton and line-item placeholder rows.

Empty state

- Default list:
  - `No purchase orders found.`
  - Action: `Create Purchase Order`
- Filtered view:
  - `No purchase orders match the selected filters.`

Error state

- Load error: `Purchase orders could not be loaded.`
- Save/amend/cancel errors preserve all draft line items.

Success state

- `Purchase order saved successfully.`
- `Purchase order canceled successfully.` as applicable.

No-permission state

- View-only users can inspect orders without edit actions.
- No-access users see a blocked-state panel.

UI-013 Purchase Requisition Management

Loading state

- Show requisition queue skeleton and detail pane placeholder.

Empty state

- `No requisitions are available in this view.`
- If user has create access, show `Create Requisition`.

Error state

- `Requisitions could not be loaded.`
- Approval/rejection/conversion failures preserve decision notes.

Success state

- `Requisition submitted successfully.`
- `Requisition approved successfully.`
- `Purchase order created successfully.` when converted.

No-permission state

- `You do not have permission to manage requisitions.`

UI-014 Inbound Delivery Planning

Loading state

- Show inbound schedule table skeleton and selected delivery detail placeholder.

Empty state

- `No inbound deliveries are scheduled for this view.`
- Suggest selecting a different period or open PO set.

Error state

- `Inbound delivery plans could not be loaded.`
- Save errors preserve entered schedule values.

Success state

- `Inbound delivery plan updated successfully.`

No-permission state

- `You do not have permission to plan inbound deliveries.`

UI-015 Goods Receipt Workbench

Loading state

- Show receipt queue skeleton and receipt-line placeholders.

Empty state

- `No inbound receipts are waiting at this location.`

- If location-scoped, confirm the active location context.

Error state

- Load error: `Goods receipts could not be loaded.`
- Verify error: preserve scanned/entered quantities and line context.

Success state

- `Receipt verified successfully.`
- Task updates to put-away-ready state.

No-permission state

- `You do not have permission to receive inventory at this location.`

UI-016 Receipt Discrepancy Resolution

Loading state

- Show discrepancy queue skeleton and detail placeholder.

Empty state

- `No receipt discrepancies are waiting for review.`

Error state

- `Discrepancies could not be loaded.`
- Resolution errors preserve notes, reason code, and evidence references.

Success state

- `Discrepancy resolved successfully.`
- Queue count updates immediately.

No-permission state

- `You do not have permission to resolve receipt discrepancies.`

UI-017 Put-Away Workbench

Loading state

- Show pending-task skeleton rows and destination placeholders.

Empty state

- `No put-away tasks are waiting at this location.`

Error state

- `Put-away tasks could not be loaded.`
- Completion errors preserve selected location and quantities.

Success state

- `Put-away completed successfully.`
- Completed task disappears from queue or moves to completed section.

No-permission state

- `You do not have permission to perform put-away at this location.`

UI-018 Non-PO and Return Receipt Intake

Loading state

- Show intake form shell and disabled submit while authorization/reference data loads.

Empty state

- Neutral form state:
  - `Select a receipt type to begin.`

Error state

- Show invalid authorization/reference errors inline and at form level.
- Preserve all entered values.

Success state

- `Receipt recorded successfully.`
- Route the user to the next applicable status or reset form for next intake.

No-permission state

- `You do not have permission to record this type of inbound receipt.`

UI-019 Inbound Supply Tracking

Loading state

- Show PO pipeline table skeleton and delay-status placeholders.

Empty state

- `No inbound supply records are available for this view.`

Error state

- `Inbound supply information could not be loaded.`
- Preserve filters and open detail state if possible.

Success state

- After a delivery update:
  - `Inbound status updated successfully.`

No-permission state

- `You do not have permission to view inbound supply tracking.`

UI-020 Inventory Status Monitor

Loading state

- Show status cards, board, and detail table skeletons.

Empty state

- If no inventory exists in scope:
  - `No inventory is available in this view.`
- If filters remove all results:
  - `No stock records match these filters.`

Error state

- `Inventory status could not be loaded.`
- Keep current filters and scope visible.

Success state

- After refresh or action return:
  - `Inventory view updated.`

No-permission state

- `You do not have permission to view inventory for this scope.`

UI-021 Inventory Availability Inquiry

Loading state

- Show search results placeholder and network availability skeleton after search.

Empty state

- Before search: `Search for a product to view availability.`
- After no match: `No matching product was found.`

Error state

- `Availability could not be retrieved.`
- Preserve search input and location scope.

Success state

- Results load with a subtle confirmation:
  - `Availability updated.`

No-permission state

- `You do not have permission to view availability for this location or product scope.`

UI-022 Inventory Movement History

Loading state

- Show query bar and movement row skeletons.

Empty state

- Before search: `Enter filters to view movement history.`
- After no results: `No movement records match this search.`

Error state

- `Movement history could not be loaded.`
- Preserve entered filters.

Success state

- Results refresh in place after query or export preparation.

No-permission state

- `You do not have permission to review movement history in this scope.`

UI-023 Cycle Count Workbench

Loading state

- Show assigned-task skeleton list and count-entry placeholder.

Empty state

- `No cycle counts are assigned to you right now.`

Error state

- `Cycle count assignments could not be loaded.`
- Submission errors preserve entered count quantity.

Success state

- `Count submitted successfully.`
- Move user to next assignment where applicable.

No-permission state

- `You do not have permission to perform cycle counts for this location.`

UI-024 Physical Inventory Count Management

Loading state

- Show count-session status placeholders and zone progress skeletons.

Empty state

- `No physical inventory count session is active.`
- Action: `Start Count` if permitted.

Error state

- `Physical count information could not be loaded.`
- Certification failure preserves review context and unresolved-item list.

Success state

- `Physical count started successfully.`
- `Physical count certified successfully.`

No-permission state

- `You do not have permission to start or certify physical counts.`

UI-025 Variance Investigation and Approval

Loading state

- Show pending-variance table skeleton and detail pane placeholder.

Empty state

- `No variances are waiting for review.`

Error state

- `Variance cases could not be loaded.`
- Approval/rejection errors preserve findings and notes.

Success state

- `Variance decision saved successfully.`
- Case exits queue or updates status immediately.

No-permission state

- `You do not have permission to approve variance cases.`

UI-026 Stock Adjustment Management

Loading state

- Show adjustment list skeleton and adjustment-form shell.

Empty state

- List: `No adjustment cases are available in this view.`
- Create mode remains available if permitted.

Error state

- `The adjustment could not be processed.`
- Preserve all fields, notes, and evidence references.

Success state

- `Adjustment submitted successfully.`
- Or `Adjustment applied successfully.` depending on workflow state.

No-permission state

- `You do not have permission to create or approve stock adjustments.`

UI-027 Write-Off Authorization

Loading state

- Show pending write-off skeleton rows and detail placeholder.

Empty state

- `No write-off requests are waiting for review.`

Error state

- `Write-off requests could not be loaded.`
- Decision errors preserve notes and acknowledgment state.

Success state

- `Write-off decision recorded successfully.`

No-permission state

- `You do not have permission to authorize write-offs.`

UI-028 Quarantine and Hold Register

Loading state

- Show hold register skeleton and detail card placeholder.

Empty state

- `No stock is currently on hold or in quarantine.`

Error state

- `Hold and quarantine records could not be loaded.`
- Apply/release errors preserve reason and quantity.

Success state

- `Hold updated successfully.`
- Availability metrics refresh immediately.

No-permission state

- `You do not have permission to manage quarantined or held stock.`

UI-029 Shrinkage, Loss, and Obsolescence Analysis

Loading state

- Show chart/card skeletons and analysis table placeholders.

Empty state

- `No shrinkage or obsolescence data is available for the selected period.`

Error state

- `The analysis could not be loaded.`
- Keep selected filters.

Success state

- `Analysis updated.`
- If export succeeds, show `Analysis exported successfully.`

No-permission state

- `You do not have permission to view shrinkage or obsolescence analysis.`

UI-030 Order Reservation Workbench

Loading state

- Show order queue skeleton, line-item placeholders, and disabled reserve actions until ATP loads.

Empty state

- `No orders are waiting for reservation in this view.`

Error state

- `Order reservation data could not be loaded.`
- Reservation failures preserve selected lines and quantities.

Success state

- `Reservation completed successfully.`
- Queue and order status update immediately.

No-permission state

- `You do not have permission to reserve or release inventory for orders.`

UI-031 Pick, Pack, and Ship Workbench

Loading state

- Show pick queue skeleton and task detail placeholder.

Empty state

- `No orders are ready for pick, pack, or ship at this location.`

Error state

- `Fulfillment tasks could not be loaded.`
- Verification/pack/ship errors preserve scanned values and task context.

Success state

- `Pick verified successfully.`
- `Order packed successfully.`
- `Shipment confirmed successfully.`

No-permission state

- `You do not have permission to perform fulfillment tasks at this location.`

UI-032 Partial Fulfillment and Backorder Management

Loading state

- Show affected-order skeletons and quantity comparison placeholders.

Empty state

- `No partial fulfillment or backorder cases require attention.`

Error state

- `The partial fulfillment workflow could not be completed.`
- Preserve quantities and customer acknowledgment reference.

Success state

- `Partial fulfillment recorded successfully.`
- `Backorder created successfully.`

No-permission state

- `You do not have permission to manage partial fulfillment or backorders.`

UI-033 Fulfillment Exception Management

Loading state

- Show exception queue skeleton and resolution panel placeholder.

Empty state

- `No fulfillment exceptions are open right now.`

Error state

- `Exceptions could not be loaded.`
- Resolution failures preserve selected path and notes.

Success state

- `Exception resolved successfully.`
- Item exits queue or changes status.

No-permission state

- `You do not have permission to resolve fulfillment exceptions.`

UI-034 Order Fulfillment Status

Loading state

- Show order search results skeleton and line-status placeholders.

Empty state

- Before search: `Search for an order to view fulfillment status.`
- No match: `No order matched your search.`

Error state

- `Order status could not be loaded.`
- Keep search/filter context intact.

Success state

- Refreshed view updates line and shipment status with subtle confirmation.

No-permission state

- `You do not have permission to view this order or its fulfillment details.`

UI-035 Multi-Location Split Fulfillment Planner

Loading state

- Show allocation-plan skeleton and network-availability placeholders.

Empty state

- `This order does not currently require multi-location fulfillment.`

Error state

- `The split fulfillment plan could not be generated.`
- Preserve manual allocations if user edited them.

Success state

- `Split fulfillment plan created successfully.`
- Location queues or reservation summaries refresh.

No-permission state

- `You do not have permission to create multi-location fulfillment plans.`

UI-036 Low-Stock Alert Review

Loading state

- Show alert summary cards and alert list skeletons.

Empty state

- `There are no active low-stock alerts in this view.`

Error state

- `Low-stock alerts could not be loaded.`
- Acknowledge errors preserve selected action and notes.

Success state

- `Alert acknowledged successfully.`
- Alert queue updates immediately.

No-permission state

- `You do not have permission to review stock alerts.`

UI-037 Replenishment Planning

Loading state

- Show recommendation cards/table skeleton and supplier/open-PO placeholders.

Empty state

- `No replenishment recommendations are available for this view.`

Error state

- `Replenishment recommendations could not be loaded.`
- Requisition-creation errors preserve entered values.

Success state

- `Requisition created successfully.`
- Recommendation status updates to reflect action taken.

No-permission state

- `You do not have permission to create replenishment actions.`

UI-038 Transfer Request

Loading state

- Show request-form shell and transfer-capacity placeholder.

Empty state

- Neutral form state with helper text:
  - `Enter transfer details to begin.`

Error state

- Validation and capacity errors remain inline.
- Business-rule failures such as source-minimum breach show a form-level banner.

Success state

- `Transfer request submitted successfully.`
- Show request reference and new status.

No-permission state

- `You do not have permission to request stock transfers.`

UI-039 Transfer Approval Queue

Loading state

- Show approval queue skeleton and impact-comparison placeholder.

Empty state

- `No transfer requests are waiting for approval.`

Error state

- `Transfer approvals could not be loaded.`
- Decision errors preserve decision note and selection.

Success state

- `Transfer decision saved successfully.`
- Queue count updates immediately.

No-permission state

- `You do not have permission to approve or reject transfers.`

UI-040 Transfer Execution and In-Transit Tracking

Loading state

- Show dispatch/receipt queue skeletons and in-transit placeholders.

Empty state

- Dispatch area: `No approved transfers are ready for dispatch.`
- In-transit area: `No stock is currently in transit.`

Error state

- `Transfer execution data could not be loaded.`
- Dispatch/receipt failures preserve entered quantities and notes.

Success state

- `Transfer dispatched successfully.`
- `Transfer received successfully.`

No-permission state

- `You do not have permission to dispatch or receive transfers for this location.`

UI-041 Network Rebalancing Analysis

Loading state

- Show imbalance-summary skeletons and recommendation placeholders.

Empty state

- `No network imbalances were identified for the selected scope.`

Error state

- `Network rebalancing analysis could not be loaded.`
- Preserve filters.

Success state

- `Recommendation generated successfully.` or `Transfer request created successfully.`

No-permission state

- `You do not have permission to view network rebalancing analysis.`

UI-042 Return Authorization Management

Loading state

- Show order-search placeholder and return-line eligibility skeleton.

Empty state

- Before search: `Search for an order to begin a return authorization.`
- No eligible lines: `No return-eligible items were found for this order.`

Error state

- `The return authorization could not be completed.`
- Preserve selected order, lines, and quantities.

Success state

- `Return authorization issued successfully.`
- Show authorization reference clearly.

No-permission state

- `You do not have permission to authorize returns.`

UI-043 Return Inspection and Disposition

Loading state

- Show pending-return queue skeleton and receipt/inspection/disposition placeholders.

Empty state

- `No returns are waiting to be received or inspected.`

Error state

- `Return processing could not be completed.`
- Preserve received quantities, condition entries, and disposition choices.

Success state

- `Return received successfully.`
- `Inspection recorded successfully.`
- `Disposition completed successfully.`

No-permission state

- `You do not have permission to process returns at this location.`

UI-044 Operational Reports and Analytics

Loading state

- Show report filter shell and table/chart skeletons.

Empty state

- Before run: `Select filters and run a report.`
- After no data: `No report data is available for the selected filters.`

Error state

- `The report could not be generated.`
- Preserve filters and chosen report type.

Success state

- `Report updated successfully.`
- `Export completed successfully.` when applicable.

No-permission state

- `You do not have permission to view this report.`

UI-045 Inventory Valuation Reconciliation

Loading state

- Show valuation summary skeleton, comparison placeholders, and note area shell.

Empty state

- `No valuation data is available for the selected period.`

Error state

- `Valuation reconciliation could not be loaded.`
- Preserve selected period and unsaved notes.

Success state

- `Reconciliation notes saved successfully.`
- `Reconciliation finalized successfully.`

No-permission state

- `You do not have permission to view inventory valuation data.`

UI-046 Promotional Stock Readiness

Loading state

- Show campaign summary skeleton, readiness rows placeholder, and gap panel shell.

Empty state

- `No promotional campaigns are available in this view.`

Error state

- `Campaign readiness data could not be loaded.`
- Preserve selected campaign and unsaved notes.

Success state

- `Readiness confirmed successfully.`
- `Allocation updated successfully.` as applicable.

No-permission state

- `You do not have permission to view or manage promotional stock readiness.`

UI-047 Audit and Control Review Workspace

Loading state

- Show scope filter shell, transaction list skeleton, and findings placeholder.

Empty state

- Before scope run: `Select an audit scope to begin review.`
- After no results: `No records matched this audit scope.`

Error state

- `Audit review data could not be loaded.`
- Preserve scope filters and unsaved findings if possible.

Success state

- `Finding recorded successfully.`
- `Audit review completed successfully.`

No-permission state

- `You do not have permission to access the audit workspace.`

UI-048 Audit Trail Query

Loading state

- Show query filter shell and result-table skeleton.

Empty state

- Before query: `Enter filters to search the audit trail.`
- No result: `No audit records match this query.`

Error state

- `Audit trail results could not be loaded.`
- Preserve query filters and selected tab/detail context.

Success state

- `Query completed successfully.`
- `Audit results exported successfully.` when applicable.

No-permission state

- `You do not have permission to query audit trail data.`

UI-049 User Account Management

Loading state

- Show user-directory skeleton and account-detail placeholder.

Empty state

- `No user accounts are available in this view.`
- If create allowed, show `Create User`.

Error state

- `User accounts could not be loaded.`
- Save/suspend/deactivate errors preserve entered role assignments and notes.

Success state

- `User account saved successfully.`
- `User suspended successfully.` or `User deactivated successfully.`

No-permission state

- `You do not have permission to manage user accounts.`

UI-050 Role and Access Control Administration

Loading state

- Show role list skeleton and permission-matrix placeholder.

Empty state

- `No role configuration is available in this view.`

Error state

- `Role and access settings could not be loaded.`
- Save failures preserve matrix edits when possible.

Success state

- `Permissions updated successfully.`
- `Segregation-of-duties validation completed.`

No-permission state

- `You do not have permission to manage roles and access controls.`

Notes

- These states should be implemented consistently with the API error model in `SRS/API_Design/Error-Response-Specification.md`.
- Field-level validation and transactional error handling for forms are further detailed in `SRS/UI_UX/IMS-Form-Specifications-and-Validation.md`.
- Detailed component behavior for banners, empty states, tables, skeletons, and dialogs should align with `SRS/UI_UX/IMS-Component-Inventory.md`.

Inventory Management System UI Guidelines

Purpose

This document defines UI guidelines for the Inventory Management System covering responsiveness, accessibility, keyboard interaction, and mobile behavior. The recommendations are aligned to WCAG principles of being:

- **Perceivable**
- **Operable**
- **Understandable**
- **Robust**

The guidance is intended for UX, UI, frontend engineering, QA, and accessibility review.

Accessibility Target

The recommended baseline target for the IMS is:

- **WCAG 2.2 AA** for all core user journeys

This is especially important because the system is used for:

- Time-sensitive warehouse operations
- Data-entry-heavy workflows
- Multi-role enterprise access
- Reporting and audit use cases

---

1. Responsive Breakpoints

Design Intent

The IMS is primarily a business web application used on desktop and laptop devices, but it should remain usable on tablets and selectively on mobile for read-heavy and lightweight task flows. Complex data-entry and table-heavy workflows should remain optimized for larger screens.

Recommended Breakpoints

Breakpoint	Width	Primary Usage
`xs`	`0-479px`	Small mobile phones
`sm`	`480-767px`	Large phones / small mobile landscape
`md`	`768-1023px`	Tablets / small laptops

Breakpoint	Width	Primary Usage
`lg`	`1024-1439px`	Standard laptops / desktop
`xl`	`1440px+`	Large desktop / wide operational monitors

Layout Guidance by Breakpoint

`xs` and `sm`

- Use a single-column layout.
- Collapse secondary navigation into drawers, accordions, or segmented tabs.
- Move filter bars into expandable panels.
- Stack form fields vertically.
- Replace dense multi-column tables with card-list or priority-column responsive tables.
- Limit simultaneous pane layouts; avoid side-by-side master-detail patterns unless converted to drill-in navigation.

`md`

- Allow two-panel layouts selectively.
- Keep primary navigation collapsible.
- Use horizontally scrollable tables only when card conversion would lose too much structure.
- Preserve important workbench workflows where possible, but simplify dense controls.

`lg` and `xl`

- Use full enterprise layout with:
  - persistent global header
  - primary navigation
  - secondary navigation
  - list-detail and workbench layouts
- Allow wider tables, multi-column forms, and parallel information panels.

Responsive Rules

- Never rely on hover-only interaction at any breakpoint.
- Do not hide critical actions inside unstable overflow patterns without a clear alternative.
- Avoid fixed-height panels that trap content at smaller sizes.
- Preserve task order when stacking columns vertically.
- Ensure zoom to 200% does not break essential workflows.

WCAG Alignment

- Supports **Perceivable** and **Operable** principles by maintaining content access across viewport sizes and zoom states.

---

2. Keyboard Navigation

Design Intent

All essential tasks must be operable using a keyboard alone, especially because IMS users may:

- perform repetitive data entry
- use barcode/scanner-assisted input
- work quickly in operational contexts
- need non-pointer accessibility support

Core Requirements

- All interactive controls must be reachable with `Tab`.

- Use `Shift + Tab` for backward movement in a predictable order.
- Ensure visible focus on every interactive element.
- All actions available by mouse must also be available by keyboard.
- Avoid keyboard traps.

Expected Keyboard Behavior

Global

- `Tab` moves through interactive elements in reading/task order.
- `Enter` activates primary buttons and selected menu items.
- `Space` toggles checkboxes, switches, and some buttons where appropriate.
- `Esc` closes dialogs, menus, drawers, and temporary overlays.

Navigation

- Menus, tabs, accordions, and dropdowns must support arrow-key navigation where standard patterns apply.
- Breadcrumb links must be keyboard reachable.
- Skip link should be available to jump to main content.

Forms

- Entering data should not unexpectedly submit multi-field forms.
- Inline validation must not steal keyboard focus unless the user submits and an error summary intentionally redirects focus.

Tables and Grids

- Row actions must be reachable via keyboard.
- Sortable headers must be operable by keyboard.
- Expandable rows must support `Enter` or `Space`.
- If grid-like interactions are used, arrow-key behavior must be consistent and documented.

Keyboard Priority Areas in IMS

Special attention is required for:

- `Goods Receipt Workbench`
- `Put-Away Workbench`
- `Cycle Count Workbench`
- `Pick, Pack, and Ship Workbench`
- `Stock Adjustment Management`
- `User Account Management`

WCAG Alignment

- Supports **Operable** principle, especially keyboard accessibility and predictable interaction.

---

3. Focus Management

Design Intent

Focus must move intentionally so users always know where they are, what changed, and what to do next.

Global Focus Rules

- On page load, focus should move to the page heading or main content container, not remain ambiguously in the browser chrome.
- Do not move focus unexpectedly during normal typing or filtering.

- When opening a modal, drawer, or popup, move focus into it.
- When closing a modal, return focus to the triggering control.
- When a page update occurs after a submit, direct focus to the most relevant next element:
  - success banner
  - first validation error
  - updated record heading

Error Focus Rules

- On failed submit, move focus to the error summary if present.
- The error summary should include links or relationships to invalid fields.
- Focus should then allow the user to move naturally into each field error.

Success Focus Rules

- For save/update actions that remain on the same page, focus should move to:
  - success banner, then
  - updated record title or first relevant next action
- For create flows that navigate to a detail page, focus should land on the new page heading.

Dynamic Content Rules

- When filters refresh results asynchronously, do not steal focus from the current control unless the user explicitly triggers navigation into results.
- If an item disappears from a queue after success, move focus to:
  - the next logical row, or
  - the list heading if the queue becomes empty

Dialog and Overlay Rules

- Focus must be trapped within active modals.
- Background content must not be keyboard reachable while modal is open.
- Closing the overlay restores prior focus context.

WCAG Alignment

- Supports **Operable** and **Understandable** principles through predictable navigation and state transitions.

---

4. Color Contrast

Design Intent

Color must support meaning, not carry meaning alone. IMS screens use many statuses, alerts, thresholds, and workflow states, so contrast and redundant cues are critical.

Minimum Contrast Recommendations

- Normal text: **4.5:1** minimum
- Large text: **3:1** minimum
- Essential UI components and icons: **3:1** minimum against adjacent colors
- Focus indicators: clearly visible with **3:1** minimum contrast against surrounding colors

Status and Semantic Guidance

Status indicators such as:

- low stock
- out of stock

- quarantined
- approved
- rejected
- delayed
- at risk

must not rely on color alone.

Use at least one of:

- text labels
- icons
- patterns
- badges
- helper text

Example:

- Not just red dot
- Use: `Out of Stock` badge with icon and accessible text

Background and Surface Guidance

- Avoid placing light text on low-contrast tinted cards.
- Avoid thin gray text in metadata areas if it drops below accessibility thresholds.
- Ensure disabled state still remains readable enough to understand context, while clearly differentiated from enabled state.

Charts and Data Visualization

- Use more than color to distinguish series:
  - labels
  - patterns
  - markers
  - direct annotations
- Ensure chart legends are keyboard and screen-reader understandable where interactive.

WCAG Alignment

- Supports **Perceivable** principle through sufficient visual contrast and non-color-dependent communication.

---

5. Form Accessibility

Design Intent

Forms are central to the IMS, especially for inventory transactions, approvals, receiving, fulfillment, and administration. They must be accessible, efficient, and forgiving.

Labeling Rules

- Every input must have a persistent visible label.
- Placeholder text must not replace labels.
- Required fields must be clearly identified visually and programmatically.
- Optional fields may be marked explicitly if that reduces ambiguity.

Instruction and Help Rules

- Put concise helper text near the field when needed.
- Use examples only if they clarify format.
- Keep field guidance short and specific.

Validation Rules

- Field errors must be programmatically associated with the field.
- Error text must explain:
  - what is wrong
  - how to fix it
- Validation must not depend only on color.
- Use clear language like:
  - `Quantity must be greater than 0`
  - not: `Invalid value`

Grouping Rules

- Related controls should be grouped with clear section headings.
- Radio groups and checkbox groups should be wrapped in accessible group semantics.
- Repeating line-item forms must expose row and field context clearly.

Input Behavior Rules

- Use appropriate input types:
  - email for email
  - numeric fields for quantities
  - date pickers for dates
- Support manual entry in addition to picker-only interactions where practical.
- Avoid requiring drag-only or pointer-only actions.

Required Accessibility Features

- Labels linked to controls
- Error text linked to controls
- Required state announced to assistive technology
- Disabled/read-only states distinguishable programmatically
- Autosave or draft patterns must not interrupt screen reader flow unexpectedly

Form-Specific IMS Priorities

Special care is needed for:

- line-item editors
- barcode/scanner-assisted fields
- conditional fields
- approval decision sections
- multi-step receiving and return workflows

WCAG Alignment

- Supports **Perceivable**, **Operable**, **Understandable**, and **Robust** principles.

---

6. Table Accessibility

Design Intent

Tables are one of the most heavily used patterns in the IMS. They must support scanning, sorting, selection, and row actions without becoming inaccessible.

Structural Rules

- Use true table semantics for tabular data.
- Header cells must be properly associated with their columns.

- Complex tables with grouped headers should preserve readable relationships.

Interaction Rules

- Sortable columns must indicate current sort state.
- Row selection checkboxes must have clear labels.
- Row action menus must be keyboard and screen-reader accessible.
- Expandable rows must announce expanded/collapsed state.

Content Rules

- Keep column names concise and descriptive.
- Do not truncate essential meaning without a way to access full content.
- Numeric values should be aligned consistently for readability, but semantic structure must remain intact.

Responsive Table Rules

On smaller screens:

- prioritize critical columns
- allow horizontal scrolling only when necessary
- consider card transformation for highly dense operational lists
- preserve access to row actions and identifiers

Empty and Error Rules

- Empty tables should clearly explain that there are no records, not simply show blank space.
- Loading tables should use skeleton rows.
- Error states should appear above the table and allow retry.

High-Priority IMS Table Types

Special care is needed for:

- inventory status tables
- purchase order line tables
- receipt line tables
- order line status tables
- transfer and approval queues
- audit logs
- permission matrices

WCAG Alignment

- Supports **Perceivable**, **Operable**, and **Understandable** principles for structured data access.

---

7. Mobile Considerations

Design Intent

The IMS should support mobile selectively. Not every desktop workflow should be forced into a full mobile experience. The mobile strategy should prioritize:

- lookup tasks
- lightweight approvals
- notifications
- simple updates
- selected warehouse actions where practical

Suitable Mobile-First or Mobile-Friendly Tasks

- notifications
- approvals
- stock lookup
- order status inquiry
- low-stock alert review
- basic return receipt steps
- simple transfer confirmation

Desktop-Preferred Tasks

These should remain optimized for tablet/desktop due to complexity:

- purchase order editing with many lines
- large discrepancy resolution
- detailed physical counts
- complex inventory adjustment cases
- valuation reconciliation
- permission matrix editing
- audit review with large datasets

Mobile Interaction Rules

- Use larger tap targets.
- Avoid hover-only actions.
- Keep one primary action visible at a time.
- Use sticky bottom action areas sparingly and only when they do not cover content.
- Favor drill-down flows over multi-pane layouts.
- Use clear back navigation and preserve entered data when moving between screens.

Mobile Form Rules

- Keep forms short and segmented.
- Reduce side-by-side fields.
- Use mobile-friendly input controls where appropriate.
- Ensure scanners or hardware keyboards do not break focus order.

Mobile Table Rules

- Convert dense tables into stacked cards where needed.
- Keep record ID, status, and primary action visible.
- Hide secondary metadata behind expand/collapse patterns if necessary.

Orientation and Zoom

- Support portrait-first for phones.
- Avoid requiring landscape for essential tasks unless the task is explicitly desktop/tablet-only.
- Ensure users can zoom without clipping essential controls.

WCAG Alignment

- Supports **Operable** and **Perceivable** principles by preserving access under touch, zoom, and small-screen constraints.

---

8. Additional WCAG-Aligned Recommendations

Language and Clarity

- Use plain, task-oriented labels.

- Keep system messages specific and actionable.
- Avoid unexplained abbreviations unless they are common in the user domain and defined consistently.

Time and Session

- Warn users before session timeout where data entry may be lost.
- Provide recovery options for long-running forms.

Motion and Animation

- Avoid unnecessary motion in operational workflows.
- Any animation should be subtle, short, and never required to understand state changes.

Error Prevention

- Confirm destructive actions such as:
  - write-offs
  - deactivation
  - cancel order
  - cancel PO
  - release promotional allocation
- Allow review before finalizing important inventory transactions.

Assistive Technology Robustness

- Use semantic HTML or equivalent accessible roles.
- Ensure labels, status messages, dialogs, tables, and menus are announced correctly by assistive technologies.
- Test with keyboard-only navigation and screen readers on core flows.

---

9. Recommended Testing Checklist

The following should be tested on all critical flows:

- Can the entire task be completed using keyboard only?
- Is focus always visible and predictable?
- Are all field errors announced and understandable?
- Are status colors understandable without color perception?
- Do tables remain usable at 200% zoom?
- Do mobile layouts preserve primary tasks clearly?
- Are dialogs and overlays focus-trapped correctly?
- Are success and error messages exposed to assistive technology?
- Are no-permission states understandable without exposing restricted data?

---

10. Priority Screens for Accessibility Review

The following screens should receive first-pass accessibility testing because they combine complexity, frequency, and business criticality:

1. `UI-015 Goods Receipt Workbench`
2. `UI-017 Put-Away Workbench`
3. `UI-023 Cycle Count Workbench`
4. `UI-026 Stock Adjustment Management`
5. `UI-030 Order Reservation Workbench`
6. `UI-031 Pick, Pack, and Ship Workbench`
7. `UI-034 Order Fulfillment Status`
8. `UI-040 Transfer Execution and In-Transit Tracking`
9. `UI-043 Return Inspection and Disposition`

10. `UI-049 User Account Management`
11. `UI-050 Role and Access Control Administration`

Summary

These guidelines establish the accessibility and responsive foundation for the IMS UI. The system should prioritize:

- desktop excellence for complex operations
- controlled mobile support for lightweight tasks
- complete keyboard operability
- predictable focus behavior
- strong color contrast and non-color status communication
- accessible forms and tables

This guidance should be used alongside:

- `IMS-Component-Inventory.md`
- `IMS-Form-Specifications-and-Validation.md`
- `IMS-Screen-State-Specifications.md`
- `IMS-Low-Fidelity-Wireframe-Specifications.md`

Inventory Management System UI Specification

Purpose

This document consolidates the completed `UI_UX` analysis package into a single UI specification optimized for AI-assisted design tools such as Figma AI. It is intended to describe the interface clearly enough that an AI can generate faithful screen concepts, layouts, and component structures without relying on implementation code.

This specification is derived from:

- `IMS-Screen-Inventory.md`
- `IMS-Navigation-Structure.md`
- `IMS-User-Journeys.md`
- `IMS-User-Flows.md`
- `IMS-Low-Fidelity-Wireframe-Specifications.md`
- `IMS-Component-Inventory.md`
- `IMS-Form-Specifications-and-Validation.md`
- `IMS-Screen-State-Specifications.md`
- `IMS-UI-Guidelines-WCAG.md`

Scope and Modeling Rules

- This UI specification follows the official 50-screen model from the screen inventory.
- Detail views that appear inside list-detail workspaces are treated as subviews, tabs, drawers, panels, or workflow states within the parent screen unless explicitly promoted to a separate screen in the source package.
- The product is a desktop-first enterprise web application for inventory, warehouse, procurement, fulfillment, finance, audit, and administration workflows.
- The baseline accessibility target is `WCAG 2.2 AA`.
- The baseline design intent is:
  - desktop-first for complex operational workflows
  - tablet-capable for selected workbenches and approvals
  - selective mobile support for lookup, approvals, notifications, and light status actions

Global Design Foundations

Shared Visual Structure

Use a consistent enterprise application shell:

- persistent global header
- role-aware primary navigation
- module-level secondary navigation
- breadcrumbs on all non-home authenticated screens
- page title bar with context, primary action, and optional status badge
- content area using either:
  - dashboard
  - list-detail workspace
  - work queue + detail panel
  - report viewer
  - multi-step transaction flow
- metadata footer or timestamp strip where useful

Shared Navigation

Primary modules:

1. `Home`
2. `Inventory`
3. `Products`
4. `Locations`
5. `Procurement`
6. `Inbound`
7. `Fulfillment`
8. `Transfers`
9. `Returns`
10. `Reports & Analytics`
11. `Governance & Audit`
12. `Administration`

Persistent utilities:

- `Notifications`
- `Global Search`
- `My Tasks / Approvals`
- `User Menu / Session`

Breadcrumb pattern:

```
Home > Module > Section > Record / Action
```

Shared Components

The following reusable components should appear throughout the system:

- global header
- main navigation bar
- secondary navigation
- breadcrumb trail
- page title bar
- filter bar
- KPI summary cards
- data tables with row actions
- detail panel or drawer

- standard form sections
- validation message blocks
- success, warning, and error banners
- queue list
- approval decision panel
- status timeline/history
- empty-state panel
- loading skeletons

Shared Interaction Patterns

- Prefer list-detail layouts for data management.
- Prefer queue + detail layouts for operational execution and approvals.
- Prefer step indicators for import, receiving, packing, and return workflows.
- Confirm sensitive actions such as cancel, deactivate, write-off, approve, reject, dispatch, and ship.
- Preserve user-entered values on validation, conflict, and temporary system errors whenever possible.

Shared Validation Rules

- Validate required fields on blur and on submit.
- Validate code formats such as SKU, location code, supplier code, and reason code on blur.
- Validate cross-field rules whenever dependent fields change.
- Show field-level messages inline and show a form-level summary when submit fails.
- Keep user-entered values intact on `409` conflicts and `500/503` system failures where possible.

Shared State Patterns

- Loading: preserve page structure with skeletons.
- Empty: explain why nothing is shown and what to do next.
- Error: use clear language, preserve work, offer retry.
- Success: show a visible but non-blocking confirmation and refresh affected panels immediately.
- No-permission: show access limits without exposing restricted data.

Shared Responsive Behavior

- `xs` and `sm`: single-column layout, collapsible secondary navigation, stacked forms, cardified tables where needed.
- `md`: allow simplified two-panel layouts and horizontally scrollable tables when necessary.
- `lg` and `xl`: full enterprise layout with persistent navigation, list-detail workspaces, wide tables, and multi-column forms.

Shared Accessibility Rules

- Full keyboard operation for all essential tasks.
- Visible focus on all interactive elements.
- Skip link to main content.
- Do not rely on hover-only behavior.
- Do not rely on color alone for status.
- Associate all labels, help text, and error messages programmatically with inputs.
- Trap focus in dialogs and return focus to the invoking control on close.
- Use live announcements for success, error, and dynamic queue updates.

Screen Specifications

UI-001 Sign In / Session Entry

1. Screen Name
Sign In / Session Entry

2. Purpose
Authenticate the user and establish an authorized IMS session before protected application access.

3. Layout Structure
Unauthenticated layout with minimal header, centered sign-in card, optional help link, and lightweight footer containing version and privacy/security note.

4. Navigation
No primary application navigation before login. Secondary actions: `Forgot Password` or `Need Help`.

5. Components
Unauthenticated header, card container, text inputs, password input, checkbox, banner message, primary CTA.

6. Tables
None.

7. Forms
Username/email, password, optional remember-session checkbox. Clear, compact, single-column form.

8. Buttons
Primary: `Sign In`. Secondary text action: `Forgot Password` or `Need Help`.

9. Icons
Lock icon near title, help icon for support link, spinner inside sign-in button while authenticating.

10. Validation
Require username/email and password. Preserve username on failure, clear password. Show form-level authentication error for invalid credentials, suspended account, deactivated account, or session failure.

11. Empty State
Neutral default state is the initial screen; no data-driven empty state.

12. Loading State
Disable inputs after submit, show button spinner and message `Signing you in...`.

13. Error State
Show a top-of-form error banner, keep username visible, clear password, keep help link accessible.

14. Success State
Brief success transition, then redirect to role-appropriate home screen.

15. Responsive Behavior
Mobile-safe single-column card. Keep form centered with wide touch targets. No horizontal scroll.

16. Accessibility Notes
Focus should land on page heading or username field. Use proper input labels, password masking, keyboard submit, and error banner announced to assistive tech.

UI-002 Operational Home Dashboard

1. Screen Name
Operational Home Dashboard

2. Purpose
Provide the operational landing view with KPIs, stock health, alerts, tasks, and shortcuts to core workflows.

3. Layout Structure
Authenticated shell with global header, main navigation, optional quick links, dashboard filter bar, KPI row, alert/task tables, and workflow shortcut tiles.

4. Navigation
Top-level `Home`. Links to `Notifications`, `My Tasks / Approvals`, `Inventory`, `Procurement`, and role-priority modules.

5. Components
Global header, search, notifications trigger, tasks widget, KPI cards, status summary strip, recent alerts table, pending tasks table, quick action tiles.

6. Tables
Recent alerts table and pending tasks table with row actions to open the related record.

7. Forms
Filter bar with location, date range, and role scope.

8. Buttons
`View Alerts`, `Open Approvals`, `Go to Inventory`, `Refresh`.

9. Icons
Notification bell, task/checklist icon, inventory box icon, warning/alert icon, trend arrows on KPI cards.

10. Validation
Filter validation is lightweight; validate location scope and date range if used.

11. Empty State
If no alerts or tasks, show `You are up to date.` with reassuring explanatory text.

12. Loading State
Skeleton KPI cards, placeholder table rows, placeholder shortcut tiles while header and navigation stay visible.

13. Error State
Banner: `We could not load the dashboard right now.` Keep shell visible and offer `Retry`.

14. Success State
After refresh or task completion, show subtle success banner: `Dashboard updated successfully.`

15. Responsive Behavior
Desktop: multi-row dashboard grid. Tablet: stack cards and collapse less-critical widgets. Mobile: summary cards and tasks only, with tables converted to compact lists.

16. Accessibility Notes
KPI cards need descriptive labels, not color only. Tables require proper headers. Shortcut tiles and widgets must be keyboard reachable and announced clearly.

UI-003 Executive KPI Dashboard

1. Screen Name
Executive KPI Dashboard

2. Purpose
Present strategic KPIs, targets, trend signals, and drill-downs for executive review.

3. Layout Structure
Wide analytics layout with page title, period filter bar, KPI cards, trend summary blocks, target variance region, and KPI summary table.

4. Navigation
Accessible from `Home` and `Reports & Analytics`, with breadcrumbs and secondary report navigation.

5. Components
KPI summary cards, trend indicators, variance blocks, date range picker, drill-down link group, export action group.

6. Tables
KPI summary table showing value, target, variance, trend, and status.

7. Forms
Filters for period, business unit, and location group.

8. Buttons
`Apply Filters`, `Export Dashboard`, `View Detail`.

9. Icons
Trend arrows, target icon, warning triangle for underperformance, export icon.

10. Validation
Validate period and any required filter combinations. Preserve filters on failed loads.

11. Empty State
`No KPI data is available for the selected period.` Suggest changing period or scope.

12. Loading State
Skeleton KPI cards, chart placeholders, and table skeleton while selected period remains visible.

13. Error State
Banner: `KPI data could not be loaded.` Keep filters intact and provide retry.

14. Success State
On filter change, update the dashboard and show subtle confirmation: `Dashboard updated.`

15. Responsive Behavior
Desktop-first analytics screen. Tablet keeps cards and a simplified table. Mobile should be read-only summary view with KPI cards stacked vertically.

16. Accessibility Notes
Charts and trends must not rely on color alone. Provide text labels, target deltas, and accessible drill-down actions.

UI-004 Notification Center

1. Screen Name
Notification Center

2. Purpose
Centralize alerts, pending approvals, delays, readiness issues, and operational notices.

3. Layout Structure
Inbox-style list-detail workspace with filter area, notification list, and selected notification detail pane.

4. Navigation
Reachable from global notification icon, `Home`, and task/approval utilities. Breadcrumbs should remain stable.

5. Components
Notification list, filter bar, saved views, detail panel, alert detail card, status badge, action buttons.

6. Tables
Notification list table with columns for type, priority, source module, age, and status.

7. Forms
Filters for type, priority, module, status, and date.

8. Buttons
`Open Related Record`, `Mark as Handled`, `Acknowledge`, `Refresh`.

9. Icons
Bell, approval/check icon, warning icon, delay/clock icon, unread dot indicator.

10. Validation
Validate date range and any required note when dismissing or deferring alerts.

11. Empty State
`No notifications to review.` Explain that alerts and approvals will appear here.

12. Loading State
Skeleton rows on the left and a shimmer detail pane on the right.

13. Error State
`Notifications could not be loaded.` Preserve filters and active saved view.

14. Success State
After acknowledge/handle, update the item in place or remove it from the active queue and show `Notification updated.`

15. Responsive Behavior
Desktop uses split inbox. Tablet may stack list over detail. Mobile uses filter drawer and drill-in detail screen.

16. Accessibility Notes
Unread, priority, and severity require text labels plus icons. List rows and row actions must be keyboard and screen-reader accessible.

UI-005 Product Catalog Management

1. Screen Name
Product Catalog Management

2. Purpose
Search, create, update, and maintain product master records.

3. Layout Structure
List-detail workspace with top action bar, search/filter toolbar, product table, and detail panel or full detail region.

4. Navigation
Module path: `Products > Product Catalog`. Context links to stock by location, ABC classification, stock policy, movement history, and duplicate resolution.

5. Components
Filter bar, product table, detail panel, standard form sections, status badges, metadata block, validation banner.

6. Tables
Product catalog table with SKU, name, category, lifecycle status, ABC class, and row actions.

7. Forms
Search inputs and a product form for SKU, name, description, UOM, category, barcode, traceability settings, shelf-life settings, and lifecycle status.

8. Buttons
`Create Product`, `Edit`, `Save`, `Change Status`, `Resolve Duplicate`.

9. Icons
Product/box icon, edit icon, duplicate/warning icon, status chip icons.

10. Validation
Uppercase SKU pattern, unique SKU and barcode, conditional rotation and shelf-life fields, lifecycle-state rules.

11. Empty State
Catalog empty: `No products have been created yet.` Show `Create Product`. Search empty: `No products match these filters.`

12. Loading State
Keep filter bar visible, show table skeleton and detail placeholder.

13. Error State
Load error banner for catalog, inline and summary errors for save, and conflict handling for duplicates.

14. Success State
`Product saved successfully.` Refresh selected row and detail panel without leaving context.

15. Responsive Behavior
Desktop keeps table + detail. Tablet may use tabbed detail. Mobile should focus on search and read-heavy detail, not full master-data editing.

16. Accessibility Notes
Form sections need clear headings and labels. Duplicate resolution should be announced clearly and not rely on color-only warnings.

UI-006 Product Import Workspace

1. Screen Name
Product Import Workspace

2. Purpose
Support bulk product and opening data import with validation and controlled confirmation.

3. Layout Structure
Multi-step workspace with step indicator, template instructions area, upload zone, validation results table, and completion summary.

4. Navigation
`Products > Product Import`, with breadcrumbs. Users remain inside one guided workflow.

5. Components
Step progress indicator, file upload control, validation table, summary cards, banner messages.

6. Tables
Validation errors table and import summary table showing created, updated, skipped, and failed rows.

7. Forms
Import mode selector, source reference, file upload field.

8. Buttons
`Download Template`, `Upload File`, `Validate`, `Confirm Import`, `Cancel`.

9. Icons
Upload, download, warning, success check, file/document icon.

10. Validation
Require file and valid mode. Validate type, size, duplicate rows, empty batch, and row-level blocking errors.

11. Empty State
`No import file uploaded yet.` Show instructions and template guidance.

12. Loading State
Keep step shell visible, skeletonize results area, show processing state after upload/validate.

13. Error State
`The import could not be processed.` Show row-level errors in the grid and keep file summary visible.

14. Success State
`Import completed successfully.` Show created/updated counts and non-blocking warnings.

15. Responsive Behavior
Desktop/tablet supported. Mobile should allow template download and status review only, not large-file operational imports.

16. Accessibility Notes
Upload zone must work without drag-and-drop. Validation tables need accessible row/column relationships and clear error descriptions.

UI-007 Location and Warehouse Management

1. Screen Name
Location and Warehouse Management

2. Purpose
Configure locations, hierarchy, storage positions, and operating status.

3. Layout Structure
Three-part workspace: location list, hierarchy view, and location detail area with storage positions subsection.

4. Navigation
`Locations > Location Directory`, with contextual navigation to hierarchy, storage positions, activation, deactivation, and closure actions.

5. Components
Location table, hierarchy tree, detail panel, storage position editor, metadata strip, confirmation dialog.

6. Tables
Location list table and storage positions table.

7. Forms
Location form for code, name, type, parent, status, dates, and operating metadata; storage position form for zone, aisle, bin, and status.

8. Buttons
`Create Location`, `Edit`, `Save`, `Activate`, `Deactivate`, `Close Location`.

9. Icons
Warehouse/building icon, hierarchy/tree icon, location pin, activation/deactivation status icons.

10. Validation
Unique location code, parent cannot equal current location, activation prerequisites, closure requirements if stock remains.

11. Empty State
`No locations have been configured yet.` Primary action: `Create Location`.

12. Loading State
List skeleton, tree placeholders, empty detail shell.

13. Error State
Load error for directory; preserve form values on save, activate, deactivate, and close failures.

14. Success State
Action-specific messages such as `Location saved successfully.` or `Location activated successfully.`

15. Responsive Behavior
Desktop uses full three-zone view. Tablet collapses hierarchy into tabs or accordion. Mobile should be mostly read-only.

16. Accessibility Notes
Hierarchy tree must be keyboard navigable. Status actions should use clear confirmation messaging and focus management.

UI-008 Stock Policy and Threshold Management

1. Screen Name
Stock Policy and Threshold Management

2. Purpose
Define reorder level, minimum threshold, safety stock, and related effective dates by product and location.

3. Layout Structure
Filter and selector area above a policy summary and editable threshold panel.

4. Navigation
`Products > Stock Policies`, with contextual links back to product detail and inventory status.

5. Components
Filter bar, policy summary card, numeric form fields, validation block, metadata footer.

6. Tables
Existing policy table by product and location.

7. Forms
Location selector, SKU selector, reorder level, minimum threshold, safety stock, maximum threshold, effective-from date.

8. Buttons
`Search`, `Edit Policy`, `Save`, `Reset`.

9. Icons
Threshold/slider icon, alert icon, policy/document icon.

10. Validation
Cross-field numeric relationships, non-negative values, required effective date, inactive product/location restrictions.

11. Empty State
`No stock policy exists for this product and location.` Show `Create Policy`.

12. Loading State
Policy selector remains visible while summary and edit area skeletonize.

13. Error State
Show inline threshold errors and preserve filter context on load error.

14. Success State
`Stock policy saved successfully.` Update summary values and metadata.

15. Responsive Behavior
Desktop and tablet supported for editing. Mobile should only support lookup or limited single-record edits.

16. Accessibility Notes
Numeric fields need unit context and explicit error text. Cross-field relationships should be described in helper text, not only after failure.

UI-009 ABC Classification Management

1. Screen Name
ABC Classification Management

2. Purpose
Review and assign ABC classes to products for differentiated inventory control.

3. Layout Structure
Summary strip at top, classification filter bar, product list, and detail/edit panel.

4. Navigation
`Products > ABC Classification`, with contextual links to product detail and inventory movement insight.

5. Components
Filter bar, classification table, segmented choice or dropdown, bulk action toolbar, metadata strip.

6. Tables
Product classification table with current class, category, and activity indicators.

7. Forms
Category filter, current class filter, activity status filter, class assignment, reason, effective date.

8. Buttons
`Assign Class`, `Save`, `Bulk Update`.

9. Icons
A/B/C badge iconography, product icon, warning icon for unclassified items.

10. Validation
Require product and class; only allow `A`, `B`, or `C`. Preserve selection on save failure.

11. Empty State
`No products are available for classification in this view.` Suggest adjusting filters.

12. Loading State
Skeleton summary strip and table rows.

13. Error State
`Classifications could not be loaded.` Save errors should preserve selected class values.

14. Success State
`Classification updated successfully.` Update the selected row immediately.

15. Responsive Behavior
Desktop/tablet supported. Mobile should use simplified filters and drill-in editing, not batch classification.

16. Accessibility Notes
ABC meaning must be available in text. Bulk selection and row actions must be operable by keyboard.

UI-010 Governance Policies and Reason Codes

1. Screen Name
Governance Policies and Reason Codes

2. Purpose
Maintain governance policy versions, thresholds, tolerances, and standardized reason codes.

3. Layout Structure
Tabbed screen with `Policies` and `Reason Codes`, each using list-detail layout with forms and metadata.

4. Navigation
`Governance & Audit > Governance Policies` or `Reason Codes`, with breadcrumbs.

5. Components
Tabs, policy table, reason-code table, form sections, publish actions, status badges, metadata block.

6. Tables
Governance policies table and reason codes table.

7. Forms
Policy form: name, version, thresholds, tolerances, dates, status. Reason code form: code, description, transaction types, active state, retirement info.

8. Buttons
`Create Policy`, `Publish`, `Create Reason Code`, `Retire`, `Save`.

9. Icons
Policy/document icon, approval shield, reason-code tag, version/history icon.

10. Validation
Validate overlapping effective dates, threshold pairs, tolerances, required transaction types, and retirement reason when retiring codes.

11. Empty State
Policies: `No governance policies have been published yet.` Reason codes: `No reason codes are available yet.`

12. Loading State
Show tabs immediately, skeletonize tables and detail forms.

13. Error State
`Governance data could not be loaded.` Preserve draft values on publish and save failures.

14. Success State
`Policy published successfully.` or `Reason code saved successfully.`

15. Responsive Behavior
Desktop-first due to dense governance forms. Tablet acceptable with stacked sections. Mobile read-only at most.

16. Accessibility Notes
Tabs need proper semantics. Threshold and tolerance relationships need explicit helper text. Status cannot rely on color alone.

UI-011 Supplier Management

1. Screen Name
Supplier Management

2. Purpose
Create, update, and deactivate supplier records used in procurement.

3. Layout Structure
List-detail workspace with supplier table on the left and supplier detail/edit form on the right.

4. Navigation
`Procurement > Suppliers`, with links to purchase orders and inbound supply tracking.

5. Components
Supplier table, detail panel, standard form sections, status badge, metadata strip.

6. Tables
Supplier table with code, name, payment terms, lead time, active state, and row actions.

7. Forms
Supplier code, name, email, phone, payment terms, lead time, currency, consignment toggle, active toggle, deactivation reason.

8. Buttons
`Create Supplier`, `Edit`, `Save`, `Deactivate`.

9. Icons
Supplier/building icon, contact icon, deactivate/warning icon.

10. Validation
Unique supplier code, required name and payment terms, email format, required deactivation reason.

11. Empty State
`No suppliers have been added yet.` Show `Create Supplier`.

12. Loading State
Supplier rows skeletonize while detail pane shows placeholder.

13. Error State
`Suppliers could not be loaded.` Save/deactivate errors keep form values intact.

14. Success State
`Supplier saved successfully.` Use action-specific messaging for deactivation.

15. Responsive Behavior
Desktop and tablet editing supported. Mobile better suited for lookup only.

16. Accessibility Notes
Detail pane headings and field labels must be explicit. Deactivate flow should use confirm dialog with focus trap.

UI-012 Purchase Order Management

1. Screen Name
Purchase Order Management

2. Purpose
Create, review, amend, and cancel purchase orders.

3. Layout Structure
Search and filter area above PO list; selected PO opens a detail region with header fields and line items.

4. Navigation
`Procurement > Purchase Orders`, with links to requisitions, delivery planning, and receipt validation.

5. Components
Filter bar, PO table, line-item editor, status timeline, metadata block, confirmation dialog.

6. Tables
Purchase orders table and PO line items table.

7. Forms
PO header: supplier, destination, expected date, status. Lines: SKU, quantity, UOM, price, notes. Action-specific amendment/cancellation reasons.

8. Buttons
`Create PO`, `Save Draft`, `Submit`, `Amend`, `Cancel PO`.

9. Icons
PO/document icon, supplier icon, line-item/add-row icon, cancel/warning icon.

10. Validation
Require supplier, location, delivery date, at least one line, valid quantities, active supplier, action-specific reasons, received-quantity constraints on amendments.

11. Empty State
Default list: `No purchase orders found.` Filtered: `No purchase orders match the selected filters.` Offer `Create Purchase Order` when relevant.

12. Loading State
PO table and line rows display skeletons while header context remains.

13. Error State
`Purchase orders could not be loaded.` Preserve draft lines and notes on amend, save, or cancel failure.

14. Success State
`Purchase order saved successfully.` or `Purchase order canceled successfully.`

15. Responsive Behavior
Desktop-first. Tablet supported with stacked header and horizontal line table scroll. Mobile should be read-only or limited.

16. Accessibility Notes
Line-item editor requires clear row/field labeling, keyboard reachability, and row-level validation messaging.

UI-013 Purchase Requisition Management

1. Screen Name
Purchase Requisition Management

2. Purpose
Review replenishment-driven requisitions, approve or reject them, and convert approved records to purchase orders.

3. Layout Structure
Queue on the left, requisition detail in the center/right, and status history below or in a side section.

4. Navigation
`Procurement > Purchase Requisitions`, plus contextual path into `Purchase Orders`.

5. Components
Queue list, requisition detail panel, line-item editor, approval decision panel, status timeline.

6. Tables
Requisition table with status, age, destination, and justification summary.

7. Forms
Receiving location, justification, line items, required date, supplier for conversion, expected delivery date, rejection reason.

8. Buttons
`Create Requisition`, `Approve`, `Reject`, `Convert to PO`, `Save`.

9. Icons
Requisition/document icon, approval icons, conversion arrow icon, warning icon.

10. Validation
At least one valid line, justification required, conditional conversion fields, rejection reason required on reject.

11. Empty State
`No requisitions are available in this view.` Offer `Create Requisition` for authorized users.

12. Loading State
Skeleton queue and detail panes.

13. Error State
`Requisitions could not be loaded.` Preserve decision notes and draft edits.

14. Success State
`Requisition submitted successfully.` `Requisition approved successfully.` `Purchase order created successfully.` depending on action.

15. Responsive Behavior
Desktop/tablet supported. Mobile may support approval review only, not full line-item creation.

16. Accessibility Notes
Approval decisions must be clearly grouped and keyboard reachable. Queue changes after action should announce the next item or empty state.

UI-014 Inbound Delivery Planning

1. Screen Name
Inbound Delivery Planning

2. Purpose
Plan and update expected supplier deliveries to receiving locations.

3. Layout Structure
Calendar-or-list hybrid with filter bar, inbound deliveries table, and selected delivery detail panel.

4. Navigation
`Procurement > Inbound Delivery Planning`, with drill-in to related purchase orders.

5. Components
Date filters, deliveries table, detail panel, update form, status badges.

6. Tables
Inbound deliveries table with PO reference, supplier, ETA, carrier, destination, and delay state.

7. Forms
PO reference, shipment date, ETA, carrier, destination, expected quantities by line.

8. Buttons
`Update Delivery`, `Save`, `Notify Receiving`.

9. Icons
Calendar icon, truck/shipping icon, delay/clock icon, notification icon.

10. Validation
Require expected arrival date. Quantities cannot exceed remaining open PO quantity.

11. Empty State
`No inbound deliveries are scheduled for this view.` Suggest changing period or open-PO scope.

12. Loading State
Schedule and selected detail panel show skeletons.

13. Error State
`Inbound delivery plans could not be loaded.` Preserve entered schedule changes on save failure.

14. Success State
`Inbound delivery plan updated successfully.`

15. Responsive Behavior
Desktop and tablet supported. Mobile should show a simpler list view rather than a dense schedule editor.

16. Accessibility Notes
Calendar/list alternatives should be equally accessible. Delay status must use text plus icon, not color only.

UI-015 Goods Receipt Workbench

1. Screen Name
Goods Receipt Workbench

2. Purpose
Receive supplier goods, verify products and quantities, capture traceability data, and stage items for put-away.

3. Layout Structure
Operational workbench with location context badge, pending receipt queue, receipt detail, line verification grid, and next-step actions.

4. Navigation
`Inbound > Goods Receipt Workbench`, with context links to PO detail, discrepancy handling, and put-away.

5. Components
Queue list, line-item editor, scanner-ready inputs, step indicator, status strip, metadata block.

6. Tables
Pending receipts table and receipt line items table.

7. Forms
Receipt identification plus line-level accepted quantity, rejected quantity, condition, lot number, expiration date, and manufacture date.

8. Buttons
`Open Receipt`, `Verify`, `Log Discrepancy`, `Confirm Receipt`, `Send to Put-Away`.

9. Icons
Receiving/dock icon, scan icon, warning/discrepancy icon, success check, put-away arrow icon.

10. Validation
Require valid receiving location, valid PO when needed, accepted quantity rules, lot requirement for traceable products, expiration rules for shelf-life products, MRSL enforcement.

11. Empty State
`No inbound receipts are waiting at this location.`

12. Loading State
Queue skeleton plus detail placeholder rows; keep location context visible.

13. Error State
`Goods receipts could not be loaded.` Preserve scanned values, quantities, and loaded line context when verify fails.

14. Success State
`Receipt verified successfully.` Mark the task as ready for put-away.

15. Responsive Behavior
Desktop and tablet operational support required. Mobile should be limited to light verification or lookup only unless a scanner-optimized experience is designed.

16. Accessibility Notes
High-priority accessibility screen. Scanner fields need predictable focus retention. Line-item grid and dynamic validation must support keyboard-only operation and live announcements.

UI-016 Receipt Discrepancy Resolution

1. Screen Name
Receipt Discrepancy Resolution

2. Purpose
Document, investigate, and resolve inbound discrepancies such as shortage, overage, damage, or wrong item.

3. Layout Structure
Queue-detail workspace with discrepancy list, discrepancy detail panel, comparison area, and evidence/notes section.

4. Navigation
`Inbound > Receipt Discrepancies`, with return path to related receipt.

5. Components
Queue list, variance comparison panel, evidence attachment panel, form sections, status timeline.

6. Tables
Discrepancies table with type, receipt reference, quantity variance, age, and status.

7. Forms
Discrepancy type, variance quantity, reason code, notes, attachment/evidence, resolution outcome, resolution notes.

8. Buttons
`Save Discrepancy`, `Resolve`, `Escalate`, `Close`.

9. Icons
Warning triangle, damaged-item icon, attachment icon, escalation icon.

10. Validation
Require discrepancy type, variance quantity, reason code, and notes. Resolution fields become required only when resolving.

11. Empty State
`No receipt discrepancies are waiting for review.`

12. Loading State
Queue and detail area show skeletons.

13. Error State
`Discrepancies could not be loaded.` Preserve notes, reason code, and attachments on failed resolution.

14. Success State
`Discrepancy resolved successfully.` Update queue counts immediately.

15. Responsive Behavior
Desktop/tablet supported. Mobile should support viewing and simple escalation only.

16. Accessibility Notes
Evidence attachments require accessible file labeling. Comparison panels should use clear headings like `Expected`, `Received`, and `Accepted`.

UI-017 Put-Away Workbench

1. Screen Name
Put-Away Workbench

2. Purpose
Guide warehouse staff through storing received goods in the correct locations.

3. Layout Structure
Task queue on the left, task detail and storage destination confirmation on the right.

4. Navigation
`Inbound > Put-Away Workbench`, with context from goods receipt and storage positions.

5. Components
Queue list, task detail panel, storage position selector, status strip, metadata footer.

6. Tables
Pending put-away tasks table.

7. Forms
Receipt reference, product, quantity, suggested location, actual location.

8. Buttons
`Open Task`, `Confirm Location`, `Complete Put-Away`, `Reassign Location`.

9. Icons
Shelf/bin icon, location marker, move arrow, completion check.

10. Validation
Require quantity and valid storage position. Quantity cannot exceed remaining unplaced quantity.

11. Empty State
`No put-away tasks are waiting at this location.`

12. Loading State
Task rows and destination area skeletonize.

13. Error State
`Put-away tasks could not be loaded.` Preserve selected location and quantities on failure.

14. Success State
`Put-away completed successfully.` Remove the task from the active queue or move it to completed history.

15. Responsive Behavior
Desktop/tablet operational support. Mobile should only support simple confirm workflows if fully optimized.

16. Accessibility Notes
High-priority workbench. Focus should move to the next logical task after completion. Location suggestions and overrides must be readable and keyboard operable.

UI-018 Non-PO and Return Receipt Intake

1. Screen Name
Non-PO and Return Receipt Intake

2. Purpose
Record inbound receipts that arrive outside the standard supplier PO flow.

3. Layout Structure
Single-screen workflow with receipt type selector, intake form, line-entry area, and verification/confirmation section.

4. Navigation
`Inbound > Non-PO Receipt Intake`, with possible return-related linkage.

5. Components
Segmented receipt-type selector, form sections, line-item editor, banner messages.

6. Tables
Optional received line items table for multi-line intake.

7. Forms
Receipt type, authorization reference or return authorization, product, quantity, condition, notes, reason code, source description.

8. Buttons
`Start Intake`, `Verify`, `Confirm Receipt`, `Cancel`.

9. Icons
Return icon, intake/inbox icon, approval reference icon, warning icon.

10. Validation
Require valid reference, reason code where applicable, line items, and positive quantities.

11. Empty State
Neutral state: `Select a receipt type to begin.`

12. Loading State
Show form shell; disable submit while authorization or reference data is being checked.

13. Error State
Show invalid reference errors inline and at form level. Preserve all typed values.

14. Success State
`Receipt recorded successfully.` Reset for next intake or route forward.

15. Responsive Behavior
Tablet supported. Mobile may support simple single-line intake only. Desktop preferred for multi-line receiving.

16. Accessibility Notes
Receipt-type choice should be a clear radio/segmented control. Conditional fields need correct live announcements when they appear.

UI-019 Inbound Supply Tracking

1. Screen Name
Inbound Supply Tracking

2. Purpose
Monitor open POs, expected arrivals, delays, and outstanding inbound quantities.

3. Layout Structure
Overview page with filters, delay indicators, open supply table, and PO drill-in panel.

4. Navigation
`Procurement > Inbound Supply Tracking`, with links to PO details and delivery planning.

5. Components
Filter bar, open-supply table, status badges, detail panel, refresh controls.

6. Tables
Open PO and inbound supply table.

7. Forms
Filters for supplier, location, status, and expected date range.

8. Buttons
`View PO`, `Update Delivery Status`, `Refresh`.

9. Icons
Truck icon, delay/clock icon, PO/document icon.

10. Validation
Validate date range and location scope where relevant.

11. Empty State
`No inbound supply records are available for this view.`

12. Loading State
Table skeleton with delay-status placeholders.

13. Error State
`Inbound supply information could not be loaded.` Preserve filters and open detail state if possible.

14. Success State
After status update, show `Inbound status updated successfully.`

15. Responsive Behavior
Desktop and tablet supported. Mobile should provide summarized read-only tracking.

16. Accessibility Notes
Delay severity needs text labels and icons. Table sorting and drill-in actions must be keyboard accessible.

UI-020 Inventory Status Monitor

1. Screen Name
Inventory Status Monitor

2. Purpose
Give network-wide visibility into stock health and route users to the right follow-up action.

3. Layout Structure
Filter bar on top, summary cards, stock board/table, and drill-down detail area.

4. Navigation
`Inventory > Inventory Status Monitor`, with contextual actions to transfer, replenish, adjust, or inspect movement history.

5. Components
KPI/status cards, filter bar, inventory status table, detail panel, drill-down links, action menu.

6. Tables
Inventory status table with SKU, location, on-hand, reserved, ATP, and stock status.

7. Forms
Filters for location, category, product, status, and ABC class.

8. Buttons
`View Detail`, `Request Transfer`, `Open Replenishment`, `Create Adjustment`, `Refresh`.

9. Icons
Inventory box icon, low-stock warning, out-of-stock icon, quarantine icon, transfer arrows.

10. Validation
Validate location scope and any search constraints. Preserve filters on failed queries.

11. Empty State
If no inventory: `No inventory is available in this view.` If filters remove all rows: `No stock records match these filters.`

12. Loading State
Skeleton cards, skeleton board/table, and detail placeholder.

13. Error State
`Inventory status could not be loaded.` Keep current filters and scope visible.

14. Success State
`Inventory view updated.` after refresh or action return.

15. Responsive Behavior
Desktop-first analytics/workbench hybrid. Tablet supported with fewer columns. Mobile should use summary + drill-in cards.

16. Accessibility Notes
High-importance table screen. Status must use badges with text. Drill-down actions need descriptive names like `Open movement history for SKU`.

UI-021 Inventory Availability Inquiry

1. Screen Name
Inventory Availability Inquiry

2. Purpose
Allow sales and service users to quickly answer availability questions using ATP and network visibility.

3. Layout Structure
Search panel at top, search results below, and a selected product availability detail view with optional network table.

4. Navigation
`Inventory > Availability Inquiry`, with context to order reservation and network availability detail.

5. Components
Search bar, results list, availability summary, location availability table, optional action shortcuts.

6. Tables
Product search results table and availability-by-location table.

7. Forms
Search by SKU, name, barcode, category, and optional location scope.

8. Buttons
`Search`, `View Network Availability`, `Reserve from Order`.

9. Icons
Search icon, ATP/info icon, network icon, reserve icon.

10. Validation
Require at least one meaningful product search input. Validate selected product and location scope.

11. Empty State
Before search: `Search for a product to view availability.` After no match: `No matching product was found.`

12. Loading State
Search results and network availability show placeholders after query.

13. Error State
`Availability could not be retrieved.` Preserve search inputs and scope.

14. Success State
`Availability updated.` displayed subtly when results refresh.

15. Responsive Behavior
Good candidate for mobile-friendly support. Use stacked product/result cards and a compact location list on smaller screens.

16. Accessibility Notes
Differentiate `On-hand`, `Reserved`, and `ATP` with visible labels and help text. Search must be keyboard efficient for live customer conversations.

UI-022 Inventory Movement History

1. Screen Name
Inventory Movement History

2. Purpose
Trace inventory transactions by product, location, type, user, and time.

3. Layout Structure
Query form above a movement history table with an adjacent or lower transaction detail panel.

4. Navigation
`Inventory > Movement History`, with drill-in from inventory status, products, and audit workflows.

5. Components
Filter bar, date range picker, movement table, detail panel, export actions.

6. Tables
Movement history table with transaction date, type, product, location, quantity, actor, and reference.

7. Forms
Filters for SKU, location, transaction type, user, and date range.

8. Buttons
`Search`, `Open Transaction`, `Export`.

9. Icons
History/clock icon, movement arrows, export icon, audit/detail icon.

10. Validation
Date range required, with optional narrowing guidance if results may be too broad.

11. Empty State
Before search: `Enter filters to view movement history.` After no results: `No movement records match this search.`

12. Loading State
Query controls remain visible while rows skeletonize.

13. Error State
`Movement history could not be loaded.` Keep entered filters and detail context where possible.

14. Success State
Results refresh in place and export preparation shows confirmation when successful.

15. Responsive Behavior
Desktop/tablet supported. Mobile should be limited to simplified search and detail lookup.

16. Accessibility Notes
Tables require true header semantics. Export options need clear labels and keyboard accessibility.

UI-023 Cycle Count Workbench

1. Screen Name
Cycle Count Workbench

2. Purpose
Present assigned blind counts and collect count results without exposing system quantities.

3. Layout Structure
Assigned counts queue plus count-entry panel and confirmation area.

4. Navigation
`Inventory > Cycle Counts`, scoped to the assigned worker and location context.

5. Components
Queue list, count-entry form, status strip, metadata footer.

6. Tables
Assigned cycle counts table.

7. Forms
Counted quantity and optional notes for the selected assignment.

8. Buttons
`Open Count`, `Submit Count`, `Save Draft`.

9. Icons
Count/checklist icon, location icon, completion check.

10. Validation
Require non-negative counted quantity. Do not reveal expected system quantity.

11. Empty State
`No cycle counts are assigned to you right now.`

12. Loading State
Assigned-task list and count-entry area skeletonize.

13. Error State
`Cycle count assignments could not be loaded.` Preserve entered count values on submit failure.

14. Success State
`Count submitted successfully.` Move focus to the next assignment when available.

15. Responsive Behavior
Tablet supported and useful for warehouse work. Mobile can support single-task submission if optimized.

16. Accessibility Notes
High-priority warehouse screen. Keep focus stable for scanner or keyboard use. Announce movement to the next task.

UI-024 Physical Inventory Count Management

1. Screen Name
Physical Inventory Count Management

2. Purpose
Manage wall-to-wall count sessions, freezing, progress monitoring, and certification.

3. Layout Structure
Count session summary at top, progress widgets, zone/team assignment table, and certification panel.

4. Navigation
`Inventory > Physical Counts`, with links to freeze status and certification.

5. Components
Session summary cards, progress table, assignment table, certification form, status timeline.

6. Tables
Count progress table and zone/team assignments table.

7. Forms
Location, planned start/end, count scope, certification notes, certifier information.

8. Buttons
`Start Count`, `Assign Teams`, `Monitor Progress`, `Certify Count`, `Release Freeze`.

9. Icons
Inventory audit icon, freeze/lock icon, certification checkmark, team icon.

10. Validation
Required location and dates, end date after start date, all required count work complete before certification.

11. Empty State
`No physical inventory count session is active.` Offer `Start Count` if permitted.

12. Loading State
Session status and zone progress show placeholders.

13. Error State
`Physical count information could not be loaded.` Preserve unresolved-item context on failed certification.

14. Success State
`Physical count started successfully.` and `Physical count certified successfully.`

15. Responsive Behavior
Desktop-first. Tablet review is acceptable. Mobile should be read-only for session oversight.

16. Accessibility Notes
Progress states need text labels and not color alone. Freeze and certification are critical actions and require strong confirmation and focus handling.

UI-025 Variance Investigation and Approval

1. Screen Name
Variance Investigation and Approval

2. Purpose
Investigate count variances and approve only justified adjustments.

3. Layout Structure
Variance queue, side-by-side expected vs counted comparison panel, investigation notes area, and approval controls.

4. Navigation
`Inventory > Variance Approvals`, with contextual path to stock adjustments.

5. Components
Queue list, comparison panel, evidence panel, approval decision panel, status history.

6. Tables
Variance cases table with age, product, location, variance amount, and status.

7. Forms
Root cause category, investigation findings, approved adjustment quantity, approver note.

8. Buttons
`Investigate`, `Approve`, `Reject`, `Escalate`.

9. Icons
Variance/delta icon, evidence attachment icon, approval/reject icons.

10. Validation
Require root cause and findings. Approval requires approved quantity. Enforce segregation-of-duties restrictions.

11. Empty State
`No variances are waiting for review.`

12. Loading State
Pending-variance table and detail pane show skeletons.

13. Error State
`Variance cases could not be loaded.` Preserve findings and notes on failed decision.

14. Success State
`Variance decision saved successfully.` Remove or update the case in queue immediately.

15. Responsive Behavior
Desktop/tablet supported. Mobile should support review only, not full investigation and approval.

16. Accessibility Notes
Comparison panels must clearly label values and units. Approval controls should be grouped semantically and fully keyboard accessible.

UI-026 Stock Adjustment Management

1. Screen Name
Stock Adjustment Management

2. Purpose
Submit, approve-route, and finalize stock adjustments with reason codes and auditability.

3. Layout Structure
Adjustment cases list plus detail form and approval/status summary.

4. Navigation
`Inventory > Stock Adjustments`, with links from inventory status and variance workflows.

5. Components
Adjustment table, standard form sections, reason-code selector, evidence panel, approval badges, banner messages.

6. Tables
Adjustment cases table.

7. Forms
Origin type, location, SKU, adjustment type, quantity, reason code, notes, evidence, and rejection reason.

8. Buttons
`Create Adjustment`, `Submit`, `Approve`, `Reject`, `Apply`.

9. Icons
Adjustment plus/minus icon, warning icon, approval icon, attachment icon.

10. Validation
Require valid origin, location, product, type, quantity, reason code; prevent excessive decrease; require evidence for write-offs; enforce threshold and SoD rules.

11. Empty State
`No adjustment cases are available in this view.`

12. Loading State
Skeleton case list and form shell.

13. Error State
`The adjustment could not be processed.` Preserve all fields, notes, and evidence references.

14. Success State
`Adjustment submitted successfully.` or `Adjustment applied successfully.`

15. Responsive Behavior
Desktop/tablet supported. Mobile should be limited to light approval review or status check.

16. Accessibility Notes
High-priority screen. Quantities and direction must be explicit. Evidence uploads and approval state changes need clear announcements.

UI-027 Write-Off Authorization

1. Screen Name
Write-Off Authorization

2. Purpose
Review write-off requests, supporting evidence, and finance acknowledgment before permanent inventory removal.

3. Layout Structure
Pending write-off list, selected request detail, and evidence/finance acknowledgment section.

4. Navigation
`Inventory > Write-Offs`, with links from stock adjustment and returns disposition contexts.

5. Components
Queue list, read-only review panel, approval decision panel, evidence attachment panel, confirmation dialog.

6. Tables
Pending write-offs table.

7. Forms
Approval decision, finance acknowledgment, approver note, rejection reason.

8. Buttons
`Review`, `Approve`, `Reject`, `Request More Info`.

9. Icons
Write-off/trash or strike-through inventory icon, finance/ledger icon, attachment icon, warning icon.

10. Validation
Require approval decision; require finance acknowledgment when policy requires it; require rejection reason when rejecting.

11. Empty State
`No write-off requests are waiting for review.`

12. Loading State
Show queue skeleton rows and detail placeholders.

13. Error State
`Write-off requests could not be loaded.` Preserve decision note and acknowledgment state.

14. Success State
`Write-off decision recorded successfully.`

15. Responsive Behavior
Desktop/tablet review supported. Mobile approval review is possible if simplified.

16. Accessibility Notes
Destructive action requires a strong confirmation pattern and clear, non-color-only risk communication.

UI-028 Quarantine and Hold Register

1. Screen Name
Quarantine and Hold Register

2. Purpose
Apply, review, and release hold or quarantine status while showing affected non-sellable quantities.

3. Layout Structure
Register table with selected hold detail and an apply/release action panel.

4. Navigation
`Inventory > Quarantine & Holds`, with context from returns and fulfillment exceptions.

5. Components
Hold register table, detail card, status badges, apply/release form, metadata strip.

6. Tables
Quarantine and hold table with product, location, quantity, hold type, reason, duration, and status.

7. Forms
Quantity, hold reason, reason code, lot identity, disposition, inspection reference.

8. Buttons
`Apply Hold`, `Release Hold`, `View Product`.

9. Icons
Quarantine icon, lock icon, release/unlock icon, warning badge.

10. Validation
Require quantity, hold reason, and reason code. Enforce lot rules for traceable products and release-specific disposition requirements.

11. Empty State
`No stock is currently on hold or in quarantine.`

12. Loading State
Register and selected detail card skeletonize.

13. Error State
`Hold and quarantine records could not be loaded.` Preserve quantity and reason fields on failure.

14. Success State
`Hold updated successfully.` Refresh availability metrics immediately.

15. Responsive Behavior
Desktop/tablet supported. Mobile may support lookup and simple release with strict controls.

16. Accessibility Notes
Hold type and status require text labels. Apply/release modes should be clearly distinguished for screen-reader users.

UI-029 Shrinkage, Loss, and Obsolescence Analysis

1. Screen Name
Shrinkage, Loss, and Obsolescence Analysis

2. Purpose
Analyze shrinkage, obsolete stock, and slow-moving inventory for corrective action and working-capital decisions.

3. Layout Structure
Analytics layout with filters, summary cards, chart zones, recommendations section, and supporting tables.

4. Navigation
`Inventory > Shrinkage & Obsolescence`, with drill-down to adjustment cases and reports.

5. Components
KPI cards, trend indicators, report viewer sections, recommendation links, export action group.

6. Tables
Shrinkage summary table and aging/obsolete items table.

7. Forms
Filters for date range, location, category, and reason code.

8. Buttons
`Apply Filters`, `Export`, `Open Adjustment Cases`.

9. Icons
Trend icons, loss/warning icon, aging/hourglass icon, export icon.

10. Validation
Validate date range and scoped filters where needed.

11. Empty State
`No shrinkage or obsolescence data is available for the selected period.`

12. Loading State
Chart, card, and table skeletons preserve layout.

13. Error State
`The analysis could not be loaded.` Keep filters selected.

14. Success State
`Analysis updated.` and `Analysis exported successfully.` when applicable.

15. Responsive Behavior
Desktop-first analytics. Tablet summary view acceptable. Mobile should be read-only high-level cards only.

16. Accessibility Notes
Charts must use labels, markers, and direct annotations. Recommendations should be actionable links, not ambiguous chart-only interactions.

UI-030 Order Reservation Workbench

1. Screen Name
Order Reservation Workbench

2. Purpose
Confirm order readiness, reserve inventory, and release reservations when orders change.

3. Layout Structure
Order queue/search panel, order line availability panel, and reservation summary area.

4. Navigation
`Fulfillment > Order Reservation`, with context links to availability inquiry, split fulfillment, and backorder management.

5. Components
Orders table, line-item table, availability summary, reservation form, status timeline.

6. Tables
Orders table and order line table.

7. Forms
Order search, selected fulfillment location, reservation quantity by line, release reason, cancellation reason.

8. Buttons
`Check Availability`, `Reserve Stock`, `Release Reservation`, `Open Split Fulfillment`.

9. Icons
Order icon, reserve/lock icon, availability/info icon, split-fulfillment network icon.

10. Validation
Validate selected location, quantity versus ATP, safety stock exception handling, and release/cancellation reasons.

11. Empty State
`No orders are waiting for reservation in this view.`

12. Loading State
Queue and line placeholders, disable reserve actions until ATP is loaded.

13. Error State
`Order reservation data could not be loaded.` Preserve selected lines and quantities.

14. Success State
`Reservation completed successfully.` Update order and queue status immediately.

15. Responsive Behavior
Desktop/tablet supported. Mobile may support order lookup and release only, not full allocation management.

16. Accessibility Notes
High-priority operational screen. ATP, reserved, and requested quantities must be clearly labeled. Reservation outcomes should be announced live.

UI-031 Pick, Pack, and Ship Workbench

1. Screen Name
Pick, Pack, and Ship Workbench

2. Purpose
Guide warehouse fulfillment from pick queue through verification, packing, and shipment handoff.

3. Layout Structure
Task queue, pick verification panel, packing section, and shipment confirmation section within one workbench.

4. Navigation
`Fulfillment > Pick / Pack / Ship`, with links to lot detail and exception handling.

5. Components
Queue list, line-item pick grid, scanner inputs, packing editor, shipment confirmation block, exception route actions.

6. Tables
Pick queue table and pick line items table.

7. Forms
Verification fields, lot/batch data, picked quantity, package references, carrier, notes, tracking reference.

8. Buttons
`Open Pick`, `Verify Pick`, `Pack`, `Raise Exception`, `Confirm Shipment`.

9. Icons
Pick/cart icon, barcode/scan icon, package icon, shipment/truck icon, exception warning icon.

10. Validation
Validate picked quantity versus reserved quantity, lot requirements, substitution references, packed quantity versus picked quantity, and ship-only-after-verification rules.

11. Empty State
`No orders are ready for pick, pack, or ship at this location.`

12. Loading State
Pick queue and task detail use skeletons while location/shift context remains visible.

13. Error State
`Fulfillment tasks could not be loaded.` Preserve scanned values and packing structure on failure.

14. Success State
Use step-specific messages: `Pick verified successfully.`, `Order packed successfully.`, `Shipment confirmed successfully.`

15. Responsive Behavior
Desktop/tablet operational support is required. Mobile support should be limited to carefully optimized warehouse flows only.

16. Accessibility Notes
High-priority screen. Scanner workflows, editable line grids, and dynamic queue movement require explicit keyboard behavior and live region announcements.

UI-032 Partial Fulfillment and Backorder Management

1. Screen Name
Partial Fulfillment and Backorder Management

2. Purpose
Manage partial shipment plans and create backorders for the unfulfilled remainder.

3. Layout Structure
Affected orders list, quantity comparison panel, and backorder plan section.

4. Navigation
`Fulfillment > Partial Fulfillment & Backorders`, often linked from order reservation and fulfillment exceptions.

5. Components
Cases table, comparison panel, line-item editor, acknowledgment section, status history.

6. Tables
Partial/backorder cases table.

7. Forms
Fulfill-now quantity, backorder quantity, customer acknowledgment reference, expected date, priority override reference.

8. Buttons
`Confirm Partial Fulfillment`, `Create Backorder`, `Cancel Remainder`.

9. Icons
Split shipment icon, backorder/clock icon, customer acknowledgment icon.

10. Validation
Fulfill quantity must be positive and less than full quantity for partial flow; backorder quantity must match remainder; acknowledgment may be required.

11. Empty State
`No partial fulfillment or backorder cases require attention.`

12. Loading State
Order skeletons and comparison placeholders.

13. Error State
`The partial fulfillment workflow could not be completed.` Preserve quantities and acknowledgment notes.

14. Success State
`Partial fulfillment recorded successfully.` and `Backorder created successfully.`

15. Responsive Behavior
Desktop/tablet supported. Mobile may support customer-service review with simplified line detail.

16. Accessibility Notes
Quantity relationships should be communicated in text, not only visually. Customer acknowledgment requirements need explicit labels and descriptions.

UI-033 Fulfillment Exception Management

1. Screen Name
Fulfillment Exception Management

2. Purpose
Resolve pick failures, quantity mismatches, damaged picks, and other outbound exceptions.

3. Layout Structure
Exception queue, exception detail/context panel, and resolution options panel.

4. Navigation
`Fulfillment > Fulfillment Exceptions`, with routes back to pick queue, backorders, quarantine, or customer service follow-up.

5. Components
Queue list, detail panel, resolution form, status timeline, contextual navigation links.

6. Tables
Fulfillment exceptions table.

7. Forms
Resolution type, notes, reassignment, customer follow-up flag.

8. Buttons
`Resolve`, `Re-Pick`, `Partial Ship`, `Quarantine`, `Route to Customer Service`.

9. Icons
Warning triangle, repick/retry icon, partial ship icon, quarantine icon, customer-service icon.

10. Validation
Require exception type and description. Resolution-specific fields should validate when selected.

11. Empty State
`No fulfillment exceptions are open right now.`

12. Loading State
Queue skeleton and resolution panel placeholders.

13. Error State
`Exceptions could not be loaded.` Preserve chosen path and notes.

14. Success State
`Exception resolved successfully.` Update or remove queue item immediately.

15. Responsive Behavior
Desktop/tablet supported. Mobile review may be possible, but routing actions should remain simplified.

16. Accessibility Notes
Resolution options should be clearly labeled as outcomes, not ambiguous buttons. Queue updates should be announced.

UI-034 Order Fulfillment Status

1. Screen Name
Order Fulfillment Status

2. Purpose
Provide current reservation, picking, packing, shipment, exception, and backorder status for order inquiries.

3. Layout Structure
Order search bar, order status summary header, line-level progress table, and tabbed shipment/backorder/exception details.

4. Navigation
`Fulfillment > Order Status`, with drill-ins to shipment, exception, or backorder details.

5. Components
Search bar, order summary, status badges, line-status table, tabs/segmented content switcher, metadata strip.

6. Tables
Orders table and order lines status table.

7. Forms
Search fields for order ID, customer reference, date, and status.

8. Buttons
`Search`, `View Shipment`, `View Exception`, `View Backorder`.

9. Icons
Order icon, shipment/truck icon, warning icon, backorder clock icon.

10. Validation
Validate search inputs lightly; preserve search/filter state on failures.

11. Empty State
Before search: `Search for an order to view fulfillment status.` No match: `No order matched your search.`

12. Loading State
Search results skeleton and line-status placeholders.

13. Error State
`Order status could not be loaded.` Keep search context intact.

14. Success State
Refreshed status view updates line and shipment states with subtle confirmation.

15. Responsive Behavior
Good candidate for mobile-friendly read-heavy use. Use stacked order cards and collapsible line details on small screens.

16. Accessibility Notes
Status progression must be understandable in text. Tabs require correct semantics and keyboard support.

UI-035 Multi-Location Split Fulfillment Planner

1. Screen Name
Multi-Location Split Fulfillment Planner

2. Purpose
Plan fulfillment of one order from multiple locations when no single site can satisfy it alone.

3. Layout Structure
Order shortage summary, proposed split allocation table, and location-specific reservation section.

4. Navigation
`Fulfillment > Multi-Location Fulfillment`, often entered from reservation or availability workflows.

5. Components
Allocation table, network availability summary, recommendation card, confirmation banner.

6. Tables
Split allocation table with location, quantity, ATP, and shipping grouping.

7. Forms
Allocations by location, quantity, shipping grouping, notes.

8. Buttons
`Generate Plan`, `Adjust Allocation`, `Confirm Split`, `Release to Queues`.

9. Icons
Network/split icon, location icons, allocation arrows, queue release icon.

10. Validation
Allocated quantity cannot exceed ATP at selected location, and total allocated quantity must equal the required line quantity.

11. Empty State
`This order does not currently require multi-location fulfillment.`

12. Loading State
Allocation skeleton and network-availability placeholders.

13. Error State
`The split fulfillment plan could not be generated.` Preserve manual allocations.

14. Success State
`Split fulfillment plan created successfully.` Refresh location queues or reservation summaries.

15. Responsive Behavior
Desktop/tablet planning screen. Mobile should be read-only or excluded.

16. Accessibility Notes
Allocation totals and remaining quantity must be communicated textually. Multi-row edits require row context and keyboard support.

UI-036 Low-Stock Alert Review

1. Screen Name
Low-Stock Alert Review

2. Purpose
Review, prioritize, and acknowledge active replenishment alerts.

3. Layout Structure
Alert summary strip, alert list, and selected alert detail with recommended action panel.

4. Navigation
`Procurement > Low-Stock Alerts`, with direct routes to replenishment planning and transfer request.

5. Components
Alert summary cards, alert table, detail card, filter bar, acknowledgment form.

6. Tables
Low-stock alerts table.

7. Forms
Filters for priority, location, category, and status; acknowledgment action and notes.

8. Buttons
`Acknowledge`, `Open Replenishment`, `Create Transfer Request`, `Refresh`.

9. Icons
Low-stock icon, severity badges, transfer icon, requisition/planning icon.

10. Validation
Validate acknowledgment action and notes when dismissing or deferring.

11. Empty State
`There are no active low-stock alerts in this view.`

12. Loading State
Summary cards and alert rows skeletonize.

13. Error State
`Low-stock alerts could not be loaded.` Preserve selected action and notes.

14. Success State
`Alert acknowledged successfully.` Update queue counts immediately.

15. Responsive Behavior
Suitable for mobile-friendly review with card-based alerts. Desktop keeps list-detail layout.

16. Accessibility Notes
Severity and priority need explicit text and iconography. Recommended actions should have clear labels tied to business meaning.

UI-037 Replenishment Planning

1. Screen Name
Replenishment Planning

2. Purpose
Convert stock alerts and recommendations into requisitions or transfer actions.

3. Layout Structure
Recommendations table, network availability comparison, and requisition creation panel.

4. Navigation
`Procurement > Replenishment Planning`, with branches to transfer request and requisition detail.

5. Components
Recommendation cards, recommendations table, comparison panel, requisition form, status badges.

6. Tables
Recommendations table and open supply/network availability table.

7. Forms
Product, suggested quantity, final quantity, required date, supplier, destination, and justification.

8. Buttons
`Create Requisition`, `Adjust Quantity`, `View Open PO`, `Open Transfer Option`.

9. Icons
Recommendation icon, replenishment arrow, transfer icon, supplier/PO icon.

10. Validation
Validate final quantity, required date, destination, and transfer-before-buy decision requirements.

11. Empty State
`No replenishment recommendations are available for this view.`

12. Loading State
Recommendation cards/tables and comparison area show skeletons.

13. Error State
`Replenishment recommendations could not be loaded.` Preserve entered requisition values.

14. Success State
`Requisition created successfully.` Update recommendation status to reflect action taken.

15. Responsive Behavior
Desktop/tablet supported. Mobile should be limited to alert review and recommendation lookup.

16. Accessibility Notes
Recommendation rationale should be written out in text. Action choices must be explicit and not depend on visual hierarchy alone.

UI-038 Transfer Request

1. Screen Name
Transfer Request

2. Purpose
Submit inter-location stock transfer requests with quantity and business justification.

3. Layout Structure
Primary request form with a side or lower source-impact summary and a recent requests table.

4. Navigation
`Transfers > Transfer Requests`, with entry from inventory status, low-stock alerts, or network rebalancing.

5. Components
Form sections, capacity preview, recent requests table, status badges, confirmation dialog.

6. Tables
Recent transfer requests table.

7. Forms
Source location, destination location, justification, SKU, quantity, and line items where multi-line requests are allowed.

8. Buttons
`Check Capacity`, `Submit Request`, `Save Draft`.

9. Icons
Transfer arrows, source/destination location icons, capacity/info icon.

10. Validation
Source and destination cannot be the same, quantity must be positive, justification required, quantity cannot exceed transferable stock or breach source minimums.

11. Empty State
Neutral create state: `Enter transfer details to begin.`

12. Loading State
Form shell and capacity placeholder while capacity is being checked.

13. Error State
Show inline quantity and location errors plus form-level business-rule banner for threshold breaches.

14. Success State
`Transfer request submitted successfully.` Show request reference and new status.

15. Responsive Behavior
Tablet-supported form flow. Mobile could support simple single-line requests if capacity preview is simplified.

16. Accessibility Notes
Source and destination fields must be distinctly labeled. Capacity warnings must use text, not only color.

UI-039 Transfer Approval Queue

1. Screen Name
Transfer Approval Queue

2. Purpose
Allow management to approve or reject transfer requests after reviewing source and destination impact.

3. Layout Structure
Approval queue list, selected request detail, and source/destination comparison panel.

4. Navigation
`Transfers > Transfer Approvals`, also reachable from `My Tasks / Approvals`.

5. Components
Queue list, comparison panel, decision form, status history, badges.

6. Tables
Transfer approvals table.

7. Forms
Approve/reject decision and approver note or rejection reason.

8. Buttons
`Approve`, `Reject`, `View Source Capacity`.

9. Icons
Approval check icon, reject icon, capacity/info icon, aging clock icon.

10. Validation
Require approval decision; require rejection reason when rejecting; revalidate request state before final approval.

11. Empty State
`No transfer requests are waiting for approval.`

12. Loading State
Queue and comparison panel display skeletons.

13. Error State
`Transfer approvals could not be loaded.` Preserve decision note and selection.

14. Success State
`Transfer decision saved successfully.` Update queue count instantly.

15. Responsive Behavior
Suitable for tablet and simplified mobile approval review, but desktop is best for full impact comparison.

16. Accessibility Notes
Source vs destination comparison must be labeled clearly. After approval, move focus to the next request or list heading.

UI-040 Transfer Execution and In-Transit Tracking

1. Screen Name
Transfer Execution and In-Transit Tracking

2. Purpose
Dispatch approved transfers, receive them at the destination, and monitor in-transit movement.

3. Layout Structure
Three operational zones: dispatch queue, in-transit tracking section, and destination receipt confirmation section.

4. Navigation
`Transfers > Transfer Execution`, with supporting `In-Transit Tracking` context and links to transfer detail.

5. Components
Queue list, in-transit table, receipt confirmation form, status timeline, location context badge.

6. Tables
Transfer queue table and in-transit transfers table.

7. Forms
Dispatch fields for quantity and lot/batch; receipt fields for received quantity and variance note; cancellation reason when needed.

8. Buttons
`Dispatch`, `Confirm Receipt`, `Cancel Transfer`.

9. Icons
Dispatch arrow, in-transit truck, receive/inbox icon, variance warning icon.

10. Validation
Dispatched quantity cannot exceed approved quantity; received quantity is required; variance note required when received differs from dispatched; cancellation needs a reason.

11. Empty State
Dispatch area: `No approved transfers are ready for dispatch.` In-transit area: `No stock is currently in transit.`

12. Loading State
Dispatch and transit tables skeletonize along with receipt placeholders.

13. Error State
`Transfer execution data could not be loaded.` Preserve entered quantities and variance notes.

14. Success State
`Transfer dispatched successfully.` and `Transfer received successfully.`

15. Responsive Behavior
Desktop/tablet operational support required. Mobile may support receipt confirmation only if carefully simplified.

16. Accessibility Notes
High-priority operational screen. Dynamic state changes between dispatch and receipt need live announcements and predictable focus transitions.

UI-041 Network Rebalancing Analysis

1. Screen Name
Network Rebalancing Analysis

2. Purpose
Analyze stock imbalances across locations and propose redistribution actions.

3. Layout Structure
Analytics layout with filter bar, imbalance summary, recommendations table, and impact preview panel.

4. Navigation
`Transfers > Network Rebalancing`, with a route into transfer request creation.

5. Components
Summary cards, recommendation cards, recommendations table, impact comparison panel, export action group.

6. Tables
Rebalancing recommendations table.

7. Forms
Filters for location pair, product category, and date range.

8. Buttons
`Generate Recommendation`, `Create Transfer`, `Export`.

9. Icons
Network icon, rebalance arrows, impact/compare icon, export icon.

10. Validation
Validate filter combinations and date ranges.

11. Empty State
`No network imbalances were identified for the selected scope.`

12. Loading State
Summary and recommendation placeholders preserve page structure.

13. Error State
`Network rebalancing analysis could not be loaded.` Preserve filters.

14. Success State
`Recommendation generated successfully.` or `Transfer request created successfully.`

15. Responsive Behavior
Desktop-first planning surface. Tablet summary view acceptable. Mobile should be read-only.

16. Accessibility Notes
Impact previews require text-based explanation of tradeoffs. Recommendation confidence or urgency must not rely on color alone.

UI-042 Return Authorization Management

1. Screen Name
Return Authorization Management

2. Purpose
Validate return eligibility and issue customer return authorizations.

3. Layout Structure
Order search at top, eligible return lines in the center, and authorization detail panel below or beside.

4. Navigation
`Returns > Return Authorizations`, with links to original order and downstream return processing.

5. Components
Search bar, eligible lines table, authorization detail section, status timeline.

6. Tables
Eligible return lines table and return authorizations table/history.

7. Forms
Search fields, selected item/quantity, reason, authorization notes, cancellation reason.

8. Buttons
`Validate Eligibility`, `Issue Authorization`, `Cancel Authorization`.

9. Icons
Return arrow icon, order lookup icon, eligibility/check icon.

10. Validation
Require original order, return reason, line selection, valid quantity not exceeding shipped quantity, and cancellation reason when canceling.

11. Empty State
Before search: `Search for an order to begin a return authorization.` If no eligible lines: `No return-eligible items were found for this order.`

12. Loading State
Search panel stays visible while eligibility results skeletonize.

13. Error State
`The return authorization could not be completed.` Preserve order, lines, and quantities.

14. Success State
`Return authorization issued successfully.` Display the authorization reference prominently.

15. Responsive Behavior
Good candidate for tablet and mobile-friendly customer-service use with simplified line selection.

16. Accessibility Notes
Eligibility outcomes need explicit text. Quantity rules and return policy failures should be clearly explained.

UI-043 Return Inspection and Disposition

1. Screen Name
Return Inspection and Disposition

2. Purpose
Receive authorized returns, inspect condition, and route them to restock, quarantine, or write-off.

3. Layout Structure
Pending returns queue and a step-based workspace for receipt, inspection, and disposition.

4. Navigation
`Returns > Return Inspection & Disposition`, with contextual routes to quarantine and write-off.

5. Components
Queue list, step progress indicator, line-item editor, condition selectors, disposition controls, status timeline.

6. Tables
Pending returns table.

7. Forms
Receiving location, received quantities, condition, disposition action, restock location, write-off reference/evidence.

8. Buttons
`Receive Return`, `Record Inspection`, `Confirm Disposition`.

9. Icons
Return icon, inspection/check icon, quarantine icon, write-off icon, restock arrow icon.

10. Validation
Require receipt before inspection, inspection before disposition, quantity balancing, disposition action, and any conditional restock/write-off fields.

11. Empty State
`No returns are waiting to be received or inspected.`

12. Loading State
Queue and step content show placeholders.

13. Error State
`Return processing could not be completed.` Preserve received quantities, condition entries, and disposition choices.

14. Success State
Use step-specific messages: `Return received successfully.`, `Inspection recorded successfully.`, `Disposition completed successfully.`

15. Responsive Behavior
Tablet support is valuable for warehouse return handling. Mobile may support simplified receipt and inspection only.

16. Accessibility Notes
High-priority workflow. Step changes and conditional fields need live announcements. Disposition choices must be explicit and not color coded only.

UI-044 Operational Reports and Analytics

1. Screen Name
Operational Reports and Analytics

2. Purpose
Provide operational reporting for stock movement, stockouts, overstock, accuracy, location performance, and related analysis.

3. Layout Structure
Report selector and filters at top, report output region in the center, and drill-down panel or export area.

4. Navigation
`Reports & Analytics > Operational Reports`, with connections to inventory dashboard views and source records.

5. Components
Report selector, filter bar, report viewer, export action group, drill-down links.

6. Tables
Report output table, optionally complemented by charts and summaries.

7. Forms
Report type, location, category, date range, status, KPI codes, and other context-sensitive filters.

8. Buttons
`Run Report`, `Export`, `Open Detail`.

9. Icons
Report/document icon, filter icon, chart icon, export icon.

10. Validation
Validate required filters per report type, especially date ranges and period-sensitive selections.

11. Empty State
Before run: `Select filters and run a report.` After no data: `No report data is available for the selected filters.`

12. Loading State
Filter shell remains visible while result table and chart zones skeletonize.

13. Error State
`The report could not be generated.` Preserve report type and filters.

14. Success State
`Report updated successfully.` and `Export completed successfully.` when relevant.

15. Responsive Behavior
Desktop-first analytics surface. Tablet can support summary and simplified tables. Mobile should be read-only summary only.

16. Accessibility Notes
Charts require labels and patterns. Export actions need descriptive names, not generic icon-only controls.

UI-045 Inventory Valuation Reconciliation

1. Screen Name
Inventory Valuation Reconciliation

2. Purpose
Compare operational valuation with finance records, investigate differences, and finalize reconciliation status.

3. Layout Structure
Period selector and summary cards at top, side-by-side valuation comparison area, notes panel, and supporting variance tables.

4. Navigation
`Reports & Analytics > Valuation Reconciliation`, with contextual links to variance drill-down and finance evidence.

5. Components
Summary cards, comparison panel, notes form, report viewer, export actions, status badges.

6. Tables
Valuation comparison table and adjustment/variance detail table.

7. Forms
Period end date, variance threshold percent, reconciliation notes, finalization status.

8. Buttons
`Load Period`, `Add Note`, `Finalize Reconciliation`, `Export`.

9. Icons
Finance/ledger icon, compare icon, variance warning icon, finalize/check icon.

10. Validation
Require period end date; validate threshold range and preserve notes on failures.

11. Empty State
`No valuation data is available for the selected period.`

12. Loading State
Summary cards, comparison blocks, and note shell skeletonize.

13. Error State
`Valuation reconciliation could not be loaded.` Keep period and unsaved notes.

14. Success State
`Reconciliation notes saved successfully.` and `Reconciliation finalized successfully.`

15. Responsive Behavior
Desktop-first finance review screen. Tablet supports summary only. Mobile should be read-only or unsupported.

16. Accessibility Notes
Comparison tables need explicit headings and units. Variance severity cannot rely on red/green alone.

UI-046 Promotional Stock Readiness

1. Screen Name
Promotional Stock Readiness

2. Purpose
Assess promotional stock readiness, ring-fenced allocations, and launch risk.

3. Layout Structure
Campaign selector, readiness summary cards, campaign product readiness grid, and gap/escalation section.

4. Navigation
`Reports & Analytics > Promotional Readiness`, with connections to campaign allocation records.

5. Components
Campaign selector, summary cards, readiness table, gap panel, escalation actions, status badges.

6. Tables
Campaign product readiness table with allocated quantity, target quantity, ATP, inbound support, and readiness state.

7. Forms
Campaign, launch date, location scope, readiness status, escalation reason, release or override inputs.

8. Buttons
`Check Readiness`, `Confirm Readiness`, `Escalate Gap`, `Release Allocation`.

9. Icons
Campaign/megaphone icon, readiness check, at-risk warning, release icon.

10. Validation
Validate dates, allocated quantity against stock, and required authorization/justification for overrides.

11. Empty State
`No promotional campaigns are available in this view.`

12. Loading State
Campaign summary and readiness rows show skeletons.

13. Error State
`Campaign readiness data could not be loaded.` Preserve campaign selection and unsaved notes.

14. Success State
`Readiness confirmed successfully.` or `Allocation updated successfully.`

15. Responsive Behavior
Desktop/tablet supported. Mobile should be summary-only.

16. Accessibility Notes
Readiness levels need text labels and icons. Gaps and overrides should be explained in plain language.

UI-047 Audit and Control Review Workspace

1. Screen Name
Audit and Control Review Workspace

2. Purpose
Allow auditors to define scope, inspect transactions and approvals, test controls, document findings, and export evidence.

3. Layout Structure
Scope setup area, sample/review workspace, transaction table, findings panel, and export actions.

4. Navigation
`Governance & Audit > Audit Workspace`, with contextual path to audit trail query and exports.

5. Components
Scope form, reviewed-transactions table, findings table, evidence panel, export action group, status strip.

6. Tables
Reviewed transactions table and findings table.

7. Forms
Scope dates, location, transaction type, reviewer, finding type, severity, note, and evidence reference.

8. Buttons
`Start Review`, `Record Finding`, `Export Evidence`, `Complete Review`.

9. Icons
Audit/search icon, evidence/attachment icon, finding/warning icon, export icon.

10. Validation
Date range required; finding fields required when recording a finding; preserve scope on export errors.

11. Empty State
Before scope run: `Select an audit scope to begin review.` After no results: `No records matched this audit scope.`

12. Loading State
Scope shell, transactions list, and findings area show placeholders.

13. Error State
`Audit review data could not be loaded.` Preserve scope filters and unsaved findings where possible.

14. Success State
`Finding recorded successfully.` and `Audit review completed successfully.`

15. Responsive Behavior
Desktop-first due to dataset density. Tablet summary review possible. Mobile should be unsupported or read-only.

16. Accessibility Notes
Audit evidence and findings need explicit structure. Tables and exports must be fully keyboard accessible.

UI-048 Audit Trail Query

1. Screen Name
Audit Trail Query

2. Purpose
Search immutable audit records and inspect before/after values by entity, actor, date, and action.

3. Layout Structure
Query filter panel at top or left, results table in the main region, and selected record before/after detail panel.

4. Navigation
`Governance & Audit > Audit Trail Query`, with drill-in to related records when allowed.

5. Components
Query form, results table, audit record viewer, export actions, metadata footer.

6. Tables
Audit trail results table.

7. Forms
Entity type, entity ID, user, date range, action type.

8. Buttons
`Run Query`, `Open Record`, `Export Results`.

9. Icons
Audit/log icon, compare/before-after icon, export icon, filter icon.

10. Validation
Date range required and may be limited for export size. Preserve query filters on failure.

11. Empty State
Before query: `Enter filters to search the audit trail.` No result: `No audit records match this query.`

12. Loading State
Filter shell stays visible while results skeletonize.

13. Error State
`Audit trail results could not be loaded.` Keep query filters and selected context.

14. Success State
`Query completed successfully.` and `Audit results exported successfully.` when relevant.

15. Responsive Behavior
Desktop-first investigative screen. Tablet summary review acceptable. Mobile should be read-only or unsupported.

16. Accessibility Notes
Before/after views must use explicit labels and a logical reading order. Exports and record-opening controls must be clearly named.

UI-049 User Account Management

1. Screen Name
User Account Management

2. Purpose
Provision, update, suspend, deactivate, and review user accounts and role assignments.

3. Layout Structure
User directory list, account detail panel, and role assignment section with status and audit metadata.

4. Navigation
`Administration > User Accounts`, with contextual links to access scope and role assignment details.

5. Components
Users table, detail panel, role assignment widget, status badges, confirmation dialogs, metadata strip.

6. Tables
Users table and role assignments table.

7. Forms
User ID, email, display name, initial roles, role code, location codes, effective dates, suspension reason, deactivation reason.

8. Buttons
`Create User`, `Save`, `Suspend`, `Deactivate`, `Sync Users`.

9. Icons
User/profile icon, role/shield icon, suspend/pause icon, deactivate/warning icon, sync icon.

10. Validation
Validate user ID pattern, email format, required display name, at least one role, effective dates, SoD conflicts, and action-specific reasons.

11. Empty State
`No user accounts are available in this view.` Offer `Create User` if allowed.

12. Loading State
Directory skeleton and account-detail placeholder.

13. Error State
`User accounts could not be loaded.` Preserve assignments and notes on save, suspend, or deactivate failure.

14. Success State
`User account saved successfully.` `User suspended successfully.` or `User deactivated successfully.`

15. Responsive Behavior
Desktop/tablet supported. Mobile should be limited to lookup and simple status actions if any.

16. Accessibility Notes
High-priority admin screen. Role assignments must be grouped accessibly, and confirmation dialogs for suspension or deactivation must manage focus correctly.

UI-050 Role and Access Control Administration

1. Screen Name
Role and Access Control Administration

2. Purpose
Configure role permissions, location scope rules, and access enforcement behavior.

3. Layout Structure
Role list on the left, permission matrix/detail panel in the main region, location scope rules section, and conflict warning area.

4. Navigation
`Administration > Roles & Access Control`, with links to user accounts and SoD evaluation context.

5. Components
Roles table, permission matrix, scope selector, conflict warning panel, save/reset actions, validation banner.

6. Tables
Roles table and permissions matrix table.

7. Forms
Role name or selected role context, permission group assignments, location scope rules, initiator/approver fields for SoD evaluation, transaction type.

8. Buttons
`Edit Role`, `Save Permissions`, `Validate SoD`, `Reset Changes`.

9. Icons
Shield/permissions icon, matrix/grid icon, warning/conflict icon, validation/check icon.

10. Validation
At least one permission required, SoD rule checks, transaction type validation for evaluation, role assignment authorization rules.

11. Empty State
`No role configuration is available in this view.`

12. Loading State
Role list and permission matrix show skeletons.

13. Error State
`Role and access settings could not be loaded.` Preserve matrix edits where possible.

14. Success State
`Permissions updated successfully.` and `Segregation-of-duties validation completed.`

15. Responsive Behavior
Desktop-only or desktop-first. Tablet should be read-only at most. Mobile not recommended due to matrix density.

16. Accessibility Notes
High-priority accessibility screen. Permission matrix requires explicit row and column headers, keyboard traversal model, and non-color-only conflict indicators.

Device Support Matrix

This matrix is included to make the responsive intent explicit for AI-generated design concepts.

Screen Group	Desktop	Tablet	Mobile
Authentication	Full	Full	Full
Dashboards and notifications	Full	Full	Summary-friendly
Master data management	Full	Limited editing	Mostly read-only
Procurement planning and PO editing	Full	Moderate	Limited
Warehouse workbenches	Full	Full for selected tasks	Selective only
Inventory monitoring and lookup	Full	Full	Lookup-friendly
Analytics and finance	Full	Summary review	Read-only or unsupported
Audit and admin configuration	Full	Limited review	Unsupported

AI Design Output Guidance

When using this specification with Figma AI or similar tools:

- Generate desktop-first layouts at `1440px` width.
- Also generate a tablet adaptation at `1024px`.
- Only generate mobile versions for:
  - `UI-001`
  - `UI-002`
  - `UI-004`
  - `UI-021`
  - `UI-034`
  - `UI-036`
  - `UI-039` simplified
  - `UI-042`
- Use a consistent design system with enterprise data tables, dense but readable forms, clear badges, and accessible status cues.
- Favor practical enterprise UX over marketing-style layouts.
- Keep interaction patterns consistent across queue screens, list-detail screens, and analytics screens.

Final Design Intent Summary

The IMS UI should feel:

- operational
- structured
- trustworthy
- fast to scan
- keyboard efficient
- role-aware
- audit-friendly
- accessible

It should prioritize clarity, speed, and controlled transactional accuracy over decorative complexity.

Inventory Management System Design System

Purpose

This document defines the design system for the Inventory Management System (IMS). It is intended to support product design, UI design, frontend implementation, QA review, and AI-assisted design generation.

The system should reflect a modern enterprise dashboard style similar to ERP platforms: structured, dense but readable, highly scannable, role-aware, operationally efficient, and accessible.

Design Principles

- Clarity first: prioritize operational readability over decoration.
- Enterprise density: support complex workflows without visual clutter.
- Consistency at scale: shared patterns across inventory, procurement, warehouse, fulfillment, audit, and admin.
- Action hierarchy: always make the next action obvious.
- Accessible by default: target `WCAG 2.2 AA`.
- Desktop-first, responsive second: optimize heavy workflows for `lg` and `xl`, simplify intelligently on smaller screens.

Foundations

Colors

Color Strategy

Use a restrained enterprise palette built around cool neutrals, clear semantic states, and strong contrast. Most screens should feel calm and structured, with status colors used sparingly for alerts, approvals, risks, and workflow states.

Core Brand and Interface Palette

Token	Hex	Usage
`color-brand-900`	`#0F2A43`	strongest brand surfaces, sidebar dark areas
`color-brand-800`	`#163A5F`	primary navigation active areas
`color-brand-700`	`#1D4E89`	primary buttons, links, selected states
`color-brand-600`	`#2563A6`	hover states, active accents
`color-brand-500`	`#2F7CC1`	secondary brand emphasis
`color-brand-100`	`#DCEBFA`	selected row background, light info surfaces

Token	Hex	Usage
color-brand-050	#F4F8FD	app background accents

Neutral Palette

Token	Hex	Usage
color-neutral-950	#0B1220	highest emphasis text
color-neutral-900	#111827	headings, strong text
color-neutral-800	#1F2937	primary body text
color-neutral-700	#374151	secondary text
color-neutral-600	#4B5563	metadata, helper text
color-neutral-500	#6B7280	muted labels, icons
color-neutral-400	#9CA3AF	disabled text, placeholders
color-neutral-300	#D1D5DB	borders, dividers
color-neutral-200	#E5E7EB	light borders, table separators
color-neutral-100	#F3F4F6	card backgrounds, hover fills
color-neutral-050	#F9FAFB	app canvas
color-white	#FFFFFF	main surface

Semantic Palette

Token	Hex	Usage
color-success-700	#15803D	success text, strong success states
color-success-100	#DCFCE7	success backgrounds
color-warning-700	#B45309	warnings, stock risk text
color-warning-100	#FEF3C7	warning backgrounds
color-danger-700	#B91C1C	destructive, errors, rejections
color-danger-100	#FEE2E2	error backgrounds
color-info-700	#1D4ED8	info messages, help states
color-info-100	#DBEAFE	info backgrounds
color-purple-700	#6D28D9	audit/governance special accents
color-purple-100	#EDE9FE	audit tinted backgrounds

Operational Status Colors

State	Color	Notes
In Stock / Available	success	always pair with text label
Low Stock / At Risk	warning	use icon + badge text
Out of Stock / Blocked	danger	use icon + explicit label
In Transit / Pending	info	neutral-progress state
Quarantined / Hold	purple	use lock/quarantine icon
Draft / In Review	neutral	low emphasis but visible
Approved	success	for approvals and certifications

State	Color	Notes
Rejected / Failed	danger	for rejected decisions and blocking failures

Surface Model

Surface Token	Hex	Usage
`surface-app`	`#F9FAFB`	overall application background
`surface-primary`	`#FFFFFF`	cards, forms, tables
`surface-secondary`	`#F3F4F6`	filter bars, subpanels
`surface-tertiary`	`#E5E7EB`	disabled areas, subtle contrast sections
`surface-sidebar`	`#0F2A43`	primary sidebar background
`surface-header`	`#FFFFFF`	top header
`surface-overlay`	`rgba(17, 24, 39, 0.48)`	modal backdrop

Color Usage Rules

- Never use color as the only indicator of state.
- Text on primary surfaces should meet a minimum `4.5:1` contrast ratio.
- Use semantic backgrounds in banners, chips, badges, and alert cards, not full-page saturation.
- Dark sidebar + light content is the default enterprise shell.

Typography

Typeface Recommendation

Use a neutral, modern UI sans-serif such as:

- `Inter`
- fallback: `Segoe UI`, `Roboto`, `Arial`, `sans-serif`

Typography Scale

Token	Size	Weight	Line Height	Usage
`type-display-lg`	32px	700	40px	executive dashboard title, rare
`type-heading-1`	28px	700	36px	top-level page headings
`type-heading-2`	24px	700	32px	major workspace headings
`type-heading-3`	20px	600	28px	section headings
`type-heading-4`	18px	600	26px	card titles, panel titles
`type-body-lg`	16px	400	24px	primary body text
`type-body-md`	14px	400	20px	forms, tables, standard UI text
`type-body-sm`	13px	400	18px	helper text, metadata
`type-label-md`	14px	500	20px	form labels, nav labels
`type-label-sm`	12px	600	16px	badge labels, overlines
`type-mono-sm`	13px	500	18px	IDs, SKU codes, transaction refs

Typography Rules

- Use `14px` as the default interface text size.

- Use `13px` for dense tables only when readability remains strong.
- Use monospaced styling selectively for record IDs, SKUs, PO numbers, transfer IDs, and audit references.
- Avoid all-caps for large content blocks; reserve uppercase styling for small labels and metadata only.

Spacing

Base Spacing Unit

Use a `4px` base unit.

Spacing Scale

Token	Value	Typical Usage
`space-1`	4px	icon gap, small offsets
`space-2`	8px	inline gaps, chip spacing
`space-3`	12px	compact form spacing
`space-4`	16px	default control spacing
`space-5`	20px	section gutters
`space-6`	24px	card padding, block spacing
`space-8`	32px	page sections
`space-10`	40px	large dashboard separation
`space-12`	48px	major layout spacing

Spacing Rules

- Page outer padding on desktop: `24px`.
- Card padding: `16px` or `24px` depending on density.
- Form field vertical spacing: `16px`.
- Label-to-input spacing: `6px` to `8px`.
- Table cell horizontal padding: `12px` to `16px`.

Grid

Breakpoints

Breakpoint	Width
`xs`	`0-479px`
`sm`	`480-767px`
`md`	`768-1023px`
`lg`	`1024-1439px`
`xl`	`1440px+`

Layout Grid

Breakpoint	Columns	Margin	Gutter
`xs`	4	16px	16px
`sm`	4	16px	16px
`md`	8	20px	20px

Breakpoint	Columns	Margin	Gutter
`lg`	12	24px	24px
`xl`	12	32px	24px

Enterprise Shell Layout

- Sidebar width expanded: `248px`
- Sidebar width collapsed: `72px`
- Header height: `64px`
- Subheader or page title bar height: `56px`
- Content max width: fluid
- Default content column behavior:
  - dashboards: multi-column responsive grid
  - list-detail: 35/65 or 40/60 split
  - forms: two columns on desktop, one on small screens

Components

Buttons

Button Variants

1. `Primary`
2. `Secondary`
3. `Tertiary`
4. `Destructive`
5. `Ghost`
6. `Icon Only`

Primary Button

- Background: `color-brand-700`
- Text: white
- Border: none
- Radius: `8px`
- Height: `40px`
- Horizontal padding: `16px`
- Hover: `color-brand-600`
- Active: `color-brand-800`

Secondary Button

- Background: white
- Text: `color-brand-700`
- Border: `1px solid color-neutral-300`
- Hover background: `color-brand-050`

Tertiary Button

- Background: transparent
- Text: `color-neutral-800`
- Border: transparent
- Hover background: `color-neutral-100`

Destructive Button

- Background: `color-danger-700`
- Text: white
- Hover: slightly darker destructive tone

Sizes

Size	Height	Text
`sm`	32px	13px
`md`	40px	14px
`lg`	48px	16px

Button Rules

- One primary button per action cluster.
- In dense tables, use tertiary or icon actions first.
- Use destructive only for irreversible actions.
- Loading buttons show spinner and retain width.

Inputs

Input Types

- text input
- searchable select
- dropdown
- textarea
- number input
- date/date-time input
- checkbox
- radio group
- switch
- file upload

Text Input Specification

- Height: `40px`
- Padding: `0 12px`
- Border: `1px solid color-neutral-300`
- Radius: `8px`
- Background: white
- Text: `color-neutral-900`
- Placeholder: `color-neutral-400`

States

- Default
- Hover
- Focus
- Filled
- Disabled
- Read-only
- Error
- Success

Focus Style

- Border color: `color-brand-700`
- Outer ring: `2px` or `3px` translucent brand ring
- Must remain visible on white and tinted surfaces

Error Style

- Border color: `color-danger-700`
- Helper text: `color-danger-700`
- Optional error icon inside field for complex forms

Textarea

- Min height: `88px`
- Resize vertically only
- Use for notes, rationale, and comments

Selects and Searchable Inputs

- Same height and radius as text input
- Searchable select is the default for product, location, supplier, and user lookup
- Use grouped options where domain structure matters

Form Layout Rules

- Label above input
- Helper text below input
- Error message replaces or sits below helper text
- Group related fields into titled sections

Tables

Table Design Intent

Tables are the backbone of the IMS and should feel highly scannable, compact, and action-oriented without becoming visually noisy.

Table Specification

- Header height: `44px`
- Row height:
  - compact: `40px`
  - default: `48px`
  - expanded: `56px`
- Cell padding: `12px 16px`
- Header text: `12px` to `13px`, semibold
- Body text: `13px` to `14px`
- Border color: `color-neutral-200`
- Zebra striping: optional and subtle only

Table Features

- sortable columns
- sticky header for long tables
- row selection checkboxes
- row action menu
- inline status badges
- pagination

- empty state row/panel
- skeleton loading rows

Row States

- default
- hover
- selected
- focused
- warning-highlighted
- disabled

Table Rules

- Keep key identifiers pinned left where practical.
- Use badges for status columns.
- Numeric values should align consistently, usually right-aligned.
- Avoid more than 8 to 10 visible columns before introducing prioritization or horizontal scroll.

Cards

Card Types

1. KPI card
2. summary card
3. detail card
4. recommendation card
5. alert card
6. empty state card

Card Specification

- Background: white
- Border: `1px solid color-neutral-200`
- Radius: `12px`
- Padding: `16px` or `24px`
- Shadow: minimal, soft, enterprise-light

KPI Card Structure

- label
- primary value
- trend indicator
- optional subtitle
- optional target delta

Recommendation Card Structure

- title
- short context sentence
- impact summary
- recommended action
- CTA

Dialogs

Dialog Types

1. confirmation dialog
2. destructive confirmation dialog
3. form dialog
4. review dialog
5. system alert dialog

Dialog Specification

- Width:
  - small: `400px`
  - medium: `560px`
  - large: `720px`
- Radius: `16px`
- Padding: `24px`
- Overlay: dark translucent scrim

Dialog Structure

- title
- explanatory body
- optional highlighted warning section
- primary and secondary actions
- close icon top-right when appropriate

Dialog Rules

- Use dialogs for confirmations and small supporting forms.
- Use full-screen pages for high-risk, multi-step, or line-item-heavy flows.
- Destructive dialogs must name the object and consequence clearly.

Icons

Icon Style

- Outline-first enterprise icon set
- Consistent stroke width
- Simple, geometric, legible at small sizes

Recommended Sizes

Size	Use
`16px`	inline metadata, compact actions
`18px`	standard control icons
`20px`	header, buttons, cards
`24px`	section or feature icons

Usage Rules

- Pair status icons with text labels.
- Use icon-only buttons only when the label is obvious and accessible.
- In ERP-like dashboards, icons should support scanning, not dominate the interface.

Core Icon Categories

- navigation
- inventory and warehouse
- procurement and supplier
- transfers and logistics
- returns
- reports and analytics
- governance and audit
- administration
- alerts and statuses

Alerts

Alert Types

- success
- info
- warning
- error

Alert Specification

- Radius: `10px`
- Padding: `12px 16px`
- Border left or full border for emphasis
- Optional icon leading
- Optional action link/button trailing

Example Styles

- Success: green tint background, dark green text
- Info: blue tint background, dark blue text
- Warning: amber tint background, dark amber text
- Error: red tint background, dark red text

Usage Rules

- Use page-level alerts for save, submit, conflict, permission, and system errors.
- Use inline alerts inside cards or panels for contextual warnings.
- Keep message copy concise and action-oriented.

Badges

Badge Purpose

Badges communicate short, high-signal status labels.

Badge Specification

- Height: `22px` to `24px`
- Padding: `0 8px`
- Radius: pill or `999px`
- Text: `12px`, semibold

Badge Types

- stock status
- workflow status
- approval state
- inventory condition
- audit severity

Example Badge Mapping

- `Available` -> success
- `Low Stock` -> warning
- `Out of Stock` -> danger
- `Draft` -> neutral
- `Pending Approval` -> info
- `Quarantine` -> purple

Chips

Chip Purpose

Chips represent filters, tags, lightweight states, and selectable attributes.

Chip Variants

- static chip
- selectable chip
- removable chip
- filter chip

Chip Specification

- Height: `28px` to `32px`
- Radius: `999px`
- Horizontal padding: `10px` to `12px`
- Background:
  - neutral by default
  - tinted for active/filter state

Usage Rules

- Use chips for active filters and quick categorization.
- Do not replace badges with chips where the purpose is status display only.

Navigation

Primary Navigation

The main navigation should be module-based and persist across authenticated screens.

Modules:

1. `Home`
2. `Inventory`
3. `Products`
4. `Locations`
5. `Procurement`
6. `Inbound`

7. `Fulfillment`
8. `Transfers`
9. `Returns`
10. `Reports & Analytics`
11. `Governance & Audit`
12. `Administration`

Navigation Style

- Vertical left sidebar by default
- Dark background, light text
- Current module highlighted with stronger surface and left accent bar
- Nested items shown in the secondary navigation region or local sidebar

Navigation Behavior

- Expanded by default on desktop
- Collapsible to icon-only mode
- Drawer on tablet/mobile
- Role-based visibility: show only relevant modules

Sidebar

Sidebar Specification

- Width expanded: `248px`
- Width collapsed: `72px`
- Background: `surface-sidebar`
- Text color: white or near-white with muted variants for secondary items

Sidebar Sections

- product / system brand area
- main nav links
- optional pinned shortcuts
- bottom area for collapse toggle

Sidebar Item States

- default
- hover
- active
- keyboard focus
- collapsed tooltip state

Sidebar Rules

- Keep icons aligned and consistent.
- Show item labels clearly in expanded state.
- Use concise enterprise labels.

Header

Header Specification

- Height: `64px`
- Background: white

- Border bottom: `1px solid color-neutral-200`
- Horizontal padding: `16px` to `24px`

Header Content

- page or module context
- global search
- notifications
- tasks / approvals shortcut
- session context badge
- user menu

Header Variants

- unauthenticated header
- standard operational header
- executive header
- admin header

Header Rules

- Search should remain visible on desktop.
- Utility controls should remain grouped on the right.
- Use concise indicators for unread counts and role/location context.

Footer

Footer Role

The footer is lightweight and informational, not promotional.

Footer Types

- global app footer
- page metadata strip
- record metadata strip
- report footer

Footer Content

- last updated timestamp
- data as-of timestamp
- record created/updated metadata
- version or environment when needed

Footer Style

- Small text
- light divider
- low visual weight

Pagination

Pagination Purpose

Pagination supports large enterprise tables and reports without overwhelming users.

Pagination Specification

- Position: bottom-right of tables by default
- Include:
  - previous/next
  - page numbers
  - page size selector
  - total items summary

Recommended Page Sizes

- `10`
- `25`
- `50`
- `100`

Pagination Rules

- Default to `25` for standard list screens.
- Default to `50` for operational queues where scanning speed matters.
- Keep pagination aligned with table width and visible after filtering.

Layout Patterns

Dashboard Pattern

Use for executive and operational summaries.

Structure:

- page title and filter bar
- KPI cards row
- summary widgets
- task or alert table
- quick action area

List-Detail Pattern

Use for products, suppliers, users, locations, and many admin screens.

Structure:

- search/filter bar
- results table or list
- detail panel or detail page region
- metadata footer

Queue Workbench Pattern

Use for receiving, put-away, counting, fulfillment, approvals, and returns.

Structure:

- task queue
- current task detail
- action panel
- status summary

Report Viewer Pattern

Use for analytics, movement history, valuation, and audit.

Structure:

- filter bar
- result area
- detail/drill-down
- export actions

Motion and Interaction

Motion Principles

- Minimal and purposeful
- Fast response for operational workflows
- No decorative animation in task-heavy screens

Recommended Motion

- 120ms to 180ms hover/focus transitions
- 180ms to 220ms panel open/close
- subtle fade/slide for banners and dialogs

Avoid

- long easing animations
- bouncing or playful effects
- transitions that slow repeated data entry

Elevation, Borders, and Radius

Elevation

Token	Use
`elevation-0`	flat app background
`elevation-1`	cards, panels
`elevation-2`	dropdowns, menus
`elevation-3`	dialogs

Use shadow sparingly. Borders should carry most structural separation.

Border Radius

Token	Value	Use
`radius-sm`	6px	tags, compact controls
`radius-md`	8px	inputs, buttons
`radius-lg`	12px	cards
`radius-xl`	16px	dialogs, large panels
`radius-pill`	999px	badges, chips

Accessibility Standards

Core Standards

- Target `WCAG 2.2 AA`
- All critical flows keyboard operable
- Visible focus on all controls
- Clear heading hierarchy
- Error messages programmatically associated with fields
- Tables use true table semantics
- Dialogs trap focus correctly
- Statuses never depend on color alone

High-Priority Accessible Components

- queue workbenches
- data tables
- editable line-item grids
- approval panels
- permission matrix
- global navigation

Enterprise Dashboard Style Notes

To keep the IMS visually aligned with modern ERP systems:

- Use generous structure, not excessive decoration.
- Keep surfaces mostly neutral with brand color used as an accent.
- Favor bordered cards and clean panel separation over strong shadows.
- Use clear iconography and visible labels for operational scanning.
- Maintain a compact but readable density model.
- Prioritize data clarity, state clarity, and action clarity over visual novelty.

Recommended Design Tokens Summary

Core Tokens

- colors: brand, neutral, semantic, surface
- typography: headings, body, label, mono
- spacing: 4px-based scale
- radius: 6 / 8 / 12 / 16 / pill
- layout: 12-column desktop grid
- shell: 248px sidebar, 64px header

Default Component Sizes

- button height: `40px`
- input height: `40px`
- badge height: `24px`
- chip height: `28px` to `32px`
- table row height: `48px`
- card padding: `16px` or `24px`

Usage Guidance for AI Design Tools

When using this design system in Figma AI or similar tools:

- Generate a desktop-first ERP dashboard design.
- Use a dark left sidebar and light content area.
- Use white cards on a soft neutral app background.
- Use blue as the main action color.
- Use green, amber, red, and purple for semantic enterprise states.
- Use Inter or a similar neutral enterprise sans-serif.
- Favor bordered cards, compact tables, structured forms, and high-scannability layouts.
- Avoid consumer-app aesthetics such as oversized hero sections, playful gradients, oversized rounded blobs, or excessive illustration.

Final Design Intent

The IMS design system should feel like a modern, dependable enterprise operating environment:

- precise
- calm
- efficient
- informative
- role-aware
- scalable
- accessible

It should support fast daily execution for warehouse and operations users while still feeling robust enough for finance, audit, and administrative control workflows.

Part X — Frontend Architecture

Inventory Management System Frontend Architecture Mapping

Purpose

This document maps the approved UI design, API design, domain model, and system architecture into a clean Angular frontend architecture. It focuses on feature decomposition, module boundaries, routing, services, DTO usage, and state ownership. It intentionally does not generate Angular code.

Frontend Architecture Principles

- Organize the Angular app by business capability, not by technical artifact type.
- Mirror the approved bounded contexts and system modules in the frontend feature structure.
- Keep route-level pages thin; move orchestration into facades and application services.
- Prefer standalone routed features or lazy-loaded feature areas with clear public entry points.
- Separate `core`, `shared`, and `features` concerns:
  - `core`: app shell, auth/session, guards, interceptors, global layout, permission evaluation.
  - `shared`: reusable UI components, pipes, directives, presentational widgets, common DTO mappers.
  - `features`: business workflows aligned to modules such as inventory, procurement, inbound, fulfillment, and administration.
- Use a facade-first UI architecture:
  - components handle rendering and user interaction
  - facades coordinate page state and commands
  - API services encapsulate HTTP contracts

    - mappers adapt DTOs into frontend view models
- Keep state ownership explicit:
    - global state only for session, navigation context, notification counts, and app-wide preferences
    - feature state for work queues, filters, selected records, wizard progress, and long-running workflows
    - transient form state stays local to the page or child form components

Recommended State Management Strategy

- Use Angular Signals for simple shared UI state and session context.
- Use `ComponentStore` or a feature-scoped signal store for page/workbench state.
- Reserve NgRx-style global state only for truly cross-application concerns:
    - authenticated session and user context
    - permission catalog and location scope
    - notification/task counts
    - shell navigation state
- Treat server data as query-driven state behind facades with caching per feature.
- Avoid a single monolithic global store for all CRUD screens.

Recommended Angular Project Structure

```
src/app/
  core/
    application/
      auth/
      authorization/
      session/
      navigation/
      notifications/
    guards/
    http/
    layout/
    shell/
    config/
  shared/
    ui/
      layout/
      tables/
      filters/
      forms/
      feedback/
      workflow/
      audit/
    directives/
    pipes/
    utils/
    models/
      common/
      view-models/
  features/
    home/
    products/
      catalog/
      import/
      abc-classification/
      stock-policies/
    locations/
    governance/
    procurement/
      suppliers/
      requisitions/
      purchase-orders/
      delivery-planning/
      supply-tracking/
      low-stock-alerts/
      replenishment/
    inbound/
      goods-receipts/
      discrepancies/
      put-away/
      ad-hoc-receipts/
    inventory/
      status-monitor/
      availability/
      movement-history/
      cycle-counts/
      physical-counts/
      variances/
      adjustments/
      write-offs/
      quarantine/
      shrinkage/
    fulfillment/
      reservations/
      pick-pack-ship/
      partial-fulfillment/
      exceptions/
      order-status/
      split-fulfillment/
```

```
    transfers/
      requests/
      approvals/
      execution/
      rebalancing/
    returns/
      authorizations/
      inspection/
    reports/
      operational/
      executive-kpis/
      valuation/
      promotional-readiness/
      audit/
    administration/
      users/
      roles-access/
  app.routes.ts
```

Cross-Cutting Shared Components

Use these shared components consistently across the application:

- App shell and navigation: `GlobalHeader`, `MainNavigationBar`, `SecondaryNavigationPanel`, `BreadcrumbTrail`, `PageTitleBar`, `SessionContextBadge`
- Querying and list management: `FilterBar`, `DateRangePicker`, `SavedViewSelector`, `DataTable`, `RowActionMenu`, `DetailPanel`, `QueueList`
- Workflow and feedback: `StepProgressIndicator`, `ApprovalDecisionPanel`, `StatusTimeline`, `ConfirmationDialog`, `SuccessWarningErrorBanner`, `EmptyStatePanel`, `LoadingSkeleton`
- Forms and auditability: `StandardFormSection`, `LineItemEditor`, `ValidationMessageBlock`, `RecordMetadataBlock`, `EvidenceAttachmentPanel`, `AuditRecordViewer`, `PermissionMatrix`

Feature Module Map

- `home`
- `notifications`
- `products`
- `locations`
- `governance`
- `procurement`
- `inbound`
- `inventory`
- `fulfillment`
- `transfers`
- `returns`
- `reports`
- `administration`

Screen-to-Architecture Mapping

UI-001 Sign In / Session Entry

- Angular feature module: `core/auth`
- Main page component: `SignInPageComponent`
- Child components: `SignInFormComponent`, `SessionContextPanelComponent`
- Shared components: `GlobalHeader`, `ValidationMessageBlock`, `SuccessWarningErrorBanner`
- Required Angular services: `AuthFacade`, `SessionService`, `SessionContextService`
- Required DTOs: `SessionResponse`, `LogoutCommandResultResponse`
- Related API endpoints: `GET /api/v1/auth/session`, `POST /api/v1/auth/logout`
- Required permissions: authenticated session bootstrap only
- Application routes: `/auth/sign-in`

- State management responsibility: local form state plus global session state in `core/session`

UI-002 Operational Home Dashboard

- Angular feature module: `home`
- Main page component: `OperationalDashboardPageComponent`
- Child components: `DashboardKpiGridComponent`, `StockHealthBoardComponent`, `AlertSummaryWidgetComponent`, `QuickActionsPanelComponent`, `MyTasksWidgetComponent`
- Shared components: `GlobalHeader`, `MainNavigationBar`, `KpiSummaryCard`, `SummaryStatisticStrip`, `QuickActionTile`, `TaskApprovalShortcutWidget`, `FilterBar`
- Required Angular services: `DashboardFacade`, `DashboardApiService`, `AlertApiService`, `TaskInboxService`
- Required DTOs: dashboard and stock-status summary responses from APIs `150`, `151`, `152`, `158`
- Related API endpoints: `GET /api/v1/intelligence/dashboard`, `GET /api/v1/intelligence/stock-status/summary`, `GET /api/v1/intelligence/stock-status/board`, `GET /api/v1/intelligence/alerts`
- Required permissions: `PERM-083`, `PERM-085`, `PERM-064`
- Application routes: `/home/operations`
- State management responsibility: feature dashboard state for filters, selected widgets, and refresh cadence

UI-003 Executive KPI Dashboard

- Angular feature module: `reports`
- Main page component: `ExecutiveKpiDashboardPageComponent`
- Child components: `ExecutiveKpiHeaderComponent`, `KpiCardGridComponent`, `TrendAnalysisPanelComponent`, `KpiDrilldownPanelComponent`
- Shared components: `KpiSummaryCard`, `TrendIndicator`, `DateRangePicker`, `ExportActionGroup`, `DrillDownLinkGroup`
- Required Angular services: `ExecutiveKpiFacade`, `ExecutiveKpiApiService`
- Required DTOs: KPI dashboard query DTOs and `KpiReportResponse`
- Related API endpoints: `GET /api/v1/intelligence/kpis`
- Required permissions: `PERM-088`
- Application routes: `/reports/executive-kpis`
- State management responsibility: feature state for date range, KPI comparisons, and drill-down selection

UI-004 Notification Center

- Angular feature module: `notifications`
- Main page component: `NotificationCenterPageComponent`
- Child components: `NotificationQueueComponent`, `AlertDetailDrawerComponent`, `NotificationFiltersComponent`, `PendingApprovalsSummaryComponent`
- Shared components: `NotificationIconInboxTrigger`, `QueueList`, `AlertDetailCard`, `BulkSelectionToolbar`, `SavedViewSelector`
- Required Angular services: `NotificationFacade`, `AlertApiService`, `TransferApprovalApiService`, `AdjustmentApprovalApiService`
- Required DTOs: alert list response, `CommandResultResponse`, pending approvals list responses
- Related API endpoints: `GET /api/v1/intelligence/alerts`, `POST /api/v1/intelligence/alerts/{alertId}/acknowledge`, `GET /api/v1/transfers/pending-approvals`, `GET /api/v1/accuracy/pending-approvals`, `GET /api/v1/intelligence/procurement/inbound-delays`, `GET /api/v1/intelligence/campaigns/readiness`
- Required permissions: `PERM-064`, `PERM-065`, plus queue permissions `PERM-072`, `PERM-044`, `PERM-046`, `PERM-036`, `PERM-092`
- Application routes: `/notifications`
- State management responsibility: global unread count in shell; feature queue/filter state in module store

UI-005 Product Catalog Management

- Angular feature module: `products`
- Main page component: `ProductCatalogPageComponent`
- Child components: `ProductSearchPanelComponent`, `ProductListComponent`, `ProductDetailPanelComponent`, `ProductEditorComponent`, `ProductLifecycleActionsComponent`
- Shared components: `GlobalSearchBar`, `DataTable`, `DetailPanel`, `StandardFormSection`, `StatusBadge`, `RecordMetadataBlock`
- Required Angular services: `ProductCatalogFacade`, `ProductApiService`, `ProductFormService`
- Required DTOs: `SearchProductsQuery`, `GetProductBySkuRequest`, `CreateProductRequest`, `UpdateProductAttributesRequest`, `ChangeProductLifecycleStatusRequest`, `ProductSummaryResponse`, `ProductDetailResponse`, `ProductStatusChangeResponse`

- Related API endpoints: `POST /api/v1/products`, `GET /api/v1/products`, `GET /api/v1/products/{sku}`, `PATCH /api/v1/products/{sku}`, `POST /api/v1/products/{sku}/lifecycle-status`, `GET /api/v1/products/categories/{categoryCode}/products`, `POST /api/v1/products/duplicate-resolutions`
- Required permissions: `PERM-001`, `PERM-002`, `PERM-003`, `PERM-004`, `PERM-005`
- Application routes: `/products/catalog`, `/products/catalog/new`, `/products/catalog/:sku`
- State management responsibility: feature state for search filters, selected SKU, and edit/view mode; form state remains local

UI-006 Product Import Workspace

- Angular feature module: `products`
- Main page component: `ProductImportWorkspacePageComponent`
- Child components: `ImportUploadStepComponent`, `ImportValidationResultsComponent`, `ImportReviewStepComponent`, `ImportCompletionStepComponent`
- Shared components: `StepProgressIndicator`, `FileUploadControl`, `ReadOnlyReviewForm`, `ValidationMessageBlock`, `SuccessWarningErrorBanner`
- Required Angular services: `ProductImportFacade`, `ProductImportApiService`, `OpeningBalanceApiService`, `LocationActivationApiService`
- Required DTOs: `BulkProductImportRequest`, opening-balance request DTOs, import result responses
- Related API endpoints: `POST /api/v1/products/import`, `POST /api/v1/inventory/opening-balances`, `POST /api/v1/locations/{locationCode}/activate`
- Required permissions: `PERM-015`, `PERM-010`, `PERM-007`
- Application routes: `/products/import`
- State management responsibility: workflow store for file, validation issues, approved rows, and completion outcome

UI-007 Location and Warehouse Management

- Angular feature module: `locations`
- Main page component: `LocationManagementPageComponent`
- Child components: `LocationDirectoryComponent`, `LocationDetailComponent`, `LocationHierarchyEditorComponent`, `StoragePositionManagerComponent`, `LocationStatusActionsComponent`
- Shared components: `DataTable`, `DetailPanel`, `StandardFormSection`, `ConfirmationDialog`, `ToggleSwitchControl`, `RecordMetadataBlock`
- Required Angular services: `LocationFacade`, `LocationApiService`, `StoragePositionApiService`
- Required DTOs: location create/update requests, storage-position requests, `LocationSummaryResponse`, location detail responses
- Related API endpoints: `POST /api/v1/locations`, `GET /api/v1/locations`, `GET /api/v1/locations/{locationCode}`, `PATCH /api/v1/locations/{locationCode}/hierarchy`, `POST /api/v1/locations/{locationCode}/storage-positions`, `GET /api/v1/locations/{locationCode}/storage-positions`, `POST /api/v1/locations/{locationCode}/activate`, `POST /api/v1/locations/{locationCode}/deactivate`, `POST /api/v1/locations/{locationCode}/close`, `GET /api/v1/locations/{locationCode}/operational-status`
- Required permissions: `PERM-006`, `PERM-007`, `PERM-008`, `PERM-009`, `PERM-004`
- Application routes: `/locations`, `/locations/new`, `/locations/:locationCode`
- State management responsibility: feature state for selected location, hierarchy mode, storage-position tab, and status transition dialogs

UI-008 Stock Policy and Threshold Management

- Angular feature module: `products`
- Main page component: `StockPolicyManagementPageComponent`
- Child components: `PolicySearchFormComponent`, `PolicyDetailComponent`, `ThresholdEditorComponent`
- Shared components: `FilterBar`, `NumberInput`, `DateField`, `ValidationMessageBlock`, `ReadOnlyReviewForm`
- Required Angular services: `StockPolicyFacade`, `StockPolicyApiService`
- Required DTOs: stock-policy create/get/update DTOs and stock-policy response DTOs
- Related API endpoints: `POST /api/v1/inventory/stock-policies`, `GET /api/v1/inventory/stock-policies/{locationCode}/{sku}`, `PATCH /api/v1/inventory/stock-policies/{locationCode}/{sku}`
- Required permissions: `PERM-011`
- Application routes: `/products/stock-policies`, `/products/stock-policies/:locationCode/:sku`
- State management responsibility: local form state plus feature filter state

UI-009 ABC Classification Management

- Angular feature module: `products`

- Main page component: `AbcClassificationPageComponent`
- Child components: `ClassificationReviewGridComponent`, `ClassificationEditorComponent`, `BulkClassificationToolbarComponent`
- Shared components: `DataTable`, `BulkSelectionToolbar`, `StatusBadge`, `SuccessWarningErrorBanner`
- Required Angular services: `AbcClassificationFacade`, `ProductApiService`
- Required DTOs: `SearchProductsQuery`, `AssignAbcClassificationRequest`, `PagedResponse<ProductSummaryResponse>`
- Related API endpoints: `PUT /api/v1/products/{sku}/abc-classification`, `GET /api/v1/products`
- Required permissions: `PERM-012`, `PERM-004`
- Application routes: `/products/abc-classification`
- State management responsibility: feature table selection and batch-edit state

UI-010 Governance Policies and Reason Codes

- Angular feature module: `governance`
- Main page component: `GovernancePoliciesPageComponent`
- Child components: `PolicyVersionListComponent`, `PolicyEditorComponent`, `ReasonCodeListComponent`, `ReasonCodeEditorComponent`, `ThresholdsTabComponent`, `TolerancesTabComponent`
- Shared components: `TabsSegmentedContentSwitcher`, `StandardFormSection`, `TextAreaNotesField`, `ToggleSwitchControl`, `ReadOnlyReviewForm`
- Required Angular services: `GovernanceFacade`, `GovernancePolicyApiService`, `ReasonCodeApiService`
- Required DTOs: governance policy request/response DTOs, reason-code request/response DTOs, threshold and tolerance response DTOs
- Related API endpoints: `POST /api/v1/governance/reason-codes`, `GET /api/v1/governance/reason-codes`, `POST /api/v1/governance/reason-codes/{code}/retire`, `GET /api/v1/governance/reason-codes/{code}/validate`, `POST /api/v1/governance/policies`, `PATCH /api/v1/governance/policies/{policyId}/approval-thresholds`, `PATCH /api/v1/governance/policies/{policyId}/receipt-tolerances`, `GET /api/v1/governance/approval-thresholds`, `GET /api/v1/governance/receipt-tolerances`
- Required permissions: `PERM-013`, `PERM-014`, `PERM-032`
- Application routes: `/governance/policies`, `/governance/reason-codes`
- State management responsibility: feature state for active tab, draft policy version, and selected reason code

UI-011 Supplier Management

- Angular feature module: `procurement`
- Main page component: `SupplierManagementPageComponent`
- Child components: `SupplierListComponent`, `SupplierDetailComponent`, `SupplierEditorComponent`
- Shared components: `DataTable`, `DetailPanel`, `StandardFormSection`, `StatusBadge`
- Required Angular services: `SupplierFacade`, `SupplierApiService`
- Required DTOs: supplier create/get/update DTOs and supplier summary/detail responses
- Related API endpoints: `POST /api/v1/suppliers`, `GET /api/v1/suppliers`, `GET /api/v1/suppliers/{supplierCode}`, `PATCH /api/v1/suppliers/{supplierCode}`, `POST /api/v1/suppliers/{supplierCode}/deactivate`
- Required permissions: `PERM-024`, `PERM-025`, `PERM-026`, `PERM-035`
- Application routes: `/procurement/suppliers`, `/procurement/suppliers/new`, `/procurement/suppliers/:supplierCode`
- State management responsibility: feature list/detail state; edit form local

UI-012 Purchase Order Management

- Angular feature module: `procurement`
- Main page component: `PurchaseOrderManagementPageComponent`
- Child components: `PurchaseOrderListComponent`, `PurchaseOrderEditorComponent`, `PurchaseOrderDetailComponent`, `ExpectedDeliveriesPanelComponent`
- Shared components: `LineItemEditor`, `DataTable`, `RecordMetadataBlock`, `ConfirmationDialog`, `StatusTimeline`
- Required Angular services: `PurchaseOrderFacade`, `PurchaseOrderApiService`, `OpenPurchaseOrdersApiService`
- Required DTOs: PO create/get/update/cancel DTOs, PO summary/detail responses, open-PO board responses
- Related API endpoints: `POST /api/v1/procurement/purchase-orders`, `GET /api/v1/procurement/purchase-orders`, `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}`, `PATCH /api/v1/procurement/purchase-orders/{purchaseOrderId}`, `POST /api/v1/procurement/purchase-orders/{purchaseOrderId}/cancel`, `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}/validate-for-receipt`, `GET /api/v1/intelligence/procurement/open-pos`
- Required permissions: `PERM-024`, `PERM-025`, `PERM-026`, `PERM-027`, `PERM-035`
- Application routes: `/procurement/purchase-orders`, `/procurement/purchase-orders/new`, `/procurement/purchase-orders/:purchaseOrderId`

- State management responsibility: feature list/detail state, draft PO state for multi-line editing

UI-013 Purchase Requisition Management

- Angular feature module: `procurement`
- Main page component: `PurchaseRequisitionManagementPageComponent`
- Child components: `RequisitionQueueComponent`, `RequisitionDetailComponent`, `RequisitionApprovalPanelComponent`, `ConvertToPoPanelComponent`
- Shared components: `QueueList`, `ApprovalDecisionPanel`, `ReadOnlyReviewForm`, `StatusTimeline`
- Required Angular services: `PurchaseRequisitionFacade`, `PurchaseRequisitionApiService`, `PurchaseOrderApiService`
- Required DTOs: requisition create/get/approve/reject/convert DTOs and requisition detail response
- Related API endpoints: `POST /api/v1/procurement/requisitions`, `GET /api/v1/procurement/requisitions/{requisitionId}`, `POST /api/v1/procurement/requisitions/{requisitionId}/approve`, `POST /api/v1/procurement/requisitions/{requisitionId}/reject`, `POST /api/v1/procurement/requisitions/{requisitionId}/convert-to-po`
- Required permissions: `PERM-067`, `PERM-068`, `PERM-024`
- Application routes: `/procurement/requisitions`, `/procurement/requisitions/:requisitionId`
- State management responsibility: feature queue and approval state; note missing list endpoint requires a temporary query-model workaround or deferred browse capability

UI-014 Inbound Delivery Planning

- Angular feature module: `procurement`
- Main page component: `InboundDeliveryPlanningPageComponent`
- Child components: `InboundCalendarComponent`, `DeliveryScheduleEditorComponent`, `InboundDelayPanelComponent`
- Shared components: `DateField`, `FilterBar`, `AlertDetailCard`, `StatusBadge`
- Required Angular services: `InboundDeliveryPlanningFacade`, `PurchaseOrderApiService`, `InboundCalendarApiService`, `InboundDelayApiService`
- Required DTOs: expected-delivery request DTOs, inbound calendar response, inbound delay response
- Related API endpoints: `POST /api/v1/procurement/purchase-orders/{purchaseOrderId}/expected-deliveries`, `GET /api/v1/intelligence/procurement/inbound-calendar`, `GET /api/v1/intelligence/procurement/inbound-delays`
- Required permissions: `PERM-027`, `PERM-036`
- Application routes: `/procurement/inbound-delivery-planning`
- State management responsibility: feature calendar/filter state

UI-015 Goods Receipt Workbench

- Angular feature module: `inbound`
- Main page component: `GoodsReceiptWorkbenchPageComponent`
- Child components: `ReceiptQueueComponent`, `ReceiptHeaderComponent`, `ReceiptLineVerificationComponent`, `ReceiptLotCaptureComponent`, `ReceiptVerificationActionsComponent`
- Shared components: `QueueList`, `LineItemEditor`, `StepProgressIndicator`, `SummaryStatisticStrip`, `SessionContextBadge`
- Required Angular services: `GoodsReceiptFacade`, `GoodsReceiptApiService`, `PurchaseOrderValidationApiService`
- Required DTOs: receipt create/get/list/verify DTOs, goods receipt detail/list responses, lot identity DTOs
- Related API endpoints: `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}/validate-for-receipt`, `POST /api/v1/inbound/goods-receipts`, `GET /api/v1/inbound/goods-receipts`, `GET /api/v1/inbound/goods-receipts/{goodsReceiptId}`, `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/verify`
- Required permissions: `PERM-028`
- Application routes: `/inbound/goods-receipts`, `/inbound/goods-receipts/:goodsReceiptId`
- State management responsibility: workflow store for current receipt, verified lines, captured lots, and verification progress

UI-016 Receipt Discrepancy Resolution

- Angular feature module: `inbound`
- Main page component: `ReceiptDiscrepancyResolutionPageComponent`
- Child components: `DiscrepancyListComponent`, `DiscrepancyEditorComponent`, `DiscrepancyResolutionPanelComponent`, `ReceiptVsExpectedComparisonComponent`
- Shared components: `ComparisonPanel`, `VarianceHighlightBlock`, `EvidenceAttachmentPanel`, `ApprovalDecisionPanel`, `AlertDetailCard`
- Required Angular services: `ReceiptDiscrepancyFacade`, `ReceiptDiscrepancyApiService`, `ReasonCodeLookupService`, `ReceiptToleranceApiService`

- Required DTOs: discrepancy create/list/resolve DTOs, reason-code validation DTOs, receipt tolerance responses
- Related API endpoints: `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies`, `GET /api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies`, `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/discrepancies/{discrepancyId}/resolve`, `GET /api/v1/governance/reason-codes/{code}/validate`, `GET /api/v1/governance/receipt-tolerances`
- Required permissions: `PERM-029`, `PERM-030`, `PERM-014`, `PERM-032`
- Application routes: `/inbound/discrepancies`, `/inbound/goods-receipts/:goodsReceiptId/discrepancies`
- State management responsibility: feature case state with selected discrepancy and attached evidence metadata

UI-017 Put-Away Workbench

- Angular feature module: `inbound`
- Main page component: `PutAwayWorkbenchPageComponent`
- Child components: `PendingPutAwayQueueComponent`, `StoragePositionSelectorComponent`, `PutAwayConfirmationPanelComponent`
- Shared components: `QueueList`, `DropdownSelectControl`, `ConfirmationDialog`, `SessionContextBadge`
- Required Angular services: `PutAwayFacade`, `PutAwayApiService`, `StoragePositionApiService`
- Required DTOs: pending-put-away list responses, put-away command DTOs, storage-position list responses
- Related API endpoints: `GET /api/v1/inbound/pending-put-away`, `POST /api/v1/inbound/goods-receipts/{goodsReceiptId}/put-away`, `POST /api/v1/locations/{locationCode}/storage-positions`, `GET /api/v1/locations/{locationCode}/storage-positions`
- Required permissions: `PERM-033`, `PERM-004`
- Application routes: `/inbound/put-away`, `/inbound/put-away/:goodsReceiptId`
- State management responsibility: feature work queue and current put-away task state

UI-018 Non-PO and Return Receipt Intake

- Angular feature module: `inbound`
- Main page component: `NonPoReturnReceiptIntakePageComponent`
- Child components: `ReceiptTypeSelectorComponent`, `AdHocReceiptFormComponent`, `ReturnReceiptFormComponent`
- Shared components: `RadioGroupSegmentedChoice`, `StandardFormSection`, `ValidationMessageBlock`
- Required Angular services: `AdHocReceiptFacade`, `AdHocReceiptApiService`, `ReturnReceiptApiService`
- Required DTOs: ad-hoc receipt request DTOs, return-receipt request DTOs, command result responses
- Related API endpoints: `POST /api/v1/inbound/ad-hoc-receipts`, `POST /api/v1/inbound/return-receipts`
- Required permissions: `PERM-031`, `PERM-034`
- Application routes: `/inbound/non-po-receipts`
- State management responsibility: local form state only; no durable feature store needed

UI-019 Inbound Supply Tracking

- Angular feature module: `procurement`
- Main page component: `InboundSupplyTrackingPageComponent`
- Child components: `OpenPurchaseOrdersBoardComponent`, `InboundCalendarPanelComponent`, `InboundDelaysPanelComponent`, `InTransitStockPanelComponent`
- Shared components: `FilterBar`, `DateRangePicker`, `DataTable`, `StatusBadge`
- Required Angular services: `InboundSupplyTrackingFacade`, `PurchaseOrderApiService`, `InboundCalendarApiService`, `InboundDelayApiService`, `TransferInTransitApiService`
- Required DTOs: PO list/detail responses, expected-delivery DTOs, in-transit transfer responses
- Related API endpoints: `GET /api/v1/procurement/purchase-orders`, `GET /api/v1/procurement/purchase-orders/{purchaseOrderId}`, `POST /api/v1/procurement/purchase-orders/{purchaseOrderId}/expected-deliveries`, `GET /api/v1/transfers/in-transit`, `GET /api/v1/intelligence/procurement/open-pos`, `GET /api/v1/intelligence/procurement/inbound-calendar`, `GET /api/v1/intelligence/procurement/inbound-delays`
- Required permissions: `PERM-027`, `PERM-035`, `PERM-036`, `PERM-075`
- Application routes: `/procurement/inbound-supply-tracking`
- State management responsibility: feature dashboard/query state

UI-020 Inventory Status Monitor

- Angular feature module: `inventory`
- Main page component: `InventoryStatusMonitorPageComponent`

- Child components: `StockStatusSummaryComponent`, `StockStatusBoardComponent`, `InventoryFiltersComponent`, `StockPositionDetailDrawerComponent`
- Shared components: `FilterBar`, `SavedViewSelector`, `DataTable`, `KpiSummaryCard`, `DrillDownLinkGroup`
- Required Angular services: `InventoryStatusFacade`, `StockPositionApiService`, `StockStatusApiService`, `LocationLookupService`
- Required DTOs: stock-position list/detail responses, stock-status indicator responses, summary/board responses
- Related API endpoints: `GET /api/v1/locations`, `GET /api/v1/inventory/stock-positions`, `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}`, `GET /api/v1/inventory/stock-status/{locationCode}/{sku}`, `GET /api/v1/intelligence/stock-status/summary`, `GET /api/v1/intelligence/stock-status/board`
- Required permissions: `PERM-017`, `PERM-020`, `PERM-083`, `PERM-085`, `PERM-004`
- Application routes: `/inventory/status-monitor`
- State management responsibility: feature filter, saved-view, and selected stock-position state

UI-021 Inventory Availability Inquiry

- Angular feature module: `inventory`
- Main page component: `InventoryAvailabilityInquiryPageComponent`
- Child components: `AvailabilitySearchFormComponent`, `AvailabilityResultComponent`, `NetworkAvailabilityPanelComponent`, `InquiryHistoryPanelComponent`
- Shared components: `GlobalSearchBar`, `SearchResultsList`, `DetailPanel`, `StatusBadge`
- Required Angular services: `AvailabilityFacade`, `ProductApiService`, `AvailabilityApiService`, `AvailabilityInquiryLogApiService`
- Required DTOs: `SearchProductsQuery`, `GetProductBySkuRequest`, availability query/request DTOs, availability response DTOs
- Related API endpoints: `GET /api/v1/products`, `GET /api/v1/products/{sku}`, `GET /api/v1/inventory/availability/{sku}`, `GET /api/v1/inventory/availability/{sku}/network`, `POST /api/v1/fulfillment/orders/{orderId}/availability`, `GET /api/v1/intelligence/availability/inquiries`
- Required permissions: `PERM-004`, `PERM-018`, `PERM-022`, `PERM-051`
- Application routes: `/inventory/availability`
- State management responsibility: feature query/result state; search form local

UI-022 Inventory Movement History

- Angular feature module: `inventory`
- Main page component: `InventoryMovementHistoryPageComponent`
- Child components: `MovementFiltersComponent`, `MovementResultsTableComponent`, `MovementDetailDrawerComponent`
- Shared components: `FilterBar`, `DateRangePicker`, `DataTable`, `ExportActionGroup`, `AuditRecordViewer`
- Required Angular services: `MovementHistoryFacade`, `MovementHistoryApiService`
- Required DTOs: movement-history query DTOs and movement record response DTOs
- Related API endpoints: `GET /api/v1/intelligence/movements`
- Required permissions: `PERM-023`, `PERM-084`
- Application routes: `/inventory/movement-history`
- State management responsibility: feature query and export state

UI-023 Cycle Count Workbench

- Angular feature module: `inventory`
- Main page component: `CycleCountWorkbenchPageComponent`
- Child components: `CycleCountQueueComponent`, `CycleCountTaskComponent`, `CycleCountSubmissionComponent`
- Shared components: `QueueList`, `NumberInput`, `SessionContextBadge`, `SuccessWarningErrorBanner`
- Required Angular services: `CycleCountFacade`, `CycleCountApiService`, `CountPerformanceApiService`
- Required DTOs: cycle-count create/list/get/result DTOs, count performance responses
- Related API endpoints: `POST /api/v1/accuracy/cycle-counts`, `GET /api/v1/accuracy/cycle-counts`, `GET /api/v1/accuracy/cycle-counts/{cycleCountId}`, `POST /api/v1/accuracy/cycle-counts/{cycleCountId}/result`, `GET /api/v1/intelligence/accuracy/count-performance`
- Required permissions: `PERM-037`, `PERM-038`, `PERM-086`
- Application routes: `/inventory/cycle-counts`, `/inventory/cycle-counts/:cycleCountId`
- State management responsibility: feature queue and active count task state

UI-024 Physical Inventory Count Management

- Angular feature module: `inventory`

- Main page component: `PhysicalInventoryCountManagementPageComponent`
- Child components: `PhysicalCountSessionComponent`, `FreezeStatusBannerComponent`, `CountProgressBoardComponent`, `CertificationPanelComponent`
- Shared components: `SummaryStatisticStrip`, `KpiSummaryCard`, `CheckboxGroup`, `ConfirmationDialog`, `StatusTimeline`
- Required Angular services: `PhysicalCountFacade`, `PhysicalCountApiService`, `LocationFreezeApiService`, `CountPerformanceApiService`
- Required DTOs: physical-count create/get/results/certify DTOs, freeze-status response, count-performance response
- Related API endpoints: `GET /api/v1/locations/{locationCode}/freeze-status`, `POST /api/v1/accuracy/physical-counts`, `GET /api/v1/accuracy/physical-counts/{physicalCountId}`, `POST /api/v1/accuracy/physical-counts/{physicalCountId}/results`, `POST /api/v1/accuracy/physical-counts/{physicalCountId}/certify`, `GET /api/v1/intelligence/accuracy/count-performance`
- Required permissions: `PERM-039`, `PERM-040`, `PERM-086`
- Application routes: `/inventory/physical-counts`, `/inventory/physical-counts/:physicalCountId`
- State management responsibility: workflow store for count session, freeze status, progress, and certification checklist

UI-025 Variance Investigation and Approval

- Angular feature module: `inventory`
- Main page component: `VarianceInvestigationApprovalPageComponent`
- Child components: `VarianceCaseQueueComponent`, `VarianceInvestigationFormComponent`, `VarianceComparisonPanelComponent`, `VarianceApprovalPanelComponent`
- Shared components: `QueueList`, `ComparisonPanel`, `ApprovalDecisionPanel`, `EvidenceAttachmentPanel`, `AuditRecordViewer`
- Required Angular services: `VarianceCaseFacade`, `AdjustmentCaseApiService`, `PendingApprovalsApiService`
- Required DTOs: adjustment-case create/get/investigate/variance-approve DTOs, pending-approvals responses
- Related API endpoints: `POST /api/v1/accuracy/adjustment-cases`, `GET /api/v1/accuracy/adjustment-cases/{caseId}`, `POST /api/v1/accuracy/adjustment-cases/{caseId}/investigate`, `POST /api/v1/accuracy/adjustment-cases/{caseId}/variances/{varianceId}/approve`, `GET /api/v1/accuracy/pending-approvals`
- Required permissions: `PERM-043`, `PERM-041`, `PERM-042`, `PERM-044`, `PERM-046`
- Application routes: `/inventory/variances`, `/inventory/variances/:caseId`
- State management responsibility: feature case-state and approval queue state

UI-026 Stock Adjustment Management

- Angular feature module: `inventory`
- Main page component: `StockAdjustmentManagementPageComponent`
- Child components: `AdjustmentRequestFormComponent`, `AdjustmentCaseDetailComponent`, `AdjustmentSubmissionPanelComponent`
- Shared components: `StandardFormSection`, `LineItemEditor`, `TextAreaNotesField`, `ConfirmationDialog`, `RecordMetadataBlock`
- Required Angular services: `StockAdjustmentFacade`, `AdjustmentCaseApiService`, `ReasonCodeLookupService`, `ApprovalThresholdApiService`
- Required DTOs: adjustment-case create/get/submit/approve/reject DTOs, reason-code responses, threshold responses
- Related API endpoints: `GET /api/v1/governance/reason-codes`, `GET /api/v1/governance/approval-thresholds`, `POST /api/v1/accuracy/adjustment-cases`, `GET /api/v1/accuracy/adjustment-cases/{caseId}`, `POST /api/v1/accuracy/adjustment-cases/{caseId}/submit`, `POST /api/v1/accuracy/adjustment-cases/{caseId}/approve`, `POST /api/v1/accuracy/adjustment-cases/{caseId}/reject`
- Required permissions: `PERM-014`, `PERM-013`, `PERM-043`, `PERM-045`, `PERM-044`
- Application routes: `/inventory/adjustments`, `/inventory/adjustments/:caseId`
- State management responsibility: draft adjustment form local; feature case workflow state for submission and approval path

UI-027 Write-Off Authorization

- Angular feature module: `inventory`
- Main page component: `WriteOffAuthorizationPageComponent`
- Child components: `WriteOffCaseQueueComponent`, `WriteOffReviewComponent`, `FinanceAcknowledgementPanelComponent`, `WriteOffDecisionPanelComponent`
- Shared components: `ReadOnlyReviewForm`, `ApprovalDecisionPanel`, `EvidenceAttachmentPanel`, `ConfirmationDialog`
- Required Angular services: `WriteOffFacade`, `AdjustmentCaseApiService`
- Required DTOs: adjustment-case detail, submit, approve, reject DTOs and command results
- Related API endpoints: `GET /api/v1/accuracy/adjustment-cases/{caseId}`, `POST /api/v1/accuracy/adjustment-cases/{caseId}/submit`, `POST /api/v1/accuracy/adjustment-cases/{caseId}/approve`, `POST /api/v1/accuracy/adjustment-cases/{caseId}/reject`
- Required permissions: `PERM-045`, `PERM-046`, `PERM-044`
- Application routes: `/inventory/write-offs`, `/inventory/write-offs/:caseId`

- State management responsibility: feature approval queue and case decision state

UI-028 Quarantine and Hold Register

- Angular feature module: `inventory`
- Main page component: `QuarantineHoldRegisterPageComponent`
- Child components: `QuarantineRegisterTableComponent`, `HoldActionPanelComponent`, `ReleaseHoldDialogComponent`
- Shared components: `DataTable`, `RadioGroupSegmentedChoice`, `StatusBadge`, `ConfirmationDialog`
- Required Angular services: `QuarantineHoldFacade`, `QuarantineHoldApiService`, `QuarantineRegisterApiService`
- Required DTOs: quarantine-hold apply/release DTOs, quarantine register response DTOs
- Related API endpoints: `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/quarantine-holds`, `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/quarantine-holds/{holdId}/release`, `GET /api/v1/intelligence/quarantine-register`
- Required permissions: `PERM-047`, `PERM-048`
- Application routes: `/inventory/quarantine`
- State management responsibility: feature filter and selected-hold state

UI-029 Shrinkage, Loss, and Obsolescence Analysis

- Angular feature module: `inventory`
- Main page component: `ShrinkageAnalysisPageComponent`
- Child components: `ShrinkageKpiPanelComponent`, `TrendAnalysisTableComponent`, `ObsolescenceBreakdownComponent`
- Shared components: `KpiSummaryCard`, `DateRangePicker`, `ReportViewer`, `TrendIndicator`, `ExportActionGroup`
- Required Angular services: `ShrinkageAnalysisFacade`, `ShrinkageAnalysisApiService`
- Required DTOs: shrinkage analysis query/response DTOs
- Related API endpoints: `GET /api/v1/intelligence/accuracy/shrinkage`
- Required permissions: `PERM-049`
- Application routes: `/inventory/shrinkage-obsolescence`
- State management responsibility: feature analytical filter state

UI-030 Order Reservation Workbench

- Angular feature module: `fulfillment`
- Main page component: `OrderReservationWorkbenchPageComponent`
- Child components: `OrderSearchQueueComponent`, `OrderReservationDetailComponent`, `AvailabilityCheckPanelComponent`, `ReservationActionsPanelComponent`
- Shared components: `QueueList`, `LineItemEditor`, `SummaryStatisticStrip`, `StatusTimeline`
- Required Angular services: `OrderReservationFacade`, `OrderApiService`, `AvailabilityApiService`, `ReservationApiService`, `CampaignRingFenceApiService`
- Required DTOs: order create/detail responses, order-availability request/response DTOs, reservation create/release DTOs, ring-fence responses
- Related API endpoints: `POST /api/v1/fulfillment/orders`, `POST /api/v1/fulfillment/orders/{orderId}/availability`, `POST /api/v1/fulfillment/orders/{orderId}/confirm`, `POST /api/v1/fulfillment/orders/{orderId}/cancel`, `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/reservations`, `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/reservations/{reservationId}/release`, `GET /api/v1/campaigns/ring-fence/{locationCode}/{sku}`
- Required permissions: `PERM-054`, `PERM-051`, `PERM-052`, `PERM-053`, `PERM-092`
- Application routes: `/fulfillment/reservations`, `/fulfillment/reservations/:orderId`
- State management responsibility: workflow store for availability result, reservation status, and order confirmation saga state

UI-031 Pick, Pack, and Ship Workbench

- Angular feature module: `fulfillment`
- Main page component: `PickPackShipWorkbenchPageComponent`
- Child components: `PickQueueComponent`, `PickVerificationComponent`, `PackingPanelComponent`, `ShipmentConfirmationPanelComponent`
- Shared components: `QueueList`, `StepProgressIndicator`, `ConfirmationDialog`, `SessionContextBadge`
- Required Angular services: `PickPackShipFacade`, `PickListApiService`, `PickVerificationApiService`, `ShipmentApiService`, `LotBreakdownApiService`
- Required DTOs: pick-list create/awaiting DTOs, pick verification DTOs, pack command DTOs, shipment command DTOs, lot-breakdown responses

- Related API endpoints: `POST /api/v1/fulfillment/orders/{orderId}/pick-lists`, `GET /api/v1/fulfillment/orders/{orderId}/pick-lists/awaiting`, `POST /api/v1/fulfillment/orders/{orderId}/picks/{pickId}/verify`, `POST /api/v1/fulfillment/orders/{orderId}/pack`, `POST /api/v1/fulfillment/orders/{orderId}/shipments`, `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/lots`
- Required permissions: `PERM-055`, `PERM-056`, `PERM-017`
- Application routes: `/fulfillment/pick-pack-ship`, `/fulfillment/pick-pack-ship/:orderId`
- State management responsibility: feature workbench state for active pick, pack completion, and shipment confirmation; note location-level pick queue API gap should be reflected as a backend dependency

UI-032 Partial Fulfillment and Backorder Management

- Angular feature module: `fulfillment`
- Main page component: `PartialFulfillmentBackorderPageComponent`
- Child components: `PartialFulfillmentEditorComponent`, `BackorderDecisionPanelComponent`, `BackorderListComponent`
- Shared components: `ComparisonPanel`, `LineItemEditor`, `StatusTimeline`, `ValidationMessageBlock`
- Required Angular services: `BackorderFacade`, `PartialFulfillmentApiService`, `BackorderApiService`
- Required DTOs: partial-fulfillment command DTOs, backorder create/fulfill DTOs, backorder pipeline responses
- Related API endpoints: `POST /api/v1/fulfillment/orders/{orderId}/partial-fulfillment`, `POST /api/v1/fulfillment/orders/{orderId}/backorders`, `POST /api/v1/fulfillment/backorders/{backorderId}/fulfill`, `GET /api/v1/intelligence/fulfillment/backorders`
- Required permissions: `PERM-057`, `PERM-058`
- Application routes: `/fulfillment/partial-backorders`, `/fulfillment/partial-backorders/:orderId`
- State management responsibility: workflow store for fulfillment split decision and backorder lifecycle

UI-033 Fulfillment Exception Management

- Angular feature module: `fulfillment`
- Main page component: `FulfillmentExceptionManagementPageComponent`
- Child components: `ExceptionQueueComponent`, `ExceptionDetailComponent`, `ExceptionResolutionFormComponent`
- Shared components: `QueueList`, `AlertDetailCard`, `TextAreaNotesField`, `SuccessWarningErrorBanner`
- Required Angular services: `FulfillmentExceptionFacade`, `FulfillmentExceptionApiService`, `QuarantineHoldApiService`
- Required DTOs: fulfillment exception create DTOs, exception log responses, quarantine-hold command DTOs
- Related API endpoints: `POST /api/v1/fulfillment/orders/{orderId}/exceptions`, `GET /api/v1/intelligence/fulfillment/exceptions`, `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/quarantine-holds`
- Required permissions: `PERM-059`, `PERM-047`
- Application routes: `/fulfillment/exceptions`, `/fulfillment/exceptions/:orderId`
- State management responsibility: feature exception queue state

UI-034 Order Fulfillment Status

- Angular feature module: `fulfillment`
- Main page component: `OrderFulfillmentStatusPageComponent`
- Child components: `OrderStatusSearchComponent`, `OrderStatusDetailComponent`, `ReservationShipmentTimelineComponent`, `BackorderSummaryTabComponent`
- Shared components: `SearchResultsList`, `RecordMetadataBlock`, `StatusTimeline`, `TabsSegmentedContentSwitcher`
- Required Angular services: `OrderStatusFacade`, `OrderStatusApiService`, `FulfillmentStatusFeedService`
- Required DTOs: order list/detail/status DTOs and fulfillment status feed responses
- Related API endpoints: `GET /api/v1/fulfillment/orders`, `GET /api/v1/fulfillment/orders/{orderId}`, `GET /api/v1/fulfillment/orders/{orderId}/status`, `GET /api/v1/integrations/sales/fulfillment-status`
- Required permissions: `PERM-062`
- Application routes: `/fulfillment/order-status`, `/fulfillment/order-status/:orderId`
- State management responsibility: feature search and selected order status state

UI-035 Multi-Location Split Fulfillment Planner

- Angular feature module: `fulfillment`
- Main page component: `MultiLocationSplitFulfillmentPlannerPageComponent`
- Child components: `NetworkAvailabilityPanelComponent`, `SplitPlanEditorComponent`, `SourceLocationAllocationGridComponent`
- Shared components: `RadioGroupSegmentedChoice`, `ComparisonPanel`, `ReadOnlyReviewForm`
- Required Angular services: `SplitFulfillmentFacade`, `NetworkAvailabilityApiService`, `MultiLocationPlanApiService`

- Required DTOs: network availability responses, multi-location plan request/response DTOs
- Related API endpoints: `GET /api/v1/inventory/availability/{sku}/network`, `POST /api/v1/fulfillment/orders/{orderId}/multi-location-plan`
- Required permissions: `PERM-022`, `PERM-061`
- Application routes: `/fulfillment/split-planner`, `/fulfillment/split-planner/:orderId`
- State management responsibility: workflow store for multi-source allocation plan

UI-036 Low-Stock Alert Review

- Angular feature module: `procurement`
- Main page component: `LowStockAlertReviewPageComponent`
- Child components: `AlertQueueComponent`, `AlertDetailPanelComponent`, `RecommendationPanelComponent`
- Shared components: `QueueList`, `AlertDetailCard`, `SavedViewSelector`, `BulkSelectionToolbar`
- Required Angular services: `LowStockAlertFacade`, `AlertApiService`, `StockStatusApiService`, `StockPolicyApiService`, `ReplenishmentRecommendationApiService`
- Required DTOs: alert responses, stock-status responses, stock-policy responses, replenishment recommendation responses
- Related API endpoints: `GET /api/v1/inventory/stock-status/{locationCode}/{sku}`, `GET /api/v1/inventory/stock-policies/{locationCode}/{sku}`, `GET /api/v1/intelligence/alerts`, `POST /api/v1/intelligence/alerts/{alertId}/acknowledge`, `GET /api/v1/intelligence/replenishment/recommendations`
- Required permissions: `PERM-020`, `PERM-011`, `PERM-064`, `PERM-065`, `PERM-066`
- Application routes: `/procurement/low-stock-alerts`
- State management responsibility: feature alert queue and acknowledgement state

UI-037 Replenishment Planning

- Angular feature module: `procurement`
- Main page component: `ReplenishmentPlanningPageComponent`
- Child components: `RecommendationListComponent`, `TransferBeforeBuyPanelComponent`, `RequisitionDraftPanelComponent`
- Shared components: `RecommendationCard`, `AlertDetailCard`, `ReadOnlyReviewForm`, `DateField`
- Required Angular services: `ReplenishmentPlanningFacade`, `ReplenishmentRecommendationApiService`, `NetworkImbalanceApiService`, `PurchaseRequisitionApiService`
- Required DTOs: replenishment recommendation responses, requisition create/detail DTOs, network imbalance responses
- Related API endpoints: `POST /api/v1/procurement/requisitions`, `GET /api/v1/procurement/requisitions/{requisitionId}`, `POST /api/v1/procurement/requisitions/{requisitionId}/approve`, `POST /api/v1/procurement/requisitions/{requisitionId}/reject`, `POST /api/v1/procurement/requisitions/{requisitionId}/convert-to-po`, `GET /api/v1/intelligence/replenishment/recommendations`, `GET /api/v1/intelligence/network/imbalance`
- Required permissions: `PERM-067`, `PERM-068`, `PERM-066`, `PERM-076`
- Application routes: `/procurement/replenishment-planning`
- State management responsibility: feature planning state; note list-browsing remains constrained by missing requisition list endpoint

UI-038 Transfer Request

- Angular feature module: `transfers`
- Main page component: `TransferRequestPageComponent`
- Child components: `TransferRequestFormComponent`, `TransferCapacityPanelComponent`, `TransferRequestDetailComponent`
- Shared components: `LineItemEditor`, `NumberInput`, `ReadOnlyReviewForm`, `StatusTimeline`
- Required Angular services: `TransferRequestFacade`, `TransferApiService`, `TransferCapacityApiService`
- Required DTOs: transfer create/list/detail DTOs, transfer-capacity responses
- Related API endpoints: `POST /api/v1/transfers`, `GET /api/v1/transfers`, `GET /api/v1/transfers/{transferId}`, `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/transfer-capacity`
- Required permissions: `PERM-071`, `PERM-075`, `PERM-076`
- Application routes: `/transfers/requests`, `/transfers/requests/new`, `/transfers/requests/:transferId`
- State management responsibility: feature draft transfer and detail state

UI-039 Transfer Approval Queue

- Angular feature module: `transfers`
- Main page component: `TransferApprovalQueuePageComponent`

- Child components: `PendingTransferQueueComponent`, `TransferApprovalDetailComponent`, `TransferDecisionPanelComponent`
- Shared components: `QueueList`, `ApprovalDecisionPanel`, `ComparisonPanel`, `SavedViewSelector`
- Required Angular services: `TransferApprovalFacade`, `TransferApprovalApiService`, `PendingTransferApprovalApiService`, `TransferCapacityApiService`
- Required DTOs: approve/reject transfer DTOs, pending approval responses, transfer-capacity responses
- Related API endpoints: `POST /api/v1/transfers/{transferId}/approve`, `POST /api/v1/transfers/{transferId}/reject`, `GET /api/v1/transfers/pending-approvals`, `GET /api/v1/inventory/stock-positions/{locationCode}/{sku}/transfer-capacity`
- Required permissions: `PERM-072`, `PERM-076`
- Application routes: `/transfers/approvals`, `/transfers/approvals/:transferId`
- State management responsibility: feature approval queue and selected transfer decision state

UI-040 Transfer Execution and In-Transit Tracking

- Angular feature module: `transfers`
- Main page component: `TransferExecutionTrackingPageComponent`
- Child components: `TransferExecutionQueueComponent`, `DispatchPanelComponent`, `ReceivePanelComponent`, `InTransitBoardComponent`
- Shared components: `QueueList`, `SummaryStatisticStrip`, `ConfirmationDialog`, `SessionContextBadge`
- Required Angular services: `TransferExecutionFacade`, `TransferDispatchApiService`, `TransferReceiveApiService`, `TransferInTransitApiService`
- Required DTOs: transfer list/detail responses, dispatch/receive/cancel command DTOs, in-transit responses
- Related API endpoints: `GET /api/v1/transfers`, `GET /api/v1/transfers/{transferId}`, `POST /api/v1/transfers/{transferId}/dispatch`, `POST /api/v1/transfers/{transferId}/receive`, `POST /api/v1/transfers/{transferId}/cancel`, `GET /api/v1/transfers/in-transit`
- Required permissions: `PERM-073`, `PERM-074`, `PERM-075`
- Application routes: `/transfers/execution`, `/transfers/execution/:transferId`
- State management responsibility: feature queue and execution-step state

UI-041 Network Rebalancing Analysis

- Angular feature module: `transfers`
- Main page component: `NetworkRebalancingAnalysisPageComponent`
- Child components: `ImbalanceHeatmapComponent`, `RecommendationListComponent`, `RebalanceActionPanelComponent`
- Shared components: `RecommendationCard`, `FilterBar`, `DrillDownLinkGroup`
- Required Angular services: `NetworkRebalancingFacade`, `NetworkAvailabilityApiService`, `ReplenishmentRecommendationApiService`, `NetworkImbalanceApiService`
- Required DTOs: network availability responses, replenishment recommendation responses, network imbalance responses
- Related API endpoints: `GET /api/v1/inventory/availability/{sku}/network`, `GET /api/v1/intelligence/replenishment/recommendations`, `GET /api/v1/intelligence/network/imbalance`
- Required permissions: `PERM-022`, `PERM-066`, `PERM-076`
- Application routes: `/transfers/network-rebalancing`
- State management responsibility: feature analytics state

UI-042 Return Authorization Management

- Angular feature module: `returns`
- Main page component: `ReturnAuthorizationManagementPageComponent`
- Child components: `ReturnAuthorizationSearchComponent`, `ReturnAuthorizationFormComponent`, `ReturnAuthorizationDetailComponent`
- Shared components: `SearchResultsList`, `LineItemEditor`, `ReadOnlyReviewForm`, `StatusTimeline`
- Required Angular services: `ReturnAuthorizationFacade`, `ReturnAuthorizationApiService`
- Required DTOs: return authorization create/list/get/cancel DTOs and return authorization response DTOs
- Related API endpoints: `POST /api/v1/returns/authorizations`, `GET /api/v1/returns/authorizations`, `GET /api/v1/returns/authorizations/{returnAuthorizationId}`, `POST /api/v1/returns/authorizations/{returnAuthorizationId}/cancel`
- Required permissions: `PERM-077`, `PERM-082`
- Application routes: `/returns/authorizations`, `/returns/authorizations/new`, `/returns/authorizations/:returnAuthorizationId`
- State management responsibility: feature list/detail state; authorization form local

UI-043 Return Inspection and Disposition

- Angular feature module: `returns`
- Main page component: `ReturnInspectionDispositionPageComponent`

- Child components: `PendingReturnReceiptsQueueComponent`, `ReturnInspectionFormComponent`, `DispositionDecisionPanelComponent`, `ReturnDispositionSummaryComponent`
- Shared components: `QueueList`, `StepProgressIndicator`, `RadioGroupSegmentedChoice`, `FieldGroupHeader`, `StatusTimeline`
- Required Angular services: `ReturnProcessingFacade`, `ReturnReceiptApiService`, `ReturnInspectionApiService`, `ReturnDispositionApiService`
- Required DTOs: return receive/inspect/disposition DTOs, pending-return list responses, return-receipt command results
- Related API endpoints: `POST /api/v1/returns/{returnId}/receive`, `POST /api/v1/returns/{returnId}/inspect`, `POST /api/v1/returns/{returnId}/disposition`, `GET /api/v1/returns/pending-receipts`, `POST /api/v1/inbound/return-receipts`
- Required permissions: `PERM-078`, `PERM-079`, `PERM-080`, `PERM-081`, `PERM-034`
- Application routes: `/returns/inspection-disposition`, `/returns/inspection-disposition/:returnId`
- State management responsibility: workflow store for receive -> inspect -> disposition progression

UI-044 Operational Reports and Analytics

- Angular feature module: `reports`
- Main page component: `OperationalReportsAnalyticsPageComponent`
- Child components: `ReportsCatalogComponent`, `ReportFiltersComponent`, `ReportViewerPanelComponent`, `DrilldownResultsPanelComponent`
- Shared components: `ReportViewer`, `FilterBar`, `DateRangePicker`, `ExportActionGroup`, `TrendIndicator`
- Required Angular services: `OperationalReportsFacade`, `DashboardApiService`, `StockStatusApiService`, `QuarantineRegisterApiService`, `MovementHistoryApiService`, `ShrinkageAnalysisApiService`, `CountPerformanceApiService`
- Required DTOs: dashboard responses, stock-status responses, quarantine register responses, movement history responses, shrinkage responses, count-performance responses
- Related API endpoints: `GET /api/v1/intelligence/dashboard`, `GET /api/v1/intelligence/stock-status/summary`, `GET /api/v1/intelligence/stock-status/board`, `GET /api/v1/intelligence/quarantine-register`, `GET /api/v1/intelligence/movements`, `GET /api/v1/intelligence/accuracy/shrinkage`, `GET /api/v1/intelligence/accuracy/count-performance`
- Required permissions: `PERM-083`, `PERM-085`, `PERM-047`, `PERM-023`, `PERM-084`, `PERM-049`, `PERM-086`
- Application routes: `/reports/operational`
- State management responsibility: feature report catalog, active report definition, filters, and export state

UI-045 Inventory Valuation Reconciliation

- Angular feature module: `reports`
- Main page component: `InventoryValuationReconciliationPageComponent`
- Child components: `ValuationSnapshotPanelComponent`, `ReconciliationComparisonPanelComponent`, `ThreeWayMatchStatusPanelComponent`, `InvoiceAuthorizationPanelComponent`
- Shared components: `ComparisonPanel`, `VarianceHighlightBlock`, `DateRangePicker`, `ReportViewer`, `ExportActionGroup`
- Required Angular services: `ValuationReconciliationFacade`, `SupplierInvoiceApiService`, `ValuationSnapshotApiService`, `ValuationReconciliationApiService`, `ConsignmentOwnershipApiService`
- Required DTOs: supplier invoice register/get/match/pay DTOs, valuation snapshot responses, reconciliation responses, consignment-ownership command DTOs
- Related API endpoints: `POST /api/v1/inbound/supplier-invoices`, `GET /api/v1/inbound/supplier-invoices/{invoiceId}`, `POST /api/v1/inbound/supplier-invoices/{invoiceId}/three-way-match`, `POST /api/v1/inbound/supplier-invoices/{invoiceId}/authorize-payment`, `POST /api/v1/inventory/stock-positions/{locationCode}/{sku}/consignment-ownership`, `GET /api/v1/intelligence/valuation/snapshots`, `GET /api/v1/intelligence/valuation/reconciliation`
- Required permissions: `PERM-094`, `PERM-017`, `PERM-090`
- Application routes: `/reports/valuation-reconciliation`
- State management responsibility: feature reconciliation workspace state

UI-046 Promotional Stock Readiness

- Angular feature module: `reports`
- Main page component: `PromotionalStockReadinessPageComponent`
- Child components: `CampaignAllocationListComponent`, `CampaignAllocationEditorComponent`, `ReadinessAssessmentPanelComponent`, `RingFenceStatusPanelComponent`
- Shared components: `KpiSummaryCard`, `TrendIndicator`, `AlertDetailCard`, `SummaryStatisticStrip`
- Required Angular services: `PromotionalReadinessFacade`, `CampaignAllocationApiService`, `CampaignReadinessApiService`, `CampaignPlanIntegrationService`

- Required DTOs: campaign allocation create/list/get/update/readiness/release/override DTOs, ring-fence responses, readiness responses
- Related API endpoints: `POST /api/v1/campaigns/allocations`, `GET /api/v1/campaigns/allocations`, `GET /api/v1/campaigns/allocations/{allocationId}`, `PATCH /api/v1/campaigns/allocations/{allocationId}`, `POST /api/v1/campaigns/allocations/{allocationId}/readiness`, `POST /api/v1/campaigns/allocations/{allocationId}/release`, `POST /api/v1/campaigns/allocations/{allocationId}/override`, `GET /api/v1/campaigns/ring-fence/{locationCode}/{sku}`, `GET /api/v1/intelligence/campaigns/readiness`, `POST /api/v1/integrations/marketing/campaign-plans`
- Required permissions: `PERM-092`, `PERM-093`
- Application routes: `/reports/promotional-readiness`
- State management responsibility: feature campaign-allocation state and readiness workflow state

UI-047 Audit and Control Review Workspace

- Angular feature module: `reports`
- Main page component: `AuditControlReviewWorkspacePageComponent`
- Child components: `AuditScopeFormComponent`, `AuditPopulationResultsComponent`, `ControlTestChecklistComponent`, `AuditFindingsEditorComponent`
- Shared components: `StepProgressIndicator`, `ReportViewer`, `EvidenceAttachmentPanel`, `AuditRecordViewer`
- Required Angular services: `AuditWorkspaceFacade`, `AuditTrailApiService`, `AuditEvidenceExportService`
- Required DTOs: audit-trail query responses and audit export responses
- Related API endpoints: `GET /api/v1/intelligence/audit-trail`, `GET /api/v1/intelligence/audit-trail/export`
- Required permissions: `PERM-098`, `PERM-099`
- Application routes: `/reports/audit-workspace`
- State management responsibility: workflow store for audit scope, selected evidence set, and findings draft

UI-048 Audit Trail Query

- Angular feature module: `reports`
- Main page component: `AuditTrailQueryPageComponent`
- Child components: `AuditQueryFiltersComponent`, `AuditResultsTableComponent`, `AuditRecordDetailDrawerComponent`
- Shared components: `DateRangePicker`, `DataTable`, `AuditRecordViewer`, `ExportActionGroup`
- Required Angular services: `AuditTrailQueryFacade`, `AuditTrailApiService`
- Required DTOs: audit-trail query DTOs, audit record responses, export responses
- Related API endpoints: `GET /api/v1/intelligence/audit-trail`, `GET /api/v1/intelligence/audit-trail/export`
- Required permissions: `PERM-098`, `PERM-099`
- Application routes: `/reports/audit-trail`
- State management responsibility: feature query, selected audit record, and export state

UI-049 User Account Management

- Angular feature module: `administration`
- Main page component: `UserAccountManagementPageComponent`
- Child components: `UserAccountListComponent`, `UserAccountDetailComponent`, `UserProvisioningFormComponent`, `RoleAssignmentPanelComponent`
- Shared components: `GlobalSearchBar`, `DataTable`, `RoleAssignmentWidget`, `AccessScopeSelector`, `ConfirmationDialog`
- Required Angular services: `UserAccountFacade`, `UserAccountApiService`, `RoleAssignmentApiService`, `UserDirectorySyncService`
- Required DTOs: user create/list/get/suspend/deactivate/sync DTOs, `RoleAssignmentDto`, user account responses
- Related API endpoints: `POST /api/v1/users`, `GET /api/v1/users`, `GET /api/v1/users/{userId}`, `POST /api/v1/users/{userId}/roles`, `DELETE /api/v1/users/{userId}/roles/{roleAssignmentId}`, `POST /api/v1/users/{userId}/suspend`, `POST /api/v1/users/{userId}/deactivate`, `POST /api/v1/users/sync`
- Required permissions: `PERM-095`, `PERM-096`
- Application routes: `/admin/users`, `/admin/users/new`, `/admin/users/:userId`
- State management responsibility: feature list/detail state; provisioning form local

UI-050 Role and Access Control Administration

- Angular feature module: `administration`
- Main page component: `RoleAccessControlAdministrationPageComponent`

- Child components: `PermissionMatrixViewComponent`, `RoleScopeRulesPanelComponent`, `AuthorizationEvaluationPanelComponent`, `LocationScopeValidationPanelComponent`
- Shared components: `PermissionMatrix`, `AccessScopeSelector`, `TabsSegmentedContentSwitcher`, `ReadOnlyReviewForm`
- Required Angular services: `AuthorizationAdminFacade`, `AuthorizationEvaluationApiService`, `LocationScopeValidationApiService`, `SessionService`
- Required DTOs: segregation-of-duties evaluation DTOs, action-evaluate DTOs, location-scope validation DTOs, session response DTO
- Related API endpoints: `POST /api/v1/governance/segregation-of-duties/evaluate`, `POST /api/v1/authorization/actions/evaluate`, `POST /api/v1/authorization/location-scope/validate`, `GET /api/v1/auth/session`
- Required permissions: `PERM-013`, `PERM-062`, authenticated session context
- Application routes: `/admin/roles-access`
- State management responsibility: feature policy-evaluation state, no long-lived workflow state required

Routing Recommendation

- Top-level route groups should mirror business navigation:
  - `/home`
  - `/inventory`
  - `/products`
  - `/locations`
  - `/procurement`
  - `/inbound`
  - `/fulfillment`
  - `/transfers`
  - `/returns`
  - `/reports`
  - `/governance`
  - `/admin`
- All route groups except `/auth` should be lazy loaded.
- Use route guards for:
  - authenticated session
  - permission-based access
  - location-scope validation where a route carries `locationCode`
- Prefer route resolvers only for lightweight reference data; main business data should load through facades to preserve retry, skeleton, and refresh behavior.

Service Layer Recommendation

- `ApiService` per business sub-domain for HTTP communication.
- `Facade` per screen or tightly related screen cluster for orchestration and view-model assembly.
- `LookupService` for reusable lists such as locations, reason codes, suppliers, and users.
- `Mapper` layer to convert backend DTOs into:
  - list row models
  - detail view models
  - command payloads
  - workflow step summaries

Shared DTO Organization Recommendation

Organize frontend DTO contracts under feature folders, not a single giant shared folder:

```
features/
  inventory/
    data-access/
      dto/
      mappers/
      services/
  procurement/
    data-access/
      dto/
      mappers/
      services/
```

Use `shared/models/common` only for:

- pagination wrappers
- command result wrappers
- audit metadata
- quantity and monetary primitives
- session and permission primitives

Clean Architecture Alignment for Angular

- `feature/ui`: page components and presentational components
- `feature/application`: facades, workflow coordinators, permission orchestration
- `feature/domain`: business-oriented frontend models, enums, policy helpers, display rules
- `feature/data-access`: HTTP services, DTOs, mappers, repository-style adapters

This keeps Angular aligned with the approved backend architecture while remaining idiomatic for a SPA.

Frontend Risks and Notes

- Requisition browsing depends on the missing requisition list endpoint identified in API review item `C-03`. Screens `UI-013` and `UI-037` should be planned with an API dependency note.
- Pick/pack/ship workbench completeness depends on the missing location-level pick queue endpoint identified in API review item `C-01`.
- Returns receipt should standardize on `POST /api/v1/returns/{returnId}/receive` as the canonical user-facing path, with `POST /api/v1/inbound/return-receipts` treated as infrastructure or transitional support.
- Permission checks should be enforced both at route entry and action level. A visible screen does not imply all commands on that screen are available.

Recommendation Summary

- Build the Angular app as a capability-aligned modular SPA.
- Use lazy-loaded feature areas that mirror the 13 approved backend modules, consolidated into practical frontend feature groups.
- Keep components focused on rendering, facades focused on orchestration, and API services focused on contracts.
- Use local and feature-scoped state by default, with small global state reserved for session, permissions, shell, and notifications.
- Standardize shared workflow components for queues, approvals, timelines, filters, and auditability to reduce duplication across the 50 screens.